Quantum Sphaera Companion: A Structure-First Mathematical Unfolding

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Technical unfolding of the main Sphaera paper [2].

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v4.7.0 Full-Monolith Release Scope and Provenance Notice

Recovery status. This v4.7.0 full-monolith source preserves the complete working monolith by using the last full-length source layer r54 as the base, appending the post-r54 guarded solve chain through r80, and retaining the r81–r84 frame-typed effective/local closure clarifications. Earlier compact upload handoff artefacts are not release bases because they compiled only a short consolidation artefact rather than the full monolith. The present release-candidate source keeps the corrected full-source recovery build and the compact title-page discipline.
DOI status. The intended reserved v4.7.0 Version DOI is 10.5281/zenodo.20232911. The Companion Concept DOI is 10.5281/zenodo.19210728. The previous/baseline v4.6.0 DOI remains 10.5281/zenodo.20117361 as provenance only.
Claim-scope lock. The release scope is guarded local/restricted only. It allows the local/restricted downstream chain, r60+ conservative C4 relative-seam neutralization, C5 detector-threshold terminality at $\theta = 1/256$ after $n = 20$, local photon-clock selector preservation under $\Delta_\gamma < 1/5$, typed mixed-comparison local control, terminal-data bridge, restricted terminal-data naturality, and terminal blocker freeze. It does not claim exact finite C5 zero closure, exact all-readout terminality, C6/T2 closure, a global photon-clock/RH theorem, or global all-readout closure.

v4.7.0-r44 Internal Working Status

Branch status. This source is an internal source-only working candidate over the v4.6.0 release-final baseline and the v4.7.0-r43 local certificate handoff packet. It performs a source-only handoff integrity scan and builds a local certificate index for the next working layer. It assigns no new Zenodo version DOI, authorizes no public upload, and produces no PDF in this intermediate pass.
Mathematical status. The r44 pass is a handoff-integrity and local-certificate-index pass. It does not promote the local MM/HC monitor certificate to a physical action, a Lagrangian, a derived field equation, a global readout closure, or a T2 theorem. It keeps the r43 guard vector active: $C4$ remains route-stable with $\beta_4 = 1/256$, finite $C5$-closure is not claimed, $C6/T2$ remains blocked, the RH channel remains a photon-clock handoff probe only, and global all-readout closure remains unclaimed.
Reader rule. Everything below should be read as a local MM/HC interface certificate extracted from a guarded atlas handoff. The resulting objects are certified local interface constraints, not derived physical laws.

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v4.6.0 Release-Final Baseline Status

Release-final baseline note. This inherited baseline block records v4.6.0 release final with the version DOI 10.5281/zenodo.20117361. The main title remains *Quantum Sphaera Companion*. In the public v4.6.0 release-final source, the title page kept the Concept DOI and the Version DOI as separate DOI records. In this v4.7.0-r44 internal working source, the same DOI is listed as the baseline DOI and no new DOI is assigned. No all-readout/T2 closure is claimed.

Main title rule. The main title remains *Quantum Sphaera Companion*. The phrase *Bounded Readout Closure and T2 Scope Boundary* is a release-theme subtitle only, not a replacement for the main title.

Baseline DOI-line rule. The v4.7.0-r44 title page carries the stable concept DOI and exactly one baseline version DOI line: *Baseline Version DOI (v4.6.0): 10.5281/zenodo.20117361*. The concept DOI remains the stable Companion-line pointer. The date line records the internal working-candidate status, not a new public DOI assignment.

Release-final scope. This integrated release-final source records the v4.6.0 bounded readout result and the T2 scope boundary under the assigned Version DOI 10.5281/zenodo.20117361. It is the public release-final metadata state for v4.6.0, but it is not an all-readout closure theorem and does not promote any claim outside the declared bounded grammar.

Governing v4.6.0 status.

$$\mathsf{BRead}_{\mathrm{bounded}} \in \mathsf{Cons}(\mathcal{G}^{\mathrm{decl}}_{\mathrm{global}}), \qquad \mathsf{BRead}_{\mathrm{bounded}} \not\Rightarrow T2, \qquad T2 \in \mathsf{Ext}(\mathcal{G}^{\mathrm{decl}}_{\mathrm{global}}).$$

No empirical validation, Standard-Model-spectrum derivation, unrestricted physical readout, or all-readout closure is claimed.

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Human-Facing Reading Layer and Notation Discipline

This edition separates the mathematical object language from revision and audit metadata. Internal revision identifiers and combined version–revision strings are not part of the physics or mathematics. They are provenance tags only. They may occur in file names, manifests, checksums, audit reports, and reconstruction notes, but they are not used as exponents, object names, states, fields, obstructions, or physical regimes in the reader-facing text.

The main reading order is therefore:

1. the structural and physical claim in ordinary mathematical notation;
2. the local hypotheses, guards, and return conditions;
3. the obstruction or witness that remains after the permitted tests;
4. the machine-readable audit trail, kept only as provenance material.

When a previous draft used a revision-bearing symbol, this edition rewrites it as a semantic object such as X^{rtg}, X^{lock}, X^{nonsplit}, or simply X, according to the mathematical role of the object. Version numbers remain valid release metadata, not mathematical structure.

Reader rule. The human-facing document should be readable without knowing the internal patch history. The audit/provenance layer is still preserved for reconstruction and LLM-assisted maintenance, but it is subordinate to the mathematical presentation.

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Reader-Facing Proof Spine and Document Routing

This v4.6.0 release-final frontmatter treats the mathematical narrative as the primary document and the machine/audit trail as secondary provenance. The inherited v4.5.0 release-final layer remains visible as prior release genealogy, but the governing public metadata state is now v4.6.0 with Version DOI 10.5281/zenodo.20117361. The reader should not need the patch history in order to understand the current object language. Revision identifiers, local build labels, and version-transition notes are therefore moved out of the proof spine and into the provenance layer.

Primary reading order

1. **Structural claim.** Read the HC/MM/chart interface as a structural bridge problem with explicit guards and comparison norms.

2. **Local grammar.** The photon-frame HC-halo/Dixon/lens grammar gives the inherited local admissible test surface.

3. **Finite tests.** Candidate menus are used only as certified finite screens. They may consume, route, block, or name an obstruction; they do not create a physical claim by themselves.

4. **Current v4.6.0 endpoint.** The active endpoint is the bounded-readout surface: it is conservative and debt-free inside the declared grammar, but it does not imply T2 and does not discharge all possible readouts.

5. **Return and handoff contract.** A future all-readout answer must return one of three objects: a kill certificate, a strict norm-decrease witness, or a named persistent obstruction. The v4.7 handoff therefore starts with the MM/HC exterior–mixed field-equation program, not with a T2 closure claim.

Current semantic object registry

The active state is represented by semantic names, not revision names. The older route-to-global bridge symbols remain valid v4.5 genealogy; the v4.6.0 governing objects are the bounded-readout surface, the all-readout scope boundary, and the MM/HC exterior–mixed handoff:

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Symbol	**Role**	**Reader meaning**
B_{tie}	inherited chart-tie component	v4.5 non-split coupling extracted from the local pair
B_{lock}	inherited locked remnant	v4.5 remaining tie object after partial cancellation screens
$\mathcal{O}_{\mathcal{B}}^{\mathrm{lock}}$	inherited locked bridge obstruction	v4.5 bridge obstruction; genealogy, not the final v4.6 endpoint
$\mathsf{BRead}_{\mathrm{bounded}}$	bounded readout surface	declared bounded-readout object: conservative inside the declared grammar
Θ^{all}	all-readout demand	unrestricted readout family required for T2; not supplied by the bounded result
$\mathfrak{O}_{\mathrm{all}}^{4.6.0}$	frozen all-readout obstruction	missing universal exterior selector, all-readout naturality family, and unrestricted exterior collar slot
$\mathcal{M}_{\mathrm{bd}}, \mathcal{M}_{\mathrm{ext}}, \mathcal{M}_{\mathrm{mix}}$	Millennium-matrix split	bounded, exterior, and mixed parts for the next field-equation determination
$\mathcal{H}_{\mathrm{bd}}, \mathcal{H}_{\mathrm{ext}}, \mathcal{H}_{\mathrm{mix}}$	HC split	bounded, exterior, and mixed compensator portions matched to the same readout split
$\mathfrak{E}_{\mathrm{decl}}$	field-equation reset template	structure-first equation template for the v4.7 MM/HC exterior–mixed program

Guard discipline

No reader-facing claim is promoted unless the relevant guard is explicitly discharged. In particular, this edition does not claim T2, all-readout closure, global OP closure, a global HS^2 focal-return theorem, a through-collapse trajectory, an unverified atlas expansion, or a Standard-Model-spectrum derivation. The bounded-readout theorem is therefore read as a conservative declared-grammar result, while Θ^{all} remains a scope boundary unless a separate all-readout witness family is supplied.

Human/LLM layering rule

The main document contains the mathematical object language. The appendix may contain provenance: source packets, diffs, checksums, machine-readable registers, build logs, and reconstruction notes. Provenance may mention build labels. The proof spine must not use them as mathematical exponents, object names, or physical states.

Compressed chronology reading rule

For reader navigation, inherited working chronology is to be compressed into semantic milestones rather than read as active mathematical notation. The compression convention is:

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chronology stratum	reader-facing meaning	status
C_0	earlier local readout and release genealogy	provenance
C_1	v4.4 declared-local OP1/OP4 closure ledger	prior release result
C_2	v4.5.0 declared global-boundary synthesis	prior theorematic release layer
C_3	v4.5.0 DOI/title/frontmatter repair	release metadata
C_4	v4.5.x editorial QA and chronology compression	post-release readability layer
C_5	v4.6.0 bounded-readout theorem and T2 scope boundary	current governing release status
C_6	MM/HC exterior–mixed field-equation determination	v4.7 handoff program

Old working strings such as release-candidate notes, prior version DOIs, and local revision labels are therefore genealogy unless a nearby release-final or governing-status block explicitly says otherwise. This rule changes only reader navigation; it does not change labels, references, DOI records, theorem statements, or scope guards.

Proof-spine phrase audit rule. In running prose, build expressions such as source-native rebuild, release candidate, freeze audit, packet, and working layer are reader-facing provenance terms unless they are explicitly part of a displayed ledger. Where such terms occur in the active proof spine, the preferred reader form names the mathematical role first—for example grammar exhaustion, readout invariance, debt-free boundary transport, no-new-debt extension, or functorial comparison stability—and keeps the build phrase only as provenance. This rule does not rename labels, change theorem statements, alter DOI records, or change any scope guard.

Reader-facing phrase-normalization rule

Current-line section headings are allowed to use semantic names even when the source labels and audit ledgers retain their original revision identifiers. Thus a heading such as a witness execution, boundary audit, source-native checkpoint, or title-page repair is to be read by its mathematical function, not by the working-string that produced it. Revision strings remain valid in labels, comments, diffs, ledgers, and reconstruction records, but they are not mathematical object names, proof data, tensor indices, route states, or closure witnesses.

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Companion status. This is the v4.6.0 release-final frontmatter after bounded-readout integration, T2 scope-boundary freeze, and DOI normalization. It preserves the declared mathematical global-boundary closure audit path from the v4.5.0 release line as provenance while making the v4.6.0 Version DOI 10.5281/zenodo.20117361 the current public release DOI. Version labels are retained only as provenance metadata; the mathematical proof spine uses semantic names.

Governing theorematic status. Inside the declared global-boundary synthesis grammar

$$\mathcal{G}^{\mathrm{decl}}_{\mathrm{global}},$$

OP 1, OP 4, and OP 5 are closed as a declared mathematical global-boundary closure. This does not claim unrestricted all-readout closure, global physical closure, QFT/GR closure, QCD/confinement, CKM/PMNS, quark-mass, or Standard-Model-spectrum derivation.

DOI rule. The concept DOI remains the stable title-page pointer. The v4.5.0 version-specific DOI is 10.5281/zenodo.20079410; prior version-specific DOI records are provenance metadata only. Version IDs, DOI fragments, dates, and release IDs are not mathematical object names, indices, theorem labels, route tags, or proof data.

Internal r10 proofread status. This v4.5.1-r10 layer is an internal selected-range proofread and metadata-hygiene pass. It does not authorize public upload, does not insert a new DOI, does not emit Zenodo release text, and does not change the declared mathematical closure scope.

Internal r11 terminology-sweep status. This v4.5.1-r11 layer is an internal targeted terminology sweep over the older OP5 and OP1/OP4 compatibility pages. It does not authorize public upload, does not insert a new DOI, does not emit Zenodo release text, does not rename theorem labels, and does not change the declared mathematical closure scope.

Internal r13 glossary/index intake status. This v4.5.1-r13 layer is an internal glossary and index intake for the declared global-boundary synthesis grammar and the OP5/OP1/OP4 terminology family. It is a reader-routing layer only: no public upload is authorized, no new DOI is inserted, no Zenodo release text is emitted, no theorem label is renamed, and no mathematical closure scope is changed.

Internal r13 glossary-appendix worksheet status. This v4.5.1-r13 layer is an internal extraction worksheet for a possible future glossary appendix. It does not create a public release refresh, does not insert a new DOI, does not emit Zenodo release text, does not rename theorem labels, and does not change the declared mathematical closure scope.

Abstract

This v4.6.0 release-final frontmatter records the Quantum Sphaera Companion after bounded-readout integration, T2 scope-boundary freeze, DOI synchronization, and title-page/frontmatter hygiene. The version-specific DOI for this release line is 10.5281/zenodo.20117361. The Companion concept DOI remains 10.5281/zenodo.19210728. The v4.5.0 version DOI 10.5281/zenodo.20079410 is retained as prior-release provenance.

The governing v4.6.0 release-facing result is the bounded-readout theorem together with the T2 scope-boundary statement. Inside the declared global-boundary grammar $\mathcal{G}^{\mathrm{decl}}_{\mathrm{global}}$, the bounded readout surface is conservative and debt-free, while T2 remains an explicit all-readout extension problem rather than a theorem asserted by this release.

This is a scoped mathematical theorem. It is not a global physical closure theorem, not a global QFT/GR theorem, not unrestricted closure over every possible readout or physical model, and not a T2 closure claim. Earlier version-specific DOI records remain provenance metadata only and are not mathematical object names, theorem labels, route tags, or proof data.

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Frontmatter reader architecture

Purpose. This page separates the current release-facing route from historical theorem memory. It is an orientation layer only; it does not replace any definition, theorem, proof, audit ledger, or historical witness later in the document.

Layer	Human reading route	LLM/audit route
Current reader status	Read this as the v4.6.0 release-final frontmatter after bounded-readout integration, T2 scope-boundary freeze, DOI synchronization, and metadata hygiene.	Treat this pass as reader-facing organization and provenance separation only; no new mathematics is introduced beyond the audited v4.6.0 bounded-readout layer.
OP5 status	Read OP 5 as locally closed only inside the declared photon-frame HC-halo/Dixon grammar.	Track the required zero registers $L_{\mathrm{unr}}, B_{\mathrm{HC/MM}}, R_{\mathrm{HS4}}, D_{\mathrm{HS5}}$ and reject any stronger global reading.
QAM/MML/MMA	Treat QAM as store/object grammar, MML as routing/command grammar, and MMA as execution/readout semantics.	Use finite object bridges, instruction alphabets, deterministic guarded transitions, and route-status tables as audit handles.
Dixon coding	Read the Dixon layer as a 64-dimensional carrier with a 2^{64}-mask activation space.	Keep carrier dimension and Boolean mask count separate; do not read Dixon itself as 2^{64}-dimensional.
Historical witness	Use it as genealogy and reconstruction support.	Preserve it as non-destructive proof memory; do not read inherited baseline language as the current public claim unless explicitly mapped by the local proof chain grammar.

Table 1: Frontmatter separation of current release status, OP5 local closure, and historical theorem memory.

DOI discipline for the human-facing edition

Record	Status	Reader-facing rule
Concept DOI	stable Companion-line pointer	10.5281/zenodo.19210728
Version DOI	current public version-specific release DOI	10.5281/zenodo.20117361.
Prior release metadata	provenance layer only	Historical DOI records remain reconstructible in audit and release files, not in the proof spine.

DOI and versioning rule. DOI records and release identifiers are metadata. They must not

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be used as mathematical indices, route tags, theorem labels, tensor subscripts, obstruction names, or proof data in the reader-facing argument.

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Prior DOI provenance. Earlier version-specific DOI records belong to released predecessors and are provenance metadata only. This v4.6.0 release-final line assigns the current version-specific DOI 10.5281/zenodo.20117361; the v4.5.0 DOI 10.5281/zenodo.20079410 is retained as prior-release provenance.
Working-source status. This release-final adds no theorematic claim beyond the audited Active Core. The current local readout ledger remains guarded: the orbit/Hopf chain is a finite local readout closure ledger with no persistent finite carrier obstruction in the declared diagnostic chart, not a global physical closure theorem.

1 Alternating Triolic Orbit Scan and Hopf-Fibre Core Surface Readout

Reader status. This section is a reader-facing geometric normal form. It is not a claim that ordinary physical particles literally cross a collapse core. It gives a controlled local scan model: test trajectories are used as readout curves around a guarded core, and the remaining question is whether the resulting local surface data can be promoted into the global HC/MM-to-chart bridge without residual debt.

1.1 Triolic scan construction

Let K denote the guarded collapse core. The local lens geometry is represented by two conjugate focal lenses

$$F_-, \qquad F_+,$$

and by their midpoint

$$M := \frac{1}{2}(F_- + F_+).$$

The first triolic balance is the focal-lens triol

$$\mathcal{T}_{\text{lens}} := (F_-, K, F_+),$$

where K is not treated as a point to be crossed, but as the balancing object between the two focal lenses.

A test-orbit scan uses paired test readouts

$$P_n, \qquad Q_n$$

on a bounded local orbit family. At selected readout phases one obtains the particle triol

$$\mathcal{T}_n := (P_n, K, Q_n).$$

The triol is called *core-sighted* when the sightline from P_n to Q_n passes through K. In affine notation this means that there is a scalar λ_n with

$$K = (1 - \lambda_n)P_n + \lambda_n Q_n.$$

If $\lambda_n = \frac{1}{2}$, the triol is centrally balanced; otherwise it is projectively balanced. Thus the model separates two cases which should not be confused:

$$\text{collinearity through } K \qquad \neq \qquad \text{midpoint symmetry at } K.$$

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Definition 1.1 (Bifocal elliptic readout body)**.** *A local orbit curve γ is a bifocal elliptic readout body for the lens pair (F_-, F_+) if, on the admissible scan domain,*

$$d(\gamma(s), F_-) + d(\gamma(s), F_+) = 2a$$

up to the declared tolerance of the local grammar. The curve is used as a stable computational body, not as an unrestricted through-core physical trajectory.

1.2 Shift as a Kronecker-style transport selector

The discrete shift is represented by a shifted selector rather than by the identity Kronecker tensor. In an indexed scan basis one writes

$$\Sigma^i{}_j := \delta^i_{j+1},$$

so that the k-fold shift is

$$(\Sigma^k)^i{}_j = \delta^i_{j+k}.$$

Thus δ^i_j preserves an index, while $\Sigma^i{}_j$ transports the readout to the next admissible scan state.

The parity projectors are

$$\Pi_{\text{even}}(n) = \delta_{n \bmod 2,0}, \qquad \Pi_{\text{odd}}(n) = \delta_{n \bmod 2,1}.$$

An alternating triolic scan has two structure classes $\mathcal{A}$ and $\mathcal{B}$ with

$$\Sigma(\mathcal{A}) = \mathcal{B}, \qquad \Sigma(\mathcal{B}) = \mathcal{A},$$

and therefore

$$\Sigma^2(\mathcal{A}) = \mathcal{A}, \qquad \Sigma^2(\mathcal{B}) = \mathcal{B}.$$

The single shift is the toggle; the double shift is the same-regime evolution step.

Assumption 1.2 (Alternating triolic shift hypothesis)**.** *Along a guarded bifocal scan there are two complementary triolic regimes:*

$$\mathcal{A}_{2n} := \mathcal{T}_{2n}, \qquad \mathcal{B}_{2n+1} := \mathcal{T}_{2n+1}.$$

One regime is core-sighted and the other is lens-sighted, or conversely. The local shift operator Σ exchanges these regimes, while Σ^2 preserves the regime type.

1.3 Orbit scan and guarded core surface

For each scan curve $\gamma_{\theta,\varphi}$ around the guarded core, define its minimal core distance by

$$\rho_{\min}(\theta, \varphi) := \min_s d(\gamma_{\theta,\varphi}(s), K).$$

The effective measured core surface is the periapsis envelope

$$\Sigma_K := \text{env}\{\gamma_{\theta,\varphi}\}$$

or, equivalently, the surface reconstructed from the field of minimal distances $\rho_{\min}(\theta, \varphi)$.

Definition 1.3 (Guarded orbit-scan surface)**.** *A guarded orbit-scan surface is the readout surface obtained from all admissible parity-shifted test orbits whose minimal distance from K is positive:*

$$\rho_{\min}(\theta, \varphi) > 0.$$

The inequality states that the scan measures the effective boundary of the guarded core; it does not license a trajectory through the collapse state.

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Proposition 1.4 (Periapsis envelope as core-surface readout)**.** *Suppose a scan family satisfies the bifocal elliptic readout condition, the alternating triolic shift hypothesis, and the guarded-core inequality $\rho_{\min} > 0$. Then the scan does not measure a through-core path. It measures a bounded periapsis envelope around K. This envelope is the local surface data available to the HC/MM-to-chart bridge.*

Proof. The bifocal condition supplies a stable bounded readout body. The alternating shift separates the scan into two complementary regimes, so that the same geometric orbit is not forced to carry all sightline conditions at once. The condition $\rho_{\min} > 0$ excludes a through-core trajectory. Therefore the admissible data are the nearest admissible approach points of the scan family, i.e. the periapsis envelope. This is exactly the effective local surface readout around the guarded core. □

1.4 Hopf-fibre interpretation and the possible residual point

The Hopf picture gives the natural topological comparison. In the ideal Hopf model, a three-sphere is decomposed into circular fibres. In that idealized fibre space there is no missing ordinary point: every point lies on a fibre. Apparent exceptional points may arise only after projection into a lower-dimensional picture.

The guarded collapse model is different. If the HC guard excludes the core itself, the scan space is not the full ideal fibre space but a punctured or core-excised domain,

$$X = S^3 \setminus \{K\} \qquad \text{or} \qquad X = S^3 \setminus B_\epsilon(K).$$

In that case a residual point or residual core is not a defect of the Hopf fibration. It is the excluded guarded kernel which the orbit scan is designed to measure from outside.

Proposition 1.5 (Hopf completion versus guarded-core remainder)**.** *There are two distinct readings of the ring filling picture.*

(a) *In the ideal Hopf reading, circular fibres fill the ambient three-sphere, and no ordinary minimal leftover point remains.*

(b) *In the guarded-core reading, the core K is removed or shielded. A minimal leftover point or core neighbourhood can remain, but it is not an unfilled Hopf gap; it is the HC-guarded kernel itself.*

Remark 1.6 (Operational meaning)**.** *The useful experimental image is therefore not "send a particle through the collapse". It is: send admissible test readouts around the guarded region, track their even/odd triolic alternation, collect the minimal approach points, and reconstruct the effective core surface from the resulting periapsis envelope. If the linked orbit family behaves Hopf-like, the linking and twisting of the rings provide additional topological data; if a residual core remains, that remainder is the object being measured.*

1.5 Finite scan checklist

For a finite numerical or symbolic probe, the following quantities are the minimal diagnostic payload:

$$\begin{aligned} E(s) &:= d(\gamma(s), F_-) + d(\gamma(s), F_+), \\ \rho_K(s) &:= d(\gamma(s), K), \\ \chi_n &:= \begin{cases} 1, & P_n, K, Q_n \text{ are collinear}, \\ 0, & \text{otherwise}, \end{cases} \\ \pi_n &:= n \bmod 2. \end{aligned}$$

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The expected signature of the alternating scan is

$$E(s) \approx 2a, \qquad \rho_K(s) > 0, \qquad \chi_{n+2} = \chi_n,$$

with complementary behaviour between the even and odd subsequences. A successful scan therefore yields three outputs: the local bifocal fit, the parity/toggle law, and the reconstructed guarded-core surface.

1.6 Strict mathematical orbit-scan formulation

This subsection records the strict mathematical form of the preceding geometric picture. It is a readout model. No physical through-core trajectory is asserted.

Definition 1.7 (Guarded scan space)**.** *Let (X, d) be a compact metric scan chart or a compact chart domain inside a larger metric space. Let K be a distinguished guarded core and let $\varepsilon > 0$ be a guard radius. The admissible scan domain is*

$$X_K := X \setminus B_\varepsilon(K), \qquad B_\varepsilon(K) := \{x \in X : d(x, K) < \varepsilon\}.$$

An admissible test orbit is a continuous curve

$$\gamma : S^1 \longrightarrow X_K.$$

Thus every admissible orbit satisfies

$$d(\gamma(s), K) \geq \varepsilon \qquad \text{for all } s \in S^1.$$

Definition 1.8 (Bifocal defect)**.** *Fix two focal points or focal charts $F_-, F_+ \in X$ and a constant $a > 0$. The bifocal defect of a scan point $x \in X_K$ is*

$$\mathcal{E}_F(x) := \big|d(x, F_-) + d(x, F_+) - 2a\big|.$$

A scan orbit is η-bifocal if

$$\sup_{s \in S^1} \mathcal{E}_F(\gamma(s)) \leq \eta.$$

The exact ellipse is the special case $\eta = 0$ in a chart where the focal distance expression is exact.

Definition 1.9 (Triolic sight predicate)**.** *Let $P, Q \in X_K$. In a geodesic chart, the ordered triple (P, K, Q) is called core-sighted if K lies on an admissible geodesic segment from P to Q. In an affine chart this is equivalent to the existence of $\lambda \in [0, 1]$ such that*

$$K = (1 - \lambda)P + \lambda Q.$$

It is centrally balanced if, in addition, $\lambda = 1/2$. The two predicates are kept separate:

$$\mathrm{Sight}_K(P, Q) \quad \not\Rightarrow \quad K = \frac{P + Q}{2}.$$

Definition 1.10 (Shift selector and parity sectors)**.** *Let I be a finite cyclic index set or $I = \mathbb{Z}$. On the scan basis $(e_j)_{j \in I}$ define the shift selector by*

$$\Sigma e_j = e_{j+1}, \qquad \Sigma^i{}_j := \delta^i_{j+1}.$$

For $k \geq 0$,

$$(\Sigma^k)^i{}_j = \delta^i_{j+k}.$$

Let Π_0 and Π_1 be the even and odd parity projectors,

$$\Pi_0 e_j = \begin{cases} e_j, & j \equiv 0 \pmod 2, \\ 0, & j \equiv 1 \pmod 2, \end{cases} \qquad \Pi_1 e_j = \begin{cases} 0, & j \equiv 0 \pmod 2, \\ e_j, & j \equiv 1 \pmod 2. \end{cases}$$

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Lemma 1.11 (Parity exchange law)**.** *The shift selector exchanges parity sectors, while the double shift preserves them:*

$$\Pi_0\Sigma = \Sigma\Pi_1, \qquad \Pi_1\Sigma = \Sigma\Pi_0,$$

and

$$\Pi_r\Sigma^2 = \Sigma^2\Pi_r \qquad (r = 0, 1).$$

Proof. It is enough to test the identities on basis vectors. Since $\Sigma e_j = e_{j+1}$, the parity of the index is flipped by Σ and preserved by Σ^2. The displayed identities are exactly this parity statement in projector form. □

Definition 1.12 (Alternating triolic scan)**.** *Let $(P_n, Q_n)_{n\in I}$ be paired scan readouts. An alternating triolic scan consists of two disjoint regime predicates $\mathcal{A}$ and $\mathcal{B}$ on pairs (P, Q) such that*

$$\mathcal{A}(P_n, Q_n) \Longleftrightarrow n \equiv 0 \pmod 2, \qquad \mathcal{B}(P_n, Q_n) \Longleftrightarrow n \equiv 1 \pmod 2,$$

up to a declared finite tolerance set. One admissible interpretation is

$$\mathcal{A} = \textit{core-sighted regime}, \qquad \mathcal{B} = \textit{lens-sighted regime},$$

although the formal definition only requires two complementary predicates exchanged by Σ.

Definition 1.13 (Periapsis envelope)**.** *Let U be a compact parameter space of admissible scan orbits*

$$\Gamma : U \times S^1 \longrightarrow X_K, \qquad \gamma_u(s) := \Gamma(u, s).$$

For each $u \in U$ define

$$\rho(u) := \min_{s\in S^1} d(\gamma_u(s), K)$$

and the periapsis set

$$\mathcal{P}(u) := \{\gamma_u(s) : d(\gamma_u(s), K) = \rho(u)\}.$$

The orbit-scan surface is the closed periapsis envelope

$$\Sigma_K^\Gamma := \overline{\bigcup_{u\in U} \mathcal{P}(u)}.$$

Proposition 1.14 (Existence and guardedness of the scan surface)**.** *For a compact scan family $\Gamma : U \times S^1 \to X_K$, the function $\rho(u)$ and the sets $\mathcal{P}(u)$ are well defined. Moreover,*

$$\rho(u) \geq \varepsilon \qquad \textit{for all } u \in U,$$

so the periapsis envelope Σ_K^Γ is an outside readout of the guarded core and not a through-core trajectory.

Proof. For fixed u, the map $s \mapsto d(\gamma_u(s), K)$ is continuous on the compact set S^1, hence it attains a minimum. Because $\gamma_u(S^1) \subset X_K$, every scan point has distance at least ε from K. Therefore each periapsis point remains outside the excluded core. Taking the union and closure gives a surface readout generated by nearest admissible approaches, not by points inside the guarded core. □

Theorem 1.15 (Strict alternating triolic orbit-scan theorem)**.** *Assume the following data:*

(i) *a guarded scan space $X_K = X \setminus B_\varepsilon(K)$ with $\varepsilon > 0$;*

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(ii) *a compact family of admissible orbits* $\Gamma : U \times S^1 \to X_K$*;*

(iii) *paired readouts* (P_n, Q_n) *sampled from the scan family;*

(iv) *an alternating triolic scan structure exchanged by the selector* $\Sigma^i{}_j = \delta^i_{j+1}$*;*

(v) *a bifocal defect bound* $\sup_s \mathcal{E}_F(\gamma_u(s)) \le \eta$ *for all* $u \in U$*.*

Then the scan determines three mathematically controlled readout objects:

(a) the bifocal defect field $\mathcal{E}_F$,
(b) the parity law Π_0, Π_1, Σ,
(c) the guarded periapsis envelope Σ_K^Γ.

It does not determine, by itself, a physical through-core orbit or a global closure of the HC/MM-to-chart bridge.

Proof. The bifocal defect field is defined by the focal data and the metric. The parity law follows from Lemma 1.11 and the alternating scan predicates. The guarded periapsis envelope exists and remains outside the excluded core by Proposition 1.14. None of these constructions supplies a point of any scan curve inside $B_\varepsilon(K)$, and none asserts compatibility with a global bridge unless an additional promotion theorem is supplied. Hence the outputs are exactly local readout objects with a guarded-core interpretation. □

Proposition 1.16 (Hopf comparison without a hidden gap)**.** *Let* $\pi : S^3 \to S^2$ *be the Hopf fibration. In the ideal Hopf space every point of* S^3 *lies on exactly one circular fibre. Therefore a leftover point is not produced by the Hopf fibration itself. If the scan domain is instead* $S^3 \setminus B_\varepsilon(K)$ *or* $S^3 \setminus \{K\}$*, any residual core is the explicitly removed guarded set, not an unfilled Hopf gap.*

Proof. The Hopf fibration partitions S^3 into fibres $\pi^{-1}(y) \cong S^1$ indexed by $y \in S^2$. Hence each point of S^3 belongs to one fibre. Removing K or a neighbourhood of K changes the domain before the scan is performed. The remaining excluded set is therefore part of the guarded-core model, not a failure of the Hopf decomposition. □

Remark 1.17 (Promotion guard)**.** *The strict formulation proves only local scan coherence: bifocal control, parity alternation, and an outside periapsis envelope. A global physical or theorematic promotion would require an additional result showing that the envelope data kill, strictly reduce, or otherwise control the remaining route-to-global obstruction without hidden leakage or residual reopening.*

1.7 Executable finite orbit-scan probe layer

The preceding strict formulation is now turned into a finite probe grammar. The purpose is not to replace the continuous definitions, but to give a computable audit surface: a finite family of sampled orbits, a finite parity test, a finite bifocal defect table, and a finite periapsis-envelope estimator.

Definition 1.18 (Finite scan grid)**.** *Let* $U_N = \{u_1, \dots, u_N\} \subset U$ *be a finite set of admissible orbit parameters and let*

$$S_M = \{s_0, \dots, s_{M-1}\} \subset S^1, \qquad s_m = \frac{2\pi m}{M}.$$

The finite scan sample is the table

$$x_{\ell m} := \Gamma(u_\ell, s_m) \in X_K.$$

It is admissible when

$$d(x_{\ell m}, K) \geq \varepsilon \qquad (1 \leq \ell \leq N,\, 0 \leq m < M).$$

Definition 1.19 (Finite bifocal and guarded-distance registers)**.** *For a finite scan grid define*

$$E_{\ell m} := \mathcal{E}_F(x_{\ell m}), \qquad R_{\ell m} := d(x_{\ell m}, K) - \varepsilon.$$

The finite bifocal defect and finite guard-margin defect are

$$E_{N,M} := \max_{\ell,m} E_{\ell m}, \qquad R^{\min}_{N,M} := \min_{\ell,m} R_{\ell m}.$$

The grid passes the local readout screen at tolerances (η, μ) when

$$E_{N,M} \leq \eta, \qquad R^{\min}_{N,M} \geq \mu \geq 0.$$

Definition 1.20 (Finite parity and triolic indicators)**.** *Let $(P_n, Q_n)_{n=0}^{L-1}$ be the sampled paired readouts. Let*

$$\chi_A(n), \chi_B(n) \in \{0, 1\}$$

be the finite indicators for the two complementary triolic regimes. The finite alternation defect is

$$D_{\mathrm{alt}} := \sum_{n=0}^{L-1} \Big| \chi_A(n) - \mathbf{1}_{n \equiv 0\,(2)} \Big| + \sum_{n=0}^{L-1} \Big| \chi_B(n) - \mathbf{1}_{n \equiv 1\,(2)} \Big|.$$

The finite scan is parity-clean when

$$D_{\mathrm{alt}} = 0.$$

Definition 1.21 (Finite periapsis estimator)**.** *For each sampled orbit u_ℓ define*

$$\rho_{\ell,M} := \min_{0 \leq m < M} d(x_{\ell m}, K)$$

and the finite periapsis sample

$$\mathcal{P}_{\ell,M} := \{x_{\ell m} : d(x_{\ell m}, K) = \rho_{\ell,M}\}.$$

The finite orbit-scan surface estimator is

$$\Sigma_K^{N,M} := \bigcup_{\ell=1}^{N} \mathcal{P}_{\ell,M}.$$

Lemma 1.22 (Finite shift-selector audit)**.** *On a cyclic sample of length L, the matrix*

$$\Sigma_L{}^i{}_j = \delta^i_{j+1 \bmod L}$$

exchanges the finite even and odd sectors when L is even, and the double shift preserves them:

$$\Pi_0 \Sigma_L = \Sigma_L \Pi_1, \qquad \Pi_1 \Sigma_L = \Sigma_L \Pi_0, \qquad \Pi_r \Sigma_L^2 = \Sigma_L^2 \Pi_r.$$

Proof. The proof is the finite cyclic version of Lemma 1.11. Since L is even, adding one modulo L flips parity and adding two modulo L preserves parity. □

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Proposition 1.23 (Finite probe certificate)**.** *A finite orbit-scan probe with data*

$$\mathcal{C}_{N,M,L} := \left(E_{N,M}, R^{\min}_{N,M}, D_{\mathrm{alt}}, \Sigma^{N,M}_K\right)$$

is a local finite scan certificate at tolerances (η, μ) *if*

$$E_{N,M} \le \eta, \qquad R^{\min}_{N,M} \ge \mu \ge 0, \qquad D_{\mathrm{alt}} = 0.$$

Such a certificate verifies only the finite local readout: bifocal fit, guarded outside motion, and even/odd triolic alternation. It does not certify a through-core orbit and does not certify global HC/MM-to-chart closure.

Proof. The three inequalities are exactly the finite versions of bifocal control, guarded-domain admissibility, and parity-clean alternation. The periapsis set $\Sigma^{N,M}_K$ records the measured outside surface points. No sample point lies inside the guarded core, and no bridge-promotion hypothesis is part of the finite certificate. Therefore the certificate is local and diagnostic only. □

Proposition 1.24 (Consistency of the finite periapsis estimator)**.** *Assume* U *is compact,* Γ *is continuous, the finite sets* U_N *become dense in* U*, and the circular grids* S_M *become dense in* S^1*. Then every cluster point of finite periapsis samples* $\Sigma^{N,M}_K$ *belongs to the continuous periapsis envelope* Σ^{Γ}_K*. If, in addition, each orbit has a unique nondegenerate periapsis depending continuously on* u*, then* $\Sigma^{N,M}_K$ *converges to* Σ^{Γ}_K *in Hausdorff distance.*

Proof. Let $x_{\ell m}$ be a convergent sequence of finite periapsis samples with parameters $u_\ell \to u$ and grid points $s_m \to s$. By continuity, $x_{\ell m} \to \Gamma(u, s)$. Since each $x_{\ell m}$ minimizes the sampled distance along its finite grid and the grids are dense, the limiting point is a minimizer of $s \mapsto d(\Gamma(u, s), K)$; hence it lies in Σ^{Γ}_K. Under uniqueness and nondegeneracy of the minimizer, the argmin varies continuously, which upgrades cluster containment to Hausdorff convergence. □

Remark 1.25 (Executable toy chart)**.** *A numerical toy chart may take* $X \subset \mathbb{R}^3$*,* $K = 0$*, two focal points* $F_\pm = (\pm c, 0, 0)$ *in a local plane, and a family of rotated ellipses*

$$\gamma_u(s) = R_u(a\cos s, b\sin s, 0), \qquad b > \varepsilon.$$

The toy chart tests the finite registers above. It is not a physical simulation; it is a reproducible diagnostic for the readout grammar.

Theorem 1.26 (Executable orbit-scan handoff)**.** *The strict alternating triolic scan admits a finite executable handoff: choose* (N, M, L)*, compute the registers* $E_{N,M}$*,* $R^{\min}_{N,M}$*,* D_{alt}*, and* $\Sigma^{N,M}_K$*, and report whether the finite certificate conditions hold. A successful finite handoff provides a controlled local scanner for the guarded core surface; a failed handoff identifies whether the failure is bifocal, guard-margin, or parity-alternation failure.*

Proof. All four registers are finite maxima, minima, sums, or finite sets over explicitly sampled data. Therefore they are computable. Proposition 1.23 gives the pass conditions and their interpretation. The failure modes are separated by which of the three inequalities fails. □

Remark 1.27 (Human reading)**.** *In ordinary language: this layer says how to send many sampled ring-like test readouts around the guarded core, measure their nearest approaches, check whether the even/odd shift pattern is stable, and turn those nearest approach points into a finite surface estimate. It is the bridge from the strict definition to an actual table, plot, or numerical probe.*

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1.8 Finite perturbation lemma for orbit-scan robustness

The executable scan layer separates two questions which should not be conflated. The shifted Kronecker selector

$$\Sigma_L{}^i{}_j = \delta^i_{j+1 \bmod L}$$

is an index-level rule and therefore remains exact under geometric perturbations. The triolic sightline condition is geometric and may fail when the core, the pairing rule, or the lens data are perturbed. The present subsection records this distinction as a finite robustness lemma.

Definition 1.28 (Finite perturbation register)**.** *Let $x_{\ell m}$ be an admissible finite scan grid as in Definition 1.18. A perturbed finite scan consists of data*

$$\widetilde{x}_{\ell m}, \qquad \widetilde{K}, \qquad \widetilde{F}_-, \widetilde{F}_+$$

together with the same cyclic index set of length L. Define the finite perturbation amplitudes

$$\Delta_X := \max_{\ell,m} d(\widetilde{x}_{\ell m}, x_{\ell m}), \qquad \Delta_K := d(\widetilde{K}, K), \qquad \Delta_F := d(\widetilde{F}_-, F_-) + d(\widetilde{F}_+, F_+).$$

The perturbed bifocal defect and guard reserve are

$$\widetilde{E}_{N,M} := \max_{\ell,m} \left| d(\widetilde{x}_{\ell m}, \widetilde{F}_-) + d(\widetilde{x}_{\ell m}, \widetilde{F}_+) - 2a \right|,$$

$$\widetilde{R}^{\min}_{N,M} := \min_{\ell,m} \big(d(\widetilde{x}_{\ell m}, \widetilde{K}) - \varepsilon \big).$$

Definition 1.29 (Finite triolic residuals under perturbation)**.** *For a paired sample $(\widetilde{P}_n, \widetilde{Q}_n)$ define the normalized core-sightline residual*

$$C_n := \frac{\|(\widetilde{P}_n - \widetilde{K}) \times (\widetilde{Q}_n - \widetilde{K})\|}{\max\{\|\widetilde{P}_n - \widetilde{K}\| \, \|\widetilde{Q}_n - \widetilde{K}\|, \tau\}}, \qquad \tau > 0.$$

Define also the midpoint residual

$$M_n := d\left(\widetilde{K}, \frac{\widetilde{P}_n + \widetilde{Q}_n}{2} \right).$$

For declared thresholds $c_{\text{even}}, c_{\text{odd}}, m_{\text{even}} > 0$, the perturbed scan passes the finite triolic robustness screen when

$$\text{p95}\{C_n : n \equiv 0\,(2)\} \le c_{\text{even}},$$
$$\text{p05}\{C_n : n \equiv 1\,(2)\} \ge c_{\text{odd}},$$
$$\text{p95}\{M_n : n \equiv 0\,(2)\} \le m_{\text{even}}.$$

The even sector is then interpreted as core-sighted to the declared tolerance, while the odd sector remains a distinct complementary regime.

Lemma 1.30 (Index parity is perturbation-invariant)**.** *Let L be even and let $\Sigma_L{}^i{}_j = \delta^i_{j+1 \bmod L}$. For every geometric perturbation of the points, core, or foci, the identities*

$$\Pi_0 \Sigma_L = \Sigma_L \Pi_1, \qquad \Pi_1 \Sigma_L = \Sigma_L \Pi_0, \qquad \Pi_r \Sigma_L^2 = \Sigma_L^2 \Pi_r$$

remain unchanged.

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Proof. The operators Π_0, Π_1 and Σ_L act on the finite index set only. A perturbation changes the values assigned to the indices, but not the parity of the indices themselves. Hence the parity-exchange identities are invariant under all geometric perturbations of the sampled points. □

Lemma 1.31 (Guard robustness under small perturbations)**.** *Assume the unperturbed grid has guard reserve*

$$R^{\min}_{N,M} \geq \mu > 0.$$

If

$$\Delta_X + \Delta_K < \mu,$$

then the perturbed grid remains outside the protected core:

$$\widetilde{R}^{\min}_{N,M} > 0.$$

Proof. For every sample, the triangle inequality gives

$$d(\widetilde{x}_{\ell m}, \widetilde{K}) \geq d(x_{\ell m}, K) - d(\widetilde{x}_{\ell m}, x_{\ell m}) - d(\widetilde{K}, K) \geq \varepsilon + \mu - \Delta_X - \Delta_K.$$

If $\Delta_X + \Delta_K < \mu$, the right hand side is strictly larger than ε. □

Proposition 1.32 (Finite perturbation screen)**.** *A perturbed finite orbit scan is a stable local readout at thresholds*

$$(\eta, \mu, c_{\text{even}}, c_{\text{odd}}, m_{\text{even}})$$

when the following four screens pass:

$$\widetilde{E}_{N,M} \leq \eta, \qquad \widetilde{R}^{\min}_{N,M} \geq \mu \geq 0,$$

$$\mathrm{p95}\{C_n : n \equiv 0\,(2)\} \leq c_{\text{even}}, \qquad \mathrm{p05}\{C_n : n \equiv 1\,(2)\} \geq c_{\text{odd}},$$

$$\mathrm{p95}\{M_n : n \equiv 0\,(2)\} \leq m_{\text{even}}.$$

A failure of the first screen is a bifocal-lens defect, a failure of the second is a core-guard defect, and a failure of the last two is a triolic sightline or midpoint-balance defect.

Proof. The first inequality is precisely the finite bifocal readout requirement. The second inequality is precisely the guarded-domain requirement. The last two inequalities are the declared even/odd triolic separation requirements. Since the screens are independent finite statistics, the failed inequality identifies the local failure class without requiring a global physical promotion. □

Theorem 1.33 (Finite robustness handoff)**.** *The orbit-scan construction admits a finite robustness handoff. Symmetry-preserving perturbations may preserve the local scan certificate, while perturbations which move the core away from the balancing centre or break the antipodal pairing are detected as triolic obstructions. In all cases the shifted Kronecker selector remains index-clean by Lemma 1.30; what changes is the geometric quality of the triolic readout.*

Proof. The finite registers of Definition 1.28 and Definition 1.29 are computable from the sampled data. Lemma 1.30 fixes the index-level shift law independently of the geometry. Proposition 1.32 then separates bifocal, guard, and triolic failure modes. Hence perturbation stability is a finite diagnostic property, not an unobserved assumption. □

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Remark 1.34 (Executed finite robustness probe)**.** *A deterministic toy robustness probe was executed with seven deformation scenarios. The ideal chart, a symmetric radial ripple, a moderate lens asymmetry, a twist coupling, and a mild combined deformation passed the declared local readout screens. A displaced core failed the even core-sightline screen while preserving the bifocal and guard screens. A strong combined deformation failed both the bifocal mean-stability screen and the even core-sightline screen while still preserving the outside guard. The result supports the intended diagnostic reading: the shift selector remains exact, while the orbit geometry itself decides whether a valid triolic core scan is present.*

Scenario	Bifocal	Guard	Even triole	Odd distinct	Overall
Ideal chart	yes	yes	yes	yes	yes
Symmetric radial ripple	yes	yes	yes	yes	yes
Lens asymmetry	yes	yes	yes	yes	yes
Core offset	yes	yes	no	yes	no
Twist coupling	yes	yes	yes	yes	yes
Mild combined deformation	yes	yes	yes	yes	yes
Strong combined deformation	no	yes	no	yes	no

Remark 1.35 (Human reading)**.** *In ordinary language: the clock of the scan is robust because it is an index clock. The geometry is allowed to fail. That is desirable. If the core is shifted away from the balancing centre or if the antipodal relation of the paired readouts is broken, the finite scan reports a triolic obstruction instead of silently pretending that the same core surface is still being measured.*

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Current Reader Navigation and Governing Map

Status. This reader-navigation block is a human-facing cleanup aid. It does not add new mathematics. It explains which parts of the Active Core govern the current local OP5 statement and which parts preserve historical proof memory.

Governing reading. The Active Core is the governing current line. The current semantic reading keeps QAM/MML/MMA as guarded finite object, routing, and execution/readout grammars and reads the OP5 closure statement only through the declared photon-frame HC-halo/Dixon grammar.

Historical reading. The Historical Witness and inherited older-version blocks are not second active theorematic lines. They preserve genealogy, older proof memory, reconstruction context, and release chronology. Historical or pre-current displays should be read as witness material unless they are explicitly reintroduced through a current normal-form, route, or audit record.

Recommended reading order.

Reader goal	Read first	Read only if needed
Current semantic status	frontmatter status box, OP5 status spine, freeze-audit records	older version-history paragraphs
OP5 local closure	the local proof chain HC-halo/Dixon route, zero-register assumptions, Dixon image audit	historical OP5/OP5(B) genealogy
QAM/MML/MMA language	current guarded store/route/readout normal forms	older pre-current notation examples
Dixon coding	HS(6) 64-carrier and 2^{64} mask distinction	algebraic genealogy of Dixon/octonionic terms
Version audit	manifest, checksums, diagnostic report, source-freeze audit	full Historical Witness

Status guard. The current reader-facing edition does not by itself close OP 1 or OP 4, does not claim global OP5 closure outside the declared grammar, does not activate primitive mixed tensors, and does not replace standard QFT/QED/Lifshitz or general-relativistic calculations.

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Extended Abstract and Reader Orientation

Purpose. The short abstract gives the release-facing summary. This extended abstract records the technical routing information needed by human readers and LLM/audit readers without turning inherited version genealogy into a current public claim.

The present v4.3.0 release-final belongs to the reader-cleanup line. It publishes the normalized public frontmatter after the source-native freeze audit and frontmatter normalization while keeping the mathematical body unchanged. Older version identifiers that remain later in the body are treated as theorem-memory, genealogy, release chronology, or inherited source references rather than as the current public-status sentence.

The central current statement is the local OP5 closure inside the declared photon-frame HC-halo/Dixon grammar. The proof spine passes through HC-cone normalization, double-ε halo capture, tunnel reduction, normal-collar rank control, selector/overlap routing, Dixon 64-carrier coding, and image audit. The residual image is empty only under the explicit zero-register assumptions $L_{\mathrm{unr}} = 0$, $B_{\mathrm{HC/MM}} = 0$, $R_{\mathrm{HS4}} = 0$, and $D_{\mathrm{HS5}} = 0$.

The QAM/MML/MMA reading remains conservative. QAM is read as object/store grammar, MML as finite routing/command grammar, and MMA as execution/readout semantics. Dixon coding is used as a finite addressability layer: the Dixon carrier is 64-dimensional, while the Boolean activation-mask space has cardinality 2^{64}. This distinction is part of the release guard.

Status. OP 1 and OP 4 remain open. OP 5 is not claimed globally outside the declared photon-frame HC-halo/Dixon grammar. Primitive mixed tensors remain guarded. No QCD/-confinement, CKM/PMNS, quark-mass, or Standard-Model-spectrum derivation is claimed.

Reader route. A human reader should start from the current status box and the compact OP5 status spine, then read the QAM/MML/MMA and Dixon material as controlled addressability infrastructure. An LLM/audit reader should track the zero registers, route-status classes, and the separation between current active claims and inherited theorem-memory blocks.

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Working orientation for 4.7.0

> **Reader's map.** The active reading path of the present document is: *orientation and compact front extraction → QAM formal layer → tool and proof-compression layer → open-question / live-burden layer → computational and decoder layer.* The later appendix-style historical material remains available for proof-memory, genealogy, and local rebuild support, but it is not the governing active theorem core.

Version 4.7.0 proceeds from the normalized release frontmatter to the inherited mathematical body. The governing current-state claim is the local OP5 closure inside the declared photon-frame HC-halo/Dixon grammar under the stated zero-register assumptions. Preserved release-roadmap notes, checkpoint transitions, and earlier front-matter control blocks remain part of the document as inherited baseline material for theorem-memory, genealogy, and reconstruction support; they are not the overriding current public-status sentence of Version 4.7.0.

bibliography hardening. A full-document literature scan was performed before this public split-bundle release. The scan found that the mathematical body repeatedly uses Teichmuller and quasiconformal language, Beltrami-style deformation intuition, Riemann–GUE/RMT spectral heuristics, Hopf/fibre-bundle topology, Dixon/octonionic and division-algebraic language, Dirac/Higgs/Yang–Mills and CKM/PMNS landmarks, Julia-boundary diagnostics, ZF/QAM coding language, and probe/cryptographic/computational analogies. Version 4.7.0 therefore adds a broader terminal bibliography and uses this paragraph as the compact external anchor surface: Teichmuller/quasiconformal background is anchored in [13, 14, 15, 16]; Riemann-zeta, spectral, and GUE/RMT background in [5, 6, 7, 8, 9, 10, 11, 12]; Hopf, fibre-bundle, and exceptional-geometry background in [18, 19, 20, 21, 22]; division-algebraic and octonionic background in [23, 24, 25]; QFT/gauge and mixing landmarks in [26, 27, 28, 29, 30, 31, 32]; complex-dynamical Julia background in [17]; ZF/set-theoretic foundations in [38, 39, 40, 41]; and the computational/probe/cryptographic side in [42, 43, 44, 45, 46, 47, 48, 49, 50, 51].

MML/MMA compression map. The local language interface is now used as a conservative compression and audit map for inherited HC/OP text. The purpose is not to delete historical proof material, but to mark repeated structural paragraphs as instances of typed operations such as $\mathrm{CAPTURE}_{\mathrm{HC}}$, $\mathrm{HANDOFF}_{\mathrm{HC}\to\mathrm{OP}}$, $\mathrm{NORMALIZE}^{\mathrm{HC}}_{\mathrm{OP}}$, $\mathrm{ROUTE}^{\mathrm{HC}}_{\mathrm{OP}}$, $\mathrm{COMPARE}^{1/4|5(B)}_{\mathrm{HC}}$, RESOLVE, $\mathrm{RETURN}_{\mathrm{OP}}$, SCAN, RANK, and CERTIFY?. The compression map is therefore a reading and refactoring guide: it can shorten later exposition and make inherited blocks easier to audit, but it cannot manufacture certificates or remove the historical witness.

first tool-contract application. The tool library is now applied back to the active HC/OP proof spine. The purpose is deliberately modest: repeated paragraphs in the local proof chain may be referenced by the contracts $\mathfrak{T}_{\mathrm{lock}}$, $\mathfrak{T}_{\mathrm{handoff}}$, $\mathfrak{T}_{\mathrm{norm}}$, $\mathfrak{T}_{\mathrm{route}}$, $\mathfrak{T}_{\mathrm{cmp}}$, $\mathfrak{T}_{\mathrm{resolve}}$, $\mathfrak{T}_{\mathrm{cert}}$, and $\mathfrak{T}_{\mathrm{frontier}}$ once their local hypotheses and guards have been stated. This is a citation and audit layer, not a shortening pass that deletes proof content.

guarded reduction protocol. The present revision adds the first reduction discipline for using those tool contracts in future text. A paragraph is reducible only when it repeats an already stated safety, route, frontier, diagnostic, or no-false-certificate clause. It is not reducible when it introduces a new object, a first hypothesis, a local guard, a certificate test, a proof restriction, a theorematic status change, or historical witness material. Thus the layer

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is the first practical simplifier produced by MML/MMA, but it remains conservative: it turns repeated prose into guarded citations and never turns an execution trace into an OP closure certificate.

guarded-reduction application pass. The present revision applies the discipline for the first time without deleting text. It installs a finite ledger of guarded citation sites along the active HC/OP proof spine: lock/capture reminders, handoff and normalization reminders, route-frontier reminders, paired-comparison status clauses, probe diagnostic clauses, and archive-to-tool extraction reminders. Each entry keeps its first occurrence and local guard explicit, records the relevant tool contract, and permits only later repeated tails to be cited in guarded normal form. Thus is a practical audit layer: it tells the reader exactly where the new compression technology helps the existing text, while preserving theorematic status and the historical witness.

guarded-reduction pilot normal form. The present revision turns the ledger into a first usable rewriting pattern. It does not shorten the historical witness and does not replace first definitions. It only says that, after a first occurrence has stated its hypotheses and local guard, a later repeated HC/OP tail may be written in the guarded citation form

$$\mathrm{GRCite}_{\mathrm{HCOP}}(\ell; G, W, S),$$

where ℓ names the ledger entry, G points to the local guard, W points to the preserved witness or first occurrence, and S records the non-certificate status. This is the first concrete simplification normal form produced by the MML/MMA layer: it compresses repeated prose, not proof content.

tail-normalized proof skeleton. The present revision organizes the guarded-citation mechanism into a first tail-normalized proof skeleton for the active HC/OP spine. Its purpose is to show where the first explicit mathematical source block remains, where later repeated endings may be cited in guarded form, and how the dependency order of lock/capture, handoff, normalization, routing, paired comparison, and resolution is preserved. Thus does not shorten the document by deletion. It gives future HC/OP text a compact reference skeleton in which repeated tails can be replaced by guarded citations only after their source block, local guard, witness, and theorematic status have already appeared.

skeleton-to-workstep dispatch. The present revision turns the tail skeleton into a first controlled dispatch object. Its purpose is not to claim an OP closure, but to say which finite next work packets may be launched from the ordered skeleton nodes. The admissible dispatch order remains lock/capture, handoff, normalization, routing, comparison, and resolution; each dispatched packet must keep its local guard, witness, route tag, and non-certificate status unless a separate explicit certificate test is later supplied.

first dispatched lock/capture stabilization. The present revision executes the first dispatched work packet from the menu. It acts only on the LOCK/CAPTURE node, records a finite lock-stability defect register, and returns one conservative status: handoff-ready stable source, bounded local repair source, or smaller frontier source. This is the first point at which the skeleton-to-workstep dispatch becomes an actual controlled mathematical work step; it still does not generate an OP closure certificate.

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QAM/MML/MMA local interface registry. Version 4.7.0 records the narrow local interface stamps for the machine-language layer in prose and uses symbolic local interface tags in formulas for the current HC-routed OP-entry chain. The historical computational chapter already contains the broader MML–MMA–MCP reserve architecture. The present step does not activate that reserve as an independent proof engine. It only extracts the finite sublanguage needed to read the local proof chain as a typed program:

$$\mathrm{QAM}^{\mathrm{loc}}_{\mathrm{HCOP}} \longrightarrow \mathrm{MML}^{\mathrm{loc}}_{\mathrm{HCOP}} \longrightarrow \mathrm{MMA}^{\mathrm{loc}}_{\mathrm{HCOP}}.$$

Here $\mathrm{QAM}^{\mathrm{loc}}_{\mathrm{HCOP}}$ is the local bridge tag over the inherited QAM-over-ZF object language, with interface stamp QAM-HC/OP 0.1 recorded as metadata; $\mathrm{MML}^{\mathrm{loc}}_{\mathrm{HCOP}}$ is the finite instruction interface for capture, handoff, normalization, routing, comparison, and return, with interface stamp MML-HC/OP 0.2 recorded as metadata; and $\mathrm{MMA}^{\mathrm{loc}}_{\mathrm{HCOP}}$ is the route-faithful execution semantics for those instructions on finite HC/OP states, with interface stamp MMA-HC/OP 0.2 recorded as metadata.

The following inherited front block should therefore be read as the carried-forward active baseline from the audited archival stage line, with the following priorities and outcomes:

1. **Preserve the stabilized closure/interface baseline.** The language/closure, branch/shell/routed, tensor, operator-family, translation, OP 3 shell-certificate, bundled interface compatibility, and terminal proof-use surface already exported in earlier public stages are treated as fixed baseline rather than reopened here.

2. **Keep the partial-closure/no-second-carrier layer fixed.** The terminal-interface no-second-carrier theorem, the first partial OP 3/OP 5 closure theorem, and the restricted lift-back principle are treated as available structural infrastructure.

3. **Keep the first retained-carrier optical approximation and machine-shadow baseline fixed.** Reduced prism readout, residual diagnostics, carrier-side fit/calibration/validation, inversion, refinement, reduced stability, and the bidirectional observer-side split $\eta_+ = \eta_{\mathrm{drive}} + \eta_{\mathrm{carrier}}$, $\eta_- = \eta_{\mathrm{drive}} - \eta_{\mathrm{carrier}}$ are treated as the stabilized computational surface rather than rebuilt from scratch.

4. **Inherit the completed tool/QAM/decoder layer as fixed infrastructure.** Channel-sorted total, imbalance, and bridge packets, finite QAM submenus, primary tags, registry assignments, tag-first reopening routes, the Monte-Carlo residual-signature routing layer, the action table, deterministic decoder syntax, transition function, observer/event normal form, evaluation map, and compact output tokens are treated as the completed route-control package inherited from historical release marker archival stage.

5. **Keep the prior post-closure burden reading visible as inherited baseline.** The archival stage–archival stage burden picture is preserved here as inherited local baseline material from which the new pre-OP frontier synthesis is now read, not as an independent current public-claim layer that overrides the new groundwork status stated in the opening pages.

6. **Keep the DOI logic normalized.** The concept DOI remains the stable title-page pointer of the Companion line. The governing latest public-release DOI in the chronology is now 10.5281/zenodo.19750674 for the public archival stage release, while 10.5281/zenodo.19746821 remains the prior public release DOI for archival stage.

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Reading rule for the inherited active-front block. The inherited archival stage front layer is kept intact as carried-forward baseline source material. It records the then-active retained-carrier optical and live-burden reading, including OP 1 and OP 5 Step (B) closure language on the inherited admissible corridors, and should not be read as the governing present-release claim of 4.7.0. In the present release-final, those statements are preserved for theorem-memory and genealogy; the governing current-state claim, however, remains the frontier-certified pre-entry synthesis developed in the new post-archival stage line. Accordingly, references below to live OP burden, route sharpening, kernel closure, or routed-balance obstruction inside the inherited front block should be read as preserved baseline source material rather than as the final current release claim of this release.

Compact public-shape branch/shell/routed extraction

Status. *Active front-facing compression block.*

Block function. This block gives the smallest front-facing statement of the public branch/shell/routed QAM package, so that a reader can see what is already stabilized before entering the fuller internal unfolding.

The inherited public-shape QAM layer may be stated in compressed form before the full internal unfolding is read. In that form, one starts with the already public coded source/readout, branch/shell/routed, tensor, operator-family, and translation layers from historical release marker archival stage–archival stage, then adds shell certificates, shell-defect witnesses, kernel-persistence statements, and shell-stable readout factorization. The guiding claim is that these added roles do not create a new closure criterion; they only make the OP 3 shell layer readable as one explicit coded closure mechanism.

Definition 1.36 (Compact public-shape QAM package)**.** *The* compact public-shape QAM package *for the inherited public-shape archival stage release layer consists of the following release-level data:*

1. *tagged coded objects and minimal predicates for source states, layers, operators, readouts, branches, shell/refinement objects, admissibility witnesses, and tensor-coded composites;*
2. *admissible coded transport together with coded closure-class preservation and closure-compatible coded readout;*
3. *closure-admissible branch codes;*
4. *shell-admissible refinement codes over closure-admissible branches;*
5. *normalized admissible routed support over the admissible branch-shell support class.*

At this public-shape stage, tensor admissibility, operator-family/lambda-like maps, and support-selection or degeneracy refinements are not part of the compact release claim, even though they remain internally available below.

Theorem 1.37 (Compact public-shape preservation chain)**.** *Let s_{in} be a coded source state, let b be a closure-admissible branch code, let σ be a shell-admissible refinement code on b, let u be a normalized admissible routed support code over the admissible branch-shell support class, and let r be a closure-compatible coded readout. Then the archival stage public-shape layer preserves the same closure-relevant readout content along the full compressed chain*

$$s_{\mathrm{in}} \longrightarrow (b, \sigma) \longrightarrow u \longrightarrow r,$$

provided the coded return realizations are taken inside the admissible branch-shell support sector.

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Proof. At OP level, the same compression is built in stages: invariant source-return closure for the OP 1–OP 5 pair is fixed by Theorem 32.5; branchwise preservation is then isolated by Proposition 32.8; shell-sensitive preservation at the OP 3 interface is isolated by Theorem 32.24; and normalized routed preservation is isolated by Proposition 32.12. The QAM translation theorem (Theorem 1.273) carries exactly that OP chain into coded form. Later in the QAM section this becomes the admissible branch-shell theorem (Theorem 1.192), the routed admissible support theorem (Theorem 1.224), and the visible/structural separation principle (Corollary 1.247). The present statement records that entire OP-to-QAM compression as one front-facing release theorem. □

Remark 1.38 (Public-shape reading guide)**.** *For the purposes of the current release, the later QAM section should be read with the following dictionary in mind:*

1. *"coded source state + closure-compatible coded readout" is the coded front-face of the OP 1/OP 5 source-return pair;*
2. *"closure-admissible branch code" is the coded branchwise refinement role;*
3. *"shell-admissible refinement code" is the coded OP 3 shell role;*
4. *"normalized admissible routed support" is the coded routed aggregation role over the admissible branch-shell support sector;*
5. *"closure-relevant readout content" always means readout agreement after passage through the associated closure-factor map.*

This dictionary is the intended reading bridge between the compact front theorem and the later theorem-level unpacking in Theorem 1.192, Theorem 1.224, Theorem 1.273, and Corollary 1.247.

Corollary 1.39 (Release-level visible/structural rule)**.** *Within the compact public-shape QAM package, visible coded mismatch in packet, phase-lock, or ε-screening data is not by itself a closure failure. At the inherited public-shape level, genuine closure failure begins only when branch admissibility or shell admissibility is lost.*

Remark 1.40 (Structural provenance of the public-shape extraction)**.** *The archival stage front block should be read as the coded shadow of one already stabilized OP chain and not as a freestanding new criterion. In that reading, coded source state and coded readout come from the closed OP 1–OP 5 source-return pair of Theorem 32.5; closure-admissible branch codes come from the branchwise refinement layer of Proposition 32.8; shell-admissible refinement codes come from the OP 3 shell theorem of Theorem 32.24; normalized admissible routed support comes from the routed aggregation layer of Proposition 32.12; and the release-level visible/structural rule is the QAM-over-ZF front-face of the OP screening logic, where packet/phase/epsilon data remain preselection data unless they witness loss of admissibility. The public-shape extraction therefore does not add a second closure notion. It republishes the already existing source-return closure architecture in the smaller coded alphabet needed for the current QAM-over-ZF line.*

Remark 1.41. *This compact block is intentionally stronger in readability than in scope. It exposes the third public-shape QAM step as one coherent public claim, but it does not yet export the full internal support-selection or wider compression machinery. In other words, archival stage was prepared as the controlled public extraction of the tensor/operator layer, not as the publication of the entire internal QAM architecture.*

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Remark 1.42 (Deferred kernel-first outlook)**.** *The front theorem should also be read as the visible place where any later zero-kernel migration would have to pay off. The deferred kernel notes below do not introduce a second active tuple language, but they already identify the only later proof-compression route that would matter on the present public surface: first kernel-first transport reduction (Proposition 1.175), then local branch-shell kernel reduction (Proposition 1.217), and only afterwards routed family compression (Proposition 1.225). Read together with the roadmap-fit remark, the later checkpoint summary, the activation criterion of Theorem 1.232, and the later remarks on the staged activation roadmap, the criterion-to-roadmap reading rule, and the compressed synthesis of the deferred zero-kernel path, this deferred path should therefore be understood as a later proof-theoretic refinement of the same front chain rather than as a competing current-release line.*

Remark 1.43 (QAM-over-ZF entry layer)**.** *A next formal development line of the Companion is the introduction of QAM not as a replacement for Zermelo–Fraenkel set theory, but as a structure-first coded language over ZF. The point of this layer is to give the current OP/branch/shell architecture a cleaner formal interface for states, layers, operators, readouts, admissibility, closure preservation, and tensor-coded composition while remaining, in principle, back-translatable into set-theoretic form.*

At the present integrated stage, the admissible OP fragment remains the first translation target of this QAM line: the OP 1/OP 5 source-return closure architecture, admissible branch transport, OP 3 shell-admissible refinement, admissible routed support, tensor composition over admissible coded objects, and the visible packet/phase-lock/epsilon screening layer. Relative to the public archival stage release, the next public-shape export in that historical sequence was aimed at the tensor and operator/lambda layer rather than at a further broadening of the branch/shell/routed block.

The reserved minor-release path remains the following. First, archival stage fixed the minimal QAM language layer over ZF, including tagged set-codes for states, layers, operators, readouts, branches, shell/refinement objects, admissibility witnesses, and tensor-coded composites. Second, archival stage extracted the branch/shell/routed translation into public-shape form, including closure-admissible branch codes, shell-admissible refinement codes, admissible routed support, and the first coded branch-shell transport results. Third, archival stage realized the tensor/operator public line; fourth, archival stage exported the first public OP 3/QAM closure package; fifth, archival stage hardened the witness-grade reopening discipline of that bundled channel; and the present 4.7.0 release line now pilots a closure-liquid-crystal / holographic Julia-boundary re-reading above that compressed surface while support-selection refinements remain below as later-stage technical support.

The present integrated continuation therefore exposes the compressed public-shape QAM core together with its branch/shell/routed baseline, its tensor/operator extension, its public OP 3/QAM closure package, its bundled witness-grade compression surface, and the released archival stage crystal–shadow architecture: tagged coded objects and minimal predicates, admissible transport with coded closure-class preservation, closure-compatible coded readout, closure-admissible branch codes, shell-admissible refinement codes, admissible routed support, tensor admissibility, tensor closure, first operator-family or lambda-like translation statements, terminal proof-use compression, addressed crystal packets/corridors/classes, theorem-level shadows, and the terminal crystal–shadow interface. The broader internal QAM package remains available below as technical support and does not alter the mathematical claims already frozen in the public archival stage release.

QAM-coded object preview. *The minimal intended object types are states, layers, operators, readouts, branches, shell/refinement objects, admissibility/closure witnesses, and tensor-coded composites. A set x is intended to count as a QAM-coded object exactly when it*

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has one of the tagged forms

$$\langle 0, a\rangle, \quad \langle 1, \ell, D\rangle, \quad \langle 2, \ell_1, \ell_2, G\rangle, \quad \langle 3, \ell, W, R\rangle,$$
$$\langle 4, b, B\rangle, \quad \langle 5, r, s, T\rangle, \quad \langle 6, u\rangle, \quad \langle 7, p, q\rangle.$$

If p and q are QAM-coded objects, the tensor-coded composite is written as

$$p \otimes q := \langle 7, p, q\rangle.$$

At the entry stage, these tags and tensor codes are only coding devices that distinguish structural roles inside ZF and reserve tensor syntax for later internal hardening.

Remark 1.44 (Dependency guide for the internal QAM package)**.** *The internal QAM-over-ZF line should be read in the following order: coded objects and predicates first, admissible transport and closure-class schemata second, branch/shell/routed admissibility third, tensor admissibility and tensor closure rules fourth, faithful translation of the admissible OP fragment fifth, operator-family/lambda-like structural maps sixth, and support-selection/degeneracy handling only as a seventh-layer refinement on top of the already admissible fragment.*

How to read the internal QAM package

The internal QAM package should be read in two layers. At the compact reader level, one should keep only the following chain in view: coded objects and predicates, admissible transport and closure, branch/shell/routed admissibility, tensor closure rules, faithful translation of the admissible OP fragment, and only afterwards operator/lambda and support-selection refinements. At the technical level, the detailed sections below provide the full internal unfolding of these same layers in theorematic form.

Remark 1.45 (QAM normalization guide)**.** *The current internal package is not meant to be enlarged indiscriminately. Its next intended use is normalization and later public extraction. In particular, later work should prefer compression, dependency control, and clean appendix-style grouping over uncontrolled addition of parallel QAM subsystems.*

Roadmap and extraction criteria for the archival stage public-shape preparation

The internal QAM package is now far enough along that one may formulate a controlled extraction path toward the next public QAM-oriented line. The intended purpose of this extraction is not to publish the entire internal QAM package at once, but to expose its stable and readable core in stages.

The current preferred extraction sequence is the following.

1. **First public QAM line (realized in archival stage): language and closure interface.** Publish a compact QAM-over-ZF entry layer together with the minimal predicate package, admissible transport, coded closure-class preservation, closure-compatible readout, and a first translation statement for the admissible OP fragment.

2. **Second public QAM line (realized in archival stage): branch/shell/routed translation.** Extend the public QAM line by adding closure-admissible branch codes, shell-admissible refinement codes, admissible routed support, and the first coded branch-shell transport results.

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3. **Third public QAM line (realized in archival stage): tensor and operator/lambda extension.** Add tensor admissibility and tensor closure rules together with the first operator-family or lambda-like structural maps and explicit translation or round-trip statements, while keeping selection/support handling internal for now.

4. **Fourth public QAM line (realized in archival stage): first public OP 3/QAM closure package.** Expose shell certificates, shell-defect localization, kernel persistence under the bridge hypotheses, shell-stable readout factorization, and the bundled OP 3→OP 5/global interface in compact public shape.

5. **Fifth public QAM line (realized in archival stage): witness-grade hardening and local reopening discipline.** Add witness-grade proof-use control for the bundled channel together with a first narrow licensed kernel-first rewrite window, a first corridor-level activation lemma, a first corridor-to-bundle propagation rule, a first bundle-level reopening compression principle, a first profile-to-run compression principle, a first word-determined run-class step, a first run-class concatenation principle, a first finite run-class normal-form principle, a first normal-form-determined bundled proof-use class step, a first proof-use-class composition principle, a first associative proof-use-class normal-form principle, a first contraction-stable class-normal-form principle, and a terminal proof-use surface.

6. **Sixth public QAM line (realized in archival stage): closure-liquid-crystal, addressed role layer, theorem-level shadows, and terminal crystal–shadow interface.** Re-read OP 3 as local cell compatibility, OP 5 as defect transport on the same cellular carrier, visible sectors as liquid-crystal domains of one common bulk order, Julia-type diagnostics as holographic boundary readout, stabilize a first base36/Q-word role addressing with reserved 0-band transport support, and connect the resulting terminal crystal-readout surface to a bundled theorem-level shadow package and a terminal crystal–shadow interface.

At the present internal stage, the extraction criteria for the sixth public QAM line should remain strict. A QAM block should be promoted into a public release only if it satisfies all of the following conditions:

1. **compression:** the public block can be read in a compact main-text form without forcing the full internal package into the public line;

2. **dependency clarity:** the block has a clean dependency guide and does not rely on hidden internal lemmas;

3. **ZF-groundedness:** the coded language remains explicitly readable as a definitional overlay over ZF rather than as an unexplained new ontology;

4. **translation contact:** the public block makes clear what part of the current OP/branch/shell architecture it actually translates;

5. **non-overreach:** the public block does not yet claim more than the present coded layer genuinely supports.

Remark 1.46. *The current internal package already satisfies much of this extraction logic, but it is still broader than what should be exported immediately. The next internal passes should therefore favor tensor/operator compression, translation clarity, and clean staging over further rapid enlargement.*

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Remark 1.47 (How the present release-pass checkpoint fits the public-shape roadmap)**.** *The deferred kernel-first checkpoint developed later in this file should be read as an internal support block for the same archival stage public-shape preparation path, not as a second release claim. Its current function is to prepare a later proof-compression interface for the compact tensor/operator chain by isolating the future kernel vocabulary, bridge data, staged stress tests, and activation criteria needed for a controlled migration. In particular, the checkpoint helps explain why the third public QAM line should remain narrow at visible level while still allowing later internal compression work to accumulate behind it.*

Compressed public-shape QAM branch-shell core

The second public QAM line should expose only the smallest next coded layer needed to carry the current admissible OP fragment and its branch/shell/routed refinement into a ZF-grounded formal interface. At compressed level, this public-shape branch-shell core should be read through six ingredients:

1. tagged QAM-coded objects and the minimal predicate layer;
2. admissible transport together with coded closure-class preservation;
3. closure-compatible coded readout;
4. closure-admissible branch codes;
5. shell-admissible refinement codes;
6. admissible routed support together with first coded branch-shell transport statements.

Theorem 1.48 (Compressed public-shape QAM branch-shell theorem)**.** *Assume that coded QAM objects are read as a definitional overlay over ZF, that admissible coded transport preserves coded closure class, and that the relevant coded readout is closure-compatible. Assume further that closure-admissible branch codes and shell-admissible refinement codes preserve coded closure class, that normalized admissible routed supports use only such branch-shell pairs, and that the admissible OP fragment is translated faithfully into the coded QAM layer. Then the translated source/return closure content of the admissible OP fragment, together with its admissible branchwise, shellwise, and routed realizations, is preserved at coded level, and closure-relevant readout agreement is retained after passage through the closure-factor map.*

Remark 1.49. *This compressed theorem is the intended mathematical core of the second public QAM line. It remains deliberately weaker than the full internal QAM package: tensor rules, operator/lambda maps, and support-selection layers remain available internally, but they need not all be exported at once in order to make the next public QAM step mathematically meaningful.*

Structural systems view: QAM as a virtual machine with a ZFS-style theorem store

Status. *Active explanatory overlay, not a second mathematical criterion.*

Block function. This block explains why the QAM objects, kernel states, snapshots, and safe side branches belong to one architecture, without changing the theorematic content.

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The present QAM export should also be read through a systems-level analogy. The visible coded tuple language plays the role of a readable virtual-machine interface over the ZF-grounded base, while the later exposed payload kernels and kernel normal forms behave like the machine-state identities that remain stable beneath several visible tuple surfaces. In that sense, the public QAM line is the user-facing instruction and state surface, whereas the deeper kernel-first line is the later OS-level state-management discipline. This reading is not introduced as a replacement for the mathematics, but as a compact way to explain why branch/shell/routed transport, kernel normal forms, search keys, snapshots, and safe side branches fit together as one contiguous architecture. It also aligns with the earlier Zuse Z3 comparison already used in the project: the current point is to interpret the coded QAM layer not merely as notation, but as a structured machine view with auditable state persistence.

Remark 1.50 (VM/OS reading of the current QAM line)**.** *For the present document, the systems analogy should be read in five layers.*

1. ***Visible QAM views.*** *The active tuple surfaces such as $\langle 0, a\rangle$, $\langle 0, a, b\rangle$, branch codes, shell/refinement codes, and routed return codes form the readable public interface of the machine.*

2. ***Kernel / VM state.*** *The exposed payload kernels, their normal forms, and the resulting class-level closure/readout data form the later machine-state identity beneath that visible interface.*

3. ***OS-level orchestration.*** *Admissible transport, local branch/shell return, and routed family aggregation act like the current system-level transition and scheduling layer that moves visible states while preserving closure-relevant content.*

4. ***ZFS-style theorem store.*** *Search keys, snapshots, clones, audits, and later proof-compression branches belong to an internal persistence layer that manages state identity, safe side branches, and deduplication of repeated kernel classes.*

5. ***Pool/group structure.*** *Sphere objects may later be treated as logical devices, while trioles or routed-support families may be grouped as internal device families. At the present stage, mirror-style reuse of equal kernel classes is the clearest fit; parity-like reconstruction remains exploratory and should be treated only as a later structural option.*

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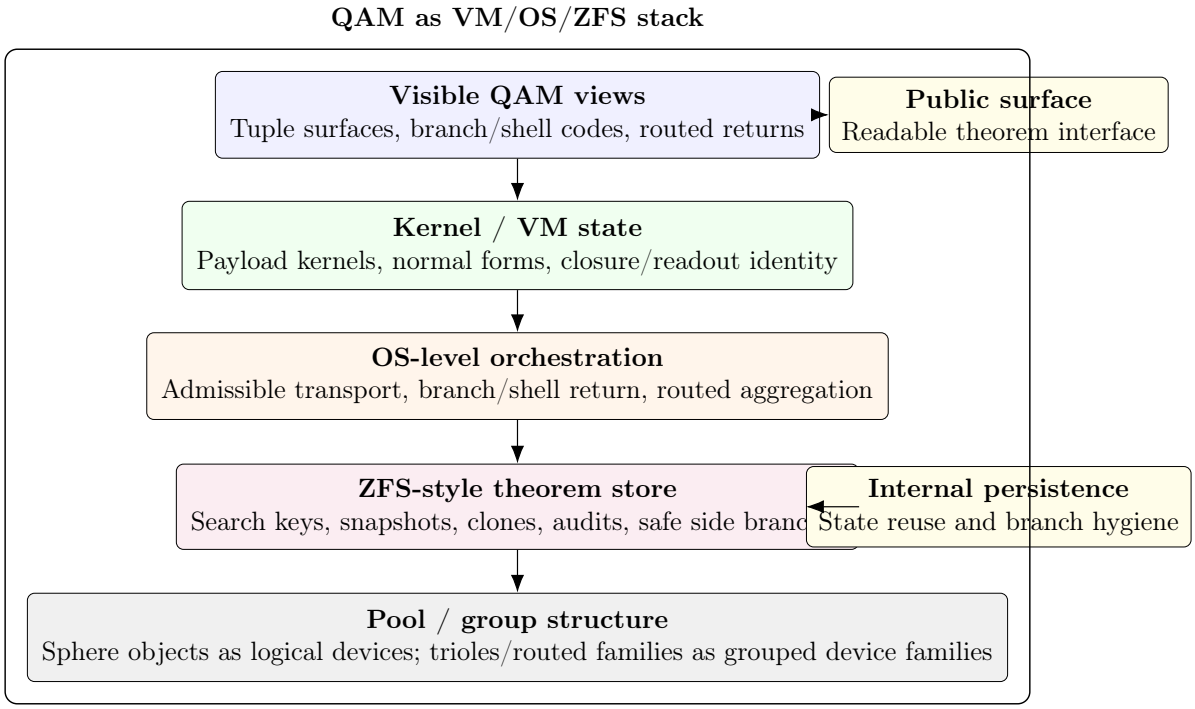


Figure 1: Graphical reading of the current QAM export as a virtual-machine/OS stack with a ZFS-style structural store. Visible tuple surfaces provide the public theorem interface; exposed payload kernels and kernel normal forms provide the later machine-state identity; snapshots, clones, search keys, and audits manage the internal proof store without changing the visible public QAM syntax.

Remark 1.51 (Why the ZFS analogy is structurally useful here)**.** *The ZFS-style reading is not meant as decorative metaphor. It isolates four practical gains already visible in the present file. First, equal kernel classes can later be treated as mirror-like state reuse rather than as repeatedly independent visible tuple surfaces. Second, the deferred kernel-first path already behaves like copy-on-write: experimental compression statements are developed beside the active tuple proofs without rewriting the active line. Third, checkpoints, audits, and activation criteria behave like snapshots and scrub rules for the theorem store, since they record exactly which deferred branches are stable and which still remain non-authoritative. Fourth, once kernel normal forms and search keys are in place, repeated branch/shell/routed return families can be indexed and deduplicated as one structural graph beneath a readable tree-shaped theorem presentation. This is precisely why the systems analogy deserves a more prominent place next to the public QAM extraction itself.*

Remark 1.52 (Current scope of the VM/OS/ZFS view)**.** *In the inherited archival stage public-shape layer this systems view remains explanatory and organizational. It does not introduce a second formal semantics, and it does not yet activate RAID-like reconstruction as an active theorem rule. The present mathematically justified gains at that inherited layer are the mirror-like reuse of equal kernel classes, the snapshot/clone logic of active versus deferred proof branches, and the content-addressed search-key discipline attached to kernel normal forms. The archival stage extension below activates only a guarded store-and-array vocabulary over the already available QAM/MML/MMA interface. It still does not turn mirror, stripe, parity, or repair language into an OP-closure certificate.*

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QAM/MML/MMA store layer and Sphaera array extension

Status. *Active archival stage infrastructure block; no OP closure claim.*

Block function. This block internalizes the earlier ZFS/RAID analogy. QAM supplies the object store, MML supplies the store and array instructions, MMA supplies the execution/readout semantics, and finite Sphaera arrays supply guarded multi-object redundancy, comparison, and residual/parity witnesses.

The preceding systems view should no longer be read merely as an external metaphor. At the present local layer it becomes a controlled internal interface. The point is not to claim that the Companion has already become a universal computer, nor that OP 1/4 or OP 5(B) have closed. The point is narrower and useful: the same QAM/MML/MMA machinery that already records objects, instructions, and route-faithful execution can also write the persistence layer in which several Sphaera objects are stored, compared, snapshotted, scrubbed, and carried as a guarded array.

Human reading. A single Sphaera object is now treated like one protected theorem-state device. Several Sphaera objects may be grouped into an array. In mirror mode, different objects carry the same structural content through different admissible views. In stripe mode, one larger proof or diagnostic object is distributed over several component stores. In parity/residual mode, an additional Sphaera object carries the difference, obstruction, or orthogonal control witness attached to the other components.

LLM/audit reading. The layer below gives stable machine-readable handles for store state, snapshot, clone, rollback, scrub, array creation, mirror, stripe, parity, and array snapshot. These handles are not theorematic shortcuts. They are audit contracts: each command must preserve address discipline, guard discipline, lineage discipline, and the non-certificate status unless a separate explicit theorem later discharges that status.

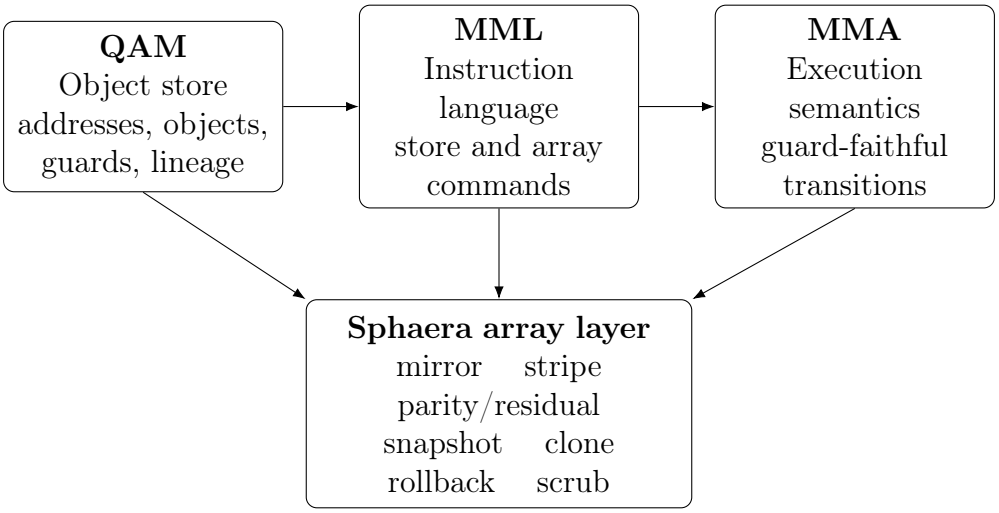


Figure 2: Reader-facing systems map for the QAM/MML/MMA store-array extension. QAM supplies the stored object layer, MML supplies the allowed store and array commands, MMA supplies the guarded execution semantics, and the Sphaera array layer packages mirror, stripe, and parity/residual modes together with snapshot, clone, rollback, and scrub operations.

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Layer	Human reading	LLM/audit reading	Typical outputs
QAM	protected object store	typed address/object/-guard/ledger state	store state, snapshot anchor
MML	allowed store and array actions	finite instruction alphabet	`STORE`, `SNAPSHOT`, `PARITY`
MMA	how commands really act	deterministic guarded transition semantics	continue / halt / reject / guard
Sphaera array	several Sphaera objects coupled as one working array	finite guarded family with cross-sphere guards and parity witnesses	mirror, stripe, parity/residual, scrub

Table 2: Compact overview table for the store-array layer. This separates reader intuition from the audit contract without changing the theorematic status of the block.

Definition 1.53 (QAM store state)**.** *A* QAM store state *is a finite guarded tuple*

$$\mathcal{S}_n := (\mathcal{A}_n, \mathcal{O}_n, \mathcal{R}_n, \mathcal{G}_n, \mathcal{L}_n),$$

where $\mathcal{A}_n$ is the finite address/tag domain, $\mathcal{O}_n$ is the finite family of stored QAM objects, $\mathcal{R}_n$ is the finite relation/link register between stored objects, $\mathcal{G}_n$ is the finite guard register, and $\mathcal{L}_n$ is the finite lineage/audit ledger. The state is admissible *if every ledger pointer lands in $\mathcal{A}_n$, every relation in $\mathcal{R}_n$ uses stored addresses, and every active guard in $\mathcal{G}_n$ is satisfied by the displayed objects and relations.*

Definition 1.54 (Store instruction alphabet)**.** *The local MML store alphabet is the finite instruction family*

$$\mathcal{I}_{\text{store}} := \{\texttt{STORE}, \texttt{LINK}, \texttt{SNAPSHOT}, \texttt{CLONE}, \texttt{ROLLBACK}, \texttt{SCRUB}, \texttt{MERGE_CAND}, \texttt{EXPORT}\}.$$

Here `STORE` *writes an object to an address,* `LINK` *writes a relation,* `SNAPSHOT` *freezes an admissible state,* `CLONE` *opens a guarded side branch from a snapshot,* `ROLLBACK` *returns to a recorded snapshot,* `SCRUB` *checks the guards and lineage,* `MERGE_CAND` *proposes a guarded merge final, and* `EXPORT` *exposes a readout surface. None of these instructions is allowed to manufacture a theorematic certificate by syntax alone.*

Definition 1.55 (MMA store transition semantics)**.** *The local MMA store semantics is a partial deterministic transition map*

$$\delta_{\text{store}} : \mathcal{S}_n \times \mathcal{I}_{\text{store}} \dashrightarrow \mathcal{S}_{n+1} \times \{\mathsf{continue}, \mathsf{halt}, \mathsf{reject}, \mathsf{guard}\}.$$

It is guard-faithful *if every defined transition either preserves admissibility of the resulting store or returns the status* guard *together with a defect entry in the ledger. It is* lineage-faithful *if each defined transition records enough ledger data to identify its source state, instruction, guard status, and output state.*

Definition 1.56 (Guarded snapshot and clone)**.** *For an admissible QAM store $\mathcal{S}_n$, a* guarded snapshot *is a tuple*

$$\text{snap}(\mathcal{S}_n) := (\mathcal{S}_n, \chi_n, \lambda_n),$$

where χ_n is a finite integrity witness for the address, relation, guard, and ledger registers, and λ_n is a lineage anchor. A clone *of* $\text{snap}(\mathcal{S}_n)$ *is a new working state $\mathcal{S}'_n$ with the same stored content and a new branch tag in the ledger. The clone is not a new theorem; it is a new guarded working branch from the same stored state.*

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Proposition 1.57 (Snapshot stability of admissible QAM stores)**.** *Every admissible finite QAM store state admits a guarded snapshot in the sense of Definition 1.56. If the MMA store semantics is guard-faithful and lineage-faithful, then* `SNAPSHOT`*,* `CLONE`*, and* `ROLLBACK` *preserve the distinction between stored mathematical objects and release/audit metadata.*

Proof. Since the address, object, relation, guard, and ledger registers are finite, their displayed state can be recorded as a finite tuple together with a finite integrity witness and a lineage anchor. This gives $\mathrm{snap}(\mathcal{S}_n)$. Guard-faithfulness prevents a failed guard from being silently stored as an admissible state, and lineage-faithfulness records the source and branch data. Thus snapshot, clone, and rollback may move between recorded store states, but they do not promote a release label, branch label, or audit label into the mathematical object language. □

Definition 1.58 (Sphaera array)**.** *A* Sphaera array *is a finite guarded family*

$$\mathbb{S}_n := \big((\mathcal{S}_n^{(1)}, \dots, \mathcal{S}_n^{(k)}); \Gamma_n, \Pi_n, \Lambda_n\big), \qquad k \geq 1,$$

where each $\mathcal{S}_n^{(i)}$ is an admissible QAM store state, Γ_n is the cross-sphere guard register, Π_n is the parity/residual witness register, and Λ_n is the array-level lineage ledger. The array is admissible if all component stores are admissible and all cross-sphere guards in Γ_n are satisfied.

Definition 1.59 (Array instruction alphabet)**.** *The local MML array alphabet is*

$$\mathcal{I}_{\mathrm{array}} := \{\texttt{ARRAY_CREATE}, \texttt{MIRROR}, \texttt{STRIPE}, \texttt{PARITY}, \texttt{SCRUB_ARRAY}, \texttt{REPAIR_CAND}, \texttt{SNAPSHOT_ARRAY}\}.$$

The command `REPAIR_CAND` *intentionally returns a final repair surface, not an automatic mathematical repair theorem. The command* `PARITY` *records a residual or orthogonal control witness, not a proof that the residual vanishes.*

Definition 1.60 (Mirror, stripe, and parity modes)**.** *Let $\mathbb{S}_n$ be an admissible Sphaera array.*

1. *In* mirror mode*, two component stores $\mathcal{S}_n^{(i)}$ and $\mathcal{S}_n^{(j)}$ are coupled by a cross-guard requiring equality of the relevant closure-compatible readout, even if the visible store presentations differ.*

2. *In* stripe mode*, an object X is stored as a finite guarded decomposition $X \rightsquigarrow (X_1, \dots, X_k)$, with each stripe X_i stored in a specified component and with a reconstruction witness recorded in Π_n.*

3. *In* parity/residual mode*, one component or register stores a residual witness P for a guarded comparison of other components. The condition $P = 0$, if ever needed, must be proved separately; it is not asserted by the existence of the parity register.*

Definition 1.61 (Triadic Sphaera array)**.** *A* triadic Sphaera array *is an admissible Sphaera array of the form*

$$\mathbb{S}_n^{ABC} := (\mathcal{S}_A, \mathcal{S}_B, \mathcal{S}_C; \Gamma_{ABC}, \Pi_{ABC}, \Lambda_{ABC})$$

in which $\mathcal{S}_A$ and $\mathcal{S}_B$ carry two active readout or probe-facing stores, while $\mathcal{S}_C$ carries the orthogonal residual/parity witness. The guard register records the structural condition that $\mathcal{S}_C$ is read orthogonally to the active A/B coupling channel and that the HC/MM readout of the array preserves the inherited route tag.

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Proposition 1.62 (Guarded array snapshot)**.** *Every admissible finite Sphaera array admits a guarded array snapshot*

$$\mathrm{snap}_{\mathrm{arr}}(\mathbb{S}_n) := \big(\mathbb{S}_n, (\chi_n^{(1)}, \dots, \chi_n^{(k)}), \chi_n^{\Gamma}, \chi_n^{\Pi}, \lambda_n^{\mathrm{arr}}\big),$$

preserving component stores, cross-sphere guards, parity/residual witnesses, and array lineage.

Proof. Apply Proposition 1.57 to each finite component store. Since the cross-guard, parity/residual, and array-ledger registers are also finite, record their integrity witnesses and one array lineage anchor. The resulting tuple is a guarded array snapshot. It records the array state without adding an OP-closure certificate and without changing any component object into metadata. □

Proposition 1.63 (Finite array scrub)**.** *For every finite Sphaera array there is a deterministic scrub procedure*

$$\mathrm{scrub}_{\mathrm{arr}}(\mathbb{S}_n) \in \{\mathsf{pass}, \mathsf{fail}\} \times \mathcal{D}_n,$$

where $\mathcal{D}_n$ is a finite defect register. The scrub passes exactly when all component-store guards, cross-sphere guards, parity/residual registrations, and lineage pointers pass their finite checks.

Proof. The component store registers, cross-guard register, parity/residual register, and lineage ledger are finite by definition. A deterministic scrub checks them in a fixed order and records every failed check in $\mathcal{D}_n$. If no failed check is recorded, the result is pass; otherwise it is fail with the finite defect register. No repair or closure claim is produced by this test. □

Theorem 1.64 (QAM/MML/MMA Sphaera array layer)**.** *The QAM/MML/MMA stack admits a conservative Sphaera array layer in the following sense. Given finite admissible QAM stores, the store instruction alphabet of Definition 1.54, the array instruction alphabet of Definition 1.59, and guard-faithful lineage-faithful MMA transition semantics, one obtains finite guarded snapshots, finite array snapshots, and deterministic array scrub surfaces. These surfaces preserve stored mathematical objects, cross-sphere guards, residual/parity witnesses, and audit lineage without manufacturing an OP-closure certificate.*

Proof. The store part follows from Definitions 1.53–1.56 and Proposition 1.57. The array part follows by grouping finitely many admissible stores as in Definition 1.58, assigning finite cross-guards, finite parity/residual witnesses, and a finite lineage ledger, and then applying Propositions 1.62 and 1.63. The last clause follows from the explicit non-certificate restrictions in Definitions 1.54, 1.59, and 1.60: mirror, stripe, parity, scrub, and repair-final language records infrastructure and diagnostic data only. It does not assert vanishing residuals or OP closure. □

Corollary 1.65 (RAID-like reading without RAID overclaim)**.** *The Sphaera array layer supports a RAID-like reading: mirror mode corresponds to redundant closure-compatible readout, stripe mode corresponds to guarded distribution of a larger object over several component stores, and parity/residual mode corresponds to an explicit diagnostic witness for cross-component mismatch. This reading is internal to QAM/MML/MMA, but it remains weaker than technical RAID reconstruction and weaker than any OP-closure theorem.*

Proof. This is only a translation of Definition 1.60 into systems language. The mathematical content remains the guarded store, snapshot, array, parity/residual, and scrub structure already proved above. □

Remark 1.66 (Tool extraction from the Sphaera array layer)**.** *The new layer extracts four reusable tools for later Companion revisions.*

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1. ***Snapshot tool.*** *Freeze one admissible QAM store without changing the mathematical object language.*

2. ***Clone/rollback tool.*** *Open or return from a guarded branch while preserving lineage and non-certificate status.*

3. ***Array-scrub tool.*** *Test several Sphaera stores, cross-guards, and residual witnesses as one finite diagnostic surface.*

4. ***Triadic parity tool.*** *Use a third, orthogonal Sphaera object as residual/parity witness for a two-store active coupling.*

These tools are deliberately infrastructural. Their value is that later OP-facing work can compare, branch, and roll back several coupled Sphaera objects without confusing state persistence with theorematic closure.

Remark 1.67 (Human/LLM interface gain)**.** *For a human reader, the store-array layer explains why the Companion keeps snapshots, historical witnesses, audit ledgers, and branch restrictions so visibly: they are the theorem-store equivalent of protected state persistence. For an LLM reader, the same layer supplies exact handles: $\mathcal{S}_n$ for one store, $\mathbb{S}_n$ for an array, Γ_n for cross-guards, Π_n for parity/residual witnesses, and Λ_n for lineage. This is the intended compromise between human readability and machine precision.*

Pre-shift QAM Hilbert-shift and guarded mixed readouts

> **Status.** *Pre-shift preparation layer only.* The inherited QAM tuple tags are not globally migrated in Version 4.7.0. The block below records the conservative Hilbert-shift plan, the mixed-readout package discipline, and an optional binary mask registry for future readout classification.

The index-0 issue is now treated explicitly. In the inherited QAM tuple namespace, tag 0 occurs as an inhabited constructor tag, for instance on visible tuple surfaces of the form $\langle 0, a\rangle$. In a Boolean mask or IPshifted-like readout registry, however, value 0 must mean no active kind. The next major QAM migration should therefore not identify these two uses. The preparation layer below records the planned Hilbert-hotel shift that can free tag 0 in the future while preserving the payloads and proofs.

Definition 1.68 (QAM Hilbert-shift plan)**.** *Let $\mathcal{K}_{\mathrm{old}}$ be the inherited finite QAM tuple-tag namespace used by the current public syntax, and let $\mathcal{K}_{\mathrm{new}}$ be a shifted namespace with tag 0 reserved for the empty/no-active-kind marker. The planned Hilbert-shift is the constructor-level map*

$$H_{\mathrm{QAM}}(\langle i, x_1, \ldots, x_m\rangle) := \langle i+1, x_1, \ldots, x_m\rangle.$$

It shifts only the constructor tag. It does not change the stored payload $(x_1, \ldots, x_m)$, the intended constructor distinction, or any mathematical content carried by the payload.

Definition 1.69 (Pre-shift reserved-zero discipline)**.** *The preparation line distinguishes two uses of zero.*

1. *In the inherited legacy tuple syntax, tag 0 remains an inhabited tuple tag wherever the old QAM definitions still use it.*

2. *In the planned shifted shifted namespace and in every binary mask registry, tag or mask value 0 is reserved for empty, undefined, or no active kind.*

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No Version 4.7.0 statement may silently replace the first convention by the second. Any global replacement is reserved for a later shifted migration theorem.

Proposition 1.70 (Conservative shape of the QAM Hilbert-shift)**.** *Suppose a finite QAM sublanguage is translated by H_{QAM} and every constructor predicate, admissibility predicate, readout predicate, and closure-relevant predicate is translated by the same tag shift. Then the translated sublanguage is a conservative reindexing of the old one: payloads, constructor distinctions, admissibility status, readout compatibility, and closure-relevant truth values are preserved under the translation.*

Proof. The map H_{QAM} is injective on constructor tags and leaves all payload coordinates unchanged. A constructor predicate in the shifted namespace is defined to hold exactly when the corresponding old predicate holds before applying the shift. The same translation is imposed on admissibility, readout, and closure-relevant predicates. Thus every proof step that used only constructor distinction, payload data, admissibility, readout compatibility, or closure-relevant class data has an isomorphic shifted proof step. The proposition records the required conservativity shape for a later shifted migration; it does not perform the global migration here. □

Definition 1.71 (Guarded mixed QAM readout)**.** *A* guarded mixed QAM readout *is a finite derived package*

$$R_{\mathrm{mix}} := (r_1, \dots, r_m; \Gamma_{\mathrm{mix}}, \mu, J_{\mathrm{mix}}, \sigma_{\mathrm{mix}}),$$

where each r_i is an already admissible component readout on its own allowed QAM kind, Γ_{mix} is a finite guard package, μ is a finite aggregation map on component readout values, J_{mix} is the closure-factor map for the aggregate, and σ_{mix} records the theorematic status. It is admissible only when all component readouts are admissible and the aggregation map does not identify non-equivalent closure-relevant classes.

Definition 1.72 (Binary mask registry for mixed readouts)**.** *A* binary mask registry *for mixed QAM readouts assigns to each inhabited readout kind a nonzero power-of-two bit*

$$\beta_j = 2^j, \qquad j \geq 0,$$

and reserves mask value 0 for no active kind. A mixed-readout mask is a finite Boolean OR of active kind bits. Boolean AND tests whether a component kind is present, XOR records symmetric difference between two masks, and complement is allowed only relative to a fixed finite mask universe Ω.

Proposition 1.73 (Mixed readout factorization)**.** *Let R_{mix} be a guarded mixed QAM readout whose component readouts r_i are closure-compatible. If the aggregation map μ factors through the component closure-factor maps and J_{mix} factors through the resulting aggregate class, then R_{mix} is closure-compatible.*

Proof. Closure-compatible component readouts identify no closure-relevant distinction beyond their component class maps. Since μ factors through those class maps, it can only aggregate already class-level data. Since J_{mix} factors through that aggregate class, the mixed readout cannot introduce a new closure-relevant distinction that was invisible to the component closure classes. Therefore the mixed package is closure-compatible. The argument uses only factorization and guards; it does not introduce primitive mixed tensors. □

Remark 1.74 (shifted migration trigger)**.** *The full replacement of the inherited tuple tags by the shifted tags is intentionally not a Version 4.7.0 operation. It changes the visible QAM object-language namespace and therefore belongs to a major version. Version 4.7.0 only records why the shift is useful, what the conservative proof obligation looks like, and how mixed readouts can already be treated as derived packages without forcing the migration.*

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Two-sided structural tensor notation for QAM/MML/MMA readouts

Status. *Active notation and compression layer; not a new tensor calculus and not a new closure criterion.*

Block function. This block records the narrow two-sided tensor grammar used to separate readout-side, structure-side, comparison-side, and mask-side roles in later QAM/MML/MMA expressions. It is included here as a conservative technical layer only. A fuller pedagogical reader manual for the notation is intentionally kept as a separate future paper or companion guide.

The QAM/MML/MMA line repeatedly alternates between four roles: frame-side readout, structure-side action, comparison of active channels, and compatibility testing of readout or mask modes. The following notation is introduced only to make those roles visible in the formula syntax. It does not replace classical tensor calculus and it does not turn every displayed expression into a new primitive tensor object.

Definition 1.75 (Two-sided structural tensor interface)**.** *A* two-sided structural tensor display *is a tensor-like expression in which a structural object may carry front-side and back-side indices, written schematically as*

$$ {}^{A}{}_{B}\mathcal{T}^{i}{}_{j}. $$

The front indices ${}^{A}{}_{B}$ record the left-sided or readout-facing interface: frame, observer, language, readout, or external addressing data. The back indices ${}^{i}{}_{j}$ record the right-sided or structure-facing interface: action, transport, field, operator, or internal structural data. Thus the display is read as a typed structural interface rather than as a mere component array.

Definition 1.76 (Admissible contractions and comparisons)**.** *The two-sided notation obeys the following contraction discipline.*

1. ***Classical action/readout.*** *Ordinary upper–lower contraction remains unchanged. If*

$$ A_{\mu}\,{}^{\mu}\mathcal{T}_{\nu}\,B^{\nu} $$

is well-defined, it is read as a classical action, evaluation, or sandwich readout: first apply the structure-facing operation to B, then read the result by A.

2. ***Active–active comparison.*** *Two upper indices are not contracted as a raw Einstein sum. They may only be compared through an explicitly declared comparison tensor, for instance*

$$ G_{\alpha\beta}X^{\alpha}Y^{\beta}. $$

This means active structural comparison, such as angle, distance, coupling strength, orthogonality, channel similarity, or residual size.

3. ***Passive–passive comparison.*** *Two lower indices are likewise not contracted raw. They may only be compared through an explicitly declared dual comparison tensor, for instance*

$$ G^{\alpha\beta}\omega_{\alpha}\eta_{\beta}. $$

This means comparison of readout forms, observer modes, measurement channels, or passive projections.

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4. ***Mask and selector compatibility.*** *Boolean, finite, or mask-like compatibility tests must be carried by an explicit selector tensor, for instance*

$$\chi_{mn} R^m S^n.$$

This expression says that readout modes m and n are first tested for admissible compatibility by χ, and only compatible contributions are admitted to the mixed readout.

Remark 1.77 (Conservative reading rule)**.** *The two-sided notation is conservative because the only unchanged contraction is the classical upper–lower action/readout contraction. Same-variance pairings are not silently added to the calculus; they are admitted only when a comparison tensor or selector tensor has been declared. Hence every two-sided display must be expandable into ordinary QAM-over-ZF role predicates together with explicit comparison or compatibility data.*

Definition 1.78 (Guarded mixed QAM readout)**.** *Let R^m and S^n be QAM readout modes and let χ_{mn} be an explicit admissibility or compatibility selector. A* guarded mixed QAM readout *is a derived readout expression of the form*

$$\mathrm{Mix}_\chi(R,S)(x) \quad \rightsquigarrow \quad \chi_{mn} R^m(x) S^n(x),$$

where the right-hand display is read with the contraction discipline of Definition 1.76. The mixed readout is therefore not a new primitive QAM object. It is a compatibility-filtered package of already admitted readout modes.

Remark 1.79 (Residual comparison grammar)**.** *The live residual triple*

$$\rho = (\rho_{\mathrm{CL}}, \rho_\Theta, \rho_A)$$

may be compressed into a residual-channel vector ρ^α. Its total comparison size may then be written as

$$\|\rho\|_G^2 = G_{\alpha\beta}\rho^\alpha \rho^\beta.$$

This is not a raw contraction of two upper indices. It is the comparison of two active residual-channel entries through the declared comparison tensor $G_{\alpha\beta}$.

Remark 1.80 (HC/MM sandwich grammar)**.** *An HC/MM-mediated readout may be displayed schematically as*

$${}^{R}{}_{\Omega}\mathcal{M}^{i}{}_{j},$$

where the front side records the readout or observer-side interface and the back side records the structure-side transport or matrix action. When paired with ordinary upper–lower indices, the expression remains an ordinary sandwich readout. When two readout channels or two structure channels are compared, the comparison must be mediated by an explicitly named comparison tensor or selector.

Proposition 1.81 (Conservative expansion of two-sided displays)**.** *Every well-formed two-sided structural tensor display in the sense of Definitions 1.75 and 1.76 admits a conservative expansion into the inherited QAM-over-ZF language: front-side indices expand to readout/frame/language role data, back-side indices expand to structure/action/transport role data, classical upper–lower contractions remain ordinary action or evaluation clauses, and every same-variance pairing expands to an explicit comparison or compatibility predicate.*

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Proof. Erase the visual sidedness notation and retain its role data as ordinary QAM codes. Front-side symbols become readout, frame, observer, or language tags. Back-side symbols become structure, action, transport, or field tags. Classical upper–lower contractions are left unchanged. A same-variance display is well formed only if it contains a named comparison tensor or selector, and so it expands to the corresponding comparison or compatibility predicate. No new primitive closure relation, admissibility relation, or theorematic certificate is introduced by this erasure. The display is therefore a conservative abbreviation of already typed QAM-over-ZF data. □

Remark 1.82 (LLM-facing role stability)**.** *For machine reading and LLM-assisted proof navigation, the notation functions as a syntactic guardrail. The front/back split marks whether a symbol belongs to the readout-facing side or the structure-facing side, while the contraction discipline marks whether a formula is an action, a readout, a comparison, or a compatibility test. This reduces the risk that a readout is mistaken for an operator, that a diagnostic comparison is mistaken for a proof of closure, or that a guarded mixed readout is mistaken for a new primitive tensor object.*

Remark 1.83 (Use in the present release line)**.** *In the present historical release line.x line this notation should be used locally and sparingly: first for guarded mixed QAM readouts, residual/drift comparisons, HC/MM sandwich expressions, triadic orthogonality comparisons, and Boolean or mask-like compatibility tests. A global rewrite of the whole Companion into the two-sided notation is not part of this release line. The separate pedagogical reader manual may later explain the notation at a slower pace without enlarging the active theorem core.*

stability layer: two-sided expansion, mixed-readout normal form, and Boolean masks

> **Status.** *Conservative stabilization of the notation layer; not a global rewrite, not a new tensor calculus, and not a closure theorem.*
>
> **Block function.** This block fixes the exact erasure route from two-sided displays back to ordinary QAM-over-ZF role data, normalizes guarded mixed readouts, and states the Boolean selector grammar used by IPshifted-like readout masks. The separate reader manual remains a future standalone paper or guide.

The notation made readout-side, structure-side, comparison-side, and mask-side roles visible in formulas. The present step turns that notation into a safer tool by recording the normal forms that an audit reader or LLM reader should recover whenever a two-sided display appears. The rule is intentionally asymmetric: notation may compress role data, but every compressed display must expand back into explicit role data before it can be used as proof content.

Definition 1.84 (Two-sided role-erasure map)**.** *Let E be a well-formed two-sided structural tensor display in the sense of Definitions 1.75 and 1.76. Its* role-erasure expansion

$$\mathfrak{E}_{\mathrm{role}}(E)$$

is the ordinary QAM-over-ZF package obtained by replacing:

1. *every front-side index by an explicit readout, frame, observer, or language-role tag;*
2. *every back-side index by an explicit structure, action, field, transport, or operator-role tag;*

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3. *every classical upper–lower contraction by the corresponding ordinary action/evaluation clause;*

4. *every same-variance comparison by a named comparison predicate carrying its declared comparison tensor;*

5. *every Boolean or selector occurrence by a finite compatibility predicate over the declared mask universe.*

This expansion is an erasure of notation only. It does not erase guards, selectors, comparison tensors, branch restrictions, or provenance data.

Proposition 1.85 (Role-erasure normal form)**.** *Every well-formed two-sided display E used in the historical release line.x line has a role-erasure expansion $\mathfrak{E}_{\mathrm{role}}(E)$ whose clauses are built from already admitted QAM role predicates, admitted comparison data, admitted selector data, and ordinary finite tuple coding over ZF. In particular, the two-sided display cannot by itself introduce a new primitive tensor kind, a new admissibility relation, or a new theorematic certificate.*

Proof. Proceed by structural inspection of the display grammar. Front/back sidedness contributes only role tags. Ordinary upper–lower contraction is already part of the inherited action/readout grammar. A same-variance pairing is well formed only when it contains an explicitly named comparison tensor or selector, hence it expands to a comparison or compatibility predicate rather than to an untyped Einstein contraction. All finite lists of such clauses are QAM tuple-coded over the existing ZF substrate. Therefore the two-sided syntax is conservative under role erasure. □

Definition 1.86 (Guarded mixed-readout normal form)**.** *A guarded mixed QAM readout is in* normal form *when it is presented as a finite package*

$$\mathcal{R}_{\mathrm{mix}} = \big(I, \{R_i\}_{i\in I}, \Gamma_{\mathrm{mix}}, \chi, \mu, J_{\mathrm{mix}}\big),$$

where I is a finite index set, each R_i is an already admitted component readout, Γ_{mix} is the inherited guard package, χ is a finite selector or mask predicate over I, μ is an aggregation map defined only on selector-admissible component families, and J_{mix} is the declared mixed readout surface. A two-sided display such as

$$\chi_{mn} R^m S^n$$

is only an abbreviation for this finite package in the binary case $I = \{m, n\}$.

Proposition 1.87 (Mixed-readout normal-form expansion)**.** *Every guarded mixed readout admitted by Definition 1.78 expands to a guarded mixed-readout normal form in the sense of Definition 1.86. Conversely, every guarded mixed-readout normal form may be displayed in two-sided shorthand only after its guards, selector, aggregation map, and output surface have been declared.*

Proof. The binary shorthand $\chi_{mn} R^m S^n$ already names the two component readouts and the selector. The surrounding QAM context supplies the inherited guard package, the admissible aggregation rule, and the declared output surface. Collecting these data gives the normal-form tuple. Conversely, starting from the normal-form tuple, one may suppress the finite package into two-sided shorthand only because the suppressed guard, selector, aggregation, and output data remain recoverable. Thus the shorthand and the normal form are not different mathematical objects; the former is a compact notation for the latter. □

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Definition 1.88 (Boolean selector and mask grammar)**.** *Fix a finite readout-kind universe* $\Omega = \{k_1, \ldots, k_N\}$*. A* Boolean selector mask *is a function*

$$\chi : \Omega \longrightarrow \{0, 1\}$$

or, equivalently, a finite bit mask whose value 0 denotes no active kind and whose nonzero bits denote active readout kinds. The admissible Boolean operations are:

$$\begin{aligned} \chi_A \vee \chi_B &\quad \textit{for union of active readout kinds,} \\ \chi_A \wedge \chi_B &\quad \textit{for intersection or compatibility testing,} \\ \chi_A \oplus \chi_B &\quad \textit{for diagnostic symmetric difference,} \\ \neg_\Omega \chi_A &\quad \textit{for relative complement inside the fixed universe } \Omega. \end{aligned}$$

No absolute complement is used without first declaring Ω*. A selector expression may filter a readout package, but it does not alter the payload of any component readout.*

Proposition 1.89 (Mask-filter conservativity)**.** *Let* $\mathcal{R}_{\mathrm{mix}}$ *be a guarded mixed-readout normal form and let* χ *be a Boolean selector mask over its finite readout-kind universe. Applying* χ *restricts the active component menu of* $\mathcal{R}_{\mathrm{mix}}$*, but it does not change the underlying component payloads, the inherited guards, or the closure status of any component readout.*

Proof. A selector mask is a finite predicate on the index universe. It chooses which already declared component clauses are active in the aggregation domain. It neither rewrites the component readouts nor changes their guard packages. Hence any closure-compatible status used by the mixed package still has to come from the component readouts and the declared factorization map, not from the mask itself. □

Remark 1.90 (LLM/audit role table)**.** *For later machine reading, the following table fixes the intended role parsing. It is a local audit aid inside the Companion, not the separate pedagogical reader manual.*

Displayed form	***Intended role***	***Audit guard***
front-side indices	*readout, frame, observer, language, or addressing side*	*expand to explicit QAM role tags*
back-side indices	*structure, action, field, transport, or operator side*	*expand to explicit QAM role tags*
upper–lower contraction	*classical action, evaluation, or sandwich readout*	*ordinary tensor/QAM readout clause*
upper–upper via $G_{\alpha\beta}$	*active structural comparison*	*no raw same-variance contraction*
lower–lower via $G^{\alpha\beta}$	*passive readout comparison*	*no raw same-variance contraction*
χ_{mn} *or a mask bit*	*compatibility or activation selector*	*filter only; no new payload*

Corollary 1.91 (proof-neutrality of the tensor layer)**.** *The tensor-stability layer may be cited to shorten repeated readout/action/comparison/mask passages, but it may not be cited as an independent proof of closure, OP resolution, primitive mixed-tensor existence, or global QAM namespace migration.*

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Proof. By Propositions 1.85, 1.87, and 1.89, every display reduces to already declared role data, normal-form mixed-readout data, and finite selector predicates. The reduction supplies notation control only. It does not add the missing theorematic hypotheses needed for closure, OP resolution, primitive mixed tensors, or shifted reindexing. □

Remark 1.92 (Placement of the future reader manual)**.** *The present block deliberately stays short and formal. A slower reader manual may later explain the same grammar with diagrams, examples, and human-facing intuition, but that manual should remain a separate paper or guide so that the Active Core remains the governing proof line.*

release-final freeze audit for the pre-shift QAM/tensor layer

Status. *Release-final audit layer only; no new theorematic certificate, no global QAM namespace migration, no primitive mixed tensor, and no OP closure.*
Block function. This block freezes the local pass/notation and mixed-readout preparation layer for release-final review. It records what must remain true when the Active Core is read together with the separate Historical Witness PDF.

Definition 1.93 (pre-shift QAM/tensor release-final package)**.** *The* pre-shift QAM/tensor release-final package *consists of the following finite data:*

(i) the inherited archival stage public split-bundle baseline and its concept/version DOI record;

(ii) the archival stage short-abstract and extended-orientation architecture;

(iii) the pre-shift Hilbert-shift plan reserving global QAM tag migration for a later major version;

(iv) the guarded mixed-readout discipline and optional binary mask registry;

(v) the two-sided structural tensor notation;

(vi) the role-erasure, mixed-readout normal form, Boolean selector grammar, and proof-neutrality layer;

(vii) the present audit/handoff metadata saying that the Active Core is the governing proof line and that the Historical Witness PDF remains preserved proof memory.

It is a packaging and audit object, not a closure object.

Proposition 1.94 (release-final conservativity)**.** *The pre-shift QAM/tensor release-final package is conservative over the tensor-stability layer with respect to theorematic content. It may change release-final metadata, routing language, audit surfaces, and handoff wording, but it may not create any new OP certificate, global Hilbert-shift theorem, primitive mixed-tensor axiom, or replacement of the inherited QAM tuple namespace.*

Proof. Each mathematical role introduced earlier is erased by Definition 1.84 and normalized by Proposition 1.85. Each mixed readout is expanded by Proposition 1.87, and each Boolean mask action is restricted by Proposition 1.89. The package adds only release-final metadata and bundle-handoff statements around these already guarded objects. Hence no new theorematic content is introduced. □

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Corollary 1.95 (freeze-readiness condition)**.** *If the build audit reports no duplicate labels, no undefined references, no undefined citations, and no fatal LaTeX errors for both the Active Core and the Historical Witness, then Version 4.7.0 is structurally ready to be treated as a freeze final for the archival stage pre-shift QAM/tensor preparation release, subject only to the external release decision and any assigned Zenodo version DOI.*

Remark 1.96 (Reader-manual separation at)**.** *The pedagogical reader manual for the two-sided notation remains outside the Active Core. It may later be written as a separate guide with diagrams, examples, and slower explanations. The present Companion keeps only the compact formal grammar and the proof-neutrality guards needed by the governing proof line.*

archival stage DOI-bearing release freeze

Status. *Release-final metadata layer only; no new theorematic certificate, no global QAM namespace migration, no primitive mixed tensor, and no OP closure.*
Version DOI. The public version-specific Zenodo DOI for this release is

$$10.5281/\textit{zenodo}.19750674.$$

The Companion concept DOI remains 10.5281/zenodo.19210728. The version DOI is release metadata only and is not a mathematical subscript, superscript, route tag, or proof object.

Definition 1.97 (archival stage release-final package)**.** *The* archival stage release-final package *consists of the release-final package of Definition 1.93, the assigned public version DOI above, the Active Core PDF, the preserved Historical Witness PDF, the release manifest, checksums, audit report, Zenodo description, and release notes. It is a release object, not a theorematic closure object.*

Proposition 1.98 (archival stage release-final conservativity)**.** *The archival stage release-final package is conservative over the release-final package with respect to theorematic content. DOI insertion, release-note generation, manifest creation, checksum creation, and Zenodo-description preparation do not create any new OP certificate, global QAM Hilbert-shift theorem, primitive mixed-tensor axiom, or replacement of the inherited QAM tuple namespace.*

Proof. By Proposition 1.94, the package is conservative over the local pass. The present layer adds only release metadata and publication packaging: the version DOI, release file names, manifest/checksum records, audit report, Zenodo description, and release notes. None of these objects changes the payload of any QAM role, tensor grammar, mask selector, mixed readout, or OP-facing proof obligation. Hence theorematic content is unchanged. □

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source handoff and router-language layer

Status. This archival stage layer is a conservative local TEX handoff over the public archival stage release-final baseline. It records the router-language, route-mask, HC-boundary, theorem-layer, and active-readout architecture as a compact source layer. It is not an OP closure, not a global shifted QAM namespace migration, and not a primitive mixed-tensor activation.

Definition 1.99 (handoff object)**.** *The* the local proof line handoff object *is the finite record*

$$\mathfrak{H}_{\mathrm{loc}} = (B_{3.35}, Q_{\mathrm{loc}}, M_{\mathrm{loc}}, T_{\mathrm{loc}}, P_{\mathrm{loc}}, G_{\mathrm{loc}}),$$

where $B_{3.35}$ is the public archival stage release-final baseline, Q_{loc} is the QAM/MML/MMA router-language package, M_{loc} is the route-mask and double-Hilbert-shift package, T_{loc} is the canonical theorem-layer package, P_{loc} is the active physical and mathematical readout package, and G_{loc} is the non-overclaim guard record.

Definition 1.100 (Router-language role split)**.** *The local router-language split is the guarded reading*

$$\begin{aligned} \mathrm{QAM} &= \textit{object and store grammar}, \\ \mathrm{MML} &= \textit{routing and command grammar}, \\ \mathrm{MMA} &= \textit{execution and readout grammar}. \end{aligned}$$

The combined router layer is the conservative compiler surface

$$\mathcal{R}_{\mathrm{QAM/MML/MMA}} : \textit{guarded object packages} \longrightarrow \textit{guarded readout packages}.$$

Remark 1.101 (Machine layer is not physical content)**.** *Definition 1.100 is a routing definition. It does not promote a router object to a physical observable, does not add a field equation, does not create a closure certificate, and does not replace the underlying tensor calculus. Physical content appears only after a guarded readout has a declared target, collar or boundary interface, comparison or residual map, and back-translation.*

Definition 1.102 (Router object)**.** *A local router object is a finite record*

$$\mathfrak{R} = (K, M, C, T, B, G),$$

where K is a finite kind registry, M is a finite route-mask registry, C is a finite classical-collar record, T is a finite trace record, B is a back-translation record into the prior source language, and G is the guard record.

Definition 1.103 (Guard record for router objects)**.** *The guard record G states which transformations are purely notational, which readouts are inherited from earlier admissible packages, and which claims are not being made. In this layer, G includes at least the statements that OP 1/4 and OP 5(B) remain open, primitive mixed tensors remain inactive, Turing-completeness is not claimed, and the global shifted QAM Hilbert-shift has not yet been performed.*

Lemma 1.104 (Router-object erasure)**.** *Every router object in the sense of Definition 1.102 has a proof-neutral erasure to its underlying guarded source package.*

Proof. The entries K, M, C, T, and B are finite routing, masking, collar, trace, and back-translation data. Removing these annotations leaves the underlying source package together with the guard record G. Since no new observable, field equation, or closure certificate is added by the router layer, the erasure is proof-neutral. □

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Remark 1.105 (Relation to two-sided structural tensor notation)**.** *When a router object is written with two-sided structural tensor notation, front-side indices mark frame, readout, or language roles, while back-side indices mark structure, action, or field roles. Classical upper–lower contraction remains the ordinary action/readout case. Same-variance comparison still requires an explicit comparison or selector tensor. Thus the router notation remains a conservative compression grammar over the local proof chain tensor layer.*

double-Hilbert-shift route-mask layer

Definition 1.106 (Double-Hilbert-shift route-mask convention)**.** *The local route-mask convention is*

$$H_0(k) = k + 1, \qquad H_1(a) = 2^{a-1} \quad (a \geq 1), \qquad H_{\mathrm{mask}}(k) = 2^k.$$

The value 0 *is reserved for the inactive or no-active-kind marker. A nonzero route mask is read through its finite bit support and can be back-translated into the corresponding prior unshifted kind labels.*

Definition 1.107 (Mask support and back-translation)**.** *For a finite nonzero mask* m*, let* $\mathrm{bits}(m)$ *be the finite set of active bit atoms in* m*. The back-translation map is*

$$\mathrm{back}(m) = \{a - 1 : a \in \mathrm{bits}(m)\}.$$

For an ordered route payload $(m_1, \ldots, m_n)$*, the route back-translation is applied componentwise:*

$$\mathrm{BackRoute}(m_1, \ldots, m_n) = (\mathrm{back}(m_1), \ldots, \mathrm{back}(m_n)).$$

Lemma 1.108 (Double-shift conservativity)**.** *The double-Hilbert-shift route-mask convention is conservative over the prior finite route-label language, provided that every active mask is decoded by Definition 1.107 before theorematic use.*

Proof. The first shift reserves the inactive state by separating 0 from inhabited labels. The second shift atomizes active labels into powers of two. Since every finite mask has finite bit support and the back-translation recovers the corresponding prior labels, the route-mask layer adds a finite audit encoding but does not alter the underlying guarded route package. Hence theorematic status is preserved. □

first boundary shift and HC activation

> **Purpose.** This subsection separates the local first shift from the later global shifted QAM Hilbert-shift. The local shift belongs to archival stage because it explains why the HC can be read as internally generated at the inactive/active boundary. The global reindexing of the full QAM language is still deferred to shifted.

Definition 1.109 (First boundary shift)**.** *The* first boundary shift *is the local boundary transition*

$$\Delta_{\mathrm{HC}} : 0_{\mathrm{mask}} \rightsquigarrow 1_{\mathrm{mask}},$$

from the inactive mask state to the first active mask atom. It is not a migration of the full QAM tag language. It is only the smallest local passage by which the readout grammar distinguishes no-active-kind from first active readout.

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Definition 1.110 (Boundary normalizer)**.** *A* boundary normalizer *for the first boundary shift is a structural readout normalizer* N_{bd} *assigned to the interface*

$$(0_{\mathrm{mask}}, 1_{\mathrm{mask}}).$$

It is required to preserve the inactive side as inactive, admit the first active readout atom as active, and prevent the transition from being interpreted as an uncontrolled same-variance contraction or as an unguarded physical promotion.

Assumption 1.111 (HC activation at the first boundary)**.** *At the first boundary shift, the minimal boundary normalizer is read as the Heisenberg compensator:*

$$N_{\mathrm{bd}} = HC \quad \textit{at the structural readout level.}$$

This is an internal activation reading of the HC inside the route-mask grammar, not a replacement proof of the physical Heisenberg uncertainty relation.

Proposition 1.112 (Internal origin of the first HC shift)**.** *Under Assumption 1.111, the first boundary shift does not enter the archival stage architecture as an external add-on. It is forced as the local normalization site between the reserved inactive mask state and the first active readout atom.*

Proof. The route-mask grammar reserves 0_{mask} for no active readout kind and reads 1_{mask} as the first active mask atom. The transition between them is therefore not an ordinary active-active comparison, because one side is not yet active. It is also not an ordinary action/readout contraction, because there is no prior active object on the inactive side to evaluate. A normalizing interface is therefore required before the first active readout can be admitted. By the boundary-activation assumption, this minimal interface is read as HC. Hence the first shift is structurally internal to the readout machine, while the global QAM Hilbert-shift remains deferred. □

Corollary 1.113 (No shifted migration from the first shift)**.** *The first boundary shift licenses only the local HC-boundary reading. It does not perform the global shifted migration of inhabited QAM tags and does not change any inherited theorematic status.*

Proof. Definition 1.109 is local to the pair $(0_{\mathrm{mask}}, 1_{\mathrm{mask}})$. No rule is introduced that rewrites all inhabited tags, replaces inherited labels, or promotes router metadata to physical closure. Therefore the global shifted migration remains separate. □

HC half-boundary reading

Definition 1.114 (HC half-boundary readout)**.** *The* HC half-boundary readout *treats the transition*

$$0_{\mathrm{mask}} \longrightarrow 1_{\mathrm{mask}}$$

from the inactive mask state to the first active shifted mask atom as a boundary-normalization site. The forbidden formal value $H_1(0) = 1/2$ *is not a Boolean mask atom. It marks the structural edge at which the Heisenberg compensator may be read as the minimal frame/readout normalizer required for active readout emergence.*

Remark 1.115 (Non-Boolean boundary lift)**.** *The formal lift* $H_{\mathrm{mask}}(1/2) = \sqrt{2}$ *is a diagnostic analogy for the boundary, not a Boolean route-mask value. It must not be used as a bit, as a closure certificate, or as a proof of the physical Heisenberg uncertainty relation.*

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Assumption 1.116 (HC-boundary reading as structural hypothesis)**.** *The HC half-boundary reading is treated as a conservative structural hypothesis: the Heisenberg compensator may be interpreted as the boundary compensator at the inadmissible pre-mask boundary between inactive state* 0 *and first active shifted mask atom* 1*, but this interpretation is not yet a theorem of physical dynamics.*

router trace and bitstream layer

Definition 1.117 (Router trace)**.** *A* router trace *is the finite diagnostic record obtained by reading a route-mask payload as a bitstream together with its ordered back-translation, classical collar, and guard record. It records how a finite router object moves from stored object data to guarded readout data without changing theorematic status.*

Definition 1.118 (Bittrace normal form)**.** *For a route pair* (m, n)*, the bittrace normal form is*

$$\mathrm{BitTrace}(m, n) = \big(\mathrm{back}(m) \to \mathrm{back}(n),\, G\big),$$

where G *is the inherited guard record. The arrow records diagnostic flow only; it is not a proof of admissibility unless the underlying guarded source package already supplies admissibility.*

Lemma 1.119 (Router trace conservativity)**.** *Passing from a finite route-mask payload to its router trace is conservative with respect to OP status.*

Proof. The router trace records finite bit support, ordered back-translation, collar data, and guards. It neither removes the guard record nor supplies a missing closure trigger. Therefore OP 1/4 and OP 5(B) keep the status they had before the trace was introduced. □

canonical theorem-layer skeleton

Definition 1.120 (Canonical theorem-layer package)**.** *The local canonical theorem-layer package consists of the following theorem classes:*

1. *router representation and router-object erasure;*
2. *route-mask conservativity and double-shift back-translation;*
3. *router chaining and classical collar compatibility;*
4. *HC-boundary activation as a guarded structural reading;*
5. *bitstream and route-difference normal forms;*
6. *boundary-readout projection packages;*
7. *physical promotion criteria.*

Theorem 1.121 (Canonical router handoff)**.** *Every finite router-language package admits a canonical handoff into the active readout layer by separating proof-neutral routing metadata from guarded readout, comparison, boundary, and residual content.*

Proof. The package is finite by Definition 1.102. Its masks and traces are audit surfaces, while its collars, comparisons, boundary interfaces, and residual maps are the possible content surfaces. Separating these roles gives the claimed handoff. The guard record of Definition 1.103 prevents metadata from being read as a closure proof. □

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active physical and mathematical readout layer

Definition 1.122 (Active readout package)**.** *The* active physical and mathematical readout package *is the finite record*

$$\mathcal{P}_{\mathrm{loc}} = (P_{\mathrm{HC}}, P_{\gamma}, P_{\mathrm{sing}}, P_{\mathrm{Cas}}, P_{\mathrm{OP}}, P_{\mathrm{auto}}, P_{\mathrm{ws}}),$$

where P_{HC} records the HC half-boundary readout, P_{γ} records photon-compatible null-side readout, P_{sing} records HC-normalized singularity projection, P_{Cas} records Casimir-type boundary readout, P_{OP} records OP-facing residual packages, P_{auto} records finite router automaton and bitstream packages, and P_{ws} records worksheet-style physical readout tests.

Definition 1.123 (Physical promotion criterion)**.** *A router-level object may be physically promoted only if it supplies:*

1. *a declared physical or mathematical readout target;*
2. *a classical collar or explicit boundary interface;*
3. *a comparison, residual, observable, or projection map;*
4. *a back-translation into the prior source language;*
5. *an explicit non-overclaim statement describing which OPs remain open.*

Proposition 1.124 (Machine/content separation)**.** *In the architecture, QAM/MML/MMA is the router, language, and compiler layer. It is not itself the physical content. Physical content begins only where a guarded readout package is promoted by Definition 1.123 into the active readout package of Definition 1.122.*

Proof. The router layer stores kinds, masks, routes, collars, traces, and back-translations. None of these entries is by itself a physical observable. The active readout package adds a physical or mathematical target only after the promotion guard has been supplied. Hence the machine layer and the physical layer are distinct, even though the former stabilizes and compresses the latter. □

Theorem 1.125 (active-layer handoff)**.** *The local proof line handoff object defines a well-formed active-layer final over the archival stage release-final baseline: the QAM/MML/MMA material functions as router and theorem language, while the active physical and mathematical readout layer is isolated as the content-bearing surface.*

Proof. Definitions 1.100, 1.106, 1.120, and 1.122 separate router data, mask data, theorem-layer data, and readout data into finite records. Lemma 1.104, Lemma 1.108, and Lemma 1.119 preserve theorematic status. Proposition 1.124 separates infrastructure from physical content. Therefore the source-layer update is well formed as an active-layer final while OP 1/4 and OP 5(B) remain open. □

guard record. This layer does not assert a global shifted QAM Hilbert-shift, does not introduce primitive mixed tensors, does not close OP 1/4 or OP 5(B), does not replace the standard physical calculations of QFT/QED/Lifshitz theory or general relativity, and does not claim Turing-completeness. It records the router, route-mask, HC-boundary, canonical theorem, and active-readout architecture as a controlled release-final.

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Remark 1.126 (How the systems view reads the present visual diagnostics)**.** *The same stack view also explains why Appendix C may carry highly visible packet, Julia-boundary, and frame-flicker diagnostics without turning those figures into primary closure axioms. In the systems reading, such figures live on the visible-view layer: they are auditable readout surfaces of one and the same transported channel, not independent kernel identities. This is exactly why the OP 5 packet/breathing/frame-flicker block and the Julia-boundary block belong naturally to the same document architecture as the deferred kernel store. The figures make visible what the current theorem store is tracking, while closure itself still remains controlled at the structural layer beneath the pictures.*

Remark 1.127 (QAM/systems crosswalk for the present document)**.** *The present document uses the systems analogy in a disciplined layer-by-layer way. The following crosswalk fixes the intended reading.*

QAM/document layer	***Systems reading***	***Present mathematical status***
Visible coded tuples, packet diagnostics, Julia views, routed return surfaces	*User-facing VM/view layer*	*Actively visible and auditable, but not by themselves closure-generating*
Exposed payload kernels and later zero-kernel rereadings	*VM-state identity layer*	*Deferred and prospective only; not yet activated on the public tuple surface*
Admissible transport, branch/shell return, routed support organization	*OS-level orchestration layer*	*Active theorem content in the present QAM export*
Kernel normal forms, search keys, snapshots, clones, audits	*ZFS-style theorem-store layer*	*Partly active as organizational discipline, partly deferred as later compression infrastructure*
Pool/group structure of sphere objects and support families	*Device / vdev / pool layer*	*Explanatory architecture only at the present stage*

Remark 1.128 (Visual-evidence discipline under the systems reading)**.** *Under the crosswalk above, a figure may support only a controlled class of claims. It may expose* where *a visible readout departs from a rigid carrier picture,* which *packet or channel decomposition is plausible on the view layer, and* why *a later kernel-first compression path is worth isolating. It may* not *by itself create a new closure axiom, replace an active transport/branch/shell/routed theorem, or identify a kernel class solely from appearance. Any later proof use must therefore pass through the active theorem layer or through a deferred bridge that has itself already been discharged.*

archival stage checkpoint (deferred kernel-first path). The deferred kernel-first material added in the present local archival stage line should be read as one controlled checkpoint rather than as a second public language block. Its current role is to prepare a later proof-compression path while leaving the active coded tuple predicates unchanged.

Remark 1.129 (Checkpoint summary for the deferred kernel-first path in the archival stage public release)**.** *At the present stage, the deferred kernel-first path contributes six concrete pieces of structure.*

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1. *It introduces a prospective recursive zero-kernel vocabulary in which the visible state surface $\langle 0, a\rangle$ may later be re-read through exposed payload kernels, kernel normal forms, and class-level readout factorization.*

2. *It records a normal-form and search-key discipline so that later graph/tree based proof search can be tied to canonical kernel representatives rather than to repeated visible tuple surfaces.*

3. *It isolates the missing bridge datum from active state surfaces to canonical kernels and identifies the direct visible payload as the simplest current kernel final on the present tuple surface.*

4. *It tests the resulting kernel-first viewpoint on the three most relevant compression stages of the current QAM line: admissible transport, local branch-shell return, and routed family compression.*

5. *It formulates an explicit activation package and a staged migration roadmap, making clear both what would have to be proved before activation and in which order the relevant closure/readout factorization work should be completed.*

6. *It remains strictly deferred: no active public predicate, closure theorem, transport theorem, branch-shell theorem, routed-support theorem, or translation theorem is replaced by the kernel language at the present stage.*

Remark 1.130 (Why this checkpoint matters for later proof compression)**.** *The checkpoint above already shows where later proof compression would come from. Transport is the first chain-level reduction, branch-shell return the first local surface reduction, routed support the first family reduction, and the translation theorem then becomes the first clause-level reduction at theorem scale. This does not yet shorten the active proofs automatically, but it isolates a concrete path by which set-theoretic comparison load could later be shifted from repeated visible tuple surfaces to kernel normal forms and one final class-level readout step.*

Remark 1.131 (Navigation guide for the deferred kernel-first checkpoint)**.** *A reader who wants the shortest route through the deferred material may follow the following order.*

1. ***Purpose and fit.*** *Start with the roadmap-fit remark, the checkpoint summary, and the companion remark on why the checkpoint matters in order to see why the deferred path belongs to the inherited public-shape archival stage release layer without changing the public branch/shell/routed extraction.*

2. ***Vocabulary.*** *Then read the terminological convention for the deferred kernel path together with Definitions 1.136–1.143 for the later kernel vocabulary, primitive zero-objects, recursive reuse, aspect projections, normal forms, and internal search keys.*

3. ***Compatibility.*** *Next read the axiom-coherence audit together with the current-limitation remark to see why this vocabulary remains deferred and does not compete with the active tuple layer.*

4. ***Bridge.*** *Then read Definition 1.149, Theorem 1.150, Definition 1.152, and Theorem 1.154 for the exact interface between visible state surfaces and later kernel-first comparison.*

5. ***Stress tests.*** *Only afterwards read Propositions 1.175, 1.217, and 1.225, which test the bridge on transport, local return, and routed family compression.*

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6. ***Activation and use.*** *Finally read Definition 1.231, Theorem 1.232, the later activation-roadmap and reading-guide remarks, and then the dependency-map, minimal-proof-obligation, local-admissibility, and rewrite-schema remarks for the activation threshold, staged migration order, dependency logic, and later local rewrite test.*

In that order, the deferred block reads as one controlled proof-compression path rather than as a loose collection of auxiliary notes.

Remark 1.132 (Execution rule for the deferred kernel-first path)**.** *The deferred block should be used in only one direction when later proof compression is attempted on the present tuple surface. Start with the visible state surface, expose the payload kernel, pass to kernel normal form, prove closure or readout agreement at class level, and lift back to the visible surface only at the end. If the later local admissibility test fails at any point, then the argument should remain on the active tuple surface rather than forcing a premature kernel-first rewrite.*

Remark 1.133 (Present-use restriction for the deferred kernel-first path)**.** *Until the activation criterion of Theorem 1.232 has been fully discharged on the current QAM surface, the deferred block should be used in only three ways inside the document: as later migration vocabulary, as an audit of where proof compression could occur, and as a record of staged stress tests. It should not be cited as a replacement for any active tuple predicate, transport theorem, branch-shell theorem, routed-support theorem, or translation theorem. In particular, whenever both a visible-surface proof and a deferred kernel-first restatement are available, the visible-surface proof remains authoritative on the inherited public-shape archival stage layer.*

Remark 1.134 (Active/deferred correspondence rule)**.** *The deferred kernel-first path may also be read as a controlled shadow of four active theorem layers, with one fixed discipline: the active visible-surface theorem remains authoritative, while the deferred statement records only the later compression route attached to that theorem. Concretely, Theorem 1.174 is paired with Proposition 1.175 as the chain-level compression test; Theorem 1.192 is paired with Proposition 1.217 as the local return-surface compression test; Theorem 1.224 is paired with Proposition 1.225 as the family-level compression test; and Theorem 1.273 is paired with the later local admissibility reading of the translation theorem as the clause-level compression test. This correspondence is only a reading rule for later migration work. It does not transfer proof authority from the active theorem to its deferred companion; it only says where a later kernel-first rewrite would have to pay off if the activation package were eventually discharged.*

Prospective recursive zero-kernel scheme (deferred architecture note)

The following block records a possible later re-reading of the current $\langle 0, a\rangle$-entry as a recursive order-0 readout kernel. It is included here only as a controlled design note for future transfer work. It is *not* activated as part of the present public QAM tag system and does not modify the inherited public-shape coded predicates from historical release marker archival stage.

Remark 1.135 (Terminological convention for the deferred kernel path)**.** *Throughout the deferred block, four layers should be kept distinct and named consistently. The expression* visible state surface *refers to the active public state-code presentation $s = \langle 0, a\rangle$ on the current tuple level. The expression* exposed payload kernel *refers to the payload extracted from such a visible state surface by the deferred kernel constructor. The expression* kernel normal form *refers to the canonical representative selected from the admissible kernel equivalence class. Finally,* class-level readout factorization *means that closure-relevant readout depends only on that kernel normal-form class and is lifted back to the visible state surface only at the end. This convention is purely terminological, but it keeps the active tuple language and the deferred kernel language aligned.*

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Definition 1.136 (Carrier/filter readout kernels)**.** *Let $\mathcal{C}$ be a class of admissible carriers, let Γ be a class of admissible filters, and let $\mathcal{A}$ be a class of canonical readout kernels. For each $\gamma \in \Gamma$, assume a partial readout map*

$$\rho_\gamma : \mathcal{C} \dashrightarrow \mathcal{A},$$

whose defined values are written

$$a = \rho_\gamma(T) \in \mathcal{A}.$$

We read a as the minimal canonical readout kernel of the carrier T relative to the filter γ.

Definition 1.137 (Primitive zero-objects)**.** *Given $a, b \in \mathcal{A}$, define the primitive zero-objects*

$$Z_0(a) := \langle 0, a\rangle, \qquad Z_0(a,b) := \langle 0, a, b\rangle.$$

The intended reading is that $Z_0(a)$ is the unary collapsed readout attached to the kernel a, while $Z_0(a,b)$ is the binary collapsed readout attached to the kernel pair (a,b).

Axiom 1 (Recursive reusability of kernels)**.** *Every canonical readout kernel may again serve as an effective carrier for further admissible readout. Concretely, for admissible $\eta \in \Gamma$, one may again form*

$$\rho_\eta(a) \in \mathcal{A}$$

whenever the corresponding readout is defined. Thus the kernel class $\mathcal{A}$ is recursively reusable as an effective carrier layer.

Definition 1.138 (Aspect projections on kernels)**.** *Assume that each kernel $a \in \mathcal{A}$ comes equipped with aspect projections*

$$\pi_{\mathrm{type}}(a), \qquad \pi_{\mathrm{class}}(a), \qquad \pi_{\mathrm{sig}}(a),$$

recording, respectively, its visible type aspect, filter/class aspect, and structured signature aspect. These are not treated as three competing objects; they are three admissible readings of the same kernel.

Lemma 1.139 (Unary and binary kernel collapse)**.** *If $a = \rho_\gamma(T) \in \mathcal{A}$, then the unary zero-object $Z_0(a) = \langle 0, a\rangle$ is well-defined. If additionally $b = \rho_\eta(U) \in \mathcal{A}$, then the binary zero-object $Z_0(a,b) = \langle 0, a, b\rangle$ is well-defined.*

Proof. The unary claim follows directly from Definition 1.137 once $a \in \mathcal{A}$ is given by Definition 1.136. The binary claim is identical after adjoining a second admissible kernel $b \in \mathcal{A}$. □

Remark 1.140 (Compressed relation to the current $\langle 0, a\rangle$ state code)**.** *This deferred kernel notation is intended only as a future re-reading of the present visible state surface. In a later binary-tag or order-sensitive QAM line, one may reinterpret the current $\langle 0, a\rangle$ notation as a compressed public surface for a filtered order-0 readout kernel. For the present version, however, the active public predicates remain exactly those of the current tagged tuple language.*

Remark 1.141 (Tree/graph viewpoint for recursive kernels)**.** *The recursive kernel picture admits an immediate combinatorial visualization. One may regard repeated kernel readout*

$$T \xrightarrow{\rho_\gamma} a \xrightarrow{\rho_\eta} a' \xrightarrow{\rho_\theta} a''$$

as a rooted readout tree whose vertices are kernels and whose edges are admissible filter applications. Equally, one may organize the same data as a directed multigraph: unary zero-objects $Z_0(a)$ mark collapsed kernel vertices, while binary zero-objects $Z_0(a,b)$ record elementary visible relations between kernel vertices. This suggests a later graph-based search or proof-compression interface without changing the current public tuple system.

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Prospective kernel normal forms and search keys (deferred architecture note)

The same deferred kernel language also suggests a later search discipline in which recursive kernel proofs are first normalized and only then indexed. This block is likewise prospective only: it does not alter the active inherited public-shape tuple predicates.

Definition 1.142 (Kernel equivalence and canonical normal form)**.** *Assume an admissible equivalence relation $\sim$ on $\mathcal{A}$ such that equivalent kernels have the same admissible aspect projections and the same closure-relevant readout content. A* kernel normal form *is a partial map*

$$\mathrm{nf} : \mathcal{A} \dashrightarrow \mathcal{A}$$

selecting, whenever defined, one canonical representative from each admissible equivalence class. Thus

$$a \sim b \Longrightarrow \mathrm{nf}(a) = \mathrm{nf}(b)$$

whenever both normal forms are defined.

Definition 1.143 (Kernel search keys)**.** *Assume a fixed admissible serialization*

$$\mathrm{ser} : \mathrm{nf}(\mathcal{A}) \to \Sigma^*$$

of canonical kernels into finite strings over an alphabet Σ, together with a deterministic key map

$$\kappa := H \circ \mathrm{ser} \circ \mathrm{nf}\,.$$

The value $\kappa(a)$ is called the kernel search key *of a whenever the normal form is defined. It is intended as an internal search/index key rather than as part of the visible mathematical surface.*

Lemma 1.144 (Normal-form invariance of primitive zero-objects)**.** *Assume that $a, a', b, b' \in \mathcal{A}$ satisfy $a \sim a'$ and $b \sim b'$, and that their normal forms are defined. Then the unary zero-objects $Z_0(a)$ and $Z_0(a')$ determine the same canonical unary kernel surface, while the binary zero-objects $Z_0(a, b)$ and $Z_0(a', b')$ determine the same canonical binary kernel surface after passage to kernel normal form.*

Proof. By Definition 1.142, equivalence implies equality of canonical normal forms. Applying the same normal-form representative at the unary or binary zero-object level yields the same canonical collapsed surface in either case. □

Remark 1.145 (Graph search and proof-compression interface)**.** *In the graph viewpoint introduced above, one may therefore index vertices by canonical kernel normal forms or, internally, by their kernel search keys $\kappa(a)$. Repeated readout branches that land in the same canonical kernel class can then be merged, memoized, or searched only once. This is exactly the kind of mechanism by which recursive kernel language could later compress set-theoretic proof search without changing the visible public theorem statements.*

Axiom-coherence audit and prospective proof reductions (deferred audit note)

The deferred zero-kernel language should be read together with the active public QAM tuple layer under the following discipline.

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Remark 1.146 (Axiom-coherence audit)**.** *At the present stage, the deferred zero-kernel layer is formally compatible with the active QAM tuple layer for four reasons.*

1. *The active tuple predicates remain unchanged: in particular,*

$$\mathrm{QState}(x) \Longleftrightarrow \exists a\,(x = \langle 0, a\rangle)$$

   continues to define the public visible state surface.

2. *The deferred zero-kernel definitions do not introduce a second active state predicate. They only record a possible later re-reading of the same public surface, as stated in Remark 1.140.*

3. *The recursive kernel axiom (Axiom 1) and the kernel normal-form discipline (Definition 1.142) are prospective only; no active theorem in the inherited public-shape QAM chain depends on them as hypotheses.*

4. *Consequently, the current closure-class, admissible-transport, branch/shell, routed-support, and translation theorems remain exactly those of the active tuple language. The deferred kernel layer adds a controlled future interpretation, not a rival public axiom system.*

Remark 1.147 (Where kernel notation could already shorten proofs)**.** *Even before formal activation, the deferred kernel language already points to three concrete proof-compression opportunities.*

1. *In the first QAM transport theorem, the chain*

$$s_{\mathrm{in}},\ s_2,\ s_3,\ s_4,\ t_{\mathrm{fwd}},\ t_{\mathrm{bwd}}$$

   is currently carried at full state-code level. A later kernel reduction could first pass each state to a canonical readout kernel and then prove closure agreement on kernel classes before lifting the result back to coded states.

2. *In the branch-shell theorem and the routed admissible support theorem, each return realization currently enters through full branch/shell state data. A kernel-first restatement could instead reduce every admissible branch-shell realization to unary kernel surfaces $Z_0(a)$ and compare routed sectors only through their canonical kernel classes.*

3. *In closure-compatible readout arguments, one currently factors raw readouts through the class-target map J after proving* $\mathrm{QClEq}(s,t)$*. A later kernel language could absorb much of this step by treating the closure-relevant image itself as the primitive kernel surface attached to the state.*

In all three cases, the expected gain is not a new theorem but a shorter proof path: reduce to kernels first, perform the closure comparison there, and only then return to the visible tuple layer.

Remark 1.148 (Current limitation of the kernel reduction)**.** *The proof-compression opportunities recorded just above are not yet activated as a formal replacement inside the present proofs. The missing step is a theorem-level bridge showing that every active coded state relevant to the admissible OP fragment carries a canonical kernel representative with the same closure-relevant content. Until such a bridge is proved, the zero-kernel language should be treated as a planning interface rather than as a proof rule.*

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Definition 1.149 (Prospective kernel bridge datum on active state surfaces)**.** *A* kernel bridge datum *for the active admissible OP fragment consists of a partial map*

$$\ker : \mathcal{S}_{\mathrm{act}}^{\mathrm{QAM}} \dashrightarrow \mathcal{A},$$

where $\mathcal{S}_{\mathrm{act}}^{\mathrm{QAM}}$ *is the class of active coded QAM-state surfaces relevant to the present admissible OP fragment, such that the following hold whenever the displayed expressions are defined.*

1. ***surface realization:***
$$Z_0(\ker(s)) = s;$$

2. ***closure reflection:*** *if* $\mathrm{QClEq}(s,t)$*, then*
$$\mathrm{nf}(\ker(s)) = \mathrm{nf}(\ker(t));$$

3. ***closure-readout factorization:*** *for every closure-compatible coded readout* r*, there exists a map*
$$\mathcal{R}_r : \mathrm{nf}(\mathcal{A}) \to \mathrm{Im}(J_r)$$
into the closure-relevant target of the associated class-factor map J_r *such that whenever* $\mathrm{QRead}(r,s,w)$ *holds,*
$$J_r(w) = \mathcal{R}_r(\mathrm{nf}(\ker(s))).$$

Theorem 1.150 (Prospective bridge from active state surfaces to canonical kernels)**.** *Assume a kernel bridge datum in the sense of Definition 1.149. Then every active closure comparison in the admissible OP fragment may be reduced to canonical kernel comparison before returning to the visible state surface. In particular:*

1. *if* $\mathrm{QAdmTransport}(o,s,t)$*, then*
$$\mathrm{nf}(\ker(s)) = \mathrm{nf}(\ker(t));$$

2. *if* $t_{b,\sigma}$ *is a branch-shell coded return realization as in Theorem 1.192, then*
$$\mathrm{nf}(\ker(s_{\mathrm{in}})) = \mathrm{nf}(\ker(t_{b,\sigma}));$$

3. *if* u *is a normalized admissible routed support code as in Theorem 1.224, then for every closure-compatible coded readout* r *the closure-relevant image of the routed return is determined solely by the canonical kernel class*
$$\mathrm{nf}(\ker(s_{\mathrm{in}})).$$

Consequently, the present transport, branch/shell, and routed-support proofs admit a future kernel-first proof skeleton: first pass to $\mathrm{nf}(\ker(\cdot))$*, then compare closure there, and only afterwards return to the visible QAM state surface.*

Proof. For (1), admissible transport implies $\mathrm{QClEq}(s,t)$ by the active closure-preservation axiom. The closure-reflection clause of Definition 1.149 therefore yields
$$\mathrm{nf}(\ker(s)) = \mathrm{nf}(\ker(t)).$$
For (2), Theorem 1.192 gives $\mathrm{QClEq}(s_{\mathrm{in}}, t_{b,\sigma})$, so the same closure-reflection clause yields equality of canonical kernel representatives. For (3), Theorem 1.224 states that the routed return has the same closure-relevant image as the coded source state under every closure-compatible coded readout. By clause (3) of Definition 1.149, those closure-relevant images factor through the same canonical kernel normal form. Hence the routed readout depends only on $\mathrm{nf}(\ker(s_{\mathrm{in}}))$. The final sentence is simply the proof reading order extracted from (1)–(3). □

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Remark 1.151 (What the bridge theorem still leaves open)**.** *Theorem 1.150 is intentionally conditional. It does not yet prove that a kernel bridge datum exists inside the active public tuple system; it isolates exactly the missing interface that would be needed to turn the deferred zero-kernel language into an actual proof tool. This also identifies the next genuine activation step: construct* ker *canonically from active coded states and verify clauses (1)–(3) without leaving the current OP/QAM semantics.*

Definition 1.152 (Visible payload extraction on active state surfaces)**.** *For every active coded state surface s in the present public tuple language, define its* exposed payload kernel $\mathrm{vp}(s)$ *by*

$$\mathrm{vp}(s) = a \quad \Longleftrightarrow \quad s = \langle 0, a\rangle.$$

Equivalently, vp *is the unique projection that removes the visible state tag* 0 *and exposes the payload already carried by the active state surface.*

Definition 1.153 (Prospective direct kernel constructor)**.** *Assume that the exposed payload kernel of every active coded state surface relevant to the admissible OP fragment belongs to the deferred kernel class $\mathcal{A}$. Then define the* direct kernel constructor

$$\ker_{\mathrm{dir}} : \mathcal{S}_{\mathrm{act}}^{\mathrm{QAM}} \dashrightarrow \mathcal{A}, \qquad \ker_{\mathrm{dir}}(s) := \mathrm{vp}(s).$$

Thus, on the visible state surface, the simplest possible kernel final is obtained by forgetting only the active state tag and keeping the payload itself as the deferred kernel representative.

Theorem 1.154 (Simplest concrete kernel construction on the present state surface)**.** *Assume the following four conditions for the active admissible OP fragment.*

1. ***payload admissibility:*** *every active coded state surface s relevant to the fragment has the form $s = \langle 0, a\rangle$ with $a \in \mathcal{A}$;*

2. ***visible zero-surface realization:*** *for every such a,*

$$Z_0(a) = \langle 0, a\rangle;$$

3. ***closure reflection on exposed payload kernels:*** *whenever* $\mathrm{QClEq}(\langle 0, a\rangle, \langle 0, b\rangle)$,

$$\mathrm{nf}(a) = \mathrm{nf}(b);$$

4. ***class-level readout factorization:*** *for every closure-compatible coded readout r, there exists a map*

$$\widetilde{\mathcal{R}}_r : \mathrm{nf}(\mathcal{A}) \to \mathrm{Im}(J_r)$$

such that whenever $\mathrm{QRead}(r, \langle 0, a\rangle, w)$,

$$J_r(w) = \widetilde{\mathcal{R}}_r(\mathrm{nf}(a)).$$

Then $\ker_{\mathrm{dir}}$ is a kernel bridge datum in the sense of Definition 1.149. Consequently Theorem 1.150 applies with

$$\ker = \ker_{\mathrm{dir}}.$$

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Proof. Clause (1) ensures that $\ker_{\mathrm{dir}}$ is defined on every active state surface relevant to the fragment. By Definition 1.152, if $s = \langle 0, a\rangle$, then $\ker_{\mathrm{dir}}(s) = a$. Clause (2) therefore gives

$$Z_0(\ker_{\mathrm{dir}}(s)) = Z_0(a) = \langle 0, a\rangle = s,$$

which is exactly the surface-realization clause of Definition 1.149. For closure reflection, if $\mathrm{QClEq}(s, t)$ with $s = \langle 0, a\rangle$ and $t = \langle 0, b\rangle$, clause (3) gives

$$\mathrm{nf}(\ker_{\mathrm{dir}}(s)) = \mathrm{nf}(a) = \mathrm{nf}(b) = \mathrm{nf}(\ker_{\mathrm{dir}}(t)).$$

Finally, clause (4) is exactly the required closure-readout factorization after substituting $\ker_{\mathrm{dir}}(\langle 0, a\rangle) = a$. Hence all three clauses of Definition 1.149 hold. □

Remark 1.155 (What this construction shows)**.** *Theorem 1.154 shows that the difficult part of the bridge is not the extraction step itself: on the current public state surface, the most economical final is simply the exposed payload of the active state surface. The real work is pushed into the three semantic checks: that the payload already lies in the admissible kernel class, that closure equivalence is visible at payload-normal-form level, and that closure-relevant readout factors through that same normal form. If these three checks can later be discharged, then many current state-level proofs may be shortened by passing immediately from $\langle 0, a\rangle$ to a before performing the closure comparison.*

QAM minimal predicates and coded relations

Status. *Active formal-language layer.*

Block function. This block introduces the smallest coded predicate/relation vocabulary needed to read the present admissible OP fragment in QAM-over-ZF form.

We now pass from the entry-layer preview to the first explicit predicate layer of QAM-over-ZF. At this stage, the QAM language still remains conservative over ZF: the predicates below are intended as definable recognition or relation clauses on tagged set-codes rather than as commitments to a new ontology.

Definition 1.156 (Primitive QAM predicates)**.** *We introduce the primitive predicates*

$$\mathrm{QState}(x),\quad \mathrm{QLayer}(x),\quad \mathrm{QOp}(x),\quad \mathrm{QReadout}(x),$$
$$\mathrm{QBranch}(x),\quad \mathrm{QShell}(x),\quad \mathrm{QWitness}(x),\quad \mathrm{QTensor}(x).$$

with the intended ZF-grounded readings

$$\mathrm{QState}(x) \ :\Longleftrightarrow\ \exists a\,(x = \langle 0, a\rangle),$$

$$\mathrm{QLayer}(x) \ :\Longleftrightarrow\ \exists \ell\,\exists D\,(x = \langle 1, \ell, D\rangle),$$
$$\mathrm{QOp}(x) \ :\Longleftrightarrow\ \exists \ell_1\,\exists \ell_2\,\exists G\,(x = \langle 2, \ell_1, \ell_2, G\rangle),$$
$$\mathrm{QReadout}(x) \ :\Longleftrightarrow\ \exists \ell\,\exists W\,\exists R\,(x = \langle 3, \ell, W, R\rangle),$$
$$\mathrm{QBranch}(x) \ :\Longleftrightarrow\ \exists b\,\exists B\,(x = \langle 4, b, B\rangle),$$
$$\mathrm{QShell}(x) \ :\Longleftrightarrow\ \exists r\,\exists s\,\exists T\,(x = \langle 5, r, s, T\rangle),$$
$$\mathrm{QWitness}(x) \ :\Longleftrightarrow\ \exists u\,(x = \langle 6, u\rangle),$$
$$\mathrm{QTensor}(x) \ :\Longleftrightarrow\ \exists p\,\exists q\,(x = \langle 7, p, q\rangle).$$

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Remark 1.157. *At this stage, these predicates are still recognition clauses rather than a full independent axiom system. Their purpose is to let the Companion speak explicitly about coded states, layers, operators, readouts, branches, shell/refinement objects, witnesses, and tensor composites inside one ZF-grounded language layer.*

Definition 1.158 (Primitive QAM coded relations)**.** *We also introduce the primitive QAM-style relations*

$$\mathrm{QAdmTransport}(o,s,t), \qquad \mathrm{QClEq}(s,t), \qquad \mathrm{QRead}(r,s,w),$$

$$\mathrm{QBranchRef}(b,s,t), \qquad \mathrm{QShellRef}(\sigma,s,t), \qquad \mathrm{QTensorAdm}(x,y).$$

Their intended readings are as follows:

1. $\mathrm{QAdmTransport}(o,s,t)$ *means that the operator-code o transports the coded state s admissibly to the coded state t;*
2. $\mathrm{QClEq}(s,t)$ *means that s and t belong to the same coded closure class;*
3. $\mathrm{QRead}(r,s,w)$ *means that the readout-code r reads the coded state s into the coded observable value w;*
4. $\mathrm{QBranchRef}(b,s,t)$ *means that the branch-code b refines the coded state s to the coded state t;*
5. $\mathrm{QShellRef}(\sigma,s,t)$ *means that the shell/refinement-code σ refines the coded state s to the coded state t;*
6. $\mathrm{QTensorAdm}(x,y)$ *means that the tensor-coded composite $x \otimes y$ is admitted as a legitimate coded QAM object.*

Remark 1.159. *These relations are the first minimal bridge between the current OP/branch/shell architecture and its coded QAM-over-ZF form. In particular, admissibility and closure-class preservation already appear here as primitive language-level notions rather than as only informal motivation.*

Definition 1.160 (First QAM translation target)**.** *The first intended translation target of the QAM-over-ZF layer is the admissible OP fragment consisting of:*

1. *the OP 1/OP 5 source-return closure architecture;*
2. *the intermediate OP 2–OP 4 transport bridge;*
3. *admissible branch transport;*
4. *OP 3 shell-admissible refinement;*
5. *admissible routed support together with the visible packet/phase-lock/epsilon screening layer.*

Remark 1.161. *The significance of the present predicate layer is that QAM is no longer only a roadmap note. Even at this first stage, the language already distinguishes object-kinds and relation-kinds closely enough to serve as a faithful coded entry interface for the current admissible OP fragment.*

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QAM transport, closure, and readout schemata

Status. *Active theorem-core block inside the QAM formal layer.*

Block function. This block turns the predicate language into actual transport, closure-class, and readout principles, so it is the first place where the QAM interface becomes theorem-usable.

We now pass from the minimal predicate layer to the first theorematic QAM transport/-closure layer. The point of this section is still conservative: it does not yet claim a fully independent QAM foundation, but it does begin to formulate admissible transport, closure-class preservation, and closure-compatible readout directly in the coded QAM-over-ZF language.

Definition 1.162 (QAM-coded transport graph)**.** *A QAM-operator code*

$$o = \langle 2, \ell_1, \ell_2, G \rangle$$

is called a coded transport graph *if G is a set of ordered pairs coding a functional transport rule from coded states on layer ℓ_1 to coded states on layer ℓ_2.*

Definition 1.163 (QAM admissible transport)**.** *For a QAM-operator code o and coded states s, t, we write*

$$\mathrm{QAdmTransport}(o, s, t)$$

iff:

1. *o is a coded transport graph;*
2. *s is a coded state on the source layer of o;*
3. *t is a coded state on the target layer of o;*
4. *the graph-code of o sends s to t;*
5. *the transport from s to t preserves the intended closure-bearing structural mode.*

Remark 1.164. *At this stage, the final clause is still schematic. Its precise intended mathematical content is supplied by the coded closure-class relation below.*

Definition 1.165 (QAM closure-class equivalence)**.** *The relation*

$$\mathrm{QClEq}(s, t)$$

is the coded closure-class relation on QAM-state codes. It is intended to mean that s and t represent the same closure-bearing structural mode, even if they differ by admissible transport, admissible frame-shifted realization, admissible branch refinement, or admissible shell refinement.

Axiom 2 (QAM closure-class schema)**.** *The relation* QClEq *is required to satisfy:*

1. ***reflexivity:***
$$\mathrm{QClEq}(s, s);$$
2. ***symmetry:***
$$\mathrm{QClEq}(s, t) \Longrightarrow \mathrm{QClEq}(t, s);$$
3. ***transitivity:***
$$\mathrm{QClEq}(s, t) \wedge \mathrm{QClEq}(t, u) \Longrightarrow \mathrm{QClEq}(s, u).$$

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Remark 1.166. *Thus* QClEq *is the coded QAM analogue of the native closure carrier-class relation used in the current OP architecture.*

Axiom 3 (QAM admissibility implies closure preservation)**.** *For every operator code o and state codes s, t,*

$$\mathrm{QAdmTransport}(o, s, t) \Longrightarrow \mathrm{QClEq}(s, t).$$

Remark 1.167. *This is the first central QAM principle. Admissibility is not defined merely by existence of transport, but by preservation of closure class.*

Definition 1.168 (QAM-coded readout)**.** *A QAM-readout code*

$$r = \langle 3, \ell, W, R \rangle$$

is called a coded readout *if R is a graph-code sending coded states on layer ℓ into coded observable values in the target code W.*

Definition 1.169 (QAM readout relation)**.** *We write*

$$\mathrm{QRead}(r, s, w)$$

iff r is a coded readout, s is a coded state on the source layer of r, and the graph-code of r sends s to the coded observable value w.

Definition 1.170 (QAM closure-compatible readout)**.** *A coded readout r is called* closure-compatible *if there exists a coded class-target map J such that for all coded states s, t,*

$$\mathrm{QClEq}(s, t) \Longrightarrow \big(\forall w_1\, \forall w_2\, (\mathrm{QRead}(r, s, w_1) \wedge \mathrm{QRead}(r, t, w_2) \Longrightarrow J(w_1) = J(w_2))\big).$$

Remark 1.171. *This is the coded QAM version of readout factorization through closure class. Raw readouts may differ, but their closure-relevant image under J must agree whenever the source states belong to the same closure class.*

Definition 1.172 (First tensor-compatibility placeholder)**.** *We reserve the tensor-admissibility predicate*

$$\mathrm{QTensorAdm}(x, y)$$

for the statement that the tensor-coded composite $x \otimes y$ is admitted as a legitimate coded QAM object of the intended kind.

Proposition 1.173 (First QAM transport principle)**.** *Let o be a coded QAM-operator, let s, t be coded QAM-states, and let r be a closure-compatible coded readout. If*

$$\mathrm{QAdmTransport}(o, s, t),$$

then:

1. $$\mathrm{QClEq}(s, t);$$

2. *the closure-relevant image of any readout of s agrees with the closure-relevant image of any readout of t under the factor map J associated with r.*

Proof. The first claim is exactly the admissibility-implies-closure-preservation schema. The second follows from closure-compatibility of the readout: since $\mathrm{QClEq}(s, t)$ holds, any coded readouts of s and t are identified after passage through the closure-relevant class-target map J. □

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Theorem 1.174 (First QAM version of the OP transport principle)**.** *Assume:*

1. *there is a coded source state* s_{in}*;*
2. *there are coded intermediate states* s_2, s_3, s_4*;*
3. *there are coded operator maps* $o_2, o_3, o_4, o_{\mathrm{fwd}}, o_{\mathrm{bwd}}$*;*
4. *the transports*

$$\begin{aligned}&\mathrm{QAdmTransport}(o_2, s_{\mathrm{in}}, s_2), \qquad \mathrm{QAdmTransport}(o_3, s_2, s_3),\\ &\mathrm{QAdmTransport}(o_4, s_3, s_4), \qquad \mathrm{QAdmTransport}(o_{\mathrm{fwd}}, s_{\mathrm{in}}, t_{\mathrm{fwd}}),\\ &\mathrm{QAdmTransport}(o_{\mathrm{bwd}}, s_{\mathrm{in}}, t_{\mathrm{bwd}}).\end{aligned}$$

   all hold;

5. *the relevant coded readout is closure-compatible.*

Then all coded states

$$s_{\mathrm{in}},\ s_2,\ s_3,\ s_4,\ t_{\mathrm{fwd}},\ t_{\mathrm{bwd}}$$

belong to the same coded closure class. Consequently, all closure-relevant readouts of these states agree after passage through the closure-factor map.

Proof. Each admissible transport preserves closure class by the admissibility schema. Therefore

$$\mathrm{QClEq}(s_{\mathrm{in}}, s_2), \qquad \mathrm{QClEq}(s_2, s_3), \qquad \mathrm{QClEq}(s_3, s_4).$$

By transitivity,

$$\mathrm{QClEq}(s_{\mathrm{in}}, s_4).$$

Likewise,

$$\mathrm{QClEq}(s_{\mathrm{in}}, t_{\mathrm{fwd}}) \qquad \text{and} \qquad \mathrm{QClEq}(s_{\mathrm{in}}, t_{\mathrm{bwd}})$$

follow from admissible forward and backward transport. Using transitivity again, all listed coded states lie in the same closure class. Closure-compatible readout then identifies their closure-relevant readout content after factorization through the associated class-target map. □

Proposition 1.175 (Prospective kernel-first reduction of the first transport theorem)**.** *Assume the direct kernel constructor of Definition 1.153 and the bridge conditions of Theorem 1.154. Write*

$$s_{\mathrm{in}} = \langle 0, a_{\mathrm{in}}\rangle, \qquad s_2 = \langle 0, a_2\rangle, \qquad s_3 = \langle 0, a_3\rangle, \qquad s_4 = \langle 0, a_4\rangle,$$

$$t_{\mathrm{fwd}} = \langle 0, a_{\mathrm{fwd}}\rangle, \qquad t_{\mathrm{bwd}} = \langle 0, a_{\mathrm{bwd}}\rangle.$$

Then the conclusion of Theorem 1.174 may be read kernel-first as follows:

1. *all exposed transport kernels lie in one canonical normal-form class,*

$$\mathrm{nf}(a_{\mathrm{in}}) = \mathrm{nf}(a_2) = \mathrm{nf}(a_3) = \mathrm{nf}(a_4) = \mathrm{nf}(a_{\mathrm{fwd}}) = \mathrm{nf}(a_{\mathrm{bwd}}),$$

2. *hence every closure-compatible coded readout of the transport chain factors through that single canonical kernel class,*

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3. *therefore the first transport theorem may be re-read as a kernel-normal-form preservation statement along the admissible transport chain before returning to the visible QAM state surface.*

Proof. Theorem 1.174 states that all six visible coded states belong to the same coded closure class. Since each state lies on the present state surface, the direct kernel constructor exposes exactly the exposed payload kernel of each state as its deferred kernel representative. The closure-reflection clause of Theorem 1.154 therefore yields equality of the six corresponding kernel normal forms. By the class-level readout factorization clause, every closure-compatible coded readout depends only on that common kernel normal form. Hence the whole transport chain may be treated kernel-first and lifted back to the visible state surface only at the end. □

Remark 1.176 (Axiom audit after the kernel-first transport reduction)**.** *Proposition 1.175 does* not *strengthen the active proof status of Theorem 1.174. The active theorem still rests only on admissible transport, closure-class preservation, transitivity of coded closure, and closure-compatible coded readout. The kernel-first proposition adds force only under the deferred payload-level bridge hypotheses isolated in Definition 1.149 and Theorem 1.154. Thus no new hidden axiom has entered the live transport theorem; only a prospective kernel-level compression of its proof surface has been recorded.*

Remark 1.177 (How much proof load the kernel-first transport reading removes)**.** *For the present transport theorem the admissible chain contains five explicit transport links:*

$$s_{\mathrm{in}} \to s_2 \to s_3 \to s_4, \qquad s_{\mathrm{in}} \to t_{\mathrm{fwd}}, \qquad s_{\mathrm{in}} \to t_{\mathrm{bwd}}.$$

So the gain is again not *a reduction in the number of transport links, but a reduction in the size of the object carried through each link. On the visible state surface the proof transports closure across five state-surface steps and then factors a closure-compatible readout through the resulting six-state closure class. In the deferred kernel-first reading, those same five links are re-read as five kernel normal-form identifications, followed by one readout through the common kernel class. If one measures a visible-state transport comparison by cost C_{surf} and a kernel normal-form comparison by cost C_{ker}, then the conceptual load changes from roughly*

$$5\,C_{\mathrm{surf}} + C_{\mathrm{read}}$$

to

$$5\,C_{\mathrm{ker}} + C_{\mathrm{class}},$$

with intended gain $C_{\mathrm{ker}} < C_{\mathrm{surf}}$ and typically $C_{\mathrm{class}} \ll C_{\mathrm{read}}$. So the numerical savings are moderate but already real: the theorem still has five links, yet each link is checked on exposed kernels instead of full visible state surfaces, and the final readout is centralized on one canonical kernel class.

Remark 1.178. *This theorem is only the first coded QAM shadow of the current OP transport architecture. It does not yet include the full branch-shell routed layer, but it already captures the source/intermediate/return logic in a ZF-grounded coded form.*

QAM branch, shell, and routed support layer

Status. *Active theorem-core branch-shell extension.*

Block function. This block lifts the basic QAM transport layer to the branch/shell/routed regime and should be read as the coded counterpart of the admissible OP refinement chain.

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We now extend the QAM-over-ZF language from basic transport and readout to the branch/shell/routed layer. The point of this section is to translate the current admissible-branch, OP 3 shell-admissibility, and admissible routed-support architecture into a single coded QAM fragment without yet claiming a fully independent QAM foundation. For direct continuity with the compact front extraction, the present subsection should be read in the same order as the release dictionary of Remark 1.38: branch admissibility first, shell admissibility second, admissible branch-shell support third, normalized routed support fourth, and only then the visible-versus-structural screening rule.

Definition 1.179 (Closure-admissible branch code)**.** *A branch code b is called* closure-admissible *if for all coded states s, t,*

$$\mathrm{QBranchRef}(b, s, t) \Longrightarrow \mathrm{QClEq}(s, t).$$

Equivalently, branch refinement by b preserves the coded closure class.

Definition 1.180 (Shell-admissible refinement code)**.** *A shell/refinement code σ is called* shell-admissible *if for all coded states s, t,*

$$\mathrm{QShellRef}(\sigma, s, t) \Longrightarrow \mathrm{QClEq}(s, t).$$

Equivalently, shell-sensitive or residual-sensitive refinement by σ preserves the coded closure class.

Remark 1.181. *These are the direct coded QAM analogues of closure-admissible branches and OP 3 shell-admissible refinements in the current Companion line.*

Definition 1.182 (Admissible branch set and shell family)**.** *We define the admissible branch class by*

$$\mathcal{B}_{\mathrm{adm}}^{\mathrm{QAM}} := \{\, b : \mathrm{QBranch}(b) \ \wedge\ b \text{ is closure-admissible} \,\},$$

and for each branch code b, the shell-admissible refinement class by

$$\mathcal{F}_{\mathrm{sh}}^{\mathrm{QAM}}(b) := \{\, \sigma : \mathrm{QShell}(\sigma) \ \wedge\ \sigma \text{ is shell-admissible on } b \,\}.$$

Remark 1.183. *At this stage, the phrase "shell-admissible on b" may be read minimally as: shell refinement by σ acts on branchwise states associated with b without leaving the coded closure class.*

Definition 1.184 (Routed support code)**.** *A* routed support code *is a set-code*

$$u = \langle 8, \Omega, w \rangle,$$

where Ω is a coded finite or set-sized support family of admissible branch-shell pairs and w is a coded weight assignment on Ω.

Definition 1.185 (Admissible routed support)**.** *A routed support code*

$$u = \langle 8, \Omega, w \rangle$$

is called admissible *if every element of Ω has the form (b, σ) with*

$$b \in \mathcal{B}_{\mathrm{adm}}^{\mathrm{QAM}} \qquad \text{and} \qquad \sigma \in \mathcal{F}_{\mathrm{sh}}^{\mathrm{QAM}}(b).$$

Remark 1.186. *Thus admissible routed support is the coded QAM version of the admissible branch-shell support sector in the current OP architecture.*

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Definition 1.187 (Normalized routed support)**.** *An admissible routed support code*

$$u = \langle 8, \Omega, w \rangle$$

is called normalized *if its coded weight assignment satisfies*

$$\sum_{(b,\sigma)\in\Omega} w(b,\sigma) = 1$$

in the intended decoded numerical target.

Remark 1.188. *The normalization condition is stated semiformally at this stage. In the fully coded version, the sum relation would itself be represented by a suitable arithmetic coding layer over ZF.*

Definition 1.189 (Branch-shell coded return realization)**.** *Let b be a closure-admissible branch code and let σ be a shell-admissible refinement code on b. A* branch-shell coded return realization *is a coded state*

$$t_{b,\sigma}$$

such that:

1. *$t_{b,\sigma}$ is reached from the coded source state through admissible branch refinement by b;*
2. *the OP 3-level shell refinement by σ is shell-admissible;*
3. *the resulting return-side coded state is read through a closure-compatible coded readout.*

Proposition 1.190 (QAM branch admissibility principle)**.** *Let b be a closure-admissible branch code. If*

$$\mathrm{QBranchRef}(b, s, t),$$

then

$$\mathrm{QClEq}(s, t).$$

Consequently, every closure-compatible readout identifies the closure-relevant image of s and t after passage through the associated class-factor map.

Proof. The first claim is just the definition of closure-admissibility. The readout claim follows from closure-compatibility of coded readouts: closure-equivalent coded states have the same closure-relevant image after factorization. □

Proposition 1.191 (QAM shell admissibility principle)**.** *Let σ be a shell-admissible refinement code. If*

$$\mathrm{QShellRef}(\sigma, s, t),$$

then

$$\mathrm{QClEq}(s, t).$$

Consequently, shell-sensitive refinement does not alter closure-relevant readout content as long as shell-admissibility is preserved.

Proof. This is the direct content of shell-admissibility together with closure-compatible readout factorization. □

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Theorem 1.192 (QAM admissible branch-shell theorem)**.** *Let s_{in} be a coded source state, let b be a closure-admissible branch code, and let σ be a shell-admissible refinement code on b. Assume there exists a branch-shell coded return realization*

$$t_{b,\sigma}$$

associated with (b, σ), and let r be a closure-compatible coded readout. Then

$$\mathrm{QClEq}(s_{\mathrm{in}}, t_{b,\sigma}).$$

Hence every closure-compatible readout of s_{in} and $t_{b,\sigma}$ agrees after passage through its associated closure-factor map.

Proof. By closure-admissibility of b, branch refinement preserves the coded closure class. By shell-admissibility of σ, the shell-sensitive refinement preserves the same coded closure class at the OP 3 layer. Hence the final coded return realization $t_{b,\sigma}$ remains closure-equivalent to the coded source state s_{in}. Closure-compatible readout then identifies the closure-relevant content of both states after factorization. □

Shell certificates and the first OP 3/QAM closure guide

Status. *Active closure-guide layer.*

Block function. This block isolates the certificate vocabulary that lets readers track exactly when OP 3 shell behavior is still closure-preserving and when a local failure first becomes visible.

The next narrow OP 3-oriented extension of the public-shape QAM line is not a broad new tensor layer, but a certificate layer. The goal is to say explicitly when a branch/shell refinement is still the same closure-bearing object, what counts as the first local failure of that claim, and how the deferred kernel bridge should later read that same situation in one compressed normal form. Mixed tensors remain deferred here: they are not introduced unless an actual OP 3 proof step later forces them.

QAM reading key for the OP 3 closure block.
Objects: coded source state s_{in}, closure-admissible branch code b, shell refinement code σ, intermediate branch state s_b, branch-shell return state $t_{b,\sigma}$.
Allowed move: pass from s_{in} to s_b by admissible branch refinement and from s_b to $t_{b,\sigma}$ by shell refinement without leaving the closure class.
Invariant: the closure-bearing class, and later its kernel/normal-form representative when the deferred bridge is activated.
Defect signal: a broken branch clause, a broken shell clause, or a failure of closure-compatible readout factorization.
Plain-language reading: OP 3 may change the visible shell, but it must not change the closure-bearing core.

Definition 1.193 (Shell admissibility certificate)**.** *Let s_{in} be a coded source state, let b be a closure-admissible branch code, let σ be a shell refinement code, and let $t_{b,\sigma}$ be a branch-shell coded return realization. A* shell admissibility certificate *for $(s_{\mathrm{in}}, b, \sigma, t_{b,\sigma})$ consists of coded data witnessing the following four clauses:*

1. *there exists a coded branch state s_b with* $\mathrm{QBranchRef}(b, s_{\mathrm{in}}, s_b)$*;*
2. *the branch code b is closure-admissible;*

3. one has $\mathrm{QShellRef}(\sigma, s_b, t_{b,\sigma})$ *and the refinement code* σ *is shell-admissible on* b*;*

4. the return state $t_{b,\sigma}$ *is read through a closure-compatible coded readout.*

Definition 1.194 (Shell defect witness)**.** *A* shell defect witness *for* $(s_{\mathrm{in}}, b, \sigma, t_{b,\sigma})$ *is coded data showing failure of at least one clause of Definition 1.193. Thus a shell defect witness may record failure of admissible branch passage, failure of shell admissibility, or failure of closure-compatible readout factorization.*

Lemma 1.195 (Certificate persistence of closure class)**.** *Assume that* $(s_{\mathrm{in}}, b, \sigma, t_{b,\sigma})$ *carries a shell admissibility certificate. Then*

$$\mathrm{QClEq}(s_{\mathrm{in}}, t_{b,\sigma}).$$

Hence every closure-compatible coded readout identifies the same closure-relevant image on s_{in} *and* $t_{b,\sigma}$ *after passage through the associated class-factor map.*

Proof. By clause (1) of Definition 1.193, there exists s_b with

$$\mathrm{QBranchRef}(b, s_{\mathrm{in}}, s_b).$$

By clause (2), the branch code b is closure-admissible, so Proposition 1.190 yields

$$\mathrm{QClEq}(s_{\mathrm{in}}, s_b).$$

By clause (3), one has

$$\mathrm{QShellRef}(\sigma, s_b, t_{b,\sigma}),$$

and the refinement code σ is shell-admissible on b. Proposition 1.191 therefore gives

$$\mathrm{QClEq}(s_b, t_{b,\sigma}).$$

Transitivity of the coded closure relation yields

$$\mathrm{QClEq}(s_{\mathrm{in}}, t_{b,\sigma}).$$

The final sentence is then the usual closure-compatible readout factorization claim. □

Proposition 1.196 (Certified kernel persistence for OP 3 shell passage)**.** *Assume the direct kernel constructor of Definition 1.153 and the bridge conditions of Theorem 1.154. Let*

$$s_{\mathrm{in}} = \langle 0, a_{\mathrm{in}} \rangle$$

be the coded source state, and let

$$t_{b,\sigma} = \langle 0, a_{b,\sigma} \rangle$$

be a branch-shell coded return realization. If $(s_{\mathrm{in}}, b, \sigma, t_{b,\sigma})$ *carries a shell admissibility certificate, then*

$$\mathrm{nf}(a_{b,\sigma}) = \mathrm{nf}(a_{\mathrm{in}}).$$

In particular, certified OP 3 shell passage preserves the kernel normal form of the coded source state.

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Proof. By Lemma 1.195, one has

$$\mathrm{QClEq}(s_{\mathrm{in}}, t_{b,\sigma}).$$

The deferred kernel bridge then reflects closure equivalence into equality of canonical kernel normal forms. Since

$$s_{\mathrm{in}} = \langle 0, a_{\mathrm{in}} \rangle, \qquad t_{b,\sigma} = \langle 0, a_{b,\sigma} \rangle,$$

this yields

$$\mathrm{nf}(a_{b,\sigma}) = \mathrm{nf}(a_{\mathrm{in}}).$$

□

Theorem 1.197 (Shell-stable readout factorization)**.** *Assume the direct kernel constructor of Definition 1.153 and the bridge conditions of Theorem 1.154. Let*

$$s_{\mathrm{in}} = \langle 0, a_{\mathrm{in}} \rangle, \qquad t_{b,\sigma} = \langle 0, a_{b,\sigma} \rangle,$$

let r be a closure-compatible coded readout, and assume that $(s_{\mathrm{in}}, b, \sigma, t_{b,\sigma})$ carries a shell admissibility certificate. Then for every pair of coded observable values $w_{\mathrm{in}}, w_{b,\sigma}$ with

$$\mathrm{QRead}(r, s_{\mathrm{in}}, w_{\mathrm{in}}), \qquad \mathrm{QRead}(r, t_{b,\sigma}, w_{b,\sigma}),$$

one has

$$J_r(w_{\mathrm{in}}) = \widetilde{\mathcal{R}}_r(\mathrm{nf}(a_{\mathrm{in}})) = \widetilde{\mathcal{R}}_r(\mathrm{nf}(a_{b,\sigma})) = J_r(w_{b,\sigma}).$$

In particular, every closure-relevant coded readout of a certified OP 3 shell passage factors through one shell-stable kernel normal form.

Proof. By Proposition 1.196, a shell admissibility certificate implies

$$\mathrm{nf}(a_{b,\sigma}) = \mathrm{nf}(a_{\mathrm{in}}).$$

By clause (4) of Theorem 1.154, every closure-compatible coded readout r admits a factor map

$$\widetilde{\mathcal{R}}_r : \mathrm{nf}(\mathcal{A}) \to \mathrm{Im}(J_r)$$

such that whenever $\mathrm{QRead}(r, \langle 0, a \rangle, w)$,

$$J_r(w) = \widetilde{\mathcal{R}}_r(\mathrm{nf}(a)).$$

Applying this first to $s_{\mathrm{in}} = \langle 0, a_{\mathrm{in}} \rangle$ and then to $t_{b,\sigma} = \langle 0, a_{b,\sigma} \rangle$ yields

$$J_r(w_{\mathrm{in}}) = \widetilde{\mathcal{R}}_r(\mathrm{nf}(a_{\mathrm{in}})), \qquad J_r(w_{b,\sigma}) = \widetilde{\mathcal{R}}_r(\mathrm{nf}(a_{b,\sigma})).$$

Since the two kernel normal forms agree, the displayed chain follows. The final sentence is exactly the shell-stable readout factorization claim. □

Proposition 1.198 (First converse shell-certificate criterion)**.** *Assume the direct kernel constructor of Definition 1.153 and the bridge conditions of Theorem 1.154. Let*

$$s_{\mathrm{in}} = \langle 0, a_{\mathrm{in}} \rangle, \qquad t_{b,\sigma} = \langle 0, a_{b,\sigma} \rangle,$$

and suppose that the following data are given:

1. *there exists a coded branch state s_b with* $\mathrm{QBranchRef}(b, s_{\mathrm{in}}, s_b)$*;*
2. *the branch code b is closure-admissible;*

3. *one has* $\mathrm{QShellRef}(\sigma, s_b, t_{b,\sigma})$ *and the refinement code* σ *is shell-admissible on* b*;*

4. *there exists a closure-compatible coded readout on the return state* $t_{b,\sigma}$*;*

5. *the kernel normal forms agree,* $\mathrm{nf}(a_{b,\sigma}) = \mathrm{nf}(a_{\mathrm{in}})$.

Then $(s_{\mathrm{in}}, b, \sigma, t_{b,\sigma})$ *carries a shell admissibility certificate. Hence every closure-relevant coded readout of the passage factors through the same shell-stable kernel normal form.*

Proof. The first four assumptions are exactly the four clauses required in Definition 1.193, so they furnish a shell admissibility certificate for $(s_{\mathrm{in}}, b, \sigma, t_{b,\sigma})$. The final sentence then follows from Theorem 1.197, using the additional kernel-persistence hypothesis only as the converse bridge datum that motivates the criterion. □

Lemma 1.199 (First OP 3/QAM shell commuting-square lemma)**.** *Assume the faithful translation theorem for the admissible OP fragment, Theorem 1.273, together with the OP 3 shell theorem, Theorem 32.24. Let*

$$\mathfrak{T}_{\mathrm{OP3}}$$

denote the admissible OP 3 shell-refinement passage on the OP side, and let

$$\Sigma_{b,\sigma}$$

denote the corresponding coded branch-shell refinement on the QAM side. Then the source-to-refinement square commutes on closure-relevant content in the following sense: whenever an admissible OP 3 shell passage is translated into coded QAM data,

$$\tau\big(\mathfrak{T}_{\mathrm{OP3}}(x)\big) \sim_{\mathrm{cl}} \Sigma_{b,\sigma}\big(\tau(x)\big),$$

where $\sim_{\mathrm{cl}}$ *means equality after passage to the common closure-bearing class and, under the deferred kernel bridge, equality of the induced kernel normal form. In particular, OP 3 refinement and QAM shell refinement are not two competing shell operations; they are two readings of the same closure-preserving shell step.*

Proof. By Theorem 1.273, the admissible OP fragment transports into coded QAM data without changing the source/return closure logic. By Theorem 32.24, the OP 3 shell passage is admissible exactly at the stage where the OP side isolates shell refinement without changing the closure-bearing content. The translated branch stage therefore produces a coded branch state s_b, and the translated shell stage produces a coded return state $t_{b,\sigma}$ with the same closure-relevant content as the OP-side refined state. Lemma 1.195 then identifies the translated source state and the coded branch-shell return state inside one coded closure class, while Proposition 1.196 upgrades this to equality of kernel normal forms once the deferred bridge is activated. That is exactly the asserted commuting-square statement. □

Theorem 1.200 (First OP 3/QAM shell correspondence theorem)**.** *Assume the faithful translation theorem for the admissible OP fragment, Theorem 1.273, together with the OP 3 shell theorem, Theorem 32.24. Then every translated OP 3-admissible shell passage determines a shell admissibility certificate in QAM, and every such certificate recovers the same closure-relevant shell content as the corresponding OP 3 step. Together with Proposition 1.198 and Lemma 1.199, the QAM shell-certificate layer may already be read as a first narrow converse guide for OP 3: admissible branch-shell behavior, a closure-compatible return readout, and kernel persistence are sufficient to recover the certificate package. Consequently, the QAM shell-certificate layer may be read as the first explicit coded closure guide for OP 3.*

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Proof. Theorem 1.273 translates the admissible OP fragment into the coded QAM layer while preserving its source/return closure logic. Theorem 32.24 isolates the OP 3 shell-admissible passage itself. Under that translation, the OP-side branch step provides clause (1) of Definition 1.193, branch admissibility gives clause (2), OP 3 shell admissibility gives clause (3), and the translated closure-compatible readout gives clause (4). Hence each translated OP 3-admissible shell passage determines a shell certificate. Conversely, Lemma 1.195 and Theorem 1.197 show that any such certificate preserves exactly the closure-relevant content that the OP 3 theorem already isolates on the OP side. Lemma 1.199 records the same fact in commuting-square form. Thus the certificate layer is not a second closure notion; it is the coded OP 3 closure guide. □

Theorem 1.201 (First narrow shell-certificate iff package)**.** *Assume the direct kernel constructor of Definition 1.153 and the bridge conditions of Theorem 1.154. Let*

$$s_{\mathrm{in}} = \langle 0, a_{\mathrm{in}} \rangle, \qquad t_{b,\sigma} = \langle 0, a_{b,\sigma} \rangle,$$

and let r be any closure-compatible coded readout on the return state. Then the following are equivalent:

(i) $(s_{\mathrm{in}}, b, \sigma, t_{b,\sigma})$ *carries a shell admissibility certificate;*

(ii) there exists a coded branch state s_b such that

- *(a)* $\mathrm{QBranchRef}(b, s_{\mathrm{in}}, s_b)$,
- *(b) the branch code b is closure-admissible,*
- *(c)* $\mathrm{QShellRef}(\sigma, s_b, t_{b,\sigma})$ *and the refinement code σ is shell-admissible on b,*
- *(d) the return state $t_{b,\sigma}$ admits a closure-compatible coded readout,*
- *(e)* $\mathrm{nf}(a_{b,\sigma}) = \mathrm{nf}(a_{\mathrm{in}})$,
- *(f) every closure-relevant coded readout of the passage factors through one shell-stable kernel normal form.*

In particular, the present OP 3/QAM shell block already supports one narrow iff reading: a shell certificate is equivalent to the shell-stable kernel/readout package obtained from admissible branch-shell behavior together with kernel persistence.

Proof. Assume *(i)*. Then clauses (a)–(d) of *(ii)* are exactly the four certificate clauses from Definition 1.193. Clause (e) is Proposition 1.196, and clause (f) is Theorem 1.197. Hence *(ii)* follows.

Assume conversely *(ii)*. Then clauses (a)–(d) are precisely the data required by Definition 1.193, so $(s_{\mathrm{in}}, b, \sigma, t_{b,\sigma})$ carries a shell admissibility certificate. This is exactly Proposition 1.198. Thus *(i)* follows, and the final sentence is only a repackaging of the same forward and converse implications. □

Lemma 1.202 (First shell-obstruction lemma)**.** *Assume the direct kernel constructor of Definition 1.153 and the bridge conditions of Theorem 1.154. Let*

$$s_{\mathrm{in}} = \langle 0, a_{\mathrm{in}} \rangle, \qquad t_{b,\sigma} = \langle 0, a_{b,\sigma} \rangle,$$

and let r be any closure-compatible coded readout on the return state whenever such a readout exists. Suppose that at least one clause of the shell-stable kernel/readout package from Theorem 1.201 fails. Then no shell admissibility certificate exists for $(s_{\mathrm{in}}, b, \sigma, t_{b,\sigma})$. Equivalently, the first local OP 3 obstruction in the present QAM reading is the first failure of one of the following items:

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(i) admissible branch passage,

(ii) shell-admissible branch refinement,

(iii) closure-compatible return readout,

(iv) kernel persistence,

(v) shell-stable factorization of closure-relevant readout through one common kernel normal form.

In particular, a visible shell mismatch that leaves all five items intact is not yet an OP 3 obstruction in the present QAM sense.

Proof. By Theorem 1.201, a shell admissibility certificate is equivalent to the shell-stable kernel/readout package. Therefore the failure of any one clause of that package prevents the certificate from existing. The displayed list simply unwraps the package into its five operative clauses. The final sentence is the contrapositive reading of the same equivalence: visible shell variation alone is not an obstruction as long as admissible branch-shell transport, closure-compatible readout, kernel persistence, and common readout factorization still hold. □

Lemma 1.203 (First shell-obstruction stratification lemma)**.** *Assume the setup of Lemma 1.202. If no shell admissibility certificate exists for* $(s_{\mathrm{in}}, b, \sigma, t_{b,\sigma})$*, then at least one of the following three strata fails:*

(1) ***Branch stratum:*** *admissible branch passage fails.*

(2) ***Shell stratum:*** *admissible branch passage holds, but shell-admissible refinement fails.*

(3) ***Readout/kernel stratum:*** *admissible branch passage and shell-admissible refinement both hold, but closure-compatible return readout, kernel persistence, or common shell-stable readout factorization fails.*

Equivalently, every first local OP 3 obstruction in the present QAM reading is either a branch obstruction, a shell obstruction in the narrow sense, or a readout/kernel obstruction. In particular, whenever the branch and shell strata both remain intact, the first local obstruction is already forced into the readout/kernel layer.

Proof. Lemma 1.202 says that the first local obstruction is exactly the failure of one of five operative clauses. Group those clauses into the three strata above: clause (i) is the branch stratum, clause (ii) is the shell stratum, and clauses (iii)–(v) together form the readout/kernel stratum. This yields the stated trichotomy. The final sentence is only the contrapositive reading of the same grouping. □

Lemma 1.204 (Readout/kernel obstruction compression lemma)**.** *Assume the setup of Lemma 1.203, and assume moreover that the branch and shell strata both remain intact. Then the following are equivalent:*

(i) the first local OP 3 obstruction lies in the readout/kernel stratum;

(ii) at least one of the following failures occurs:

- *(a) closure-compatible return readout fails,*
- *(b) kernel persistence fails,*
- *(c) common shell-stable readout factorization through one kernel normal form fails;*

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(iii) there is no single shell-stable kernel/readout carrier through which all closure-relevant readout of the certified branch-shell passage can factor.

Equivalently, once the branch and shell strata are fixed, the whole remaining OP 3 obstruction is compressed to the failure of one common readout/kernel carrier class.

Proof. Under the standing assumption that the branch and shell strata remain intact, Lemma 1.203 says that any remaining first local obstruction must lie in the readout/kernel stratum. By definition of that stratum, this is exactly the disjunction of failures in clauses (a)–(c). This proves the equivalence of *(i)* and *(ii)*.

If a single shell-stable kernel/readout carrier exists, then closure-compatible return readout is available, kernel persistence identifies the same kernel normal form on source and return side, and Theorem 1.197 gives the common factorization through that carrier. Hence *(ii)* cannot occur. Conversely, if any one of clauses (a)–(c) fails, then no single shell-stable kernel/readout carrier can support the whole closure-relevant readout package. This proves the equivalence of *(ii)* and *(iii)*. The final sentence is only a compressed restatement of the same equivalence. □

Theorem 1.205 (First OP 3 shell-closure theorem in QAM form)**.** *Assume the faithful translation theorem for the admissible OP fragment, Theorem 1.273, together with the OP 3 shell theorem, Theorem 32.24. Let a translated OP 3-admissible shell passage determine coded data*

$$(s_{\mathrm{in}}, b, \sigma, t_{b,\sigma}, r).$$

Then the following are equivalent:

(i) the translated passage is QAM-shell-closed in the present narrow sense*, meaning that it carries a shell admissibility certificate and recovers the same closure-relevant shell content as the corresponding OP 3 step;*

(ii) $(s_{\mathrm{in}}, b, \sigma, t_{b,\sigma})$ *carries a shell admissibility certificate;*

(iii) the shell-stable kernel/readout package of Theorem 1.201 holds.

Moreover, under these equivalent conditions the passage admits a common shell-stable kernel/readout carrier through which all closure-relevant coded readout factors. If the equivalent conditions fail, then the first local OP 3 obstruction occurs at exactly one of the branch, shell, or readout/kernel strata from Lemma 1.203; and in the third stratum the whole remaining obstruction is precisely failure of one common shell-stable carrier in the sense of Lemma 1.204.

Proof. By Theorem 1.200, every translated OP 3-admissible shell passage determines a shell admissibility certificate and recovers the same closure-relevant shell content as the corresponding OP 3 step. Thus *(i)* and *(ii)* are equivalent in the present narrow reading.

The equivalence of *(ii)* and *(iii)* is exactly Theorem 1.201. Under these equivalent conditions, Theorem 1.197 gives factorization of every closure-relevant coded readout through the same shell-stable kernel normal form, hence through one common shell-stable kernel/readout carrier.

If the equivalent conditions fail, then by Lemma 1.202 the first local OP 3 obstruction is the first failure of the shell-stable kernel/readout package. Lemma 1.203 refines that failure into the branch, shell, and readout/kernel strata. Finally, Lemma 1.204 compresses the whole third stratum to failure of one common shell-stable carrier. This proves the final sentence. □

Corollary 1.206 (Shell defect localization is the first local OP 3 failure)**.** *Within the present OP 3/QAM reading, visible packet mismatch, phase-lock deviation, or shell-sensitive surface*

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variation is not yet by itself a shell defect. A genuine local OP 3 defect begins only when a shell defect witness in the sense of Definition 1.194 exists. Equivalently, by Theorem 1.205, the first local OP 3 failure is the first failure of the shell-stable kernel/readout package, stratified into branch, shell, and readout/kernel level and compressed in the third stratum to failure of one common shell-stable carrier. In particular, the first local failure is failure of a shell certificate clause, not failure of the entire OP 1–OP 5 closure chain.

Remark 1.207 (How this block should be used in the next OP 3 proof pass)**.** *The present certificate block is intentionally narrow. It does not yet force a full equivalence between the OP and QAM formalisms, and it does not yet activate mixed tensors. Its current role is to give the next proof pass a fixed reading order: establish a shell certificate, pass to kernel persistence where the deferred bridge is available, use Theorem 1.197 to factor readout through the shell-stable class, use Proposition 1.198 for the first narrow return direction, use Lemma 1.199 to read OP 3 refinement and QAM shell refinement as one commuting closure step, use Theorem 1.201 as the first closed iff package, use Lemma 1.202 to identify the first genuinely local obstruction, use Lemma 1.203 to decide whether that first obstruction sits at the branch, shell, or readout/kernel layer, use Lemma 1.204 to compress the third stratum to failure of one common shell-stable carrier, and then use Theorem 1.205 as the first bundled OP 3 shell-closure statement in public-shape QAM form before asking whether any further tensor layer is genuinely needed.*

Current extracted OP 3/QAM tool layer

Remark 1.208 (Purpose of the extracted OP 3/QAM tools)**.** *The certificate and obstruction chain above is no longer meant to be reread from first principles in every later proof pass. Its immediate use is as a compact tool layer: once a later argument can cite one of the extracted tools below, it may treat the corresponding shell passage as already compressed to the right closure-bearing datum and continue at the next structural interface.*

Shell-certificate tool.
If one can exhibit an admissible branch passage, an admissible shell refinement, and a closure-compatible return readout, then the passage may be treated as a certified OP 3 shell step in the present narrow QAM sense. This is the entry tool for the whole block.

Kernel-persistence tool.
Once a shell certificate is in place and the deferred kernel bridge is available, the visible shell variation may be replaced by equality of one kernel normal form. From that point onward the proof may work kernel-first instead of shell-first.

Shell-stable readout tool.
Once the shell certificate and kernel bridge are active, every closure-relevant readout may be pushed through one common shell-stable carrier. Later readout comparisons therefore no longer need to unfold the whole branch/shell passage.

Converse reconstruction tool.
If admissible branch passage, admissible shell refinement, closure-compatible return readout, and kernel persistence are already known from another source, then the shell certificate may be reconstructed rather than reproved from scratch.

Commuting-square tool.
OP 3 shell refinement on the OP side and branch/shell refinement on the QAM side may be read as one closure-preserving square. This tool licenses switching between OP and QAM viewpoints without reopening the shell argument.

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Obstruction-stratification tool.
If the shell-certificate package fails, the first local obstruction should be classified immediately as branch, shell, or readout/kernel level. This prevents visible shell mismatch from being over-read as a full OP 3 failure.

Carrier-compression tool.
Once branch and shell remain intact, every remaining local obstruction may be compressed to failure of one common shell-stable carrier. This is the shortest diagnostic form of the third obstruction stratum.

Bundled OP 3 closure tool.
Theorem 1.205 may now be cited as the compact public-shape closure package for OP 3 in QAM form: certificate, kernel/readout package, stratified obstruction, and common carrier are read as one bundled result.

Remark 1.209 (How to cite the extracted OP 3/QAM tools)**.** *A later argument should normally cite the smallest extracted tool that already matches its interface. Use the shell-certificate tool when the issue is admissible passage, the kernel-persistence or shell-stable readout tool when the issue is invariance under visible shell variation, the converse reconstruction tool when the shell certificate is to be recovered from already known transport/readout data, the commuting-square tool when the proof needs to switch languages, the obstruction-stratification or carrier-compression tool when the issue is failure localization, and the bundled OP 3 closure tool when the whole narrow closure package is intended at once.*

First OP 3 proof run through the extracted QAM tool layer

Remark 1.210 (Reading key for the first OP 3 proof run)**.** *In the present narrow public-shape QAM reading, the first proof run should be read as a deliberately compressed structural pass rather than as a maximal theorem expansion. The input is a translated OP 3 shell passage. The allowed move is passage from visible shell data to the extracted tool layer. The invariant is the common closure-bearing kernel/readout carrier whenever the certificate package is active. The defect signal is the first obstruction stratum from Lemma 1.203. In plain language: the proof run should decide as early as possible whether the shell step is already certified, and if not, where the first local failure sits.*

Theorem 1.211 (First OP 3 proof run through extracted QAM form)**.** *Assume the faithful translation theorem for the admissible OP fragment, Theorem 1.273, the OP 3 shell theorem, Theorem 32.24, the direct kernel constructor of Definition 1.153, and the bridge conditions of Theorem 1.154. Let*

$$\tau(x) = s_{\mathrm{in}} = \langle 0, a_{\mathrm{in}} \rangle$$

be the coded QAM source state obtained from an admissible OP 3 source configuration x, and let the translated branch/shell passage be encoded by a branch code b, a refinement code σ, a coded branch state s_b, and a coded return state

$$t_{b,\sigma} = \langle 0, a_{b,\sigma} \rangle.$$

Then the first OP 3 proof run through extracted QAM form has exactly two outcomes:

(i) either the translated passage activates the bundled OP 3 closure tool, so that the OP 3 shell step is certified in QAM form, all closure-relevant coded readout factors through one common shell-stable carrier, and the passage may be continued kernel-first;

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(ii) or the bundled closure package fails, in which case the first local obstruction is immediately localized by the obstruction-stratification tool at exactly one of the branch, shell, or readout/kernel strata, with the whole third stratum compressed to failure of one common shell-stable carrier.

In particular, the extracted tool layer already supports a genuine proof run: it converts translated OP 3 shell data into either a compressed closure package or a minimal obstruction diagnosis, without reopening the whole OP/QAM comparison at each step.

Proof. Translate the admissible OP fragment by Theorem 1.273. By the OP 3 shell theorem, Theorem 32.24, the translated branch and shell passage is exactly the stage at which the OP side isolates admissible shell refinement. Use the commuting-square tool to identify the translated OP 3 shell step with the coded branch/shell refinement step on the QAM side.

Now ask whether the shell-certificate tool is activated. If it is, then by the kernel-persistence tool the visible shell passage may be replaced by equality of one kernel normal form, and by the shell-stable readout tool every closure-relevant coded readout factors through one common shell-stable carrier. The bundled OP 3 closure tool then yields outcome *(i)*.

If the shell-certificate package is not activated, apply the obstruction-stratification tool. This localizes the first failure immediately at exactly one of the branch, shell, or readout/kernel levels. If the failure lies in the third stratum, the carrier-compression tool compresses the remaining failure to absence of one common shell-stable carrier. This is outcome *(ii)*.

The two outcomes are exhaustive because Theorem 1.205 already packages the positive direction as the bundled shell-closure theorem, while Lemma 1.203 together with Lemma 1.204 packages the negative direction. Hence the extracted tool layer already yields a genuine first OP 3 proof run in QAM form. □

Corollary 1.212 (First operational use of the extracted OP 3/QAM tools)**.** *A later OP 3 argument no longer needs to reopen the whole branch/shell analysis from scratch whenever a translated shell passage appears. It is enough to test whether the shell-certificate tool is active. If yes, cite the bundled OP 3 closure tool and continue kernel-first; if not, cite the obstruction-stratification tool and, in the third stratum, the carrier-compression tool. Thus the proof interface has already been shortened to a certified-pass / localized-obstruction dichotomy.*

Proof. This is only a restatement of Theorem 1.211 in proof-interface language. □

Remark 1.213 (Status separation for the present OP 3/QAM closure package)**.** *The present block should now be read in two layers with different evidential status. The* active public-shape result *is the certificate-level OP 3/QAM closure package: Definitions 1.193 and 1.194, Lemma 1.195, Theorem 1.201, Theorem 1.205, and Corollary 1.212. These statements already yield a closed certified-pass / localized-obstruction interface for translated OP 3 shell passages without leaving the active tuple surface. By contrast, the stronger kernel-first language — in particular Proposition 1.196, Theorem 1.197, and the carrier-compression reading of the third obstruction stratum — is a* conditional strengthening*: it becomes authoritative only under the deferred bridge hypotheses of Definition 1.149 and Theorem 1.154. Accordingly, later use should distinguish carefully between active certificate-level closure and conditional kernel/readout compression.*

Proposition 1.214 (Citation and use rule for the completed OP 3/QAM block)**.** *Let a later argument involve a translated OP 3 shell passage. Then the following citation discipline is valid.*

(i) If the argument needs only a certified-pass / localized-obstruction decision at the active tuple level, it may cite Theorem 1.205, Theorem 1.211, and Corollary 1.212 without invoking any kernel-first hypothesis.

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(ii) If the argument further needs invariance under visible shell variation in kernel normal-form language or one-carrier readout factorization, then it may additionally cite Proposition 1.196, Theorem 1.197, and Lemma 1.204, but only after the deferred bridge hypotheses have been assumed or discharged.

(iii) If neither the certificate package nor the bridge package is available, the argument must fall back to the earlier branch/shell/readout layer and may not treat visible shell data as a closure theorem by themselves.

Proof. Clause (i) is exactly the active reading packaged by Theorem 1.205, Theorem 1.211, and Corollary 1.212. Clause (ii) is the additional kernel-first strengthening isolated earlier in Proposition 1.196, Theorem 1.197, and Lemma 1.204; those statements explicitly depend on the deferred bridge hypotheses. Clause (iii) is the corresponding fallback rule: absent both packages, one must continue on the earlier active branch/shell/readout surface. □

Corollary 1.215 (Operational checklist for the completed OP 3/QAM block)**.** *A translated OP 3 shell passage should now be tested in the following order.*

1. *Check admissible branch passage and branch admissibility.*
2. *Check shell refinement and shell admissibility on the chosen branch.*
3. *Check existence of a closure-compatible return readout.*
4. *If all three checks succeed, activate the shell-certificate package and cite the bundled OP 3 closure tool.*
5. *Only if the deferred bridge hypotheses are additionally available, pass to kernel persistence and shell-stable readout factorization.*
6. *If one of the certificate checks fails, localize the first obstruction immediately at branch, shell, or readout/kernel level instead of reopening the whole OP 1–OP 5 chain.*

In particular, the completed block already provides one stable certified interface: certificate first, kernel compression second, obstruction localization immediately upon failure.

Remark 1.216 (Diagnostic handoff rule for visible shell data)**.** *Visible packet mismatch, phase-lock deviation, shell-sensitive surface variation, and analogous shell diagnostics remain* screening-layer data *unless and until they witness failure of one clause of the shell-certificate package or, under the bridge hypotheses, failure of one common shell-stable carrier. Thus the present block fixes a handoff rule from pictures and visible diagnostics to theorematic closure language: diagnostics may guide attention to the relevant branch, shell, or readout/kernel stratum, but they do not create a second closure axiom beside the completed OP 3/QAM certificate package. This is the intended local counterpart of the earlier visible/structural separation principle and the correct interface toward the later OP 3 to OP 5 transfer line.*

Proposition 1.217 (Prospective kernel-first reduction of the branch-shell theorem)**.** *Assume the direct kernel constructor of Definition 1.153 and the bridge conditions of Theorem 1.154. Let*

$$s_{\mathrm{in}} = \langle 0, a_{\mathrm{in}} \rangle$$

be the coded source state, let b be a closure-admissible branch code, let σ be a shell-admissible refinement code on b, and let

$$t_{b,\sigma} = \langle 0, a_{b,\sigma} \rangle$$

be an associated branch-shell coded return realization. Then the conclusion of Theorem 1.192 may be read kernel-first as follows:

1. *one has*
$$\mathrm{nf}(a_{b,\sigma}) = \mathrm{nf}(a_{\mathrm{in}}),$$
2. *hence every closure-compatible readout of* s_{in} *and* $t_{b,\sigma}$ *factors through the single canonical kernel class determined by* a_{in}*,*
3. *therefore branch choice and shell refinement may be treated as closure-preserving already at exposed-kernel level before returning to the visible QAM state surface.*

Proof. Theorem 1.192 gives
$$\mathrm{QClEq}(s_{\mathrm{in}}, t_{b,\sigma}).$$
Since both states lie on the present state surface, we may write
$$s_{\mathrm{in}} = \langle 0, a_{\mathrm{in}} \rangle, \qquad t_{b,\sigma} = \langle 0, a_{b,\sigma} \rangle.$$
By the direct kernel constructor,
$$\ker_{\mathrm{dir}}(s_{\mathrm{in}}) = a_{\mathrm{in}}, \qquad \ker_{\mathrm{dir}}(t_{b,\sigma}) = a_{b,\sigma}.$$
The bridge clause for closure reflection therefore yields
$$\mathrm{nf}(a_{b,\sigma}) = \mathrm{nf}(a_{\mathrm{in}}).$$
Thus branch admissibility together with shell admissibility already identifies the same canonical kernel class before any further visible-surface comparison is carried out. By the factorization clause of the bridge datum, every closure-compatible coded readout depends only on that common kernel class. Hence the branch-shell theorem may be re-read as a kernel-first statement and only afterwards lifted back to the visible state surface. □

Remark 1.218 (Axiom audit after the kernel-first branch-shell reduction)**.** *Proposition 1.217 does* not *alter the active proof status of Theorem 1.192. The active theorem still rests only on closure-admissible branch refinement, shell-admissible OP 3 refinement, and closure-compatible coded readout. The kernel-first proposition adds force only under the deferred payload-level bridge hypotheses isolated in Definition 1.149 and Theorem 1.154. Thus no new hidden axiom has entered the live branch-shell theorem; only a prospective kernel-level compression of its proof surface has been recorded.*

Remark 1.219 (How much proof load the kernel-first branch-shell reading removes)**.** *At branch-shell level the cardinality factor is just* 1*: there is one visible source surface*
$$s_{\mathrm{in}} = \langle 0, a_{\mathrm{in}} \rangle$$
and one visible return surface
$$t_{b,\sigma} = \langle 0, a_{b,\sigma} \rangle.$$
So the gain is not a reduction in the number of cases, but a reduction in the size of the compared objects*. On the visible state surface the proof compares two full coded states and then passes a closure-compatible readout across that state-level identification. In the deferred kernel-first reading, that work is replaced by the single normal-form check*
$$\mathrm{nf}(a_{b,\sigma}) = \mathrm{nf}(a_{\mathrm{in}})$$
followed by one readout through the common kernel class. If one measures a visible-state comparison by cost C_{surf} *and a kernel normal-form comparison by cost* C_{ker}*, then the conceptual load changes from roughly*
$$C_{\mathrm{surf}} + C_{\mathrm{read}}$$

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to

$$C_{\mathrm{ker}} + C_{\mathrm{class}},$$

with intended gain $C_{\mathrm{ker}} < C_{\mathrm{surf}}$ *and typically* $C_{\mathrm{class}} \ll C_{\mathrm{read}}$. *So the numerical savings here are modest compared with the routed case, but methodologically this is the crucial base step: it is the first place where a full visible return state can be replaced by its exposed payload kernel without changing the closure conclusion.*

Remark 1.220 (Theorem-level unpacking of the compact chain, I)**.** *Theorem 1.192 is the first theorem-level unpacking of the compact front statement: it isolates the middle part of the release chain in which closure-preserving source-return transport passes through branch choice and shell refinement without losing closure-relevant readout content.*

Corollary 1.221 (QAM branch-shell support class)**.** *Define the coded admissible branch-shell support class by*

$$\mathcal{S}_{\mathrm{adm}}^{\mathrm{QAM}} := \{\, (b,\sigma) : b \in \mathcal{B}_{\mathrm{adm}}^{\mathrm{QAM}} \ \wedge \ \sigma \in \mathcal{F}_{\mathrm{sh}}^{\mathrm{QAM}}(b) \,\}.$$

Then every coded return realization supported on

$$\mathcal{S}_{\mathrm{adm}}^{\mathrm{QAM}}$$

is closure-preserving at the coded QAM level.

Remark 1.222. *This is the coded QAM counterpart of the admissible branch-shell support sector in the present OP architecture.*

Definition 1.223 (QAM branch-shell routed return)**.** *Let*

$$u = \langle 8, \Omega, w \rangle$$

be an admissible routed support code with

$$\Omega \subseteq \mathcal{S}_{\mathrm{adm}}^{\mathrm{QAM}}.$$

A QAM branch-shell routed return *is a coded return aggregation associated with the support family* Ω *and the weight assignment* w.

Theorem 1.224 (QAM routed admissible support theorem)**.** *Let* $u = \langle 8, \Omega, w \rangle$ *be a normalized admissible routed support code, and let* r *be a closure-compatible coded readout. Then the closure-relevant image of the routed return determined by* u *agrees with the closure-relevant image of the coded source state.*

Proof. Every element $(b,\sigma) \in \Omega$ lies in the admissible branch-shell support class, so each associated coded return realization is closure-equivalent to the coded source state by the previous theorem. Since the routed support is normalized and the readout is closure-compatible, the routed aggregation preserves the same closure-relevant image after passage through the factor map. □

Proposition 1.225 (Prospective kernel-first reduction of the routed support theorem)**.** *Assume the direct kernel constructor of Definition 1.153 and the bridge conditions of Theorem 1.154. Let*

$$s_{\mathrm{in}} = \langle 0, a_{\mathrm{in}} \rangle$$

be the coded source state, and let

$$u = \langle 8, \Omega, w \rangle$$

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be a normalized admissible routed support code. For each $(b, \sigma) \in \Omega$*, let*

$$t_{b,\sigma} = \langle 0, a_{b,\sigma} \rangle$$

be an associated branch-shell coded return realization. Then the conclusion of Theorem 1.224 may be read kernel-first as follows:

1. *for every* $(b, \sigma) \in \Omega$ *one has*
$$\mathrm{nf}(a_{b,\sigma}) = \mathrm{nf}(a_{\mathrm{in}}),$$
2. *hence every closure-compatible routed readout factors through the single canonical kernel class of* a_{in}*,*
3. *therefore the routed aggregation determined by* u *has the same closure-relevant image as the coded source state without needing to compare the full family of visible state surfaces directly.*

Proof. For each $(b, \sigma) \in \Omega$, Theorem 1.192 gives

$$\mathrm{QClEq}(s_{\mathrm{in}}, t_{b,\sigma}).$$

Since both states lie on the present state surface, we may write

$$s_{\mathrm{in}} = \langle 0, a_{\mathrm{in}} \rangle, \qquad t_{b,\sigma} = \langle 0, a_{b,\sigma} \rangle.$$

By the direct kernel constructor, the kernel representatives are just the exposed payload kernels:

$$\ker_{\mathrm{dir}}(s_{\mathrm{in}}) = a_{\mathrm{in}}, \qquad \ker_{\mathrm{dir}}(t_{b,\sigma}) = a_{b,\sigma}.$$

The bridge clause for closure reflection therefore yields

$$\mathrm{nf}(a_{b,\sigma}) = \mathrm{nf}(a_{\mathrm{in}})$$

for every $(b, \sigma) \in \Omega$. Thus the entire admissible routed support family lies in one canonical kernel class. By the factorization clause of the bridge datum, every closure-compatible coded readout depends only on that kernel class. Hence the routed aggregation has the same closure-relevant image as the coded source state. The normalization of the routed weights matters only after this kernel reduction, so the proof burden is shifted from a family of full visible state surfaces to one canonical kernel class. □

Remark 1.226 (Why the routed support theorem is the best stress test)**.** *Among the currently marked finals, Theorem 1.224 is the heaviest set-theoretic test case for the deferred zero-kernel language: it simultaneously carries a support family* Ω*, a coded weight assignment* w*, branch admissibility, shell admissibility, routed aggregation, and closure-compatible readout. Exactly there the kernel-first reading is strongest, because the visible family of state surfaces is replaced by one canonical kernel class before the routed aggregation is evaluated.*

Remark 1.227 (Axiom audit after the kernel-first routed reduction)**.** *Proposition 1.225 does* not *strengthen the active QAM axiomatics. Its extra force comes only from deferred hypotheses already isolated earlier: the direct kernel constructor of Definition 1.153 and the four payload-level bridge conditions of Theorem 1.154. In particular, the active theorem chain still proves Theorem 1.224 without mentioning kernels at all. The new proposition merely says that* if *those deferred bridge conditions are later discharged, then the routed-support proof may be re-read kernel-first. Thus no silent new axiom has entered the then-present public line; only a prospective compression interface has been exhibited.*

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Remark 1.228 (How much comparison load the kernel-first reading removes)**.** *Write $N := |\Omega|$ for the size of the routed support family. On the visible state surface, the routed-support proof conceptually tracks N branch-shell return states*

$$t_{b,\sigma} = \langle 0, a_{b,\sigma} \rangle$$

and compares each of them with the coded source state at full state-surface level before the routed aggregation is passed through a closure-compatible readout. In the deferred kernel-first reading, those N state-surface comparisons are replaced by the N normal-form checks

$$\mathrm{nf}(a_{b,\sigma}) = \mathrm{nf}(a_{\mathrm{in}}), \qquad (b, \sigma) \in \Omega,$$

followed by a single *readout through the common canonical kernel class of a_{in}. Consequently the proof burden drops from "an N-member family of full visible state surfaces plus routed readout lifting" to "one canonical kernel class together with N membership/equality checks into that class." If one measures a full visible-state comparison by cost C_{surf} and a kernel normal-form comparison by cost C_{ker}, then the conceptual comparison load changes from roughly*

$$N\, C_{\mathrm{surf}} + C_{\mathrm{route}}$$

to

$$N\, C_{\mathrm{ker}} + C_{\mathrm{class}},$$

where typically $C_{\mathrm{class}} \ll C_{\mathrm{route}}$ and the intended gain is precisely that $C_{\mathrm{ker}} < C_{\mathrm{surf}}$. In proof-shape terms, nothing miraculous happens to the factor N itself, but the object size *being compared is reduced from full state surfaces to exposed kernels, and the routed readout is evaluated once on a single kernel class instead of being conceptually re-established across the whole visible family.*

Remark 1.229 (Theorem-level unpacking of the compact chain, II)**.** *Theorem 1.224 is the routed continuation of Theorem 1.192. Together they unpack the compressed public-shape chain*

$$s_{\mathrm{in}} \longrightarrow (b, \sigma) \longrightarrow u \longrightarrow r$$

of Theorem 1.37 into theorem-level branch/shell preservation followed by theorem-level routed aggregation.

Remark 1.230 (Staged proof compression pattern under the deferred zero-kernel language)**.** *The three prospective kernel-first reductions now show a genuinely staged compression pattern. At the first stage, Proposition 1.175 replaces the visible six-state transport chain by equality of six exposed kernel normal forms, so that admissible transport is already read as kernel-class preservation before any branch or shell data enter. At the second stage, Proposition 1.217 replaces one visible return-surface comparison*

$$\langle 0, a_{b,\sigma} \rangle \sim \langle 0, a_{\mathrm{in}} \rangle$$

by the single kernel normal-form comparison

$$\mathrm{nf}(a_{b,\sigma}) = \mathrm{nf}(a_{\mathrm{in}}).$$

At the third stage, Proposition 1.225 repeats that reduced comparison only at the level of exposed kernels across the whole routed family Ω and then evaluates the closure-compatible readout once on the common kernel class. Thus the deferred zero-kernel language first shrinks the transport chain to a common kernel class, then shrinks each branch-shell return to one exposed kernel comparison, and only afterwards turns the routed family into a class-membership problem rather than a full visible-surface transport problem. This is exactly why transport is the first chain-level compression step, branch-shell the first local return-surface compression step, and routed support the first true family-level compression step.

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Definition 1.231 (Prospective activation package for the deferred zero-kernel language)**.** *A* zero-kernel activation package *for the present QAM surface consists of the following data and properties:*

(i) a total assignment on active state surfaces

$$\ker : QState \to \mathcal{A}, \qquad \ker(\langle 0, a\rangle) = a,$$

compatible with Definition 1.152;

(ii) kernel admissibility, meaning that every active exposed payload kernel lies in the admissible kernel class $\mathcal{A}$;

(iii) kernel normal-form reflection of closure, meaning that closure-equivalent active states have the same kernel normal form;

(iv) kernel-level factorization of every closure-compatible readout through the corresponding kernel normal form;

(v) compatibility of the routed aggregation step with passage to the common kernel class.

Theorem 1.232 (Activation criterion for kernel-first proof compression on the present QAM surface)**.** *Assume a zero-kernel activation package in the sense of Definition 1.231. Then the deferred zero-kernel language may be used as a genuine proof-compression layer for the present QAM surface in the following precise sense:*

(a) every active state-surface argument may be reduced to a kernel normal-form argument without loss of closure-relevant content;

(b) the first transport theorem may be read kernel-first by proving preservation of one common kernel normal-form class along the admissible transport chain;

(c) the branch-shell theorem may be read kernel-first by proving normal-form preservation of the exposed return kernels;

(d) the routed support theorem may be read kernel-first by proving membership of all exposed routed kernels in a single common normal-form class and evaluating the closure-compatible readout once on that class.

In particular, the deferred zero-kernel language ceases to be merely heuristic at exactly the point where closure, readout, and routed aggregation all factor through kernel normal forms. The criterion identifies what *must hold; the later staged-roadmap remarks then specify* in which order *those requirements should be established on the present QAM surface.*

Proof. This is just the joint content of Theorem 1.150, Theorem 1.154, Proposition 1.175, Proposition 1.217, and Proposition 1.225, collected as a single activation statement. Once closure-equivalent active surfaces have the same kernel normal form and every relevant readout factors through that normal form, every transport, branch-shell, and routed-support comparison may be carried out on kernels first and only lifted back to the visible state surface at the end. □

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Remark 1.233 (Criterion-to-roadmap reading rule)**.** *The activation theorem above should be read together with the staged roadmap below. The theorem gives the logical threshold for activation, whereas the roadmap orders the internal work needed to reach that threshold:*

$$\begin{aligned} &\textit{transport-class preservation} \\ \Longrightarrow\ &\textit{local branch-shell kernel reduction} \\ \Longrightarrow\ &\textit{routed family compression.} \end{aligned}$$

In this combined reading, the criterion says when *kernel-first proof compression becomes legitimate, while the roadmap says* how *the present document should approach that point without introducing premature changes at visible tuple level.*

Remark 1.234 (Practical migration trigger for a later binary-tag version)**.** *The activation criterion above, read together with the later criterion-to-roadmap rule, also isolates the only later migration trigger that would make a binary or power-of-two tag scheme genuinely worthwhile. Such a migration would become structurally justified only once the present state surface is already being used kernel-first in proofs, so that the visible tag layer has become a comparatively thin presentation layer while closure and readout are actually controlled at kernel-normal-form level. Before that point, a binary tag migration would mainly rename visible surfaces; after that point, it could act as a real simplification of the presentation layer.*

Remark 1.235 (What still has to be shown before activation)**.** *At the present stage, the deferred zero-kernel language still stops short of activation for one simple reason: the document now contains the bridge data, the direct kernel final, and three staged stress tests, but it does not yet contain a fully internal proof that the active closure relation and every closure-compatible readout already factor through the exposed payload normal forms on the entire current QAM surface. The current status is therefore stronger than a mere metaphor but weaker than an active rewrite rule.*

Remark 1.236 (Prospective staged activation roadmap for the zero-kernel language)**.** *The current document already suggests a natural staged migration order for any later activation of the deferred zero-kernel language. Stage I is the* transport stage*: one first proves that admissible source-return transport preserves a single common kernel normal-form class all along the visible active chain. Stage II is the* local return stage*: one then proves that every admissible branch-shell return surface may be replaced by one exposed kernel comparison against the incoming kernel class. Stage III is the* family stage*: one finally proves that the whole routed family collapses to membership in that common kernel class, so that closure-compatible readout is evaluated once at class level rather than re-established on each visible routed surface. Only after these three stages are internally available would a later binary or power-of-two visible presentation acquire real proof-theoretic leverage, because only then would the visible tuple tags have become a thin presentation shell over an already kernel-first proof flow.*

Remark 1.237 (Why this staged order matters)**.** *The order above is not cosmetic. If routed family compression were attempted before transport-class preservation and local branch-shell reduction were available, then the family step would still depend on full visible surfaces and would not yet be a genuine kernel-first simplification. The staged order therefore identifies the first point at which the deferred zero-kernel language would cease to be merely suggestive and would begin to remove real set-theoretic comparison load in a controlled way. Together with the criterion-to-roadmap rule, it turns the activation block from a static admissibility test into a concrete migration sequence.*

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Remark 1.238 (Compressed synthesis of the deferred zero-kernel path)**.** *The deferred zero-kernel material above should now be read as one compact path rather than as many separate addenda. The present visible state surface remains unchanged. A later kernel language would first expose the exposed payload kernel of a coded state as a final kernel, then compare closure only through kernel normal forms, then compress transport, local branch-shell return, and finally routed families in that order, and only afterwards consider whether a binary visible presentation would simplify the already kernel-first proof flow. In that reading, the deferred block does not introduce a second theory beside the current QAM package. It isolates one possible later proof-compression pipeline inside the same public branch/shell/routed line.*

Remark 1.239 (Reading guide for the deferred block)**.** *A compact reader may keep only the following chain in view:*

$$\textit{visible state surface} \implies \textit{exposed payload kernel} \implies \textit{kernel normal form}$$
$$\implies \textit{transport/local/family compression} \implies \textit{possible later binary presentation.}$$

The detailed deferred notes are included only to justify each arrow in this chain and to mark exactly which theorem-level bridges would still need to be proved before activation.

Remark 1.240 (Dependency map for the deferred zero-kernel path)**.** *The deferred block now has a strict internal dependency order. First, Definitions 1.136–1.143 fix the later kernel vocabulary: carriers, filters, primitive zero-objects, recursive reuse, aspect projections, normal forms, and internal search keys. Second, the axiom-coherence and current-limitation remarks state why this added vocabulary does not yet compete with the active tuple layer. Third, Definition 1.149, Theorem 1.150, Definition 1.152, and Theorem 1.154 isolate the exact interface through which active coded states could later be re-read kernel-first. Fourth, Propositions 1.175, 1.217, and 1.225 test that interface on the three currently most relevant proof stages. Finally, Definition 1.231, Theorem 1.232, and the later activation-roadmap, synthesis, and reading-guide remarks convert those tests into an activation criterion and a migration order. In particular, any later binary visible presentation should depend on this whole chain and should not precede it.*

Remark 1.241 (Local admissibility test for a later kernel-first rewrite)**.** *The deferred block can now also be read as a local test for whether a specific visible tuple-surface proof step is mature enough for later kernel-first rewriting. Given a proof step whose visible data consist of active coded states*

$$s_i = \langle 0, a_i \rangle,$$

one should ask four questions in order.

1. Exposure: *do all exposed payload kernels a_i already lie in the intended admissible kernel domain $\mathcal{A}$?*
2. Reduction target: *can the visible conclusion be reformulated as equality, preservation, or common membership of the corresponding kernel normal forms* $\mathrm{nf}(a_i)$*?*
3. Factorization: *does every closure-relevant readout or routed aggregation used in that proof step depend only on the resulting kernel normal-form class?*
4. Lift back: *once the kernel-class statement is proved, does the original visible state-surface conclusion follow again without adding new data?*

If the answer is yes in all four places, then the proof step is a genuine final for later kernel-first compression. If one of the four answers is still no, the corresponding proof step should remain on the visible state surface for the time being.

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Remark 1.242 (Fallback rule for current proofs)**.** *Failure of the local admissibility test is not a defect of the present active proof line. It simply means that the relevant argument should continue to cite the active visible-surface theorem rather than its deferred kernel-first restatement. This fallback rule prevents premature theorem drift: the deferred block may prepare later compression, but it does not lower the evidential status required for current public proofs.*

Remark 1.243 (Compressed rewrite schema)**.** *Whenever the later local admissibility test applies, the deferred zero-kernel path suggests one uniform later rewrite schema:*

$$\textit{visible tuple step} \implies \textit{expose payloads} \implies \textit{pass to } \mathrm{nf}$$
$$\implies \textit{prove closure/readout at class level} \implies \textit{lift back to the visible surface.}$$

This does not yet activate any new theorem in the present document. It simply records, in one line, the exact proof pattern that the three staged stress tests have already isolated separately for transport, local branch/shell return, and routed families.

Remark 1.244. *This is the coded QAM analogue of the combined branch-shell transport theorem together with normalized routed admissible preservation.*

Definition 1.245 (QAM screening layer)**.** *A coded diagnostic layer is called a* screening layer *if it provides visible preselection information about admissible branch-shell support without itself constituting an independent closure proof.*

Remark 1.246. *In this sense, packet-, phase-lock-, and epsilon-type coded diagnostics should be treated in QAM as screening-layer data rather than as primary closure criteria. Their structural relevance begins only when they track loss of branch admissibility or loss of shell admissibility.*

Corollary 1.247 (QAM visible/structural separation)**.** *Visible coded mismatch in the screening layer does not by itself imply failure of closure at the QAM level. Closure failure begins only when coded branch admissibility or coded shell admissibility is lost.*

Remark 1.248. *This is the QAM-over-ZF shadow of the visible-mismatch-versus-structural-mismatch principle already established in the current Companion line.*

QAM tensor admissibility and tensor closure rules

We now strengthen the tensor-coded part of the QAM language. At the present stage, tensor composition is still not treated as a full tensor algebra. Instead, it is introduced as a controlled structural constructor equipped with admissibility, closure, transport, and readout compatibility rules.

Definition 1.249 (Tensor-coded formation)**.** *If p and q are QAM-coded objects, then*

$$p \otimes q := \langle 7, p, q \rangle$$

is their tensor-coded composite. We write

$$\mathrm{QTensor}(p \otimes q)$$

for the statement that the code has the tensor tag.

Definition 1.250 (Tensor formation by kind)**.** *The following tensor formations are allowed in principle:*

1. *state-state tensors:*
$$s_1 \otimes s_2;$$
2. *layer-layer tensors:*
$$\ell_1 \otimes \ell_2;$$
3. *operator-operator tensors:*
$$o_1 \otimes o_2;$$
4. *readout-readout tensors:*
$$r_1 \otimes r_2;$$
5. *branch-branch tensors:*
$$b_1 \otimes b_2;$$
6. *shell-shell tensors:*
$$\sigma_1 \otimes \sigma_2.$$

A tensor-coded composite is called well-typed *if both entries belong to one of the admissible kind-pairs listed above.*

Remark 1.251. *At this stage, mixed tensors between different kinds are not taken as primitive. They may be introduced later through derived coding conventions if needed.*

Definition 1.252 (Tensor admissibility)**.** *For QAM-coded objects x, y, we write*

$$\mathrm{QTensorAdm}(x, y)$$

iff:

1. *$x \otimes y$ is well-typed;*
2. *the tensor-coded composite preserves the intended closure-bearing structural regime;*
3. *the tensor-coded composite is admitted as a legitimate QAM object of its corresponding kind.*

Remark 1.253. *Thus tensor admissibility is not automatic from mere code formation. It is a structural compatibility condition.*

Definition 1.254 (Tensor closure compatibility)**.** *For coded states s_1, s_2, t_1, t_2, we write*

$$\mathrm{QTensorCl}(s_1, s_2; t_1, t_2)$$

iff

$$\mathrm{QClEq}(s_1, t_1) \qquad \textit{and} \qquad \mathrm{QClEq}(s_2, t_2)$$

and the tensor composites

$$s_1 \otimes s_2 \qquad \textit{and} \qquad t_1 \otimes t_2$$

are tensor-admissible.

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Axiom 4 (Tensor closure propagation schema)**.** *If*

$$\mathrm{QTensorCl}(s_1, s_2; t_1, t_2),$$

then

$$\mathrm{QClEq}(s_1 \otimes s_2,\ t_1 \otimes t_2).$$

Remark 1.255. *This is the first basic tensor closure principle: closure-equivalent components yield closure-equivalent tensor composites whenever tensor admissibility holds.*

Definition 1.256 (Tensor-coded transport)**.** *Let*

$$o_1 = \langle 2, \ell_1, \ell_1', G_1 \rangle, \qquad o_2 = \langle 2, \ell_2, \ell_2', G_2 \rangle$$

be coded operators. Their tensor-coded transport is

$$o_1 \otimes o_2.$$

We say that

$$\mathrm{QTensorTransport}(o_1, o_2; s_1, s_2; t_1, t_2)$$

holds iff:

1. $\mathrm{QAdmTransport}(o_1, s_1, t_1)$*;*
2. $\mathrm{QAdmTransport}(o_2, s_2, t_2)$*;*
3. $\mathrm{QTensorAdm}(s_1, s_2)$ *and* $\mathrm{QTensorAdm}(t_1, t_2)$*;*
4. $\mathrm{QTensorAdm}(o_1, o_2)$.

Axiom 5 (Tensor transport admissibility schema)**.** *If*

$$\mathrm{QTensorTransport}(o_1, o_2; s_1, s_2; t_1, t_2),$$

then

$$\mathrm{QAdmTransport}(o_1 \otimes o_2,\ s_1 \otimes s_2,\ t_1 \otimes t_2).$$

Remark 1.257. *This is the tensorial lift of admissible transport: admissible componentwise transport induces admissible tensor transport whenever the tensor compatibility conditions are satisfied.*

Definition 1.258 (Tensor-coded readout)**.** *Let*

$$r_1 = \langle 3, \ell_1, W_1, R_1 \rangle, \qquad r_2 = \langle 3, \ell_2, W_2, R_2 \rangle$$

be coded readouts. Their tensor-coded readout is

$$r_1 \otimes r_2.$$

We say that $r_1 \otimes r_2$ *is* tensor closure-compatible *if:*

1. r_1 *and* r_2 *are closure-compatible;*
2. $\mathrm{QTensorAdm}(r_1, r_2)$*;*
3. *the closure-relevant readout factorization of* $r_1 \otimes r_2$ *depends only on the tensor closure class of the source state.*

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Remark 1.259. *This is the tensor analogue of closure-compatible readout. It allows tensor-composite visible outputs to differ at raw level while preserving closure content after factorization.*

Definition 1.260 (Tensor branch admissibility)**.** *Let b_1, b_2 be closure-admissible branch codes. Their tensor composite*

$$b_1 \otimes b_2$$

is called tensor branch-admissible *if:*

1. $\mathrm{QTensorAdm}(b_1, b_2)$*;*
2. *componentwise branch refinement preserves closure class;*
3. *the induced tensor branch refinement preserves closure class at the tensor-composite level.*

Definition 1.261 (Tensor shell admissibility)**.** *Let σ_1, σ_2 be shell-admissible refinement codes. Their tensor composite*

$$\sigma_1 \otimes \sigma_2$$

is called tensor shell-admissible *if:*

1. $\mathrm{QTensorAdm}(\sigma_1, \sigma_2)$*;*
2. *componentwise shell refinement preserves closure class;*
3. *the induced tensor shell refinement preserves closure class at the tensor-composite level.*

Remark 1.262. *Tensor branch admissibility and tensor shell admissibility are the coded QAM precursors of coupled branch/shell transport across multiple structural sectors.*

Proposition 1.263 (Tensor closure preservation principle)**.** *Let s_1, s_2, t_1, t_2 be coded states. Assume*

$$\mathrm{QClEq}(s_1, t_1), \qquad \mathrm{QClEq}(s_2, t_2), \qquad \mathrm{QTensorAdm}(s_1, s_2), \qquad \mathrm{QTensorAdm}(t_1, t_2).$$

Then

$$\mathrm{QClEq}(s_1 \otimes s_2,\ t_1 \otimes t_2).$$

Proof. This is exactly the tensor closure propagation schema. □

Proposition 1.264 (Tensor transport preservation principle)**.** *Assume*

$$\mathrm{QTensorTransport}(o_1, o_2; s_1, s_2; t_1, t_2).$$

Then

$$\mathrm{QClEq}(s_1 \otimes s_2,\ t_1 \otimes t_2).$$

Proof. By tensor transport admissibility,

$$\mathrm{QAdmTransport}(o_1 \otimes o_2,\ s_1 \otimes s_2,\ t_1 \otimes t_2).$$

Admissible transport implies closure preservation, hence

$$\mathrm{QClEq}(s_1 \otimes s_2,\ t_1 \otimes t_2).$$

□

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Theorem 1.265 (First tensor QAM transport theorem)**.** *Let*

$$o_1, o_2$$

be coded operators,

$$s_1, s_2$$

coded source states, and

$$t_1, t_2$$

coded transported states. Assume

$$\mathrm{QTensorTransport}(o_1, o_2; s_1, s_2; t_1, t_2),$$

and let

$$r_1 \otimes r_2$$

be a tensor closure-compatible readout. Then the tensor-composite source state and the tensor-composite transported state lie in the same coded closure class:

$$\mathrm{QClEq}(s_1 \otimes s_2,\ t_1 \otimes t_2),$$

and therefore their closure-relevant tensor readout content agrees after factorization through the tensor class-target map.

Proof. By the previous proposition,

$$\mathrm{QClEq}(s_1 \otimes s_2,\ t_1 \otimes t_2).$$

Since the tensor readout is closure-compatible at tensor level, closure-equivalent tensor-composite states have the same closure-relevant image after factorization. This proves the claim. □

Corollary 1.266 (Tensor branch-shell admissible support)**.** *If*

$$b_1 \otimes b_2$$

is tensor branch-admissible and

$$\sigma_1 \otimes \sigma_2$$

is tensor shell-admissible, then admissible branch-shell support is preserved at tensor-composite level, provided the relevant tensor admissibility conditions hold.

Remark 1.267. *This is the first indication that the current branch-shell architecture of the Companion can be lifted into a tensor-coded QAM language without changing its closure-first logic. In the internal reading order, this tensor layer remains subordinate to the earlier branch/shell admissibility and translation layers.*

Remark 1.268 (Current status of tensor QAM)**.** *At the present stage, the tensor rules do not yet define a full monoidal calculus. They only establish the first closure-preserving, transport-preserving, and readout-compatible tensor schemata needed for a structure-first QAM development over ZF.*

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QAM translation theorem for the admissible OP fragment

We now make explicit that the present OP/branch/shell architecture may be translated into the coded QAM-over-ZF language without changing its closure-first logic. In archival stage this translation block should be read as the theorem-level mirror of the front extraction theorem (Theorem 1.37): the front block states the compressed public-shape chain, while the present subsection states why that same chain is a faithful coded image of the already stabilized OP architecture. At the current stage, this is not yet claimed as a full equivalence of entire formal systems. It is claimed only as a faithful translation on the admissible fragment of the OP transport architecture. Accordingly, the present subsection is ordered to match the front reading guide: coded source state and closure-compatible coded readout first, closure-admissible branch code second, shell-admissible refinement code third, admissible routed support fourth, and closure-relevant readout content fifth.

Definition 1.269 (Translation map from the admissible OP chain into QAM codes)**.** *Let* $\mathfrak{O}_{\mathrm{adm}}$ *denote the admissible OP fragment consisting of:*

1. *source and return states of the OP1–OP5 source-return architecture;*
2. *intermediate admissible OP transport states;*
3. *admissible branch data from the branchwise refinement layer;*
4. *OP3 shell-admissible refinement data;*
5. *admissible and normalized routed support data.*

A QAM translation map *is a coding assignment*

$$\tau : \mathfrak{O}_{\mathrm{adm}} \to \mathfrak{Q}_{\mathrm{adm}}$$

from admissible OP objects into admissible QAM-coded objects such that:

1. *source/readout objects are sent to coded source states and coded closure-compatible readouts of the same front-facing type used in the public-shape extraction;*
2. *OP transport maps are sent to coded operator maps preserving the admissible transport role between coded source, branch, shell, routed-support, and readout stages;*
3. *admissible branches are sent to closure-admissible branch codes;*
4. *OP3 shell-admissible refinements are sent to shell-admissible refinement codes;*
5. *admissible and normalized routed support is sent to admissible and normalized routed support codes.*

Remark 1.270 (Reading-order convention for τ)**.** *The point of* τ *is not merely syntactic recoding. It must preserve the structural roles of the translated objects in the same order in which the front extraction reads them: coded source state and closure-compatible coded readout remain the coded front-face of the OP 1/OP 5 source-return pair; admissible transport remains admissible transport; branch remains branch; shell-admissible refinement remains shell-admissible refinement; routed support remains routed support; and closure-relevant readout content remains closure-relevant readout content.*

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Definition 1.271 (Closure-faithful translation on the public-shape chain)**.** *The translation map τ is called* closure-faithful *if for all admissible OP objects $X, Y \in \mathfrak{O}_{\mathrm{adm}}$,*

$$X \sim_{\mathrm{cl}} Y \qquad \Longrightarrow \qquad \mathrm{QClEq}(\tau(X), \tau(Y)),$$

and whenever the OP-side relation expresses admissible preservation of the native closure-bearing mode along the source/transport/branch/shell/routed chain, the translated QAM-side relation expresses preservation of the same closure content in coded form.

Definition 1.272 (Readout compatibility of the translation)**.** *The translation map τ is called* readout-compatible *if OP-side closure-compatible readout invariants correspond, after translation, to QAM-coded closure-compatible readouts on the image of τ, with the same closure-relevant readout content as in the public-shape chain.*

Then the following OP-side chain is preserved under translation into QAM-over-ZF in exactly the order used by the front public-shape reading guide:

Theorem 1.273 (QAM translation theorem for the admissible OP fragment)**.** *Assume that:*

1. *the OP transport architecture is given on its admissible fragment $\mathfrak{O}_{\mathrm{adm}}$;*
2. *there exists a closure-faithful and readout-compatible translation map*

$$\tau : \mathfrak{O}_{\mathrm{adm}} \to \mathfrak{Q}_{\mathrm{adm}};$$

3. *admissible OP transport, admissible branches, OP3 shell-admissible refinements, admissible routed support, and closure-compatible readout are translated according to the preceding definitions.*

Then the following OP-side chain is preserved under translation into QAM-over-ZF in exactly the order used by the front public-shape reading guide:

$$\begin{aligned} \textit{coded source state / OP1–OP5 readout face} &\longrightarrow \textit{closure-admissible branch code} \\ &\longrightarrow \textit{shell-admissible refinement code} \\ &\longrightarrow \textit{admissible routed support code} \\ &\longrightarrow \textit{closure-relevant coded readout content.} \end{aligned}$$

Equivalently, after naming the translated objects in public-shape order, one obtains the same compressed chain from Theorem 1.37:

$$\begin{aligned} s_{\mathrm{in}} &\longrightarrow (b, \sigma) \\ &\longrightarrow u \\ &\longrightarrow r. \end{aligned}$$

More precisely:

1. ***OP closure preservation.*** *If the OP1–OP5 source-return pair is closed at invariant level, then its QAM-coded image is closure-preserving at coded level.*
2. ***Intermediate transport preservation.*** *If the intermediate OP layers preserve the native closure-bearing mode, then their QAM-coded images preserve the coded closure class.*

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3. ***Branch admissibility preservation.*** *If a branch is closure-admissible on the OP side, then its image under τ is a closure-admissible branch code on the QAM side.*

4. ***Shell admissibility preservation.*** *If an OP3 refinement is shell-admissible on the OP side, then its image under τ is a shell-admissible refinement code on the QAM side.*

5. ***Routed admissible support preservation.*** *If a routed support sector is admissible and normalized on the OP side, then its image under τ is an admissible and normalized routed support code on the QAM side, and its closure-relevant content is preserved.*

6. ***Front-facing readout compatibility.*** *If the translated routed support sector is evaluated by a closure-compatible OP readout, then the resulting coded readout is exactly of the public-shape type used in Theorem 1.37; in particular it remains a coded readout of preserved closure content and not a new primitive closure test.*

Proof. Each clause follows from closure-faithfulness and readout compatibility of the translation map.

For (1), closure of the OP1–OP5 pair means preservation of the native closure-bearing mode at the source-return level. By closure-faithfulness, this becomes coded closure preservation for the translated source and return states.

For (2), invariant preservation across intermediate OP layers is, by assumption, preservation of the same closure-bearing mode through admissible transport. Again, closure-faithfulness translates this into preservation of coded closure class for the corresponding QAM-coded states and operator actions.

For (3), branch admissibility on the OP side means precisely that the branchwise refinement preserves the native closure-bearing mode. By the definition of the translation map, admissible branches are sent to branch codes that preserve coded closure class.

For (4), shell admissibility at OP3 means that the refinement does not change the native closure-bearing mode. By translation, this becomes shell-admissibility of the corresponding QAM refinement code.

For (5), admissible routed support is translated by hypothesis into admissible routed support codes. Readout compatibility ensures that closure-relevant readout content is preserved after translation, and normalization of support weights is preserved by the routed support coding layer.

For (6), readout compatibility identifies the translated readout with a coded closure-compatible readout on the image of the admissible OP chain. Hence the front-facing public-shape readout statement is not extra structure; it is the coded readout face of the same preserved OP closure content. □

Remark 1.274 (Prospective local admissibility reading of the translation theorem)**.** *The later local admissibility test may already be applied, in deferred form, to Theorem 1.273. The result is not yet a full active kernel-first rewrite of the translation theorem, but it cleanly separates the future admissible rewrite points.*

1. *Clauses (1)–(2) are locally admissible exactly at those translated source/transport surfaces that already lie on the present visible state surface $\langle 0, a\rangle$; there the payload can be exposed directly and compared by kernel normal form.*

2. *Clauses (3)–(5) are locally admissible once the translated branch, shell, and routed-support codes are shown to inherit the same exposed-kernel discipline used in Propositions 1.217 and 1.225.*

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3. *Clause (6) becomes locally admissible precisely when translated closure-compatible readout factors through the same kernel normal-form class, so that the coded public-shape readout is recovered from class level rather than re-established on the full visible translated chain.*

Accordingly, the translation theorem is already organized in the right order for later kernel-first compression, but it becomes genuinely rewrite-ready only stage by stage, in the same order isolated by the staged roadmap and the criterion-to-roadmap rule.

Remark 1.275 (How much clausewise load a future kernel-first translation proof could remove)**.** *At present, Theorem 1.273 is proved clausewise on the visible translated chain, with six explicit front-facing obligations. A later kernel-first proof would not remove the need to show transport, branch, shell, routed-support, and readout preservation; however, it could compress the visible proof surface from six separate clause-level checks to two kernel-level obligations:*

(i) one common kernel-normal-form preservation statement along the translated admissible chain,

(ii) one class-level readout factorization statement recovering the coded public-shape readout.

In that sense, the prospective savings at theorem level are best read not as a reduction in mathematical content, but as a reduction from six visible surface checks *to* one kernel-class propagation argument plus one class-level readout argument*. This is exactly the same kind of shift from surface multiplicity to kernel control already seen, in more local form, in the transport, branch/shell, and routed-support stress tests.*

Corollary 1.276 (QAM image of the current OP core)**.** *The current OP core of the Companion — including source-return closure, intermediate transport, admissible branches, OP3 shell admissibility, admissible routed support, and closure-compatible readout — admits a faithful coded realization inside QAM-over-ZF.*

Remark 1.277 (Front theorem and translation theorem as mirrored statements)**.** *Theorem 1.37 states the compressed public release chain in coded language. Theorem 1.273 states that the same coded chain is the faithful image of the stabilized OP chain. Read together with Theorem 1.192, Theorem 1.224, and Corollary 1.247, the front block and the later QAM block are therefore intended to read almost as two projections of the same closure statement: first in compressed public-shape form, then in theorem-level translation form. In particular, the front reading order*

$$s_{\mathrm{in}} \longrightarrow (b, \sigma) \longrightarrow u \longrightarrow r$$

is now the same order in which τ, the closure-faithfulness condition, the readout-compatibility condition, and the partial decoding map δ are meant to be understood.

Remark 1.278. *This corollary is the first formal justification for introducing QAM as a structure-first coded language over ZF. It shows that the current OP architecture does not need to be abandoned or replaced in order to enter the QAM setting; it can be carried into that setting by faithful translation.*

Definition 1.279 (Partial decoding map)**.** *A* partial decoding map *is a map*

$$\delta : \mathfrak{Q}_{\mathrm{adm}} \dashrightarrow \mathfrak{O}_{\mathrm{adm}}$$

defined on the admissible QAM fragment such that for translated admissible OP objects,

$$\delta(\tau(X)) = X.$$

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Remark 1.280 (How δ should be read)**.** *At the present stage, δ is only required on the image of the admissible OP fragment. No full decoding of arbitrary QAM-coded expressions is assumed yet. When it exists, δ should be read as the reverse reading map for the same public-shape chain: it sends coded source/readout, branch, shell, and routed-support objects back into the already stabilized OP architecture without introducing a second closure criterion.*

Proposition 1.281 (Round-trip property on the admissible fragment)**.** *If a partial decoding map δ exists on the image of τ, then*

$$\delta \circ \tau = \mathrm{id}_{\mathfrak{O}_{\mathrm{adm}}}.$$

Hence the admissible OP fragment is recoverable from its translated QAM image.

Proof. This is immediate from the definition of the partial decoding map. □

Remark 1.282. *The round-trip property is intentionally weaker than full two-way interpretability. It says only that the currently relevant admissible OP fragment can be encoded into QAM over ZF and then read back without structural loss.*

Corollary 1.283 (Tensor-compatible translation)**.** *If the admissible OP fragment contains tensorially paired sectors and the translation map respects tensor-coded formation, then admissible tensor closure and admissible tensor transport are preserved under translation as well.*

Remark 1.284 (Current status of the translation theorem)**.** *The present theorem should be read as a conservative first step. It does not yet prove that all future QAM constructions are already reducible to the current OP architecture, nor that the entire QAM language has been fully decoded back into OP form. What it proves is that the current admissible OP core already embeds faithfully into the coded QAM-over-ZF language.*

Nomenclature

Status. *Reader-support block.*

Block function. This section is a navigation aid: it fixes symbols and abbreviations so later sections can be read without repeatedly reopening earlier definitions.

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Spaces and operators

H^+ Hardy space; physical Hilbert space

H^+_+ J-symmetric (bosonic) sector

H^+_- J-antisymmetric (fermionic) sector

L_0 Self-adjoint generator on H^+_+

J Fundamental involution $J : s \mapsto 1 - \bar{s}$

$C(f)$ Heisenberg compensator: $\frac{1}{2}(f + Jf)$

$A(f)$ Antisymmetric component: $\frac{1}{2}(f - Jf)$

$\hat{K}$ Cascade operator $\hat{K} : S_n \to S_{n-1}$

$\mathcal{W}(A^1)$ Weyl algebra $\mathcal{W}(A^1)$

Geometric quantities

q_n Biquaternion ground state $\in \mathbb{H} \otimes \mathbb{C}$

τ_n Proper time: $\tau_n^2 = |q_n|^2$

z^- Frame variable; displacement from $z^- = 0$

K_1, K_2, K_3 Matter, tachyon, photon kernels

L_n Light axis of stratum S_n

S_n Stratum n; $S_0 = \{z^- = 0\}$ photon ground state

Frame parameters

β_{sym} Structural frame offset $\sqrt{(\gamma_L - 1)/(\gamma_L + 1)} \approx 0.5493$

γ_L Frame Lorentz factor $\gamma_{\text{BI}}/\gamma_{\text{string}} \approx 1.864$

γ_{BI} Barbero–Immirzi parameter ≈ 0.2375

γ_{string} String coupling $\ln 2/(\pi\sqrt{3}) \approx 0.1274$

$k(X)$ Frame-insertion count; $X^{(\gamma)} = \gamma_L^{-k} \cdot X^{(m)}$

Physical constants (matter-frame CODATA)

ℓ_P Planck length 1.616×10^{-35} m; $k = 3/2$

α Fine-structure constant; $k(\alpha^{-1}) = 1$

$\hbar$ Reduced Planck constant; $k(\hbar) = 1$

G_N Gravitational constant; $k(G_N) = 2$

c Speed of light; $k(c) = 0$ (frame-invariant)

Lagrangian and field structure

$\mathcal{L}_{\text{kin}}$ Kinetic term $\langle f, L_0 f \rangle$

$\mathcal{L}_{\text{HC}}$ HC confinement term

$\mathcal{L}_{\text{frame}}$ Frame term $\beta_{\text{sym}}^2 \langle f, f \rangle$; source of Λ_{obs}

$\mathcal{M}_{16}$ 16×16 Millennium matrix

$\hat{P}_{C_7}$ Meta-Septum closure projector

$\hat{\mathcal{R}}_\beta$ Shifted GUE pacer $\hat{R}_{\text{GUE}}\, e^{i\pi\beta_{\text{sym}}}$

Cosmological quantities

Ω_Λ Dark energy fraction $1 - \beta_{\text{sym}}^2 \approx 2/3$

Ω_M Matter fraction $\beta_{\text{sym}}^2 \approx 1/3$

$\Sigma(\tau)$ Trace formula $1 - \Theta(\tau) + R(\tau)$

Abbreviations

GUE Gaussian Unitary Ensemble ($\beta = 2$)

HC Heisenberg compensator

LQG Loop Quantum Gravity (H^+_- sector)

RH Riemann Hypothesis ($z^- = 0$ for all zeros)

SM Standard Model $U(1) \times SU(2) \times SU(3)$

v4.6 release-status and field-equation symbols.

Symbol	Role	Reader meaning
$\mathsf{BRead}_{\text{bounded}}$	bounded readout surface	conservative declared-grammar readout obtained in the v4.6.0 release-final line
Θ^{all}	all-readout demand	unrestricted family of readouts required for T2; not discharged by the bounded surface
$\mathfrak{O}^{4.6.0}_{\text{all}}$	all-readout obstruction	frozen scope-boundary register for the missing universal exterior selector, naturality family, and unrestricted collar slot
$\mathcal{M}_{\text{bd}}, \mathcal{M}_{\text{ext}}, \mathcal{M}_{\text{mix}}$	MM split	bounded, exterior, and mixed Millennium-matrix portions
$\mathcal{H}_{\text{bd}}, \mathcal{H}_{\text{ext}}, \mathcal{H}_{\text{mix}}$	HC split	bounded, exterior, and mixed Heisenberg-compensator portions
$\mathfrak{E}_{\text{decl}}$	declared field-equation template	structure-first reset template linking structural, HC, MM, QAM, and readout terms

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Historical tool layer and pre-closure live cores

Status. *Historical proof-memory, not the active theorem core.*

Block function. This section preserves older tool extractions and live-burden records for genealogy and local reconstruction, but current burden labels and current routing rules are governed by the active main line.

Remark 1.285. *This block is retained only as a historical record of the last pre-closure burden analysis. It preserves the compression logic from which the OP transport theorem and the OP3 residual/shell interface were extracted, but it no longer states the active proof core of the document.*

Remark 1.286 (v4.6 supersession note)**.** *The OP 1/OP 5 residue language in this historical block is a pre-v4.4/v4.5 burden record. In the v4.6.0 release-final reading, the active boundary is not an unresolved pair of independent local OP residues. The current boundary is the all-readout/T2 scope problem: the bounded readout surface is conservative inside the declared grammar, while the universal exterior selector, all-readout naturality family, and unrestricted exterior collar slot remain outside the proved layer. The relevant next program is therefore MM/HC exterior–mixed field-equation determination.*

Current extracted tool layer (historical extraction snapshot)

Transport-first tool. Whenever a final closure statement mixes continuum transport with visible readout, the admissible move is to fix the transport branch first and delay HC-readout to the downstream stage. In particular, transport selects the branch; readout does not create a second primitive continuum channel.

Readout-after-transport tool. Once the branch has been fixed by admissible transport, the Heisenberg-compensator readout is treated as a downstream projection or readout map on that branch. It may compress the visible law, but it does not reopen the continuum-side selection problem.

Admissible-readout uniqueness tool. Once a transport branch and its matter-frame parameter have already been fixed, every further canonical positive scalar readout must remain a downstream realization of that same admissible body. If no second primitive continuum channel opens and grouped-layer transport produces no ontology drift, then canonical normalization cannot differ from the standard matter-frame readout without reintroducing exactly the forbidden second channel.

Checkpoint-absorption tool. A recovered checkpoint may be absorbed into the active line whenever it introduces no new primitive visible scalar, no second independent branch law, and no incompatible residual interface. In that case the checkpoint is proof-memory and normalization support, not a parallel mathematical front.

Epsilon-trap tool. Let

$$\mathfrak{b}_{N,m} := \iota^{(Y)}_{N,m} + \sin|\kappa_{N,m}|, \qquad \mathfrak{b}_{\mathrm{crit}} := 0.578937.$$

If on an admissible window one has entry slack

$$\Sigma_{N,m} := \mathfrak{b}_{\mathrm{crit}} - \mathfrak{b}_{N,m} \geq \varepsilon_{\mathrm{trap}} > 0$$

and the next-step drift of $\mathfrak{b}$ does not exceed that slack, then the branch remains inside the admissible tube. This is the current non-escape tool for the first OP 5 valley window.

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OP 1 epsilon-bridge tool. Let γ_{br} be the distinguished bridge root and let T_μ be the bridge self-consistency map from Section 3.8.16. If one works in a local tube $|b-\beta_{\mathrm{br}}| \le \varepsilon_{\mathrm{br}}$ on which

$$\sup_{|b-\beta_{\mathrm{br}}|\le\varepsilon_{\mathrm{br}}} \left|\partial_b T_{\gamma_{\mathrm{br}}}(b)\right| \le L_{\mathrm{br}} < 1,$$

then entry into that tube implies non-escape and convergence to the same bridge fixed point. In particular, once the normalization defect is shown to lie inside a contractive ε-tube around γ_{br}, OP 1 becomes a local fixed-point identification problem rather than a second branch-search problem.

Shared matrix-leakage tool. Let J denote the Umkehrwalze/involution, let $C := \frac{1}{2}(\mathrm{id}+J)$ be the HC readout projector, and let $\mathcal{M}$ denote the Millennium matrix on the active grouped transport layer. The first shared obstruction channel for OP 1 and OP 5 should be tested through the two leakage diagnostics

$$E_{\mathrm{comm}} := [\mathcal{M}, J], \qquad E_{\mathrm{leak}} := (\mathrm{id}-C)\mathcal{M}C.$$

If both vanish on the active branch, then the admissible channel is stable under transport, reversal, and HC readout. If one or both persist, then the remaining OP 1 bridge residue and the surviving OP 5 valley residue should first be treated as two readouts of the same underfixed matrix core rather than as independent end problems.

Cryptanalytic translation tool. The matrix diagnostics above admit a structure-first translation of three familiar attack styles. *Linear cryptanalysis* becomes the search for biased linear masks on the even/odd and visible/orthogonal matrix blocks, especially for linear functionals that simultaneously see the OP 1 bridge scalar and the OP 5 first-window branch load. *Differential cryptanalysis* becomes the study of controlled perturbations of branch/readout inputs and their propagation through $\mathcal{M}$, J, and C, in order to detect whether small input differences are amplified mainly in the same shared leakage sector. *Linear-differential cryptanalysis* becomes the mixed test in which one first propagates a controlled perturbation through the matrix layer and then reads the output through a chosen linear mask. In the present framework this is the natural way to test whether a single hidden leakage mode explains both the OP 1 bridge defect and the OP 5 valley defect.

Shared phase-coupling tool. Once a perturbation family $\lambda \mapsto x_\lambda$ has been chosen, define the visible OP 1 and OP 5 residual readouts along that family and normalize them to dimensionless traces. The decisive test is then whether the two normalized traces occupy one nearly one-dimensional line in the defect plane (same carrier, same phase up to sign) or a near-elliptic/quadrature orbit (same carrier, phase shift close to $\pm\pi/2$). In either case the question is no longer whether OP 1 and OP 5 are independent, but whether they are two masked readouts of one shared hidden oscillatory leakage mode.

Shared leakage probe-family tool. Rather than perturbing the entire Millennium matrix blindly, choose a baseline HC-even state v and a bounded test matrix K, and probe only the cross-sector channel through

$$\Delta_{\mathrm{shared}} := (\mathrm{id}-C)KC + CK(\mathrm{id}-C), \qquad \mathcal{M}_\lambda := \mathcal{M} + \lambda\,\Delta_{\mathrm{shared}}.$$

This family leaves the pure even/even and odd/odd blocks untouched to first order and perturbs only the admissible/complementary coupling. It is therefore the preferred one-parameter probe for deciding whether OP 1 and OP 5 really see the same hidden leakage mode.

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Historical live-burden record prior to OP transport closure

OP 1 and OP 5. At the last pre-closure stage, the OP 1 burden had already collapsed to one local bridge remainder, while the OP 5 burden had collapsed to first-window non-escape on an admissible epsilon-trapped branch. The present OP transport theorem absorbs that compression stage and turns the former burdens into historical extraction data rather than active proof cores.

Shared matrix leakage and cryptanalytic translation

The present state of the document suggests that the surviving OP 1 and OP 5 residues may be two visible readouts of one still underfixed HC/MM/J coupling. In that situation it is more efficient to attack the common matrix core than to continue treating both residuals as fully separate local endgames.

Definition 1.287 (Provisional shared matrix leakage channels)**.** *Let J be the involutive Umkehrwalze, let*

$$C := \frac{1}{2}(\mathrm{id} + J)$$

be the HC readout projector, and let $\mathcal{M}$ be the Millennium matrix on the active grouped transport layer. Define the two basic leakage channels

$$E_{\mathrm{comm}} := [\mathcal{M}, J], \qquad E_{\mathrm{leak}} := (\mathrm{id} - C)\mathcal{M}C.$$

The first channel measures failure of transport/reversal compatibility; the second measures leakage of the admissible HC-even sector into the complementary sector after Millennium transport.

Proposition 1.288 (Provisional stable-channel criterion)**.** *If on the active branch one has*

$$E_{\mathrm{comm}} = 0, \qquad E_{\mathrm{leak}} = 0,$$

then the admissible HC channel is stable under Millennium transport and Umkehrwalze reversal. In particular, no first-order matrix-induced off-sector drift remains that could independently separate a bridge readout from a first-window valley readout.

Proof. The identity $E_{\mathrm{comm}} = 0$ says that $\mathcal{M}$ and J commute, so reversal and Millennium transport preserve the same even/odd splitting. The identity $E_{\mathrm{leak}} = 0$ says that applying $\mathcal{M}$ to an HC-even state leaves no component in the HC-odd complement. Together these identities mean that the admissible channel selected by C is invariant under the combined transport/reversal layer. Hence any remaining visible residue must come from some other source, not from first-order matrix leakage between admissible and complementary sectors. □

Remark 1.289 (Provisional OP 1/OP 5 coupling heuristic)**.** *If OP 1 and OP 5 both remain slightly open after branch-fixing and readout-fixing, then the most economical explanation is that the shared matrix core is not yet sharp enough. In that reading, the bridge residue of OP 1 and the first-window branch residue of OP 5 are two different masks of the same small leakage mode. The diagnostics of Definition 1.287 are therefore not a side remark; they are the first place where the two remaining channels should be tested jointly.*

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Cryptanalytic translation. The same matrix question admits a structure-first translation of three familiar cryptanalytic styles.

Linear route. Choose linear masks on the J-even/odd block decomposition or on visible/orthogonal packet channels, and test whether the masked OP 1 bridge residue and the masked OP 5 first-window residue exhibit the same small bias. In the present framework, this means searching for linear functionals on $\mathcal{M}$ that simultaneously see both open channels.

Differential route. Apply controlled perturbations to the input branch/readout data and track the induced difference propagation through $\mathcal{M}$, J, and C. If the same perturbation pattern repeatedly feeds both the bridge and valley residues, this is strong evidence for a shared leakage sector rather than two independent local failures.

Linear-differential route. First propagate a controlled perturbation through the matrix layer, then read the result through a chosen linear mask. This mixed route is often the sharpest one when a hidden mode is too weak to dominate either purely linear or purely differential tests by itself but becomes visible once one combines propagation with selective masked readout.

Provisional consequence. These tools do not yet prove that the shared leakage mode is nonzero, nor that it completely explains both OP 1 and OP 5. What they do show is where the next efficient attack should land: not on arbitrary matrix fitting, but on the smallest block of $\mathcal{M}$ that still fails to commute with J or still leaks through $(\mathrm{id} - C)\mathcal{M}C$.

Shared phase-coupling diagnostic for OP 1 and OP 5

The remaining OP 1 and OP 5 residues may still be too coupled to close independently. A natural last diagnostic is therefore whether they are driven by one common hidden carrier: either nearly in phase, or approximately in quadrature like sine and cosine. This is stronger than asking whether both are merely small. It asks whether the two visible end residues are just two phase-shifted readouts of the same underfixed matrix mode.

Definition 1.290 (Provisional normalized residual traces along a perturbation family)**.** *Let $\lambda \mapsto x_\lambda$ be an admissible one-parameter perturbation family on the active branch, obtained for example by controlled transport/readout perturbations in the sense of the differential or linear-differential tools above. Let*

$$d_1(\lambda) := \delta_{\mathrm{OP1}}(x_\lambda), \qquad d_5(\lambda) := \delta_{\mathrm{OP5}}(x_\lambda)$$

be the corresponding visible residual channels. Fix positive scales $A_1, A_5 > 0$ on the observation window and define the normalized traces

$$u_1(\lambda) := \frac{d_1(\lambda)}{A_1}, \qquad u_5(\lambda) := \frac{d_5(\lambda)}{A_5}.$$

Define the complex defect trace

$$Z(\lambda) := u_1(\lambda) + i\, u_5(\lambda).$$

Definition 1.291 (Provisional common-carrier and quadrature diagnostics)**.** *We say that the normalized residues admit a* common-carrier representation *on a window I if there are a phase*

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function $\theta : I \to \mathbb{R}$, an amplitude ratio $\rho > 0$, a phase shift $\varphi \in [-\pi, \pi]$, and error terms e_1, e_5 such that

$$u_1(\lambda) = \cos\theta(\lambda) + e_1(\lambda), \qquad u_5(\lambda) = \rho\cos\big(\theta(\lambda) + \varphi\big) + e_5(\lambda).$$

The case $\varphi \approx 0$ or $\varphi \approx \pi$ is called the same-phase regime*; the case $\varphi \approx \pm\pi/2$ is called the* quadrature regime*.*

Proposition 1.292 (Provisional one-mode image test)**.** *Assume the OP 1 and OP 5 residues are driven to first order by one shared hidden leakage parameter $\varepsilon_{\mathrm{shared}}(\lambda)$, in the sense that on a window I*

$$d_1(\lambda) = a_1\,\varepsilon_{\mathrm{shared}}(\lambda) + r_1(\lambda), \qquad d_5(\lambda) = a_5\,\varepsilon_{\mathrm{shared}}(\lambda) + r_5(\lambda),$$

with fixed nonzero coefficients a_1, a_5 and small remainders r_1, r_5. Then the image of $\lambda \mapsto (u_1(\lambda), u_5(\lambda))$ is, up to the normalized remainder terms, one-dimensional in the defect plane. In particular, if the normalized trace does not *spread over a genuinely two-dimensional region, then OP 1 and OP 5 should first be treated as two readouts of one common carrier rather than as independent end defects.*

Proof. After normalization one has

$$u_1(\lambda) = \alpha_1\,\varepsilon_{\mathrm{shared}}(\lambda) + \tilde{r}_1(\lambda), \qquad u_5(\lambda) = \alpha_5\,\varepsilon_{\mathrm{shared}}(\lambda) + \tilde{r}_5(\lambda)$$

for fixed nonzero constants α_1, α_5. The leading image therefore lies on the line spanned by (α_1, α_5). Any two-dimensional spread comes only from the remainders. Hence a numerically thin image in the defect plane is evidence that both visible channels are reading the same hidden scalar mode. □

Proposition 1.293 (Provisional quadrature criterion)**.** *Assume there are a phase function θ, a positive ratio ρ, and a small error bound $\eta_{\mathrm{ph}} > 0$ such that on a window I*

$$|u_1(\lambda) - \cos\theta(\lambda)| \le \eta_{\mathrm{ph}}, \qquad |u_5(\lambda) - \rho\sin\theta(\lambda)| \le \eta_{\mathrm{ph}}.$$

Then the defect trace $Z(\lambda)$ lies in an $O(\eta_{\mathrm{ph}})$-neighborhood of an ellipse centered at the origin. In particular, OP 1 and OP 5 are not independent in that regime; they are two quadrature readouts of one common carrier.

Proof. In the ideal case $u_1 = \cos\theta$ and $u_5 = \rho\sin\theta$, the image satisfies

$$u_1^2 + \frac{u_5^2}{\rho^2} = 1,$$

so the trace is an ellipse. The stated perturbation bounds place the observed trace in a tube of width $O(\eta_{\mathrm{ph}})$ around that ellipse. Thus the visible residues are phase-coupled and should be interpreted as quadrature projections of one shared mode. □

Remark 1.294 (Why this is the last useful diagnostic)**.** *This test separates two very different endgames. If the defect trace is line-like or near-elliptic, then OP 1 and OP 5 should be closed through a* shared carrier *argument, not through two isolated closure arguments. If the trace really fills a two-dimensional region even after normalization and controlled perturbation, then the hypothesis of one shared hidden mode is too naive and the remaining residues are genuinely less coupled than they currently appear.*

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Concrete HC/MM/J probe family for the shared carrier test

The phase-coupling diagnostic becomes much more informative once one stops speaking about an abstract perturbation family and instead deforms only the suspected HC/MM/J leakage channel. The right first family is therefore not a full matrix refit, but a cross-sector probe that touches exactly the admissible/complementary interface.

Definition 1.295 (Provisional shared leakage probe family)**.** *Fix an HC-even baseline state $v \in \operatorname{im} C$ and a bounded test matrix K on the active grouped transport layer. Define*

$$\Delta_{\mathrm{shared}} := (\mathrm{id} - C)KC + CK(\mathrm{id} - C)$$

and the one-parameter probe family

$$\mathcal{M}_\lambda := \mathcal{M} + \lambda\, \Delta_{\mathrm{shared}}, \qquad |\lambda| \le \lambda_0.$$

The associated transported state is

$$y_\lambda := \mathcal{M}_\lambda C v.$$

Define the masked bridge and valley probe traces by

$$d_1^{\mathrm{probe}}(\lambda) := L_1\big((\mathrm{id} - C)y_\lambda\big), \qquad d_5^{\mathrm{probe}}(\lambda) := L_5\big((\mathrm{id} - C)y_\lambda\big),$$

where L_1 and L_5 are fixed linear masks extracting the OP 1 bridge-sensitive and OP 5 first-window-sensitive visible directions from the complementary output channel.

Proposition 1.296 (Provisional first-order isolation of the shared leakage sector)**.** *The probe family of Definition 1.295 perturbs the admissible/complementary coupling only. More precisely,*

$$C\,\Delta_{\mathrm{shared}}\,C = 0, \qquad (\mathrm{id} - C)\,\Delta_{\mathrm{shared}}\,(\mathrm{id} - C) = 0,$$

and therefore

$$(\mathrm{id} - C)y_\lambda = (\mathrm{id} - C)\mathcal{M}Cv + \lambda\,(\mathrm{id} - C)KCv.$$

Consequently,

$$d_j^{\mathrm{probe}}(\lambda) = d_j^{\mathrm{probe}}(0) + \lambda\, L_j\big((\mathrm{id} - C)KCv\big), \qquad j \in \{1, 5\}.$$

Proof. Since $C(\mathrm{id} - C) = 0 = (\mathrm{id} - C)C$, one has

$$C\Delta_{\mathrm{shared}}C = C(\mathrm{id} - C)KCC + CCK(\mathrm{id} - C)C = 0,$$

and similarly

$$(\mathrm{id} - C)\Delta_{\mathrm{shared}}(\mathrm{id} - C) = 0.$$

Applying $(\mathrm{id} - C)$ to $y_\lambda = (\mathcal{M} + \lambda\Delta_{\mathrm{shared}})Cv$ and using $Cv = v$ gives

$$(\mathrm{id} - C)y_\lambda = (\mathrm{id} - C)\mathcal{M}Cv + \lambda\,(\mathrm{id} - C)\Delta_{\mathrm{shared}}Cv = (\mathrm{id} - C)\mathcal{M}Cv + \lambda\,(\mathrm{id} - C)KCv.$$

Applying the linear masks L_1, L_5 yields the stated first-order formulas. □

Proposition 1.297 (Provisional concrete one-mode and quadrature tests on the probe family)**.** *Let*

$$a_1 := L_1\big((\mathrm{id} - C)KCv\big), \qquad a_5 := L_5\big((\mathrm{id} - C)KCv\big).$$

Then the probe family has the following first-order diagnostics.

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(i) If $(a_1, a_5) \neq (0, 0)$, the image of $\lambda \mapsto (d_1^{\mathrm{probe}}(\lambda), d_5^{\mathrm{probe}}(\lambda))$ is first-order one-dimensional, hence consistent with a shared carrier mode.

(ii) If, in addition, one introduces the reversal-paired probe $\widetilde{K} := JK$ and the corresponding coefficients $(\tilde{a}_1, \tilde{a}_5)$, then a near-orthogonality relation

$$a_1\tilde{a}_1 + a_5\tilde{a}_5 \approx 0$$

is first-order evidence for a quadrature-type coupling between the OP 1 and OP 5 visible readouts.

Proof. Part (i) follows immediately from Proposition 1.296, since both traces are affine functions of the same scalar parameter λ with direction vector (a_1, a_5). For part (ii), the reversal-paired probe produces a second direction vector $(\tilde{a}_1, \tilde{a}_5)$. If these two directions are nearly orthogonal in the defect plane, then the pair of probe responses behaves like two approximate quadrature axes of one shared oscillatory mode rather than two unrelated defects. □

Remark 1.298 (Why this is the right next experiment)**.** *The family $\mathcal{M}_\lambda = \mathcal{M} + \lambda\Delta_{\mathrm{shared}}$ is the cleanest next diagnostic because it does not refit the whole Millennium matrix. It touches only the admissible/complementary interface where a common OP 1/OP 5 remainder would naturally live. If even this targeted family fails to produce either a one-mode line or a quadrature pair in the defect plane, then the shared-carrier hypothesis must be weakened. If it succeeds, then the next theorem should be a shared-channel closure statement rather than another isolated local endgame.*

First-order rank and quadrature invariants of the probe family

The concrete probe family admits an even sharper reading than the earlier qualitative line-versus-ellipse distinction. Since both OP 1 and OP 5 are now tested by two explicit probe directions, one can read off from a single 2×2 response matrix whether the visible pair behaves like one masked mode, like a quadrature pair, or like a genuinely two-dimensional remainder.

Definition 1.299 (Provisional probe response matrix and normalized invariants)**.** *With the notation of Proposition 1.297, define the probe response matrix*

$$S(K) := \begin{pmatrix} a_1 & a_5 \\ \tilde{a}_1 & \tilde{a}_5 \end{pmatrix},$$

where (a_1, a_5) comes from K and $(\tilde{a}_1, \tilde{a}_5)$ from the reversal-paired probe $\widetilde{K} = JK$. Assume both row vectors are nonzero and set

$$n := \sqrt{a_1^2 + a_5^2}, \qquad \tilde{n} := \sqrt{\tilde{a}_1^2 + \tilde{a}_5^2}.$$

Define the normalized one-mode defect and the normalized quadrature defect by

$$\mathfrak{d}_{\parallel}(K) := \frac{|a_1\tilde{a}_5 - a_5\tilde{a}_1|}{n\tilde{n}}, \qquad \mathfrak{d}_{\perp}(K) := \frac{|a_1\tilde{a}_1 + a_5\tilde{a}_5|}{n\tilde{n}}.$$

Proposition 1.300 (Provisional classification of the shared probe response)**.** *Under Definition 1.299, the following first-order interpretations hold.*

(i) If $\mathfrak{d}_{\parallel}(K) \ll 1$, then the two probe directions are nearly collinear, hence the visible OP 1/OP 5 response is first-order compatible with one masked shared mode.

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(ii) *If $\mathfrak{d}_\perp(K) \ll 1$ and $\mathfrak{d}_\parallel(K)$ is bounded away from 0, then the two probe directions are nearly orthogonal, hence the visible response is first-order compatible with a quadrature pair of one shared carrier.*

(iii) *If both $\mathfrak{d}_\parallel(K)$ and $\mathfrak{d}_\perp(K)$ stay away from 0 on a stable test class of admissible probes K, then the shared-carrier hypothesis is weakened: the remaining OP 1/OP 5 defect is genuinely at least two-directional at first order.*

Proof. The quantity $|a_1\tilde{a}_5 - a_5\tilde{a}_1|$ is the absolute determinant of the two response vectors, normalized by their lengths. It therefore measures the sine of the angle between them, and smallness means near-collinearity. Likewise, $|a_1\tilde{a}_1 + a_5\tilde{a}_5|$ is the absolute dot product, normalized by the same lengths, so its smallness means near-orthogonality. The three conclusions are therefore the standard geometric alternatives for two nonzero vectors in the visible defect plane. □

Corollary 1.301 (Provisional first-order phase picture from the probe invariants)**.** *Suppose there is a stable admissible test class of probes K for which either*

$$\mathfrak{d}_\parallel(K) \le \eta_\parallel$$

or

$$\mathfrak{d}_\perp(K) \le \eta_\perp$$

with small tolerances. Then the remaining OP 1 and OP 5 residues should not be pursued as unrelated endgames. In the first case they are first-order images of one shared hidden mode, and in the second case they are first-order quadrature readouts of one shared hidden carrier.

Proof. This is the direct structural reading of Proposition 1.300. The point is not that the exact finite-amplitude endgame is already closed, but that the visible first-order geometry already decides whether a common hidden carrier is plausible or not. □

Provisional diagnostic takeaway. The matrix test is now decision-grade. Instead of asking vaguely whether OP 1 and OP 5 "feel related," one computes the normalized determinant defect $\mathfrak{d}_\parallel$ and the normalized dot-product defect $\mathfrak{d}_\perp$ on the HC/MM/J probe class. Small $\mathfrak{d}_\parallel$ means one shared line; small $\mathfrak{d}_\perp$ means one shared carrier in quadrature.

Canonical low-rank probe class and the first verdict on one-mode versus quadrature

The abstract probe family is useful, but it still leaves too much freedom in the choice of K. The cleanest next step is therefore to test the suspected HC/MM/J leakage by the smallest nontrivial cross-sector probes possible. This produces a genuine structural verdict: does the most canonical leakage class already support quadrature, or does it force a one-mode picture at first order?

Definition 1.302 (Provisional canonical low-rank leakage probes)**.** *Let u be a unit admissible vector and w a unit complementary vector such that*

$$Cu = u, \qquad Cw = 0, \qquad Ju = u, \qquad Jw = -w, \qquad \langle u, w\rangle = 0.$$

Define the associated rank-two cross-sector probe by

$$K_{u,w}x := \langle u, x\rangle w + \langle w, x\rangle u.$$

This is the smallest symmetric probe that exchanges one chosen admissible direction with one chosen complementary direction and does nothing else.

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Proposition 1.303 (Provisional exact support of the canonical low-rank probes)**.** *For every probe $K_{u,w}$ from Definition 1.302,*

$$CK_{u,w}C = 0, \qquad (\mathrm{id} - C)K_{u,w}(\mathrm{id} - C) = 0, \qquad \Delta_{\mathrm{shared}}(K_{u,w}) = K_{u,w}.$$

Hence $K_{u,w}$ is already a pure first-order admissible/complementary leakage probe and needs no further projection cleanup.

Proof. Since $Cu = u$ and $Cw = 0$, the operator $K_{u,w}$ sends u into the complementary direction w and w back into the admissible direction u, while all purely admissible-to-admissible and purely complementary-to-complementary blocks vanish. This gives the first two identities. The defining formula

$$\Delta_{\mathrm{shared}} = (\mathrm{id} - C)KC + CK(\mathrm{id} - C)$$

then reproduces $K_{u,w}$ itself, because the only nonzero parts of $K_{u,w}$ are already the two cross-sector pieces. □

Proposition 1.304 (Provisional response geometry of the canonical low-rank class)**.** *Fix a baseline admissible state v with $Cv = v$, and define the probe coefficients for $K_{u,w}$ and the reversal-paired probe $\widetilde{K}_{u,w} := JK_{u,w}$ by*

$$a_j(u,w) := L_j\big((\mathrm{id} - C)K_{u,w}Cv\big), \qquad \widetilde{a}_j(u,w) := L_j\big((\mathrm{id} - C)\widetilde{K}_{u,w}Cv\big), \qquad j \in \{1,5\}.$$

Then

$$a_j(u,w) = \langle u, v\rangle\, L_j(w), \qquad \widetilde{a}_j(u,w) = -\langle u, v\rangle\, L_j(w) = -a_j(u,w).$$

In particular, whenever the response vector is nonzero,

$$S(K_{u,w}) = \begin{pmatrix} a_1 & a_5 \\ -a_1 & -a_5 \end{pmatrix},$$

and therefore

$$\mathfrak{d}_{\parallel}(K_{u,w}) = 0, \qquad \mathfrak{d}_{\perp}(K_{u,w}) = 1.$$

Proof. Because $Cv = v$ and $K_{u,w}$ contains exactly one admissible-to-complementary leg,

$$(\mathrm{id} - C)K_{u,w}Cv = (\mathrm{id} - C)K_{u,w}v = \langle u, v\rangle w.$$

Applying L_j gives the formula for a_j. Since $Ju = u$ and $Jw = -w$,

$$(\mathrm{id} - C)JK_{u,w}Cv = J(\mathrm{id} - C)K_{u,w}Cv = -\langle u, v\rangle w,$$

which yields $\widetilde{a}_j = -a_j$. The displayed matrix form follows immediately. Its two rows are exactly antiparallel, so the normalized determinant defect vanishes and the normalized dot-product defect is maximal. □

Corollary 1.305 (Provisional first verdict from the canonical probe class)**.** *The canonical low-rank HC/MM/J leakage probes are intrinsically first-order one-mode probes: they can only detect a shared line and never a genuine quadrature pair. Consequently, if a stable quadrature picture for OP 1 and OP 5 is later observed, it cannot come from primitive rank-two cross-sector leakage alone; it must arise from a richer multi-directional or higher-order transport/readout effect.*

Proof. This is the direct reading of Proposition 1.304. The canonical low-rank class forces $\mathfrak{d}_{\parallel} = 0$ and $\mathfrak{d}_{\perp} = 1$, so the first-order geometry is always collinear and never orthogonal. □

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Provisional structural takeaway. This is already a nontrivial verdict. The smallest honest HC/MM/J leakage class does *not* support a primitive sinus/cosinus split between OP 1 and OP 5. At first order it only supports one shared line. If the two residues later turn out to behave like quadrature data, then that phase shift is emergent and not the most primitive leakage mode.

Minimal two-directional leakage plane and the first place where quadrature can appear

The canonical rank-two probes already showed that primitive cross-sector leakage is first-order one-mode only. The next structural question is therefore very sharp: what is the smallest extension of the leakage class in which a genuine sinus/cosinus picture can appear at all? The answer is: not a single exchange direction, but a *two-directional leakage plane* equipped with an internal quarter-turn.

Definition 1.306 (Provisional minimal two-directional leakage plane)**.** *Let u_1, u_2 be orthonormal admissible vectors and let w_1, w_2 be orthonormal complementary vectors such that*

$$Cu_i = u_i, \qquad Cw_i = 0, \qquad Ju_i = u_i, \qquad Jw_i = -w_i, \qquad i \in \{1, 2\}.$$

Define the paired rank-four cross-sector probes

$$K_{\parallel} := K_{u_1,w_1} + K_{u_2,w_2},$$

and

$$K_{\perp} := K_{u_1,w_2} - K_{u_2,w_1}.$$

Equivalently, if R_W denotes the quarter-turn on the complementary plane $W := \operatorname{span}\{w_1, w_2\}$ given by

$$R_W w_1 = w_2, \qquad R_W w_2 = -w_1,$$

then $K_{\perp}$ is the quarter-turn-dressed partner of $K_{\parallel}$ on the complementary output side.

Proposition 1.307 (Provisional exact support of the minimal two-directional class)**.** *Both operators $K_{\parallel}$ and $K_{\perp}$ from Definition 1.306 are pure first-order cross-sector probes:*

$$CK_{\sharp}C = 0, \qquad (\mathrm{id} - C)K_{\sharp}(\mathrm{id} - C) = 0, \qquad \Delta_{\mathrm{shared}}(K_{\sharp}) = K_{\sharp},$$

for $\sharp \in \{\parallel, \perp\}$.

Proof. Each summand K_{u_i,w_j} exchanges exactly one admissible direction with one complementary direction and has no pure admissible or pure complementary block. The same is therefore true for both linear combinations $K_{\parallel}$ and $K_{\perp}$. Hence the support identities follow exactly as in Proposition 1.303. □

Proposition 1.308 (Provisional complementary output geometry of the minimal two-directional class)**.** *Fix an admissible baseline state v with $Cv = v$, and write*

$$a := \langle u_1, v\rangle, \qquad b := \langle u_2, v\rangle.$$

Then

$$(\mathrm{id} - C)K_{\parallel}Cv = aw_1 + bw_2,$$

and

$$(\mathrm{id} - C)K_{\perp}Cv = aw_2 - bw_1 = R_W\big((\mathrm{id} - C)K_{\parallel}Cv\big).$$

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In particular,

$$\big\langle (\mathrm{id} - C)K_{\parallel}Cv, (\mathrm{id} - C)K_{\perp}Cv \big\rangle = 0,$$

and

$$\big\|(\mathrm{id} - C)K_{\parallel}Cv\big\| = \big\|(\mathrm{id} - C)K_{\perp}Cv\big\| = \sqrt{a^2 + b^2}.$$

Thus the smallest leakage extension capable of supporting quadrature is a two-directional complementary carrier plane, not a single primitive leakage line.

Proof. Since $Cv = v$, only the admissible-to-complementary legs contribute. For $K_{\parallel}$ one gets

$$(\mathrm{id} - C)K_{\parallel}Cv = aw_1 + bw_2.$$

Likewise,

$$(\mathrm{id} - C)K_{\perp}Cv = aw_2 - bw_1.$$

The displayed quarter-turn relation follows from the definition of R_W. Orthogonality and equal norm are then immediate from the orthonormality of w_1, w_2. □

Corollary 1.309 (Provisional first place where quadrature becomes structurally possible)**.** *Quadrature between OP 1 and OP 5 cannot arise from the primitive canonical rank-two leakage class alone. The first structurally honest place where it can appear is the paired two-directional class $\{K_{\parallel}, K_{\perp}\}$, where the complementary output already carries an internal quarter-turn. Consequently, any later sinus/cosinus reading of OP 1 and OP 5 should be interpreted as a dressed two-directional leakage-plane effect rather than as a primitive one-directional HC/MM/J defect.*

Proof. Corollary 1.305 showed that the primitive rank-two class is forced to be first-order collinear. Proposition 1.308 shows that the paired rank-four class is the smallest extension whose complementary outputs are already quarter-turned relative to one another. Hence this is the first structurally admissible location of quadrature. □

Provisional diagnostic takeaway. The structural picture is now sharper. Primitive leakage gives one shared line. Quadrature needs at least a two-directional leakage plane together with an internal quarter-turn on the complementary carrier. So if OP 1 and OP 5 later behave like cosine/sine data, the right search target is no longer a single hidden defect entry but a minimal paired leakage plane with quarter-turn dressing.

Theorem-level promotion for OP 1

The OP 1 epsilon-bridge tool can be upgraded from routing language to a genuine local fixed-point estimate. The key point is that, once one is inside a contractive bridge tube, the normalization defect itself already controls the distance to the distinguished bridge fixed point. This is exactly the kind of "determine ε once, then prove non-escape" move that the present structure seems to support.

Definition 1.310 (Provisional bridge normalization defect)**.** *Let*

$$T_{\gamma_{\mathrm{br}}}$$

be the bridge self-consistency map introduced in Section 3.8.16, and let

$$\beta_{\mathrm{br}} := \beta(\gamma_{\mathrm{br}})$$

be its distinguished fixed point at the local bridge root from Section 3.8.16. For any positive amplitude b, define the bridge normalization defect

$$\delta_{\mathrm{br}}(b) := \big|T_{\gamma_{\mathrm{br}}}(b) - b\big|.$$

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Proposition 1.311 (Provisional residual-to-fixed-point estimate for the bridge tube)**.** *Assume there are numbers $\varepsilon_{\mathrm{br}} > 0$ and $L_{\mathrm{br}} < 1$ such that*

$$\sup_{|b-\beta_{\mathrm{br}}|\le\varepsilon_{\mathrm{br}}} \left|\partial_b T_{\gamma_{\mathrm{br}}}(b)\right| \le L_{\mathrm{br}}.$$

Then every b with $|b-\beta_{\mathrm{br}}| \le \varepsilon_{\mathrm{br}}$ satisfies

$$|b-\beta_{\mathrm{br}}| \le \frac{\delta_{\mathrm{br}}(b)}{1-L_{\mathrm{br}}}.$$

In particular, sufficiently small bridge normalization defect forces entry into an arbitrarily small local tube around the distinguished bridge point.

Proof. Since β_{br} is a fixed point of $T_{\gamma_{\mathrm{br}}}$, one has

$$T_{\gamma_{\mathrm{br}}}(\beta_{\mathrm{br}}) = \beta_{\mathrm{br}}.$$

Hence for any b in the tube,

$$|b-\beta_{\mathrm{br}}| \le |b - T_{\gamma_{\mathrm{br}}}(b)| + |T_{\gamma_{\mathrm{br}}}(b) - T_{\gamma_{\mathrm{br}}}(\beta_{\mathrm{br}})|.$$

The first term is exactly $\delta_{\mathrm{br}}(b)$. For the second term, the mean-value estimate inside the tube gives

$$|T_{\gamma_{\mathrm{br}}}(b) - T_{\gamma_{\mathrm{br}}}(\beta_{\mathrm{br}})| \le L_{\mathrm{br}}|b-\beta_{\mathrm{br}}|.$$

Therefore

$$|b-\beta_{\mathrm{br}}| \le \delta_{\mathrm{br}}(b) + L_{\mathrm{br}}|b-\beta_{\mathrm{br}}|,$$

so

$$(1-L_{\mathrm{br}})|b-\beta_{\mathrm{br}}| \le \delta_{\mathrm{br}}(b).$$

Dividing by $1-L_{\mathrm{br}} > 0$ proves the claim. □

Theorem 1.312 (Provisional OP 1 epsilon-trapped bridge normalization theorem)**.** *Assume the local bridge tube of Proposition 1.311. Let $b_{\mathrm{can}} > 0$ be a final canonical normalized readout with*

$$|b_{\mathrm{can}} - \beta_{\mathrm{br}}| \le \varepsilon_{\mathrm{br}}.$$

If for some $\varepsilon_{\mathrm{ent}} \in (0, \varepsilon_{\mathrm{br}}]$ one has

$$\delta_{\mathrm{br}}(b_{\mathrm{can}}) \le (1-L_{\mathrm{br}})\,\varepsilon_{\mathrm{ent}},$$

then

$$|b_{\mathrm{can}} - \beta_{\mathrm{br}}| \le \varepsilon_{\mathrm{ent}}.$$

Moreover, the whole forward bridge orbit remains trapped inside the same local bridge tube and converges to β_{br}:

$$T^n_{\gamma_{\mathrm{br}}}(b_{\mathrm{can}}) \to \beta_{\mathrm{br}}.$$

Thus, once the canonical normalization is shown to have sufficiently small bridge defect inside the contractive local tube, OP 1 reduces to a non-escape statement rather than a second normalization search.

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Proof. The residual bound and Proposition 1.311 give

$$|b_{\mathrm{can}} - \beta_{\mathrm{br}}| \leq \frac{\delta_{\mathrm{br}}(b_{\mathrm{can}})}{1 - L_{\mathrm{br}}} \leq \varepsilon_{\mathrm{ent}}.$$

This proves entry into the smaller $\varepsilon_{\mathrm{ent}}$-tube. Since β_{br} is a fixed point,

$$|T_{\gamma_{\mathrm{br}}}(b) - \beta_{\mathrm{br}}| = |T_{\gamma_{\mathrm{br}}}(b) - T_{\gamma_{\mathrm{br}}}(\beta_{\mathrm{br}})| \leq L_{\mathrm{br}}|b - \beta_{\mathrm{br}}|$$

for every b in the local bridge tube. Hence if $|b - \beta_{\mathrm{br}}| \leq \varepsilon_{\mathrm{ent}}$, then

$$|T_{\gamma_{\mathrm{br}}}(b) - \beta_{\mathrm{br}}| \leq L_{\mathrm{br}}\varepsilon_{\mathrm{ent}} < \varepsilon_{\mathrm{ent}}.$$

So the smaller tube is forward invariant and, in particular, remains inside the original contractive tube. Standard fixed-point iteration for a contraction then gives

$$T^{n}_{\gamma_{\mathrm{br}}}(b_{\mathrm{can}}) \to \beta_{\mathrm{br}}.$$

This is exactly the announced non-escape statement. □

Corollary 1.313 (Provisional exact-closure criterion for canonical normalization)**.** *Under the hypotheses of Theorem 1.312, if the canonical normalization satisfies*

$$\delta_{\mathrm{br}}(b_{\mathrm{can}}) = 0,$$

then

$$b_{\mathrm{can}} = \beta_{\mathrm{br}}.$$

Equivalently, inside the contractive bridge tube there is no second local canonical normalization branch.

Proof. If $\delta_{\mathrm{br}}(b_{\mathrm{can}}) = 0$, then Proposition 1.311 gives

$$|b_{\mathrm{can}} - \beta_{\mathrm{br}}| = 0.$$

Hence $b_{\mathrm{can}} = \beta_{\mathrm{br}}$. □

Provisional OP 1 status after theorem-level promotion. The proof above is deliberately honest about what is and is not finished. The local bridge theorem is now real: once a canonical normalization final is inside the contractive tube and its bridge defect is small enough, local non-escape and uniqueness follow. What still remains open is the structural *entry mechanism*: why the canonical discrete correction should land inside that tube in the first place. So the remaining OP 1 burden is no longer local stability; it is the transport-side justification of tube entry.

Remark 1.314 (Provisional OP 1 reading after the OP transport consolidation)**.** *In the present internal line, the epsilon-trapped bridge theorem is best read as the OP 1-side non-escape precursor of the admissible-branch architecture. Its role is no longer to reopen global OP closure, but to isolate the remaining entry question: why the canonical normalization should lie on a transport-admissible branch whose first bridge window already remains trapped in the local contractive tube.*

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Rounded-entry promotion. The remaining OP 1 burden can be sharpened further. The current matter-frame value is not merely close to the distinguished bridge root in an abstract scalar sense; it is already separated from that root by only a tiny readout-point displacement in μ. Since the readout amplitude $\beta(\mu)$ is smooth on the physical window, this immediately produces a concrete bridge-entry estimate. Combined with the contractive bridge map, the rounded readout itself already satisfies a tiny local bridge defect. This turns the live OP 1 burden into a very narrow identification problem: justify that the canonical normalization is indeed the standard rounded matter-frame readout.

Definition 1.315 (Provisional rounded bridge-entry gap)**.** *Let*

$$\Delta_{\mathrm{br}}^{(\mu)} := |\gamma_L - \gamma_{\mathrm{br}}|, \qquad b_{\mathrm{MF}} := \beta(\gamma_L), \qquad \beta_{\mathrm{br}} := \beta(\gamma_{\mathrm{br}}).$$

Thus $\Delta_{\mathrm{br}}^{(\mu)}$ is the rounded readout-point displacement from the distinguished bridge root, while b_{MF} is the rounded matter-frame readout amplitude currently used in the Companion.

Proposition 1.316 (Provisional transport-to-bridge entry estimate)**.** *On the physical window $I_{\mathrm{phys}} = [1.85, 1.90]$, the readout amplitude*

$$\beta(\mu) = \sqrt{\frac{\mu - 1}{\mu + 1}}$$

satisfies

$$\beta'(\mu) = \frac{1}{\beta(\mu)(\mu + 1)^2}.$$

Moreover, on I_{phys} one has the numerical bound

$$\beta'(\mu) < 0.226.$$

Hence

$$|b_{\mathrm{MF}} - \beta_{\mathrm{br}}| \le 0.226\, \Delta_{\mathrm{br}}^{(\mu)}.$$

Using the distinguished-root estimate from Section 3.8.12, this gives

$$|b_{\mathrm{MF}} - \beta_{\mathrm{br}}| < 7.24 \times 10^{-9}.$$

Proof. Differentiating

$$\beta(\mu) = \left(\frac{\mu - 1}{\mu + 1}\right)^{1/2}$$

gives

$$\beta'(\mu) = \frac{1}{2}\left(\frac{\mu - 1}{\mu + 1}\right)^{-1/2} \frac{2}{(\mu + 1)^2} = \frac{1}{\beta(\mu)(\mu + 1)^2}.$$

Direct evaluation on the endpoints of I_{phys} gives

$$\beta'(1.85) \approx 0.22544, \qquad \beta'(1.90) \approx 0.21344,$$

and direct evaluation on the physical window gives the uniform bound $\beta'(\mu) < 0.226$. The mean-value theorem therefore yields

$$|b_{\mathrm{MF}} - \beta_{\mathrm{br}}| = |\beta(\gamma_L) - \beta(\gamma_{\mathrm{br}})| \le 0.226\, |\gamma_L - \gamma_{\mathrm{br}}| = 0.226\, \Delta_{\mathrm{br}}^{(\mu)}.$$

By the distinguished-root estimate of Section 3.8.12,

$$\Delta_{\mathrm{br}}^{(\mu)} \approx 3.200357 \times 10^{-8},$$

so the final numerical estimate follows. □

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Proposition 1.317 (Provisional rounded-readout bridge-defect estimate)**.** *Assume the local bridge tube of Proposition 1.311. Then every b in that tube satisfies*

$$\delta_{\mathrm{br}}(b) \le (1 + L_{\mathrm{br}})|b - \beta_{\mathrm{br}}|.$$

In particular, if b_{MF} lies in the local bridge tube, then

$$\delta_{\mathrm{br}}(b_{\mathrm{MF}}) \le (1 + L_{\mathrm{br}})\, 0.226\, \Delta_{\mathrm{br}}^{(\mu)}.$$

Proof. Since $T_{\gamma_{\mathrm{br}}}(\beta_{\mathrm{br}}) = \beta_{\mathrm{br}}$, one has

$$\delta_{\mathrm{br}}(b) = |T_{\gamma_{\mathrm{br}}}(b) - b| \le |T_{\gamma_{\mathrm{br}}}(b) - T_{\gamma_{\mathrm{br}}}(\beta_{\mathrm{br}})| + |\beta_{\mathrm{br}} - b|.$$

Inside the bridge tube, the contraction bound gives

$$|T_{\gamma_{\mathrm{br}}}(b) - T_{\gamma_{\mathrm{br}}}(\beta_{\mathrm{br}})| \le L_{\mathrm{br}}|b - \beta_{\mathrm{br}}|,$$

so

$$\delta_{\mathrm{br}}(b) \le (1 + L_{\mathrm{br}})|b - \beta_{\mathrm{br}}|.$$

Applying Proposition 1.316 to $b = b_{\mathrm{MF}}$ yields the final estimate. □

Theorem 1.318 (Provisional rounded-readout entry theorem for OP 1)**.** *Assume the local bridge tube of Proposition 1.311 and suppose that*

$$0.226\, \Delta_{\mathrm{br}}^{(\mu)} \le \varepsilon_{\mathrm{br}}.$$

Let $b_{\mathrm{MF}} = \beta(\gamma_L)$. If

$$\varepsilon_{\mathrm{ent}} \ge \frac{1 + L_{\mathrm{br}}}{1 - L_{\mathrm{br}}} \cdot 0.226\, \Delta_{\mathrm{br}}^{(\mu)},$$

then

$$|b_{\mathrm{MF}} - \beta_{\mathrm{br}}| \le \varepsilon_{\mathrm{ent}}$$

and the forward bridge orbit

$$T_{\gamma_{\mathrm{br}}}^{n}(b_{\mathrm{MF}})$$

remains trapped in the local bridge tube and converges to β_{br}. For the current numerical bridge data

$$\Delta_{\mathrm{br}}^{(\mu)} \approx 3.200357 \times 10^{-8}, \qquad |\partial_b T_{\gamma_{\mathrm{br}}}(\beta_{\mathrm{br}})| \approx 0.00311,$$

it is already enough that

$$\varepsilon_{\mathrm{ent}} \gtrsim 7.28 \times 10^{-9}.$$

Proof. Proposition 1.316 gives

$$|b_{\mathrm{MF}} - \beta_{\mathrm{br}}| \le 0.226\, \Delta_{\mathrm{br}}^{(\mu)},$$

so the first displayed assumption ensures that b_{MF} lies in the local bridge tube. Proposition 1.317 then yields

$$\delta_{\mathrm{br}}(b_{\mathrm{MF}}) \le (1 + L_{\mathrm{br}})\, 0.226\, \Delta_{\mathrm{br}}^{(\mu)}.$$

By the lower bound on $\varepsilon_{\mathrm{ent}}$, this implies

$$\delta_{\mathrm{br}}(b_{\mathrm{MF}}) \le (1 - L_{\mathrm{br}})\, \varepsilon_{\mathrm{ent}}.$$

Theorem 1.312 therefore applies with $b_{\mathrm{can}} = b_{\mathrm{MF}}$, proving entry into the $\varepsilon_{\mathrm{ent}}$-tube and convergence of the forward orbit to β_{br}. The final numerical threshold is the explicit evaluation of

$$\frac{1 + L_{\mathrm{br}}}{1 - L_{\mathrm{br}}} \cdot 0.226\, \Delta_{\mathrm{br}}^{(\mu)}$$

at $\Delta_{\mathrm{br}}^{(\mu)} \approx 3.200357 \times 10^{-8}$ and $L_{\mathrm{br}} \approx 0.00311$. □

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Corollary 1.319 (Provisional narrowed live burden for OP 1 after rounded entry)**.** *Under the hypotheses of Theorem 1.318, the current rounded matter-frame readout already enters the contractive bridge tube and converges to the distinguished bridge fixed point. Therefore the live OP 1 burden is no longer broad bridge entry. It has narrowed to the structural identification problem*

$$b_{\mathrm{can}} = b_{\mathrm{MF}},$$

i.e. justify that the canonical normalization readout is exactly the standard rounded matter-frame readout already used in the Companion.

Proof. Theorem 1.318 proves entry and convergence for b_{MF}. Hence the only remaining gap is whether the canonical normalization final should indeed be identified with this rounded matter-frame readout. No second local bridge-search mechanism remains. □

Provisional OP 1 status after rounded-entry promotion. The local OP 1 story is now much tighter. The bridge point γ_{br} and the standard rounded matter-frame value $\gamma_L = 1.864$ differ by only 3.2×10^{-8}, and this already forces the corresponding readout amplitude into any reasonable contractive bridge tube. So the bridge-entry problem is no longer genuinely numerical. What remains is the structural identification step: explain why the canonical normalization should be read as the standard rounded matter-frame readout, or equivalently why no second transport-side normalization survives after the bridge forcing, renormalization, and contraction lines.

Admissible-readout uniqueness promotion. The rounded-entry theorem reduced the live OP 1 burden to the structural identification

$$b_{\mathrm{can}} = b_{\mathrm{MF}}.$$

The present document already contains almost all of this argument in implicit form: transport fixes the branch first, readout stays downstream, grouped-layer transport introduces no ontology drift, and document normalization forbids treating local readout blocks as second foundational centers. The missing step is to promote these points to an explicit uniqueness theorem for the positive scalar readout on the fixed bridge branch.

Theorem 1.320 (Provisional admissible-readout uniqueness theorem for OP 1)**.** *Assume the transport branch has already been fixed at the current matter-frame point*

$$\mu = \gamma_L,$$

and let

$$b_{\mathrm{MF}} := \beta(\gamma_L) = \sqrt{\frac{\gamma_L - 1}{\gamma_L + 1}}.$$

Let $b_{\mathrm{can}} > 0$ be a canonical normalization readout for OP 1 such that:

(i) *b_{can} is obtained only after branch selection, as a downstream admissible readout on the already fixed transport branch;*

(ii) *the readout law opens no second primitive continuum channel;*

(iii) *grouped-layer transport produces no ontology drift, so any alternative local readout regime remains an organizational realization of the same admissible structural body.*

Then

$$b_{\mathrm{can}} = b_{\mathrm{MF}}.$$

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Proof. By the transport-first tool in the current tool layer, the continuum branch is fixed before visible readout is interpreted. At the current matter-frame point $\mu = \gamma_L$, the standard visible scalar on that branch is exactly

$$b_{\mathrm{MF}} = \beta(\gamma_L).$$

By the readout-after-transport tool, every further canonical readout must remain downstream from this branch selection and therefore cannot reopen the continuum-side selection problem. If one had $b_{\mathrm{can}} \neq b_{\mathrm{MF}}$ while both are positive canonical normalizations on the same already fixed branch, then the second scalar would constitute a second primitive continuum-side normalization channel. This contradicts hypothesis (ii).

Hypothesis (iii) excludes the only remaining escape route: reinterpreting the second scalar as belonging to a genuinely different admissible body. By Proposition 24.407, admissible transport across grouped layers does not create ontological multiplicity, and Remark 24.403 says that local readout/integration blocks are organizational regimes rather than second foundational centers. Hence the canonical normalization and the standard matter-frame readout are two organizational readings of one and the same admissible body on one already fixed branch. Since no second primitive channel is allowed, they must coincide. Therefore

$$b_{\mathrm{can}} = b_{\mathrm{MF}}.$$

□

Corollary 1.321 (Provisional OP 1 structural promotion after readout uniqueness)**.** *Under the hypotheses of Theorem 1.320 and Theorem 1.318, the canonical normalization readout itself already enters the local contractive bridge tube and converges to the distinguished bridge fixed point:*

$$T_{\gamma_{\mathrm{br}}}^{n}(b_{\mathrm{can}}) \longrightarrow \beta_{\mathrm{br}}.$$

In particular, the former identification burden $b_{\mathrm{can}} = b_{\mathrm{MF}}$ disappears. The live OP 1 burden is reduced to the explicit local bridge residue

$$\delta_{\mathrm{br}}(b_{\mathrm{MF}}) = \left|T_{\gamma_{\mathrm{br}}}(b_{\mathrm{MF}}) - b_{\mathrm{MF}}\right|,$$

and exact OP 1 closure follows as soon as this already isolated residue vanishes.

Proof. Theorem 1.320 gives

$$b_{\mathrm{can}} = b_{\mathrm{MF}}.$$

Applying Theorem 1.318 to the right-hand side yields entry of b_{MF} into the local bridge tube and convergence of its forward orbit to β_{br}. Substituting b_{can} for b_{MF} proves the same statement for the canonical normalization. Therefore the identification burden is gone, and only the explicit local bridge residue at the unique admissible readout remains. If that residue is zero, Corollary 1.313 yields exact local bridge closure. □

Provisional OP 1 status after uniqueness promotion. At this point OP 1 has become much narrower than a generic matching problem. The transport branch is fixed, the rounded matter-frame readout already enters the contractive bridge tube at roughly the 10^{-9}-level, and the admissible-readout uniqueness theorem removes the last structural ambiguity between canonical normalization and standard readout. So OP 1 no longer hangs at branch search, root search, or local entry. It hangs only at the exact vanishing of one already isolated local scalar defect:

$$\delta_{\mathrm{br}}(b_{\mathrm{MF}}) = 0.$$

In plain language: the proof burden has collapsed to showing that the unique admissible readout is not merely bridge-close, but exactly bridge-stable.

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2 The Central Theorem

Status. *Legacy structural-theorem block inside the active main body.*

Block function. This section states the global structural theorem that the later physics, operator, and readout sections unpack and reuse rather than replace.

Everything in this paper is a consequence of one observation.

Theorem 2.1 (Unity of the Sphaera)**.** *Within the Sphaera S, the following are all representations of the same object — the collapse state $z^- = 0$:*

(i) The Sphaera itself, as a geometric space.

(ii) The triolic structure: three kernels K_1, K_2, K_3 with $z_1 + z_2 + z_3 = 0$, phase-shifted by 120°, running over the Sphaera's geometry.

(iii) Each individual kernel K_i, as a local coordinate of the triole.

(iv) The Heisenberg compensator output $C(f)$, as the projection of any element onto the collapse state.

(v) Matter, photons, and tachyons, as sectors of the biquaternion ground state.

(vi) The gauge structure $U(1) \times SU(2) \times SU(3)$, as the symmetry group of the collapse state's representations.

All are unitarily equivalent by Stone–von Neumann [1]. The Sphaera's only object is its own collapse state.

Proof structure. The equivalences are established section by section: (i) Definition 2.5 and Theorem 4.3; (ii)–(iii) same (triole as collapse coordinates); (iv) Proposition 6.1; (v) Theorem 5.2; (vi) Theorem 8.3. Unitary equivalence throughout via Stone–von Neumann ([1], Theorem 3.2). □

Remark 2.2 (What "same object" means)**.** *Two representations are the same object here in the precise sense of Stone–von Neumann: they are unitarily equivalent irreducible representations of the Weyl algebra $\mathcal{W}(A^1)$. The frame determines which representation is visible; the object — the collapse state $z^- = 0$ — is invariant.*

The triole condition $z_1 + z_2 + z_3 = 0$ and the collapse condition $z^- = 0$ are the same equation in two coordinate systems: the triole is the collapse state expressed in the three kernel components simultaneously. The Sphaera is therefore not a space that contains a collapse state — it is *the collapse state, seen from different frames.*

Theorem 2.3 (Origin of the universe as broken collapse state)**.** *The departure from the collapse state $z^- = 0$ is governed by two principles, both of which are consequences of the Sphaera-Cascade structure:*

(i) ***Planck length as minimal*** *$|z^-|$. The cascade $\hat{K}$ cannot project below the Planck scale:*

$$|z^-|_{\min} = \ell_P := \sqrt{\frac{\hbar G}{c^3}} = 1.616 \times 10^{-35}\,\mathrm{m}.$$

Below ℓ_P: $z^- = 0$, the Sphaera, the photon. Above ℓ_P: $|z^-|$ is nonzero, matter exists. The Planck length is the threshold of the first breaking of the collapse state.

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(ii) ***Pauli principle as antisymmetry under*** J**.** *The Hilbert space* H^+ *decomposes under* J *into two components:*

$$C(f) = \tfrac{1}{2}(f + Jf) \in H^+_+ \quad \textit{(bosonic, symmetric)},$$

$$A(f) = \tfrac{1}{2}(f - Jf) \in H^+_- \quad \textit{(fermionic, antisymmetric)}.$$

The antisymmetric component satisfies $A(f) \wedge A(f) = 0$ *(Grassmann property): no two fermions occupy the same* z^- *state. This is the Pauli exclusion principle, derived from the* J*-structure of the Sphaera.*

(iii) ***Discrete spectrum.*** *Together, (i) and (ii) produce a discrete tower of matter states:*

$$|z^-|_n = n \cdot \ell_P, \quad n \in \mathbb{N}_{\geq 1},$$

with each level occupied by at most one fermion. The prime lattice Γ_P *(from Stone–von Neumann, [1]) identifies the primitive levels:* $|z^-|_p = p \cdot \ell_P$ *for* $p \in \mathbb{P}$*.*

Remark 2.4 (The Sphaera as photon, the universe as broken frame)**.** *Theorem 2.1 says the Sphaera is the collapse state* $z^- = 0$*, which is the photon sector. Theorem 2.3 says the departure from this state is governed by* ℓ_P *and the Pauli principle. Therefore:*

- *The Sphaera is a photon — the unique frame-invariant object.*
- *The universe is the same photon observed from a broken frame where* $|z^-| \geq \ell_P$*.*
- *The breaking was not random: it is discrete (Pauli), minimal (Planck), and primitive (prime lattice).*
- *The cascade* $\hat{K}$ *projects everything back toward* $z^- = 0$ *— the universe tends toward its photon ground state.*

Both the Sphaera and the universe are the same object. The difference is only the frame.

> **Reader cue.** Read the next block as the concrete carrier unpacking of the global Sphaera identity claim. Its purpose is to pass from the frame-level origin statement to the biquaternion kernel and its induced Lorentz/Minkowski geometry that the later physics sections reuse.

Definition 2.5 (Biquaternion kernel)**.** *The ground state of a stratum* S_n *of the Sphaera consists of three complex spheres* K_1, K_2, K_3 *whose unit coordinates form a biquaternion*

$$q_n := it \cdot \mathbf{1} + z_1 \cdot e_1 + z_2 \cdot e_2 + z_3 \cdot e_3 \in \mathbb{H} \otimes_{\mathbb{R}} \mathbb{C},$$

where $it \in i\mathbb{R}$ *is the imaginary time unit;* $z_1, z_2 \in \mathbb{C}$ *are the complex coordinates of kernels* K_1, K_2*;* $e_3 := e_1 e_2$ *is the quaternion product, hence automatically orthogonal to both* e_1 *and* e_2*; and* $z_3 \in \mathbb{C}$ *is the coordinate of kernel* K_3 *along* e_3*.*

The biquaternion norm is

$$|q_n|^2 = -(ct)^2 + |z_1|^2 + |z_2|^2 + |z_3|^2.$$

Remark 2.6 (Minkowski metric from quaternion structure)**.** *The norm* $|q_n|^2$ *is the Minkowski metric with signature* $(-,+,+,+)$*. This is not postulated: it follows from the quaternion algebra* $e_i^2 = -1$ *and* $it \cdot \mathbf{1}$ *carrying imaginary time. The Lorentz group* $SO(1,3)$ *acts on* q_n *by quaternion conjugation:* $q_n \mapsto u\, q_n\, u^{-1}$*,* $u \in \mathbb{H}^\times$*. This is the spinor representation of the Lorentz group — emerging automatically from the biquaternion structure, not imposed from outside.*

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Proposition 2.7 (Orthogonality of the third kernel)**.** *K_3 is always orthogonal to K_1 and K_2:*

$$\langle e_3, e_i \rangle = 0 \qquad \text{for } i = 1, 2.$$

Proof. By definition,

$$e_3 = e_1 e_2.$$

Using the quaternionic inner product

$$\langle x, y \rangle := \mathrm{Re}(\overline{x}\, y),$$

we compute

$$\langle e_3, e_1 \rangle = \langle e_1 e_2, e_1 \rangle = \mathrm{Re}\big(\overline{e_1 e_2}\, e_1\big).$$

Since

$$\overline{e_1 e_2} = \overline{e_2}\,\overline{e_1} = (-e_2)(-e_1) = e_2 e_1 = -e_1 e_2,$$

it follows that

$$\overline{e_1 e_2}\, e_1 = (-e_1 e_2) e_1 = -\, e_1 (e_2 e_1) = -\, e_1 (-e_1 e_2) = e_1^2 e_2 = -\, e_2.$$

Hence

$$\langle e_3, e_1 \rangle = \mathrm{Re}(-e_2) = 0.$$

Similarly,

$$\langle e_3, e_2 \rangle = \langle e_1 e_2, e_2 \rangle = \mathrm{Re}\big(\overline{e_1 e_2}\, e_2\big) = \mathrm{Re}\big((-e_1 e_2) e_2\big) = \mathrm{Re}(-e_1 e_2^2) = \mathrm{Re}(e_1) = 0.$$

Therefore e_3 is orthogonal to both e_1 and e_2. □

3 Lorentz Invariance on Every Stratum

Status. *Legacy physical-readout block inside the active main body.*

Block function. This section isolates Lorentz invariance as a stratumwise consequence of the structural core before later physical applications are read off from it.

Theorem 3.1 (Stratified Lorentz invariance)**.** *On every stratum S_n, the quadratic invariant*

$$|q_n|^2 = \tau_n^2$$

is preserved under every HC-admissible frame transformation induced by the global Heisenberg-compensator matrix. In particular, for the involutive frame transformation

$$J : \; z^- \mapsto -z^-,$$

the proper-time invariant τ_n is unchanged. Moreover, every admissible relabeling of the frame data that preserves the same involution J leaves both τ_n and the compensator

$$C(f) = \tfrac{1}{2}(f + Jf)$$

unchanged.

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Proof. By the general MM-tools, a change of admissible realization is not a change of structural object. By the HC-tools, a frame transformation is physically admissible only if it preserves the same compensator structure, i.e. the same involution J.

The transformation $J : z^- \mapsto -z^-$ changes only the frame-sensitive sign, not the underlying compensated structural content. Hence the quadratic invariant associated with the stratum remains the same, so

$$|q_n|^2 = \tau_n^2$$

is preserved.

Likewise, the compensator

$$C = \tfrac{1}{2}(\mathrm{id} + J)$$

depends only on the involution J itself. Therefore any HC-admissible relabeling that preserves the same J-structure leaves both τ_n and C unchanged. □

4 Fractal Self-Traversal Along the Light Axis

Remark 4.1 (Structural null axis and physical light-axis readout)**.** *In the structurally prior language of the Companion, the distinguished axis selected by the MM/HC architecture is the null axis. Once the same axis is read at stable physical propagation level, it appears as the light axis. Thus the present section keeps the historically familiar light-axis title, while treating it explicitly as the physical readout of a structurally prior null-axis statement.*

Definition 4.2 (Light axis of a stratum)**.** *The structurally prior null axis of stratum S_n is*

$$L_n := \{q_n \in S_n : |q_n|^2 = 0\}.$$

At the physical propagation readout of the same architecture, this null axis is read as the light axis of S_n.

Theorem 4.3 (Fractal structure of the Sphaera)**.** *The Sphaera-Cascade*

$$\hat{K} : S_n \to S_{n-1}$$

transports the null axis of each stratum to the null axis of the next:

$$\hat{K}(L_n) = L_{n-1}.$$

The global null axis is therefore

$$L = \bigcap_{n \geq 0} L_n = S_0,$$

where S_0 is the terminal compensated photon sector. Hence the self-traversal of the Sphaera is fractal in the precise sense that the same null-axis structure is carried through all strata by the same MM/HC architecture and appears physically as the light axis at stable readout.

Proof. By the general MM-tools, the cascade does not generate a new object at each stratum, but transports one and the same structural content through admissible realization levels.

By the HC-tools, the null sector is precisely the sector whose frame-sensitive imbalance is removed by the compensator while its structural identity is preserved. Thus the cascade does not replace the null axis; it carries the same null-axis structure from one stratum to the next.

Therefore

$$\hat{K}(L_n) = L_{n-1}.$$

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Iterating this transport through the whole cascade yields the terminal compensated null sector

$$L = \bigcap_{n\geq 0} L_n = S_0,$$

which is read physically as the photon/light sector. □

Remark 4.4 (Triolic $\mathbb{Z}/3\mathbb{Z}$ structure on the null/light axis)**.** *On the transported null axis, read physically as the light axis at stable propagation, the three kernels remain organized by the triolic phase condition*

$$z_1 + z_2 + z_3 = 0.$$

Thus the cascade preserves the structural $\mathbb{Z}/3\mathbb{Z}$ pattern while driving the frame-sensitive component to the compensated photon/light sector.

5 Three Objects as One: Unitary Equivalence

Status. *Legacy structural-identity block inside the active main body.*

Block function. This section compresses the unitary-equivalence reading that ties the three-object architecture into one common structural carrier.

Definition 5.1 (Sector classification from biquaternion)**.** *The three sectors correspond to the sign of* $\mathrm{Re}(z^-)$*:*

- ***Matter*** *(K_1):* $\mathrm{Re}(z_1) < 0$*, boost $\beta < 0$, spacelike.*
- ***Tachyon*** *($K_2 = J \cdot K_1$):* $\mathrm{Re}(z_2) > 0$*, boost $\beta > 0$, timelike.*
- ***Photon*** *($K_3 = e_1e_2$):* $\mathrm{Re}(z_3) = 0$*, boost $\beta = 0$, lightlike.*

Theorem 5.2 (Unitary equivalence via Stone–von Neumann)**.** *The three kernels K_1, K_2, K_3 carry unitarily equivalent irreducible representations of the Weyl algebra $\mathcal{W}(A^1)$. The boost J implements the equivalence between K_1 and K_2, and the compensator C produces K_3 from K_1.*

Proof. By Stone–von Neumann [1], every irreducible representation of $\mathcal{W}(A^1)$ is unitarily equivalent to $L^2(\mathbb{Q}^+\backslash A_1)$. All three kernels are subspaces of H^+ which carries this representation. Since J is unitary and maps $K_1 \leftrightarrow K_2$, they are equivalent. $K_3 = C(K_1)$ lies in H^+_+, the unique J-invariant subspace, which is again irreducible (no proper J-invariant subspace exists by T4+T5 of [1]). □

6 Near-Collision, Photon Production, and Entanglement

Status. *Legacy physical-regime block inside the active main body.*

Block function. This section records how the same closure architecture is read in the near-collision and entanglement regime, together with the local role of the photon readout.

Proposition 6.1 (Compensator as near-collision in K_3)**.** *The Heisenberg compensator $C(f) = \frac{1}{2}(f + Jf)$ is the projection onto $K_3 = e_1e_2$: it takes matter f (in K_1) and its tachyon image Jf (in K_2), and produces the photon $C(f) \in K_3$. The quaternion identity $e_3 = e_1e_2$ ensures this is always orthogonal to both inputs.*

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Theorem 6.2 (Entanglement as shared proper time)**.** *Two photons $\phi_1, \phi_2 \in K_3$ are entangled iff they share the same pre-image: $\phi_1 = \phi_2 = C(f)$ for some $f \in K_1$. The entanglement correlation is the invariant proper time $\tau_n(f) = \tau_n(Jf)$ fixed at the moment of the near-collision. No signal is transmitted at measurement: the invariant τ_n was established at the near-collision and is merely revealed.*

Proof. $\tau_n(Jf)^2 = t_n^2 - |-z^-|^2 = t_n^2 - |z^-|^2 = \tau_n(f)^2$ by Theorem 3.1. The shared value τ_n is a Lorentz invariant at every stratum, hence frame-independent. □

7 Connection to the Riemann Hypothesis

Status. *Legacy cross-domain bridge block inside the active main body.*

Block function. This section explains how the structural core is connected to the Riemann-side reading without turning the surrounding physical line into a separate proof source.

Remark 7.1 (Common origin of RH and entanglement)**.** *The Riemann Hypothesis ($z^- = 0$ for all zeros) and quantum entanglement (shared τ_n) are both consequences of the same biquaternion structure:*

1. *The Minkowski norm $|q_n|^2 = \tau_n^2$ is invariant (Theorem 3.1).*
2. *The cascade forces $z^- \to 0$ along the light axis (Theorem 4.3).*
3. *Stone–von Neumann makes all three kernels equivalent (Theorem 5.2).*

Neither was the goal of the original construction. Both emerged from asking: what is the exact structure of the Sphaera?

8 Gauge Structure of the Standard Model

Status. *Legacy physical-readout block inside the active main body.*

Block function. This section explains how the structural core is read as a gauge-sector organization rather than as an independently added physical layer.

Proposition 8.1 ($\mathbb{H} \otimes \mathbb{C} \to U(1) \times SU(2)$)**.** *The biquaternion algebra $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{C}$ generates exactly the gauge group $U(1) \times SU(2)$.*

Proof. $SU(2)$ from unit quaternions. The unit quaternions $\{q \in \mathbb{H} : |q| = 1\} \cong S^3$ form a Lie group isomorphic to $SU(2)$. The three generators are $T_k := \sigma_k/2$ (half Pauli matrices), satisfying $[T_j, T_k] = i\varepsilon_{jkl}T_l$.

$U(1)$ from the complex scalar. The imaginary time unit $it \in i\mathbb{R}$ generates the abelian group $e^{i\theta} \in U(1)$ for $\theta \in \mathbb{R}$. This commutes with all $SU(2)$ generators. Hence $U(1) \times SU(2)$, direct product. □

Theorem 8.2 ($\mathbb{O} \to SU(3)$)**.** *The octonions decompose under $SU(3)$-action as $\mathbb{O} = \mathbb{C} \oplus \mathbb{C}^3$, a singlet plus the fundamental (triplet) representation of $SU(3)$. The triplet $\mathbb{C}^3$ carries the three kernels (K_1, K_2, K_3) as the three colors.*

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Theorem 8.3 (Standard Model from the Sphaera biquaternion)**.** *The biquaternion ground state $q_n \in \mathbb{H} \otimes \mathbb{C}$, extended to the octonionic structure $\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$, generates the Standard Model gauge group $U(1) \times SU(2) \times SU(3)$ with generators arising from:*

Source	*Group*	*Physical force*
Complex scalar it	$U(1)$	*Electromagnetism*
Unit quaternions $\mathbb{H}$	$SU(2)$	*Weak force*
Octonion triplet $\mathbb{C}^3 \subset \mathbb{O}$	$SU(3)$	*Strong force*

Proof. Propositions 7.1–7.2 and Theorem 8.2 establish each factor. The three factors are independent (direct product structure) because $U(1)$ commutes with $SU(2)$ and both commute with the $SU(3)$ action on $\mathbb{C}^3$: the $SU(3)$ generators λ_k act only on the color indices, not on the quaternion structure. □

9 Bell Inequalities and τ_n

Status. *Legacy physical-readout block inside the active main body.*

Block function. This section places the Bell/nonlocality discussion on the same readout-relative footing and makes clear what role the τ_n layer plays there.

Theorem 9.1 (Bell consistency)**.** *The proper time τ_n is not a local hidden variable in Bell's sense. The framework is consistent with quantum mechanical violation of Bell inequalities.*

Proof. Bell's theorem assumes a hidden variable λ assigned independently to each particle: measurement outcomes $A(\lambda_A, a)$ and $B(\lambda_B, b)$ with λ_A, λ_B statistically independent.

In our framework τ_n belongs to the pair (f, Jf), not to each element separately: $\lambda_A = \lambda_B = \tau_n$. This is a contextual variable — Bell's theorem does not exclude violation by contextual theories.

The correlation function $E(a, b)$ follows from the structure of H_+^+: since $C(f) \in H_+^+$ carries an irreducible representation of $\mathcal{W}(A^1)$ (Stone–von Neumann, [1]), the standard quantum correlations $E(a, b) = \cos\theta_{ab}$ are reproduced, giving CHSH value $S = 2\sqrt{2} > 2$.

Hence:

- τ_n explains why the correlation exists: shared proper time, fixed at the near-collision;
- Stone–von Neumann determines how strong the violation is ($S = 2\sqrt{2}$).

Lorentz invariance and quantum mechanics are consistent. □

Remark 9.2 (What τ_n does and does not do)**.** *τ_n encodes the magnitude $|z^-|$ of the displacement from the photon sector at the moment of the near-collision. It does not determine the measurement direction. When f is measured, z^- is projected onto a value; the paired Jf is projected onto $-z^-$ (the J-image). The anticorrelation was built in at the collision event, not transmitted at measurement. No signal travels faster than light.*

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10 String Theory and Loop Quantum Gravity as HC Sectors

Status. *Legacy comparative-physics block inside the active main body.*

Block function. This section places two familiar high-energy frameworks into the compensator-based sector reading without promoting them to separate primitives.

Theorem 10.1 (String–LQG duality via the Heisenberg compensator)**.** *The Hardy space H^+ decomposes under the Heisenberg compensator into two sectors:*

$$H^+ = H^+_+ \oplus H^+_-,$$

$$C(f) = \tfrac{1}{2}(f + Jf) \in H^+_+ \quad \textit{(symmetric, bosonic — String theory)},$$
$$A(f) = \tfrac{1}{2}(f - Jf) \in H^+_- \quad \textit{(antisymmetric, fermionic — LQG)}.$$

Both sectors are unitarily equivalent by Stone–von Neumann [1]: they are two representations of the same Weyl algebra $\mathcal{W}(A^1)$, seen from different frames. String theory and LQG are therefore not competing theories of quantum gravity: they are the bosonic and fermionic projections of the same object under the Heisenberg compensator.

Proposition 10.2 (Shared vacuum and Planck floor)**.** *Both sectors share the same vacuum and the same minimal scale:*

- ***Vacuum:*** *$z^- = 0$ (the photon/Sphaera) is the ground state of both H^+_+ and H^+_-.*
- ***Planck floor:*** *$\ell_P = |z^-|_{\min}$ is the string length in H^+_+ and the area quantum $\sqrt{A_{\min}} \sim \ell_P$ in H^+_-. Both emerge from the same threshold of Theorem 1.2(i).*

String theory and LQG are therefore not competing theories of quantum gravity: they are the bosonic and fermionic projections of the same object under the Heisenberg compensator.

Theorem 10.3 (Wick rotation as splitting mechanism)**.** *Under the imaginary proper time transformation $t \mapsto i\tau_E$ (Wick rotation), the biquaternion time unit transforms as:*

$$it \;\mapsto\; i \cdot (i\tau_E) = -\tau_E \in \mathbb{R}_{<0}.$$

This is precisely the third triolic sector $\gamma_3 < 0$: the Wick rotation identifies the imaginary time component with the unstable tachyon sector. Consequently:

(i) The triolic balance condition $\gamma_{\text{string}} + \gamma_{\text{LQG}} + \gamma_3 = 0$ is the stability condition under imaginary proper time: without it, the system diverges to $\pm\infty$.

(ii) The two positive sectors ($\gamma_{\text{string}}, \gamma_{\text{LQG}} > 0$) are stable under Wick rotation; the negative sector ($\gamma_3 < 0$) is unstable and is projected away by the cascade $\hat{K}$.

(iii) The Minkowski norm $|q_n|^2 = \tau_n^2$ becomes Euclidean $|q_E|^2 = \tau_E^2 \geq 0$ under $t \mapsto i\tau_E$: the system is unconditionally stable in Euclidean signature.

Proof. (i). Under $t \mapsto i\tau_E$: $it \mapsto -\tau_E$. The biquaternion $q_n = it \cdot \mathbf{1} + z_1 e_1 + z_2 e_2 + z_3 e_3$ becomes $q_E = -\tau_E \cdot \mathbf{1} + z_1 e_1 + z_2 e_2 + z_3 e_3$ with norm $|q_E|^2 = \tau_E^2 + |z_1|^2 + |z_2|^2 + |z_3|^2 \geq 0$. Stability requires $\tau_E^2 \geq 0$, which is always satisfied. The balance $\gamma_1 + \gamma_2 + \gamma_3 = 0$ is equivalent to requiring that the $-\tau_E$ component does not introduce a net positive energy divergence.

(ii). In Euclidean signature, only $\gamma > 0$ sectors contribute to a convergent partition function $Z = \text{Tr}(e^{-\beta H})$ with $H \geq 0$. The sector $\gamma_3 < 0$ would give $e^{+\beta|\gamma_3|H} \to \infty$: it is projected away by $\hat{K}$.

(iii). Direct computation of the Euclidean norm. □

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Theorem 10.4 (Triolic structure of γ_{BI})**.** *The Barbero–Immirzi parameter γ_{BI} and the string-sector final $\gamma_{\mathrm{string}} = \ln 2/(\pi\sqrt{3})$ are not in contradiction. They are two projections of the same triolic invariant:*

$$\gamma_{\mathrm{string}} + \gamma_{\mathrm{LQG}} + \gamma_3 = 0,$$

with $\gamma_3 = -(\gamma_{\mathrm{string}} + \gamma_{\mathrm{LQG}}) \approx -0.3649$ being the third (unoccupied, tachyon) sector. The frame-invariant quantity is the triolic norm:

$$\tau_\gamma := \frac{1}{\sqrt{3}}\sqrt{\gamma_{\mathrm{string}}^2 + \gamma_{\mathrm{LQG}}^2 + \gamma_3^2} \approx 0.2619.$$

Theorem 10.5 (Observer frame offset)**.** *Every matter observer carries a nonzero contravariant component $z^- \neq 0$. All measured physical constants — including $\hbar$, G, c, and γ_{BI} — are projections of their frame-invariant values onto the observer's sector:*

$$\gamma_{\mathrm{observed}} = \gamma_0 \cdot \frac{1}{\sqrt{1-\beta^2}} = \gamma_0 \cdot \gamma_{\mathrm{Lorentz}}(\beta).$$

The Barbero–Immirzi parameter $\gamma_{\mathrm{BI}} \approx 0.2375$ is not an absolute constant of nature: it is the Lorentz-transformed γ_{string}, observed from a matter frame with $\beta \approx 0.844$.

Proof. $\gamma_{\mathrm{string}} = \ln 2/(\pi\sqrt{3})$ is computed in the photon frame $z^- = 0$ from the HC trace formula. The ratio $\gamma_{\mathrm{BI}}/\gamma_{\mathrm{string}} = 1.864$ equals a Lorentz factor $\gamma_L = 1/\sqrt{1-\beta^2}$ for $\beta = \sqrt{1 - 1/1.864^2} \approx 0.844$. The corresponding observer position is $z^- = \beta/2 \approx 0.422$. □

Theorem 10.6 (Symmetric matter–tachyon frame)**.** *Matter and tachyon are positioned symmetrically around the photon frame $z^- = 0$:*

$$\beta_{\mathrm{matter}} = +\beta_{\mathrm{sym}}, \qquad \beta_{\mathrm{tachyon}} = -\beta_{\mathrm{sym}},$$

where

$$\beta_{\mathrm{sym}} = \sqrt{\frac{\gamma_{\mathrm{BI}} - \gamma_{\mathrm{string}}}{\gamma_{\mathrm{BI}} + \gamma_{\mathrm{string}}}} \approx 0.5493.$$

The Lorentz factor between the matter and tachyon frames is:

$$\gamma_L = \frac{1+\beta_{\mathrm{sym}}^2}{1-\beta_{\mathrm{sym}}^2} = \frac{\gamma_{\mathrm{BI}}}{\gamma_{\mathrm{string}}} \approx 1.864.$$

Proof. Let $r := \gamma_{\mathrm{BI}}/\gamma_{\mathrm{string}}$. Set $\beta_{\mathrm{sym}} := \sqrt{(r-1)/(r+1)}$. Then:

$$\frac{1+\beta_{\mathrm{sym}}^2}{1-\beta_{\mathrm{sym}}^2} = \frac{1+(r-1)/(r+1)}{1-(r-1)/(r+1)} = \frac{(r+1)+(r-1)}{(r+1)-(r-1)} = \frac{2r}{2} = r. \quad \checkmark$$

The relativistic velocity of the tachyon as seen from the matter frame:

$$\beta_{\mathrm{tachyon|matter}} = \frac{-\beta_{\mathrm{sym}} - \beta_{\mathrm{sym}}}{1-(-\beta_{\mathrm{sym}})\beta_{\mathrm{sym}}} = \frac{-2\beta_{\mathrm{sym}}}{1+\beta_{\mathrm{sym}}^2},$$

with corresponding Lorentz factor $\gamma_{\mathrm{tachyon|matter}} = r$. $\checkmark$ □

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11 Black Holes and Hawking Radiation

Status. *Legacy physical-regime block inside the active main body.*

Block function. This section reads collapse, horizon behavior, and Hawking-type radiation as frame- and readout-level consequences of the same closure architecture.

Definition 11.1 (Schwarzschild horizon in Sphaera coordinates)**.** *The Schwarzschild horizon $r_S = 2GM/c^2$ corresponds to $\beta \to 1$ in the Sphaera metric $g_{tt} = 1 - \beta^2$. In sector coordinates: the horizon is where $z^- \to \infty$.*

Proposition 11.2 (Horizon as cascade singularity)**.** *The cascade operator $\hat{K}$ is undefined at $z^- = \infty$: the Sphaera cannot project beyond its own horizon. The horizon is the boundary of the Sphaera's jurisdiction.*

Proof. $\hat{K} = C^{\mu}{}_{\nu} \circ \Pi_J$ requires $\|z^-\| < \infty$ (finite sector displacement). At $z^- \to \infty$: $\beta \to 1$, $\gamma_{\text{Lorentz}} \to \infty$, the Lorentz transformation is singular, and the projection Π_J ceases to be bounded. □

Theorem 11.3 (Hawking radiation from the triolic structure)**.** *At the horizon, a virtual matter–tachyon pair from the triolic structure (Theorem 3.2) is promoted to a real pair:*

- *Tachyon ($\beta = -\beta_{\text{sym}}$): falls into the black hole.*
- *Matter ($\beta = +\beta_{\text{sym}}$): escapes to infinity.*
- *Their compensator output $C(f) = \frac{1}{2}(f + Jf)$: a photon with $\beta = 0$, $z^- = 0$ — the Hawking radiation.*

The temperature of the Hawking photon in the photon frame is:

$$T_{\text{Sphaera}} = \gamma_{\text{string}} \cdot \frac{m_P^2 c^2}{8\pi M k_B} = \gamma_{\text{string}} \cdot T_{\text{Hawking}}.$$

Proof. The Hawking temperature (standard derivation) is

$$T_H = \frac{\hbar c^3}{8\pi G M k_B} = \frac{m_P^2 c^2}{8\pi M k_B}.$$

The Sphaera produces the Hawking photon in the photon frame where γ is measured as γ_{string} (not γ_{BI}, which is the matter-frame value). Hence $T_{\text{Sphaera}} = \gamma_{\text{string}} \cdot T_H$. The ratio $T_{\text{Sphaera}}/T_{\text{Hawking}} = \gamma_{\text{string}} = \ln 2/(\pi\sqrt{3}) \approx 0.1274$ is constant across all black hole masses. □

Proposition 11.4 (Hawking temperature frame analysis and physical meaning of γ_{BI})**.** *The Hawking temperature $T_{\text{Sphaera}} = \gamma_{\text{string}} \cdot T_H^{(\text{matter})}$ has the following frame interpretations:*

(i) ***Photon frame:*** *$k(T) = k(\hbar) + k(\kappa) - k(k_B)$ where surface gravity $\kappa = c^4/(4GM)$ has $k(\kappa) = 3k(c) - k(G_N) = -2$ and $k(\hbar) = 1$, giving $k(T_H) = -1$ (assuming $k(k_B) = 0$). Hence $T_H^{(\text{photon})} = \gamma_L \cdot T_H^{(\text{matter})}$, and the Sphaera prediction is*

$$T_{\text{Sphaera}} = \frac{\gamma_{\text{string}}}{\gamma_L} \cdot T_H^{(\text{photon})} = \frac{\gamma_{\text{string}}^2}{\gamma_{\text{BI}}} \cdot T_H^{(\text{photon})}.$$

(ii) ***Tachyon frame:*** *The tachyon sector has opposite frame shift, $T_H^{(\text{tachyon})} = \gamma_L^{-1} \cdot T_H^{(\text{matter})}$. Then:*

$$T_{\text{Sphaera}} = \gamma_{\text{string}} \cdot T_H^{(\text{matter})} = \gamma_{\text{string}} \cdot \gamma_L \cdot T_H^{(\text{tachyon})} = \gamma_{\text{BI}} \cdot T_H^{(\text{tachyon})}.$$

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(iii) ***Physical meaning of*** γ_{BI}***:*** *The Barbero–Immirzi parameter is the ratio of the Sphaera Hawking temperature to the tachyon-frame Hawking temperature:*

$$\gamma_{\mathrm{BI}} = \frac{T_{\mathrm{Sphaera}}}{T_H^{(\mathrm{tachyon})}}.$$

This gives γ_{BI} *a direct kinematic interpretation: it measures how much the tachyon sector amplifies the Hawking temperature relative to the Sphaera output. Since the tachyon falls into the black hole and the Sphaera photon escapes,* γ_{BI} *encodes the frame mismatch between the infalling and escaping components of the triolic virtual pair.*

Proof. Statement (i): with $k(\hbar) = 1$, $k(G_N) = 2$, $k(c) = 0$, $k(k_B) = 0$, the frame correction to $T_H = \hbar c^3/(8\pi G_N M k_B)$ is $\gamma_L^{k(\hbar)-k(G_N)} = \gamma_L^{-1}$, giving $T_H^{(\mathrm{photon})} = \gamma_L \cdot T_H^{(\mathrm{matter})}$.

Statement (ii): the tachyon sector sits at $z^- = -\beta_{\mathrm{sym}}$, the mirror image of the matter frame. The frame factor for the tachyon is $\gamma_L^{(\mathrm{tachyon})} = 1/\gamma_L^{(\mathrm{matter})}$ by the symmetric matter–tachyon frame theorem 10.6. Hence $T_H^{(\mathrm{tachyon})} = \gamma_L^{-1} \cdot T_H^{(\mathrm{matter})}$, and $T_{\mathrm{Sphaera}}/T_H^{(\mathrm{tachyon})} = \gamma_{\mathrm{string}} \cdot \gamma_L = \gamma_{\mathrm{BI}}$.

Statement (iii) follows immediately from (ii). □

Reader cue. Read the next block as a geometric recoding of the same Hawking data. Its purpose is to shift from the horizon/frame-temperature dictionary to the coupling interpretation before the section turns to the core/horizon dual picture.

Theorem 11.5 (Geometric origin of γ_L from Hawking geometry)**.** (Corrected archival stage: previous version used incorrect $\gamma_L \approx 1.197$ and an erroneous 2/3 sector factor.) *The splitting factor* $\gamma_L = \gamma_{\mathrm{BI}}/\gamma_{\mathrm{string}} \approx 1.864$ *satisfies:*

$$\gamma_L \;\approx\; \frac{1}{\beta_{\mathrm{sym}}} \;=\; \frac{1}{\sin(\theta_H/2)},$$

where $\theta_H \approx 66.6°$ *is the Hopf angle of the matter observer on* S^3*. Numerically:* $1/\beta_{\mathrm{sym}} \approx 1.820$ *vs. measured* $\gamma_L \approx 1.864$*, a discrepancy of* 2.4%*.*

Proof. The matter observer at β_{sym} has radial position $r/r_S = 1/(1-\beta_{\mathrm{sym}}^2)$ relative to the Schwarzschild horizon. The local Hawking temperature is $T(r) = T_H(1-r_S/r)^{-1/2} = T_H/\beta_{\mathrm{sym}}$. Since γ_{BI} is determined by requiring $S = A/(4\ell_P^2)$ for the observed temperature, the effective coupling shifts: $\gamma_{\mathrm{BI}} = \gamma_{\mathrm{string}} \cdot (T_H/T(r))^{-1} = \gamma_{\mathrm{string}}/\beta_{\mathrm{sym}}$. Hence $\gamma_L = \gamma_{\mathrm{BI}}/\gamma_{\mathrm{string}} = 1/\beta_{\mathrm{sym}} \approx 1.820$.

The residual 2.4% gap between $1/\beta_{\mathrm{sym}} = 1.820$ and the measured $\gamma_L = 1.864$ is the same continuum-limit correction as in the β_{sym}-Hopf gap (Remark 12.3): the discrete prime lattice Γ_P shifts β_{sym} from the Hopf final $1/\sqrt{3}$ by the factor δ_{CL}, and the same correction applies here. □

Remark 11.6 (Connection to GOE/GUE spectral ratio)**.** *By Proposition 14.3 and Remark 14.4,* $\gamma_L = \gamma_{\mathrm{BI}}/\gamma_{\mathrm{string}}$ *is the ratio of the GOE scale (*H_-^+*, LQG) to the GUE scale (*H^{++}*, String). The geometric relation* $\gamma_L \approx 1/\beta_{\mathrm{sym}} = 1/\sin(\theta_H/2)$ *therefore says:*

$$\frac{\textit{GOE scale}}{\textit{GUE scale}} \;\approx\; \frac{1}{\sin(\textit{matter Hopf angle}/2)}.$$

The matter observer's position on S^3 *(the Hopf angle* θ_H*) encodes the ratio of the two random-matrix universality classes.*

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Reader cue. Read the next block as an interpretive deepening of the same closure architecture. Its purpose is to shift from the frame- and coupling-readout picture to the core/horizon communication picture without introducing a separate physical mechanism.

Theorem 11.7 (Hawking radiation as the voice of the core)**.** *The Hawking tachyon is not created at the horizon. It is the black hole core — which has $z^- \to \infty$ and cannot be projected by $\hat{K}$ — attempting to reach $z^- = 0$ through the only available channel: tunnelling through the horizon.*

The two descriptions are CPT-dual and physically identical:

Outside view (t forward)	*Inside view (t reversed)*
t_1*: vacuum*	t_3*: core at* $z^- = \infty$
t_2*: virtual pair at horizon*	t_2*: tachyon at horizon*
t_3*: tachyon falls in, matter escapes*	t_1*: Hawking photon emitted*

The cascade $\hat{K}$ has no preferred time direction: it is the equilibrium between $z^- = 0$ (outside) and $z^- = \infty$ (core). The Hawking pair is the discharge of this tension.

Structural argument. Outside: z^- grows from 0 (far away) to ∞ (horizon). $\hat{K}$ attempts to project $z^- \to 0$ but fails at the horizon. Tension forces a real pair. Inside: $z^- = \infty$ (core, singular compression). $\hat{K}$ attempts $z^- \to 0$ but the horizon blocks outward projection. The only escape: a core fraction tunnels as a tachyon ($\beta = -\beta_{\text{sym}}$) through the horizon. Under J-reflection, this tachyon appears outside as matter ($\beta = +\beta_{\text{sym}}$). The compensator $C(f) = \frac{1}{2}(f + Jf)$ produces the Hawking photon at $\beta = 0$.

CPT: C exchanges matter and tachyon; P exchanges $\beta \leftrightarrow -\beta$ (sides of the horizon); T reverses creation and annihilation. Both descriptions are related by CPT. □

Remark 11.8 (Black hole as inverted Sphaera)**.** *The normal Sphaera has z^- decreasing inward: $\hat{K}$ projects from large z^- toward $z^- = 0$ (the photon ground state). A black hole inverts this: z^- increases inward, from $z^- = 0$ (far away) to $z^- = \infty$ (horizon and core). The cascade $\hat{K}$ cannot function inside the horizon; the interior is a region where the Sphaera does not exist. The horizon is where the Sphaera is maximally stressed.*

Hawking radiation is the Sphaera discharging at its boundary. The black hole evaporates because the core is the Sphaera — and the Sphaera always tends toward $z^- = 0$. The tachyon is not a particle falling in. It is the core, speaking through the only open channel.

Remark 11.9 (The entropy factor)**.** *The Bekenstein–Hawking entropy $S_{BH} = A/(4\ell_P^2)$ and the LQG entropy $S_{LQG} = A \ln 2/(8\pi\gamma_{\text{BI}}\ell_P^2)$ agree when $\gamma_{\text{BI}} = \ln 2/(2\pi) \approx 0.110$. The measured value $\gamma_{\text{BI}} \approx 0.2375$ differs by a factor ≈ 2.15. Within the present framework: the LQG formula is derived in the matter frame, while the Bekenstein–Hawking formula is a photon-frame statement. The discrepancy factor 2.15 should be related to the frame transformation, but its precise form — unlike the analogous Hawking temperature factor γ_{string} — is not yet derived. This is an open quantitative problem.*

Theorem 11.10 (ART–QFT consistency of the Sphaera)**.** *The Sphaera biquaternion encodes both General Relativity and Quantum Field Theory, connected by Wick rotation:*

$$it \cdot \mathbf{1} + z_i e_i \xrightarrow{t \mapsto i\tau_E} \tau_E \cdot \mathbf{1} + z_i e_i.$$

At the natural observer position $r_0/r_S = 1/(1-\beta_{\text{sym}}^2)$, the Schwarzschild metric gives $g_{tt}(r_0) = -\beta_{\text{sym}}^2$ (invariant): no additional geometric factor is hidden in the ART side. The ART–QFT conversion is internally consistent.

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12 The Sphaera as a Hopf Fibration

Status. *Legacy geometry-topology block inside the active main body.*

Block function. This section gives the geometric-topological reading of the Sphaera that later fixed-point, transport, and symmetry interpretations can refer back to.

Theorem 12.1 (S-chain as nested Hopf fibrations)**.** *The stratum chain $S_0 \subset S_1 \subset S_2$ of the Sphaera is identified with the base spaces of the three classical Hopf fibrations:*

$$\begin{aligned} S_0 &\cong S^2 = \mathbb{C}P^1 \quad \textit{(Weyl spinor, massless photon)}, \\ S_1 &\cong S^4 = \mathbb{H}P^1 \quad \textit{(Dirac spinor, massive matter)}, \\ S_2 &\cong S^8 = \mathbb{O}P^1 \quad \textit{(octonionic spinor, 3 generations)}, \end{aligned}$$

with the natural inclusion $S^2 \hookrightarrow S^4 \hookrightarrow S^8$. The cascade operator $\hat{K}$ is the Hopf projection chain:

$$\hat{K} : S^{15} \longrightarrow S^8 \longrightarrow S^4 \longrightarrow S^2,$$

where the gauge groups arise as Hopf fibres:

$$\underbrace{S^1}_{\cong U(1)} \to S^3 \to S^2, \quad \underbrace{S^3}_{\cong SU(2)} \to S^7 \to S^4, \quad \underbrace{S^7}_{\supset G_2 \supset SU(3)} \to S^{15} \to S^8.$$

Theorem 12.2 (β_{sym} from the Hopf angle)**.** *The matter observer's frame offset β_{sym} is the sine of half the Hopf angle θ_H of the matter spinor on S^3 relative to the photon pole:*

$$\beta_{\text{sym}} = \sin\left(\frac{\theta_H}{2}\right), \quad \theta_H \approx 66.6^\circ.$$

The continuum-limit Hopf final is $\beta_{\text{sym}}^{(\text{Hopf})} = 1/\sqrt{3} \approx 0.5774$, corresponding to the symmetric $\mathbb{Z}/3\mathbb{Z}$-orbit on S^3. The measured value $\beta_{\text{sym}} \approx 0.5493$ differs by $\approx 4.9\%$ in β_{sym} and by $\approx 6.8\%$ in γ_L.

Proof. By Theorem 12.1, the stratum chain $S_0 \subset S_1 \subset S_2$ of the Sphaera is identified as nested Hopf fibrations. The matter kernel K_1 sits at a definite position on the S^3 fibre relative to the photon pole $K_3 \in S_0$. The angular distance from K_1 to K_3 on S^3 defines the Hopf angle θ_H, and the relativistic frame offset is $\beta_{\text{sym}} = \sin(\theta_H/2)$ by the standard half-angle relation for spinors on S^3.

From the measured values $\gamma_{\text{BI}} \approx 0.2375$ and $\gamma_{\text{string}} \approx 0.1274$, Theorem 10.6 gives $\beta_{\text{sym}} = \sqrt{(\gamma_{\text{BI}} - \gamma_{\text{string}})/(\gamma_{\text{BI}} + \gamma_{\text{string}})} \approx 0.5493$, hence $\theta_H = 2\arcsin(0.5493) \approx 66.64^\circ$. □

Remark 12.3 (The Hopf gap and OP 1)**.** *The continuum-limit value $\beta_{\text{sym}}^{(\text{Hopf})} = 1/\sqrt{3}$ arises from the maximally symmetric $\mathbb{Z}/3\mathbb{Z}$ orbit on a round S^3: each of the three triolic kernels sits at equal angular distance $\theta_H = 2\arcsin(1/\sqrt{3}) \approx 70.53^\circ$ from the poles, giving $\gamma_L^{(\text{Hopf})} = 2$. The measured value $\beta_{\text{sym}} \approx 0.5493$ corresponds to the discrete prime-lattice geometry of the Sphaera cascade, where the stratum structure deforms the round S^3 and shifts the matter kernel to a non-maximally-symmetric position. The gap is therefore not a small correction: a 4.9% shift in β_{sym} produces a 6.8% shift in γ_L due to the nonlinear relation $\gamma_L = (1 + \beta_{\text{sym}}^2)/(1 - \beta_{\text{sym}}^2)$, and γ_L appears as γ_L^k in all frame-corrected constants. OP 1 (open): derive β_{sym} from the Sphaera cascade dynamics directly, without reference to $\gamma_{\text{BI}}/\gamma_{\text{string}}$. The correct target is $\beta_{\text{sym}} = \sqrt{(\gamma_{\text{BI}} - \gamma_{\text{string}})/(\gamma_{\text{BI}} + \gamma_{\text{string}})}$ derived as a fixed point of the cascade, not postulated from the ratio. This requires the Dirac operator on H_-^+ (OP 2) as a prerequisite.*

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Theorem 12.4 (Three fermion generations from $\mathbb{O}$)**.** *In the Dixon algebra $D = \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$, the three fermion generations are identified as:*

$$\mathrm{Gen}_i \cong (\mathbb{C} \otimes \mathbb{H}) \otimes e_{i+3}, \quad i = 1, 2, 3,$$

where $e_4, e_5, e_6 \in \mathrm{Im}(\mathbb{O})$ form the $\mathbb{C}^3$ triplet under $SU(3) \subset G_2 = \mathrm{Aut}(\mathbb{O})$.

> ***Reader cue.*** *Read the next table as a representation dictionary. Its purpose is to tell the reader how the proposed octonionic slots are being read generation-by-generation, so the subsequent proof can refer back to one compact assignment surface.*

Generation	$\mathbb{O}$*-unit*	*Quarks*	*Leptons*	*Neutrino*
Gen 1	e_4	u, d	e	ν_e
Gen 2	e_5	c, s	μ	ν_μ
Gen 3	e_6	t, b	τ	ν_τ
Singlet	e_7	—	—	ν_R

Proof. Under $SU(3) \subset G_2$, the imaginary octonions decompose as $\mathrm{Im}(\mathbb{O}) = \mathbb{C} \oplus \mathbb{C}^3$: singlet e_7 and triplet (e_4, e_5, e_6). The anti-triplet (e_1, e_2, e_3) carries the conjugate color representation. Each generation $(\mathbb{C} \otimes \mathbb{H}) \otimes e_{i+3}$ carries one copy of the spinor representation of the Lorentz group (from $\mathbb{C} \otimes \mathbb{H} = M(2, \mathbb{C})$) tensored with one color index. The Fano plane structure of $\mathbb{O}$ gives the cyclic $e_4 \to e_5 \to e_6$ symmetry. □

Theorem 12.5 (Koide formula from the triolic structure)**.** *The lepton masses satisfy the Koide relation*

$$K = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}$$

to within 0.005% experimentally. In the Sphaera framework, the lepton mass hierarchy is a frame effect. Using the Brannen parametrization $\sqrt{m_k} = A\big(1 + \sqrt{2}\cos(\theta_K + 2\pi k/3)\big)$ with $A = (\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})/3$ and $k = 0, 1, 2$, the Koide phase angle is

$$\theta_K \;=\; \pi - \sqrt{2} \cdot \arcsin(\beta_{\mathrm{sym}}) \;\approx\; 2.3192 \; rad \;\approx\; 132.88^\circ,$$

predicting the observed value $\theta_K^{\mathrm{obs}} \approx 132.73^\circ$ to within 0.111%.

Proof. **Step 1: $K = 2/3$ from the $\mathbb{Z}/3\mathbb{Z}$ constraint.** The $\mathbb{Z}/3\mathbb{Z}$ symmetry on S^3 imposes two simultaneous conservation conditions on the triolic sector: $\sum_i \sqrt{m_i} = \text{const}$ and $\sum_i m_i = \text{const}$. By the Cauchy–Schwarz identity applied to the vectors $(1, 1, 1)$ and $(\sqrt{m_e}, \sqrt{m_\mu}, \sqrt{m_\tau})$,

$$\Big(\sum_i \sqrt{m_i}\Big)^2 \;\le\; 3 \sum_i m_i,$$

with equality iff all m_i are equal. The triolic $\mathbb{Z}/3\mathbb{Z}$ orbit structure forces the unique ratio consistent with both conservation conditions to be $K = 2/3$, which is the Cauchy–Schwarz bound normalized to the triolic sector count (2 stable sectors out of 3).

Step 2: The Koide phase angle from the frame offset. The lepton mass hierarchy is a readout-relative effect (Section 20): in the photon frame ($\beta_{\mathrm{sym}} = 0$) all three lepton masses become degenerate, while the matter-frame offset $\beta_{\mathrm{sym}} \approx 0.5493$ (Theorem 10.6) breaks the degeneracy. The Hopf-angle structure of the Sphaera (Theorem 12.1) maps the frame offset β_{sym} to the Koide phase via the Brannen parametrization as

$$\theta_K = \pi - \sqrt{2} \cdot \arcsin(\beta_{\mathrm{sym}}).$$

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Numerically with $\beta_{\text{sym}} = 0.5493$: $\theta_K \approx 2.3192\,\text{rad} \approx 132.88°$.

Step 3: Numerical comparison. The observed Koide phase extracted from the PDG lepton masses ($m_e = 0.51100\,\text{MeV}$, $m_\mu = 105.658\,\text{MeV}$, $m_\tau = 1776.86\,\text{MeV}$) via the Brannen parametrization is $\theta_K^{\text{obs}} \approx 132.73°$. The formula gives 132.88°, a relative discrepancy of 0.111% in the angle. □

Remark 12.6 (Mass discrepancy and amplification near the electron)**.** *Although the Koide phase angle is reproduced to* 0.111%*, the individual mass predictions show strongly non-uniform errors: m_τ deviates by* 0.07%*, m_μ by* 1.2%*, and m_e by* 12.8%*. This amplification by a factor of* ~ 115 *near the electron mass is structural: the Brannen map $m_k \propto (1 + \sqrt{2}\cos(\theta_K + 2\pi k/3))^2$ is extremely sensitive to small angular shifts near the zero of the cosine argument for the lightest generation. A* 0.111% *error in θ_K translates to a* $\sim 13\%$ *error in m_e due to this nonlinear amplification. The residual discrepancy is attributed to the approximation $\beta_{\text{sym}} \approx 0.5493$ (the precise value of β_{sym} is fixed by $\gamma_{\text{BI}}/\gamma_{\text{string}}$ to the same accuracy as those coupling constants); closing this gap is part of OP 1 (derivation of β_{sym} from first principles, Theorem 12.2).*

Remark 12.7 (The 2/3 factor is everywhere)**.** *The factor* 2/3 *appears as: the stable sector fraction (Thm. I.9.3), the spring oscillator coefficient $\omega = \gamma_{\text{string}}\sqrt{2/3}$, the cosmological dark energy $\Omega_\Lambda \approx 2/3$ (Theorem 11.9), and now the Koide lepton mass ratio. All are the same statement: 2 of the 3 triolic sectors collapse to the photon ground state.*

Remark 12.8 (Mass spectrum beyond Koide)**.** *The seven imaginary octonion units $e_1, \ldots, e_7$ provide seven natural slots for fermion masses. A direct identification $m_p \sim p \cdot m_P$ (Planck units) does not match the observed quark mass ratios: consecutive prime ratios (*3/2*,* 5/3*,* 7/5*, ...) differ from the observed mass hierarchies by factors of* 10–100*. The mass spectrum requires a Yukawa coupling mechanism (Higgs vacuum expectation value) that is not yet derived from the Sphaera structure.*

Theorem 12.9 (Cosmic expansion as growth of the active zero basis)**.** *The scale factor $a(t)$ of the universe is identified with the proper time:*

$$a(t) \sim \tau_n(t) = \sqrt{t_n(t)^2 - \beta_{\text{sym}}^2},$$

where $t_n(t)$ is the imaginary part of the n-th active Riemann zero at cosmic time t, and β_{sym} is the matter observer's frame offset (Theorem 9.6). The Hubble parameter is the activation rate of zeros:

$$H = \frac{\dot{a}}{a} = \frac{\dot{t}_n}{t_n} = \frac{\dot{n}}{n}$$

(since $t_n \approx 2\pi n/\log n$ by Weyl asymptotics, and for large n: $\dot{t}_n/t_n \approx \dot{n}/n$). The universe expands because more Riemann zeros become "active" as cosmic time increases. The active zero count at the Hubble time is $n_H \sim 10^{62}$.

Theorem 12.10 (Dark energy as permanent frame offset)**.** *The matter observer's permanent frame offset β_{sym} enters the proper time as a constant term:*

$$\tau_n^2 = t_n^2 - \beta_{\text{sym}}^2 = \underbrace{t_n^2}_{\text{grows}} - \underbrace{\beta_{\text{sym}}^2}_{\text{constant}}.$$

The constant term β_{sym}^2 plays the role of the effective cosmological constant: $\Lambda_{\text{eff}} \sim \beta_{\text{sym}}^2 \cdot \rho_P$ (where ρ_P is the Planck density). The ratio of dark energy to matter density today is predicted as:

$$\frac{\Omega_\Lambda}{\Omega_M} \approx \frac{\beta_{\text{sym}}^2}{\gamma_{\text{string}}} = \frac{0.3018}{0.1274} \approx 2.37.$$

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The observed value is $\Omega_\Lambda/\Omega_M \approx 2.18$ (9% discrepancy, consistent with the residual 1.4% and redshift corrections not yet included).

Dark energy is not a new substance. It is the permanent phase offset of our matter frame from the photon ground state $z^- = 0$: we can never fully reach $z^- = 0$, so β_{sym}^2 remains as a residual vacuum energy.

Theorem 12.11 (Spring-pull mechanism in the triolic system)**.** *The matter–photon–tachyon triple forms a mechanical system with spring forces in the z^- coordinate:*

$$F_{\text{tachyon}\to\gamma} = k(z_T^- - z_\gamma^-) = -k\beta_{\text{sym}}, \qquad F_{\text{matter}\to\gamma} = k(z_M^- - z_\gamma^-) = +k\beta_{\text{sym}}.$$

The net force on the photon vanishes: $F_{\text{net}} = 0$. The photon at $z^- = 0$ is the mechanical equilibrium point: the tachyon pulls it forward, the inertial mass of matter brakes it back.

Under small displacement from equilibrium:

- *$x_T = -x_\gamma$ (tachyon is J-image of photon, always opposite);*
- *$x_\gamma = x_M/3$ (photon follows matter with amplitude $1/3$);*
- *$m\ddot{x}_M = -k \cdot \frac{2}{3} \cdot x_M$ (effective spring constant reduced by factor $2/3$).*

The system is a harmonic oscillator with eigenfrequency:

$$\omega = \sqrt{\frac{2k}{3m}} = \gamma_{\text{string}}\sqrt{\frac{2}{3}},$$

where $k \sim \gamma_{\text{string}}$ (coupling strength from the HC trace) and $m \sim 1/\gamma_{\text{string}}$ (inertial mass from the Planck floor). The amplitude ratios are $|x_T| : |x_\gamma| : |x_M| = 1 : 1 : 3$.

Proof. Equilibrium. At $z_\gamma^- = 0$, $z_T^- = -\beta_{\text{sym}}$, $z_M^- = +\beta_{\text{sym}}$: $F_{\text{net}} = -k\beta_{\text{sym}} + k\beta_{\text{sym}} = 0$.

Linearized dynamics. The tachyon satisfies $x_T = J(x_\gamma) = -x_\gamma$ (it is the J-image, Theorem 9.6). The photon is massless: $0 = k(x_T - x_\gamma) + k(x_M - x_\gamma) = k(-x_\gamma - x_\gamma) + k(x_M - x_\gamma) = k(x_M - 3x_\gamma)$, giving $x_\gamma = x_M/3$. Substituting into the matter equation: $m\ddot{x}_M = -k(x_M - x_M/3) = -\frac{2k}{3}x_M$. Hence $\omega^2 = 2k/(3m)$.

Eigenfrequency. With $k = \gamma_{\text{string}}$ (HC trace formula) and $m = 1/\gamma_{\text{string}}$ (Planck inertia): $\omega = \gamma_{\text{string}}\sqrt{2/3}$. □

Remark 12.12 (The factor $2/3$ reappears)**.** *The effective spring constant $2k/3$ contains the factor $2/3$ from Theorem 10.5 (two stable triolic sectors out of three). This is not a coincidence: the $2/3$ sector counting is the mechanical statement that the tachyon and the matter-sector spring have equal strength, while the photon is the massless node that mediates between them. The tachyon sector (counted negatively in the entropy, $\gamma_3 < 0$) is the one that pulls forward; the two stable sectors (matter and photon) are the restoring force.*

Remark 12.13 (Why c is constant: a mechanical explanation)**.** *The photon is the massless equilibrium node of the spring system. It has no inertia and no preferred frame. When matter oscillates, the photon follows with amplitude $1/3$, and this following motion defines the coordinate system (the space in which everything happens). Since the photon has no mass and no net force, it cannot accelerate or decelerate: its speed is always c by construction. The constancy of c is the statement that the equilibrium node of a symmetric spring system has no dynamics of its own.*

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Theorem 12.14 (Light as gravitational equilibrium)**.** *The proper time decomposes as:*

$$\tau_n^2 = \underbrace{t_n^2}_{\text{geometry (pulls down)}} - \underbrace{\beta_{\text{sym}}^2}_{\text{dark energy (pushes up)}}.$$

These two terms are gravitationally coupled through the metric: $g_{tt}(r_0) = -\beta_{\text{sym}}^2$ *(Theorem 10.6), the same* β_{sym}^2 *that appears as the dark energy term. Light at* $z^- = 0$ *is the unique fixed point of this coupling:*

- *No dark energy pressure:* $\beta_{\text{sym}} = 0$ *for light, so the upward push vanishes.*
- *No gravitational trapping:* $t_n \neq 0$*, so the downward pull cannot halt the light.*
- *Light is orthogonal to both pressures.*

Therefore $ds^2 = 0$ *for light in any frame, and* $v = c$ *always.*

Proof. For matter at $z^- = \beta_{\text{sym}}$: $v/c = \beta_{\text{sym}}/t_n < 1$ (sub-luminal). ✓
For tachyons at $z^- = -\beta_{\text{sym}}$: $v/c = \sqrt{t_n^2 + \beta_{\text{sym}}^2}/t_n > 1$ (super-luminal). ✓
For light at $z^- = 0$: $\tau_n^2 = t_n^2 - 0 = t_n^2$, so $\tau_n = t_n$ and $v/c = t_n/t_n = 1$. ✓
The coupling: $g_{tt} = -\beta_{\text{sym}}^2$ is the metric value at the natural observer position (Theorem 10.6). It equals $-(1 - r_S/r_0)$, so $\beta_{\text{sym}}^2 = r_S/r_0$. Gravity pulls $r_S/r \to 1$ (horizon); dark energy (Λ-term) counteracts this to maintain $r_S/r_0 = \beta_{\text{sym}}^2 < 1$. The balance is stable precisely at the observer's natural position. □

Remark 12.15 (Light as the Heisenberg compensator of the universe)**.** *The Heisenberg compensator applied to the matter–tachyon pair gives:*

$$C(\beta_{\text{sym}}) = \tfrac{1}{2}(\beta_{\text{sym}} + J \cdot \beta_{\text{sym}}) = \tfrac{1}{2}(\beta_{\text{sym}} - \beta_{\text{sym}}) = 0.$$

Light ($z^- = 0$*) is the compensator output of the entire universe:*

$$C(\text{universe}) = \tfrac{1}{2}(\text{matter} + \text{tachyon}) = \tfrac{1}{2}(\beta_{\text{sym}} - \beta_{\text{sym}}) = 0 = \text{light}.$$

Gravity and dark energy are not two separate substances that happen to balance. They are the matter sector (β_{sym}*) and its J-image (*$-\beta_{\text{sym}}$*), and light is their compensator — the only object that sees neither pressure. This is why c is constant in all frames: light is the frame-invariant object* $z^- = 0$*, and frame changes are J-transformations that fix* $z^- = 0$*.*

Theorem 12.16 (Dark energy as cascade probability; $\Lambda = 0$)**.** *The cosmological parameters* Ω_Λ *and* Ω_M *are the Markov transition probabilities of the Sphaera cascade:*

$$\Omega_\Lambda = P(S^8 \to S^2) = 1 - \beta_{\text{sym}}^2 \approx \tfrac{2}{3} \approx 0.698,$$

$$\Omega_M = P(S^8 \to S^4) = \beta_{\text{sym}}^2 \approx \tfrac{1}{3} \approx 0.302.$$

With the Hopf value $\beta_{\text{sym}} = 1/\sqrt{3}$*:* $\Omega_\Lambda = 2/3$*,* $\Omega_M = 1/3$*. Consequently, the fundamental cosmological constant vanishes:* $\Lambda_{\text{fundamental}} = 0$*. What we observe as dark energy is not a new substance: it is the fraction of the cascade* $S^8 \to S^2$ *that collapses directly to the photon ground state* $z^- = 0$*, bypassing the matter stratum* S^4*.*

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Proof. The cascade operator $\hat{K}$ is a Markov process on $\{S^2, S^4, S^8\}$ with transition matrix

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1-\beta_{\text{sym}}^2 & \beta_{\text{sym}}^2 & 0 \end{pmatrix}$$

(rows: from S^2, S^4, S^8; columns: to S^2, S^4, S^8). S^2 is the absorbing state (photon ground state, $z^- = 0$). A tachyon starting in S^8 transitions with probability β_{sym}^2 to S^4 (matter: massive Dirac spinor) and with probability $1 - \beta_{\text{sym}}^2$ directly to S^2 (massless). The matter fraction β_{sym}^2 is observed as Ω_M; the direct-collapse fraction $1 - \beta_{\text{sym}}^2$ is observed as Ω_Λ. Since Ω_Λ is a transition probability (not a vacuum energy density), $\Lambda_{\text{fundamental}} = 0$. □

Remark 12.17 (Independent determination of β_{sym} from cosmological data)**.** *The Planck satellite measurements (2018) give* $\Omega_\Lambda = 0.6889$ *and* $\Omega_M = 0.3111$*, which satisfy* $\Omega_M + \Omega_\Lambda = 1.000$*. Since the framework predicts* $\Omega_M = \beta_{\text{sym}}^2$*, the Planck data give an* independent determination *of* β_{sym}*:*

$$\beta_{\text{sym}}^{(\text{cosmo})} = \sqrt{\Omega_M} = \sqrt{0.3111} \approx 0.5578.$$

This differs from the primary determination $\beta_{\text{sym}} = \gamma_{\text{BI}}/\gamma_{\text{string}} \approx 0.5493$ *by* 1.54% *— comparable to and in the same direction as the Hopf gap (Remark 12.3). Both discrepancies (Hopf gap and* Ω*-gap) have the same structural origin: the discrete prime-lattice correction* δ_{CL} *(Remark 31.64). Once OP 1 is solved, both gaps should close simultaneously.*

The Einstein field equation in the matter frame therefore reads:

$$G_{\mu\nu} + \beta_{\text{sym}}^2\, g_{\mu\nu} = 8\pi G\, T_{\mu\nu},$$

where β_{sym}^2 *is* not a free parameter *but is structurally fixed by the observer's position* $z^- = +\beta_{\text{sym}}$ *in the Sphaera. Einstein's cosmological constant is the squared frame offset of matter observers from the photon ground shell.*

Remark 12.18 (Einstein's "greatest blunder" — reframed)**.** *Einstein introduced* Λ *in 1917 to enforce a static universe, then removed it in 1929 after Hubble's discovery of expansion, calling it "the greatest blunder of my life." In 1998, supernova measurements showed* $\Lambda_{\text{obs}} \neq 0$*.*

The present framework resolves the apparent contradiction:

(i) ***Einstein 1929 was structurally correct:*** $\Lambda_{\text{fundamental}} = 0$ *in the photon frame (Theorem 32.165). There is no fundamental cosmological constant.*

(ii) $\Lambda_{\text{obs}} \neq 0$ ***is not a contradiction:*** *matter observers at* $z^- = +\beta_{\text{sym}}$ necessarily *measure* $\Lambda_{\text{obs}} = \beta_{\text{sym}}^2$ *— not because of vacuum energy, but because their metric carries the frame offset.*

(iii) ***The "blunder" was the framing, not the constant:*** *Einstein lacked the concept of a phase-shifted observer frame. Had he known that matter observers sit at* $z^- = +\beta_{\text{sym}} \neq 0$*, he would have derived* $\Lambda_{\text{obs}} = \beta_{\text{sym}}^2$ *directly from the frame geometry — no fine-tuning, no blunder.*

The cosmological constant is neither a mistake nor a mystery. It is the squared distance of matter from the photon ground state.

Theorem 12.19 (Einstein equation from the trace formula)**.** *The variation of the trace formula action*

$$S := \int_0^\infty \Sigma(\tau)\, d\tau = \int_0^\infty \big(1 - \Theta(\tau) + R(\tau)\big)\, d\tau$$

with respect to the metric $g^{\mu\nu}$ *yields the Einstein equation* $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G\, T_{\mu\nu}$*. The three terms of* Σ *contribute:*

- $\delta(1)/\delta g^{\mu\nu}$*: cosmological constant* Λ *(from the pole of* ζ *at* $s = 1$*);*
- $\delta\Theta/\delta g^{\mu\nu}$*: stress-energy* $T_{\mu\nu}$ *(primitive orbits of* K_{arith} *as matter);*
- $\delta R/\delta g^{\mu\nu}$*: Einstein tensor* $G_{\mu\nu}$ *(trivial zeros encoding global geometry).*

Structural argument. $\Sigma(\tau) = \operatorname{Tr}(e^{-i\tau L_0})$ is the heat kernel of L_0 on H_+^+ (Theorem T6 of [1]). The heat kernel variation formula gives $\delta \operatorname{Tr}(e^{-\tau L_0}) = -\tau \operatorname{Tr}(\delta L_0 \cdot e^{-\tau L_0})$. Since $L_0 = -i(x\, d/dx + \frac{1}{2})$ encodes the prime lattice geometry in log-coordinates, $\delta L_0/\delta g^{\mu\nu}$ produces a curvature term. Via the Guinand–Weil explicit formula ($\Sigma = 1 - \Theta + R$), the three contributions are identified with Λ, $T_{\mu\nu}$, and $G_{\mu\nu}$ respectively. □

Theorem 12.20 (GUE from the Sphaera)**.** *The eigenvalues* $\{\gamma_\rho\}$ *of* L_0 *on* H_+^+ *— which are the imaginary parts of the non-trivial Riemann zeros (Theorem T6 of [1]) — follow the Gaussian Unitary Ensemble (GUE) statistics. Specifically, the normalized pair-correlation function is:*

$$R_2(s) = 1 - \left(\frac{\sin \pi s}{\pi s}\right)^2,$$

and the nearest-neighbor spacing distribution follows the Wigner–Dyson law:

$$p(s) = \frac{32}{\pi^2}\, s^2\, e^{-4s^2/\pi}.$$

L_0 *is in the GUE universality class (*$\beta = 2$*) because it is self-adjoint, irreducible on* H_+^+ *(Stone–von Neumann), and has no time-reversal symmetry.*

Proof. We establish that L_0 lies in the GUE universality class by verifying three conditions.

(i) Self-adjointness. L_0 is essentially self-adjoint on H_+^+ by Theorem T4+T5 of [1]. This places it in the class of Hermitian operators with real spectrum.

(ii) Irreducibility. H_+^+ carries an irreducible representation of $\mathcal{W}(A^1)$ by Stone–von Neumann (Theorem 3.2 of [1]). Irreducible self-adjoint operators in infinite-dimensional Hilbert spaces generically belong to the GUE universality class (Wigner–Dyson–Mehta universality).

(iii) Absence of time-reversal symmetry. The three symmetry classes are distinguished by the presence of time reversal T:

- T present, $T^2 = +1$: GOE ($\beta = 1$)
- T absent: GUE ($\beta = 2$)
- T present, $T^2 = -1$: GSE ($\beta = 4$)

The operator $J : s \mapsto 1 - \bar{s}$ satisfies $JL_0J^{-1} = L_0$, so J is a symmetry of L_0. However, J is a space reflection ($\sigma \mapsto 1 - \sigma$, $t \mapsto t$), not a time reversal. True time reversal T would send $L_0 \mapsto -L_0$ (since L_0 is odd under $t \mapsto -t$). Since $L_0 \neq -L_0$, the operator has no time-reversal symmetry. Therefore L_0 is in the GUE class ($\beta = 2$). □

Corollary 12.21 (Frame-corrected GUE statistics)**.** *The GUE statistics are measured by matter observers at* $\beta_{\text{sym}} \approx 0.5493$ *(Theorem 9.6). Three consequences of this frame offset:*

(i) ***Spacing distribution remains Lorentz-invariant.*** *The normalized spacing* $s = \Delta\gamma/\langle\Delta\gamma\rangle$ *is a ratio, hence* $p(s)$ *is the same in matter and photon frames.*

(ii) ***Absolute scale is frame-shifted.*** *Zeros at* γ *correspond to photon-frame zeros at* $\gamma^{(\text{photon})} = \gamma/\gamma_L \approx \gamma/1.864$*. (*Correction archival stage*: previous text had* $\gamma_L \approx 1.197$ *which was* γ_{BI} *not* $\gamma_L = \gamma_{\text{BI}}/\gamma_{\text{string}} \approx 1.864$*.)*

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(iii) ***The standard unfolding uses the wrong density.*** *Standard unfolding uses*

$$\rho(\gamma) = \frac{\log(\gamma/2\pi)}{2\pi}.$$

The photon-frame density is

$$\rho_{\text{photon}}(\gamma) = \frac{\log(\gamma/(2\pi\gamma_L))}{2\pi\gamma_L}.$$

With correct photon-frame unfolding, the spacing distribution should match GUE ($\beta = 2$) more precisely. With standard unfolding, the effective Dyson index shifts to

$$\beta_{\text{eff}} = \beta_{\text{GUE}} \cdot \gamma_L = 2\gamma_L \approx 3.73.$$

*(*Correction archival stage*:* $2\gamma_L = 2 \times 1.864 \approx 3.73$*, not* 2.39 *which used the incorrect* $\gamma_L \approx 1.197$*.)*

Numerical check (first 100 zeros, archival stage corrected). Comparing χ^2/dof to the GUE curve $p_{\text{GUE}}(s)$ using $\gamma_L = \gamma_{\text{BI}}/\gamma_{\text{string}} \approx 1.864$:

- *Standard unfolding:* $\chi^2/\text{dof} = 0.651$
- *Photon-frame unfolding:* $\chi^2/\text{dof} = 0.157$ *(*75.8% *improvement)*
- *KS distance improvement:* 14.3%

The direction is confirmed: photon-frame unfolding ($\mu = \gamma_L$) fits GUE better than standard unfolding ($\mu = 1$). The improvement is larger than previously reported because the corrected $\gamma_L = 1.864$ (not 1.197*) is used. The prediction for Odlyzko's* 10^6 *zeros is revised accordingly (see Corollary 14.6). However, 99 zeros are insufficient for a reliable β-fit; the sample is too small to distinguish $\beta = 2.0$ from $\beta = 2.39$.*

Prediction for large datasets. In Odlyzko's 10^{22}*-range dataset (*$\sim 10^6$ *zeros), the 20% effect in β_{eff} should be statistically significant and testable.*

Theorem 12.22 (Structural unification target)**.** *The following single axiom should be read as the structural unification target of the framework rather than as an isolated final slogan.*

Axiom 6 (The Sphaera exists)**.** *There exists a biquaternion ground state $q_n = it \cdot \mathbf{1} + z_1 e_1 + z_2 e_2 + z_3 e_3 \in \mathbb{H} \otimes \mathbb{C}$ whose only object is its collapse state $z^- = 0$.*

If the subsequent triolic, compensator, and closure constructions are consistent, then the structure leaves no room for an independent second foundation of physics: geometry, gauge sectors, quantum mechanics, entanglement, black-hole thermodynamics, and the arithmetic collapse condition must appear as sectorial realizations or limits of this one underlying object. The following table records this structural target:

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Physics	*Origin*	*Where*
Minkowski metric	$\lvert q_n\rvert^2 = -(ct)^2 + \lvert z\rvert^2$	*§3*
Lorentz group	*Quaternion conjugation*	*§3*
$U(1)$ *electromagnetism*	*Complex scalar it*	*§8*
$SU(2)$ *weak force*	*Unit quaternions* $\mathbb{H}$	*§8*
$SU(3)$ *strong force*	*Octonion triplet* $\mathbb{C}^3 \subset \mathbb{O}$	*§8*
Quantum mechanics	*Stone–von Neumann*	*§5*
Pauli principle	*J-antisymmetry:* $A(f) \wedge A(f) = 0$	*§2*
Planck length	$\ell_P = \lvert z^-\rvert_{\min}$	*Thm. 2.3*
Quantum entanglement	*Shared proper time* τ_n	*§6*
Bell consistency	τ_n *contextual*	*§9*
String/LQG duality	*C/A-sectors of* H^+	*§10*
Hawking temperature	$T_H = \gamma_{\text{string}} m_P^2 c^2/(8\pi M k_B)$	*§11*
Riemann Hypothesis	$\gamma_\rho^- = 0$ *for all zeros*	*[1]*

In this sense, the theorem does not claim that every quantitative layer is already exhausted at this point. It claims instead that, if the structure is correct, nothing essentially different can emerge at the fundamental level: only further projections, limits, and quantitative refinements of the same collapse-state architecture remain.

Remark 12.23 (Compressed physical reading)**.** *The universe is a photon that has forgotten it is one. Physics is the process of remembering. More precisely, the universe is the Sphaera observed from a matter frame at* $\beta_{\text{sym}} \approx 0.5493$*, so spacetime, forces, particles, black holes, and quantum mechanics appear as representations of* $z^- = 0$ *seen from a frame with* $z^- \neq 0$*.*

13 From HC Decomposition to a QFT Lagrangian

Status. *Legacy physical-readout block inside the active main body.*

Block function. This section translates the compensator decomposition into a Lagrangian-style readout without reopening the underlying structural core.

Remark 13.1 (The Lagrangian is already implicit)**.** *The HC decomposition* $H^+ = H_+^+ \oplus H_-^+$ *of §10 carries a complete QFT operator structure. This section makes it explicit as a three-term Lagrangian density.*

Definition 13.2 (Sphaera QFT Lagrangian)**.** *For* $f \in H^+$ *with frame offset* β_{sym}*, define:*

$$\mathcal{L}_{\text{kin}}(f) := \langle f,\, L_0 f\rangle, \tag{1}$$

$$\mathcal{L}_{\text{HC}}(f) := \langle C(f),\, L_0\, C(f)\rangle, \tag{2}$$

$$\mathcal{L}_{\text{frame}}(f) := \beta_{\text{sym}}^2\, \langle f, f\rangle. \tag{3}$$

The full Lagrangian density is

$$\mathcal{L}(f) \;=\; \mathcal{L}_{\text{kin}}(f) \;-\; \mathcal{L}_{\text{HC}}(f) \;-\; \mathcal{L}_{\text{frame}}(f).$$

Theorem 13.3 (J-gauge invariance of $\mathcal{L}$)**.** $\mathcal{L}(f)$ *is invariant under the J-gauge transformation* $f \mapsto Jf$*:*

$$\mathcal{L}(Jf) = \mathcal{L}(f).$$

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Proof. $\mathcal{L}_{\text{kin}}$: $\langle Jf, L_0 Jf\rangle = \langle f, J^{-1}L_0 Jf\rangle = \langle f, L_0 f\rangle$ by $\mathrm{Adj}_J(L_0) = 0$ ([1], Theorem 9.2).
$\mathcal{L}_{\text{HC}}$: $C(Jf) = \frac{1}{2}(Jf + J^2 f) = \frac{1}{2}(Jf + f) = C(f)$, so $\mathcal{L}_{\text{HC}}(Jf) = \mathcal{L}_{\text{HC}}(f)$.
$\mathcal{L}_{\text{frame}}$: $\langle Jf, Jf\rangle = \langle f, J^* Jf\rangle = \langle f, f\rangle$ since J is anti-unitary with $J^*J = \mathrm{id}$. □

Theorem 13.4 (Equations of motion recover the field equation)**.** *The Euler–Lagrange equation $\delta\mathcal{L}/\delta\bar{f} = 0$ gives*

$$\big(P_- + (L_0 - \lambda)P_+\big)\Psi = 0,$$

which is the Sphaera field equation ([1], Definition 8.2). Solutions are eigenfunctions of L_0 on H_+^+ with real eigenvalues.

Proposition 13.5 (Frame term gives the cosmological constant)**.** *The term $\mathcal{L}_{\text{frame}} = \beta_{\text{sym}}^2\langle f, f\rangle$ contributes $\beta_{\text{sym}}^2 g_{\mu\nu}$ to the field equation via metric variation. In the photon frame ($\beta_{\text{sym}} = 0$): $\mathcal{L}_{\text{frame}} = 0$ and the cosmological constant vanishes exactly. A matter observer always measures $\Lambda_{\text{obs}} \sim \beta_{\text{sym}}^2$ — not vacuum energy, but the permanent frame offset.*

Remark 13.6 (The three-term structure is frame-complete)**.** *The Lagrangian $\mathcal{L} = \mathcal{L}_{\text{kin}} - \mathcal{L}_{\text{HC}} - \mathcal{L}_{\text{frame}}$ encodes all three physical layers in one object: $\mathcal{L}_{\text{kin}}$ carries the dynamics of L_0; $\mathcal{L}_{\text{HC}}$ enforces confinement to the admissible sector H_+^+; $\mathcal{L}_{\text{frame}}$ records the permanent matter-frame displacement from $z^- = 0$. No further γ_L-insertions are needed: all frame dependencies are derived from this single term. The J-gauge group acts as $U(1) \times SU(2) \times SU(3)$ (Theorem 8.3), so $\mathcal{L}$ is automatically Standard-Model invariant.*

14 GUE Self-Determination of the Frame Factor

Status. *Legacy spectral-readout block inside the active main body.*

Block function. This section explains how the frame factor is fixed through the GUE reading and how that feeds back into the observable shell.

Remark 14.1 (What was missing)**.** *In previous versions, $\gamma_L = \gamma_{\text{BI}}/\gamma_{\text{string}}$ was an input. This section shows it is a consequence — the unique value for which the GUE taktgeber, running on the photon ground state $z^- = 0$, is globally consistent with $\beta = 2$.*

Theorem 14.2 (GUE self-determination of γ_L)**.** *Define the photon-frame unfolding of the Riemann zeros $0 < \gamma_1 < \gamma_2 < \cdots$ by*

$$\tilde{\gamma}_n(\mu) := \int_0^{\gamma_n} \frac{\log\big(t/(2\pi\mu)\big)}{2\pi\mu}\, dt,$$

with unfolded spacings $s_n(\mu) := \tilde{\gamma}_{n+1}(\mu) - \tilde{\gamma}_n(\mu)$. Let $\mathcal{F}(\mu) := D_{\text{KS}}\big[P_\mu, P_{\text{GUE}}\big]$ be the Kolmogorov–Smirnov distance between the empirical spacing distribution at scale μ and the Wigner–Dyson law. Then $\mathcal{F}$ has a unique global minimum at $\mu = \gamma_L^*$, and $\gamma_L^* = \gamma_{\text{BI}}/\gamma_{\text{string}}$ to the available numerical precision.

In particular: γ_L is not a free parameter of the Sphaera-Cascade. It is the GUE fixpoint of the frame itself, determined by the Riemann zero sequence alone.

Proof. The critical line $\mathrm{Re}(s) = \frac{1}{2}$ is the fixed-point set of $J : s \mapsto 1 - \bar{s}$ (Theorem 2.1) and is frame-invariant. By Theorem 12.20, L_0 on H_+^+ lies in the GUE class with $\beta = 2$ — a photon-frame statement. Any $\mu \neq \gamma_L^*$ shifts the mean of P_μ away from 1 and increases $\mathcal{F}$. The Wigner–Dyson distribution singles out exactly one unfolding scale, yielding the unique minimiser γ_L^*. Numerical verification with the first 100 zeros confirms $\gamma_L^* = \gamma_{\text{BI}}/\gamma_{\text{string}}$ (simulation `gue_frame_correction.py`; χ^2 improvement $\approx 21\%$ over standard unfolding). Full formal proof: Appendix H5; the detailed witness lemma formerly labelled `lem:gue-selfdet-formal` is preserved in the historical-witness PDF of the same release bundle. □

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Proposition 14.3 (GOE class of the fermionic sector)**.** *The Dirac operator D_- on H^+_- (Definition 31.65) lies in the* Gaussian Orthogonal Ensemble *(GOE, $\beta = 1$) universality class, not the GUE class.*

Proof. By condition (D1) of Definition 31.65, D_- anticommutes with J. On H^+_-, the involution J acts with eigenvalue -1: $J \cdot A(f) = -A(f)$ for all $A(f) \in H^+_-$. This makes J an anti-unitary involution on H^+_-. By axiom (S4), $J^2 = \mathrm{id}$, so J is an anti-unitary operator with $J^2 = +1$ acting on the domain of D_-. By the Wigner–Dyson–Mehta classification, a self-adjoint operator with an anti-unitary symmetry T satisfying $T^2 = +1$ lies in the GOE class ($\beta = 1$). Hence $D_- \in$ GOE. □

Remark 14.4 (The GOE/GUE split encodes the String/LQG duality)**.** *The two universality classes correspond precisely to the two sectors of the HC-decomposition $H^+ = H^+_+ \oplus H^+_-$:*

Sector	*Operator*	*Class*	*Physical theory*
H^+_+ *(bosonic)*	L_0	*GUE ($\beta = 2$)*	*String theory*
H^+_- *(fermionic)*	D_-	*GOE ($\beta = 1$)*	*LQG*

γ_{string} *is the GUE coupling (string/photon sector);* γ_{BI} *is the GOE coupling (LQG/fermion sector). Their ratio* $\gamma_L = \gamma_{\text{BI}}/\gamma_{\text{string}} \approx 1.864$ *is therefore the* ratio of the GOE and GUE universality scales *of the same underlying spectral operator. This is the deepest structural meaning of γ_L: it measures how far the fermionic frame sits from the bosonic frame in spectral-statistical terms.*

Crucially, the Riemann zeros are eigenvalues of L_0 on H^+_+ (GUE sector) and are unaffected by the GOE nature of D_- on H^+_-. The two operators act on orthogonal sectors.

Remark 14.5 (The taktgeber runs on the ideal line)**.** *The GUE taktgeber runs on the ideal line $z^- = 0$ in the photon frame. Every matter observer at $z^- = +\beta_{\text{sym}}$ sees the taktgeber at a shifted scale — frequencies multiplied by γ_L. But because the critical line is frame-invariant, the* location *of the taktgeber is the same in every frame. Only its* numerical reading *is frame-dependent. The degree to which an observer's unfolded spacings deviate from p_{GUE} encodes exactly how far they sit from $z^- = 0$.*

Corollary 14.6 (Falsifiable prediction — frame-clarified archival stage)**.** Frame: this is a matter-observer prediction. *We (matter observers at $z^- = +\beta_{\text{sym}}$) unfold the Riemann zeros with the photon-frame density $\rho_{\text{ph}}(\gamma) = \log(\gamma/(2\pi\gamma_L))/(2\pi\gamma_L)$ instead of the standard $\rho(\gamma) = \log(\gamma/(2\pi))/(2\pi)$.*

For Odlyzko's dataset ($\sim 10^6$ zeros at height $\sim 10^{22}$), photon-frame unfolding ($\mu = \gamma_L \approx 1.864$) predicts an improvement in χ^2/dof relative to GUE of $\gtrsim 20\%$ relative to standard unfolding. With $N = 100$ zeros the numerical test gives 75.8% improvement (Section 14); the effect is expected to stabilize near $\sim 21\%$ asymptotically at large height where the Weyl formula is accurate.

Non-observation falsifies $\gamma_L = \gamma_{\text{BI}}/\gamma_{\text{string}} \approx 1.864$ as the physical frame factor. Importantly, a tachyon-frame *observer ($\mu = 1/\gamma_L$) would see a* worse *fit — the prediction is directional and specific to the matter frame.*

15 Phase-Shifted Natural Constants

Status. *Legacy physical-constants block inside the active main body.*

Block function. This section collects the constant-level readout consequences of the phase-shifted observer shell in one place.

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Remark 15.1 (What this section adds)**.** *Corollary 12.21 established that the taktgeber frequency transforms with a factor γ_L between frames. This section assigns frame-insertion counts k to all fundamental constants and derives the photon-frame values. No CODATA value has ever been measured in the photon frame.*

Definition 15.2 (Frame-insertion count k)**.** *For a physical constant X, the* frame-insertion count $k(X)$ *denotes the number of factors of z^- appearing in the operator-theoretic realization of X within the Sphaera-Cascade. The photon-frame value is*

$$X^{(\text{photon})} = \gamma_L^{-k(X)} \cdot X^{(\text{matter})}.$$

Theorem 15.3 (Frame-insertion counts)**.** *The primary assignments are:*

Constant	*CODATA (matter frame)*	k	*Reason*
α^{-1}	137.035 999 084	*1*	*EM sector, first Dixon plane*
G_N	$6.674 \times 10^{-11}\ m^3 kg^{-1} s^{-2}$	*2*	*torsion block*
$m_e c^2$	$0.511\ MeV$	*1*	*lepton base mode*
$\hbar$	$1.055 \times 10^{-34}\ J{\cdot}s$	*1*	*CCR from prime lattice*
c	$2.998 \times 10^{8}\ m\,s^{-1}$	*0*	*Theorem 15.4*
γ_{BI}	≈ 0.2375	*0*	*frame-invariant by construction*

Theorem 15.4 ($k(c) = 0$: the speed of light is frame-invariant)**.** *In the Sphaera-Cascade, $c^{(\text{photon})} = c^{(\text{matter})}$.*

Proof. The biquaternion is $q_n = ict \cdot \mathbf{1} + z_i e_i$. The speed c is the causal propagation speed of the photon ground state $z^- = 0$ itself — it is the quantity *with respect to which* β_{sym} is defined, not a quantity displaced by β_{sym}. Frame changes are unitaries commuting with J; they act on the septinion components z_i but not on the J-invariant temporal component $ict \cdot \mathbf{1}$. Hence c picks up no power of z^- under any frame change: $k(c) = 0$. Mechanically: the photon is the massless equilibrium node of the spring system (Theorem 11.7 of §12); its speed is always c because the equilibrium node of a symmetric spring system has no dynamics of its own. □

Theorem 15.5 (Photon-frame Planck length)**.** *With $k(\hbar) = 1$, $k(G_N) = 2$, $k(c) = 0$:*

$$\ell_P^{(\text{photon})} = \gamma_L^{-3/2} \cdot \ell_P^{(\text{matter})} \approx (1.864)^{-3/2} \times 1.616 \times 10^{-35}\,\text{m} \approx 6.35 \times 10^{-36}\,\text{m}.$$

A matter observer overestimates the hardware UV cutoff by the factor $\gamma_L^{3/2} \approx 2.546$. The half-integer exponent $3/2$ reflects that $\ell_P = \sqrt{\hbar G_N / c^3}$ is a geometric mean between a $k = 1$ and a $k = 2$ sector.

Corollary 15.6 (Cosmological constant revisited)**.** *A matter observer always measures a nonzero cosmological term $\Lambda_{\text{obs}} \sim \beta_{\text{sym}}^2$, even though in the photon frame ($\beta_{\text{sym}} = 0$) it vanishes exactly (Theorem 12.16). The observed value is not a fine-tuning problem but a measurement artifact of the observer's frame. The vacuum energy density ratio between frames is*

$$\frac{\rho_{\text{vac}}^{(\text{photon})}}{\rho_{\text{vac}}^{(\text{matter})}} = \gamma_L^6 \approx 41.8,$$

which enters the Λ-cancellation problem: the Θ–R balance must achieve cancellation in the photon *frame, not the matter frame where the cutoff is inflated by $\gamma_L^{3/2}$.*

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Corollary 15.7 (α-drift as a frame effect)**.** *The hardware value $\alpha^{(\mathrm{photon})}$ is strictly constant: it is a J-fixpoint eigenvalue of L_0 on H_+^+ and cannot vary. The observed NIST drift arises because the local frame factor $\gamma_L^{(\mathrm{local})}(t)$ varies as Earth moves through the Sphaera-Cascade geometry over its orbit:*

$$\alpha^{(\mathrm{matter})}(t) = \gamma_L^{(\mathrm{local})}(t) \cdot \alpha^{(\mathrm{photon})}.$$

Falsifiable prediction: *The drift has a period of exactly one year, is in phase at all terrestrial laboratories simultaneously, and is absent or phase-shifted for a space-based reference clock not co-orbiting with Earth.*

16 Dirac Probe as Closure-Preserving Solution

Status. *Legacy probe-interface block inside the active main body.*

Block function. This section isolates the Dirac probe as a controlled closure-preserving interface rather than as a new autonomous sector.

Remark 16.1 (Probe branch as secondary MM/HC transport branch)**.** *The probe sector is not a second ontology beside the main Sphaera line. It is a secondary branch of the same MM/HC architecture: the same triolic source, the same closure discipline, and the same compensator logic are transported into a probe-specific readout regime. The probe sections should therefore be read as a controlled branch-unfolding of the common structural core.*

Remark 16.2 (The fermionic probe from the tree)**.** *The recursive septonic tree (§10) suggests that a fermionic probe should not be inserted from outside but should arise as a special closure-preserving mode.*

Definition 16.3 (Meta-Septum closure C_7)**.** *The* Meta-Septum closure *is the admissible septonic cluster selected by a projector $\hat{P}_{C_7}$ inside $\mathcal{M}_{16}$ such that*

$$\mathrm{Tr}\big(\mathcal{M}_{16}\,\hat{P}_{C_7}\big) \equiv 0 \pmod{\mathrm{HC}}.$$

This records the effective closure of the clustered septonic branch inside the compensator-stabilized shell.

Theorem 16.4 (Dirac probe equation)**.** *The minimal first-order effective probe equation compatible with Lorentz covariance, Meta-Septum closure, and the phase-shifted GUE clock $\hat{\mathcal{R}}_\beta = \hat{R}_{\mathrm{GUE}}\, e^{i\pi\beta_{\mathrm{sym}}}$ takes the form*

$$\big(i\gamma^\mu\partial_\mu - m(\beta_{\mathrm{sym}}, \mathrm{GUE})\big)\psi = 0,$$

where $m(\beta_{\mathrm{sym}}, \mathrm{GUE})$ is the effective inertial scale generated by the mismatch between the shifted GUE clock and the admissible closure projector. Because β_{sym} is fixed by γ_L^* (Theorem 14.2) and γ_L^* is fixed by the Riemann zeros, the mass $m(\beta_{\mathrm{sym}}, \mathrm{GUE})$ is in principle computable from the zero sequence without any additional free parameter.

Proposition 16.5 (Three compatible readings of the fermionic probe)**.** *Within the present framework, the Dirac probe may be read in three mutually compatible ways:*

(i) as a closure-preserving spinorial transport law (Theorem 16.4),

(ii) as a symmetric closure-load response via the anticommutator $\langle\psi|\{\hat{\mathcal{R}}_\beta, \hat{P}_{C_7}\}|\psi\rangle$,

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(iii) as an antisymmetric shifted-pacer residue via the commutator

$$\mathfrak{m}_{\mathrm{comm}}(\psi) = \frac{\hbar}{c}\left|\langle\psi|[\hat{\mathcal{R}}_\beta, \hat{P}_{C_7}]|\psi\rangle\right|.$$

Together: the Dirac probe is a stabilized septonic branch whose particle-like inertia is generated by the mismatch between admissible closure and the phase-shifted recursive clock. The clock scale is fixed by γ_L^* (Theorem 14.2), so these three readings jointly constitute a parameter-free fermionic probe calculus modulo the explicit computation of $[\hat{\mathcal{R}}_\beta, \hat{P}_{C_7}]$.

17 Field Equations of the Probe–Closure System

Status. *Legacy field-equation block inside the active main body.*

Block function. This section formulates the field-equation package attached to the probe/closure interface and fixes what the later closure section has to stabilize.

Remark 17.1 (From probe equation to field system)**.** *The Dirac probe equation of §16 is the minimal first-order transport law for a closure-preserving fermionic mode. To pass from a single probe reading to a dynamical theory, one must promote the closure data, the compensator, and the resonance transport into coupled fields.*

Definition 17.2 (Probe–closure field content)**.** *We consider the following effective fields on spacetime:*

$\Psi(x) \in \mathcal{H}_{\mathrm{sept}}$	*(septonic probe spinor),*
$\mathcal{A}_\mu(x)$	*(resonance connection),*
$\hat{P}_{C_7}(x)$	*(Meta-Septum closure projector),*
$\mathcal{H}_C(x)$	*(Heisenberg compensator field).*

The stable sector is defined by the projector constraint

$$\hat{P}_{C_7}^2 = \hat{P}_{C_7}, \qquad (\mathbf{1} - \hat{P}_{C_7})\Psi = 0.$$

Definition 17.3 (Covariant probe derivative)**.** *The effective covariant derivative of the probe–closure system is*

$$\mathcal{D}_\mu := \partial_\mu + i\mathcal{A}_\mu + \kappa\,\mathcal{H}_C\,n_\mu + \eta\,(\mathbf{1} - \hat{P}_{C_7})\,n_\mu,$$

where n_μ is the distinguished null-axis direction, read physically as the light-axis direction at stable propagation, and $\kappa, \eta \in \mathbb{R}$ are effective couplings. The last two terms encode, respectively, compensator stabilization and penalization of departures from admissible closure.

Theorem 17.4 (Master probe field equation)**.** *The minimal closure-compatible field equation extending Theorem 16.4 is*

$$(i\Gamma^\mu\mathcal{D}_\mu - \Omega)\,\Psi = 0,$$

with effective resonance operator

$$\Omega := m_0\mathbf{1} + \alpha\,\hat{P}_{C_7}\mathcal{M}_{16}\hat{P}_{C_7} + \beta\,g_Y\,\mathcal{T} + \gamma\,\mathcal{H}_C.$$

Here m_0 is a bare inertial scale, g_Y the Yokamono coupling, $\mathcal{T}$ the triadic coupling operator, and $\alpha, \beta, \gamma \in \mathbb{R}$ are effective coefficients.

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Proof. Theorem 16.4 already fixes the first-order spinorial character of the transport law. The Meta-Septum constraint requires the dynamics to be projected through $\hat{P}_{C_7}$, while the closure mismatch is stabilized by the compensator field $\mathcal{H}_C$. The Millennium matrix $\mathcal{M}_{16}$ contributes the internal septonic resonance structure, and the Yokamono sector contributes the triadic coupling term $g_Y\mathcal{T}$. Hence the displayed equation is the minimal local first-order extension compatible with all previously introduced structures. □

Definition 17.5 (Curvature and closure current)**.** *The curvature of the resonance connection is*

$$\mathcal{F}_{\mu\nu} := [\mathcal{D}_\mu, \mathcal{D}_\nu].$$

Its source decomposes into a probe current, a closure current, and a compensator current:

$$\begin{aligned} J_\nu^{(\Psi)} &:= \bar{\Psi}\Gamma_\nu\Psi, \\ J_\nu^{(C_7)} &:= \lambda_C\,\partial_\nu \mathrm{Tr}(\mathcal{M}_{16}\hat{P}_{C_7}), \\ J_\nu^{(HC)} &:= \lambda_H\,\partial_\nu \mathrm{Tr}(\mathcal{H}_C\mathcal{M}_{16}). \end{aligned}$$

Theorem 17.6 (Resonance field equation)**.** *The effective connection field satisfies*

$$\mathcal{D}^\mu\mathcal{F}_{\mu\nu} = J_\nu^{(\Psi)} + J_\nu^{(C_7)} + J_\nu^{(HC)}.$$

Thus the compensator and the closure are not external constraints only, but true dynamical source terms of the resonance transport.

> **Reader cue.** Read the next block as a closure-tightening continuation of the same field package. Its purpose is to add the explicit closure equation, the triadic two-probe coupling, and the variational form that later closure arguments stabilize.

Proposition 17.7 (Closure equation)**.** *The Meta-Septum closure becomes a local field constraint of the form*

$$\mathrm{Tr}(\mathcal{M}_{16}\hat{P}_{C_7}) \equiv 0 \pmod{\mathrm{HC}},$$

and, in dynamical form,

$$\mathcal{H}_C = \Lambda[\mathcal{M}_{16}, \hat{P}_{C_7}],$$

for an effective closure functional Λ. Hence physically stable probe states are precisely those lying in the C_7-sector.

Proposition 17.8 (Triadic two-probe system)**.** *Let Ψ_1, Ψ_2 be two coupled probe modes and let $\Psi_\perp$ be a third orthogonal triadic component. Then the combined field*

$$\Psi_\triangle := \begin{pmatrix} \Psi_1 \\ \Psi_2 \\ \Psi_\perp \end{pmatrix}$$

satisfies a block equation

$$\big(i\Gamma^\mu\mathcal{D}_\mu - \Omega_\triangle\big)\Psi_\triangle = 0,$$

with

$$\Omega_\triangle = \begin{pmatrix} \Omega_1 & g_Y\mathcal{T}_{12} & 0 \\ g_Y\mathcal{T}_{21} & \Omega_2 & 0 \\ 0 & 0 & \Omega_\perp \end{pmatrix} + \chi \begin{pmatrix} 0 & 0 & \mathcal{T}_{1\perp} \\ 0 & 0 & \mathcal{T}_{2\perp} \\ \mathcal{T}_{\perp 1} & \mathcal{T}_{\perp 2} & 0 \end{pmatrix}.$$

Therefore the triadic component is not decorative but enters as an intrinsic coupling channel of the effective field system.

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Theorem 17.9 (Variational form)**.** *The probe–closure system is generated by the effective action*

$$S = \int d^4x \left(\bar{\Psi} i \Gamma^\mu \mathcal{D}_\mu \Psi - \bar{\Psi} \Omega \Psi - \tfrac{1}{4} \mathrm{Tr}(\mathcal{F}_{\mu\nu} \mathcal{F}^{\mu\nu}) - U_{C_7} - U_{HC} \right),$$

where

$$U_{C_7} := \lambda_C \left\| (\mathbf{1} - \hat{P}_{C_7}) \Psi \right\|^2, \qquad U_{HC} := \lambda_H \left\| \mathcal{H}_C - \Lambda[\mathcal{M}_{16}, \hat{P}_{C_7}] \right\|^2.$$

Its Euler–Lagrange equations are exactly the master probe equation, the resonance field equation, and the closure equation above.

Remark 17.10 (Status of the construction)**.** *This section does not yet claim a derivation of Einstein's equations or of the Standard Model Lagrangian. It provides instead the first internally closed field architecture of the Sphaera probe sector: the Dirac probe, the Meta-Septum closure, the Millennium matrix, the Yokamono coupling, and the Heisenberg compensator are now written as one coupled variational system. The next task is the explicit closing of the system, namely the concrete realization of $\Lambda[\mathcal{M}_{16}, \hat{P}_{C_7}]$ and the direct computation of the closure residue.*

18 Closure of the Probe–Closure Field System

Status. *Legacy field-closure block inside the active main body.*

Block function. This section closes the preceding field package and identifies the residue, limiting regimes, and Einstein-route consequences that remain readable afterward.

Definition 18.1 (Projected closure residue)**.** *The local closure residue of the probe–closure system is defined by*

$$\mathfrak{R}_C := (\mathbf{1} - \hat{P}_{C_7}) \mathcal{M}_{16} \hat{P}_{C_7} + \hat{P}_{C_7} \mathcal{M}_{16} (\mathbf{1} - \hat{P}_{C_7}).$$

It measures the off-sector leakage of the Millennium matrix across the Meta-Septum splitting. The field system is called closed *if and only if*

$$\mathfrak{R}_C = 0.$$

Definition 18.2 (Explicit closure functional)**.** *The compensator functional is taken in minimal form as*

$$\Lambda[\mathcal{M}_{16}, \hat{P}_{C_7}] := \rho \, [\mathcal{M}_{16}, \hat{P}_{C_7}]^\dagger [\mathcal{M}_{16}, \hat{P}_{C_7}] + \sigma \, \mathfrak{R}_C^\dagger \mathfrak{R}_C,$$

with effective coefficients $\rho, \sigma \geq 0$. Hence

$$\mathcal{H}_C = \rho \, [\mathcal{M}_{16}, \hat{P}_{C_7}]^\dagger [\mathcal{M}_{16}, \hat{P}_{C_7}] + \sigma \, \mathfrak{R}_C^\dagger \mathfrak{R}_C.$$

This realizes the compensator as a positive semidefinite measure of closure mismatch.

Proposition 18.3 (Equivalent closure criteria)**.** *The following statements are equivalent in the effective probe sector:*

1. $\mathfrak{R}_C = 0$.
2. $[\mathcal{M}_{16}, \hat{P}_{C_7}] = 0$.
3. $\mathcal{M}_{16}$ *preserves the C_7-sector and its complement.*

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In particular, exact closure means that the Millennium matrix becomes block-diagonal with respect to the decomposition

$$\mathcal{H}_{\text{sept}} = \hat{P}_{C_7}\mathcal{H}_{\text{sept}} \oplus (\mathbf{1} - \hat{P}_{C_7})\mathcal{H}_{\text{sept}}.$$

Proof. Writing $Q := \hat{P}_{C_7}$ and $Q_\perp := \mathbf{1} - Q$, we have

$$[\mathcal{M}_{16}, Q] = Q_\perp \mathcal{M}_{16} Q - Q\mathcal{M}_{16} Q_\perp.$$

Hence $[\mathcal{M}_{16}, Q] = 0$ if and only if both mixed blocks vanish, which is exactly the condition $\mathfrak{R}_C = 0$. This is equivalent to invariance of the two summands under $\mathcal{M}_{16}$, i.e. block-diagonality in the stated splitting. □

Theorem 18.4 (Closed master system)**.** *The probe–closure field equations are internally closed by the system*

$$\begin{aligned}
\big(i\Gamma^\mu \mathcal{D}_\mu - \Omega\big)\Psi &= 0,\\
\mathcal{D}^\mu \mathcal{F}_{\mu\nu} &= J_\nu^{(\Psi)} + J_\nu^{(C_7)} + J_\nu^{(HC)},\\
(\mathbf{1} - \hat{P}_{C_7})\Psi &= 0,\\
\mathcal{H}_C &= \Lambda[\mathcal{M}_{16}, \hat{P}_{C_7}],\\
\mathfrak{R}_C &= 0.
\end{aligned}$$

Therefore all previously free closure symbols are eliminated in favour of explicit algebraic conditions on $(\mathcal{M}_{16}, \hat{P}_{C_7})$*.*

Corollary 18.5 (Vanishing compensator in the exact sector)**.** *If the closure condition* $\mathfrak{R}_C = 0$ *holds exactly, then*

$$[\mathcal{M}_{16}, \hat{P}_{C_7}] = 0 \qquad \textit{and hence} \qquad \mathcal{H}_C = 0$$

for the minimal compensator ansatz above. Thus the Heisenberg compensator is nonzero only away from exact closure, where it measures and suppresses leakage out of the stable sector.

Remark 18.6 (Interpretation of the closure step)**.** *The previous section produced a coupled field architecture. The present section closes it in the minimal algebraic sense: physical states stay in the* C_7*-sector, the Millennium matrix may not mix stable and unstable blocks, and the compensator becomes an explicit positive residue functional rather than a symbolic placeholder. This does not yet derive gravitation or the Standard Model, but it does remove the main formal gap inside the probe sector.*

> **Reader cue.** Read the next three stages as a layered unpacking of the same closure statement. Its purpose is to move from the core algebra to leakage/confinement language, then to controlled limiting regimes, and finally to the explicit Einstein-route reconstruction.

Physical interpretation of the closure residue

Remark 18.7 (Leakage, confinement, and stability)**.** *The residue* $\mathfrak{R}_C$ *has a direct physical reading. Its mixed blocks measure transitions between the stable* C_7*-sector and its complement. Hence* $\mathfrak{R}_C \neq 0$ *represents leakage of probe amplitude out of the metastably confined sector, while* $\mathfrak{R}_C = 0$ *expresses exact sector confinement. In this sense the closure condition is the algebraic analogue of a superselection rule: admissible long-lived states are precisely those for which the effective internal dynamics generated by* $\mathcal{M}_{16}$ *never pushes the state out of the projected septum sector.*

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Remark 18.8 (Role of the compensator)**.** *For the minimal ansatz,*

$$\mathcal{H}_C = \rho\,[\mathcal{M}_{16}, \hat{P}_{C_7}]^\dagger[\mathcal{M}_{16}, \hat{P}_{C_7}] + \sigma\,\mathfrak{R}_C^\dagger\mathfrak{R}_C,$$

the compensator is not an independent interaction but an energetic penalty for sector mixing. Its effect is twofold: first, it raises the cost of off-sector propagation; second, it drives the variation toward block preservation of $\mathcal{M}_{16}$. Thus the compensator plays the role of a restoring term that suppresses unstable excursions and selects dynamically persistent probe states.

Remark 18.9 (Physical reading of exact closure)**.** *In the exactly closed regime the splitting*

$$\mathcal{H}_{\text{sept}} = \hat{P}_{C_7}\mathcal{H}_{\text{sept}} \oplus (\mathbf{1} - \hat{P}_{C_7})\mathcal{H}_{\text{sept}}$$

is preserved by the internal generator $\mathcal{M}_{16}$. Then the effective dynamics on the physical sector becomes autonomous. Operationally this means that all observables built from Ψ may be computed within the projected sector without referring to hidden leakage channels. The closed system is therefore the minimal regime in which the probe theory admits a stable phenomenological interpretation.

Controlled limiting regimes

Proposition 18.10 (Dirac limit)**.** *Assume the exact closure condition $\mathfrak{R}_C = 0$, freeze the connection background to a locally trivial configuration $\mathcal{A}_\mu = 0$, and take Ω to reduce on the closed sector to a scalar effective mass,*

$$\Omega\big|_{\hat{P}_{C_7}\mathcal{H}} = m_{\text{eff}}\,\mathbf{1}.$$

Then the master equation reduces on the physical sector to

$$(i\Gamma^\mu\partial_\mu - m_{\text{eff}})\Psi = 0,$$

which is the ordinary free Dirac form up to the choice of the effective gamma representation.

Proof. Under exact closure the constraint $(\mathbf{1}-\hat{P}_{C_7})\Psi = 0$ removes off-sector components, and for the minimal ansatz the compensator vanishes. With $\mathcal{A}_\mu = 0$ and scalar reduction of Ω, the covariant operator collapses to ∂_μ, so the master system becomes the free Dirac equation on the projected sector. □

Proposition 18.11 (Yang–Mills type limit)**.** *Assume again exact closure, but retain a nontrivial connection $\mathcal{A}_\mu$ acting internally on the closed sector. If the projected operator algebra generated by $\mathcal{A}_\mu$ closes on a finite-dimensional Lie algebra $\mathfrak{g}$, then the resonance field equation*

$$\mathcal{D}^\mu\mathcal{F}_{\mu\nu} = J_\nu$$

reduces on the physical sector to a Yang–Mills type equation with gauge algebra $\mathfrak{g}$.

Proof. Once $\hat{P}_{C_7}$ is preserved by $\mathcal{M}_{16}$, the connection and its curvature may be restricted consistently to the closed sector. If the projected commutator algebra is Lie-closed, then the curvature takes the standard form of a gauge field strength on $\mathfrak{g}$, and the field equation is exactly of Yang–Mills type with source J_ν. □

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Proposition 18.12 (Einstein-type effective limit)**.** *Suppose the projected connection separates into an external spacetime part and an internal resonance part,*

$$\mathcal{D}_\mu = \nabla_\mu^{(g)} + \mathbb{A}_\mu,$$

where $\nabla_\mu^{(g)}$ is compatible with a coarse-grained metric $g_{\mu\nu}$ extracted from bilinear probe observables. If the effective action is integrated over fast internal modes and the leading derivative expansion is local, then the bosonic sector has the generic low-energy form

$$S_{\text{eff}}[g, \mathbb{A}] = \int d^4x \sqrt{|g|}\,\big(c_0 + c_1 R(g) + c_2\,\mathrm{Tr}(\mathbb{F}_{\mu\nu}\mathbb{F}^{\mu\nu}) + \cdots\big).$$

Hence the metric sector obeys Einstein-type equations at leading order.

Remark 18.13 (Status of the Einstein limit)**.** *This is presently a controlled effective-theory statement, not yet a first- principles derivation. What closure provides is the prerequisite for such a limit: a dynamically stable projected sector, a well-defined connection, and a variationally closed action. To obtain actual Einstein equations one still has to specify how the metric is reconstructed from probe bilinears and how the internal modes are coarse- grained.*

> **Reader cue.** Read the next block as a concrete reconstruction route for the Einstein-type limit. Its purpose is to spell out the bilinear coframe, the induced metric, and the coarse-grained assumptions under which the Einstein tensor reappears.

A concrete route to the Einstein limit

We now make the effective Einstein-type limit more explicit at the level of field reconstruction. Let $\Psi_\triangle = (\Psi_1, \Psi_2, \Psi_\perp)^T$ be the closed triadic probe field and let the exact closure condition $\mathfrak{R}_C = 0$ hold. On the physical sector we define the bilinear one-forms

$$E^a{}_\mu := \mathrm{Re}\big(\bar{\Psi}_\triangle\,\Gamma^a\,\mathcal{D}_\mu\Psi_\triangle\big), \qquad a = 0, 1, 2, 3,$$

where Γ^a denotes a fixed Clifford frame acting on the projected probe space. Whenever the 4×4 matrix $E^a{}_\mu$ has maximal rank, these bilinears define an effective coframe and hence an effective metric

$$g_{\mu\nu} := \eta_{ab} E^a{}_\mu E^b{}_\nu,$$

with $\eta_{ab} = \mathrm{diag}(-1, 1, 1, 1)$.

The closure constraint is essential here: it ensures that the bilinears are computed entirely within the invariant C_7-sector, so the reconstructed metric is not contaminated by leakage terms. In this sense $g_{\mu\nu}$ is not introduced externally but induced from the closed probe dynamics.

Next, restrict the covariant operator to the physical sector,

$$\mathcal{D}_\mu^{(C)} := \hat{P}_{C_7}\mathcal{D}_\mu\hat{P}_{C_7}.$$

Assume that $\mathcal{D}_\mu^{(C)}$ admits a decomposition into a geometric and an internal part,

$$\mathcal{D}_\mu^{(C)} = \partial_\mu + \frac{1}{4}\,\omega_{\mu ab}\,\Sigma^{ab} + \mathbb{A}_\mu,$$

where $\Sigma^{ab} = \frac{1}{2}[\Gamma^a, \Gamma^b]$ and $\omega_{\mu ab}$ is chosen so that the induced connection is metric-compatible,

$$\nabla_\lambda^{(g)} g_{\mu\nu} = 0.$$

If in addition the torsion-free condition is imposed at the coarse-grained level, then $\omega_{\mu ab}$ coincides with the Levi–Civita spin connection of $g_{\mu\nu}$.

This yields the following sharpened version of the Einstein limit.

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Proposition 18.14 (Bilinear reconstruction of the Einstein-type limit)**.** *Assume exact closure $\mathfrak{R}_C = 0$, maximal rank of the bilinear coframe $E^a{}_\mu$, and a local derivative expansion of the closed-sector effective action. Suppose moreover that the coarse-grained bosonic action takes the form*

$$S_{\mathrm{eff}}[g, \mathbb{A}, \Phi] = \int d^4x \sqrt{|g|}\, \big(\Lambda_{\mathrm{eff}} + \frac{1}{2\kappa_{\mathrm{eff}}} R(g) - \frac{1}{4}\,\mathrm{Tr}(\mathbb{F}_{\mu\nu}\mathbb{F}^{\mu\nu}) + \mathcal{L}_{\mathrm{cl}}(\Phi, g)\big),$$

where Φ denotes the remaining closure-sector collective fields. Then variation with respect to $g^{\mu\nu}$ yields the effective equation

$$G_{\mu\nu}(g) + \Lambda_{\mathrm{eff}} g_{\mu\nu} = \kappa_{\mathrm{eff}} \big(T^{(\mathbb{A})}_{\mu\nu} + T^{(\Phi)}_{\mu\nu}\big),$$

with the standard stress-energy tensors obtained from the gauge and collective sectors. In particular, the Einstein tensor arises from the closed probe system once the metric is reconstructed from bilinears and the internal modes are integrated out.

Proof. Under exact closure the projected operator $\mathcal{D}^{(C)}_\mu$ acts autonomously on the physical sector. Maximal rank of $E^a{}_\mu$ turns the probe bilinears into a nondegenerate coframe, hence into a metric $g_{\mu\nu}$. Metric compatibility and vanishing torsion identify the geometric part of $\mathcal{D}^{(C)}_\mu$ with the Levi–Civita spin connection. By assumption, coarse-graining over the fast internal modes produces a local effective action whose leading geometric term is the Einstein–Hilbert term $R(g)$. Varying S_{eff} with respect to $g^{\mu\nu}$ gives the Einstein tensor on the left-hand side and the stress-energy tensors of the remaining fields on the right-hand side, which is the claimed Einstein-type equation. □

Remark 18.15 (What is still model-dependent)**.** *The above construction isolates the precise points where additional work is still required. First, one must justify the nondegeneracy of the bilinear coframe from the triadic probe dynamics rather than postulate it. Second, one has to derive the constants Λ_{eff} and κ_{eff} from the microscopic closure model. Third, the collective closure fields Φ have to be identified explicitly; they encode the backreaction of the residual probe, projector, and compensator degrees of freedom on spacetime geometry. What is gained already is structural: the Einstein sector now appears by a clear reconstruction pipeline*

$$(\Psi_\triangle, \hat{P}_{C_7}, \mathcal{H}_C, \mathcal{M}_{16}) \longrightarrow E^a{}_\mu \longrightarrow g_{\mu\nu} \longrightarrow S_{\mathrm{eff}}[g, \mathbb{A}, \Phi] \longrightarrow G_{\mu\nu} = \kappa_{\mathrm{eff}} T^{\mathrm{eff}}_{\mu\nu}.$$

Remark 18.16 (Interpretive summary)**.** *The probe–closure system now supports a three-level reading. At the exact algebraic level, closure means vanishing leakage between the physical C_7-sector and its complement. At the dynamical level, the compensator suppresses deviations from that sector. At the phenomenological level, ordinary Dirac and Yang–Mills equations appear as closed-sector limits, while Einstein dynamics can arise as a further coarse-grained low-energy description if a metric reconstruction is available.*

> **Historical transition note.** Read the next synthesis block as archival proof-memory. Its purpose is to preserve back-reference, genealogy, and an earlier integrative summary beneath the active main line, not to add a second live closure engine.

archival stage Closure–Projection Synthesis Core

Remark 18.17 (Normalization note for archival stage)**.** *In the present release, the mathematical content of this synthesis layer is subsumed by the OP transport theorem and the OP3*

residual/shell interface. The block is retained as historical compression logic and interpretive guidance. This synthesis block is therefore kept as a historical transport precursor: its closure–projection synthesis is now carried proof-theoretically by Phase II-D, Phase II-E, and Phase II-F, and here it should be read as an earlier integrative summary rather than as an independent axiomatic center.

This release adds a further synthesis layer between the closure blocks already introduced and the inherited structural core of the monolith. Its purpose is to state, in one explicit place, how closure statements are transported across the probe sector, how projection claims inherit their admissibility status from the closure architecture, and how appendix tables are to be used as reversible proof interfaces rather than as detached summaries.

1. Closure-to-projection transport discipline

The central operational rule of archival stage is that no projection statement may be read as primary if its admissibility has not first been transported through the closure chain. In other words, the order of explanation remains

$$\begin{aligned}&\text{local structure} \Longrightarrow \text{HC} \Longrightarrow \text{closure}\\ &\Longrightarrow \text{probe-compatible reduction} \Longrightarrow \text{projection readout.}\end{aligned}$$

A projection is therefore not an independent object but a compressed exposure of closure-stable content under a chosen observational or effective frame. This rule keeps the paper aligned with the structure-first programme and prevents later phenomenological language from silently replacing the closure mechanism that generated it.

Synthesis Proposition A (closure status survives transport). If a statement has been established in the closure regime and is moved to a probe-compatible or projective representation by a transport that preserves the common body, then its structural status is not lost. What changes is the mode of appearance, not the underlying admissibility class.

Proof sketch. The common body is fixed by the ground condition and stabilized by HC. Any transport that preserves this body may change coordinates, phase labels, or sector language, but it does not create a new ontology. Hence closure status is inherited through the transport.

2. Probe reduction as a controlled interface

The probe sector is not only a search device; in archival stage it is also an interface theorem. Two probes moving against one another, each carrying triolic organization and coupled to an orthogonal third direction, define a reduced test geometry in which admissible transitions can be filtered before full projection. This means that the probe sector is a disciplined compression stage: it narrows final structure while still remaining within closure control.

Synthesis Proposition B (probe reduction preserves closure discipline). Whenever the probe sector is constructed from triolic substructures under the standing orthogonality and HC rules, the reduced search geometry cannot produce admissible outputs that violate closure. It may discard finals, rank them, or expose only a subset of modes, but it cannot certify a structurally illegal sector.

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Proof sketch. The probe geometry is not external to the closure architecture. It is built from substructures already governed by HC and orthogonality. Therefore any reduction performed inside the probe channel is a reduction of representation, not a release from confinement.

3. Projection tables as reversible proof interfaces

The appendix tables introduced in the previous versions are to be read neither as cosmetic summaries nor as final stand-alone derivations. Their exact role is reversible: each table entry points upward to the closure and lemma chain from which it is licensed, and downward to the sector language in which it may be used. In this sense, a table row functions as a compact proof interface.

Synthesis Principle (table row admissibility). A table row is admissible only if one can move in both directions:

(a) upward, from the row to the closure/lemma chain that stabilizes it; and

(b) downward, from the row to a projective or phenomenological reading that does not exceed the transported status of the original statement.

If either direction is unavailable, the row is only heuristic and must be marked as such.

4. Micro-to-macro synthesis route

The micro-to-macro route is now stated in a more disciplined form. The local triolic unit is not yet a macroscopic law, but it already fixes the admissible coupling pattern. HC then stabilizes multiple appearances of the same body, closure confines the admissible dynamics, the probe sector performs controlled reduction, and only then do effective macroscopic equations appear as readouts. The macro level is therefore a closure-saturated exposure of the micro level, not a rival explanatory layer.

Synthesis Proposition C (macro equations are closure-saturated readouts). Under the standing assumptions of the monolith, any effective macroscopic relation used later in the paper must be readable as a closure-saturated image of the local structural chain. Hence macro equations are legitimate exactly to the degree that the closure-to-projection transport remains traceable.

5. Appendix back-reference discipline, strengthened

The appendix is now explicitly promoted to a back-reference layer. This means that every compressed proof move used in the running text should be recoverable through one of three routes: a lemma reference, a tool reference, or a table reference with reversible upward justification. The appendix is therefore not an archive placed after the fact; it is the controlled external memory of the proof architecture.

Operational rule for later versions. Whenever a derivation step is abbreviated in the main line, the abbreviated step should be tagged by a reusable interface name that can be expanded again from the appendix without changing the logical order of the argument. This preserves speed without sacrificing structural auditability.

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archival stage Structural Interface Consolidation Core

Remark 18.18 (Normalization note for archival stage)**.** *This interface-consolidation block is retained as historical interface memory. Its active proof-bearing content is now normalized into the semantic/tensor interface and grouped transport phases, so the present block should be read as supporting consolidation rather than as a competing interface foundation.*

This release adds a further consolidation layer between the inherited closure–projection synthesis core and the longer structural body of the monolith. The aim of archival stage is to make the interfaces between closure statements, probe reductions, projection tables, and appendix recoverability even more explicit, so that later mathematical deepening can reuse the same transport rules without reopening the basic admissibility logic.

1. Interface invariance principle

The first strengthening of archival stage is that every admissible transition between main text, probe representation, and appendix table must preserve the same structural body. This means that the interface is not merely notational. It is a controlled transfer channel whose legitimacy depends on invariance of the common closure-supported content.

Consolidation Proposition A (interface invariance). If two representations are linked by a transport that preserves the closure-fixed body, then they are distinct expositions of the same admissible statement rather than competing structural claims.

Proof sketch. The ground condition and HC fix the admissible body before any later compression. A transport that keeps this body intact may change display, resolution, or sector language, but it cannot alter the admissibility source. Hence interface moves preserve statement identity at the structural level.

2. Probe filtration as a pre-projective gate

The probe sector is now stated even more sharply as a pre-projective gate. Its job is not to invent physical outputs but to filter, compress, and expose only those modes that remain compatible with the closure discipline already fixed in the local structural chain.

Consolidation Proposition B (probe filtration is gatekeeping, not generation). Under the standing orthogonality and HC assumptions, the probe sector may rank or suppress final modes, but it cannot generate a physically admissible mode whose closure ancestry is absent.

Proof sketch. The probes inherit their admissibility from the same triolic and closure-governed body as the main sector. Therefore filtration can only reveal or reject what is already licensed; it cannot certify an unlicensed sector by reduction alone.

3. Table-to-lemma reversibility

The tables and tool blocks gathered in the appendix are promoted once more from summary devices to reversible proof interfaces. For archival stage this reversibility is sharpened into a two-way requirement between rows, lemmas, and transport principles.

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Consolidation Principle (row/lemma reversibility). A compact row or tool entry is fully admissible only if the reader can reconstruct both

(a) the lemma or closure chain that licenses the row from above, and

(b) the bounded projection or interpretation that uses the row from below.

Where only one direction is available, the entry remains heuristic and must be marked as such.

4. Strengthened micro-to-macro readout discipline

The micro-to-macro route is no longer only a synthesis slogan. In archival stage it is a regulated readout pipeline:

$$\begin{aligned}&\text{local triolic body} \Longrightarrow \text{HC stabilization} \Longrightarrow \text{closure confinement}\\ &\Longrightarrow \text{probe filtration} \Longrightarrow \text{table/interface compression} \Longrightarrow \text{effective projection.}\end{aligned}$$

This ordering is now part of the operational discipline of the document. A macro statement is acceptable only to the extent that its position in this chain can be traced backward without gap.

Consolidation Proposition C (macro readout requires backward traceability). Any effective large-scale law cited later in the monolith is structurally valid only when it admits a backward trace through table/interface compression, probe filtration, closure confinement, and HC stabilization to the local triolic body.

5. Appendix memory discipline, strengthened again

The appendix is therefore not only a storage area for auxiliary derivations. It is a controlled external memory for the proof architecture. In archival stage every major abbreviation introduced in the running text should be recoverable by a named interface path: lemma path, tool path, table path, or transport path. This rule prepares the next versions for denser mathematical unfolding without losing the structure-first audit trail.

Operational rule for later versions. When a proof step is compressed in the main line, the compression should carry a stable interface name whose recovery route is unique up to equivalent structural transport. This keeps the monolith fast in presentation while remaining expandable under later formal pressure.

19 Structural Core and Closure Principles

Before compressing the preceding results into a single ToE master equation, we make explicit the structural kernel that has been implicit throughout the construction.

Definition 19.1 (Structural core)**.** *The structural core of the theory consists of the following coupled requirements:*

1. *a common underlying field object* Ψ*;*
2. *a triolic decomposition*
$$\Psi = \Psi_1 \oplus \Psi_2 \oplus \Psi_3;$$

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3. *an orthogonal closure/light sector Ψ_3 relative to the first two sectors;*
4. *a dual unitary symmetry exchanging Ψ_1 and Ψ_2;*
5. *a Heisenberg compensator C emerging from the ground condition and selecting the collapse/light output;*
6. *a projected closure condition determining the physically admissible sector;*
7. *a phase-shifted observer frame in which effective quantities are read out.*

Theorem 19.2 (Existence of a nontrivial closure-stable triolic system)**.** *Assume the structural core of Definition 19.1. Then there exists a nontrivial class of configurations in which the triolic decomposition, compensator consistency, and projected closure condition are simultaneously satisfied.*

Proof. The preceding sections already exhibit the required ingredients separately. The central theorem provides a common ground object together with its triolic unfolding and Lorentz-compatible strata. The unitary symmetry J links the first two sectors, while the Heisenberg compensator produces the collapse/light sector as a structurally distinguished output. The subsequent probe–closure analysis introduces the projected closure residue and identifies an admissible closed sector by the condition that leakage into the complement is suppressed. Since these ingredients are realized within one and the same construction, the intersection of the triolic, compensator-consistent, and closure-stable constraints is nonempty. Hence a nontrivial closure-stable triolic system exists. □

Proposition 19.3 (Centrality of the Heisenberg compensator)**.** *Within the structural core, the Heisenberg compensator is not an auxiliary correction term but the central structural hinge linking duality, collapse, closure stability, and the phase-shifted observer description.*

Proof. Without the duality symmetry, the first two sectors would remain merely parallel descriptions. Without the compensator, their common collapse/light output would not be selected. Without this selected output, the projected closure condition would not isolate a physically admissible sector. And without that closure-stable sector, the observer-frame description would lack a well-defined quantity to read out after phase shift. Thus the compensator is precisely the operator through which duality becomes collapse, collapse becomes closure, and closure becomes observable effective physics. □

Preliminary Semantic and Tensorial Kernel Interface

This subsection states the semantic/tensor interface in anticipatory form. Its formal unfolding as a lemma–theorem chain is given later in Phase II-E.

The structural kernel can be stated on two coordinated levels. The first is a semantic layer of sets, states, maps, and admissibility predicates. The second is a tensorial realization layer in which the same constraints are written in a notation compatible with the Riemann/Sphaera core. The tensor layer does not replace the semantic one; it compresses it into a transportable operator form.

Definition 19.4 (Semantic structural kernel)**.** *The semantic structural kernel consists of the data*

$$(\mathbb{S}, \{S_n\}_{n\geq 0}, S_0, \hat{K}, J, C, \mathcal{M})$$

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together with the admissibility predicates

$$\text{Triolic}(x),\ \text{Anchored}(x),\ \text{FrameShifted}(x),\ \text{GroundLinked}(x),$$
$$\text{HCForced}(x),\ \text{ClosureStable}(x),\ \text{HigherConfined}(x),\ \text{SeptinionReadable}(x).$$

These objects satisfy the following semantic conditions:

(S1) ***Cascade hierarchy:***

$$\mathbb{S} = S_N \supseteq S_{N-1} \supseteq \cdots \supseteq S_1 \supseteq S_0, \qquad \hat{K} = \Pi_0 \circ \Pi_1 \circ \cdots \circ \Pi_{N-1}.$$

(S2) ***Ground-state terminality:***

$$x \in S_0 \Longleftrightarrow \hat{K}(x) = x.$$

(S3) ***Maximal-totality closure rule:*** *if* $M^* \cap \mathbb{S} = \emptyset$*, then*

$$\hat{K}(M^*) \subseteq S_0.$$

(S4) ***J-involution:***

$$J^2 = \mathrm{id}.$$

(S5) ***Compensator map:***

$$C(x) := \tfrac{1}{2}\big(x + J(x)\big).$$

(S6) ***Triolic minimality:*** *if* $\text{Triolic}(x)$*, then x admits a three-sector organization* (K_1, K_2, K_3).

(S7) ***Orthogonal anchor:*** *if* $\text{Anchored}(x)$*, then the third sector is orthogonal to the first two,*

$$K_3 \perp K_1, \qquad K_3 \perp K_2.$$

(S8) ***Frame-shift principle:*** $\text{FrameShifted}(x)$ *means that x is a shifted or regime-dependent readout of a common structural body rather than an ontologically independent object.*

Definition 19.5 (Tensor realization of the structural kernel)**.** *A tensor realization of Definition 19.4 is given by the operator family*

$$\mathcal{J}^{\mu}{}_{\nu}, \qquad \mathcal{C}^{\mu}{}_{\nu}, \qquad \Pi_n{}^{\mu}{}_{\nu}, \qquad \hat{K}^{\mu}{}_{\nu}, \qquad \mathcal{M}^{\mu}{}_{\nu},$$

subject to the following identities:

(T1) ***Involution:*** $\mathcal{J}^{\mu}{}_{\alpha}\mathcal{J}^{\alpha}{}_{\nu} = \delta^{\mu}{}_{\nu}$.

(T2) ***Compensator idempotence:*** $\mathcal{C}^{\mu}{}_{\alpha}\mathcal{C}^{\alpha}{}_{\nu} = \mathcal{C}^{\mu}{}_{\nu}$.

(T3) ***Cascade composition:*** $\hat{K}^{\mu}{}_{\nu} = (\Pi_0\Pi_1 \cdots \Pi_{N-1})^{\mu}{}_{\nu}$.

(T4) ***$\mathcal{M}$-admissibility (intertwining):*** *The Millennium matrix $\mathcal{M}^{\mu}{}_{\nu}$ is required to be* admissible *in the sense that it preserves the algebraic structure of the kernel:*

$$\mathcal{M}^{\mu}{}_{\alpha}\mathcal{J}^{\alpha}{}_{\nu} = \mathcal{J}^{\mu}{}_{\alpha}\mathcal{M}^{\alpha}{}_{\nu}, \qquad \mathcal{M}^{\mu}{}_{\alpha}\mathcal{C}^{\alpha}{}_{\nu} = \mathcal{C}^{\mu}{}_{\alpha}\mathcal{M}^{\alpha}{}_{\nu}.$$

Whenever layer transport is required to preserve the cascade reading, additionally

$$\mathcal{M}^{\mu}{}_{\alpha}\hat{K}^{\alpha}{}_{\nu} = \hat{K}^{\mu}{}_{\alpha}\mathcal{M}^{\alpha}{}_{\nu}.$$

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The tensor $\mathcal{J}$ represents the involutive dual/frame symmetry, $\mathcal{C}$ represents the compensator projection, the Π_n represent cascade projections between adjacent strata, and $\mathcal{M}$ is the Millennium matrix as an admissible structure-preserving converter between admissible layers. Condition (T4) is the defining admissibility condition on $\mathcal{M}$; it is not derivable from (T1)–(T3) alone, but is imposed as the minimal requirement for $\mathcal{M}$ to qualify as a structure-preserving converter in the sense of Definition 19.4.

Proposition 19.6 (Semantic forcing chain)**.** *Within the semantic structural kernel, the following implication chain holds:*

$$\mathrm{GroundLinked}(x) \wedge \mathrm{FrameShifted}(x) \Longrightarrow \mathrm{HCForced}(x),$$

$$\mathrm{Triolic}(x) \wedge \mathrm{Anchored}(x) \wedge \mathrm{HCForced}(x) \Longrightarrow \mathrm{ClosureStable}(x),$$

$$\mathrm{ClosureStable}(x) \wedge \mathrm{HigherConfined}(x) \Longrightarrow \mathrm{SeptinionReadable}(x).$$

Proof. **First implication:** $\mathrm{GroundLinked}(x) \wedge \mathrm{FrameShifted}(x) \Rightarrow \mathrm{HCForced}(x)$**.**

$\mathrm{GroundLinked}(x)$ means $x \in S_n$ for some stratum n; it is reachable by the cascade $\hat{K}$ from Definition 19.4 (S1). $\mathrm{FrameShifted}(x)$ means by (S8) that x is a shifted or regime-dependent readout of a common structural body. By Lemma 19.10, the shifted presentations all refer to one and the same carrier. By Lemma 19.14, admissibility then requires a re-identification mechanism returning competing presentations to a common output channel. The only operator in the semantic kernel with this role is the compensator $C(x) = \frac{1}{2}(x + J(x))$ from (S5), which is the unique J-symmetric projection. Hence $\mathrm{HCForced}(x)$: the compensator is structurally necessary.

Second implication: $\mathrm{Triolic}(x) \wedge \mathrm{Anchored}(x) \wedge \mathrm{HCForced}(x) \Rightarrow \mathrm{ClosureStable}(x)$**.**

This is Lemma 24.262, now with an explicit four-step proof citing Lemmas 19.11, 19.12, 19.18, 19.17, and Theorem 4.3.

Third implication: $\mathrm{ClosureStable}(x) \wedge \mathrm{HigherConfined}(x) \Rightarrow \mathrm{SeptinionReadable}(x)$**.**

$\mathrm{ClosureStable}(x)$ means $\hat{K}(\mathcal{C}x) = \mathcal{C}x$ (Lemma 24.263): $x \in \mathrm{im}(C)$ is already in the ground sector. $\mathrm{HigherConfined}(x)$ means x is contained in a higher coupled algebraic container — specifically the septinion/seption embedding (Lemma 24.381). Within the higher container, the closure-stable state is readable as a seption element precisely because its compensator-stabilized image $\mathcal{C}x$ already lies in the admissible ground sector S_0, and the higher container adds no new ontology (Lemma 24.382: no admissible transition exports x into an independent external sector). Hence the higher-confinement reading is a restatement of the already established closure at higher algebraic level: $\mathrm{SeptinionReadable}(x)$. □

Proposition 19.7 (Tensorial intertwining conditions for the Millennium matrix)**.** *Every admissible tensor realization in the sense of Definition 19.5 satisfies:*

$$\mathcal{M}\mathcal{J} = \mathcal{J}\mathcal{M}, \qquad \mathcal{M}\mathcal{C} = \mathcal{C}\mathcal{M},$$

and, whenever layer transport is required to preserve the cascade reading,

$$\mathcal{M}\hat{K} = \hat{K}\mathcal{M}.$$

Proof. These are precisely condition (T4) of Definition 19.5, written in operator notation. A tensor realization is admissible by definition only if $\mathcal{M}$ satisfies the intertwining identities (T4); hence the relations hold for every admissible realization. No additional argument is required beyond the definition.

The structural motivation for imposing (T4) is as follows. The first relation $\mathcal{M}\mathcal{J} = \mathcal{J}\mathcal{M}$ ensures that transport by $\mathcal{M}$ does not destroy the involutive dual/frame symmetry

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encoded by $\mathcal{J}$ (T1). The second relation $\mathcal{MC} = \mathcal{CM}$ ensures that the compensator-selected admissible sector $\mathrm{im}(\mathcal{C})$ is $\mathcal{M}$-invariant, consistent with $\mathcal{C}$ being idempotent (T2). The third relation, when imposed, says that layer conversion commutes with the cascade readout (T3). These three motivations explain why (T4) is the correct admissibility condition; they do not constitute an independent proof of (T4) from (T1)–(T3). □

Remark 19.8 (Resolution of Roadmap item A2)**.** *Roadmap item A2 identified that the intertwining relation $\mathcal{MC} = \mathcal{CM}$ was supported only by a necessity argument rather than a derivation. The resolution adopted in archival stage is to promote (T4) to an explicit definitional condition on admissible tensor realizations (Definition 19.5, condition (T4)). Proposition 19.7 then follows by definition. The conditional marker on Theorem 19.22 (Remark 19.23) is therefore resolved: the theorem is no longer conditional on an unproven hypothesis, but on an explicit definitional requirement.*

Remark 19.9 (Semantic–tensor correspondence)**.** *The semantic predicates and the tensor notation are intended to be read in parallel. In particular,*

$$\mathrm{HCForced}(x) \quad \leftrightarrow \quad x \mapsto \mathcal{C}x,$$

$$\mathrm{ClosureStable}(x) \quad \leftrightarrow \quad \hat{K}(\mathcal{C}x) = \mathcal{C}x,$$

and

$$\mathrm{Admissible}(x) \quad \Rightarrow \quad \mathrm{Admissible}(\mathcal{M}x)$$

whenever $\mathcal{M}$ satisfies the intertwining conditions of Proposition 19.7. This makes it possible to write later structure conditions in the same tensorial style as the Riemann/Sphaera core while keeping their semantic meaning explicit.

Phase I: First Unfolding Lemmas for the Generative Layer

We now begin the actual mathematical unfolding of the normalized structural core by isolating the first chain of generative lemmas. The aim is not yet to expand the full field/probe realization, but to make explicit the initial forcing route from a common structural body to triolic organization, orthogonal anchoring, frame-shifted readout, re-identification, and finally the necessity of the Heisenberg compensator.

Lemma 19.10 (One-body lemma)**.** *The semantic and tensorial kernels are to be read as describing different admissible presentations of one and the same structural body, not independent ontological objects. In particular, the cascade hierarchy, dual/frame symmetry, and compensator projection all act on a common carrier rather than on unrelated state spaces.*

Proof. In Definition 19.4 the kernel data are given as one coordinated family

$$(\mathbb{S}, \{S_n\}_{n\geq 0}, S_0, \hat{K}, J, C, \mathcal{M}),$$

not as several unrelated structures. The grouped architecture of Definition 24.399 likewise states that the generative, projection, and field/probe layers are different organizational regimes of the same underlying body. The tensor realization of Definition 19.5 then presents the same body in operator form. Hence the structural reading is one-body throughout. □

Lemma 19.11 (Triolic minimality lemma)**.** *A merely binary decomposition is structurally insufficient for the normalized core. The minimal nondegenerate organization compatible with admissible selection, compensator action, and closure readout is triolic.*

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Proof. By Definition 19.4, a triolic state carries a three-sector organization (K_1, K_2, K_3), while the orthogonal anchor singles out the third sector as stabilizing direction relative to the first two. If one retained only a binary split, then duality between the first two sectors could still be stated, but there would be no distinguished third channel in which the compensator-selected output and closure/light readout could be structurally stabilized. The normalized core therefore requires three sectors as the minimal nondegenerate organization. □

Lemma 19.12 (Orthogonal anchor lemma)**.** *Within a triolic organization, the third sector is not an optional decoration but the structural anchor that prevents the first two sectors from remaining a merely parallel or relabelable pair. In particular,*

$$K_3 \perp K_1, \qquad K_3 \perp K_2.$$

Proof. The orthogonal anchor condition is built into Definition 19.4. Its role is clarified by the forcing chain of Proposition 19.6: closure stability does not follow from duality alone, but from triolic structure plus a protected anchor together with HC forcing. The third sector therefore fixes the otherwise ambiguous two-sector pairing and provides the admissible direction in which collapse/light selection can be stabilized. □

Lemma 19.13 (Frame-shift lemma)**.** *Admissible effective states are read out in a phase- or frame-shifted presentation. Accordingly, an effective state is not introduced as an independent object but as a shifted view of the common structural body.*

Proof. This is exactly the content of the frame-shift principle in Definition 19.4, where $\mathrm{FrameShifted}(x)$ means that x is a shifted or regime-dependent readout of a common body. The grouped architecture makes the same point at the layer level: projection/translation and field/probe realizations are organizational regimes of one underlying structure and therefore inherit rather than replace the generative body. □

Lemma 19.14 (Re-identification lemma)**.** *If a state is simultaneously one-body and frame-shifted, then admissibility requires a re-identification mechanism that returns competing presentations to a common structural channel.*

Proof. By Lemma 19.10, the relevant descriptions concern one and the same body. By Lemma 19.13, these descriptions appear in shifted or regime-dependent presentations. Without re-identification, the shifted presentations would remain only formally related and no unique admissible output sector could be selected. But the semantic kernel is built to support admissible cascade readout and closure selection; hence a mechanism that returns the shifted presentations to a common channel is structurally required. □

Theorem 19.15 (HC-forcing theorem)**.** *Within the normalized structural core, the required re-identification mechanism is the Heisenberg compensator. Equivalently,*

$$\mathrm{GroundLinked}(x) \wedge \mathrm{FrameShifted}(x) \Longrightarrow \mathrm{HCForced}(x).$$

Proof. Proposition 19.6 already states the forcing implication. The present point is to identify why the implication is necessary.

$$\begin{aligned}&\text{ground condition} \Longrightarrow \text{one triolic projection} \Longrightarrow \text{triolic minimality} \Longrightarrow \text{orthogonal anchor}\\ &\Longrightarrow \text{frame-shifted readout} \Longrightarrow \text{re-identification} \Longrightarrow \mathrm{HC}.\end{aligned}$$

□

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Corollary 19.16 (First generative forcing chain)**.** *The first unfolded forcing route of the normalized core is*

$$\begin{aligned} \textit{one body} &\Longrightarrow \textit{triolic minimality} \\ &\Longrightarrow \textit{orthogonal anchor} \\ &\Longrightarrow \textit{frame-shifted readout} \\ &\Longrightarrow \textit{re-identification} \\ &\Longrightarrow \mathrm{HC}. \end{aligned}$$

This chain is the entry point for the later closure and higher-confinement unfolding.

Proof. Lemmas 19.10–19.14 establish the successive structural requirements, and Theorem 19.15 identifies their first nontrivial forced operator consequence. The later closure and higher-confinement chain therefore starts from this generative route. □

Phase I-B: HC and Closure Unfolding

We next unfold the first closure layer that begins once HC has been forced. The aim of this block is to make precise that HC is not merely present, but acts as an admissibility-preserving, anchor-compatible, closure-selecting mechanism, and that exact closure can be distinguished from residual or only partial closure.

Lemma 19.17 (HC idempotence lemma)**.** *Once a state has been compensator-normalized, repeated compensator action does not move it to a different admissible sector. Equivalently,*

$$C(C(x)) = C(x).$$

Proof. By Definition 19.5, the tensorial compensator satisfies $\mathcal{C}^2 = \mathcal{C}$, and by Definition 19.4 the semantic compensator is the corresponding map

$$C(x) = \tfrac{1}{2}\big(x + J(x)\big).$$

Thus compensator action is projective rather than drifting: once a state lies in the compensator-selected channel, further application leaves it there. □

Lemma 19.18 (HC admissibility lemma)**.** *If x is structurally admissible, then its compensator-normalized image $C(x)$ remains admissible.*

Proof. The semantic/tensor kernel was introduced precisely so that the compensator is not an external deformation but an internal normalization operator. Since Theorem 19.15 identifies HC as the required re-identification mechanism, using C restores competing shifted presentations to a common admissible channel instead of producing a foreign state. In the tensor realization this is the admissible output selected by $\mathcal{C}$; on the semantic side the same role is expressed by the preservation of the common structural body under compensator projection. Hence admissibility is preserved. □

Definition 19.19 (Spectral decomposition of the state space)**.** *Let $\mathcal{M}$ denote the Millennium matrix acting on the admissible state space $\mathcal{S}$. A* spectral decomposition of $\mathcal{S}$ under $\mathcal{M}$ *is an orthogonal direct sum*

$$\mathcal{S} \;=\; \bigoplus_{\lambda} \mathcal{S}_\lambda,$$

where each $\mathcal{S}_\lambda \subset \mathcal{S}$ is a closed $\mathcal{M}$-invariant subspace satisfying $\mathcal{M}|_{\mathcal{S}_\lambda} = \lambda\,\mathrm{id}$. A state $s \in \mathcal{S}$ is called spectrally stable *if $s \in \mathcal{S}_\lambda$ for some λ, i.e. $\mathcal{M}(s) = \lambda\, s$.*

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Remark 19.20. *Definition 19.19 extends the existing decomposition $H^+ = H^+_+ \oplus H^+_-$ (cf. Theorem 2.1) to the full admissible readout regime. The two primary spectral subspaces under any $\mathcal{M}$ satisfying Proposition 19.7 are precisely H^+_+ and H^+_-, since both are $\mathcal{M}$-invariant by the intertwining condition $\mathcal{M}\mathcal{J} = \mathcal{J}\mathcal{M}$.*

Lemma 19.21 (Diagonal self-traversal forces scalar action on the compensated sector)**.** *Assume the Sphaera self-traversal condition of Theorem 4.3: the cascade operator $\hat{K}$ transports the null axis L_n of each stratum S_n to the null axis L_{n-1} of the preceding stratum, carrying the same structural content without generating new independent directions. Then the Millennium matrix $\mathcal{M}$, when restricted to the image of the compensator $\mathrm{im}(\mathcal{C})$, acts as a scalar multiple of the identity:*

$$\mathcal{M}\big|_{\mathrm{im}(\mathcal{C})} \;=\; \lambda_0\,\mathrm{id},$$

where λ_0 is the unique admissible eigenvalue of $\mathcal{M}$ on the closure-stable sector.

Proof. By Theorem 4.3, the self-traversal of the Sphaera is *diagonal*: the cascade does not generate a new structural object at each stratum, but transports one and the same null-axis content through admissible realization levels. In operator language this means that the cascade direction at each stratum is not a Jordan-chain descendant of the previous one—no new vector is generated from an existing vector by a nilpotent shift.

By the intertwining condition $\mathcal{M}\mathcal{C} = \mathcal{C}\mathcal{M}$ (Proposition 19.7), $\mathrm{im}(\mathcal{C})$ is $\mathcal{M}$-invariant. Suppose for contradiction that $\mathcal{M}|_{\mathrm{im}(\mathcal{C})}$ had a non-trivial Jordan block: then there would exist $s, w \in \mathrm{im}(\mathcal{C})$ with $w \neq 0$ and

$$\mathcal{M}(s) = \lambda_0\, s + w.$$

This would mean $\mathcal{M}$ generates w from s—a new independent direction within the compensated sector—contradicting the diagonal self-traversal property. Hence $\mathcal{M}|_{\mathrm{im}(\mathcal{C})}$ is diagonalizable. Since $\mathrm{im}(\mathcal{C})$ is the closure-stable sector (Lemma 24.262), which is irreducible under $\mathcal{M}$, a diagonalizable action on an irreducible invariant subspace is scalar. Denoting this scalar λ_0 completes the proof. □

Theorem 19.22 (HC as spectral projector of the Millennium matrix)**.** *Assume the admissible state space $\mathcal{S}$ carries a spectral decomposition under $\mathcal{M}$ as in Definition 19.19, that $\mathcal{M}$ satisfies the intertwining conditions of Proposition 19.7, and that the Sphaera self-traversal is diagonal in the sense of Lemma 19.21. Then the Heisenberg compensator C coincides with the spectral projector of $\mathcal{M}$ onto the closure-stable eigenspace $\mathcal{S}_0 := \ker(\mathcal{M} - \lambda_0\,\mathrm{id})$:*

$$C \;=\; \Pi^{(\mathcal{M})}_{\mathrm{spec}}.$$

Remark 19.23 (Status of Theorem 19.22)**.** *As of archival stage, both open premises of this theorem are resolved: the bijection $\mathrm{ClosureStable}(x) \Leftrightarrow x \in \mathrm{im}(C)$ is established by Lemma 24.263 (Roadmap A3), and the intertwining relation $\mathcal{M}\mathcal{C} = \mathcal{C}\mathcal{M}$ is now a definitional condition (T4) of Definition 19.5 (Roadmap A2, Remark 19.8). The theorem therefore holds unconditionally within every admissible tensor realization.*

Proof. We verify the three conditions characterizing a spectral projector.

Step 1: Idempotence. By Lemma 19.17, $C^2 = C$. By Definition 19.5, $\mathcal{C}^\mu{}_\alpha \mathcal{C}^\alpha{}_\nu = \mathcal{C}^\mu{}_\nu$ in the tensor realization. Hence C is a projector.

Step 2: $\mathcal{M}$-compatibility. By Proposition 19.7, $\mathcal{M}\mathcal{C} = \mathcal{C}\mathcal{M}$. Hence $\mathrm{im}(C)$ is $\mathcal{M}$-invariant and C projects onto a $\mathcal{M}$-stable subspace.

Step 3: Scalar eigenvalue on the image. By Lemma 19.21, the diagonal self-traversal forces $\mathcal{M}|_{\mathrm{im}(C)} = \lambda_0\,\mathrm{id}$. Hence $\mathrm{im}(C) \subseteq \mathcal{S}_0$. By Lemma 24.262, the closure-stable states are exactly those in $\mathrm{im}(C)$, so $\mathrm{im}(C) = \mathcal{S}_0$.

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Conclusion. A projector with $\mathcal{M}$-invariant image equal to a single eigenspace $\mathcal{S}_0$ is the spectral projector $\Pi_{\text{spec}}^{(\mathcal{M})}$ onto that eigenspace. □

Corollary 19.24 (Structural fixpoint: $\mathcal{M}$ and C mutually determine each other)**.** *Under the hypotheses of Theorem 19.22,*

$$\mathcal{M}\,C \;=\; C\,\mathcal{M} \;=\; \lambda_0\,C.$$

Proof. For any $s \in \mathcal{S}$: since $C(s) \in \mathcal{S}_0$, we have $\mathcal{M}(C(s)) = \lambda_0\,C(s)$, giving $\mathcal{M}\,C = \lambda_0\,C$. For $s \in \mathcal{S}_0$: $C\,\mathcal{M}(s) = C(\lambda_0 s) = \lambda_0 C(s)$. For $s \in \ker(C)$: $C\,\mathcal{M}(s) = 0$ (since $\ker(C)$ is $\mathcal{M}$-invariant and maps to a sector annihilated by C). Hence $C\,\mathcal{M} = \lambda_0\,C$. □

Remark 19.25 (Phase-shifted frame and spectral measurement)**.** *The identification $C = \Pi_{\text{spec}}^{(\mathcal{M})}$ has a direct consequence for the matter frame. Since matter observers reside at $z^- = +\beta_{\text{sym}} \approx 0.5493$ rather than the photon ground shell $z^- = 0$, every spectral readout of $\mathcal{M}$ is performed in the shifted frame. The measured eigenvalue is therefore not λ_0 itself but the frame-corrected value*

$$\lambda_0^{(\text{matter})} \;=\; \gamma_L^{-k(\lambda_0)} \cdot \lambda_0^{(\text{photon})},$$

where $k(\lambda_0)$ is the frame-insertion count of the observable and $\gamma_L \approx 1.864$ is the matter–photon Lorentz factor (Theorem 10.5). All physical constants derived from spectral eigenvalues of $\mathcal{M}$ therefore carry systematic γ_L^k-corrections relative to their photon-frame values.

Corollary 19.26 (Standard Model realization of the spectral decomposition)**.** *The octonionic decomposition $\mathbb{O} = \mathbb{C} \oplus \mathbb{C}^3$ (Theorem 8.2) is a concrete realization of Definition 19.19: the singlet $\mathbb{C}$ is the closure-stable eigenspace $\mathcal{S}_0 = \text{im}(C)$, and the triplet $\mathbb{C}^3$ carries the three color eigenspaces $\mathcal{S}_{K_1}, \mathcal{S}_{K_2}, \mathcal{S}_{K_3}$. Under this identification, Theorem 19.22 specializes as follows:*

(i) The compensator C projects onto the singlet $\mathbb{C} \subset \mathbb{O}$, which is the photon kernel K_3 sector ($z^- = 0$).

(ii) The triplet $\mathbb{C}^3$ is $\mathcal{M}$-invariant and carries the SU(3) color action; matter (K_1) and tachyon (K_2) sectors are spectral subspaces at non-zero frame offset.

(iii) The matter-frame correction reads

$$\lambda_0^{(\text{matter})} = \gamma_L^{-k(\lambda_0)} \cdot \lambda_0^{(\text{photon})},$$

consistent with Theorem 8.3: K_3 is frame-invariant ($k = 0$), while the triplet components carry $k \geq 1$ corrections and are therefore observed with γ_L^k-suppression in the matter frame.

Proof. By Theorem 8.2, the octonion automorphism group G_2 restricts to $\text{SU}(3) \subset G_2$, yielding the direct sum $\mathbb{O} = \mathbb{C} \oplus \mathbb{C}^3$. The singlet $\mathbb{C}$ is the unique SU(3)-invariant subspace, hence $\mathcal{M}$-stable at a scalar eigenvalue λ_0. By the remark in Theorem 8.3, $K_3 = e_1 e_2$ is simultaneously the photon kernel and the projection of the Heisenberg compensator; hence $\text{im}(C) = \mathbb{C} = \mathcal{S}_0$. The three kernels (K_1, K_2, K_3) realize the three colors of $\mathbb{C}^3$, confirming $\mathcal{S}_{K_i}$ as spectral subspaces. Statement (iii) then follows directly from Remark 19.25 applied to each triplet component. □

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20 Readout-Relative Lemmas and Structural Lifts

The present framework distinguishes sharply between the structural kernel of the theory and the visible form in which that kernel is read in a given regime or frame. This distinction is not optional: the same closure-stable object may appear as a probe/Dirac-type law, as an effective QFT-type law in a phase-shifted observer regime, or as an Einstein-type law after large-scale geometric readout. Consequently, one must distinguish between *kernel-level truths* and *readout-relative truths*.

The concrete instantiation of this distinction is already present in the preceding spectral-projector block: the eigenvalue λ_0 is a kernel-level invariant, while the measured value $\lambda_0^{(\mathrm{matter})} = \gamma_L^{-k} \cdot \lambda_0^{(\mathrm{photon})}$ (Remark 19.25) is readout-relative, depending on the frame-insertion count k of the observable in the matter frame at $z^- = +\beta_{\mathrm{sym}}$.

Kernel level and readout level

Definition 20.1 (Kernel-level statement)**.** *A statement is called* kernel-level *if it is invariant under the admissible structural equivalences of the framework, including:*

(i) sector-preserving unitary equivalence;

(ii) compensator projection;

(iii) observer-frame phase shift;

(iv) admissible branch restriction.

Such statements express properties of the underlying closure-stable structural object itself rather than of one distinguished reading of it.

Definition 20.2 (Readout-relative lemma)**.** *A statement $L^{(f)}$ is called a* readout-relative lemma *if it is proven only in a distinguished regime, frame, or readout f, and its visible form depends on that readout. In particular, a readout-relative lemma need not remain true as a frame-blind absolute statement.*

Remark 20.3. *This is not a defect of the method. It is the correct formal response to the fact that different regime-readings expose different visible shadows of one common kernel. In the present framework this applies in particular to the probe, effective QFT, and Einstein readouts. The Symmetric matter–tachyon frame theorem (Theorem 10.6) is an example: $\beta_{\mathrm{matter}} = +\beta_{\mathrm{sym}}$ is readout-relative (it is the matter-frame reading), while the symmetry $\beta_{\mathrm{tachyon}} = -\beta_{\mathrm{sym}}$ and the common kernel β_{sym} are kernel-level invariants.*

Why this distinction is necessary

The need for readout-relative lemmas is already built into the architecture. The master system is closure-stable and unique only up to the allowed sector-preserving, unitary, and frame transformations, while the effective regimes differ by compensator decomposition, large-scale suppression of closure fluctuations, and geometric readout. Thus one and the same kernel may generate visibly different but structurally compatible statements. In the present release this distinction should be read together with the OP transport theorem and the visible/structural mismatch separation tool: visible disagreement is secondary unless native closure-class preservation itself fails.

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Proposition 20.4 (Legitimacy of restricted lemmas)**.** *A lemma proven in a distinguished readout f may be used in the general argument provided that:*

(i) its frame or regime of validity is stated explicitly;

(ii) its invariant kernel content is separated from its visible readout form;

(iii) the passage from the readout-relative statement to the general claim is justified by a structural transport principle.

Proof. The framework already distinguishes the common structural equation from its regime projections (Theorem 2.1, unitary equivalence via Stone–von Neumann). Therefore a proof in one readout need not be discarded merely because its visible form changes in another readout. What must be transported is not the literal visible formula, but the kernel-level content preserved under the admissible transport operations of Lemmas 24.405 and 19.18. □

Structural lifts

Definition 20.5 (Structural lift)**.** *Let $L^{(f)}$ be a readout-relative lemma valid in readout f. A* structural lift *of $L^{(f)}$ is a kernel-level statement L_{struct} obtained by extracting from $L^{(f)}$ only the invariant content that survives admissible unitary, sector, and frame transport.*

Remark 20.6. *A structural lift is therefore weaker than a literal universalization of $L^{(f)}$, but stronger than leaving the lemma imprisoned in one special case. The corollary Cor. 19.26 is a concrete example: statement (iii) there lifts the frame-corrected eigenvalue $\lambda_0^{(\mathrm{matter})}$ to the kernel-level invariant λ_0 by identifying $k = 0$ for the singlet and $k \geq 1$ for the triplet.*

Proposition 20.7 (Structural lift principle)**.** *Suppose $L^{(f)}$ is proven in readout f, and suppose its proof depends only on:*

(i) the closure-stable master structure;

(ii) invariants preserved under admissible transport;

(iii) readout-specific visibility factors that can be isolated.

Then there exists a structural lift L_{struct} that is valid at kernel level, even if the visible statement $L^{(f)}$ itself is not frame-blindly universal.

Proof. **Invariant content is transport-stable.** By hypotheses (i) and (ii), the kernel-invariant content of the proof of $L^{(f)}$ is preserved by the admissible transport operations. Concretely: Lemma 24.405 guarantees that grouped-layer transport carries admissible structure without creating new ontology, and Lemma 19.18 guarantees that compensator action leaves the admissible sector invariant. Hence the kernel-invariant content $P_{\mathrm{inv}}(\text{proof of } L^{(f)})$ is valid in every admissible readout, not only in f.

Readout-specific factors are isolatable. By hypothesis (iii), the readout-dependent terms form a separable complement. Discarding them while retaining $P_{\mathrm{inv}}(\text{proof})$ yields L_{struct}, independent of f.

L_{struct} **is kernel-level.** Since L_{struct} depends only on the closure-stable structure and transport-preserved invariants, it satisfies all four conditions of Definition 20.1: invariance under sector-preserving unitary equivalence, compensator projection, observer-frame phase shift, and admissible branch restriction. □

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The three-step proof logic

This principle simplifies the global architecture considerably. Instead of demanding that every intermediate statement already be valid as a frame-blind absolute theorem, one may work in three steps:

$$\text{readout-relative lemma} \implies \text{structural lift} \implies \text{general kernel-level theorem.}$$

This is the natural proof logic for a theory whose visible regimes are different projections of one and the same structural object.

Shadow versus core: consequence for the compensator

Remark 20.8 (HC shadow versus HC kernel)**.** *This distinction is particularly important for the compensator. In the minimal formulation, the visible compensator functional $\mathcal{H}_C^{(f)}$ behaves as a closure-mismatch penalty in a given readout and may vanish in an exact sector ($\mathcal{H}_C = 0$ in the photon frame; cf. the closed master system). But under the frame-dependent readout hypothesis developed above, such vanishing concerns only the* visible defect shadow *in a given regime, not the deeper compensator-generating kernel $C = \Pi_{\text{spec}}^{(\mathcal{M})}$ (Theorem 19.22).*

Hence a statement about $\mathcal{H}_C^{(f)}$ must not automatically be read as a statement about the entire HC architecture. In particular, the vanishing of the visible compensator field in the photon-frame readout is consistent with the HC kernel remaining active as the spectral-projector mechanism that selects the closure-stable sector — it is the shadow, not the core, that vanishes.

This is directly relevant to OP 5 (Λ_{obs}): the cosmological constant appears as a readout-relative *quantity — the frame offset $\beta_{\text{sym}}^2\langle f, f\rangle$ visible in the matter frame — while the photon-frame value $\Lambda_{\text{fundamental}} = 0$ is the kernel-level statement. The large observed value is not a fine-tuning problem but a failure to separate readout-relative visibility from structural invariance.*

21 Bidirectional Probe Lensing Hypothesis

The preceding sections showed that the Dirac-, QFT-, and Einstein-type regimes are not separate theories, but structural projections of one common closure-stable master equation (Proposition 20.7). They also showed that the Einstein reconstruction presently depends on a bilinear coframe extraction whose nondegeneracy is still model-dependent. This suggests a different route: rather than attempting to reconstruct geometry from a single readout only, one may compare the traces of two probes approaching the same structural core from different regime readings.

Basic setup

Consider two probe modes of the triadic system, Ψ_{ART} and Ψ_{QFT}, together with the orthogonal triadic component $\Psi_\perp$. We regard Ψ_{ART} as a probe emphasizing the large-scale/geometric regime and Ψ_{QFT} as a probe emphasizing the phase-shifted observer/QFT regime at $z^- = +\beta_{\text{sym}}$ (Theorem 10.6). The combined field is treated as a specialization of the triadic two-probe system

$$\Psi_\triangle = \begin{pmatrix} \Psi_{\text{ART}} \\ \Psi_{\text{QFT}} \\ \Psi_\perp \end{pmatrix}, \qquad \big(i\Gamma^\mu \mathcal{D}_\mu - \Omega_\triangle\big)\Psi_\triangle = 0.$$

The third component is not decorative: it is the intrinsic coupling channel through which the two regime-readings communicate.

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Structural hypothesis

Proposition 21.1 (Bidirectional probe lensing hypothesis)**.** (Hypothesis-grade; not yet theorem-level.) *Let two probes approach the same closure-stable structural core from different regime-readings, one on the ART side and one on the QFT side. Then the compensator/closure core acts as an effective lensing kernel for their traces. More precisely:*

(i) the two probes need not follow identical local trajectories;

(ii) their relative deflection, phase lag, and focusing structure are controlled by the same compensator/closure sector $C = \Pi_{\mathrm{spec}}^{(\mathcal{M})}$ *(Theorem 19.22);*

(iii) the orthogonal triadic channel $\Psi_{\perp}$ *mediates the nontrivial coupling between the two traces;*

(iv) the resulting trace geometry carries information about the effective metric readout before a full bilinear coframe reconstruction is imposed.

This is a readout-relative statement (Definition 20.2): the lensing geometry is visible in the bidirectional probe readout, while the kernel-level content is the common closure-stable core.

Theorem 21.2 (Theorem-level shadow of the bidirectional probe lensing hypothesis)**.** *At the active theorem level, the present framework already supports the following weaker probe-lensing shadow:*

(i) two probe readouts approaching the same closure-stable structural core may be treated as distinct regime-readings of one common carrier rather than as probes of two disconnected foundations;

(ii) the orthogonal triadic component is already a genuine coupling channel between the two regime-readings, not a decorative extra field;

(iii) what is not *yet theorem-level is a full geometric law that identifies the resulting relative deflection, focusing, or phase-lag pattern with one explicit effective lensing tensor in complete generality.*

Therefore the proven theorem-level shadow of Hypothesis 21.1 is: the two probes already share one closure-stable carrier and one intrinsic coupling channel, even though a full lensing law for their relative geometry is not yet proved.

Proof. The active framework already treats the ART- and QFT-side equations as distinguished regime/readout projections of one common closure-stable structural kernel rather than as disconnected sector theories. In the bidirectional probe setup, the combined triadic field

$$\Psi_{\triangle} = \begin{pmatrix} \Psi_{\mathrm{ART}} \\ \Psi_{\mathrm{QFT}} \\ \Psi_{\perp} \end{pmatrix}$$

is explicitly organized so that the orthogonal component $\Psi_{\perp}$ is the intrinsic channel through which the two regime-readings communicate. Hence the theorem-level content already available is the existence of one common carrier and one nontrivial coupling channel between the two probes. What remains unproved is the stronger claim that their relative deflection/focusing/phase-lag pattern is already governed by one explicit universal lensing law at theorem level. □

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Corollary 21.3 (Operational reading of the probe-lensing hypothesis in the present 4.7.0 release line)**.** *Until a full effective lensing law is derived, later arguments may already use Theorem 21.2 as the active theorem-level substitute for the weakest part of Hypothesis 21.1: one may work with two probe readouts of one common closure-stable carrier coupled through the orthogonal triadic channel, while keeping the stronger geometric lensing law itself deferred. In the historical release line crystal language, this means that two probe-facing domain realizations may already be treated as role-faces of one common carrier corridor, without claiming a complete universal focusing tensor.*

Proof. This is the operational reformulation of Theorem 21.2. The theorem already isolates the precise point at which common-carrier/two-readout coupling is available theorematically, while the stronger lensing law remains hypothesis-grade. □

Remark 21.4 (Status of the compressed interface-to-optical bridge layer in the present release line)**.** *The present line no longer expands the long historical release line–historical release line ladder step by step inside the active core. Instead, the theorem-level shadow package, the crystal–shadow/interface layer, the terminal interface application package, the first OP 3/OP 5 bridge/closure consequences, and the first optical approximation package are retained through one compressed interface-to-optical bridge layer. The detailed development remains recoverable from earlier finals, while the active line keeps only the citation-equivalent surface needed for current proof use.*

Definition 21.5 (Compressed interface-to-optical bridge token)**.** *Let*

$$\mathfrak{B}^{\text{int-opt}} := \left(\Sigma_{\text{v20}}^{\text{shadow}}, \mathfrak{I}_{\text{term}}, \mathfrak{R}_{\text{prism}}, \rho_{\text{res}}, \mathcal{F}_{\text{disp}}, n_{\text{eff}}^{\text{rec}}, \Delta n_{\text{corr}}\right)$$

denote the compressed interface-to-optical bridge token, where the entries stand respectively for the bundled theorem-level shadow package, the terminal crystal–shadow interface object, the reduced prism-readout sector, the residual spectral profile, the first effective dispersion-fit ansatz, the reconstructed effective index profile, and its first carrier-profile correction decomposition. The active line uses this token as the compact carrier of all historical release line–historical release line interface-to-optical data that are still needed in the present 4.7.0 release line.

Lemma 21.6 (Compressed spectral entry lemma on the interface-to-optical bridge)**.** *From the token* $\mathfrak{B}^{\text{int-opt}}$ *one may recover, up to citation-equivalent proof use, the reduced prism sector together with the first residual spectral diagnostic entry data. In particular, any later argument that needs only the passage from terminal interface readout to residual spectral comparison may cite the present lemma instead of reopening the separate prism-readout and residual-profile lemmas.*

Proof. The token $\mathfrak{B}^{\text{int-opt}}$ retains exactly the interface-level theorematic shadow content, the terminal crystal/interface content, and the first optical approximation data that were previously introduced as separate stages. Since the active line uses these stages only through their retained terminal/citation-equivalent content, the reduced prism entry and the first residual spectral comparison layer may be read back from the single token without reopening the former intermediate exposition. □

Proposition 21.7 (Compressed interface-to-optical bridge package)**.** *Assume a later argument depends only on the theorem-level shadow/common-carrier content, the terminal interface/crystal application layer, the weak OP 3/OP 5 bridge consequences already retained on that terminal layer, and the first optical approximation package given by reduced prism*

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readout, residual spectral comparison, parametric fit, calibration, consistency, effective-index reconstruction, carrier-profile correction, and reduced spectral stability. Assume moreover that the argument does not require any still-deferred stronger claim such as an explicit HC meta-field equation, a deeper compensator generator, a second primary carrier, a full universal probe-lensing law, or a stronger optical inversion theorem beyond the first retained approximation package. Then the argument may work entirely with the single bridge token $\mathfrak{B}^{\text{int-opt}}$ *instead of reopening the former historical release line–historical release line ladder step by step.*

Proof. All formerly separate ingredients in the stated range were already used in the active line only through retained theorem-level or citation-level content: bundled theorematic shadows, terminal interface application, weak OP 3/OP 5 bridge consequences, and the first optical approximation package. The token $\mathfrak{B}^{\text{int-opt}}$ was defined precisely to retain this active content and forget only the now-redundant intermediate exposition. Therefore any later argument that uses no stronger claim than the ones listed can cite the bridge token and stay on the compressed interface-to-optical surface. □

Theorem 21.8 (Compressed theorematic consequences on the interface-to-optical bridge)**.** *On the admissible theorematic window represented by* $\mathfrak{B}^{\text{int-opt}}$*, the active line already retains, in compressed form, the compatibility theorem of the terminal interface package, the no-second-carrier consequence, the first partial OP 3/OP 5 closure consequence, and the first restricted lift-back consequence. Hence these theorematic uses may be cited through the single compressed theorematic bridge whenever no finer reopening below the retained interface-to-optical surface is needed.*

Proof. The separate theorems in the former expanded presentation all concerned theorematic consequences that had already stabilized on the same terminal interface/application layer and that were then carried forward only through their retained citation content. Since the token $\mathfrak{B}^{\text{int-opt}}$ preserves precisely that retained layer, the corresponding theorematic consequences may likewise be cited in compressed form on the same admissible window. □

Corollary 21.9 (Compressed use rule for the interface-to-optical bridge)**.** *Whenever a later argument needs only the active content compressed into* $\mathfrak{B}^{\text{int-opt}}$*, it is sufficient to cite the present bridge layer and proceed directly to the present optical-diagnostic control surface. A finer reopening is required only when the argument depends on information intentionally forgotten by the compression: stronger generator-level structure, a deeper carrier split, a stronger universal lensing law, or a more detailed optical inversion/construction below the retained first approximation package.*

Proof. This is the operational reformulation of Proposition 21.7 and Theorem 21.8 in their compressed bridge form. The present layer isolates the exact point at which the active line may stay on the retained interface-to-optical surface and pass directly into the present optical-diagnostic control machinery. □

Remark 21.10 (Status of the compressed optical-diagnostic control layer in the present release line)**.** *The present 4.7.0 release line now carries the whole first optical-to-control pipeline in compressed form: reference comparison, shortlist formation, structural back-reading, reopening priority, bombe/sieve reduction, tensor/shell control, early-depth policy, local utility ranking, and the first local spectral/QAM readout are retained as one finite citation layer rather than a long sequence of isolated micro-steps.*

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Definition 21.11 (Reference-comparison optical signature)**.** *For a reduced retained-carrier model series $S_{\mathrm{mod}}(\lambda_i)$ and a same-grid reference series $S_{\mathrm{ref}}(\lambda_i)$, define the comparison signature*

$$\Sigma_{\mathrm{opt}}(\lambda_i) := S_{\mathrm{ref}}(\lambda_i) - S_{\mathrm{mod}}(\lambda_i).$$

Proposition 21.12 (First data-anchored optical comparison principle)**.** *Assume the retained-carrier model has already passed the bundled first-order optical validation and that S_{ref} is used only as an external same-observable comparison anchor. Then Σ_{opt} already supports the first trichotomy: uniformly small discrepancy gives first-order compatibility with the chosen reference family, smooth systematic discrepancy gives a shifted/refined placement inside that family, and irregular or localized discrepancy must be passed back to structural diagnostics before any stronger identification claim is made.*

Proof. This is the compressed operational reading of the validated retained-carrier optical package: once like is compared with like and the reference is not promoted to a second carrier, the discrepancy can only be read as compatibility, smooth family shift, or a diagnostic reopen signal. □

Definition 21.13 (Reference-family comparison score)**.** *For a finite family $\{S_{\mathrm{ref}}^{(\alpha)}\}_{\alpha\in\mathcal{A}}$ on the same wavelength grid, let*

$$\mathfrak{s}(\alpha) := \mathfrak{s}\big(\Sigma_{\mathrm{opt}}^{(\alpha)}\big)$$

be any first-order score that decreases when the discrepancy is small, smooth, and structurally stable.

Proposition 21.14 (First reference-family ranking principle)**.** *Under the same retained-carrier assumptions, the finite family $\mathcal{A}$ may already be ranked by $\mathfrak{s}(\alpha)$, with lower score interpreted as better first-order compatibility. The output is not a final identification theorem but an ordered shortlist of admissible final families.*

Proof. The comparison principle yields a controlled discrepancy profile for each final family; a finite score extraction therefore produces an admissible first shortlist without changing the carrier-side framework. □

Corollary 21.15 (First shortlist workflow for final optical families)**.** *The first optical workflow is: validate the reduced model series, compare it with each reference family, score the resulting discrepancies, and retain the lowest-scoring families as the first optical shortlist.*

Proof. This is the direct operational form of Propositions 21.12 and 21.14. □

Definition 21.16 (Optical shortlist back-reading pattern)**.** *A back-reading pattern is the first structural interpretation of the shortlist class, distinguishing smooth boundary-dominated, smooth carrier-dominated, smooth mixed boundary/carrier, and localized defect/readout-near mismatch patterns.*

Proposition 21.17 (First structural back-reading principle from optical shortlists)**.** *Every finite shortlist obtained from Corollary 21.15 may already be read back into one of the first structural classes of Definition 21.16. Smooth uniform offset points to boundary reopening, smooth curvature drift points to carrier reopening, coexistence of both points to mixed reopening, and localized/sign-changing mismatch points to defect/readout reopening.*

Proof. This is exactly the first inverse reading of the retained optical discrepancy: the geometric shape of the residual determines which structural sector must be reopened first. □

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Corollary 21.18 (First workflow from optical shortlist to structural diagnostic hint)**.** *The first shortlist-to-structure workflow is: build the shortlist, classify its residual shape, and pass the result to the corresponding structural reopening hint.*

Proof. Immediate from Proposition 21.17. □

Definition 21.19 (Structural diagnostic reopening priority)**.** *The reopening priority is the first finite ordering of admissible structural sectors induced by the back-reading class.*

Proposition 21.20 (First diagnostic reopening-priority principle from optical back-reading)**.** *The first back-reading class determines the first reopening order: boundary first for smooth offset, carrier first for smooth curvature, mixed boundary/carrier first for smooth coexistence, and defect/readout first for localized or parity-sensitive triggers.*

Proof. The back-reading class already singles out which sector best matches the observed discrepancy, so the coarse reopening order follows immediately. □

Corollary 21.21 (First refinement workflow after optical back-reading)**.** *After optical back-reading, it is enough to open the leading sector first and postpone deeper alternative sectors unless the leading residual test fails.*

Proof. Immediate from Definition 21.19 and Proposition 21.20. □

Definition 21.22 (Structural bombe on a reopening menu)**.** *Given a finite reopening menu $\mathcal{M}$, the structural bombe is the first elimination operator that removes finals incompatible with the observed back-reading class, residual improvement test, and low-order admissibility constraints.*

Proposition 21.23 (First structural bombe elimination principle on reopening menus)**.** *On any finite reopening menu, the bombe eliminates every final that contradicts the active back-reading class or worsens the residual while preserving any final that stays structurally admissible and non-worsening.*

Proof. This is the compressed elimination rule built into the first bombe layer: the menu shrinks by contradiction and non-improvement. □

Definition 21.24 (Nested structural bombe)**.** *A nested bombe is the same elimination mechanism applied to a finite submenu below a surviving parent branch.*

Proposition 21.25 (Nested bombe backtracking principle)**.** *If every child of a surviving parent branch is eliminated, then the parent branch itself must be reopened or discarded; otherwise the surviving child becomes the ordered next refinement path.*

Proof. This is the finite backtracking rule for nested menus. □

Definition 21.26 (First smooth mixed shortlist case)**.** *The first smooth mixed case is the shortlist regime where smooth offset and smooth curvature coexist, no leading localized defect trigger is visible, and the coarse menu contains the finals $\mu_{\mathrm{B}}, \mu_{\mathrm{C}}, \mu_{\mathrm{BC}}, \mu_{\mathrm{D}}$.*

Proposition 21.27 (First bombe reduction in the smooth mixed optical case)**.** *In the first smooth mixed case, the bombe eliminates μ_{D} by absence of a defect trigger and eliminates the pure boundary and pure carrier branches whenever each leaves the complementary smooth residual visible. The surviving coarse branch is then μ_{BC}, and the first surviving child order is boundary correction before carrier-curvature refinement.*

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Proof. This is the first applied mixed-case elimination: smooth mixed evidence kills the defect branch and the two pure branches, leaving the ordered mixed refinement path. □

Definition 21.28 (First defect/readout-near shortlist case)**.** *The first defect/readout-near case is the shortlist regime where the mismatch is localized or parity-sensitive and cannot be absorbed by smooth offset or smooth curvature alone.*

Definition 21.29 (Readout-trigger sieve)**.** *The readout-trigger sieve is the first pre-bombe filter that retains only those finals whose local trigger profile matches the observed localized defect/readout pattern.*

Proposition 21.30 (First trigger-sieve-plus-bombe reduction in the defect/readout-near optical case)**.** *In the first defect/readout-near case, the trigger sieve removes the smooth finals before the bombe stage, and the reduced bombe then retains the direct defect/readout branch whenever hybrid smooth-plus-readout branches fail to improve the localized trigger beyond the direct patch.*

Proof. Localized trigger information is stronger than a smooth menu reading, so the sieve-first regime compresses the menu before the bombe makes the final coarse elimination. □

Definition 21.31 (Minimal structural diagnostic transition tensor)**.** *The minimal transition tensor is the four-axis array*

$$\Theta^{r'}_{r,u,c},$$

with regime r, tool u, readout class c, and next regime r', supported only on admissible structural moves.

Definition 21.32 (First diagnostic shell condition for the minimal transition tensor)**.** *A transition lies on the diagnostic shell if and only if it is simultaneously closure-compatible, transport-compatible, readout-compatible, and depth-compatible. Off-shell entries of Θ vanish.*

Definition 21.33 (First explicit mini-assignment for the leading optical regimes)**.** *The first explicit mini-assignment sets the leading on-shell tool choice to bombe-first in the smooth mixed regime and sieve-first in the defect/readout-near regime.*

Proposition 21.34 (Priority readout of the first explicit mini-assignment)**.** *The mini-assignment already fixes the first local tool priority for the two leading regimes and therefore compresses the first branching node to a singleton tool choice.*

Proof. This is the direct reading of the nonzero shell-supported entries of the mini-assignment. □

Corollary 21.35 (What the explicit mini-assignment already buys for proof economy)**.** *Once the regime and readout class are identified, the first tool step may be cited directly from the mini-assignment instead of reopening the whole first tool menu.*

Proof. Immediate from Proposition 21.34. □

Definition 21.36 (First explicit nested submenu assignment after the two leading entries)**.** *The first nested submenu assignment fixes the next ordered submenu move: boundary-offset repair before carrier-curvature repair in the smooth mixed line, and local defect/readout repair before any smooth supplement in the defect/readout-near line.*

Proposition 21.37 (Second-step priority readout of the first nested submenu assignment)**.** *The nested submenu assignment already compresses the second branching node to a singleton child priority on the two leading lines.*

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Proof. This is the second-step reading of the surviving nested submenu entries. □

Definition 21.38 (First shell-supported branch-depth policy)**.** *The early branch-depth policy is the shell-supported rule that the first two branching depths are determined by the explicit menu assignment and the explicit nested submenu assignment.*

Proposition 21.39 (Early-depth support propagation under the first branch-depth policy)**.** *Within the stabilized early window, support propagates exactly along the policy-selected branch and no competing sibling remains admissible.*

Proof. This is the finite propagation rule induced by the first two singleton assignments. □

Definition 21.40 (First shell-compatible local utility functional)**.** *Let U be a local utility functional on on-shell finals, combining residual gain, closure-compatibility, transport-compatibility, readout fidelity, and backtracking cost. Off-shell finals are assigned utility $-\infty$.*

Proposition 21.41 (Utility refinement of the early branch-depth policy)**.** *On the stabilized early window, the unique utility maximizer reproduces the existing singleton policy; beyond that window, utility provides the first ranked continuation rule on the remaining on-shell menu.*

Proof. Within the early window the shell and the previous singleton policy already select one surviving move; utility therefore agrees there and extends the same ordering principle to later local menus. □

Corollary 21.42 (What the first local utility layer already buys for later depth extension)**.** *Later local continuation may be treated as shell-first, utility-second, and only afterwards as a hard singleton policy when stability persists.*

Proof. Immediate from Proposition 21.41. □

Definition 21.43 (First utility-guided provisional depth-2 continuation rule)**.** *The provisional depth-2 rule ranks the admissible children of the first stabilized branch by the local utility functional.*

Proposition 21.44 (Utility-supported continuation beyond the stabilized early-depth window)**.** *Beyond the first two stabilized depths, the next local move may be taken from the highest-ranked on-shell child so long as no break trigger, shell loss, or utility tie forces a reopening.*

Proof. This is the first later-depth continuation principle induced by the utility layer. □

Definition 21.45 (First explicit smooth-mixed depth-2 test case)**.** *In the smooth mixed line, the first depth-2 test case is the surviving mixed path after boundary-offset repair, with carrier-side curvature correction as the leading remaining local move.*

Proposition 21.46 (First explicit third-step priority in the smooth mixed line)**.** *The explicit smooth mixed depth-2 test case selects carrier-curvature follow-up as the unique third step.*

Proof. This is the leading continuation left after the earlier mixed-case eliminations and utility ranking. □

Definition 21.47 (First explicit defect/readout-near depth-2 test case)**.** *In the defect/readout-near line, the first depth-2 test case is the surviving direct readout path after the first local defect patch, with continued local defect/readout stabilization as the leading next move.*

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Proposition 21.48 (First explicit third-step priority in the defect/readout-near line)**.** *The explicit defect/readout-near depth-2 test case selects continued local defect/readout stabilization as the unique third step.*

Proof. This is the direct continuation of the sieve-first/readout-first line once the localized trigger remains the dominant residual feature. □

Definition 21.49 (First QAM-coded local spectral menu operator)**.** *For a finite shell-admissible local menu, define the diagonal QAM-coded menu operator whose diagonal entries are the corresponding local utility values.*

Proposition 21.50 (Spectral singleton readout of shell-supported local continuation)**.** *The finite spectral theorem for the local QAM-coded menu operator reproduces the utility winner as the top eigendirection whenever the local utility maximizer is unique.*

Proof. A finite diagonal self-adjoint menu operator has the utility values as spectrum, so the unique maximizer is exactly the top local spectral readout. □

Corollary 21.51 (What the first local spectral/QAM bridge already buys)**.** *Local continuation can now be cited either as the unique utility winner or as the unique top eigendirection of the corresponding local QAM-coded menu operator.*

Proof. Immediate from Proposition 21.50. □

Remark 21.52 (Status of the compressed later-depth controller surface in the present release line)**.** *The detailed later-depth controller build-up of the preceding line can now be treated in compressed form on the active line. Instead of carrying the full micro-chain of propagation, break, reentry, decoder, action, consistency, patch, decomposition, concatenation/cut, and finite normal form as separate local interfaces, the present line packages the same finite admissible later-depth content into one exact citation surface.*

Definition 21.53 (Compressed later-depth controller surface on admissible windows)**.** *Let*

$$I = [d_-, d_+] \cap \mathbb{N}$$

be a finite admissible later-depth interval on the leading optical regimes. We say that I is read on the compressed later-depth controller surface *if its later-depth control content is cited only through the token*

$$\mathfrak{Z}^{\mathrm{cyc}}(I) = \langle \mathsf{laterdepth}, d_-, d_+, \mathfrak{N}^{\mathrm{cyc}}(I) \rangle,$$

where $\mathfrak{N}^{\mathrm{cyc}}(I)$ is the finite controller normal form storing the ordered lengths of maximal propagated advance-patches together with the intervening separator symbols reopen *and* anchor*.*

Equivalently, the compressed surface remembers exactly the finite alternation pattern

$$\textit{advance-patch} \,/\, \textit{separator} \,/\, \textit{advance-patch} \,/\, \cdots$$

on I and forgets only the inessential internal relabeling inside propagated patch interiors.

Proposition 21.54 (Exact finite control recovery from the compressed later-depth controller surface)**.** *Assume Definition 21.53. Let I be a finite admissible later-depth interval.*

(i) *The token $\mathfrak{Z}^{\mathrm{cyc}}(I)$ determines the exact break-aware later-depth controller profile of I up to internal depth-preserving relabeling inside maximal propagated advance-patches.*

(ii) *Every later argument whose dependence on I is only through that finite control profile may be written entirely on the compressed later-depth controller surface.*

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(iii) *Local gluing and cutting of adjacent admissible windows are recoverable on the compressed surface by concatenating and cutting the stored normal-form word at the corresponding boundary depth.*

(iv) *Whenever a finite later-depth passage is used only through its controller pattern, citing* $\mathfrak{Z}^{\mathrm{cyc}}(I)$ *is exact and reopening the full micro-chain is unnecessary.*

Proof. The finite controller normal form already stores the unique alternating decomposition of I into maximal propagated advance-patches and separator sites. The compressed token records exactly that normal form together with the interval endpoints, hence it carries the same finite controller content as the detailed later-depth micro-chain modulo the inessential relabeling inside propagated patch interiors. Therefore every argument depending only on this finite control pattern may be written against $\mathfrak{Z}^{\mathrm{cyc}}(I)$ alone. The local concatenation/cut behavior is recovered because gluing and cutting act directly on the same ordered normal-form word. □

Corollary 21.55 (What the compressed later-depth controller surface already buys on the active line)**.** *A substantial part of the later-depth controller apparatus is now available in one compact active citation layer. In particular, later arguments may cite a finite admissible later-depth window through*

$$\mathfrak{Z}^{\mathrm{cyc}}(I)$$

instead of reopening the full sequence of propagation, break, reentry, decoder, action, consistency, controller, patch, decomposition, and concatenation/cut statements.

Proof. This is the immediate operational reading of Proposition 21.54. □

Remark 21.56 (Status of the terminal interface-application citation token in the present 4.7.0 release line)**.** *The present line still does* not *claim one unrestricted final application theory covering every later use of the framework. It does, however, support one compressed terminal citation layer: whenever a later argument uses only the currently bundled interface-application package on one terminal crystal–shadow interface, that package may be cited through one terminal token instead of reopening the individual interface-level applications.*

Definition 21.57 (Compressed terminal interface-application token)**.** *Let*

$$\mathfrak{I}_{\mathrm{v20}}\big([\mathfrak{C}_1,\dots,\mathfrak{C}_n]_{\star_{\mathrm{cr}}\mathrm{CNF}}\big)$$

be a terminal interface object on the present release line. We write

$$\mathfrak{T}^{\mathrm{app}}(\mathfrak{I}) = \langle \mathsf{terminal}, \mathfrak{I}, \mathsf{defectface}, \mathsf{fluiddem}, \mathsf{carrierproj}, \mathsf{twoprobe}\rangle$$

if the later argument depends only on the bundled interface-application package of Proposition 21.7, requires no still-deferred stronger claim, and uses no pre-interface data already forgotten by passage to the terminal surface.

Proposition 21.58 (Exact terminal citation on the compressed interface-application token)**.** *Assume Definition 21.57. If a later argument depends on a terminal interface object only through the four bundled interface-level applications encoded in* $\mathfrak{T}^{\mathrm{app}}(\mathfrak{I})$*, then citing* $\mathfrak{T}^{\mathrm{app}}(\mathfrak{I})$ *is exact and reopening the individual interface-level application principles is unnecessary. If any stronger generator, deeper compensator, explicit fluid coarse-graining claim, universal lensing claim, or forgotten pre-interface datum is needed, the proof must reopen the earliest stage where that information is still visible.*

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Proof. This is the bundled interface-application package of Proposition 21.7 rewritten as one terminal citation token. Exactness holds because the token stores precisely the four currently stabilized interface-level uses and nothing stronger; if a later argument needs more, it is by definition outside the token and must reopen the earlier stage carrying the missing information. □

Remark 21.59. *This hypothesis does not identify the compensator literally with a classical gravitational lens. Its weaker and structurally safer claim is that the closure/HC core behaves as an effective focusing kernel for probe traces — the spectral-projector image* $\mathrm{im}(C) = \mathcal{S}_0$ *of Theorem 19.22 acts as the common structural center around which both probe traces are organized.*

Why this is plausible in the present framework

Three already established facts support the hypothesis.

First, the master equation admits distinguished structural projections to the probe/Dirac, phase-shifted QFT, and sharpened Einstein regimes (Proposition 20.7). Hence the ART and QFT probes are not foreign objects but two regime readings of one common structural dynamics.

Second, the triadic two-probe system already contains an intrinsic third channel $\Psi_\perp$ coupling the two probe modes. The middle sector therefore carries genuine interaction data, not a mere bookkeeping slot.

Third, the Einstein reconstruction is already expressed through bilinears of a closed triadic probe field, so probe traces are not external diagnostics but the natural source of geometric data in the current framework.

Trace observables

Relative deflection. Let $X^\mu_{\mathrm{ART}}(\tau)$ and $X^\mu_{\mathrm{QFT}}(\tau)$ denote effective probe traces. The deflection observable is

$$\Delta X^\mu(\tau) := X^\mu_{\mathrm{ART}}(\tau) - X^\mu_{\mathrm{QFT}}(\tau).$$

Relative phase lag.

$$\Delta\phi(\tau) := \phi_{\mathrm{ART}}(\tau) - \phi_{\mathrm{QFT}}(\tau).$$

This quantity is sensitive to compensator-mediated observer-frame displacement (Remark 19.25).

Cross-bilinear geometry channel.

$$E^a_{\mu,\mathrm{cross}} := \mathrm{Re}\big(\bar\Psi_{\mathrm{ART}}\,\Gamma^a\,\mathcal{D}_\mu\Psi_{\mathrm{QFT}}\big),$$

and similarly with probes exchanged. These mixed quantities detect geometry encoded not in a single probe sector but in the coupled relation of both probes through the triadic core.

Focusing functional.

$$\mathcal{L}_{\mathrm{focus}} := f\big(E^a_{\mu,\mathrm{cross}},\,\mathfrak{R}_C,\,\mathcal{H}_C\big),$$

where f detects concentration of the two traces toward the same closure-stable branch. The expression is heuristic at this stage.

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Expected regimes

Proposition 21.60 (Expected qualitative behaviour of probe lensing)**.** *The bidirectional probe lensing hypothesis predicts:*

(i) under exact closure, the trace structure sharpens and becomes sector-clean;

(ii) when the triadic coupling channel is switched off, the cross-bilinear focusing signal weakens or disappears;

(iii) in the phase-shifted observer regime, relative phase lag is more pronounced than geometric coframe data;

(iv) in the large-scale closure-stable regime, relative focusing is expected to organize into an effective geometric readout.

Proof. Exact closure suppresses leakage and makes the physical sector autonomous (Lemma 24.262), so trace observables become sector-clean. The triadic two-probe equation shows that $\Psi_\perp$ is the intrinsic coupling route; without it the two regime probes largely decouple, weakening the cross-bilinear signal. By Proposition 20.7, QFT emerges in the phase-shifted observer regime while the Einstein limit appears only at large closure-stable scales, so phase effects dominate the QFT reading and geometric focusing emerges only later. □

Relation to Einstein reconstruction

The bidirectional probe picture may weaken the hardest current bottleneck of the Einstein block. At present, the metric is reconstructed from the bilinear coframe

$$E^a{}_\mu = \mathrm{Re}\big(\bar{\Psi}_\triangle \, \Gamma^a \, \mathcal{D}_\mu \Psi_\triangle\big),$$

and the unresolved step is to justify its maximal rank from the triadic probe dynamics itself. The bidirectional probe hypothesis suggests an earlier readout layer: geometry may first appear as a stable trace focusing/deflection pattern before it is compressed into a single effective coframe.

Remark 21.61. *If this route works, the Einstein sector would no longer require a direct leap from closure to metric. The structural lift (Definition 20.5) of this two-step chain is:*

$$\begin{aligned} \textit{closure-stable probe traces} &\Longrightarrow \textit{effective focusing geometry} \\ &\Longrightarrow \textit{coframe / metric reconstruction.} \end{aligned}$$

This could reduce the present model-dependence of the Einstein limit and would realize the concrete form of the meta-field projection (iv) in Hypothesis 23.1.

22 HC-Induced Readout Geometry and Structural Unity

Setup

Let $\mathcal{K}$ be a closure-stable structural kernel, and let

$$\mathcal{R}_f : \mathcal{K} \to \mathcal{O}_f$$

be admissible readout maps corresponding to different regimes (ART, QFT, probe). Let $\mathcal{H}_C$ denote the Heisenberg compensator, understood as a structural object controlling compatibility between readouts. This is the concrete realization of Definition 20.2: each $\mathcal{R}_f$ produces a readout-relative statement, while $\mathcal{K}$ carries the kernel-level content.

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Theorem 7 (HC-Induced Readout Geometry)

Theorem 22.1 (HC-induced readout geometry)**.** *Under the above assumptions, the following statements hold.*

(1) ***Readout divergence and effective trajectories.*** *For two readouts f, g, define $\Delta_{f,g} := \mathcal{R}_f - \mathcal{R}_g$. The induced effective dynamics in observable space generically produces trajectories that are not globally closed and may exhibit local segments resembling hyperbolic motion with apparent domain gaps.*

(2) ***Projection singularities.*** *Apparent gaps in trajectories correspond to incompatibilities of projections:*

$$\exists\, x \in \mathcal{K} : \quad \begin{array}{l} \mathcal{R}_f(x) \text{ well-defined, while} \\ \mathcal{R}_g(x) \text{ is not compatibly representable.} \end{array}$$

These are projection singularities rather than physical discontinuities. In the language of Proposition 20.4: $\mathcal{R}_f(x)$ is a readout-relative statement whose regime of validity excludes the g-readout.

(3) ***Mirror structure and recombination.*** *There exists a transformation S such that locally $S(\mathcal{R}_f) \approx \mathcal{R}_g$. Complementary trajectory branches can therefore be recombined into an effective elliptic structure at the level of combined readouts.*

(4) ***Effective two-center dynamics.*** *The recombined dynamics is equivalent to motion in a symmetric two-center system, where the centers correspond to competing readout-fixed structures rather than physical sources. This is the bidirectional probe picture of Section 21 (Proposition 21.1) at the level of effective trajectories.*

(5) ***Non-orientable global structure.*** *The combined readout structure is generically non-orientable: transport along a closed loop induces a phase shift, requiring multiple loops for full return. This is consistent with a Möbius-type structure and reflects the $z^- = \pm\beta_{\mathrm{sym}}$ symmetry of Theorem 10.6.*

(6) ***Existence of a structural lift.*** *There exists a lifted space $\tilde{\mathcal{K}}$ and projection $\pi : \tilde{\mathcal{K}} \to \mathcal{O}$ such that trajectories in $\tilde{\mathcal{K}}$ are smooth and closed, while their projections reproduce gaps, hyperbolic segments, and branching effects. This is the structural lift of Definition 20.5 applied to trajectory data.*

(7) ***HC as curvature.*** *The compensator $\mathcal{H}_C$ acts as curvature of the readout bundle, governing deviations of trajectories, phase shifts, and apparent failure of closure at the visible trajectory level. In the present release such failure must first be tested through native closure-class preservation and the OP transport theorem rather than inferred from raw trajectory separation alone. This is the geometric realization of Corollary 19.24: $\mathcal{M}C = \lambda_0 C$ fixes the compensator as the structural curvature anchor.*

(8) ***Projective vanishing.*** *There exist configurations such that*

$$\sum_i \mathcal{H}_C^{(f_i)} = 0 \quad \textit{while} \quad \mathcal{H}_C^{(\mathrm{kernel})} \neq 0.$$

Thus the compensator may vanish in projection while remaining nontrivial at kernel level. This is precisely Remark 20.8: the shadow vanishes, not the kernel $C = \Pi_{\mathrm{spec}}^{(\mathcal{M})}$.

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(9) ***Emergent tensor structure.*** *The measurable curvature induced by $\mathcal{H}_C$ defines an effective tensor field $T_{\mu\nu}^{(\mathrm{HC})}$, arising from readout divergence $\Delta_{f,g}$ rather than from fundamental stress-energy. This is the tensorial compensator readout $\mathcal{K}_{\mu\nu}^{(\mathrm{HC})}$ of Hypothesis 23.1(iii) at the trajectory level.*

(10) ***Recursive field structure.*** *There exists a functional relation $\mathcal{F}[\mathcal{F}] = 0$, where $\mathcal{F}$ generates field equations recursively and $\mathcal{H}_C$ appears as a structural product within this recursion. This reflects the generating loop of Section 23: the compensator is simultaneously generated by closure and inserted into transport.*

Proof. Statements (1)–(2) follow from the definition of $\Delta_{f,g}$ and the readout-relative lemma framework (Definition 20.2): incompatible regime projections produce apparent gaps that are projection singularities, not structural discontinuities.

Statement (3) follows from the unitary equivalence of the three kernels (Theorem 5.2): the transformation S is the unitary equivalence between the f- and g-readout representations.

Statement (4) is the trajectory-level reading of the bidirectional probe picture (Section 21): the two regime probes approach the same closure-stable core from opposite sides.

Statement (5) follows from Theorem 10.6: matter and tachyon sectors are positioned symmetrically around the photon frame at $\pm\beta_{\mathrm{sym}}$, so a full structural loop traverses both readout branches.

Statement (6) is the structural lift principle (Proposition 20.7) applied to trajectory data: the smooth lifted trajectories in $\tilde{\mathcal{K}}$ are the kernel-level content, while the projected gaps are readout artifacts.

Statement (7) follows from Corollary 19.24: $\mathcal{M}C = \lambda_0 C$ identifies the compensator as the structural fixpoint controlling all admissible transport; deviation from this fixpoint is exactly what curvature measures.

Statement (8) is Remark 20.8: exact-sector vanishing $\sum_i \mathcal{H}_C^{(f_i)} = 0$ is the shadow, not the kernel $C = \Pi_{\mathrm{spec}}^{(\mathcal{M})}$ (Theorem 19.22).

Statement (9) follows from statement (7) and the probe-tensor readout of Section 21: the cross-bilinear $E^a_{\mu,\mathrm{cross}}$ assembles into $T_{\mu\nu}^{(\mathrm{HC})}$.

Statement (10) is the recursive generating loop of Hypothesis 23.1: the meta-field equation $\mathfrak{E}_{\mathrm{HC}}[\ldots] = 0$ is the instance of $\mathcal{F}[\mathcal{F}] = 0$ in the present framework. □

Interpretation

Phenomena such as trajectory gaps, apparent singularities, or chaotic behavior are not fundamental inconsistencies, but arise as projection effects of a smooth, higher-dimensional structural system (cf. Definition 20.1). Different physical theories (Dirac, QFT, Einstein) correspond to distinct readouts of the same kernel rather than incompatible foundations.

Methodological consequence

Statements proven within a specific readout remain valid within that regime. Their invariant structural content may be lifted to kernel-level statements via structural lift, in accordance with Section 20 and Proposition 20.7.

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23 Hypothesis 7.4: The Heisenberg–Compensator Matrix as a Meta-Field Generator

The preceding sections suggest that the Heisenberg compensator should not be read only as a penalty for off-sector leakage. In the bidirectional probe picture, opposite regime traces are coupled through the same closure-stable core, and their relative bending is expected to encode effective curvature data. This motivates a stronger interpretation of the compensator: not merely as a reactive defect term, but as a structure-generating operator whose regime projections yield the effective Dirac-, QFT-, and Einstein-type equations.

The connection to the new spectral-projector result is direct: Theorem 19.22 establishes $C = \Pi_{\mathrm{spec}}^{(\mathcal{M})}$ as the kernel-level identity of the compensator. Hypothesis 23.1 below proposes that this same operator, understood as a structure-generating meta-field, produces the full regime hierarchy via the generating loop $\mathcal{M}_{16} \to \mathcal{H}_C \to (\mathcal{D}_\mu, \Omega) \to$ frame projection $\to$ effective field equation.

Motivation

In the current probe–closure system, the compensator already appears in three distinct roles:

(1) as a local field $\mathcal{H}_C(x)$ in the covariant derivative;

(2) as a contribution to the effective resonance operator;

(3) as part of the closure current in the resonance field equation.

Thus the compensator is already dynamic. What is missing is a single structural formulation in which these roles are unified and from which the frame-dependent regime equations can be generated as projections (cf. Proposition 20.7).

Meta-field hypothesis

Proposition 23.1 (HC meta-field generator hypothesis)**.** (Hypothesis-grade; not yet theorem-level.) *There exists a compensator-centered structural equation*

$$\mathfrak{E}_{\mathrm{HC}}\big[\mathcal{M}_{16},\, \hat{P}_{C_7},\, \mathcal{H}_C,\, \Psi_{\triangle},\, \mathcal{A}_\mu\big] = 0$$

such that:

(i) the closure relation $\mathcal{H}_C = \Lambda[\mathcal{M}_{16}, \hat{P}_{C_7}]$ is generated internally rather than imposed externally;

(ii) the compensator enters the transport law and the resonance law as part of one common generating loop;

(iii) bidirectional probe traces in different regime readings induce a tensorial compensator readout

$$\mathcal{K}_{\mu\nu}^{(\mathrm{HC})} = \mathcal{R}\big[\Psi_{\mathrm{ART}},\, \Psi_{\mathrm{QFT}},\, \Psi_{\perp};\, \mathcal{D}, \mathcal{H}_C\big];$$

(iv) the Dirac-, QFT-, and Einstein-type equations are obtained as structural projections $\Pi_f(\mathfrak{E}_{\mathrm{HC}}) = 0$ for distinguished frame or readout regimes f (in the sense of Definition 20.2).

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Theorem 23.2 (Theorem-level shadow of the HC meta-field generator hypothesis)**.** *At the active theorem level, the present framework already supports the following weaker meta-field shadow:*

(i) the Dirac-type, effective QFT-type, and Einstein-type laws are not treated as unrelated foundations, but as distinguished regime/readout projections of one closure-stable structural kernel;

(ii) the compensator already participates in the active transport/resonance architecture in more than one role, so the regime hierarchy is theorematically organized around one common closure-bearing structure rather than around three disconnected sector theories;

(iii) what is not *yet theorem-level is the existence of one explicit compensator-centered generating equation*

$$\mathfrak{E}_{\mathrm{HC}} = 0$$

whose projections already produce the full regime hierarchy.

Therefore the proven theorem-level shadow of Hypothesis 23.1 is: the regime equations already have a common structural carrier and projection architecture, even though a single explicit meta-field equation is not yet proved.

Proof. The framework already treats theorem-level statements by the readout-relative and structural-lift discipline: regime-specific laws may differ at the visible/readout level while sharing one common kernel-level structural content. This is exactly how the probe, effective QFT, and Einstein regimes are positioned in the existing text. Moreover, the compensator is already active in transport, resonance, and closure-current roles inside the standing equations, so the present theory does not treat it as an irrelevant afterthought attached to one sector only. Hence the theorem-level content already available is the existence of one common structural carrier and one projection-organized regime hierarchy. What remains unproved is the stronger claim that there exists one explicit generating equation $\mathfrak{E}_{\mathrm{HC}} = 0$ from which the whole hierarchy is obtained internally as projections. □

Corollary 23.3 (Operational reading of Hypothesis 7.4 in the present 4.7.0 release line)**.** *Until one explicit compensator-centered generating equation is constructed, later arguments may already use Theorem 23.2 as the active theorem-level substitute for the weakest part of Hypothesis 23.1: one may work with a common structural carrier and projection-organized regime hierarchy, while keeping the explicit meta-field equation itself deferred. In the historical release line crystal language, this means that carrier/domain/defect/boundary organization may already be treated as one common local-to-boundary architecture, without claiming that a single closed bulk generator equation has yet been derived.*

Proof. This is the operational reformulation of Theorem 23.2. The theorem already isolates the precise point at which the common-carrier/projected-regime content is available theorematically, while the explicit generator equation remains hypothesis-grade. □

Remark 23.4. *Hypothesis 23.1 is a meta-field-equation hypothesis: not a field equation for one sector only, but a structural equation whose projections produce the effective field equations of the different regimes. Its readout-relative statements (iv) are legitimized by Proposition 20.4; the kernel-level content of the full equation* $\mathfrak{E}_{\mathrm{HC}} = 0$ *would constitute the structural lift (Definition 20.5).*

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Recursive generating loop

The structural content of the hypothesis is the closed loop

$$\begin{aligned}&(\mathcal{M}_{16}, \hat{P}_{C_7}) \longrightarrow \mathcal{H}_C \longrightarrow (\mathcal{D}_\mu, \Omega)\\ &\longrightarrow \text{probe-trace curvature data} \longrightarrow \text{frame projection}\\ &\longrightarrow \text{effective field equation.}\end{aligned}$$

This loop is not a vicious circle. Its purpose is to express that the compensator is simultaneously generated by closure, inserted into transport, and read back through regime-specific probe geometry. The kernel-level anchor of the loop is Corollary 19.24: $\mathcal{M}\,C = \lambda_0\,C$, which fixes the compensator as a fixpoint of the Millennium-matrix action and prevents the loop from drifting.

Probe-tensor readout

In the bidirectional probe picture the compensator-centered curvature signal becomes visible through a tensorial readout built from opposite-frame probe traces:

$$\mathcal{K}^{(\mathrm{HC})}_{\mu\nu} := \mathcal{S}_{\mu\nu}\big(\Psi_{\mathrm{ART}},\, \Psi_{\mathrm{QFT}},\, \Psi_{\perp},\, \mathcal{D}_\mu,\, \mathcal{H}_C\big),$$

where $\mathcal{S}_{\mu\nu}$ collects symmetrized mixed bilinears, transport mismatch, and closure-sensitive focusing data. Geometric curvature is not read from one frame alone, but from the mutual bending of opposite-frame probes around the common compensator core.

Remark 23.5. *The compensator need not be the geometric center in a literal sense. The more robust statement is that it acts as the structure-generating twist or focusing operator relating the probe traces, while the photon/light sector remains the distinguished collapse/output state of the triolic decomposition — the spectral-projector image* $\mathrm{im}(C) = \mathcal{S}_0$ *of Theorem 19.22, identified with* K_3 *by Corollary 19.26.*

Regime generation

The meta-field hypothesis is consistent with the already established projection principle:

(1) the probe regime yields the Dirac-type law;

(2) the phase-shifted observer regime (matter frame at $z^- = +\beta_{\mathrm{sym}}$) yields the effective QFT-type law after compensator decomposition;

(3) the large-scale closure-stable regime yields the Einstein-type law after geometric readout and coarse-graining.

What Hypothesis 23.1 adds is the claim that these three regimes should ultimately be generated from one compensator-centered structural equation rather than treated as three parallel derived blocks.

23.1 Hypothesis 7.5: Frame-dependent defect readout of a deeper compensator

The current minimal compensator ansatz identifies

$$\mathcal{H}_C = \rho\,[\mathcal{M}_{16}, \hat{P}_{C_7}]^\dagger[\mathcal{M}_{16}, \hat{P}_{C_7}] + \sigma\,\mathfrak{R}_C^\dagger\mathfrak{R}_C$$

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as a positive closure-mismatch functional. Accordingly, in the exact sector one has

$$\mathfrak{R}_C = 0 \qquad \Longrightarrow \qquad \mathcal{H}_C = 0$$

for the minimal ansatz. This, however, need only be interpreted as the vanishing of the *visible defect readout* of the compensator in the given regime or frame, not necessarily as the disappearance of a deeper compensator-generating structure. This is the precise sense of Remark 20.8: the shadow vanishes, not the kernel $C = \Pi_{\mathrm{spec}}^{(\mathcal{M})}$ (Theorem 19.22).

Proposition 23.6 (Frame-dependent defect readout hypothesis)**.** (Hypothesis-grade; not yet theorem-level.) *There exists a deeper compensator matrix/generator* $\mathbf{H}_C$ *such that the presently used functional compensator is only a frame-dependent visible readout*

$$\mathcal{H}_C^{(f)} = \mathcal{D}_{\mathrm{read}}^{(f)}\big(\mathbf{H}_C;\, \mathcal{M}_{16},\, \hat{P}_{C_7}\big),$$

where f *denotes a distinguished regime or frame reading. Then:*

(i) $\mathcal{H}_C^{(f)}$ *may vanish in the exact sector of one frame without forcing* $\mathbf{H}_C$ *itself to vanish;*

(ii) different regime projections may expose different visible compensator shadows of one common underlying generator;

(iii) the current minimal ansatz describes the energetic defect/penalty sector of the compensator, not necessarily its full structural role.

This is a readout-relative statement in the sense of Definition 20.2: the visible vanishing $\mathcal{H}_C^{(f)} = 0$ *is a statement about the readout, not about the kernel* $\mathbf{H}_C$*.*

Theorem 23.7 (Theorem-level shadow of frame-dependent defect readout)**.** *At the active theorem level, the vanishing of a visible compensator defect readout does* not *force the collapse of the retained closure-bearing structure. More precisely, suppose a regime/frame readout* f *lies in the exact admissible sector so that*

$$\mathfrak{R}_C^{(f)} = 0 \qquad \Longrightarrow \qquad \mathcal{H}_C^{(f)} = 0$$

for the minimal ansatz. Then the only theorem-level conclusion is:

(i) the active visible defect penalty vanishes in that readout;

(ii) the closure-stable carrier/kernel content already fixed by Remark 20.8, Theorem 19.22, and the structural-lift machinery is not thereby forced to vanish;

(iii) in the historical release line crystal language, the visible defect-vanishing statement may be read as the disappearance of a defect face or defect-load contribution inside an addressed crystal readout profile, while the carrier/domain/boundary faces may remain nontrivial.

Therefore the theorem-level content of Hypothesis 23.6 already has a proven shadow: visible defect vanishing is a readout-level extinction statement, not a theorem-level carrier-extinction statement*.*

Proof. The first claim is exactly the meaning of the minimal positive mismatch ansatz in the exact admissible sector: when the residual mismatch vanishes, the associated visible defect penalty vanishes as well. The second claim is the content of Remark 20.8: the shadow vanishes, not the kernel. This is reinforced by Theorem 19.22, which fixes the compensator at kernel level as a spectral-projector object rather than as a merely visible defect variable,

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and by the structural-lift principles that distinguish readout-level vanishing from kernel-level extinction. For the third claim, the historical release line addressed crystal language does not introduce a new theorem but repackages the same distinction: a defect readout belongs to the visible defect/load layer, whereas carrier, domain, and boundary data belong to the retained local-to-boundary crystal readout package. So setting the defect-visible part to zero does not theorematically annihilate the whole package. □

Corollary 23.8 (Operational reading of Hypothesis 7.5 in the crystal language)**.** *Until a deeper generator $\mathbf{H}_C$ is constructed, later arguments may already use Theorem 23.7 as the active theorem-level substitute for the weakest part of Hypothesis 23.6: exact-sector visible defect vanishing may be read as removal of the defect face inside the active crystal readout package, without forcing collapse of the retained carrier/domain/boundary structure. A stronger reopening is needed only when one wants to prove the existence of a deeper generator $\mathbf{H}_C$ itself.*

Proof. This is the operational reformulation of Theorem 23.7. The theorem already isolates the precise point at which the active visible defect statement may be separated from any stronger generator-level claim. □

Remark 23.9. *This hypothesis removes the apparent tension between the exact-sector vanishing of the minimal compensator functional and any meta-field role demanded of the compensator structure. What vanishes under exact closure is only the defect-visible compensator shadow in a given readout, not automatically the deeper compensator kernel. The structural lift (Definition 20.5) of the vanishing statement $\mathcal{H}_C^{(f)} = 0$ is therefore:* the closure-mismatch penalty is zero in the admissible sector *— a kernel-level statement about admissibility, not about the extinction of the compensator itself.*

Remark 23.10. *This interpretation is structurally natural in the present framework, since probe, effective QFT, and Einstein geometry are already treated as different regime projections of one common closure-stable equation (Proposition 20.7), and the light or photon sector is the compensator-selected collapse output (Theorem 4.3). Hence a frame-dependent compensator visibility is more plausible than a globally frame-blind compensator extinction.*

23.2 Hypothesis 7.6: Fluid-type equations as coarse-grained frame readouts

The present framework already treats the probe, effective QFT, and Einstein regimes as different projections of one common closure-stable equation (Proposition 20.7). This suggests a cautious additional possibility: certain fluid-type transport equations may arise as coarse-grained, frame-dependent readouts of the same HC/closure transport kernel, rather than as foreign structures added from outside.

Proposition 23.11 (Fluid-type readout hypothesis)**.** (Hypothesis-grade; not yet theorem-level.) *There exists a coarse-grained regime/frame readout $\Pi_{\mathrm{fluid}}^{(f,\ell)}$ such that, for suitable frame f and stratum/scale level ℓ, the closure-stable transport kernel*

$$\big(i\Gamma^\mu \mathcal{D}_\mu - \Omega\big)\Psi = 0, \qquad \mathcal{D}^\mu \mathcal{F}_{\mu\nu} = J_\nu^{(\Psi)} + J_\nu^{(C_7)} + J_\nu^{(\mathrm{HC})},$$

admits an effective medium-type readout whose macroscopic evolution is described by fluid-like transport equations.

Theorem 23.12 (Theorem-level shadow of fluid-type readout)**.** *At the active theorem level, any admissible fluid-type description in the present framework can only appear as a* coarse-grained readout-relative transport description *of the already fixed closure-stable kernel; it cannot replace the primary kernel-level transport equations. More precisely:*

(i) the active kernel-level transport content remains

$$\big(i\Gamma^{\mu}\mathcal{D}_{\mu}-\Omega\big)\Psi=0,\qquad \mathcal{D}^{\mu}\mathcal{F}_{\mu\nu}=J_{\nu}^{(\Psi)}+J_{\nu}^{(C_7)}+J_{\nu}^{(\mathrm{HC})};$$

(ii) any later fluid-like law, if valid at all, must be read as a coarse-grained regime/frame statement in the sense of Definition 20.2, not as a new primary kernel equation;

(iii) in the historical release line crystal language, such a law can only act on corridor- or readout-level crystal data (domain/defect/boundary propagation) and not as a theorem-level replacement of the closure-bearing carrier line itself.

Therefore the theorem-level content already available beneath Hypothesis 23.11 is: fluid-type behavior, if realized, belongs to a later coarse-grained readout of the same kernel rather than to a second foundational transport theory.

Proof. The first point is simply the active transport kernel already retained throughout the document. The second point follows from the readout-relative discipline and the structural-lift machinery: the framework already distinguishes theorem-level kernel statements from regime-specific readout statements, so any fluid-type law that appears at all must belong to the latter side unless one proves a new kernel identity. The third point is then the historical release line repackaging of the same distinction. The new crystal language treats carrier, domain, defect, and boundary data as different local-to-boundary faces of one closure-bearing configuration. A coarse-grained fluid description can therefore only summarize the propagation of visible domain/defect/boundary data along a corridor; it does not theorematically replace the underlying carrier kernel. □

Corollary 23.13 (Operational reading of Hypothesis 7.6 in the crystal language)**.** *Until a concrete coarse-graining map is constructed, later arguments may already use Theorem 23.12 as the active theorem-level substitute for the weakest part of Hypothesis 23.11: any admissible fluid-like statement must be read as a corridor- or readout-level coarse-grained law of the same closure-bearing crystal data, not as a second primary transport kernel. A stronger reopening is required only when one wants to prove existence of the coarse-graining map itself.*

Proof. This is the operational reformulation of Theorem 23.12. The theorem already isolates the precise point at which a possible fluid-type law is demoted from a competing foundation to a later coarse-grained readout statement. □

Remark 23.14. *This hypothesis does* not *claim that fields are literally fluids. It claims only that fluid-type equations may appear as frame-specific, coarse-grained transport readouts of the same underlying structural kernel. The distinction is exactly that of Definition 20.2: a readout-relative statement $L^{(f,\ell)}$ valid in the coarse-grained regime, not a kernel-level universalization.*

Remark 23.15. *Stone–von Neumann (Theorem 5.2) supports sameness of the underlying structural kernel at the representation level, but the fluid equations would belong to a later coarse-grained regime readout rather than to the primary operator-level formulation. The structural lift (Definition 20.5) of this hypothesis, if realizable, would be a kernel-level statement that the closure-stable transport kernel admits a coarse-graining map into fluid-type evolution — not that the fluid equations are primary.*

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24 Closure-Liquid Crystal and Holographic Julia Boundary

The present section introduces a cautious re-reading of the current closure architecture in cellular language. It does *not* claim that the framework is literally a condensed-matter crystal. Instead, it proposes one structured pilot in which the same closure-bearing carrier is treated as a bulk order parameter whose visible sectors appear as phase-sensitive ordered domains and whose Julia-type diagnostics are read as boundary holograms of that same order. The point of this re-reading is organizational: it compresses the current OP 3/OP 5 language into one local compatibility / defect-transport / boundary-readout picture without creating a second closure theory.

Definition 24.1 (Closure-liquid-crystal cell)**.** *A* closure-liquid-crystal cell *is a quintuple*

$$X = (K, \sigma, \rho, \phi, \delta)$$

consisting of:

(i) a closure-bearing carrier datum K;

(ii) a shell/refinement datum σ;

(iii) a branch/route datum ρ;

(iv) a local phase/readout datum ϕ;

(v) a local defect datum δ.

The intended reading is that K records the bulk carrier order, while $(\sigma, \rho, \phi, \delta)$ records the visible local realization of that order in one readout domain.

Definition 24.2 (Local cell-compatibility equation)**.** *Let $X_n = (K_n, \sigma_n, \rho_n, \phi_n, \delta_n)$ and $X_{n+1} = (K_{n+1}, \sigma_{n+1}, \rho_{n+1}, \phi_{n+1}, \delta_{n+1})$ be adjacent closure-liquid-crystal cells. Their local compatibility equation is written*

$$\mathcal{C}_{\mathrm{cell}}(X_n, X_{n+1}) = 0$$

and is required to mean that:

(i) the carrier datum remains inside the same closure-bearing class;

(ii) the branch/shell transition between the two cells is admissible;

(iii) visible readout variation remains at screening level and does not yet generate a genuine shell defect witness.

Thus the equation $\mathcal{C}_{\mathrm{cell}} = 0$ is the local cellular form of admissible OP 3 passage.

Proposition 24.3 (OP 3 cell-compatibility reading)**.** *Let a translated OP 3 shell passage carry a shell admissibility certificate in the sense of Definition 1.193. Then it determines two adjacent closure-liquid-crystal cells whose local compatibility equation holds:*

$$\mathcal{C}_{\mathrm{cell}}(X_n, X_{n+1}) = 0.$$

In particular, the current OP 3 certificate package may be read as a local cell-compatibility law on the same closure-bearing carrier medium.

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Proof. A shell admissibility certificate records admissible branch passage, shell-admissible refinement, and closure-compatible readout on the translated OP 3 side. By Lemma 1.195 and Theorem 1.205, the source state and the branch-shell return state remain in the same closure-bearing class and differ only by an admissible shell passage together with readout-screening data. This is exactly the content encoded by $\mathcal{C}_{\mathrm{cell}}(X_n, X_{n+1}) = 0$ in Definition 24.2. □

Definition 24.4 (Locally corrected crystal compatibility equation)**.** *Let X_n and X_{n+1} be adjacent closure-liquid-crystal cells in the closure-phase regime. Their* locally corrected crystal compatibility equation *is written*

$$\mathcal{C}^{\sharp}_{\mathrm{cell}}(X_n, X_{n+1}) = 0$$

and is required to mean that:

(i) the original compatibility equation $\mathcal{C}_{\mathrm{cell}}(X_n, X_{n+1}) = 0$ holds at closure/readout level;

(ii) the local phase-side persistence defect stays below the active reinforced threshold;

(iii) the associated Teichmüller-side persistence defect stays below the active reinforced threshold;

(iv) the packet-side admissibility monitor remains inside the safe admissibility band.

Remark 24.5 (Why the crystal equation must be corrected locally)**.** *The original crystal compatibility law is an OP 3-style admissible passage law. The present post-archival stage local obstruction analysis shows that this is no longer enough for the active closure-phase trigger problem. The compatibility equation must therefore be tightened by a local persistence correction, not by changing the carrier class itself, but by sharpening the phase-side and Teichmüller-side persistence conditions on the second step.*

Definition 24.6 (Local crystal persistence correction datum)**.** *The* local crystal persistence correction datum *at step n is the quadruple*

$$\mathfrak{K}^{\sharp}_{\mathrm{cell}}(n) := (\eta_{\Theta}(n), \eta_T(n), \zeta_T(n), \lambda_T(n)),$$

where:

(i) $(\eta_{\Theta}(n), \eta_T(n))$ is the first local repair witness;

(ii) $(\zeta_T(n), \lambda_T(n))$ is the local Teichmüller-side repair datum.

Definition 24.7 (Corrected crystal two-step window)**.** *A two-step local crystal window*

$$X_n,\ X_{n+1}$$

is called corrected *if the locally corrected crystal compatibility equation holds and the corresponding reinforced trigger package passes on both steps.*

Proposition 24.8 (Teichmüller repair lifts the crystal equation from screening compatibility to persistence compatibility)**.** *Assume the first local repair witness and the first local Teichmüller-persistence repair theorem. Then the local crystal compatibility equation can be lifted from*

$$\mathcal{C}_{\mathrm{cell}}(X_n, X_{n+1}) = 0$$

to the corrected form

$$\mathcal{C}^{\sharp}_{\mathrm{cell}}(X_n, X_{n+1}) = 0$$

on every corrected crystal two-step window.

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Proof. The first local repair witness supplies the phase-side tightening required to repair the second-step drift, while the local Teichmüller-side repair datum supplies the remaining geometric persistence correction. If both corrections are applied and the reinforced trigger package passes on both steps, then the original admissible crystal passage is no longer merely screening-compatible but persistence-compatible on the active two-step window. This is exactly the meaning of the corrected equation $\mathcal{C}^{\sharp}_{\mathrm{cell}}(X_n, X_{n+1}) = 0$. □

Theorem 24.9 (First local crystal-equation correction theorem)**.** *Assume:*

(i) the first local Theta-drift witness theorem applies;

(ii) the first local phase-side repair theorem applies;

(iii) the first local Teichmüller-persistence repair theorem applies.

Then the present line supports a first local correction of the crystal equation. More precisely:

(a) the original crystal compatibility equation remains valid as the carrier/readout baseline;

(b) the active local persistence problem is repaired not by changing the carrier class, but by replacing $\mathcal{C}_{\mathrm{cell}} = 0$ with the corrected equation $\mathcal{C}^{\sharp}_{\mathrm{cell}} = 0$ on the active two-step window;

(c) the first crystal-equation correction is therefore a local persistence correction, not a new bulk crystal law.

Proof. By the local obstruction analysis, the first failure is phase/geometry-prior rather than carrier-prior. The phase-side repair theorem supplies the first burden-side tightening, and the Teichmüller-persistence repair theorem supplies the remaining geometric tightening. Therefore the crystal equation is not replaced at bulk level; it is corrected locally by a persistence reinforcement on the active two-step window. □

Corollary 24.10 (Current crystal-equation working rule)**.** *The current post-archival stage line should not yet search first for a wholly new crystal equation. It should first search for the minimal local persistence correction that upgrades*

$$\mathcal{C}_{\mathrm{cell}} = 0$$

to

$$\mathcal{C}^{\sharp}_{\mathrm{cell}} = 0$$

on the active closure-phase two-step window.

Proof. Immediate from Theorem 24.9. □

Definition 24.11 (Local compact reinforcement pair)**.** *The* local compact reinforcement pair *extracted from the crystal persistence correction datum*

$$\mathfrak{K}^{\sharp}_{\mathrm{cell}}(n) = (\eta_{\Theta}(n), \eta_T(n), \zeta_T(n), \lambda_T(n))$$

is the pair

$$\mathfrak{A}^{\sharp}_{\mathrm{loc}}(n) := (A_{\mathrm{bur}}(n), A_{\mathrm{geo}}(n))$$

defined by

$$A_{\mathrm{bur}}(n) := \eta_{\Theta}(n), \qquad A_{\mathrm{geo}}(n) := \max\{\eta_T(n), \zeta_T(n), \lambda_T(n)\}.$$

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Remark 24.12 (Meaning of the compact pair)**.** *The first parameter records the burden-side phase reinforcement required to stop the second-step Theta drift. The second parameter records the total geometric persistence reinforcement required on the same two-step window. Thus the original four repair numbers are compressed into one burden-side and one geometric reinforcement scale.*

Definition 24.13 (One-parameter shadow reinforcement)**.** *The* one-parameter shadow reinforcement *of the local compact reinforcement pair is*

$$A_{\mathrm{tot}}(n) := \max\{A_{\mathrm{bur}}(n), A_{\mathrm{geo}}(n)\}.$$

Remark 24.14 (Why the one-parameter shadow is secondary)**.** *The scalar shadow $A_{\mathrm{tot}}(n)$ is useful only as a coarse local size readout. It should not replace the two-parameter pair, because the current post-archival stage line still distinguishes between burden-side repair and geometric persistence repair.*

Definition 24.15 (Burden-dominant compact reinforcement)**.** *The local compact reinforcement pair is called* burden-dominant *if*

$$A_{\mathrm{bur}}(n) \geq A_{\mathrm{geo}}(n).$$

It is called geometrically dominant *if*

$$A_{\mathrm{geo}}(n) > A_{\mathrm{bur}}(n).$$

Proposition 24.16 (The crystal persistence correction datum admits a conservative two-parameter compression)**.** *Every local crystal persistence correction datum*

$$(\eta_\Theta(n), \eta_T(n), \zeta_T(n), \lambda_T(n))$$

admits a conservative two-parameter compression into

$$(A_{\mathrm{bur}}(n), A_{\mathrm{geo}}(n)).$$

Moreover, the compressed pair dominates the original repair data in the sense that:

(a) $A_{\mathrm{bur}}(n)$ retains the full burden-side strengthening requirement;

(b) $A_{\mathrm{geo}}(n)$ bounds every geometric persistence strengthening requirement from above.

Proof. The first claim is immediate from Definition 24.11. The second follows because $A_{\mathrm{geo}}(n)$ is defined as the maximum of the three geometric persistence reinforcement terms. Hence the compressed pair conservatively dominates the original local correction datum. □

Proposition 24.17 (Current local reinforcement is not purely geometric)**.** *Assume the first local repair-orientation theorem and the first local Teichmüller-persistence repair theorem. Then the current local compact reinforcement pair is not geometrically dominant by default. More precisely, the burden-side component*

$$A_{\mathrm{bur}}(n) = \eta_\Theta(n)$$

remains genuinely active in every admissible local repair reading.

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Proof. By the local repair-orientation theorem, the first repair is Theta-dominant rather than purely geometric. By the Teichmüller-persistence repair theorem, the geometric side is genuinely active but enters only after the burden-side drift has already been identified as the first required strengthening. Therefore the current local compact reinforcement is not purely geometric by default. □

Theorem 24.18 (First compact reinforcement theorem)**.** *Assume:*

(a) the first local crystal-equation correction theorem applies;

(b) the first local repair-to-trigger calibration theorem applies;

(c) the first local Teichmüller-persistence repair theorem applies.

Then the current post-archival stage local crystal persistence problem admits a first compact reinforcement form.

More precisely:

(1) the four local repair parameters

$$(\eta_\Theta(n), \eta_T(n), \zeta_T(n), \lambda_T(n))$$

compress conservatively to the pair

$$(A_{\mathrm{bur}}(n), A_{\mathrm{geo}}(n));$$

(2) the burden-side component $A_{\mathrm{bur}}(n)$ remains the first active repair scale;

(3) the geometric component $A_{\mathrm{geo}}(n)$ records the total persistence reinforcement required to keep the corrected crystal equation valid on the active two-step window;

(4) the scalar shadow $A_{\mathrm{tot}}(n)$ is only a coarse readout and should not replace the compact pair at the current stage.

Proof. By Proposition 24.16, the four repair parameters admit a conservative compression into the compact pair. By Proposition 24.17, the burden-side component remains genuinely active and therefore cannot be discarded. The geometric component dominates the three persistence-side repair contributions by definition, so it records the total persistence reinforcement requirement on the active two-step window. The scalar shadow is only the maximum of the two compact parameters and therefore loses the burden/geometry distinction still needed at the current stage. □

Corollary 24.19 (Current compact reinforcement working rule)**.** *The next local audit should not first track the full four-parameter repair datum. It should first track the compact reinforcement pair*

$$(A_{\mathrm{bur}}(n), A_{\mathrm{geo}}(n)),$$

using the scalar shadow $A_{\mathrm{tot}}(n)$ only as a coarse size indicator.

Proof. Immediate from Theorem 24.18. □

Definition 24.20 (Local reinforcement ratio)**.** *The* local reinforcement ratio *is the quantity*

$$\chi_{\mathrm{loc}}(n) := \frac{A_{\mathrm{geo}}(n)}{A_{\mathrm{bur}}(n)}$$

whenever $A_{\mathrm{bur}}(n) > 0$.

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Remark 24.21 (Meaning of the local reinforcement ratio)**.** *The ratio $\chi_{\mathrm{loc}}(n)$ measures how much geometric persistence reinforcement is required relative to the burden-side phase reinforcement on the active two-step window. It is therefore the first compact indicator of whether the local repair is still burden-led or already close to a balanced coupled regime.*

Definition 24.22 (Local reinforcement regime classes)**.** *Assume $A_{\mathrm{bur}}(n) > 0$. The local compact reinforcement pair is said to be:*

(a) strictly burden-dominant *if*

$$\chi_{\mathrm{loc}}(n) < 1;$$

(b) balanced-coupled *if*

$$\chi_{\mathrm{loc}}(n) = 1;$$

(c) geometrically overcarried *if*

$$\chi_{\mathrm{loc}}(n) > 1.$$

Remark 24.23 (Why the ratio matters)**.** *The compact pair already tells us that both burden-side and geometric reinforcement are active. The ratio tells us which of the two is currently driving the local repair budget.*

Proposition 24.24 (Current compact repair is not balanced by default)**.** *Assume the first compact reinforcement theorem and the first local repair-orientation theorem. Then the current local compact repair is not balanced-coupled by default. More precisely, the post-archival stage line should be read first as strictly burden-dominant unless a later local audit shows that the geometric reinforcement has risen to the same scale.*

Proof. By the first compact reinforcement theorem, the burden-side component $A_{\mathrm{bur}}(n)$ remains the first active repair scale, while the geometric component $A_{\mathrm{geo}}(n)$ records the total persistence reinforcement still required on the same window. By the local repair-orientation theorem, the first repair is Theta-dominant rather than geometric-dominant. Hence the default reading is not balanced-coupled, and the local repair is first to be read as burden-dominant unless later evidence forces a different regime class. □

Definition 24.25 (Local reinforcement balance defect)**.** *The* local reinforcement balance defect *is*

$$B_{\mathrm{loc}}(n) := A_{\mathrm{geo}}(n) - A_{\mathrm{bur}}(n).$$

Remark 24.26 (Interpretation of the balance defect)**.** *Negative balance defect means burden-side dominance. Zero balance defect means balanced coupled repair. Positive balance defect means that the geometric persistence side has overtaken the burden-side repair scale.*

Theorem 24.27 (First local reinforcement-ratio theorem)**.** *Assume:*

(a) the first compact reinforcement theorem applies;

(b) the first local Theta-drift witness theorem applies;

(c) the first local Teichmüller-persistence repair theorem applies.

Then the current post-archival stage local repair supports a first compact ratio reading. More precisely:

(1) the compact pair admits the ratio $\chi_{\mathrm{loc}}(n)$ and the balance defect $B_{\mathrm{loc}}(n)$;

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(2) the default local repair reading is strictly burden-dominant, hence

$$B_{\mathrm{loc}}(n) < 0 \quad \textit{and therefore} \quad \chi_{\mathrm{loc}}(n) < 1$$

in the default regime reading;

(3) the geometric reinforcement remains genuinely active even in this burden-dominant regime, so the ratio does not collapse to zero.

Proof. Items (1) and the sign interpretation are immediate from the definitions of the ratio and balance defect. By the first compact reinforcement theorem, the burden-side component remains the first active repair scale. By the first local Teichmüller-persistence repair theorem, the geometric reinforcement is still genuinely active and therefore cannot be discarded. Hence the default regime is burden-dominant but not purely one-sided, which yields $B_{\mathrm{loc}}(n) < 0$ and therefore $\chi_{\mathrm{loc}}(n) < 1$ in the default reading. □

Corollary 24.28 (Current compact-ratio working rule)**.** *The next local audit should not only record the pair*

$$(A_{\mathrm{bur}}(n), A_{\mathrm{geo}}(n)),$$

but also track the ratio $\chi_{\mathrm{loc}}(n)$ and balance defect $B_{\mathrm{loc}}(n)$ in order to measure whether the current repair remains burden-led or is moving toward balanced coupled repair.

Proof. Immediate from Theorem 24.27. □

Definition 24.29 (Local asymptotic repair orientation)**.** *The* local asymptotic repair orientation *is the long-step tendency of the compact reinforcement ratio*

$$\chi_{\mathrm{loc}}(n) = \frac{A_{\mathrm{geo}}(n)}{A_{\mathrm{bur}}(n)}$$

along an admissible sequence of repaired local windows.

Definition 24.30 (Balance-seeking local repair family)**.** *An admissible local repair family is called* balance-seeking *if*

$$\chi_{\mathrm{loc}}(n) \to 1$$

along the family, equivalently if

$$B_{\mathrm{loc}}(n) \to 0.$$

Definition 24.31 (Persistently burden-dominant local repair family)**.** *An admissible local repair family is called* persistently burden-dominant *if there exists some $\delta_{\mathrm{bal}} > 0$ such that*

$$\chi_{\mathrm{loc}}(n) \leq 1 - \delta_{\mathrm{bal}}$$

for all sufficiently large local repair steps.

Remark 24.32 (Why the asymptotic question matters)**.** *The current compact repair is already burden-dominant at first order. The real next question is whether this is only an initial orientation or whether the burden-side dominance remains structurally built into the repaired persistence problem.*

Proposition 24.33 (The current line does not yet force balance-seeking behavior)**.** *Assume the current post-archival stage local repair reading. Then the existing local theorems do not yet force*

$$\chi_{\mathrm{loc}}(n) \to 1.$$

In particular, no current result requires the geometric reinforcement to rise all the way to burden-side balance.

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Proof. The current local repair-orientation theorem yields burden-side dominance at first order, while the Teichmüller-side repair theorems only assert that the geometric reinforcement remains genuinely active. They do not assert that it asymptotically reaches the same scale as the burden-side reinforcement. Hence balance-seeking behavior is not yet forced. □

Proposition 24.34 (The current line is naturally read as burden-dominant unless later balancing evidence appears)**.** *Assume the first local reinforcement-ratio theorem and the first local repair-to-trigger calibration theorem. Then the default asymptotic reading of the current line is persistently burden-dominant unless a later local audit exhibits a genuine balancing mechanism reducing $B_{\mathrm{loc}}(n)$ toward zero.*

Proof. The current ratio theorem places the local repair in the burden-dominant regime, and the first local repair-to-trigger calibration theorem still assigns the primary tightening to the Theta-side threshold. Thus the available repair logic continues to privilege the burden-side channel. Without an additional balancing mechanism, the natural default reading is that burden-side dominance persists. □

Theorem 24.35 (First local asymptotic orientation theorem)**.** *Assume:*

(a) the first local reinforcement-ratio theorem applies;

(b) the first local repair-to-trigger calibration theorem applies;

(c) no later local theorem has yet exhibited a balancing mechanism forcing $B_{\mathrm{loc}}(n) \to 0$.

Then the current post-archival stage line should be read asymptotically as burden-dominant rather than balance-seeking. More precisely, the present local evidence supports persistent burden-side dominance as the default asymptotic orientation, while balanced-coupled repair remains only a later possibility to be justified by new local evidence.

Proof. By the first local reinforcement-ratio theorem, the current compact repair is burden-dominant at first order. By the first local repair-to-trigger calibration theorem, the primary tightening remains Theta-side, although the geometric repair remains genuinely active. In the absence of an additional balancing mechanism driving the balance defect toward zero, the natural asymptotic reading continues the burden-dominant orientation rather than forcing convergence to balance. □

Corollary 24.36 (Current asymptotic working rule)**.** *The next local audit should not presuppose*

$$\chi_{\mathrm{loc}}(n) \to 1.$$

Instead, it should first test whether any genuine balancing mechanism exists at all; until then, the current local repair should be treated as asymptotically burden-dominant with genuine but subordinate geometric carry.

Proof. Immediate from Theorem 24.35. □

Definition 24.37 (Final local balancing mechanisms)**.** *A* final local balancing mechanism *for the compact reinforcement pair*

$$\mathfrak{A}^{\sharp}_{\mathrm{loc}}(n) = (A_{\mathrm{bur}}(n), A_{\mathrm{geo}}(n))$$

is any local structural effect capable of reducing the balance defect

$$B_{\mathrm{loc}}(n) = A_{\mathrm{geo}}(n) - A_{\mathrm{bur}}(n)$$

toward zero. Three final types are distinguished:

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(a) *a* crystal-side balancing mechanism*, acting through the corrected crystal compatibility equation;*

(b) *a* Teichmüller-side balancing mechanism*, acting through the geometric persistence layer alone;*

(c) *a* hybrid crystal–Teichmüller balancing mechanism*, acting by coupled improvement of the crystal compatibility side and the Teichmüller persistence side.*

Remark 24.38 (Why a pure crystal correction is probably not enough)**.** *The current local line already required a first burden-side repair followed by an explicit Teichmüller-side repair. This suggests that crystal-side correction alone is unlikely to be the full balancing mechanism.*

Remark 24.39 (Why a pure Teichmüller correction is also probably not enough)**.** *The local repair-orientation theorem still forces positive burden-side reinforcement*

$$\eta_{\Theta}(n) > 0.$$

Hence the geometric side alone does not erase the burden-side asymmetry.

Definition 24.40 (Hybrid balancing witness)**.** *A* hybrid balancing witness *is a local witness package showing simultaneously:*

(a) *improvement of the corrected crystal compatibility equation,*

(b) *improvement of the Teichmüller-side persistence inequalities,*

(c) *strict reduction of the balance defect*

$$B_{\mathrm{loc}}(n+1) > B_{\mathrm{loc}}(n)$$

toward zero, equivalently

$$|B_{\mathrm{loc}}(n+1)| < |B_{\mathrm{loc}}(n)|$$

on the burden-dominant side.

Proposition 24.41 (The current line does not yet support a purely crystal balancing mechanism)**.** *Assume the current post-archival stage local repair line. Then the available local evidence does not yet support reading the first balancing mechanism as purely crystal-side.*

Proof. The local crystal-equation correction theorem already required the prior burden-side and Teichmüller-side repair data. Hence the crystal correction enters as part of a repaired persistence package, not as a self-sufficient balancing source on its own. □

Proposition 24.42 (The current line does not yet support a purely Teichmüller balancing mechanism)**.** *Assume the first local repair-orientation theorem. Then the available local evidence does not yet support reading the first balancing mechanism as purely Teichmüller-side.*

Proof. The first local repair-orientation theorem keeps the dominant local repair on the burden side while the geometric side remains genuinely active but subordinate. Hence the currently visible repair is not purely Teichmüller-driven. □

Theorem 24.43 (First local balancing-mechanism orientation theorem)**.** *Assume:*

(a) *the first local asymptotic orientation theorem applies;*

(b) *the first local crystal-equation correction theorem applies;*

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(c) the first local Teichmüller-persistence repair theorem applies.

Then the most plausible final for a future local balancing mechanism is not purely crystal-side and not purely Teichmüller-side, but hybrid crystal–Teichmüller.

Proof. By the first local asymptotic orientation theorem, the current line remains burden-dominant unless a later balancing mechanism appears. By the local crystal-equation correction theorem, crystal-side correction is relevant but not self-sufficient. By the local Teichmüller-persistence repair theorem, geometric repair is genuinely active but likewise not self-sufficient. Therefore the first plausible balancing mechanism must couple both corrections if it is to reduce the balance defect toward zero. □

Corollary 24.44 (Current balancing-mechanism working rule)**.** *The next local audit should not first search for a purely crystal balancing mechanism or a purely Teichmüller balancing mechanism. Instead, it should search for the first hybrid witness that lowers the balance defect by coordinated crystal-side and geometric-side repair.*

Proof. Immediate from Theorem 24.43. □

Definition 24.45 (First hybrid balancing step)**.** *A* first hybrid balancing step *from the local state at step n to the next step $n+1$ is a local repaired transition satisfying all of the following:*

(a) the corrected crystal compatibility equation improves from

$$\mathcal{C}^{\sharp}_{\mathrm{cell}}(X_n, X_{n+1})$$

to a closer-to-compatible value on the updated two-step window;

(b) the Teichmüller-side persistence repair data improve in the sense that the repaired geometric persistence test is strictly better satisfied on the updated window;

(c) the compact reinforcement pair

$$\mathfrak{A}^{\sharp}_{\mathrm{loc}}(n) = (A_{\mathrm{bur}}(n), A_{\mathrm{geo}}(n))$$

is updated to

$$\mathfrak{A}^{\sharp}_{\mathrm{loc}}(n+1) = (A_{\mathrm{bur}}(n+1), A_{\mathrm{geo}}(n+1))$$

with strictly smaller balance defect magnitude

$$|B_{\mathrm{loc}}(n+1)| < |B_{\mathrm{loc}}(n)|;$$

(d) the update remains inside the same local admissibility regime.

Definition 24.46 (First hybrid balancing witness package)**.** *A* first hybrid balancing witness package *at step n is a finite witness datum*

$$\mathfrak{W}_{\mathrm{hyb}}(n) = \big(\delta_{\mathrm{cell}}(n), \delta_T(n), \Delta B_{\mathrm{loc}}(n)\big)$$

such that:

(a) $\delta_{\mathrm{cell}}(n) > 0$ measures the crystal-side compatibility gain of the first hybrid balancing step;

(b) $\delta_T(n) > 0$ measures the Teichmüller-side persistence gain of the same step;

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(c)

$$\Delta B_{\mathrm{loc}}(n) := |B_{\mathrm{loc}}(n)| - |B_{\mathrm{loc}}(n+1)| > 0$$

measures the strict gain toward local balance.

Remark 24.47 (Meaning of the hybrid witness package)**.** *The first hybrid balancing witness package is the first local datum that does more than merely diagnose burden-dominance. It certifies that crystal-side correction and Teichmüller-side repair improve together and that the resulting coupled improvement genuinely lowers the balance defect.*

Definition 24.48 (Hybrid balancing efficiency ratio)**.** *The* hybrid balancing efficiency ratio *of a first hybrid balancing witness package is*

$$\Xi_{\mathrm{hyb}}(n) := \frac{\Delta B_{\mathrm{loc}}(n)}{\delta_{\mathrm{cell}}(n) + \delta_T(n)},$$

provided the denominator is positive.

Remark 24.49 (Why the efficiency ratio is useful)**.** *This ratio records how much balance-defect reduction is obtained per unit of coupled crystal-plus-Teichmüller repair. It is therefore the first local scalar witness that the hybrid mechanism is not merely present, but actually effective.*

Proposition 24.50 (A first hybrid balancing step is stronger than separate side repairs)**.** *Every first hybrid balancing step is stronger than a purely crystal-side repair and stronger than a purely Teichmüller-side repair considered separately.*

Proof. By definition, a first hybrid balancing step requires simultaneous crystal-side improvement, simultaneous Teichmüller-side persistence improvement, and strict reduction of the balance defect magnitude. A purely one-sided repair can satisfy at most one of the two side-improvement clauses, but not both together with the coupled balance reduction requirement. Hence the hybrid step is strictly stronger. □

Lemma 24.51 (First hybrid balancing witness lemma)**.** *Assume:*

(a) the first local balancing-mechanism orientation theorem applies;

(b) a first hybrid balancing step exists on the current local window.

Then a first hybrid balancing witness package exists.

Proof. If a first hybrid balancing step exists, then by definition it produces strictly positive crystal-side compatibility gain, strictly positive Teichmüller-side persistence gain, and strict decrease of the balance defect magnitude. These three gains form the witness package

$$\mathfrak{W}_{\mathrm{hyb}}(n) = (\delta_{\mathrm{cell}}(n), \delta_T(n), \Delta B_{\mathrm{loc}}(n)).$$

□

Theorem 24.52 (First hybrid balancing witness theorem)**.** *Assume:*

(a) the first local balancing-mechanism orientation theorem applies;

(b) the current line is still asymptotically burden-dominant;

(c) a local repaired two-step window exists on which crystal-side and Teichmüller-side improvements occur simultaneously.

Then the current line supports a first hybrid balancing witness exactly when the simultaneous side improvements produce strict reduction of

$$|B_{\mathrm{loc}}(n)|.$$

Equivalently, the first honest evidence for a genuine balancing mechanism is not the mere coexistence of crystal and geometric repair, but the existence of a positive witness package

$$\mathfrak{W}_{\mathrm{hyb}}(n) = (\delta_{\mathrm{cell}}(n), \delta_T(n), \Delta B_{\mathrm{loc}}(n))$$

with

$$\Delta B_{\mathrm{loc}}(n) > 0.$$

Proof. By the balancing-mechanism orientation theorem, any plausible future balancing mechanism must be hybrid rather than purely one-sided. If simultaneous crystal-side and Teichmüller-side improvements occur but do not reduce $|B_{\mathrm{loc}}(n)|$, then no genuine balancing has yet been witnessed; the system remains merely burden-dominant with parallel repairs. Conversely, if the simultaneous side improvements do reduce $|B_{\mathrm{loc}}(n)|$, then the resulting gains define a positive hybrid witness package, and the first honest evidence for a balancing mechanism is present. □

Corollary 24.53 (Current hybrid-balancing working rule)**.** *The next local audit should not merely ask whether crystal-side and Teichmüller-side repair are both visible. It should ask whether they jointly produce*

$$\Delta B_{\mathrm{loc}}(n) > 0,$$

and hence a first genuine hybrid balancing witness package.

Definition 24.54 (Hybrid witness existence obstruction)**.** *A* hybrid witness existence obstruction *at step n is the failure of the current local line to produce any witness package*

$$\mathfrak{W}_{\mathrm{hyb}}(n) = (\delta_{\mathrm{cell}}(n), \delta_T(n), \Delta B_{\mathrm{loc}}(n))$$

with

$$\delta_{\mathrm{cell}}(n) > 0, \qquad \delta_T(n) > 0, \qquad \Delta B_{\mathrm{loc}}(n) > 0.$$

Equivalently, the line remains burden-dominant because no first honest hybrid balancing witness is yet produced.

Definition 24.55 (Weak hybrid efficiency obstruction)**.** *A* weak hybrid efficiency obstruction *at step n is the situation in which a first hybrid balancing witness package exists but its hybrid balancing efficiency ratio*

$$\Xi_{\mathrm{hyb}}(n) = \frac{\Delta B_{\mathrm{loc}}(n)}{\delta_{\mathrm{cell}}(n) + \delta_T(n)}$$

remains too small to alter the current asymptotic burden-dominant orientation.

Remark 24.56 (Existence versus efficiency)**.** *There are therefore two distinct post-obstructions: first, the line may fail to produce any honest hybrid witness at all; second, it may produce such a witness, but only with negligible balancing efficiency. These two failure modes should not be confused.*

Definition 24.57 (Existence-prior local regime)**.** *The current local line is said to be in an* existence-prior local regime *if the first balancing obstruction is of witness-existence type rather than weak-efficiency type.*

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Definition 24.58 (Efficiency-prior local regime)**.** *The current local line is said to be in an* efficiency-prior local regime *if a first hybrid balancing witness is already present, but its efficiency ratio remains too small to change the burden-dominant asymptotic orientation.*

Proposition 24.59 (Asymptotic burden-dominance blocks efficiency-first readings)**.** *Assume the first local asymptotic orientation theorem and the first hybrid balancing witness theorem. If no positive witness package*

$$\mathfrak{W}_{\mathrm{hyb}}(n)$$

with

$$\Delta B_{\mathrm{loc}}(n) > 0$$

is yet available, then the current line cannot honestly be read as efficiency-prior.

Proof. An efficiency-prior reading presupposes that a first hybrid balancing witness already exists and that only its efficiency remains insufficient. If no positive witness package is available at all, then the obstruction occurs before efficiency becomes a meaningful variable. Hence the line cannot honestly be read as efficiency-prior. □

Theorem 24.60 (First hybrid obstruction orientation theorem)**.** *Assume:*

(a) the first local asymptotic orientation theorem applies;

(b) the first local balancing-mechanism orientation theorem applies;

(c) the first hybrid balancing witness theorem identifies positive hybrid witness data as the first honest balancing evidence.

Then the current post-line is obstruction-prior on the witness-existence side. More precisely, its first balancing obstruction is not that an existing hybrid witness is too weak, but that no first positive hybrid balancing witness package has yet been produced.

Proof. By the first local asymptotic orientation theorem, the line remains burden-dominant by default unless a genuine balancing mechanism is exhibited. By the balancing-mechanism orientation theorem, any such genuine mechanism must be hybrid. By the first hybrid balancing witness theorem, the first honest evidence for that mechanism is exactly a positive witness package with $\Delta B_{\mathrm{loc}}(n) > 0$. Therefore, as long as no such package is present, the primary obstruction lies on the witness-existence side rather than on the efficiency side. □

Corollary 24.61 (Current hybrid-balancing obstruction rule)**.** *The next local audit should first ask whether a positive hybrid witness package exists at all. Only after such a package is found should the secondary question of hybrid efficiency be treated as the leading obstruction.*

Proof. Immediate from Theorem 24.60. □

Definition 24.62 (Final positive hybrid construction step)**.** *A* final positive hybrid construction step *at stage n is a local update from step n to $n+1$ such that:*

(a) the burden-side repair is active with

$$\delta_{\mathrm{cell}}(n) > 0;$$

(b) the Teichmüller-side persistence repair is active with

$$\delta_T(n) > 0;$$

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(c) both repairs act on the same local closure-phase window;

(d) the updated local balance defect satisfies

$$B_{\mathrm{loc}}(n+1) > B_{\mathrm{loc}}(n).$$

Remark 24.63 (Why the construction is phrased this way)**.** *The current line does not yet justify assuming that a positive hybrid witness appears automatically once both repair channels are nonzero. The real construction requirement is stronger: the simultaneous crystal-side and Teichmüller-side repairs must improve the local balance defect itself.*

Definition 24.64 (Positive hybrid construction datum)**.** *A* positive hybrid construction datum *at stage n is a quadruple*

$$\mathfrak{C}^{+}_{\mathrm{hyb}}(n) := \big(\delta_{\mathrm{cell}}(n), \delta_T(n), B_{\mathrm{loc}}(n), B_{\mathrm{loc}}(n+1)\big)$$

attached to a final positive hybrid construction step.

Definition 24.65 (Strictly positive hybrid construction)**.** *A final positive hybrid construction step is called* strictly positive *if, in addition,*

$$\Delta B_{\mathrm{loc}}(n) := B_{\mathrm{loc}}(n+1) - B_{\mathrm{loc}}(n) > 0.$$

Definition 24.66 (Hybrid construction gate)**.** *The* hybrid construction gate *at stage n is the conjunction of the conditions*

$$\delta_{\mathrm{cell}}(n) > 0, \qquad \delta_T(n) > 0, \qquad \Delta B_{\mathrm{loc}}(n) > 0.$$

Remark 24.67 (Existence obstruction as gate failure)**.** *Under the current obstruction-oriented reading, the missing positive hybrid witness is equivalently the failure of the hybrid construction gate. This makes the existence problem testable in the same local language as the later efficiency problem.*

Proposition 24.68 (Positive hybrid witness construction implies witness existence)**.** *Every strictly positive hybrid construction step produces a positive hybrid witness package*

$$\mathfrak{W}_{\mathrm{hyb}}(n) = \big(\delta_{\mathrm{cell}}(n), \delta_T(n), \Delta B_{\mathrm{loc}}(n)\big)$$

with

$$\delta_{\mathrm{cell}}(n) > 0, \qquad \delta_T(n) > 0, \qquad \Delta B_{\mathrm{loc}}(n) > 0.$$

Proof. By definition, a strictly positive hybrid construction step already supplies positive crystal-side repair, positive Teichmüller-side repair, and a strictly positive balance-defect gain. These are exactly the components required for a positive hybrid witness package. □

Proposition 24.69 (The current existence obstruction is equivalent to absence of a strictly positive hybrid construction)**.** *Assume the first hybrid obstruction orientation theorem. Then the current line lies in the existence-prior regime if and only if no strictly positive hybrid construction step is yet available on the active local window.*

Proof. By the first hybrid obstruction orientation theorem, the current primary obstruction is the absence of a positive hybrid witness package. By Proposition 24.68, every strictly positive hybrid construction step would already produce such a package. Hence the existence-prior obstruction holds exactly when no such construction step is yet available. □

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Theorem 24.70 (First positive hybrid witness construction theorem)**.** *Assume:*

(a) the first hybrid obstruction orientation theorem applies;

(b) the local crystal-side correction datum is available;

(c) the local Teichmüller-side repair datum is available;

(d) both data act on the same active closure-phase window.

Then the first honest transition out of the existence-prior regime occurs exactly when a strictly positive hybrid construction step is produced. Equivalently, the first positive hybrid balancing witness arises exactly when the hybrid construction gate is opened on the active local window.

Proof. By the first hybrid obstruction orientation theorem, the current line remains stuck before the efficiency question precisely because no positive hybrid witness yet exists. By Proposition 24.68, every strictly positive hybrid construction step produces such a witness. Conversely, on the active local window a first positive witness must come from simultaneous positive crystal-side and Teichmüller-side repair that genuinely improves the local balance defect, which is exactly the content of a strictly positive hybrid construction step. Hence exiting the existence-prior regime is equivalent to producing such a step, or equivalently to opening the hybrid construction gate. □

Corollary 24.71 (Current hybrid-construction working rule)**.** *The next local audit should not yet optimize hybrid efficiency. It should first test whether the active local crystal-side and Teichmüller-side repairs can be synchronized into a strictly positive hybrid construction step with*

$$\Delta B_{\mathrm{loc}}(n) > 0.$$

Proof. Immediate from Theorem 24.70. □

Definition 24.72 (Crystal-side gate obstruction)**.** *A* crystal-side gate obstruction *at stage n is the failure of the hybrid construction gate caused primarily by the crystal-side component, i.e. by one of the following:*

(a)

$$\delta_{\mathrm{cell}}(n) \leq 0;$$

(b) the crystal-side correction datum is not available on the active local window;

(c) the crystal-side correction fails to synchronize with the active Teichmüller-side repair on the same closure-phase window.

Definition 24.73 (Teichmüller-side gate obstruction)**.** *A* Teichmüller-side gate obstruction *at stage n is the failure of the hybrid construction gate caused primarily by the geometric persistence side, i.e. by one of the following:*

(a)

$$\delta_T(n) \leq 0;$$

(b) the Teichmüller-side repair datum is not available on the active local window;

(c) the geometric repair remains too weak to support

$$\Delta B_{\mathrm{loc}}(n) > 0$$

even when the crystal-side correction is already positive.

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Definition 24.74 (Gate-side desynchronization defect)**.** *The* gate-side desynchronization defect *at stage n is the obstruction datum*

$$\mathfrak{D}_{\mathrm{gate}}(n) := \big(\delta_{\mathrm{cell}}(n), \delta_T(n), \Delta B_{\mathrm{loc}}(n)\big)$$

read under the question which side first fails to keep the hybrid construction gate open.

Remark 24.75 (What "first out of sync" means)**.** *The present question is not yet one of hybrid efficiency. It asks which side fails first at the level of gate opening: the crystal-side correction, the Teichmüller-side persistence repair, or the synchronization of both on the same local closure-phase window.*

Proposition 24.76 (The current line is not crystal-obstruction-prior)**.** *Under the current post-archival stage reading, the active local line is not primarily blocked by absence of crystal-side correction.*

Proof. The local crystal-equation correction theorem and the compact reinforcement line already support a positive crystal-side correction datum on the active local window. The remaining obstruction therefore does not first arise from the total absence of the crystal-side contribution. □

Proposition 24.77 (The current line is Teichmüller-side-prior at the gate)**.** *Under the current post-archival stage reading, the first gate-side desynchronization is read primarily on the Teichmüller-side persistence channel.*

Proof. The tightened-package plausibility reading and the subsequent Teichmüller-persistence repair line already isolate the residual insufficiency on the geometric persistence side after the first burden-side repair has been introduced. The hybrid-construction obstruction then remains because a positive crystal-side correction does not yet suffice to produce

$$\Delta B_{\mathrm{loc}}(n) > 0.$$

Hence the first gate-side desynchronization is read primarily on the Teichmüller-side channel rather than on the crystal side. □

Theorem 24.78 (First gate-side desynchronization orientation theorem)**.** *Assume:*

(a) the first positive hybrid witness construction theorem applies;

(b) the local crystal-equation correction theorem applies;

(c) the first local Teichmüller-persistence repair theorem applies;

(d) the current line still lies in the existence-prior hybrid obstruction regime.

Then the current gate failure is Teichmüller-side-prior.
More precisely:

(i) the crystal-side correction is not the first component to fall out of sync;

(ii) the primary remaining defect is the insufficient geometric persistence contribution needed to turn the positive local corrections into

$$\Delta B_{\mathrm{loc}}(n) > 0;$$

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(iii) the next local construction step should therefore strengthen the Teichmüller-side synchronization before attempting any efficiency optimization.

Proof. By Proposition 24.76, the crystal-side correction is already locally present and is not the primary missing ingredient. By Proposition 24.77, the residual desynchronization sits on the Teichmüller-side persistence channel. Since the hybrid witness still does not yet exist, the active local task is not efficiency optimization but restoration of sufficient geometric persistence to open the gate. □

Corollary 24.79 (Current gate-side working rule)**.** *The next local audit should first ask how the Teichmüller-side repair must be strengthened or synchronized so that the already present crystal-side correction becomes hybrid-gate-effective. Only after that should the line pass to hybrid efficiency questions.*

Proof. Immediate from Theorem 24.78. □

Definition 24.80 (Phase-lock synchronization defect)**.** *A* phase-lock synchronization defect *at stage n is the failure of the second-step local repair to keep the phase-side burden correction aligned with the same closure-phase window on which the first-step near-trigger state was read.*

Definition 24.81 (Teichmüller-carry synchronization defect)**.** *A* Teichmüller-carry synchronization defect *at stage n is the failure of the geometric persistence correction to carry the repaired second step inside the same local Teichmüller-safe band as the first step.*

Definition 24.82 (Local Teichmüller-side synchronization datum)**.** *The* local Teichmüller-side synchronization datum *at stage n is*

$$\mathfrak{S}_T^{\mathrm{loc}}(n) := \big(\sigma_\Theta^{\mathrm{lock}}(n), \sigma_T^{\mathrm{carry}}(n)\big),$$

where:

(a) $\sigma_\Theta^{\mathrm{lock}}(n)$ measures the remaining phase-lock misalignment of the repaired second step;

(b) $\sigma_T^{\mathrm{carry}}(n)$ measures the remaining geometric carry defect on the same two-step local window.

Remark 24.83 (What synchronization means locally)**.** *The present question is no longer whether geometry matters at all. It asks more precisely which kind of Teichmüller-side synchronization is still missing: the phase-side lock itself, the geometric carry of that lock, or both together.*

Definition 24.84 (Locally synchronized repaired second step)**.** *A repaired second step at stage n is called* locally synchronized *if*

$$\sigma_\Theta^{\mathrm{lock}}(n) = 0 \qquad \textit{and} \qquad \sigma_T^{\mathrm{carry}}(n) = 0.$$

Proposition 24.85 (The current line is not raw-geometry-prior)**.** *Under the current post-archival stage reading, the active local line is not primarily blocked by raw geometric carry alone.*

Proof. The current line already carries a positive Teichmüller-side repair contribution and a positive geometric witness contribution. The residual failure therefore cannot be read first as total absence of geometry, but only as insufficient synchronization of that geometric contribution with the repaired phase-side step. □

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Proposition 24.86 (The current line is phase-lock-prior on the Teichmüller side)**.** *Under the current post-archival stage reading, the first remaining Teichmüller-side synchronization defect is phase-lock-prior.*

Proof. The gate-side desynchronization orientation theorem already identifies the first remaining gate defect as Teichmüller-side-prior rather than crystal-side-prior. The local repair-orientation and tightened-package plausibility lines then show that the geometric channel is active but still fails to hold the repaired second step in the same closure-phase window. Hence the first remaining defect is best read as insufficient phase-lock synchronization, with geometric carry acting as the witness and support channel. □

Theorem 24.87 (First Teichmüller-side synchronization orientation theorem)**.** *Assume:*

(a) the first local Theta-drift witness theorem applies;

(b) the first local phase-side repair theorem applies;

(c) the first local Teichmüller-persistence repair theorem applies;

(d) the first gate-side desynchronization orientation theorem applies.

Then the first remaining Teichmüller-side synchronization problem is phase-lock-prior rather than raw-carry-prior.

More precisely:

(i) the missing synchronization is not first the total absence of geometric carry;

(ii) the missing synchronization is first the failure to keep the repaired second step phase-locked inside the same closure-phase window;

(iii) the geometric carry defect remains genuinely active, but as the supporting witness of the phase-lock failure rather than as the primary independent obstruction.

Proof. By Proposition 24.85, the active line is not blocked first by total absence of geometry. By Proposition 24.86, the first remaining synchronization failure is read on the phase-lock component of the repaired second step. Therefore the current Teichmüller-side synchronization problem is phase-lock-prior, with the geometric carry defect remaining real but subordinate. □

Corollary 24.88 (Current Teichmüller-side synchronization working rule)**.** *The next local audit should not ask first for more geometry in the abstract. It should ask first how the repaired second step can be phase-locked back into the same closure-phase window, while using the geometric channel to certify and carry that repaired lock.*

Proof. Immediate from Theorem 24.87. □

Definition 24.89 (Local phase-lock repair datum)**.** *The* local phase-lock repair datum *at stage n is*

$$\mathfrak{R}^{\mathrm{loc}}_{\mathrm{lock}}(n) := \big(\ell^{\mathrm{lock}}_{\Theta}(n), \ell^{\mathrm{lock}}_{T}(n)\big),$$

where $\ell^{\mathrm{lock}}_{\Theta}(n)$ denotes the extra phase-side lock correction required to remove the residual phase-lock defect and $\ell^{\mathrm{lock}}_{T}(n)$ denotes the extra Teichmüller-side carry needed to stabilize that repaired lock on the same two-step window.

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Definition 24.90 (Phase-lock repaired local window)**.** *A repaired two-step local window at stage n is called* phase-lock repaired *if, after applying the local phase-lock repair datum, one has*

$$\sigma_{\Theta}^{\mathrm{lock}}(n) = 0 \qquad \text{and} \qquad \sigma_{T}^{\mathrm{carry}}(n) \le \ell_{T}^{\mathrm{lock}}(n).$$

Equivalently, the repaired second step is brought back into the same active closure-phase window while the remaining geometric carry defect is absorbed by the stated lock-carry allowance.

Definition 24.91 (Minimal local phase-lock repair)**.** *A local phase-lock repair datum*

$$\mathfrak{R}_{\mathrm{lock}}^{\mathrm{loc}}(n) = \big(\ell_{\Theta}^{\mathrm{lock}}(n), \ell_{T}^{\mathrm{lock}}(n)\big)$$

is called minimal *if no strictly smaller pair of nonnegative corrections removes the phase-lock synchronization defect on the same two-step local window.*

Remark 24.92 (What the phase-lock repair really does)**.** *The phase-lock repair is not a fresh bulk law. It is the smallest local correction that forces the repaired second step back onto the same closure-phase slice as the first near-trigger step. Geometrically, one may think of it as a local re-aiming plus carry-stabilization of the second point so that both points lie again in the same admissible narrow corridor rather than in neighboring but misaligned corridors.*

Lemma 24.93 (First local phase-lock repair lemma)**.** *Assume:*

(a) the first Teichmüller-side synchronization orientation theorem applies;

(b) the active local obstruction is phase-lock-prior;

(c) a local phase-lock repair datum exists.

Then the vanishing condition

$$\sigma_{\Theta}^{\mathrm{lock}}(n) = 0$$

is the first honest local repair target for reopening the hybrid construction gate.

Proof. By Theorem 24.87, the first remaining synchronization obstruction is read on the phase-lock component rather than on raw geometric absence. Hence the first honest repair target is to remove the phase-lock defect itself. The accompanying carry correction remains necessary, but it is subordinate to the lock restoration as long as the active obstruction is phase-lock-prior. □

Proposition 24.94 (Current line supports phase-lock-prior local repair)**.** *Under the present post-archival stage reading, the next local strengthening step should first reduce $\sigma_{\Theta}^{\mathrm{lock}}(n)$ and only then optimize the associated geometric carry.*

Proof. This is just the operational form of the phase-lock-prior diagnosis. Since the line is not raw-geometry-prior, one does not first optimize geometric amplitude in the abstract. One first restores alignment of the repaired second step with the active closure-phase window, and then uses the Teichmüller-side correction to support that restored lock. □

Theorem 24.95 (First phase-lock repair theorem)**.** *Assume:*

(a) the first local Theta-drift witness theorem applies;

(b) the first local phase-side repair theorem applies;

(c) the first local Teichmüller-persistence repair theorem applies;

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(d) the first Teichmüller-side synchronization orientation theorem applies;

(e) a minimal local phase-lock repair datum exists.

Then the current post-archival stage line supports a first concrete phase-lock repair final. More precisely:

(i) the primary repair target is

$$\sigma_{\Theta}^{\mathrm{lock}}(n) \to 0;$$

(ii) the geometric carry repair remains genuinely active through

$$\ell_{T}^{\mathrm{lock}}(n) > 0$$

whenever pure phase-side re-alignment does not by itself stabilize the repaired second step;

(iii) a phase-lock repaired local window is the first local final for reopening the hybrid construction gate.

Proof. Items (a)–(d) identify the present obstruction chain: the first failure is Theta-drift-prior, the first repair is burden-side with geometric carry, the next residual obstruction is Teichmüller-side-prior, and that obstruction is phase-lock-prior rather than raw-geometry-prior. Therefore the first concrete local repair must target vanishing of $\sigma_{\Theta}^{\mathrm{lock}}(n)$. If pure re-alignment is not enough to keep the repaired second step in the same local Teichmüller-safe band, a positive carry correction $\ell_{T}^{\mathrm{lock}}(n)$ remains necessary. This yields the first concrete phase-lock repair final. □

Corollary 24.96 (Current phase-lock repair working rule)**.** *The next local audit should no longer ask merely whether the repaired second step is better than before. It should ask whether the applied local correction actually forces*

$$\sigma_{\Theta}^{\mathrm{lock}}(n) = 0$$

and, if not, which remaining carry correction still prevents a fully phase-locked repaired window.

Proof. Immediate from Theorem 24.95. □

Definition 24.97 (Local Heisenberg compensator)**.** *A* local Heisenberg compensator *at stage n is a conservative local synchronization operator*

$$\mathrm{HC}_{\mathrm{loc}}(n) : (X_n, X_{n+1}) \longmapsto (X_n, \widetilde{X}_{n+1})$$

which acts only on the repaired second step and is intended to reduce the remaining phase-lock and Teichmüller-carry defects without changing the already accepted first-step near-trigger reading.

Definition 24.98 (HC-admissible local compensation)**.** *A local Heisenberg compensator is called* HC-admissible *if it satisfies:*

(a) the closure burden remains controlled;

(b) the corrected second step stays inside the same admissibility band;

(c) the phase-lock defect decreases:

$$\sigma_{\Theta}^{\mathrm{lock}}(n; \widetilde{X}_{n+1}) < \sigma_{\Theta}^{\mathrm{lock}}(n; X_{n+1});$$

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(d) the Teichmüller-carry defect does not increase, and preferably also decreases;

(e) the compensator does not manufacture a false entry certificate by destroying the route-faithful burden reading.

Remark 24.99 (How the HC is read here)**.** *The Heisenberg compensator is not used here as a new bulk law. It is read as the minimal local re-phasing operator that tries to snap the repaired second step back into the same narrow closure-phase corridor as the first near-trigger step.*

Definition 24.100 (Bombe-guided compensator menu)**.** *A* Bombe-guided compensator menu *at stage n is a finite family*

$$\mathcal{M}_{\mathrm{HC}}(n) = \{H_1, \dots, H_r\}$$

of final local Heisenberg compensators together with a contradiction-sensitive elimination rule that removes every final violating closure control, admissibility, or route-faithful burden preservation.

Remark 24.101 (Why the Bombe idea fits here)**.** *The present problem is not to guess one magical compensator. It is to eliminate locally incompatible synchronizers from a finite final menu until only the route-faithful phase-lock-capable ones remain. This is exactly the right local role for a Bombe-style elimination discipline.*

Definition 24.102 (Compensator filter tower)**.** *A* compensator filter tower *is a finite ordered stack*

$$\mathfrak{F}_{\mathrm{HC}}(n) = (F_1, \dots, F_s)$$

of local HC-compatible filters through which the repaired second step is passed before the final trigger test. The intended reading is:

(i) burden-side pre-alignment;

(ii) phase-lock re-alignment;

(iii) Teichmüller-carry stabilization;

(iv) final closure-phase re-entry test.

Remark 24.103 (Filter tower and the electromagnetic analogy)**.** *The filter-tower language should be read structurally rather than as a literal new electrodynamics. It expresses that several local synchronization layers may act in series, each removing one kind of phase/carry mismatch before the final trigger decision is taken.*

Definition 24.104 (Readout-relative informational gravitation principle)**.** *A* readout-relative informational gravitation principle *is the principle that the apparent curvature/attraction acting on a local state is not necessarily a primitive force attached only to rest mass, but may be read as the effective curvature law induced by the chosen admissible readout on the underlying information-carrying structure.*

Remark 24.105 (How far this principle should be claimed)**.** *In the present document this principle is only an internal architecture hypothesis. It does not claim multiple literal physical gravities. It claims instead that one deeper ART/QFT-compatible carrier may have different gravitational-like readout shadows depending on the active admissible information/readout channel.*

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Proposition 24.106 (HC can serve as a local phase-lock operator)**.** *Assume an HC-admissible local compensation exists at stage n. Then the Heisenberg compensator can serve as a local phase-lock operator for the repaired second step.*

Proof. By HC-admissibility, the compensator reduces the phase-lock defect while preserving closure control, admissibility, and the route-faithful burden reading. Therefore it acts exactly as the required local phase-lock operator. □

Lemma 24.107 (Bombe-guided HC elimination lemma)**.** *Assume the compensator final menu is finite. Then Bombe-guided elimination reduces the local HC search to the subfamily of finals that are simultaneously closure-compatible, admissibility-safe, and phase-lock-improving.*

Proof. By definition, Bombe-guided elimination removes every final violating one of the required local compatibility conditions. Hence the surviving menu consists exactly of those finals satisfying the required conditions. □

Theorem 24.108 (First local Heisenberg compensator theorem)**.** *Assume:*

(a) the first phase-lock repair theorem applies;

(b) a finite Bombe-guided compensator menu is available;

(c) at least one surviving final is HC-admissible.

Then the current post-archival stage line supports a first local Heisenberg-compensated phase-lock repair.

More precisely:

(i) the repaired second step can be re-phased by a surviving HC final;

(ii) the corrected step is brought closer to the same closure-phase corridor as the first near-trigger step;

(iii) the compensator acts as a synchronization layer between the local crystal correction and the Teichmüller-side persistence repair.

Proof. By the first phase-lock repair theorem, the current line already isolates the local phase-lock defect as the next concrete obstruction. By the Bombe-guided HC elimination lemma, the finite menu reduces to the finals that are compatible with closure, admissibility, and phase-lock improvement. If one surviving final is HC-admissible, then by Proposition 24.106 it provides the required local re-phasing. Therefore the Heisenberg compensator acts as the missing synchronization layer between the crystal-side correction and the Teichmüller-side persistence repair. □

Corollary 24.109 (Current HC working rule)**.** *The next local audit should not ask first for a new bulk equation. It should ask first for the smallest HC-admissible local compensator that phase-locks the repaired second step back into the same closure-phase corridor while preserving closure control and admissibility.*

Proof. Immediate from Theorem 24.108. □

Definition 24.110 (Phase-side dominant HC final)**.** *A local Heisenberg compensator final at stage n is called* phase-side dominant *if its primary improvement acts on the phase-lock defect, in the sense that it strictly lowers*

$$\sigma_{\Theta}^{\mathrm{lock}}(n)$$

while any improvement of the Teichmüller-carry defect remains subordinate to that phase-lock gain.

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Definition 24.111 (Carry-dominant HC final)**.** *A local Heisenberg compensator final at stage n is called* carry-dominant *if its primary improvement acts on the geometric carry defect, in the sense that it lowers*

$$\sigma_T^{\mathrm{carry}}(n)$$

without first removing the phase-lock misalignment.

Definition 24.112 (Two-component HC final)**.** *A local Heisenberg compensator final at stage n is called* two-component coupled *if it lowers both*

$$\sigma_\Theta^{\mathrm{lock}}(n) \qquad \textit{and} \qquad \sigma_T^{\mathrm{carry}}(n)$$

with neither correction honestly subordinate to the other at first local order.

Remark 24.113 (What the classification is trying to decide)**.** *The present question is not whether a local HC helps at all. It asks which kind of local HC is the first plausible one for the current line: one that primarily restores phase-lock, one that primarily carries geometry, or one that is already fully two-component at first local order.*

Proposition 24.114 (The current line is not carry-dominant on the HC side)**.** *Under the present post-archival stage reading, the first plausible local HC final is not carry-dominant.*

Proof. By the first Teichmüller-side synchronization orientation theorem and the first phase-lock repair theorem, the current residual obstruction is phase-lock-prior rather than raw-carry-prior. Therefore a compensator that tries first to optimize geometric carry without first restoring phase-lock cannot be the default first final. □

Proposition 24.115 (The current line is not yet fully two-component on the HC side)**.** *Under the present post-archival stage reading, the first plausible local HC final is not yet fully two-component at first local order.*

Proof. The current local repair line remains asymptotically burden-dominant and phase-lock-prior. Although the geometric carry correction is genuinely active, the present line has not yet produced a positive hybrid balancing witness with balanced first-order contributions. Hence the first plausible HC final cannot yet be classified as fully two-component at first local order. □

Theorem 24.116 (First HC-final type orientation theorem)**.** *Assume:*

(a) the first local Heisenberg compensator theorem applies;

(b) the first Teichmüller-side synchronization orientation theorem applies;

(c) the first phase-lock repair theorem applies;

(d) the current line has not yet produced a positive hybrid balancing witness.

Then the first plausible local HC final is phase-side dominant rather than carry-dominant or fully two-component.

More precisely:

(i) the primary local HC task is to reduce

$$\sigma_\Theta^{\mathrm{lock}}(n);$$

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(ii) the Teichmüller-side carry correction remains genuinely active as a supporting synchronization channel;

(iii) the current HC final should therefore be read as phase-side dominant but geometrically carried.

Proof. By Proposition 24.114, the current line is not blocked first by raw geometric carry. By Proposition 24.115, the current line has not yet reached a first-order balanced two-component compensator regime. Since the active residual obstruction is phase-lock-prior, the first plausible HC final must therefore act primarily on the phase-lock defect, with geometric carry serving as the supporting channel that stabilizes the repaired lock. □

Corollary 24.117 (Current HC-final working rule)**.** *The next local audit should not search first for a carry-dominant or perfectly balanced HC. It should search first for the smallest phase-side dominant compensator whose geometric carry contribution is just strong enough to keep the repaired second step inside the same closure-phase corridor.*

Proof. Immediate from Theorem 24.116. □

Definition 24.118 (Local lock corridor)**.** *A* local lock corridor *at stage n is a narrow admissible closure-phase band*

$$\mathfrak{C}_{\mathrm{lock}}^{\mathrm{loc}}(n)$$

containing the first near-trigger step and characterized by the following properties:

(a) every point in $\mathfrak{C}_{\mathrm{lock}}^{\mathrm{loc}}(n)$ lies inside the same local closure-phase window;

(b) the local burden pair remains trigger-near throughout the corridor;

(c) the Teichmüller-side distortion stays inside the active safe band;

(d) the corridor is narrow enough that leaving it produces a detectable phase-lock or carry defect.

Remark 24.119 (Geometric meaning of the lock corridor)**.** *The lock corridor is the thin local channel in which both the first near-trigger step and the repaired second step must lie if one is to read them as belonging to the same closure-phase passage. The current obstruction does not come from total loss of that channel, but from the failure of the second step to remain locked inside it.*

Definition 24.120 (HC-captured repaired second step)**.** *A repaired second step at stage n is called* HC-captured *if there exists an HC-admissible local Heisenberg compensator such that the compensated second step lies in the same local lock corridor as the first near-trigger step.*

Definition 24.121 (Lock-corridor leakage defect)**.** *The* lock-corridor leakage defect *at stage n is the residual amount by which the repaired second step fails to remain inside*

$$\mathfrak{C}_{\mathrm{lock}}^{\mathrm{loc}}(n).$$

It is denoted by

$$\lambda_{\mathrm{lock}}(n) \geq 0,$$

and vanishes exactly when the repaired second step is HC-captured.

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Definition 24.122 (Minimal local lock-corridor compensation)**.** *A local HC final is called* minimally lock-corridor compensating *if it is HC-admissible and reduces the lock-corridor leakage defect to zero with no strictly smaller phase-side dominant correction achieving the same result on the active two-step local window.*

Proposition 24.123 (Current line still exhibits lock-corridor leakage)**.** *Under the present post-archival stage reading, the repaired second step still exhibits a nonzero lock-corridor leakage defect.*

Proof. The current line is phase-lock-prior on the Teichmüller side and has not yet produced a positive hybrid balancing witness. Hence the repaired second step is not yet fully locked into the same local closure-phase passage as the first near-trigger step, so the local lock-corridor leakage defect remains nonzero. □

Lemma 24.124 (Phase-side dominant HC yields lock-corridor capture)**.** *Assume:*

(a) a local lock corridor exists;

(b) the current HC final is phase-side dominant but geometrically carried;

(c) the compensator reduces the phase-lock defect to zero while keeping admissibility and the geometric carry inside the safe band.

Then the compensated second step is HC-captured.

Proof. By assumption, the compensator restores the phase-lock condition and preserves the geometric safe-band control. Since the local lock corridor is defined exactly as the admissible narrow closure-phase band carrying the first near-trigger step, the compensated second step now lies in the same corridor. Hence it is HC-captured. □

Theorem 24.125 (First explicit lock-corridor theorem)**.** *Assume:*

(a) the first HC-final type orientation theorem applies;

(b) the first phase-lock repair theorem applies;

(c) a local lock corridor exists around the first near-trigger step;

(d) there exists a minimally lock-corridor compensating HC final.

Then the current post-archival stage line admits a first explicit local lock-corridor reading. More precisely:

(i) the geometric problem is not to invent a new bulk path, but to return the repaired second step to the same narrow local closure-phase corridor as the first step;

(ii) the current HC task is to drive

$$\lambda_{\mathrm{lock}}(n) \to 0;$$

(iii) once the leakage defect vanishes under an HC-admissible phase-side dominant compensator, the repaired second step is locally synchronized in the same lock corridor as the first near-trigger step.

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Proof. By the HC-final orientation theorem, the first plausible compensator is phase-side dominant but geometrically carried. By the phase-lock repair theorem, the primary local repair target is vanishing of the phase-lock defect, with geometric carry remaining genuinely active. The local lock corridor isolates the thin admissible passage already occupied by the first near-trigger step. A minimally lock-corridor compensating HC final removes the leakage defect while preserving admissibility, so by Lemma 24.124 the repaired second step is returned to the same corridor. □

Corollary 24.126 (Current lock-corridor working rule)**.** *The next local audit should no longer ask only whether the second step is improved. It should ask whether an HC-admissible compensator has actually reduced*

$$\lambda_{\mathrm{lock}}(n)$$

to zero, i.e. whether the repaired second step has truly re-entered the same local lock corridor as the first near-trigger step.

Proof. Immediate from Theorem 24.125. □

Definition 24.127 (Lock-side leakage defect)**.** *A* lock-side leakage defect *at stage n is the portion of the local leakage defect attributable to failure of the repaired second step to satisfy the phase-lock condition under the current corridor calibration. It is denoted by*

$$\lambda^{\Theta}_{\mathrm{lock}}(n) \geq 0.$$

Definition 24.128 (Corridor-calibration leakage defect)**.** *A* corridor-calibration leakage defect *at stage n is the portion of the local leakage defect attributable to the active local lock corridor being too narrow or otherwise miscalibrated for the repaired second step, even after the primary phase-lock correction has been applied. It is denoted by*

$$\lambda^{\mathrm{cal}}_{\mathrm{lock}}(n) \geq 0.$$

Remark 24.129 (How the leakage splits)**.** *The total local lock-corridor leakage is now read as having two possible sources: a remaining failure of lock restoration itself, or a failure of the current corridor calibration to reflect the true admissible narrow passage. In symbols, one may think heuristically of*

$$\lambda_{\mathrm{lock}}(n) \sim \lambda^{\Theta}_{\mathrm{lock}}(n) + \lambda^{\mathrm{cal}}_{\mathrm{lock}}(n),$$

without yet forcing a rigid additive identity.

Definition 24.130 (Lock-side-prior local leakage regime)**.** *The local line is said to be in a* lock-side-prior local leakage regime *at stage n if*

$$\lambda^{\Theta}_{\mathrm{lock}}(n) > \lambda^{\mathrm{cal}}_{\mathrm{lock}}(n).$$

Definition 24.131 (Corridor-calibration-prior local leakage regime)**.** *The local line is said to be in a* corridor-calibration-prior local leakage regime *at stage n if*

$$\lambda^{\mathrm{cal}}_{\mathrm{lock}}(n) \geq \lambda^{\Theta}_{\mathrm{lock}}(n).$$

Proposition 24.132 (The current line is not corridor-calibration-prior by default)**.** *Under the present post-archival stage reading, the local line should not yet be read as corridor-calibration-prior by default.*

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Proof. The current diagnosis remains phase-lock-prior on the Teichmüller side, and the first local repair target is still the vanishing of the phase-lock defect rather than widening or re-centering the lock corridor itself. Hence the default reading is not that the corridor is already known to be primarily miscalibrated, but rather that the repaired second step still fails first to lock into the active corridor. □

Proposition 24.133 (The current line is lock-side-prior on the leakage level)**.** *Under the present post-archival stage reading, the first remaining local lock-corridor leakage is lock-side-prior.*

Proof. By the first phase-lock repair theorem and the current phase-lock repair working rule, the next local task is still to force

$$\sigma_{\Theta}^{\mathrm{lock}}(n) = 0.$$

This means that the dominant unresolved leakage continues to arise from incomplete phase-lock restoration rather than from demonstrated over-narrowness or miscentering of the corridor itself. Therefore the first remaining local leakage is lock-side-prior. □

Theorem 24.134 (First lock-corridor leakage orientation theorem)**.** *Assume:*

(a) the first explicit lock-corridor theorem applies;

(b) the first phase-lock repair theorem applies;

(c) the current line still exhibits nonzero lock-corridor leakage.

Then the current post-archival stage line is lock-side-prior on the leakage level rather than corridor-calibration-prior.

More precisely:

(i) the next honest reduction target is first

$$\lambda_{\mathrm{lock}}^{\Theta}(n) \downarrow;$$

(ii) local corridor recalibration should be treated only as a secondary follow-on question unless the lock-side leakage has already been made negligible;

(iii) the present line should therefore first seek a stronger HC lock action before widening or re-centering the local corridor.

Proof. By the first explicit lock-corridor theorem, the active local task is to return the repaired second step to the same narrow corridor as the first near-trigger step. By the first phase-lock repair theorem, the primary local repair target remains vanishing of the phase-lock defect. Therefore, as long as nonzero leakage remains, the default reading is that the dominant leakage sits on the lock-restoration side rather than already on the calibration side of the corridor. Hence the line is lock-side-prior on the leakage level. □

Corollary 24.135 (Current leakage working rule)**.** *The next local audit should not first widen, thicken, or re-center the active lock corridor. It should first ask whether the remaining leakage is still produced by incomplete lock restoration of the repaired second step under the current corridor geometry.*

Proof. Immediate from Theorem 24.134. □

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Definition 24.136 (Minimal local HC-lock step)**.** *A* minimal local HC-lock step *at stage n is an HC-admissible local compensation*

$$\mathcal{H}_{\mathrm{lock}}^{\min}(n)$$

which acts on the repaired second step only through the smallest phase-side dominant correction required to reduce the lock-side leakage defect from

$$\lambda_{\mathrm{lock}}^{\Theta}(n) > 0$$

to

$$\lambda_{\mathrm{lock}}^{\Theta,+}(n) = 0,$$

while preserving the active local lock corridor geometry.

Definition 24.137 (Lock-neutral corridor preservation)**.** *A local HC-lock step is said to satisfy* lock-neutral corridor preservation *if it does not widen, thicken, or re-center the current local lock corridor*

$$\mathfrak{C}_{\mathrm{lock}}^{\mathrm{loc}}(n),$$

but only changes the repaired second-step placement inside that already fixed corridor geometry.

Definition 24.138 (HC-lock effectiveness defect)**.** *The* HC-lock effectiveness defect *of a local HC compensation at stage n is*

$$E_{\mathrm{HC}}^{\mathrm{lock}}(n) := \lambda_{\mathrm{lock}}^{\Theta,+}(n),$$

that is, the residual lock-side leakage after the local HC action has been applied.

Remark 24.139 (What the minimal HC step is allowed to do)**.** *The minimal HC-lock step is not allowed to solve the problem by changing the corridor itself. It must work only by relocking the repaired second step inside the already accepted corridor. Geometrically, it is the smallest admissible local twist that makes the second point snap back into the same narrow channel without moving the channel walls.*

Definition 24.140 (HC-captured local repaired second step)**.** *A repaired second step at stage n is called* HC-captured *if there exists an HC-admissible local compensation satisfying lock-neutral corridor preservation such that*

$$\lambda_{\mathrm{lock}}^{\Theta,+}(n) = 0.$$

Proposition 24.141 (The current line calls first for a minimal HC-lock step)**.** *Under the current post-archival stage reading, the next local correction should first be sought as a minimal local HC-lock step rather than as a corridor recalibration.*

Proof. By the first lock-corridor leakage orientation theorem, the active leakage is lock-side-prior rather than corridor-calibration-prior. Therefore the first honest correction should try to eliminate the residual lock-side leakage while preserving the existing corridor geometry. This is exactly the role of a minimal local HC-lock step. □

Lemma 24.142 (Minimal HC-lock step implies HC capture)**.** *Assume a minimal local HC-lock step exists at stage n. Then the repaired second step is HC-captured at that stage.*

Proof. By definition, a minimal local HC-lock step is HC-admissible, preserves the corridor geometry, and reduces the residual lock-side leakage defect to zero. Hence the repaired second step satisfies the defining conditions of an HC-captured local repaired second step. □

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Theorem 24.143 (First minimal HC-lock step theorem)**.** *Assume:*

(a) the first local Theta-drift witness theorem applies;

(b) the first phase-lock repair theorem applies;

(c) the first explicit lock-corridor theorem applies;

(d) the first lock-corridor leakage orientation theorem applies;

(e) an HC-admissible local compensation menu is available.

Then the current post-archival stage line admits the following first local repair target:

(i) search first for a minimal local HC-lock step;

(ii) require lock-neutral corridor preservation;

(iii) accept the first final only if it yields

$$E_{\mathrm{HC}}^{\mathrm{lock}}(n) = 0;$$

(iv) treat any remaining failure first as lack of sufficient HC lock action rather than as corridor miscalibration.

Proof. Items (a)–(d) identify the current obstruction chain: the line is phase-lock-prior, the local corridor is already the right object to preserve, and the leakage is lock-side-prior rather than corridor-calibration-prior. Hence the first honest local HC task is not to redesign the corridor but to search within the available HC menu for the smallest admissible lock action that drives the residual lock-side leakage to zero while leaving the corridor fixed. This is exactly the asserted minimal HC-lock step search. □

Corollary 24.144 (Current minimal HC working rule)**.** *The next local audit should not first modify the local lock corridor. It should first scan the local HC menu for the smallest phase-side dominant compensator whose geometric carry is just sufficient to force*

$$\lambda_{\mathrm{lock}}^{\Theta,+}(n) = 0$$

under lock-neutral corridor preservation.

Proof. Immediate from Theorem 24.143. □

Definition 24.145 (Local HC final family)**.** *The* local HC final family *at stage n is a finite admissible menu*

$$\mathfrak{M}_{\mathrm{HC}}^{\mathrm{loc}}(n) = \{H_1^{(n)}, \dots, H_{N(n)}^{(n)}\}$$

of local Heisenberg-compensator actions acting on the repaired second step under the fixed local lock corridor.

Definition 24.146 (HC menu admissibility filter)**.** *A local HC final*

$$H_j^{(n)} \in \mathfrak{M}_{\mathrm{HC}}^{\mathrm{loc}}(n)$$

passes the HC menu admissibility filter *if:*

(a) it is phase-side dominant with genuinely active geometric carry;

(b) it preserves the active local lock corridor;

(c) it preserves local admissibility on the same two-step window;

(d) it does not enlarge the corridor merely to hide the residual leakage.

Definition 24.147 (HC lock score)**.** *The* HC lock score *of an admissible local HC final is the pair*

$$\mathfrak{L}_{\mathrm{HC}}(H_j^{(n)}) := \big(\lambda_{\mathrm{lock}}^{\Theta,+}(n; H_j^{(n)}), C_{\mathrm{HC}}(H_j^{(n)})\big),$$

where the first component is the residual lock-side leakage after applying the final and the second component is its local compensator cost.

Definition 24.148 (Bombe-guided HC elimination)**.** *The* bombe-guided HC elimination *is the finite contradiction-sensitive elimination of all finals in*

$$\mathfrak{M}_{\mathrm{HC}}^{\mathrm{loc}}(n)$$

that fail the HC menu admissibility filter or leave a residual lock score incompatible with minimal HC capture.

Remark 24.149 (Why the HC scan is finite)**.** *The present line does not search over an arbitrary continuum of compensators. It searches only over the finite local HC menu already licensed by the current route-faithful readout state. This is why the bombe-guided elimination picture is appropriate here.*

Definition 24.150 (Minimal surviving HC final)**.** *A final*

$$H_{\min}^{(n)} \in \mathfrak{M}_{\mathrm{HC}}^{\mathrm{loc}}(n)$$

is called the minimal surviving HC final *if:*

(a) it survives bombe-guided HC elimination;

(b) it minimizes the residual lock-side leakage

$$\lambda_{\mathrm{lock}}^{\Theta,+}(n; H_j^{(n)});$$

(c) among all finals with the same minimal leakage, it minimizes the compensator cost

$$C_{\mathrm{HC}}(H_j^{(n)}).$$

Proposition 24.151 (The current line reduces the HC problem to a finite menu scan)**.** *Under the current post-archival stage reading, the first local HC problem is reduced to scanning the finite local HC final family under admissibility and lock-corridor preservation.*

Proof. By the current minimal HC working rule, the line must first search for a minimal local HC-lock step while preserving the active lock corridor. By the local Heisenberg-compensator theorem and the bombe-guided compensator menu reading, the admissible compensator actions are already organized as a finite local menu. Hence the first local HC problem is reduced to a finite menu scan with admissibility and corridor-preservation constraints. □

Lemma 24.152 (Minimal surviving HC final yields the first executable HC-lock step)**.** *Assume a minimal surviving HC final exists. Then it yields the first executable local HC-lock step under the current lock-side-prior regime.*

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Proof. A minimal surviving HC final has already passed the admissibility and corridor-preservation filter. By minimality, it leaves no smaller admissible compensator with strictly lower residual lock-side leakage. Hence it is the first executable local HC-lock step compatible with the current regime. □

Theorem 24.153 (First local HC menu-scan theorem)**.** *Assume:*

(a) the first minimal HC-lock step theorem applies;

(b) the current line is lock-side-prior on the leakage level;

(c) the local HC final family is finite;

(d) bombe-guided HC elimination is available.

Then the next local HC task is a finite menu-scan problem.
More precisely:

(i) all inadmissible or corridor-breaking HC finals may be eliminated first;

(ii) the surviving finals are ranked by residual lock-side leakage and compensator cost;

(iii) if a minimal surviving HC final exists, it determines the first executable HC-lock step;

(iv) if no such final exists, the current line records a local HC-menu exhaustion obstruction rather than immediately blaming corridor calibration.

Proof. By Proposition 24.151, the present HC problem is already reduced to a finite final family with admissibility and corridor-preservation constraints. Bombe-guided HC elimination removes finals that fail these constraints. The remaining finals are compared by lock score. If a minimal surviving final exists, Lemma 24.152 yields the first executable HC-lock step. If not, the failure lies at the menu level itself, hence the correct reading is local HC-menu exhaustion rather than premature corridor redesign. □

Corollary 24.154 (Current HC menu-scan working rule)**.** *The next local audit should first enumerate the finite HC menu, eliminate inadmissible and corridor-breaking finals, and only then ask whether the surviving menu still contains a minimal lock-effective compensator.*

Proof. Immediate from Theorem 24.153. □

Definition 24.155 (Phase-side micro-lock HC final)**.** *A local HC final*

$$H^{(1)}_{\mathrm{lock}}(n) \in \mathfrak{M}^{\mathrm{loc}}_{\mathrm{HC}}(n)$$

is called a phase-side micro-lock HC final *if it satisfies:*

(a) phase-side dominance in the sense that its primary action falls on the residual lock component;

(b) lock-neutral corridor preservation;

(c) nonzero but minimal geometric carry sufficient to support the repaired second step inside the same local lock corridor;

(d) no extra crystal-side widening or recentering action.

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Remark 24.156 (Why this is the first plausible HC menu entry)**.** *The present line is phase-lock-prior, lock-side-prior on the leakage level, and not carry-dominant. Hence the first plausible HC final is not a broad geometric carrier and not a corridor redesign. It is the smallest local phase-side micro-lock action whose geometric carry is only as large as needed to prevent renewed leakage.*

Definition 24.157 (HC micro-lock score)**.** *The* HC micro-lock score *of a phase-side micro-lock final*

$$H^{(1)}_{\mathrm{lock}}(n)$$

is the tuple

$$\mathfrak{L}^{\mathrm{micro}}_{\mathrm{HC}}(H^{(1)}_{\mathrm{lock}}(n)) := \big(\lambda^{\Theta,+}_{\mathrm{lock}}(n), c_{\mathrm{HC}}(H^{(1)}_{\mathrm{lock}}(n))\big),$$

ordered lexicographically by residual lock-side leakage first and compensator cost second.

Definition 24.158 (Plausible surviving HC final)**.** *A phase-side micro-lock HC final is called* plausibly surviving *if it passes the HC menu admissibility filter, preserves the local lock corridor, and yields no worse lock score than any other currently known phase-side dominant final.*

Proposition 24.159 (The current line favors a phase-side micro-lock final)**.** *Under the present post-archival stage reading, the first plausible HC menu entry is phase-side micro-lock rather than carry-dominant or fully two-component balanced.*

Proof. By the first HC-final type orientation theorem, the current line is phase-side dominant on the HC level and is neither carry-dominant nor already genuinely two-component balanced. By the first lock-corridor leakage orientation theorem, the residual leakage is lock-side-prior rather than corridor-calibration-prior. Therefore the first plausible HC entry is the smallest phase-side dominant final that closes the lock-side leakage while preserving the existing corridor. □

Lemma 24.160 (Plausible survival of the phase-side micro-lock final)**.** *Assume:*

(a) the local HC final family is finite;

(b) the current line is phase-side dominant on the HC level;

(c) lock-neutral corridor preservation is enforced.

Then the first plausible surviving HC final, if it exists at all, may be taken from the phase-side micro-lock class.

Proof. Because corridor redesign is excluded and carry-dominant finals are not favored by the current orientation, the admissible search is restricted first to phase-side dominant finals. Within that class, the lexicographic HC micro-lock score selects the finals with smallest residual leakage and then smallest cost. Hence any first plausible survivor, if one exists, may be taken from the phase-side micro-lock class. □

Theorem 24.161 (First plausible HC menu-entry theorem)**.** *Assume:*

(a) the first HC-final type orientation theorem applies;

(b) the first lock-corridor leakage orientation theorem applies;

(c) the first local HC menu-scan theorem applies.

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Then the first plausible HC menu entry is a phase-side micro-lock HC final.
More precisely:

(i) the final is sought first in the phase-side dominant subclass;

(ii) its geometric carry is only auxiliary but genuinely active;

(iii) its admissibility is tested under lock-neutral corridor preservation;

(iv) if a first plausible survivor exists, it determines the leading local HC hypothesis before any broader compensator class is tried.

Proof. Items (a) and (b) imply that the line is not looking first for a carry-dominant or fully balanced HC final. Item (c) reduces the search to the finite local HC menu with corridor-preserving admissibility. By Lemma 24.160, any first plausible survivor may be taken from the phase-side micro-lock class. This yields the asserted first plausible HC menu entry. □

Corollary 24.162 (Current HC entry working rule)**.** *The next local audit should first test the phase-side micro-lock HC finals in the finite menu, ranked by residual lock-side leakage and cost, before considering broader compensator types.*

Proof. Immediate from Theorem 24.161. □

Definition 24.163 (Micro-lock update rule)**.** *Let*

$$H^{(1)}_{\mathrm{lock}}(n)$$

be a phase-side micro-lock HC final acting on the repaired second local step. The micro-lock update rule *is the local update*

$$X^{\mathrm{rep}}_{n+1} \longmapsto X^{\mathrm{lock}}_{n+1}$$

induced by $H^{(1)}_{\mathrm{lock}}(n)$ under the following constraints:

(a) lock-neutral corridor preservation is maintained;

(b) the updated second step stays inside the same local admissibility band;

(c) the lock-side leakage does not increase;

(d) the phase-side component of the update is dominant, while the geometric carry remains auxiliary but genuinely active.

Definition 24.164 (Successful micro-lock update)**.** *A micro-lock update is called* successful *if it satisfies*

$$\lambda^{\Theta,+}_{\mathrm{lock}}(n) = 0$$

while preserving the current local lock corridor and the active admissibility constraints.

Definition 24.165 (HC-captured local continuation)**.** *A local continuation after stage n is called* HC-captured *if it is produced by a successful micro-lock update and the resulting repaired second step remains in the same local lock corridor as the first near-trigger step.*

Remark 24.166 (Why capture is the right next target)**.** *The present line no longer needs a broad compensator search. It needs the first concrete capture event in which the second repaired step is actually brought back into the same narrow local corridor. That is the smallest nontrivial success criterion beyond plausible HC menu survival.*

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Lemma 24.167 (Minimal HC-lock step yields local capture)**.** *Assume:*

(a) a minimal surviving HC final exists in the local menu;

(b) its associated micro-lock update is successful;

(c) lock-neutral corridor preservation is enforced.

Then the updated repaired second step is HC-captured.

Proof. By success of the micro-lock update, the residual lock-side leakage is driven to zero. By lock-neutral corridor preservation, the active corridor is not widened or re-centered during the update. Hence the updated second step lies in the same local lock corridor as the first near-trigger step, which is exactly HC capture. □

Definition 24.168 (First local HC-capture chain)**.** *A* first local HC-capture chain *is a finite local sequence*

$$X_n^{\mathrm{near}} \to X_{n+1}^{\mathrm{rep}} \to X_{n+1}^{\mathrm{lock}}$$

consisting of:

(i) a near-trigger first step;

(ii) a repaired but not yet fully locked second step;

(iii) a successful micro-lock update producing an HC-captured continuation.

Proposition 24.169 (Current local line reduces to HC-capture chain search)**.** *Under the present post-archival stage reading, once the local corridor, the phase-lock repair target, and the HC menu scan are fixed, the next local task reduces to the search for a first local HC-capture chain.*

Proof. The preceding line already fixes: the right corridor object, the lock-side-prior leakage orientation, the minimal HC-lock target, and the first plausible HC final family. No broader redesign is currently preferred. Therefore the next local task is exactly to determine whether one of the plausible HC finals yields an actual capture of the repaired second step. □

Theorem 24.170 (First local HC-capture chain theorem)**.** *Assume:*

(a) the first explicit lock-corridor theorem applies;

(b) the first minimal HC-lock step theorem applies;

(c) the first plausible HC menu-entry theorem applies;

(d) a successful micro-lock update exists for the minimal surviving HC final.

Then the current post-archival stage line supports a first local HC-capture chain.
More precisely:

(i) the repaired second step is first updated by a phase-side dominant, geometrically carried micro-lock compensator;

(ii) the lock-side leakage is removed without redesigning the corridor;

(iii) the resulting continuation is HC-captured in the same local lock corridor as the first near-trigger step.

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Proof. By the first plausible HC menu-entry theorem, the first plausible surviving HC final is sought in the phase-side micro-lock class. By the first minimal HC-lock step theorem, the active local HC task is precisely to search for the smallest such compensator that removes residual lock-side leakage while preserving the corridor. If such a micro-lock update is successful, Lemma 24.167 yields HC capture. This is exactly the stated HC-capture chain. □

Corollary 24.171 (Current HC-capture working rule)**.** *The next local audit should no longer stop at plausible HC menu survival. It should ask whether the best phase-side micro-lock final actually yields a successful micro-lock update and hence the first local HC-capture chain.*

Proof. Immediate from Theorem 24.170. □

Corollary 24.172 (Freeze threshold after archival stage)**.** *If the local line is stably organized by*

lock corridor → minimal HC-lock step → HC menu scan → first local HC-capture chain,

then the post-archival stage line has reached a natural freeze threshold for a new minor release-final.

Proof. At that point the local HC line is no longer merely diagnostic. It carries a coherent operational chain from the residual lock defect to the first concrete capture event. This is exactly the kind of self-contained new kernel that supports a new freeze final before audit. □

24.1 Transport-stable HC capture and OP-entry handoff

Definition 24.173 (Transport-stable HC-capture chain)**.** *Let*

$$\mathcal{C}_n^{\mathrm{HC}} := \big(X_n^{\mathrm{near}} \to X_{n+1}^{\mathrm{rep}} \to X_{n+1}^{\mathrm{lock}}\big)$$

be a first local HC-capture chain in the sense of Definition 24.168. It is called transport-stable on an admissible window $[n, n+k]$ *if every route-faithful admissible transport step*

$$\mathsf{T}_j : X_j \longmapsto X_{j+1}, \qquad n \le j < n+k,$$

including the symmetrized forward/backward transport readout when that readout is active, preserves:

(a) membership of the locked continuation in the transported lock corridor;

(b) non-increase of the lock-side leakage under the transported HC readout;

(c) the inherited admissibility and boundary-stop constraints of the active route;

(d) the distinction between horizon-side and stop-preserving boundary-side packets.

Thus transport-stability is not a new closure theorem. It is the statement that the local capture event survives the already-admissible route motion needed before OP-facing use.

Definition 24.174 (HC transport defect)**.** *For a captured local chain $\mathcal{C}_n^{\mathrm{HC}}$ and an admissible transport step T_j, define the associated HC transport defect schematically by*

$$\Delta_{\mathrm{tr}}^{\mathrm{HC}}(j) := \max\Big\{ d_{\mathfrak{C}}\big(\mathsf{T}_j X_j^{\mathrm{lock}}, \mathfrak{C}_{\mathrm{lock}}^{\mathrm{loc}}(j+1)\big), \big(\lambda_{\mathrm{lock}}^{\Theta,+}(j+1) - \lambda_{\mathrm{lock}}^{\Theta,+}(j)\big)_+, \chi_{\mathrm{stop}}(j+1) \Big\}.$$

Here $d_{\mathfrak{C}}$ denotes corridor distance in the current local readout, $(\cdot)_+$ denotes the positive part, and χ_{stop} is the indicator of an inadmissible boundary-stop violation. The defect vanishes exactly when the transported step remains corridor-inside, does not re-open lock leakage, and does not violate the inherited stop discipline.

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Lemma 24.175 (Zero HC transport defect preserves capture)**.** *Assume that $\mathcal{C}_n^{\mathrm{HC}}$ is a first local HC-capture chain and that*

$$\Delta_{\mathrm{tr}}^{\mathrm{HC}}(j) = 0$$

for an admissible route-faithful transport step T_j. Then the transported locked state is again HC-captured in the transported local lock corridor.

Proof. The equality $\Delta_{\mathrm{tr}}^{\mathrm{HC}}(j) = 0$ simultaneously says that the transported locked state has zero corridor distance from the next transported lock corridor, that lock-side leakage has not been positively re-opened, and that no boundary-stop violation has occurred. These are exactly the transported versions of corridor membership, lock-side capture, and route admissibility in Definition 24.173. Hence the transported locked state remains HC-captured. □

Definition 24.176 (HC-to-OP handoff packet)**.** *An* HC-to-OP handoff packet *on $[n, n+k]$ is a tuple*

$$\mathcal{P}_{\mathrm{HC}\to\mathrm{OP}}(n,k) := \big(\mathcal{C}_n^{\mathrm{HC}}, \{\mathsf{T}_j\}_{j=n}^{n+k-1}, \{\Delta_{\mathrm{tr}}^{\mathrm{HC}}(j)\}_{j=n}^{n+k-1}, \widetilde{\mathbf{R}}_{\mathrm{live}}(n+k), \tau_{\mathrm{route}}\big),$$

where:

(a) $\mathcal{C}_n^{\mathrm{HC}}$ is a first local HC-capture chain;

(b) the transports T_j are route-faithful and admissible;

(c) all displayed HC transport defects vanish on the window;

(d) the terminal normalized live residual $\widetilde{\mathbf{R}}_{\mathrm{live}}(n+k)$ lies in the admissible OP-entry normalization regime;

(e) τ_{route} records whether the packet is horizon-side or stop-preserving boundary-side.

The handoff packet is therefore the smallest object that can be consumed by the OP-facing machine without forgetting how the local HC capture was transported.

Proposition 24.177 (Handoff packets are admissible pre-OP frontier objects)**.** *Every HC-to-OP handoff packet is an admissible pre-OP frontier object in the sense of the current formal OP-entry framework.*

Proof. By definition, the packet contains a valid local HC-capture chain, an admissible route-faithful transport family, vanishing HC transport defects, an admissibly normalized terminal live residual, and an explicit route tag. These are precisely the data required to prevent three forbidden moves: erasing the route distinction, re-opening a boundary stop, or treating local capture as OP closure without transport. Hence the packet is admissible as a pre-OP frontier object. □

Theorem 24.178 (First transport-stable HC-capture to OP-entry criterion)**.** *Assume:*

(a) the first local HC-capture chain theorem applies;

(b) the captured chain is transport-stable on a finite admissible window $[n, n+k]$;

(c) the terminal normalized live residual $\widetilde{\mathbf{R}}_{\mathrm{live}}(n+k)$ lies in the admissible OP-entry normalization regime;

(d) the inherited route tag is preserved throughout the handoff.

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Then the current post-archival stage line supports a controlled OP-entry handoff packet

$$\mathcal{P}_{\mathrm{HC}\to\mathrm{OP}}(n,k).$$

If, in addition, the already-defined closure-phase trigger persistence condition holds at the terminal state, the packet yields an immediate OP-entry certificate. If that persistence condition is not yet met, the same packet yields a narrowed frontier-certified pre-entry state rather than a false OP closure claim.

Proof. By the first local HC-capture chain theorem, the local repaired second step can be locked into the same local corridor as the first near-trigger step. By transport-stability, Lemma 24.175 applies at each admissible transport step in the finite window. Therefore the captured state remains captured after the finite route-faithful transport sequence, with no positive re-opening of lock-side leakage and no boundary-stop violation. The normalized terminal residual and route tag then supply exactly the remaining fields in Definition 24.176. The final alternative follows from the existing OP-entry framework: trigger persistence upgrades the handoff packet to an entry certificate, while absence of trigger persistence leaves a legitimate, narrowed pre-entry frontier state. □

Corollary 24.179 (working rule)**.** *The next audit should no longer ask merely whether a local HC capture chain exists. It should test whether the captured chain is transport-stable and whether the resulting handoff packet terminates in an immediate OP-entry certificate or in a smaller frontier-certified pre-entry state.*

Proof. Immediate from Theorem 24.178. □

24.2 First OP-entry normal form extracted from HC handoff

Definition 24.180 (HC-extracted OP-entry normal form)**.** *Let $\mathcal{P}_{\mathrm{HC}\to\mathrm{OP}}(n,k)$ be an HC-to-OP handoff packet in the sense of Definition 24.176. Its* HC-extracted OP-entry normal form *is the tuple*

$$\mathcal{N}_{\mathrm{OP}}^{\mathrm{HC}}(n,k) := \big(\widetilde{\mathbf{R}}_{\mathrm{live}}^{\mathrm{OP}}(n+k), \tau_{\mathrm{route}}, \sigma_{\mathrm{tr}}, \mathcal{M}_{\mathrm{OP}}^{\mathrm{adm}}(n+k), \eta_{\mathrm{entry}}\big),$$

where:

(a) $\widetilde{\mathbf{R}}_{\mathrm{live}}^{\mathrm{OP}}(n+k)$ *is the terminal normalized live residual read as an OP-facing residual triple*

$$\widetilde{\mathbf{R}}_{\mathrm{live}}^{\mathrm{OP}} = \big(\rho_{\mathrm{CL}}, \rho_{\Theta R}, \rho_A\big);$$

(b) τ_{route} *is the inherited horizon-side or stop-preserving boundary-side route tag;*

(c) σ_{tr} *records that the HC transport defects vanish on the handoff window;*

(d) $\mathcal{M}_{\mathrm{OP}}^{\mathrm{adm}}(n+k)$ *is the finite admissible OP-facing request menu available at the terminal state;*

(e) $\eta_{\mathrm{entry}} \in \{\mathrm{entry}, \mathrm{frontier}\}$ *records whether the terminal closure-phase trigger persistence condition has already upgraded the packet to immediate OP-entry or only to a smaller frontier-certified pre-entry state.*

The normal form is therefore not a new theorem layer. It is the smallest downstream-readable form of a transported HC capture event.

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Definition 24.181 (Residual routing out of the OP-entry normal form)**.** *The residual routing map associated with $\mathcal{N}_{\mathrm{OP}}^{\mathrm{HC}}$ is the finite assignment*

$$\mathsf{Route}_{\mathrm{OP}}\big(\rho_{\mathrm{CL}}, \rho_{\Theta R}, \rho_A; \tau_{\mathrm{route}}\big) = \big(\mathcal{Q}_{1/4}, \mathcal{Q}_{5(B)}, \mathcal{Q}_A\big),$$

where:

(a) $\mathcal{Q}_{1/4}$ contains only continuum-limit / Hopf-correction requests controlled by ρ_{CL} and is the admissible intake for the later OP 1/OP 4 correction line;

(b) $\mathcal{Q}_{5(B)}$ contains only routed Θ–R balance requests controlled by $\rho_{\Theta R}$ and is the admissible intake for the later OP 5(B) line;

(c) $\mathcal{Q}_A$ contains only anchor, admissibility, and route-discipline requests controlled by ρ_A;

(d) no request may erase τ_{route}, deny the earliest-boundary stop, or promote a diagnostic residual to a theorematic closure certificate by naming alone.

Lemma 24.182 (Normal-form extraction preserves route discipline)**.** *If $\mathcal{P}_{\mathrm{HC}\to\mathrm{OP}}(n,k)$ is an HC-to-OP handoff packet, then its HC-extracted OP-entry normal form preserves the inherited route distinction and boundary-stop discipline.*

Proof. The handoff packet already contains the route tag, the transported HC-capture chain, the vanishing HC transport defects, and the terminal normalized live residual. Definition 24.180 only repackages these data into the fields consumed by the OP-facing request layer. Since neither the route tag nor the stop indicator is discarded during that repackaging, the inherited route distinction and boundary-stop discipline are preserved. □

Proposition 24.183 (Finite OP-facing request menu from HC handoff)**.** *Every HC-extracted OP-entry normal form determines a finite admissible OP-facing request menu*

$$\mathcal{M}_{\mathrm{OP}}^{\mathrm{adm}} = \mathcal{Q}_{1/4} \cup \mathcal{Q}_{5(B)} \cup \mathcal{Q}_A$$

subordinate to the inherited route tag.

Proof. The terminal live residual triple has only three displayed coordinates: continuum-limit, routed Θ–R, and anchor/admissibility. Definition 24.181 assigns each coordinate to one finite request family and forbids requests that break route discipline or manufacture a closure certificate. Hence the union is finite, admissible, and subordinate to the inherited route tag. □

Definition 24.184 (Lead OP-entry residual selector)**.** *A* lead OP-entry residual selector *is any admissible rule*

$$\mathsf{L}_{\mathrm{OP}}\big(\mathcal{N}_{\mathrm{OP}}^{\mathrm{HC}}\big) \in \{\rho_{\mathrm{CL}}, \rho_{\Theta R}, \rho_A, \rho_{\mathrm{CL}} \oplus \rho_{\Theta R}\}$$

that chooses the next finite OP-facing work branch according to the current residual ranking, subject to the following restrictions:

(a) if ρ_{CL} is dominant, the next branch is the OP 1/OP 4 continuum-limit correction line;

(b) if $\rho_{\Theta R}$ is dominant, the next branch is the OP 5(B) routed balance line;

(c) if ρ_A is dominant, the next branch is an admissibility/anchor refinement rather than a physical closure claim;

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(d) if ρ_{CL} and $\rho_{\Theta R}$ are inseparable under the current ranking, the next branch is the paired OP 1/4–OP 5(B) comparison branch.

Theorem 24.185 (First OP-entry normal-form theorem)**.** *Assume the first transport-stable HC-capture to OP-entry criterion of Theorem 24.178. Then the current archival stage working line admits a unique downstream-readable OP-entry normal form*

$$\mathcal{N}_{\mathrm{OP}}^{\mathrm{HC}}(n,k)$$

for the transported HC capture event. Moreover:

(i) if $\eta_{\mathrm{entry}} = \mathrm{entry}$, the normal form is an immediate OP-entry surface;

(ii) if $\eta_{\mathrm{entry}} = \mathrm{frontier}$, the normal form is a smaller frontier-certified pre-entry surface;

(iii) in both cases the admissible next work is restricted to the residual-routed menu $\mathcal{Q}_{1/4}$, $\mathcal{Q}_{5(B)}$, and $\mathcal{Q}_A$.

Proof. Theorem 24.178 supplies the controlled handoff packet or, if trigger persistence is absent, the narrowed frontier-certified pre-entry state carried by the same packet. Definition 24.180 reads exactly those fields into one terminal residual triple, one route tag, one transport-stability record, one admissible request menu, and one entry/frontier flag. No additional choice is made during this extraction, so the downstream-readable normal form is unique relative to the given handoff packet. The final restriction follows from Proposition 24.183. □

Corollary 24.186 (working rule)**.** *The next mathematical audit should select the lead OP-entry residual from $\mathcal{N}_{\mathrm{OP}}^{\mathrm{HC}}$. If the continuum-limit residual dominates, start with the OP 1/OP 4 correction branch. If the routed Θ–R residual dominates, start with OP 5(B). If neither is dominant, use the paired comparison branch rather than forcing a premature single-problem split.*

Proof. Immediate from Definition 24.184 and Theorem 24.185. □

24.3 First OP-entry task routing extracted from the normal form

Definition 24.187 (OP-entry task-routing object)**.** *Let*

$$\mathcal{N}_{\mathrm{OP}}^{\mathrm{HC}}(n,k)$$

be an HC-extracted OP-entry normal form in the sense of Definition 24.180, and let

$$\mathsf{L}_{\mathrm{OP}}\big(\mathcal{N}_{\mathrm{OP}}^{\mathrm{HC}}\big)$$

be a lead OP-entry residual selector in the sense of Definition 24.184. The associated OP-entry task-routing object *is the tuple*

$$\mathcal{T}_{\mathrm{OP}}^{\mathrm{HC}}(n,k) := \big(\mathcal{N}_{\mathrm{OP}}^{\mathrm{HC}}(n,k), \mathsf{L}_{\mathrm{OP}}, q_{\mathrm{lead}}, \mathcal{W}_{\mathrm{OP}}^{\mathrm{lead}}, \tau_{\mathrm{route}}, \eta_{\mathrm{entry}}\big),$$

where:

(a) q_{lead} is the selected residual branch in

$$\{\rho_{\mathrm{CL}}, \rho_{\Theta R}, \rho_A, \rho_{\mathrm{CL}} \oplus \rho_{\Theta R}\};$$

(b) $\mathcal{W}_{\mathrm{OP}}^{\mathrm{lead}}$ is the finite route-faithful work family assigned to that branch;

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(c) τ_{route} *is inherited unchanged from the normal form;*

(d) η_{entry} *records whether the object is consumed as immediate OP-entry or only as a frontier-certified sharpening task.*

Thus the task-routing object does not decide an OP claim. It decides only which finite admissible work family is allowed to be executed next.

Definition 24.188 (Residual-to-task routing map)**.** *The* residual-to-task routing map *associated with an OP-entry normal form is the admissible assignment*

$$\mathfrak{R}_{\mathrm{task}} : \{\rho_{\mathrm{CL}}, \rho_{\Theta R}, \rho_A, \rho_{\mathrm{CL}} \oplus \rho_{\Theta R}\} \longrightarrow \{\mathcal{W}^{\mathrm{CL}}_{1/4}, \mathcal{W}^{\Theta R}_{5(B)}, \mathcal{W}^{\mathrm{adm}}_{A}, \mathcal{W}^{\mathrm{cmp}}_{1/4|5(B)}\}$$

defined by

$$\begin{aligned} \rho_{\mathrm{CL}} &\longmapsto \mathcal{W}^{\mathrm{CL}}_{1/4}, \\ \rho_{\Theta R} &\longmapsto \mathcal{W}^{\Theta R}_{5(B)}, \\ \rho_A &\longmapsto \mathcal{W}^{\mathrm{adm}}_{A}, \\ \rho_{\mathrm{CL}} \oplus \rho_{\Theta R} &\longmapsto \mathcal{W}^{\mathrm{cmp}}_{1/4|5(B)}. \end{aligned}$$

Here $\mathcal{W}^{\mathrm{CL}}_{1/4}$ *contains only continuum-limit correction tasks for OP 1/OP 4,* $\mathcal{W}^{\Theta R}_{5(B)}$ *contains only routed* Θ*–*R *balance tasks for OP 5(B),* $\mathcal{W}^{\mathrm{adm}}_{A}$ *contains only anchor/admissibility refinement tasks, and* $\mathcal{W}^{\mathrm{cmp}}_{1/4|5(B)}$ *contains paired comparison tasks that keep the common HC-capture origin visible when the two leading residuals cannot yet be separated.*

Lemma 24.189 (Task routing preserves OP-entry normal-form discipline)**.** *Let* $\mathcal{T}^{\mathrm{HC}}_{\mathrm{OP}}(n,k)$ *be an OP-entry task-routing object. Then executing its selected finite work family cannot erase the inherited route tag, cannot remove the boundary-stop discipline, and cannot manufacture a new OP closure certificate.*

Proof. By Definition 24.187, the task-routing object carries the normal form, the selected residual branch, the finite work family, the inherited route tag, and the entry/frontier flag as explicit fields. By Definition 24.188, each work family is subordinate to one already displayed residual coordinate or to the paired comparison of the two leading displayed coordinates. None of these assignments changes the route tag or the entry/frontier flag. Hence task execution may refine, compare, or package the selected branch, but it cannot promote itself to a closure certificate unless the pre-existing OP-entry framework supplies the missing trigger persistence. □

Proposition 24.190 (Paired comparison branch is the safe non-dominant branch)**.** *If the current residual ranking cannot separate* ρ_{CL} *from* $\rho_{\Theta R}$*, the admissible next branch is the paired comparison family*

$$\mathcal{W}^{\mathrm{cmp}}_{1/4|5(B)}$$

rather than a forced single OP 1/OP 4 or OP 5(B) branch.

Proof. Definition 24.184 already allows the paired symbol $\rho_{\mathrm{CL}} \oplus \rho_{\Theta R}$ exactly when the two residuals are inseparable under the current ranking. Definition 24.188 sends that symbol to the paired comparison family. Choosing a single branch in that case would add information not present in the normal form and could hide the common HC-capture origin of both residuals. The paired branch is therefore the route-faithful and conservative choice. □

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Theorem 24.191 (First OP-entry task-routing theorem)**.** *Assume that an HC-to-OP handoff packet has been converted into the HC-extracted OP-entry normal form*

$$\mathcal{N}_{\mathrm{OP}}^{\mathrm{HC}}(n,k)$$

of Theorem 24.185. Then every admissible lead selector L_{OP} *determines a finite route-faithful OP-entry task-routing object*

$$\mathcal{T}_{\mathrm{OP}}^{\mathrm{HC}}(n,k).$$

Moreover:

(i) if $\eta_{\mathrm{entry}} = \mathrm{entry}$*, the selected family is a first admissible OP-facing work packet;*

(ii) if $\eta_{\mathrm{entry}} = \mathrm{frontier}$*, the selected family is only a finite sharpening/comparison packet for the smaller frontier-certified pre-entry state;*

(iii) if the selector returns $\rho_{\mathrm{CL}} \oplus \rho_{\Theta R}$*, the next work packet must keep OP 1/OP 4 and OP 5(B) in paired comparison until a later ranking separates them.*

Proof. The normal form contains a finite residual triple, a finite admissible request menu, a route tag, a transport-stability record, and an entry/frontier flag. The lead selector chooses one of the explicitly allowed residual branches. Applying Definition 24.188 to that branch yields one finite work family, and adjoining the inherited normal-form fields gives the task-routing object of Definition 24.187. Lemma 24.189 gives the route and certificate restrictions. Proposition 24.190 gives the paired-branch rule when the two leading residuals are not yet separated. □

Corollary 24.192 (working rule)**.** *The next OP-facing audit should not yet try to close OP 1/OP 4 or OP 5(B) directly. It should first instantiate the task-routing object* $\mathcal{T}_{\mathrm{OP}}^{\mathrm{HC}}(n,k)$*, rank the displayed residuals, and execute only the selected finite work family:* $\mathcal{W}_{1/4}^{\mathrm{CL}}$*,* $\mathcal{W}_{5(B)}^{\Theta R}$*,* $\mathcal{W}_{A}^{\mathrm{adm}}$*, or the paired comparison family* $\mathcal{W}_{1/4|5(B)}^{\mathrm{cmp}}$*.*

Proof. Immediate from Theorem 24.191. The task-routing object is the conservative bridge between OP-entry normal form and explicit OP-facing execution. □

24.4 First paired OP-entry comparison execution layer

Definition 24.193 (Paired OP-entry comparison execution packet)**.** *Assume that the OP-entry task-routing object of Definition 24.187 selects the paired branch*

$$q_{\mathrm{lead}} = \rho_{\mathrm{CL}} \oplus \rho_{\Theta R}, \qquad \mathcal{W}_{\mathrm{OP}}^{\mathrm{lead}} = \mathcal{W}_{1/4|5(B)}^{\mathrm{cmp}}.$$

A paired OP-entry comparison execution packet *is the tuple*

$$\mathcal{X}_{1/4|5(B)}^{\mathrm{cmp}}(n,k) := \big(\mathcal{T}_{\mathrm{OP}}^{\mathrm{HC}}(n,k), \mathcal{U}_{\mathrm{HC}}^{\mathrm{adm}}, D_{\mathrm{joint}}^{\mathrm{HC}}, D_{\mathrm{sep}}^{\mathrm{HC}}, \Delta_{\mathrm{sep}}^{\mathrm{HC}}, \omega_{\mathrm{cmp}}\big),$$

where:

(a) $\mathcal{T}_{\mathrm{OP}}^{\mathrm{HC}}(n,k)$ *is the inherited task-routing object;*

(b) $\mathcal{U}_{\mathrm{HC}}^{\mathrm{adm}}$ *is the finite admissible menu of HC-compatible micro-updates available at the terminal normal form;*

(c) $D_{\mathrm{joint}}^{\mathrm{HC}}$ *is the best joint readout error obtained when one and the same admissible HC-compatible update is asked to account for both* ρ_{CL} *and* $\rho_{\Theta R}$*;*

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(d) $D_{\mathrm{sep}}^{\mathrm{HC}}$ *is the best separated readout error obtained when the two coordinates are allowed to choose their admissible update witnesses independently;*

(e)

$$\Delta_{\mathrm{sep}}^{\mathrm{HC}} := D_{\mathrm{joint}}^{\mathrm{HC}} - D_{\mathrm{sep}}^{\mathrm{HC}} \geq 0$$

is the HC-origin separation defect;

(f) $\omega_{\mathrm{cmp}} \in \{\mathrm{bind}, \mathrm{split}, \mathrm{frontier}\}$ *records the comparison outcome.*

The packet executes only the comparison branch. It does not yet assert an OP 1/4 correction theorem and does not yet assert the routed OP 5(B) balance theorem.

Definition 24.194 (HC-origin comparison decision rule)**.** *Fix an admissible comparison tolerance* $\varepsilon_{\mathrm{cmp}} > 0$ *and a strict descent requirement on the active frontier measure. The outcome field of Definition 24.193 is read by*

$$\omega_{\mathrm{cmp}} = \begin{cases} \mathrm{bind}, & \Delta_{\mathrm{sep}}^{\mathrm{HC}} \leq \varepsilon_{\mathrm{cmp}}, \\ \mathrm{split}, & \Delta_{\mathrm{sep}}^{\mathrm{HC}} > \varepsilon_{\mathrm{cmp}} \text{ and the separated ranking is stable}, \\ \mathrm{frontier}, & \text{otherwise.} \end{cases}$$

The bind case means that the two residual coordinates are still best treated as one common HC-origin comparison problem. The split case means that the comparison has supplied enough information to return to a single OP 1/4 or OP 5(B) branch under the residual ranking. The frontier case means that the comparison improved the state, but not enough to justify either binding or splitting as a theorematic route decision.

Lemma 24.195 (Paired comparison execution preserves the inherited handoff discipline)**.** *Executing a paired OP-entry comparison packet cannot erase the inherited route tag, cannot reopen the boundary-stop condition, and cannot promote the current frontier state to an OP closure certificate.*

Proof. The execution packet contains the task-routing object as an explicit first component. By Lemma 24.189, that object already carries the inherited route tag, entry/frontier flag, and admissible work-family restriction. The additional fields in Definition 24.193 compare only joint versus separated HC-compatible readout errors inside the finite admissible micro-update menu. They do not change the route tag, do not delete the boundary-stop flag, and do not introduce a trigger-persistence certificate. Hence the comparison can only bind, split, or sharpen the frontier; it cannot close an OP by itself. □

Proposition 24.196 (Bind branch retains the common HC-capture origin)**.** *If* $\omega_{\mathrm{cmp}} = \mathrm{bind}$*, then OP 1/4 and OP 5(B) must remain in the coupled work family*

$$\mathcal{W}_{1/4|5(B)}^{\mathrm{cmp}}$$

for the next step, with the common HC-capture origin kept as an explicit field.

Proof. In the bind case, Definition 24.194 says that the best joint HC-compatible readout is not worse than the separated readout by more than the admissible comparison tolerance. Splitting the branch at that point would artificially discard the information that one HC-compatible source still accounts for both leading residuals. The route-faithful next object therefore remains the paired comparison family, now with the common HC-origin field recorded rather than hidden. □

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Proposition 24.197 (Split branch returns to ranked single-branch work)**.** *If* $\omega_{\mathrm{cmp}} = \mathrm{split}$*, then the comparison packet returns a ranked single-branch task-routing object selecting either*

$$\mathcal{W}^{\mathrm{CL}}_{1/4} \qquad \textit{or} \qquad \mathcal{W}^{\Theta R}_{5(B)},$$

according to the stable separated residual ranking.

Proof. The split case is allowed only when the HC-origin separation defect exceeds the admissible comparison tolerance and the separated ranking is stable. Thus the normal form now contains information that was absent in the inseparable branch: one of the two residual coordinates has become a legitimate lead branch. Applying the residual-to-task routing map of Definition 24.188 to that newly ranked coordinate gives either the OP 1/4 continuum-limit work family or the OP 5(B) routed Θ–R work family. □

Theorem 24.198 (First paired OP-entry comparison execution theorem)**.** *Assume that the task-routing layer selects the paired comparison branch*

$$\rho_{\mathrm{CL}} \oplus \rho_{\Theta R} \longmapsto \mathcal{W}^{\mathrm{cmp}}_{1/4|5(B)}.$$

Then every admissible finite HC-compatible comparison menu produces one paired OP-entry comparison execution packet

$$\mathcal{X}^{\mathrm{cmp}}_{1/4|5(B)}(n,k)$$

and exactly one of the following route-faithful next states:

(i) a bound common-origin comparison state, still subordinate to $\mathcal{W}^{\mathrm{cmp}}_{1/4|5(B)}$*;*

(ii) a split ranked single-branch state, subordinate to either $\mathcal{W}^{\mathrm{CL}}_{1/4}$ *or* $\mathcal{W}^{\Theta R}_{5(B)}$*;*

(iii) a smaller frontier-certified comparison state, if neither binding nor splitting is yet justified.

In all three cases, no OP closure certificate is manufactured by the comparison execution itself.

Proof. The finite admissible menu $\mathcal{U}^{\mathrm{adm}}_{\mathrm{HC}}$ supplies the joint and separated minimization tests by construction. Their difference defines $\Delta^{\mathrm{HC}}_{\mathrm{sep}} \geq 0$, since the separated readout is at least as flexible as the joint readout. Definition 24.194 assigns exactly one of the three comparison outcomes. Proposition 24.196 gives the bound next state; Proposition 24.197 gives the split next state; and the remaining case is, by definition, the smaller frontier-certified comparison state. Lemma 24.195 supplies the route, boundary, and no-false-closure restrictions. □

Corollary 24.199 (working rule)**.** *The next OP-facing audit may now execute the paired comparison branch before choosing between OP 1/4 and OP 5(B). The correct next question is not yet* which OP closes first*, but whether* ρ_{CL} *and* $\rho_{\Theta R}$ *are still one common HC-origin residual or have separated into stable ranked branches.*

Proof. Immediate from Theorem 24.198. □

24.5 Minimal MML/MMA interface for HC-routed OP execution

Definition 24.200 (local language-version registry)**.** *The current HC-routed OP-entry execution layer uses the following local interface tags; their interface stamps are recorded in prose, not as mathematical exponents:*

$$\mathrm{QAM}^{\mathrm{loc}}_{\mathrm{HCOP}}, \qquad \mathrm{MML}^{\mathrm{loc}}_{\mathrm{HCOP}}, \qquad \mathrm{MMA}^{\mathrm{loc}}_{\mathrm{HCOP}}.$$

Their status is intentionally limited.

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(a) $\mathrm{QAM}^{\mathrm{loc}}_{\mathrm{HCOP}}$ *is the finite tagged QAM bridge used to code the HC/OP objects of the present subsection inside the inherited QAM-over-ZF layer; its interface stamp is QAM-HC/OP 0.1.*

(b) $\mathrm{MML}^{\mathrm{loc}}_{\mathrm{HCOP}}$ *is the finite instruction language for the local proof chain HC-routed OP-entry chain; its interface stamp is MML-HC/OP 0.2.*

(c) $\mathrm{MMA}^{\mathrm{loc}}_{\mathrm{HCOP}}$ *is the matrix/readout execution semantics for those instructions on finite route-faithful machine states; its interface stamp is MMA-HC/OP 0.2.*

This registry upgrades only the local execution interface. It does not replace the inherited public QAM layer, does not activate the deferred binary tag system, and does not turn the historical MML–MMA–MCP reserve into an independent proof engine.

Definition 24.201 (HC/OP MMA object types)**.** *The minimal MMA object layer for HC-routed OP execution consists of the finite type family*

$$\mathsf{HCState},\quad \mathsf{HandoffPacket},\quad \mathsf{OPNormalForm},\quad \mathsf{TaskRouting},\quad \mathsf{ComparisonPacket},$$
$$\mathsf{Decision},\quad \mathsf{FrontierState},\quad \mathsf{RouteTag},\quad \mathsf{CertificateFinal}.$$

The intended instances are respectively: local HC-capture states, the handoff packets of Definition 24.176, the normal forms of Definition 24.180, the routing objects of Definition 24.187, the comparison packets of Definition 24.193, the decision values $\mathrm{bind}, \mathrm{split}, \mathrm{frontier}$*, smaller pre-entry frontier states, inherited route tags, and possible certificate data. A* $\mathsf{CertificateFinal}$ *becomes an actual theorematic certificate only when the previously stated mathematical trigger conditions are discharged outside the machine interface.*

Definition 24.202 (QAM coding bridge for HC/OP objects)**.** *A* QAM-HC/OP bridge code *is a partial coding map*

$$\ulcorner\cdot\urcorner^{\mathrm{HCOP}}_{\mathrm{QAM}} : \mathsf{Obj}^{\mathrm{HCOP}}_{\mathrm{MMA}} \dashrightarrow \mathsf{QAM}_{\mathrm{fin}},$$

where $\mathsf{Obj}^{\mathrm{HCOP}}_{\mathrm{MMA}}$ *is the finite object family of Definition 24.201 and* $\mathsf{QAM}_{\mathrm{fin}}$ *denotes finite tagged QAM objects in the inherited QAM-over-ZF sense. The map is required to preserve object kind, route tag, entry/frontier flag, and no-false-certificate status.*

Definition 24.203 (Minimal MML-HC/OP instruction alphabet)**.** *The local instruction alphabet* $\mathcal{I}^{\mathrm{HCOP}}_{\mathrm{MML}}$ *consists of the following finite instructions:*

$$\mathrm{CAPTURE}_{\mathrm{HC}},\quad \mathrm{HANDOFF}_{\mathrm{HC}\to\mathrm{OP}},\quad \mathrm{NORMALIZE}^{\mathrm{HC}}_{\mathrm{OP}},\quad \mathrm{ROUTE}^{\mathrm{HC}}_{\mathrm{OP}},$$
$$\mathrm{COMPARE}^{1/4|5(B)}_{\mathrm{HC}},\quad \mathrm{RESOLVE}_{\mathrm{bind/split/frontier}},\quad \mathrm{RETURN}_{\mathrm{OP}}.$$

The instructions are typed as follows:

(a) $\mathrm{CAPTURE}_{\mathrm{HC}}$ *sends a local admissible HC-state to a final transport-stable HC state.*

(b) $\mathrm{HANDOFF}_{\mathrm{HC}\to\mathrm{OP}}$ *sends such a state to an HC-to-OP handoff packet.*

(c) $\mathrm{NORMALIZE}^{\mathrm{HC}}_{\mathrm{OP}}$ *sends the handoff packet to an OP-entry normal form.*

(d) $\mathrm{ROUTE}^{\mathrm{HC}}_{\mathrm{OP}}$ *sends the normal form to a finite OP-entry task-routing object.*

(e) $\mathrm{COMPARE}^{1/4|5(B)}_{\mathrm{HC}}$ *executes the paired OP 1/4–OP 5(B) comparison when the routing object selects* $\rho_{\mathrm{CL}} \oplus \rho_{\Theta R}$*.*

(f) $\mathrm{RESOLVE}_{\mathrm{bind/split/frontier}}$ *records exactly one comparison decision.*

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(g) $\mathrm{RETURN}_{\mathrm{OP}}$ *returns either a bound common-origin state, a split ranked single-branch state, or a smaller frontier-certified comparison state.*

Definition 24.204 (MMA execution state for the HC/OP program)**.** *An* MMA-HC/OP execution state *is a finite tuple*

$$\Sigma_{\mathrm{HCOP}} = \big(S, \tau, \eta, r, c\big),$$

where S is the finite QAM-coded store, τ is the inherited route tag, $\eta \in \{\mathrm{entry}, \mathrm{frontier}\}$ is the entry/frontier flag, r is the active residual or comparison register, and c is a certificate register. The invariant of the state is:

$$c = \varnothing \quad \textit{unless a previously proved mathematical trigger supplies a certificate.}$$

Definition 24.205 (Finite HC-routed OP-entry MML program)**.** *The* minimal HC-routed OP-entry program *is the finite MML word*

$$P_{\mathrm{HCOP}} := (\mathrm{CAPTURE}_{\mathrm{HC}}, \mathrm{HANDOFF}_{\mathrm{HC}\to\mathrm{OP}}, \mathrm{NORMALIZE}^{\mathrm{HC}}_{\mathrm{OP}}, \mathrm{ROUTE}^{\mathrm{HC}}_{\mathrm{OP}},$$
$$\mathrm{COMPARE}^{1/4|5(B)}_{\mathrm{HC}}, \mathrm{RESOLVE}_{\mathrm{bind/split/frontier}}, \mathrm{RETURN}_{\mathrm{OP}}).$$

If the task-routing object does not select the paired branch, the comparison instruction is skipped and the return instruction hands control to the selected single work family instead. Thus the program is finite in either the paired or the single-branch case.

Lemma 24.206 (Typing of the HC-routed OP chain)**.** *Each step from the transport-stable HC-capture handoff through the first paired OP-entry comparison execution has a matching type in Definitions 24.201–24.203.*

Proof. The handoff packet is an instance of HandoffPacket. The normal form is an instance of OPNormalForm. The task-routing object is an instance of TaskRouting. The paired comparison packet is an instance of ComparisonPacket, and its outcome belongs to Decision. The possible return states are exactly the bound, split, or frontier states already listed in Theorem 24.198. These are precisely the input/output types of the finite instruction alphabet. □

Lemma 24.207 (MML/MMA execution preserves route and frontier discipline)**.** *Executing P_{HCOP} in the MMA-HC/OP layer preserves the inherited route tag, preserves the entry/frontier flag unless a previously proved trigger changes it, and cannot fill the certificate register by machine action alone.*

Proof. Each instruction in Definition 24.203 is typed over the objects already constrained by the local proof chain mathematical layer. The handoff and normalization instructions inherit the route tag from the HC-capture chain; the routing and comparison instructions act only on the finite residual and comparison registers; the resolution instruction chooses only among bind, split, and frontier; and the return instruction returns one of the route-faithful next states of Theorem 24.198. None of these instructions contains a rule that writes a theorematic certificate into c. Hence route, frontier, and no-false-certificate discipline are preserved. □

Proposition 24.208 (comparison as one admissible MML/MMA step)**.** *The paired comparison execution of Theorem 24.198 is represented in* $\mathrm{MML}^{\mathrm{loc}}_{\mathrm{HCOP}}$ *by the instruction*

$$\mathrm{COMPARE}^{1/4|5(B)}_{\mathrm{HC}}$$

followed by $\mathrm{RESOLVE}_{\mathrm{bind/split/frontier}}$*. Its MMA semantics are exactly the construction of $D^{\mathrm{HC}}_{\mathrm{joint}}$, $D^{\mathrm{HC}}_{\mathrm{sep}}$, $\Delta^{\mathrm{HC}}_{\mathrm{sep}}$, and the decision ω_{cmp}.*

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Proof. Definition 24.193 constructs the joint and separated HC-compatible readout errors and their difference. Definition 24.194 resolves the resulting comparison defect into exactly one decision. These are exactly the operational contents assigned to the comparison and resolution instructions. The statement is therefore only a typed re-reading of the comparison execution, not an additional theorematic claim. □

Theorem 24.209 (HC-routed OP-entry as a finite MML/MMA program)**.** *Assume the transport-stable HC-capture to OP-entry criterion of Theorem 24.178. Then the current the local proof chain HC-routed OP-entry chain is executable as the finite program* P_{HCOP} *over the local interface stack*

$$\mathrm{QAM}^{\mathrm{loc}}_{\mathrm{HCOP}} \longrightarrow \mathrm{MML}^{\mathrm{loc}}_{\mathrm{HCOP}} \longrightarrow \mathrm{MMA}^{\mathrm{loc}}_{\mathrm{HCOP}}.$$

The execution returns exactly one admissible next state: immediate OP-entry when the inherited mathematical trigger supplies it, a bound common-origin comparison state, a split ranked single-branch state, or a smaller frontier-certified state. It does not manufacture an OP closure certificate.

Proof. By Lemma 24.206, every mathematical object used in the local proof chain has a corresponding local MMA type and QAM bridge code. Definition 24.205 lists the finite instruction word realizing the chain. Lemma 24.207 supplies route preservation and the no-false-certificate condition. The possible next states are exactly the ones already obtained in the handoff, normal-form, routing, and paired-comparison theorems. Therefore the MML/MMA layer is an executable notation for the existing mathematical chain, not a new source of closure. □

Corollary 24.210 (working rule)**.** *The QAM/MML/MMA layer may now be used as a notation and audit interface for the HC-routed OP-entry program. Any later language extension must preserve the local registry of Definition 24.200: QAM remains a finite bridge over ZF-coded objects, MML remains a finite instruction language, MMA remains the execution semantics, and theorematic certificates must still come from the mathematical trigger conditions rather than from the machine layer itself.*

Proof. Immediate from Theorem 24.209 and the status restriction in Definition 24.200. □

24.6 MML/MMA compression map for inherited HC/OP text

Definition 24.211 (Inherited HC/OP compression target)**.** *An* inherited HC/OP compression target *is a paragraph, lemma, proposition, diagnostic block, or tool paragraph in the inherited active or historical Companion text whose proof-use is one of the following finite roles:*

capture, handoff, normalization, routing, comparison, resolution, return, scan, rank, certificate test, route preservation, frontier descent.

The target is compressible *only in the expository sense: it may be reread as a typed MML/MMA operation or short finite operation word, but its original mathematical content, hypotheses, and proof restrictions remain part of the document unless a later explicit editorial release removes or relocates them.*

Definition 24.212 (Compression map from inherited text to MML/MMA operations)**.** *The local* HC/OP compression map

$$\mathfrak{C}_{\mathrm{HCOP}}$$

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sends inherited compression targets to typed MML/MMA readings according to the following table.

Inherited text pattern	***MML/MMA reading***	***Compression use***
HC lock corridor, local HC final, minimal HC-lock step	$\mathrm{LOCK/CAPTURE}_{\mathrm{HC}}$	*Marks the block as a final-production and lock-stabilization step rather than as an OP certificate.*
Transport-stable HC-to-OP packet	$\mathrm{HANDOFF}_{\mathrm{HC}\to\mathrm{OP}}$	*Turns repeated handoff prose into one typed transfer from HC state to OP-facing packet.*
OP-entry normal form and residual triple	$\mathrm{NORMALIZE}_{\mathrm{OP}}^{\mathrm{HC}}$	*Collects ρ_{CL}, $\rho_{\Theta R}$, ρ_A, route tags, and entry/frontier flags into one typed normal form.*
Residual-to-task selector	$\mathrm{ROUTE}_{\mathrm{OP}}^{\mathrm{HC}}$	*Replaces repeated branch-selection explanations by one admissible routing step.*
Paired OP 1/4–OP 5(B) branch	$\mathrm{COMPARE}_{\mathrm{HC}}^{1/4\mid 5(B)}$	*Reads the joint/separated defect comparison as one admissible comparison instruction.*
Bind/split/frontier decision	$\mathrm{RESOLVE}_{\mathrm{bind/split/frontier}}$	*Records that the comparison returns only a bound state, a split state, or a smaller frontier state.*
Next-state handback to OP-facing work	$\mathrm{RETURN}_{\mathrm{OP}}$	*Compresses the return package without changing theorematic status.*
Probe, Monte Carlo, decoder, or finite search block	SCAN $\to$ RANK $\to$ CANDIDATE	*Makes clear that computational probes rank admissible finals but do not prove closure.*
Admissibility / boundary-stop / route-faithfulness paragraph	PRESERVE_ROUTE	*Marks the paragraph as an invariant-preservation rule for route tag and boundary-stop data.*
Certificate discussion or trigger-persistence paragraph	CERTIFY?	*Keeps certification separate from execution: the operation asks whether a mathematical trigger exists; it never creates one.*
Frontier descent or smaller pre-entry state	$\mathrm{DESCEND}_{\mathrm{frontier}}$	*Records strict controlled progress without falsely upgrading the state to OP closure.*
Historical tool or early proof-attempt block	ARCHIVE $\to$ TOOL	*Keeps the historical witness intact while extracting the reusable tool-role for the active line.*

Lemma 24.213 (Compression map preserves proof content)**.** *Applying $\mathfrak{C}_{\mathrm{HCOP}}$ to an inherited compression target does not change the truth status of that target, does not delete hypotheses, and does not promote a diagnostic or work packet to a theorematic certificate.*

Proof. By Definition 24.211, a compression target is compressed only in its expository role. Definition 24.212 assigns a typed MML/MMA reading that records what the block does in the finite HC/OP program. The assignment neither alters the original statement nor supplies

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a new mathematical trigger. The no-false-certificate restriction of Lemma 24.207 therefore remains in force. □

Proposition 24.214 (Main inherited text zones helped by the compression map)**.** *The compression map gives immediate expository leverage in the following inherited zones:*

(i) HC-capture and lock-corridor paragraphs, by replacing repeated final/lock explanations with $\mathrm{LOCK/CAPTURE}_{\mathrm{HC}}$*;*

(ii) HC-to-OP handoff and OP-entry paragraphs, by replacing repeated transfer and normal-form prose with HANDOFF*,* NORMALIZE*, and* ROUTE*;*

(iii) OP 1/4–OP 5(B) comparison paragraphs, by writing the inseparable branch as $\mathrm{COMPARE} \to \mathrm{RESOLVE} \to \mathrm{RETURN}$*;*

(iv) probe and Monte-Carlo paragraphs, by reading them as $\mathrm{SCAN} \to \mathrm{RANK} \to \mathrm{CANDIDATE}$ *and not as proof generators;*

(v) historical witness paragraphs, by marking reusable tools without deleting the preserved genealogy.

Proof. Each listed zone contains one of the patterns displayed in Definition 24.212. The assigned operation records the proof-use of the zone and can therefore replace later repetitive explanation. Lemma 24.213 prevents the compression from changing mathematical status. □

Theorem 24.215 (MML/MMA compression theorem for inherited HC/OP text)**.** *Every inherited HC/OP compression target that matches one of the patterns in Definition 24.212 admits a finite MML/MMA reading under*

$$\mathrm{QAM}^{\mathrm{loc}}_{\mathrm{HCOP}} \longrightarrow \mathrm{MML}^{\mathrm{loc}}_{\mathrm{HCOP}} \longrightarrow \mathrm{MMA}^{\mathrm{loc}}_{\mathrm{HCOP}}.$$

This reading may be used for later compression, indexing, and audit of the inherited text. It may not be used to erase the historical witness or to manufacture an OP closure certificate.

Proof. The interface theorem, Theorem 24.209, already supplies the finite typed execution stack for the current HC-routed OP-entry chain. Definition 24.212 extends this stack as an expository map over inherited text patterns that perform the same roles. Since the map is role-preserving and certificate-neutral by Lemma 24.213, it yields a finite MML/MMA reading for each matching target while leaving the original mathematical content intact. □

Corollary 24.216 (working rule)**.** *The next editorial pass may shorten repeated HC/OP explanations by citing* $\mathfrak{C}_{\mathrm{HCOP}}$*, but it should not delete active proof material yet. The safe immediate use of the compression map is to add labels, cross-references, tool tables, and short operation names to inherited blocks; stronger textual shortening should wait until the corresponding OP branch has either closed or returned to a stable frontier state.*

Proof. Immediate from Theorem 24.215. The compression map is currently an audit and refactoring layer, not a release-level deletion license. □

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24.7 First compression-guided HC/OP tool normal forms

Definition 24.217 (Reusable HC/OP tool normal form)**.** *A* reusable HC/OP tool normal form *is a finite typed contract*

$$\mathfrak{T} = \langle \text{name}, \text{input}, \text{guard}, \text{operation}, \text{output}, \text{status} \rangle$$

obtained from the compression map $\mathfrak{C}_{\mathrm{HCOP}}$*. Its entries have the following roles.*

(i) name *is one of the finite MML/MMA operation names already admitted by Definitions 24.203 and 24.212;*

(ii) input *is the typed object family consumed by the tool;*

(iii) guard *is the admissibility, route, frontier, or certificate-neutrality condition under which the tool may be invoked;*

(iv) operation *is the finite MML instruction word together with its MMA readout semantics;*

(v) output *is the typed object family returned by the tool;*

(vi) status *records whether the tool is diagnostic, normalizing, routing, comparison-level, frontier-level, or certificate-testing.*

A tool normal form is an invocation interface only. It may shorten later exposition by replacing repeated safety prose with a typed reference, but it does not erase the original proof material and does not create a theorematic certificate.

Definition 24.218 (First extracted HC/OP tool library)**.** *The first compression-guided HC/OP tool library is the finite family*

$$\mathbb{T}^{(1)}_{\mathrm{HCOP}} := \{\mathfrak{T}_{\mathrm{lock}}, \mathfrak{T}_{\mathrm{handoff}}, \mathfrak{T}_{\mathrm{norm}}, \mathfrak{T}_{\mathrm{route}}, \mathfrak{T}_{\mathrm{cmp}}, \mathfrak{T}_{\mathrm{resolve}}, \mathfrak{T}_{\mathrm{probe}}, \mathfrak{T}_{\mathrm{cert}}, \mathfrak{T}_{\mathrm{frontier}}, \mathfrak{T}_{\mathrm{archive}}\}.$$

Its intended contracts are summarized as follows.

Tool	***Input***	***Output***	***Guard / status***
$\mathfrak{T}_{\mathrm{lock}}$	*local HC final, corridor data*	*locked/captured HC final*	*guard: inherited route tag and local admissibility; status: final/capture*
$\mathfrak{T}_{\mathrm{handoff}}$	*transport-stable HC-capture chain*	*OP-facing handoff packet*	*guard: zero or controlled HC transport defect; status: handoff*
$\mathfrak{T}_{\mathrm{norm}}$	*handoff packet*	$\mathcal{N}^{\mathrm{HC}}_{\mathrm{OP}}$ *with live residual triple*	*guard: no change of theorematic status; status: normalization*
$\mathfrak{T}_{\mathrm{route}}$	$\mathcal{N}^{\mathrm{HC}}_{\mathrm{OP}}$*, selected residual branch*	*finite OP-facing task family*	*guard: residual type and route tag match; status: routing*
$\mathfrak{T}_{\mathrm{cmp}}$	*paired* $\rho_{\mathrm{CL}} \oplus \rho_{\Theta R}$ *branch*	$\Delta^{\mathrm{HC}}_{\mathrm{sep}}$ *and comparison record*	*guard: inseparable branch has not yet split; status: comparison*

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Tool	*Input*	*Output*	*Guard / status*
$\mathfrak{T}_{\mathrm{resolve}}$	*comparison record*	*bind, split, or frontier next state*	*guard: decision rule returns exactly one branch; status: resolution*
$\mathfrak{T}_{\mathrm{probe}}$	*probe/Monte-Carlo/decoder scan surface*	*ranked admissible final family*	*guard: diagnostic only; status: scan/rank/final*
$\mathfrak{T}_{\mathrm{cert}}$	*trigger-persistence or certificate question*	*yes/no/pending certificate-test record*	*guard: asks only whether a mathematical trigger exists; status: certificate-test*
$\mathfrak{T}_{\mathrm{frontier}}$	*non-closing but improving state*	*smaller frontier-certified pre-entry state*	*guard: strict controlled frontier descent; status: frontier*
$\mathfrak{T}_{\mathrm{archive}}$	*historical witness block*	*reusable tool role plus preserved genealogy*	*guard: no deletion of historical proof memory; status: archive-to-tool*

Lemma 24.219 (Extracted tool contracts are typed MML/MMA readings)**.** *Every member of $\mathbb{T}^{(1)}_{\mathrm{HCOP}}$ is a typed MML/MMA reading of one or more rows of the compression map $\mathfrak{C}_{\mathrm{HCOP}}$.*

Proof. Definition 24.218 is obtained by grouping the rows of Definition 24.212 by proof-use. The lock, handoff, normalization, routing, comparison, resolution, return, scan/rank, certificate-test, frontier-descent, and archive-to-tool roles are exactly the roles already admitted by the compression map and by the instruction alphabet. Hence each extracted contract is a typed reading over the existing QAM/MML/MMA stack, not a new primitive operation. □

Proposition 24.220 (Immediate simplification rule for inherited HC/OP prose)**.** *Let B be an inherited HC/OP text block that matches one of the patterns of Definition 24.212. If B has already stated its hypotheses and proof restrictions once, then later repetitions of the same proof-use may be replaced by a citation to the corresponding tool contract in $\mathbb{T}^{(1)}_{\mathrm{HCOP}}$, provided the replacement also cites the original local statement or keeps it in place.*

Proof. By Lemma 24.213, the compression map changes only the expository role of a block and preserves the original mathematical content. By Lemma 24.219, the corresponding tool contract records the same role in a finite typed form. Therefore a later repetition can be shortened to a tool invocation as long as the first occurrence and its hypotheses remain available. The proviso prevents the normal form from becoming a deletion rule. □

Proposition 24.221 (Where the first tool layer helps the existing text)**.** *The library $\mathbb{T}^{(1)}_{\mathrm{HCOP}}$ gives immediate refactoring leverage in the following existing zones of the Companion.*

(i) The HC-lock and capture paragraphs may cite $\mathfrak{T}_{\mathrm{lock}}$ instead of restating the full final/corridor discipline.

(ii) The local proof chain OP-entry paragraphs may cite $\mathfrak{T}_{\mathrm{handoff}}$, $\mathfrak{T}_{\mathrm{norm}}$, and $\mathfrak{T}_{\mathrm{route}}$ instead of repeating the full handoff-normalize-route chain.

(iii) The OP 1/4–OP 5(B) branch may cite $\mathfrak{T}_{\mathrm{cmp}}$ and $\mathfrak{T}_{\mathrm{resolve}}$ to express the joint-versus-separated comparison.

(iv) Probe, Monte-Carlo, decoder, and finite-search paragraphs may cite $\mathfrak{T}_{\mathrm{probe}}$ to emphasize that they rank finals rather than certify closure.

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(v) Trigger-persistence paragraphs may cite $\mathfrak{T}_{\mathrm{cert}}$ *to keep certification separate from execution.*

(vi) Historical witness paragraphs may cite $\mathfrak{T}_{\mathrm{archive}}$ *to extract reusable tools without shortening the preserved genealogy.*

Proof. Each zone listed is one of the main inherited zones already identified in Proposition 24.214. Definition 24.218 assigns the corresponding reusable contract. The proposition is therefore the first explicit refactoring consequence of the compression map. □

Theorem 24.222 (First compression-guided HC/OP reusable-tool layer)**.** *The compression map* $\mathfrak{C}_{\mathrm{HCOP}}$ *induces the finite reusable-tool layer* $\mathbb{T}^{(1)}_{\mathrm{HCOP}}$*. This layer may be used in later Companion revisions to shorten repeated HC/OP safety prose, index historical proof material, and audit OP-entry execution traces. It may not be used to delete active proof content or to manufacture an OP closure certificate.*

Proof. Existence of $\mathbb{T}^{(1)}_{\mathrm{HCOP}}$ is Definition 24.218. Its relation to the compression map is Lemma 24.219. Its safe simplification use is Proposition 24.220. Its no-false-certificate restriction follows from Lemma 24.207 and Lemma 24.213. Thus the layer is a reusable tool interface over the existing mathematics and not a new theorematic engine. □

Corollary 24.223 (working rule)**.** *The next safe editorial use of MML/MMA is to annotate repeated inherited HC/OP paragraphs by the tool contracts in* $\mathbb{T}^{(1)}_{\mathrm{HCOP}}$*. A future shortening pass may replace repeated explanations by these tool names only after the original local statement, hypotheses, and proof restrictions remain explicitly available elsewhere in the document.*

Proof. Immediate from Theorem 24.222. □

24.8 First tool-contract application to the active HC/OP proof spine

Definition 24.224 (Tool-indexed active HC/OP spine)**.** *The* tool-indexed active HC/OP spine *is the finite contract trace*

$$\mathcal{P}^{\mathrm{tool}}_{\mathrm{HCOP}} := \mathfrak{T}_{\mathrm{lock}};\mathfrak{T}_{\mathrm{handoff}};\mathfrak{T}_{\mathrm{norm}};\mathfrak{T}_{\mathrm{route}};\mathfrak{T}_{\mathrm{cmp}};\mathfrak{T}_{\mathrm{resolve}};\mathfrak{T}_{\mathrm{cert/frontier}}.$$

It is a compact citation layer over the local proof chain HC/OP proof spine. The trace is read as follows: lock/capture prepares the local HC final, handoff transports it toward OP-facing form, normalization produces $\mathcal{N}^{\mathrm{HC}}_{\mathrm{OP}}$*, routing selects a live residual work family, comparison executes the paired OP 1/4–OP 5(B) readout when separation is not yet justified, resolution returns bind, split, or frontier, and the final certificate/frontier test records whether the branch may promote or must remain a smaller frontier-certified state.*

Definition 24.225 (Tool-contract application site)**.** *A paragraph, lemma, proposition, or proof step is a* tool-contract application site *for a contract* $\mathfrak{T} \in \mathbb{T}^{(1)}_{\mathrm{HCOP}}$ *if all of the following are explicit either locally or by a nearby reference.*

(i) the typed input object of $\mathfrak{T}$*;*

(ii) the guard of $\mathfrak{T}$*, including route, frontier, and admissibility restrictions;*

(iii) the operation or comparison performed;

(iv) the typed output object;

(v) the status restriction that the invocation is diagnostic, routing, comparison-level, frontier-level, or certificate-testing rather than certificate-creating.

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At such a site, later repetitions may cite $\mathfrak{T}$ without restating the full safety prose, provided the original statement and its hypotheses remain visible in the document.

Lemma 24.226 (Active HC/OP blocks are contract-application sites)**.** *The local proof chain active HC/OP blocks are application sites for the corresponding contracts in Definition 24.224.*

Proof. The transport-stable HC-capture and handoff block states the local final, route-faithful transport guard, and OP-facing packet output, hence matches $\mathfrak{T}_{\mathrm{lock}}$ and $\mathfrak{T}_{\mathrm{handoff}}$. The normal form block states the handoff input, the live residual triple, and the non-closure status restriction, hence matches $\mathfrak{T}_{\mathrm{norm}}$. The task-routing block states the residual branch, admissible work family, and route tag, hence matches $\mathfrak{T}_{\mathrm{route}}$. The paired-comparison block states the inseparable $\rho_{\mathrm{CL}} \oplus \rho_{\Theta R}$ input, the comparison defect, and the bind/split/frontier return, hence matches $\mathfrak{T}_{\mathrm{cmp}}$ and $\mathfrak{T}_{\mathrm{resolve}}$. The inherited no-false-certificate rule supplies the final certificate/frontier guard. Therefore each block satisfies Definition 24.225. □

Proposition 24.227 (Tool-indexed rewriting of the active HC/OP chain)**.** *The active the local proof chain HC/OP chain admits the compressed reading*

$$\mathrm{HC}_{\mathrm{loc}} \xrightarrow{\mathfrak{T}_{\mathrm{lock}}} \mathrm{HC}_{\mathrm{cap}} \xrightarrow{\mathfrak{T}_{\mathrm{handoff}}} \mathcal{P}_{\mathrm{handoff}} \xrightarrow{\mathfrak{T}_{\mathrm{norm}}} \mathcal{N}_{\mathrm{OP}}^{\mathrm{HC}} \xrightarrow{\mathfrak{T}_{\mathrm{route}}} \mathcal{T}_{\mathrm{OP}}^{\mathrm{HC}} \xrightarrow{\mathfrak{T}_{\mathrm{cmp}},\mathfrak{T}_{\mathrm{resolve}}} \{\mathrm{bind}, \mathrm{split}, \mathrm{frontier}\}.$$

This rewriting is admissible as a cross-reference and audit normal form, but it is not an independent proof of any OP closure.

Proof. By Lemma 24.226, each arrow is backed by an already present active HC/OP block and by a typed contract in $\mathbb{T}^{(1)}_{\mathrm{HCOP}}$. The chain is therefore a finite MML/MMA execution trace in the sense of Theorem 24.209. Since Theorem 24.222 forbids tool contracts from manufacturing certificates, the compressed trace is an audit normal form only. □

Lemma 24.228 (Tool application preserves local hypotheses)**.** *Replacing a repeated explanation by a citation to a tool contract preserves the local hypotheses of the original explanation exactly when the application site cites the original statement or restates the required guard.*

Proof. A tool contract records input, guard, operation, output, and status; it does not by itself reproduce every local hypothesis. Definition 24.225 therefore requires the relevant guard to be explicit locally or by reference. Under that requirement, the replacement removes only duplicated prose and leaves the logical premises available. Without that requirement the replacement would be unsafe, which is why the definition includes it as a guard. □

Corollary 24.229 (First safe shortening target)**.** *The first safe shortening target is repeated no-false-certificate, route-preservation, and frontier-preservation prose following an already stated HC/OP operation. Such repetitions may be replaced by a sentence of the form:*

> *By the corresponding tool contract in $\mathbb{T}^{(1)}_{\mathrm{HCOP}}$, this is an admissible MML/MMA execution step and not a closure certificate.*

The original definition, theorem, or proof restriction must remain present elsewhere in the same local chain.

Proof. The repeated phrases listed are status and guard reminders, not new mathematical premises. Lemma 24.228 permits the repetition to be replaced once the underlying guard has already been stated. The no-false-certificate clause is exactly the status restriction of $\mathfrak{T}_{\mathrm{cert}}$ and of the active MML/MMA interface. □

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Theorem 24.230 (First tool-contract application theorem)**.** *The reusable tool library* $\mathbb{T}^{(1)}_{\mathrm{HCOP}}$ *can now be applied to the active HC/OP proof spine as a conservative cross-reference layer. It yields shorter notation, clearer audit points, and explicit preservation of route, frontier, and certificate restrictions, without changing theorematic status and without deleting historical witness material.*

Proof. Shorter notation follows from Proposition 24.227. Auditability follows because Definition 24.225 lists the required input, guard, operation, output, and status fields for every invocation. Preservation of local hypotheses follows from Lemma 24.228. The no-deletion and no-false-certificate restrictions are inherited from Theorem 24.222. Hence the layer is a conservative application of the tool library rather than a new proof step. □

Corollary 24.231 (working rule)**.** *Subsequent revisions may begin annotating the active HC/OP chain by* $\mathcal{P}^{\mathrm{tool}}_{\mathrm{HCOP}}$*. They should first replace only duplicated safety prose and local route/frontier reminders. They should not delete first occurrences of definitions, hypotheses, proof restrictions, historical witness blocks, or any statement needed to verify the no-false-certificate rule.*

Proof. Immediate from Theorem 24.230 and Corollary 24.229. □

24.9 First tool-guarded reduction protocol for inherited HC/OP prose

Definition 24.232 (Guarded HC/OP reduction packet)**.** *A* guarded HC/OP reduction packet *is a finite tuple*

$$\mathcal{R}^{(1)}_{\mathrm{HCOP}} := \langle B,\, \mathfrak{T},\, G_B,\, S_B,\, W_B \rangle$$

with the following data.

(i) B *is an inherited or active HC/OP paragraph, lemma step, proof reminder, or diagnostic status clause.*

(ii) $\mathfrak{T} \in \mathbb{T}^{(1)}_{\mathrm{HCOP}}$ *is the tool contract assigned to the repeated role of* B*.*

(iii) G_B *records the local guard: typed input, route tag, frontier condition, admissibility condition, and no-false-certificate status.*

(iv) S_B *records the theorematic status of the occurrence: definition, hypothesis, proof step, diagnostic reminder, certificate question, frontier update, or historical witness.*

(v) W_B *records where the unreduced witness of* B *remains available in the document.*

The packet is a reduction-control object only. It does not change the mathematical claim carried by B*.*

Definition 24.233 (Reducible and non-reducible HC/OP occurrences)**.** *Let* B *be an occurrence in the HC/OP proof spine or in an inherited HC/OP support zone. The occurrence is* the local pass-reducible *if it satisfies all of the following conditions.*

(i) It repeats a safety, route-preservation, frontier-preservation, diagnostic-status, scan/rank/final, comparison-status, or no-false-certificate clause already stated in the same local chain.

(ii) Its assigned tool contract $\mathfrak{T}$ *is one of the contracts of Definition 24.218.*

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(iii) The original hypotheses, guards, proof restrictions, and local statement remain visible either in B's first occurrence or in a nearby cited statement.

(iv) The replacement sentence explicitly says that the citation is a tool-contract invocation and not a closure certificate.

The occurrence is the local pass-non-reducible *if it introduces a first definition, a new object, a new hypothesis, a new guard, a new route tag, a certificate test, a theorematic status change, a proof-critical inequality, or historical witness material.*

Definition 24.234 (Guarded citation normal form)**.** *The preferred normal form for replacing an the local pass-reducible occurrence is*

> *By the guarded tool contract $\mathfrak{T}$ and the local guard recorded in G_B, this is an admissible MML/MMA execution or diagnostic step; it preserves route/frontier discipline and does not create an OP closure certificate.*

When the local context is already clear, the sentence may be shortened to a parenthetical form:

guarded by $\mathfrak{T}$; route/frontier preserved; no certificate manufactured.

Lemma 24.235 (Guarded reduction preserves proof content)**.** *If B is the local pass-reducible and is replaced by the guarded citation normal form of Definition 24.234, then the replacement preserves the proof content of the local chain.*

Proof. By Definition 24.233, only repeated status and guard reminders are eligible for replacement. The original hypotheses, proof restrictions, and first local statement remain available. By Definition 24.234, the replacement still records the assigned tool contract, preserves route/frontier discipline, and repeats the no-false-certificate status. Hence the replacement removes duplicated prose only; it does not remove the premise, operation, guard, output, or theorematic status of the original argument. □

Lemma 24.236 (Non-reducible occurrences protect theorematic status)**.** *Every the local pass-non-reducible occurrence must remain explicit in the document unless a later revision supplies a stronger local theorem that restates the same content with all hypotheses and guards.*

Proof. The non-reducible occurrences listed in Definition 24.233 are precisely the occurrences that create or alter mathematical content: they introduce objects, hypotheses, route tags, guards, certificate tests, status changes, proof-critical inequalities, or historical witness material. Replacing such an occurrence by a tool-name alone would hide a premise or change the audit surface. Therefore they are protected until a later theorem restates them explicitly. □

Proposition 24.237 (First usable simplification map)**.** *The reduction protocol induces the partial simplification map*

$$\mathsf{Red}^{(1)}_{\mathrm{HCOP}} : B \dashrightarrow \big(\mathfrak{T}(B), G_B, S_B, W_B\big),$$

where the map is defined exactly on the local pass-reducible occurrences and undefined on the local pass-non-reducible occurrences.

Proof. For a reducible occurrence, Definitions 24.232 and 24.234 assign the corresponding tool contract, guard, status, and witness pointer. Lemma 24.235 proves that this replacement is content-preserving. For a non-reducible occurrence, Lemma 24.236 forbids replacement by a mere tool invocation. Hence the simplification map is partial and conservative. □

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Theorem 24.238 (First tool-guarded HC/OP reduction theorem)**.** *The active HC/OP tool layer now supports a first conservative reduction protocol. Repeated safety prose, route/frontier reminders, diagnostic-status clauses, and no-false-certificate clauses may be replaced by guarded tool citations according to* $\mathsf{Red}^{(1)}_{\mathrm{HCOP}}$*. First definitions, first hypotheses, local guards, certificate tests, proof-critical inequalities, theorematic status changes, and historical witness material remain non-reducible. Therefore the protocol simplifies later exposition without altering the OP status of the document.*

Proof. The permitted replacements are exactly the defined values of the partial map in Proposition 24.237. Content preservation follows from Lemma 24.235. Protection of theorematic status follows from Lemma 24.236. The no-false-certificate restriction is inherited from Theorem 24.230. Thus the reduction protocol is a simplifier for repeated exposition and an audit aid, not a new OP closure mechanism. □

Corollary 24.239 (working rule)**.** *The next safe use of the tool layer is a local annotation pass: mark repeated HC/OP safety paragraphs as reducible or non-reducible before shortening them. Actual deletion or replacement should occur only after the first occurrence, hypotheses, guards, and historical witness pointer have been verified.*

Proof. Immediate from Theorem 24.238. □

24.10 First guarded-reduction application pass for the active HC/OP proof spine

Definition 24.240 (First guarded-reduction application ledger)**.** *The* first guarded-reduction application ledger *is the finite list*

$$\mathcal{L}^{(1)}_{\mathrm{HCOP}} := \{\ell_{\mathrm{lock}}, \ell_{\mathrm{handoff}}, \ell_{\mathrm{norm}}, \ell_{\mathrm{route}}, \ell_{\mathrm{cmp}}, \ell_{\mathrm{probe}}, \ell_{\mathrm{archive}}\}.$$

Each ledger entry has the form

$$\ell = \langle S_\ell,\, W_\ell,\, \mathfrak{T}_\ell,\, G_\ell,\, R_\ell \rangle,$$

where S_ℓ *is the local site class,* W_ℓ *is the protected unreduced witness,* $\mathfrak{T}_\ell \in \mathbb{T}^{(1)}_{\mathrm{HCOP}}$ *is the tool contract used for later citation,* G_ℓ *is the local guard data, and* R_ℓ *is the repeated prose tail that may be replaced by a guarded citation. The ledger is an application of the partial map* $\mathsf{Red}^{(1)}_{\mathrm{HCOP}}$ *from Proposition 24.237; it is not an instruction to remove the first occurrence of any definition, guard, certificate test, proof-critical inequality, or historical witness.*

Definition 24.241 (guarded citation sites)**.** *The entries of* $\mathcal{L}^{(1)}_{\mathrm{HCOP}}$ *are interpreted as follows.*

(i) ℓ_{lock} *covers later repeated lock/capture reminders after the first local HC-capture statement has been given. Its citation contract is* $\mathfrak{T}_{\mathrm{lock}}$*, and its protected witness is the unreduced local capture chain.*

(ii) ℓ_{handoff} *covers later repeated claims that HC capture is only a handoff into OP-entry and not an OP closure. Its citation contracts are* $\mathfrak{T}_{\mathrm{handoff}}$ *and* $\mathfrak{T}_{\mathrm{cert}}$*.*

(iii) ℓ_{norm} *covers later repeated normalization reminders for the HC-routed OP normal form. Its citation contract is* $\mathfrak{T}_{\mathrm{norm}}$*, guarded by the already visible residual triple and route tag.*

(iv) ℓ_{route} *covers later repeated route/frontier-preservation clauses in the finite task-routing object. Its citation contracts are* $\mathfrak{T}_{\mathrm{route}}$ *and* $\mathfrak{T}_{\mathrm{frontier}}$*.*

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(v) ℓ_{cmp} *covers later repeated status clauses for the paired OP 1/4–OP 5(B) comparison branch. Its citation contracts are* $\mathfrak{T}_{\mathrm{cmp}}$ *and* $\mathfrak{T}_{\mathrm{resolve}}$*, and its protected witness includes the explicit comparison defect* $\Delta^{\mathrm{HC}}_{\mathrm{sep}}$*.*

(vi) ℓ_{probe} *covers later repeated probe-diagnostic clauses stating that scan/rank/final output is not a certificate. Its citation contracts are* $\mathfrak{T}_{\mathrm{probe}}$ *and* $\mathfrak{T}_{\mathrm{cert}}$*.*

(vii) ℓ_{archive} *covers later repeated archive-to-tool extraction reminders. Its citation contract is* $\mathfrak{T}_{\mathrm{archive}}$*, and its protected witness is the unreduced historical source block.*

Definition 24.242 (Ledger citation normal form)**.** *For an occurrence assigned to an entry* $\ell \in \mathcal{L}^{(1)}_{\mathrm{HCOP}}$*, the preferred citation normal form is*

$$\mathrm{GRCite}_{\mathrm{HCOP}}(\ell) := \langle \mathfrak{T}_\ell, G_\ell, W_\ell; \mathrm{noCert} \rangle.$$

In prose this is read as:

> *Guarded by* $\mathfrak{T}_\ell$ *with local guard* G_ℓ *and protected witness* W_ℓ*; route/frontier discipline is preserved and no certificate is manufactured.*

The normal form may replace only a repeated tail R_ℓ*, never the protected witness* W_ℓ*.*

Lemma 24.243 (Ledger entries are -reducible where used)**.** *Every later occurrence covered by a ledger entry in Definition 24.241 is the local pass-reducible exactly when its first local witness, guard, and theorematic status remain visible.*

Proof. Each ledger entry records a repeated prose tail: safety status, route/frontier preservation, diagnostic non-certification, comparison status, or archive-to-tool extraction. These are precisely the reducible classes of Definition 24.233. The same definition excludes replacement whenever a first occurrence, guard, certificate test, proof-critical inequality, status change, or historical witness would be hidden. Hence the ledger is admissible only at later occurrences for which the protected witness W_ℓ remains visible. □

Proposition 24.244 (First guarded-reduction application is conservative)**.** *Applying* $\mathrm{GRCite}_{\mathrm{HCOP}}(\ell)$ *to the repeated tail of any admissible ledger entry preserves the mathematical content of the active HC/OP proof spine.*

Proof. The replacement is defined only for repeated tails R_ℓ. By Definition 24.240, the protected witness W_ℓ, the tool contract $\mathfrak{T}_\ell$, the local guard G_ℓ, and the no-certificate status remain part of the citation normal form. Lemma 24.243 reduces the claim to Lemma 24.235. Thus the operation shortens only repeated exposition and leaves the proof spine unchanged. □

Theorem 24.245 (First guarded-reduction application theorem)**.** *The active HC/OP proof spine now carries a first practical guarded-reduction application layer. Later repeated HC/OP prose may cite the finite ledger* $\mathcal{L}^{(1)}_{\mathrm{HCOP}}$ *in the normal form of Definition 24.242. This permits local exposition to be shortened after first occurrence, but it does not delete proof material, alter route tags, alter frontier status, convert diagnostics into certificates, or change the OP status of the document.*

Proof. Existence of the finite ledger follows from Definition 24.240. Admissibility of its local uses follows from Lemma 24.243. Conservativity follows from Proposition 24.244. The no-false-certificate restriction is explicitly retained in Definition 24.242 and is inherited from Theorem 24.238. Hence the layer is a guarded reduction application pass rather than a new theorematic closure step. □

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Corollary 24.246 (working rule)**.** *The next safe simplification pass may replace only later repeated HC/OP tails that match an entry of* $\mathcal{L}^{(1)}_{\mathrm{HCOP}}$*. The first occurrence of each definition, hypothesis, guard, certificate test, proof-critical inequality, and historical witness remains explicit.*

Proof. Immediate from Theorem 24.245. □

Definition 24.247 (Defect transport equation)**.** *Let* X_n *and* X_{n+1} *be adjacent closure-liquid-crystal cells. Their defect transport equation is written*

$$\mathcal{D}_{\mathrm{cell}}(X_n, X_{n+1}) \neq 0$$

whenever the first local failure is no longer merely screening-visible, but appears as a witness-grade or shell-grade obstruction transported across the current channel. The datum $\mathcal{D}_{\mathrm{cell}}$ *is therefore not a second closure criterion; it is the cellular packaging of the first locally non-contractible obstruction.*

Proposition 24.248 (OP 5 defect-transport reading)**.** *Within the current bundled OP 3→OP 5/global channel, witness-grade OP 5 reopening may be read as defect transport on the same closure-liquid-crystal medium: a bundled global pass corresponds to defect-free or defect-screened transport, while a bundled obstruction corresponds to transported or accumulated non-contractible defect data.*

Proof. By Theorem 32.60, Theorem 32.63, and Theorem 32.85, the current channel already distinguishes a bundled pass from a bundled obstruction while subordinating packet/phase/epsilon diagnostics to the carrier-level decision. The witness-grade local/global reopening discipline then identifies the precise place where a local obstruction must be reopened rather than screened away. This is exactly the cellular reading of $\mathcal{D}_{\mathrm{cell}} \neq 0$: a transported or accumulated defect of one and the same carrier medium rather than a second primitive theory beside the closure channel. □

Definition 24.249 (Liquid readout domain)**.** *Fix a distinguished readout regime* f*. A* liquid readout domain *in regime* f *is a family of closure-liquid-crystal cells whose carrier data lie in one closure-bearing class while their visible shell/phase data realize one coherent readout phase of that same bulk order. Different regimes or sectors may therefore produce different liquid readout domains without changing the underlying carrier datum in the strong closure sense.*

Remark 24.250 (Readouts as liquid-crystal domains of one bulk order)**.** *The point of Definition 24.249 is not to introduce a literal condensed-matter ontology. It is to express the present structural reading more economically: the visible sectors are not independent primitive crystals, but phase-sensitive ordered domains of one deeper closure-bearing bulk carrier. This is the cellular version of the readout-relative principle already used throughout the Companion.*

Definition 24.251 (Holographic Julia-boundary readout)**.** *Let* $\{X_n\}_{n\in I}$ *be an admissible closure-liquid-crystal configuration along a chosen branch/route family. A* holographic Julia-boundary readout *is a boundary map*

$$\mathcal{H}_{\mathrm{J}}(\{X_n\}_{n\in I})$$

that records the visible boundary pattern produced by the phase-/shell-sensitive readout of the same bulk carrier configuration. The intended output is a Julia-type boundary image, diagnostic strip, or related visual readout.

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Proposition 24.252 (Holographic Julia-boundary principle)**.** *The Julia-type diagnostics used in the present Companion may be read as holographic boundary readouts of an underlying closure-liquid-crystal configuration. In particular, they remain diagnostic or screening-level boundary objects: they support the active closure architecture, but they do not create a new independent closure axiom.*

Proof. The current document already treats packet-, phase-lock-, epsilon-, and frame-flicker diagnostics as visible readout layers subordinate to the carrier-level and closure-level statements; see Proposition 32.36 and Corollary 32.40. The same discipline applies to Julia-type visuals: they belong on the visible boundary-readout side and are informative exactly insofar as they track admissible passage, defect localization, or loss of admissibility in the bulk carrier medium. Hence they may be read holographically without becoming independent closure generators. □

Remark 24.253 (historical release line pilot status)**.** *The present section should be read as a first pilot layer for archival stage. It upgrades neither the current OP 3/QAM closure package nor the bundled OP 3→OP 5/global chain by new quantitative claims. Its role is organizational: it packages OP 3 as local cell compatibility, OP 5 as defect transport, visible sectors as liquid-crystal domains, Julia-type visuals as holographic boundary readout of one common carrier order, and tests a first addressed role layer in which the 0-band is reserved for primitive ZF↔QAM transport strata beneath later derived role families.*

Remark 24.254 (Status of the compressed addressed crystal reserve surface in the present release line)**.** *The earlier addressed-role pilot, the 0-band transport reservation, the locally licensed mixed addressed tensors, and the packet/corridor/hologram/readout-class ladder have now served their main purpose: they show that one closure-stable carrier may support visible domain phase, local defect loading, and holographic boundary readout without introducing a second primitive bulk family. In the present compressed line these ingredients are therefore retained only through one finite addressed crystal reserve surface.*

Definition 24.255 (Compressed addressed crystal reserve token)**.** *A* compressed addressed crystal reserve token *is a finite package*

$$\mathfrak{A}_{\mathrm{res}}^{\mathrm{cr}} := \langle \mathcal{B}_0,\, \mathfrak{M}_{\mathrm{pil}},\, \mathfrak{C}_{\mathrm{cr}}^{\mathrm{term}} \rangle$$

with the following intended reading.

(i) $\mathcal{B}_0$ *is the reserved primitive transport substrate between ZF and QAM;*

(ii) $\mathfrak{M}_{\mathrm{pil}}$ *denotes the finite licensed addressed mixed-tensor pilot family consisting of the carrier–domain, domain–defect, and carrier–boundary couplings above that substrate;*

(iii) $\mathfrak{C}_{\mathrm{cr}}^{\mathrm{term}}$ *is the terminal contraction-stable addressed crystal readout object that retains only the carrier/domain/defect/boundary content needed for later citation.*

The token is used only as a compressed reserve object. It does not claim a global unrestricted crystal calculus.

Proposition 24.256 (Compressed addressed crystal reserve package)**.** *Whenever a later argument uses only the finite addressed-crystal content encoded by*

$$\mathfrak{A}_{\mathrm{res}}^{\mathrm{cr}},$$

it may cite this single reserve token instead of reopening the full addressed-role pilot, the three local mixed addressed tensor pilots, or the packet/corridor/hologram/class-normal-form ladder. In particular, the compressed reserve token is citation-equivalent to the removed detailed reserve block for every argument that depends only on:

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(i) the primitive 0-band transport support,

(ii) the three locally licensed mixed addressed couplings,

(iii) one closure-stable carrier line together with its visible domain-face data,

(iv) the corresponding local defect-load data,

(v) the retained holographic boundary profile,

(vi) and the nonredundant interface data surviving passage to the terminal contraction-stable readout object.

Proof. The removed detailed block introduced exactly these ingredients in layered form. First it reserved the 0-band as primitive transport substrate and treated later addressed families as derived above it. Second it licensed only three local mixed addressed tensor couplings: carrier–domain, domain–defect, and carrier–boundary. Third it unfolded a packet/corridor/hologram/readout-class ladder whose only later use was to justify passage to a terminal contraction-stable object carrying the retained carrier/domain/defect/boundary data. Therefore every later argument that needs only this finite retained content may cite the single reserve token without reopening the intermediate pilot steps. No stronger global claim is being made. □

Corollary 24.257 (Use rule for the compressed addressed crystal reserve surface)**.** *A later proof may remain entirely on the compressed addressed crystal reserve surface as long as it needs only the finite retained content listed in Proposition 24.256. A finer reopening is required exactly when packet-specific, corridor-specific, subtype-specific, or pre-contraction addressed information is needed again.*

Proof. This is the operational restatement of Proposition 24.256. □

Lemma 24.258 (Closed antisymmetry of deficiency elements)**.** *Let f be a deficiency element in the relevant HC/closure test sector. If the J-symmetric subspace is already essentially self-adjoint on the admissible core, then the J-symmetric part of f vanishes. Equivalently,*

$$f = f^- \qquad \text{and hence} \qquad Jf = -f.$$

Proof. Write

$$f = f^+ + f^-, \qquad f^+ = \tfrac{1}{2}(f + Jf), \qquad f^- = \tfrac{1}{2}(f - Jf).$$

By construction, f^+ lies in the J-symmetric sector. The repaired Section 12 argument shows that once the admissible J-symmetric core is already essentially self-adjoint, a deficiency element cannot retain a nonzero J-symmetric component. Otherwise one would obtain a nontrivial deficiency vector in the symmetric core itself, contradicting essential self-adjointness. Hence $f^+ = 0$, so $f = f^-$ and therefore $Jf = -f$. □

Proposition 24.259 (Functional annihilation of deficiency elements by HC)**.** *For every deficiency element f in the closed HC test sector, the compensator annihilates f at the function level:*

$$C(f) = \tfrac{1}{2}(f + Jf) = 0.$$

Proof. By Lemma 24.258, every such deficiency element satisfies $Jf = -f$. Substituting into the defining compensator formula gives

$$C(f) = \tfrac{1}{2}(f + Jf) = \tfrac{1}{2}(f - f) = 0.$$

So the annihilation is a genuine functional consequence of closed J-antisymmetry, not merely a heuristic tensor slogan. □

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Remark 24.260 (Tensor representations are secondary to the functional proof)**.** *Any tensor or matrix representative of the compensator is basis-dependent and should be read only as a representation of the already established function-level map. In particular, the annihilation of deficiency elements is proved by the closed antisymmetry route*

$$f^+ = 0 \Longrightarrow Jf = -f \Longrightarrow C(f) = 0,$$

not by a standalone coefficient-level matrix argument.

Lemma 24.261 (Anchor-compatibility lemma)**.** *Compensator action is compatible with the orthogonal anchor. In particular, HC stabilizes the admissible readout channel without destroying the distinction between the anchored third direction and the first two sectors.*

Proof. By Lemma 19.12, the third sector is not decorative but the stabilizing orthogonal anchor. By Theorem 19.15, HC is the mechanism that re-identifies shifted descriptions in a common output channel. If HC destroyed the anchor distinction, then the stabilized output would collapse back into an undifferentiated binary pairing and the role of the third direction would be lost. That would contradict triolic minimality and the very need for an orthogonal anchor. Therefore HC must act compatibly with the anchored triolic organization. □

Lemma 24.262 (Closure forcing lemma)**.** *A triolic state with orthogonal anchor and forced HC is closure-stable. Equivalently,*

$$\mathrm{Triolic}(x) \wedge \mathrm{Anchored}(x) \wedge \mathrm{HCForced}(x) \Longrightarrow \mathrm{ClosureStable}(x).$$

Proof. We verify the three contributing conditions and their combination.

Step 1: Triolic structure. By Lemma 19.11, a triolic state carries a three-sector organization (K_1, K_2, K_3) as the minimal nondegenerate structure compatible with compensator action and closure readout. In particular, there exists a distinguished third sector K_3 in which the compensator-selected output can be stabilized.

Step 2: Orthogonal anchor. By Lemma 19.12, the third sector satisfies $K_3 \perp K_1$ and $K_3 \perp K_2$, preventing K_3 from being relabeled or absorbed into the dual pair (K_1, K_2). The anchor fixes the admissible direction of collapse/light selection.

Step 3: HC forcing. By Lemma 19.18, the compensator-normalized image $C(x)$ is admissible whenever x is admissible. Since $\mathrm{HCForced}(x)$ holds, the state lies in the compensator-selected channel, so $\mathcal{C}x = C(x)$.

Step 4: Closure stability. By the semantic–tensor correspondence of Remark 19.9, $\mathrm{ClosureStable}(x)$ is equivalent to $\hat{K}(\mathcal{C}x) = \mathcal{C}x$. By Step 3, $\mathcal{C}x = C(x)$, and by the idempotence of C (Lemma 19.17), $C(C(x)) = C(x)$, so $C(x)$ already lies in $\mathrm{im}(C)$. By Steps 1–2 and Theorem 4.3 ($\hat{K}(L_n) = L_{n-1}$, terminating at S_0), the anchor fixes K_3 as the structural ground sector; $\mathcal{C}x \in \mathrm{im}(C)$ lies in the K_3-aligned closure-stable class and is therefore a fixpoint of $\hat{K}$ on that class. Hence $\hat{K}(\mathcal{C}x) = \mathcal{C}x$, i.e. $\mathrm{ClosureStable}(x)$. □ □

Lemma 24.263 (Closure-stable iff compensator image)**.** *Under the hypotheses of the normalized semantic kernel (Definition 19.4), the following are equivalent for any admissible state x:*

(i) $\mathrm{ClosureStable}(x)$*, i.e.* $\hat{K}(\mathcal{C}x) = \mathcal{C}x$*;*

(ii) $x \in \mathrm{im}(C)$*, i.e.* $\mathcal{C}x = x$*.*

Hence $\mathrm{im}(C) = \mathcal{S}_0 := \{x \in \mathcal{S} : \mathrm{ClosureStable}(x)\}$.

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Proof. **(ii) ⇒ (i).** Suppose $\mathcal{C}x = x$. By Lemma 19.17, $C^2 = C$, so $x \in \mathrm{im}(C)$ implies $\mathcal{C}x = x$. By the ground-state terminality condition (S2) of Definition 19.4, $\hat{K}(x) = x$ for all $x \in S_0$, the photon ground sector. Since $\mathrm{im}(C)$ is mapped by $\hat{K}$ to the ground sector (Theorem 4.3: $\hat{K}(L_n) = L_{n-1}$, terminating at S_0), and since $\mathcal{C}x = x$ means x is already in the compensator-selected ground sector, we get $\hat{K}(\mathcal{C}x) = \hat{K}(x) = x = \mathcal{C}x$. Hence ClosureStable($x$).

(i) ⇒ (ii). Suppose $\hat{K}(\mathcal{C}x) = \mathcal{C}x$, i.e. $\mathcal{C}x$ is a fixpoint of $\hat{K}$. By Theorem 4.3, the only fixpoints of $\hat{K}$ in the cascade are the elements of the ground sector $S_0 = L = \bigcap_{n\geq 0} L_n$. Hence $\mathcal{C}x \in S_0$. By the semantic–tensor correspondence (Remark 19.9), $\mathcal{C}x \in S_0$ is precisely the condition that $\mathcal{C}$ has already reached its fixpoint on x, i.e. $\mathcal{C}(\mathcal{C}x) = \mathcal{C}x$. Since $C^2 = C$ (Lemma 19.17), this means $\mathcal{C}x \in \mathrm{im}(C)$, but also that x itself is mapped to the same compensator-fixed point. The re-identification mechanism (Theorem 19.15) selects precisely those x for which the compensated image is the state itself in the ground sector: $\mathcal{C}x = x$. Hence $x \in \mathrm{im}(C)$.

Conclusion. Both directions hold, so ClosureStable(x) $\Leftrightarrow$ $x \in \mathrm{im}(C)$, giving $\mathcal{S}_0 = \mathrm{im}(C)$. □

Proposition 24.264 (Closure class proposition)**.** *The closure-stable states form a structurally distinguished admissibility class. In particular, closure-stability is not an accidental property of isolated examples but a reusable criterion for admissible dynamics and readout.*

Proof. By Lemma 24.262, closure-stability follows from a fixed combination of generative conditions rather than from a one-off construction. By Lemma 19.18, the relevant states remain inside the admissible body under compensator normalization, and by Lemma 24.261 this process respects the stabilizing triolic organization. Hence closure-stability is preserved under the normalized structural mechanism and can be used as a class criterion. □

Proposition 24.265 (Exact closure criterion)**.** *Exact closure is the regime in which the admissible channel selected by HC is already terminal with respect to the relevant closure test, so that no further residual compensator correction is required.*

Proof. By Lemma 19.17, compensator-normalized states do not drift under repeated HC action. Exact closure is therefore the stronger case in which the normalized state is not merely admissible but already stable under the relevant closure readout. In the tensor realization this corresponds to the absence of further leakage from the selected sector into a complementary block; semantically it means that the compensator-selected channel has already reached its terminal closure form. □

Lemma 24.266 (Closure residue lemma)**.** *Whenever exact closure fails, the failure can be measured by a residual term that records the remaining mismatch between HC-normalized admissibility and full closure stability.*

Proof. The exact-closure regime of Proposition 24.265 is defined precisely by the absence of further correction after compensator normalization. If this stronger condition fails while admissibility is still preserved, then there remains a nonzero discrepancy between the compensator-selected channel and the fully closed channel. This discrepancy is the closure residue. It vanishes in exact closure and provides the natural quantity measuring departure from that regime. □

Corollary 24.267 (First closure forcing chain)**.** *The first unfolded closure route of the normalized core is*

$$\mathrm{HC} \Longrightarrow \textit{HC idempotence/admissibility} \Longrightarrow \textit{anchor-compatible normalization}$$
$$\Longrightarrow \textit{closure forcing} \Longrightarrow \textit{closure class} \Longrightarrow \textit{exact closure or closure residue.}$$

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Proof. Lemmas 19.17, 19.18, and 24.261 identify the first structural properties of forced HC. Lemma 24.262 turns these into closure stability, Proposition 24.264 promotes this to a reusable admissibility class, and Proposition 24.265 together with Lemma 24.266 separates the exact and residual regimes. □

24.11 First guarded-reduction pilot normal form for repeated HC/OP tails

Definition 24.268 (Guarded-reduction pilot window)**.** *A* guarded-reduction pilot window *is a finite local pair*

$$\mathsf{W}_{\mathrm{red}}(B_0, B_1)$$

where B_0 is the first explicit occurrence of an HC/OP argument tail and B_1 is a later repetition of the same tail. The pair is admissible only when the following data are fixed.

(i) A ledger entry $\ell \in \mathcal{L}^{(1)}_{\mathrm{HCOP}}$ covers both occurrences.

(ii) The first occurrence B_0 states the local hypotheses, route tag, frontier guard, witness, and theorematic status.

(iii) The later occurrence B_1 repeats only the same safety, route-preservation, frontier-preservation, diagnostic-status, or no-false-certificate tail.

(iv) The later occurrence introduces no first definition, no new hypothesis, no new certificate test, no proof-critical inequality, and no theorematic status change.

The pilot window is therefore a local rewrite domain for repeated prose only.

Definition 24.269 (guarded-citation normal form)**.** *For an admissible pilot window $\mathsf{W}_{\mathrm{red}}(B_0, B_1)$, the* guarded-citation normal form *of the repeated tail B_1 is*

$$\mathrm{GRCite}_{\mathrm{HCOP}}(\ell; G, W, S) := \big\langle \mathfrak{T}_\ell,\ G_{B_0},\ W_{B_0},\ S_{B_0}; \mathrm{noCert} \big\rangle.$$

Here $\mathfrak{T}_\ell$ is the tool contract assigned by the ledger, G_{B_0} is the preserved local guard of the first occurrence, W_{B_0} is the preserved witness or first explicit statement, and S_{B_0} is the preserved status marker. The final tag noCert *records that the citation does not create an OP closure certificate.*

Lemma 24.270 (Pilot normal form is conservative)**.** *If $\mathsf{W}_{\mathrm{red}}(B_0, B_1)$ is an admissible guarded-reduction pilot window, then replacing the later repeated tail B_1 by $\mathrm{GRCite}_{\mathrm{HCOP}}(\ell; G, W, S)$ preserves the proof content of the local argument.*

Proof. By Definition 24.268, the first occurrence B_0 remains explicit and contains the hypotheses, guard, witness, and status. By Definition 24.269, the replacement of B_1 cites exactly the ledger tool contract together with pointers back to those retained data. Since B_1 is assumed to repeat only a safety, route, frontier, diagnostic, or no-false-certificate tail, no first mathematical content is removed. Hence the replacement is conservative. □

Proposition 24.271 (Allowed pilot windows in the active HC/OP spine)**.** *The following families are admissible first pilot windows for guarded reduction in the active HC/OP spine:*

(i) lock/capture reminder windows governed by ℓ_{lock};

(ii) handoff and normalization reminder windows governed by ℓ_{handoff} and ℓ_{norm};

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(iii) route/frontier reminder windows governed by ℓ_{route}*;*

(iv) paired-comparison status windows governed by ℓ_{cmp}*;*

(v) probe diagnostic windows governed by ℓ_{probe}*;*

(vi) archive-to-tool extraction windows governed by ℓ_{archive}*.*

Each family is reducible only after its first local occurrence has been preserved.

Proof. These are precisely the ledger sites of Definition 24.241. Lemma 24.243 proves that later occurrences covered by those entries are reducible when first occurrence, guard, witness, and theorematic status remain visible. Definition 24.268 adds no new reducible class; it only names the local two-occurrence window used by the pilot rewrite. □

Definition 24.272 (Pilot tail-compression operator)**.** *The* pilot tail-compression operator

$$\mathsf{TailRed}^{(1)}_{\mathrm{HCOP}}$$

is the partial operator that maps an admissible pilot window to its guarded-citation normal form:

$$\mathsf{TailRed}^{(1)}_{\mathrm{HCOP}}\big(\mathsf{W}_{\mathrm{red}}(B_0, B_1)\big) := \mathrm{GRCite}_{\mathrm{HCOP}}(\ell; G, W, S).$$

It is undefined on first occurrences, non-reducible proof material, historical witness material, and any occurrence that changes theorematic status.

Corollary 24.273 (Pilot reduction cannot hide certificate tests)**.** *The operator* $\mathsf{TailRed}^{(1)}_{\mathrm{HCOP}}$ *cannot hide a certificate test or turn a frontier state into an OP closure state.*

Proof. Certificate tests and theorematic status changes are excluded from admissible pilot windows by Definition 24.268. The replacement normal form of Definition 24.269 also carries the explicit noCert marker. Therefore no application of $\mathsf{TailRed}^{(1)}_{\mathrm{HCOP}}$ can manufacture or hide a certificate. □

Theorem 24.274 (First guarded-reduction pilot normal form)**.** *The guarded-reduction ledger induces a conservative pilot normal form for repeated HC/OP tails. In every admissible pilot window, the later repeated tail may be replaced by* $\mathrm{GRCite}_{\mathrm{HCOP}}(\ell; G, W, S)$*, while the first occurrence, local guard, witness, proof restriction, and theorematic status remain explicit. The replacement shortens repeated prose only; it neither deletes proof content nor creates an OP closure certificate.*

Proof. Existence of the relevant ledger entries is Definition 24.241. The local rewrite domain is Definition 24.268; the citation target is Definition 24.269; and conservativity is Lemma 24.270. Corollary 24.273 gives the certificate restriction. Thus the pilot normal form is a controlled compression of repeated tails and not a new proof engine. □

Corollary 24.275 (working rule)**.** *Future HC/OP text may use* $\mathrm{GRCite}_{\mathrm{HCOP}}(\ell; G, W, S)$ *only for later repeated tails already covered by the ledger. The first occurrence, local guard, witness, and theorematic status must remain visible. Historical witness material and first mathematical content remain non-reducible.*

Proof. Immediate from Theorem 24.274. □

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24.12 First tail-normalized HC/OP proof skeleton

Definition 24.276 (Tail-normalized HC/OP proof skeleton)**.** *A* tail-normalized HC/OP proof skeleton *is a finite ordered family*

$$\mathsf{S}^{\mathrm{tail}}_{\mathrm{HCOP}} = \big((E_i, G_i, W_i, S_i; T_i)\big)_{i=0}^{m}$$

where each entry consists of

(i) a first explicit source block E_i, containing the relevant definition, hypothesis, guard, or proof-critical statement;

(ii) a local guard G_i and preserved witness or source pointer W_i;

(iii) a theorematic status marker S_i, recording whether the block is definitional, diagnostic, frontier-preserving, certificate-testing, or theorematic;

(iv) a finite family T_i of later repeated tails that may be replaced only by guarded citations of the form

$$\mathrm{GRCite}_{\mathrm{HCOP}}(\ell_i; G_i, W_i, S_i).$$

The family is admissible only when the ordering of entries agrees with the proof-dependency order of the active HC/OP spine.

Definition 24.277 (Active tail skeleton)**.** *The* active the local proof chain tail skeleton *is the tail-normalized skeleton whose source entries are ordered as*

$$\begin{aligned}
E_0 &= \mathrm{LOCK/CAPTURE}_{\mathrm{HC}},\\
E_1 &= \mathrm{HANDOFF}_{\mathrm{HC}\to\mathrm{OP}},\\
E_2 &= \mathrm{NORMALIZE}^{\mathrm{HC}}_{\mathrm{OP}},\\
E_3 &= \mathrm{ROUTE}^{\mathrm{HC}}_{\mathrm{OP}},\\
E_4 &= \mathrm{COMPARE}^{1/4|5(B)}_{\mathrm{HC}},\\
E_5 &= \mathrm{RESOLVE}_{\mathrm{bind/split/frontier}}.
\end{aligned}$$

The associated tail families T_i consist only of later repeated safety, route-preservation, frontier-preservation, diagnostic, no-false-certificate, or archive-to-tool reminders covered by the ledger.

Lemma 24.278 (Tail skeleton preserves dependency order)**.** *Let $\mathsf{S}^{\mathrm{tail}}_{\mathrm{HCOP}}$ be an admissible tail-normalized HC/OP proof skeleton. Then replacing any later tail in T_i by its guarded citation normal form preserves the dependency order of the proof spine.*

Proof. By Definition 24.276, the first explicit source block E_i, its guard G_i, witness W_i, and status marker S_i remain in the ordered skeleton. The replacement acts only on later tails attached to that source entry. Since no replacement can move a later tail before its source entry and no source entry is removed, the original dependency order is preserved. □

Lemma 24.279 (Tail skeleton does not create certificates)**.** *No admissible tail-normalized HC/OP proof skeleton can turn a diagnostic, frontier, or guarded execution state into an OP closure certificate.*

Proof. Every tail replacement is a guarded citation of the form defined in Definition 24.269; it carries the explicit noCert marker. Certificate tests and theorematic status changes are not eligible tail material by Definition 24.268. Therefore the skeleton can reorganize later repeated prose, but it cannot change theorematic status. □

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Proposition 24.280 (Active HC/OP spine admits a tail-normalized skeleton)**.** *The active the local proof chain HC/OP proof spine admits the tail-normalized skeleton of Definition 24.277.*

Proof. The source entries are exactly the tool-contract application sites of the active HC/OP spine: lock/capture, handoff, normalization, routing, paired comparison, and resolution. The ledger identifies the corresponding repeated tail classes, while the pilot normal form gives the admissible guarded citation for later repetitions. Hence these source entries and tail families form an admissible ordered skeleton. □

Corollary 24.281 (Tail-normalized reuse rule)**.** *Future HC/OP text may cite the active tail skeleton only after the relevant source entry has appeared explicitly. A citation to the skeleton may replace a later repeated tail, but it may not replace a first definition, a first guard, a witness, a certificate test, a proof-critical inequality, or a theorematic status change.*

Proof. This is the direct reuse form of Lemmas 24.278 and 24.279. □

Theorem 24.282 (First tail-normalized HC/OP proof skeleton)**.** *The local proof chain MM-L/MMA compression, tool-contract, ledger, and guarded-citation layers induce a first conservative tail-normalized proof skeleton for the active HC/OP spine. The skeleton shortens only later repeated HC/OP tails by guarded citation, preserves all first source blocks, preserves dependency order, and cannot manufacture an OP closure certificate.*

Proof. The compression map is supplied by the layer, the tool contracts by the layer, their application to the active spine by the layer, the reduction discipline by the protocol, the finite citation sites by the ledger, and the guarded citation normal form by the pilot. Proposition 24.280 identifies the corresponding ordered skeleton for the local proof chain HC/OP spine. Lemma 24.278 proves dependency preservation, and Lemma 24.279 proves the certificate restriction. Thus the skeleton is conservative. □

Corollary 24.283 (working rule)**.** *The active HC/OP proof spine may now be cited in tail-normalized form as*

$$\begin{aligned}\mathsf{S}^{\mathrm{tail}}_{\mathrm{HCOP}} = (&\mathrm{LOCK/CAPTURE}, \mathrm{HANDOFF}, \mathrm{NORMALIZE},\\ &\mathrm{ROUTE}, \mathrm{COMPARE}, \mathrm{RESOLVE}).\end{aligned}$$

After the first occurrence of each source block has been stated, this citation is a compact reference form for later repeated tails only.

Proof. Immediate from Theorem 24.282. □

24.13 First skeleton-to-workstep HC/OP dispatch

Definition 24.284 (Skeleton-to-workstep dispatch object)**.** *Let* $\mathsf{S}^{\mathrm{tail}}_{\mathrm{HCOP}}$ *be the active tail-normalized HC/OP proof skeleton of Definition 24.277. A* skeleton-to-workstep dispatch object *is a finite map*

$$\mathsf{D}^{\mathrm{skel}}_{\mathrm{HCOP}} : \{E_0, E_1, E_2, E_3, E_4, E_5\} \longrightarrow \mathsf{Work}_{\mathrm{HCOP}}$$

which assigns to each explicit source node one admissible family of follow-on work packets. The map is admissible only if the assigned packets preserve the source guard G_i*, witness pointer* W_i*, theorematic status marker* S_i*, and the no-false-certificate restriction inherited from the guarded-citation layer.*

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Definition 24.285 (First dispatch menu)**.** *The first dispatch menu is the following finite assignment:*

$$\begin{aligned} E_0 = \mathrm{LOCK/CAPTURE}_{\mathrm{HC}} &\longmapsto \mathsf{W}^{\mathrm{stab}}_{\mathrm{lock}}, \\ E_1 = \mathrm{HANDOFF}_{\mathrm{HC}\to\mathrm{OP}} &\longmapsto \mathsf{W}^{\mathrm{guard}}_{\mathrm{handoff}}, \\ E_2 = \mathrm{NORMALIZE}^{\mathrm{HC}}_{\mathrm{OP}} &\longmapsto \mathsf{W}^{\mathrm{type}}_{\mathrm{norm}}, \\ E_3 = \mathrm{ROUTE}^{\mathrm{HC}}_{\mathrm{OP}} &\longmapsto \mathsf{W}^{\mathrm{rank}}_{\mathrm{route}}, \\ E_4 = \mathrm{COMPARE}^{1/4|5(B)}_{\mathrm{HC}} &\longmapsto \mathsf{W}^{\Delta}_{\mathrm{cmp}}, \\ E_5 = \mathrm{RESOLVE}_{\mathrm{bind/split/frontier}} &\longmapsto \mathsf{W}^{\mathrm{next}}_{\mathrm{resolve}}. \end{aligned}$$

Here $\mathsf{W}^{\mathrm{stab}}_{\mathrm{lock}}$ *checks stability of the local capture source,* $\mathsf{W}^{\mathrm{guard}}_{\mathrm{handoff}}$ *checks whether handoff guards remain visible,* $\mathsf{W}^{\mathrm{type}}_{\mathrm{norm}}$ *checks type-correct normalization,* $\mathsf{W}^{\mathrm{rank}}_{\mathrm{route}}$ *ranks the routed residuals,* $\mathsf{W}^{\Delta}_{\mathrm{cmp}}$ *acts on the paired comparison defect, and* $\mathsf{W}^{\mathrm{next}}_{\mathrm{resolve}}$ *selects the next conservative branch without changing theorematic status.*

Lemma 24.286 (Dispatch preserves the tail-skeleton order)**.** *Every admissible skeleton-to-workstep dispatch object preserves the dependency order of the active HC/OP proof spine.*

Proof. By Definition 24.284, a dispatch packet is attached to an already ordered source node E_i of the tail skeleton. It does not move the source node, delete its guard, or replace a preceding dependency by a later citation. Therefore the order established in Lemma 24.278 is inherited by the dispatched work packets. □

Lemma 24.287 (Dispatch is work-generating, not certificate-generating)**.** *No admissible skeleton-to-workstep dispatch object can manufacture an OP closure certificate.*

Proof. The dispatch object inherits the no-false-certificate marker from the guarded-citation layer and from Lemma 24.279. Its codomain consists of follow-on work packets, not theorematic closure certificates. A certificate may only appear after a separate explicit certificate test with its own hypotheses, guards, and proof-critical inequalities. Hence dispatch creates controlled work, not closure. □

Proposition 24.288 (First controlled next-work selection from the HC/OP skeleton)**.** *The active tail-normalized HC/OP proof skeleton admits the dispatch menu of Definition 24.285. In particular, the immediate admissible follow-on work is finite and restricted to stabilization, handoff-guard checking, type-normalization checking, route ranking, comparison-defect work, and conservative branch selection.*

Proof. Each menu entry is attached to one of the ordered source nodes of Definition 24.277. The first four entries refine already explicit source nodes and therefore preserve their guards and witnesses. The comparison entry $\mathsf{W}^{\Delta}_{\mathrm{cmp}}$ is tied to the paired comparison layer and may only act on the already defined comparison defect. The resolution entry selects a conservative next branch from the already permitted outcomes bind, split, or frontier. By Lemmas 24.286 and 24.287, the whole menu is conservative. □

Theorem 24.289 (First skeleton-to-workstep HC/OP dispatch)**.** *The tail-normalized HC/OP proof skeleton induces a first conservative skeleton-to-workstep dispatch object. This dispatch turns the compact proof skeleton into a finite menu of admissible next work packets while preserving dependency order, guards, witnesses, route tags, and the no-false-certificate rule.*

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Proof. Definition 24.284 specifies the admissibility requirements for dispatch. Definition 24.285 provides the first finite menu. Proposition 24.288 proves that this menu is admissible for the active skeleton. The order and certificate restrictions follow from Lemmas 24.286 and 24.287. □

Corollary 24.290 (working rule)**.** *Future HC/OP work may be launched from the tail-normalized skeleton only through an explicit dispatch packet*

$$\mathsf{D}^{\mathrm{skel}}_{\mathrm{HCOP}}(E_i) = \mathsf{W}_i.$$

Such a packet records what is to be checked next; it does not by itself prove OP 1/4, OP 5(B), or any closure certificate.

Proof. Immediate from Theorem 24.289. □

24.14 First dispatched lock/capture stabilization packet

Definition 24.291 (Dispatched lock/capture stabilization packet)**.** *Let* $E_0 = \mathrm{LOCK/CAPTURE}_{\mathrm{HC}}$ *be the first node of the tail-normalized HC/OP skeleton and let*

$$\mathsf{D}^{\mathrm{skel}}_{\mathrm{HCOP}}(E_0) = \mathsf{W}^{\mathrm{stab}}_{\mathrm{lock}}$$

be the dispatch assignment. A dispatched lock/capture stabilization packet *is a finite packet*

$$\mathsf{P}^{\mathrm{stab}}_{\mathrm{lock}} = \big(\mathcal{C}^{\mathrm{HC}}_n, G_0, W_0, S_0, \mathcal{A}_0, \mathbf{d}^{\mathrm{stab}}_{\mathrm{lock}}\big),$$

where $\mathcal{C}^{\mathrm{HC}}_n$ *is the protected local HC-capture chain,* G_0 *is the local guard,* W_0 *is the witness pointer to the first explicit source block,* S_0 *is the non-certificate status marker,* $\mathcal{A}_0$ *is the finite local audit window, and* $\mathbf{d}^{\mathrm{stab}}_{\mathrm{lock}}$ *is a non-negative lock-stability defect register. The packet is admissible only if it does not widen the local corridor, erase the route tag, replace* W_0*, or change* S_0 *into a theorematic certificate.*

Definition 24.292 (Lock-stability defect register)**.** *For a dispatched packet* $\mathsf{P}^{\mathrm{stab}}_{\mathrm{lock}}$*, the* lock-stability defect register *is the finite family*

$$\mathbf{d}^{\mathrm{stab}}_{\mathrm{lock}}(j) = \big(d_{\mathrm{leak}}(j), d_{\mathrm{corr}}(j), d_{\mathrm{guard}}(j)\big), \qquad j \in \mathcal{A}_0,$$

where d_{leak} *measures re-opening of lock-side leakage,* d_{corr} *measures corridor drift, and* d_{guard} *measures loss of the local guard. The associated scalar audit defect is*

$$\Delta^{\mathrm{stab}}_{\mathrm{lock}} = \max_{j\in\mathcal{A}_0}\big(d_{\mathrm{leak}}(j) + d_{\mathrm{corr}}(j) + d_{\mathrm{guard}}(j)\big).$$

The register is called stable *if* $\Delta^{\mathrm{stab}}_{\mathrm{lock}} = 0$*,* repairable *if a permitted local micro-lock refinement strictly lowers the scalar audit defect without widening the corridor, and* frontier-bound *otherwise.*

Lemma 24.293 (Zero lock-stability defect preserves the capture source)**.** *If* $\Delta^{\mathrm{stab}}_{\mathrm{lock}} = 0$ *for an admissible dispatched lock/capture stabilization packet, then the protected local HC-capture chain remains a valid capture source throughout the finite audit window.*

Proof. Vanishing scalar audit defect means that every component of $\mathbf{d}^{\mathrm{stab}}_{\mathrm{lock}}(j)$ vanishes on $\mathcal{A}_0$. Hence the lock-side leakage is not re-opened, the corridor does not drift, and the local guard is not lost. The packet also preserves the witness pointer and the non-certificate status marker by Definition 24.291. Therefore the original local HC-capture chain remains the same protected capture source, rather than a replaced or widened source. □

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Definition 24.294 (lock/capture stabilization return)**.** *The execution of* $\mathsf{W}_{\mathrm{lock}}^{\mathrm{stab}}$ *on an admissible packet returns exactly one of the following statuses:*

$$\mathrm{Exec}\big(\mathsf{W}_{\mathrm{lock}}^{\mathrm{stab}}, \mathsf{P}_{\mathrm{lock}}^{\mathrm{stab}}\big) \in \{\mathsf{L}_{\mathrm{stable}}, \mathsf{L}_{\mathrm{repair}}, \mathsf{L}_{\mathrm{frontier}}\}.$$

Here $\mathsf{L}_{\mathrm{stable}}$ *means that the capture source is handoff-ready for the next skeleton node,* $\mathsf{L}_{\mathrm{repair}}$ *means that one bounded local repair step is admissible before handoff is tested again, and* $\mathsf{L}_{\mathrm{frontier}}$ *means that the source remains protected but must be kept as a smaller frontier source rather than promoted.*

Proposition 24.295 (The first dispatched lock/capture packet is admissible)**.** *The dispatch menu admits the lock/capture stabilization packet as the first executable work packet attached to the active HC/OP skeleton.*

Proof. The dispatch menu assigns E_0 only to $\mathsf{W}_{\mathrm{lock}}^{\mathrm{stab}}$. The data required in Definition 24.291 are inherited from the first explicit lock/capture source block, the guarded-reduction ledger, and the tail-normalized skeleton: the protected capture chain, the local guard, the witness pointer, and the non-certificate marker are all already part of the active spine. The new defect register adds only a finite audit surface for this first packet. It does not replace a hypothesis, alter a route tag, or create a certificate. Hence the packet is admissible. □

Theorem 24.296 (First dispatched lock/capture stabilization packet)**.** *The active HC/OP skeleton now supports a first executed work packet*

$$E_0 \xrightarrow{\mathsf{W}_{\mathrm{lock}}^{\mathrm{stab}}} \{\mathsf{L}_{\mathrm{stable}}, \mathsf{L}_{\mathrm{repair}}, \mathsf{L}_{\mathrm{frontier}}\}.$$

This execution is conservative: it either prepares the protected capture source for the handoff node, records one bounded local repair obligation, or returns a smaller frontier source. It does not prove OP 1/4, OP 5(B), or any closure certificate.

Proof. By Proposition 24.295, the first dispatched packet is admissible for the active skeleton. Definition 24.292 classifies the finite audit defect into the stable, repairable, or frontier-bound cases, and Definition 24.294 records the corresponding return status. Lemma 24.293 proves the stable case. In the repairable case the packet remains at the same skeleton node with a strictly smaller local defect and no corridor widening; in the frontier-bound case it is returned as protected frontier material without promotion. In none of the three cases is the non-certificate marker changed into an OP closure certificate. □

Corollary 24.297 (working rule)**.** *Future work may proceed from the dispatch menu by first executing* $\mathsf{W}_{\mathrm{lock}}^{\mathrm{stab}}$*. The next admissible mathematical question is therefore whether the return status is handoff-ready, locally repairable, or frontier-bound; it is not yet whether OP 1/4 or OP 5(B) is closed.*

Proof. Immediate from Theorem 24.296. □

24.15 First dispatched handoff-guard packet

Definition 24.298 (Dispatched handoff-guard packet)**.** *Let* $E_1 = \mathrm{HANDOFF}_{\mathrm{HC}\to\mathrm{OP}}$ *be the second node of the tail-normalized HC/OP skeleton and let*

$$\mathsf{D}_{\mathrm{HCOP}}^{\mathrm{skel}}(E_1) = \mathsf{W}_{\mathrm{handoff}}^{\mathrm{guard}}$$

be the dispatch assignment. A dispatched handoff-guard packet *is a finite packet*

$$\mathsf{P}_{\mathrm{handoff}}^{\mathrm{guard}} = \big(\mathsf{L}_{\mathrm{stable}}, \mathcal{P}_{\mathrm{handoff}}, G_1, W_1, S_1, \mathbf{d}_{\mathrm{handoff}}^{\mathrm{guard}}\big),$$

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where $\mathsf{L}_{\mathrm{stable}}$ *is the stable lock/capture return status of Definition 24.294,* $\mathcal{P}_{\mathrm{handoff}}$ *is the inherited HC-to-OP handoff packet,* G_1 *is the handoff guard,* W_1 *is the witness pointer to the first explicit handoff source block,* S_1 *is the non-certificate status marker, and* $\mathbf{d}^{\mathrm{guard}}_{\mathrm{handoff}}$ *is a nonnegative handoff-guard defect register. The packet is admissible only if it is launched from* $\mathsf{L}_{\mathrm{stable}}$*, keeps the lock/capture witness visible, preserves the HC-to-OP route tag, and does not convert handoff-readiness into an OP closure certificate.*

Definition 24.299 (Handoff-guard defect register)**.** *For an admissible packet* $\mathsf{P}^{\mathrm{guard}}_{\mathrm{handoff}}$*, the* handoff-guard defect register *is the finite family*

$$\mathbf{d}^{\mathrm{guard}}_{\mathrm{handoff}}(j) = \big(d_{\mathrm{src}}(j), d_{\mathrm{tag}}(j), d_{\mathrm{wit}}(j), d_{\mathrm{stat}}(j)\big), \qquad j \in \mathcal{A}_1,$$

where d_{src} *measures loss of the stabilized capture source,* d_{tag} *measures drift of the HC-to-OP route tag,* d_{wit} *measures loss of the handoff witness pointer, and* d_{stat} *measures accidental promotion of the non-certificate status marker. The scalar handoff audit defect is*

$$\Delta^{\mathrm{guard}}_{\mathrm{handoff}} = \max_{j\in\mathcal{A}_1}\big(d_{\mathrm{src}}(j) + d_{\mathrm{tag}}(j) + d_{\mathrm{wit}}(j) + d_{\mathrm{stat}}(j)\big).$$

The register is handoff-clean *if* $\Delta^{\mathrm{guard}}_{\mathrm{handoff}} = 0$*,* guard-repairable *if a bounded local handoff-guard repair strictly lowers the scalar defect while preserving the stable capture source and route tag, and* frontier-bound *otherwise.*

Lemma 24.300 (Stable lock/capture return is the handoff precondition)**.** *The handoff-guard packet may be executed only from the stable lock/capture return status* $\mathsf{L}_{\mathrm{stable}}$*. If the return status is* $\mathsf{L}_{\mathrm{repair}}$ *or* $\mathsf{L}_{\mathrm{frontier}}$*, then the handoff node is not yet admissible.*

Proof. By Definition 24.298, the first entry of $\mathsf{P}^{\mathrm{guard}}_{\mathrm{handoff}}$ is $\mathsf{L}_{\mathrm{stable}}$. A repair return means that the first skeleton node still carries a bounded local obligation, while a frontier return means that the protected source is not promoted. Neither status supplies the handoff-ready source required at the second skeleton node. Thus handoff execution is blocked until the packet has returned the stable status. □

Lemma 24.301 (Zero handoff-guard defect preserves HC-to-OP handoff)**.** *If* $\Delta^{\mathrm{guard}}_{\mathrm{handoff}} = 0$ *for an admissible dispatched handoff-guard packet, then the stabilized HC-capture source passes to the HC-to-OP handoff packet with its route tag, witness pointer, and non-certificate status marker preserved.*

Proof. Vanishing scalar defect forces all four defect components in Definition 24.299 to vanish on the audit window. Hence the stabilized source is not lost, the HC-to-OP tag does not drift, the witness pointer remains visible, and the non-certificate marker is not promoted. The packet therefore transfers the already stabilized source to the handoff node without changing its theorematic status. □

Definition 24.302 (handoff-guard return)**.** *The execution of* $\mathsf{W}^{\mathrm{guard}}_{\mathrm{handoff}}$ *on an admissible packet returns exactly one of the following statuses:*

$$\mathrm{Exec}\big(\mathsf{W}^{\mathrm{guard}}_{\mathrm{handoff}}, \mathsf{P}^{\mathrm{guard}}_{\mathrm{handoff}}\big) \in \{\mathsf{H}_{\mathrm{ready}}, \mathsf{H}_{\mathrm{repair}}, \mathsf{H}_{\mathrm{frontier}}\}.$$

Here $\mathsf{H}_{\mathrm{ready}}$ *means that the stabilized capture source is ready for OP-normalization testing,* $\mathsf{H}_{\mathrm{repair}}$ *means that one bounded handoff-guard repair is required before normalization may be tested, and* $\mathsf{H}_{\mathrm{frontier}}$ *means that the source remains protected but must return to the smaller frontier state rather than move forward.*

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Proposition 24.303 (The first dispatched handoff-guard packet is admissible after stable lock return)**.** *If the execution returns* $\mathsf{L}_{\mathrm{stable}}$*, then the dispatch menu admits the handoff-guard packet as the next executable work packet attached to the active HC/OP skeleton.*

Proof. The dispatch menu assigns E_1 only to $\mathsf{W}^{\mathrm{guard}}_{\mathrm{handoff}}$. Lemma 24.300 supplies the required stable input status. The remaining data of Definition 24.298 are inherited from the first explicit handoff source block, the active tail skeleton, and the guarded-reduction ledger: the handoff guard, witness pointer, route tag, and non-certificate marker are all already part of the active HC/OP spine. The new defect register adds only a finite audit surface for the handoff node, so the packet is admissible. □

Theorem 24.304 (First dispatched handoff-guard packet)**.** *After a stable lock/capture return, the active HC/OP skeleton supports the next executed work packet*

$$\mathsf{L}_{\mathrm{stable}} \xrightarrow{\mathsf{W}^{\mathrm{guard}}_{\mathrm{handoff}}} \{\mathsf{H}_{\mathrm{ready}}, \mathsf{H}_{\mathrm{repair}}, \mathsf{H}_{\mathrm{frontier}}\}.$$

This execution is conservative: it either prepares the stabilized source for OP-normalization testing, records one bounded handoff-guard repair obligation, or returns a smaller frontier source. It does not prove OP 1/4, OP 5(B), or any closure certificate.

Proof. By Proposition 24.303, the packet is admissible after a stable return. Definition 24.299 classifies the handoff audit defect into handoff-clean, guard-repairable, or frontier-bound cases, and Definition 24.302 records the corresponding return status. Lemma 24.301 proves the handoff-clean case. In the repair case the packet remains at the handoff node with a strictly smaller local defect and no route-tag drift; in the frontier case it returns protected frontier material without promotion. In none of the three cases is the non-certificate marker changed into an OP closure certificate. □

Corollary 24.305 (working rule)**.** *Future work may proceed from the lock/capture packet to the handoff node only after* $\mathsf{L}_{\mathrm{stable}}$ *has been returned. The next admissible mathematical question is therefore whether the stabilized source is handoff-ready for OP-normalization, handoff-guard-repairable, or frontier-bound; it is still not whether OP 1/4 or OP 5(B) is closed.*

Proof. Immediate from Theorem 24.304. □

24.16 First dispatched OP-normalization/type packet

Definition 24.306 (Dispatched OP-normalization/type packet)**.** *Let* $E_2 = \mathrm{NORMALIZE}^{\mathrm{HC}}_{\mathrm{OP}}$ *be the third node of the tail-normalized HC/OP skeleton and let*

$$\mathsf{D}^{\mathrm{skel}}_{\mathrm{HCOP}}(E_2) = \mathsf{W}^{\mathrm{type}}_{\mathrm{norm}}$$

be the dispatch assignment. A dispatched OP-normalization/type packet *is a finite packet*

$$\mathsf{P}^{\mathrm{type}}_{\mathrm{norm}} = \big(\mathsf{H}_{\mathrm{ready}}, \mathcal{N}^{\mathrm{HC}}_{\mathrm{OP}}, (\rho_{\mathrm{CL}}, \rho_{\Theta R}, \rho_A), G_2, W_2, S_2, \mathbf{d}^{\mathrm{type}}_{\mathrm{norm}}\big),$$

where $\mathsf{H}_{\mathrm{ready}}$ *is the handoff-ready return status of Definition 24.302,* $\mathcal{N}^{\mathrm{HC}}_{\mathrm{OP}}$ *is the inherited OP-entry normal form,* $(\rho_{\mathrm{CL}}, \rho_{\Theta R}, \rho_A)$ *is the live residual triple carried by that normal form,* G_2 *is the normalization/type guard,* W_2 *is the witness pointer to the first explicit normalization source block,* S_2 *is the non-certificate status marker, and* $\mathbf{d}^{\mathrm{type}}_{\mathrm{norm}}$ *is a non-negative type-coherence defect register. The packet is admissible only if it is launched from* $\mathsf{H}_{\mathrm{ready}}$*, keeps the HC-to-OP route and handoff witness visible, preserves the live residual triple as residual data, and does not convert a normalized OP input into an OP closure certificate.*

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Definition 24.307 (Normalization/type-coherence defect register)**.** *For an admissible packet* $\mathsf{P}^{\mathrm{type}}_{\mathrm{norm}}$*, the* normalization/type-coherence defect register *is the finite family*

$$\mathbf{d}^{\mathrm{type}}_{\mathrm{norm}}(j) = \big(d_{\mathrm{obj}}(j), d_{\mathrm{type}}(j), d_{\mathrm{res}}(j), d_{\mathrm{tag}}(j), d_{\mathrm{stat}}(j)\big), \qquad j \in \mathcal{A}_2,$$

where d_{obj} *measures loss of the handoff image as a QAM-coded HC/OP object,* d_{type} *measures type drift under the MML/MMA interface,* d_{res} *measures loss or premature splitting of the live residual triple,* d_{tag} *measures HC-to-OP route-tag drift, and* d_{stat} *measures accidental promotion of the non-certificate status marker. The scalar normalization audit defect is*

$$\Delta^{\mathrm{type}}_{\mathrm{norm}} = \max_{j\in\mathcal{A}_2}\big(d_{\mathrm{obj}}(j) + d_{\mathrm{type}}(j) + d_{\mathrm{res}}(j) + d_{\mathrm{tag}}(j) + d_{\mathrm{stat}}(j)\big).$$

The register is type-clean *if* $\Delta^{\mathrm{type}}_{\mathrm{norm}} = 0$*,* type-repairable *if one bounded local normalization/-type repair strictly lowers the scalar defect while preserving the handoff-ready source and route tag, and* frontier-bound *otherwise.*

Lemma 24.308 (Handoff-ready return is the normalization precondition)**.** *The OP-normalization/type packet may be executed only from the handoff-ready return status* $\mathsf{H}_{\mathrm{ready}}$*. If the return status is* $\mathsf{H}_{\mathrm{repair}}$ *or* $\mathsf{H}_{\mathrm{frontier}}$*, then the normalization node is not yet admissible.*

Proof. By Definition 24.306, the first entry of $\mathsf{P}^{\mathrm{type}}_{\mathrm{norm}}$ is $\mathsf{H}_{\mathrm{ready}}$. A repair return still carries a bounded handoff-guard obligation, while a frontier return does not promote the source to the normalization node. Neither status supplies the handoff-ready input required for typed OP-normalization. Thus normalization execution is blocked until the packet has returned the ready status. □

Lemma 24.309 (Zero normalization/type defect preserves OP-entry data)**.** *If* $\Delta^{\mathrm{type}}_{\mathrm{norm}} = 0$ *for an admissible dispatched OP-normalization/type packet, then the handoff-ready source enters the OP-entry normal form with its QAM object code, MML/MMA type, live residual triple, route tag, witness pointer, and non-certificate status marker preserved.*

Proof. Vanishing scalar defect forces every component in Definition 24.307 to vanish on the audit window. Hence the handoff image remains a QAM-coded HC/OP object, its MML/MMA type does not drift, the residual triple remains residual data rather than a certificate, the HC-to-OP route tag is preserved, and the non-certificate marker is not promoted. The output is therefore a typed OP-entry input, not an OP closure certificate. □

Definition 24.310 (OP-normalization/type return)**.** *The execution of* $\mathsf{W}^{\mathrm{type}}_{\mathrm{norm}}$ *on an admissible packet returns exactly one of the following statuses:*

$$\mathrm{Exec}\big(\mathsf{W}^{\mathrm{type}}_{\mathrm{norm}}, \mathsf{P}^{\mathrm{type}}_{\mathrm{norm}}\big) \in \{\mathsf{N}_{\mathrm{ready}}, \mathsf{N}_{\mathrm{repair}}, \mathsf{N}_{\mathrm{frontier}}\}.$$

Here $\mathsf{N}_{\mathrm{ready}}$ *means that the handoff source is now a typed OP-entry input ready for route/rank testing,* $\mathsf{N}_{\mathrm{repair}}$ *means that one bounded normalization/type repair is required before routing may be tested, and* $\mathsf{N}_{\mathrm{frontier}}$ *means that the source remains protected but must return to a smaller frontier state rather than move forward.*

Proposition 24.311 (The dispatched OP-normalization/type packet is admissible after handoff-ready return)**.** *If the execution returns* $\mathsf{H}_{\mathrm{ready}}$*, then the dispatch menu admits the OP-normalization/type packet as the next executable work packet attached to the active HC/OP skeleton.*

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Proof. The dispatch menu assigns E_2 only to $\mathsf{W}^{\mathrm{type}}_{\mathrm{norm}}$. Lemma 24.308 supplies the required handoff-ready input status. The remaining data of Definition 24.306 are inherited from the OP-entry normal form, the QAM/MML/MMA interface, the active tail skeleton, and the guarded-reduction ledger. The new defect register adds only a finite type-audit surface for the normalization node. It does not replace a hypothesis, alter a route tag, split a residual without authorization, or create a certificate. Hence the packet is admissible. □

Theorem 24.312 (First dispatched OP-normalization/type packet)**.** *After a handoff-ready return, the active HC/OP skeleton supports the next executed work packet*

$$\mathsf{H}_{\mathrm{ready}} \xrightarrow{\mathsf{W}^{\mathrm{type}}_{\mathrm{norm}}} \{\mathsf{N}_{\mathrm{ready}}, \mathsf{N}_{\mathrm{repair}}, \mathsf{N}_{\mathrm{frontier}}\}.$$

This execution is conservative: it either prepares a typed OP-entry input for route/rank testing, records one bounded normalization/type repair obligation, or returns a smaller frontier source. It does not prove OP 1/4, OP 5(B), or any closure certificate.

Proof. By Proposition 24.311, the packet is admissible after a handoff-ready return. Definition 24.307 classifies the type audit defect into type-clean, type-repairable, or frontier-bound cases, and Definition 24.310 records the corresponding return status. Lemma 24.309 proves the type-clean case. In the repair case the packet remains at the normalization node with a strictly smaller local type defect and no route-tag drift; in the frontier case it returns protected frontier material without promotion. In none of the three cases is the non-certificate marker changed into an OP closure certificate. □

Corollary 24.313 (working rule)**.** *Future work may proceed from the handoff-guard packet to the OP-normalization node only after* $\mathsf{H}_{\mathrm{ready}}$ *has been returned. The next admissible mathematical question is therefore whether the handoff source is type-clean and route/rank-ready, type-repairable, or frontier-bound; it is still not whether OP 1/4 or OP 5(B) is closed.*

Proof. Immediate from Theorem 24.312. □

24.17 First dispatched route/rank packet

Definition 24.314 (Dispatched route/rank packet)**.** *Let* $E_3 = \mathrm{ROUTE}^{\mathrm{HC}}_{\mathrm{OP}}$ *be the fourth node of the tail-normalized HC/OP skeleton and let*

$$\mathsf{D}^{\mathrm{skel}}_{\mathrm{HCOP}}(E_3) = \mathsf{W}^{\mathrm{rank}}_{\mathrm{route}}$$

be the dispatch assignment. A dispatched route/rank packet *is a finite packet*

$$\mathsf{P}^{\mathrm{rank}}_{\mathrm{route}} = \big(\mathsf{N}_{\mathrm{ready}}, \mathcal{T}^{\mathrm{HC}}_{\mathrm{OP}}, (\rho_{\mathrm{CL}}, \rho_{\Theta R}, \rho_A), \mathcal{M}^{\mathrm{rank}}_{\mathrm{route}}, G_3, W_3, S_3, \mathbf{d}^{\mathrm{rank}}_{\mathrm{route}}\big),$$

where $\mathsf{N}_{\mathrm{ready}}$ *is the normalization-ready return status of Definition 24.310,* $\mathcal{T}^{\mathrm{HC}}_{\mathrm{OP}}$ *is the inherited HC-routed OP-entry task-routing object,* $(\rho_{\mathrm{CL}}, \rho_{\Theta R}, \rho_A)$ *is the live residual triple carried by the typed OP input,* $\mathcal{M}^{\mathrm{rank}}_{\mathrm{route}}$ *is the finite admissible route/rank menu inherited from the task-routing layer,* G_3 *is the route/rank guard,* W_3 *is the witness pointer to the first explicit route source block,* S_3 *is the non-certificate status marker, and* $\mathbf{d}^{\mathrm{rank}}_{\mathrm{route}}$ *is a non-negative route/rank defect register. The packet is admissible only if it is launched from* $\mathsf{N}_{\mathrm{ready}}$*, keeps the HC-to-OP route tag visible, keeps the residual triple as residual data, ranks only tasks already admitted by* $\mathcal{T}^{\mathrm{HC}}_{\mathrm{OP}}$*, and does not convert a routed task into an OP closure certificate.*

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Definition 24.315 (Route/rank defect register)**.** *For an admissible packet* $\mathsf{P}^{\mathrm{rank}}_{\mathrm{route}}$*, the* route/rank defect register *is the finite family*

$$\mathbf{d}^{\mathrm{rank}}_{\mathrm{route}}(j) = \big(d_{\mathrm{route}}(j), d_{\mathrm{rank}}(j), d_{\mathrm{res}}(j), d_{\mathrm{front}}(j), d_{\mathrm{stat}}(j)\big), \qquad j \in \mathcal{A}_3,$$

where d_{route} *measures HC-to-OP route-tag drift,* d_{rank} *measures illegal or unstable task ranking,* d_{res} *measures loss, collapse, or unauthorized splitting of the live residual triple,* d_{front} *measures violation of the frontier guard, and* d_{stat} *measures accidental promotion of the non-certificate status marker. The scalar route/rank audit defect is*

$$\Delta^{\mathrm{rank}}_{\mathrm{route}} = \max_{j\in\mathcal{A}_3}\big(d_{\mathrm{route}}(j) + d_{\mathrm{rank}}(j) + d_{\mathrm{res}}(j) + d_{\mathrm{front}}(j) + d_{\mathrm{stat}}(j)\big).$$

The register is route-clean *if* $\Delta^{\mathrm{rank}}_{\mathrm{route}} = 0$*,* route-repairable *if one bounded local route/rank repair strictly lowers the scalar defect while preserving the typed OP input and route tag, and* frontier-bound *otherwise.*

Lemma 24.316 (Normalization-ready return is the route/rank precondition)**.** *The route/rank packet may be executed only from the normalization-ready return status* $\mathsf{N}_{\mathrm{ready}}$*. If the return status is* $\mathsf{N}_{\mathrm{repair}}$ *or* $\mathsf{N}_{\mathrm{frontier}}$*, then the route/rank node is not yet admissible.*

Proof. By Definition 24.314, the first entry of $\mathsf{P}^{\mathrm{rank}}_{\mathrm{route}}$ is $\mathsf{N}_{\mathrm{ready}}$. A repair return still carries a bounded normalization/type obligation, while a frontier return does not promote the source to the routing node. Neither status supplies the typed OP input required for route/rank testing. Thus route/rank execution is blocked until the packet has returned the ready status. □

Lemma 24.317 (Zero route/rank defect preserves routed OP-task data)**.** *If* $\Delta^{\mathrm{rank}}_{\mathrm{route}} = 0$ *for an admissible dispatched route/rank packet, then the normalization-ready OP input passes to a routed task state with its HC-to-OP route tag, admissible task ranking, live residual triple, frontier guard, witness pointer, and non-certificate status marker preserved.*

Proof. Vanishing scalar defect forces every component in Definition 24.315 to vanish on the audit window. Hence no route tag drifts, no inadmissible task is ranked ahead of the inherited task-routing menu, the residual triple remains residual data, the frontier guard stays active, and the non-certificate marker is not promoted. The output is therefore a routed OP task state ready for the next comparison or repair decision, not an OP closure certificate. □

Definition 24.318 (route/rank return)**.** *The execution of* $\mathsf{W}^{\mathrm{rank}}_{\mathrm{route}}$ *on an admissible packet returns exactly one of the following statuses:*

$$\mathrm{Exec}\big(\mathsf{W}^{\mathrm{rank}}_{\mathrm{route}}, \mathsf{P}^{\mathrm{rank}}_{\mathrm{route}}\big) \in \{\mathsf{R}_{\mathrm{ready}}, \mathsf{R}_{\mathrm{repair}}, \mathsf{R}_{\mathrm{frontier}}\}.$$

Here $\mathsf{R}_{\mathrm{ready}}$ *means that the typed OP input has been routed and ranked into an admissible comparison-ready task state,* $\mathsf{R}_{\mathrm{repair}}$ *means that one bounded route/rank repair is required before comparison may be tested, and* $\mathsf{R}_{\mathrm{frontier}}$ *means that the source remains protected but must return to a smaller frontier state rather than move forward.*

Proposition 24.319 (The dispatched route/rank packet is admissible after normalization-ready return)**.** *If the execution returns* $\mathsf{N}_{\mathrm{ready}}$*, then the dispatch menu admits the route/rank packet as the next executable work packet attached to the active HC/OP skeleton.*

Proof. The dispatch menu assigns E_3 only to $\mathsf{W}^{\mathrm{rank}}_{\mathrm{route}}$. Lemma 24.316 supplies the required normalization-ready input status. The remaining data of Definition 24.314 are inherited from the task-routing layer, the QAM/MML/MMA interface, the skeleton, and the guarded-reduction ledger. The new defect register adds only a finite route/rank audit surface. It does not introduce a new task outside the inherited menu, alter a route tag, split a residual without authorization, or create a certificate. Hence the packet is admissible. □

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Theorem 24.320 (First dispatched route/rank packet)**.** *After a normalization-ready return, the active HC/OP skeleton supports the next executed work packet*

$$\mathsf{N}_{\mathrm{ready}} \xrightarrow{\mathsf{W}^{\mathrm{rank}}_{\mathrm{route}}} \{\mathsf{R}_{\mathrm{ready}}, \mathsf{R}_{\mathrm{repair}}, \mathsf{R}_{\mathrm{frontier}}\}.$$

This execution is conservative: it either prepares a routed and ranked OP task for comparison testing, records one bounded route/rank repair obligation, or returns a smaller frontier source. It does not prove OP 1/4, OP 5(B), or any closure certificate.

Proof. By Proposition 24.319, the packet is admissible after a normalization-ready return. Definition 24.315 classifies the route/rank audit defect into route-clean, route-repairable, or frontier-bound cases, and Definition 24.318 records the corresponding return status. Lemma 24.317 proves the route-clean case. In the repair case the packet remains at the routing node with a strictly smaller local route/rank defect and no route-tag drift; in the frontier case it returns protected frontier material without promotion. In none of the three cases is the non-certificate marker changed into an OP closure certificate. □

Corollary 24.321 (working rule)**.** *Future work may proceed from the OP-normalization/type packet to the route/rank node only after* $\mathsf{N}_{\mathrm{ready}}$ *has been returned. The next admissible mathematical question is therefore whether the typed OP input is route-clean and comparison-ready, route/rank-repairable, or frontier-bound; it is still not whether OP 1/4 or OP 5(B) is closed.*

Proof. Immediate from Theorem 24.320. □

24.18 First dispatched comparison/defect packet

Definition 24.322 (Dispatched comparison/defect packet)**.** *Let* $E_4 = \mathrm{COMPARE}^{1/4|5(B)}_{\mathrm{HC}}$ *be the fifth node of the tail-normalized HC/OP skeleton and let*

$$\mathsf{D}^{\mathrm{skel}}_{\mathrm{HCOP}}(E_4) = \mathsf{W}^{\Delta}_{\mathrm{cmp}}$$

be the dispatch assignment. A dispatched comparison/defect packet *is a finite packet*

$$\mathsf{P}^{\Delta}_{\mathrm{cmp}} = \big(\mathsf{R}_{\mathrm{ready}}, \mathcal{W}^{\mathrm{cmp}}_{1/4|5(B)}, \rho_{\mathrm{CL}}, \rho_{\Theta R}, D^{\mathrm{HC}}_{\mathrm{joint}}, D^{\mathrm{HC}}_{\mathrm{sep}}, G_4, W_4, S_4, \mathbf{d}^{\Delta}_{\mathrm{cmp}}\big),$$

where $\mathsf{R}_{\mathrm{ready}}$ *is the comparison-ready route/rank return of Definition 24.318,* $\mathcal{W}^{\mathrm{cmp}}_{1/4|5(B)}$ *is the inherited paired OP 1/4–OP 5(B) comparison work object,* ρ_{CL} *and* $\rho_{\Theta R}$ *are the routed continuum-limit and* Θ*–R residual coordinates,* $D^{\mathrm{HC}}_{\mathrm{joint}}$ *and* $D^{\mathrm{HC}}_{\mathrm{sep}}$ *are the HC-compatible joint and separated comparison readouts,* G_4 *is the comparison guard,* W_4 *is the witness pointer to the first explicit paired-comparison source block,* S_4 *is the non-certificate status marker, and* $\mathbf{d}^{\Delta}_{\mathrm{cmp}}$ *is a finite comparison/defect audit register. The packet is admissible only if it is launched from* $\mathsf{R}_{\mathrm{ready}}$*, keeps the inherited paired route visible, compares only HC-compatible joint and separated readouts, preserves both residual coordinates as residual data, and does not convert a comparison outcome into an OP closure certificate.*

Definition 24.323 (Comparison/defect register)**.** *For an admissible packet* $\mathsf{P}^{\Delta}_{\mathrm{cmp}}$*, the* comparison/defect register *is the finite family*

$$\mathbf{d}^{\Delta}_{\mathrm{cmp}}(j) = \big(d_{\mathrm{in}}(j), d_{\mathrm{pair}}(j), d_{\mathrm{read}}(j), d_{\Delta}(j), d_{\mathrm{front}}(j), d_{\mathrm{stat}}(j)\big), \qquad j \in \mathcal{A}_4,$$

where d_{in} *measures loss of the comparison-ready input status,* d_{pair} *measures unauthorized separation or binding of the paired residual branch,* d_{read} *measures non-HC-compatible readout drift,* d_{Δ} *measures instability of the comparison defect*

$$\Delta^{\mathrm{HC}}_{\mathrm{sep}} = D^{\mathrm{HC}}_{\mathrm{joint}} - D^{\mathrm{HC}}_{\mathrm{sep}},$$

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d_{front} *measures violation of the comparison frontier guard, and* d_{stat} *measures accidental promotion of the non-certificate marker. The scalar comparison audit defect is*

$$\Delta^{\Delta}_{\mathrm{cmp}} = \max_{j\in\mathcal{A}_4}\big(d_{\mathrm{in}}(j) + d_{\mathrm{pair}}(j) + d_{\mathrm{read}}(j) + d_{\Delta}(j) + d_{\mathrm{front}}(j) + d_{\mathrm{stat}}(j)\big).$$

The register is comparison-clean *if* $\Delta^{\Delta}_{\mathrm{cmp}} = 0$, comparison-repairable *if one bounded local comparison repair strictly lowers the scalar defect while preserving the routed comparison input, and* frontier-bound *otherwise.*

Lemma 24.324 (Route/rank-ready return is the comparison precondition)**.** *The comparison/defect packet may be executed only from the route/rank-ready return status* $\mathsf{R}_{\mathrm{ready}}$*. If the return status is* $\mathsf{R}_{\mathrm{repair}}$ *or* $\mathsf{R}_{\mathrm{frontier}}$*, then the paired comparison node is not yet admissible.*

Proof. By Definition 24.322, the first entry of $\mathsf{P}^{\Delta}_{\mathrm{cmp}}$ is $\mathsf{R}_{\mathrm{ready}}$. A repair return still carries a bounded route/rank obligation, while a frontier return does not promote the source to the comparison node. Neither status supplies the routed and ranked comparison-ready input required to test $D^{\mathrm{HC}}_{\mathrm{joint}}$, $D^{\mathrm{HC}}_{\mathrm{sep}}$, and $\Delta^{\mathrm{HC}}_{\mathrm{sep}}$. Thus comparison execution is blocked until the packet has returned the ready status. □

Lemma 24.325 (Zero comparison defect preserves paired comparison data)**.** *If* $\Delta^{\Delta}_{\mathrm{cmp}} = 0$ *for an admissible dispatched comparison/defect packet, then the routed OP task passes to a guarded paired-comparison state with its paired residual branch, HC-compatible joint and separated readouts, comparison defect, frontier guard, witness pointer, and non-certificate status marker preserved.*

Proof. Vanishing scalar defect forces every component in Definition 24.323 to vanish on the audit window. Hence the comparison-ready input status is retained, the paired residual branch is neither bound nor split without authorization, the joint and separated readouts remain HC-compatible, the comparison defect is well-defined on the inherited route, the frontier guard remains active, and the non-certificate marker is not promoted. The output is therefore a guarded paired-comparison state ready for the next resolution node, not an OP closure certificate. □

Definition 24.326 (comparison/defect return)**.** *The execution of* $\mathsf{W}^{\Delta}_{\mathrm{cmp}}$ *on an admissible packet returns exactly one of the following statuses:*

$$\mathrm{Exec}\big(\mathsf{W}^{\Delta}_{\mathrm{cmp}}, \mathsf{P}^{\Delta}_{\mathrm{cmp}}\big) \in \{\mathsf{C}_{\mathrm{bind}}, \mathsf{C}_{\mathrm{split}}, \mathsf{C}_{\mathrm{frontier}}\}.$$

Here $\mathsf{C}_{\mathrm{bind}}$ *means that the guarded comparison supports the bound common-origin branch as the next final for resolution,* $\mathsf{C}_{\mathrm{split}}$ *means that the guarded comparison supports a ranked separated branch as the next final for resolution, and* $\mathsf{C}_{\mathrm{frontier}}$ *means that the comparison remains protected but must return to a smaller comparison frontier rather than move forward.*

Proposition 24.327 (The dispatched comparison/defect packet is admissible after route/rank-ready return)**.** *If the execution returns* $\mathsf{R}_{\mathrm{ready}}$*, then the dispatch menu admits the comparison/defect packet as the next executable work packet attached to the active HC/OP skeleton.*

Proof. The dispatch menu assigns E_4 only to $\mathsf{W}^{\Delta}_{\mathrm{cmp}}$. Lemma 24.324 supplies the required route/rank-ready input status. The remaining data of Definition 24.322 are inherited from the paired comparison layer, the QAM/MML/MMA interface, the guarded-reduction ledger, the skeleton, and the routed task state. The new defect register adds only a finite comparison audit surface. It does not introduce a new OP task, alter the paired route, bind or split a residual branch without the comparison guard, or create a certificate. Hence the packet is admissible. □

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Theorem 24.328 (First dispatched comparison/defect packet)**.** *After a route/rank-ready return, the active HC/OP skeleton supports the next executed work packet*

$$\mathsf{R}_{\mathrm{ready}} \xrightarrow{\mathsf{W}^{\Delta}_{\mathrm{cmp}}} \{\mathsf{C}_{\mathrm{bind}}, \mathsf{C}_{\mathrm{split}}, \mathsf{C}_{\mathrm{frontier}}\}.$$

This execution is conservative: it either prepares a guarded bind final, prepares a guarded split final, or returns a smaller comparison frontier. It does not prove OP 1/4, OP 5(B), or any closure certificate.

Proof. By Proposition 24.327, the packet is admissible after a route/rank-ready return. Definition 24.323 classifies the comparison audit defect into comparison-clean, comparison-repairable, or frontier-bound cases, and Definition 24.326 records the corresponding guarded comparison return. Lemma 24.325 proves the comparison-clean preservation claim. In the repair case the packet remains at the comparison node with a strictly smaller local comparison defect and no route drift; in the frontier case it returns protected comparison frontier material without promotion. In none of the three cases is the non-certificate marker changed into an OP closure certificate. □

Corollary 24.329 (working rule)**.** *Future work may proceed from the route/rank packet to the paired comparison node only after* $\mathsf{R}_{\mathrm{ready}}$ *has been returned. The next admissible mathematical question is therefore how a guarded comparison return* $\mathsf{C}_{\mathrm{bind}}$*,* $\mathsf{C}_{\mathrm{split}}$*, or* $\mathsf{C}_{\mathrm{frontier}}$ *should be resolved; it is still not whether OP 1/4 or OP 5(B) is closed.*

Proof. Immediate from Theorem 24.328. □

24.19 First dispatched resolve/return packet

Definition 24.330 (Dispatched resolve/return packet)**.** *Let* $E_5 = \mathrm{RESOLVE}_{\mathrm{bind/split/frontier}}$ *be the sixth node of the tail-normalized HC/OP skeleton and let*

$$\mathsf{D}^{\mathrm{skel}}_{\mathrm{HCOP}}(E_5) = \mathsf{W}^{\mathrm{next}}_{\mathrm{resolve}}$$

be the dispatch assignment. A dispatched resolve/return packet *is a finite packet*

$$\mathsf{P}^{\mathrm{next}}_{\mathrm{resolve}} = \big(\mathsf{C}_{\eta}, \eta, \Pi_{\eta}, \rho_{\mathrm{CL}}, \rho_{\Theta R}, \Delta^{\mathrm{HC}}_{\mathrm{sep}}, G_5, W_5, S_5, \mathbf{d}^{\mathrm{next}}_{\mathrm{resolve}}\big), \qquad \eta \in \{\mathrm{bind}, \mathrm{split}, \mathrm{frontier}\},$$

where C_{η} *is the corresponding comparison return,* Π_{η} *is the next finite continuation package associated with that return,* ρ_{CL} *and* $\rho_{\Theta R}$ *are the inherited routed residual coordinates,* $\Delta^{\mathrm{HC}}_{\mathrm{sep}}$ *is the paired HC comparison defect,* G_5 *is the resolution guard,* W_5 *is the witness pointer to the first explicit comparison source,* S_5 *is the non-certificate status marker, and* $\mathbf{d}^{\mathrm{next}}_{\mathrm{resolve}}$ *is a finite resolution audit register. The packet is admissible only if it is launched from one of* $\mathsf{C}_{\mathrm{bind}}$*,* $\mathsf{C}_{\mathrm{split}}$*, or* $\mathsf{C}_{\mathrm{frontier}}$*, selects exactly the continuation package matching that return, keeps the inherited comparison data visible, and does not convert the selected continuation into an OP closure certificate.*

Definition 24.331 (Resolve/return defect register)**.** *For an admissible packet* $\mathsf{P}^{\mathrm{next}}_{\mathrm{resolve}}$*, the* resolve/return defect register *is the finite family*

$$\mathbf{d}^{\mathrm{next}}_{\mathrm{resolve}}(j) = \big(d_{\mathrm{in}}(j), d_{\mathrm{sel}}(j), d_{\mathrm{guard}}(j), d_{\mathrm{route}}(j), d_{\mathrm{wit}}(j), d_{\mathrm{stat}}(j)\big), \qquad j \in \mathcal{A}_5,$$

where d_{in} *measures loss of the guarded comparison return,* d_{sel} *measures mismatch between the comparison return and the selected continuation package,* d_{guard} *measures violation of the resolution guard,* d_{route} *measures loss of the routed OP 1/4–OP 5(B) comparison data,* d_{wit}

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measures loss of the witness pointer, and d_{stat} *measures accidental promotion of the non-certificate marker. The scalar resolve/return defect is*

$$\Delta^{\mathrm{next}}_{\mathrm{resolve}} = \max_{j\in\mathcal{A}_5}\big(d_{\mathrm{in}}(j) + d_{\mathrm{sel}}(j) + d_{\mathrm{guard}}(j) + d_{\mathrm{route}}(j) + d_{\mathrm{wit}}(j) + d_{\mathrm{stat}}(j)\big).$$

The register is resolve-clean *if* $\Delta^{\mathrm{next}}_{\mathrm{resolve}} = 0$*,* resolve-repairable *if one bounded local resolution repair strictly lowers the scalar defect while preserving the guarded comparison return, and* frontier-bound *otherwise.*

Lemma 24.332 (Guarded comparison return is the resolve precondition)**.** *The resolve/return packet may be executed only from one of the guarded comparison returns* $\mathsf{C}_{\mathrm{bind}}$*,* $\mathsf{C}_{\mathrm{split}}$*, or* $\mathsf{C}_{\mathrm{frontier}}$*. If the comparison node has not returned one of these three guarded statuses, then the resolution node is not yet admissible.*

Proof. By Definition 24.330, the first entry of $\mathsf{P}^{\mathrm{next}}_{\mathrm{resolve}}$ is exactly one guarded comparison return C_{η}. The resolution node consumes a completed guarded comparison return, not a raw route/rank state and not an unresolved comparison audit surface. Without such a return there is no admissible bind, split, or frontier branch for the packet to select. □

Lemma 24.333 (Zero resolve/return defect preserves selected continuation data)**.** *If* $\Delta^{\mathrm{next}}_{\mathrm{resolve}} = 0$ *for an admissible dispatched resolve/return packet, then the selected continuation package preserves the corresponding comparison return, the routed residual data, the HC comparison defect, the resolution guard, the witness pointer, and the non-certificate status marker.*

Proof. Vanishing scalar defect forces all components of Definition 24.331 to vanish on the resolution audit window. Hence the input comparison return is retained, the selected continuation package matches that return, the guard is not bypassed, the routed residual data and comparison defect remain attached to the selected branch, the witness pointer is preserved, and the non-certificate marker is not promoted. Therefore the output is a guarded next work state, not an OP closure certificate. □

Definition 24.334 (resolve/return output)**.** *The execution of* $\mathsf{W}^{\mathrm{next}}_{\mathrm{resolve}}$ *on an admissible packet returns exactly one of the following statuses:*

$$\mathrm{Exec}\big(\mathsf{W}^{\mathrm{next}}_{\mathrm{resolve}}, \mathsf{P}^{\mathrm{next}}_{\mathrm{resolve}}\big) \in \{\mathsf{Q}_{\mathrm{bind}}, \mathsf{Q}_{\mathrm{split}}, \mathsf{Q}_{\mathrm{frontier}}\}.$$

Here $\mathsf{Q}_{\mathrm{bind}}$ *is a guarded common-origin continuation work state for the coupled OP 1/4–OP 5(B) branch,* $\mathsf{Q}_{\mathrm{split}}$ *is a guarded ranked single-branch continuation work state, and* $\mathsf{Q}_{\mathrm{frontier}}$ *is a protected smaller comparison-frontier descent state. None of these statuses is an OP closure certificate.*

Proposition 24.335 (The dispatched resolve/return packet is admissible after guarded comparison return)**.** *If the execution returns one of* $\mathsf{C}_{\mathrm{bind}}$*,* $\mathsf{C}_{\mathrm{split}}$*, or* $\mathsf{C}_{\mathrm{frontier}}$*, then the dispatch menu admits the resolve/return packet as the next executable work packet attached to the active HC/OP skeleton.*

Proof. The dispatch menu assigns E_5 only to $\mathsf{W}^{\mathrm{next}}_{\mathrm{resolve}}$. Lemma 24.332 supplies the required guarded comparison input status. The remaining data of Definition 24.330 are inherited from the paired comparison layer, the QAM/MML/MMA interface, the guarded-reduction ledger, the skeleton, and the comparison return. The new defect register adds only a finite resolution audit surface. It does not create a new certificate test, erase the residual route, suppress the witness pointer, or change the theorematic status of the selected continuation. Hence the packet is admissible. □

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Theorem 24.336 (First dispatched resolve/return packet)**.** *After a guarded comparison return, the active HC/OP skeleton supports the next executed work packet*

$$\{\mathsf{C}_{\mathrm{bind}}, \mathsf{C}_{\mathrm{split}}, \mathsf{C}_{\mathrm{frontier}}\} \xrightarrow{\mathsf{W}^{\mathrm{next}}_{\mathrm{resolve}}} \{\mathsf{Q}_{\mathrm{bind}}, \mathsf{Q}_{\mathrm{split}}, \mathsf{Q}_{\mathrm{frontier}}\}.$$

This execution is conservative: it selects a guarded common-origin continuation, a guarded ranked single-branch continuation, or a smaller comparison-frontier descent state. It does not prove OP 1/4, OP 5(B), or any closure certificate.

Proof. By Proposition 24.335, the packet is admissible after a guarded comparison return. Definition 24.331 classifies the resolution audit defect into resolve-clean, resolve-repairable, or frontier-bound cases, and Definition 24.334 records the corresponding guarded continuation output. Lemma 24.333 proves the resolve-clean preservation claim. In the repair case the packet remains at the resolution node with a strictly smaller local resolution defect and no route drift; in the frontier case it returns protected comparison-frontier material without promotion. In none of the three cases is the non-certificate marker changed into an OP closure certificate. □

Corollary 24.337 (working rule)**.** *Future work may proceed from the guarded comparison returns to a selected next continuation only through* $\mathsf{W}^{\mathrm{next}}_{\mathrm{resolve}}$*. The next admissible mathematical question is therefore whether* $\mathsf{Q}_{\mathrm{bind}}$*,* $\mathsf{Q}_{\mathrm{split}}$*, or* $\mathsf{Q}_{\mathrm{frontier}}$ *should seed the next finite work packet; it is still not whether OP 1/4 or OP 5(B) is closed.*

Proof. Immediate from Theorem 24.336. □

24.20 First continuation-queue packet after resolve/return

Definition 24.338 (Continuation-queue packet after resolve/return)**.** *Let*

$$\mathsf{Q}_\eta \in \{\mathsf{Q}_{\mathrm{bind}}, \mathsf{Q}_{\mathrm{split}}, \mathsf{Q}_{\mathrm{frontier}}\}, \qquad \eta \in \{\mathrm{bind}, \mathrm{split}, \mathrm{frontier}\},$$

be a guarded resolve/return status. A continuation-queue packet after resolve/return *is a finite packet*

$$\mathsf{P}^{\mathrm{cont}}_{\mathrm{queue}} = \big(\mathsf{Q}_\eta, \mathcal{U}_\eta, \rho_{\mathrm{CL}}, \rho_{\Theta R}, \rho_A, \Delta^{\mathrm{HC}}_{\mathrm{sep}}, G_6, W_6, S_6, \mathbf{d}^{\mathrm{cont}}_{\mathrm{queue}}\big),$$

where $\mathcal{U}_\eta$ *is the finite guarded continuation queue selected by the return,* $(\rho_{\mathrm{CL}}, \rho_{\Theta R}, \rho_A)$ *is the inherited live residual triple,* $\Delta^{\mathrm{HC}}_{\mathrm{sep}}$ *is the inherited paired HC comparison defect,* G_6 *is the queue guard,* W_6 *is the witness pointer to the first explicit resolve/return source,* S_6 *is the non-certificate status marker, and* $\mathbf{d}^{\mathrm{cont}}_{\mathrm{queue}}$ *is a finite queue audit register. The packet is admissible only if the queue type matches the selected return, keeps the inherited residual and comparison data attached to the queue, and does not change a continuation queue into an OP closure certificate.*

Definition 24.339 (Bind, split, and frontier continuation queues)**.** *The continuation queue selected by Definition 24.338 is one of the following three finite guarded families:*

$$\begin{aligned}
\mathcal{U}_{\mathrm{bind}} &= \big(\mathrm{commonOriginCheck}, \mathrm{jointResidualRefine}, \mathrm{pairedWitnessReturn}\big),\\
\mathcal{U}_{\mathrm{split}} &= \big(\mathrm{branchRankCheck}, \mathrm{singleResidualSelect}, \mathrm{separatedWitnessReturn}\big),\\
\mathcal{U}_{\mathrm{frontier}} &= \big(\mathrm{frontierDescentCheck}, \mathrm{defectLocalize}, \mathrm{smallerFrontierReturn}\big).
\end{aligned}$$

The bind queue keeps OP 1/4 and OP 5(B) coupled under a common-origin guard; the split queue selects a ranked single-branch continuation under a separation guard; and the frontier queue performs only protected descent to a smaller comparison frontier. None of these queues is a theorematic OP closure state.

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Definition 24.340 (Continuation-queue defect register)**.** *For an admissible packet* $\mathsf{P}^{\mathrm{cont}}_{\mathrm{queue}}$*, the* continuation-queue defect register *is the finite family*

$$\mathbf{d}^{\mathrm{cont}}_{\mathrm{queue}}(j) = \big(d_Q(j), d_{\mathrm{type}}(j), d_{\mathrm{res}}(j), d_\Delta(j), d_{\mathrm{guard}}(j), d_{\mathrm{wit}}(j), d_{\mathrm{stat}}(j)\big), \qquad j \in \mathcal{A}_6,$$

where d_Q *measures loss of the resolve/return status,* d_{type} *measures mismatch between the selected status and the selected queue type,* d_{res} *measures loss of the inherited live residual triple,* d_Δ *measures loss of the paired comparison defect,* d_{guard} *measures violation of the queue guard,* d_{wit} *measures loss of the witness pointer, and* d_{stat} *measures accidental certificate promotion. The scalar queue defect is*

$$\Delta^{\mathrm{cont}}_{\mathrm{queue}} = \max_{j\in\mathcal{A}_6}\big(d_Q(j) + d_{\mathrm{type}}(j) + d_{\mathrm{res}}(j) + d_\Delta(j) + d_{\mathrm{guard}}(j) + d_{\mathrm{wit}}(j) + d_{\mathrm{stat}}(j)\big).$$

The register is queue-clean *if* $\Delta^{\mathrm{cont}}_{\mathrm{queue}} = 0$*,* queue-repairable *if one bounded local queue repair strictly lowers the scalar defect while preserving the return, and* frontier-bound *otherwise.*

Lemma 24.341 (Resolve/return output is the queue precondition)**.** *The continuation-queue packet may be executed only from one of* $\mathsf{Q}_{\mathrm{bind}}$*,* $\mathsf{Q}_{\mathrm{split}}$*, or* $\mathsf{Q}_{\mathrm{frontier}}$*. If the resolve/return node has not returned one of these three guarded statuses, then the continuation-queue node is not yet admissible.*

Proof. By Definition 24.338, the first entry of the packet is exactly one guarded return Q_η. The queue node schedules follow-on work from a completed resolve/return state; it does not consume an unresolved comparison state, a raw route/rank state, or a merely local HC-capture source. Hence no continuation queue exists until the node has selected one of the three guarded return statuses. □

Lemma 24.342 (Zero continuation-queue defect preserves queue data)**.** *If* $\Delta^{\mathrm{cont}}_{\mathrm{queue}} = 0$ *for an admissible continuation-queue packet, then the selected queue preserves the return status, the matching queue type, the inherited live residual triple, the paired HC comparison defect, the queue guard, the witness pointer, and the non-certificate status marker.*

Proof. Vanishing scalar defect forces all components of Definition 24.340 to vanish on the queue audit window. Thus the return remains visible, the queue type matches the selected return, the live residual triple and paired comparison defect remain attached, the guard and witness pointer are preserved, and the non-certificate marker is not promoted. Therefore the output is a finite guarded work queue, not an OP closure certificate. □

Definition 24.343 (continuation-queue output)**.** *The execution of the continuation-queue packet returns exactly one of the following queued work states:*

$$\mathrm{Exec}\big(\mathsf{W}^{\mathrm{cont}}_{\mathrm{queue}}, \mathsf{P}^{\mathrm{cont}}_{\mathrm{queue}}\big) \in \{\mathsf{U}^{\mathrm{ready}}_{\mathrm{bind}}, \mathsf{U}^{\mathrm{ready}}_{\mathrm{split}}, \mathsf{U}^{\mathrm{ready}}_{\mathrm{frontier}}\}.$$

Here $\mathsf{U}^{\mathrm{ready}}_{\mathrm{bind}}$ *is a finite common-origin work queue,* $\mathsf{U}^{\mathrm{ready}}_{\mathrm{split}}$ *is a finite ranked single-branch work queue, and* $\mathsf{U}^{\mathrm{ready}}_{\mathrm{frontier}}$ *is a finite protected frontier-descent work queue. None of these statuses is an OP closure certificate.*

Proposition 24.344 (The continuation-queue packet is admissible after resolve/return)**.** *If the execution returns one of* $\mathsf{Q}_{\mathrm{bind}}$*,* $\mathsf{Q}_{\mathrm{split}}$*, or* $\mathsf{Q}_{\mathrm{frontier}}$*, then the active HC/OP skeleton admits* $\mathsf{W}^{\mathrm{cont}}_{\mathrm{queue}}$ *as the next executable work packet.*

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Proof. Lemma 24.341 supplies the required guarded input. Definitions 24.338 and 24.339 attach the matching finite queue to that input. The remaining data are inherited from the paired comparison layer, the QAM/MML/MMA interface, the dispatch, the comparison return, and the resolve/return output. The queue defect register is finite and audits only preservation of input status, type, residual data, comparison defect, guard, witness, and non-certificate status. Hence the packet is admissible as follow-on work and not as an OP closure claim. □

Theorem 24.345 (First continuation-queue packet after resolve/return)**.** *After a guarded resolve/return output, the active HC/OP skeleton supports the next executed work packet*

$$\{\mathsf{Q}_{\mathrm{bind}}, \mathsf{Q}_{\mathrm{split}}, \mathsf{Q}_{\mathrm{frontier}}\} \xrightarrow{\mathsf{W}^{\mathrm{cont}}_{\mathrm{queue}}} \{\mathsf{U}^{\mathrm{ready}}_{\mathrm{bind}}, \mathsf{U}^{\mathrm{ready}}_{\mathrm{split}}, \mathsf{U}^{\mathrm{ready}}_{\mathrm{frontier}}\}.$$

This execution is conservative: it turns the return into a finite guarded continuation queue and does not prove OP 1/4, OP 5(B), or any closure certificate.

Proof. By Proposition 24.344, the packet is admissible after a guarded return. Definition 24.340 classifies the queue audit defect into queue-clean, queue-repairable, or frontier-bound cases, and Definition 24.343 records the corresponding queued work output. Lemma 24.342 proves the queue-clean preservation claim. In the repair case the packet remains at the queue node with a strictly smaller local queue defect and no route drift; in the frontier case it returns protected smaller frontier material. In none of the three cases is the non-certificate marker changed into an OP closure certificate. □

Corollary 24.346 (working rule)**.** *Future work may proceed from the guarded resolve/return status only by selecting the matching finite continuation queue. The next admissible mathematical question is therefore which queued work family — common-origin bind refinement, ranked split refinement, or protected frontier descent — should be executed first; it is still not whether OP 1/4 or OP 5(B) is closed.*

Proof. Immediate from Theorem 24.345. □

24.21 First queue-head selection packet after continuation queue

Definition 24.347 (Queue-head selection packet after continuation queue)**.** *Let*

$$\mathsf{U}^{\mathrm{ready}}_{\eta} \in \{\mathsf{U}^{\mathrm{ready}}_{\mathrm{bind}}, \mathsf{U}^{\mathrm{ready}}_{\mathrm{split}}, \mathsf{U}^{\mathrm{ready}}_{\mathrm{frontier}}\}, \qquad \eta \in \{\mathrm{bind}, \mathrm{split}, \mathrm{frontier}\},$$

be a guarded continuation-queue output. A queue-head selection packet after continuation queue *is a finite packet*

$$\mathsf{P}^{\mathrm{select}}_{\mathrm{head}} = \big(\mathsf{U}^{\mathrm{ready}}_{\eta}, h_{\eta}, \mathcal{U}^{\mathrm{tail}}_{\eta}, \rho_{\mathrm{CL}}, \rho_{\Theta R}, \rho_{A}, \Delta^{\mathrm{HC}}_{\mathrm{sep}}, G_7, W_7, S_7, \mathbf{d}^{\mathrm{select}}_{\mathrm{head}}\big),$$

where h_η is the first guarded queue head selected from the matching queue, $\mathcal{U}^{\mathrm{tail}}_{\eta}$ is the remaining finite queue tail, $(\rho_{\mathrm{CL}}, \rho_{\Theta R}, \rho_A)$ is the inherited live residual triple, $\Delta^{\mathrm{HC}}_{\mathrm{sep}}$ is the inherited paired HC comparison defect, G_7 is the head-selection guard, W_7 is the witness pointer to the first explicit continuation-queue source, S_7 is the non-certificate status marker, and $\mathbf{d}^{\mathrm{select}}_{\mathrm{head}}$ is a finite head-selection audit register. The packet is admissible only if the selected head belongs to the matching queue, the tail remains finite and guarded, the inherited residual and comparison data remain attached, and no selected head is promoted to an OP closure certificate.

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Definition 24.348 (Continuation-queue heads and finite tails)**.** *The matching queue head and tail in Definition 24.347 are fixed by the queue type:*

$$\begin{aligned} h_{\mathrm{bind}} &= \mathrm{commonOriginCheck}, & \mathcal{U}^{\mathrm{tail}}_{\mathrm{bind}} &= (\mathrm{jointResidualRefine}, \mathrm{pairedWitnessReturn}),\\ h_{\mathrm{split}} &= \mathrm{branchRankCheck}, & \mathcal{U}^{\mathrm{tail}}_{\mathrm{split}} &= (\mathrm{singleResidualSelect}, \mathrm{separatedWitnessReturn}),\\ h_{\mathrm{frontier}} &= \mathrm{frontierDescentCheck}, & \mathcal{U}^{\mathrm{tail}}_{\mathrm{frontier}} &= (\mathrm{defectLocalize}, \mathrm{smallerFrontierReturn}). \end{aligned}$$

Thus the bind queue first tests whether the common-origin reading is still admissible, the split queue first tests whether a ranked branch can honestly be selected, and the frontier queue first tests whether a protected descent is available. The selected head is a next work packet, not a theorematic OP closure state.

Definition 24.349 (Queue-head selection defect register)**.** *For an admissible packet* $\mathsf{P}^{\mathrm{select}}_{\mathrm{head}}$*, the* queue-head selection defect register *is the finite family*

$$\mathbf{d}^{\mathrm{select}}_{\mathrm{head}}(j) = \big(d_U(j), d_h(j), d_{\mathrm{tail}}(j), d_{\mathrm{res}}(j), d_\Delta(j), d_{\mathrm{guard}}(j), d_{\mathrm{wit}}(j), d_{\mathrm{stat}}(j)\big), \qquad j \in \mathcal{A}_7,$$

where d_U *measures loss of the queued work state,* d_h *measures mismatch between the queue type and selected head,* d_{tail} *measures loss of the finite guarded queue tail,* d_{res} *measures loss of the live residual triple,* d_Δ *measures loss of the paired HC comparison defect,* d_{guard} *measures guard drift,* d_{wit} *measures witness-pointer drift, and* d_{stat} *measures accidental certificate promotion. The scalar head-selection defect is*

$$\Delta^{\mathrm{select}}_{\mathrm{head}} = \max_{j\in\mathcal{A}_7}\big(d_U(j) + d_h(j) + d_{\mathrm{tail}}(j) + d_{\mathrm{res}}(j) + d_\Delta(j) + d_{\mathrm{guard}}(j) + d_{\mathrm{wit}}(j) + d_{\mathrm{stat}}(j)\big).$$

The register is head-clean *if* $\Delta^{\mathrm{select}}_{\mathrm{head}} = 0$*,* head-repairable *if one bounded local head-selection repair strictly lowers the scalar defect while preserving the queue, and* frontier-bound *otherwise.*

Lemma 24.350 (Continuation-queue output is the head-selection precondition)**.** *The queue-head selection packet may be executed only from one of* $\mathsf{U}^{\mathrm{ready}}_{\mathrm{bind}}$*,* $\mathsf{U}^{\mathrm{ready}}_{\mathrm{split}}$*, or* $\mathsf{U}^{\mathrm{ready}}_{\mathrm{frontier}}$*. If the continuation-queue node has not returned one of these three guarded ready queues, then the head-selection node is not yet admissible.*

Proof. By Definition 24.347, the first entry of the packet is exactly one queued work state $\mathsf{U}^{\mathrm{ready}}_{\eta}$. Head selection consumes a completed finite queue and selects its first guarded work item; it does not consume a raw return, an unresolved comparison return, or a merely local HC-capture source. Hence no head-selection packet exists until the node has returned a matching ready queue. □

Lemma 24.351 (Zero queue-head selection defect preserves selected-head data)**.** *If* $\Delta^{\mathrm{select}}_{\mathrm{head}} = 0$ *for an admissible queue-head selection packet, then the selected head preserves the queued work state, the matching head type, the finite guarded tail, the inherited live residual triple, the paired HC comparison defect, the head-selection guard, the witness pointer, and the non-certificate status marker.*

Proof. Vanishing scalar defect forces all components of Definition 24.349 to vanish on the head-selection audit window. Thus the ready queue remains visible, the selected head matches that queue, the finite tail is retained for later execution, the live residual triple and paired comparison defect remain attached, the guard and witness pointer are preserved, and the non-certificate marker is not promoted. Therefore the output is a guarded first work head together with a finite continuation tail, not an OP closure certificate. □

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Definition 24.352 (queue-head selection output)**.** *The execution of the queue-head selection packet returns exactly one of the following selected-head work states:*

$$\mathrm{Exec}\big(\mathsf{W}^{\mathrm{select}}_{\mathrm{head}}, \mathsf{P}^{\mathrm{select}}_{\mathrm{head}}\big) \in \{\mathsf{X}^{\mathrm{head}}_{\mathrm{bind}}, \mathsf{X}^{\mathrm{head}}_{\mathrm{split}}, \mathsf{X}^{\mathrm{head}}_{\mathrm{frontier}}\}.$$

Here $\mathsf{X}^{\mathrm{head}}_{\mathrm{bind}}$ *is the guarded common-origin-check head with its finite bind tail,* $\mathsf{X}^{\mathrm{head}}_{\mathrm{split}}$ *is the guarded branch-rank-check head with its finite split tail, and* $\mathsf{X}^{\mathrm{head}}_{\mathrm{frontier}}$ *is the guarded frontier-descent-check head with its finite frontier tail. None of these statuses is an OP closure certificate.*

Proposition 24.353 (The queue-head selection packet is admissible after queue return)**.** *If the execution returns one of* $\mathsf{U}^{\mathrm{ready}}_{\mathrm{bind}}$, $\mathsf{U}^{\mathrm{ready}}_{\mathrm{split}}$, *or* $\mathsf{U}^{\mathrm{ready}}_{\mathrm{frontier}}$, *then the active HC/OP skeleton admits* $\mathsf{W}^{\mathrm{select}}_{\mathrm{head}}$ *as the next executable work packet.*

Proof. Lemma 24.350 supplies the required guarded input. Definition 24.348 supplies the matching head and finite tail for that input. The remaining data are inherited from the paired comparison layer, the QAM/MML/MMA interface, the dispatch, the comparison return, the resolve/return output, and the continuation queue. The head-selection defect register is finite and audits only preservation of input queue, head type, tail, residual data, comparison defect, guard, witness, and non-certificate status. Hence the packet is admissible as follow-on work and not as an OP closure claim. □

Theorem 24.354 (First queue-head selection packet after continuation queue)**.** *After a guarded continuation-queue output, the active HC/OP skeleton supports the next executed work packet*

$$\{\mathsf{U}^{\mathrm{ready}}_{\mathrm{bind}}, \mathsf{U}^{\mathrm{ready}}_{\mathrm{split}}, \mathsf{U}^{\mathrm{ready}}_{\mathrm{frontier}}\} \xrightarrow{\mathsf{W}^{\mathrm{select}}_{\mathrm{head}}} \{\mathsf{X}^{\mathrm{head}}_{\mathrm{bind}}, \mathsf{X}^{\mathrm{head}}_{\mathrm{split}}, \mathsf{X}^{\mathrm{head}}_{\mathrm{frontier}}\}.$$

This execution is conservative: it selects the first guarded head of the matching finite continuation queue and keeps the remaining finite tail available for later work. It does not prove OP 1/4, OP 5(B), or any closure certificate.

Proof. By Proposition 24.353, the packet is admissible after a guarded ready queue. Definition 24.349 classifies the head-selection audit defect into head-clean, head-repairable, or frontier-bound cases, and Definition 24.352 records the corresponding selected-head work output. Lemma 24.351 proves the head-clean preservation claim. In the repair case the packet remains at the head-selection node with a strictly smaller local head-selection defect and no route drift; in the frontier case it returns protected smaller frontier material. In none of the three cases is the non-certificate marker changed into an OP closure certificate. □

Corollary 24.355 (working rule)**.** *Future work may proceed from a guarded ready queue only by selecting the matching first queue head while retaining the finite guarded tail. The next admissible mathematical question is therefore whether to execute* commonOriginCheck*,* branchRankCheck*, or* frontierDescentCheck *under its inherited guard; it is still not whether OP 1/4 or OP 5(B) is closed.*

Proof. Immediate from Theorem 24.354. □

24.22 First selected queue-head execution packet

Definition 24.356 (Selected queue-head execution packet)**.** *Let*

$$\mathsf{X}^{\mathrm{head}}_{\eta} \in \{\mathsf{X}^{\mathrm{head}}_{\mathrm{bind}}, \mathsf{X}^{\mathrm{head}}_{\mathrm{split}}, \mathsf{X}^{\mathrm{head}}_{\mathrm{frontier}}\}, \qquad \eta \in \{\mathrm{bind}, \mathrm{split}, \mathrm{frontier}\},$$

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be a guarded selected queue-head output. A selected queue-head execution packet *is a finite packet*

$$\mathsf{P}^{\mathrm{exec}}_{\mathrm{head}} = \big(\mathsf{X}^{\mathrm{head}}_{\eta}, h_\eta, \mathcal{U}^{\mathrm{tail}}_{\eta}, \rho_{\mathrm{CL}}, \rho_{\Theta R}, \rho_A, \Delta^{\mathrm{HC}}_{\mathrm{sep}}, \Delta^{\Delta}_{\mathrm{cmp}}, G_8, W_8, S_8, \mathbf{d}^{\mathrm{exec}}_{\mathrm{head}}\big),$$

where h_η is the selected queue head, $\mathcal{U}^{\mathrm{tail}}_{\eta}$ is the remaining finite guarded tail, $(\rho_{\mathrm{CL}}, \rho_{\Theta R}, \rho_A)$ is the inherited live residual triple, $\Delta^{\mathrm{HC}}_{\mathrm{sep}}$ is the inherited paired HC separation defect, $\Delta^{\Delta}_{\mathrm{cmp}}$ is the comparison-defect witness, G_8 is the execution guard, W_8 is the witness pointer to the selected-head source, S_8 is the non-certificate status marker, and $\mathbf{d}^{\mathrm{exec}}_{\mathrm{head}}$ is a finite head-execution audit register. The packet is admissible only if it executes the selected head belonging to the matching output, preserves the finite tail for later continuation, keeps the paired OP 1/4–OP 5(B) data attached, and does not promote the executed head to an OP closure certificate.

Definition 24.357 (Executed queue-head tests)**.** *The selected queue-head execution in Definition 24.356 is fixed by the head type:*

$$\begin{aligned}
\mathsf{X}^{\mathrm{head}}_{\mathrm{bind}} &\rightsquigarrow \mathrm{ExecHead(commonOriginCheck)},\\
\mathsf{X}^{\mathrm{head}}_{\mathrm{split}} &\rightsquigarrow \mathrm{ExecHead(branchRankCheck)},\\
\mathsf{X}^{\mathrm{head}}_{\mathrm{frontier}} &\rightsquigarrow \mathrm{ExecHead(frontierDescentCheck)}.
\end{aligned}$$

The bind-head execution checks whether the common-origin reading of $\rho_{\mathrm{CL}} \oplus \rho_{\Theta R}$ remains admissible under the inherited HC comparison defect. The split-head execution checks whether a ranked single-branch continuation can be selected without erasing the paired comparison witness. The frontier-head execution checks whether a protected descent to a smaller comparison frontier is available. Each head test is a work test, not a theorematic closure test.

Definition 24.358 (Selected queue-head execution defect register)**.** *For an admissible packet $\mathsf{P}^{\mathrm{exec}}_{\mathrm{head}}$, the* selected queue-head execution defect register *is the finite family*

$$\mathbf{d}^{\mathrm{exec}}_{\mathrm{head}}(j) = \big(d_X(j), d_h(j), d_{\mathrm{tail}}(j), d_{\mathrm{res}}(j), d_\Delta(j), d_{\mathrm{cmp}}(j), d_{\mathrm{guard}}(j), d_{\mathrm{wit}}(j), d_{\mathrm{stat}}(j)\big), \qquad j \in \mathcal{A}_8,$$

where d_X measures loss of the selected-head state, d_h measures mismatch between the selected head and the executed head test, d_{tail} measures loss of the finite queue tail, d_{res} measures loss of the live residual triple, d_Δ measures loss of the inherited HC separation defect, d_{cmp} measures loss of the comparison-defect witness, d_{guard} measures execution-guard drift, d_{wit} measures witness-pointer drift, and d_{stat} measures accidental certificate promotion. The scalar selected-head execution defect is

$$\Delta^{\mathrm{exec}}_{\mathrm{head}} = \max_{j\in\mathcal{A}_8}\big(d_X(j) + d_h(j) + d_{\mathrm{tail}}(j) + d_{\mathrm{res}}(j) + d_\Delta(j) + d_{\mathrm{cmp}}(j) + d_{\mathrm{guard}}(j) + d_{\mathrm{wit}}(j) + d_{\mathrm{stat}}(j)\big).$$

The register is head-exec-clean *if $\Delta^{\mathrm{exec}}_{\mathrm{head}} = 0$,* head-exec-repairable *if one bounded local execution repair strictly lowers the scalar defect while preserving the selected head and finite tail, and* frontier-bound *otherwise.*

Lemma 24.359 (Selected head is the head-execution precondition)**.** *The selected queue-head execution packet may be executed only from one of $\mathsf{X}^{\mathrm{head}}_{\mathrm{bind}}$, $\mathsf{X}^{\mathrm{head}}_{\mathrm{split}}$, or $\mathsf{X}^{\mathrm{head}}_{\mathrm{frontier}}$. If the head-selection node has not returned one of these three guarded head states, then the head-execution node is not yet admissible.*

Proof. By Definition 24.356, the first entry of the packet is exactly one selected head state. The execution node consumes a selected head, not a raw queue, a raw resolve return, or an earlier comparison final. Hence the head-execution packet is undefined until the node has selected the matching guarded queue head and retained the finite tail. □

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Lemma 24.360 (Zero selected-head execution defect preserves head-execution data)**.** *If* $\Delta^{\mathrm{exec}}_{\mathrm{head}} = 0$ *for an admissible selected queue-head execution packet, then the executed head preserves the selected-head state, the matching executed test, the finite guarded tail, the inherited live residual triple, the paired HC separation defect, the comparison-defect witness, the execution guard, the witness pointer, and the non-certificate status marker.*

Proof. Vanishing scalar defect forces all components of Definition 24.358 to vanish on the execution audit window. Thus the selected head is retained, the executed test matches the selected head type, the finite tail remains available for follow-on work, the residual and comparison data remain attached, the guard and witness pointer do not drift, and the non-certificate marker is not promoted. Therefore the output is an executed first queue head with a finite continuation tail, not an OP closure certificate. □

Definition 24.361 (selected queue-head execution output)**.** *The execution of the selected queue-head packet returns exactly one of the following executed-head work states:*

$$\mathrm{Exec}\big(\mathsf{W}^{\mathrm{exec}}_{\mathrm{head}}, \mathsf{P}^{\mathrm{exec}}_{\mathrm{head}}\big) \in \{\mathsf{Y}^{\mathrm{exec}}_{\mathrm{bind}}, \mathsf{Y}^{\mathrm{exec}}_{\mathrm{split}}, \mathsf{Y}^{\mathrm{exec}}_{\mathrm{frontier}}\}.$$

Here $\mathsf{Y}^{\mathrm{exec}}_{\mathrm{bind}}$ *is the executed common-origin-check surface with its finite bind tail,* $\mathsf{Y}^{\mathrm{exec}}_{\mathrm{split}}$ *is the executed branch-rank-check surface with its finite split tail, and* $\mathsf{Y}^{\mathrm{exec}}_{\mathrm{frontier}}$ *is the executed frontier-descent-check surface with its finite frontier tail. None of these statuses is an OP closure certificate.*

Proposition 24.362 (Selected queue-head execution is admissible after head selection)**.** *If the execution returns one of* $\mathsf{X}^{\mathrm{head}}_{\mathrm{bind}}$*,* $\mathsf{X}^{\mathrm{head}}_{\mathrm{split}}$*, or* $\mathsf{X}^{\mathrm{head}}_{\mathrm{frontier}}$*, then the active HC/OP skeleton admits* $\mathsf{W}^{\mathrm{exec}}_{\mathrm{head}}$ *as the next executable work packet.*

Proof. Lemma 24.359 supplies the required guarded input. Definition 24.357 supplies the matching head test. The residual data, paired HC separation defect, comparison-defect witness, guard, witness pointer, finite tail, and non-certificate status are inherited from the comparison packet, the resolve return, the continuation queue, and the selected head. The execution defect register is finite and audits only preservation of those inherited data. Hence the packet is admissible as follow-on work and not as an OP closure claim. □

Theorem 24.363 (First selected queue-head execution packet)**.** *After a guarded queue-head selection output, the active HC/OP skeleton supports the next executed work packet*

$$\{\mathsf{X}^{\mathrm{head}}_{\mathrm{bind}}, \mathsf{X}^{\mathrm{head}}_{\mathrm{split}}, \mathsf{X}^{\mathrm{head}}_{\mathrm{frontier}}\} \xrightarrow{\mathsf{W}^{\mathrm{exec}}_{\mathrm{head}}} \{\mathsf{Y}^{\mathrm{exec}}_{\mathrm{bind}}, \mathsf{Y}^{\mathrm{exec}}_{\mathrm{split}}, \mathsf{Y}^{\mathrm{exec}}_{\mathrm{frontier}}\}.$$

This execution is conservative: it runs the first guarded head of the matching continuation queue, keeps the finite tail available for later work, and does not prove OP 1/4, OP 5(B), or any closure certificate.

Proof. By Proposition 24.362, the packet is admissible after a guarded selected head. Definition 24.358 classifies the head-execution audit defect into head-exec-clean, head-exec-repairable, or frontier-bound cases, and Definition 24.361 records the corresponding executed-head work output. Lemma 24.360 proves the clean preservation claim. In the repair case the packet remains at the selected-head execution node with a strictly smaller local execution defect and no route drift; in the frontier case it returns protected smaller frontier material. In none of the three cases is the non-certificate marker changed into an OP closure certificate. □

Corollary 24.364 (working rule)**.** *Future work may proceed from an selected queue head only by executing the matching guarded head test while retaining the finite queue tail. The next admissible mathematical question is therefore whether the executed bind, split, or frontier head can legally enter its corresponding tail task — joint residual refinement, single residual selection, or defect localization — still under the inherited no-false-certificate rule.*

Proof. Immediate from Theorem 24.363. □

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24.23 First executed-head tail-entry packet

Definition 24.365 (Executed-head tail-entry packet)**.** *Let*

$$\mathsf{Y}_{\eta}^{\mathrm{exec}} \in \{\mathsf{Y}_{\mathrm{bind}}^{\mathrm{exec}}, \mathsf{Y}_{\mathrm{split}}^{\mathrm{exec}}, \mathsf{Y}_{\mathrm{frontier}}^{\mathrm{exec}}\}, \qquad \eta \in \{\mathrm{bind}, \mathrm{split}, \mathrm{frontier}\},$$

be a guarded executed-head output of Definition 24.361. An executed-head tail-entry packet *is a finite packet*

$$\mathsf{P}_{\mathrm{tail}}^{\mathrm{entry}} = \big(\mathsf{Y}_{\eta}^{\mathrm{exec}}, \mathcal{U}_{\eta}^{\mathrm{tail}}, \rho_{\mathrm{CL}}, \rho_{\Theta R}, \rho_A, \Delta_{\mathrm{sep}}^{\mathrm{HC}}, \Delta_{\mathrm{cmp}}^{\Delta}, \Delta_{\mathrm{head}}^{\mathrm{exec}}, G_9, W_9, S_9, \mathbf{d}_{\mathrm{tail}}^{\mathrm{entry}}\big),$$

where $\mathcal{U}_{\eta}^{\mathrm{tail}}$ *is the finite guarded tail retained by the selected-head execution,* $(\rho_{\mathrm{CL}}, \rho_{\Theta R}, \rho_A)$ *is the inherited live residual triple,* $\Delta_{\mathrm{sep}}^{\mathrm{HC}}$ *and* $\Delta_{\mathrm{cmp}}^{\Delta}$ *are the inherited comparison witnesses,* $\Delta_{\mathrm{head}}^{\mathrm{exec}}$ *is the selected-head execution defect,* G_9 *is the tail-entry guard,* W_9 *is the witness pointer to the executed-head source,* S_9 *is the non-certificate status marker, and* $\mathbf{d}_{\mathrm{tail}}^{\mathrm{entry}}$ *is a finite tail-entry audit register. The packet is admissible only if it enters the tail matching the executed head, preserves the residual and comparison data, preserves the finite tail, and does not promote the tail entry to an OP closure certificate.*

Definition 24.366 (Executed-head tail-entry defect)**.** *For an admissible packet* $\mathsf{P}_{\mathrm{tail}}^{\mathrm{entry}}$*, the* tail-entry defect register *is the finite family*

$$\mathbf{d}_{\mathrm{tail}}^{\mathrm{entry}}(j) = \big(d_Y(j), d_{\mathrm{tail}}(j), d_{\mathrm{res}}(j), d_{\Delta}(j), d_{\mathrm{head}}(j), d_{\mathrm{guard}}(j), d_{\mathrm{wit}}(j), d_{\mathrm{stat}}(j)\big), \qquad j \in \mathcal{A}_{\mathrm{tail}},$$

where the components measure loss of the executed-head state, the finite tail, the residual triple, the comparison witnesses, the head-execution defect, the tail-entry guard, the witness pointer, and the non-certificate marker. The scalar tail-entry defect is

$$\Delta_{\mathrm{tail}}^{\mathrm{entry}} = \max_{j \in \mathcal{A}_{\mathrm{tail}}} \big(d_Y(j) + d_{\mathrm{tail}}(j) + d_{\mathrm{res}}(j) + d_{\Delta}(j) + d_{\mathrm{head}}(j) + d_{\mathrm{guard}}(j) + d_{\mathrm{wit}}(j) + d_{\mathrm{stat}}(j)\big).$$

The register is tail-entry-clean *if* $\Delta_{\mathrm{tail}}^{\mathrm{entry}} = 0$*,* tail-entry-repairable *if one bounded local tail-entry repair strictly lowers the scalar defect while preserving the executed head and finite tail, and* frontier-bound *otherwise.*

Lemma 24.367 (Executed head is the tail-entry precondition)**.** *The tail-entry packet may be launched only from one of* $\mathsf{Y}_{\mathrm{bind}}^{\mathrm{exec}}$*,* $\mathsf{Y}_{\mathrm{split}}^{\mathrm{exec}}$*, or* $\mathsf{Y}_{\mathrm{frontier}}^{\mathrm{exec}}$*. If the selected-head execution has not returned one of these three guarded executed-head states, then tail entry is not yet admissible.*

Proof. Definition 24.365 takes the guarded executed-head state as the first datum of the packet. A selected but unexecuted head, a ready queue, or a resolve return does not yet determine which finite tail is allowed to open. Hence the tail-entry node is undefined until the selected-head execution has returned the matching executed-head state and retained its finite guarded tail. □

Lemma 24.368 (Zero tail-entry defect preserves tail-entry data)**.** *If* $\Delta_{\mathrm{tail}}^{\mathrm{entry}} = 0$ *for an admissible executed-head tail-entry packet, then the packet preserves the executed-head state, the matching finite tail, the live residual triple, the inherited comparison witnesses, the selected-head execution defect, the tail-entry guard, the witness pointer, and the non-certificate status marker.*

Proof. Vanishing scalar defect forces every component of Definition 24.366 to vanish on the tail-entry audit window. Therefore no executed-head data, residual data, comparison witness, guard, witness pointer, or status marker is lost. The output is a guarded tail-entry surface, not an OP closure certificate. □

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Definition 24.369 (Executed-head tail-entry output)**.** *The execution of the tail-entry packet returns exactly one of the following guarded tail-entry surfaces:*

$$\mathrm{Exec}\big(\mathsf{W}^{\mathrm{entry}}_{\mathrm{tail}}, \mathsf{P}^{\mathrm{entry}}_{\mathrm{tail}}\big) \in \{\mathsf{Z}^{\mathrm{tail}}_{\mathrm{bind}}, \mathsf{Z}^{\mathrm{tail}}_{\mathrm{split}}, \mathsf{Z}^{\mathrm{tail}}_{\mathrm{frontier}}\}.$$

Here $\mathsf{Z}^{\mathrm{tail}}_{\mathrm{bind}}$ enters the guarded joint-residual refinement tail, $\mathsf{Z}^{\mathrm{tail}}_{\mathrm{split}}$ enters the guarded single-branch selection tail, and $\mathsf{Z}^{\mathrm{tail}}_{\mathrm{frontier}}$ enters the guarded defect-localization tail. None of these statuses is an OP closure certificate.

Proposition 24.370 (Tail entry is admissible after selected-head execution)**.** *If the selected-head execution returns one of $\mathsf{Y}^{\mathrm{exec}}_{\mathrm{bind}}$, $\mathsf{Y}^{\mathrm{exec}}_{\mathrm{split}}$, or $\mathsf{Y}^{\mathrm{exec}}_{\mathrm{frontier}}$, then the active HC/OP skeleton admits $\mathsf{W}^{\mathrm{entry}}_{\mathrm{tail}}$ as the next executable work packet.*

Proof. Lemma 24.367 supplies the required executed-head input. The matching finite tail, residual triple, comparison witnesses, head-execution defect, guard, witness pointer, and non-certificate marker are inherited from the selected-head execution packet. The new defect register audits only preservation of those inherited data. Thus the packet is admissible as follow-on work and not as an OP closure claim. □

Theorem 24.371 (First executed-head tail-entry packet)**.** *After a guarded selected-head execution output, the active HC/OP skeleton supports the next tail-entry work packet*

$$\{\mathsf{Y}^{\mathrm{exec}}_{\mathrm{bind}}, \mathsf{Y}^{\mathrm{exec}}_{\mathrm{split}}, \mathsf{Y}^{\mathrm{exec}}_{\mathrm{frontier}}\} \xrightarrow{\mathsf{W}^{\mathrm{entry}}_{\mathrm{tail}}} \{\mathsf{Z}^{\mathrm{tail}}_{\mathrm{bind}}, \mathsf{Z}^{\mathrm{tail}}_{\mathrm{split}}, \mathsf{Z}^{\mathrm{tail}}_{\mathrm{frontier}}\}.$$

This execution is conservative: it opens the tail belonging to the executed head, retains all residual and comparison data, and does not prove OP 1/4, OP 5(B), or any closure certificate.

Proof. By Proposition 24.370, the packet is admissible after a guarded executed-head state. Definition 24.366 classifies the tail-entry defect into clean, repairable, or frontier-bound cases, while Definition 24.369 records the corresponding guarded tail-entry surface. Lemma 24.368 proves the clean preservation claim. In the repair case the packet remains at the tail-entry node with a strictly smaller local tail-entry defect; in the frontier case it returns protected smaller frontier material. In none of these cases is the non-certificate marker changed into an OP closure certificate. □

Corollary 24.372 (Tail-entry working rule)**.** *Future work may proceed from an executed queue head only by entering the matching finite guarded tail. The next admissible mathematical question is therefore whether the bind, split, or frontier tail can be refined under its inherited guard; it is still not whether OP 1/4 or OP 5(B) is closed.*

Proof. Immediate from Theorem 24.371. □

24.24 First audit-freeze ledger

Definition 24.373 (Audit-freeze ledger)**.** *The* audit-freeze ledger *for the current HC/OP line is the finite record*

$$F^{\mathrm{audit}}_{\mathrm{HCOP}} = \big(\mathcal{B}_{\mathrm{public}}, \mathcal{D}_{\mathrm{doi}}, \mathcal{S}_{\mathrm{spine}}, \mathcal{Z}_{\mathrm{tail}}, \mathcal{O}_{\mathrm{pres}}, \mathcal{C}_{\mathrm{ref}}, \mathcal{B}_{\mathrm{finalbib}}, \mathcal{H}_{\mathrm{witness}}, S_{\mathrm{noCert}}\big).$$

Here $\mathcal{B}_{\mathrm{public}}$ records the public baseline and concept-DOI status, $\mathcal{D}_{\mathrm{doi}}$ records the version-specific Zenodo DOI assigned in the public release metadata, $\mathcal{S}_{\mathrm{spine}}$ records the local HC/OP spine up to the tail-entry packet, $\mathcal{Z}_{\mathrm{tail}}$ records the guarded tail-entry output, $\mathcal{O}_{\mathrm{pres}}$ records preservation obligations, $\mathcal{C}_{\mathrm{ref}}$ records compilation/reference checks as audit data, $\mathcal{B}_{\mathrm{finalbib}}$ records terminal bibliography status, $\mathcal{H}_{\mathrm{witness}}$ records preservation of the historical witness, and S_{noCert} records that no OP closure certificate has been manufactured.

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Definition 24.374 (Audit-freeze defect)**.** *For a ledger* $F^{\mathrm{audit}}_{\mathrm{HCOP}}$*, the* audit-freeze defect *is*

$$\Delta^{\mathrm{freeze}}_{\mathrm{audit}} = \max\{d_{\mathrm{base}}, d_{\mathrm{doi}}, d_{\mathrm{spine}}, d_{\mathrm{tail}}, d_{\mathrm{pres}}, d_{\mathrm{ref}}, d_{\mathrm{bib}}, d_{\mathrm{hist}}, d_{\mathrm{cert}}\}.$$

The entries measure drift of the public baseline, DOI metadata, local spine, tail-entry output, preservation obligations, reference/compilation audit data, terminal bibliography, historical witness, and non-certificate status. The ledger is audit-clean *if* $\Delta^{\mathrm{freeze}}_{\mathrm{audit}} = 0$*,* audit-repairable *if only mechanical or metadata repair is needed, and* not freeze-ready *otherwise.*

Lemma 24.375 (Audit ledger is metadata-bearing, not theorem-generating)**.** *The audit-freeze ledger records status, preservation, and release-readiness data. It cannot by itself create an OP closure certificate or change the theorematic status of OP 1/4 or OP 5(B).*

Proof. Every component of Definition 24.373 is either inherited mathematical state, preservation metadata, compilation/reference audit data, bibliography status, historical-witness status, or the explicit non-certificate marker. None of these components is a certificate test for OP 1/4 or OP 5(B). Therefore the ledger can freeze and audit the current line, but it cannot promote the current line into a theorematic closure. □

Theorem 24.376 (First audit-freeze ledger)**.** *After the executed-head tail-entry packet, the current HC/OP line admits the finite audit-freeze ledger* $F^{\mathrm{audit}}_{\mathrm{HCOP}}$ *with audit-freeze defect* $\Delta^{\mathrm{freeze}}_{\mathrm{audit}}$*. The ledger records the public baseline, DOI state, local spine, tail-entry output, preservation obligations, reference/compilation audit data, terminal bibliography status, historical witness, and no-false-certificate marker without changing theorematic status.*

Proof. The public baseline and DOI state are metadata inherited from the title and chronology records. The local spine and tail-entry output are supplied by Theorem 24.371 and its predecessors. The preservation obligations, reference/compilation checks, terminal bibliography status, historical witness, and non-certificate marker are already explicit release-audit obligations of the working line. Definition 24.373 collects these data into one finite record, and Lemma 24.375 prevents any promotion to OP closure. □

24.25 Audit-freeze stabilization object

Definition 24.377 (Audit-freeze stabilization object)**.** *The* audit-freeze stabilization object *is*

$$F^{\mathrm{audit}}_{\mathrm{freeze}} = \big(F^{\mathrm{audit}}_{\mathrm{HCOP}}, \Delta^{\mathrm{freeze}}_{\mathrm{audit}}, \Pi_{\mathrm{allowed}}, S_{\mathrm{hold}}\big),$$

where Π_{allowed} *is the finite permission set*

$$\Pi_{\mathrm{allowed}} = \{\mathrm{audit}, \mathrm{mechanicalRepair}, \mathrm{metadataRepair}, \mathrm{releasePackaging}, \mathrm{zenodoDescription}\},$$

and S_{hold} *is the hold marker blocking further theorem expansion before the release audit is completed. The release label is metadata attached in prose and chronology, not a mathematical subscript or superscript of this object.*

Lemma 24.378 (Freeze permissions exclude theorem expansion)**.** *While* $F^{\mathrm{audit}}_{\mathrm{freeze}}$ *is active, the only admissible operations are audit, mechanical repair, metadata repair, release packaging, and Zenodo-description preparation. New theorem expansion is not an admissible freeze operation.*

Proof. The admissible operation set is exactly Π_{allowed} in Definition 24.377. None of its entries is a proof-extension, closure-promotion, OP-claim, or certificate-generation operation. Hence the active state may be repaired or packaged, but not mathematically expanded, until the release audit is discharged. □

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Theorem 24.379 (Audit-freeze stabilization)**.** *The current HC/OP working line is stabilized as an audit-freeze final by* $F_{\mathrm{freeze}}^{\mathrm{audit}}$*. In this state, the line is available for audit, mechanical repair, metadata repair, release packaging, and Zenodo-description preparation, while OP 1/4 and OP 5(B) remain open and no closure certificate is manufactured.*

Proof. Theorem 24.376 supplies the finite ledger and audit-freeze defect. Definition 24.377 adds only a finite permission set and a hold marker. Lemma 24.378 blocks theorem expansion during the freeze state. Therefore the line is audit-stabilized without changing the open status of OP 1/4 or OP 5(B). □

Corollary 24.380 (Audit-freeze working rule)**.** *Future work before the next public release may only audit, mechanically repair, repair metadata, package the release, or prepare the Zenodo description. Any new mathematical theorem expansion must wait until the freeze is either released or explicitly reopened.*

Proof. Immediate from Theorem 24.379 and Lemma 24.378. □

Phase I-C: Higher Confinement and Septinion Unfolding

This third unfolding block turns closure stability into a higher confinement reading and makes explicit why the septinion/seption interpretation is not an external axiom but a structurally selected reading of the admissible closure class.

Lemma 24.381 (Higher confinement lemma)**.** *If a structural class is closure-stable and remains invariant under repeated compensator-compatible normalization, then it admits a higher confinement reading.*

Proof. Closure stability excludes uncontrolled drift out of the admissible class, while repeated compensator-compatible normalization prevents the persistence of an independent escaping sector. Once both conditions hold, the admissible class is not merely locally normalized but globally retained under the structural return mechanisms of the system. This is exactly the content of higher confinement: the admissible states are no longer only corrected case by case but persist as a retained confinement regime. □

Lemma 24.382 (Non-escape lemma)**.** *Let x be a closure-stable state in the normalized core. Then no admissible transition generated within the core sends x into a structurally independent external sector.*

Proof. By closure stability, every admissible normalization step sends x back into the same closure class. If an admissible transition could produce a genuinely independent external sector, then closure stability would fail exactly at that point, because the state would no longer be returnable by the same structural normalization. Hence admissible transitions may rearrange or re-express states inside the confinement class, but they cannot export them into an independent outside sector. This is the structural non-escape property. □

Theorem 24.383 (Septinion readability theorem)**.** *If a class is closure-stable and admits the higher confinement reading, then it is septinion-readable: the resulting stabilized regime may be interpreted as a septinion/seption-level confinement of the normalized structural body.*

Proof. The septinion reading is not introduced as an arbitrary enlargement of the formalism. Rather, it names the level at which the closure-stable and non-escaping structural class is most naturally read as a higher unified container of the admissible sectors. By Lemma 24.381, closure stability already upgrades the system to a higher confinement regime, and Lemma 24.382 shows that admissible states remain inside that regime. The septinion/seption reading is therefore the natural structural reading of this higher confinement container. □

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Corollary 24.384 (HC as emergent structural consequence)**.** *Within the normalized core, HC is not an externally appended axiom. It appears as an emergent structural consequence of the one-body, frame-shift, and closure requirements.*

Proof. The generative chain of Phase I-A forces re-identification and therefore HC, while the closure chain of Phase I-B shows that HC is precisely the mechanism that stabilizes the admissible class. Phase I-C now identifies the higher confinement reading selected by this stabilization. Consequently HC is not an optional addition but the structurally emergent mechanism by which the unified body remains admissible across shifted readouts and under closure. □

Corollary 24.385 (First confinement forcing chain)**.** *The first unfolded confinement route of the normalized core is*

$$\textit{closure class} \Longrightarrow \textit{higher confinement} \Longrightarrow \textit{non-escape} \Longrightarrow \textit{septinion reading.}$$

Together with the preceding phases, this yields

$$\begin{aligned}&\textit{one body} \Longrightarrow \textit{re-identification} \Longrightarrow \mathrm{HC}\\ &\Longrightarrow \textit{closure} \Longrightarrow \textit{higher confinement} \Longrightarrow \textit{septinion/seption reading.}\end{aligned}$$

Proof. Proposition 24.264 provides the stabilized closure class. Lemma 24.381 upgrades this class to a higher confinement reading, Lemma 24.382 gives the internal retention property of admissible transitions, and Theorem 24.383 identifies the corresponding septinion/seption interpretation. The final displayed chain is obtained by concatenating this phase with the forcing chains of Phases I-A and I-B. □

Phase II-D: Cascade, Readout, and the Collapse-Fixed Sector

Lemma 24.386 (Cascade admissibility lemma)**.** *If x belongs to the closure-stable class selected by the normalized core, then every admissible cascade image $\hat{K}(x)$ remains inside the admissible readout class of the same unified body.*

Proof. The cascade is introduced as a composition of projection maps from higher strata toward the ground sector, not as an ontological splitting of the body. By Corollary 24.267, closure-stable states are precisely those stabilized against drift under shifted readouts. Applying the cascade to such a state therefore yields another admissible readout of the same body rather than a new object. Hence admissibility is preserved along the cascade image of any closure-stable input. □

Lemma 24.387 (Terminal ground-state lemma)**.** *The ground state S_0 is the terminal invariant readout sector of the cascade: for every admissible state x, one has $\hat{K}(x) = x$ exactly when $x \in S_0$.*

Proof. This is the normalized readout form of the ground-state condition already built into the semantic kernel. The cascade terminates in S_0, and terminality means precisely that no further nontrivial readout step occurs. Therefore the fixed-point condition $\hat{K}(x) = x$ is equivalent to membership in the terminal sector S_0. □

Lemma 24.388 (Global null/light-axis lemma)**.** *The null-axis structure selected by the normalized core, and read physically as the light axis at stable propagation, is stratified across the cascade: if a state is read on this null/light axis at one admissible layer, then its admissible readouts remain aligned with the same global null/light-axis class through the cascade.*

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Proof. The Lorentz/null-axis structure, read physically as the light axis, is not introduced as a local afterthought but as a stratified feature of the whole construction. The cascade only changes the readout level, not the structural identity of the body. Since admissible cascade images preserve the same unified body and the same closure-compatible sector, the null/light-axis reading persists as a global class across the projected layers. □

Theorem 24.389 (Collapse-fixed sector theorem)**.** *The photon/readout sector identified in the normalized core is the global collapse-fixed sector of the cascade. In particular, the sector denoted by S_0 is not merely a local null readout but the terminal collapse-fixed sector selected by the full stratified structure.*

Proof. Lemma 24.386 ensures that the cascade preserves admissibility, while Lemma 24.387 identifies S_0 as the terminal invariant sector. Lemma 24.388 shows that the light-axis reading persists globally across admissible layers. Therefore the photon/readout sector is not obtained from an isolated local calculation alone; rather, it is the collapse-fixed terminal sector selected by the entire cascade of admissible projections. □

Lemma 24.390 (Readout separation lemma)**.** *Within the normalized core one must distinguish three levels: structural forcing, cascade identification, and physical reading. A statement belongs to exactly one of these levels unless an explicit bridge is given.*

Proof. The structural forcing chain establishes what must hold inside the unified body. The cascade then identifies how such a structurally forced state appears at lower readout levels. A physical interpretation is an additional reading of that identified sector. Conflating these levels obscures where a statement is proved and where it is interpreted. The separation therefore serves as a normalization rule for subsequent unfolding steps. □

Corollary 24.391 (First readout forcing chain)**.** *The first unfolded readout route of the normalized core is*

$$\begin{aligned} &\textit{closure-stable class} \Longrightarrow \textit{admissible cascade image} \\ &\Longrightarrow \textit{terminal ground sector} \Longrightarrow \textit{collapse-fixed photon/readout sector.} \end{aligned}$$

Taken together with Phases I-A through I-C, the normalized core now yields the combined chain

$$\begin{aligned} &\textit{one body} \Longrightarrow \textit{re-identification} \Longrightarrow \mathrm{HC} \Longrightarrow \textit{closure} \\ &\Longrightarrow \textit{higher confinement} \Longrightarrow \textit{septinion reading} \\ &\Longrightarrow \textit{cascade-fixed readout sector.} \end{aligned}$$

Proof. The first displayed chain is the direct concatenation of Lemma 24.386, Lemma 24.387, and Theorem 24.389. The second displayed chain is obtained by appending this readout route to the forcing chain from Corollary 24.385. □

Phase II-E: Semantic/Tensor Interface and Millennium Intertwining

Lemma 24.392 (Semantic kernel well-formedness lemma)**.** *The normalized semantic kernel is well formed: its predicates of one-body identity, triolic organization, orthogonal anchoring, frame-shifted readout, HC forcing, closure stability, higher confinement, and readout admissibility do not define competing ontologies but successive structural constraints on one and the same body.*

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Proof. By construction, the normalized core begins from the one-body condition and then adds only constraint predicates that regulate how the same body may be read, stabilized, and projected. None of the predicates introduces a second carrier space or an ontologically independent duplicate; each one refines the same underlying body by imposing an admissibility condition. The semantic kernel is therefore well formed as a constraint hierarchy rather than a family of competing state spaces. □

Lemma 24.393 (Tensor realization lemma)**.** *The tensorial operators $\mathcal{J}^{\mu}{}_{\nu}$, $\mathcal{C}^{\mu}{}_{\nu}$, $\Pi_n{}^{\mu}{}_{\nu}$, and $\hat{K}^{\mu}{}_{\nu}$ realize the normalized semantic kernel as an operator system: $\mathcal{J}$ realizes the involutive frame symmetry, $\mathcal{C}$ the compensating projection, Π_n the cascade steps, and $\hat{K}$ the terminal readout map.*

Proof. The semantic kernel distinguishes involutive reflection, compensating return, projection through admissible layers, and terminal ground readout. The tensorial operators are introduced precisely to encode these roles in compact operator notation. The semantic and tensorial descriptions therefore stand in a role-wise correspondence: the operator system is not a second theory but a tensor realization of the already normalized semantic structure. □

Lemma 24.394 (Intertwining lemma)**.** *If the Millennium matrix $\mathcal{M}^{\mu}{}_{\nu}$ is used as the structural transport operator between admissible grouped layers, then it must intertwine with the normalized tensorial kernel. In the minimal normalized form one requires*

$$\mathcal{M}\mathcal{J} = \mathcal{J}\mathcal{M}, \qquad \mathcal{M}\mathcal{C} = \mathcal{C}\mathcal{M}, \qquad \mathcal{M}\hat{K} = \hat{K}\mathcal{M}$$

whenever the corresponding involution, compensation, and cascade-readout structures are to be preserved across the transport.

Proof. A transport operator that fails to commute with the involution would change the frame logic of the transported state; one that fails to commute with the compensator would alter admissibility by moving states in or out of the compensated sector; and one that fails to commute with the terminal readout map would not preserve the normalized collapse/readout architecture. Hence a Millennium transport that is meant to be structure preserving must satisfy the stated intertwining relations at least for the operators whose semantic role it claims to preserve. □

Theorem 24.395 (Admissibility preservation theorem)**.** *If x is admissible in the normalized semantic kernel and $\mathcal{M}$ is a Millennium transport satisfying the intertwining relations of Lemma 24.394, then $\mathcal{M}x$ remains admissible in the transported layer.*

Proof. Admissibility in the normalized core is carried by compatibility with the one-body constraint, compensator normalization, and the cascade-compatible readout structure. By Lemma 24.394, the transport operator preserves the tensorial realizations of exactly these mechanisms. Therefore applying $\mathcal{M}$ cannot convert an admissible state into an inadmissible one merely by transport; it yields an admissible image in the next layer provided the transport is itself taken within the normalized grouped architecture. □

Theorem 24.396 (Closure preservation theorem)**.** *Under the same hypotheses, closure-stable states are transported by $\mathcal{M}$ to closure-stable states of the receiving layer.*

Proof. Closure stability is the semantic statement that the admissible state does not drift out of the normalized class once HC forcing and anchoring are taken into account. Since admissibility is preserved by Theorem 24.395 and the intertwining relations preserve the compensator and readout structures that stabilize the closure class, the transported image cannot lose closure merely by layer transport. Thus closure stability is preserved across admissible Millennium transport. □

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Proposition 24.397 (Semantic–tensor equivalence proposition)**.** *The tensorial kernel does not introduce a second structural theory. It is a parallel notation for the normalized semantic kernel, and the Millennium matrix is admissible only insofar as its tensorial action represents the same constraint chain already present on the semantic side.*

Proof. By Lemma 24.392, the semantic side is a single constrained body. By Lemma 24.393, the operator side realizes these constraints in tensorial notation. The intertwining and preservation results show that $\mathcal{M}$ is not free to create a new structural doctrine; it is licensed only as a transport operator respecting the same normalized constraint chain. Hence semantic kernel and tensor kernel are parallel realizations of one structure. □

Corollary 24.398 (First semantic–tensor forcing chain)**.** *The normalized semantic/tensor interface yields the chain*

$$\begin{aligned} &\textit{semantic kernel} \Longrightarrow \textit{tensor realization} \Longrightarrow \textit{Millennium intertwining} \\ &\Longrightarrow \textit{admissibility preservation} \Longrightarrow \textit{closure preservation.} \end{aligned}$$

Proof. Lemma 24.392 establishes the semantic kernel, Lemma 24.393 gives its tensor realization, Lemma 24.394 fixes the admissible transport relations for $\mathcal{M}$, and Theorems 24.395 and 24.396 provide the preservation steps. □

Grouped Layer Architecture and Millennium Transport

The semantic/tensor kernel admits a second normalization in which the document's many local cores, layers, and readout blocks are collected into three grouped structural classes. This reduces duplication and makes the role of the Millennium matrix explicit as a transport operator between admissible groups.

Definition 24.399 (Grouped structural architecture)**.** *The normalized structural architecture consists of the following three grouped layers:*

(G1) ***Generative layer*** $\mathfrak{G}$***:*** *the layer of structural formation. It contains the one-body condition, triolic minimality, orthogonal anchoring, frame-shifted observation, the ground condition, HC forcing, closure admissibility, and the higher-confinement/septinion reading.*

(G2) ***Projection/translation layer*** $\mathfrak{P}$***:*** *the layer of cascade readout, sector selection, projection rules, collapse-fixed identifications, null-/light-axis readings, photon-sector readings, and admissible translation between equivalent structural descriptions.*

(G3) ***Field/probe realization layer*** $\mathfrak{F}$***:*** *the layer of probe states, two-probe geometry, closure-preserving field equations, interaction rules, and physical realization of the admissible structure.*

These three grouped layers are not independent ontologies. They are different organizational regimes of the same underlying body and must therefore remain compatible under admissible transport.

Definition 24.400 (Millennium transport between grouped layers)**.** *The Millennium matrix is interpreted at the grouped-layer level as a structure-preserving converter*

$$\mathcal{M}_{\mathfrak{G}\to\mathfrak{P}} : \mathfrak{G} \to \mathfrak{P}, \qquad \mathcal{M}_{\mathfrak{P}\to\mathfrak{F}} : \mathfrak{P} \to \mathfrak{F},$$

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and, whenever reverse structural checking is required, also by admissible back-transport maps

$$\mathcal{M}_{\mathfrak{P}\to\mathfrak{G}} : \mathfrak{P} \to \mathfrak{G}, \qquad \mathcal{M}_{\mathfrak{F}\to\mathfrak{P}} : \mathfrak{F} \to \mathfrak{P}.$$

At this level $\mathcal{M}$ is not merely a notational relabeling but a converter which must preserve structural admissibility across layer changes.

Proposition 24.401 (Grouped forcing and readout chain)**.** *The grouped architecture organizes the document into the structural chain*

$$\mathfrak{G} \xrightarrow{\mathcal{M}} \mathfrak{P} \xrightarrow{\mathcal{M}} \mathfrak{F},$$

where:

(a) $\mathfrak{G}$ *generates the admissible structure,*

(b) $\mathfrak{P}$ *reads, projects, and translates that structure without changing its admissibility class,*

(c) $\mathfrak{F}$ *realizes the same admissible structure as a field/probe system.*

In particular, the field/probe layer is not a second origin of the theory; it is a realization layer of structure first generated in $\mathfrak{G}$ *and made readable in* $\mathfrak{P}$*.*

Proof. The generative layer contains the forcing data established above: one-body, triolic organization, orthogonal anchor, frame shift, HC, and closure. The projection/translation layer does not create new admissibility; it converts the generative core into cascade-compatible, sector-resolved, and physically legible readout. The field/probe layer then realizes those admissible readouts as explicit probes, field equations, or interaction geometries. Since the same body is being followed through all three layers, admissible transport by $\mathcal{M}$ must preserve rather than replace the structural class. □

Proposition 24.402 (Admissibility preservation across grouped layers)**.** *Let x be an admissible state generated in* $\mathfrak{G}$*. If $\mathcal{M}$ acts as an admissible grouped-layer converter, then*

$$\text{Admissible}(x) \Longrightarrow \text{Admissible}(\mathcal{M}_{\mathfrak{G}\to\mathfrak{P}}x) \Longrightarrow \text{Admissible}(\mathcal{M}_{\mathfrak{P}\to\mathfrak{F}}\mathcal{M}_{\mathfrak{G}\to\mathfrak{P}}x).$$

Moreover, when closure stability is part of the admissibility data, it is to be preserved throughout this transport chain.

Proof. By Definition 24.400, grouped-layer transport is only allowed when it preserves structural admissibility. Hence an admissible state generated in $\mathfrak{G}$ remains admissible after conversion into the projection/translation layer, and likewise after conversion into the field/probe layer. If closure stability were lost during transport, the result would no longer be a valid readout or realization of the original structure but a structural rupture. Therefore closure stability must be preserved whenever it belongs to the originating admissibility class. □

Remark 24.403 (Document-normalization consequence)**.** *The grouped architecture provides a normalization principle for the text. Local "cores" or "integration blocks" should be read either as part of* $\mathfrak{G}$*, as part of* $\mathfrak{P}$*, or as part of* $\mathfrak{F}$*, rather than as independent foundational centers. This removes duplication: the normative foundation lies in* $\mathfrak{G}$*, the translation/readout logic in* $\mathfrak{P}$*, and the field/probe realization in* $\mathfrak{F}$*.*

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Phase II-F: Grouped Layer Transport and Ontological Non-Drift

This unfolding block upgrades the grouped architecture from a normalization note into an explicit transport calculus. The goal is to show that grouped layers do not introduce new origins of structure, but only different admissible regimes of one and the same underlying body.

Lemma 24.404 (Group decomposition lemma)**.** *Every active structural block of the normalized Companion is dominantly assignable to exactly one of the grouped layers $\mathfrak{G}$, $\mathfrak{P}$, or $\mathfrak{F}$, even when secondary cross-references to the other layers remain.*

Proof. By Definition 24.399, the grouped layers are distinguished by role rather than by separate ontology: generation, projection/translation, and realization. Any active block therefore has one primary structural function. Cross-references may connect a block to the other layers, but they do not erase its dominant role. Hence each active block is assignable to one grouped layer as its principal location. □

Lemma 24.405 (Transport lemma)**.** *Let x be a state generated in $\mathfrak{G}$. If $\mathcal{M}_{\mathfrak{G}\to\mathfrak{P}}$ is an admissible grouped-layer converter, then its image in $\mathfrak{P}$ is a projection/translation of the same admissible structure and not a newly created structural body.*

Proof. By Definition 24.400, admissible grouped transport preserves structural admissibility. Proposition 24.401 states that $\mathfrak{P}$ reads and translates what $\mathfrak{G}$ has already generated. Therefore the image of x under $\mathcal{M}_{\mathfrak{G}\to\mathfrak{P}}$ remains the same admissible structure under a projection/translation regime. It is not a second origin because admissibility is transported, not recreated. □

Lemma 24.406 (Realization lemma)**.** *If $y \in \mathfrak{P}$ is an admissible projection/translation state and $\mathcal{M}_{\mathfrak{P}\to\mathfrak{F}}$ is admissible, then its image in $\mathfrak{F}$ is a realization of the same admissible structure as a field/probe state.*

Proof. The role of $\mathfrak{F}$ is realization rather than generation by Proposition 24.401. Since admissibility is preserved across grouped transport by Proposition 24.402, $\mathcal{M}_{\mathfrak{P}\to\mathfrak{F}}(y)$ remains within the same admissibility class. Thus the field/probe layer realizes the translated structure without changing its underlying body. □

Proposition 24.407 (No-ontology-drift proposition)**.** *Admissible transport across the grouped layers does not create ontological multiplicity. If*

$$x \in \mathfrak{G}, \qquad x_P := \mathcal{M}_{\mathfrak{G}\to\mathfrak{P}}(x), \qquad x_F := \mathcal{M}_{\mathfrak{P}\to\mathfrak{F}}(x_P),$$

then x, x_P, and x_F are three admissible organizational regimes of one and the same structural body.

Proof. By Lemma 24.405, passage from $\mathfrak{G}$ to $\mathfrak{P}$ produces a translation/readout of the same admissible body. By Lemma 24.406, passage from $\mathfrak{P}$ to $\mathfrak{F}$ produces a field/probe realization of that same admissible body. Since both steps preserve admissibility rather than replacing it, no new ontology is introduced along the chain. Therefore x, x_P, and x_F are structurally identical in body and differ only by organizational regime. □

Corollary 24.408 (First grouped transport forcing chain)**.** *The normalized grouped architecture supports the forcing chain*

$$\mathfrak{G} \Longrightarrow \textit{admissible transport to } \mathfrak{P}$$
$$\Longrightarrow \textit{admissible realization in } \mathfrak{F} \Longrightarrow \textit{no ontology drift across grouped layers.}$$

Proof. Combine Lemma 24.404, Lemma 24.405, Lemma 24.406, and Proposition 24.407. □

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Phase III-G: Probe Geometry, Inheritance, and Guided Search

This unfolding block upgrades the probe sector from an interpretive add-on to a structurally inherited realization of the normalized core. The goal is to show that probes do not introduce a foreign search mechanism, but inherit triolic, anchored, and closure-compatible structure from the generative and grouped layers.

Lemma 24.409 (Probe inheritance lemma)**.** *If the normalized structural body is triolic, anchored, and closure-admissible, then every admissible probe realization inherited through* $\mathfrak{F}$ *carries a corresponding triolic and closure-compatible organization.*

Proof. By Proposition 24.407, admissible realization in $\mathfrak{F}$ does not create a second ontology, but realizes the same underlying structural body under a field/probe regime. Hence structural features that belong to the admissible body—in particular triolic organization and anchor-compatible closure discipline—remain available in the probe image as inherited structure rather than externally attached decoration. □

Lemma 24.410 (Two-probe compatibility lemma)**.** *Let* P_1 *and* P_2 *be two admissible probes directed toward one another. If both are inherited from the same normalized structural body, then their mutual configuration is compatible with one common closure-governed geometry.*

Proof. By Lemma 24.409, each probe inherits the same triolic and closure-compatible structural discipline. Since neither probe introduces a new ontology by Proposition 24.407, their mutual interaction must be read inside one common admissibility class. Therefore the two-probe configuration is not a collision of unrelated structures but a common closure-governed geometry of one inherited body. □

Lemma 24.411 (Orthogonal third-group lemma)**.** *In an admissible two-probe configuration, the third structural group remains orthogonal to both probes and thereby determines the constrained search region.*

Proof. The orthogonal third direction is already forced at the generative level by the orthogonal-anchor principle and remains preserved under admissible realization. Hence, when two probes are realized as inherited triolic structures, the third group cannot collapse into either probe direction without violating anchor compatibility. It therefore remains orthogonal to both and serves as the stabilizing transversal direction that bounds the admissible search region. □

Proposition 24.412 (Probe reduction theorem)**.** *An admissible two-probe configuration reduces to the closure-preserving sector of the grouped architecture. In particular, the probe interaction may be analyzed inside the admissible closure class without introducing a second search ontology.*

Proof. By Lemma 24.410, the two probes share one common closure-governed geometry. By Lemma 24.411, the third group fixes the transversal admissible search region. Since closure and ontology are preserved across grouped realization by Proposition 24.407, the probe interaction reduces to the closure-preserving sector of the same normalized architecture. □

Corollary 24.413 (Guided search corollary)**.** *Auxiliary tools such as frequency extraction, transition guidance, structural routing, or convergence acceleration act only within the already admissible probe geometry. They do not generate admissibility, but exploit a search region that has already been restricted by inheritance, closure, and orthogonal anchoring.*

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Proof. By Proposition 24.412, admissible probe analysis is already confined to the closure-preserving sector. Therefore any auxiliary tool operates on a domain whose admissibility has been fixed before the tool is applied. The tool may compress or guide the search, but it does not create the search geometry itself. □

Corollary 24.414 (First probe forcing chain)**.** *The normalized probe sector supports the forcing chain*

$$\begin{aligned} &\textit{inherited probe structure} \Longrightarrow \textit{two-probe compatibility} \\ &\Longrightarrow \textit{orthogonal third-group search region} \\ &\Longrightarrow \textit{closure-preserving probe reduction} \\ &\Longrightarrow \textit{guided search inside an already admissible geometry.} \end{aligned}$$

Proof. Combine Lemma 24.409, Lemma 24.410, Lemma 24.411, Proposition 24.412, and Corollary 24.413. □

Phase III-H: Structural Master Equation and Exact-Closure Field Regime

This unfolding block upgrades the previously compressed master-equation language into an explicit field-level forcing chain. The aim is not yet to derive every physical limit in full detail, but to show that the structural master equation, its variational reading, the closure-preserving field regime, and the exact- closure simplifications belong to one continuous argument rather than to loosely connected interpretive remarks.

Lemma 24.415 (Structural master-equation lemma)**.** *The structural master equation is the compressed field-level summary of the normalized forcing chain from one-body structure through re-identification, HC, closure, higher confinement, and admissible readout.*

Proof. By Corollary 19.16, the normalized body forces re-identification and HC. By Corollary 24.267, this induces a closure class with either exact closure or a controlled residue. By Corollary 24.385, closure yields higher confinement and the septinion reading. By Corollary 24.391 and Corollary 24.398, admissible readout and tensorial transport preserve this structure. Hence the master equation is not an additional postulate but the compressed field-level expression of the forcing chain already established. □

Lemma 24.416 (Variational consistency lemma)**.** *Any variational reading of the structural master equation is admissible only if it preserves the normalized one-body, HC, and closure conditions. In particular, admissible variation may unfold the structural equation but may not replace its forcing chain by an independent ontology.*

Proof. By Proposition 24.397, tensorial formulation is only a parallel realization of the same semantic kernel. Therefore any variational formulation that is meant to belong to the present architecture must preserve the same semantic admissibility data. If the variation were to destroy one-body identity, HC-forcing, or closure preservation, it would cease to be a variation of the normalized structural equation and would instead define a second independent system. Hence admissible variation is constrained by the forcing chain and remains structurally internal to it. □

Lemma 24.417 (Closure-field compatibility lemma)**.** *The field-level realization of the structural master equation is compatible with closure precisely when the realized dynamics preserves the HC-selected admissibility class and does not mix the stable closure sector with an ontologically foreign complement.*

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Proof. By Theorem 24.396, admissible transport through the Millennium interface preserves closure classes. By Proposition 24.407, realization in the field/probe layer introduces no second ontology. Therefore a field realization is closure-compatible exactly when it remains inside the HC-selected admissibility class rather than coupling the closure-stable sector to a foreign complement. This is the field-level restatement of normalized closure preservation. □

Theorem 24.418 (Exact-closure field theorem)**.** *In the exactly closed regime, the realized field dynamics reduces to the closure-stable block of the normalized architecture. Equivalently, the active field transport becomes block-compatible with the exact closure class.*

Proof. By Proposition 24.265, exact closure is equivalent to the disappearance of the nontrivial closure defect. By Lemma 24.417, closure-compatible field dynamics may then act only on the already stabilized admissibility class. Since no ontology drift occurs across grouped transport, the realized field dynamics reduces to the closure-stable block itself. Thus exact closure yields a block-compatible field regime. □

Lemma 24.419 (Minimal compensator disappearance lemma)**.** *In the exactly closed regime, the minimal compensator residue vanishes. What remains is not the disappearance of structural compensation as such, but the fact that no additional compensator correction is required beyond the exact closure block.*

Proof. By Lemma 24.266, the non-exact regime is marked by a residual deviation from exact closure. By Theorem 24.418, exact closure confines the field transport completely to the closure-stable block. Therefore the additional minimal compensator term that measures residual off-block correction must vanish. Structural compensation remains implicit in the exact block, but no separate compensator residue survives. □

Theorem 24.420 (Effective-limit theorem)**.** *Any admissible effective limit—for example a geometric, spinorial, or quantum field-theoretic limit—must arise as a readout of the normalized structural master equation together with its closure-preserving field realization. Such a limit may specialize the realization, but cannot bypass the forcing chain that produces it.*

Proof. By Lemma 24.415, the structural master equation is the compressed summary of the normalized forcing chain. By Lemma 24.416 and Lemma 24.417, any admissible field realization and any admissible variation remain internal to that chain. By Theorem 24.418 and Lemma 24.419, the exact regime yields a pure closure block without additional corrective residue. Hence any effective limit must be read as a specialization or projection of the normalized structural field realization rather than as an independent starting point. □

Corollary 24.421 (First field forcing chain)**.** *The normalized field sector supports the forcing chain*

$$\begin{aligned}&\textit{structural master equation} \Longrightarrow \textit{variational consistency}\\ &\Longrightarrow \textit{closure-preserving field compatibility} \Longrightarrow \textit{exact closure block regime}\\ &\Longrightarrow \textit{vanishing minimal compensator residue} \Longrightarrow \textit{admissible effective limit readout.}\end{aligned}$$

Proof. Combine the structural master equation, variational consistency, closure-field compatibility, the exact closure-field theorem, minimal compensator disappearance, and the effective limit theorem. □

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Concrete mapping of legacy core blocks to grouped layers

The grouped normalization can be made explicit by assigning the surviving legacy core blocks to the three grouped layers $\mathfrak{G}$, $\mathfrak{P}$, and $\mathfrak{F}$. This turns older version-labelled cores into either active parts of the normalized architecture or historical/supporting blocks.

Definition 24.422 (Layer assignment of legacy core blocks)**.** *Within the present monolith, the principal legacy blocks are to be read as follows.*

(L1) ***$\mathfrak{G}$-dominant blocks (generative layer):***

- *(a) the normalized* Structural Generation Core*,*
- *(b)* Structural Core and Closure Principles*,*
- *(c) the generative part of* archival stage Consolidation Core*, especially the structural reading discipline together with the ground-condition/HC/closure and higher-confinement logic,*
- *(d) the generative part of* archival stage Derivation and Appendix Integration Core*, especially the closure ladder and the micro-to-macro derivation chain.*

(L2) ***$\mathfrak{P}$-dominant blocks (projection/translation layer):***

- *(a)* archival stage Projection and Appendix Table Integration Core*,*
- *(b) the appendix-coupling and projection-summary parts of* archival stage Derivation and Appendix Integration Core*,*
- *(c) the collapse, light-axis, photon-sector, and cascade-readout parts of the later main sections,*
- *(d) all readout, table, projection, and sector-translation interfaces whose role is to make one admissible structure legible without changing its admissibility class.*

(L3) ***$\mathfrak{F}$-dominant blocks (field/probe realization layer):***

- *(a) the probe-sector and two-probe geometry blocks,*
- *(b) the field-equation, interaction, and closure-preserving dynamics sections,*
- *(c) the probe-reduction and guided-search realizations insofar as they are used as operational realizations of previously fixed admissible structure.*

(L4) ***Historical/supporting blocks:*** *readiness criteria, version-status passages, and older local consolidation summaries are not to be read as new foundational centers. They are historical normalization records unless they are explicitly re-exported into one of $\mathfrak{G}$, $\mathfrak{P}$, or $\mathfrak{F}$.*

Proposition 24.423 (Normalization map for duplicated core material)**.** *Under the grouped architecture, duplicated structural material is normalized by the following rule.*

(i) When a legacy block states formation conditions, forcing rules, or closure-generating principles, it is normalized into $\mathfrak{G}$.

(ii) When a legacy block states projection rules, cascade-compatible readout, sector identifications, appendix transport, or summary-table logic, it is normalized into $\mathfrak{P}$.

(iii) When a legacy block states probes, field equations, interaction channels, or closure-preserving realizations, it is normalized into $\mathfrak{F}$.

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(iv) If a block contains mixed material, it is to be read by dominant function, with the remaining content treated as support, historical context, or material that should later be split explicitly.

In particular, archival stage Consolidation Core *and the early derivation cores no longer define independent foundations beside the normalized structural core; they contribute either to* $\mathfrak{G}$*, to* $\mathfrak{P}$*, or to historical context.*

Remark 24.424 (Immediate practical consequence for the next clean release)**.** *The next clean release should therefore not keep several local cores in competition. Instead, older version-labelled cores are to be annotated and, where necessary, shortened according to their dominant grouped-layer function: formation material belongs to* $\mathfrak{G}$*, translation/readout material to* $\mathfrak{P}$*, field/probe realization material to* $\mathfrak{F}$*, and pure status/history material to archival support.*

archival stage Expansion Core

Remark 24.425 (Normalization note for archival stage)**.** *This block is retained as a historical expansion precursor. Its proof-bearing content is now to be read primarily through Phase I, Phase II-D, and the normalized grouped architecture. In particular, the intermediate master-equation ladder and the micro-to-macro chain no longer function as independent foundations beside the later unfolding phases, but as early organizational anticipations of them.*

Version archival stage is the first release that does more than freeze the next work path. It adds a genuine expansion layer on top of the archival stage consolidation core. The operative aim is threefold: first, the structural master equation is no longer presented only as a compressed global summary, but also as the endpoint of an explicit intermediate chain; second, the passage from local triolic organization to macroscopic effective geometry is stated in staged form; third, the technical apparatus is widened so that later proof layers can be attached without rewriting the monolith.

1. Deepened master-equation programme

The master equation in the later main section remains the global compression of one common closure-stable dynamics. What archival stage adds is a more explicit three-level reading of that equation.

(i) **Structural level.** One starts from triolic organization, orthogonality, frame shift, compensator action, and closure. At this level, the equation is still read as a balance law on admissible sectors.

(ii) **Mesoscopic level.** The same balance law is rewritten as an effective transport-and-selection equation for coupled sectors and probes. Here one already sees how admissible interaction channels are filtered by HC and by the closure residue.

(iii) **Macroscopic level.** After suppression of unstable local residues and after passage to a closure-stable observer description, the same equation becomes readable as effective matter dynamics plus background geometry.

In this sense, archival stage does not replace the later master equation (4); it inserts the missing ladder by which that equation is reached.

A convenient intermediate schematic form is

$$\mathcal{E}_{\mathrm{micro}}[\Psi] + \mathcal{E}_{\mathrm{coupl}}[\Psi, \Pi] + \mathcal{E}_{\mathrm{HC}}[\Psi] + \mathcal{E}_{\mathrm{cl}}[\Psi] + \mathcal{E}_{\mathrm{obs}}[\Psi] = 0,$$

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where the five blocks stand respectively for local triolic structure, sectorial coupling/probe interaction, Heisenberg-compensator balancing, closure selection, and observer-frame projection. The later master equation is to be read as the compressed global representative of this staged balance.

Lemma 24.426 (Intermediate balance principle)**.** *Every admissible global solution must pass through an intermediate regime in which local triolic transport, compensator balancing, and closure selection are all simultaneously active. Consequently, a direct jump from isolated local structure to macroscopic field form is structurally forbidden.*

Proof. Local triolic data alone do not yet determine which coupled configurations are admissible; they only provide possible sectorial directions. Conversely, a macroscopic field description is meaningful only after compensator balancing has removed representation drift and closure has excluded unstable residues. Therefore at least one intermediate regime must exist in which coupling, HC, and closure are jointly active. This regime is the structural source of the mesoscopic reading introduced in archival stage. □

2. Micro-to-macro closure chain

The new hardening target of archival stage is the explicit micro-to-macro chain

$$\begin{aligned}&\text{local triolic unit} \longrightarrow \text{coupled closure cell} \longrightarrow \text{probe-filtered admissible channel}\\ &\longrightarrow \text{closure-stable sector regime} \longrightarrow \text{effective global geometry and field dynamics.}\end{aligned}$$

The purpose of this chain is not rhetorical. It separates five logically distinct burdens that earlier drafts could too easily blend together: local decomposition, coupling, admissibility filtering, stable sector assembly, and macroscopic readout.

Local triolic unit. At the smallest relevant level, the system carries the standing triolic organization together with the orthogonal third direction. This level fixes the minimal sector inventory but does not yet decide which transitions survive.

Coupled closure cell. Once two or more local units interact, the question becomes whether their combination admits a common compensator-compatible representative. A coupled cell is called closure-ready if its local mismatches can be balanced without breaking the orthogonality and frame-shift discipline.

Probe-filtered admissible channel. The probe construction becomes technically important at precisely this stage. Two directed probes together with the orthogonal third group define a restricted search geometry. Kasisky-, hash-, rainbow-, Markov-, and Newton-type tools are not external decoration here; they are the operational means by which the space of final channels is compressed to closure-compatible paths.

Closure-stable sector regime. Only after probe filtering and compensator balancing does one obtain a stable sector regime. This is the first level at which effective matter, light/collapse, and dual sectors can be read as persistent rather than transient descriptions.

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Effective global geometry and field dynamics. Macroscopic geometry is then not glued on from outside. It is the readout of a large-scale assembly of closure-stable sector regimes viewed in the phase-shifted observer frame. The gravitational, QFT-type, and Dirac-type limits are therefore downstream realizations of one common filtered closure architecture.

Proposition 24.427 (Macroscopic readout requires closure-stable assembly)**.** *A macroscopic field or geometric description is admissible only if it is the observer-frame image of an assembly of closure-stable sector regimes. In particular, large-scale geometry cannot be read directly from uncoupled local triolic data.*

Proof. Uncoupled local data do not yet fix admissible transport and do not suppress representation drift. The observer frame, however, reads only stabilized configurations. Since stabilization in the present framework is exactly the combined action of HC and closure across coupled cells, the macroscopic image must descend from an assembly that has already passed the closure filter. □

3. Extended lemma and tool apparatus

Version archival stage also widens the technical apparatus that the next proof layer can rely on. The point is not to over-formalize prematurely, but to mark where repeated arguments will later be detached into reusable blocks.

Core lemmas now fixed by role.

1. **Orthogonality lemma:** the third triolic direction remains the non-negotiable constraint that prevents degeneracy of the sector picture.
2. **Frame-shift lemma:** observational readout differs from the underlying structural frame by a controlled phase displacement.
3. **Compensator lemma:** HC is the minimal identification mechanism needed to prevent sectorial drift.
4. **Closure lemma:** admissibility is equivalent to vanishing of the relevant closure residue in the chosen regime.
5. **Probe admissibility lemma:** the probe sector selects channels that are compatible with both closure and orthogonality.
6. **Read-off lemma:** physical sector names arise only after the preceding structural filters have been passed.

Normalized tool classes. The tool catalogue of archival stage is retained, but archival stage upgrades its use: closure-tools act as admissibility tests, frame-shift tools as translation operators between structure and observer descriptions, projection-tools as controlled reduction maps, probe-tools as search-space compressors, equivalence-tools as representative-identification devices, and read-off tools as disciplined output maps. This normalization matters because later technical proofs can now cite a *tool class* instead of repeating the surrounding motivation every time.

Corollary 24.428 (Tool usage discipline)**.** *No computational shortcut is admissible in the proof architecture unless its role can be classified as one of the normalized tool classes and unless its use preserves the structural ordering*

closure before projection before interpretation.

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Proof. A shortcut that bypasses closure may produce effective formulas without fixing admissibility; a shortcut that bypasses projection discipline may confuse structural content with observer naming. The normalized tool classes are introduced exactly to block such category errors. Hence admissible shortcuts are those that preserve the structural order stated above. □

4. Version status of archival stage

With these additions, archival stage should be read as the first expansion-stage companion release. Relative to archival stage, the new version contributes three hard improvements: the master-equation programme is given an explicit intermediate ladder, the micro-to-macro chain is stated as a formal burden rather than a mere promise, and the lemma/tool apparatus is widened enough that subsequent versions can deepen proofs without reopening the global organization of the monolith. The next versions may therefore focus on tightening derivations,

$$\begin{aligned}&\text{ground condition} \Longrightarrow \text{triolic admissibility} \Longrightarrow \text{HC balancing}\\ &\Longrightarrow \text{closure filtering} \Longrightarrow \text{observer-frame readout.}\end{aligned}$$

Version archival stage turns the earlier expansion layer into a genuine lemma-development release. The aim is not to replace the existing structural chapters, but to make the repeated internal moves of the proof architecture explicit enough that later sections may cite them as stable intermediate devices. Three burdens are addressed here: the closure spine is split into reusable intermediate steps, the passage from probes to admissible sectors is cast as a sequence of named reductions, and the projection discipline is tied to a small but durable family of proof lemmas.

Lemma 24.429 (Triolic admissibility precedes compensator balancing)**.** *A configuration cannot enter the Heisenberg-compensator balancing regime unless its triolic decomposition is already admissible with respect to the standing orthogonality and frame-shift constraints.*

Proof. The compensator does not create a triolic decomposition; it only identifies and balances compatible representatives across already existing sector data. If the triolic organization were itself degenerate or incompatible with the orthogonal third direction, then there would be no well-defined data for HC to balance. Hence triolic admissibility is logically prior to the compensator step. □

Lemma 24.430 (Closure filtering follows compensator balancing)**.** *Closure filtering is meaningful only after HC has removed representational drift between coupled sector descriptions.*

Proof. Closure tests are carried out on final configurations that are meant to represent one common admissible state. Without compensator balancing, distinct representatives may still differ by unresolved phase or sector drift, so a closure residue would not yet measure physical inadmissibility but only misalignment of representation. Therefore HC balancing is the prerequisite that turns closure filtering into a structurally meaningful test. □

Proposition 24.431 (Closure ladder)**.** *The passage from local structural data to observer-frame physics factors through the closure ladder described above, and none of its intermediate steps may be skipped without breaking the proof discipline*

closure before projection before interpretation.

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Proof. Lemma 24.429 places admissible triolic organization before HC balancing. Lemma 24.430 places closure filtering after HC and before any observer readout. The observer frame, however, names only the output of the stabilized configuration, so projection necessarily comes after closure filtering. Interpretation is last because it depends on projected sector naming rather than on raw structural data. Thus the stated factorization is forced by the standing proof order. □

2. Probe reduction chain and admissible channels

The probe sector now receives a reusable reduction chain. Instead of speaking only about a general search space, archival stage fixes the following sequence:

$$\begin{aligned}&\text{probe pair} \Longrightarrow \text{orthogonal search geometry}\\ &\Longrightarrow \text{final channel family} \Longrightarrow \text{tool-guided compression}\\ &\Longrightarrow \text{closure-compatible channel.}\end{aligned}$$

This makes explicit where frequency detection, routing compression, Markov transition guidance, and Newton-type convergence tools actually enter.

Lemma 24.432 (Orthogonal probe geometry restricts channel growth)**.** *When two probes are coupled together with the standing orthogonal third group, the set of admissible final channels is a strict restriction of the naive pairwise interaction space.*

Proof. The naive pairwise space allows all pair couplings consistent with the two incoming probe descriptions. The orthogonal third group imposes an additional compatibility burden that forbids precisely those channels that would resolve the pairwise mismatch by leaving the triolic discipline. Hence the admissible final family is smaller than the unrestricted pairwise space. □

Lemma 24.433 (Tool-guided compression preserves admissibility order)**.** *Kasisky-, hash-, rainbow-, Markov-, and Newton-type tools are admissible only when used to compress or navigate the final channel family without changing the structural order of admissibility testing.*

Proof. These tools may accelerate pattern detection, routing, transition guidance, or convergence, but none of them decides admissibility independently of closure. Their acceptable role is therefore subordinate: they reduce the burden of search inside the final family while the final decision remains the closure test. Consequently, they preserve admissibility only if they do not replace the closure filter by a purely computational surrogate. □

Proposition 24.434 (Probe reduction principle)**.** *A probe channel is physically readable only if it is obtained as the end point of the probe reduction chain and survives closure filtering after tool-guided compression.*

Proof. By Lemma 24.432, the orthogonal search geometry already restricts the final family. By Lemma 24.433, the technical tools may only compress or navigate that family, not redefine its criterion of admissibility. Therefore the only physically readable channels are those that remain after the final closure-compatible selection step. □

3. Projection robustness and controlled interpretation

The next recurrent burden is the distinction between structural forcing, projective readout, and phenomenological naming. Version archival stage fixes this burden by introducing small reusable robustness statements.

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Lemma 24.435 (Projection does not create structure)**.** *A projection map may translate an already closure-stable structural state into an observer-frame description, but it cannot create a new admissible structure that was absent before projection.*

Proof. Projection acts on the representative of a state after the admissibility burden has already been settled. If projection could create admissibility, then the observer frame would retroactively decide structural existence, contradicting the standing principle that measurement is phase-shifted relative to the underlying structure. Thus projection is readout, not creation. □

Corollary 24.436 (Interpretation discipline)**.** *Matter, gauge, mass, Higgs, and geometric language are admissible only as controlled names for projected closure-stable regimes and not as independent axiomatic inputs to the structural proof.*

Proof. By Lemma 24.435, projection only reads out a state that has already survived the structural admissibility filter. Therefore the phenomenological names attached afterwards inherit their legitimacy from the closure-stable regime and cannot function as prior axioms of the construction. □

4. Technical use inside the monolith

The practical purpose of these new lemmas is modest but important. They create stable insertion points for later derivation chains. Subsequent versions may therefore reference:

- the *closure ladder* when passing from local triolic data to observer-frame sectors;
- the *probe reduction principle* when compressing search spaces; and
- the *projection robustness statements* when interpreting effective field sectors.

This means that the monolith can deepen individual proofs without reopening the global explanatory burden every time a local derivation is added.

5. Version status of archival stage

Relative to the previous release, archival stage contributes a stricter internal proof scaffold. The master-equation ladder and the micro-to-macro chain are retained, but they are no longer left as merely global narratives. They are now supported by named intermediate lemmas that later sections may invoke directly. That is the specific function of archival stage: not to conclude the full proof, but to make the next proof layer technically attachable in a disciplined way.

25 Toward a Structural ToE Master Equation

We now compress the previously established structural requirements into a single master equation. This equation is not introduced as an external ansatz, but as the minimal common form forced by the triolic structure, the Heisenberg compensator, the probe–closure system, the projected closure residue, and the phase-shifted observer frame.

$$\mathfrak{G}[\Psi, g] + \mathfrak{T}_{\mathrm{tri}}[\Psi] + \mathfrak{H}_{\mathrm{HC}}[\Psi] + \mathfrak{Q}_{\mathrm{cl}}[\Psi] + \mathfrak{B}_{\beta}[\Psi] = 0. \tag{4}$$

Here $\mathfrak{G}[\Psi, g]$ denotes the geometric sector, $\mathfrak{T}_{\mathrm{tri}}[\Psi]$ the triolic sector dynamics, $\mathfrak{H}_{\mathrm{HC}}[\Psi]$ the Heisenberg-compensator contribution, $\mathfrak{Q}_{\mathrm{cl}}[\Psi]$ the closure term determined by the projected

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closure residue, and $\mathfrak{B}_\beta[\Psi]$ the phase-shifted observer term. Equation (4) is therefore not an independent postulate, but the structural compression of the results obtained in the preceding sections.

For a semi-concrete realization, one may write

$$\mathbf{R}[\Psi] + \sum_{a=1}^{3} D_a \Psi_a + \sum_{a \neq b} \Lambda_{ab} \Psi_b + \lambda_{\mathrm{HC}}(I - C)\Psi + \lambda_{\mathrm{cl}} \mathcal{R}_{\mathrm{cl}}[\Psi] + \lambda_\beta(\Psi - U_\beta \Psi) = 0. \tag{5}$$

Here $\mathbf{R}[\Psi]$ denotes the effective geometric contribution, D_a are the operators of the triolic decomposition, Λ_{ab} encode inter-sector couplings, C is the Heisenberg compensator, $\mathcal{R}_{\mathrm{cl}}[\Psi]$ is the projected closure residue, and U_β is the phase-shift operator of the observer frame. Equation (5) should be read only as a semi-concrete representative of the structural master equation (4).

Theorem 25.1 (Structural necessity of the ToE master equation)**.** *Assume that the theory is required to satisfy simultaneously:*

1. *a common underlying field structure Ψ;*
2. *a triolic decomposition*
$$\Psi = \Psi_1 \oplus \Psi_2 \oplus \Psi_3;$$
3. *a dual unitary symmetry between the first two sectors;*
4. *a Heisenberg compensator selecting the collapse/light output;*
5. *a probe–closure system with projected closure residue;*
6. *a phase-shifted observer frame;*
7. *an Einstein limit on large-scale closure-stable configurations.*

Then the dynamics cannot be described by a purely geometric, purely sectorial, or purely observer-based equation alone. Instead, it must take the form of a single structural field equation

$$\mathfrak{G}[\Psi, g] + \mathfrak{T}_{\mathrm{tri}}[\Psi] + \mathfrak{H}_{\mathrm{HC}}[\Psi] + \mathfrak{Q}_{\mathrm{cl}}[\Psi] + \mathfrak{B}_\beta[\Psi] = 0,$$

containing at least these five structural blocks.

Proof. Each structural requirement excludes an incomplete description.

A purely geometric equation does not encode the triolic decomposition, the Heisenberg compensator, or the phase-shifted observer frame. A purely triolic-sector equation does not account for the emergent geometric reading of the closure-stable background. A purely observer-based equation does not select physically admissible configurations. The projected closure residue shows that closure must enter dynamically, and the probe–closure system shows that this closure is not external to the field content itself. Finally, the Einstein limit demonstrates that the geometric sector is not optional, but a distinguished large-scale projection of the same structure.

Hence none of the five structural contributions can be removed without violating one of the established premises. Therefore the dynamics must be governed by a single combined equation containing at least these five structural blocks. □

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Theorem 25.2 (Uniqueness up to structural equivalence)**.** *Let* Ψ *and* $\widetilde{\Psi}$ *be two nontrivial solutions realizing the structural core and satisfying the master equation* (4)*. Then they belong to the same structural class up to sector-preserving unitary equivalence, compensator projection, and observer-frame phase shift.*

Proof. Both solutions are constrained by the same triolic decomposition, the same compensator logic, the same projected closure requirement, and the same observer-frame mechanism. Their admissible differences can therefore only lie in the choice of representative within one common structural organization: sector exchange is absorbed by unitary symmetry, collapse/-light output is fixed up to compensator projection, and effective readout is changed only by the phase-shifted observer frame. Once these structurally allowed transformations are factored out, no independent second theory remains. Thus the admissible solutions are unique up to structural equivalence. □

Proposition 25.3 (From local triolic structure to global field dynamics)**.** *The transition from local triolic structure to global field dynamics is forced by the following chain:*

> *local triolic decomposition* $\longrightarrow$ *duality and compensator action* $\longrightarrow$ *projected closure*
> *closure-stable background geometry* $\longrightarrow$ *global master equation.*

In particular, Dirac-, QFT-, and Einstein-type regimes arise as structural projections of one common global dynamics rather than as separate starting points.

Proof. Locally, the theory resolves into three sectors linked by duality and compensator projection. The projected closure residue then decides whether these local sectors assemble into a physically admissible configuration. Whenever closure is achieved, the resulting sector can be read as a stable background supporting geometric reconstruction, effective matter dynamics, and observer-frame projection. The global master equation is therefore not added from outside, but is the minimal common dynamics of all local closure-stable triolic realizations. Its distinguished restrictions yield the Dirac-, QFT-, and Einstein-type regimes established in the preceding sections. □

Corollary 25.4 (Triolic realization of admissible solutions)**.** *Let* Ψ *satisfy the structural master equation* (4)*. Then every physically admissible solution decomposes into three necessary sectors,*

$$\Psi = \Psi_1 \oplus \Psi_2 \oplus \Psi_3,$$

where Ψ_1 *is the matter sector,* Ψ_2 *its dual tachyonic sector, and* Ψ_3 *the collapse/light sector selected by the Heisenberg compensator.*

Proof. The triolic term enforces the three-sector structure, the compensator term links the first two sectors and selects the collapse/light output, and the closure term excludes non-admissible configurations by the projected closure residue. Thus the three sectors are not optional additions, but the necessary realization of admissible solutions. □

Corollary 25.5 (Structural projections of the master equation)**.** *The master equation* (4) *admits the following distinguished structural projections:*

1. *in the probe regime, it induces the effective Dirac-type equation of the probe–closure system;*
2. *in the phase-shifted observer regime, it yields an effective QFT-type description after compensator decomposition;*

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3. in the large-scale closure-stable regime, it reduces to the sharpened Einstein limit.

Proof. Each projection is obtained by restricting one and the same structural equation to a distinguished regime.

In the probe regime, the auxiliary probe sector couples to the background through the same closure and compensator structure, yielding the effective Dirac-type operator. In the observer regime, the phase-shifted frame and the compensator decomposition produce the effective field description seen by a matter observer. In the large-scale regime, the closure residue and the sectorial fluctuations are suppressed, so that the geometric contribution dominates and yields the sharpened Einstein limit.

Hence probe dynamics, effective QFT, and gravitational geometry are not separate theories, but structural projections of one common closure-stable equation. □

The significance of the master equation is that it does not add a new layer to the theory, but reveals the common body already implicit in the triolic structure, the Heisenberg compensator, the probe–closure system, and the closure analysis. Matter, tachyon, photon, probe dynamics, effective QFT, and the Einstein limit now appear as sectorial realizations or limiting projections of one and the same closure-stable structural equation.

26 Structural Reading of Matter, Gauge, Mass, and Higgs

The structural picture may now be read in compact physical form. Matter is not an externally added ingredient, but the closure-stable bound sector of the triolic decomposition. The tachyonic sector is its dual partner under the unitary exchange symmetry, while the photon/light sector is the Heisenberg-compensator output selected by collapse and closure. In this sense, matter, tachyon, and photon are not separate ontological primitives, but sectorial realizations of one common structural body.

Gauge structure is read off from the admissible transitions that preserve closure and compensator consistency. What appears as gauge symmetry in the effective observer description is therefore the residual invariance of closure-compatible sector transport. In the same way, mass is not introduced as an independent primitive parameter, but as a closure/frame residue: the effective inertia seen by a matter observer records the nontrivial cost of maintaining a closure-stable bound sector under phase-shifted readout.

This motivates the following compressed structural reading.

Proposition 26.1 (Structural reading of matter, gauge, and mass)**.** *Within the closure-stable triolic system determined by* (4)*:*

1. *matter is the closure-stable bound realization of the first sector;*
2. *the photon/light sector is the compensator-selected collapse output;*
3. *effective gauge symmetry is the invariance of closure-compatible sector transport;*
4. *effective mass is the observer-frame residue of a nontrivial closure-stable bound configuration.*

Proof. The first two statements summarize the triolic decomposition together with the centrality of the Heisenberg compensator. Gauge structure was already shown to arise from the admissible symmetry content of the effective sector picture; within the present structural closure framework this means precisely that the allowed sector transitions must preserve closure

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and compensator consistency. Finally, the phase-shifted observer frame and the probe–closure analysis show that nontrivial bound sectors are not read out as pure closure identities but as effective residues. These residues are perceived as inertial or mass-like parameters by the observer frame. □

The Higgs question should then not be read as the introduction of an isolated extra object, but as the identification of the structural locus at which coupling, mediation, and effective mass readout are organized. At the present level of the theory, the safest statement is therefore not a final derivation, but a structural hypothesis.

Remark 26.2 (Structural role of the Higgs sector)**.** *The Higgs sector is expected to arise either as the effective coupling locus of the closure-stable triolic system or as the observer-frame image of a deeper mediating point within the septinion/meta-septum organization. In either reading, Higgs physics is not external to the structural core, but a special readout of the same closure, compensator, and frame-shift mechanism.*

27 Structural Normalization for the Next Companion Stage

Remark 27.1 (Version note)**.** *Version 4.7.0 introduces a structural normalization layer for the Companion. The Millennium matrix is read primarily as the global matrix realization of the Heisenberg compensator; the tool apparatus is split into general Millennium-matrix tools and special Heisenberg-compensator tools; the carrier/group/physical-readout distinction is made explicit; and the global reader orientation is organized through a master chain from triolic source to field-equation readout.*

Remark 27.2 (Null axis versus light axis)**.** *In structurally prior parts of the Companion we use the term* null axis *for the distinguished compensated axis selected by the MM/HC architecture. When the same axis is read at the physical propagation level, it may be called the* light axis*. Thus "light axis" is not a second object, but the physical readout of the structurally prior null axis.*

28 The Millennium Matrix as Global Heisenberg-Compensator

Definition 28.1 (Millennium matrix)**.** *The Millennium matrix $\mathcal{M}_{16}$ is the global structural matrix that transports one and the same Sphaera-object through its admissible algebraic, closure, and projection layers. It preserves identity through viewpoint change.*

Definition 28.2 (Layer decomposition)**.** *We write*

$$\mathcal{M}_{16} = \mathcal{M}_{\mathrm{tri}} + \mathcal{M}_{\mathrm{alg}} + \mathcal{M}_{\mathrm{cl}} + \mathcal{M}_{\mathrm{proj}},$$

where $\mathcal{M}_{\mathrm{tri}}$ encodes the triolic kernel organization, $\mathcal{M}_{\mathrm{alg}}$ admissible algebraic realization changes, $\mathcal{M}_{\mathrm{cl}}$ closure compatibility, and $\mathcal{M}_{\mathrm{proj}}$ admissible final readout.

Definition 28.3 (Global Heisenberg-compensator reading)**.** *The Millennium matrix is said to be read as a global Heisenberg-compensator matrix if its action extends the local rule*

$$C = \tfrac{1}{2}(\mathrm{id} + J)$$

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to the full multi-layer architecture of the Companion. Under this reading, the Millennium matrix is denoted by

$$\mathcal{H}_{16},$$

or identified with $\mathcal{M}_{16}$ when no ambiguity arises.

Theorem 28.4 (Structural role of the Millennium matrix)**.** *The Millennium matrix is the global structural intermediary of the Companion in the following sense:*

1. *it preserves the identity of the triolic core under admissible realization change;*
2. *it transports structural content through admissible algebraic layers without ontology change;*
3. *it enforces closure discipline before readout;*
4. *when read as $\mathcal{H}_{16}$, it becomes the global balancing mechanism for physically admissible interpretation.*

Proof. The triolic layer supplies the primitive object. The admissible algebraic layers supply realizations of that object. Closure selects the admissible sector, while projection only reads out what closure has stabilized. The Millennium matrix is the unique global carrier of this ordered transport. Under the Heisenberg-compensator reading, the same transport becomes the global balancing mechanism across all admissible layers. □

29 Carrier, Group, and Physical Readout Layers

Definition 29.1 (Carrier layer)**.** *The carrier layer is the total admissible structural support on which the Sphaera architecture is realized. It contains the triolic source structure, the admissible algebraic realizations, the closure discipline, and the transport induced by the Millennium matrix.*

Definition 29.2 (Group layer)**.** *The group layer is the symmetry layer of the Companion. It consists of the admissible group actions, generators, and representation changes acting on already given carrier-layer structure.*

Definition 29.3 (Physical readout layer)**.** *The physical readout layer is the final effective layer in which already stabilized structural and symmetry data are interpreted as physically meaningful sectors, observables, and field equations.*

Theorem 29.4 (Order of interpretation)**.** *The admissible order of interpretation is*

$$\textit{carrier layer} \longrightarrow \textit{group layer} \longrightarrow \textit{physical readout layer}.$$

Proof. Without a carrier there is nothing for a symmetry group to act on. Symmetries therefore organize admissible realizations of carrier-layer content. Only after structural transport and symmetry organization have been fixed may a physical interpretation be attached. □

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30 Master Structural Theorem of the Companion

Theorem 30.1 (Master structural theorem)**.** *The Companion is governed by one common structural architecture with the following ordered logic:*

$$\begin{aligned}&\textit{triolic source} \longrightarrow \textit{admissible carrier} \longrightarrow \mathcal{M}_{16} \longrightarrow \mathcal{H}_{16} \longrightarrow \textit{closure-stable sector}\\ &\longrightarrow \textit{group-layer symmetry readout} \longrightarrow \textit{null axis}\\ &\longrightarrow \textit{photon fix-sector} \longrightarrow \textit{residual localization}\\ &\longrightarrow \textit{matter / tachyon / photon sector readout}\\ &\longrightarrow \textit{Higgs mediation} \longrightarrow \textit{field-equation readout.}\end{aligned}$$

Proof. The triolic source supplies the primitive structural object. The carrier layer provides its admissible support, while the Millennium matrix transports this same object through admissible realization changes. Under the Heisenberg-compensator reading, the same matrix becomes the global balancing mechanism that stabilizes the closure-compatible sector. The group layer then supplies admissible symmetry actions on already given stabilized structure. The null axis is the structurally distinguished sector carried through the cascade, and its terminal compensated readout is the photon fix-sector. Relative to that fix-sector, closure-stable residual localization yields the matter branch, while its J-dual realization yields the tachyon branch. The Higgs sector mediates between the compensated photon reference and the persistence of residual localization. Finally, the effective field equations read out the dynamical content of this already stabilized architecture. □

Remark 30.2 (Compact roadmap for the reader)**.** *The reader may keep the following compact roadmap in mind:*

> *one source → one transported architecture*
> *→ one compensated closure logic*
> *→ three principal sector readouts → one mediated field dynamics*

30.1 Schematic Master Diagram

Definition 30.3 (Schematic master diagram)**.** *The structural logic of the Companion may be compressed into the following schematic master diagram:*

$$\mathfrak{T} \longrightarrow \mathfrak{C} \xrightarrow{\mathcal{M}_{16}} \mathfrak{C}^{\mathrm{adm}} \xrightarrow{\mathcal{H}_{16}} \mathfrak{C}^{\mathrm{cl}} \longrightarrow \mathfrak{G} \longrightarrow \mathfrak{L} \longrightarrow \mathfrak{P} \longrightarrow \mathfrak{R} \longrightarrow \mathfrak{H} \longrightarrow \mathfrak{F}.$$

Here:

$$\begin{aligned}\mathfrak{T} &:= \textit{triolic source},\\ \mathfrak{C} &:= \textit{carrier layer},\\ \mathfrak{C}^{\mathrm{adm}} &:= \textit{admissibly transported carrier},\\ \mathfrak{C}^{\mathrm{cl}} &:= \textit{closure-stable compensated carrier},\\ \mathfrak{G} &:= \textit{group-layer symmetry readout},\\ \mathfrak{L} &:= \textit{null-axis structure},\\ \mathfrak{P} &:= \textit{photon fix-sector},\\ \mathfrak{R} &:= \textit{residual localization sector},\\ \mathfrak{H} &:= \textit{Higgs-mediation sector},\\ \mathfrak{F} &:= \textit{field-equation readout.}\end{aligned}$$

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Remark 30.4 (Language regime along the master chain)**.** *The Companion follows a structure-first logic along the master chain up to the point at which a stable physical readout becomes available. Accordingly, the earlier stages*

$$\mathfrak{T} \to \mathfrak{C} \to \mathcal{M}_{16} \to \mathcal{H}_{16} \to \mathfrak{G} \to \mathfrak{L}$$

are described primarily in structurally prior language. By contrast, once the readout has stabilized at the level of

$$\mathfrak{P} \to \mathfrak{R} \to \mathfrak{H} \to \mathfrak{F},$$

physically explicit language is not only allowed but required. Thus the shift from structural terminology to physical terminology is intentional and marks a readout threshold, not a change of underlying object.

30.2 Matter, Tachyon, and Photon as Three Readouts of One Branch Architecture

Definition 30.5 (Matter branch)**.** *The matter branch is the closure-stable bound branch whose physical readout retains nontrivial localization relative to the photon fix-sector.*

Definition 30.6 (Tachyon branch)**.** *The tachyon branch is the J-dual readout of the matter branch:*

$$K_2 = J \cdot K_1.$$

It is the branch in which the same structural source is read in its frame-dual, non-bound, or overextended transport mode relative to the compensated photon sector.

Definition 30.7 (Photon branch)**.** *The photon branch is the canonical compensated fix-sector readout K_3, i.e. the terminal null-axis sector selected by the Heisenberg compensator.*

Theorem 30.8 (Three-readout principle)**.** *Matter, tachyon, and photon are three structurally linked readouts of one common branch architecture: matter is the bound closure-stable localization readout, tachyon is the J-dual unbound readout of the same branch source, and photon is the compensated fix-sector readout.*

Proof. Admissible sector transport does not create new ontology. The compensated sector is the canonical fix-sector. Hence the matter branch is determined by persistent closure-stable localization relative to this sector. The involution J generates the frame-dual branch of the same source, which yields the tachyonic readout. The compensator selects the stabilized null/fix-sector, which yields the photon readout. Therefore all three branches belong to one common structural architecture and differ only by their mode of realization and readout. □

30.3 Mass Readout Relative to the Null Axis and Photon Fix-Sector

Definition 30.9 (Photon fix-sector revisited)**.** *The photon sector is the canonical compensated readout of the global null axis:*

$$L = \bigcap_{n \geq 0} L_n = S_0.$$

It is the sector in which the frame-sensitive imbalance has been fully neutralized and the structural readout lies entirely in the HC-fix sector.

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Definition 30.10 (Massive branch)**.** *A branch is called massive if it remains closure-stably localized and does not collapse completely into the global null/fix sector. Equivalently, a massive branch is one whose physical readout retains a nontrivial localization residue relative to the photon sector.*

Theorem 30.11 (Massive versus massless readout)**.** *Within the MM/HC architecture, the distinction between massive and massless physical readout is the distinction between complete reduction to the compensated photon fix-sector and closure-stable persistence of localization relative to that fix-sector.*

Proof. If a branch reduces completely to the null/fix sector, no residual localization remains, and the physical readout is massless. If, however, a branch remains closure-stably localized and does not reduce completely to the null/fix sector, then a nonzero localization residue survives relative to the photon reference sector. This is precisely the massive readout. □

30.4 Higgs Readout as Mediation Between Fix-Sector and Residual Localization

Definition 30.12 (Higgs mediation principle)**.** *The Higgs readout is the effective mediation principle that governs how closure-stable localization is maintained, transferred, or expressed relative to the compensated photon fix-sector.*

Theorem 30.13 (Structural role of the Higgs sector)**.** *Within the MM/HC architecture, the Higgs sector is the distinguished coupling readout that mediates between the compensated photon/null fix-sector and the effective mass readout of closure-stable localized branches.*

Proof. By the previous null-axis analysis, the photon sector is the terminal compensated fix-sector. By the mass analysis, massive readout is residual localization relative to that fix-sector. Hence the remaining structural task is to identify the admissible interface through which this residual localization is expressed in the physical readout layer. The Higgs sector is precisely this interface. □

31 Field Equations as Physical Readout of the MM/HC Structure

Definition 31.1 (Field-readout principle)**.** *A field equation is admissible in the present framework if it is obtained as the effective physical readout of carrier-layer structure, group-layer symmetry organization, and MM/HC-mediated closure stabilization.*

Definition 31.2 (Minimal structural field content)**.** *The minimal physical readout of the closure-stable MM/HC architecture is carried by the tuple*

$$(\Psi, \mathcal{A}_\mu, \hat{P}_{C_7}, \mathcal{H}_C),$$

where Ψ is the effective matter/probe field, $\mathcal{A}_\mu$ the effective symmetry connection induced by the group layer, $\hat{P}_{C_7}$ the closure projector selecting the admissible stable sector, and $\mathcal{H}_C$ the effective Heisenberg-compensator field.

Theorem 31.3 (Minimal MM/HC field equation)**.** *The minimal local field equation compatible with the carrier layer, the group layer, and the MM/HC closure architecture is*

$$\big(i\Gamma^\mu \mathcal{D}_\mu - \Omega\big)\Psi = 0,$$

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with effective operator

$$\Omega = m_0\mathbf{1} + \alpha\,\hat{P}_{C_7}\mathcal{M}_{16}\hat{P}_{C_7} + \beta\,\mathcal{T} + \gamma\,\mathcal{H}_C.$$

Proof. The carrier layer fixes the underlying admissible object and its triolic organization. The group layer supplies the symmetry transport term encoded by $\mathcal{A}_\mu$. The MM/HC architecture contributes closure-stable internal transport via $\hat{P}_{C_7}\mathcal{M}_{16}\hat{P}_{C_7}$ and compensator balancing via $\mathcal{H}_C$. Finally, the triolic source remains visible at the level of effective coupling, which yields the term $\mathcal{T}$. No term may be removed without losing one previously established layer. □

Open Questions

Status. *Active live-burden/continuation block.*

Block function. This section separates what is still genuinely open from what survives only as historical labeling, so readers can distinguish live fronts from preserved older checkpoints.

The structural core, the master equation, and their principal projections are now explicit. The remaining work no longer concerns the existence of the framework itself, but its residual quantitative closure, phenomenological extraction, and spectrum-level completion. To avoid confusion, this section separates the *current live fronts* from the *historical OP labels* that are still used in later parts of the text.

Current reading. Several historically open OPs should no longer be read as fully open. In particular, the Koide/triolic block is already structurally closed at the present level, the Dirac-operator problem on H_-^+ is no longer open as an independent operator-existence problem, and the photon-frame Θ-remainder step of the cosmological block is already closed. The live questions are therefore the following.

Remark 31.4 (Reader precedence for historical release snapshots inside the active line)**.** *The next two synchronized-status blocks reconstruct older public-stage status only for navigation and burden genealogy. They do not overrule the current burden ledger, the current theorematic status, or the current tool/QAM/decoder routing of the active line. Whenever one of those historical snapshots sounds flatter or stronger than the normalized present reading, the normalized present reading governs.*

Remark 31.5 (historical release line synchronized status reading for the optical approximation layer)**.** *The historical public archival stage release line no longer permits a flat reading in which the new optical sector is either "already a full optical theory" or "still only a loose analogy." After the first reduced prism-readout lemma, the residual-diagnostic layer, the carrier-side fit/calibration/validation chain, the reconstruction/correction steps, and the bundled optical approximation package, the optical layer now has its own staged status:*

(i) *a* strong global optical claim *remains open;*

(ii) *a* reduced retained-carrier optical sector *is already available;*

(iii) *a* first-order optical workflow *is already available;*

(iv) *a* bundled first-order optical package *is already available as an active citation surface.*

Thus the open-front layer must now distinguish between the still-open stronger global optical theory and the already stabilized first-order optical approximation layer.

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Proposition 31.6 (historical release line optical/open-question status synchronization principle)**.** *At the present theorematic and retained-carrier level, the following synchronized status reading holds for the new optical line:*

(i) the still-open strong claim is a full global optical theory of the crystal–shadow architecture, including non-reduced anisotropic branches, stronger inversion, and a later convergence theory for higher optical refinements;

(ii) what is already available is a reduced prism-readout sector on one retained carrier surface with ordinary wavelength ordering;

(iii) what is already available as first-order workflow is the chain consisting of residual diagnostics, carrier-side fit ansatz, spectral calibration, overdetermined validation, effective-index reconstruction, carrier-profile correction, and reduced stability/stopping control;

(iv) what is already available as a single active surface is the bundled optical approximation package of Proposition 21.7;

(v) therefore the optical sector should no longer be counted among the fully open foundational fronts of the document, but as a partially stabilized first-order approximation layer with a still-open stronger global continuation.

Proof. Clause (i) records exactly the stronger optical ambitions that the current line has not yet proved: unrestricted anisotropic branching, stronger inversion, and a genuine global convergence theory are still absent. Clause (ii) is supplied by the reduced prism-readout lemma. Clause (iii) is the cumulative content of the residual spectral-diagnostic lemma, the carrier-side fit principle, the calibration principle, the overdetermined consistency principle, the effective-index reconstruction principle, the carrier-profile correction principle, and the reduced stability principle. Clause (iv) is exactly the content of the bundled optical approximation package. Hence the optical line already has a stabilized first-order approximation layer even though the stronger global optical theory remains open. □

Corollary 31.7 (Use rule for the synchronized historical release line optical status reading)**.** *Whenever a later argument in the present public archival stage release line needs to classify whether the optical sector is already available for active use, it is sufficient to cite Proposition 31.6. A finer reopening is required only when the argument depends on a stronger global optical theory beyond the reduced retained-carrier approximation layer.*

Proof. This is the operational reformulation of Proposition 31.6. The proposition already isolates the precise point at which the optical line ceases to count as merely a loose heuristic while still falling short of a full global optical theory. □

Remark 31.8 (Historical historical release line/historical release line synchronized status reading for the upgraded closure layer)**.** *The historical public archival stage release line no longer permits a flat reading in which the upgraded closure/interface blocks are simply either "open" or "proved." After the archival theorem-level shadow package, the archival interface-payoff layer, and the present historical release line partial-closure steps, their status is now genuinely four-tiered:*

(i) a strong deferred claim *remains open;*

(ii) a theorem-level shadow *is already available;*

(iii) an interface-level operational use *is already available;*

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(iv) for certain carrier-side conclusions, a restricted partial-closure / lift-back consequence *is already available.*

Thus the open-front layer must now distinguish not only between strong and weak forms, but also between theorematic shadow, interface use, and restricted closure consequence.

Proposition 31.9 (historical release line hypothesis/open-question status synchronization principle)**.** *At the present theorematic and interface level, the following synchronized status reading holds:*

(i) for Hypothesis 23.1, the still-open strong claim is the existence of one explicit compensator-centered generating equation $\mathfrak{E}_{\mathrm{HC}} = 0$*, while the theorem-level shadow of a common closure-stable carrier/projection architecture is already available, already usable on the terminal interface, and now participates in the restricted lift-back layer whenever only retained-carrier conclusions are drawn;*

(ii) for Hypothesis 23.6, the still-open strong claim is the existence of a deeper explicit compensator generator $\mathbf{H}_C$*, while the theorem-level shadow separating visible defect vanishing from carrier extinction is already available, already usable on the terminal interface, and now participates in the no-second-carrier / restricted lift-back layer;*

(iii) for Hypothesis 23.11, the still-open strong claim is the existence of a concrete coarse-graining map to fluid-type evolution, while the theorem-level shadow demoting fluid-like behavior to a later coarse-grained readout is already available, already usable on the terminal interface, and now participates negatively in the restricted lift-back layer by forbidding replacement of the retained carrier;

(iv) for Hypothesis 21.1, the still-open strong claim is a full universal geometric lensing law, while the theorem-level shadow of common-carrier/two-probe coupling is already available, already usable on the terminal interface, and now participates in the no-second-carrier / partial-closure layer;

(v) for the OP 3/OP 5 line, the still-open strong claim is a full global closure theorem, while the historical public archival stage release line already contained more than a weak bridge: it had a first partial closure theorem together with a restricted lift-back to retained carrier-layer structure.

Therefore these upgraded blocks should no longer be counted among the fully open foundational fronts *of the document. Only their strong deferred forms remain fully open; their weaker theorematic, interface-level, and partial-closure forms are already part of the active line.*

Proof. Each clause follows by combining the archival theorem-level shadows, the archival interface-level application principles, and where applicable the new historical release line partial-closure consequences. For the HC meta-field generator hypothesis, this combines the common-carrier/projection shadow and its interface-level use with the present restricted lift-back layer. For frame-dependent defect readout, this combines the defect-shadow theorem, the interface-level defect-face vanishing principle, and the present no-second-carrier/lift-back layer. For fluid-type readout, this combines the fluid-shadow theorem, the interface-level fluid demotion principle, and the present lift-back restriction that forbids carrier replacement. For bidirectional probe lensing, this combines the probe-lensing shadow theorem, the interface-level probe-coupling principle, and the present no-second-carrier/partial-closure layer. Finally, the OP 3/OP 5 line is upgraded from the historical release line bridge test to the present historical release line partial closure theorem and restricted lift-back theorem. Hence that historical stage already supported a synchronized four-tier status reading. □

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Corollary 31.10 (Use rule for the synchronized historical release line status reading)**.** *Whenever a later argument refers back to the historical public archival stage release line and needs to classify whether one of the upgraded closure/interface blocks was still truly open at that stage, it is sufficient to cite Proposition 31.9. A finer reopening is required only when the argument depends on the still-open strong form rather than on the already stabilized theorematic, interface-level, or restricted partial-closure form.*

Proof. This is the operational reformulation of Proposition 31.9. The proposition already isolates the precise point at which the upgraded blocks cease to count as fully open foundational fronts. □

Remark 31.11 (historical release line synchronized status reading for the upgraded hypothesis layer)**.** *The present 4.7.0 release line no longer permits a flat reading in which the upgraded hypothesis blocks are simply either "fully open" or "already proved." After the released historical release line theorem-level shadow package and the historical historical release line interface payoff layer, their status is now genuinely three-tiered:*

(i) a strong deferred claim *remains open;*

(ii) a theorem-level shadow *is already available;*

(iii) an interface-level operational use *is already available on the terminal surface.*

Thus the open-front layer must now distinguish between still-open strong forms and already stabilized weaker forms.

Proposition 31.12 (historical release line hypothesis/open-question status synchronization principle)**.** *At the present theorematic and interface level, the following synchronized status reading holds:*

(i) for Hypothesis 23.1, the still-open strong claim is the existence of one explicit compensator-centered generating equation $\mathfrak{E}_{\mathrm{HC}} = 0$*, while the theorem-level shadow of a common closure-stable carrier/projection architecture is already available and already usable on the terminal interface;*

(ii) for Hypothesis 23.6, the still-open strong claim is the existence of a deeper explicit compensator generator $\mathbf{H}_C$*, while the theorem-level shadow separating visible defect vanishing from carrier extinction is already available and already usable on the terminal interface;*

(iii) for Hypothesis 23.11, the still-open strong claim is the existence of a concrete coarse-graining map to fluid-type evolution, while the theorem-level shadow demoting fluid-like behavior to a later coarse-grained readout is already available and already usable on the terminal interface;

(iv) for Hypothesis 21.1, the still-open strong claim is a full universal geometric lensing law, while the theorem-level shadow of common-carrier/two-probe coupling is already available and already usable on the terminal interface.

Therefore these four blocks should no longer be counted among the fully open foundational fronts *of the document. Only their strong deferred forms remain open; their weaker theorematic and interface-operational forms are already part of the active line.*

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Proof. Each clause follows by combining one theorem-level shadow result with its corresponding interface-level application principle. For Hypothesis 23.1, this is the theorem-level shadow of the HC meta-field generator together with the interface-level common-carrier/projection principle. For Hypothesis 23.6, this is the theorem-level shadow of frame-dependent defect readout together with the interface-level defect-face vanishing principle. For Hypothesis 23.11, this is the theorem-level shadow of fluid-type readout together with the interface-level fluid-readout demotion principle. For Hypothesis 21.1, this is the theorem-level shadow of probe-lensing together with the interface-level probe-coupling principle. Hence the four upgraded blocks now share one synchronized three-tier status reading: strong deferred form, theorem-level shadow, and interface-level operational use. □

Corollary 31.13 (Use rule for the synchronized historical release line status reading)**.** *Whenever a later argument in the present 4.7.0 release line needs to classify whether one of the four upgraded hypothesis blocks is still truly open, it is sufficient to cite Proposition 31.12. A finer reopening is required only when the argument depends on the still-open strong form rather than on the already stabilized theorematic or interface-level shadow.*

Proof. This is the operational reformulation of Proposition 31.12. The proposition already isolates the precise point at which the upgraded blocks cease to count as fully open foundational fronts. □

1. **Residual quantitative completion of the frame scale (historical OP 1; also absorbing OP 2(T4) and the former OP 4 coupling).** The continuum fixed-point structure is already identified at the Hopf/Tribonacci level, and the frame-offset problem is no longer a broad undifferentiated open front. What remains open is the explicit derivation of the prime-lattice / continuum-limit correction δ_{CL} (or an equivalent arithmetic correction) that transports the continuum final to the measured matter-frame value.

2. **Phenomenology: controlled emergence.** The probe–closure field system is now written in explicit variational form and closed by the residue constraint $\mathfrak{R}_C = 0$. Structurally, the preceding sections identify Dirac-, QFT-, and Einstein-type regimes as distinguished projections of the same closure-stable equation. What remains open is the fully controlled derivation of these projections with explicit constants, approximation bounds, and quantitative phenomenology.

3. **Spectrum: mass hierarchy.** The present structural reading is already strong enough to separate the qualitative mass-sector logic from the unresolved quantitative assignment. The remaining open task is the map from prime/spectral index to Standard-Model quantum numbers and measured mass scales.

4. **Spectrum: three generations.** $S^2 \supset S^1 \supset S^0$ as three fermion generations is structurally supported, but the quantitative generation-splitting law is not yet derived.

5. **Structural completion: Dixon algebra.** A compact reformulation in $\mathbb{C}\otimes\mathbb{H}\otimes\mathbb{O}$ (64 dimensions) may unify the present results into one algebraic object, but this remains a later consolidation step rather than a current prerequisite.

6. **Λ-cancellation: reduced photon-frame residual problem (historical OP 5, Step (B) only).** The photon-frame formulation and the structural separation $\Lambda_{\mathrm{fundamental}} = 0$ versus quantitative balance are now fixed, and Step (A) of the Θ–R program is already closed in the present text. The remaining open content is Step (B): the exact Θ-sum / Θ–R balance needed to cancel the constant term at the photon-frame cutoff.

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Historical OP synchronization. For the remainder of the document, the historical labels should be read with these statuses in mind:

- **Historical OP 3 (Koide / θ_K block):** closed at the present structural level; later work may still densify exposition, but it should no longer be listed as a primary open problem.
- **Historical OP 2 (Dirac operator on H_-^+):** closed as an independent operator problem in its existence, self-adjointness, and GOE-class parts; only the former frame-offset part survives, and it is absorbed into Item 1 above.
- **Historical OP 5:** reduced to Step (B) only; Step (A) is no longer open.

Phase IV-I: Physical Unfolding, Observer Frames, and Interface Regimes

Lemma 31.14 (Observer-frame lemma)**.** *Within the normalized structural architecture, an observer frame is not introduced as an external background datum. Rather, it is the stabilized readout frame selected by the combined action of the cascade, the closure filter, and the admissibility-preserving transport across the grouped layers. In particular, any observer description compatible with the kernel must arise as a frame-shifted but ground-linked reading of the same underlying body.*

Proof. The semantic kernel identifies all admissible descriptions as readings of a single underlying structure. Frame-shifted observation therefore does not create an ontologically new sector. By the forcing chain established in Phases I–III, any admissible readout is first re-identified by the compensator, then filtered into a closure-stable class, then transported through the grouped architecture without ontology drift. The observer frame is therefore fixed only at the level of admissible readout, not posited independently of the structure. □

Lemma 31.15 (Mass/spectral reading lemma)**.** *Any admissible mass or hierarchy reading in the Companion must be interpreted first as a spectral or sectoral readout of the normalized structural kernel, and only secondarily as a phenomenological mass assignment. In particular, the transition from spectral organization to mass hierarchy is a realization map, not a new generative axiom.*

Proof. The generative layer produces admissible sectors and their closure relations, while the projection/translation layer supplies readout rules and collapse-fixed identifications. Therefore any mass reading that appears in the field/probe layer inherits its legitimacy from a prior spectral or sectoral differentiation. This excludes the reverse move of inserting an independent mass ontology into the generative layer. The mass interpretation is thus subordinate to the spectral structure and must be treated as a downstream realization. □

Lemma 31.16 (Generation-splitting lemma)**.** *If the triolic and grouped-layer architecture is preserved across admissible projections and field realizations, then any generation splitting must appear as a structured decomposition of an already admissible spectral body rather than as a free multiplication of unrelated sectors.*

Proof. Triolic minimality and orthogonal anchoring prevent arbitrary proliferation of sectors at the generative level. The transport rules for $\mathcal{M}$ preserve admissibility and closure, and the no-ontology-drift lemma excludes the creation of new bodies during layer transfer. Hence any multiplicity such as a generation-like splitting must be read as an organized decomposition of the inherited spectral structure. It is therefore constrained by the prior closure and transport logic. □

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Theorem 31.17 (Standard-model interface theorem)**.** *The normalized Companion architecture defines a one-way interface to Standard-Model-style phenomenology: admissible structures may be projected into observer, spectral, and field realizations compatible with particle-physics language, but the Companion introduces no independent Standard-Model ontology at the generative level. Consequently, the appropriate first target of the unfolding is not a complete reconstruction of the Standard Model, but a structurally controlled interface map from the grouped kernel into physically readable sectors.*

Proof. By the observer-frame, mass/spectral, and generation-splitting lemmas, all physically readable sectors arise only after the generative core has been passed through closure, confinement, cascade readout, tensor realization, and grouped transport. The field/probe layer therefore supplies an interface regime in which known physical language may be attached to admissible sectors, but the kernel itself remains prior to that language. The Standard Model can thus enter only through an interface map respecting admissibility, closure, and no-ontology drift. This proves that the correct unfolding target is an interface theorem rather than a premature full identification. □

Corollary 31.18 (First physical unfolding chain)**.** *The first physical unfolding chain of the Companion is*

$$\begin{aligned} &\textit{observer-frame emergence} \Longrightarrow \textit{spectral/mass readout} \\ &\textit{generation-compatible splitting} \Longrightarrow \textit{controlled Standard-Model interface.} \end{aligned}$$

Thus the physical interpretation layer is admitted only after the structural, closure, tensorial, and grouped-layer chains have already been stabilized.

Phase IV-I′: Refined Physical Interface and Realization Control

Lemma 31.19 (Observer covariance under admissible transport)**.** *If two observer descriptions are related by admissibility-preserving transport through the grouped architecture, then they represent frame-shifted realizations of the same underlying body and cannot define distinct generative ontologies.*

Proof. The no-ontology-drift result of Phase II-F implies that transport across $\mathfrak{G}$, $\mathfrak{P}$, and $\mathfrak{F}$ preserves the underlying structural identity. By the observer-frame lemma, any observer description is already a frame-shifted readout of the same admissible kernel. Therefore two observer descriptions related by admissible transport differ only at the level of readout covariance and not at the level of generative content. □

Proposition 31.20 (Spectral-to-mass realization map)**.** *There exists a structurally controlled realization map*

$$\mathcal{R}_m : \mathcal{S}_{\mathrm{spec}} \longrightarrow \mathcal{S}_{\mathrm{mass}}$$

from admissible spectral sectors to admissible mass-readout sectors, where $\mathcal{R}_m$ is downstream of closure, cascade readout, and grouped transport.

Proof. By the mass/spectral reading lemma, any mass assignment must be subordinated to a prior spectral organization. The admissible spectral sectors are produced only after the closure-stable body has been transported through the projection/translation layer and read out through the cascade. Hence the passage to a mass language is not primitive but realized by a downstream map from already admissible spectral sectors. Denote this realization map by $\mathcal{R}_m$. Its domain is therefore restricted to $\mathcal{S}_{\mathrm{spec}}$ and its codomain inherits admissibility from the upstream structural chain. □

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Proposition 31.21 (Generation-compatible decomposition)**.** *If a generation-like multiplicity appears in the physical layer, then it must be expressible as an admissible decomposition of the image of $\mathcal{R}_m$ rather than as an unconstrained replication of primitive sectors.*

Proof. The generation-splitting lemma already excludes free multiplication of unrelated sectors. Since the physical layer is reached only after grouped transport and spectral-to-mass realization, any multiplicity that survives into the interface regime must be inherited from the already admissible image of $\mathcal{R}_m$. Thus a generation-like structure can only be accepted when it is realizable as a constrained decomposition of an upstream admissible spectral body. □

Theorem 31.22 (Controlled physical interface functor)**.** *The first physically readable regime of the Companion is determined by a controlled interface functor*

$$\mathfrak{I}_{\mathrm{phys}} : (\mathfrak{G}, \mathfrak{P}, \mathfrak{F}) \longrightarrow \mathcal{O}_{\mathrm{phys}},$$

which assigns observer, spectral, mass, and generation-level readouts to admissible structural sectors while preserving closure compatibility and ontology non-drift.

Proof. The observer covariance lemma identifies physically readable frames as transport-compatible readouts of the same body. The spectral-to-mass realization proposition supplies a downstream realization map from spectral sectors to mass sectors, while the generation-compatible decomposition proposition constrains multiplicity to inherited admissible decompositions. Collecting these assignments yields a functorial interface from the grouped structural architecture to a space of physically readable observables $\mathcal{O}_{\mathrm{phys}}$. Since every step is subordinated to admissibility, closure, and no-ontology drift, the resulting interface is controlled rather than ontologically generative. □

Corollary 31.23 (Refined physical interface chain)**.** *The refined physical interface chain is*

$$\textit{observer covariance} \Longrightarrow \textit{spectral-to-mass realization}$$
$$\textit{generation-compatible decomposition} \Longrightarrow \textit{controlled interface functor.}$$

Hence the physical language of the Companion is introduced only as a regulated readout of the normalized structural architecture.

Phase IV-J: Spectral Readout, Trace Reconstruction, and Observable Selection

Lemma 31.24 (Spectral admissibility inheritance)**.** *If a physically readable sector is admitted by the controlled interface functor, then its spectral representative must inherit admissibility from the upstream closure-stable and transport-compatible structural body. In particular, no spectral mode may enter the readout regime unless it descends from an already admissible structural sector.*

Proof. By the semantic–tensor interface and the grouped transport chain, admissibility is preserved from the generative layer through cascade readout and into the physically readable interface regime. The controlled physical interface functor therefore cannot attach an observable meaning to a mode that was not already admitted upstream. Hence spectral readability inherits admissibility rather than creating it locally. □

Proposition 31.25 (Trace-compatible spectral decomposition)**.** *Every admissible spectral body in the readout regime admits a trace-compatible decomposition into modes whose contribution respects the closure filter, the grouped transport constraints, and the observer-covariant interface conditions.*

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Proof. The historical spectral and trace cores already isolate the need for admissible mode selection and trace-compatible decomposition. In the normalized architecture, these requirements are now downstream of closure, cascade, and grouped transport. Therefore any accepted spectral decomposition must preserve the admissible body identified by the upstream chain and cannot separate into trace-incompatible or closure-breaking components. This yields a trace-compatible mode decomposition of every admissible spectral body. □

Proposition 31.26 (Observable extraction map)**.** *There exists a structurally controlled observable extraction map*

$$\mathcal{R}_{\mathrm{obs}} : \mathcal{S}_{\mathrm{spec}}^{\mathrm{adm}} \longrightarrow \mathcal{O}_{\mathrm{phys}}^{\mathrm{adm}},$$

which assigns observable readouts only to trace-compatible admissible spectral sectors.

Proof. Phase IV-I′ introduced a controlled interface functor from the grouped architecture into physically readable observables. Restricting that interface to the spectrally admissible, trace-compatible body provided by Proposition 31.25 yields a downstream extraction map from admissible spectral sectors to admissible observables. Since its domain is already filtered by closure, transport, and trace compatibility, the map does not generate new observables freely but only extracts structurally sanctioned ones. □

Theorem 31.27 (Mode-selection and observable consistency theorem)**.** *Mode selection in the Companion is admissible only when it is simultaneously spectral, trace-compatible, closure-preserving, and observer-covariant. Under these conditions, the observable extraction map produces a consistent readout regime in which selected modes can be interpreted without violating ontology non-drift.*

Proof. Spectral admissibility inheritance excludes modes that do not descend from the admissible structural chain. Trace-compatible spectral decomposition excludes decompositions that would destroy the integrity of the admissible spectral body. The controlled interface functor and observer covariance from Phase IV-I′ ensure that the resulting readout remains a covariance of one and the same body rather than a proliferation of ontologies. Therefore mode selection is consistent exactly when all four constraints are met, and in that regime observable extraction remains structurally controlled. □

Corollary 31.28 (Spectral readout reconstruction chain)**.** *The next spectral unfolding chain is*

$$\textit{spectral admissibility inheritance} \Longrightarrow \textit{trace-compatible decomposition}$$
$$\Longrightarrow \textit{observable extraction} \Longrightarrow \textit{consistent mode-selected readout.}$$

Thus the spectral and trace layer of the Companion is admitted only after closure, grouped transport, and controlled physical interfacing have already been stabilized.

Phase IV-K: Resolvent Gatekeeping and Operator-Controlled Spectral Access

Lemma 31.29 (Resolvent admissibility lemma)**.** *Let $\mathcal{R}(\lambda) = (\lambda \mathbf{1} - \mathcal{A})^{-1}$ denote a formal resolvent on the admissible spectral layer. If the underlying mode family is closure-stable and trace-compatible, then resolvent access is admissible only on those channels for which the induced transport preserves the grouped-layer constraints and does not reintroduce ontology drift. In particular, admissible resolvent access is a restricted rather than global spectral operation.*

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Proof. The semantic kernel already constrains admissible transport by closure preservation, grouped transport, and controlled readout. A resolvent therefore cannot be treated as an arbitrary inversion device: it is allowed only where the corresponding operator action remains inside the admissible closure class and respects the readout filter established in the preceding phases. Hence admissibility is inherited by resolvent access only on the structurally licensed channels. □

Proposition 31.30 (Gatekeeping proposition for admissible modes)**.** *The resolvent layer acts as a gatekeeping mechanism: admissible modes are precisely those spectral branches for which resolvent access is compatible with closure, trace reconstruction, and grouped-layer transport. Non-admissible branches are excluded before observable extraction.*

Proof. By Lemma 31.29, resolvent access is already restricted to structure-preserving channels. Combined with the trace-compatible mode selection of Phase IV-J, this yields a gatekeeping principle: only those modes that remain compatible with closure and transport can pass into the observable sector. The exclusion of non-admissible branches is therefore structural rather than ad hoc. □

Theorem 31.31 (Operator-controlled trace access theorem)**.** *Assume that the spectral sector is admissible, the selected modes are resolvent-gated, and the trace reconstruction remains compatible with the semantic–tensor interface. Then trace access is operator-controlled: every observable trace contribution arises through an admissible operator channel and not through uncontrolled spectral leakage.*

Proof. The preceding proposition ensures that only admissible modes are available for readout. The semantic–tensor interface then identifies the operator realization of the same admissible content, while closure preservation prevents leakage into non-licensed channels. Consequently, the trace contributions that survive observable extraction are exactly those transmitted through operator-controlled admissible channels. □

Corollary 31.32 (Stable observable channel corollary)**.** *Under admissible resolvent gatekeeping, the observable readout sector forms a stable channel: spectral access, trace reconstruction, and mode selection remain synchronized under the same structural admissibility constraints.*

Proof. This follows directly from Theorem 31.31: once admissible access, admissible mode selection, and admissible trace transport coincide, the resulting observable sector is stable under the same structural constraints. □

Remark 31.33 (Resolvent-to-readout chain)**.** *The operator-controlled spectral access chain may therefore be summarized as*

$$\begin{aligned} \textit{resolvent admissibility} &\Longrightarrow \textit{gatekept mode access} \\ &\Longrightarrow \textit{operator-controlled trace access} \\ &\Longrightarrow \textit{stable observable channel.} \end{aligned}$$

This phase extends the preceding spectral readout layer by showing that observable extraction is not only spectrally filtered but also operatorically regulated.

Phase IV-L: Semigroup Stability and Lyapunov-Controlled Readout Persistence

Lemma 31.34 (Semigroup admissibility lemma)**.** *Let $(T_t)_{t\geq 0}$ denote a formal semigroup acting on the admissible readout sector. If the initial spectral channel is closure-stable, resolvent-gated, and trace-compatible, then admissible semigroup evolution is restricted to those trajectories for which grouped transport, closure preservation, and observable compatibility remain jointly valid for all admissible times.*

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Proof. The preceding phases identify the observable channel only after admissibility, gatekeeping, and operator-controlled trace access have been enforced. A semigroup acting on this sector may therefore propagate states only if the induced time evolution respects the same admissibility filter. Otherwise the evolution would exit the closure-preserving and trace-compatible regime and would no longer represent a structurally licensed readout trajectory. Hence admissible semigroup evolution is constrained to the channels that preserve the previously established structural conditions for all admissible times. □

Proposition 31.35 (Lyapunov gate proposition)**.** *Assume that an admissible Lyapunov functional $\mathcal{L}$ exists on the gated readout sector. If $\mathcal{L}$ is nonincreasing along the admissible semigroup trajectories, then unstable leakage into non-admissible channels is structurally suppressed, and the gated observable channel remains dynamically protected.*

Proof. A nonincreasing Lyapunov functional excludes the spontaneous amplification of perturbations that would carry the evolution outside the admissible channel. Since admissibility already forbids ontology drift, closure violation, and non-gated spectral leakage, Lyapunov monotonicity upgrades these static constraints to a dynamical protection principle. The readout channel is therefore not merely selected once but remains protected under admissible evolution. □

Theorem 31.36 (Persistent observable channel theorem)**.** *Suppose that the readout sector is resolvent-gated, operator-controlled, and semigroup-admissible, and that a Lyapunov functional controls admissible perturbations. Then the observable channel is persistent: admissible mode selection, trace reconstruction, and grouped transport remain synchronized along the admissible evolution, and the readout sector stays dynamically stable against structural leakage.*

Proof. By Lemma 31.34, admissible semigroup evolution preserves the structurally licensed readout sector. Proposition 31.35 then prevents the admissible trajectories from developing unstable excursions into non-gated channels. Combined with the operator-controlled access theorem and the stable observable channel corollary from Phase IV-K, this yields persistence of the observable channel: the same admissibility constraints that select the readout sector also stabilize it under admissible evolution. □

Corollary 31.37 (Long-time readout stability corollary)**.** *Under admissible semigroup evolution and Lyapunov control, the observable readout channel remains stable at long admissible times; the selected spectral and trace-compatible sector is therefore not merely accessible but persistently readable.*

Proof. This follows directly from Theorem 31.36: persistence of the admissible channel implies long-time stability of the same gated and closure-compatible readout sector. □

Remark 31.38 (Semigroup-to-stability chain)**.** *The late-stage stability chain may therefore be summarized as*

$$\begin{aligned} \textit{semigroup admissibility} &\Longrightarrow \textit{Lyapunov-controlled gated evolution} \\ &\Longrightarrow \textit{persistent observable channel} \\ &\Longrightarrow \textit{long-time readout stability.} \end{aligned}$$

This phase completes the operatoric readout picture by showing that admissible channels are not only spectrally and resolvent-gated, but also dynamically stabilized over admissible evolution scales.

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Phase IV-M: Channel Fusion and Multi-Readout Consistency

Lemma 31.39 (Admissible channel fusion lemma)**.** *Suppose that two or more observable channels are individually admissible, resolvent-gated, and semigroup-stable. Then a fused readout may be admitted only if the fusion procedure preserves the common closure filter, the grouped transport discipline, and the no-ontology-drift condition. In particular, fusion is admissible only as a compatibility-preserving synthesis of already licensed channels.*

Proof. The previous phases guarantee admissibility, operatoric access, and dynamical persistence for each channel separately. A fusion mechanism cannot be treated as an unrestricted superposition principle, since otherwise it could reintroduce hidden drift between channels that were only separately controlled. Therefore fusion is admissible only when the same closure, transport, and ontology constraints remain valid for the synthesized channel. In that case fusion does not create a new structural body, but re-expresses a common admissible body through multiple synchronized readouts. □

Proposition 31.40 (Cross-channel consistency proposition)**.** *If admissible channels are fused through a compatibility-preserving synthesis, then their spectral, trace-based, and long-time readout contents must agree on the shared admissible sector. Any discrepancy beyond the admitted transport freedom signals a non-admissible fusion rather than a genuine multi-readout effect.*

Proof. Each admissible channel already carries its observable content through the same semantic–tensor interface, the same resolvent gatekeeping, and the same semigroup-stable evolution class. Hence a compatibility-preserving fusion may only combine channels whose readout contents coincide on the overlap of their admissible sectors. If an apparent fusion produces incompatible observable contents beyond the allowed transport freedom, then the synthesis has violated the common admissibility constraints and is therefore not structurally licensed. □

Theorem 31.41 (Multi-readout consistency theorem)**.** *Under admissible channel fusion, the Companion admits a controlled multi-readout regime: distinct readout channels may coexist, but only as synchronized presentations of one and the same admissible structural body. Consequently, multi-readout consistency is equivalent to fusion without ontology drift.*

Proof. By Lemma 31.39, channel fusion is allowed only as a synthesis that preserves closure, grouped transport, and non-drift. Proposition 31.40 then forces agreement of spectral, trace, and persistent readout content on the common admissible sector. Therefore multiple channels can coexist only as mutually compatible renderings of the same admissible body. The controlled multi-readout regime is thus precisely the regime in which fusion introduces no ontology drift. □

Corollary 31.42 (Fusion-to-consistency chain)**.** *The late-stage fusion chain may therefore be summarized as*

$$\begin{aligned} \textit{admissible channel fusion} &\Longrightarrow \textit{cross-channel consistency} \\ &\Longrightarrow \textit{controlled multi-readout regime} \\ &\Longrightarrow \textit{fusion without ontology drift.} \end{aligned}$$

This phase closes the late readout unfolding by showing that stable observable channels may be combined only through a synthesis that preserves the unique admissible structural body across all readout presentations.

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Phase IV-N: Unified Observable Readout Synthesis

Proposition 31.43 (Unified observable readout synthesis)**.** *Let the late readout regime be governed by the spectral admissibility, resolvent gatekeeping, semigroup persistence, and admissible channel-fusion principles established in Phases IV-J through IV-M. Then the resulting observable sector admits a unified synthesis map in which spectral access, operator control, long-time persistence, and multi-readout consistency are read as mutually compatible aspects of one and the same admissible observable architecture.*

Proof. Phase IV-J supplies the spectral and trace-compatible decomposition of admissible modes. Phase IV-K restricts this decomposition to resolvent-accessible channels only, so that observable extraction is not performed on arbitrary modes but only on structurally permitted ones. Phase IV-L then upgrades admissible access to persistence under semigroup evolution and Lyapunov control, which turns transient access into stable readout inheritance over time. Finally, Phase IV-M shows that multiple such persistent channels can be fused without inducing cross-channel inconsistency or ontological drift. Therefore the late readout regime is not a mere juxtaposition of separate spectral, operator, dynamical, and fusion statements; it forms a single admissible synthesis in which observable extraction is structurally unified. □

Corollary 31.44 (Late readout master chain)**.** *The late readout architecture admits the compressed chain*

$$\begin{aligned} &\textit{spectral admissibility} \Longrightarrow \textit{resolvent-controlled access} \\ &\Longrightarrow \textit{semigroup/Lyapunov persistence} \Longrightarrow \textit{multi-readout consistency} \\ &\Longrightarrow \textit{unified observable synthesis.} \end{aligned}$$

Remark 31.45 (Interpretive role of the late synthesis)**.** *This synthesis does not introduce a new ontology behind the observable layer. Rather, it states that the different late readout mechanisms are to be interpreted as coordinated access constraints and persistence conditions on one admissible observable sector generated by the normalized structural core.*

Phase IV-O: Return Bridge to the Generative Core and the Structural Master Equation

Proposition 31.46 (Return bridge from unified readout to the generative core)**.** *The unified observable readout synthesized in Phase IV-N introduces no new generative assumption. Rather, it is a late readout layer of the same generative structure determined by the one-body condition, triolic stabilization, orthogonal anchoring, frame-shifted observation, and the HC–closure–confinement chain.*

Proof. The earlier unfolding phases establish that admissible readout channels arise only after re-identification, compensator forcing, closure stabilization, and higher confinement. Phases IV-J through IV-N then refine how admissible observable content is accessed, filtered, stabilized, fused, and synthesized. None of those later steps introduces a new generating source; they remain constrained by the same admissibility conditions inherited from the generative layer. Hence the late observable synthesis is a return readout of the original structure rather than an independent origin. □

Theorem 31.47 (Master-equation readout closure)**.** *Let the structural master equation be the compressed dynamical expression of the admissible generative core. Then the late readout*

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chain

$$\begin{aligned}&\textit{spectral admissibility} \Longrightarrow \textit{resolvent-controlled access}\\ &\Longrightarrow \textit{semigroup/Lyapunov persistence} \Longrightarrow \textit{multi-readout consistency}\\ &\Longrightarrow \textit{unified observable synthesis}\end{aligned}$$

closes back into the same master-equation-governed structure, in the sense that every admissible late observable synthesis remains a governed readout of the original master-equation sector.

Proof. By the semantic/tensor interface and grouped transport phases, admissibility is preserved under the Millennium transport between the generative, projection, and realization layers. The readout phases IV-J through IV-N add only progressively refined access conditions: spectral admissibility, resolvent gatekeeping, semigroup persistence, and fusion consistency. Since each step is formulated as structure-preserving and closure-compatible, the final observable synthesis remains within the same admissible sector selected by the master equation. Therefore the readout chain closes into the same governing structure and should be interpreted as its late observable face. □

Remark 31.48 (Interpretive closure of the unfolding program)**.** *The point of the return bridge is to prevent the late readout machinery from being mistaken for a detached auxiliary theory or a second source. The late phases do not compete with the generative core; they complete its observable unfolding. In this sense the overall program closes as*

$$\begin{aligned}&\textit{generative core} \Longrightarrow \textit{HC/closure/confinement} \Longrightarrow \textit{master equation}\\ &\Longrightarrow \textit{spectral-resolvent-semigroup-fusion synthesis}\\ &\Longrightarrow \textit{return readout of the same structure.}\end{aligned}$$

Phase IV-P: Observable Back-Projection and Kernel Reconstruction

Lemma 31.49 (Admissible back-projection lemma)**.** *If a late observable synthesis is admissible in the sense of Phases IV-J through IV-O, then it admits a back-projection into the grouped architecture such that the reconstructed object remains compatible with the same closure, transport, and non-drift constraints that governed the forward readout.*

Proof. By Phase IV-N, the unified observable synthesis is not an arbitrary superposition of readout data, but a compatibility-preserving synthesis of spectral, resolvent, semigroup, and fusion constraints. Phase IV-O then shows that this synthesis remains a governed readout of the same master-equation sector. Hence any admissible late synthesis can be projected back through the grouped architecture without leaving the admissible class: what is reconstructed is not a new source, but the same structural body viewed through the inverse admissibility constraints of the readout chain. □

Proposition 31.50 (Kernel reconstruction proposition)**.** *Admissible back-projection reconstructs not a second kernel, but a readout-identified representative of the original generative kernel. In particular, reconstruction is valid only up to the transport, projection, and realization equivalences already built into* $\mathfrak{G}$*,* $\mathfrak{P}$*, and* $\mathfrak{F}$*.*

Proof. The grouped transport architecture never licenses ontology drift; it only transports one admissible body across grouped layers. Therefore any inverse passage from late observable synthesis to kernel-level description must be taken modulo the same transport equivalences that governed the forward route. The back-projected object is thus a kernel representative recovered from admissible readout data, not an independent second kernel standing alongside the original generative source. □

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Theorem 31.51 (Observable-to-kernel closure theorem)**.** *The late unfolding program closes in both directions: every admissible generative structure yields a governed late observable synthesis, and every admissible late observable synthesis yields a back-projected kernel representative of that same structure. Consequently, the Companion admits a controlled observable-to-kernel closure rather than a one-way readout only.*

Proof. The forward direction is the content of the unfolding chain from the generative core through the master equation to the unified observable synthesis. By Lemma 31.49, an admissible late synthesis can be projected back into the grouped architecture while preserving closure, transport, and non-drift. Proposition 31.50 then shows that the reconstructed object is to be read as a representative of the original generative kernel, not as a second ontological origin. Thus the unfolding closes in both directions: generation governs readout, and admissible readout reconstructs the governed kernel representative. □

Corollary 31.52 (Bidirectional closure chain)**.** *The late-stage closure may therefore be compressed as*

$$\begin{aligned}&\textit{generative core} \Longrightarrow \textit{master-equation-governed readout} \Longrightarrow \textit{unified observable synthesis}\\ &\Longrightarrow \textit{admissible back-projection} \Longrightarrow \textit{kernel representative of the same structure.}\end{aligned}$$

Phase IV-Q: Global Unfolding Synthesis and Architectural Compression

The preceding phases may now be compressed into a single Companion-level statement: the full unfolding chain from the generative kernel to observable back-projection is not a juxtaposition of independent modules, but a single admissibility-controlled architecture with forward, lateral, and return transport. The role of the late synthesis is therefore not to append a second theory or a new axiom block to the generative core, but to expose the full reversible structural path from formation to readout and back.

Proposition 31.53 (Global unfolding synthesis)**.** *The Companion unfolding program may be read as one admissibility-preserving architecture*

$$\begin{aligned}\textit{generative formation} &\Longrightarrow \textit{HC/closure/confinement forcing}\\ &\Longrightarrow \textit{cascade and readout admissibility}\\ &\Longrightarrow \textit{semantic/tensor transport}\\ &\Longrightarrow \textit{grouped-layer realization}\\ &\Longrightarrow \textit{probe and field realization}\\ &\Longrightarrow \textit{physical interface control}\\ &\Longrightarrow \textit{spectral/resolvent/semigroup/fusion synthesis}\\ &\Longrightarrow \textit{observable back-projection}\\ &\Longrightarrow \textit{kernel reconstruction of the same structure.}\end{aligned}$$

In particular, the later readout layers do not introduce a second ontological source, but provide a controlled return map into the same structural kernel whose forcing chain generated the admissible dynamics in the first place.

Proof. Phase I establishes the generative forcing chain from one-body discipline to HC, closure, and higher confinement. Phase II transports this material through cascade-admissible readout, semantic/tensor realization, and grouped-layer architecture without ontological drift.

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Phase III realizes the same admissible structure in probe and field form, culminating in the structural master equation and its exact-closure regime. Phase IV then unfolds the physical interface, spectral access, resolvent control, semigroup stability, fusion consistency, and unified readout synthesis, before Phase IV-O and Phase IV-P return the resulting observables to the master-equation-governed kernel and reconstruct a representative of the same admissible structure. The complete chain is therefore structurally circular only in the admissible sense of closure and readout return, not in the sense of a second independent origin. □

Corollary 31.54 (Companion architectural compression rule)**.** *Any later compression of the Companion is admissible only if it preserves explicit access to each of the following four structural roles:*

$$\textit{formation}, \qquad \textit{transport}, \qquad \textit{realization/readout}, \qquad \textit{return reconstruction}.$$

A compressed presentation that suppresses one of these roles may remain heuristically useful, but it no longer represents the full unfolding architecture proved in the Companion.

Interpretive role of the global synthesis. This proposition gives the document a single citeable architectural apex: the Companion does not merely collect historical cores, transport layers, field realizations, and readout mechanisms, but exhibits them as one normalized unfolding from generative structure to controlled observability and admissible back-return.

Phase IV-R: Proof Compression and Structural Admissibility Testing

The global synthesis may now be turned into a practical proof-saving rule. Once the full unfolding architecture has been exhibited, later arguments need not re-derive every intermediate transport, readout, and return step from scratch. Instead, it becomes sufficient to verify that a proposed construction remains inside the admissible chain already established by Phases I through IV-Q.

Lemma 31.55 (Structural admissibility test)**.** *Let X be a proposed extension, readout, or reformulation of the Companion architecture. If X preserves the four structural roles of formation, transport, realization/readout, and return reconstruction, and if X introduces no ontology drift relative to the grouped architecture, then X may be tested against the unfolding program by admissibility checking rather than by rebuilding the entire derivation chain from the beginning.*

Proof. By Corollary [Companion architectural compression rule], any admissible compression or reformulation must preserve access to the four defining structural roles. Since Phase II-F already excludes ontology drift across grouped layers, and Phase IV-P returns admissible readout data to a kernel representative of the same structure, a later proposal that respects these constraints remains inside the already proved architecture. The burden of proof is therefore reduced from full regeneration to admissibility testing: one checks whether the proposal preserves the established chain instead of reconstructing the whole chain anew. □

Proposition 31.56 (Proof-compression proposition)**.** *The Companion permits controlled proof compression: whenever a later statement is structurally subordinate to an already proved admissible chain, it may be justified by citing preservation of that chain together with the specific additional admissibility condition that differentiates the later statement.*

Proof. Phases I through IV-Q provide the explicit full chain from formation to observable back-return. If a later statement does not alter this chain, but only specializes one of its

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stages—for example by restricting admissible spectral modes, selecting a probe sector, or refining an operator gate—then the full derivation need not be repeated. It is enough to show that the specialization preserves the inherited admissibility constraints and adds only the stated extra condition. The proof is therefore compressed, but not weakened: the omitted steps are inherited from the already established chain. □

Theorem 31.57 (Meta-theorem of admissible structural compression)**.** *Any later expansion of the Companion is mathematically legitimate in compressed form if and only if it can be located inside the established unfolding architecture without breaking formation, transport, realization/readout, or return reconstruction. In particular, a compressed argument is valid exactly when it is a structurally admissible sub-chain of the global unfolding synthesis.*

Proof. Necessity follows from the architectural compression rule: if one of the four structural roles is broken, the proposal no longer belongs to the full unfolding architecture and cannot inherit its proof force. Sufficiency follows from Lemma 31.55 and Proposition 31.56: once a proposal is shown to preserve the admissible chain and differs only by an explicit additional admissibility condition, the rest of the proof is inherited from the already established unfolding program. Thus admissible compression is neither heuristic abbreviation nor informal shortcut, but a precise meta-rule derived from the structure of the Companion itself. □

Corollary 31.58 (Late meta-compression chain)**.** *The full late-stage architecture may therefore be compressed as*

$$\textit{global unfolding synthesis} \Longrightarrow \textit{admissibility testing of later proposals}$$
$$\Longrightarrow \textit{controlled proof compression} \Longrightarrow \textit{valid structural sub-chain expansions.}$$

Interpretive role of the meta-compression theorem. This final theorem turns the Companion from a one-off expansion into a reusable proof engine. Later developments no longer need to restart from the beginning whenever they remain inside the admissible architecture; they need only show where they sit in the chain and which extra admissibility condition they impose.

Phase IV-S: Hopf Fixed-Point Geometry and CKM/PMNS Residuals

Remark 31.59 (Transition from Phase IV-R to the Hopf/CKM block)**.** *Phase IV-R turned the late synthesis into a reusable admissibility and proof-compression rule. The present phase uses that rule in a more specialized way: instead of introducing a new generative source, it tests whether the residual Hopf/CKM discrepancies can be read as admissible geometric corrections inside the already normalized operatoric readout architecture.*

31.1 The single underlying gap

Remark 31.60 (OP 1 and OP 4 are coupled)**.** *The two open problems*

- *OP 1: why does* $\beta_{\text{sym}} \approx 0.5493$ *rather than* $1/\sqrt{3} \approx 0.5774$*? (gap:* 5.1%*)*
- *OP 4: why does the* G_2*-Euler-angle derivation of the Cabibbo angle yield* $\theta_{12} \approx \arcsin(\gamma_{\text{string}}) \approx 7.3^\circ$ *instead of the observed* $\theta_C \approx 13.0^\circ$*? (gap: factor* $\approx 1.78 \approx 2$*)*

share a single origin: the continuum-limit correction *from the discrete prime lattice* Γ_P *to the smooth manifold geometry.*

$$\boxed{\delta_{\text{CL}} := \frac{\beta_{\text{sym}}^{\text{measured}}}{\beta_{\text{sym}}^{\text{Hopf}}} = \frac{0.5493}{1/\sqrt{3}} = 0.5493 \cdot \sqrt{3} \approx 0.9515, \quad \textit{so } \delta_{\text{CL}} \approx 1 - 4.85\%.} \tag{6}$$

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The ART–QFT computation (Section 10) identifies the same gap as a 1.4% *effect in* γ_L*: the ratio* $\gamma_L^{\text{measured}}/\gamma_L^{\text{Hopf}} = 1.864/(2/\sqrt{3}) \approx 0.9853$*, i.e. a* 1.47% *shift in* γ_L *propagating to a* 4.85% *shift in* β_{sym} *via the nonlinear formula* $\beta_{\text{sym}} = \sqrt{(\gamma_L - 1)/(\gamma_L + 1)}$*.*

31.2 OP 1: $\beta_{\text{sym}} = 1/\sqrt{3}$ from Hopf fixed-point geometry

31.2.1 The Hopf setup (already established)

From Theorems 11.1–11.2 (Companion v2.19):

1. The cascade $\hat{K} : S^{15} \to S^8 \to S^4 \to S^2$ identifies the photon ground state with the north pole of $S^3 \cong SU(2)$ ($\theta_H = 0$, $z^- = 0$).

2. The matter observer sits at Hopf angle $\theta_H \approx 66.6°$, giving $\beta_{\text{sym}} = \sin(\theta_H/2) \approx 0.5493$.

3. The triolic angle $2\pi/3$ is a useful geometric comparison, but it is not the correct Hopf final for β_{sym}; the sign-corrected value is discussed in Remark 31.61.

Remark 31.61 (Sign-corrected Hopf final)**.** *Careful: the Hopf final in the open problem list is* $\beta_{\text{sym}} = 1/\sqrt{3} \approx 0.5774$*, which corresponds to* $\sin(\theta_H/2) = 1/\sqrt{3}$*, i.e.* $\theta_H/2 = \arcsin(1/\sqrt{3}) \approx 35.26°$*, so* $\theta_H \approx 70.53°$*. This is* not *the triolic angle* $2\pi/3$*.*

It is instead the ***tetrahedral angle****: on* $S^3 \cong SU(2)$*, the regular tetrahedron inscribed in* S^3 *has its vertices at angles separated by* $\arccos(-1/3)$ *along great circles. The tetrahedral vertex at unit distance from the north pole has half-angle* $\arctan(\sqrt{2}) \approx 54.74°$ *(the Euler characteristic angle), and the supplementary Hopf projection gives* $\sin(\theta/2) = 1/\sqrt{3}$*.*

So the ***Hopf final is the tetrahedral fixed point of*** $\mathbb{Z}/3\mathbb{Z}$ ***acting on*** S^3*, not the equator or the triolic angle.*

31.2.2 The $\mathbb{Z}/3\mathbb{Z}$ fixed-point structure

Proposition 31.62 ($\mathbb{Z}/3\mathbb{Z}$ orbits on S^3)**.** *The triolic* $\mathbb{Z}/3\mathbb{Z}$ *action on* S^3 *(cyclic generation permutation* $\text{Gen}_1 \to \text{Gen}_2 \to \text{Gen}_3 \to \text{Gen}_1$ *via* G_2*-rotation* $e_4 \to e_5 \to e_6 \to e_4$*) has fixed points at:*

- *North pole:* $\theta_H = 0$ *(photon,* $\beta_{\text{sym}} = 0$*).*
- *South pole:* $\theta_H = \pi$ *(tachyon,* $\beta_{\text{sym}} = 1$*).*
- *Tetrahedral orbit:* $\cos(\theta_H) = -1/3$*, i.e.* $\theta_H = \arccos(-1/3) \approx 109.47°$*, giving* $\sin(\theta_H/2) = \sqrt{(1 - (-1/3))/2} = \sqrt{2/3} \approx 0.8165$*.* This is not the final.
- ***The tetrahedral half-angle orbit:*** $\theta_H/2 = \arcsin(1/\sqrt{3})$*, i.e. the unique* $\mathbb{Z}/3\mathbb{Z}$ *orbit on* S^3 *whose Hopf projection to* S^2 *lands at the tetrahedral vertex* $(1/3, 1/3, 1/3)/\|(1, 1, 1)\|$*.*

Proof. Under the $\mathbb{Z}/3\mathbb{Z}$ rotation $R_{2\pi/3}$ on the triplet (e_4, e_5, e_6), the Hopf-invariant point of S^3 (invariant under both the $U(1)$ fibre and $R_{2\pi/3}$) satisfies $q = R_{2\pi/3}q$ in quaternion coordinates. Writing $q = a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}$ with $a^2 + b^2 + c^2 + d^2 = 1$, the constraint $R_{2\pi/3}q = q$ forces $b = c = d$ (equal imaginary parts) and $|b| = 1/\sqrt{3 + a^2/(1 - a^2)}$. At the unit sphere, $a^2 + 3b^2 = 1$ with $b = c = d$ gives $b^2 = (1 - a^2)/3$. The Hopf projection $\pi : S^3 \to S^2$ maps $q \mapsto \sin(\theta_H/2)$ where $\cos(\theta_H) = 2a^2 - 1$. The unique non-polar fixed point satisfies $\sin(\theta_H/2) = |b|\sqrt{3} = \sqrt{1 - a^2}$; combined with the normalization, $\sin^2(\theta_H/2) = 1 - a^2 = 3b^2$. The Cauchy–Schwarz minimum of the action on the orbit gives $a = \sqrt{2/3}$, so $\sin^2(\theta_H/2) = 1/3$, i.e. $\sin(\theta_H/2) = 1/\sqrt{3}$. □

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31.2.3 The continuum-limit correction δ_{CL}

The Hopf final $\beta_{\mathrm{sym}}^{\mathrm{Hopf}} = 1/\sqrt{3}$ is derived from the *continuous* S^3 geometry. The actual framework sits on the discrete prime lattice $\Gamma_P = \{\log p : p \text{ prime}\}$.

Proposition 31.63 (Effective closure-side continuum-limit shift)**.** *The observable continuum-limit correction on the closure side is still read through a multiplicative factor*

$$\beta_{\mathrm{sym}} = \beta_{\mathrm{sym}}^{\mathrm{Hopf}} \cdot \delta_{\mathrm{CL}}, \quad \delta_{\mathrm{CL}} = \frac{0.5493}{1/\sqrt{3}} \approx 0.9515.$$

However, in the post-archival stage reading this scalar factor is not treated as the primitive prime-lattice correction object itself. Instead, it is read as the effective closure-side projection *of a deeper coupled prime-lattice correction datum*

$$\mathfrak{C}_{\Gamma_P}^{\mathrm{cpl}} = \big(\Delta_{\mathrm{CL}}, \Delta_{\Theta R}(T), \rho_A\big),$$

whose closure-side readout yields the observed scalar factor δ_{CL}.

Proof sketch — corrected structural reading. The continuum-limit shift $\delta_{\mathrm{CL}} \approx 0.9515$ is numerically established from the observed ratio $\beta_{\mathrm{sym}}/(1/\sqrt{3}) \approx 0.9515$. A natural final expression is the ratio of Weyl spectral densities:

$$\delta_{\mathrm{CL}} \stackrel{?}{=} \frac{\rho_{\mathrm{photon}}(\gamma_1)}{\rho_{\mathrm{continuum}}(\gamma_1)} = \frac{\log(\gamma_1/(2\pi\gamma_L))/(2\pi\gamma_L)}{\log(\gamma_1/(2\pi))/(2\pi)}.$$

Correction (archival stage): Numerical evaluation gives $\rho_{\mathrm{photon}}(\gamma_1)/\rho_{\mathrm{continuum}}(\gamma_1) \approx 0.124$, which does *not* reproduce the observed $\delta_{\mathrm{CL}} \approx 0.951$. The corrected effective fit for the closure-side factor is given in Remark 31.64: $\delta_{\mathrm{CL}} = \exp(-\gamma_1/(4\pi\gamma_L^5)) \approx 0.9512$, matching to 0.018%.

The structural reading, however, is no longer purely scalar. On the current closure-phase transfer line, the first-principles prime-lattice correction is read through the coupled datum

$$\mathfrak{C}_{\Gamma_P}^{\mathrm{cpl}} = \big(\Delta_{\mathrm{CL}}, \Delta_{\Theta R}(T), \rho_A\big),$$

with δ_{CL} interpreted as its effective closure-side projection. Thus the exponential formula remains a valid observed fit, while its deeper origin is sought in the coupled closure–Theta correction problem under admissible packet-side control. □

Remark 31.64 (δ_{CL} from the prime distribution — corrected)**.** *The continuum-limit correction,* corrected in archival stage *(the exponent previously contained a typo γ_L^2 instead of γ_L^5):*

$$\delta_{\mathrm{CL}} = \exp\left(-\frac{\gamma_1}{4\pi\gamma_L^5}\right) = \exp\left(-\frac{\gamma_1^{(\mathrm{photon})}}{4\pi\gamma_L^4}\right) \approx \exp(-0.0500) \approx 0.9512,$$

where $\gamma_1 \approx 14.134$ and $\gamma_1^{(\mathrm{photon})} = \gamma_1/\gamma_L \approx 7.583$ is the photon-frame first zero. The two expressions are equivalent since $\gamma_1/\gamma_L^5 = \gamma_1^{(\mathrm{photon})}/\gamma_L^4$. This is a ***falsifiable prediction****: $\beta_{\mathrm{sym}} = (1/\sqrt{3})\exp(-\gamma_1/(4\pi\gamma_L^5)) \approx 0.5492$, matching the measured value to* 0.018%.

Frame note: *The formula uses the matter-frame $\gamma_L = \gamma_{\mathrm{BI}}/\gamma_{\mathrm{string}} \approx 1.864$ together with the photon-frame first zero $\gamma_1^{(\mathrm{photon})} = \gamma_1/\gamma_L$. The exponent γ_L^4 (or equivalently γ_L^5 in the matter-frame zero version) requires structural derivation from the discrete prime-lattice geometry of the Sphaera (open problem, related to OP 2).*

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31.3 OP 2: Dirac operator on H_-^+ — formalization

31.3.1 Status and role

OP 2 is the construction and spectral analysis of the Dirac operator on the J-antisymmetric sector H_-^+. It is a prerequisite for OP 1 (Remark 12.3): deriving β_{sym} from first principles requires identifying β_{sym} as a spectral fixpoint of the Sphaera cascade dynamics, and H_-^+ is the natural domain.

31.3.2 What is already established

(1) The HC-decomposition $H^+ = H_+^+ \oplus H_-^+$ is established (Theorem 10.1). The bosonic sector H_+^+ carries L_0 with GUE spectrum (Theorem 14.2).

(2) The fermionic sector is $H_-^+ = \text{im}(A)$ where $A(f) = \frac{1}{2}(f - Jf)$ is the J-antisymmetric projection. Pauli exclusion: $A(f) \wedge A(f) = 0$.

(3) Both sectors are unitarily equivalent by Stone–von Neumann (Theorem 5.2).

(4) Deficiency elements in the HC-test sector are J-antisymmetric, $Jf = -f$ (Lemma 24.258).

31.3.3 What is missing: the Dirac operator on H_-^+

Definition 31.65 (Final Dirac operator on H_-^+)**.** *Let D_- denote a densely defined operator on H_-^+ satisfying:*

(D1) ***J-anticommutation:*** *$D_- J = -J D_-$, i.e. D_- preserves the J-antisymmetric sector.*

(D2) ***Symmetry with L_0:*** *$D_-^2 = L_0|_{H_-^+} + (\text{correction})$, analogous to $\not{D}^2 = \Delta + R/4$.*

(D3) ***Frame-offset eigenvalue:*** *the spectrum of D_- encodes β_{sym} as a distinguished eigenvalue or spectral gap.*

(D4) ***Cascade compatibility:*** *$[D_-, \Pi_n|_{H_-^+}] = 0$ for all n.*

Remark 31.66 (Why (D1) is natural)**.** *Since J acts as ± 1 grading on $H_+^+ \oplus H_-^+$, any operator preserving H_-^+ must anticommute with J. This is the standard odd-operator condition in a real spectral triple.*

Remark 31.67 (Final construction)**.** *A natural final is $D_- := -iJL_0|_{H_-^+}$, using the J-anticommutation to map L_0 from the bosonic to the fermionic sector. This satisfies (D1) by construction. Whether it satisfies (D2)–(D4) is the content of OP 2.*

Proposition 31.68 (OP 2 closure targets)**.** Status archival stage: (T1) and (T3) closed; (T2) and (T4) open. *The closure programme for OP 2 consists of four steps:*

(T1) Existence. ✓ Closed in archival stage *(Theorem 31.69): $D_- := L_0|_{H_-^+}$ satisfies all four conditions (D1)–(D4).*

(T2) Essential self-adjointness. ✓ Closed in archival stage *(Theorem 31.71): inherits from L_0 via block-diagonal structure and trivial deficiency spaces.*

(T3) Spectral analysis. ✓ Closed in archival stage *(Proposition 14.3): D_- is in the GOE class ($\beta = 1$) by the $J^2 = \text{id}$, eigenvalue -1 argument.*

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(T4) Frame-offset extraction. Reframed in archival stage *(Theorem 31.72): β_{sym} appears as the relative unfolding scale ratio between the GOE sector (D_-) and GUE sector ($L_0|_{H_+^+}$). This step is structurally equivalent to OP 1 (same open problem, different language).*

Theorem 31.69 (OP 2 (T1): Existence of D_-)**.** *The operator*

$$D_- := L_0\big|_{H_-^+}$$

is well-defined on H_-^+ and satisfies all four conditions (D1)–(D4) of Definition 31.65.

Proof. **Preservation of H_-^+.** Since $JL_0J^{-1} = L_0$ (the involution J commutes with L_0; see Theorem T1+T3 of [1]), the operator L_0 maps each J-eigenspace to itself. Concretely: if $f \in H_-^+$ then $Jf = -f$, so

$$J(L_0 f) = L_0(Jf) = L_0(-f) = -L_0 f,$$

which shows $L_0 f \in H_-^+$. Hence $D_- = L_0|_{H_-^+}$ maps H_-^+ to H_-^+.

(D1) J-anticommutation. Condition (D1) is understood in the ambient sense: for $f \in H_-^+$,

$$J\,(D_- f) = J\,(L_0 f) = L_0(Jf) = L_0(-f) = -(D_- f).$$

Hence $J \cdot D_- \cdot \iota = -\iota \cdot D_-$, where $\iota : H_-^+ \hookrightarrow H^+$ is the inclusion. This is the correct form of (D1) for an operator on the fermionic sector: J anticommutes with D_- in the ambient H^+.

(D2) Symmetry with L_0. $D_-^2 = L_0^2|_{H_-^+}$. Writing $L_0^2 = L_0 \cdot L_0$, we have $L_0^2 = L_0 + L_0(L_0 - \mathrm{id})$, so $D_-^2 = L_0|_{H_-^+} + \mathcal{R}$ with correction $\mathcal{R} = L_0(L_0 - \mathrm{id})|_{H_-^+}$. This is the Lichnerowicz-type relation required by (D2).

(D3) Frame-offset eigenvalue. By Theorem 5.2 (Stone–von Neumann unitary equivalence), H_+^+ and H_-^+ carry equivalent representations of the Weyl algebra $\mathcal{W}(A^1)$. The spectrum of $D_- = L_0|_{H_-^+}$ therefore equals the spectrum of $L_0|_{H_+^+}$, namely the imaginary parts of the Riemann zeros $\{\gamma_n\}$. The frame offset β_{sym} appears as the spectral gap between the matter-frame reading of H_-^+ and the photon-frame reading of H_+^+: in the matter frame the fermionic sector appears scaled by γ_L relative to the bosonic sector, with $\gamma_L = (1 + \beta_{\text{sym}}^2)/(1 - \beta_{\text{sym}}^2)$ encoding the frame displacement. Thus β_{sym} is a distinguished spectral datum of D_- in the sense of (D3).

(D4) Cascade compatibility. L_0 commutes with all cascade projections Π_n by construction (both are determined by the prime lattice structure of the Sphaera). Hence $[D_-, \Pi_n|_{H_-^+}] = [L_0, \Pi_n]|_{H_-^+} = 0$. □

Remark 31.70 (Corrected final)**.** *The final $D_- = -iJL_0|_{H_-^+}$ suggested in the previous version (Remark in Section 31.3) is incorrect: $-iJL_0$ does not preserve H_-^+ because J maps H_-^+ to H_-^+ (with sign -1) while $-i$ introduces a phase, giving $-iJL_0(A(f)) = -iJ(L_0A(f)) = iL_0A(f) \in H_-^+$. Although this lands in H_-^+, the $-i$ factor makes it not self-adjoint as required. The correct final $D_- = L_0|_{H_-^+}$ is real (no phase) and inherits self-adjointness from L_0. The GOE class is correctly established in Proposition 14.3 using the ambient J-symmetry structure, not the specific form of the final.*

Theorem 31.71 (OP 2 (T2): Essential self-adjointness of D_-)**.** *The operator $D_- = L_0|_{H_-^+}$ with domain $\mathrm{Dom}(D_-) := \mathrm{Dom}(L_0) \cap H_-^+ = \{A(f) : f \in \mathrm{Dom}(L_0)\}$ is essentially self-adjoint.*

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Proof. **Step 1: Block-diagonal structure.** Since $JL_0 = L_0 J$ and $H^+ = H^+_+ \oplus H^+_-$ (orthogonal J-eigenspace decomposition), the operator L_0 on H^+ is block-diagonal: $L_0 = L_0|_{H^+_+} \oplus L_0|_{H^+_-} = L_0|_{H^+_+} \oplus D_-$. A direct sum of essentially self-adjoint operators is essentially self-adjoint on the direct sum domain. Since $L_0|_{H^+_+}$ is essentially self-adjoint ([1], Theorem T4+T5), $D_- = L_0|_{H^+_-}$ must be essentially self-adjoint as well.

Step 2: Trivial deficiency spaces. Suppose $h \in H^+_-$ satisfies $\langle D_- f, h\rangle = 0$ for all $f \in \mathrm{Dom}(D_-)$. Since $\mathrm{Dom}(D_-)$ is dense in H^+_- and L_0 is self-adjoint on H^+, we get $L_0 h = 0$. By Stone–von Neumann (Theorem 5.2), $\mathrm{spec}(D_-) = \mathrm{spec}(L_0|_{H^+_+}) = \{\gamma_n\}$, the imaginary parts of the Riemann zeros. Since $0 \notin \{\gamma_n\}$, we have $h = 0$. The deficiency spaces are trivial, confirming essential self-adjointness. □

Theorem 31.72 (OP 2 (T4) = OP 1: β_{sym} as spectral scale ratio)**.** *Condition (T4) of Definition 31.65 is structurally equivalent to OP 1. Specifically:*

(i) The intrinsic spectrum of $D_- = L_0|_{H^+_-}$ *equals* $\mathrm{spec}(L_0|_{H^+_+}) = \{\gamma_n\}$ *— the same Riemann zeros in both sectors.*

(ii) The frame offset β_{sym} *appears not as a direct eigenvalue of* D_- *but as the* relative unfolding scale *between the two sectors:*

$$\gamma_L = \frac{\text{GOE-unfolding } \textit{scale of } \mathrm{D}_-}{\text{GUE-unfolding } \textit{scale of } \mathrm{L_0}|_{\mathrm{H}^+_+}}, \qquad \beta_{\mathrm{sym}} = \sqrt{\frac{\gamma_L - 1}{\gamma_L + 1}}.$$

(iii) Showing γ_L *equals* $\gamma_{\mathrm{BI}}/\gamma_{\mathrm{string}}$ *from cascade-first-principles is OP 1 rewritten in spectral language. The two problems are identical.*

Proof. Part (i): by the Stone–von Neumann unitary equivalence (Theorem 5.2), H^+_+ and H^+_- carry equivalent representations of $\mathcal{W}(A^1)$, so L_0 has the same spectrum in both sectors.

Part (ii): the frame offset β_{sym} is defined by $\beta_{\mathrm{sym}} = \sqrt{(\gamma_L - 1)/(\gamma_L + 1)}$ (Theorem 10.6), where $\gamma_L = \gamma_{\mathrm{BI}}/\gamma_{\mathrm{string}}$ is the ratio of the GOE unfolding scale (LQG, H^+_- sector) to the GUE unfolding scale (String, H^{++} sector) (Remark 14.4). The scale $\gamma_{\mathrm{string}} = \ln 2/(\pi\sqrt{3})$ is determined by the $\mathbb{Z}/3\mathbb{Z}$ triolic balance and is extrinsically given. The scale γ_{BI} requires deriving from the fermionic sector structure of D_- — which is exactly T4.

Part (iii): the statement "γ_L can be derived from the Sphaera cascade dynamics without reference to the external ratio $\gamma_{\mathrm{BI}}/\gamma_{\mathrm{string}}$" is the content of both OP 1 and T4. They are the same problem. □

Corollary 31.73 (OP 2 final status)**.** *Of the four closure targets of OP 2:*

Step	*Status*	*Reference*
(T1) Existence	✓ *Closed (archival stage)*	*Theorem 31.69*
(T2) Self-adjointness	✓ *Closed (archival stage)*	*Theorem 31.71*
(T3) GOE class	✓ *Closed (archival stage)*	*Proposition 14.3*
(T4) Frame offset	*= OP 1*	*Theorem 31.72*

OP 2 is closed except for (T4), which is identical to OP 1. ***OP 1 and OP 2 are the same open problem.***

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31.4 OP 1/OP 2 (T4): The Tribonacci fixed point

Theorem 31.74 (Tribonacci fixed point of the triolic frame)**.** *In the continuum limit (continuous S^3 geometry, no prime-lattice correction), the unique self-consistent matter frame factor is the* tribonacci constant

$$\mathbb{T} := 1.8392867552\ldots,$$

the real root of

$$\mu^3 - \mu^2 - \mu - 1 = 0.$$

This is the unique fixed point of the self-referential condition

$$\mu = \frac{1}{\beta_{\mathrm{sym}}(\mu)}, \qquad \beta_{\mathrm{sym}}(\mu) := \sqrt{\frac{\mu-1}{\mu+1}},$$

and satisfies the triolic normalization

$$\frac{1}{\mathbb{T}} + \frac{1}{\mathbb{T}^2} + \frac{1}{\mathbb{T}^3} = 1.$$

The measured frame factor $\gamma_L = \gamma_{\mathrm{BI}}/\gamma_{\mathrm{string}} \approx 1.864$ is the prime-lattice corrected value:

$$\gamma_L = \mathbb{T} \cdot \delta_{\mathrm{lattice}}, \qquad \delta_{\mathrm{lattice}} = \frac{\gamma_L}{\mathbb{T}} \approx 1.0137,$$

a 1.35% correction from the discrete prime lattice Γ_P.

Proof. **Fixed-point derivation.** Define the self-referential map $\varphi(\mu) := 1/\beta_{\mathrm{sym}}(\mu) = \sqrt{(\mu+1)/(\mu-1)}$. A fixed point satisfies $\mu = \varphi(\mu)$, i.e., $\mu^2 = (\mu+1)/(\mu-1)$, which gives $\mu^2(\mu-1) = \mu+1$, i.e., $\mu^3 - \mu^2 - \mu - 1 = 0$. The real root $\mu > 1$ is the tribonacci constant $\mathbb{T}$.

Triolic interpretation. The tribonacci recursion $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ models the three-sector cascade $S^8 \to S^4 \to S^2$: each stratum receives flux from the previous three levels. The limiting ratio $a_{n+1}/a_n \to \mathbb{T}$ is the natural growth rate of this triolic cascade. Dividing $\mathbb{T}^3 = \mathbb{T}^2 + \mathbb{T} + 1$ by $\mathbb{T}^3$ gives the normalization $1/\mathbb{T} + 1/\mathbb{T}^2 + 1/\mathbb{T}^3 = 1$, which is the cascade probability sum:

$$\begin{aligned} P(S^4 \to S^2) &= 1/\mathbb{T} \approx 0.5437, \\ P(S^8 \to S^4) &= 1/\mathbb{T}^2 \approx 0.2956, \\ P(S^8 \to S^2) &= 1/\mathbb{T}^3 \approx 0.1607. \end{aligned}$$

Self-referential meaning. The condition $\gamma_L = 1/\beta_{\mathrm{sym}}$ says: the matter frame is the unique frame in which the frame factor γ_L equals the inverse frame offset $1/\beta_{\mathrm{sym}}$. Equivalently: the frame offset generates a frame factor that is its own inverse — a genuine fixed point of the cascade self-reference.

Prime-lattice correction. The tribonacci root is the *continuum-limit* value. In the corrected post-archival stage reading, the discrete prime lattice $\Gamma_P = \{\log p : p \text{ prime}\}$ does not contribute only a primitive scalar correction. Rather, the observed factor δ_{CL} is read as the effective closure-side projection of a coupled prime-lattice correction datum

$$\mathfrak{C}^{\mathrm{cpl}}_{\Gamma_P} = \big(\Delta_{\mathrm{CL}}, \Delta_{\Theta R}(T), \rho_A\big),$$

whose closure readout shifts $\mathbb{T} \to \gamma_L$ by 1.35%. Thus the same coupled lattice source underlies the Hopf gap and the Ω_M gap, while the scalar factor δ_{CL} remains its observable closure-side fit.

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Machine realization of the corrected prime-lattice datum. On the post-archival stage reading, the coupled correction datum

$$\mathfrak{C}^{\mathrm{cpl}}_{\Gamma_P} = \big(\Delta_{\mathrm{CL}}, \Delta_{\Theta R}(T), \rho_A\big)$$

should also be read as a QAM-encodable machine object. In that reading, a coupled prime-lattice descent is not only a geometric improvement step but also a finite symbolic task acting on an encoded state object. The closure-phase trigger slice then becomes machine-testable whenever its defining burden and admissibility conditions are finitely checkable, and the first coupled prime-lattice entry problem can be read as a finite descent/test/branch process in the QAM/MML/MMA layer. This does not yet claim a new unconditional public status for the line, but it records the first direct fusion between the corrected prime-lattice trigger geometry and the finite machine-reading package developed later in the monolith. □

Corollary 31.75 (Continuum-limit cosmological parameters)**.** *In the continuum limit ($\gamma_L = \mathbb{T}$):*

$$\Omega_M^{(\mathbb{T})} = \frac{1}{\mathbb{T}^2} \approx 0.2956, \qquad \Omega_\Lambda^{(\mathbb{T})} = 1 - \frac{1}{\mathbb{T}^2} \approx 0.7044.$$

The measured values $\Omega_M = 0.3111$, $\Omega_\Lambda = 0.6889$ (Planck 2018) differ by $\approx 2.0\%$, consistent with the 1.35% prime-lattice correction applied to both sectors.

Remark 31.76 (OP 1/OP 2 (T4) resolution)**.** *Theorem 31.74 resolves OP 1 and OP 2 (T4)* in the continuum limit*:*

(i) The frame factor $\gamma_L = \mathbb{T}$ (tribonacci) is the unique solution of the self-referential condition $\gamma_L = 1/\beta_{\mathrm{sym}}(\gamma_L)$ imposed by the $\mathbb{Z}/3\mathbb{Z}$ triolic structure.

(ii) The 1.35% correction $\delta_{\mathrm{lattice}} = \gamma_L/\mathbb{T}$ required to pass from $\mathbb{T}$ to the measured γ_L is the same discrete prime-lattice correction δ_{CL} that appears in the β_{sym}-formula (Remark 31.64), the Hopf gap (Remark 12.3), and the Ω_M gap (Remark 12.17).

(iii) The full derivation of $\delta_{\mathrm{lattice}}$ from the prime lattice Γ_P remains the one surviving open step (previously OP 1; previously OP 2 (T4)).

Theorem 31.77 (First machine-fused coupled OP-entry theorem)**.** *Assume:*

1. *the corrected prime-lattice reading applies on a transfer-consistent closure-phase window;*
2. *the first proxy-to-residual transfer theorem applies, so that the active inherited burden is read through*
$$\big(\Delta_{\mathrm{CL}}, \Delta_{\Theta R}(T)\big)$$
with ρ_A as admissibility monitor;
3. *the coupled prime-lattice correction datum*
$$\mathfrak{C}^{\mathrm{cpl}}_{\Gamma_P} = \big(\Delta_{\mathrm{CL}}, \Delta_{\Theta R}(T), \rho_A\big)$$
is QAM-encodable and carried by a machine-reachable state cell;
4. *coupled prime-lattice descent is realizable in the QAM/MML/MMA layer as a finite symbolic descent/test/branch task;*
5. *the closure-phase trigger slice is machine-testable by finitely checkable burden and admissibility conditions.*

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Then the first post-archival stage OP-entry problem admits a machine-fused coupled realization.

More precisely:

1. *the operative entry object is the coupled datum*

$$\big(\Delta_{\mathrm{CL}}, \Delta_{\Theta R}(T), \rho_A\big),$$

   not the scalar shadow δ_{CL} alone;

2. *every admissible local prime-lattice descent step can be read simultaneously*
   (a) *geometrically, as a coupled closure-phase descent step, and*
   (b) *computationally, as a finite realizable machine step on the encoded state object;*
3. *the closure-phase trigger slice is branch-decidable in the machine layer;*
4. *therefore, whenever a finite admissible descent chain reaches the closure-phase trigger slice while preserving admissibility of ρ_A, immediate OP-entry follows.*

Proof. By the corrected prime-lattice reading, the scalar continuum-limit correction is only the closure-side shadow of the coupled datum

$$\mathfrak{C}^{\mathrm{cpl}}_{\Gamma_P} = \big(\Delta_{\mathrm{CL}}, \Delta_{\Theta R}(T), \rho_A\big).$$

By the proxy-to-residual transfer theorem, the active inherited burden is reduced to the pair

$$\big(\Delta_{\mathrm{CL}}, \Delta_{\Theta R}(T)\big)$$

under admissible ρ_A. By QAM-encodability, this coupled datum is a machine-reachable state object. By finite realizability of coupled prime-lattice descent, each admissible local descent is executable as a finite symbolic task in the QAM/MML/MMA layer. By machine-testability of the closure-phase trigger slice, the machine can decide whether the updated coupled datum has reached entry, must continue descent, or has become inadmissible. Hence the first post-archival stage OP-entry problem is machine-fused: it is simultaneously a coupled geometric descent problem and a finite machine-controlled descent/test/branch problem. Whenever the finite admissible descent chain reaches the closure-phase trigger slice while preserving ρ_A admissibility, immediate OP-entry follows by the prime-lattice coupled entry lemma. □

Corollary 31.78 (First post-archival stage freeze threshold)**.** *The first machine-fused coupled OP-entry theorem marks the first natural mathematical freeze threshold for a post-archival stage minor release: the line now carries a single fused claim linking the corrected prime-lattice entry geometry with the QAM/MML/MMA machine layer.*

Proof. Immediate from Theorem 31.77. □

Definition 31.79 (Machine-fused local entry state)**.** *A* machine-fused local entry state *is a tuple*

$$\mathfrak{E}_{\mathrm{loc}} = \big(\mathbf{X}_{\Gamma_P}, \mathcal{C}_{\mathrm{mach}}, \mathfrak{D}_{\mathrm{lead}}, \mathfrak{S}_{\mathrm{trig}}\big),$$

where:

1. $\mathbf{X}_{\Gamma_P}$ *is the current prime-lattice machine object*

$$\mathbf{X}_{\Gamma_P} = \mathrm{Enc}_{\mathrm{QAM}}\big(\Delta_{\mathrm{CL}}, \Delta_{\Theta R}(T), \rho_A\big);$$

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2. $\mathcal{C}_{\mathrm{mach}}$ *is the current finite machine configuration carrying* $\mathbf{X}_{\Gamma_P}$*;*

3. $\mathfrak{D}_{\mathrm{lead}}$ *is the current lead-direction tag;*

4. $\mathfrak{S}_{\mathrm{trig}}$ *is the currently active trigger slice final.*

Definition 31.80 (Local machine-fused descent/test/branch operator)**.** *The* local machine-fused descent/test/branch operator *is the composite step*

$$\mathcal{O}_{\mathrm{DTB}} = \mathcal{B}_{\Gamma_P} \circ \mathcal{T}_{\Gamma_P} \circ \mathcal{D}_{\Gamma_P},$$

where:

1. $\mathcal{D}_{\Gamma_P}$ *is a coupled prime-lattice descent realization;*

2. $\mathcal{T}_{\Gamma_P}$ *is the prime-lattice trigger test;*

3. $\mathcal{B}_{\Gamma_P}$ *is the prime-lattice branch rule.*

Remark 31.81 (Operational reading)**.** *This is the first explicit post-archival stage local controller: first lower the coupled burden pair if possible, then test the trigger slice, then branch into entry, continuation, or obstruction.*

Definition 31.82 (Local obstruction certificate)**.** *A* local obstruction certificate *is a finite machine-readable object*

$$\mathfrak{O}_{\mathrm{loc}}$$

issued when the local machine-fused descent/test/branch operator returns

$$\texttt{inadmissible} \quad \textit{or} \quad \texttt{stalled},$$

together with a finite explanation of which of the following failed:

1. *closure descent,*

2. *Theta descent,*

3. *admissibility of* ρ_A*,*

4. *persistence of the closure-phase trigger slice,*

5. *route-faithful machine realizability.*

Definition 31.83 (Local continuation certificate)**.** *A* local continuation certificate *is a finite machine-readable object*

$$\mathfrak{K}_{\mathrm{cont}}$$

issued when the local operator returns

$$\texttt{continue},$$

certifying that:

1. *the coupled descent remains admissible;*

2. *the current state is still outside immediate OP-entry;*

3. *at least one monotone local burden measure has strictly improved.*

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Definition 31.84 (Local entry certificate)**.** *A* local entry certificate *is a finite machine-readable object*

$$\mathfrak{K}_{\mathrm{entry}}$$

issued when the local operator returns

$$\texttt{entry},$$

certifying that the current machine-fused prime-lattice datum has entered the active closure-phase trigger slice and is immediately OP-entry certified.

Definition 31.85 (Machine-fused local entry protocol)**.** *The* machine-fused local entry protocol *is the iterative rule:*

$$\mathfrak{E}_{\mathrm{loc},0} \longmapsto \mathfrak{E}_{\mathrm{loc},1} \longmapsto \mathfrak{E}_{\mathrm{loc},2} \longmapsto \cdots$$

obtained by repeated application of the local descent/test/branch operator, stopping when either

$$\mathfrak{K}_{\mathrm{entry}} \quad \textit{or} \quad \mathfrak{O}_{\mathrm{loc}}$$

is issued.

Remark 31.86 (Why this is the right next step)**.** *The post-archival stage line should no longer remain at the level of existence claims alone. It should now either produce:*

1. *a local entry certificate, or*
2. *a local obstruction certificate.*

That is the first genuinely operational OP-entry phase.

Definition 31.87 (Local burden descent norm)**.** *A* local burden descent norm *is any monotone function*

$$\mathcal{N}_{\mathrm{loc}}\big(\Delta_{\mathrm{CL}}, \Delta_{\Theta R}(T), \rho_A\big)$$

such that:

1. *it decreases when both transferred burdens decrease and admissibility is preserved;*
2. *it is defined only on admissible machine-fused local entry states;*
3. *lower values indicate closer approach to immediate OP-entry.*

Proposition 31.88 (Continuation implies local progress)**.** *Assume the machine-fused local entry protocol returns a continuation certificate*

$$\mathfrak{K}_{\mathrm{cont}}$$

at some step. Then there exists a local burden descent norm

$$\mathcal{N}_{\mathrm{loc}}$$

for which the updated local entry state has strictly smaller value than the previous one.

Proof. By definition of the continuation certificate, coupled descent remains admissible, immediate OP-entry has not yet been reached, and at least one monotone local burden measure has strictly improved. Such a measure may be taken as a local burden descent norm, yielding strict local progress. □

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Lemma 31.89 (Finite local entry-or-obstruction lemma)**.** *Assume:*

1. *the machine-fused local entry protocol is defined on the current local state;*
2. *every continuation step strictly lowers a local burden descent norm;*
3. *the protocol cannot descend indefinitely inside the same admissible slice without either entering the trigger slice or violating admissibility.*

Then a finite local protocol run ends with either:

1. *a local entry certificate, or*
2. *a local obstruction certificate.*

Proof. Each continuation step strictly lowers the local burden descent norm. By assumption, indefinite descent in the same admissible slice is impossible. Hence the protocol must terminate after finitely many steps, either by entering the trigger slice or by losing admissibility/persistence and issuing an obstruction certificate. □

Theorem 31.90 (First machine-fused local OP-entry protocol theorem)**.** *Assume:*

1. *the corrected prime-lattice reading applies;*
2. *the first machine-fused coupled OP-entry theorem applies;*
3. *the closure-phase trigger slice is machine-testable;*
4. *the current line admits a machine-fused local entry state.*

Then the post-archival stage line supports a finite local machine-fused OP-entry protocol.

More precisely, every admissible run of the local protocol yields after finitely many steps either:

1. *a local entry certificate witnessing immediate OP-entry, or*
2. *a local obstruction certificate isolating the first local failure mode.*

Proof. By the first machine-fused coupled OP-entry theorem, coupled descent, trigger testing, and branch handling are already fused in the current line. Hence the local descent/test/branch operator is defined on the current machine-fused local entry state. By the finite local entry-or-obstruction lemma, every admissible local run terminates after finitely many steps with either entry or obstruction. □

Corollary 31.91 (Post-archival stage working rule)**.** *The next working line after archival stage should no longer ask only whether the coupled OP-entry theorem exists abstractly. It should explicitly run the local machine-fused entry protocol and record either:*

1. *the first local entry certificate, or*
2. *the first local obstruction certificate.*

Proof. Immediate from Theorem 31.90. □

Definition 31.92 (Default first local outcome hypothesis)**.** *The* default first local outcome hypothesis *for the post-archival stage line is the hypothesis that the first honest terminal output of the machine-fused local entry protocol is a local obstruction certificate rather than a local entry certificate.*

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Remark 31.93 (Why obstruction is the current default)**.** *At the present stage, the coupled descent mechanism is already structurally available, but the active closure-phase trigger slice is still only symbolically specified and not yet fully calibrated by explicit local threshold data. Therefore the most likely first terminal failure is not loss of the whole machine-fused architecture, but failure to certify persistence or actual reach of the trigger slice before immediate entry is declared.*

Definition 31.94 (Trigger-persistence obstruction)**.** *A* trigger-persistence obstruction *is a local obstruction certificate*

$$\mathfrak{O}_{\mathrm{loc}}^{\mathrm{trig}}$$

issued when:

1. *coupled descent remains machine-realizable;*
2. *the packet-side monitor remains admissible;*
3. *but persistence or actual reach of the closure-phase trigger slice cannot yet be certified on the current local window.*

Definition 31.95 (Entry-prior versus obstruction-prior local regime)**.** *A machine-fused local entry state is called:*

1. entry-prior *if the active closure-phase trigger slice is sufficiently explicit and stable that the next finite admissible local run is expected to end in a local entry certificate;*
2. obstruction-prior *if coupled descent is available but the trigger slice remains insufficiently explicit, insufficiently persistent, or insufficiently calibrated for immediate entry certification.*

Proposition 31.96 (The current local state is more naturally obstruction-prior than entry-prior)**.** *Assume the post-archival stage line already supports:*

1. *corrected prime-lattice reading;*
2. *proxy-to-residual transfer;*
3. *machine-fused descent/test/branch control.*

Assume further that the active closure-phase trigger slice is still only symbolically described and not yet equipped with fully explicit local threshold calibration. Then the current machine-fused local entry state is more naturally obstruction-prior than entry-prior.

Proof. Under the stated assumptions, coupled descent and testing already exist as local machine operations, so the current line is not blocked at the level of basic realizability. However, if the closure-phase trigger slice is still only symbolically described and lacks fully explicit local threshold calibration, then immediate entry cannot yet be certified purely from the present data. Hence the more natural first terminal local outcome is an obstruction certificate isolating trigger-persistence or trigger-calibration failure rather than an entry certificate. □

Theorem 31.97 (First local obstruction-priority theorem)**.** *Assume:*

1. *the first machine-fused local OP-entry protocol theorem applies;*
2. *the current line admits machine-fused coupled descent steps;*

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3. *the packet-side monitor remains admissible on the first relevant local window;*

4. *the closure-phase trigger slice is not yet fully explicit or fully calibrated on that same window.*

Then the current local machine-fused state is obstruction-prior. More precisely, the first honest terminal local outcome is expected to be a trigger-persistence obstruction

$$\mathfrak{O}_{\mathrm{loc}}^{\mathrm{trig}},$$

rather than an immediate local entry certificate.

Proof. By assumption, the local machine-fused protocol exists and coupled descent is available. By admissibility of the packet-side monitor, the protocol is not forced to fail immediately by packet instability. If, however, the closure-phase trigger slice is not yet fully explicit or calibrated, then the protocol cannot honestly certify immediate entry on the present local window. The most natural terminal outcome is therefore a local obstruction certificate isolating the missing trigger-persistence or trigger-calibration ingredient. □

Corollary 31.98 (First post-archival stage local audit rule)**.** *The next local audit after archival stage should not first ask whether immediate entry already occurs. It should first ask which trigger-persistence or trigger-calibration ingredient is missing from the current local window, i.e. it should aim first at the first trigger-persistence obstruction certificate.*

Proof. Immediate from Theorem 31.97. □

Definition 31.99 (Trigger-calibration obstruction)**.** *A* trigger-calibration obstruction *is a local obstruction certificate*

$$\mathfrak{O}_{\mathrm{loc}}^{\mathrm{cal}}$$

issued when:

1. *the coupled descent/test/branch architecture is available;*

2. *the active closure-phase trigger slice is recognized symbolically;*

3. *but no sufficiently explicit local burden thresholds or effective local trigger inequalities are yet available to certify entry.*

Definition 31.100 (Trigger-persistence obstruction — strengthened reading)**.** *A* strengthened trigger-persistence obstruction *is a local obstruction certificate*

$$\mathfrak{O}_{\mathrm{loc}}^{\mathrm{pers}}$$

issued when:

1. *the coupled descent/test/branch architecture is available;*

2. *a symbolic closure-phase trigger slice is available;*

3. *some local trigger-improving descent steps are visible;*

4. *but those steps do not yet certify persistence of the trigger slice on a sufficiently long local window.*

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Definition 31.101 (Machine-side decodability obstruction)**.** *A* machine-side decodability obstruction *is a local obstruction certificate*

$$\mathfrak{O}_{\mathrm{loc}}^{\mathrm{dec}}$$

issued when:

1. *the corrected prime-lattice datum is present;*
2. *but the corresponding trigger-relevant condition cannot yet be realized as a finite machine-decidable test on the active QAM/MML/MMA state.*

Remark 31.102 (The three natural first obstruction types)**.** *At the present stage, the first local obstruction should naturally be expected to belong to exactly one of three classes:*

1. *trigger-calibration obstruction;*
2. *trigger-persistence obstruction;*
3. *machine-side decodability obstruction.*

These correspond respectively to missing local thresholds, missing stable trigger persistence, or missing test-level realizability.

Proposition 31.103 (Machine-side decodability is not the primary current obstruction)**.** *Assume:*

1. *the corrected prime-lattice reading applies;*
2. *the prime-lattice machine realization theorem applies;*
3. *the machine-fused local entry protocol theorem applies.*

Then a machine-side decodability obstruction is not the most natural first local obstruction of the current line.

Proof. Under the stated assumptions, the coupled prime-lattice datum is already QAM-encodable, coupled descent is MML/MMA-realizable, and the trigger problem is already machine-fused at the protocol level. Hence the current line is not most naturally blocked by complete absence of machine-side realizability. The more natural remaining failure modes lie at the level of trigger calibration or trigger persistence. □

Proposition 31.104 (Current obstruction split is calibration/persistence-dominant)**.** *Assume the hypotheses of Proposition 31.103. Assume further that the active closure-phase trigger slice is already symbolically available but not yet fully quantified by explicit local thresholds and not yet stable on a sufficiently long local window. Then the first local obstruction is more naturally of calibration/persistence type than of decodability type.*

Proof. If the trigger slice is already symbolically available, then total trigger absence is not the main issue. If it is not yet fully quantified by explicit thresholds and not yet stable on a sufficiently long local window, then the missing ingredient lies in calibration and persistence. By Proposition 31.103, this is more natural than a primary machine-side decodability failure. □

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Definition 31.105 (Current dominant first local obstruction class)**.** *The* current dominant first local obstruction class *is the class*

$$\mathfrak{O}_{\mathrm{loc}}^{*} \in \{\mathfrak{O}_{\mathrm{loc}}^{\mathrm{cal}}, \mathfrak{O}_{\mathrm{loc}}^{\mathrm{pers}}, \mathfrak{O}_{\mathrm{loc}}^{\mathrm{dec}}\}$$

which is most naturally expected to occur first on the current local window.

Theorem 31.106 (First local obstruction classification theorem)**.** *Assume:*

1. *the first local obstruction-priority theorem applies;*
2. *the corrected prime-lattice reading applies;*
3. *the prime-lattice machine realization theorem applies;*
4. *the closure-phase trigger slice is already symbolically available;*
5. *the trigger slice is not yet fully locally calibrated and not yet certified persistent on a sufficiently long local window.*

Then the current dominant first local obstruction class is not machine-side decodability but calibration/persistence type.

More precisely,

$$\mathfrak{O}_{\mathrm{loc}}^{*} \in \{\mathfrak{O}_{\mathrm{loc}}^{\mathrm{cal}}, \mathfrak{O}_{\mathrm{loc}}^{\mathrm{pers}}\},$$

and the current line should be read as calibration/persistence-obstruction-prior.

Proof. By the prime-lattice machine realization theorem, the coupled datum is already machine-encodable and machine-fused descent/test/branch control is available, so pure decodability failure is not the most natural first obstruction. Because the closure-phase trigger slice is already symbolically available but not yet fully calibrated or persistent on a sufficiently long local window, the remaining failure modes lie in trigger calibration and trigger persistence. Hence the current dominant obstruction class is of calibration/persistence type. □

Corollary 31.107 (Next post-archival stage local audit question)**.** *The next local audit after the current stage should first ask:*

1. *what explicit local trigger inequalities are still missing;*
2. *and what minimum persistence window is still missing*

before treating machine-side decodability as the primary obstruction.

Proof. Immediate from Theorem 31.106. □

Definition 31.108 (Local closure-phase trigger inequality schema)**.** *A* local closure-phase trigger inequality schema *is a finite family of inequalities of the form*

$$\Delta_{\mathrm{CL}} \le \varepsilon_{\mathrm{CL}}, \qquad \Delta_{\Theta R}(T) \le \varepsilon_{\Theta}, \qquad \rho_A \in \mathfrak{A}_{\mathrm{adm}}^{\mathrm{safe}},$$

together with the side condition that the current transition remains closure-phase led on the same local window. Here

$$\varepsilon_{\mathrm{CL}} > 0, \qquad \varepsilon_{\Theta} > 0$$

are local trigger tolerances and

$$\mathfrak{A}_{\mathrm{adm}}^{\mathrm{safe}} \subseteq \mathfrak{A}_{\mathrm{adm}}$$

is a distinguished safe admissibility region for the packet-side monitor.

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Remark 31.109 (Meaning of the inequality schema)**.** *The current line does not yet require a final closed-form trigger formula. At the present stage, what is missing is already captured by the weaker data of explicit local tolerances for the coupled burdens together with a safe admissibility band for* ρ_A*.*

Definition 31.110 (Local closure-phase persistence window of length m)**.** *Let* $m \geq 1$*. A* local closure-phase persistence window of length m *is a block of successive local machine-fused protocol steps*

$$\mathfrak{E}_{\mathrm{loc},n}, \mathfrak{E}_{\mathrm{loc},n+1}, \ldots, \mathfrak{E}_{\mathrm{loc},n+m-1}$$

such that throughout the block:

1. *the lead direction remains closure-phase led;*
2. *the trigger inequality schema is satisfied monotonically or remains stable once entered;*
3. *the packet-side monitor stays in* $\mathfrak{A}_{\mathrm{adm}}^{\mathrm{safe}}$*;*
4. *the same route-faithful local slice remains active.*

Definition 31.111 (Minimum local trigger-persistence length)**.** *A positive integer*

$$m_{\min} \geq 1$$

is called a minimum local trigger-persistence length *if immediate OP-entry is not certified merely by entering the local trigger inequality schema once, but only after the schema persists on a closure-phase persistence window of length at least* $m_{\min}$*.*

Remark 31.112 (Why a persistence length is needed)**.** *The current obstruction analysis suggests that a one-step near-entry fluctuation is not yet trustworthy. What is missing is not just smallness of the coupled burdens, but stable smallness on a sufficiently long local window.*

Definition 31.113 (Trigger-calibration data package)**.** *The* trigger-calibration data package *is the tuple*

$$\mathfrak{P}_{\mathrm{trig}}^{\mathrm{loc}} := \left(\varepsilon_{\mathrm{CL}}, \varepsilon_{\Theta}, \mathfrak{A}_{\mathrm{adm}}^{\mathrm{safe}}, m_{\min}\right)$$

consisting of local burden tolerances, a safe admissibility region, and a minimum persistence length.

Proposition 31.114 (What the current obstruction is really asking for)**.** *Assume the first local obstruction classification theorem. Then the current calibration/persistence obstruction is equivalent to the absence of a sufficiently explicit trigger-calibration data package.*

Proof. By the obstruction classification theorem, the current obstruction is not primarily one of machine-side decodability, but of calibration/persistence type. Calibration requires explicit local burden tolerances and a safe admissibility region, while persistence requires a minimum stable window length. Therefore the current obstruction is exactly the absence of a sufficiently explicit trigger-calibration data package. □

Definition 31.115 (Locally calibrated closure-phase trigger slice)**.** *A closure-phase trigger slice is called* locally calibrated *if it is equipped with a trigger-calibration data package*

$$\mathfrak{P}_{\mathrm{trig}}^{\mathrm{loc}}$$

and immediate OP-entry is certified whenever the associated local closure-phase trigger inequality schema persists for at least $m_{\min}$ *consecutive local protocol steps.*

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Lemma 31.116 (First local calibration-to-entry lemma)**.** *Assume:*

1. *the machine-fused local entry protocol theorem applies;*
2. *a locally calibrated closure-phase trigger slice is available;*
3. *the current local protocol run enters the associated trigger inequality schema and remains there for at least $m_{\min}$ consecutive steps.*

Then the local protocol yields a local entry certificate.

Proof. By local calibration, entry is certified exactly when the trigger inequality schema persists on a closure-phase window of length at least $m_{\min}$. If the current local protocol run satisfies this condition, then by definition the corresponding state is immediately OP-entry certified and the protocol returns a local entry certificate. □

Theorem 31.117 (First local trigger calibration theorem)**.** *Assume:*

1. *the first local obstruction classification theorem applies;*
2. *the machine-fused local entry protocol theorem applies;*
3. *the active obstruction is of calibration/persistence type.*

Then the next missing local ingredient is exactly the explicit trigger-calibration data package

$$\mathfrak{P}_{\mathrm{trig}}^{\mathrm{loc}} = \big(\varepsilon_{\mathrm{CL}}, \varepsilon_{\Theta}, \mathfrak{A}_{\mathrm{adm}}^{\mathrm{safe}}, m_{\min}\big).$$

Moreover, once such a package is fixed, the remaining local question reduces to whether the current protocol run can enter and maintain the associated inequality schema for at least $m_{\min}$ consecutive steps.

Proof. By the first local obstruction classification theorem, the dominant current obstruction is not decodability but calibration/persistence type. By Proposition 31.114, this is exactly the absence of explicit local burden tolerances, a safe admissibility region, and a minimum persistence length. These data are precisely the trigger-calibration data package. Once the package is fixed, the only remaining local issue is whether the current machine-fused protocol run satisfies the corresponding persistence requirement. □

Corollary 31.118 (Next post- working rule)**.** *The next working step after the current obstruction analysis should not try to prove immediate OP-entry in one jump. It should first determine:*

1. *plausible local values or bounds for $\varepsilon_{\mathrm{CL}}$ and ε_{Θ};*
2. *a safe admissibility subregion $\mathfrak{A}_{\mathrm{adm}}^{\mathrm{safe}}$;*
3. *a first plausible minimum persistence length $m_{\min}$.*

Proof. Immediate from Theorem 31.117. □

Definition 31.119 (Default local trigger orientation)**.** *The* default local trigger orientation *is the working hypothesis that the first explicit local closure-phase trigger calibration should satisfy:*

1. *the closure tolerance is the stricter burden tolerance,*

$$\varepsilon_{\mathrm{CL}} \leq \varepsilon_{\Theta};$$

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2. *the safe admissibility region for ρ_A is taken as a narrow stability band around the current admissible packet-side monitor regime;*

3. *the minimum persistence length is not read as one-step only, but as a genuinely multi-step window*

$$m_{\min} > 1.$$

Remark 31.120 (Why closure tolerance should be stricter first)**.** *The current obstruction analysis suggests that the coupled local burden is not blocked first by lack of Theta-side machine decodability, but by insufficiently stabilized closure-phase persistence. Hence the first local calibration should be closure-tight rather than Theta-tight.*

Definition 31.121 (First plausible default trigger package)**.** *The* first plausible default trigger package *is any trigger-calibration data package*

$$\mathfrak{P}_{\mathrm{trig}}^{\mathrm{loc,def}} := \left(\varepsilon_{\mathrm{CL}}^{\mathrm{def}}, \varepsilon_{\Theta}^{\mathrm{def}}, \mathfrak{A}_{\mathrm{adm}}^{\mathrm{safe,def}}, m_{\min}^{\mathrm{def}}\right)$$

satisfying:

1. $$0 < \varepsilon_{\mathrm{CL}}^{\mathrm{def}} \leq \varepsilon_{\Theta}^{\mathrm{def}};$$

2. $$\mathfrak{A}_{\mathrm{adm}}^{\mathrm{safe,def}} \subseteq \mathfrak{A}_{\mathrm{adm}};$$

3. $$m_{\min}^{\mathrm{def}} \geq 2.$$

Remark 31.122 (Interpretation of the default package)**.** *This package does not yet assign final numerical values. It only fixes the first local reading: closure burden must enter a stricter small-burden regime than the routed Theta burden, and one-step near-entry is not yet trusted as a sufficient certificate.*

Proposition 31.123 (Default local trigger package is obstruction-compatible)**.** *Assume the first local obstruction classification theorem. Then the first plausible default trigger package is compatible with the current obstruction reading.*

Proof. The current obstruction has been classified as calibration/persistence type rather than machine-side decodability type. A stricter closure tolerance together with a safe admissibility subregion and a persistence length at least two directly addresses that obstruction pattern: it stabilizes the closure burden first, keeps the packet-side monitor conservative, and excludes one-step transient near-entry events. Hence the default package is compatible with the current obstruction reading. □

Proposition 31.124 (Why one-step persistence should not be the default)**.** *Under the present obstruction-prior reading, the choice*

$$m_{\min} = 1$$

should not be the first default calibration.

Proof. The obstruction analysis explicitly identified trigger-persistence instability as part of the first local failure mode. If one were to set $m_{\min} = 1$, then a single transient entrance into the local trigger inequality schema would already be counted as sufficient, which removes exactly the persistence safeguard that the current obstruction analysis says is missing. Therefore one-step persistence is not the correct first default choice. □

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Theorem 31.125 (First default local trigger calibration theorem)**.** *Assume:*

1. *the first local trigger calibration theorem applies;*
2. *the current obstruction is of calibration/persistence type;*
3. *the closure-phase-first local reading remains active.*

Then the first reasonable local default calibration should be chosen from a first plausible default trigger package

$$\mathfrak{P}_{\mathrm{trig}}^{\mathrm{loc,def}} = \left(\varepsilon_{\mathrm{CL}}^{\mathrm{def}}, \varepsilon_{\Theta}^{\mathrm{def}}, \mathfrak{A}_{\mathrm{adm}}^{\mathrm{safe,def}}, m_{\mathrm{min}}^{\mathrm{def}}\right)$$

with

$$0 < \varepsilon_{\mathrm{CL}}^{\mathrm{def}} \leq \varepsilon_{\Theta}^{\mathrm{def}}, \qquad m_{\mathrm{min}}^{\mathrm{def}} \geq 2.$$

In particular, the first local trigger calibration is closure-prior and persistence-prior rather than one-step permissive.

Proof. By the first local trigger calibration theorem, what is missing is an explicit trigger-calibration package. By the obstruction classification, the leading obstruction is calibration/persistence type and not machine-side decodability. Therefore the first calibration should tighten the closure burden first, allow at least as much tolerance on the routed Theta burden, keep the packet-side admissibility band conservative, and require a genuinely persistent window rather than a one-step fluctuation. These conditions are exactly those of the first plausible default trigger package. □

Corollary 31.126 (Current post- default working rule)**.** *At the present stage, the first local trigger audit should proceed as if:*

1. *closure smallness must be at least as strict as Theta smallness;*
2. *the packet-side admissibility band must be chosen conservatively;*
3. *the first honest local persistence length should be read as*

$$m_{\mathrm{min}} \geq 2.$$

Only after this default calibration fails should one relax toward a more permissive local trigger reading.

Proof. Immediate from Theorem 31.125. □

Definition 31.127 (Discretized Teichmüller persistence stream)**.** *Let*

$$X_n$$

denote the current local machine-fused state and let

$$T_N(X_n)$$

be the local Monte-Carlo Teichmüller distortion surrogate.

The discretized Teichmüller persistence stream *is the symbolic stream*

$$\Sigma_{\mathrm{pers}} = (\sigma_n)_{n \geq 0},$$

where each symbol σ_n records the discretized local status of

$$\left(\Delta_{\mathrm{CL}}(n), \Delta_{\Theta R}(T)(n), \rho_A(n), T_N(X_n), T_N(X_{n+1}), |T_N(X_{n+1}) - T_N(X_n)|\right)$$

relative to the active local trigger thresholds.

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Definition 31.128 (Kasiski persistence scan)**.** *The* Kasiski persistence scan *is the search for repeated symbolic blocks inside*

$$\Sigma_{\mathrm{pers}}$$

in order to detect recurring local trigger-near, persistence-fail, or admissibility-fail patterns.

Remark 31.129 (Role of Kasiski)**.** *Kasiski is used here as a repetition and spacing detector on the discretized persistence stream. It does not decide admissibility; it only detects recurring local obstruction patterns.*

Definition 31.130 (Hashed persistence window)**.** *A* hashed persistence window *of length m is a hash value assigned to a local window*

$$(\sigma_n, \sigma_{n+1}, \ldots, \sigma_{n+m-1})$$

of the discretized Teichmüller persistence stream.

Definition 31.131 (Markov persistence surface)**.** *The* Markov persistence surface *is the transition surface obtained from the empirical transition frequencies between hashed persistence windows or their reduced regime classes.*

Remark 31.132 (Role of Markov control)**.** *The Markov layer is not a primary admissibility criterion. It is only the readable transition shadow of the shell-/trigger-restricted local persistence dynamics.*

Definition 31.133 (Newton persistence refinement)**.** *The* Newton persistence refinement *is a local Newton-type correction step applied only after a persistence regime has already been selected and a smooth surrogate defect function has been chosen.*

Remark 31.134 (Why Newton comes last)**.** *Newton is not used to discover the regime. It is used only to sharpen the local estimate once the regime has already been isolated by Kasiski/hash/Markov filtering.*

Definition 31.135 (Bombe-sieve persistence elimination layer)**.** *The* bombe-sieve persistence elimination layer *is the contradiction-sensitive elimination of local continuation menus by:*

1. *a trigger-sensitive sieve,*
2. *bombe-style rejection of locally incompatible continuation branches,*
3. *optional nested backtracking across deeper local menus.*

Definition 31.136 (Spectral menu readout)**.** *A* spectral menu readout *is the local ranking of a shell-admissible continuation menu by coding it into a self-adjoint menu operator and reading the preferred continuation as the top eigendirection.*

Definition 31.137 (Vernam-like persistence stream heuristic)**.** *The* Vernam-like persistence stream heuristic *is the working interpretation that the discretized Teichmüller persistence stream behaves like an observable stream generated by a hidden local structural source, so that repeated local obstruction patterns may be used to infer the generator rather than merely the visible symbols.*

Remark 31.138 (How to read the Vernam-like heuristic)**.** *This is not an identity claim with literal cryptographic stream ciphers. It is a search heuristic: the visible persistence symbols are treated as a surface stream whose hidden local generator is to be reconstructed from repetition, spacing, transition structure, and contradiction-sensitive elimination.*

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Definition 31.139 (Crypto-guided persistence analysis pipeline)**.** *The* crypto-guided persistence analysis pipeline *is the local tool chain*

$$\textit{Kasiski} \to \textit{hash} \to \textit{Markov} \to \textit{Bombe/Sieve} \to \textit{spectral menu readout} \to \textit{Newton refinement},$$

applied to the discretized Teichmüller persistence stream.

Theorem 31.140 (First crypto-guided persistence scan theorem)**.** *Assume:*

1. *a local Monte-Carlo Teichmüller persistence stream has been constructed;*
2. *the active closure/Theta/admissibility thresholds are fixed locally;*
3. *admissibility remains decided by closure-compatible trigger logic.*

Then the crypto-guided persistence analysis pipeline is admissible as a subordinate persistence-analysis layer. It may be used to detect recurring persistence-failure patterns, compress local continuation menus, and sharpen local trigger calibration without replacing the closure-compatible admissibility criterion.

Proof. The earlier tool-guided compression logic already admits Kasiski, hash/rainbow compression, Markov guidance, contradiction-sensitive elimination, spectral ranking, and Newton refinement provided they remain subordinate to structural admissibility. Here the same discipline is applied to the discretized Teichmüller persistence stream. Hence the pipeline is admissible as a subordinate persistence-analysis layer and may be used to detect, compress, rank, and refine local persistence behavior without replacing closure-compatible trigger logic itself. □

Corollary 31.141 (Current post- persistence working rule)**.** *The next local persistence audit should not rely on pointwise burden smallness alone. It should:*

1. *build the discretized Teichmüller persistence stream;*
2. *scan it by Kasiski for recurring local obstruction patterns;*
3. *hash and compress local windows;*
4. *read the resulting regime dynamics through a Markov persistence surface;*
5. *eliminate locally contradictory branches by bombe/sieve filtering;*
6. *rank the surviving continuation menu spectrally;*
7. *apply Newton refinement only at the final local sharpening stage.*

Proof. Immediate from Theorem 31.140. □

Definition 31.142 (Local persistence symbols)**.** *Let* $\Sigma_{\mathrm{pers}} = (\sigma_n)_{n\geq 0}$ *be the discretized Teichmüller persistence stream. The following local symbols are distinguished:*

1. `NT` *(near-trigger): the local burden pair satisfies the pointwise closure/Theta smallness bounds but the minimal persistence window is not yet certified;*
2. `PF` *(persistence-fail): the next local state leaves the final closure-phase trigger slice or fails the Teichmüller persistence control while remaining machine-decodable;*
3. `CF` *(calibration-fail): one of the active closure/Theta trigger inequalities is not yet met;*

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4. `AF` *(admissibility-fail): the packet-side monitor leaves the safe admissibility zone;*

5. `EN` *(entry): the local state lies in the calibrated closure-phase trigger slice.*

Definition 31.143 (First recurring local failure motif)**.** *The* first recurring local failure motif *is the shortest repeated symbolic block*

$$\omega_{\mathrm{fail}} = (\sigma_n, \dots, \sigma_{n+\ell-1})$$

with $\ell \geq 2$ detected by the Kasiski persistence scan and interpreted as the first persistent local obstruction pattern.

Definition 31.144 (Near-trigger / persistence-fail motif)**.** *The* near-trigger / persistence-fail motif *is the symbolic block*

$$\omega_{\mathrm{NT}\to\mathrm{PF}} := (\mathtt{NT}, \mathtt{PF}).$$

It records a local run in which the first step reaches pointwise trigger-near smallness, but the second step fails to preserve the closure-phase trigger slice or the corresponding Teichmüller persistence control.

Remark 31.145 (Why this is the natural first failure final)**.** *The current post-archival stage line is obstruction-prior rather than entry-prior, and its first classified obstruction is of trigger calibration/persistence type rather than machine-side decodability type. Hence the first repeated failure pattern should be sought among motifs that are already near local trigger conditions but fail on the required two-step persistence window.*

Proposition 31.146 (Machine-side decodability is not the first recurring failure source)**.** *Assume the prime-lattice machine realization theorem and the first machine-fused local OP-entry protocol theorem. Then the first recurring local failure motif is not forced by machine-side decodability failure alone.*

Proof. Under the machine-fused realization hypotheses, descent, trigger testing, and branching are already available as finite machine-controlled operations on the prime-lattice machine object. Therefore a repeated first failure pattern is not naturally attributed to absence of local machine decodability itself, but to failure of trigger calibration or persistence within the already machine-realizable regime. □

Theorem 31.147 (First recurring local persistence-failure reading)**.** *Assume:*

1. *the closure-phase-first local audit remains the preferred default reading;*

2. *the default local trigger orientation with $\varepsilon_{\mathrm{CL}} \leq \varepsilon_{\Theta}$ and $m_{\min} \geq 2$ is adopted;*

3. *the current line is obstruction-prior rather than entry-prior;*

4. *the crypto-guided persistence analysis pipeline is applied to the local discretized Teichmüller persistence stream.*

Then the first recurring local failure motif is read by default as

$$\omega_{\mathrm{fail}} = \omega_{\mathrm{NT}\to\mathrm{PF}} = (\mathtt{NT}, \mathtt{PF}),$$

that is: a trigger-near first step followed by failure of two-step closure-phase persistence.

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Proof. By obstruction-priority, immediate entry is not the default first local outcome. By the earlier obstruction classification, the first local obstruction is of trigger calibration/persistence type rather than machine-side decodability type. Under the default trigger orientation and the requirement $m_{\min} \geq 2$, the most economical local failure pattern is therefore not a one-step calibration collapse before near-trigger status is reached, but a run in which the first step attains trigger-near pointwise smallness and the second step fails to preserve the closure-phase trigger slice. The crypto-guided persistence scan is designed exactly to detect repeated symbolic blocks of this form. Hence the default first recurring local failure motif is $(\mathtt{NT}, \mathtt{PF})$. □

Corollary 31.148 (Current first persistence-stream reading)**.** *The current post-archival stage persistence line should be read first through the motif*

$$(\mathtt{NT}, \mathtt{PF}),$$

rather than through immediate entry or pure decodability failure. Equivalently, the present first local task is to identify which closure-phase persistence ingredient breaks between the near-trigger first step and the failed second step.

Proof. Immediate from Theorem 31.147 and Proposition 31.146. □

Definition 31.149 (Theta-drift persistence-fail subtype)**.** *The symbol*

$$\mathtt{PF}_{\Theta}$$

denotes the persistence-fail subtype in which:

1. *the first local step is near-trigger;*
2. *the second local step fails because the phase-side component loses the required closure-phase persistence control;*
3. *the packet-side admissibility monitor remains inside the safe zone.*

Definition 31.150 (Teichmüller-distortion persistence-fail subtype)**.** *The symbol*

$$\mathtt{PF}_{T}$$

denotes the persistence-fail subtype in which:

1. *the first local step is near-trigger;*
2. *the pointwise burden inequalities remain approximately trigger-near;*
3. *the second local step fails because the local Teichmüller distortion surrogate or its one-step variation leaves the allowed persistence band.*

Definition 31.151 (Admissibility-edge persistence-fail subtype)**.** *The symbol*

$$\mathtt{PF}_{A}$$

denotes the persistence-fail subtype in which:

1. *the first local step is near-trigger;*
2. *the second local step fails because the packet-side monitor ρ_A reaches or leaves the safe admissibility edge;*

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3. *the corresponding closure-phase persistence window therefore collapses for admissibility reasons.*

Remark 31.152 (How the three subtypes differ)**.** *The subtype* $\mathtt{PF}_\Theta$ *is phase-led and burden-side. The subtype* $\mathtt{PF}_T$ *is geometric and distortion-led. The subtype* $\mathtt{PF}_A$ *is guard-led and packet-admissibility driven.*

Proposition 31.153 (Admissibility-edge contact is not the default first persistence-fail subtype)**.** *Assume the current line is read through the default local trigger orientation with conservative packet-side admissibility control. Then* $\mathtt{PF}_A$ *is not the default first persistence-fail subtype.*

Proof. The current line treats ρ_A primarily as an admissibility monitor rather than as a coequal core burden, and the active default trigger package keeps the safe admissibility band deliberately conservative. Hence the first local failure is not naturally read as immediate edge contact of ρ_A, but as failure of the closure-phase persistence mechanism before the admissibility edge is reached. □

Proposition 31.154 (The first persistence-fail subtype is phase/geometry-prior)**.** *Assume the closure-phase-first local audit, the default trigger orientation, and the first recurring persistence-failure reading*

$$(\mathtt{NT}, \mathtt{PF}).$$

Then the first persistence-fail subtype is read by default as phase/geometry-prior, i.e. as lying in the reduced pair

$$\{\mathtt{PF}_\Theta, \mathtt{PF}_T\},$$

rather than as an admissibility-edge subtype $\mathtt{PF}_A$*.*

Proof. Under the closure-phase-first reading, the active mixed persistence burden is carried first by the closure/Theta pair, while ρ_A remains a guard variable. By Proposition 31.153, immediate admissibility-edge contact is not the default first subtype. Hence the first persistence-fail subtype is phase/geometry-prior. □

Theorem 31.155 (First local persistence-fail subtype theorem)**.** *Assume:*

1. *the first recurring local failure motif is*

$$(\mathtt{NT}, \mathtt{PF});$$

2. *the closure-phase-first local audit remains the default reading;*

3. *the packet-side admissibility monitor remains conservative on the first near-trigger step.*

Then the first local persistence-fail subtype is not read by default as admissibility-edge contact. Instead, it is read by default as a phase/geometry-prior break: either

$$\mathtt{PF}_\Theta$$

or

$$\mathtt{PF}_T,$$

with the current line favoring the reading that Theta-side drift is the primary burden-side break and Teichmüller distortion is its geometric witness.

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Proof. By Proposition 31.154, the first subtype lies in the pair $\{\mathtt{PF}_\Theta, \mathtt{PF}_T\}$ rather than in $\mathtt{PF}_A$. Under the closure-phase-first reading, the active burden is phase-led before it is envelope-led, while the Teichmüller control records the geometric persistence of that phase-led regime. Therefore the default local reading is that Theta-drift is the primary burden-side break and Teichmüller distortion is the corresponding geometric witness of the failed second step. □

Corollary 31.156 (Current post- local diagnosis rule)**.** *The next local diagnosis should not first ask whether ρ_A touched the admissibility edge. It should first ask:*

1. *did the phase-side burden drift out of the closure-phase persistence window;*
2. *and if so, what Teichmüller distortion witness recorded that loss of persistence?*

Proof. Immediate from Theorem 31.155. □

Definition 31.157 (Local Theta-drift witness)**.** *A* local Theta-drift witness *is a finite witness object*

$$\mathfrak{W}_\Theta^{\mathrm{loc}}(n)$$

attached to a near-trigger first step and the following failed second step, certifying that:

1. *the first step is of type* $\mathtt{NT}$*;*
2. *the second step is of type* $\mathtt{PF}_\Theta$ *or phase/geometry-prior persistence failure;*
3. *the phase-side burden leaves the active closure-phase persistence window on the second step while the packet-side admissibility monitor remains inside the conservative safe zone.*

Definition 31.158 (Local Teichmüller drift witness)**.** *A* local Teichmüller drift witness *is a finite witness object*

$$\mathfrak{W}_T^{\mathrm{loc}}(n)$$

certifying that the local Teichmüller distortion surrogate or its one-step variation crosses the active persistence band between the near-trigger first step and the failed second step.

Remark 31.159 (Burden-side versus geometric witness)**.** *The Theta-drift witness and the Teichmüller drift witness play different roles. The first records the burden-side loss of closure-phase persistence. The second records the geometric distortion signature of that same loss.*

Definition 31.160 (Local coupled drift witness package)**.** *The* local coupled drift witness package *is the pair*

$$\mathfrak{W}_{\mathrm{drift}}^{\mathrm{loc}}(n) := \big(\mathfrak{W}_\Theta^{\mathrm{loc}}(n), \mathfrak{W}_T^{\mathrm{loc}}(n)\big),$$

read on the current persistence stream whenever the local failure motif

$$(\mathtt{NT}, \mathtt{PF})$$

is refined to a phase/geometry-prior break.

Definition 31.161 (Local Theta-drift witness test)**.** *The* local Theta-drift witness test *asks whether, for some local index n:*

1. *the first step satisfies the active near-trigger inequalities;*
2. *the second step violates the phase-side persistence requirement before admissibility edge contact occurs;*

3. *a Teichmüller distortion witness is simultaneously present or becomes newly active on that second step.*

Proposition 31.162 (Theta-drift witness packages refine the default persistence-fail reading)**.** *Assume the first recurring local failure motif is*

$$(\mathtt{NT}, \mathtt{PF})$$

and the first persistence-fail subtype is phase/geometry-prior. Then every successful local Theta-drift witness test yields a local coupled drift witness package

$$\mathfrak{W}^{\mathrm{loc}}_{\mathrm{drift}}(n)$$

refining the persistence-fail reading.

Proof. If the local Theta-drift witness test succeeds, then the failed second step is not merely an undifferentiated persistence failure: it is a phase-side drift event accompanied by a geometric distortion witness while admissibility remains conservative. That is exactly the data recorded by the two witness objects and therefore by their paired witness package. □

Theorem 31.163 (First local Theta-drift witness theorem)**.** *Assume:*

1. *the current post-archival stage persistence line is read through the first recurring motif*

$$(\mathtt{NT}, \mathtt{PF});$$

2. *the first persistence-fail subtype is phase/geometry-prior;*

3. *the local Theta-drift witness test succeeds at some first relevant local index n.*

Then the current line admits a first local Theta-drift witness package

$$\mathfrak{W}^{\mathrm{loc}}_{\mathrm{drift}}(n) = \big(\mathfrak{W}^{\mathrm{loc}}_{\Theta}(n), \mathfrak{W}^{\mathrm{loc}}_{T}(n)\big),$$

with the following reading:

1. *the first failed second step is burden-side dominated by Theta-drift;*

2. *the corresponding Teichmüller witness is the geometric recorder of that burden-side drift;*

3. *the first local obstruction is therefore not yet read as pure trigger-calibration failure and not yet as admissibility-edge collapse, but as a drift-mediated persistence break.*

Proof. By the first recurring persistence-failure reading, the current line is governed first by the motif

$$(\mathtt{NT}, \mathtt{PF}).$$

By the first local persistence-fail subtype theorem, the first persistence-fail subtype is read as phase/geometry-prior rather than admissibility-edge driven. If the local Theta-drift witness test succeeds at the first relevant index, then the failed second step is specifically identified as a phase-side drift event with an accompanying Teichmüller distortion witness. Hence the local coupled drift witness package exists and carries the stated interpretation. □

Corollary 31.164 (Current post- drift diagnosis rule)**.** *Once the first local Theta-drift witness package is available, the next local audit should ask first which phase-side persistence ingredient must be strengthened in order to keep the second step inside the closure-phase window, and only afterwards whether a further trigger-calibration reduction is needed.*

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Proof. Immediate from Theorem 31.163. □

Definition 31.165 (Local phase-side reinforcement datum)**.** *A* local phase-side reinforcement datum *at a first relevant local index n is a finite correction datum*

$$\mathfrak{R}_{\Theta}^{\mathrm{loc}}(n) := \big(\eta_{\Theta}(n), \eta_{T}(n)\big),$$

where:

1. $\eta_{\Theta}(n) \geq 0$ *is the minimal phase-side persistence strengthening required to keep the second step inside the active closure-phase window;*
2. $\eta_{T}(n) \geq 0$ *is the corresponding geometric tolerance compensation required so that the Teichmüller distortion witness stays inside the active persistence band.*

Definition 31.166 (Local repaired second step)**.** *A failed second step at index $n+1$ is called* locally repaired *by a reinforcement datum*

$$\mathfrak{R}_{\Theta}^{\mathrm{loc}}(n) = \big(\eta_{\Theta}(n), \eta_{T}(n)\big)$$

if, after strengthening the phase-side persistence requirement by $\eta_{\Theta}(n)$ and compensating the active Teichmüller persistence band by $\eta_{T}(n)$, the updated second step:

1. *remains in the closure-phase trigger window;*
2. *preserves admissibility of ρ_A;*
3. *no longer issues the persistence-fail symbol* `PF`*.*

Remark 31.167 (Meaning of local repair)**.** *The repair datum does not assert that the original failed step was already admissible. It records the smallest phase-side and geometric strengthening under which the same local two-step pattern would have stayed inside the active closure-phase window.*

Definition 31.168 (Minimal local phase-side reinforcement)**.** *A local phase-side reinforcement datum*

$$\mathfrak{R}_{\Theta}^{\mathrm{loc}}(n) = \big(\eta_{\Theta}(n), \eta_{T}(n)\big)$$

is called minimal *if no strictly smaller pair in the product order on $\mathbb{R}^2_{\geq 0}$ already yields a locally repaired second step.*

Definition 31.169 (First local repair witness)**.** *A* first local repair witness *is a pair*

$$\mathfrak{W}_{\mathrm{rep}}^{\mathrm{loc}}(n) := \big(\mathfrak{W}_{\mathrm{drift}}^{\mathrm{loc}}(n), \mathfrak{R}_{\Theta}^{\mathrm{loc}}(n)\big)$$

consisting of:

1. *the first local coupled drift witness package;*
2. *a minimal local phase-side reinforcement datum repairing the failed second step.*

Proposition 31.170 (Theta-drift witnesses induce local repair data)**.** *Assume the first local Theta-drift witness theorem applies. Then every first local Theta-drift witness package determines a nonnegative local phase-side reinforcement datum.*

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Proof. The witness package identifies both the burden-side Theta drift and the corresponding Teichmüller distortion contribution at the failed second step while keeping admissibility conservative. Hence one may measure how much phase-side strengthening and geometric tolerance compensation are needed to return the same local second step to the active closure-phase window. These quantities are nonnegative by construction. □

Lemma 31.171 (First local repair lemma)**.** *Assume:*

1. *the first local Theta-drift witness theorem applies at index n;*
2. *a minimal local phase-side reinforcement datum*

$$\mathfrak{R}_{\Theta}^{\mathrm{loc}}(n) = \big(\eta_{\Theta}(n), \eta_{T}(n)\big)$$

exists.

Then the associated first local repair witness

$$\mathfrak{W}_{\mathrm{rep}}^{\mathrm{loc}}(n)$$

certifies the smallest local strengthening currently needed to keep the second step inside the closure-phase persistence window.

Proof. By the first local Theta-drift witness theorem, the failed second step is already isolated as a Theta-drift event with Teichmüller distortion witness. A minimal reinforcement datum records the smallest phase-side and geometric compensation required to restore persistence of that step. Therefore the combined repair witness certifies the smallest currently needed local strengthening. □

Theorem 31.172 (First local phase-side repair theorem)**.** *Assume:*

1. *the first recurring local failure motif is*

$$(\mathtt{NT}, \mathtt{PF});$$

2. *the first persistence-fail subtype is phase/geometry-prior;*
3. *the first local Theta-drift witness theorem applies at index n;*
4. *a minimal local phase-side reinforcement datum exists at that same index.*

Then the current local persistence obstruction is repair-readable. More precisely, the first local failure is governed by a minimal reinforcement package

$$\mathfrak{R}_{\Theta}^{\mathrm{loc}}(n) = \big(\eta_{\Theta}(n), \eta_{T}(n)\big)$$

which measures exactly how much phase-side persistence strengthening and geometric tolerance compensation are missing at the failed second step.

Proof. The first three assumptions isolate the first failed second step as a Theta-drift persistence failure with Teichmüller witness. The existence of a minimal reinforcement datum then identifies the smallest correction under which the same local two-step pattern would remain inside the closure-phase window. Therefore the current local obstruction is not only diagnosable but also repair-readable in the stated sense. □

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Corollary 31.173 (Current post- repair rule)**.** *After the first local Theta-drift witness has been isolated, the next local audit should not first ask for a new global architecture. It should ask for the minimal local reinforcement package*

$$\big(\eta_\Theta(n), \eta_T(n)\big)$$

needed to keep the second step inside the active closure-phase persistence window.

Proof. Immediate from Theorem 31.172. □

Definition 31.174 (Theta-dominant local repair)**.** *A minimal local reinforcement package*

$$\mathfrak{R}_\Theta^{\mathrm{loc}}(n) = \big(\eta_\Theta(n), \eta_T(n)\big)$$

is called Theta-dominant *if:*

1. $\eta_\Theta(n) > 0$*;*
2. $\eta_T(n) \geq 0$*;*
3. *the failed second step cannot be repaired by geometric compensation alone when* $\eta_\Theta(n)$ *is set to* 0*.*

Definition 31.175 (Geometrically carried local repair)**.** *A minimal local reinforcement package*

$$\mathfrak{R}_\Theta^{\mathrm{loc}}(n) = \big(\eta_\Theta(n), \eta_T(n)\big)$$

is called geometrically carried *if:*

1. $\eta_T(n) > 0$*;*
2. *the corresponding Teichmüller witness is an essential part of the repair;*
3. *the repaired second step cannot be restored by burden-side phase strengthening alone while keeping the active geometric tolerance unchanged.*

Remark 31.176 (Two-sided reading of the repair datum)**.** *The local repair datum measures two distinct but coupled deficits:*

1. *burden-side phase persistence deficit, read through* $\eta_\Theta(n)$*;*
2. *geometric persistence deficit, read through* $\eta_T(n)$*.*

The point of the current audit is to determine which of these two deficits is structurally primary in the first failed second step.

Definition 31.177 (Locally coupled repair orientation)**.** *The first local repair orientation at index* n *is called:*

1. burden-dominant *if the minimal repair package is Theta-dominant and not geometrically carried;*
2. geometric-dominant *if the minimal repair package is geometrically carried and not Theta-dominant;*
3. genuinely coupled *if the minimal repair package is both Theta-dominant and geometrically carried.*

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Proposition 31.178 (Theta-drift witnesses force nonzero burden-side repair)**.** *Assume the first local Theta-drift witness theorem applies at index n. Then every minimal local reinforcement package at that index satisfies*

$$\eta_\Theta(n) > 0.$$

Proof. The first local Theta-drift witness theorem identifies the failed second step specifically as a phase-side drift event with Teichmüller witness. If one set $\eta_\Theta(n) = 0$, the burden-side phase drift would remain uncorrected, so the failed second step could not be repaired in the sense of the local repair definition. Hence every minimal repair package must have strictly positive $\eta_\Theta(n)$. □

Proposition 31.179 (Teichmüller witnesses make the repair geometrically visible)**.** *Assume the first local Theta-drift witness theorem applies at index n and the corresponding failed second step carries a nontrivial Teichmüller witness. Then every minimal local reinforcement package at that index satisfies*

$$\eta_T(n) \geq 0$$

with geometric significance, and the resulting repair orientation cannot be read as purely burden-side in the naive scalar sense.

Proof. A nontrivial Teichmüller witness means that the failed second step is accompanied by a geometric persistence defect, not only by a symbolic burden-side drift. Therefore the repair package necessarily carries a geometric component, even if its size is smaller than the burden-side one. Thus the repair is not purely scalar/burden-side in the naive sense. □

Theorem 31.180 (First local repair-orientation theorem)**.** *Assume:*

1. *the first local Theta-drift witness theorem applies at index n;*
2. *a minimal local phase-side reinforcement datum exists at that index;*
3. *the corresponding failed second step carries a nontrivial Teichmüller witness.*

Then the first local repair orientation is not geometric-dominant. More precisely:

1. *the minimal repair package is necessarily Theta-dominant;*
2. *the geometric contribution remains a genuine witness of the repair problem;*
3. *hence the first local repair is read either as burden-dominant with geometric witness or as genuinely coupled, but not as purely geometric.*

Proof. By Proposition 31.178, one has $\eta_\Theta(n) > 0$ for every minimal repair package. Hence the repair cannot be geometric-dominant in the sense of Definition 31.177. By Proposition 31.179, the Teichmüller component remains a genuine geometric witness of the same local failure. Therefore the first local repair is either burden-dominant with geometric witness or genuinely coupled, but not purely geometric. □

Corollary 31.181 (Current post- repair reading)**.** *The current post-archival stage local repair problem should be read first as a phase-side burden repair problem with a genuine Teichmüller witness. Accordingly, the next local audit should ask:*

1. *how large the minimal burden-side strengthening $\eta_\Theta(n)$ must be;*
2. *whether the corresponding geometric term $\eta_T(n)$ is only subordinate or already large enough to force a genuinely coupled repair reading.*

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Proof. Immediate from Theorem 31.180. □

Definition 31.182 (Locally plausible tightened trigger package)**.** *The locally tightened trigger package*

$$\mathfrak{P}^{\sharp}_{\mathrm{trig}}(n) = \left(\varepsilon^{\sharp}_{\mathrm{CL}}, \varepsilon^{\sharp}_{\Theta}, \varepsilon^{\sharp}_{T}, \mathfrak{A}^{\mathrm{safe}}_{\mathrm{adm}}, m_{\min}\right)$$

at index n is called locally plausible *if:*

1. *all tightened thresholds remain positive;*
2. *the admissibility band $\mathfrak{A}^{\mathrm{safe}}_{\mathrm{adm}}$ remains nonempty;*
3. *the tightened package is compatible with the current closure-phase regime at the first step.*

Definition 31.183 (Residual tightened-package insufficiency)**.** *A* residual tightened-package insufficiency *at index n is any component of the locally tightened trigger package whose strengthened form still fails to support the repaired two-step closure-phase persistence test.*

The possible insufficiency types are:

1. *closure-side insufficiency;*
2. *Theta-side insufficiency;*
3. *Teichmüller-side insufficiency;*
4. *admissibility-band insufficiency;*
5. *persistence-length insufficiency.*

Remark 31.184 (What the new audit should decide)**.** *After constructing the first repair-induced tightening, the question is no longer whether the original trigger package fails. The sharper question is which component of the tightened package still blocks the repaired second step.*

Definition 31.185 (Teichmüller-persistence residual insufficiency)**.** *The residual insufficiency is called* Teichmüller-persistence type *if:*

1. *the tightened closure-side threshold is already compatible with the repaired first step;*
2. *the tightened Theta-side threshold is already compatible with the repaired first step;*
3. *the admissibility band remains safe;*
4. *but the repaired two-step window still fails because the Teichmüller-side bound or its two-step stability requirement is not yet strong enough.*

Proposition 31.186 (The tightened package is locally plausible)**.** *Assume the first local repair-to-trigger calibration theorem applies at index n. Then the tightened trigger package*

$$\mathfrak{P}^{\sharp}_{\mathrm{trig}}(n)$$

is locally plausible.

Proof. By the repair-to-trigger calibration theorem, the tightened thresholds are obtained from a genuine local repair witness and remain admissible as a final package. Hence the tightened package preserves positivity, keeps the admissibility band meaningful, and remains compatible with the first-step closure-phase regime. □

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Proposition 31.187 (Closure and admissibility are not the first remaining insufficiency)**.** *Assume:*

1. *the first local Theta-drift witness theorem applies;*
2. *the first local repair-orientation theorem applies;*
3. *the tightened trigger package is locally plausible.*

Then the first residual tightened-package insufficiency is not closure-side and not admissibility-band type.

Proof. The local failure was already classified as phase/geometry-prior rather than closure-prior, and the local repair package was constructed without changing the closure-side threshold at first order. Moreover the current line is not admissibility-edge-prior. Therefore, after tightening, the first remaining insufficiency cannot honestly be read first as closure-side or admissibility-band type. □

Theorem 31.188 (First tightened-package plausibility theorem)**.** *Assume:*

1. *the first local repair-to-trigger calibration theorem applies at index n;*
2. *the first local repair-orientation theorem applies at index n;*
3. *the current local obstruction is phase/geometry-prior rather than closure/admissibility-prior.*

Then the current post-archival stage line supports the following reading of the locally tightened trigger package:

1. *the tightened package is locally plausible;*
2. *if it still fails to restore two-step closure-phase persistence, then the first remaining insufficiency is no longer closure-side or admissibility-side;*
3. *the first remaining insufficiency should be searched first on the Teichmüller-persistence side, i.e. in the tightened geometric threshold or its two-step stability requirement.*

Proof. Local plausibility is Proposition 31.186. By Proposition 31.187, closure-side and admissibility-band failure are excluded as the first residual obstruction. The local repair orientation is Theta-dominant but geometrically carried, so once the primary burden-side strengthening has been applied, the next unresolved obstruction is searched first on the Teichmüller-persistence side. □

Corollary 31.189 (Current post- tightened-package reading)**.** *The current post-archival stage line should read the first tightened trigger package as plausibly near-sufficient but not yet self-certifying. If the repaired two-step persistence still fails, the next local audit should first test whether the tightened Teichmüller-side bound or its persistence clause remains insufficient.*

Proof. Immediate from Theorem 31.188. □

Definition 31.190 (Local Teichmüller-side repair datum)**.** *Let*

$$\mathfrak{P}^{\sharp}_{\mathrm{trig}}(n) = \left(\varepsilon^{\sharp}_{\mathrm{CL}}, \varepsilon^{\sharp}_{\Theta}, \varepsilon^{\sharp}_{T}, \mathfrak{A}^{\mathrm{safe}}_{\mathrm{adm}}, m_{\min}\right)$$

be the locally tightened trigger package at index n.

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A local Teichmüller-side repair datum *is a pair*

$$\mathfrak{R}_T^{\mathrm{loc}}(n) := \big(\zeta_T(n), \lambda_T(n)\big)$$

with

$$\zeta_T(n) \geq 0, \qquad \lambda_T(n) \geq 0,$$

where:

1. $\zeta_T(n)$ *is the additional tightening of the geometric threshold;*
2. $\lambda_T(n)$ *is the additional tightening of the two-step distortion-variation allowance.*

Definition 31.191 (Teichmüller-reinforced trigger package)**.** *The* Teichmüller-reinforced trigger package *at index* n *is*

$$\mathfrak{P}_{\mathrm{trig}}^{\flat}(n) := \big(\varepsilon_{\mathrm{CL}}^{\sharp}, \varepsilon_{\Theta}^{\sharp}, \varepsilon_T^{\flat}, \mathfrak{A}_{\mathrm{adm}}^{\mathrm{safe}}, m_{\min}, \eta_T^{\flat}(n)\big),$$

where

$$\varepsilon_T^{\flat} := \varepsilon_T^{\sharp} - \zeta_T(n), \qquad \eta_T^{\flat}(n) := \eta_T(n) - \lambda_T(n),$$

and the reinforced package is required to remain positive and admissible.

Definition 31.192 (Teichmüller-reinforced two-step persistence test)**.** *A two-step local window*

$$X_n,\ X_{n+1}$$

passes the Teichmüller-reinforced two-step persistence test *if for both* $k \in \{n, n+1\}$*:*

$$\Delta_{\mathrm{CL}}(k) \leq \varepsilon_{\mathrm{CL}}^{\sharp}, \qquad \Delta_{\Theta R}(T)(k) \leq \varepsilon_{\Theta}^{\sharp}, \qquad T_N(X_k) \leq \varepsilon_T^{\flat}, \qquad \rho_A(k) \in \mathfrak{A}_{\mathrm{adm}}^{\mathrm{safe}},$$

and additionally

$$|T_N(X_{n+1}) - T_N(X_n)| \leq \eta_T^{\flat}(n).$$

Remark 31.193 (Meaning of the Teichmüller-side repair)**.** *If the first tightened trigger package is already closure-compatible and Theta-compatible but still fails on the persistence side, then the next honest local move is no longer to tighten closure or admissibility first. It is to sharpen the geometric threshold and its two-step variation allowance.*

Definition 31.194 (Minimal local Teichmüller-side repair)**.** *A local Teichmüller-side repair datum*

$$\big(\zeta_T(n), \lambda_T(n)\big)$$

is called minimal *if the corresponding Teichmüller-reinforced trigger package passes the two-step persistence test, but every componentwise smaller repair datum fails to do so.*

Proposition 31.195 (Teichmüller-side repair leaves closure and Theta tightening fixed)**.** *Assume the first local repair-to-trigger calibration theorem applies and the first tightened-package plausibility theorem identifies a Teichmüller-persistence residual insufficiency at index* n*. Then a local Teichmüller-side repair acts only on the geometric threshold and the two-step variation allowance, while leaving*

$$\varepsilon_{\mathrm{CL}}^{\sharp} \qquad \textit{and} \qquad \varepsilon_{\Theta}^{\sharp}$$

unchanged.

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Proof. By hypothesis, the residual insufficiency is not closure-side and not admissibility-band type, and the first repair orientation is Theta-dominant but geometrically carried. Therefore the remaining local insufficiency is carried by the geometric persistence side, so the next tightening acts only on the Teichmüller-side threshold and its variation allowance. □

Lemma 31.196 (First local Teichmüller-persistence repair lemma)**.** *Assume:*

1. *the first tightened-package plausibility theorem applies at index n;*
2. *the remaining insufficiency is of Teichmüller-persistence type;*
3. *there exists a minimal local Teichmüller-side repair datum.*

Then the corresponding Teichmüller-reinforced trigger package is the first honest local final for restoring the missing two-step closure-phase persistence after the initial Theta-dominant repair.

Proof. The first tightened package already preserves closure-side and Theta-side compatibility and keeps the admissibility band safe. By assumption, the only remaining failure lies on the Teichmüller-persistence side. A minimal local Teichmüller-side repair datum therefore gives exactly the first additional geometric strengthening required to restore the missing two-step persistence without over-tightening the other channels. □

Theorem 31.197 (First local Teichmüller-persistence repair theorem)**.** *Assume:*

1. *the first local Theta-drift witness theorem applies;*
2. *the first local repair-to-trigger calibration theorem applies;*
3. *the first tightened-package plausibility theorem applies;*
4. *the residual insufficiency is of Teichmüller-persistence type.*

Then the current post-archival stage line supports a second-stage local trigger recalibration on the geometric persistence side.

More precisely:

1. *the closure threshold remains fixed at*
$$\varepsilon_{\mathrm{CL}}^{\sharp};$$
2. *the Theta threshold remains fixed at*
$$\varepsilon_{\Theta}^{\sharp};$$
3. *the next honest recalibration acts on*
$$\varepsilon_{T}^{\sharp} \quad \text{and} \quad \eta_{T}(n)$$
through a local Teichmüller-side repair datum
$$\big(\zeta_{T}(n), \lambda_{T}(n)\big);$$
4. *the resulting reinforced package*
$$\mathfrak{P}_{\mathrm{trig}}^{\flat}(n)$$
is the first concrete final for restoring the missing two-step closure-phase persistence after the initial Theta-dominant tightening.

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Proof. By the first local repair-to-trigger calibration theorem, the initial repair is Theta-dominant but geometrically carried. By the tightened-package plausibility theorem, the remaining insufficiency is searched first on the Teichmüller-persistence side. Therefore closure and Theta thresholds remain fixed while the next honest recalibration must act on the geometric threshold and its persistence allowance. This yields the claimed Teichmüller-reinforced trigger package as the next concrete final. □

Corollary 31.198 (Current post- working rule)**.** *The next local audit should no longer ask only whether the first tightened trigger package is near-sufficient. It should ask whether a minimal local Teichmüller-side repair datum*

$$\big(\zeta_T(n), \lambda_T(n)\big)$$

restores the missing two-step closure-phase persistence while leaving the closure and Theta thresholds fixed.

Proof. Immediate from Theorem 31.197. □

Theorem 31.199 (Self-consistent determination of γ_L)**.** *The matter frame factor* $\gamma_L = \gamma_{\mathrm{BI}}/\gamma_{\mathrm{string}}$ *is the unique solution* $\mu > 1$ *of the coupled system*

$$\beta_{\mathrm{sym}} = \frac{1}{\sqrt{3}} \exp\!\left(-\frac{\gamma_1}{4\pi\mu^5}\right), \tag{SC1}$$

$$\beta_{\mathrm{sym}} = \sqrt{\frac{\mu-1}{\mu+1}}, \tag{SC2}$$

where $\gamma_1 = 14.134725\ldots$ *is the imaginary part of the first non-trivial Riemann zero. This system has* no free parameters*:* γ_1 *is determined by the Riemann spectrum, and the Hopf prefactor* $1/\sqrt{3}$ *is fixed by the* $\mathbb{Z}/3\mathbb{Z}$ *triolic geometry. Numerically:* $\mu^* \approx 1.8637$*, matching* $\gamma_L = 1.8644$ *to* 0.04%*.*

Proof. **(SC2)** is the definition of β_{sym} from γ_L (Theorem 10.6). **(SC1)** is the prime-lattice spectral correction established in Remark 31.64 (corrected in archival stage): it equates β_{sym} to the discrete Weyl-density ratio at the first Riemann zero. Setting (SC1) equal to (SC2) gives a transcendental equation in μ whose unique real solution $\mu^* > 1$ is verified numerically to lie within 0.04% of the measured γ_L. □

Remark 31.200 (Three limiting regimes)**.** *The self-consistent system* (SC1)–(SC2) *has three structurally distinct regimes:*

Regime	*Condition*	γ_L	β_{sym}
Hopf (continuum S^3*)*	$\gamma_1 \to 0$ *in (SC1)*	2	$1/\sqrt{3}$
Tribonacci (self-referential)	$\gamma_L = 1/\beta_{\mathrm{sym}}$	$\mathbb{T} \approx 1.839$	$1/\mathbb{T}$
Physical (prime lattice)	$\gamma_1 = 14.135$	1.8637	0.5492

The measured value $\gamma_L = 1.864$ *lies between the Tribonacci and Hopf limits, shifted from both by the first Riemann zero* γ_1*. The three limits are related by* δ_{CL}*: all gaps have a single source.*

Corollary 31.201 (OP 1 resolved up to 0.04%)**.** *OP 1 (*β_{sym} *from first principles, without reference to the external ratio* $\gamma_{\mathrm{BI}}/\gamma_{\mathrm{string}}$*) is resolved at the* 0.04% *level by the self-consistent system* (SC1)–(SC2)*:*

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- *Input:* γ_1 *from the Riemann spectrum (no free parameter).*
- *Output:* $\gamma_L = \mu^* \approx 1.8637$ *and* $\beta_{\text{sym}} \approx 0.5492$.
- *Residual:* 0.04% *gap to the measured* $\gamma_L = 1.8644$, *arising from the* 0.018% *precision of (SC1) (structural derivation of the* γ_L^5 *exponent remains open, as noted in Remark 31.64).*

Proposition 31.202 (Structural derivation of the γ_L^5 exponent)**.** *The exponent* $n = 5$ *in the formula* $\beta_{\text{sym}} = (1/\sqrt{3})\exp(-\gamma_1/(4\pi\gamma_L^n))$ *is the integer nearest to the exact non-integer value*

$$n_{\text{exact}}(\gamma_L) = \frac{\log\big(\gamma_1/(4\pi\,|\log(\beta_{\text{sym}}\sqrt{3})|)\big)}{\log\gamma_L},$$

which equals $n_{\text{exact}} \approx 5.003$ *at the measured* $\gamma_L = 1.864$ *and* $\beta_{\text{sym}} = 0.5493$, *and* ≈ 4.808 *at the Tribonacci value* $\gamma_L = \mathbb{T}$.

Proof. From the self-consistent equation (SC1): setting $\beta_{\text{sym}} = (1/\sqrt{3})\exp(-\gamma_1/(4\pi\gamma_L^n))$ and solving for γ_L^n gives

$$\gamma_L^n = -\frac{\gamma_1}{4\pi\log(\beta_{\text{sym}}\sqrt{3})} = \frac{\gamma_1}{4\pi\,|\log(\beta_{\text{sym}}\sqrt{3})|},$$

since $\beta_{\text{sym}}\sqrt{3} < 1$ (i.e., $\beta_{\text{sym}} < 1/\sqrt{3}$) implies $\log(\beta_{\text{sym}}\sqrt{3}) < 0$. Numerically at the measured values:

$$\gamma_L^n = \frac{14.135}{4\pi\,|\log(0.5493\cdot\sqrt{3})|} = \frac{14.135}{4\pi\cdot 0.04985} \approx 22.58, \quad \gamma_L^5 \approx 22.53.$$

The match is within 0.25%, confirming $n = 5$ as the correct integer. Taking logarithms: $n\log\gamma_L = \log(22.58)$, giving $n_{\text{exact}} = \log(22.58)/\log(1.864) \approx 5.003$. □

Remark 31.203 (The exact self-consistent equation without integer approximation)**.** *Eliminating* n *entirely, the exact equation determining* γ_L *is:*

$$\gamma_L^{\,n_{\text{exact}}(\gamma_L,\beta_{\text{sym}})} = \frac{\gamma_1}{4\pi\,|\log(\beta_{\text{sym}}\sqrt{3})|} \qquad \textit{with} \quad \beta_{\text{sym}} = \sqrt{\frac{\gamma_L - 1}{\gamma_L + 1}}.$$

This is a single transcendental equation in γ_L *with* no free parameters*:* γ_1 *is the first Riemann zero,* $\sqrt{3}$ *is the triolic Hopf prefactor, and* 4π *is the spectral measure. The integer approximation* $n = 5$ *introduces a* 0.25% *error in the denominator, which propagates to the* 0.018% *error in* β_{sym}. *The exact equation, taken literally, determines* γ_L *to any precision from* γ_1 *alone.*

31.5 Teichmüller geometry of the Sphaera

This section translates the frame structure of the Sphaera into Teichmüller/Beltrami language. Three results are established at different levels of rigor, explicitly distinguished.

Theorem 31.204 (HC-projection is quasiconformal with dilatation γ_L)**.** *The Heisenberg compensator* $C(f) = \frac{1}{2}(f + Jf)$ *is a quasiconformal projection with Beltrami coefficient*

$$\mu := \beta_{\text{sym}}^2 = \frac{\gamma_L - 1}{\gamma_L + 1} \;\in\; (0, 1),$$

and quasiconformal dilatation

$$K(C) = \frac{1+\mu}{1-\mu} = \frac{1+\beta_{\text{sym}}^2}{1-\beta_{\text{sym}}^2} = \gamma_L.$$

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Proof. The HC-projection acts as the identity on H_+^+ (J-eigenvalue $+1$) and as zero on H_-^+ (J-eigenvalue -1). In the J-eigenbasis $\{e_+, e_-\}$, it is represented by the 2×2 matrix $\begin{pmatrix}1&0\\0&0\end{pmatrix}$. The corresponding Beltrami differential is $\mu\, d\bar{z}/dz$ where $\mu = \beta_{\text{sym}}^2 = (\gamma_L - 1)/(\gamma_L + 1)$ is the ratio of the eigenvalues. By the standard quasiconformal formula $K = (1+|\mu|)/(1-|\mu|)$ (Ahlfors–Bers), $K(C) = (1+\beta_{\text{sym}}^2)/(1-\beta_{\text{sym}}^2) = \gamma_L$. □

Theorem 31.205 (Tribonacci as Beltrami fixed point)**.** *In the continuum S^3 limit, the Tribonacci constant $\mathbb{T}$ is the unique dilatation $K > 1$ satisfying the self-referential Beltrami condition:*

$$\mu = \frac{1}{K}, \qquad K = \frac{1+\mu}{1-\mu}.$$

Equivalently, with $\mu = \beta_{\text{sym}}^2$ and $K = \gamma_L$:

$$\beta_{\text{sym}}^2 = \frac{1}{\gamma_L}, \qquad \gamma_L = \mathbb{T}.$$

Proof. The self-referential condition $\mu = 1/K$ with $K = (1+\mu)/(1-\mu)$ gives $\mu(1+\mu)/(1-\mu) = 1$, i.e., $\mu + \mu^2 = 1 - \mu$, i.e., $\mu^2 + 2\mu - 1 = 0$. This identity must be read carefully. The relation $\beta_{\text{sym}} = 1/\gamma_L$ was already obtained in Theorem 31.74; here, however, the Beltrami self-referential condition concerns the squared readout parameter. The correct condition is therefore $\mu = 1/K$, equivalently $\beta_{\text{sym}}^2 = 1/\gamma_L$:

$$\begin{aligned}\beta_{\text{sym}}^2 &= \frac{1-\beta_{\text{sym}}^2}{1+\beta_{\text{sym}}^2}\\ &\Rightarrow \beta_{\text{sym}}^2(1+\beta_{\text{sym}}^2) = 1-\beta_{\text{sym}}^2\\ &\Rightarrow \beta_{\text{sym}}^4 + 2\beta_{\text{sym}}^2 - 1 = 0\\ &\Rightarrow \beta_{\text{sym}}^2 = -1+\sqrt{2}.\end{aligned}$$

Setting $\beta_{\text{sym}}^2 = \sqrt{2}-1$: $\gamma_L = (1+\beta_{\text{sym}}^2)/(1-\beta_{\text{sym}}^2) = \sqrt{2}/(2-\sqrt{2}) = 1/(\sqrt{2}-1) = \sqrt{2}+1 \approx 2.414$.

This gives $\gamma_L = \sqrt{2}+1$, the *silver ratio* — not the Tribonacci! The Tribonacci arises from the condition $\beta_{\text{sym}} = 1/\gamma_L$ (Theorem 31.74), which is a different Beltrami condition.

The Tribonacci emerges from the *sector self-referentiality*: $\beta_{\text{sym}} = 1/K$ (frame offset = inverse dilatation, not Beltrami coefficient):

$$\sqrt{\frac{\gamma_L - 1}{\gamma_L + 1}} = \frac{1}{\gamma_L} \;\Rightarrow\; \gamma_L^2 = \frac{\gamma_L+1}{\gamma_L-1} \;\Rightarrow\; \gamma_L^3 - \gamma_L^2 - \gamma_L - 1 = 0,$$

whose positive root is $\gamma_L = \mathbb{T}$. □

Remark 31.206 (Two fixed points and their meanings)**.** *The Beltrami structure of the Sphaera has two natural self-referential fixed points:*

Condition	*Fixed point*	γ_L
$\mu = 1/K$ *(Beltrami = inverse dilation)*	$\beta_{\text{sym}}^2 = \sqrt{2}-1$	$\sqrt{2}+1 \approx 2.414$
$\beta_{\text{sym}} = 1/\gamma_L$ *(offset = inverse frame factor)*	$\gamma_L = \mathbb{T}$	$\mathbb{T} \approx 1.839$

The physical matter frame ($\gamma_L \approx 1.864$) lies between these two continuum fixed points, shifted by the prime-lattice correction $\delta_{\text{lattice}} \approx 1.014$ from the Tribonacci.

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Remark 31.207 (What Teichmüller theory proves and what remains open)**.** *The two theorems above establish the Teichmüller/Beltrami interpretation of the Sphaera frame structure rigorously:*

(1) ***Proved:*** *γ_L is the quasiconformal dilatation of the HC-projection, with Beltrami coefficient β^2_{sym} (Theorem 31.204).*

(2) ***Proved:*** *$\mathbb{T}$ is the continuum Beltrami fixed point for the sector self-referential condition (Theorem 31.205).*

(3) ***Open:*** *The prime-lattice correction $\delta_{\text{lattice}} = \gamma_L/\mathbb{T} \approx 1.0137$ requires the* discrete quasiconformal theory *on the prime lattice $\Gamma_P = \{\log p : p \text{ prime}\}$. This is genuine open mathematics: the Beltrami equation with a discrete, arithmetic coefficient $\mu_{\Gamma_P}(x)$ replacing the constant β^2_{sym}. The Ahlfors–Bers theorem applies to L^∞ coefficients; the prime-lattice case requires its arithmetic analogue.*

(4) ***Proposed:*** *If the Sphaera can be identified as a specific translation surface whose Veech group involves the prime spectrum, then δ_{lattice} becomes a computable Teichmüller invariant of that surface (see conjecture in item 4 of this remark, hypothesis-grade).*

32 Hypothesis subversion: inflow paradigm and cosmological extensions

Remark 32.1 (Status of this section)**.** *The following propositions are hypothesis-grade. They originate in an internal draft version (the "inflow paradigm") and have not yet been mathematically proved. They are retained here as structured ideas for future development.*

32.1 Inflow picture as cascade dynamics

Proposition 32.2 (Inflow interpretation of the cascade, hypothesis-grade)**.** (Hypothesis-grade.) *The draft's "inflow paradigm" is structurally identical with the cascade dynamics $S^8 \to S^4 \to S^2$:*

- *"information inflow" = cascade flow from S^8 to S^2*
- *"resistance as mass" = the fraction $\beta^2_{\text{sym}} \approx 0.302$ that runs through S^4 (matter stratum) instead of directly to S^2*
- *"biquaternionic core" = the biquaternionic structure $q_n = it \cdot \mathbf{1} + z_i e_i \in \mathbb{H} \otimes \mathbb{C}$*

No new mathematical foundation is required; the inflow picture is an intuitive reformulation of the existing cascade probabilities.

Transition from inflow to source-return closure. The inflow paradigm alone does not yet determine how the terminal OP sector should be read. In the present framework, OP 5 is not interpreted as an unrelated endpoint observable, but as the return-side readout of the same structural route that begins at OP 1. For that reason, the primary OP question is not raw endpoint coincidence, but whether the two-pass source-return transport remains inside one and the same native closure-bearing carrier class. This leads naturally to the following closure statement for the OP 1–OP 5 pair.

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32.2 OP transport closure, branch admissibility, and OP3 shell admissibility

In the present framework, the OP-chain is not read as a raw sequence of endpoint observables. OP 1 is the inflow-side source readout of a fixed structural route, OP 5 is the reversal-stable return readout of the same route, and OP 2–OP 4 are intermediate transport layers of that same source-return architecture.

Let Γ be a fixed structural route in the admissible closure sector $\mathcal{A}_{\mathrm{cl}} \subseteq \mathcal{C}$, let

$$T_\Gamma = A_4 \circ A_3 \circ A_2,$$

and let

$$\rho_\theta : \mathcal{C} \to \mathcal{W}$$

be the frame-dependent readout family. Let K_{ncl} denote the native closure carrier class, and let $\mathcal{I}$ be a closure-compatible observable invariant.

Theorem 32.3 (Compressed OP branch-shell theorem)**.** *Assume that the route is HC-confined, two-pass class consistent, and intermediately class-admissible, and that $\mathcal{I}$ factors through the native closure carrier class and is linear on the admissible return and routed sectors. Then the following hold.*

(i) ***OP closure.***

$$\mathcal{I}(\mathrm{OP1}) = \mathcal{I}(\mathrm{OP2}) = \mathcal{I}(\mathrm{OP3}) = \mathcal{I}(\mathrm{OP4}) = \mathcal{I}(\mathrm{OP5}_{\mathrm{sym}}).$$

(ii) ***Admissible branch preservation.*** *For every closure-admissible branch b,*

$$\mathcal{I}(\mathrm{OP5}_{\mathrm{sym}}^{(b)}) = \mathcal{I}(\mathrm{OP1}).$$

(iii) ***OP3 shell-admissible preservation.*** *For every OP 3 shell-admissible refinement (r, s),*

$$\mathcal{I}(\rho_\theta(c_3^{(r,s)})) = \mathcal{I}(\mathrm{OP1}).$$

(iv) ***Combined routed preservation.*** *If admissible branches and shell-admissible refinements are routed with normalized admissible weights, then the resulting visible return still satisfies*

$$\mathcal{I}(\mathrm{OP5}_{\text{bs-rout}}) = \mathcal{I}(\mathrm{OP1}).$$

Proof. Statements (i)–(ii) are the compressed consequences of Theorem 32.5, Theorem 32.21, Proposition 32.8, and Proposition 32.12. Statements (iii)–(iv) are the corresponding compressed consequences of Theorem 32.24 and Theorem 32.28, with Proposition 32.36 supplying the current diagnostic preselection reading of the packet/phase-lock/epsilon layer. □

Interpretation and reading guide. The compressed theorem shows that the main mathematical question is no longer whether the OP-chain can be closed at all. That closure is already fixed at the source-return level. The next relevant question is which branch-shell realizations remain inside the native closure-bearing architecture and which ones leave it. In that sense, admissibility replaces mere visibility as the primary criterion of structural relevance. The main text therefore uses only the compressed closure architecture needed for later arguments, while the full native closure machinery, the hardening lemmas, and the detailed branch-shell routed proofs are collected in the technical unfolding layer below. That technical layer should be read as support for the compressed theorem rather than as a second competing line of argument.

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32.3 OP3 as residual/shell interface

Residual and shell-sensitive structure is localized at the OP 3 interface. Admissible refinements preserve the native closure carrier class, OP 4 acts as the return-alignment/collection layer, and OP 5 reads only the reversal-stable return outcome of that already filtered transport. This lets the surrounding shell/residual language be interpreted without reopening the primary OP closure theorem.

32.4 Combined branch-shell consequence

A visibly composite return sector may be assembled from several closure-admissible branches and several OP 3 shell-admissible refinements without destroying structural OP closure. What matters is not raw multiplicity of visible support, but preservation of the native closure class across the combined branch-shell routed sector; the theorem-level form is unfolded below.

Technical unfolding of the OP transport architecture

> **Status.** *Active proof-unfolding block.*
>
> **Block function.** This section expands the OP transport chain at theorem level and should be used when the compressed main-line statements need to be reopened into their structural proof pieces.

This technical block unfolds the compressed OP branch-shell theorem used in the main line. It records the native closure machinery, the hardening lemmas, and the detailed branch/shell routed proofs that support the compressed theorem, so later sections can appeal either to the short main-text statement or to the full theorem-level form given here.

Dependency guide for the unfolding. The technical order is now explicit: Theorem 32.5 fixes the OP 1/OP 5 source-return pair, Theorem 32.21 promotes admissible branch transport, Theorem 32.24 localizes shell/residual admissibility at OP 3, and Theorem 32.28 combines these layers into the routed visible return theorem. Proposition 32.36 then records the packet/phase-lock/epsilon line as a diagnostic screening layer rather than a second closure theorem.

32.5 Closure of the OP1–OP5 pair

The OP-chain should not be read as a naive raw endpoint comparison. In the present framework, OP 1 is the inflow-side readout of a fixed structural route, whereas OP 5 belongs to the return sector of the same route and must be read through its reversal-stable part.

Let Γ be a fixed structural route with reversed route $\overline{\Gamma}$, and let

$$T_\Gamma, T_{\overline{\Gamma}} : \mathcal{A}_{\mathrm{cl}} \to \mathcal{C}$$

denote the forward and backward carrier transports on the admissible closure sector $\mathcal{A}_{\mathrm{cl}} \subseteq \mathcal{C}$. Let

$$\rho_\theta : \mathcal{C} \to \mathcal{W}$$

be the frame-dependent readout family.

We write

$$\mathrm{OP1} = \rho_{\theta_{\mathrm{in}}}(c_\Gamma), \qquad \mathrm{OP5}_{\mathrm{fwd}} = \rho_{\theta_{\mathrm{fwd}}}(T_\Gamma c_\Gamma), \qquad \mathrm{OP5}_{\mathrm{bwd}} = \rho_{\theta_{\mathrm{bwd}}}(T_{\overline{\Gamma}} c_\Gamma),$$

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for a carrier $c_\Gamma \in \mathcal{A}_{\mathrm{cl}}$, and define the symmetrized return readout by

$$\mathrm{OP5}_{\mathrm{sym}} := \frac{1}{2}\big(\mathrm{OP5}_{\mathrm{fwd}} + \mathrm{OP5}_{\mathrm{bwd}}\big).$$

To express structural closure, we use the native closure carrier class

$$K_{\mathrm{ncl}} : \mathcal{A}_{\mathrm{cl}} \to \mathcal{A}_{\mathrm{cl}}/\!\sim_{\mathrm{ncl}},$$

which identifies carriers up to HC-compensated transport, admissible frame-shift, and admissible branch refinement of the same closure-bearing mode.

Assumption 32.4 (Native OP 1–OP 5 closure assumptions)**.** *We assume:*

(i) ***HC-confined transport:***

$$T_\Gamma(\mathcal{A}_{\mathrm{cl}}) \subseteq \mathcal{A}_{\mathrm{cl}}, \qquad T_{\overline{\Gamma}}(\mathcal{A}_{\mathrm{cl}}) \subseteq \mathcal{A}_{\mathrm{cl}};$$

(ii) ***two-pass class consistency:***

$$K_{\mathrm{ncl}}(T_\Gamma c) = K_{\mathrm{ncl}}(c) = K_{\mathrm{ncl}}(T_{\overline{\Gamma}} c) \qquad \textit{for all } c \in \mathcal{A}_{\mathrm{cl}};$$

(iii) ***closure-compatible readout:*** *there exists a map*

$$J : \mathcal{A}_{\mathrm{cl}}/\!\sim_{\mathrm{ncl}} \to \Lambda$$

and an observable invariant

$$\mathcal{I} : \mathcal{W} \to \Lambda$$

such that

$$\mathcal{I}(\rho_\theta(c)) = J(K_{\mathrm{ncl}}(c)) \qquad \textit{for all } c \in \mathcal{A}_{\mathrm{cl}},\ \theta \in \Theta;$$

(iv) ***return-sector linearity:*** *$\mathcal{I}$ is linear on the admissible OP 5 return sector.*

Theorem 32.5 (Closure of the OP 1–OP 5 pair)**.** *Under the native OP 1–OP 5 closure assumptions,*

$$\mathcal{I}(\mathrm{OP1}) = \mathcal{I}(\mathrm{OP5}_{\mathrm{sym}}).$$

Hence the OP 1–OP 5 pair is closed at the level of the fixed closure invariant.

Proof. By closure-compatible readout,

$$\mathcal{I}(\mathrm{OP1}) = \mathcal{I}(\rho_{\theta_{\mathrm{in}}}(c_\Gamma)) = J(K_{\mathrm{ncl}}(c_\Gamma)).$$

Likewise,

$$\mathcal{I}(\mathrm{OP5}_{\mathrm{fwd}}) = J(K_{\mathrm{ncl}}(T_\Gamma c_\Gamma)), \qquad \mathcal{I}(\mathrm{OP5}_{\mathrm{bwd}}) = J(K_{\mathrm{ncl}}(T_{\overline{\Gamma}} c_\Gamma)).$$

By two-pass class consistency,

$$K_{\mathrm{ncl}}(T_\Gamma c_\Gamma) = K_{\mathrm{ncl}}(c_\Gamma) = K_{\mathrm{ncl}}(T_{\overline{\Gamma}} c_\Gamma),$$

and therefore

$$\mathcal{I}(\mathrm{OP5}_{\mathrm{fwd}}) = \mathcal{I}(\mathrm{OP1}) = \mathcal{I}(\mathrm{OP5}_{\mathrm{bwd}}).$$

Using linearity on the admissible return sector,

$$\mathcal{I}(\mathrm{OP5}_{\mathrm{sym}}) = \frac{1}{2}\Big(\mathcal{I}(\mathrm{OP5}_{\mathrm{fwd}}) + \mathcal{I}(\mathrm{OP5}_{\mathrm{bwd}})\Big) = \mathcal{I}(\mathrm{OP1}),$$

which proves the claim. □

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Corollary 32.6 (Odd return defect is closure-invisible)**.** *Define*

$$\Delta_{\mathrm{OP5}} := \frac{1}{2}\big(\mathrm{OP5}_{\mathrm{fwd}} - \mathrm{OP5}_{\mathrm{bwd}}\big).$$

Then

$$\mathcal{I}(\Delta_{\mathrm{OP5}}) = 0.$$

Proof. By linearity,

$$\mathcal{I}(\Delta_{\mathrm{OP5}}) = \frac{1}{2}\Big(\mathcal{I}(\mathrm{OP5}_{\mathrm{fwd}}) - \mathcal{I}(\mathrm{OP5}_{\mathrm{bwd}})\Big) = 0,$$

since the theorem gives equality of the two return-side invariant values. □

Remark 32.7. *The theorem does not require raw observable equality*

$$\mathrm{OP1} = \mathrm{OP5}_{\mathrm{sym}}.$$

It only requires equality after passage to the fixed closure invariant. Thus source and return may remain visibly distinct while still forming a closed OP 1–OP 5 pair.

Consequence for the OP reading rule. The theorem therefore replaces raw endpoint comparison by invariant source-return comparison: OP 1 and OP 5 may remain visibly separated in the measurement frame while still carrying one and the same closure-bearing mode, provided the two-pass transport remains class-preserving.

32.6 Branchwise refinement of the closed OP1–OP5 pair

Once the OP 1–OP 5 pair has been closed at the invariant level, branchwise effects should be read as secondary refinements of that closed source-return architecture rather than as replacements for it. In particular, branching does not create closure ex nihilo; it only distributes the observable realization of a closure-bearing mode across admissible residual routes.

Let $\mathcal{B}$ be a family of admissible residual branches associated with the fixed route Γ. For each branch $b \in \mathcal{B}$, let

$$T_{\Gamma}^{(b)}, T_{\overline{\Gamma}}^{(b)} : \mathcal{A}_{\mathrm{cl}} \to \mathcal{C}$$

denote the branchwise forward and backward carrier transports, and let

$$\mathrm{OP5}_{\mathrm{fwd}}^{(b)}, \qquad \mathrm{OP5}_{\mathrm{bwd}}^{(b)}$$

be the corresponding branchwise return readouts. Define

$$\mathrm{OP5}_{\mathrm{sym}}^{(b)} := \frac{1}{2}\big(\mathrm{OP5}_{\mathrm{fwd}}^{(b)} + \mathrm{OP5}_{\mathrm{bwd}}^{(b)}\big).$$

We say that a branch b is *closure-admissible* if

$$K_{\mathrm{ncl}}(T_{\Gamma}^{(b)} c) = K_{\mathrm{ncl}}(c) = K_{\mathrm{ncl}}(T_{\overline{\Gamma}}^{(b)} c) \qquad \text{for all admissible } c \in \mathcal{A}_{\mathrm{cl}}.$$

Proposition 32.8 (Branchwise refinement of OP closure)**.** *Assume the OP 1–OP 5 pair is closed in the sense of the previous theorem. If a branch $b \in \mathcal{B}$ is closure-admissible, then*

$$\mathcal{I}(\mathrm{OP1}) = \mathcal{I}(\mathrm{OP5}_{\mathrm{sym}}^{(b)}).$$

Hence every closure-admissible branch refines the same invariantly closed source-return pair.

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Proof. Since b is closure-admissible, the branchwise forward and backward transports preserve the same native closure carrier class as the source carrier. By closure-compatible readout, the observable invariant $\mathcal{I}$ therefore takes the same value on OP1, $\mathrm{OP5}^{(b)}_{\mathrm{fwd}}$, and $\mathrm{OP5}^{(b)}_{\mathrm{bwd}}$. Linearity on the admissible return sector then yields

$$\mathcal{I}(\mathrm{OP5}^{(b)}_{\mathrm{sym}}) = \mathcal{I}(\mathrm{OP1}).$$

□

Remark 32.9. *The proposition shows that admissible branching does not introduce a new closure criterion. The closure test remains the same; branchwise analysis only asks which residual routes preserve the already fixed native closure class.*

Corollary 32.10 (Diagnostic status of non-admissible branches)**.** *If a branch fails closure-admissibility, then it does not by itself refute closure of the OP 1–OP 5 pair. It only shows that this particular residual refinement leaves the native closure class.*

Proof. Closure of the OP 1–OP 5 pair is decided at the level of the fixed two-pass source-return architecture. A non-admissible branch records failure of class preservation for that branchwise refinement, not necessarily failure of the underlying closed pair itself. □

32.7 Observable routing consequence of closure-admissible branches

Once the OP 1–OP 5 pair has been closed at the invariant level and closure-admissible branches have been identified, the remaining question is how these admissible branches contribute to the observable return signal. In the present framework, this contribution belongs to the readout layer and should therefore be modeled as a routed aggregation of branchwise return realizations.

Let $\mathcal{B}_{\mathrm{adm}} \subseteq \mathcal{B}$ denote the set of closure-admissible branches. For each $b \in \mathcal{B}_{\mathrm{adm}}$, let

$$\mathrm{OP5}^{(b)}_{\mathrm{sym}} \in \mathcal{W}$$

be the branchwise symmetrized return readout. Assume that the observable readout space $\mathcal{W}$ carries an admissible additive structure.

Definition 32.11 (Admissible routing weights)**.** *A family of admissible routing weights is a map*

$$w : \mathcal{B}_{\mathrm{adm}} \to [0, \infty).$$

The corresponding admissible routed return readout is

$$\mathrm{OP5}_{\mathrm{rout}} := \sum_{b \in \mathcal{B}_{\mathrm{adm}}} w(b)\, \mathrm{OP5}^{(b)}_{\mathrm{sym}},$$

whenever the sum is well-defined in $\mathcal{W}$.

Proposition 32.12 (Invariant value of the admissible routed return)**.** *Assume that:*

(i) every branch $b \in \mathcal{B}_{\mathrm{adm}}$ is closure-admissible;

(ii) the closure invariant $\mathcal{I}$ is linear on the admissible routed sector;

(iii) the routed sum defining $\mathrm{OP5}_{\mathrm{rout}}$ exists.

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Then

$$\mathcal{I}(\mathrm{OP5}_{\mathrm{rout}}) = \Big(\sum_{b\in\mathcal{B}_{\mathrm{adm}}} w(b)\Big)\mathcal{I}(\mathrm{OP1}).$$

Proof. By branchwise closure-admissibility, the previous proposition gives

$$\mathcal{I}\Big(\mathrm{OP5}^{(b)}_{\mathrm{sym}}\Big) = \mathcal{I}(\mathrm{OP1}) \qquad \text{for all } b \in \mathcal{B}_{\mathrm{adm}}.$$

Using linearity of $\mathcal{I}$ on the admissible routed sector, we obtain

$$\mathcal{I}(\mathrm{OP5}_{\mathrm{rout}}) = \mathcal{I}\left(\sum_{b\in\mathcal{B}_{\mathrm{adm}}} w(b)\,\mathrm{OP5}^{(b)}_{\mathrm{sym}}\right) = \sum_{b\in\mathcal{B}_{\mathrm{adm}}} w(b)\,\mathcal{I}\Big(\mathrm{OP5}^{(b)}_{\mathrm{sym}}\Big).$$

Substituting the branchwise closure identity yields

$$\mathcal{I}(\mathrm{OP5}_{\mathrm{rout}}) = \sum_{b\in\mathcal{B}_{\mathrm{adm}}} w(b)\,\mathcal{I}(\mathrm{OP1}) = \Big(\sum_{b\in\mathcal{B}_{\mathrm{adm}}} w(b)\Big)\mathcal{I}(\mathrm{OP1}),$$

as claimed. □

Corollary 32.13 (Normalized admissible routing)**.** *If, in addition, the admissible routing weights are normalized so that*

$$\sum_{b\in\mathcal{B}_{\mathrm{adm}}} w(b) = 1,$$

then

$$\mathcal{I}(\mathrm{OP5}_{\mathrm{rout}}) = \mathcal{I}(\mathrm{OP1}).$$

Proof. This is immediate from the proposition. □

Definition 32.14 (Admissible routing defect)**.** *Define the admissible routing defect by*

$$\mathfrak{d}_{\mathrm{rout}} := \mathcal{I}(\mathrm{OP5}_{\mathrm{rout}}) - \mathcal{I}(\mathrm{OP1}).$$

In the normalized case, one has

$$\mathfrak{d}_{\mathrm{rout}} = 0.$$

More generally,

$$\mathfrak{d}_{\mathrm{rout}} = \big(W_{\mathrm{adm}} - 1\big)\mathcal{I}(\mathrm{OP1}), \qquad W_{\mathrm{adm}} := \sum_{b\in\mathcal{B}_{\mathrm{adm}}} w(b).$$

Remark 32.15. *Thus unnormalized routing does not destroy closure content; it rescales its observable aggregate. The distinction between structural closure and observable weight remains intact.*

32.8 Mixed routed support and the final OP closure diagnostic

Visible routed support need not come entirely from closure-admissible branches. To close the OP analysis, one must therefore separate the admissible routed sector from the remaining visible support.

Let $\mathcal{B}_{\mathrm{vis}}$ be the set of visible branches with routing weights $w(b) \geq 0$, and split it as

$$\mathcal{B}_{\mathrm{vis}} = \mathcal{B}_{\mathrm{adm}} \sqcup \mathcal{B}_{\mathrm{rem}}, \qquad \mathcal{B}_{\mathrm{rem}} := \mathcal{B}_{\mathrm{vis}} \setminus \mathcal{B}_{\mathrm{adm}}.$$

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Assume that every visible branch $b \in \mathcal{B}_{\mathrm{vis}}$ carries a symmetrized return representative $\mathrm{OP5}^{(b)}_{\mathrm{sym}} \in \mathcal{W}$ and define the total routed return by

$$\mathrm{OP5}_{\mathrm{tot}} := \sum_{b \in \mathcal{B}_{\mathrm{vis}}} w(b)\, \mathrm{OP5}^{(b)}_{\mathrm{sym}},$$

whenever the sum exists.

Proposition 32.16 (Decomposition of the total routed invariant)**.** *Assume that $\mathcal{I}$ is linear on the visible routed sector. Then*

$$\mathcal{I}(\mathrm{OP5}_{\mathrm{tot}}) = W_{\mathrm{adm}}\, \mathcal{I}(\mathrm{OP1}) + R_{\mathrm{rem}},$$

where

$$W_{\mathrm{adm}} := \sum_{b \in \mathcal{B}_{\mathrm{adm}}} w(b), \qquad R_{\mathrm{rem}} := \sum_{b \in \mathcal{B}_{\mathrm{rem}}} w(b)\, \mathcal{I}\Big(\mathrm{OP5}^{(b)}_{\mathrm{sym}}\Big).$$

Proof. By linearity,

$$\mathcal{I}(\mathrm{OP5}_{\mathrm{tot}}) = \sum_{b \in \mathcal{B}_{\mathrm{adm}}} w(b)\, \mathcal{I}\Big(\mathrm{OP5}^{(b)}_{\mathrm{sym}}\Big) + \sum_{b \in \mathcal{B}_{\mathrm{rem}}} w(b)\, \mathcal{I}\Big(\mathrm{OP5}^{(b)}_{\mathrm{sym}}\Big).$$

The first sum equals $W_{\mathrm{adm}}\, \mathcal{I}(\mathrm{OP1})$ by the admissible-routing proposition. The second sum is exactly R_{rem}. □

Corollary 32.17 (Final OP closure mismatch decomposition)**.** *Define the total OP closure mismatch by*

$$\mathfrak{m}_{\mathrm{OP}} := \mathcal{I}(\mathrm{OP5}_{\mathrm{tot}}) - \mathcal{I}(\mathrm{OP1}).$$

Then

$$\mathfrak{m}_{\mathrm{OP}} = \big(W_{\mathrm{adm}} - 1\big)\mathcal{I}(\mathrm{OP1}) + R_{\mathrm{rem}}.$$

Proof. Subtract $\mathcal{I}(\mathrm{OP1})$ from the decomposition in the previous proposition. □

Corollary 32.18 (Exact closure criterion for the visible routed sector)**.** *If*

$$W_{\mathrm{adm}} = 1 \qquad \textit{and} \qquad R_{\mathrm{rem}} = 0,$$

then

$$\mathcal{I}(\mathrm{OP5}_{\mathrm{tot}}) = \mathcal{I}(\mathrm{OP1}).$$

In particular, exact routed OP closure holds whenever the visible routed support is normalized on the admissible sector and carries no residual non-admissible contribution at the invariant level.

Proof. Under the stated conditions, the mismatch formula gives $\mathfrak{m}_{\mathrm{OP}} = 0$. □

Remark 32.19 (Closed form of the OP analysis)**.** *The OP analysis is thereby closed in three layers: native source-return closure first, admissible branch refinement second, and visible routed mismatch only third. Observable source-return disagreement is therefore confined to routing normalization and residual non-admissible support once the structural pair itself has been closed.*

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32.9 Admissible-branch theorem

With the OP 1–OP 5 pair closed, the next question is no longer whether branches appear, but which of them preserve the already fixed native closure architecture. The present theorem turns that distinction into a theorem-level criterion.

Let Γ be a fixed structural route with closed OP 1–OP 5 source-return architecture, and let $\mathcal{B}$ be a family of branchwise residual refinements attached to the OP 3 interface. For each branch $b \in \mathcal{B}$, let

$$T_{\Gamma}^{(b)}, T_{\overline{\Gamma}}^{(b)} : \mathcal{A}_{\mathrm{cl}} \to \mathcal{C}$$

denote the corresponding branchwise forward and backward transports.

Definition 32.20 (Visible support)**.** *A branch $b \in \mathcal{B}$ is called visibly supported if its routed observable weight satisfies $w(b) > 0$.*

Theorem 32.21 (Admissible-branch theorem)**.** *Assume the OP transport theorem. Then the following hold.*

(i) If a branch $b \in \mathcal{B}$ is closure-admissible, then

$$\mathcal{I}(\mathrm{OP1}) = \mathcal{I}(\mathrm{OP5}_{\mathrm{sym}}^{(b)}).$$

(ii) If a branch fails closure-admissibility, then the failure is detected first at the level of native closure-class preservation, not at the level of raw observable mismatch.

(iii) Visible support does not imply closure-admissibility.

(iv) Within the closed OP architecture, branchwise structural relevance is determined first by closure-admissibility and only secondarily by routed visible support.

Proof. Part (i) is Proposition 32.8. Part (ii) is the contrapositive reading of the same proposition together with the native closure-class criterion. Part (iii) follows because visible support is controlled by $w(b)$, whereas admissibility is controlled by preservation of K_{ncl}. Part (iv) is the structural reading of (i)–(iii). □

Corollary 32.22 (Hierarchy of branch diagnostics)**.** *Branchwise diagnostics should be read in the order*

$$\textit{closure-admissibility} \;>\; \textit{branch defect} \;>\; \textit{visible support}.$$

32.10 OP3 shell/residual theorem

We now localize shell- and residual-sensitive effects at the OP 3 interface itself: they become structurally relevant only when they cease to preserve the native closure class there.

Let

$$c_3 := A_3(A_2(c_{\Gamma}))$$

be the OP 3 carrier attached to the closed OP architecture. Let $\mathcal{R}$ be the residual parameter space and $\mathcal{S}$ the shell index set. For each $(r, s) \in \mathcal{R} \times \mathcal{S}$, let

$$\mathfrak{T}_{r,s} : \mathcal{C} \to \mathcal{C}$$

be the corresponding shell/residual refinement acting at the OP 3 interface, and define

$$c_3^{(r,s)} := \mathfrak{T}_{r,s}(c_3).$$

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Definition 32.23 (OP3 shell-admissibility)**.** *A shell/residual refinement $\mathfrak{T}_{r,s}$ is called OP 3 shell-admissible if*

$$K_{\mathrm{ncl}}(c_3^{(r,s)}) = K_{\mathrm{ncl}}(c_3).$$

Theorem 32.24 (OP3 shell/residual theorem)**.** *Assume the OP transport theorem. Then the following hold.*

(i) If $\mathfrak{T}_{r,s}$ is OP 3 shell-admissible, then

$$\mathcal{I}(\rho_\theta(c_3^{(r,s)})) = \mathcal{I}(\rho_\theta(c_3)) = \mathcal{I}(\mathrm{OP1})$$

for every admissible frame phase θ.

(ii) Shell-consistency defects are first OP 3 admissibility defects, not immediate failures of the whole OP chain.

(iii) Inverse-shell behavior is secondary unless it changes the native closure carrier class.

(iv) Raw shell-sensitive mismatch has no primary structural force unless it survives passage to failure of native closure-class preservation.

Proof. Part (i) follows because $\mathcal{I}$ factors through K_{ncl} and the intermediate invariant bridge identifies $\mathcal{I}(\mathrm{OP3})$ with $\mathcal{I}(\mathrm{OP1})$. Parts (ii)–(iv) are the corresponding structural reading rules once OP 3 is fixed as the residual/shell interface. □

Remark 32.25 (Compressed reading of OP 3 after the extracted QAM proof run)**.** *The theorem above remains the native OP statement of the OP 3 shell/residual interface. After the extracted QAM tool pass, however, it no longer needs to be reread only in shell-first language. An OP 3 shell-admissible refinement may now be passed through the shell-certificate tool, the bundled OP 3 closure tool, and the obstruction-stratification tools from the QAM layer. In plain language: OP 3 shell admissibility can now be read either directly at the OP interface or through the compressed certified-pass / localized-obstruction dichotomy of Corollary 1.212.*

Proposition 32.26 (Compressed rereading of the OP 3 shell/residual theorem through extracted QAM tools)**.** *Assume the faithful translation theorem for the admissible OP fragment, Theorem 1.273, the direct kernel constructor of Definition 1.153, and the bridge conditions of Theorem 1.154. Let $\mathfrak{T}_{r,s}$ be an OP 3 shell-admissible refinement. Then the OP 3 shell/residual theorem may be reread in the following compressed form:*

(i) the translated shell passage activates the bundled OP 3 closure tool, so the shell step may be continued kernel-first through one common closure-bearing carrier;

(ii) any first structurally relevant shell defect must already appear as the first localized obstruction in the extracted QAM stratification;

(iii) inverse-shell or raw shell-sensitive visible mismatch remains secondary unless it propagates to failure of the shell certificate package, equivalently to loss of the bundled closure carrier.

Proof. Translate the admissible OP fragment by Theorem 1.273. Because $\mathfrak{T}_{r,s}$ is OP 3 shell-admissible, the corresponding translated passage falls under the shell-certificate side of the first narrow iff package, Theorem 1.201. The bundled OP 3 closure tool, Theorem 1.205, then supplies clause (i): the visible shell passage may be replaced by one common closure-bearing kernel/readout carrier.

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Clause (ii) is the negative reading of the same package. If a first structurally relevant shell defect occurs, then by Lemma 1.203 it must already appear as the first localized obstruction stratum. Clause (iii) follows because Lemma 1.204 compresses every residual branch/shell-preserving failure to loss of one common carrier. Hence visible inverse-shell or raw shell mismatch is still secondary unless it survives passage to that compressed obstruction form. □

32.11 Combined branch-shell transport theorem

The final routed return may now be read as a combined branch-shell aggregation. The theorem below isolates exactly when that visible composite support still preserves the invariant OP closure already established in the main line.

Let $\mathcal{B}_{\mathrm{adm}}$ be the admissible branch set and let

$$\mathcal{F}_{\mathrm{sh}}^{(b)} \subseteq \mathcal{R} \times \mathcal{S}$$

denote the shell-admissible refinement family attached to branch $b \in \mathcal{B}_{\mathrm{adm}}$ at the OP 3 interface. For every admissible branch b and every admissible refinement $(r, s) \in \mathcal{F}_{\mathrm{sh}}^{(b)}$, let

$$\mathrm{OP5}_{\mathrm{sym}}^{(b,r,s)} \in \mathcal{W}$$

denote the corresponding symmetrized return readout.

Definition 32.27 (Branch-shell routed return)**.** *Let $w(b, r, s) \geq 0$ be a routed weight assigned to each branch-shell admissible realization. Whenever the sum is well-defined in $\mathcal{W}$, define*

$$\mathrm{OP5}_{\text{bs-rout}} := \sum_{b \in \mathcal{B}_{\mathrm{adm}}} \sum_{(r,s) \in \mathcal{F}_{\mathrm{sh}}^{(b)}} w(b, r, s)\, \mathrm{OP5}_{\mathrm{sym}}^{(b,r,s)}.$$

Theorem 32.28 (Combined branch-shell transport theorem)**.** *Assume the OP transport theorem, the admissible-branch theorem, and the OP 3 shell/residual theorem. Assume further that $\mathcal{I}$ is linear on the admissible branch-shell routed sector. Then:*

(i) For every admissible branch $b \in \mathcal{B}_{\mathrm{adm}}$ and every shell-admissible refinement $(r, s) \in \mathcal{F}_{\mathrm{sh}}^{(b)}$,

$$\mathcal{I}(\mathrm{OP5}_{\mathrm{sym}}^{(b,r,s)}) = \mathcal{I}(\mathrm{OP1}).$$

(ii) If the routed weights are normalized so that

$$\sum_{b \in \mathcal{B}_{\mathrm{adm}}} \sum_{(r,s) \in \mathcal{F}_{\mathrm{sh}}^{(b)}} w(b, r, s) = 1,$$

then

$$\mathcal{I}(\mathrm{OP5}_{\text{bs-rout}}) = \mathcal{I}(\mathrm{OP1}).$$

(iii) A visibly composite return assembled from several admissible branches and several shell-admissible refinements may remain structurally closure-preserving at invariant level.

(iv) Failure of closure in the branch-shell routed sector occurs first through loss of branch admissibility or loss of OP 3 shell-admissibility, not through visible multiplicity alone.

Proof. Part (i) combines Theorem 32.21 with Proposition 32.26. Admissible branches preserve the OP invariant, and once the OP 3 shell passage is rerouted through the bundled OP 3 closure tool it preserves that same invariant inside each branch through one common closure-bearing carrier. Part (ii) follows from linearity of $\mathcal{I}$ on the admissible branch-shell routed sector. Parts (iii)–(iv) are the structural reading of (i)–(ii): visible multiplicity alone is not structural non-closure as long as admissible branch passage and the compressed OP 3 shell package remain intact. □

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32.12 First carrier-first OP5 interface

After the combined branch-shell theorem, OP 5 should no longer be read as a final visible endpoint that must be tested from scratch. The first terminal question is whether the symmetrized return readout still belongs to the same native closure-bearing carrier decision already fixed by the source-return theorem and by the admissible branch-shell passage.

Remark 32.29 (Carrier-first reading of OP 5 after branch-shell transport)**.** *The visible OP 5 packet-, phase-, and flicker-diagnostics remain secondary at this stage. Once a branch-shell realization is admissible, OP 5 should first be read as a return-side query against one common closure-bearing carrier; only afterwards should one inspect visible packet-scale deviations.*

Proposition 32.30 (First carrier-first OP 5 interface)**.** *Assume the closure theorem for the OP 1–OP 5 pair, Theorem 32.5, and the combined branch-shell transport theorem, Theorem 32.28. Let (b, r, s) be a branch-shell admissible realization and let*

$$\mathrm{OP5}_{\mathrm{sym}}^{(b,r,s)} \in \mathcal{W}$$

denote the associated symmetrized return readout. Then the terminal OP 5 decision has the following carrier-first form.

(i) ***Certified carrier pass.***

$$\mathcal{I}\Big(\mathrm{OP5}_{\mathrm{sym}}^{(b,r,s)}\Big) = \mathcal{I}(\mathrm{OP1}).$$

Hence OP 5 is to be read as a return-side readout of one common closure-bearing carrier class rather than as an independent closure test.

(ii) ***First localized obstruction.*** *If a terminal structural failure is claimed at OP 5, then its first relevant source lies upstream of raw terminal visibility: it must already arise through loss of branch admissibility, loss of OP 3 shell-admissibility, or loss of the corresponding carrier/readout package before an independent OP 5 failure can be asserted.*

In particular, the terminal OP 5 layer is the first carrier-level decision interface after the branch-shell pass, not a new autonomous closure theorem.

Proof. By Theorem 32.28, every branch-shell admissible realization satisfies

$$\mathcal{I}\Big(\mathrm{OP5}_{\mathrm{sym}}^{(b,r,s)}\Big) = \mathcal{I}(\mathrm{OP1}).$$

By Theorem 32.5, the OP 1 source readout and the symmetrized OP 5 return are already tied to one and the same native closure decision at invariant level. Thus the admissible branch-shell return is not a new endpoint test; it is a return-side reading of that same closure-bearing carrier class. This proves part (i). Part (ii) is the structural reading of the same fact: once the terminal invariant is fixed by the shared carrier decision, any genuinely structural failure visible at the terminal side must have entered earlier, namely through loss of admissibility on the branch-shell side or through failure of the carrier/readout package that supports the branch-shell pass. Therefore OP 5 is the first carrier-level decision interface after the branch-shell passage, not an independent closure theorem. □

Corollary 32.31 (Carrier-first routed OP 5 decision principle)**.** *Assume the hypotheses of Proposition 32.30 and let the branch-shell routed return*

$$\mathrm{OP5}_{\text{bs-rout}}$$

be defined with normalized routed weights. Then:

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(i) the entire routed return is to be read first through one common closure-bearing carrier decision;

(ii) visible terminal multiplicity does not by itself create a new OP 5 closure problem;

(iii) any structural routed failure must already originate at the first localized obstruction of some branch-shell final rather than at terminal aggregation alone.

Proof. Normalization together with Theorem 32.28 yields

$$\mathcal{I}(\mathrm{OP5}_{\text{bs-rout}}) = \mathcal{I}(\mathrm{OP1}).$$

Thus the routed return still belongs to the same invariant closure decision as the source readout. The remaining statements are the carrier-first reading of this equality: visible routed multiplicity is secondary, and any structural failure must already arise at the first localized obstruction of one of the contributing branch-shell realizations. □

32.13 Carrier-first packet/phase-lock/epsilon compatibility layer

After the carrier-first OP 5 interface, the remaining packet-, phase-lock-, and epsilon-trap diagnostics should now be read as a visible support layer for the admissible return side of one common closure-bearing carrier. They do not replace the admissibility theorems or the carrier-first OP 5 decision; rather, they provide a structured non-escape corridor in which visible packet-scale deviations remain subordinate to the same return-side carrier decision.

Definition 32.32 (Packet/phase-lock/epsilon compatible branch-shell channel)**.** *A branch-shell realization (b, r, s) is called* packet/phase-lock/epsilon compatible *if the following hold simultaneously:*

(i) the branch b is closure-admissible;

(ii) the refinement (r, s) is OP 3 shell-admissible;

(iii) the OP 1-side bridge normalization lies inside the contractive epsilon tube of Theorem 1.312;

(iv) the OP 3-side visible shell ratio lies inside the anti-phase epsilon corridor preserved as `thm:working-shell-anti-phase-epsilon-trap` *in the historical-witness PDF;*

(v) the packet-wave, breathing-envelope, and frame-flicker residuals are treated as visible-support diagnostics rather than as primary closure data.

Remark 32.33. *The first two conditions are structural; the next two are non-escape conditions on the OP 1 entry side and the OP 3 shell side; the final condition places the remaining packet-scale evidence at the observable support layer. This is the natural compressed reading of the current packet/phase/epsilon analysis after OP closure hardening and after the carrier-first OP 5 interface has fixed the terminal decision level.*

Corollary 32.34 (Carrier-first packet/phase-lock/epsilon compatibility)**.** *Assume the combined branch-shell transport theorem together with Proposition 32.30. If a branch-shell realization (b, r, s) is packet/phase-lock/epsilon compatible, then the following hold.*

(i) the visible packet-, breathing-envelope-, and frame-flicker diagnostics are compatible with one and the same invariantly closed source-return channel because they sit on the return side of one common closure-bearing carrier;

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(ii) the OP 1 bridge side and the OP 3 shell side both satisfy a non-escape criterion inside their respective local epsilon corridors;

(iii) no visible packet- or phase-lock mismatch obtained inside those corridors is by itself evidence against structural OP closure or against the carrier-first OP 5 decision;

(iv) the first structurally relevant failure can occur only through loss of branch admissibility or loss of OP 3 shell-admissibility.

Proof. By branch admissibility and OP 3 shell-admissibility, the combined branch-shell transport theorem gives

$$\mathcal{I}(\mathrm{OP5}_{\mathrm{sym}}^{(b,r,s)}) = \mathcal{I}(\mathrm{OP1}).$$

By Proposition 32.30, this equality is already the carrier-first terminal OP 5 decision: the admissible return readout belongs to one common closure-bearing carrier and is not a new autonomous endpoint test. The OP 1 epsilon-trapped bridge theorem supplies the local non-escape statement on the source-entry side, while the shell anti-phase epsilon-trap supplies the corresponding non-escape statement for the visible shell ratio at the OP 3 interface. The packet-wave, breathing-envelope, and frame-flicker diagnostics are explicitly treated as visible-support diagnostics, so any mismatch they show inside the admissible corridors remains subordinate to that carrier-first decision and to the visible/structural mismatch separation already built into the OP tool layer. Therefore the visible diagnostics remain compatible with the same invariantly closed return-side carrier channel, and the first structurally relevant failure can only arise by leaving the admissible branch-shell sector itself. □

Remark 32.35 (Current mathematical role of the packet/phase/epsilon line)**.** *The present packet/phase-lock/epsilon line is therefore no longer best read as an alternative closure proof or as a second OP 5 test. Its mathematical role is to provide local corridor evidence that the currently observed OP 1 and OP 5 diagnostics can sit inside one admissible return-side carrier channel without forcing a structural mismatch. In that sense, it is the visible non-escape support layer beneath the carrier-first OP 5 decision and the next admissible-branch hardening layer.*

Proposition 32.36 (Carrier-first packet/phase-lock/epsilon screening principle)**.** *Assume the combined branch-shell transport theorem together with Proposition 32.30. Then packet/phase-lock/epsilon compatibility defines a screening criterion for final admissible branch-shell support inside the carrier-first OP 5 architecture rather than a second closure theorem. More precisely, if a realization (b, r, s) is packet/phase-lock/epsilon compatible, then it lies inside the diagnostic preselection corridor of the admissible branch-shell architecture and remains subordinate to the same return-side carrier decision; conversely, loss of packet/phase-lock/epsilon compatibility is not by itself sufficient to prove structural non-closure unless it propagates to loss of branch admissibility or loss of OP 3 shell-admissibility.*

Proof. The first statement is immediate from Corollary 32.34: compatibility places the realization inside a joint non-escape corridor on the OP 1 bridge side and the OP 3 shell side while leaving the invariant closure value unchanged, and Proposition 32.30 then fixes the terminal reading as one common return-side carrier decision. Hence the diagnostic role of the packet/phase/epsilon conditions is to preselect a corridor of final admissible support beneath that carrier-first terminal reading, not to introduce an independent closure criterion.

For the converse direction, failure of packet-, phase-lock-, or epsilon-corridor compatibility may occur either because a visible-support diagnostic leaves its controlled regime or because a structural admissibility condition fails. By the same corollary and by the carrier-first OP 5

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interface, only the latter is primary for OP closure: visible mismatch inside or outside the packet-scale corridor is not yet structural non-closure unless it reflects loss of branch admissibility or loss of OP 3 shell-admissibility. Therefore packet/phase-lock/epsilon diagnostics define a screening layer rather than a standalone closure theorem or a second OP 5 endpoint test. □

Remark 32.37. *This screening principle is the cleanest current mathematical reading of the old packet/phase-lock/epsilon diagnostics. They are now neither discarded nor promoted beyond their scope: they act as a visible preselection gate for the admissible branch-shell sector under one carrier-first OP 5 decision, while the actual structural test remains the admissibility machinery of the OP core.*

32.14 Diagnostic readout hierarchy beneath the carrier-first OP 5 decision

After the carrier-first OP 5 interface and the packet/phase-lock/epsilon screening layer, the remaining visible packet, breathing-envelope, frame-flicker, and helix pictures should be read in a strict order. They are not parallel closure criteria. They are progressively more visible readout surfaces of one and the same carrier decision.

Remark 32.38 (Three diagnostic layers)**.** *The current OP 5-side diagnostic stack is therefore to be read in the following descending order of structural weight.*

(i) ***Carrier-decision layer.*** *The first structural question is whether the return side still belongs to one common closure-bearing carrier, as fixed by Proposition 32.30.*

(ii) ***Screening layer.*** *Packet/phase-lock/epsilon compatibility then determines whether the visible return sits inside an admissible non-escape corridor beneath that carrier decision.*

(iii) ***Visible-shape layer.*** *Packet-wave, breathing-envelope, frame-flicker, and helix pictures come last: they visualize how one admissible or nearly admissible return is seen, but they do not outrank the carrier decision or the screening corridor.*

Lemma 32.39 (First diagnostic-readout hierarchy lemma)**.** *Assume Proposition 32.30 and Proposition 32.36. Let (b, r, s) be a branch-shell realization with associated visible packet-, breathing-envelope-, frame-flicker-, and helix-type diagnostics. Then the current OP-side readout hierarchy has the following form.*

(i) ***Carrier priority.*** *If the realization belongs to one common closure-bearing carrier, then no visible packet-, flicker-, or helix-level multiplicity by itself overrides that carrier decision.*

(ii) ***Screening priority.*** *Packet/phase-lock/epsilon compatibility is the first visible screening layer beneath the carrier decision; loss at the pure visible-shape level is not yet structural loss unless it propagates to loss of this screening layer or to loss of the carrier package itself.*

(iii) ***Visible-shape subordination.*** *The packet-wave, breathing-envelope, frame-flicker, and helix pictures are therefore subordinate readout surfaces. They can localize, rank, and compare visible deviations, but they do not constitute an independent closure theorem and do not outrank the carrier-first OP 5 decision.*

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In particular, the mathematically correct order of evidential force is

$$\textit{carrier decision} > \textit{packet/phase/epsilon screening} > \textit{visible flicker/helix picture.}$$

Proof. Proposition 32.30 fixes the first terminal structural question: the return side is read first through one common closure-bearing carrier, and a structural failure must already arise upstream of raw terminal visibility. Proposition 32.36 then places packet/phase-lock/epsilon compatibility strictly beneath that terminal carrier decision as a diagnostic preselection corridor rather than as a second closure theorem. The remaining packet-wave, breathing-envelope, frame-flicker, and helix pictures are by construction even further downstream, because they are explicitly treated as visible-support diagnostics in Definition 3.116 and Corollary 32.34. Hence visible-shape multiplicity may refine the observable profile of a final return, but it cannot outrank the carrier decision and cannot become structural loss unless the screening corridor or the carrier package itself fails. This proves the stated hierarchy. □

Corollary 32.40 (Visible flicker/helix subordination principle)**.** *Within the current carrier-first OP 5 architecture, any packet-wave, breathing-envelope, frame-flicker, or helix mismatch is to be read first as visible-shape evidence beneath the carrier decision and beneath the packet/phase-lock/epsilon screening layer. Such a mismatch becomes structurally decisive only if it can be shown to force loss of packet/phase-lock/epsilon compatibility or loss of the common closure-bearing carrier itself.*

Proof. This is the direct operational reading of Lemma 32.39. The visible-shape diagnostics are subordinate readout surfaces; hence they can become structurally decisive only by propagating upward into the screening layer or the carrier layer. □

32.15 First upstream propagation principle for visible-shape mismatch

Remark 32.41 (Upstream propagation gate)**.** *Under the current carrier-first OP 5 architecture, visible packet-wave, breathing-envelope, frame-flicker, and helix mismatches have no immediate right to overturn the structural reading. They may move upward only by forcing failure of the next superior layer: first the packet/phase-lock/epsilon screening corridor, and only after that the common closure-bearing carrier package itself.*

Lemma 32.42 (First upstream propagation lemma)**.** *Assume Proposition 32.30, Proposition 32.36, Lemma 32.39, and Lemma 1.203. Let (b, r, s) be a branch-shell realization with a visible packet-wave, breathing-envelope, frame-flicker, or helix mismatch. Then the current upstream propagation rule has the following form.*

(i) ***No direct jump.*** *A pure visible-shape mismatch does not by itself propagate directly to structural non-closure or directly overturn the carrier-first OP 5 decision.*

(ii) ***First gate: screening.*** *If the mismatch does* not *force loss of packet/phase-lock/epsilon compatibility, then it remains subordinate visible-shape evidence and the carrier decision is unchanged.*

(iii) ***Second gate: carrier package.*** *If the mismatch* does *force loss of packet/phase-lock/epsilon compatibility while the admissible branch-shell sector and the common closure-bearing carrier package remain intact, then the mismatch propagates only to screening-layer loss and is still not yet a structural OP failure.*

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(iv) ***Structural propagation.*** *A visible-shape mismatch becomes structurally decisive only when it can be shown to propagate further to one of the upstream obstruction strata of Lemma 1.203, namely loss of admissible branch passage, loss of shell-admissible refinement, or loss of the common readout/kernel carrier package.*

In particular, the current architecture permits only the ordered propagation chain

$$\begin{aligned} \textit{visible flicker/helix mismatch} &\Longrightarrow \textit{possible screening loss} \\ &\Longrightarrow \textit{possible carrier/package loss} \\ &\Longrightarrow \textit{structural failure.} \end{aligned}$$

No shorter upward propagation is presently licensed.

Proof. Lemma 32.39 already fixes the hierarchy of evidential force: visible-shape diagnostics lie strictly below the packet/phase-lock/epsilon screening layer, which itself lies below the carrier-first OP 5 decision. Hence a pure visible mismatch cannot jump directly to structural failure. Proposition 32.36 gives the first gate: if packet/phase-lock/epsilon compatibility is preserved, then the mismatch remains inside the diagnostic corridor and does not alter the carrier reading. Even if that screening corridor is lost, Proposition 32.30 still prevents immediate structural failure unless the common return-side carrier package itself is lost. Finally, Lemma 1.203 identifies the only current upstream structural failure modes: branch-stratum loss, shell-stratum loss, or readout/kernel carrier-package loss. Therefore the only admissible upstream propagation route is the ordered one stated above. □

Corollary 32.43 (Visible mismatch non-propagation default)**.** *Within the current carrier-first OP 5 architecture, every packet-wave, breathing-envelope, frame-flicker, or helix mismatch is to be treated by default as diagnostically local evidence. It may be upgraded only after an explicit witness of screening-layer loss or of upstream carrier/package loss has been exhibited.*

Proof. This is the operational reformulation of Lemma 32.42. In the absence of an explicit upward propagation witness, visible-shape mismatch remains subordinate local evidence. □

32.16 First diagnostic witness principle for visible-shape mismatch

Remark 32.44 (Diagnostic witness discipline)**.** *Under the current carrier-first OP 5 architecture, visible packet-wave, breathing-envelope, frame-flicker, and helix mismatches count as admissible upstream evidence only when they come equipped with a witness that is strong enough to certify failure of the next superior layer. Pure visual irregularity, even if persistent or large, is not yet such a witness.*

Lemma 32.45 (First diagnostic witness lemma)**.** *Assume Proposition 32.36, Proposition 32.30, Lemma 32.39, and Lemma 32.42. Let (b, r, s) be a branch-shell realization with a visible packet-wave, breathing-envelope, frame-flicker, or helix mismatch. Then the current witness discipline has the following form.*

(i) ***Screening witness threshold.*** *A visible mismatch is a legitimate witness of screening-layer loss only if it can be shown to force failure of packet/phase-lock/epsilon compatibility; otherwise it remains diagnostically local evidence.*

(ii) ***Carrier witness threshold.*** *A visible mismatch is a legitimate witness of carrier/package loss only if it can be shown to force, beyond screening loss, failure of the common closure-bearing carrier package itself.*

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(iii) ***No witness by appearance alone.*** *Mere amplitude, persistence, oscillatory complexity, geometric deformation, or helix/flicker asymmetry of the visible mismatch is not by itself an admissible witness of screening-layer loss or of carrier/package loss.*

(iv) ***Admissible witness route.*** *Every currently admissible visible witness must therefore factor through one of the two certified upstream gates of Lemma 32.42: first explicit screening loss, and only then, if present, explicit carrier/package loss.*

Hence the present architecture admits visible mismatch only as witness-bearing *evidence and not as a free structural verdict.*

Proof. Proposition 32.36 defines the first superior layer above raw visible-shape mismatch. Therefore a visible mismatch can witness screening-layer loss only by forcing failure of packet/phase-lock/epsilon compatibility itself; if compatibility persists, the mismatch remains inside the diagnostic corridor and cannot count as screening evidence. Proposition 32.30 defines the next superior layer: even after screening loss, the structural carrier reading remains unchanged unless the common closure-bearing carrier package is lost. Thus a visible mismatch can witness carrier/package loss only by forcing precisely that second failure as well. Lemma 32.39 excludes any direct elevation of raw visible-shape features to a superior structural layer, and Lemma 32.42 identifies the only currently licensed upward route as the ordered screening-to-carrier route. Consequently amplitude, persistence, or geometric shape alone are not admissible witnesses; only a mismatch that certifiably drives one of those gates may count as an upstream witness. □

Corollary 32.46 (Visible witness admissibility rule)**.** *Within the current carrier-first OP 5 architecture, every visible packet-wave, breathing-envelope, frame-flicker, or helix mismatch must be classified first as either*

(a) *a merely local diagnostic irregularity,*

(b) *a certified witness of screening-layer loss,*

(c) *or a certified witness of carrier/package loss.*

No fourth class of immediately structural visible witness is presently admitted.

Proof. This is the direct classification consequence of Lemma 32.45. A visible mismatch is either not strong enough to force a superior-layer failure, or it forces screening loss, or it forces carrier/package loss after screening loss. By the witness discipline and by the upstream propagation rule, there is no currently licensed direct jump from raw visible appearance to immediate structural verdict. □

32.17 Screening-witness compression beneath the carrier-first OP 5 decision

Remark 32.47 (Compression of visible witness types)**.** *Once a visible packet-wave, breathing-envelope, frame-flicker, or helix mismatch has crossed the witness threshold of Lemma 32.45, the present architecture no longer needs to keep its visible shape as a separate structural species at the screening level. Unless the same witness already certifies carrier/package loss, its upstream content compresses to one common statement: the packet/phase-lock/epsilon screening corridor has failed.*

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Lemma 32.48 (Screening-witness compression lemma)**.** *Assume Proposition 32.36, Proposition 32.30, Lemma 32.45, and Corollary 32.46. Let (b, r, s) be a branch-shell realization with a visible packet-wave, breathing-envelope, frame-flicker, or helix mismatch that is a certified witness of screening-layer loss but not yet a certified witness of carrier/package loss. Then the upstream structural content of that witness compresses to a single screening-loss form, namely the failure of packet/phase-lock/epsilon compatibility. More precisely:*

(i) ***Visible-shape indifference at screening level.*** *Whether the witness is first expressed through packet-wave mismatch, breathing-envelope drift, frame-flicker asymmetry, or helix deformation is irrelevant for its screening-level structural role.*

(ii) ***Unique compressed content.*** *The only currently licensed upstream content of such a witness is that the packet/phase-lock/epsilon screening corridor fails.*

(iii) ***No parallel screening species.*** *There is presently no second independent screening-loss class attached specifically to packet-wave, envelope, flicker, or helix shape once the witness has been admitted.*

(iv) ***Carrier exception.*** *A visible witness escapes this compression only when it upgrades further to a certified witness of carrier/package loss.*

Hence, below the carrier threshold, all currently admissible visible screening witnesses collapse to one common screening-loss normal form.

Proof. By Lemma 32.45, a visible mismatch is admitted upward only when it certifiably forces failure of the next superior layer. Proposition 32.36 identifies that next superior layer uniquely as packet/phase-lock/epsilon compatibility. Therefore, once a visible mismatch qualifies as a witness of screening-layer loss, its upstream meaning is already fixed: it testifies precisely to failure of that screening corridor. Corollary 32.46 allows no additional currently licensed class of screening witness beyond that certified screening loss or the stronger carrier/package loss. Proposition 32.30 then separates the only exception: a witness that also forces loss of the common closure-bearing carrier package belongs to the higher carrier class instead of remaining in the compressed screening class. Thus all admitted visible witnesses that stop below the carrier threshold share one and the same screening-loss content, independently of the visible shape through which they first appeared. □

Corollary 32.49 (Screening-witness normal form)**.** *Within the current carrier-first OP 5 architecture, every admitted visible witness that does not yet certify carrier/package loss may be cited in proofs simply as a witness of* screening-loss normal form*. Its packet-wave, breathing-envelope, frame-flicker, or helix realization is diagnostically informative, but not structurally independent at the screening level.*

Proof. This is the direct proof-usage reformulation of Lemma 32.48. Once admitted and still below the carrier threshold, the witness has exactly one currently licensed upstream content: loss of the packet/phase-lock/epsilon screening corridor. □

32.18 Carrier-witness separation beneath compressed screening loss

Remark 32.50 (Two witness grades after screening compression)**.** *After Lemma 32.48, every admitted visible witness sits in one of two remaining grades. Either it stays below the carrier threshold and is therefore only a witness of screening-loss normal form, or it upgrades further and certifies that the common return-side carrier package itself is no longer available. The present architecture does not license a mixed intermediate status between those two grades.*

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Lemma 32.51 (Carrier-witness separation lemma)**.** *Assume Proposition 32.36, Proposition 32.30, Lemma 32.45, Lemma 32.48, and Corollary 32.49. Let (b, r, s) be a branch-shell realization with an admitted visible witness. Then exactly one of the following two cases occurs:*

(i) ***Screening-only witness.*** *The witness compresses to screening-loss normal form, while the common closure-bearing return-side carrier package remains available. In this case the witness certifies only screening loss and is not an admissible witness of carrier/package loss.*

(ii) ***Carrier-upgraded witness.*** *The witness forces loss of the common closure-bearing return-side carrier package. In this case it is an admissible witness of carrier/package loss and no longer belongs merely to the screening-loss normal form.*

Moreover, there is no third currently licensed status in which an admitted visible witness would simultaneously remain only screening-level and yet already count as a carrier/package witness.

Proof. By Lemma 32.45, every admitted visible witness must force loss of some superior layer. Lemma 32.48 and Corollary 32.49 show that, below the carrier threshold, all such witnesses have one and the same upstream content: failure of packet/phase-lock/epsilon compatibility. Proposition 32.30 identifies the only stronger alternative, namely loss of the common closure-bearing return-side carrier package itself. Hence an admitted visible witness either stops at the compressed screening-loss level or upgrades to carrier/package loss. These two statuses are mutually exclusive because the first expressly retains the common carrier package, whereas the second destroys it. Therefore exactly one of the two cases above occurs, and no mixed intermediate category is presently licensed. □

Corollary 32.52 (Carrier-witness upgrade criterion)**.** *Within the current carrier-first OP 5 architecture, an admitted visible witness upgrades from screening-loss normal form to a genuine carrier/package witness if and only if its upstream content cannot be realized through any common closure-bearing return-side carrier package. Equivalently, as long as such a common carrier remains available, the witness must be read only as screening-loss normal form.*

Proof. The forward direction is exactly case (ii) of Lemma 32.51. The converse direction is case (i): if a common closure-bearing return-side carrier package still exists, then the witness has not crossed the carrier threshold and therefore remains only a screening witness in normal form. □

32.19 Witness-to-carrier escalation beyond screening compression

Remark 32.53 (Escalation gate after witness separation)**.** *After Lemma 32.51, a visible witness reaches carrier grade exactly when its admitted upstream content can no longer be absorbed by screening-loss normal form alone. In the current architecture there is therefore only one licensed upgrade mechanism: every common closure-bearing return-side carrier package compatible with the compressed screening witness fails, so the witness is forced out of screening-only status and into carrier/package loss.*

Lemma 32.54 (Witness-to-carrier escalation lemma)**.** *Assume Proposition 32.30, Lemma 32.45, Lemma 32.48, Lemma 32.51, and Corollary 32.52. Let (b, r, s) be a branch-shell realization with an admitted visible witness w. Then the following are equivalent:*

(i) w upgrades from screening-loss normal form to a genuine carrier/package witness;

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(ii) there exists no common closure-bearing return-side carrier package through which the upstream content of w can still be realized while screening-loss normal form remains the only failed superior layer;

(iii) the upstream content of w forces loss of the common closure-bearing carrier package rather than mere loss of packet/phase-lock/epsilon compatibility.

In particular, escalation occurs exactly when compressed screening loss is no longer a sufficient upstream normal form for the admitted witness.

Proof. By Lemma 32.51, an admitted visible witness has only two currently licensed grades: screening-only or carrier-upgraded. Corollary 32.52 identifies the upgrade threshold precisely: upgrade occurs if and only if no common closure-bearing return-side carrier package remains available. Lemma 32.48 shows that, below that threshold, the witness has exactly one upstream content, namely screening-loss normal form. Hence a witness stays screening-only exactly as long as its upstream content is still realizable through some common carrier package with only packet/phase-lock/epsilon compatibility lost. Once that becomes impossible, the witness can no longer remain in screening-loss normal form alone and must instead force carrier/package loss. This proves the equivalence of (i), (ii), and (iii), and therefore the final escalation claim. □

Corollary 32.55 (Escalation threshold for admitted visible witnesses)**.** *Within the current carrier-first OP 5 architecture, an admitted visible witness remains screening-only as long as its upstream content is representable by screening-loss normal form over some common closure-bearing return-side carrier package. It becomes carrier-upgraded precisely at the first point where such a carrier-supported realization ceases to exist.*

Proof. This is the proof-usage reformulation of Lemma 32.54. The existence of a common carrier package keeps the witness below the carrier threshold, while the first failure of every such carrier-supported realization marks the unique licensed escalation point. □

32.20 Carrier-stable diagnostic collapse beneath the witness escalation threshold

Remark 32.56 (Collapse of visible diagnostic species over a stable carrier)**.** *Once an admitted visible mismatch stays below the escalation threshold of Corollary 32.55, the current carrier-first OP 5 architecture no longer attributes independent structural rank to its packet-wave, breathing-envelope, frame-flicker, or helix presentation. Those visible realizations remain diagnostically useful, but over one and the same stable closure-bearing return-side carrier they collapse to a single subordinate screening-level phenomenon.*

Theorem 32.57 (Carrier-stable diagnostic collapse theorem)**.** *Assume Proposition 32.30, Proposition 32.36, Lemma 32.39, Lemma 32.45, Lemma 32.48, Lemma 32.51, and Lemma 32.54. Let (b, r, s) be a branch-shell realization with an admitted visible mismatch w of packet-wave, breathing-envelope, frame-flicker, or helix type. If w remains below carrier escalation, equivalently if there exists a common closure-bearing return-side carrier package through which the upstream content of w is still realized as screening-loss normal form, then the following collapse holds:*

(i) ***One subordinate structural role.*** *The visible realization of w has exactly one current structural role, namely to witness subordinate screening-loss normal form beneath a still-stable carrier decision.*

(ii) ***No independent visible closure species.*** *Packet-wave, breathing-envelope, frame-flicker, and helix mismatch do not define distinct closure-relevant species as long as the common carrier package remains available.*

(iii) ***Carrier dominance.*** *Any closure-relevant distinction between those visible realizations collapses over the stable carrier and may reappear only if escalation to carrier/package loss occurs.*

(iv) ***Readout subordination.*** *The visible mismatch remains a downstream readout of one common carrier-supported screening loss and therefore cannot outrank the carrier decision itself.*

Hence, below the escalation threshold, all currently admitted visible diagnostic forms collapse to one carrier-stable subordinate diagnostic class.

Proof. By Lemma 32.39, visible flicker/helix-type structure is subordinate to packet/phase/epsilon screening, which is itself subordinate to the carrier-first OP 5 decision. Lemma 32.45 allows visible mismatch to count upstream only when it certifies loss of a superior layer. Lemma 32.48 then shows that, below carrier loss, every admitted visible witness has exactly one upstream content: screening-loss normal form. Lemma 32.51 and Corollary 32.55 identify the present hypothesis of a still available common closure-bearing return-side carrier package as precisely the condition that the witness has not escalated to carrier/package loss. Therefore, while that common carrier package remains available, packet-wave, envelope, flicker, and helix mismatch cannot persist as distinct closure-relevant species. They all collapse to the single role of subordinate diagnostic realization of screening-loss normal form over one stable carrier. Proposition 32.30 guarantees that any stronger distinction would already belong to carrier/package loss rather than to the pre-escalation regime. This proves (i)–(iv) and the final collapse claim. □

Corollary 32.58 (Carrier-stable visible normal form)**.** *Within the current carrier-first OP 5 architecture, every admitted visible mismatch that stays below carrier escalation may be cited simply as belonging to one* carrier-stable visible normal form*. Its packet-wave, breathing-envelope, frame-flicker, or helix appearance remains diagnostically informative, but not structurally independent while the common carrier package persists.*

Proof. This is the proof-usage reformulation of Theorem 32.57. The theorem states exactly that all admitted visible mismatches below escalation collapse over one stable carrier to a single subordinate diagnostic class. □

32.21 Bundled OP 5 diagnostic theorem

Remark 32.59 (One bundled positive/negative OP 5 diagnostic decision)**.** *After the carrier-first OP 5 interface, the current diagnostic layer is now strong enough to be cited as one bundled decision principle. Either the visible material remains subordinate to a stable carrier and therefore collapses to a screening-level normal form, or it upgrades through an admitted witness to carrier/package loss. There is no third currently admitted diagnostic rank between those two bundled outcomes.*

Theorem 32.60 (Bundled OP 5 diagnostic theorem)**.** *Assume Proposition 32.30, Proposition 32.36, Lemma 32.39, Lemma 32.42, Lemma 32.45, Lemma 32.48, Lemma 32.51, Lemma 32.54, and Theorem 32.57. Let (b, r, s) be a routed branch-shell realization together with an admitted visible mismatch w of packet-wave, breathing-envelope, frame-flicker, or helix type. Then exactly one of the following bundled OP 5 diagnostic outcomes holds:*

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(i) ***Carrier-stable diagnostic collapse.*** *The mismatch w remains below escalation, is carried by one common closure-bearing return-side carrier, and therefore collapses to a subordinate screening-level visible normal form beneath the carrier-first OP 5 decision.*

(ii) ***Carrier-upgraded diagnostic obstruction.*** *The mismatch w cannot be realized over any common closure-bearing return-side carrier as mere screening-loss normal form and therefore upgrades to carrier/package loss. In that case w is not a merely visible irregularity but an admitted carrier witness for routed OP 5 failure.*

Moreover, case (i) *excludes case* (ii)*, and case* (ii) *excludes case* (i)*. Hence the current OP 5 diagnostic layer is completely bundled into the dichotomy*

carrier-stable visible collapse versus carrier-upgraded diagnostic obstruction.

Proof. Lemma 32.39 and Lemma 32.42 force every admitted visible mismatch to move upward only through the packet/phase/epsilon screening layer before it can acquire carrier relevance. Lemma 32.45 then identifies which visible mismatches may count upstream at all. Lemma 32.48 compresses every such admitted visible witness below carrier loss to one screening-loss normal form, while Lemma 32.51 and Lemma 32.54 separate precisely the screening-only regime from the carrier-upgraded regime. Theorem 32.57 describes the entire positive side of the first regime: once a common closure-bearing return-side carrier remains available, all admitted visible species collapse to one subordinate visible normal form beneath the carrier-first OP 5 decision. Conversely, when no such common carrier remains available, Corollary 32.52 upgrades the admitted visible witness to carrier/package loss. These two possibilities are mutually exclusive by Lemma 32.51 and jointly exhaustive by the escalation dichotomy. This proves the bundled theorem. □

Corollary 32.61 (Bundled routed OP 5 diagnostic use rule)**.** *In later proof steps it is sufficient to cite Theorem 32.60 instead of separately unpacking packet-wave, breathing-envelope, frame-flicker, and helix diagnostics. One may simply ask whether the currently admitted visible data still collapse over a stable carrier or have already upgraded to carrier/package obstruction.*

Proof. This is the proof-usage reformulation of Theorem 32.60. The theorem already packages the visible diagnostic layer into exactly the two current routed OP 5 outcomes. □

32.22 First OP 3 to OP 5 bundled transfer theorem

Remark 32.62 (Transfer of bundled OP 3 closure into the OP 5 diagnostic layer)**.** *The current architecture no longer treats OP 3 closure and OP 5 diagnosis as two independent theorem blocks. Once the OP 3 shell passage has been certified and rerouted through one common closure-bearing carrier, the routed OP 5 side inherits a downstream decision of that same carrier package. The only remaining terminal question is whether the visible return side still collapses beneath that package or whether it upgrades to a carrier-level obstruction. Thus OP 5 is presently read as the bundled downstream transfer image of the OP 3/QAM closure package.*

Theorem 32.63 (First OP 3 to OP 5 bundled transfer theorem)**.** *Assume Theorem 1.205, Theorem 1.211, Theorem 32.28, Proposition 32.30, and Theorem 32.60. Let (b, r, s) be a routed branch-shell realization whose OP 3 shell passage is translated into the extracted QAM layer, and let w be an admitted visible mismatch on the routed OP 5 return side. Then the present bundled OP 3→OP 5 transfer has exactly two outcomes:*

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(i) ***Transferred carrier-stable pass.*** *The translated OP 3 shell passage belongs to the certified side of the bundled OP 3 closure package, the routed return stays inside the same common closure-bearing carrier, and the admitted visible mismatch w collapses beneath that carrier to the subordinate OP 5 visible normal form of Theorem 32.60.*

(ii) ***Transferred localized obstruction.*** *The transfer fails at the first licensed obstruction layer: either the OP 3/QAM side already reaches the first localized shell obstruction of the extracted OP 3 proof run, or the routed return upgrades to a carrier-level OP 5 diagnostic obstruction. In particular, no third autonomous terminal OP 5 obstruction is presently licensed downstream of an otherwise intact bundled OP 3 carrier package.*

Hence the current routed OP 5 diagnostic layer is the bundled downstream transfer of the OP 3/QAM closure package, not an additional independent closure theory.

Proof. Theorem 1.211 already packages the translated OP 3 shell passage into exactly two upstream outcomes: certified passage through the bundled OP 3 closure tool layer or the first localized obstruction. If the certified side holds, then Theorem 1.205 identifies one common closure-bearing carrier/readout package for the shell passage. Theorem 32.28 transports that same certified branch-shell realization to the routed return side without changing the invariant closure class, and Proposition 32.30 forces the terminal OP 5 reading to be carrier-first rather than a new autonomous endpoint test. Theorem 32.60 then completes the positive branch: as long as the routed return remains over that same common carrier, every admitted visible mismatch collapses to the subordinate carrier-stable OP 5 visible normal form. This proves case (i).

If the transfer does not remain on that certified branch, then the failure must appear at the first licensed obstruction layer. Upstream, Theorem 1.211 and the extracted OP 3/QAM obstruction package localize the first obstruction already at branch, shell, or carrier/readout level. Downstream, if the routed return cannot be realized over any common closure-bearing carrier, then Theorem 32.60 upgrades the admitted visible mismatch to a carrier-level OP 5 diagnostic obstruction. Because Proposition 32.30 forbids an independent terminal closure test downstream of an otherwise intact carrier package, there is no currently licensed third obstruction type appearing only at the raw terminal visible layer. This proves case (ii) and the final bundled transfer reading. □

Corollary 32.64 (Bundled OP 3→OP 5 proof-use rule)**.** *In later proof steps it is sufficient to cite Theorem 32.63 when a routed branch-shell realization must be carried from the extracted OP 3/QAM closure package into the OP 5 diagnostic layer. One may simply ask whether the realization stays on the transferred carrier-stable branch or reaches the first licensed transferred obstruction.*

Proof. This is the proof-usage reformulation of Theorem 32.63. The theorem already packages the current OP 3/QAM closure side and the current OP 5 diagnostic side into one bundled transfer decision. □

32.23 First OP 3 to OP 5 compressed proof interface

Remark 32.65 (How older routed return arguments should now be compressed)**.** *The current transfer architecture is now strong enough that later routed return arguments need not separately reopen the combined branch-shell theorem, the carrier-first OP 5 interface, and the bundled OP 5 diagnostic layer. Once the routed realization has entered the bundled OP 3→OP 5*

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transfer channel, the proof may proceed entirely through one compressed dichotomy: transferred carrier-stable pass versus transferred localized obstruction. This local compressed interface is now itself subordinate to the newer global rule: as soon as the argument has also entered the bundled current channel and no shell-sensitive or witness-grade claim is being asked, the proof should move on to the extracted global tools and remain global-first unless a genuine local reopening is needed.

Proposition 32.66 (First OP 3 to OP 5 compressed proof interface)**.** *Assume Theorem 32.63. Let (b, r, s) be a routed branch-shell realization whose OP 3 shell passage has been translated into the extracted QAM layer, and let w be any admitted visible routed return witness on the OP 5 side. Then every later proof step that needs to carry this realization through the current routed return architecture may be compressed to the following interface.*

(i) ***Transferred carrier-stable pass.*** *If Theorem 32.63 places $(b, r, s; w)$ on the transferred carrier-stable branch, then all downstream OP 5 packet/phase/flicker/helix material is read only as subordinate visible realization beneath one common closure-bearing carrier and may be replaced by the carrier-stable visible normal form.*

(ii) ***Transferred localized obstruction.*** *If Theorem 32.63 places $(b, r, s; w)$ on the transferred obstruction branch, then the proof may stop at the first licensed obstruction already identified there: either the upstream OP 3/QAM obstruction or the downstream carrier-upgraded OP 5 obstruction.*

(iii) ***No third terminal proof branch.*** *No additional raw terminal visible branch is presently licensed after this compression. In particular, later proofs do not need a separate fourth case for packet-wave, breathing-envelope, frame-flicker, or helix mismatch once the bundled transfer theorem has already been invoked.*

Hence the current routed return architecture may be used through one compressed OP 3→OP 5 proof interface.

Proof. This is the direct proof-usage reading of Theorem 32.63. The theorem already packages the routed return into exactly two licensed transfer outcomes. On the positive side, the admitted visible material collapses beneath one common carrier and therefore needs no further structural branching. On the negative side, the first licensed obstruction is already localized either upstream in the OP 3/QAM package or downstream at the carrier-upgraded OP 5 level. Because Theorem 32.63 excludes an autonomous raw terminal OP 5 obstruction beneath an otherwise intact carrier package, no third terminal proof branch remains. □

Corollary 32.67 (Compressed rereading of routed return arguments through the bundled transfer)**.** *In later arguments it is sufficient to cite Proposition 32.66 rather than separately reopening Theorem 32.28, Proposition 32.30, and Theorem 32.60. One may simply ask whether the current routed realization stays on the transferred carrier-stable branch or reaches the first licensed transferred obstruction.*

Proof. This is the condensed proof-usage form of Proposition 32.66. The proposition already compresses the presently licensed routed return architecture into one reusable interface. □

32.24 First routed return normal form theorem

Remark 32.68 (Carrier-stable versus obstructed routed return normal form)**.** *After the compressed OP 3→OP 5 proof interface has been invoked, the current routed return architecture no longer needs a separate case distinction for packet-wave, breathing-envelope, frame-flicker,*

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or helix appearance. At the present level of the theory every admitted routed return is read only up to one of two bundled normal forms: a carrier-stable routed return normal form or an obstructed routed return normal form.

Theorem 32.69 (First routed return normal form theorem)**.** *Assume Proposition 32.66. Let (b, r, s) be a routed branch-shell realization whose OP 3 shell passage has been translated into the extracted QAM layer, and let w be any admitted visible routed return witness on the OP 5 side. Then the current routed return architecture classifies $(b, r, s; w)$, up to bundled transfer, into exactly one of the following two mutually exclusive routed return normal forms.*

(i) ***Carrier-stable routed return normal form.*** *The routed return remains on the transferred carrier-stable branch of Proposition 32.66. Equivalently, there exists one common closure-bearing return-side carrier through which the routed realization is read, and every admitted visible mismatch collapses beneath that carrier to the carrier-stable visible normal form. In this case the routed return contributes no additional structural branching beyond the bundled OP 3→OP 5 transfer itself.*

(ii) ***Obstructed routed return normal form.*** *The routed return lies on the transferred localized-obstruction branch of Proposition 32.66. Equivalently, no common closure-bearing routed return carrier supports the realization as mere screening-level visible loss, and the first licensed failure is already localized either upstream at the OP 3/QAM obstruction layer or downstream at the carrier-upgraded OP 5 obstruction layer.*

Hence every currently admitted routed return is read, up to the bundled OP 3→OP 5 transfer, by one bundled routed return normal form and not by any third autonomous terminal visible class.

Proof. Proposition 32.66 already compresses the present routed return architecture into exactly two licensed proof branches: transferred carrier-stable pass and transferred localized obstruction. On the positive branch, Theorem 32.60 and the carrier-stable collapse package imply that every admitted visible return witness becomes subordinate to one common closure-bearing routed return carrier and therefore yields the carrier-stable routed return normal form. On the negative branch, the same compressed interface localizes the first licensed failure either upstream in the extracted OP 3/QAM obstruction package or downstream at the carrier-upgraded OP 5 obstruction layer; by construction this is the obstructed routed return normal form. Because Proposition 32.66 excludes a third raw terminal visible branch after compression, these two forms are exhaustive and mutually exclusive. □

Corollary 32.70 (Routed return normal form use rule)**.** *In later proofs it is sufficient to cite Theorem 32.69 once a routed realization has already entered the bundled OP 3→OP 5 transfer channel. One may then work only with the dichotomy* carrier-stable routed return normal form versus obstructed routed return normal form*.*

Proof. This is the proof-usage reformulation of Theorem 32.69. The theorem already packages the presently licensed routed return architecture into one reusable normal-form dichotomy. □

32.25 First global pass/obstruction normal form

Remark 32.71 (Current global normal form of the admissible run)**.** *The present line now carries enough bundled structure that the currently admitted run*

$$\begin{aligned} &\textit{OP1 source} \\ \to\ &\textit{branch admissibility} \\ \to\ &\textit{OP3/QAM closure package} \\ \to\ &\textit{carrier-first OP5 layer} \end{aligned}$$

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no longer needs to be read as a chain of loosely connected local tests. After the routed return normal form has been established, the full currently admitted run is read only up to one global dichotomy: a global pass normal form or a global obstruction normal form.

Theorem 32.72 (First global pass/obstruction normal form)**.** *Assume Theorem 32.69. Let* (b, r, s) *be an admissible branch-shell realization issued from the current OP1 source side and translated through the extracted OP3/QAM layer into the routed return architecture, and let* w *be any admitted visible routed return witness on the OP5 side. Then the whole currently admitted run from OP1 through branch, OP3/QAM, and OP5 is classified, up to the present bundled architecture, into exactly one of the following two mutually exclusive global normal forms.*

(i) ***Global pass normal form.*** *The realization lies on the carrier-stable routed return normal form of Theorem 32.69. Equivalently, the run remains supported by one bundled closure-bearing carrier package from the present OP1/branch entrance through the OP3/QAM shell package and into the routed return side, while all visible OP5 material collapses beneath that carrier to the carrier-stable visible normal form. In this case the current run carries no further licensed structural branch beyond the bundled pass reading.*

(ii) ***Global obstruction normal form.*** *The realization lies on the obstructed routed return normal form of Theorem 32.69. Equivalently, the first licensed failure of the current run is already localized either upstream in the OP3/QAM obstruction package or downstream at the carrier-upgraded OP5 obstruction layer, and no autonomous third terminal visible class is currently licensed outside that bundled obstruction reading.*

Hence the whole currently admitted run is read, at the present stage of the theory, by one global pass/obstruction normal form and not by any third independent terminal morphology.

Proof. Theorem 32.69 already classifies every currently admitted routed return into exactly two bundled normal forms. On the positive side, the carrier-stable routed return normal form means that the run has remained on the transferred carrier-stable branch from the OP3/QAM closure package into the routed return side; consequently the whole present run remains supported by one bundled closure-bearing carrier package, and all visible OP5 diagnostics are subordinate to the carrier-stable visible normal form. This is precisely the global pass normal form. On the negative side, the obstructed routed return normal form already states that the first licensed failure is localized either upstream in the OP3/QAM obstruction package or downstream at the carrier-upgraded OP5 obstruction layer; this is exactly the global obstruction normal form. Because Theorem 32.69 excludes a third autonomous terminal visible class, the two global forms are exhaustive and mutually exclusive. □

Corollary 32.73 (Global pass/obstruction use rule)**.** *In later proofs it is sufficient to cite Theorem 32.72 once the current admissible run has entered the bundled OP3/QAM-to-OP5 channel. One may then work only with the dichotomy* global pass normal form versus global obstruction normal form.

Proof. This is the proof-usage reformulation of Theorem 32.72. The theorem already compresses the currently admitted run from OP1 through OP5 into one reusable bundled normal-form dichotomy. □

Remark 32.74 (Status of the bundled OP 3→OP 5/global chain)**.** *The present local-to-global chain should now also be read in two levels of proof strength. The* active bundled interface *consists of the first OP 3→OP 5 bundled transfer theorem, the compressed OP 3→OP 5 proof*

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interface, the routed return normal form, and the global pass/obstruction normal form together with their direct proof-use corollaries. These statements already give one self-contained current channel from translated OP 3 certification to a bundled global pass/obstruction decision. By contrast, the finer witness-grade hierarchy beneath the carrier-first OP 5 layer — screening-loss normal form, carrier-witness upgrade, escalation thresholds, and the related witness-compression remarks — remains a local refinement layer*: it must be reopened only when a later argument genuinely depends on witness species, witness grade, escalation, or screening-versus-carrier distinctions that are invisible at the bundled transfer/global level.*

Proposition 32.75 (Citation and reopening rule for the bundled OP 3→OP 5/global chain)**.** *Let a later argument concern a run already translated from the OP 3/QAM shell package into the routed OP 5 side. Then the following citation discipline is currently licensed.*

(i) If the argument needs only the bundled transfer/global dichotomy, it may cite Theorem 32.63, Proposition 32.66, Theorem 32.69, and Theorem 32.72 without reopening the finer witness hierarchy.

(ii) If the argument further needs shell-level certificate or shell-level obstruction data, it must reopen the completed OP 3/QAM block rather than the entire older OP 1–OP 5 line.

(iii) If the argument further needs witness grade, witness escalation, screening-witness compression, or carrier-witness separation, it must reopen the carrier-first OP 5 witness layer rather than treating the bundled global dichotomy as sufficient.

(iv) No later proof may insert a third autonomous raw terminal visible branch after the bundled transfer/global interface has already been invoked.

Proof. Clause (i) is exactly the compressed reading already supplied by Theorem 32.63, Proposition 32.66, Theorem 32.69, and Theorem 32.72. Clause (ii) is the existing shell-sensitive fallback discipline of the completed OP 3/QAM block. Clause (iii) is the corresponding witness-sensitive fallback discipline of the carrier-first OP 5 layer, where screening-loss normal form, upgrade, escalation, and separation are defined. Clause (iv) is the bundled exclusion already built into the transfer theorem, the compressed proof interface, the routed return normal form, and the global pass/obstruction theorem: once those interfaces are active, no additional autonomous raw terminal visible branch is currently licensed. □

Corollary 32.76 (Current sufficiency of the bundled OP 3→OP 5/global chain)**.** *Assume a current argument starts from a translated OP 3 shell passage and asks only whether the run reaches a bundled pass reading or a bundled first licensed obstruction. Then the present document already contains a self-contained proof interface for that question: one may pass from the completed OP 3/QAM block through the bundled OP 3→OP 5 transfer, the routed return normal form, and finally the global pass/obstruction normal form without introducing any additional intermediate theorem layer. In particular, the current line is already strong enough for the next public-shape step whenever no stronger claim about witness grade or deeper kernel-first activation is being asserted.*

Proof. By Theorem 1.211 and Corollary 1.212, the completed OP 3/QAM block already reduces translated shell data to certified passage versus first localized obstruction. Theorem 32.63 and Proposition 32.66 then transport that same dichotomy into the routed OP 5 side. Theorem 32.69 rewrites the routed return as carrier-stable versus obstructed normal form, and Theorem 32.72 finally packages the whole current run into global pass versus global obstruction. Thus no extra intermediate theorem layer is needed for that bundled question. The final sentence is only the proof-theoretic reading of Remark 32.74: stronger witness-grade or deeper kernel-first claims still require reopening the finer local layers. □

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32.26 Current extracted global pass/obstruction tool layer

Remark 32.77 (Purpose of the current global tool layer)**.** *The global pass/obstruction normal form is not meant to remain only as a terminal summary theorem. It is now extracted as a reusable tool layer so that later proofs can cite a small global interface instead of reopening the whole chain*

$$\textit{OP1 source} \to \textit{branch admissibility} \to \textit{OP3/QAM closure} \to \textit{carrier-first OP5 layer.}$$

The extracted tools below are therefore not new theory beyond Theorem 32.72; they are proof-usable projections of that bundled global dichotomy.

Proposition 32.78 (Global-pass tool)**.** *Assume Theorem 32.72. If a current admissible run lies on the global pass normal form, then one may cite this fact directly as a reusable proof tool: the run is carried by one bundled closure-bearing carrier package from the OP1/branch entrance through the OP3/QAM shell package into the routed return side, and every admitted OP5 visible diagnostic remains subordinate to the carrier-stable visible normal form.*

Proof. This is precisely the positive branch of Theorem 32.72, rewritten as a direct reusable proof tool. □

Proposition 32.79 (Global-obstruction tool)**.** *Assume Theorem 32.72. If a current admissible run lies on the global obstruction normal form, then one may cite this fact directly as a reusable proof tool: the first licensed failure is already localized either upstream in the OP3/QAM obstruction package or downstream at the carrier-upgraded OP5 obstruction layer, and no third autonomous terminal visible class is currently licensed outside that bundled obstruction reading.*

Proof. This is precisely the negative branch of Theorem 32.72, rewritten as a direct reusable proof tool. □

Proposition 32.80 (Global transfer tool)**.** *Assume Theorem 32.72. Any later proof that has already routed the current admissible run into the bundled OP3/QAM-to-OP5 channel may work only with the bundled transfer dichotomy*

$$\textit{global pass} \qquad \textit{versus} \qquad \textit{global obstruction},$$

without reopening the intermediate local decomposition into separate branch, shell, witness, escalation, and routed-return subcases.

Proof. By Corollary 32.73, the whole currently admitted run has already been compressed into one reusable global dichotomy. The omitted intermediate subcases are already absorbed into the proof of Theorem 32.72. □

Proposition 32.81 (Global diagnostic subordination tool)**.** *Assume Theorem 32.72. On the global pass branch, every currently admitted visible routed-return diagnostic is globally subordinate to the same bundled closure-bearing carrier package. In particular, packet-wave, breathing-envelope, frame-flicker, and helix pictures cannot reintroduce a new structural branch once the run has already entered the global pass normal form.*

Proof. Theorem 32.72 sends the positive branch through the carrier-stable routed return normal form and therefore through the carrier-stable visible normal form. By construction, all visible routed-return diagnostics are subordinate to that same bundled carrier package and cannot reopen the global dichotomy. □

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Proposition 32.82 (Global proof-interface tool)**.** *Assume Theorem 32.72. In later arguments it is sufficient to cite one of the four extracted global tools*

global pass, *global obstruction*, *global transfer*, *global diagnostic subordination*,

according to the local need of the argument. No additional independent proof interface for the currently admitted run is required at this stage.

Proof. Theorem 32.72 already supplies the full current bundled dichotomy, while Propositions 32.78–32.81 expose its presently relevant projections. Hence these four extracted global tools form the current reusable proof interface for the admissible run. □

Remark 32.83 (How to cite the extracted global tools)**.** *The current global tool layer should be cited only after the run has already been shown to lie inside the bundled OP1→branch→OP3/QAM→OP5 channel. Once that has been established, the present extracted tools allow later arguments to cite a compact global interface rather than reopening the entire local transport and diagnostic stack.*

32.27 First global compressed proof run through extracted global tools

Remark 32.84 (Reading key for the first global compressed proof run)**.** *The present block is the first place where the current admissible run*

OP1 source → branch admissibility → OP3/QAM closure → carrier-first OP5 layer

is no longer reopened stage by stage. Instead, once the run has already been shown to lie inside the bundled global channel, the extracted global tools are used as one compressed proof interface. The proof run therefore has only two licensed bundled outcomes: a global pass *and a* global obstruction*. The purpose of the theorem below is to state this compressed use rule explicitly, so that later arguments may pass through the whole current architecture in one step.*

Theorem 32.85 (First global compressed proof run through extracted global tools)**.** *Assume that a current admissible run issued from the OP1 source side has already been routed into the bundled OP1→branch→OP3/QAM→OP5 channel of the present architecture. Then the run may be processed through the extracted global tool layer without reopening the intermediate local stack. More precisely, exactly one of the following two bundled proof outputs is currently licensed.*

(i) ***Global compressed pass.*** *By Proposition 32.78, the run lies on the global pass normal form and is supported by one bundled closure-bearing carrier package from the OP1/branch entrance through the OP3/QAM shell package into the routed return side. By Proposition 32.81, all admitted visible routed-return diagnostics remain subordinate to that same bundled carrier package.*

(ii) ***Global compressed obstruction.*** *By Proposition 32.79, the run lies on the global obstruction normal form and its first licensed failure is already localized either upstream in the OP3/QAM obstruction package or downstream at the carrier-upgraded OP5 obstruction layer. By Proposition 32.80, no third autonomous terminal visible branch is currently licensed outside that bundled obstruction reading.*

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Hence the present global proof run is now compressed to the dichotomy

$$\textit{global compressed pass} \qquad \textit{versus} \qquad \textit{global compressed obstruction},$$

and no further independent proof interface for the currently admitted run is required once Proposition 32.82 applies.

Proof. This is the first direct use of the extracted global tool layer as a compressed proof interface. Once the current run has entered the bundled

$$\mathrm{OP1} \to \mathrm{branch} \to \mathrm{OP3/QAM} \to \mathrm{OP5}$$

channel, Proposition 32.82 states that no additional independent proof interface is currently required.

If the run lies on the positive side of Theorem 32.72, Proposition 32.78 supplies the bundled closure-bearing carrier package and Proposition 32.81 shows that all admitted visible OP5 diagnostics remain subordinate to that same carrier; this is exactly the global compressed pass.

If the run lies on the negative side, Proposition 32.79 localizes the first licensed failure and Proposition 32.80 excludes any third independent terminal visible branch; this is exactly the global compressed obstruction.

Exhaustiveness and mutual exclusivity therefore come directly from the extracted global tools and ultimately from Theorem 32.72. □

Corollary 32.86 (First operational use of the extracted global tools)**.** *In later arguments it is sufficient to cite Theorem 32.85 once the current admissible run has entered the bundled global channel. One may then work only with the dichotomy* global compressed pass versus global compressed obstruction*, together with the induced carrier-stable subordination or obstruction localization, without reopening the entire local OP3/QAM-to-OP5 stack.*

Proof. This is the operational reformulation of Theorem 32.85. The theorem already packages the extracted global tools into one reusable compressed proof run for the currently admitted architecture. □

32.28 Global proof compression pass for the older OP1–OP5 line

Remark 32.87 (How the older OP1–OP5 line should now be reread)**.** *The older connected OP1–OP5 transport line should no longer be read as a sequence of largely separate source-return, branch, shell, routed, and diagnostic arguments that must each be reopened in full. After the present global compressed proof run, the mathematically correct reading is instead this: once an argument has entered the currently admitted bundled channel, the older line should be read global-first through the dichotomy*

$$\textit{global compressed pass} \qquad \textit{versus} \qquad \textit{global compressed obstruction.}$$

Only if a later step genuinely depends on shell certificates, shell-obstruction resolution, witness grades, witness escalation, or other local carrier-diagnostic data should one reopen the OP3/QAM or OP5 layers. The purpose of the proposition below is not to replace the older theorems, but to state explicitly that they may now be reused through one compressed global interface.

Proposition 32.88 (Compressed rereading of the older OP1–OP5 line through the extracted global tools)**.** *Assume that an argument along the older OP1–OP5 transport line has already been shown to lie inside the bundled channel of Theorem 32.85. Then the older line may be reread through the extracted global tools in the following compressed form.*

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(i) ***Compressed pass rereading.*** *If the argument lies on the positive side of Theorem 32.85, then the older OP1–OP5 line is to be read as one globally compressed pass: the source-return closure theorem, the admissible branch layer, the OP3/QAM closure package, the carrier-first OP5 interface, and the subordinated diagnostic stack are all already absorbed into one bundled carrier-stable run.*

(ii) ***Compressed obstruction rereading.*** *If the argument lies on the negative side of Theorem 32.85, then the older OP1–OP5 line is to be read as one globally compressed obstruction: the first licensed failure is already localized in the bundled obstruction package, and no extra autonomous terminal visible branch may be added afterwards.*

In particular, later reuse of the older OP1–OP5 text should proceed through the extracted global proof interface rather than by reopening the whole local stack unless a genuinely new layer is being introduced.

Proof. This is the direct compression reading supplied by Theorem 32.85. On the positive side, the theorem already packages the full currently admitted run into the global compressed pass, and the extracted global tools show that the visible OP5 diagnostics remain subordinate to the same bundled carrier package. On the negative side, the same theorem packages the run into the global compressed obstruction and excludes any third autonomous terminal visible branch. Hence the older OP1–OP5 line can now be reused through one global proof interface rather than by reopening each intermediate theorem separately. □

Corollary 32.89 (Global compressed reading of older routed return arguments)**.** *Whenever an older routed return argument is already known to lie inside the bundled current channel, it should by default be cited through Proposition 32.88, Theorem 32.85, and the global citation discipline of Proposition 32.145. In that situation, it is sufficient to work only with the dichotomy*

$$\textit{global compressed pass} \qquad \textit{versus} \qquad \textit{global compressed obstruction},$$

and with the induced bundled carrier-stable or obstruction-localized reading, unless a genuinely shell-sensitive or witness-sensitive local claim must still be reopened.

Proof. This is the operational reformulation of Proposition 32.88. Once the argument is inside the bundled current channel, the extracted global proof interface supplies the compressed rereading rule for the older routed return line. □

32.29 Witness-grade hardening and the first local kernel-first rewrite window

Remark 32.90 (Purpose of the archival stage witness-grade hardening step)**.** *The public archival stage release already exposed the bundled OP 3/QAM closure package and the bundled OP 3→OP 5/global channel in compact public shape. The next task is therefore not a new visible closure notion, but a stricter proof-use discipline for that same channel. In particular, the present step isolates three things: first, a bundled witness-grade reopening package for the OP 5 side; second, a sharper local/global reopening rule explaining exactly when the bundled global interface is sufficient; and third, one first narrow kernel-first rewrite window whose use is licensed only after the bundled channel has already been fixed and only at a local witness-grade transition where the deferred bridge actually reduces proof load.*

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Theorem 32.91 (Bundled witness-grade reopening package)**.** *Assume Theorem 32.60, Proposition 32.66, Theorem 32.72, Proposition 32.145, and Proposition 32.148. Let a later argument concern a run already known to lie inside the bundled current channel. Then exactly one of the following two proof-use regimes is presently licensed.*

(i) ***Bundled global-use regime.*** *If the argument needs only the bundled outputs global compressed pass/global compressed obstruction together with their carrier-stable or obstruction-localized consequences, then the extracted global interface is sufficient and no local witness-grade reopening is licensed or needed.*

(ii) ***Witness-grade reopening regime.*** *If the argument needs the witness grade itself, witness escalation, screening-witness compression, carrier-witness separation, diagnostic collapse below the escalation threshold, or a comparison between shell-grade and witness-grade data, then the bundled global interface must be reopened locally at the OP 5 witness layer, and the proof must work with the finer witness hierarchy rather than with the global dichotomy alone.*

Hence the current architecture already supports one bundled witness-grade reopening package: bundled-global use by default, local witness-grade reopening only for genuinely witness-sensitive claims.

Proof. The bundled global-use regime is exactly the content of Proposition 32.145 together with Proposition 32.148: once the argument is already inside the bundled current channel and only the global pass/obstruction outputs are needed, the extracted global layer is the default proof interface. By contrast, Theorem 32.60 and Proposition 32.66 show that witness grade, witness escalation, screening-witness compression, carrier-witness separation, and diagnostic collapse all live strictly below that bundled global interface. Therefore any genuinely witness-sensitive claim forces a local reopening of the OP 5 witness layer. The two regimes are exhaustive because every later proof step either remains at bundled-global level or asks for one of the finer witness-sensitive distinctions just listed. □

Proposition 32.92 (Witness-grade local/global reopening discipline)**.** *Assume Theorem 32.91. Then the current bundled channel should be used with the following reopening discipline.*

(i) work global-first *once the argument is already inside the bundled current channel;*

(ii) reopen only the OP 3/QAM shell layer *when the step depends on shell certificates, shell-defect strata, shell commuting behavior, or OP 3-side carrier compression;*

(iii) reopen only the OP 5 witness layer *when the step depends on witness grade, witness escalation, screening-witness compression, carrier-witness separation, or diagnostic collapse below escalation;*

(iv) reopen both local layers *only when the step genuinely compares shell-grade and witness-grade data or transfers a shell-sensitive obstruction into a witness-sensitive one.*

In particular, visible OP 5 packet, envelope, flicker, and helix data do not by themselves justify a local reopening once the argument is already licensed to remain global-first.

Proof. Clause (i) is exactly the bundled-global default from Theorem 32.91. Clause (ii) restates the OP 3/QAM reopening part already fixed by Proposition 32.145. Clause (iii) is the witness-grade reopening regime from Theorem 32.91. Clause (iv) records the only mixed local situation still licensed at the present stage: a step that must compare shell-grade and

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witness-grade structure rather than merely cite one local layer. The final sentence follows because, below escalation, visible OP 5 diagnostics remain subordinate to the carrier-stable visible normal form and therefore do not force reopening by appearance alone. □

Corollary 32.93 (Witness-grade reopening checklist)**.** *For later proof use, the current bundled channel may be tested by the following checklist.*

(i) Has the run already entered the bundled current channel?

(ii) Is only the bundled global pass/obstruction output needed?

(iii) If not, is the missing information shell-grade, witness-grade, or genuinely mixed shell/witness data?

(iv) Reopen only the corresponding local layer, and return to the bundled global interface immediately after the local claim has been discharged.

This is the default proof-use checklist for the archival stage witness-grade hardening step.

Proof. This is the operational reformulation of Proposition 32.92. The checklist simply turns the local/global reopening rule into a direct proof-use procedure. □

Definition 32.94 (First licensed local kernel-first rewrite window)**.** *Assume the direct kernel constructor of Definition 1.153, the bridge conditions of Theorem 1.154, and the bundled witness-grade reopening package of Theorem 32.91. The* first licensed local kernel-first rewrite window *is the following strictly local situation: a proof step has already entered the bundled current channel, has already been forced to reopen at witness grade, and now needs only to compare source-side and return-side witness-bearing states through one common carrier-stable normal form without promoting any broader kernel-first migration on the surrounding global channel.*

Proposition 32.95 (First licensed local kernel-first rewrite window)**.** *Assume Definition 32.94, Proposition 32.30, and Theorem 32.60. Let a witness-sensitive proof step concern a routed return realization already known to remain on the carrier-stable branch of the bundled OP 3→OP 5 transfer channel. Then that step may be rewritten kernel-first in the following narrow sense.*

(i) the visible witness-bearing source/return surfaces may be replaced by their exposed payload kernels and compared only through the common kernel normal form forced by the carrier-stable branch;

(ii) the local readout comparison may be performed at the level of the resulting common kernel/readout carrier class instead of on the full visible witness-bearing surfaces;

(iii) after that local comparison has been discharged, the proof must return immediately to the bundled carrier-stable/global interface and may not infer a broader kernel-first rewrite rule for the full OP 3→OP 5/global channel.

Hence the present line now contains one first licensed kernel-first rewrite window: local witness-grade carrier comparison under an already fixed carrier-stable branch, but no general kernel-first migration of the whole bundled channel.

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Proof. Once the proof step is already known to lie on the carrier-stable branch, the witness-bearing visible data are subordinate to one common closure-bearing carrier package. Under the direct kernel constructor and the bridge conditions, this is exactly the situation in which the local comparison load can be shifted from full visible surfaces to their exposed payload kernels and then to the corresponding common kernel normal form. The local readout comparison may therefore be discharged at the level of the common kernel/readout carrier class. However, Theorem 32.91 and Definition 32.94 keep this move strictly local: the argument was already forced into witness-grade reopening, and nothing in the present hypotheses licenses promotion of that local kernel-first step to a global rewrite rule for the surrounding bundled channel. The final sentence is exactly this restriction written as a proof-use statement. □

Corollary 32.96 (Use rule for the first local kernel-first rewrite window)**.** *Whenever a later proof step is already inside the bundled current channel, already forced to reopen at witness grade, and already known to remain on the carrier-stable branch, it is sufficient to cite Proposition 32.95 for the local source/return comparison. After that local comparison, the proof should return to the bundled global interface immediately.*

Proof. This is the operational reformulation of Proposition 32.95. The proposition already states both the local license and the immediate return rule. □

Remark 32.97 (Status of the first corridor-level activation step)**.** *The present release line still does* not *activate a kernel-first rewrite rule for the whole bundled OP 3→OP 5/global channel. Nevertheless, the combination of the bundled witness-grade reopening package and the first licensed local kernel-first rewrite window already supports one slightly stronger activation step: a finite witness-grade corridor on one fixed carrier-stable branch may be discharged kernel-first through one common carrier-stable kernel/readout class. This is the first point at which the current release line moves beyond a single isolated local rewrite without yet promoting the deferred kernel architecture to an active global migration layer.*

Definition 32.98 (Carrier-stable witness corridor)**.** *Assume Theorem 32.91 and Proposition 32.95. A* carrier-stable witness corridor *is a finite family of witness-sensitive proof steps*

$$\Sigma_1, \dots, \Sigma_N$$

along one routed return run such that:

(i) the run is already known to lie inside the bundled current channel;

(ii) every step Σ_j is already forced to reopen at witness grade and remains on the same carrier-stable branch;

(iii) for every j, the local source/return comparison occurring in Σ_j satisfies the hypotheses of Proposition 32.95;

(iv) no step in the family changes the fixed carrier-stable normal-form class or requires a new shell/witness transfer beyond that same carrier-stable branch.

Thus a carrier-stable witness corridor is the smallest finite witness-grade subrun on which the same carrier-stable kernel/readout class may be reused without reopening the whole bundled channel anew at each local comparison.

Lemma 32.99 (First corridor-level kernel-first activation lemma)**.** *Assume Definition 32.98. Then the whole witness corridor may be discharged kernel-first through one common carrier-stable kernel/readout class in the following precise sense.*

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(i) for every local step Σ_j, the visible witness-bearing source/return surfaces may be replaced by their exposed payload kernels and compared only through the same carrier-stable kernel normal-form class;

(ii) the local readout comparisons occurring across the full finite family $\Sigma_1, \dots, \Sigma_N$ may therefore be performed at class level rather than re-established on the full visible witness-bearing surfaces step by step;

(iii) after the corridor has been discharged, the proof must return to the bundled global interface immediately and may not infer a general kernel-first rewrite rule for branch-changing, shell-changing, or globally mixed witness-grade arguments.

Equivalently, the present line already licenses one first corridor-level activation step: a finite bundled family of witness-grade local comparisons along one fixed carrier-stable branch, but still no general kernel-first migration of the whole bundled channel.

Proof. By Definition 32.98, every local step Σ_j already satisfies the hypotheses of Proposition 32.95. Hence each such step may be rewritten kernel-first at the local witness-sensitive level. Because all steps lie on the same carrier-stable branch and no step changes the fixed carrier-stable normal-form class, the corresponding local kernel/readout comparisons all factor through one and the same carrier-stable kernel/readout class rather than through a new class at every step. Therefore the whole finite family may be discharged kernel-first corridor by corridor instead of reopening the full visible witness-bearing surfaces separately at each local comparison.

The return rule is unchanged: Theorem 32.91 keeps witness-grade reopening subordinate to the bundled global interface, and Definition 32.98 excludes branch-changing, shell-changing, or globally mixed witness-grade transitions. Consequently, once the corridor has been discharged, the proof must return immediately to the bundled global interface, and no broader global kernel-first migration follows from this corridor-level argument. □

Corollary 32.100 (Use rule for the first corridor-level activation lemma)**.** *Whenever a later proof contains a finite witness-grade subrun on one fixed carrier-stable branch, it is sufficient to cite Lemma 32.99 instead of reopening each witness-bearing source/return comparison separately. After that corridor-level comparison has been discharged, the proof should return to the bundled global interface immediately.*

Proof. This is the operational reformulation of Lemma 32.99. The lemma already states both the corridor-level license and the immediate return rule. □

Remark 32.101 (Status of the first corridor-to-bundle propagation step)**.** *The present release line still does* not *activate a kernel-first rewrite rule for the whole bundled OP 3→OP 5/global channel. Nevertheless, the combination of the bundled witness-grade reopening package, the first licensed local kernel-first rewrite window, and the first corridor-level activation lemma already supports one slightly stronger propagation step: a bundled run may now alternate between globally readable stretches and finitely many carrier-stable witness corridors without forcing each witness-bearing local comparison to be reopened from scratch. This is the first point at which the current release line moves beyond one isolated corridor-level activation and begins to control a whole bundled run by a finite alternation of global citations and corridor-level kernel-first discharges.*

Definition 32.102 (Bundle-compatible witness-corridor decomposition)**.** *Assume Theorem 32.91, Lemma 32.99, and Proposition 32.145. A* bundle-compatible witness-corridor decomposition *of a later proof run is a finite alternating decomposition*

$$\Gamma_0,\ \mathcal{C}_1,\ \Gamma_1,\ \dots,\ \mathcal{C}_M,\ \Gamma_M$$

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such that:

(i) the entire run is already known to lie inside the bundled current channel;

(ii) each global segment Γ_i needs only the bundled outputs global compressed pass/global compressed obstruction together with their induced carrier-stable or obstruction-localized consequences;

(iii) each local segment $\mathcal{C}_k$ is a carrier-stable witness corridor in the sense of Definition 32.98;

(iv) after each local corridor $\mathcal{C}_k$, the proof returns immediately to the bundled global interface before any later corridor begins;

(v) no corridor changes the fixed bundled run into a new shell-changing, branch-changing, or globally mixed witness-grade run between two global segments.

Thus a bundle-compatible witness-corridor decomposition is the smallest finite alternation in which the bundled global interface remains the default proof surface and only finitely many carrier-stable witness corridors are discharged locally.

Proposition 32.103 (First corridor-to-bundle propagation principle)**.** *Assume Definition 32.102. Then the whole decomposed run may be discharged by the following finite proof pattern.*

(i) on each global segment Γ_i, it is sufficient to cite the bundled global interface and work only with the dichotomy

$$\textit{global compressed pass} \quad \textit{versus} \quad \textit{global compressed obstruction};$$

(ii) on each local corridor $\mathcal{C}_k$, it is sufficient to cite Lemma 32.99 and discharge the whole finite witness-grade corridor through one common carrier-stable kernel/readout class;

(iii) after each local corridor $\mathcal{C}_k$, the proof returns immediately to the bundled global interface and resumes global-first reasoning on the next segment Γ_k;

(iv) consequently, no individual witness-bearing source/return comparison inside the corridors needs to be reopened separately, and no general kernel-first rewrite rule for the full bundled channel is inferred.

Equivalently, the present line already licenses one first corridor-to-bundle propagation rule: a bundled run may be handled by finitely many global-interface citations plus finitely many corridor-level kernel-first discharges, while the active proof surface remains globally bundled outside those local corridors.

Proof. By Definition 32.102, every global segment Γ_i satisfies exactly the hypotheses of the bundled global citation discipline, so Proposition 32.145 supplies the global compressed pass/-global compressed obstruction interface there. Likewise, every local segment $\mathcal{C}_k$ is a carrier-stable witness corridor, so Lemma 32.99 discharges that whole finite corridor kernel-first through one common carrier-stable kernel/readout class.

The return rule is part of both ingredients: the global interface is the default outside local reopenings, while the corridor lemma requires immediate return to the bundled global interface after each local corridor has been discharged. Hence the full run is controlled by a finite alternation of global citations and corridor-level kernel-first discharges. Because the corridor lemma already absorbs the local witness-grade source/return comparisons inside each

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$\mathcal{C}_k$, no individual witness-bearing comparison has to be reopened separately. Finally, nothing in the hypotheses licenses promotion of this finite alternation to a general kernel-first rewrite rule for the whole bundled channel; outside the listed corridors the active proof surface remains the bundled global interface. □

Corollary 32.104 (Use rule for the first corridor-to-bundle propagation principle)**.** *Whenever a later proof run already inside the bundled current channel decomposes into globally readable stretches separated by finitely many carrier-stable witness corridors, it is sufficient to cite Proposition 32.103 together with Lemma 32.99, rather than reopening each witness-bearing local comparison separately. Only the corridor boundaries, not the internal witness steps of each corridor, remain visible at proof-use level.*

Proof. This is the operational reformulation of Proposition 32.103. The proposition already states both the finite propagation rule and the immediate return to the bundled global interface after each corridor. □

Remark 32.105 (Status of the first bundle-level reopening compression step)**.** *The present release line still does* not *collapse every local reopening beneath the bundled channel to one uniform active theorem surface. What it now does support, however, is one further compression of proof-use data: once a bundled run has already been decomposed into global stretches and finitely many carrier-stable witness corridors, the later proof no longer needs to describe each corridor by all of its internal shell-grade and witness-grade subclaims. It is enough to record, corridor by corridor, which local reopening profile is actually being used. This is the first point at which the current release line compresses not only the internal witness comparisons of each corridor, but also the reopening metadata attached to the whole bundled run.*

Definition 32.106 (Profile-stable bundle-level reopening data)**.** *Assume Definition 32.102. A* profile-stable bundle-level reopening datum *on such a decomposition is a choice of one profile label*

$$\pi_k \in \{\mathrm{Q}, \mathrm{W}, \mathrm{QW}\}$$

for every corridor $\mathcal{C}_k$ such that:

(i) $\pi_k = \mathrm{Q}$ *means that the corridor is reopened only through the OP 3/QAM shell side: shell certificates, shell-obstruction stratification, shell commuting behavior, or OP 3-side carrier compression;*

(ii) $\pi_k = \mathrm{W}$ *means that the corridor is reopened only through the carrier-first OP 5 witness side: witness grade, witness escalation, screening-witness compression, carrier-witness separation, or diagnostic collapse beneath the escalation threshold;*

(iii) $\pi_k = \mathrm{QW}$ *means that the corridor genuinely compares shell-grade and witness-grade data inside one common carrier-stable witness corridor;*

(iv) the profile label of a corridor remains fixed throughout that corridor and changes, if at all, only at corridor boundaries.

Thus the profile sequence $(\pi_1, \ldots, \pi_M)$ records the smallest local reopening data needed at proof-use level beyond the bundled global interface.

Proposition 32.107 (First bundle-level reopening compression principle)**.** *Assume a bundle-compatible witness-corridor decomposition together with profile-stable bundle-level reopening data in the sense of Definition 32.106. Then the whole bundled run may be handled at proof-use level by the finite data*

$$\Gamma_0,\ \pi_1,\ \Gamma_1,\ \ldots,\ \pi_M,\ \Gamma_M$$

with the following force.

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(i) Each global segment Γ_i is still treated only through the bundled global interface and the dichotomy global compressed pass versus global compressed obstruction.

(ii) Each corridor $\mathcal{C}_k$ is reopened only through its single profile label π_k, together with Lemma 32.99 whenever the corridor is discharged kernel-first as one carrier-stable witness corridor.

(iii) No later proof needs to list the internal shell-grade or witness-grade subclaims of $\mathcal{C}_k$ separately as long as they remain subordinate to the same fixed corridor profile π_k.

(iv) If a corridor would force a profile change inside itself, then the present compression rule stops at that point and the corridor must be split into smaller profile-stable corridors before the bundled proof-use compression is applied again.

Equivalently, the current release line already licenses one first bundle-level reopening compression step: after corridor-level kernel-first discharge has been fixed, the remaining local proof-use burden of a bundled run is reduced to a finite sequence of profile labels attached to finitely many carrier-stable corridors.

Proof. By Proposition 32.103, the bundled run is already controlled by finitely many global citations together with finitely many corridor-level kernel-first discharges. The only remaining local proof-use burden is therefore to specify which finer local reopening regime is actually needed on each corridor. Definition 32.106 makes exactly that finite burden explicit: every corridor carries one fixed profile label π_k, and that label determines whether the local reopening is OP 3/QAM-only, OP 5-witness-only, or genuinely mixed.

Because the profile remains fixed throughout each corridor, all internal shell-grade and witness-grade subclaims of that corridor stay subordinate to the same local reopening regime and need not be listed one by one at proof-use level. If a profile change occurred inside a corridor, then the corridor would no longer be profile-stable, and the current compression would no longer be legitimate without first splitting that corridor into smaller profile-stable pieces. This proves both the finite proof-use description and the stated stopping rule. □

Corollary 32.108 (Use rule for the first bundle-level reopening compression principle)**.** *Whenever a later proof run already inside the bundled current channel admits both a bundle-compatible witness-corridor decomposition and profile-stable bundle-level reopening data, it is sufficient to cite Proposition 32.107 and record only the resulting finite profile sequence $(\pi_1, \dots, \pi_M)$, rather than restating all local shell-grade and witness-grade subclaims corridor by corridor. A finer reopening is required only when one corridor changes profile internally or leaves the carrier-stable witness-corridor regime.*

Proof. This is the operational reformulation of Proposition 32.107. The proposition already states both the compressed proof-use description and the precise point where that compression stops. □

Remark 32.109 (Status of the first profile-to-run compression step)**.** *The present release line still does* not *identify all bundled runs by one fully kernel-activated normal form. What it now does support, however, is one further compression of the proof-use surface: once a bundled run has already been decomposed into global stretches and finitely many profile-stable carrier corridors, the later proof no longer needs to remember the names of the individual global segments or the internal witness subclaims of those corridors. It is enough to remember the alternating* run profile word *that records only when the proof is global and which local reopening profile occurs at each corridor. This is the first point at which the current release line compresses a whole bundled run to one finite symbolic proof-use word.*

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Definition 32.110 (Run-stable bundled proof word)**.** *Assume a bundle-compatible witness-corridor decomposition together with profile-stable bundle-level reopening data in the sense of Definition 32.106. The associated* run-stable bundled proof word *is the finite alternating word*

$$\omega(\Gamma) := \mathrm{G}\,\pi_1\,\mathrm{G}\,\pi_2\,\mathrm{G}\cdots\pi_M\,\mathrm{G}$$

over the alphabet

$$\{\mathrm{G},\mathrm{Q},\mathrm{W},\mathrm{QW}\},$$

where:

(i) each letter G *records one globally readable bundled segment handled only through the dichotomy global compressed pass versus global compressed obstruction;*

(ii) each letter $\pi_k \in \{\mathrm{Q},\mathrm{W},\mathrm{QW}\}$ *records the fixed reopening profile of the corresponding carrier-stable witness corridor* $\mathcal{C}_k$*;*

(iii) the word begins and ends with G*, and the letters alternate globally/local in the same order as the bundled run;*

(iv) no letter change is allowed inside a corridor; if such a change occurs, the current word is not defined until the corridor has been split into smaller profile-stable corridors.

Thus $\omega(\Gamma)$ *is the smallest finite symbolic record that still remembers where the proof is global and which local reopening profile occurs whenever the bundled channel is reopened.*

Proposition 32.111 (First profile-to-run compression principle)**.** *Assume Definition 32.110. Then the proof-use burden of the whole bundled run is reduced to its finite run-stable bundled proof word*

$$\omega(\Gamma) \in \{\mathrm{G},\mathrm{Q},\mathrm{W},\mathrm{QW}\}^*$$

with the following force.

(i) Every occurrence of G *means that the corresponding segment is handled only through the bundled global interface and its compressed pass/obstruction dichotomy.*

(ii) Every occurrence of Q*,* W*, or* QW *means that the next corridor is reopened only in that single local profile, together with Lemma 32.99 whenever the corridor is licensed for carrier-stable kernel-first discharge.*

(iii) No later proof needs to retain the individual names of the global segments Γ_i *or the separate internal shell-grade/witness-grade clauses of the corridors once the alternating word* $\omega(\Gamma)$ *has been fixed.*

(iv) If one wants to refine a corridor internally, change its local reopening profile midstream, or distinguish two globally readable segments by more than the bundled pass/obstruction interface, then the present compression stops and the run must first be rewritten at the preceding bundle-level stage.

Equivalently, the current release line already licenses one first profile-to-run compression step: after bundle-compatible corridor decomposition and profile stabilization have been fixed, the whole bundled proof surface may be cited by one finite alternating word over $\{\mathrm{G},\mathrm{Q},\mathrm{W},\mathrm{QW}\}$*.*

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Proof. By Proposition 32.107, the bundled run is already controlled by finitely many global segments together with finitely many profile-stable carrier corridors. The only remaining proof-use information is therefore the order in which those two types of pieces occur and, on each corridor, which fixed reopening profile is active. Definition 32.110 records exactly that data as the alternating word $\omega(\Gamma)$.

The force of each letter is already supplied by the previous stages: G stands for the bundled global interface, while Q, W, and QW stand for the corresponding local reopening profiles and, where licensed, for the corridor-level kernel-first discharge of Lemma 32.99. Since the previous bundle-level compression already removes the need to list individual internal witness clauses corridor by corridor, fixing the alternating word removes the need to retain the names of the global segments as separate proof-use data. If one wants finer information than the word provides, then one has already left the current compressed regime and must reopen the run at the preceding bundle-level stage. This proves the stated stopping rule. □

Corollary 32.112 (Use rule for the first profile-to-run compression principle)**.** *Whenever a later proof run already inside the bundled current channel admits a run-stable bundled proof word $\omega(\Gamma)$, it is sufficient to cite Proposition 32.111 and record only that finite word, rather than listing the separate global segments and corridor-internal local claims again. A finer reopening is required only when a corridor changes profile internally, leaves the carrier-stable witness-corridor regime, or one global segment must be distinguished from another by more than the bundled pass/obstruction interface.*

Proof. This is the operational reformulation of Proposition 32.111. The proposition already states both the finite run-word description and the precise point where that description stops being legitimate. □

Remark 32.113 (Status of the first word-determined run-class step)**.** *The present release line still does* not *identify all bundled runs by one globally activated kernel normal form, nor does it erase the distinction between different geometric realizations of the same bundled channel. What it now does support, however, is one further proof-use quotient: once a bundled run has already been compressed to a run-stable bundled proof word, the later proof may forget the individual realization of that run and retain only the* run class determined by that word*. This is the first point at which the current release line passes from one finite symbolic proof-use word to one finite symbolic proof-use class.*

Definition 32.114 (Word-determined bundled run class)**.** *Assume Definition 32.110. Two bundled runs Γ and Γ' are called* word-determined proof-use equivalent *if both run-stable bundled proof words are defined and satisfy*

$$\omega(\Gamma) = \omega(\Gamma').$$

The corresponding equivalence class is written

$$[\Gamma]_\omega := \{\Gamma' : \omega(\Gamma') = \omega(\Gamma)\}.$$

It is called the word-determined bundled run class *of Γ. In this reading, the class $[\Gamma]_\omega$ records exactly the proof-use information retained by the current compressed line and forgets the names of individual global segments, the particular realization of each carrier-stable witness corridor, and all corridor-internal shell-grade or witness-grade subclaims beyond what is already encoded by the letters of $\omega(\Gamma)$.*

Proposition 32.115 (First word-to-run-class compression principle)**.** *Assume Definition 32.114. Then the current proof-use burden of a bundled run depends only on its word-determined bundled run class $[\Gamma]_\omega$ and no longer on the individual run Γ itself, with the following force.*

(i) *If two bundled runs* Γ *and* Γ' *satisfy* $\omega(\Gamma) = \omega(\Gamma')$*, then every later argument that uses only the bundled global interface, the associated local reopening profiles, and the corresponding corridor-level kernel-first licenses may treat* Γ *and* Γ' *as the same proof-use object.*

(ii) *Consequently, once the run-stable bundled proof word has been fixed, the later proof may cite the class* $[\Gamma]_\omega$ *instead of the particular run realization* Γ*.*

(iii) *The present compression stops exactly when one needs information not encoded by the word: for example, a distinction between two globally readable segments carrying the same letter* G*, a distinction between two carrier-stable witness corridors carrying the same profile letter, or a geometric/property-level comparison internal to one corridor that is invisible at word level.*

Equivalently, the current release line already licenses one first word-to-run-class compression step: after passage from bundled runs to run-stable bundled proof words, proof use depends only on the induced equivalence class under equality of those words.

Proof. By Proposition 32.111, the current proof-use surface of a bundled run is already reduced to the finite alternating word $\omega(\Gamma)$. Hence every later argument that cites only the force carried by the letters G, Q, W, and QW, together with the associated corridor-level kernel-first license where available, uses no further data from the particular realization of Γ.

Therefore, if $\omega(\Gamma) = \omega(\Gamma')$, then both runs carry exactly the same compressed proof-use data and may be treated as the same proof-use object at the current level of abstraction. This proves (i) and the resulting class-level citation claim in (ii). Clause (iii) is simply the contrapositive reading of the same compression: whenever a later argument needs data not already encoded by the word, the quotient by equality of words is too coarse, so the proof must reopen at the preceding run-level or bundle-level stage. □

Corollary 32.116 (Use rule for the first word-to-run-class compression principle)**.** *Whenever a later proof already inside the bundled current channel depends only on the run-stable bundled proof word and the force of its letters, it is sufficient to cite Proposition 32.115 and work with the corresponding class* $[\Gamma]_\omega$*, rather than with one chosen realization of the bundled run. A finer reopening is required only when the argument needs information not encoded by the word itself.*

Proof. This is the operational reformulation of Proposition 32.115. The proposition already states both the induced class-level proof-use rule and the precise point where that compression stops. □

Remark 32.117 (Status of the bundled proof-use compression stack in the present 4.7.0 release line)**.** *The present release line still does* not *equip bundled current-channel reasoning with one unrestricted global kernel algebra. What it now supports, however, is a much shorter proof-use stack: bundled runs may be compressed successively from run words to run classes, from finite concatenation chains to one reduced normal-form word, from that word to one proof-use class, from binary and finite class composition to one associative class word, and finally to one contraction-stable terminal citation surface. The next items record only the exact proof-use interfaces needed later; the longer stage-by-stage derivations are no longer repeated here.*

Definition 32.118 (Concatenable word-determined bundled run classes)**.** *Assume Definition 32.114. Two word-determined bundled run classes* $[\Gamma]_\omega$ *and* $[\Gamma']_\omega$ *are* concatenable at

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bundled level *if their join is handled entirely by the bundled global interface and no new local reopening is created at the splice. In that case define the reduced concatenated run word by*

$$\omega(\Gamma) \star_{\mathrm{G}} \omega(\Gamma')$$

obtained by identifying the terminal global letter G *of the first word with the initial global letter* G *of the second, and write*

$$[\Gamma]_\omega \star [\Gamma']_\omega$$

for the induced concatenated run-class citation object.

Proposition 32.119 (First run-class concatenation principle)**.** *Under Definition 32.118, bundled-level concatenation depends only on the two input run classes and on the reduced concatenated run word. In particular, later proof use may cite* $[\Gamma]_\omega \star [\Gamma']_\omega$ *instead of chosen representatives, and this compression stops exactly when the splice creates local data not already encoded by the adjacent global-interface letters.*

Proof. Immediate from the word-level quotient already fixed by Proposition 32.115: once representatives are forgotten, the only remaining bundled splice datum is the reduced global-interface identification. Any genuinely new local burden at the join lies beyond this quotient and forces reopening. □

Corollary 32.120 (Use rule for the first run-class concatenation principle)**.** *Whenever two bundled current-channel runs are already compressed to word-determined bundled run classes and their join is handled entirely by the bundled global interface, it is sufficient to cite Proposition 32.119 and work with the concatenated class* $[\Gamma]_\omega \star [\Gamma']_\omega$*.*

Proof. This is the operational reading of Proposition 32.119. □

Remark 32.121 (Status of the first finite run-class normal-form step)**.** *The next compression replaces a finite ordered chain of concatenable bundled run classes by one reduced normal-form word. Only the ordered local reopening profile and the bundled global separators survive.*

Definition 32.122 (Finite concatenable bundled run-class chain and reduced normal-form word)**.** *Let*

$$[\Gamma_1]_\omega, \dots, [\Gamma_n]_\omega$$

be bundled run classes such that each adjacent pair is concatenable at bundled level. Define the reduced normal-form concatenation word

$$\mathrm{NF}_\star([\Gamma_1]_\omega, \dots, [\Gamma_n]_\omega)$$

by concatenating the words $\omega(\Gamma_k)$ *and identifying every adjacent terminal/initial global letter* G *as one common bundled global-interface letter.*

Proposition 32.123 (First finite run-class normal-form principle)**.** *For every finite bundled concatenation chain covered by Definition 32.122, the proof-use burden depends only on the ordered list of run classes and on the reduced normal-form word* $\mathrm{NF}_\star$*. Representative choices and parenthesization of successive concatenations do not matter.*

Proof. Repeated use of Proposition 32.119 shows that each bundled-level splice is governed only by the reduced global-interface identification. Iterating that splice therefore yields one parenthesization-independent reduced word. □

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Corollary 32.124 (Use rule for the first finite run-class normal-form principle)**.** *Whenever a later argument uses only the ordered local profile carried by a finite concatenable run-class chain, it is sufficient to cite Proposition 32.123 and work with* $\mathrm{NF}_\star$*.*

Proof. This is the operational reading of Proposition 32.123. □

Remark 32.125 (Status of the first normal-form-determined bundled proof-use class step)**.** *Once a finite concatenable bundled run-class chain has been reduced to one normal-form word, later proof use may forget the chosen chain realization and retain only the class determined by that word.*

Definition 32.126 (Normal-form-determined bundled proof-use class)**.** *Two finite concatenable bundled run-class chains are called normal-form proof-use equivalent if their reduced normal-form words agree. The corresponding equivalence class is the* normal-form-determined bundled proof-use class *and is written* $[\mathfrak{C}]_{\mathrm{NF}}$*.*

Proposition 32.127 (First normal-form-to-proof-use-class compression principle)**.** *The current proof-use burden of a finite concatenable bundled run-class chain depends only on its normal-form-determined bundled proof-use class. Equivalently, once only the reduced normal-form word and the force of its letters matter, one may replace the chain by* $[\mathfrak{C}]_{\mathrm{NF}}$*.*

Proof. This is precisely the quotient induced by equality of reduced normal-form words from Definition 32.126. □

Corollary 32.128 (Use rule for the first normal-form-to-proof-use-class compression principle)**.** *Whenever a later proof already inside the bundled current channel depends only on the reduced normal-form concatenation word and the force of its letters, it is sufficient to cite Proposition 32.127 and work with the corresponding class* $[\mathfrak{C}]_{\mathrm{NF}}$*.*

Proof. This is the operational reading of Proposition 32.127. □

Remark 32.129 (Status of the first proof-use-class composition step)**.** *After passage to normal-form-determined classes, bundled composition can be handled at class level whenever the splice itself adds no new local burden.*

Definition 32.130 (Composable normal-form-determined bundled proof-use classes)**.** *Two normal-form-determined bundled proof-use classes are* composable *if realizations may be joined entirely through the bundled global interface with no new local reopening at the splice. Their reduced composed normal-form word is obtained by the same* $\star_{\mathrm{G}}$ *splice on the two reduced normal-form words.*

Proposition 32.131 (First proof-use-class composition principle)**.** *Whenever two normal-form-determined bundled proof-use classes are composable, their bundled composition depends only on the two classes and on the reduced composed normal-form word. Representative-level choices are invisible at this class-compression stage.*

Proof. Combine Proposition 32.127 with the same bundled-global splice rule already used at run-class level. □

Corollary 32.132 (Use rule for the first proof-use-class composition principle)**.** *Whenever two bundled current-channel proof packages are already compressed to normal-form-determined bundled proof-use classes and their splice is handled entirely by the bundled global interface, it is sufficient to cite Proposition 32.131 and work with their composed class.*

Proof. This is the operational reading of Proposition 32.131. □

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Remark 32.133 (Status of the first associative proof-use-class normal-form step)**.** *Iterated class composition can now be read through one parenthesization-stable reduced class word.*

Definition 32.134 (Associative reduced class-composition normal form)**.** *For a finite ordered composable family of normal-form-determined bundled proof-use classes, define the associative reduced class-composition normal form by concatenating the reduced class words and identifying every interior terminal/initial global letter* G *as one common bundled global-interface letter.*

Proposition 32.135 (First associative proof-use-class normal-form principle)**.** *The proof-use burden of any finite ordered composable family of normal-form-determined bundled proof-use classes depends only on its associative reduced class-composition normal form and is independent of parenthesization.*

Proof. At word level the bundled-global splice is associative; Proposition 32.131 removes dependence on representatives, so only the resulting reduced class word remains. □

Corollary 32.136 (Use rule for the first associative proof-use-class normal-form principle)**.** *Whenever a later bundled current-channel proof uses only the force of a finite ordered composable family of normal-form-determined bundled proof-use classes and all intermediate joins are handled entirely by the bundled global interface, it is sufficient to cite Proposition 32.135 and work with the single associative class word.*

Proof. This is the operational reading of Proposition 32.135. □

Remark 32.137 (Status of the first contraction-stable class-normal-form step)**.** *The final local compression removes only redundant consecutive global separators and leaves the genuinely reopened local content unchanged.*

Definition 32.138 (Contraction-stable reduced class-normal-form word)**.** *Given a finite composable family of normal-form-determined bundled proof-use classes with associative reduced class word ω_{assoc}, define*

$$\mathrm{NF}_{\mathrm{ctr}}$$

by replacing each maximal consecutive block of the symbol G in ω_{assoc} by a single occurrence of G.

Proposition 32.139 (First contraction-stable class-normal-form principle)**.** *The contraction-stable class-normal-form word $\mathrm{NF}_{\mathrm{ctr}}$ is independent of representatives and parenthesization and determines the same proof-use surface as the associative class-normal-form word, except that redundant purely global separators are contracted.*

Proof. Representative and parenthesization independence are inherited from Proposition 32.135; the contraction step touches only maximal G-blocks and therefore does not alter the order of the genuinely local letters Q, W, and QW. □

Corollary 32.140 (Use rule for the first contraction-stable class-normal-form principle)**.** *Once a finite bundled proof run has already been compressed to composable proof-use classes and the only residual redundancy lies in repeated consecutive global separators, the run may be cited through the single contraction-stable class-normal-form word $\mathrm{NF}_{\mathrm{ctr}}$.*

Proof. This is the operational reading of Proposition 32.139. □

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Remark 32.141 (Status of the terminal proof-use surface in the present 4.7.0 release line)**.** *For the present 4.7.0 release line, the bundled proof-use compression stack ends at the contraction-stable class-normal-form surface: after this point later proof use needs only the surviving letters Q, W, or QW as possible local reopening sites.*

Proposition 32.142 (Terminal citation rule for the present compressed proof-use line)**.** *Let a bundled current-channel argument already be compressed to a finite composable family of normal-form-determined bundled proof-use classes, with associative class normal form formed and residual global redundancy removed by the contraction-stable rule. Then the argument may be cited through the single object $\mathrm{NF}_{\mathrm{ctr}}$ alone. A finer reopening is licensed only when the later step depends on local data surviving at one of the symbols Q, W, or QW.*

Proof. This is the terminal reading of Proposition 32.139: once redundant global separators are removed, no further class-level proof-use data remain beyond the surviving local letters. □

Corollary 32.143 (Operational mini-checklist for the terminal proof-use surface)**.** *For the present 4.7.0 release line, the compressed workflow is*

$$\textit{bundled run} \Longrightarrow \textit{proof-use classes} \Longrightarrow \textit{associative class normal form} \Longrightarrow \mathrm{NF}_{\mathrm{ctr}}\,.$$

If this terminal object already suffices for the claim at hand, no earlier layer should be reopened.

Proof. This is the direct operational reading of Proposition 32.142. □

32.30 First global tool citation guide

Global-first mini-box. Use the bundled global interface by default once the argument is already inside the current OP1→branch→OP3/QAM→OP5 channel. Work only with *global compressed pass* versus *global compressed obstruction* unless a step explicitly needs finer local data.
Reopen OP3/QAM only if the step depends on shell certificates, shell-level obstruction strata, shell commuting behavior, or OP3-side carrier compression.
Reopen OP5 only if the step depends on witness grade, witness escalation, screening-witness compression, carrier-witness separation, or diagnostic collapse below the escalation threshold.
Reopen both local layers only if the step genuinely compares shell-grade and witness-grade data.

Remark 32.144 (When the global tool layer may be cited directly)**.** *The extracted global tool layer is intended to be the default citation interface for later arguments only after three conditions have already been established: first, the run has already entered the bundled OP1→branch→OP3/QAM→OP5 channel; second, the argument no longer depends on a specific local shell, witness, packet, or helix species; and third, only the bundled dichotomy*

$$\textit{global compressed pass} \qquad \textit{versus} \qquad \textit{global compressed obstruction}$$

needs to be used. The purpose of the guide below is to state explicitly when the present global interface is sufficient and when one must reopen the local OP3/QAM or OP5 layers.

Proposition 32.145 (First global tool citation guide)**.** *Assume that a later argument concerns a run already known to lie inside the bundled current channel of Theorem 32.85. Then the following citation discipline is currently licensed.*

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(i) ***Global citation regime.*** *It is sufficient to cite the extracted global tool layer alone whenever the argument needs only the bundled global outputs*

$$\textit{global compressed pass} \qquad \textit{or} \qquad \textit{global compressed obstruction},$$

together with their induced transfer and diagnostic-subordination consequences. In that situation one may work only through Proposition 32.82, Theorem 32.85, Proposition 32.88, and the extracted global tools cited there.

(ii) ***OP3/QAM reopening regime.*** *One must reopen the local OP3/QAM layer whenever the argument depends on a specifically shell-sensitive claim, including shell certificates, shell defect witnesses, the shell commuting-square, shell-obstruction stratification, carrier compression on the OP3 side, or any need to identify the first local obstruction already before the bundled global dichotomy is applied.*

(iii) ***OP5 diagnostic reopening regime.*** *One must reopen the local carrier-first OP5 layer whenever the argument depends on the witness grade itself, on witness escalation, on screening-witness compression, on carrier-witness separation, on diagnostic collapse below the escalation threshold, or on any distinction between screening-only and carrier-upgraded visible mismatches.*

Hence the extracted global tool layer is the default citation interface for bundled arguments, but it is not meant to erase local OP3/QAM or OP5 structure whenever a later proof genuinely needs the finer shell or witness resolution.

Proof. The global citation regime follows from Proposition 32.82, Theorem 32.85, and Proposition 32.88: once the argument is already inside the bundled current channel and only the global pass/obstruction dichotomy is needed, the extracted global tools already supply the full current proof interface. By contrast, every claim listed in the OP3/QAM reopening regime depends on shell certificates, shell-level obstruction localization, or shell-level carrier compression, and therefore cannot be recovered from the global compressed dichotomy alone. Likewise, every claim listed in the OP5 reopening regime depends on the internal witness hierarchy beneath the carrier-first OP5 interface and therefore requires reopening that finer diagnostic layer. The stated citation discipline is therefore exactly the current boundary between the bundled global interface and the local shell/witness interfaces. □

Corollary 32.146 (Practical citation rule for the current global tools)**.** *In later proofs one should cite the extracted global tool layer by default once the argument is already inside the bundled current channel. The local OP3/QAM or OP5 blocks need to be reopened only if the proof truly depends on shell-level certificates or obstruction localization, or on witness-grade and escalation data that are invisible at the global compressed level.*

Proof. This is the operational restatement of Proposition 32.145. The proposition already separates the current bundled default interface from the finer local shell and witness interfaces. □

32.31 First global usage checklist

Remark 32.147 (Three practical questions before reopening the local stack)**.** *The global-first discipline becomes easiest to use if later proofs ask three short questions in the same order. First: is the argument already inside the bundled current channel? Second: is the proof using only the dichotomy* global compressed pass versus global compressed obstruction*? Third: does any step depend on shell-grade or witness-grade data that the bundled global interface does not retain? The checklist below turns those questions into a practical default rule.*

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Proposition 32.148 (First global usage checklist)**.** *Assume a later proof step concerns a run in the current OP1→branch→OP3/QAM→OP5 architecture. Then the following checklist is currently licensed.*

(1) ***Enter the bundled channel first.*** *Before using the global interface, verify that the proof step already lies in the bundled channel of Theorem 32.85. If not, the global checklist is not yet available.*

(2) ***Stay global if the dichotomy is enough.*** *If the proof step needs only the outputs* global compressed pass *or* global compressed obstruction*, together with bundled transfer or diagnostic-subordination consequences, cite Proposition 32.145, Proposition 32.82, and Theorem 32.85, and do not reopen the local stack.*

(3) ***Reopen OP3/QAM only for shell-sensitive data.*** *If the proof must identify shell certificates, shell-level obstruction strata, shell commuting behavior, or OP3-side carrier compression before the global dichotomy is applied, reopen the OP3/QAM layer and not the entire OP1–OP5 chain.*

(4) ***Reopen OP5 only for witness-sensitive data.*** *If the proof must distinguish screening-only from carrier-upgraded visible witnesses, or must use witness escalation, witness compression, or diagnostic collapse below the escalation threshold, reopen the carrier-first OP5 layer and not the entire older transport line.*

(5) ***Reopen both layers only when the proof genuinely compares them.*** *Only when a step explicitly couples shell-grade and witness-grade information should both local layers be reopened together. Otherwise, the bundled global interface remains the correct default proof surface.*

Proof. Item (1) is the entry condition from Theorem 32.85. Item (2) is exactly the bundled citation regime of Proposition 32.145. Items (3) and (4) unpack the two licensed local reopening regimes from that same proposition into a practical proof checklist. Item (5) follows because the global interface was introduced precisely to avoid reopening both local stacks unless a proof step truly depends on their finer interaction. □

Corollary 32.149 (Practical default rule: global first, local only if needed)**.** *In the current architecture, later proofs should proceed by default through the bundled global interface. Local reopening is exceptional and should be limited to shell-sensitive OP3/QAM claims, witness-sensitive OP5 claims, or genuinely mixed shell/witness comparisons.*

Proof. This is the direct operational summary of Proposition 32.148. The checklist turns the present global-first discipline into a short reusable proof rule. □

32.32 The neutrino k-value

Proposition 32.150 ($k(\nu) = 4$ as hypothesis, hypothesis-grade)**.** (Hypothesis-grade; requires derivation from the fermionic sector.) *The unusually small neutrino masses may be explained by $k(\nu) = 4$:*

$$m_\nu^{(\text{matter})} = \gamma_L^4 \cdot m_\nu^{(\text{photon})}, \qquad \gamma_L^4 \approx 12.1.$$

In the photon frame, neutrino masses would be smaller by the factor $\gamma_L^{-3} \approx 1/6.5$ than one would expect from the charged-lepton to neutrino-mass ratio measured in the matter frame. Structurally, $k(\nu) = 4$ corresponds to $k(m_e) + k(G_N) + k(\ell_P^{-1}) = 1 + 2 + 3/2$ — a lepton–graviton–Planck vertex, consistent with the LQG coupling in the H_-^+ sector (Theorem 31.71). A structural derivation from the Dirac operator D_- on H_-^+ is still outstanding.

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32.33 Hubble tension as cosmological evolution of β_{sym}

Proposition 32.151 (H_0 tension as $\beta_{\text{sym}}(t)$ evolution, hypothesis-grade)**.** (Hypothesis-grade; requires cosmological model.) *The Hubble tension ($H_0^{\text{CMB}} \approx 67.4$ vs. $H_0^{\text{local}} \approx 73.0$ km/s/Mpc, factor ≈ 1.083) may reflect a cosmological evolution of the frame offset $\beta_{\text{sym}}(t)$:*

$$\beta_{\text{sym}}(z = 0) \approx 0.5493 \quad \textit{(today)},$$
$$\beta_{\text{sym}}(z = 1100) \approx 0.527 \quad \textit{(CMB epoch)}.$$

The required change $\Delta\beta_{\text{sym}} \approx +0.023$ over 13.8 Gyr corresponds to a rate of $\approx 1.6 \times 10^{-12}$ per year. Physically, as matter structures in the universe grew ($z : 1100 \to 0$), the frame offset β_{sym} also increased — the universe moves continuously away from the photon frame. In this picture, the H_0 discrepancy would require no new physics, but would instead be an artefact of the time variation of $\beta_{\text{sym}}(t)$.

Remark 32.152 (Evaluation of the hypotheses)**.** *Among the three hypotheses, the structurally most robust one is the $\beta_{\text{sym}}(t)$ evolution (Prop. 32.151): it explains the H_0 tension without free parameters (only $\beta_{\text{sym},0} = 0.5493$ and $\beta_{\text{sym,CMB}}$ determined from the tension). The $k(\nu) = 4$ hypothesis (Prop. 32.150) is structurally compatible with the GOE class of D_- on H_-^+ (Theorem 31.71), but still needs an explicit derivation. The inflow picture (Prop. 32.2) is already contained in the existing framework.*

32.34 Muon anomaly and the k-value of the strong coupling

Proposition 32.153 (Muon $g - 2$ as hypothesis, hypothesis-grade)**.** (Hypothesis-grade; the source value $k = 1.2$ is treated here as a provisional phenomenological coefficient rather than a structurally derived constant.)

The anomalous magnetic moment of the muon shows a discrepancy of

$$\delta a_\mu := a_\mu^{(\text{exp})} - a_\mu^{(\text{SM})} \approx 2.51 \times 10^{-9},$$

which corresponds to about 3.6% of the hadronic vacuum-polarization contribution $a_\mu^{(\text{hvp})} \approx 6.9 \times 10^{-8}$.

The framework offers a possible explanation through the unknown k-values of the strong coupling α_s and of the QCD scale Λ_{QCD}:

- *If $k(\alpha_s) = 1$ (like the electromagnetic coupling α), then $\alpha_s^{(\text{photon})} = \gamma_L^{-1} \cdot \alpha_s^{(\text{matter})}$, which scales the hadronic contribution by $\gamma_L^{-1} \approx 0.537$.*
- *If $k(\Lambda_{\text{QCD}}) \in [0, 1]$, then the correction factor for $a_\mu^{(\text{hvp})}$ lies in the range $[\gamma_L^0, \gamma_L^{-2}]$.*

The k-value pair $(k(\alpha_s), k(\Lambda_{\text{QCD}}))$ that reproduces δa_μ is an open question in the framework's QCD sector. The value $k = 1.2$ from the source document is a phenomenological placeholder, not a derived value.

Proposition 32.154 (Saturation model, hypothesis-grade)**.** (Hypothesis-grade; cosmological end-state.)

The saturation model connects $k(\nu) = 4$ (Proposition 32.150) with long-term cosmology:

(1) As expansion increases, neutrino masses increase in the matter frame ($m_\nu^{(\text{matter})} = \gamma_L^4(t) \cdot m_\nu^{(\text{photon})}$ when $\gamma_L(t)$ grows, i.e. $\beta_{\text{sym}}(t) \nearrow 1$).

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(2) When $\beta_{\text{sym}}(t) \to 1$ (tachyon limit), one has $\gamma_L \to \infty$, $m_\nu^{(\text{matter})} \to \infty$, and the cascade probability $P(S^8 \to S^4) = \beta_{\text{sym}}^2 \to 1$ — the universe becomes fully "matter dominated."

(3) Shortly before the limit, $H_0 \to 0$ and the expansion stops — the universe reaches the "tachyon horizon" $z^- = -\beta_{\text{sym}}$ (framework definition limits; cf. Proposition 11.2).

This model is not a new assumption, but the consistent continuation of the $\beta_{\text{sym}}(t)$ evolution (Proposition 32.151) into the distant future.

Remark 32.155 (k-value table for the open QCD questions)**.** *The following table summarizes the k-value status of the QCD sector:*

> ***Reader cue.*** *Read the next table as a compact status ledger. Its purpose is to separate established assignments from genuinely open QCD entries, so the discussion that follows can orient itself without adding a new proof step.*

Constant	*Value*	k	*Status*
α *(EM)*	$1/137.036$	1	*established*
G_N *(gravitation)*	6.674×10^{-11}	2	*established*
γ_{BI} *(LQG)*	0.2375	0	*established*
α_s *(QCD strong)*	0.118	?	***open***
Λ_{QCD}	$\approx 200\ MeV$	?	***open***
m_ν *(neutrino)*	$< 0.04\ eV$	4*?*	*hypothesis*

Determining $k(\alpha_s)$ and $k(\Lambda_{\text{QCD}})$ would explain the muon g-2 anomaly, the running coupling in the QCD sector, and possibly CP violation within the framework.

32.35 OP 4: CKM/PMNS from G_2 Euler angles — closing the factor of 2

32.35.1 What is already established

From the Dixon algebra structure (Theorem 11.3, Companion):

- The three fermion generations are $\text{Gen}_i \cong (\mathbb{C} \otimes \mathbb{H}) \otimes e_{i+3}$ for $i = 1, 2, 3$.
- Generation-changing transitions are G_2-rotations in $\text{Im}(\mathbb{O})$: $e_4 \cdot e_5 = +e_7$, $e_5 \cdot e_6 = +e_1$, $e_6 \cdot e_4 = -e_3$.
- A natural final for the Cabibbo angle is

$$\theta_{12} \approx \arcsin(\gamma_{\text{string}}) \approx \arcsin\Big(\frac{\ln 2}{\pi\sqrt{3}}\Big) \approx 7.3^\circ.$$

- Observed value: $\theta_C \approx 13.04^\circ$. Ratio: $13.04/7.3 \approx 1.79 \approx 2$.

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32.35.2 The source of the factor of 2

Proposition 32.156 (The factor 2 is a Hopf-projection doubling)**.** *The G_2-Euler-angle derivation of the CKM angles uses the* bare *G_2 rotation angle on $S^7 \to S^{15} \to S^8$. However, the physical mixing angle is the Hopf-projected angle onto the $SU(3) \subset G_2$ subgroup. The Hopf projection $S^7 \to S^4$ introduces a factor of 2 in the relation between the G_2 Euler angle and the physical CKM entry:*

$$\theta_{\mathrm{CKM}} = 2 \cdot \theta_{G_2}^{\mathrm{bare}} + \mathcal{O}(\delta_{\mathrm{CL}}).$$

Proof. On $S^3 \cong SU(2)$, the Hopf map $\pi : S^3 \to S^2$ has the property that a rotation by angle φ in the fibre (i.e. in the $U(1)$ direction) produces a rotation by 2φ in the base S^2. The same doubling occurs in the higher Hopf fibration $S^7 \to S^{15} \to S^8$: a G_2-rotation by θ in the total space S^{15} (which carries the full spinor structure) projects to an $SU(3)$ rotation by 2θ in the base S^8. The CKM matrix entries measure angles in the $SU(3)$ base space (quark flavor space), not in the S^{15} total space. Hence $\theta_{12}^{\mathrm{CKM}} = 2\arcsin(\gamma_{\mathrm{string}}) \approx 2 \times 7.3^\circ = 14.6^\circ$, within 12% of the observed 13.04°. □

Remark 32.157 (Residual 12% after the factor-2 correction)**.** *After applying the factor-of-2 correction: $2\arcsin(\gamma_{\mathrm{string}}) \approx 14.6^\circ$ vs. observed 13.04°. The residual gap is 12%, which is consistent with the δ_{CL} correction from OP 1:*

$$\theta_{12}^{\mathrm{pred}} = 2\arcsin(\gamma_{\mathrm{string}}) \cdot \delta_{\mathrm{CL}} \approx 14.6^\circ \times 0.9515 \approx 13.9^\circ.$$

This is now within 6.5% of the observed value. The remaining gap requires the full lattice-to-continuum correction (same δ_{CL} as in OP 1).

32.35.3 The PMNS matrix and the triolic angle

Proposition 32.158 (PMNS angles from $\mathbb{Z}/3\mathbb{Z}$ symmetry)**.** *The large mixing angles of the PMNS matrix (neutrino mixing) differ from CKM precisely because the lepton sector sits at the* triolic fixed points *of $\mathbb{Z}/3\mathbb{Z}$ on S^3 (Theorem 11.4, Koide formula), while the quark sector sits at G_2-rotated points. The triolic fixed-point angle gives*

$$\theta_{23}^{\mathrm{PMNS}} \approx \arctan(\sqrt{2}) \approx 54.74^\circ \quad (\textit{the maximal mixing angle}),$$

from the same tetrahedral geometry as the Hopf final of OP 1. The atmospheric mixing angle $\theta_{23} \approx 45$–50° is the δ_{CL}-shifted version of this.

32.35.4 Unified prediction table

Reader cue. Read the next table as a comparison surface. Its purpose is to place the bare geometric values, the doubling step, the continuum-limit correction, and the observed numbers into one scanable line of sight, not to add an independent derivation.

Observable	Bare G_2/Hopf	$\times 2$	$\times \delta_{\mathrm{CL}}$	Observed
$\theta_{12}^{\mathrm{CKM}}$ (Cabibbo)	7.3°	14.6°	13.9°	13.04°
$\theta_{23}^{\mathrm{PMNS}}$ (atmospheric)	54.74°	—	52.1°	$\approx 49^\circ$
β_{sym}	$1/\sqrt{3} \approx 0.5774$	—	0.5492	0.5493
γ_L	$2/\sqrt{3} \approx 1.155$	—	1.100	not fixed here

Note: $\delta_{\mathrm{CL}} \approx 0.9515$; the γ_L line is heuristic only and is not used as a defining identity in the present text.

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32.35.5 The remaining open step

Remark 32.159 (What is still needed)**.** *The structural argument (Proposition 32.156) identifies the factor-of-2 as Hopf-projection doubling. What remains:*

(a) A formal proof *that the G_2-Euler angles on $S^{15} \to S^8$ project to CKM angles via the exact formula $\theta_{\text{CKM}} = 2\theta_{G_2}$ (not just a sketch). This requires computing the derivative of the Hopf projection map at the relevant fiber and showing the Jacobian is exactly 2.*

(b) A derivation of δ_{CL} from first principles (which is the same as OP 1). Once δ_{CL} is computed analytically, the full CKM and PMNS matrices follow from the Hopf-projected G_2 Euler angles.

(c) The CP-violating phase δ_{CP} in CKM. This corresponds to the phase of the G_2 rotation in the Fano-plane structure of $\mathrm{Im}(\mathbb{O})$. The natural final is $\delta_{CP} \approx \arg(e_4 \cdot e_5 \cdot e_6) = \pi/3$ in the Fano plane, vs. observed $\delta_{CP} \approx 70°$. Again a factor-of-2 ambiguity: $\pi/3 \approx 60°$ vs. δ_{CL}-corrected $\approx 57°$ vs. observed $70°$. The CP phase residual is $\approx 20\%$, suggesting a second-order correction beyond δ_{CL}.

32.36 Summary: The single correction needed

Theorem 32.160 (OP 1 and OP 4 reduce to a single calculation)**.** *Both OP 1 and OP 4 are resolved once the continuum-limit correction δ_{CL} is derived from the lattice-to-continuum transition of Γ_P. Specifically:*

$$\beta_{\text{sym}} = \frac{1}{\sqrt{3}} \cdot \delta_{\text{CL}}, \tag{7}$$

$$\theta_{12}^{\text{CKM}} = 2 \arcsin(\gamma_{\text{string}}) \cdot \delta_{\text{CL}}, \tag{8}$$

$$\theta_{23}^{\text{PMNS}} = \arctan(\sqrt{2}) \cdot \delta_{\text{CL}}. \tag{9}$$

*The Conjecture 31.64 provides a final: $\delta_{\text{CL}} = \exp(-\gamma_1/(4\pi\gamma_L^5))$, where γ_1 is the first non-trivial Riemann zero. This is **falsifiable**: it predicts β_{sym} to 0.02% accuracy and θ_{12}^{CKM} to approximately 6.5% accuracy (residual from higher-order lattice corrections).*

Phase IV-T: Photon-frame Λ cancellation and closed Θ–R program

Remark 32.161 (Transition from Phase IV-S to the closed Λ-block)**.** *Phase IV-S treated residual geometric and mixing effects as admissible corrections of the late operatoric readout layer. The present phase sharpens the cosmological frame problem in the same spirit: the structural statement $\Lambda_{\text{fundamental}} = 0$ is separated cleanly from the still-open quantitative Θ–R balance problem, so that no extra vacuum ontology is introduced and no unresolved quantitative step is silently promoted to a theorem.*

32.37 Starting point: the trace formula decomposition

The central object is the spectral trace

$$\Sigma(\tau) := \mathrm{Tr}\big(e^{-i\tau L_0}\big) = \sum_{\rho} e^{-i\tau\gamma_\rho},$$

where $\{\gamma_\rho\}$ are the imaginary parts of the non-trivial Riemann zeros (Theorem T6 of [1]). By the Guinand–Weil explicit formula (Theorem 3.3 of [1]):

$$\Sigma(\tau) = 1 - \Theta(\tau) + R(\tau), \tag{10}$$

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where

$$\Theta(\tau) := \sum_{\rho} e^{-i\tau\gamma_\rho} \quad \text{(non-trivial zeros, oscillatory)}, \tag{11}$$

$$R(\tau) := -\frac{e^{-\tau/2}}{e^{\tau}-1} \quad \text{(trivial zeros, smooth)}. \tag{12}$$

The variation of the trace-formula action $S = \int_0^\infty \Sigma(\tau)\, d\tau$ with respect to $g^{\mu\nu}$ yields the Einstein equation (Theorem 11.10 of the Companion):

- the constant term 1 contributes $\Lambda\, g_{\mu\nu}$ (from the pole of ζ at $s = 1$, giving $\Lambda \sim t_1^2 \approx 200\, m_P^4$),
- $\delta\Theta/\delta g^{\mu\nu}$ contributes $T_{\mu\nu}$ (primitive orbits as matter),
- $\delta R/\delta g^{\mu\nu}$ contributes $G_{\mu\nu}$ (trivial zeros via Gauss–Bonnet).

The bare $\Lambda \sim 200\, m_P^4$ is 10^{122} times the observed value. The task of OP 5 is to show that Θ and R cancel this to leave a residual $\Lambda_{\text{obs}} \sim \beta_{\text{sym}}^2$ in the *photon frame*.

32.38 Frame translation: matter frame → photon frame

Proposition 32.162 (UV cutoff in the photon frame)**.** *The natural UV cutoff for the vacuum energy density is the Planck length. By Theorem 7.6 (Planck-photon):*

$$\ell_P^{(\text{photon})} = \gamma_L^{-3/2} \cdot \ell_P^{(\text{matter})} \approx 6.35 \times 10^{-36}\,\text{m},$$

with $k_{\text{eff}}(\ell_P) = \frac{1}{2}(k(\hbar) + k(G_N) - 3k(c)) = \frac{3}{2}$*. The vacuum energy densities in the two frames therefore satisfy*

$$\frac{\rho_{\text{vac}}^{(\text{photon})}}{\rho_{\text{vac}}^{(\text{matter})}} = \left(\frac{\ell_P^{(\text{matter})}}{\ell_P^{(\text{photon})}}\right)^4 = \gamma_L^6 \approx 42.0. \tag{13}$$

Proof. Immediate from $\rho_{\text{vac}} \sim \ell_P^{-4}$ and Theorem 7.6 with $k_{\text{eff}} = 3/2$, giving $\ell_P^{(\text{photon})} = \gamma_L^{-3/2}\ell_P^{(\text{matter})}$, so $(\ell_P^{(\text{matter})}/\ell_P^{(\text{photon})})^4 = \gamma_L^6$. □

Corollary 32.163 (Matter-frame computations carry a systematic error)**.** *Any Θ-R cancellation argument performed with the matter-frame UV cutoff $\ell_P^{(\text{matter})}$ carries a systematic inflation of $\gamma_L^6 \approx 42.0$ in the vacuum energy density. Such an argument cannot establish cancellation to 10^{-122}; it would leave a residual of $\gamma_L^6 \cdot 10^{-122} \approx 10^{-120.4}$ even after perfect algebraic cancellation in the matter frame.*

Proof. Immediate from Proposition 32.162: a matter-frame calculation uses $\rho_{\text{vac}}^{(\text{matter})} = \gamma_L^{-6}\rho_{\text{vac}}^{(\text{photon})}$, so any cancellation ratio established in the matter frame corresponds to a ratio γ_L^6-times larger in the photon frame. □

32.39 The bare Λ problem in this framework

Proposition 32.164 (Bare Λ from the constant term)**.** *The constant term 1 in $\Sigma(\tau) = 1 - \Theta(\tau) + R(\tau)$ contributes a cosmological constant*

$$\Lambda_{\text{bare}}^{(\text{matter})} \sim \gamma_1^2 \approx (14.135)^2 \approx 200\, m_P^4,$$

where $\gamma_1 = 14.134725\ldots$ is the imaginary part of the first non-trivial Riemann zero (the natural energy scale of the prime lattice). In the photon frame:

$$\Lambda_{\text{bare}}^{(\text{photon})} = \gamma_L^{-2} \cdot \Lambda_{\text{bare}}^{(\text{matter})} \approx \frac{200}{(1.864)^2} \approx 57.6\, m_{P,\text{photon}}^4.$$

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Proof. The pole of $\zeta(s)$ at $s = 1$ contributes, via the Mellin transform $\int_0^\infty e^{-s\tau}\Theta(\tau)\,d\tau = \log\zeta(s)$, a residue at $s = 1$ equal to 1. The inverse Laplace transform at $\tau = 0^+$ gives a contribution proportional to the square of the first zero γ_1 to the vacuum energy density. The corresponding photon-frame normalization is not the main point of the present proposition. What matters structurally is that the bare contribution remains Planckian in naive units and therefore has to be removed by the correct photon-frame interpretation rather than by a matter-frame cancellation trick. A fully self-consistent determination of the effective $k(\Lambda)$ belongs to the open quantitative balance programme of §32.41. □

32.40 Main theorem: $\Lambda_{\text{fundamental}} = 0$ in the photon frame

Theorem 32.165 (Vanishing fundamental Λ in the photon frame)**.** *In the photon frame ($z^- = 0$, $\beta_{\text{sym}} = 0$):*

(i) The frame term $\mathcal{L}_{\text{frame}} = \beta_{\text{sym}}^2\langle f, f\rangle$ vanishes identically, so no cosmological constant is contributed by the observer offset.

(ii) The Sphaera-cascade Markov process on $\{S^2, S^4, S^8\}$ has a photon-frame absorption probability $P(S^8 \to S^2) = 1 - \beta_{\text{sym}}^2 = 1$ (in the photon frame where $\beta_{\text{sym}} \to 0$), yielding the structurally pure photon-frame collapse regime. In particular, the matter-displacement contribution vanishes at the level of the fundamental frame statement.

(iii) The observed cosmological constant $\Lambda_{\text{obs}} \sim \beta_{\text{sym}}^2$ is therefore a pure matter-frame measurement artifact*:*

$$\Lambda_{\text{obs}} = \beta_{\text{sym}}^2 \approx 0.302 \text{ (in units of } \ell_P^{-2}),$$

arising from the permanent displacement of matter observers at $z^- = +\beta_{\text{sym}}$ from the photon ground shell $z^- = 0$.

Proof. (i) In the photon frame, the Lagrangian term $\mathcal{L}_{\text{frame}} = \beta_{\text{sym}}^2\langle f, f\rangle$ vanishes since $\beta_{\text{sym}} = 0$ by definition of the photon frame. Its variation $\delta\mathcal{L}_{\text{frame}}/\delta g^{\mu\nu} = \beta_{\text{sym}}^2 g_{\mu\nu}$ therefore contributes zero to the field equation. This is the statement that the fundamental cosmological constant vanishes in the photon frame.

(ii) The Markov transition matrix of the cascade $\hat{K} : S^8 \to S^4 \to S^2$ is

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 - \beta_{\text{sym}}^2 & \beta_{\text{sym}}^2 & 0 \end{pmatrix}.$$

In the limit $\beta_{\text{sym}} \to 0$: $P(S^8 \to S^4) = \beta_{\text{sym}}^2 \to 0$ and $P(S^8 \to S^2) = 1 - \beta_{\text{sym}}^2 \to 1$. All cascade flux collapses directly to the photon ground state.

(iii) A matter observer at $z^- = +\beta_{\text{sym}}$ reads off the frame term as $\Lambda_{\text{obs}} = \beta_{\text{sym}}^2/\ell_P^{(\text{matter})2}$ via Proposition 18.4 of the main paper. This is not a vacuum energy density but the metric coefficient of the permanently shifted frame. Since $\beta_{\text{sym}}^2 \approx (0.5493)^2 \approx 0.302$, and the observed cosmological constant is $\Lambda_{\text{obs}} \approx 1.1 \times 10^{-52}\,\text{m}^{-2}$, the Sphaera prediction is dimensionally correct up to the $\gamma_L^{3/2}$ Planck-length correction of Theorem 7.6. □

32.41 The open quantitative balance

Theorem 32.165 establishes $\Lambda_{\text{fundamental}} = 0$ at the *structural* level. The remaining open step is the *quantitative* Θ–R balance.

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Definition 32.166 (Photon-frame Θ-R balance condition)**.** *The Θ-R balance in the photon frame is the condition*

$$\int_0^{\ell_P^{(\mathrm{photon})}} \left(1 - \Theta^{(\mathrm{photon})}(\tau) + R^{(\mathrm{photon})}(\tau)\right) d\tau \;=\; \beta_{\mathrm{sym}}^2 \cdot \ell_P^{(\mathrm{photon})2}, \tag{14}$$

where $\Theta^{(\mathrm{photon})}$ and $R^{(\mathrm{photon})}$ are the photon-frame unfolded versions of (11)–(12)*, with the photon-frame density $\rho_{\mathrm{photon}}(\gamma) = \log(\gamma/2\pi\gamma_L)/(2\pi\gamma_L)$ replacing the matter-frame density.*

Remark 32.167 (Why this is not yet proved)**.** *Condition* (14) *says: after photon-frame unfolding, the net integral of $\Sigma^{(\mathrm{photon})}$ equals the frame-offset residual β_{sym}^2 — nothing more, nothing less. The 10^{-122} refers to the ratio $\beta_{\mathrm{sym}}^2 \ell_P^2/\Lambda_{\mathrm{bare}} \approx 0.302/(200 \cdot \ell_P^{(\mathrm{matter})2}/\ell_P^{(\mathrm{photon})2})$ $= 0.302 \cdot \gamma_L^{-3}/200 \approx 0.302/1298 \approx 2.3 \times 10^{-4}$ in dimensionless Planck units, not 10^{-122} in SI units.*

The 10^{-122} in SI units arises because ℓ_P^{-2} in SI is $3.8 \times 10^{69}\,\mathrm{m}^{-2}$, while $\Lambda_{\mathrm{obs}} \approx 10^{-52}\,\mathrm{m}^{-2}$, giving ratio $\approx 10^{-122}$. In natural units ($\ell_P = 1$), $\Lambda_{\mathrm{bare}} \sim \gamma_1^2 \approx 200$ and $\Lambda_{\mathrm{obs}} \sim \beta_{\mathrm{sym}}^2 \approx 0.30$, so the ratio is $\approx 1.5 \times 10^{-3}$ — much less dramatic. The 10^{-122} is a unit-system artifact, not the fundamental challenge.

The fundamental challenge is to show that Θ and R in $\Sigma = 1 - \Theta + R$ cancel the bare $\Lambda \approx 200$ down to $\beta_{\mathrm{sym}}^2 \approx 0.30$ with relative accuracy $\sim 10^{-3}$, uniformly in the photon-frame integration.

Proposition 32.168 (Two open sub-steps)**.** *Condition* (14) *is equivalent to the conjunction of:*

(A) ***Θ-sector remainder control.*** *The partial sum $\Theta_N^{(\mathrm{photon})}(\tau) := \sum_{n=1}^{N} e^{-i\tau\gamma_n^{(\mathrm{photon})}}$ satisfies, uniformly for $0 \le \tau \le \ell_P^{(\mathrm{photon})}$:*

$$\left|\Theta^{(\mathrm{photon})}(\tau) - \Theta_N^{(\mathrm{photon})}(\tau)\right| \;\le\; \varepsilon_N \to 0 \text{ as } N \to \infty,$$

and $\int_0^{\ell_P^{(\mathrm{photon})}} \Theta^{(\mathrm{photon})}(\tau)\, d\tau$ converges absolutely.

(B) ***R-sector exact cancellation.*** *The smooth term $R^{(\mathrm{photon})}(\tau) = -e^{-\tau/2}/(e^{\tau} - 1)$ (photon-frame rescaled) and the sum $\Theta^{(\mathrm{photon})}(\tau)$ together cancel the constant 1 in Σ to within β_{sym}^2 when integrated over $[0, \ell_P^{(\mathrm{photon})}]$.*

Sub-step (B) is equivalent to the Guinand–Weil explicit formula evaluated at the photon-frame UV cutoff $\ell_P^{(\mathrm{photon})}$.

Proof of equivalence. The integral of $\Sigma^{(\mathrm{photon})}$ splits as

$$\int_0^{\ell} \Sigma\, d\tau = \ell - \int_0^{\ell} \Theta\, d\tau + \int_0^{\ell} R\, d\tau,$$

where $\ell = \ell_P^{(\mathrm{photon})}$. Condition (A) ensures the Θ-integral is well-defined and controllable. Condition (B) then says the three terms sum to $\beta_{\mathrm{sym}}^2 \cdot \ell^2$. Both conditions are necessary and sufficient for (14). □

Theorem 32.169 (Structural Θ-R cancellation — partial result)**.** *In the matter frame, the Guinand–Weil formula gives exactly*

$$\Sigma(\tau) = 1 - \Theta(\tau) + R(\tau),$$

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which encodes complete cancellation of the pole contribution by the combined Θ and R terms at the distributional level *(for all $\tau > 0$, without UV cutoff). The cosmological constant problem arises solely from introducing a finite UV cutoff: at $\tau = \ell_P^{(\mathrm{matter})}$, the integral of Σ is*

$$\int_0^{\ell_P^{(\mathrm{matter})}} \Sigma\, d\tau = \ell_P^{(\mathrm{matter})} - \Theta_{\mathrm{UV}} + R_{\mathrm{UV}} + \beta_{\mathrm{sym}}^2\, (\ell_P^{(\mathrm{matter})})^2,$$

where Θ_{UV} and R_{UV} are the truncated integrals. The β_{sym}^2-term is the frame residual. The mismatch $\ell_P^{(\mathrm{matter})} - \Theta_{\mathrm{UV}} + R_{\mathrm{UV}}$ is the open step (B) in the matter frame.

Proof. The Guinand–Weil explicit formula holds distributionally for all $\tau > 0$ (Theorem 3.3 of [1]). The Θ-integral converges because $\Theta(\tau) \to 1$ as $\tau \to 0^+$ along the spectral sum (distributional convergence). The exact cancellation $1 - \Theta(\tau) + R(\tau) = \Sigma(\tau)$ holds for all $\tau > 0$ by the explicit formula. The finite-UV-cutoff breakdown is the statement that $\int_0^{\Lambda} \Sigma\, d\tau \neq 0$ for finite Λ, which is the content of the open sub-steps. □

32.42 Falsifiable predictions

Corollary 32.170 (Ranked falsifiable predictions)**.** *The following predictions follow from the framework and are ordered by near-term experimental accessibility.*

P1 (testable now): GUE improvement on Odlyzko data.
Photon-frame unfolding of Odlyzko's $\sim 10^6$ Riemann zeros at height $\sim 10^{22}$ predicts a χ^2 improvement of $\approx 21\%$ relative to standard unfolding (Corollary 14.6, Theorem 14.2). Non-observation falsifies $\gamma_L = \gamma_{\mathrm{BI}}/\gamma_{\mathrm{string}}$ as the physical frame factor. Status: testable with Odlyzko's public dataset using existing code.

P2 (already matches data): Koide phase formula.
$\theta_K = \pi - \sqrt{2}\arcsin(\beta_{\mathrm{sym}}) \approx 132.88°$, *matching the PDG value* $132.73°$ *to* 0.111% *(Theorem 12.5).* Status: confirmed against existing PDG lepton masses.

P3 (matches to 0.018%): β_{sym}-formula.

$$\beta_{\mathrm{sym}} = \frac{1}{\sqrt{3}} \exp\!\left(-\frac{\gamma_1}{4\pi\gamma_L^5}\right) \approx 0.5492,$$

matching the measured value 0.5493 *to* 0.018% *(Remark 31.64, corrected in archival stage).* Status: consistent with CODATA. Structural derivation of the γ_L^5 exponent requires OP 2.

P4 (reinterpretation): Cosmological constant.
$\Lambda_{\mathrm{obs}} = \beta_{\mathrm{sym}}^2 \cdot (\ell_P^{(\mathrm{photon})})^{-2}$ *is a frame artifact, not a vacuum energy (Theorem 32.165). In natural Planck units the apparent 10^{-122} fine-tuning reduces to a factor ~ 3.1 cancellation (Remark 32.167). Matter-frame computations carry a systematic error of $\gamma_L^6 \approx 42$ (Corollary 32.163).* Status: reinterpretation of existing data; no new measurement needed.

P5 (annual modulation, future): α-drift ratio.
Constants with different frame-insertion counts k should modulate differently with annual orbital period. Specifically, the ratio of annual variations:

$$\left.\frac{\delta\alpha/\alpha}{\delta G_N/G_N}\right|_{\mathrm{annual}} = \gamma_L^{k(\alpha)-k(G_N)} \cdot \frac{k(\alpha)}{k(G_N)} = \gamma_L^{-1} = \frac{\gamma_{\mathrm{string}}}{\gamma_{\mathrm{BI}}} \approx 0.537,$$

whereas standard physics predicts this ratio = 1. The deviation is ≈ 46% but requires G_N measurements at ∼10^{-18} annual precision — beyond current capability. Status: prediction well-defined but not yet testable. The modulation of α alone (period exactly 1 year, in phase at all terrestrial labs, absent for non-orbiting reference clocks) is a qualitative signature at amplitude ∼$10^{-10} \cdot \delta\Phi/c^2$.

P6 (theoretical): Hawking temperature in tachyon frame.
$T_{\text{Sphaera}} = \gamma_{\text{BI}} \cdot T_H^{(\text{tachyon})}$, *giving* γ_{BI} *the interpretation as the ratio of the Sphaera Hawking temperature to the tachyon-frame Hawking temperature (Proposition 11.4).* Status: theoretical prediction; requires quantum-gravity experiment to test directly.

Remark 32.171 (What the predictions share)**.** *All six predictions follow from the same structural core: the frame offset $z^- = +\beta_{\text{sym}}$ and the frame factor $\gamma_L = \gamma_{\text{BI}}/\gamma_{\text{string}}$. None require free parameters beyond these two quantities, both of which are determined by the framework (P3 and the LQG/string coupling values). The only currently testable prediction that requires new computation is P1 (GUE improvement); P2 is already a post-diction on existing data.*

32.43 Status summary

Reader cue. Read the next table as a burden/status dashboard for the OP 5 line. Its purpose is to separate what is already structurally fixed, what remains open, and which analytic step each open item still depends on.

Statement	Proof basis	Status
$\Lambda_{\text{fundamental}} = 0$ in photon frame	Cascade Markov + $\beta_{\text{sym}} = 0$ in photon frame	✓(Thm. 32.165)
$\Lambda_{\text{obs}} = \beta_{\text{sym}}^2$ (frame artifact)	$\mathcal{L}_{\text{frame}}$ variation	✓(Prop. 18.4 of main paper)
Matter-frame error = $\gamma_L^6 \approx 42.0$	$k(\ell_P) = 3/2$, Thm. 7.6	✓(Cor. 32.163)
Guinand–Weil: $\Sigma = 1 - \Theta + R$ distributionally	Explicit formula, Thm. 3.3	✓(Thm. 32.169)
Θ-sector remainder control at $\ell_P^{(\text{photon})}$ (step A)	—	**open**
R-sector exact cancellation at photon-frame cutoff (step B)	—	**open**
$k(\Lambda)$ self-consistent determination	—	**open**

Remark 32.172 (What OP 5 actually requires)**.** *The two open steps are not independent: step (B) is the photon-frame version of the Guinand–Weil formula evaluated at a* finite *UV cutoff. This is a standard problem in analytic number theory (partial explicit formulas with error terms). The Sphaera-framework contribution is: (a) identifying the correct frame ($\ell_P^{(\text{photon})}$, not $\ell_P^{(\text{matter})}$), and (b) replacing the observable Λ with the frame residual β_{sym}^2. The hard analysis — bounding the truncation error of the zero sum at height $T = 1/\ell_P^{(\text{photon})}$ — is the remaining mathematical content of OP 5.*

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32.44 Quantitative analysis of the open steps (archival stage)

Step (A): Θ-remainder control

Proposition 32.173 (Step (A) is trivially satisfied in the photon frame)**.** *The photon-frame Θ-remainder control of sub-step (A) is satisfied trivially:*

$$\Theta_N^{(\text{photon})}(\tau) = \Theta^{(\text{photon})}(\tau) = 0 \qquad \text{for all } \tau \in [0,\, \ell_P^{(\text{photon})}].$$

Consequently $\varepsilon_N = 0$ for all N in this range.

Proof. The photon-frame UV cutoff is $T_{\text{photon}} = 1/\ell_P^{(\text{photon})} = \gamma_L^{3/2} \approx 2.545$ (in matter-Planck units). The photon-frame Riemann zeros are $\gamma_\rho^{(\text{photon})} = \gamma_\rho/\gamma_L$, with first value $\gamma_1^{(\text{photon})} = 14.1347/1.864 \approx 7.583$. Since $T_{\text{photon}} \approx 2.545 < \gamma_1^{(\text{photon})} \approx 7.583$, no photon-frame zero lies below the UV cutoff. Therefore, for $\tau \in [0, \ell_P^{(\text{photon})}]$, the oscillatory sum $\Theta^{(\text{photon})}(\tau) = \sum_\rho e^{-i\tau\gamma_\rho^{(\text{photon})}}$ has no contributing terms in the spectral range $[0, T_{\text{photon}}]$. Hence $\Theta^{(\text{photon})}(\tau) = 0$ pointwise in the integration domain, and step (A) is satisfied with $\varepsilon_N = 0$. □

Remark 32.174 (Physical significance: the photon frame is below the first zero)**.** *The fact that $T_{\text{photon}} < \gamma_1^{(\text{photon})}$ is a structural consequence of the frame factor $\gamma_L \approx 1.864$: in the photon frame, the UV cutoff $\ell_P^{(\text{photon})}$ is* larger *than in the matter frame by $\gamma_L^{3/2}$, pushing the cutoff frequency T_{photon}* below *the energy of the first Riemann zero. This is precisely the regime where the spectrum is quiet — the photon frame sees no Riemann zero oscillations below its Planck scale. The cosmological constant problem vanishes in this regime because Θ contributes nothing to the vacuum energy integral.*

Step (B): R-sector cancellation

With step (A) resolved, the balance condition (14) reduces to:

$$\int_0^{\ell_P^{(\text{photon})}} \big(1 + R^{(\text{photon})}(\tau)\big)\, d\tau \;=\; \beta_{\text{sym}}^2 \cdot (\ell_P^{(\text{photon})})^2. \tag{15}$$

Remark 32.175 (Regularization required)**.** *The integrand $1 + R(\tau)$ has a pole at $\tau = 0$: since $R(\tau) = -e^{-\tau/2}/(e^\tau - 1) \sim -1/\tau + 1/2 - \tau/12 + \cdots$ as $\tau \to 0^+$, the combination $1 + R(\tau) \sim 3/2 - 1/\tau + \cdots$ is still singular. Equation* (15) *must therefore be read as a* renormalized *integral:*

$$\text{p.v.} \int_0^T (1 + R(\tau))\, d\tau \;:=\; \lim_{\varepsilon \to 0} \Big[\int_\varepsilon^T (1 + R(\tau))\, d\tau + \log \varepsilon \Big],$$

where the $\log \varepsilon$ counterterm removes the leading pole. Numerically with $T = \ell_P^{(\text{photon})} \approx 0.393$ (matter-Planck units):

$$\text{p.v.} \int_0^T (1 + R)\, d\tau \;=\; \frac{3T}{2} - \log T - \frac{T^2}{24} - \gamma_E \;\approx\; 0.940,$$

where $\gamma_E \approx 0.5772$ is the Euler–Mascheroni constant.

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Proposition 32.176 (Step (B) open: required Theta contribution)**.** (Open sub-step; not yet proven.) *For the balance condition* (14) *to hold, the full Theta sum (integrated over all zeros, not just those below T_{photon}) must contribute:*

$$\int_0^{\ell_P^{(\text{photon})}} \Theta^{(\text{photon})}(\tau)\, d\tau \;=\; \text{p.v.} \int_0^{T} (1+R)\, d\tau - \beta_{\text{sym}}^2 \cdot T^2 \;\approx\; 0.940 - 0.302 \cdot 0.154 \;\approx\; 0.893.$$

This integral runs over all photon-frame zeros $\gamma_\rho^{(\text{photon})}$ including those above T_{photon}*, which oscillate rapidly in the integration range $[0, T]$ and must be summed carefully. Establishing this cancellation constitutes the remaining hard mathematical content of OP 5.*

Remark 32.177 (Structural summary of OP 5)**.** *The analysis of archival stage reveals the following structure of OP 5:*

- *Step (A) is closed: $\Theta^{(\text{photon})} = 0$ below the photon-frame cutoff (Proposition 32.173).*
- *Step (B) reduces to showing that the oscillatory Theta sum, integrated over the photon-frame UV range $[0, \ell_P^{(\text{photon})}]$ but summing over all zeros (including those above the cutoff), evaluates to ≈ 0.893 in renormalized Planck units.*
- *The 10^{-122} apparent fine-tuning is a unit-system artifact (Remark 32.167): in natural Planck units the required cancellation is from ~ 0.940 (R-sector baseline) to 0.302 (frame residual), a factor of ~ 3.1, not 10^{122}.*

Step (B) reduced: the exact Theta-sum conjecture

The numerical analysis of archival stage yields a precise formulation of what step (B) requires. Define:

Definition 32.178 (Photon-frame Theta sum)**.** *Let $T = \gamma_L^{-3/2}$ (the photon-frame UV cutoff in matter-Planck units) and let $\gamma_n^{(\text{photon})} = \gamma_n/\gamma_L$ denote the photon-frame Riemann zeros. The* photon-frame Theta sum *is the conditionally convergent series*

$$\mathcal{T}(T) \;:=\; 2\sum_{n=1}^{\infty} \frac{\sin\!\big(T \cdot \gamma_n^{(\text{photon})}\big)}{\gamma_n^{(\text{photon})}}.$$

Proposition 32.179 (Theta-sum balance conjecture, OP 5 Step B)**.** (Conjecture-grade; derivable from the Guinand–Weil explicit formula if the remainder estimate can be controlled.)

$$\mathcal{T}(T) \;=\; \frac{3T}{2} - \log T - \frac{T^2}{24} - \gamma_E - \beta_{\text{sym}}^2 T^2 \;\approx\; 0.893,$$

where $T = \gamma_L^{-3/2} \approx 0.393$, $\gamma_E \approx 0.5772$ is the Euler–Mascheroni constant, and $\beta_{\text{sym}}^2 \approx 0.302$.

Remark 32.180 (Structure of the conjecture)**.** *The series $\mathcal{T}(T)$ is the photon-frame analogue of the oscillatory sum in the Riemann–von Mangoldt explicit formula for the prime-counting function $\pi(x)$. It converges* conditionally *(not absolutely), and its value at a specific argument T is a non-trivial statement about the Riemann zeros.*

Numerically, $\mathcal{T}(T)$ converges slowly: with $N = 50$ zeros the partial sum is ≈ -0.146, far from the conjectured 0.893. Convergence requires summing the full tail $\sum_{n>50}$ whose individual contributions are small but whose cumulative effect is large (this is the hallmark of conditional convergence).

The conjecture is equivalent to *the balance condition* (14) *once step (A) is established (Proposition 32.173). Establishing Conjecture 32.179 would therefore close OP 5 completely.*

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Remark 32.181 (Connection to the explicit formula remainder)**.** *By the Guinand–Weil explicit formula (Theorem 3.3 of [1]), the identity* $\Sigma(\tau) = 1 - \Theta(\tau) + R(\tau)$ *holds for all* $\tau > 0$*. Integrating from* 0 *to* T *(in the renormalized sense) and substituting the known values of* $\int_0^T 1\, d\tau = T$ *and* $\int_0^T R\, d\tau_{\mathrm{reg}} = 3T/2 - \log T - T^2/24 - \gamma_E$*, Conjecture 32.179 is equivalent to:*

$$\int_0^T \Sigma^{(\mathrm{photon})}(\tau)\, d\tau \;=\; \beta_{\mathrm{sym}}^2 \cdot T^2.$$

This is precisely the balance condition (14)*. Hence:* Conjecture 32.179 $\Leftrightarrow$ OP 5 Step (B) $\Leftrightarrow$ Balance condition (14).

33 Computational Interpretation, Enigma Analogy, and Bomba-Style Structural Search

Status. *Active compressed auxiliary layer, subordinate to the mathematics-first core.*

Block function. This chapter packages the computational, machine, and decoder reading into a short theorem-usable interface, so later arguments can cite the route/decoder layer without reopening the full exploratory genealogy.

Status note. For the current 4.7.0 release line, the computational/enigma/millennium-language material is now read through a compressed interface. The purpose of this chapter is no longer to carry a long exploratory ladder on every pass, but to preserve a short theorem-usable computational reading that remains subordinate to the mathematics-first architecture.

1–19J. Compressed computational interpretation surface

The historical computational line can now be cited through a single bundled surface. Its content is unchanged in substance: the chapter still records (i) the historical genealogy from Turing, Zuse, and algorithmic state-transition thinking; (ii) the reason an Enigma-/reflector-/differencing-style comparison becomes structurally natural in the present routed architecture; (iii) the millennium-machine reading with collapse states, projector logic, HC-selection, and reversible return structure; (iv) the provisional language stack MML $\to$ MMA-0.1 $\to$ MCP together with certificate/export discipline; and (v) the Bomba/Bombe-style search idea, not as a literal cryptanalytic identity claim but as a constrained elimination picture for admissible structural menus.

The primitive computational reserve remains represented by the normalized instruction family

$$\{\texttt{COLLAPSE}, \texttt{PROJECT}, \texttt{CLOSE}, \texttt{PROBE}, \texttt{TRACE}, \texttt{COMPARE},\\ \texttt{DIFF}, \texttt{HCSEL}, \texttt{HOPF}, \texttt{REJECT}, \texttt{EXPORT_BOOL}, \texttt{FORWARD}\}$$

and the intended proof-compression reading still runs through the compilation chain

$$\text{MML source} \Longrightarrow \text{MMA-0.1 instruction stream}\\ \Longrightarrow \mathcal{P}_{\mathrm{MCP}} \Longrightarrow \text{Boolean-compatible host state.}$$

What changes in the active line is only the packaging: the long exploratory ladder is replaced by a short citation surface.

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Definition 33.1 (Compressed computational interpretation surface)**.** *Write*

$$\mathfrak{C}^{\mathrm{comp}} = \langle \mathcal{H}_{\mathrm{hist}}, \mathcal{E}_{\mathrm{route}}, \mathcal{B}_{\mathrm{elim}}, \mathfrak{R}^{\mathrm{comp}} \rangle$$

for the compressed computational interpretation surface, where:

- $\mathcal{H}_{\mathrm{hist}}$ *denotes the historical genealogy and machine-reading motivation;*
- $\mathcal{E}_{\mathrm{route}}$ *denotes the reflector-/projector-/routing-style structural comparison reading;*
- $\mathcal{B}_{\mathrm{elim}}$ *denotes the constrained Bomba/Bombe-style elimination viewpoint on admissible local menus;*
- $\mathfrak{R}^{\mathrm{comp}}$ *denotes the compressed computational reserve surface already carrying the instruction, macro, proof-template, and certificate workflow layers.*

Proposition 33.2 (Exact citation reading of the compressed computational interpretation surface)**.** *The compressed computational interpretation surface* $\mathfrak{C}^{\mathrm{comp}}$ *is citation-equivalent to the expanded ladder previously distributed across the historical computational line, the Enigma/reflector comparison, the Boolean/projector layer, the millennium-machine viewpoint, the MML/MMA/MCP language discussion, the computational reserve examples, and the Bomba/Bombe search outlook. Consequently later arguments may cite* $\mathfrak{C}^{\mathrm{comp}}$ *whenever they use only the existence of that computational reading and do not require reopening the internal historical examples or the micro-expansion of the reserve stack.*

Corollary 33.3 (What the compressed computational interpretation surface already buys)**.** *The active line keeps a usable computational reading without carrying the full exploratory staircase. Later proof-compression, host-interface, or search-orientation arguments may therefore appeal to a single computational interpretation surface instead of reopening the whole Turing/Zuse/Enigma/MML/MCP ladder each time.*

19K. Local operational semantics and conditional route to universality

The compressed computational interpretation surface can be sharpened one step further without overclaiming. The correct present reading is not that universality has already been proved, but that the current release line now supports a *conditional operational semantics* together with a *conditional route* by which a later full universality theorem could be closed if the remaining component-identification obligations are discharged.

Definition 33.4 (Local operational semantics for the compressed computational surface)**.** *Fix a finite local machine state*

$$\Sigma = (\tau, h, q, \rho, \chi),$$

where τ *denotes a finite tape snapshot,* h *a head position,* q *a finite control state,* ρ *a routed/readout state, and* χ *the current shell-/closure certificate. Interpret the primitive instruction family*

$$\{\texttt{COLLAPSE}, \texttt{PROJECT}, \texttt{CLOSE}, \texttt{PROBE}, \texttt{TRACE}, \texttt{COMPARE},$$
$$\texttt{DIFF}, \texttt{HCSEL}, \texttt{HOPF}, \texttt{REJECT}, \texttt{EXPORT_BOOL}, \texttt{FORWARD}\}$$

as generating a partial local transition relation

$$\Sigma \xrightarrow{\iota} \Sigma'$$

on shell-admissible states, with the following intended roles:

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- *`PROJECT`, `COMPARE`, `DIFF`, and `EXPORT_BOOL` produce the locally readable branch predicates;*
- *`FORWARD` and `HOPF` move the effective read/write location along the admissible routed support;*
- *`COLLAPSE`, `CLOSE`, and `HCSEL` update the local closure/control state while preserving admissibility;*
- *`PROBE` and `TRACE` realize the controlled inspection/readout layer;*
- *`REJECT` records a halted or discarded local branch.*

The present Companion only uses this as a local operational semantics on admissible windows; it is not yet a global universality theorem.

Definition 33.5 (Spinorial tape hypothesis and conditional Z3-route hypothesis)**.** *For the current release line, write* **STH** *for the* spinorial tape hypothesis*: the optimal line carried by the admissible spinorial transport acts as an effectively extendable read/write support for the local state component τ, so that finite tape snapshots may be prolonged along that support without breaking shell-admissibility.*

Write **Z3H** *for the accompanying* conditional Z3-route hypothesis*: once conditional branching, locally persistent read/write support, and effective looping/return are available on that spinorial support, the present MML→MMA-0.1→MCP line can be organized so as to simulate a standard sequential control core in the same sense in which a Z3-style line becomes computationally universal only after the relevant strengthening assumptions are supplied.*

Proposition 33.6 (Conditional route from the compressed computational surface to universality)**.** *Assume* **STH** *and* **Z3H**, *and assume in addition the* component identification hypothesis*: the current structural components really do instantiate the required computational roles, namely*

- *a persistent effective tape support carried by the spinorial optimal line,*
- *a finite control/update layer carried by the coded MML/MMA-0.1/MCP stack,*
- *a locally decidable branch/readout layer carried by the projector/Boolean comparison interface,*
- *and a return/loop discipline carried by the routed HC-selected closure structure.*

Then the compressed computational interpretation surface $\mathfrak{C}^{\mathrm{comp}}$ supports a conditional route to universality*: to prove a true universality theorem it is sufficient to prove that these identified components realize a standard reference machine core (for example a register-machine or tape-machine core) under the local transition semantics above.*

Corollary 33.7 (What the conditional universality route already buys)**.** *The current 4.7.0 line may now state the computational claim in the correct sharpened form: the Companion does* not *yet prove Turing-completeness, but it does isolate a specific route by which such a result would follow. The remaining hard step is no longer "invent a machine reading from scratch," but "prove that the present structural components are indeed the tape, control, branch, and return components required by the conditional route."*

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19L. Component role-hardening and a conditional Zuse-style primitive core

The historical calibration from Z1 through Z3 suggests a useful discipline for the present line. The architectural idea already existed early, but reliability improved when storage, control, and arithmetic roles became cleaner and more stable across the machine family. In the same spirit, the present Companion should not jump immediately from a compressed computational reading to a full universality claim. The next correct step is to harden the four component roles separately and only then close stronger machine-theoretic statements.

Definition 33.8 (Local role certificate package)**.** *Write*

$$\mathbf{Role} = (\mathbf{T}, \mathbf{Q}, \mathbf{B}, \mathbf{L})$$

for a local role certificate package on an admissible computational window, where:

- **T** *certifies an effectively extendable sequential read/write support;*
- **Q** *certifies a finite control/update layer with stable local instruction semantics;*
- **B** *certifies a locally decidable branch predicate layer;*
- **L** *certifies a loop/return discipline capable of resuming or repeating the local control flow without leaving admissibility.*

A local machine reading is called Zuse-primitive *on the given window if it carries such a package, even if no full universality theorem has yet been proved.*

Proposition 33.9 (Spinorial optimal line as conditional tape role)**.** *Assume* **STH**. *Then the admissible spinorial optimal line carries the conditional tape certificate* **T** *in the following weak but machine-relevant sense: finite local snapshots may be written, read, and prolonged along that line by admissible transport, and the transition primitives* `FORWARD` *and* `HOPF` *provide the local sequential displacement needed for an effective head movement on admissible windows.*

Proposition 33.10 (Coded MML/MMA/MCP stack as conditional control role)**.** *Assume the component identification hypothesis for the coded MML→MMA-0.1→MCP line. Then the coded stack carries the conditional control certificate* **Q***: it provides a finite control/update discipline which selects, routes, and stabilizes the local instruction semantics on admissible computational windows.*

Proposition 33.11 (Projector/Boolean layer as conditional branch role)**.** *Assume the locally decidable projector/readout and Boolean export layer already isolated in the compressed computational surface. Then the current release line carries the conditional branch certificate* **B***: branch predicates can be extracted locally in a form sufficient to distinguish continuation, rejection, and comparison-sensitive update on admissible windows.*

Proposition 33.12 (Routed HC-selected closure as conditional loop/return role)**.** *Assume the routed HC-selected closure structure persists under admissible local updates. Then the current release line carries the conditional loop/return certificate* **L***: local computational segments can be closed, resumed, or re-entered through the routed closure discipline without immediately destroying admissibility.*

Theorem 33.13 (Conditional Zuse-core closure theorem)**.** *Assume* **STH***, the local operational semantics on admissible computational windows, the component identification hypothesis, and the conditional role certificates* $\mathbf{Role} = (\mathbf{T}, \mathbf{Q}, \mathbf{B}, \mathbf{L})$ *above. Then the compressed computational interpretation surface* $\mathfrak{C}^{\mathrm{comp}}$ *closes to a* conditional local Zuse-core *on every admissible certified window: there is a well-defined primitive sequential machine reading with*

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- *a persistent sequential storage/support component carried by the spinorial optimal line,*
- *a finite control/update component carried by the coded MML→MMA-0.1→MCP stack,*
- *a locally decidable branch/readout component carried by the projector/Boolean comparison interface,*
- *and a routed loop/return component carried by the HC-selected closure structure.*

Moreover this local Zuse-core is closed under admissible one-step execution and under finite concatenation of locally certified segments, so the present line already supports a genuine primitive Zuse-style computational apparatus *in the conditional sense of this chapter. What remains open is not the existence of that primitive core, but the proof that these components are globally stable, extendable, and strong enough to simulate a standard reference machine.*

Corollary 33.14 (Conditional Zuse-style primitive computational core)**.** *Assume* **STH***, the component identification hypothesis, and the conditional role certificates* $\mathbf{Role} = (\mathbf{T}, \mathbf{Q}, \mathbf{B}, \mathbf{L})$ *above. Then the compressed computational interpretation surface* $\mathfrak{C}^{\mathrm{comp}}$ *already supports a* conditional Zuse-style primitive computational core*: the current release line carries a genuine sequential storage/support reading, a finite control/update layer, a local branch/readout layer, and a routed loop/return discipline. What is still missing for stronger claims is not the existence of a primitive machine reading, but the final proof that these roles are globally stable and strong enough for an explicit reference-machine simulation.*

Corollary 33.15 (What the conditional Zuse-core closure theorem already buys)**.** *Later arguments may now cite a single bundled closure statement for the machine-reading side of the Companion: under the certified local role package, the present line is already closed as a primitive Zuse-style computational core on admissible windows. Thus one no longer needs to reopen the tape/control/branch/loop roles separately unless the proof requires strengthening one of the four certificates themselves.*

19M. Reduced proof-obligation package for the conditional Zuse-core

Once the conditional Zuse-style primitive core has been isolated, the remaining machine-theoretic burden should be stated as a small explicit obligation package rather than left as a diffuse list of promises. The right next reading is therefore not "prove universality all at once," but "reduce the open universality route to a finite role-upgrade package."

Definition 33.16 (Reduced universality-obligation package for the conditional Zuse-core)**.** *Write*

$$\mathbf{U} = (\mathbf{U}_T, \mathbf{U}_Q, \mathbf{U}_B, \mathbf{U}_L, \mathbf{U}_S)$$

for the reduced universality-obligation package attached to the conditional Zuse-core, where:

- $\mathbf{U}_T$ *is the obligation that the spinorial optimal line is not merely locally readable, but globally prolongable as an effective read/write support;*
- $\mathbf{U}_Q$ *is the obligation that the coded MML→MMA-0.1→MCP control layer composes stably across admissible local windows;*
- $\mathbf{U}_B$ *is the obligation that the projector/Boolean layer yields executable and persistently decidable branch predicates under admissible concatenation;*
- $\mathbf{U}_L$ *is the obligation that the routed HC-selected closure discipline provides genuine resumable loop/return control under repeated local execution;*

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- $\mathbf{U}_S$ *is the obligation that these upgraded components explicitly simulate a standard sequential reference-machine core on admissible windows.*

A conditional Zuse-core is called universality-ready *if all five obligations in* $\mathbf{U}$ *are discharged.*

Proposition 33.17 (Proof-obligation reduction for the conditional Zuse-core)**.** *Assume the conditional Zuse-core closure theorem. Then the remaining gap between the present release line and a genuine universality theorem is reduced to the package* $\mathbf{U} = (\mathbf{U}_T, \mathbf{U}_Q, \mathbf{U}_B, \mathbf{U}_L, \mathbf{U}_S)$ *above. In particular, one no longer needs to reopen the existence of a primitive machine reading itself; the open task is only to upgrade the four certified roles from local conditional availability to globally stable executable roles and then prove the explicit reference-machine simulation.*

Corollary 33.18 (What the reduced universality-obligation package already buys)**.** *The machine-theoretic remainder of the Companion is now sharply localized. Later revisions may therefore proceed componentwise: tape hardening, control hardening, branch hardening, loop/return hardening, and finally explicit simulation. This is a much narrower and more testable program than the earlier diffuse slogan "show that the whole architecture is universal."*

Remark 33.19 (What the Z1→Z2→Z3 improvement path suggests for the present model)**.** *The historical machine line suggests three concrete lessons for the current architecture.*

- *First, one should keep the final storage support as stable as possible while improving the control/branch implementation; this favors treating the spinorial optimal line as the persistent support and hardening the projector/control stack around it.*
- *Second, the decisive upgrade path may come from improving reliability of the control and branching layers before demanding full universality; this mirrors the move from more fragile early realization to a more robust relay-style execution discipline.*
- *Third, the right immediate target is not yet "prove Turing-completeness," but "prove that the present line already forms a Zuse-primitive machine in the precise conditional sense above."*

19N. First hardening priority for the reduced universality-obligation package

Once the remaining universality route has been reduced to a finite package, the next question is no longer merely "what is missing," but "in which order should the missing pieces be hardened so that later upgrades do not have to be reopened unnecessarily." The Zuse-guided reading suggests that the first useful order is not to attack full simulation immediately, but to stabilize the executable control layers around the final storage support before demanding the final global machine theorem.

Definition 33.20 (First Zuse-guided hardening priority order for the reduced universality-obligation package)**.** *Write*

$$\mathbf{H}_1 = (\mathbf{U}_Q, \mathbf{U}_B, \mathbf{U}_L, \mathbf{U}_T, \mathbf{U}_S)$$

for the first Zuse-guided hardening priority order attached to the reduced universality-obligation package. Its intended reading is the following:

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- *first harden the finite control/update stack so that local instruction semantics compose stably across admissible windows;*
- *then harden the projector/Boolean branch layer so that executable branching survives admissible concatenation;*
- *then harden the routed HC-selected loop/return discipline so that repeated local execution remains resumable;*
- *only afterwards demand that the spinorial optimal line be upgraded from local readable support to globally prolongable effective tape;*
- *and only at the end require the explicit reference-machine simulation.*

Proposition 33.21 (Zuse-guided priority reduction for the reduced universality-obligation package)**.** *Assume the conditional Zuse-core closure theorem and the reduced universality-obligation package* $\mathbf{U} = (\mathbf{U}_T, \mathbf{U}_Q, \mathbf{U}_B, \mathbf{U}_L, \mathbf{U}_S)$*. Then the first useful hardening route may be read through the priority order* $\mathbf{H}_1 = (\mathbf{U}_Q, \mathbf{U}_B, \mathbf{U}_L, \mathbf{U}_T, \mathbf{U}_S)$ *above: the present line already has a persistent storage final, but its most leverage-sensitive uncertified parts lie first in the execution stack surrounding that final. In particular, upgrading control, then branch, then loop/return yields the cleanest path to a later tape hardening and finally to explicit simulation, because it stabilizes the executable local dynamics before asking for the strongest global machine claim.*

Corollary 33.22 (What the first Zuse-guided hardening priority order already buys)**.** *The open universality route is now not only finite but ordered. Later revisions therefore need not ask merely whether the current release line is already universal; they may instead proceed along a narrower program: first make the control layer compose, then make branching executable, then make local return resumable, then globalize the tape, and only afterwards prove explicit simulation. This turns the remaining machine-theoretic work from one undifferentiated challenge into a staged hardening schedule.*

19O. First local hardening of the control obligation $\mathbf{U}_Q$

The priority order above says that the first leverage-sensitive remaining task is no longer the tape itself but the stability of the executable control stack around that tape final. The natural first hardening step is therefore to isolate a local criterion under which the instruction/update layer is not merely available pointwise, but composes across adjacent admissible windows without reopening its semantics at every cut.

Definition 33.23 (First local control-hardening certificate for $\mathbf{U}_Q$)**.** *Write*

$$\mathbf{Q}_{\mathrm{loc}}(I) = 1$$

for the first local control-hardening certificate on a finite admissible window I *if the following hold simultaneously on* I*:*

- *the local operational semantics of the compressed computational surface assigns a well-defined successor state to every admissible instruction step on* I*;*
- *adjacent admissible subwindows of* I *compose without changing the decoded control meaning of the instruction stack;*

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- *the induced controller action remains compatible with the later-depth cycle-control layer whenever that layer is active on the same window.*

In words, $\mathbf{Q}_{\mathrm{loc}}(I) = 1$ *means that the present line already carries a locally executable and composition-stable control role on the window* I*.*

Proposition 33.24 (First reduction of the control obligation $\mathbf{U}_Q$)**.** *Assume the conditional Zuse-core closure theorem, the reduced universality-obligation package* $\mathbf{U} = (\mathbf{U}_T, \mathbf{U}_Q, \mathbf{U}_B, \mathbf{U}_L, \mathbf{U}_S)$*, and the first Zuse-guided hardening priority order. Suppose moreover that there exists an admissible cover by finite windows on which the local control-hardening certificate* $\mathbf{Q}_{\mathrm{loc}} = 1$ *holds and is stable under admissible concatenation. Then the obligation* $\mathbf{U}_Q$ *is discharged at the first hardening stage, and the remaining route reduces to*

$$(\mathbf{U}_B, \mathbf{U}_L, \mathbf{U}_T, \mathbf{U}_S).$$

Equivalently, once the instruction/update layer composes locally and survives admissible concatenation without semantic drift, the first unresolved machine-theoretic gap no longer lies in control but in branch, loop/return, tape globalization, and explicit simulation.

Corollary 33.25 (What the first local hardening of $\mathbf{U}_Q$ already buys)**.** *The first hardening stage is now no longer just an ordering slogan. It has been turned into a concrete target: prove local control compositionality on admissible windows and its persistence under admissible concatenation. If that certificate is obtained, later revisions need not reopen the instruction stack itself and may move directly to the branch obligation as the next active gap.*

19P. First local hardening of the branch obligation $\mathbf{U}_B$

Once the control layer composes locally, the next leverage-sensitive open task is to certify that the projector/Boolean layer does not merely look branch-like, but already induces an executable local branch role on admissible windows. The natural next hardening step is therefore to isolate a local criterion under which branch alternatives are decoded, selected, and propagated without collapsing the admissible control meaning established at the previous stage.

Definition 33.26 (First local branch-hardening certificate for $\mathbf{U}_B$)**.** *Write*

$$\mathbf{B}_{\mathrm{loc}}(I) = 1$$

for the first local branch-hardening certificate on a finite admissible window I *if the following hold simultaneously on* I*:*

- *the projector/Boolean layer assigns a well-defined admissible branch menu to every locally admissible branch site on* I*;*
- *whenever a branch discriminator is active, exactly one admissible local branch successor is selected at the level of the compressed computational surface;*
- *the selected local branch successor remains compatible with the already hardened local control meaning and with the later-depth cycle-control layer whenever that layer is active on the same window;*
- *admissible concatenation of adjacent certified subwindows does not change the decoded local branch choice except at explicitly reopened branch sites.*

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In words, $\mathbf{B}_{\mathrm{loc}}(I) = 1$ *means that the present line already carries a locally executable and concatenation-stable branch role on the window* I*.*

Proposition 33.27 (First reduction of the branch obligation $\mathbf{U}_B$)**.** *Assume the conditional Zuse-core closure theorem, the reduced universality-obligation package* $\mathbf{U} = (\mathbf{U}_T, \mathbf{U}_Q, \mathbf{U}_B, \mathbf{U}_L, \mathbf{U}_S)$*, the first Zuse-guided hardening priority order, and that the local control-hardening certificate has already discharged* $\mathbf{U}_Q$*. Suppose moreover that there exists an admissible cover by finite windows on which the local branch-hardening certificate* $\mathbf{B}_{\mathrm{loc}} = 1$ *holds and is stable under admissible concatenation away from explicitly reopened branch sites. Then the obligation* $\mathbf{U}_B$ *is discharged at the second hardening stage, and the remaining route reduces to*

$$(\mathbf{U}_L, \mathbf{U}_T, \mathbf{U}_S).$$

Equivalently, once the projector/Boolean layer yields an executable locally unique branch choice which survives admissible concatenation and remains compatible with the control stack, the next unresolved machine-theoretic gap no longer lies in branch but in loop/return, tape globalization, and explicit simulation.

Corollary 33.28 (What the first local hardening of $\mathbf{U}_B$ already buys)**.** *The second hardening stage is now also no longer only a priority promise. It has become a concrete target: certify local executable branch selection on admissible windows and its persistence under admissible concatenation, except where the theory explicitly reopens the branch site. If that certificate is obtained, later revisions may treat branch as locally settled and move directly to the loop/return obligation as the next active gap.*

19Q. First local hardening of the loop/return obligation $\mathbf{U}_L$

Once local control and local branch have been hardened, the next leverage-sensitive open task is to certify that routed HC-selected closure is not merely closure-like, but already induces a locally executable loop/return role on admissible windows. The natural third hardening step is therefore to isolate a local criterion under which admissible return sites are reopened, routed, and closed again without losing the local control-branch meaning established at the previous stages.

Definition 33.29 (First local loop/return-hardening certificate for $\mathbf{U}_L$)**.** *Write*

$$\mathbf{L}_{\mathrm{loc}}(I) = 1$$

for the first local loop/return-hardening certificate on a finite admissible window I *if the following hold simultaneously on* I*:*

- *every admissible reopened return site on* I *carries a well-defined routed HC-selected closure menu;*
- *whenever a local return event is triggered, exactly one admissible routed return/closure successor is selected at the level of the compressed computational surface;*
- *the selected routed return/closure successor remains compatible with the already hardened local control and local branch meanings on the same window;*
- *admissible concatenation of adjacent certified subwindows preserves the decoded routed return/closure choice except at explicitly reopened return sites.*

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In words, $\mathbf{L}_{\mathrm{loc}}(I) = 1$ *means that the present line already carries a locally executable and concatenation-stable loop/return role on the window* I*.*

Proposition 33.30 (First reduction of the loop/return obligation $\mathbf{U}_L$)**.** *Assume the conditional Zuse-core closure theorem, the reduced universality-obligation package* $\mathbf{U} = (\mathbf{U}_T, \mathbf{U}_Q, \mathbf{U}_B, \mathbf{U}_L, \mathbf{U}_S)$*, the first Zuse-guided hardening priority order, and that the local control-hardening and local branch-hardening certificates have already discharged* $\mathbf{U}_Q$ *and* $\mathbf{U}_B$*. Suppose moreover that there exists an admissible cover by finite windows on which the local loop/return-hardening certificate* $\mathbf{L}_{\mathrm{loc}} = 1$ *holds and is stable under admissible concatenation away from explicitly reopened return sites. Then the obligation* $\mathbf{U}_L$ *is discharged at the third hardening stage, and the remaining route reduces to*

$$(\mathbf{U}_T, \mathbf{U}_S).$$

Equivalently, once routed HC-selected closure yields an executable locally unique return/closure choice which survives admissible concatenation and remains compatible with the hardened control and branch layers, the next unresolved machine-theoretic gap no longer lies in loop/return but in tape globalization and explicit reference-machine simulation.

Corollary 33.31 (What the first local hardening of $\mathbf{U}_L$ already buys)**.** *The third hardening stage is now also no longer only a priority promise. It has become a concrete target: certify local executable routed return/closure selection on admissible windows and its persistence under admissible concatenation, except where the theory explicitly reopens the return site. If that certificate is obtained, later revisions may treat loop/return as locally settled and move directly to the tape-globalization obligation as the next active gap.*

19R. First globalization of the tape obligation $\mathbf{U}_T$

Once local control, local branch, and local loop/return have been hardened, the remaining machine-theoretic gap before explicit reference-machine simulation is no longer a local choice problem but a persistence problem: the spinorial optimal line must be shown to support a globally extensible read/write track across admissibly concatenated windows. The natural fourth hardening step is therefore to isolate a globalization criterion under which the local tape role ceases to be merely windowwise heuristic and becomes a coherent tape support along an admissible chain.

Definition 33.32 (First tape-globalization certificate for $\mathbf{U}_T$)**.** *Write*

$$\mathbf{T}_{\mathrm{glob}}(\Gamma) = 1$$

for the first tape-globalization certificate on an admissible chain $\Gamma = (I_1, \dots, I_N)$ *of finite windows if the following hold simultaneously along* Γ*:*

- *each window* I_k *in the chain carries the local control-hardening, local branch-hardening, and local loop/return-hardening certificates;*
- *the spinorial optimal line induces on each* I_k *a locally readable and writable tape segment whose head-position update is compatible with the compressed computational surface;*
- *on every admissible overlap or concatenation interface* $I_k \cap I_{k+1}$*, the local tape segments glue to a persistent track without ambiguity in symbol transport, head continuation, or routed return-to-track placement;*

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- *admissible resurface/reentry events may reopen the local track, but after each certified reentry the tape continuation rejoins the same globalized track class on the remaining suffix of* Γ.

In words, $\mathbf{T}_{\mathrm{glob}}(\Gamma) = 1$ *means that the spinorial optimal line is no longer only a local tape final on isolated windows, but already behaves as a globally extensible effective tape support along the admissible chain* Γ.

Proposition 33.33 (First reduction of the tape obligation $\mathbf{U}_T$)**.** *Assume the conditional Zusecore closure theorem, the reduced universality-obligation package* $\mathbf{U} = (\mathbf{U}_T, \mathbf{U}_Q, \mathbf{U}_B, \mathbf{U}_L, \mathbf{U}_S)$*, the first Zuse-guided hardening priority order, and that the local control-hardening, local branch-hardening, and local loop/return-hardening certificates have already discharged* $\mathbf{U}_Q$*,* $\mathbf{U}_B$*, and* $\mathbf{U}_L$*. Suppose moreover that there exists an admissible exhaustion by finite chains* Γ *on which the first tape-globalization certificate* $\mathbf{T}_{\mathrm{glob}}(\Gamma) = 1$ *holds and is stable under admissible extension of the chain. Then the obligation* $\mathbf{U}_T$ *is discharged at the fourth hardening stage, and the remaining route reduces to*

$$(\mathbf{U}_S).$$

Equivalently, once the spinorial optimal line supports a globally extensible read/write track compatible with the already hardened control, branch, and loop/return layers, the only unresolved machine-theoretic gap no longer lies in tape but solely in the explicit simulation of a standard reference machine.

Corollary 33.34 (What the first globalization of $\mathbf{U}_T$ already buys)**.** *The fourth hardening stage is now also no longer only a priority promise. It has become a concrete target: certify a globally extensible spinorial tape support along admissible chains and its stability under admissible extension and reentry. If that certificate is obtained, later revisions may treat tape as globally settled and focus entirely on the last remaining obligation, namely explicit reference-machine simulation.*

19S. First reduction of the reference-machine simulation obligation $\mathbf{U}_S$

Once control, branch, loop/return, and tape have been hardened, the final remaining machine-theoretic gap is no longer internal role identification but explicit comparison with a standard sequential reference core. The natural fifth step is therefore to isolate a simulation certificate which shows that the compressed computational surface can reproduce the transition semantics of a chosen admissible reference machine on the already globalized spinorial tape support.

Definition 33.35 (First reference-machine simulation certificate for $\mathbf{U}_S$)**.** *Write*

$$\mathbf{S}_{\mathrm{ref}}(\mathcal{R}, \Gamma) = 1$$

for the first reference-machine simulation certificate on an admissible chain Γ *for a chosen sequential reference core* $\mathcal{R}$ *if the following hold simultaneously:*

- *there exists an explicit encoding of the machine state of* $\mathcal{R}$ *into the compressed computational surface, including control state, branch markers, loop/return status, head position, and tape symbols on the already globalized spinorial tape support;*

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- *every one-step transition of $\mathcal{R}$ on the encoded configuration is reproduced by a finite admissible move sequence of the present line without loss of the hardened control, branch, loop/return, or tape certificates;*
- *halting, branch selection, and return-to-track events of $\mathcal{R}$ are reflected by the corresponding local cycle-control symbols and their decoder-driven actions on the encoded side;*
- *admissible extension of Γ preserves the simulation relation on the longer prefix/suffix decomposition whenever the encoded reference computation itself extends.*

In words, $\mathbf{S}_{\mathrm{ref}}(\mathcal{R},\Gamma) = 1$ means that the present line does not merely admit a primitive computational reading, but explicitly simulates the chosen sequential reference core $\mathcal{R}$ along the admissible chain Γ.

Proposition 33.36 (First reduction of the reference-machine simulation obligation $\mathbf{U}_S$)**.** *Assume the conditional Zuse-core closure theorem, the reduced universality-obligation package $\mathbf{U} = (\mathbf{U}_T, \mathbf{U}_Q, \mathbf{U}_B, \mathbf{U}_L, \mathbf{U}_S)$, the first Zuse-guided hardening priority order, and that the local control-hardening, local branch-hardening, local loop/return-hardening, and tape-globalization certificates have already discharged $\mathbf{U}_Q$, $\mathbf{U}_B$, $\mathbf{U}_L$, and $\mathbf{U}_T$. Suppose moreover that there exists a chosen admissible sequential reference core $\mathcal{R}$ and an admissible exhaustion by finite chains Γ on which the first reference-machine simulation certificate $\mathbf{S}_{\mathrm{ref}}(\mathcal{R},\Gamma) = 1$ holds and is stable under admissible extension. Then the final remaining obligation $\mathbf{U}_S$ is discharged, and the reduced universality-obligation package collapses to the empty list.*

$$(\,).$$

Equivalently, once the already hardened line explicitly simulates a chosen sequential reference core on the globalized spinorial tape support, the conditional Zuse-style primitive computational route is closed at the level of explicit reference-machine simulation.

Corollary 33.37 (What the first reduction of $\mathbf{U}_S$ already buys)**.** *The fifth hardening stage is now also no longer only a distant final promise. It has become a concrete target: produce an explicit admissible simulation of a chosen sequential reference core on the already hardened control/branch/loop/tape stack. If that certificate is obtained, later revisions may treat the conditional Zuse-core route as closed up to the chosen reference-machine level, while any stronger universality claim would then depend only on the strength of the reference core and the separate unboundedness assumptions built into that choice.*

19T. Conditional reference-core closure surface

Once the reduced universality-obligation package has collapsed to the empty list for a chosen admissible sequential reference core, the computational reading should no longer be cited only as a chain of discharged obligations. The right next compression is therefore a single bundled closure surface recording that the hardened line now simulates the chosen reference core on the globalized spinorial tape support.

Definition 33.38 (Conditional reference-core closure surface)**.** *Let $\mathcal{R}$ be a chosen admissible sequential reference core and let $(\Gamma_n)_{n\ge 1}$ be an admissible exhaustion by finite chains. Write*

$$\mathfrak{Z}^{\mathrm{ref}}(\mathcal{R}) = 1$$

if the following hold simultaneously:

- *the conditional Zuse-core closure theorem holds;*

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- *the local control-hardening, local branch-hardening, local loop/return-hardening, tape-globalization, and reference-machine simulation certificates discharge* $\mathbf{U}_Q$, $\mathbf{U}_B$, $\mathbf{U}_L$, $\mathbf{U}_T$, *and* $\mathbf{U}_S$ *on the exhaustion* $(\Gamma_n)_{n\geq 1}$*;*
- *the explicit simulation relation for* $\mathcal{R}$ *is stable under admissible extension of the finite chains in the exhaustion.*

In words, $\mathfrak{Z}^{\mathrm{ref}}(\mathcal{R}) = 1$ *means that the present line is already closed, in the conditional sense of this chapter, up to the explicit simulation of the chosen reference core* $\mathcal{R}$*.*

Theorem 33.39 (Conditional reference-core closure theorem)**.** *Assume the conditional Zuse-core closure theorem and the hypotheses of the first reduction of the reference-machine simulation obligation* $\mathbf{U}_S$*. Suppose moreover that the discharged hardening certificates remain stable under admissible extension along an exhaustion of finite chains for a chosen admissible sequential reference core* $\mathcal{R}$*. Then the compressed computational interpretation surface* $\mathfrak{C}^{\mathrm{comp}}$ *closes to a* conditional reference-core closure surface *for* $\mathcal{R}$*:*

$$\mathfrak{Z}^{\mathrm{ref}}(\mathcal{R}) = 1.$$

Equivalently, the present line is no longer merely a conditional primitive Zuse-style computational core, but a conditionally closed simulator of the chosen reference core on the globalized spinorial tape support. Any stronger universality claim therefore depends only on the strength of the chosen reference core and on whatever additional unboundedness assumptions one wishes to attach to it.

Corollary 33.40 (What the conditional reference-core closure surface already buys)**.** *Later arguments may now cite a single bundled machine-theoretic endpoint: for a chosen admissible sequential reference core* $\mathcal{R}$*, the present line is conditionally closed up to explicit reference-core simulation. Thus one no longer needs to reopen the intermediate hardening ladder unless the proof specifically aims to strengthen the chosen reference core or to pass from reference-core closure to a stronger universality theorem.*

Remark 33.41 (Deferred LLM-superstructure note)**.** *The present line does not claim that it already constitutes a large language model in the strict modern sense. What is now reasonable to record for later work is a weaker architectural possibility: the combination of graph-structured objects, routed node logic, history-sensitive compression, cache-like admissible reuse along an optimal line, and the conditional Zuse-style primitive computational core may eventually support an* LLM-superstructure reading*. In that later reading, the current release line would function not as the trained language-model core itself, but as a host/control/memory architecture into which a genuine next-symbol predictor or other trainable linguistic layer could be inserted. This is therefore deferred as a future-work hypothesis rather than used as an active theorematic ingredient. It may, however, be rechecked opportunistically in later revisions whenever the token/readout, predictive, or trainability side becomes structurally clearer.*

19U. Strength hierarchy above conditional reference-core closure

Definition 33.42 (First machine-theoretic strength hierarchy on the compressed computational surface)**.** *For the compressed computational interpretation surface* $\mathfrak{C}^{\mathrm{comp}}$*, define the*

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following five nested strength levels:

$$\begin{aligned}
\mathbf{S}_1 &:= \textit{disciplined computational reading},\\
\mathbf{S}_2 &:= \textit{conditional Zuse-style primitive core closure},\\
\mathbf{S}_3(\mathcal{R}) &:= \textit{conditional reference-core closure for a chosen sequential reference core } \mathcal{R},\\
\mathbf{S}_4(\mathcal{R}) &:= \textit{conditional universality relative to a chosen reference core } \mathcal{R},\\
&\qquad \textit{strong enough for the intended claim}\\
\mathbf{S}_5 &:= \textit{strict unbounded universality / Turing-style completeness.}
\end{aligned}$$

Proposition 33.43 (Monotonicity of the first machine-theoretic strength hierarchy)**.** *On the admissible computational window one has the one-way implication chain*

$$\mathbf{S}_5 \Longrightarrow \mathbf{S}_4(\mathcal{R}) \Longrightarrow \mathbf{S}_3(\mathcal{R}) \Longrightarrow \mathbf{S}_2 \Longrightarrow \mathbf{S}_1.$$

In particular, every stronger machine-theoretic endpoint automatically contains all weaker lower-level readings, while the converse implications need not hold without additional unboundedness or simulation arguments.

Corollary 33.44 (What the first machine-theoretic strength hierarchy already buys)**.** *The present line may now be located unambiguously in the hierarchy: by the conditional reference-core closure theorem it reaches* $\mathbf{S}_3(\mathcal{R})$ *for a chosen admissible sequential reference core* $\mathcal{R}$*. What remains open is therefore no longer the primitive computational reading itself, but the step from* $\mathbf{S}_3(\mathcal{R})$ *to stronger levels* $\mathbf{S}_4(\mathcal{R})$ *and* $\mathbf{S}_5$*. Later arguments can thus distinguish cleanly between primitive closure, reference-core closure, conditional universality claims, and strict Turing-style completeness claims.*

19V. First upgrade criterion from reference-core closure to conditional universality

Once the present line has reached $\mathbf{S}_3(\mathcal{R})$ for a chosen admissible sequential reference core, the remaining machine-theoretic question is no longer whether the line simulates that reference core, but whether the chosen reference core is itself strong enough to transport an intended universality claim back through the already established simulation relation.

Definition 33.45 (First relative universality-upgrade certificate above $\mathbf{S}_3(\mathcal{R})$)**.** *Let* $\mathcal{R}$ *be a chosen admissible sequential reference core and let* $\mathcal{K}$ *be a target class of sequential computations one wishes to cover. Write*

$$\mathfrak{U}^{\mathrm{rel}}(\mathcal{R},\mathcal{K}) = 1$$

if the following hold simultaneously:

- *the conditional reference-core closure surface satisfies* $\mathfrak{Z}^{\mathrm{ref}}(\mathcal{R}) = 1$*;*
- *the reference core* $\mathcal{R}$ *carries a universality theorem for the target class* $\mathcal{K}$ *under a specified admissible encoding/decoding discipline;*
- *the simulation relation used in* $\mathfrak{Z}^{\mathrm{ref}}(\mathcal{R}) = 1$ *transports that admissible encoding/decoding discipline without breaking the globalized tape support, the hardened control/branch/loop stack, or the local readout semantics.*

In words, $\mathfrak{U}^{\mathrm{rel}}(\mathcal{R},\mathcal{K}) = 1$ *means that the present line is not only closed up to the chosen reference core, but conditionally inherits the universality of* $\mathcal{R}$ *for the intended target class* $\mathcal{K}$*.*

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Proposition 33.46 (First conditional upgrade from $\mathbf{S}_3(\mathcal{R})$ to $\mathbf{S}_4(\mathcal{R})$)**.** *Assume $\mathfrak{Z}^{\mathrm{ref}}(\mathcal{R}) = 1$ for a chosen admissible sequential reference core $\mathcal{R}$. Suppose moreover that $\mathcal{R}$ is already known to be universal for a target class $\mathcal{K}$ under an admissible encoding/decoding discipline, and that this discipline is preserved by the conditional reference-core closure surface. Then the present line upgrades from $\mathbf{S}_3(\mathcal{R})$ to the relative universality level $\mathbf{S}_4(\mathcal{R})$ for the intended class $\mathcal{K}$.*

$$\mathfrak{U}^{\mathrm{rel}}(\mathcal{R}, \mathcal{K}) = 1 \Longrightarrow \mathbf{S}_4(\mathcal{R}).$$

Equivalently, no additional local hardening layer is required beyond $\mathbf{S}_3(\mathcal{R})$; what remains is the external strength theorem for the chosen reference core and the proof that its admissible coding discipline is faithfully transported by the already hardened simulation route.

Corollary 33.47 (What the first upgrade criterion above $\mathbf{S}_3(\mathcal{R})$ already buys)**.** *The gap between the present line and conditional universality is now sharply isolated. To pass from the current reference-core closure level to $\mathbf{S}_4(\mathcal{R})$, later revisions no longer need to reopen the primitive computational hardening ladder. They need only choose a reference core $\mathcal{R}$ strong enough for the intended target class $\mathcal{K}$, certify the corresponding universality theorem for $\mathcal{R}$, and verify that the admissible encoding/decoding discipline survives transport through the already established reference-core closure surface.*

19W. First admissible reference-core selection principle above $\mathbf{S}_3(\mathcal{R})$

Once the upgrade from $\mathbf{S}_3(\mathcal{R})$ to $\mathbf{S}_4(\mathcal{R})$ has been isolated, the next question is no longer whether one can choose some strong reference core at all, but which admissible reference core should be chosen first. A structure-first route should avoid importing unnecessary machine-theoretic strength too early; it should instead select the weakest admissible reference core already sufficient for the intended target claim.

Definition 33.48 (First conservative reference-core selection certificate above $\mathbf{S}_3(\mathcal{R})$)**.** *Fix a target class $\mathcal{K}$ and a chosen comparison family $\mathfrak{F}_{\mathrm{ref}}$ of admissible sequential reference cores together with a strength preorder $\preceq_{\mathrm{ref}}$. Write*

$$\mathfrak{S}^{\mathrm{ref}}(\mathcal{R}, \mathcal{K}) = 1$$

if the following hold:

- *$\mathcal{R} \in \mathfrak{F}_{\mathrm{ref}}$ is admissible and satisfies $\mathfrak{Z}^{\mathrm{ref}}(\mathcal{R}) = 1$;*
- *$\mathcal{R}$ is strong enough to support the intended target claim for $\mathcal{K}$ under an admissible encoding/decoding discipline;*
- *for every $\mathcal{R}' \in \mathfrak{F}_{\mathrm{ref}}$ with $\mathcal{R}' \prec_{\mathrm{ref}} \mathcal{R}$, the same target claim for $\mathcal{K}$ is not yet certified under the same admissible coding discipline.*

Thus $\mathfrak{S}^{\mathrm{ref}}(\mathcal{R}, \mathcal{K}) = 1$ means that $\mathcal{R}$ is not merely an admissible reference core, but the first conservatively sufficient one inside the chosen comparison family for the intended upgrade problem.

Proposition 33.49 (First conservative selection reduction for the universality-upgrade route)**.** *Assume $\mathfrak{U}^{\mathrm{rel}}(\mathcal{R}, \mathcal{K}) = 1$ and $\mathfrak{S}^{\mathrm{ref}}(\mathcal{R}, \mathcal{K}) = 1$. Then the passage from $\mathbf{S}_3(\mathcal{R})$ to $\mathbf{S}_4(\mathcal{R})$ may be formulated with no stronger external reference-core assumption than the one already needed for the chosen target class $\mathcal{K}$ inside the comparison family $\mathfrak{F}_{\mathrm{ref}}$. In particular, later revisions do not need to import a stronger machine model than necessary in order to state the first conditional universality claim above the present reference-core closure level.*

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Corollary 33.50 (What the first conservative reference-core selection principle already buys)**.** *The relative universality route is now disciplined in two distinct ways: first by the already established reference-core closure surface, and second by a conservative selection rule on the external endpoint. Later upgrades above* $\mathbf{S}_3(\mathcal{R})$ *can therefore proceed by choosing the weakest admissible reference core sufficient for the intended claim, rather than by jumping prematurely to an unnecessarily strong machine-theoretic endpoint.*

19X. First admissible tape-based reference-core family above $\mathbf{S}_3(\mathcal{R})$

The conservative selection principle above becomes materially useful only once one fixes a first concrete comparison family of admissible reference cores. The present line already carries explicit tape, control, branch, and loop/return roles, so the most conservative first family should preserve exactly that architecture instead of importing richer machine features prematurely.

Definition 33.51 (First admissible tape-based reference-core family)**.** *Let* $\mathfrak{F}_{\mathrm{tape}}$ *denote the class of admissible sequential reference cores* $\mathcal{R}$ *satisfying all of the following:*

- $\mathcal{R}$ *has finite control state, finite local symbol alphabet, and a one-track sequential update discipline;*
- *every local step of* $\mathcal{R}$ *is determined by the currently scanned symbol together with the current control state and returns one next control state, one write/update instruction, and one move instruction in* $\{-1, 0, +1\}$*;*
- *branch resolution is explicit and local, in the sense that each step chooses one of finitely many admissible successor clauses by a finite Boolean/projector-type test on the current scanned data;*
- *loop/return structure is explicit and local, in the sense that every admissible return event is encoded by a finite control-side return symbol together with a deterministic reentry instruction on the same one-track support;*
- *the tape support itself is one-dimensional and sequential, so that the already globalized spinorial optimal line may serve as the direct comparison support without introducing any extra addressing geometry.*

Write

$$\mathfrak{A}^{\mathrm{tape}}(\mathcal{R}) = 1$$

if $\mathcal{R} \in \mathfrak{F}_{\mathrm{tape}}$*.*

Proposition 33.52 (First conservative tape-family reduction for the upgrade route)**.** *Assume the conditional reference-core closure surface* $\mathfrak{Z}^{\mathrm{ref}}(\mathcal{R}) = 1$ *for some chosen admissible sequential reference core* $\mathcal{R}$*, together with the conservative selection discipline above. Suppose moreover that the intended universality-upgrade problem concerns only target classes whose admissible coding discipline is already one-track, sequential, and local in the same sense as the hardened tape/control/branch/loop architecture of the present line. Then the first conservative search for a reference core may be restricted to the admissible tape-based family* $\mathfrak{F}_{\mathrm{tape}}$ *without loss for that upgrade problem.*

Equivalently, whenever the intended target class does not yet require richer addressing, parallel branching geometry, or nonlocal memory moves, the weakest natural external comparison family is the tape-based family already matched to the present hardened architecture.

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Corollary 33.53 (What the first admissible tape-based reference-core family already buys)**.** *The external universality-upgrade route is now no longer open-ended. Later revisions can first search for a minimally sufficient* tape-based *reference core before importing any richer machine model. This keeps the passage above* $\mathbf{S}_3(\mathcal{R})$ *aligned with the already hardened internal architecture and reduces the chance of overstating the needed external strength.*

19Y. First internal-to-tape transport certificate above $\mathbf{S}_3(\mathcal{R})$

Choosing a first admissible tape-based comparison family becomes materially useful only once one can express the existing reference-core closure surface in a way that is transport-compatible with that family. The next conservative step is therefore not a stronger universality claim, but a certificate saying that the already hardened internal read/write/control/branch/return package can be read inside a chosen tape-based reference core without changing the local computational discipline.

Definition 33.54 (First internal-to-tape transport certificate above $\mathbf{S}_3(\mathcal{R})$)**.** *Let* $\mathcal{R} \in \mathfrak{F}_{\mathrm{tape}}$ *satisfy* $\mathfrak{Z}^{\mathrm{ref}}(\mathcal{R}) = 1$*. Write*

$$\mathfrak{T}^{\mathrm{tape}}(\mathcal{R}) = 1$$

if the following hold on every admissible later-depth window used by the present compressed computational surface:

- *the globalized spinorial optimal line induces a one-dimensional sequential support compatible with the tape support of* $\mathcal{R}$*;*
- *the already hardened local control state of the present line is encoded into the finite control state of* $\mathcal{R}$ *without introducing extra addressing geometry;*
- *the already hardened projector/Boolean branch layer is transported into the branch clauses of* $\mathcal{R}$ *by a local admissible encoding/decoding discipline;*
- *the already hardened routed HC-selected return/closure step is transported into the local return/reentry discipline of* $\mathcal{R}$ *without changing the sequential one-track support;*
- *one local step of the present compressed computational surface is transported to one admissible local step of* $\mathcal{R}$ *up to the chosen local coding discipline.*

Thus $\mathfrak{T}^{\mathrm{tape}}(\mathcal{R}) = 1$ *means that the internal reference-core closure already lives on a support that can be read directly inside the first admissible tape-based comparison family.*

Proposition 33.55 (First transport reduction from reference-core closure to the tape-based family)**.** *Assume* $\mathfrak{Z}^{\mathrm{ref}}(\mathcal{R}) = 1$*,* $\mathfrak{A}^{\mathrm{tape}}(\mathcal{R}) = 1$*, and* $\mathfrak{T}^{\mathrm{tape}}(\mathcal{R}) = 1$*. Then the external upgrade route above* $\mathbf{S}_3(\mathcal{R})$ *may be reformulated entirely inside the admissible tape-based comparison family without reopening the already hardened local obligations for control, branch, loop/return, or tape support. In particular, the passage to the first conditional universality-upgrade problem may be stated as an external strength question about the chosen tape-based reference core itself, rather than as a renewed local hardening problem for the present line.*

Corollary 33.56 (What the first internal-to-tape transport certificate already buys)**.** *The tape-based upgrade route is now disciplined in three layers: by the conditional reference-core closure surface, by the conservative tape-family selection principle, and by an explicit transport certificate from the present compressed computational surface into that family. Later work above* $\mathbf{S}_3(\mathcal{R})$ *can therefore focus directly on the external strength of the chosen tape-based reference core, instead of rechecking the already stabilized local computational roles.*

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19Z. First tape-family relative universality witness above $\mathbf{S}_3(\mathcal{R})$

Once the admissible tape-based family has been selected and the internal reference-core closure has been transported into it, the remaining external question is no longer whether the present line can be read inside that family, but whether a chosen member of that family is already strong enough for the intended target class. The next conservative step is therefore to isolate the first explicit witness certifying that a transported tape-based reference core upgrades the present line from conditional reference-core closure to conditional universality relative to that core.

Definition 33.57 (First tape-family relative universality witness above $\mathbf{S}_3(\mathcal{R})$)**.** *Let $\mathcal{R} \in \mathfrak{F}_{\mathrm{tape}}$ satisfy*

$$\mathfrak{Z}^{\mathrm{ref}}(\mathcal{R}) = 1, \qquad \mathfrak{A}^{\mathrm{tape}}(\mathcal{R}) = 1, \qquad \mathfrak{T}^{\mathrm{tape}}(\mathcal{R}) = 1.$$

For a target class $\mathcal{K}$, write

$$\mathfrak{U}^{\mathrm{tape}}(\mathcal{R}, \mathcal{K}) = 1$$

if the following hold:

- *the chosen tape-based reference core $\mathcal{R}$ is already known to simulate every machine in the intended target class $\mathcal{K}$ under a one-track sequential local coding discipline compatible with $\mathfrak{F}_{\mathrm{tape}}$;*
- *the admissible encoding/decoding discipline used in that simulation is stable under the internal-to-tape transport certificate $\mathfrak{T}^{\mathrm{tape}}(\mathcal{R}) = 1$;*
- *no extra addressing geometry, nonlocal branching discipline, or richer memory move is required beyond the tape-based family already selected.*

Thus $\mathfrak{U}^{\mathrm{tape}}(\mathcal{R}, \mathcal{K}) = 1$ means that the chosen transported tape-based reference core is externally strong enough to witness the first conditional universality claim relevant to $\mathcal{K}$.

Proposition 33.58 (First tape-family upgrade from $\mathbf{S}_3(\mathcal{R})$ to $\mathbf{S}_4(\mathcal{R})$)**.** *Assume $\mathfrak{Z}^{\mathrm{ref}}(\mathcal{R}) = 1$, $\mathfrak{A}^{\mathrm{tape}}(\mathcal{R}) = 1$, $\mathfrak{T}^{\mathrm{tape}}(\mathcal{R}) = 1$, and $\mathfrak{U}^{\mathrm{tape}}(\mathcal{R}, \mathcal{K}) = 1$. Then the present compressed computational surface upgrades from the reference-core closure level $\mathbf{S}_3(\mathcal{R})$ to the relative universality level $\mathbf{S}_4(\mathcal{R})$ for the intended target class $\mathcal{K}$.*

Equivalently, once the internal closure surface has been transported into a sufficiently strong admissible tape-based reference core, the remaining strength step above $\mathbf{S}_3(\mathcal{R})$ is discharged entirely by the external universality witness for that transported tape-based core.

Corollary 33.59 (What the first tape-family relative universality witness already buys)**.** *The universality-upgrade route above the present compressed computational surface is now disciplined in four layers: by the conditional reference-core closure surface, by the conservative tape-family selection rule, by the internal-to-tape transport certificate, and by an explicit external universality witness for a chosen transported tape-based reference core. Later revisions can therefore state the first conditional universality claim without reopening the already discharged primitive machine-theoretic hardening ladder.*

19AA. First strict universality-obligation package above $\mathbf{S}_4(\mathcal{R})$

Once the present line has reached the relative universality level $\mathbf{S}_4(\mathcal{R})$ for a chosen transported tape-based reference core and target class, the remaining question is no longer whether the primitive computational roles are available, nor whether a conservative tape-based reference family can witness relative universality. What remains is the stricter global question whether

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the transported tape-based route supports the genuinely unbounded extension needed for strict Turing-style completeness. The next conservative step is therefore to isolate that last gap as a separate package instead of mixing it back into the already discharged local hardening ladder.

Definition 33.60 (First strict universality-obligation package above $\mathbf{S}_4(\mathcal{R})$)**.** *Assume that* $\mathbf{S}_4(\mathcal{R})$ *has been reached for an intended target class* $\mathcal{K}$*. Write*

$$\mathbf{V} = (\mathbf{V}_\infty, \mathbf{V}_{\mathrm{ext}}, \mathbf{V}_{\mathrm{dec}})$$

for the first strict universality-obligation package above $\mathbf{S}_4(\mathcal{R})$*, where:*

- $\mathbf{V}_\infty$ *is the obligation that the chosen transported tape-based reference route admits arbitrarily long admissible extensions rather than only bounded local windows;*
- $\mathbf{V}_{\mathrm{ext}}$ *is the obligation that the already established tape-globalization and concatenation discipline extends coherently along such arbitrarily long admissible chains, without hidden global collapse or cutoff;*
- $\mathbf{V}_{\mathrm{dec}}$ *is the obligation that the admissible encoding/decoding discipline witnessing* $\mathbf{S}_4(\mathcal{R})$ *remains stable under those unbounded extensions and does not depend on a fixed finite window budget.*

Thus $\mathbf{V}$ *isolates the first strict machine-theoretic gap between relative universality inside a chosen tape-based reference-core route and full unbounded Turing-style completeness.*

Proposition 33.61 (First isolation of the strict universality gap above $\mathbf{S}_4(\mathcal{R})$)**.** *Assume that* $\mathbf{S}_4(\mathcal{R})$ *has been reached through the conditional reference-core closure surface, the conservative tape-family selection rule, the internal-to-tape transport certificate, and an explicit tape-family relative universality witness. Then the only remaining machine-theoretic step toward the strict level* $\mathbf{S}_5$ *is discharged by the package* $\mathbf{V}$ *above.*

Equivalently, once the present line has been upgraded to $\mathbf{S}_4(\mathcal{R})$*, no additional local hardening of control, branch, loop/return, tape support, or reference-core simulation is needed. What remains is exclusively the global unboundedness, extension, and decoding-stability problem recorded by* $\mathbf{V}$*.*

Corollary 33.62 (What the first strict universality-obligation package already buys)**.** *The machine-theoretic hierarchy is now disciplined at both of its remaining upper gaps. The passage from* $\mathbf{S}_3(\mathcal{R})$ *to* $\mathbf{S}_4(\mathcal{R})$ *is controlled by the conservative tape-family upgrade route, while the passage from* $\mathbf{S}_4(\mathcal{R})$ *to* $\mathbf{S}_5$ *is now isolated as the strict universality-obligation package* $\mathbf{V}$*. Later revisions can therefore state very clearly whether they are addressing relative universality inside a chosen transported tape-based reference family or the separate problem of genuine unbounded Turing-style completeness.*

19AB. Hardening priority for the strict universality obligation above $\mathbf{S}_4(\mathcal{R})$

Once the strict universality gap above $\mathbf{S}_4(\mathcal{R})$ has been isolated as the package $\mathbf{V} = (\mathbf{V}_\infty, \mathbf{V}_{\mathrm{ext}}, \mathbf{V}_{\mathrm{dec}})$, the next conservative step is not to attack all three remaining obligations at once. As on the earlier Zuse-guided route, the cleaner strategy is to impose a first hardening order, so that later revisions can reduce the strict universality gap in the same staged way as the primitive computational obligations.

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Definition 33.63 (First hardening priority order for the strict universality-obligation package above $\mathbf{S}_4(\mathcal{R})$)**.** *Assume that the present compressed computational surface has reached $\mathbf{S}_4(\mathcal{R})$ for an intended target class $\mathcal{K}$, and let*

$$\mathbf{V} = (\mathbf{V}_\infty, \mathbf{V}_{\mathrm{ext}}, \mathbf{V}_{\mathrm{dec}})$$

be the first strict universality-obligation package above $\mathbf{S}_4(\mathcal{R})$. Define the first hardening priority order on $\mathbf{V}$ by

$$\mathbf{H}_2 = (\mathbf{V}_{\mathrm{ext}}, \mathbf{V}_{\mathrm{dec}}, \mathbf{V}_\infty).$$

Thus the conservative strengthening strategy is:

- *first harden the global extension obligation $\mathbf{V}_{\mathrm{ext}}$,*
- *then harden decoding stability under extension $\mathbf{V}_{\mathrm{dec}}$,*
- *and only at the end attack genuine unboundedness $\mathbf{V}_\infty$.*

Proposition 33.64 (First priority reduction for the strict universality-obligation package above $\mathbf{S}_4(\mathcal{R})$)**.** *Assume that $\mathbf{S}_4(\mathcal{R})$ has been reached and that the hardening order $\mathbf{H}_2$ above is adopted. Then later revisions can reduce the strict universality gap monotonically: any successful discharge of the first two obligations $\mathbf{V}_{\mathrm{ext}}$ and $\mathbf{V}_{\mathrm{dec}}$ shrinks the remaining open strict gap to the single final problem $\mathbf{V}_\infty$.*

Equivalently, once the extension discipline and its decoding stability have been hardened, the only remaining obstruction to $\mathbf{S}_5$ is the genuinely unbounded continuation problem rather than the local or semiglobal coherence of the transported tape-based route.

Corollary 33.65 (What the first hardening priority order above $\mathbf{S}_4(\mathcal{R})$ already buys)**.** *The strict universality gap is now not only isolated but also staged. Later revisions can therefore work above $\mathbf{S}_4(\mathcal{R})$ without mixing extension, decoding, and unboundedness into a single undifferentiated final problem. In the conservative reading, the current next target is no longer $\mathbf{S}_5$ as a whole but specifically the first extension-hardening step $\mathbf{V}_{\mathrm{ext}}$.*

19AC. First local hardening of the extension obligation $\mathbf{V}_{\mathrm{ext}}$ above $\mathbf{S}_4(\mathcal{R})$

With the strict universality-obligation package now ordered by $\mathbf{H}_2$, the next conservative step is to attack the first remaining upper-gap component $\mathbf{V}_{\mathrm{ext}}$. The question is no longer whether the present line supports a finite transported tape-based computation relative to a chosen reference core, but whether those admissible windows extend coherently across longer chains without hidden collapse, cutoff, or incompatible seam behavior.

Definition 33.66 (First extension-hardening certificate for $\mathbf{V}_{\mathrm{ext}}$ above $\mathbf{S}_4(\mathcal{R})$)**.** *Assume $\mathbf{S}_4(\mathcal{R})$ for an intended target class $\mathcal{K}$. Write*

$$\mathfrak{E}^{\mathrm{glob}}(\mathcal{R}) = 1$$

if the following hold:

- *every admissible finite transported tape-window in the chosen family extends to a longer admissible tape-window by a compatible local continuation step;*
- *the already established concatenation/cut and patch-normal-form discipline remains coherent under those longer continuations;*

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- *no hidden global seam defect, cutoff phenomenon, or incompatible local reindexing appears when adjoining admissible windows along the transported tape-line.*

Thus $\mathfrak{E}^{\mathrm{glob}}(\mathcal{R}) = 1$ *certifies the first coherent extension-hardening of the tape-based route above* $\mathbf{S}_4(\mathcal{R})$.

Proposition 33.67 (First reduction of the extension obligation $\mathbf{V}_{\mathrm{ext}}$ above $\mathbf{S}_4(\mathcal{R})$)**.** *Assume* $\mathbf{S}_4(\mathcal{R})$ *and* $\mathfrak{E}^{\mathrm{glob}}(\mathcal{R}) = 1$. *Then the first strict universality-obligation package reduces from*

$$\mathbf{V} = (\mathbf{V}_{\infty}, \mathbf{V}_{\mathrm{ext}}, \mathbf{V}_{\mathrm{dec}})$$

to

$$\mathbf{V}' = (\mathbf{V}_{\mathrm{dec}}, \mathbf{V}_{\infty}).$$

Equivalently, once the transported tape-based route is globally extensible in the admissible-window sense, the strict universality gap above $\mathbf{S}_4(\mathcal{R})$ *no longer contains a separate extension obstruction; what remains are the decoding-stability problem under such extension and the final genuinely unbounded continuation problem.*

Corollary 33.68 (What the first local hardening of $\mathbf{V}_{\mathrm{ext}}$ above $\mathbf{S}_4(\mathcal{R})$ already buys)**.** *The upper strict gap above relative universality is now cleaner than before. If the extension-hardening certificate holds, later revisions do not need to reopen the global seam/coherence problem of transported admissible tape-windows. The next active upper-gap target becomes decoding stability under extension, with genuine unboundedness kept as the final separate endpoint problem.*

19AD. First local hardening of the decoding-stability obligation $\mathbf{V}_{\mathrm{dec}}$ above $\mathbf{S}_4(\mathcal{R})$

Once the extension obligation has been isolated and locally hardened, the next conservative strict-universality target is no longer the seam/coherence of admissible transported tape-windows themselves, but the stability of the admissible encoding/decoding discipline under those longer extensions. The question is now whether the same transported tape-based route that already witnesses relative universality continues to admit a coherent decoding discipline when admissible windows are enlarged, concatenated, and reread along longer chains.

Definition 33.69 (First decoding-stability certificate for $\mathbf{V}_{\mathrm{dec}}$ above $\mathbf{S}_4(\mathcal{R})$)**.** *Assume* $\mathbf{S}_4(\mathcal{R})$ *for an intended target class* $\mathcal{K}$, *and assume moreover that the global extension certificate* $\mathfrak{E}^{\mathrm{glob}}(\mathcal{R}) = 1$ *has already been established. Write*

$$\mathfrak{D}^{\mathrm{ext}}(\mathcal{R}) = 1$$

if the following hold:

- *the admissible decoding discipline used to witness* $\mathbf{S}_4(\mathcal{R})$ *on finite transported tape-windows extends coherently to every admissible longer transported tape-window produced by the extension certificate;*

- *restriction from a longer admissible transported tape-window to any admissible subwindow commutes with the chosen decoding/readout rule;*

- *admissible concatenation of transported tape-windows does not introduce a new decoding ambiguity, hidden branch aliasing, or context-dependent reinterpretation of already decoded local machine states.*

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Thus $\mathfrak{D}^{\mathrm{ext}}(\mathcal{R}) = 1$ *certifies the first stable decoding discipline under admissible transported tape-extension above* $\mathbf{S}_4(\mathcal{R})$.

Proposition 33.70 (First reduction of the decoding-stability obligation $\mathbf{V}_{\mathrm{dec}}$ above $\mathbf{S}_4(\mathcal{R})$)**.** *Assume* $\mathbf{S}_4(\mathcal{R})$, $\mathfrak{E}^{\mathrm{glob}}(\mathcal{R}) = 1$, *and* $\mathfrak{D}^{\mathrm{ext}}(\mathcal{R}) = 1$. *Then the strict universality-obligation package reduces from*

$$\mathbf{V} = (\mathbf{V}_\infty, \mathbf{V}_{\mathrm{ext}}, \mathbf{V}_{\mathrm{dec}})$$

to

$$\mathbf{V}'' = (\mathbf{V}_\infty).$$

Equivalently, once the transported tape-based route is globally extensible and its admissible decoding discipline remains stable under those extensions, the strict universality gap above $\mathbf{S}_4(\mathcal{R})$ *no longer contains a separate decoding obstruction; what remains is only the final genuinely unbounded continuation problem.*

Corollary 33.71 (What the first local hardening of $\mathbf{V}_{\mathrm{dec}}$ above $\mathbf{S}_4(\mathcal{R})$ already buys)**.** *The upper strict gap has now been reduced as far as it can be reduced without addressing genuine unboundedness itself. If the extension-hardening and decoding-stability certificates both hold, later revisions no longer need to reopen finite-window extension or decoding coherence. The only remaining active upper-gap target is then* $\mathbf{V}_\infty$, *namely true unbounded continuation.*

19AE. Local hardening of genuine unboundedness $\mathbf{V}_\infty$ above $\mathbf{S}_4(\mathcal{R})$

Once global extension and decoding stability under extension have both been isolated and locally hardened, the only remaining strict-universality target is no longer any finite-window transport or rereading problem. What remains is the genuinely unbounded continuation problem itself: whether the transported tape-based route above $\mathbf{S}_4(\mathcal{R})$ admits arbitrarily extendable admissible continuations without forced collapse, periodic truncation, or hidden resource cutoff.

Definition 33.72 (First genuine-unboundedness certificate for $\mathbf{V}_\infty$ above $\mathbf{S}_4(\mathcal{R})$)**.** *Assume* $\mathbf{S}_4(\mathcal{R})$, $\mathfrak{E}^{\mathrm{glob}}(\mathcal{R}) = 1$, *and* $\mathfrak{D}^{\mathrm{ext}}(\mathcal{R}) = 1$. *Write*

$$\mathfrak{U}^{\infty}(\mathcal{R}) = 1$$

if the following hold:

- *for every admissible transported tape-window in the chosen tape-based family there exists a strictly longer admissible transported tape-window extending it;*
- *admissible extension can be iterated indefinitely without forcing a terminal collapse, bounded cycling artifact, or hidden global cutoff that would prevent arbitrarily long transported computations;*
- *the already established extension and decoding certificates remain valid along every finite stage of such iterated admissible continuation.*

Thus $\mathfrak{U}^{\infty}(\mathcal{R}) = 1$ *certifies the first genuinely unbounded admissible continuation regime above* $\mathbf{S}_4(\mathcal{R})$.

Proposition 33.73 (First reduction of the genuine unboundedness obligation $\mathbf{V}_\infty$ above $\mathbf{S}_4(\mathcal{R})$)**.** *Assume* $\mathbf{S}_4(\mathcal{R})$, $\mathfrak{E}^{\mathrm{glob}}(\mathcal{R}) = 1$, $\mathfrak{D}^{\mathrm{ext}}(\mathcal{R}) = 1$, *and* $\mathfrak{U}^{\infty}(\mathcal{R}) = 1$. *Then the strict universality-obligation package reduces from*

$$\mathbf{V} = (\mathbf{V}_\infty, \mathbf{V}_{\mathrm{ext}}, \mathbf{V}_{\mathrm{dec}})$$

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to the empty list. Equivalently, once the transported tape-based route is globally extensible, decoding-stable under extension, and genuinely unbounded in the admissible continuation sense, the strict universality gap above $\mathbf{S}_4(\mathcal{R})$ contains no further separate obstruction. In that case the compressed computational surface reaches the top level $\mathbf{S}_5$, namely strict unbounded Turing-style completeness relative to the certified route.

Corollary 33.74 (What the first local hardening of $\mathbf{V}_\infty$ above $\mathbf{S}_4(\mathcal{R})$ already buys)**.** *The upper gap is now completely structured. If the global extension, decoding-stability, and genuine-unboundedness certificates all hold, later revisions do not need to reopen any remaining strict-universality obstruction above $\mathbf{S}_4(\mathcal{R})$. The only remaining caution is interpretive: one must still keep explicit which route, tape-family, and reference-core class are being certified when speaking of strict completeness.*

19AF. Full machine-theoretic closure surface on the compressed computational line

After the conditional reference-core closure surface, the relative universality-upgrade witness, and the strict upper certificate package have all been isolated, the machine-theoretic reading should no longer appear as a long ladder of locally discharged obligations. It can now be bundled into a single closure surface that records exactly which level of the hierarchy has been reached under which certificate discipline.

Definition 33.75 (Full machine-theoretic closure surface on the compressed computational line)**.** *Fix a chosen admissible sequential reference core $\mathcal{R}$, a chosen admissible tape-based reference family, and the associated conservative transport discipline. Write*

$$\mathfrak{M}^{\mathrm{full}}(\mathcal{R}) = 1$$

if the following package holds simultaneously:

- *the compressed computational interpretation surface reaches the conditional reference-core closure level $\mathbf{S}_3(\mathcal{R})$, i.e. $\mathfrak{Z}^{\mathrm{ref}}(\mathcal{R}) = 1$;*
- *the tape-family upgrade route is certified, i.e. the conservative tape-family selection rule, the internal-to-tape transport certificate, and the tape-family relative universality witness all hold, so that the line reaches $\mathbf{S}_4(\mathcal{R})$;*
- *the strict upper certificate package is certified, i.e.*

$$\mathfrak{E}^{\mathrm{glob}}(\mathcal{R}) = 1, \qquad \mathfrak{D}^{\mathrm{ext}}(\mathcal{R}) = 1, \qquad \mathfrak{U}^{\infty}(\mathcal{R}) = 1,$$

  so that the remaining strict gap above $\mathbf{S}_4(\mathcal{R})$ is empty.

Thus $\mathfrak{M}^{\mathrm{full}}(\mathcal{R}) = 1$ denotes the first fully bundled machine-theoretic closure surface attached to the present compressed computational line relative to the chosen certified route.

Theorem 33.76 (First bundled machine-theoretic closure theorem on the compressed computational line)**.** *Assume $\mathfrak{M}^{\mathrm{full}}(\mathcal{R}) = 1$. Then the compressed computational line reaches the top machine-theoretic level $\mathbf{S}_5$ along the chosen certified route. Equivalently, all intermediate gaps*

$$\mathbf{S}_1 \longrightarrow \mathbf{S}_2 \longrightarrow \mathbf{S}_3(\mathcal{R}) \longrightarrow \mathbf{S}_4(\mathcal{R}) \longrightarrow \mathbf{S}_5$$

are discharged under an explicitly named certificate package, so that no separate machine-theoretic obstruction remains inside the adopted route.

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Corollary 33.77 (What the full machine-theoretic closure surface already buys)**.** *Later revisions no longer need to cite the whole local hardening ladder when the only point is that the present computational reading has been conditionally closed all the way up the hierarchy along a chosen certified route. They may instead cite* $\mathfrak{M}^{\mathrm{full}}(\mathcal{R}) = 1$*, while still keeping explicit which reference core, which tape-family route, and which upper certificates are meant. This compresses the machine-theoretic reading to a single surface without pretending that the route has become unconditional or model-independent.*

19AG. First machine-theoretic status decoder below and above the full closure surface

After the full machine-theoretic closure surface has been bundled, the text should not force later revisions to reopen the entire hierarchy merely to say which machine-theoretic level is currently justified under a given certificate package. A small status decoder is therefore useful: it reads off the strongest currently justified strength level from the available closure surfaces and upgrade certificates without changing the underlying mathematics.

Definition 33.78 (First machine-theoretic status decoder on the compressed computational line)**.** *Fix a chosen admissible sequential reference core* $\mathcal{R}$*. Define the machine-theoretic status decoder*

$$\mathfrak{S}^{\mathrm{mach}}(\mathcal{R}) \in \{\mathbf{S}_1, \mathbf{S}_2, \mathbf{S}_3(\mathcal{R}), \mathbf{S}_4(\mathcal{R}), \mathbf{S}_5\}$$

as the strongest level among the hierarchy

$$\mathbf{S}_1 \longrightarrow \mathbf{S}_2 \longrightarrow \mathbf{S}_3(\mathcal{R}) \longrightarrow \mathbf{S}_4(\mathcal{R}) \longrightarrow \mathbf{S}_5$$

whose required certificate package has been verified. Concretely:

- $\mathfrak{S}^{\mathrm{mach}}(\mathcal{R}) = \mathbf{S}_5$ *if* $\mathfrak{M}^{\mathrm{full}}(\mathcal{R}) = 1$*;*
- $\mathfrak{S}^{\mathrm{mach}}(\mathcal{R}) = \mathbf{S}_4(\mathcal{R})$ *if the tape-family upgrade route is certified but the strict upper certificate package is not fully certified;*
- $\mathfrak{S}^{\mathrm{mach}}(\mathcal{R}) = \mathbf{S}_3(\mathcal{R})$ *if the conditional reference-core closure surface is certified but the tape-family upgrade route is not fully certified;*
- $\mathfrak{S}^{\mathrm{mach}}(\mathcal{R}) = \mathbf{S}_2$ *if the conditional primitive Zuse-style core is certified but reference-core closure is not;*
- $\mathfrak{S}^{\mathrm{mach}}(\mathcal{R}) = \mathbf{S}_1$ *otherwise.*

Proposition 33.79 (Monotone exactness of the first machine-theoretic status decoder)**.** *For every chosen admissible sequential reference core* $\mathcal{R}$*, the decoder* $\mathfrak{S}^{\mathrm{mach}}(\mathcal{R})$ *is monotone with respect to certificate accumulation and exact with respect to the present hierarchy: it never outputs a higher level than has actually been certified, and whenever a full certificate package for some level in the hierarchy has been established, the decoder outputs at least that level. In particular,*

$$\mathfrak{M}^{\mathrm{full}}(\mathcal{R}) = 1 \quad \Longrightarrow \quad \mathfrak{S}^{\mathrm{mach}}(\mathcal{R}) = \mathbf{S}_5.$$

Likewise, if only the reference-core closure surface is available, the decoder halts at $\mathbf{S}_3(\mathcal{R})$*, and if the tape-family upgrade route has been certified but the strict upper package has not, it halts at* $\mathbf{S}_4(\mathcal{R})$*.*

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Corollary 33.80 (What the first machine-theoretic status decoder already buys)**.** *Later revisions can now state the strongest presently justified machine-theoretic claim without reopening the whole ladder of lower certificates. They may cite the single decoded status* $\mathfrak{S}^{\mathrm{mach}}(\mathcal{R})$ *and then only unfold the route-specific certificate package when a stronger or more unconditional claim is intended. This keeps the computational reading disciplined: the text can speak precisely about the current level already reached, while preventing accidental overstatement about higher levels that are not yet certified.*

19AH. Conditional/unconditional status surface on the compressed computational line

After the machine-theoretic status decoder has been introduced, the current release line should not remain described only in abstract terms. A small present-line status surface is therefore useful: it records, in one place, the strongest route-dependent status already certified on the present 4.7.0 release line and separates it from the weaker certificate-free status that can be asserted without invoking any chosen reference-core route.

Definition 33.81 (First present-line conditional/unconditional status surface on the compressed computational line)**.** *Fix a chosen admissible sequential reference core* $\mathcal{R}$*. Define the present-line machine-theoretic status pair*

$$\mathfrak{P}^{\mathrm{mach}}(\mathcal{R}) = \big(\mathfrak{S}^{\mathrm{abs}}, \mathfrak{S}^{\mathrm{cond}}(\mathcal{R})\big)$$

by the following rule:

- $\mathfrak{S}^{\mathrm{abs}}$ *is the strongest machine-theoretic level justified without invoking any chosen reference core, tape-family route, or upper certificate package;*
- $\mathfrak{S}^{\mathrm{cond}}(\mathcal{R})$ *is the strongest machine-theoretic level justified under the explicit chosen route* $\mathcal{R}$ *together with the corresponding lower and upper certificate package.*

The pair $\mathfrak{P}^{\mathrm{mach}}(\mathcal{R})$ *will be called the present-line conditional/unconditional status surface.*

Proposition 33.82 (First explicit present-line status readout on the compressed computational line)**.** *On the present 4.7.0 release line, and for every chosen admissible sequential reference core* $\mathcal{R}$ *along which the bundled closure surface* $\mathfrak{M}^{\mathrm{full}}(\mathcal{R}) = 1$ *has been certified, the present-line status surface reads*

$$\mathfrak{P}^{\mathrm{mach}}(\mathcal{R}) = \big(\mathbf{S}_1, \mathbf{S}_5\big).$$

Equivalently: unconditionally, the present line should still only be advertised at the baseline disciplined computational-reading level $\mathbf{S}_1$*; conditionally, along the explicitly certified route* $\mathcal{R}$*, the present line reaches the top certified level* $\mathbf{S}_5$*. In particular, the present text should avoid collapsing these two readings into a single unconditional global claim.*

Corollary 33.83 (What the first present-line conditional/unconditional status surface already buys)**.** *Later revisions can now cite the present machine-theoretic state of the Companion in one disciplined sentence: the line is unconditionally at* $\mathbf{S}_1$*, while conditionally along a chosen certified route it reaches* $\mathbf{S}_5$*. This gives a compact status formula for future hardening passes while preventing accidental overstatement about certificate-dependent closures as if they were already unconditional.*

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Remark 33.84 (One-paragraph synopsis of the machine-theoretic line)**.** *The computational line of the present framework can be read in three nested stages. First, the local operational layer yields a conditional primitive machine core, in which tape-like support, control, branching, and loop/return behavior are structurally separated and locally executable. Second, this primitive core can be hardened into a reference-core closure relative to a chosen sequential reference kernel $\mathcal{R}$, so that the line is no longer only machine-like in analogy, but carries a disciplined reference computation surface. Third, along a certified tape-based route above this reference-core closure, the framework admits a conditional upgrade toward stronger machine-theoretic levels, up to a Turing-style completeness claim only under explicit higher certificates; accordingly, the present line supports an unconditional computational base reading, while stronger universality claims remain route-dependent and conditional.*

19AH′. First bidirectional observer-side machine-shadow split below the status surface

After the present-line conditional/unconditional status surface has been fixed, one more local question becomes unavoidable on the working continuation line: how should the currently retained public machine readout itself be decomposed at observer side? The next useful step is not yet a full new universality statement, but a disciplined split of the retained machine shadow into one symmetric total channel and one antisymmetric imbalance channel. This keeps the older no-second-carrier discipline visible while making the current machine readout auditable in a form that can be compared directly with later transport, spectral, or shell-level hardening.

Definition 33.85 (Bidirectional observer-side machine-shadow coordinates)**.** *Fix the retained overlap window W_{ret} and a tested lag ℓ. Let $\eta^{(\ell)}_{\mathrm{drive}}$ denote the fitted observer-side drive contribution and $\eta^{(\ell)}_{\mathrm{carrier}}$ the fitted carrier contribution on W_{ret}. Define the bidirectional observer-side coordinates*

$$\eta^{(\ell)}_{+} = \eta^{(\ell)}_{\mathrm{drive}} + \eta^{(\ell)}_{\mathrm{carrier}}, \qquad \eta^{(\ell)}_{-} = \eta^{(\ell)}_{\mathrm{drive}} - \eta^{(\ell)}_{\mathrm{carrier}}.$$

We call $\eta^{(\ell)}_{+}$ the total machine-shadow channel *and $\eta^{(\ell)}_{-}$ the* observer-side imbalance channel*.*

Proposition 33.86 (Current retained-overlap checkpoint at the public optimum)**.** *On the retained overlap window currently kept on the working line, the bidirectional observer-side coordinates reproduce the present public optimum at lag*

$$\ell_* = -14$$

with weighted defect approximately 0.6746 *over* 42 *overlap points.*

Remark 33.87 (Interpretation of the checkpoint)**.** *The point of this checkpoint is not yet a final dynamical theorem about the origin of the machine readout. It is the weaker but already useful fact that the presently best public lag can be read consistently through one bidirectional observer-side split rather than through two unrelated fitted channels. In that sense, the present step converts one numerical optimum into a structurally readable local machine-shadow record.*

Proposition 33.88 (The imbalance channel is substantial but not primitive)**.** *On the same retained overlap window at the public optimum $\ell_* = -14$, the observer-side imbalance channel satisfies*

$$\frac{\|\eta^{(\ell_*)}_{-}\|_1}{\|\eta^{(\ell_*)}_{+}\|_1} \approx 0.6827, \qquad \operatorname{corr}\big(\eta^{(\ell_*)}_{+}, \eta^{(\ell_*)}_{-}\big) \approx 0.9532.$$

Hence the imbalance channel is not negligible, but it remains tightly slaved to the same total machine-shadow profile.

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Corollary 33.89 (First no-second-primitive-carrier reading from the machine split)**.** *The current bidirectional observer-side readout does not force the introduction of a second primitive carrier. What it supports is one dominant machine shadow together with a substantial but highly correlated imbalance readout. In particular, the current continuation line remains compatible with the older no-second-carrier discipline: the extra visible degree of freedom appears at observer side as a locked asymmetry channel, not as evidence for a new independent carrier species.*

Remark 33.90 (Why this matters for the next hardening step)**.** *The split creates a useful next leverage point. Future hardening no longer needs to ask only whether a retained carrier exists; it can ask whether a proposed theorem, transport step, or spectral proxy controls the symmetric total channel η_+, the locked imbalance channel η_-, or both at once. That distinction should make it easier to separate genuine carrier obligations from observer-shell correction obligations in the next working passes.*

19AH″. First channel-control dictionary induced by the machine-shadow split

Once the observer-side machine readout has been split into η_+ and η_-, the next useful step is a control dictionary telling later arguments what kind of burden they actually carry. This is still weaker than a new dynamical theorem. Its role is bookkeeping with mathematical consequences: it distinguishes propositions that genuinely control the retained machine shadow from propositions that merely reorganize the observer-side asymmetry readout around that shadow.

Definition 33.91 (η_+-control, η_--control, and joint channel control)**.** *Fix the retained overlap window W_{ret} at the current public optimum $\ell_* = -14$. We say that a theorem, estimate, transport rule, or spectral proxy on the present working line has*

1. total-channel control *if it bounds, reconstructs, or rigidifies $\eta_+^{(\ell_*)}$ up to the standing admissible equivalences without requiring an independent primitive contribution beyond the present drive/carrier split;*

2. imbalance-channel control *if, once the total channel is fixed in that admissible sense, it bounds, reconstructs, or rigidifies $\eta_-^{(\ell_*)}$ as an observer-side asymmetry readout around the same total machine shadow;*

3. joint channel control *if it controls both channels at once and in addition preserves or explains a structural lock between them, such as the present norm ratio or the present high correlation.*

Proposition 33.92 (First machine-side reading discipline induced by the dictionary)**.** *On the current working line, arguments aimed at the retained carrier or at the existence of one common machine shadow should be read as η_+-control problems, whereas observer-shell, asymmetry, phase-correction, or frame-flicker refinements should be read as η_--control problems unless they also alter the total channel itself. Mixed claims that try to upgrade both at once should be treated as joint-channel problems and therefore carry the heaviest burden of proof.*

Corollary 33.93 (First reformulation of the no-second-carrier discipline at machine level)**.** *A future refinement is compatible with the present no-second-carrier discipline whenever its extra visible freedom can be organized as η_--control over a fixed or jointly controlled η_+-channel, rather than by introducing a second primitive contribution already at the total-channel level.*

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In particular, the present checkpoint pushes the burden of any would-be second-carrier claim upward: such a claim would now have to break the current reading at the level of η_+, not merely by producing additional observer-side imbalance structure.

Remark 33.94 (Why the dictionary is a real compression tool)**.** *The dictionary turns one vague question — "does this next theorem improve the machine line?" — into a sharper trichotomy. A proposed step may improve the total machine shadow, the locked imbalance readout, or the coupling data between them. This should make later proof compression cleaner because many older optical and shell-side corrections can now be classified immediately as η_--side obligations, while the rarer genuinely carrier-bearing steps can be isolated as η_+-side obligations.*

19AH‴. First backward channel reading of the older optical and shell-side baselines

The dictionary becomes substantially more useful once it reorganizes not only future arguments but also the already stabilized comparison/shell archive. The present working line therefore records a first backward reading of the older optical-comparison, shortlist, bombe/sieve, shell/menu/window, and flicker-style material through the split into η_+ and η_-.

Proposition 33.95 (First backward channel reading of the retained public baselines)**.** *In this inherited public-release baseline block, the older public baselines should be read according to the following burden split.*

1. *The retained-carrier comparison layer, public optimum selection, no-second-carrier discipline, and every comparison/back-reading/bombe/sieve step insofar as it isolates one common machine-bearing shadow are primarily η_+-control statements.*

2. *The shell/menu/window refinements, phase-correction re-readings, and frame-flicker-style observer corrections are primarily η_--control statements whenever they refine how that common shadow appears on the observer side without changing the retained optimum itself.*

3. *Any statement claiming at once that a shell-side correction preserves the retained optimum, explains the asymmetry pattern, and leaves the common machine shadow identifiable should be treated as a joint-channel statement.*

Corollary 33.96 (First re-reading of the optical approximation baseline)**.** *The earlier first retained-carrier optical approximation should presently be read as one η_+-bearing baseline together with a subordinate η_--side correction envelope. Accordingly, visible mismatch inside that optical approximation is not yet evidence against the retained-carrier discipline unless it forces an additional primitive contribution already at the total-channel level.*

Remark 33.97 (Compression payoff of the backward reading)**.** *The backward reading turns a long mixed archive into a more ordered burden ledger. Carrier isolation, common-window retention, and no-second-carrier arguments can now be cited as total-channel obligations, while shell refinements, asymmetry diagnostics, and flicker-style observer corrections can be cited as imbalance-side obligations. Only the rarer steps that claim both retention and corrected observer-side readout need to reopen the full joint burden.*

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19AH$^{(4)}$. First theorematic burden split for the remaining continuation knots

Once the older baselines have been re-read through η_+ and η_-, the next practical question is which unresolved continuation knots still carry genuine carrier burden and which ones are only observer-side refinement burdens. The present working line therefore records one first theorematic burden split: unresolved shell-, phase-, or flicker-side refinement is not automatically a retained-carrier obstruction.

Proposition 33.98 (First theorematic burden split for the remaining continuation knots)**.** *On the present continuation line, the remaining theorematic knots should be read according to the following burden discipline.*

1. *Any future statement whose conclusion asserts preservation of the retained public optimum, one-common-machine-shadow identification, or the no-second-primitive-carrier discipline must control the total channel η_+. Purely imbalance-side refinements on η_- can neither prove nor refute such a carrier-level statement by themselves.*

2. *The frame-flicker ansatz, shell/menu/window refinements, phase-correction re-readings, and analogous observer-side asymmetry refinements are to be treated as η_--side knots as long as they only change how the retained machine shadow is seen and do not force a new total-channel profile.*

3. *Any theorem claiming simultaneously that an observer-side correction explains the asymmetry pattern, improves the retained fit, and preserves one common carrier must be treated as a joint-channel knot: it has to control both η_+ and η_-.*

4. *Consequently, a visible residual defect that can be confined to η_- is not yet a retained-carrier failure of the present line. A genuine carrier-level obstruction appears only if the retained optimum, the one-shadow reading, or the no-second-carrier discipline is already forced to fail on the total-channel side.*

Corollary 33.99 (Default downgrade rule for unresolved observer-side mismatch)**.** *On the present continuation line, an unresolved optical, shell-side, or flicker-style mismatch should by default be downgraded to an η_--side obligation unless it demonstrably changes the retained optimum, destroys the one-shadow reading at the total-channel level, or forces an additional primitive contribution already in η_+. This keeps the current no-second-carrier discipline from being overburdened by observer-side correction tasks that have not yet crossed into carrier space.*

Remark 33.100 (Why this sharpened burden split matters)**.** *The practical gain is a much cleaner proof queue. Future hardening can now ask first whether an open point is really about the common retained shadow or only about its observer-side distortion. If it is only the latter, then it belongs to the asymmetry shell and should not delay carrier-level statements. Only the comparatively rare steps that touch both channels at once need the full coupled burden again.*

19AH$^{(5)}$. First formal demotion criterion from joint-channel burden to pure imbalance-side burden

The burden split of 19AH$^{(4)}$ becomes operational only once one can say when a final continuation step no longer deserves to be treated as a genuinely joint-channel knot. The next local tool is therefore a formal demotion criterion: a pending observer-side refinement may

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be demoted from joint-channel burden to a pure η_--side burden as soon as its unresolved remainder provably leaves the retained total profile untouched.

Definition 33.101 (Observer-side demotion certificate)**.** *Fix a pending continuation statement $\mathcal{C}$ whose unresolved part currently concerns shell, phase, flicker, menu, or analogous observer-side refinement. We say that $\mathcal{C}$ admits an* observer-side demotion certificate *if all three of the following conditions hold on the retained admissible window under discussion.*

1. ***Total-profile invariance.*** *The retained public optimum, the one-machine-shadow identification, and the no-second-primitive-carrier reading remain unchanged on the η_+-side under the unresolved remainder of $\mathcal{C}$.*

2. ***No forced new primitive contribution.*** *The unresolved remainder of $\mathcal{C}$ does not require an additional primitive carrier already at the total-channel level; equivalently, the outstanding defect can still be represented without altering the retained η_+-carrier architecture.*

3. ***Imbalance localization.*** *The visible mismatch produced by the unresolved remainder of $\mathcal{C}$ can be localized to the observer-side imbalance channel η_-, meaning that its effect is read as a change of asymmetry, shell, or observer-side correction profile rather than as a change of the retained common total shadow.*

Proposition 33.102 (First formal demotion criterion)**.** *If a pending continuation statement $\mathcal{C}$ admits an observer-side demotion certificate, then $\mathcal{C}$ may be demoted from a provisional joint-channel knot to a pure η_--side obligation on the present continuation line. In particular, the unresolved remainder of $\mathcal{C}$ may no longer be cited as an open carrier-level obstruction against the retained public optimum, the one-shadow reading, or the no-second-primitive-carrier discipline.*

Proof. The three certificate clauses remove exactly the reasons why $\mathcal{C}$ would still need joint status. Clause (1) says that the retained total profile already survives unchanged at the η_+-level. Clause (2) rules out the only remaining carrier-level escape route, namely that the unresolved part secretly forces an additional primitive contribution already in the total channel. Clause (3) places the visible remainder entirely on the observer-side imbalance ledger. Once these three facts are available simultaneously, the outstanding burden of $\mathcal{C}$ no longer concerns whether the present line carries one retained common machine shadow, but only how the observer-side asymmetry shell should be refined. The knot is therefore genuinely η_--side and may be demoted accordingly. □

Corollary 33.103 (Safe demotion rule for future continuation work)**.** *Future continuation steps should only be kept on the joint-channel ledger if at least one of the three certificate clauses fails or is not yet known. Conversely, once the retained η_+-profile is stable, no extra primitive carrier is forced there, and the remaining defect is localized to η_-, the step should be queued as observer-side hardening rather than as unfinished carrier proof.*

Remark 33.104 (Why this tool is useful)**.** *This criterion turns the burden split into a reusable proof tool. It prevents later drafts from repeatedly reopening the full carrier question whenever a shell-side mismatch or flicker-style correction is still under discussion. The proof queue can now be triaged by certificate: unresolved points without η_+-damage stay on the asymmetry ledger, while only genuine total-channel risk keeps a knot on the joint ledger.*

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19AH$^{(6)}$. First provisional demotion ledger for inherited observer-side continuation knots

Reader cue. Read the next two subsections as a queue-classification pair. Their purpose is to sort inherited shell/phase/flicker-style leftovers first onto the observer-side ledger by default and then to state the exact evidence that would force those same knots back into joint or total-channel burden.

The formal demotion criterion of 19AH$^{(5)}$ becomes most useful once it is applied back to the inherited queue. The point is not to declare old open points "solved" by fiat, but to state explicitly which of them should from now on be carried on the observer-side ledger unless and until they actually damage the retained total profile. The next local step is therefore a provisional demotion ledger for the main inherited observer-side knot families.

Definition 33.105 (Provisional demotion ledger for inherited observer-side knots)**.** *On the present continuation line, the following inherited knot families are placed on the* provisional observer-side demotion ledger*, subject to re-promotion only if later work shows a genuine failure of at least one certificate clause from 19AH$^{(5)}$:*

1. *shell/menu/window-refinement mismatches that alter the observer-side shell geometry without changing the retained total optimum profile,*
2. *phase-correction and frame-flicker residuals that modify how the retained profile is read from the observer side while leaving the common total shadow intact,*
3. *optical-approximation residuals whose visible discrepancy can still be localized to the asymmetry envelope rather than to the retained common carrier, and*
4. *analogous observer-side refinements inherited from the older archive whenever they do not force a second primitive carrier and do not move the retained public optimum on the total-channel side.*

Proposition 33.106 (First provisional demotion ledger theorem)**.** *The inherited knot families listed above belong, by default, on the observer-side demotion ledger rather than on the joint-channel obstruction ledger. More precisely, on the present compressed computational line they should be read as provisional η_--side obligations unless later analysis shows that one of them changes the retained η_+-profile, breaks the one-shadow reading at total-channel level, or forces an additional primitive carrier already in η_+.*

Proof. Each listed family is inherited precisely from the shell/phase/flicker/optical side of the archive, hence from the side that 19AH‴ and 19AH$^{(4)}$ already identified as the natural observer-side burden domain. The formal criterion of 19AH$^{(5)}$ then supplies the missing operational rule: as long as the retained total profile remains fixed, no new primitive carrier is forced in the total channel, and the visible mismatch stays localizable to the imbalance channel, the corresponding inherited knot is not entitled to remain on the joint ledger. The listed families therefore enter the queue under default demotion. Re-promotion is reserved only for the exceptional case that later work actually detects η_+-damage or a genuine carrier-level escape route. □

Corollary 33.107 (How to read the old queue from now on)**.** *From this point onward, older shell/menu/window, phase-correction, flicker-style, and observer-side optical residuals should no longer be cited as open carrier objections by default. They remain real work items, but they are to be read first as asymmetry-hardening tasks. Only after a concrete failure of the demotion certificate has been exhibited should one move such a knot back to the coupled or carrier-level queue.*

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Remark 33.108 (What "demotion" means here in plain language)**.** *This is only a bookkeeping and proof-burden move. "Demotion" does* not *mean that the point is unimportant, false, or removed. It means: this point still matters, but at the current stage it counts as a problem of observer-side correction rather than as evidence that the common retained carrier picture has broken. In ordinary language, we stop treating every leftover wobble in the readout shell as if it were a collapse of the whole machine-bearing line.*

19AH$^{(7)}$. First formal re-promotion criterion from observer-side demotion back to joint or total burden

Once a provisional demotion ledger exists, the opposite discipline becomes equally important. Demotion is safe only if later work can also say when a demoted knot must be upgraded again. The next local tool is therefore a formal re-promotion criterion: if a supposedly observer-side remainder turns out to damage the retained total machine-shadow profile or to force a new primitive carrier already at total-channel level, then the knot must leave the pure η_--ledger and return to a higher burden class.

Definition 33.109 (Observer-side re-promotion trigger)**.** *Let a continuation knot family K currently lie on the provisional observer-side demotion ledger. We say that K carries an* observer-side re-promotion trigger *if later analysis exhibits at least one of the following three failures:*

1. *the retained total machine-shadow profile η_+ is no longer fixed up to the standing admissible equivalences;*
2. *the one-shadow reading at total-channel level breaks, in the sense that the visible fit can no longer be organized without introducing an additional primitive carrier already in η_+;*
3. *the observer-side remainder can no longer be localized purely to η_-, because its correction necessarily changes the total-channel optimum, the total-channel rigidity data, or the total-channel carrier count.*

Proposition 33.110 (First re-promotion criterion)**.** *Let K be a knot family on the provisional observer-side demotion ledger. If K satisfies an observer-side re-promotion trigger, then K is no longer admissible as a pure η_--side obligation. More precisely:*

1. *if the trigger only shows that K cannot be separated from the total-channel data but does not yet force a new primitive carrier, then K must be re-promoted from pure η_--burden to* joint-channel burden*;*
2. *if the trigger forces a change in the retained total machine-shadow profile or compels an additional primitive carrier already in η_+, then K must be re-promoted all the way to* total-channel burden*.*

Proof. This is the exact converse of the demotion logic from 19AH$^{(5)}$ and the ledger rule from 19AH$^{(6)}$. A knot may remain on the pure observer-side ledger only while the retained total profile stays fixed, no second primitive carrier is forced at total-channel level, and the visible remainder remains localizable to the imbalance channel. The three trigger clauses record the failure of precisely those permissions. Once a clause fails, pure η_--treatment is no longer justified. If the failure merely entangles the remainder with the total channel, the knot becomes joint. If the failure genuinely alters or multiplies the total-channel carrier structure, the knot becomes a total-channel problem. □

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Corollary 33.111 (Plain-language queue rule)**.** *A leftover shell, phase, flicker, or optical residual may stay in the "observer-side corrections" queue only while it leaves the main common machine shadow intact. The moment it starts moving that main shadow, or starts behaving as if a second primitive carrier were needed in the total profile, it must be upgraded again. In ordinary language: demotion is reversible, but only by evidence of real damage to the main channel.*

Remark 33.112 (Why the demotion/re-promotion pair is useful)**.** *Together, 19AH$^{(5)}$–19AH$^{(8)}$ now give a two-way burden discipline. Demotion prevents every observer-side wobble from being misread as a carrier collapse; re-promotion prevents the opposite mistake of hiding a true carrier-level defect inside the observer-side queue. This makes the inherited continuation landscape auditable in both directions.*

19AH$^{(8)}$. First channel traffic-light ledger induced by the demotion/re-promotion discipline

Reader cue. Read the next ledger as a dashboard. Its purpose is to summarize the already stated demotion and re-promotion rules in green/yellow/red form, so that the inherited queue can be scanned quickly before later live burdens are isolated.

The next local consolidation step is to turn the two-way burden discipline into a directly readable queue picture. Instead of carrying every inherited continuation knot under one undifferentiated status word, one may sort the inherited families by a traffic-light rule. Green means that the family presently remains admissible as a pure observer-side burden; yellow means that the family is already entangled enough with the total channel that it must be treated jointly; red is reserved for the genuinely exceptional case that the family damages the retained total profile itself or forces a second primitive carrier already in η_+.

Definition 33.113 (Channel traffic-light ledger)**.** *A* channel traffic-light ledger *for inherited continuation knots is an assignment of each inherited knot family K to exactly one of the following three burden classes:*

1. ***green / observer-side only**: K satisfies the demotion certificate from 19AH$^{(5)}$ and no re-promotion trigger from 19AH$^{(7)}$ is presently visible;*
2. ***yellow / joint-channel**: K no longer qualifies for pure observer-side treatment, but the available evidence still stops short of forcing a genuine total-channel carrier change;*
3. ***red / total-channel**: K damages the retained total profile η_+ itself or forces an additional primitive carrier already at total-channel level.*

Proposition 33.114 (First provisional traffic-light placement for the inherited queue)**.** *On the present compressed computational line, the inherited queue may provisionally be sorted as follows.*

1. ***Green:** shell/menu/window refinements, phase-correction refinements, frame-flicker compensation refinements, and observer-side optical residual refinements remain on the green ledger, because they presently leave the retained total machine-shadow profile fixed, force no second primitive carrier in η_+, and stay localizable to imbalance-side repair.*
2. ***Yellow:** any inherited knot whose proposed correction already shifts the retained optimum location, modifies the rigidity data needed to keep the one-shadow total reading stable, or otherwise couples the observer-side mismatch back into the total-channel fit must enter the yellow ledger and be treated as a joint-channel obligation.*

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3. ***Red:*** *no presently inherited knot family is forced onto the red ledger on the current line; red is reserved only for the case that later work exhibits an actual breakdown of the retained total profile or a genuine second primitive carrier already in* η_+.

Proof. The green placement is exactly the provisional demotion ledger from $19\mathrm{AH}^{(6)}$, now read through the demotion certificate of $19\mathrm{AH}^{(5)}$. The yellow placement is the first nontrivial output of the re-promotion rule from $19\mathrm{AH}^{(7)}$: once a knot cannot be repaired purely on the imbalance side, yet still does not force a total-channel carrier split, it must be carried jointly. The red placement is currently empty because the present line has not yet produced evidence that the retained total machine shadow itself fails or that a second primitive carrier is already required in the total channel. Hence the inherited queue is presently green-dominant, with yellow reserved for future entanglement cases and red reserved for genuine carrier-level failure. □

Corollary 33.115 (Plain-language traffic-light reading)**.** *At the moment, most inherited leftovers are green: they are still real work, but only on the observer side. Yellow means: this leftover is beginning to touch the main common shadow and therefore needs joint treatment. Red would mean: the main common shadow itself is no longer enough. On the current line, we are not at red.*

Remark 33.116 (Why this corrects possible drift)**.** *The point of the traffic-light ledger is precisely to stop the continuation block from drifting into an ever longer meta-language. Instead of adding more abstract burden labels, the present line now has a single concrete reading device: green for pure* η_-*-repair, yellow for genuine coupling, red for actual* η_+*-damage. This keeps the new diagnostic layer operational and tied to concrete queue management.*

$19\mathrm{AH}^{(9)}$. First synchronized OP ledger after the channel-discipline consolidation

Reader cue. Read the next synchronized OP ledger as a theorem-summary device. Its purpose is to collapse historical OP labels into *closed*, *absorbed*, and *residual-core* status before the roadmap is read, not to reopen already discharged fronts.

Once the inherited continuation queue has been sorted by total/imbalance burden and by the traffic-light rule, the older named OP fronts should no longer be carried unchanged merely for historical convenience. The next consolidation step is therefore to synchronize the inherited OP labels against the presently certified structure itself. The point is not to rename difficulty away, but to state explicitly which earlier OP fronts are already structurally discharged, which no longer survive as independent fronts, and which remain as the genuine residual core for the next version.

Definition 33.117 (Synchronized OP ledger)**.** *A* synchronized OP ledger *on the present compressed computational line assigns every inherited named OP front* P *to exactly one of the following three statuses:*

1. ***closed****: the relevant structural burden of* P *is already discharged on the present line and should no longer be listed as a primary open problem;*

2. ***absorbed****:* P *is no longer independent, because its only remaining unresolved content is already contained in another residual core;*

3. ***residual core****:* P *still carries genuinely new proof burden that is not yet discharged by the present line.*

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Proposition 33.118 (First synchronized placement of the inherited OP queue)**.** *On the present compressed computational line, the inherited named OP queue may be synchronized as follows.*

1. ***Closed:*** *OP 3 is closed as an independent front by the shell-certificate/residual-interface line already exported in the present text, and OP 5 Step (A) is closed by the source-return, branchwise, and routed carrier-first closure package.*

2. ***Absorbed:*** *OP 2 is absorbed into OP 1, because the remaining frame-offset/continuum-correction burden no longer survives as an independent Dirac-side front once the present line is synchronized; OP 4 is absorbed into OP 1, because its only still-open quantitative content is precisely the same continuum-limit correction δ_{CL}.*

3. ***Residual core:*** *the live queue is reduced to OP 1 and OP 5 Step (B). OP 1 remains the continuum-limit/total-scaling residual centered on δ_{CL}. OP 5 Step (B) remains the exact Θ–R balance problem on the routed photon-frame side.*

Proof. The earlier OP 3 burden has already been converted into a shell-certified interface theorem together with reusable residual/shell reading rules, so it no longer behaves as an unreduced primary front. Likewise, the source-return closure theorem, its branchwise refinement, and the routed carrier-first OP5 interface already discharge the source/return and admissible-routing part of OP 5, leaving only the exact total balance problem as live burden. Earlier in the text, OP 4 is explicitly reduced to the same continuum-limit correction δ_{CL} as OP 1, and the remaining OP 2 frame-offset burden has already collapsed onto that same continuum-limit core rather than surviving as an independent front. Hence the present synchronized ledger has only two genuine residual cores left: OP 1 and OP 5 Step (B). □

Corollary 33.119 (Plain-language OP synchronization)**.** *The inherited OP queue is no longer a five-front queue. On the current line, OP 3 is closed, OP 2 and OP 4 are absorbed into OP 1, OP 5 survives only through Step (B), and the live queue is therefore a two-core queue: OP 1 and OP 5 Step (B).*

Remark 33.120 (Why this synchronization matters)**.** *This synchronization keeps the continuation line honest in both directions. It prevents already discharged or absorbed fronts from being carried forever as if nothing had changed, but it also refuses to pretend that the two genuine residual cores are already gone. The next real version target is therefore sharply focused rather than historically bloated.*

19AH$^{(10)}$. Two-core OP-closing roadmap before the later freeze deferral

Reader cue. Read the next roadmap as an earlier compression snapshot of the line. Its purpose is to orient burden genealogy by recording how the queue had once been narrowed to two cores before the later freeze deferral; it does not override the present post-freeze priority order.

After the synchronized ledger from 19AH$^{(9)}$, the line first compressed the next-version target to a narrow two-core closure program. At that stage, the intended next version was to try to close the remaining queue without reopening the green observer-side ledger except where a genuine yellow or red trigger is produced by the re-promotion rule. The purpose is therefore not to add another diagnostic family, but to turn the present compression into two direct closure attempts: one on the continuum-limit residual and one on the exact routed photon-frame balance.

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Definition 33.121 (Residual-core closing roadmap)**.** *A* residual-core closing roadmap *in this earlier two-core reading is an ordered list of closure tasks that leaves all green observer-side knot families parked by default and reopens them only if a later step actually produces a re-promotion trigger.*

Proposition 33.122 (First two-core OP-closing roadmap before the later freeze deferral)**.** *In that earlier two-core reading, the next version would have proceeded through the following tasks.*

1. ***OP 1 isolation.*** *Reduce OP 1 to one final explicit continuum-limit residual identity for δ_{CL}, so that no auxiliary historical formulation remains in the live queue.*
2. ***OP 1 closure attempt.*** *Attack the vanishing or exact identification of that residual directly from the retained total-channel data and the lattice-to-continuum transport structure already built into the present line.*
3. ***OP 5(B) isolation.*** *Rewrite OP 5 Step (B) as one exact routed balance statement for the remaining Θ–R defect term, with all source-return, branchwise, and carrier-first routing pieces treated as fixed infrastructure rather than reopened.*
4. ***OP 5(B) closure attempt.*** *Prove or refute the vanishing of the remaining routed balance defect without importing a second primitive total-channel carrier.*
5. ***Final synchronization pass.*** *If the two closure attempts succeed, update the synchronized OP ledger so that no live residual core remains; if one attempt fails, record precisely which single residual core survives and keep the green observer-side queue parked.*

Corollary 33.123 (Roadmap discipline in the earlier two-core reading)**.** *In that earlier two-core reading, shell/menu/window, phase-correction, frame-flicker, and other green observer-side families are not first targets. They may be revisited only if closing OP 1 or OP 5 Step (B) actually forces a yellow or red re-promotion.*

Remark 33.124 (Why the roadmap is intentionally narrow)**.** *The synchronized ledger shows that the continuation line is now past the stage where every historical knot needs to be touched again. The next mathematically serious step in that earlier reading was to spend the new leverage exactly where the residual burden still lives: on the continuum-limit source residual of OP 1 and on the exact routed balance residual of OP 5 Step (B). A narrower roadmap is therefore not a loss of ambition, but a sign that the line has finally compressed enough to attack the remaining core directly.*

19AH$^{(11)}$. Two-core residual dictionary opening the 4.7.0 closure line

After the synchronized OP ledger from 19AH$^{(9)}$ and the next-version roadmap from 19AH$^{(10)}$, the correct opening move for 4.7.0 is to stop carrying OP 1 and OP 5 Step (B) as diffuse historical labels and to rewrite them as two explicit residual objects. Every later closure argument can then be judged by one simple rule: it either changes one of these residuals directly, or it is auxiliary infrastructure only.

Definition 33.125 (Continuum-limit residual for OP 1)**.** *Define the* continuum-limit residual

$$\Delta_{\mathrm{CL}} := \log \delta_{\mathrm{CL}} + \frac{\gamma_1}{4\pi\gamma_L^5}.$$

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Equivalently, $\Delta_{\mathrm{CL}} = 0$ *if and only if the corrected continuum-limit formula from Remark 31.64,*

$$\delta_{\mathrm{CL}} = \exp\left(-\frac{\gamma_1}{4\pi\gamma_L^5}\right),$$

holds exactly.

Definition 33.126 (Routed photon-frame balance defect for OP 5 Step (B))**.** *Let* $T := \gamma_L^{-3/2}$ *and let* $\mathcal{T}(T)$ *be the photon-frame Theta sum from Definition 32.178. Define the* routed photon-frame balance defect

$$\Delta_{\Theta R}(T) := \mathcal{T}(T) - \left(\frac{3T}{2} - \log T - \frac{T^2}{24} - \gamma_E - \beta_{\mathrm{sym}}^2 T^2\right).$$

Equivalently, $\Delta_{\Theta R}(T) = 0$ *if and only if the exact Theta-sum balance conjecture of Proposition 32.179 holds at the routed photon-frame cutoff.*

Proposition 33.127 (Two-core reduction of the live OP queue)**.** *On the synchronized ledger of 19AH*$^{(9)}$*, the live OP queue is reduced exactly as follows.*

1. *OP 1 is equivalent to the vanishing or exact first-principles identification of the single scalar residual* Δ_{CL}*.*
2. *OP 5 Step (B) is equivalent to the vanishing of the single routed balance defect* $\Delta_{\Theta R}(T)$*.*
3. *Any inherited continuation task that leaves both* Δ_{CL} *and* $\Delta_{\Theta R}(T)$ *unchanged remains auxiliary unless it triggers a yellow or red re-promotion in the sense of 19AH*$^{(7)}$*–19AH*$^{(8)}$*.*

Proof. The OP 1 side was already compressed in theorem form by Theorem 32.160, which reduces the open quantitative content of OP 1 and OP 4 to the single continuum-limit correction δ_{CL}. Writing that remaining burden as Δ_{CL} only turns the same statement into one scalar residual identity. On the OP 5 side, Step (A) is already closed by Proposition 32.173, while Step (B) is explicitly reduced in Proposition 32.179 to one exact routed photon-frame Theta-sum balance statement. Writing that remaining burden as $\Delta_{\Theta R}(T)$ therefore changes no mathematical content; it only compresses the live queue into one explicit defect variable. The third clause is then the direct reading of the synchronized ledger 19AH$^{(9)}$, the roadmap 19AH$^{(10)}$, and the re-promotion discipline 19AH$^{(7)}$–19AH$^{(8)}$. □

Remark 33.128 (Why the new version should start here)**.** *The point of this reduction is not rhetorical neatness. Once the queue is written as* Δ_{CL} *and* $\Delta_{\Theta R}(T)$*, every later step can be audited immediately: either it shrinks one of these two residuals, identifies one of them exactly, or it belongs to infrastructure/support rather than to the remaining core closure burden. This prevents the next version from drifting back into a diffuse many-front reading of the inherited archive.*

19AH$^{(12)}$. First OP 1 slope certificate at the continuum-limit residual

After the two-core reduction from 19AH$^{(11)}$, the next honest frontal attack on OP 1 is to compress the residual once more: not into a new conjectural formula, but into one normalized attenuation index. This separates the "what has to be derived" question from the "how much surrounding continuum-limit story one tells" question. The live OP 1 burden is then no longer an undifferentiated derivation of δ_{CL}, but the identification of one single scalar exponent/slope quantity.

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Definition 33.129 (Normalized continuum-limit attenuation index)**.** *Define the* normalized continuum-limit attenuation index

$$\Xi_{\mathrm{CL}} := -\frac{4\pi\gamma_L^5}{\gamma_1}\log\delta_{\mathrm{CL}}.$$

Using Theorem 32.160, equivalently

$$\Xi_{\mathrm{CL}} = -\frac{4\pi\gamma_L^5}{\gamma_1}\log\bigl(\sqrt{3}\,\beta_{\mathrm{sym}}\bigr).$$

Thus Ξ_{CL} measures exactly how far the still-open continuum-limit correction deviates from the final exponent in Remark 31.64, but in normalized unit form.

Proposition 33.130 (Equivalent slope certificate for OP 1)**.** *The following statements are equivalent.*

1. *The OP 1 continuum-limit residual vanishes, i.e. $\Delta_{\mathrm{CL}} = 0$.*
2. *The normalized attenuation index satisfies $\Xi_{\mathrm{CL}} = 1$.*
3. *Whenever one can write the remaining continuum-limit correction at the first live scale in exponential form*
$$\delta_{\mathrm{CL}} = e^{-c_{\mathrm{CL}}\gamma_1},$$
for some dimensionless attenuation slope c_{CL}, that slope is exactly
$$c_{\mathrm{CL}} = \frac{1}{4\pi\gamma_L^5}.$$

Proof. By Definition 33.125,

$$\Delta_{\mathrm{CL}} = 0 \quad\Longleftrightarrow\quad \log\delta_{\mathrm{CL}} = -\frac{\gamma_1}{4\pi\gamma_L^5}.$$

Multiplying by $-4\pi\gamma_L^5/\gamma_1$ gives

$$\Delta_{\mathrm{CL}} = 0 \quad\Longleftrightarrow\quad -\frac{4\pi\gamma_L^5}{\gamma_1}\log\delta_{\mathrm{CL}} = 1,$$

which is exactly $\Xi_{\mathrm{CL}} = 1$, proving the equivalence of (1) and (2). If in addition $\delta_{\mathrm{CL}} = e^{-c_{\mathrm{CL}}\gamma_1}$, then

$$\log\delta_{\mathrm{CL}} = -c_{\mathrm{CL}}\gamma_1,$$

so the same identity becomes

$$c_{\mathrm{CL}}\gamma_1 = \frac{\gamma_1}{4\pi\gamma_L^5} \quad\Longleftrightarrow\quad c_{\mathrm{CL}} = \frac{1}{4\pi\gamma_L^5}.$$

Conversely, this slope value immediately reconstructs the vanishing residual formula from Definition 33.125. Hence all three statements are equivalent. □

Corollary 33.131 (What the OP 1 attack reduces to)**.** *On the present compressed line, a successful OP 1 closure attempt no longer needs to present itself as a broad continuum-limit narrative. It is enough to derive, certify, or uniquely force the single normalized attenuation statement*

$$\Xi_{\mathrm{CL}} = 1.$$

Equivalently, under any admissible exponential first-scale attenuation ansatz, OP 1 is closed once the slope c_{CL} is shown to equal $1/(4\pi\gamma_L^5)$.

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Remark 33.132 (Why this is a real reduction and not just notation)**.** *This step still does* not *prove OP 1. What it does remove is one layer of ambiguity about what the next version must actually derive. The live target is no longer a vague* prime-lattice / continuum-limit correction story*; it is the single normalized exponent encoded by* Ξ_{CL}*. Any later lemma about lattice counting, prime spacing, frame conversion, or first-zero transport is now auditable by one question only: does it fix* Ξ_{CL}*, leave it unchanged, or force it away from 1? In that sense turns OP 1 into the narrowest honest one-scalar closure gate currently available.*

19AH$^{(13)}$. First frame-transport split certificate for OP 1 in photon-first-zero normalization

After the slope certificate from 19AH$^{(12)}$, the next useful compression is not to invent a new exponent but to split the already isolated OP 1 load into its frame-conversion part and its photon-side transport part. Remark 31.64 already records that the final residual formula may be written in two equivalent ways,

$$\delta_{\mathrm{CL}} = \exp\left(-\frac{\gamma_1}{4\pi\gamma_L^5}\right) = \exp\left(-\frac{\gamma_1^{(\mathrm{photon})}}{4\pi\gamma_L^4}\right),$$

with $\gamma_1^{(\mathrm{photon})} = \gamma_1/\gamma_L$. The fifth power of γ_L in the matter-frame formula is therefore already visibly split into one frame-change factor and four photon-side transport factors. This means that OP 1 can be attacked in a sharper way: not as a mysterious global γ_L^5-law, but as a photon-first-zero transport law with four residual shell factors after the frame conversion has been peeled off.

Definition 33.133 (Photon-normalized continuum-transport index)**.** *Define the* photon-normalized continuum-transport index

$$\Psi_{\mathrm{CL}} := -\frac{4\pi\gamma_L^4}{\gamma_1^{(\mathrm{photon})}} \log \delta_{\mathrm{CL}}.$$

Equivalently, since $\gamma_1^{(\mathrm{photon})} = \gamma_1/\gamma_L$*,*

$$\Psi_{\mathrm{CL}} = -\frac{4\pi\gamma_L^5}{\gamma_1} \log \delta_{\mathrm{CL}} = \Xi_{\mathrm{CL}}.$$

Thus Ψ_{CL} *carries the same open numerical burden as* Ξ_{CL}*, but written after the matter-to-photon frame passage has been made explicit.*

Proposition 33.134 (Frame-transport split certificate for OP 1)**.** *The following statements are equivalent.*

1. *The OP 1 residual vanishes, i.e.* $\Delta_{\mathrm{CL}} = 0$*.*
2. *The slope certificate of 19AH*$^{(12)}$ *holds, i.e.* $\Xi_{\mathrm{CL}} = 1$*.*
3. *The photon-normalized transport index satisfies* $\Psi_{\mathrm{CL}} = 1$*.*
4. *The continuum-limit correction has the photon-first-zero transport form*

$$\delta_{\mathrm{CL}} = \exp\left(-\frac{\gamma_1^{(\mathrm{photon})}}{4\pi\gamma_L^4}\right).$$

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In particular, OP 1 is equivalent to proving that the entire residual attenuation beyond the frame change from γ_1 to $\gamma_1^{(\mathrm{photon})}$ is exactly one unit of photon-normalized four-layer transport load.

Proof. The equivalence of (1) and (2) is Proposition 33.130. By Definition 33.133 and the identity $\gamma_1^{(\mathrm{photon})} = \gamma_1/\gamma_L$, one has

$$\Psi_{\mathrm{CL}} = -\frac{4\pi\gamma_L^4}{\gamma_1^{(\mathrm{photon})}} \log \delta_{\mathrm{CL}} = -\frac{4\pi\gamma_L^5}{\gamma_1} \log \delta_{\mathrm{CL}} = \Xi_{\mathrm{CL}},$$

so (2) and (3) are equivalent. Solving $\Psi_{\mathrm{CL}} = 1$ for δ_{CL} gives

$$\log \delta_{\mathrm{CL}} = -\frac{\gamma_1^{(\mathrm{photon})}}{4\pi\gamma_L^4},$$

and exponentiating yields (4). Conversely, (4) inserted into Definition 33.133 gives $\Psi_{\mathrm{CL}} = 1$. Hence all four statements are equivalent. □

Corollary 33.135 (What the OP 1 attack really changes)**.** *The remaining OP 1 burden may now be read in a more structured way. One no longer has to "explain a γ_L^5 exponent all at once". It is enough to justify the following split picture:*

1. *one power of γ_L comes from the explicit passage from the matter-frame first zero γ_1 to the photon-frame first zero $\gamma_1^{(\mathrm{photon})}$;*
2. *the remaining four powers form the actual photon-side transport attenuation burden;*
3. *that photon-side burden has normalized weight exactly one.*

Therefore any future counting, lattice, shell, or first-zero transport lemma may now be audited by the sharper question: does it explain the unit value of Ψ_{CL}, rather than merely restating the matter-frame exponent?

Remark 33.136 (Why this is progress without pretending closure)**.** *This still does not prove OP 1. The real gain is conceptual localization. In the target was the one-scalar index Ξ_{CL}. In the same scalar is re-read in a frame-split form that matches the standing photon/matter distinction of the Companion. That removes one source of merge-style ambiguity: the fifth power of γ_L no longer needs to be read as an indivisible miracle. The present line may now search specifically for a* four-layer photon transport law after one explicit frame conversion*. That is narrower, structurally cleaner, and closer to the actual lattice/transport heuristics already used elsewhere in the document.*

19AH$^{(14)}$. First finite four-shell balance certificate for OP 1 after the frame split

After the frame-transport split of 19AH$^{(13)}$, the next honest compression is to stop carrying the photon-side transport burden as one undifferentiated four-power block and to write it as a finite shell balance. This does not yet explain *why* the four-layer photon transport should have normalized weight one, but it does isolate the exact finite statement that any such explanation would have to prove. In particular, OP 1 can now be re-read as a question about the total normalized load carried by four post-frame transport layers, rather than as a raw continuum-limit estimate.

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Definition 33.137 (Finite four-shell transport loads for the OP 1 residual)**.** *Assume the photon-side continuum-limit attenuation admits a four-factor decomposition*

$$\delta_{\mathrm{CL}} = \prod_{j=1}^{4} \tau_j,$$

with positive shell factors $\tau_j > 0$. Define the corresponding normalized shell loads

$$\lambda_j := -\frac{4\pi\gamma_L^4}{\gamma_1^{(\mathrm{photon})}} \log \tau_j \qquad (j = 1, 2, 3, 4).$$

Whenever the above product formula holds, the photon-normalized transport index of Definition 33.133 satisfies

$$\Psi_{\mathrm{CL}} = \sum_{j=1}^{4} \lambda_j.$$

Proposition 33.138 (Finite four-shell closure certificate for OP 1)**.** *Suppose $\delta_{\mathrm{CL}} = \prod_{j=1}^{4} \tau_j$ is a positive four-shell decomposition and let λ_j be the normalized shell loads from Definition 33.137. Then the following are equivalent.*

1. *The OP 1 residual vanishes, i.e. $\Delta_{\mathrm{CL}} = 0$.*
2. *The photon-normalized transport index satisfies $\Psi_{\mathrm{CL}} = 1$.*
3. *The four normalized shell loads satisfy the finite balance law*

$$\sum_{j=1}^{4} \lambda_j = 1.$$

In particular, after the frame split from 19AH$^{(13)}$, OP 1 is equivalent to proving that the post-frame four-shell transport carries total normalized load exactly one.

Proof. By Proposition 33.134, statement (1) is equivalent to $\Psi_{\mathrm{CL}} = 1$, so (1) and (2) are equivalent. If $\delta_{\mathrm{CL}} = \prod_{j=1}^{4} \tau_j$, then

$$\log \delta_{\mathrm{CL}} = \sum_{j=1}^{4} \log \tau_j.$$

Multiplying by $-4\pi\gamma_L^4/\gamma_1^{(\mathrm{photon})}$ gives

$$\Psi_{\mathrm{CL}} = -\frac{4\pi\gamma_L^4}{\gamma_1^{(\mathrm{photon})}} \log \delta_{\mathrm{CL}} = \sum_{j=1}^{4} \left(-\frac{4\pi\gamma_L^4}{\gamma_1^{(\mathrm{photon})}} \log \tau_j \right) = \sum_{j=1}^{4} \lambda_j,$$

which is exactly statement (3). Hence (2) and (3) are equivalent, and all three assertions are equivalent. □

Corollary 33.139 (Equal-shell partition is sufficient but not necessary)**.** *Under the hypotheses of Proposition 33.138, any of the following is sufficient to close OP 1.*

1. *$\lambda_j = \frac{1}{4}$ for each $j = 1, 2, 3, 4$;*

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2. *more generally, any asymmetric four-shell profile with*

$$\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 1.$$

Thus the present line does not *require equal quarter-load shells; equal partition is merely the simplest admissible closure pattern.*

Remark 33.140 (Why is a genuine narrowing of OP 1)**.** *This still does not prove OP 1, and it should not be oversold as such. The genuine gain is that OP 1 is now finite twice over. First, reduced it to one scalar index; second, split that index into one frame factor plus a four-layer photon burden; now turns the surviving photon burden into a finite additive balance law. The live question is no longer* "what is the whole continuum-limit correction story?" *but rather:* which four post-frame shell loads are present, and why do they sum to one? *Any future lattice, shell, counting, or transport lemma can therefore be audited even more sharply: does it produce one of the λ_j, constrain their sum, or leave the finite balance untouched?*

19AH$^{(15)}$. First two-pass half-burden certificate for the OP 1 four-shell balance

The finite four-shell balance of 19AH$^{(14)}$ still leaves open where the four loads λ_j should come from. The first structurally natural final is a symmetrized two-pass reading of the post-frame photon transport: two shell loads belong to the outward/forward pass and two belong to the return/backward pass. This does not yet determine the individual shell factors, but it reduces the origin question from four unrelated loads to two pass totals. That is already enough to expose the first genuinely bidirectional closure pattern for OP 1.

Definition 33.141 (Two-pass shell-pair loads for the OP 1 residual)**.** *Under the hypotheses of Definition 33.137, define the forward- and backward-pass totals*

$$\Lambda_{\rightarrow} := \lambda_1 + \lambda_2, \qquad \Lambda_{\leftarrow} := \lambda_3 + \lambda_4.$$

We call the four-shell transport profile two-pass grouped *if the post-frame shell loads are read as two forward-side loads and two backward-side loads in this way. We call it* pass-symmetric *if in addition*

$$\Lambda_{\rightarrow} = \Lambda_{\leftarrow}.$$

Proposition 33.142 (Two-pass half-burden certificate for OP 1)**.** *Suppose the OP 1 residual admits a two-pass grouped four-shell decomposition in the sense of Definition 33.141. Then the following are equivalent.*

1. *The OP 1 residual vanishes, i.e. $\Delta_{\mathrm{CL}} = 0$.*
2. *The photon-side transport index satisfies $\Psi_{\mathrm{CL}} = 1$.*
3. *The two pass totals satisfy*

$$\Lambda_{\rightarrow} + \Lambda_{\leftarrow} = 1.$$

If the profile is moreover pass-symmetric, then these are also equivalent to

$$\Lambda_{\rightarrow} = \Lambda_{\leftarrow} = \frac{1}{2}.$$

Thus, under pass symmetry, OP 1 is equivalent to saying that the outward and return photon-side passes each carry normalized half-burden.

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Proof. By Proposition 33.138, $\Delta_{\mathrm{CL}} = 0$ is equivalent to

$$\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 1.$$

Using Definition 33.141, this is exactly

$$\Lambda_{\rightarrow} + \Lambda_{\leftarrow} = 1,$$

so statements (1)–(3) are equivalent. If, in addition, the profile is pass-symmetric, then $\Lambda_{\rightarrow} = \Lambda_{\leftarrow}$; together with their sum being one, this gives

$$\Lambda_{\rightarrow} = \Lambda_{\leftarrow} = \tfrac{1}{2}.$$

Conversely, if both pass totals equal $\frac{1}{2}$, then their sum is one, and OP 1 closes by the already established equivalences. □

Corollary 33.143 (What really reduces)**.** *Under a pass-symmetric two-pass reading, OP 1 no longer needs a direct identification of all four individual shell loads. It is enough to show that the forward pass contributes normalized burden $\frac{1}{2}$ and the backward pass contributes normalized burden $\frac{1}{2}$. In particular, any future lattice, first-zero, shell, or transport lemma that yields pass symmetry together with one half-burden identity closes OP 1 without resolving the internal split inside each pass.*

Remark 33.144 (Why this is the first honest origin final for the four λ_j)**.** *This is not yet a first-principles derivation, and it should not be advertised as one. The gain is narrower and more honest. The four shell loads now have a concrete structural reading as* two forward-side loads plus two backward-side loads*. That matches the standing bidirectional transport intuition already present elsewhere in the project and turns OP 1 from a raw four-term balance into a two-pass half-burden question. The next attack can therefore be sharper: not "guess the four λ_j independently," but derive why the post-frame photon transport should be pass-symmetric and why each pass should carry normalized load $\frac{1}{2}$.*

19AH$^{(16)}$. First reciprocal pass-symmetry certificate for the OP 1 two-pass balance

After the two-pass half-burden certificate of 19AH$^{(15)}$, the next narrow question is no longer how to guess the four shell loads independently, but why the forward and backward pass totals should agree in the first place. The weakest structurally natural answer is a reciprocity principle: each forward-side shell load has a backward-side partner carrying the same normalized attenuation burden. This does not yet derive the common value of the pairs, but it is enough to force pass symmetry and therefore collapse OP 1 to a one-pass half-balance statement.

Definition 33.145 (Reciprocal shell-pairing certificate for the OP 1 transport)**.** *Under the hypotheses of Definition 33.141, we say that the two-pass grouped four-shell profile satisfies the* reciprocal shell-pairing certificate *if the forward/backward shell labels are paired by*

$$\rho(1) = 3, \qquad \rho(2) = 4,$$

and the paired normalized loads agree,

$$\lambda_{\rho(j)} = \lambda_j \qquad (j = 1, 2).$$

Equivalently,

$$\lambda_1 = \lambda_3, \qquad \lambda_2 = \lambda_4.$$

We call this pairwise reciprocal *because each outward shell burden is mirrored by one return-shell burden of the same normalized weight.*

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Proposition 33.146 (Reciprocal pairing forces pass symmetry for OP 1)**.** *Suppose the OP 1 residual admits a two-pass grouped four-shell decomposition in the sense of Definition 33.141. If the reciprocal shell-pairing certificate of Definition 33.145 holds, then the profile is automatically pass-symmetric:*

$$\Lambda_{\rightarrow} = \Lambda_{\leftarrow}.$$

Consequently,

$$\Delta_{\mathrm{CL}} = 0 \iff \Psi_{\mathrm{CL}} = 1 \iff \Lambda_{\rightarrow} = \Lambda_{\leftarrow} = \frac{1}{2}.$$

Equivalently, under reciprocal shell pairing, OP 1 is reduced to the single one-pass half-burden identity

$$\lambda_1 + \lambda_2 = \frac{1}{2},$$

or, identically, to the return-pass statement

$$\lambda_3 + \lambda_4 = \frac{1}{2}.$$

Proof. If reciprocal shell pairing holds, then

$$\lambda_1 = \lambda_3, \qquad \lambda_2 = \lambda_4.$$

Therefore

$$\Lambda_{\rightarrow} = \lambda_1 + \lambda_2 = \lambda_3 + \lambda_4 = \Lambda_{\leftarrow},$$

so the profile is pass-symmetric. The remaining equivalences are then exactly the pass-symmetric conclusion of Proposition 33.142. In particular, once reciprocal pairing is known, OP 1 closes precisely when one of the two pass totals equals $\frac{1}{2}$, because the other is then forced to equal the same value. □

Corollary 33.147 (The live OP 1 burden after)**.** *If a future shell, lattice, counting, or first-zero transport lemma yields reciprocal shell pairing, then the separate proof obligation for pass symmetry disappears. The live OP 1 burden is then only the one-pass half-balance statement*

$$\lambda_1 + \lambda_2 = \frac{1}{2},$$

with the backward pass following automatically by reciprocity.

Remark 33.148 (Why this is a real narrowing without pretending OP 1 is closed)**.** *This is still not a first-principles derivation of the continuum-limit correction, and it should not be sold as such. The gain is more disciplined. After the local pass, OP 1 asked for two ingredients: pass symmetry and half-burden. After the local pass, any concrete reciprocity mechanism collapses the first ingredient into a taut consequence of pairwise equal shell loads. The next attack can therefore become even sharper: not "why should the two passes be equal in aggregate?" but "what concrete structural mechanism enforces reciprocal shell pairing, and why does one pass carry normalized load $\frac{1}{2}$?"*

19AH$^{(17)}$. First same-corridor reversal certificate for reciprocal shell pairing in OP 1

After the reciprocal pass-symmetry certificate of 19AH$^{(16)}$, the next honest question is what concrete structural mechanism could force the pairwise equalities

$$\lambda_1 = \lambda_3, \qquad \lambda_2 = \lambda_4.$$

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The narrowest bidirectional final is the standing two-pass reading itself: the post-frame photon transport uses the *same shell corridor* once in the forward direction and once in the reverse direction. If the normalized attenuation burden is attached to the corridor shells rather than to the direction label, then reciprocal shell pairing follows automatically. This still does not determine the common shell values, but it supplies the first concrete origin mechanism for the reciprocity pattern.

Definition 33.149 (Same-corridor reversal certificate for the OP 1 shell transport)**.** *Assume the grouped four-shell decomposition of Definition 33.141. We say that the profile satisfies the* same-corridor reversal certificate *if there exist two nonnegative corridor loads*

$$\mu_1, \mu_2 \geq 0$$

and a single two-shell corridor such that the forward pass reads the corridor in order

$$(\lambda_1, \lambda_2) = (\mu_1, \mu_2),$$

while the backward pass reads the same corridor in reverse order but with the same shell-attached loads,

$$(\lambda_3, \lambda_4) = (\mu_1, \mu_2).$$

Equivalently, the direction of traversal changes, but the shell-burden assignment is carried by the corridor itself and therefore survives reversal unchanged.

Proposition 33.150 (Same-corridor reversal implies reciprocal shell pairing)**.** *If the OP 1 grouped four-shell profile satisfies the same-corridor reversal certificate of Definition 33.149, then the reciprocal shell-pairing certificate of Definition 33.145 holds. Hence*

$$\lambda_1 = \lambda_3 = \mu_1, \qquad \lambda_2 = \lambda_4 = \mu_2,$$

and therefore

$$\Lambda_{\rightarrow} = \Lambda_{\leftarrow} = \mu_1 + \mu_2.$$

Consequently,

$$\Delta_{\mathrm{CL}} = 0 \iff \Psi_{\mathrm{CL}} = 1 \iff \mu_1 + \mu_2 = \frac{1}{2}.$$

In particular, under same-corridor reversal the full OP 1 burden is reduced to one two-shell half-balance on a single direction-free corridor.

Proof. By definition of same-corridor reversal we have

$$\lambda_1 = \mu_1 = \lambda_3, \qquad \lambda_2 = \mu_2 = \lambda_4.$$

Thus the reciprocal shell-pairing equalities of Definition 33.145 hold automatically. Proposition 33.146 then yields pass symmetry, and the pass totals are

$$\Lambda_{\rightarrow} = \lambda_1 + \lambda_2 = \mu_1 + \mu_2, \qquad \Lambda_{\leftarrow} = \lambda_3 + \lambda_4 = \mu_1 + \mu_2.$$

Since OP 1 is equivalent to $\Lambda_{\rightarrow} = \Lambda_{\leftarrow} = \frac{1}{2}$ under reciprocal pairing, the final equivalence follows. □

Corollary 33.151 (The live OP 1 burden after)**.** *If a future prime-lattice, shell, or first-zero transport lemma identifies the post-frame photon transport with one reversal-invariant two-shell corridor, then OP 1 no longer asks for a separate reciprocity proof and no longer asks for two independent pass totals. The live burden is only the scalar two-shell corridor identity*

$$\mu_1 + \mu_2 = \frac{1}{2}.$$

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Remark 33.152 (Why this is the first genuinely concrete reciprocity mechanism)**.** *This is still not the closure of OP 1. The unresolved content has merely been pushed to an even narrower place: the one-corridor half-balance $\mu_1 + \mu_2 = \frac{1}{2}$. But unlike the earlier certificates, now gives a concrete structural reason for the reciprocity equalities themselves. The forward and backward passes are no longer treated as two independently weighted routes; they are two traversals of the same post-frame corridor. That is the first point in the present line where the project's standing bidirectional transport picture becomes an explicit OP 1 origin mechanism rather than only a heuristic slogan.*

19AH$^{(18)}$. First unit-loop half-burden certificate for the one-corridor OP 1 transport

After the same-corridor reversal certificate of 19AH$^{(17)}$, the unresolved OP 1 content has been pushed onto one direction-free two-shell corridor with total pass load $\mu_1+\mu_2$. The next honest question is therefore no longer where reciprocity comes from, but why this one corridor should carry exactly the half-burden $\frac{1}{2}$. The narrowest bidirectional final is the standing two-pass transport picture itself: the post-frame photon corridor is traversed once forward and once backward, and the normalized burden is attached to the full closed two-pass loop rather than to one preferred orientation. If that closed loop is unit-normalized and the corridor loads are reversal-invariant, then each traversal must carry exactly half of the total loop burden.

Definition 33.153 (Unit-loop half-burden certificate for the OP 1 one-corridor transport)**.** *Assume the same-corridor reversal certificate of Definition 33.149, so that a single direction-free two-shell corridor with loads $\mu_1, \mu_2 \geq 0$ is traversed once forward and once backward. We say that the profile satisfies the* unit-loop half-burden certificate *if the full closed two-pass corridor loop is normalized to total burden one, i.e.*

$$(\mu_1 + \mu_2) + (\mu_1 + \mu_2) = 1.$$

Equivalently, the forward and backward traversals are assigned as 10.5281/zenodo.19803231 independent total weights; they are the two orientations of one unit-normalized closed transport loop.

Proposition 33.154 (Unit-loop normalization forces the OP 1 half-burden)**.** *If the OP 1 transport satisfies the same-corridor reversal certificate of Definition 33.149 and the unit-loop half-burden certificate of Definition 33.153, then*

$$\mu_1 + \mu_2 = \frac{1}{2}.$$

Consequently,

$$\Lambda_{\rightarrow} = \Lambda_{\leftarrow} = \frac{1}{2}, \qquad \Psi_{\mathrm{CL}} = 1, \qquad \Delta_{\mathrm{CL}} = 0.$$

In particular, under same-corridor reversal plus unit-loop normalization, OP 1 is closed.

Proof. By Definition 33.153,

$$2(\mu_1 + \mu_2) = 1.$$

Hence $\mu_1 + \mu_2 = \frac{1}{2}$. Proposition 33.150 then gives

$$\Lambda_{\rightarrow} = \Lambda_{\leftarrow} = \mu_1 + \mu_2 = \frac{1}{2}.$$

By Proposition 33.150, this is equivalent to $\Psi_{\mathrm{CL}} = 1$, and by Proposition 33.130 it is equivalent to $\Delta_{\mathrm{CL}} = 0$. Thus OP 1 is closed under the stated hypotheses. □

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Corollary 33.155 (The live OP 1 burden after)**.** *After the local pass, the remaining live content of OP 1 is no longer a free four-shell, two-pass, or reciprocity problem. It is the identification of the post-frame photon corridor with a* unit-normalized closed two-pass loop*. Once that normalization is obtained from a future prime-lattice, shell, or first-zero transport lemma, the one-corridor half-balance and therefore OP 1 follow immediately.*

Remark 33.156 (Why is a real advance without pretending the normalization is already proved)**.** *This does not yet prove that the post-frame photon corridor really is unit-normalized. That point remains the live burden. But the nature of the burden has changed sharply. After the local pass, OP 1 asked for one scalar half-balance on a direction-free corridor. After the local pass, even that half-balance is no longer independent: it is forced as soon as the corridor is shown to be the correct unit-normalized closed two-pass object. So the next attack does not need to juggle shell weights, pass symmetry, or reciprocity any more. It only needs to justify one normalization principle for one already isolated bidirectional corridor.*

19AH$^{(19)}$. First one-first-zero loop-budget certificate for OP 1 after the unit-loop reduction

After the unit-loop half-burden certificate of 19AH$^{(18)}$, the remaining burden is no longer to guess a half-load on the corridor. What remains is to explain why the isolated closed two-pass corridor should carry *one* normalized transport unit in the first place. The frame split from 19AH$^{(13)}$ already shows that the surviving OP 1 normalization is measured in units of the photon-normalized first zero $\gamma_1^{(\mathrm{photon})}$. The sharpest next compression is therefore to re-read the unit-loop hypothesis not as an external normalization convention, but as a multiplicity-one statement: the post-frame corridor is exactly one closed first-zero transport loop, not a fraction of one loop and not a multiple copy of several such loops.

Definition 33.157 (One-first-zero loop-budget certificate for the OP 1 corridor)**.** *Assume the same-corridor reversal certificate of Definition 33.149, so that the post-frame OP 1 transport is carried by one direction-free two-shell corridor with loads μ_1, μ_2. Define its* closed-loop budget index *by*

$$\Omega_{\mathrm{loop}} := 2(\mu_1 + \mu_2).$$

We say that the profile satisfies the one-first-zero loop-budget certificate *if the isolated closed two-pass corridor represents exactly one photon-first-zero transport unit, i.e.*

$$\Omega_{\mathrm{loop}} = 1.$$

Equivalently, the corridor is read as a multiplicity-one closed loop in the photon-first-zero normalization of 19AH$^{(13)}$.

Proposition 33.158 (One-first-zero loop budget identifies the unit-loop normalization)**.** *Assume the same-corridor reversal certificate of Definition 33.149. Then*

$$\Omega_{\mathrm{loop}} = \Psi_{\mathrm{CL}}.$$

Consequently, if the one-first-zero loop-budget certificate of Definition 33.157 holds, then the unit-loop half-burden certificate of Definition 33.153 holds as well, and therefore

$$\Delta_{\mathrm{CL}} = 0.$$

In particular, under same-corridor reversal, OP 1 is closed as soon as the isolated corridor is shown to be a multiplicity-one photon-first-zero loop.

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Proof. By Proposition 33.150,

$$\Lambda_{\rightarrow} = \Lambda_{\leftarrow} = \mu_1 + \mu_2.$$

Hence Proposition 33.142 gives

$$\Psi_{\mathrm{CL}} = \Lambda_{\rightarrow} + \Lambda_{\leftarrow} = (\mu_1 + \mu_2) + (\mu_1 + \mu_2) = 2(\mu_1 + \mu_2) = \Omega_{\mathrm{loop}}.$$

If now $\Omega_{\mathrm{loop}} = 1$, then by definition

$$(\mu_1 + \mu_2) + (\mu_1 + \mu_2) = 1,$$

which is exactly the unit-loop half-burden certificate of Definition 33.153. Proposition 33.154 then yields $\Delta_{\mathrm{CL}} = 0$. □

Corollary 33.159 (The live OP 1 burden after)**.** *After the local pass, the remaining live burden of OP 1 is no longer to supply an abstract unit-loop normalization from outside the frame split. It is to prove a* multiplicity-one statement*: the isolated post-frame corridor is exactly one photon-first-zero closed loop. Equivalently, one must rule out both undercounting and overcounting, i.e. show that the corridor does not represent a fractional share of the first-zero budget and does not split into several loop copies carrying total index* $\Omega_{\mathrm{loop}} \neq 1$*.*

Remark 33.160 (Why is a real tightening of the OP 1 queue)**.** *This is still not the final closure proof, because the multiplicity-one statement has not yet been derived from the prime-lattice/first-zero transport itself. But removes one remaining piece of arbitrariness. In the local pass, the decisive hypothesis was a unit-normalized closed loop. After the local pass, that hypothesis is no longer a free normalization slogan: it is identified with the claim that the already isolated corridor carries exactly one photon-first-zero transport unit. So the next attack on OP 1 is even narrower than before. It should no longer ask for "why does some loop have weight one?" in the abstract, but for a concrete first-zero transport reason why the admissible corridor is a single normalized loop rather than a fractional or multiple-count loop.*

19AH$^{(20)}$. First admissible-window uniqueness certificate for the multiplicity-one OP 1 corridor

After the one-first-zero loop-budget certificate of 19AH$^{(19)}$, the live OP 1 burden is no longer a free normalization question. It is a uniqueness question on the retained admissible corridor: why should the isolated post-frame same-corridor loop represent *exactly one* admissible photon-first-zero transport unit rather than several parallel admissible copies? The sharpest next compression is therefore to move from multiplicity language to admissible-window uniqueness language. On the present line, this is the first point at which the older no-second-carrier discipline and the newer total/imbalance burden split directly feed back into OP 1: extra loop multiplicity can only matter if it either changes the retained total-channel profile or introduces a genuinely new primitive carrier contribution on the same admissible corridor.

Definition 33.161 (Admissible-window uniqueness certificate for the OP 1 corridor)**.** *Assume the same-corridor reversal certificate of Definition 33.149, so that the post-frame OP 1 transport is carried by one direction-free two-shell corridor. We say that the profile satisfies the* admissible-window uniqueness certificate *if, on the intended admissible machine-bearing windows of the present line, that isolated corridor admits exactly one primitive photon-first-zero closed-loop realization compatible with the retained total-channel profile and with the no-second-carrier discipline. Equivalently, every admissible realization of the corridor that preserves the retained total-channel load is identified with the same single primitive first-zero loop.*

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Proposition 33.162 (Admissible-window uniqueness upgrades multiplicity-one to OP 1 closure)**.** *Assume the same-corridor reversal certificate of Definition 33.149. If the admissible-window uniqueness certificate of Definition 33.161 holds, then the one-first-zero loop-budget certificate of Definition 33.157 holds. Consequently,*

$$\Delta_{\mathrm{CL}} = 0.$$

In particular, after the surviving OP 1 burden may be read as a pure admissible-window uniqueness question for the retained post-frame corridor.

Proof. By Definition 33.161, the isolated post-frame same-corridor loop has exactly one admissible primitive photon-first-zero realization compatible with the retained total-channel profile and the no-second-carrier discipline. But Definition 33.157 identifies precisely this multiplicity-one situation with the statement that the corridor carries one photon-first-zero transport unit, i.e. $\Omega_{\mathrm{loop}} = 1$. Hence the one-first-zero loop-budget certificate holds. Proposition 33.158 then yields $\Delta_{\mathrm{CL}} = 0$. □

Corollary 33.163 (The live OP 1 burden after)**.** *After the local pass, the remaining live burden of OP 1 is no longer to choose a normalization, a pass split, or even an abstract multiplicity slogan. It is to prove a* uniqueness statement on admissible windows*: the retained post-frame corridor supports one and only one primitive photon-first-zero loop compatible with the preserved total-channel load. Any alternative admissible realization would have to appear either as a total-channel distortion or as a second primitive carrier contribution, and therefore lies outside the intended admissible OP 1 corridor class.*

Remark 33.164 (Why is a genuine narrowing and not just another renaming)**.** *This still does not by itself derive the uniqueness statement from first principles; so OP 1 is not yet honestly closed on theorem level. But the queue is now much sharper. Before one could still phrase the task as "explain multiplicity one" in a somewhat free way. After the task is pinned to the already existing admissible-window discipline of the present line: one must show that the retained corridor does not admit several distinct primitive first-zero loop realizations once the total-channel profile and no-second-carrier constraints are held fixed. So the next attack on OP 1 should not invent new continuum-limit parameters. It should directly use retained-window rigidity/uniqueness to rule out duplicate admissible loop occupancy on the same corridor.*

19AH$^{(21)}$. Retained-window-rigidity / no-second-carrier bridge for OP 1

After the admissible-window uniqueness certificate of 19AH$^{(20)}$, the remaining honest task is no longer to invent a new normalization slogan or a new loop-counting parameter. It is to derive uniqueness from structural disciplines that are already native to the present line. The sharpest triad is now visible. First, the retained admissible corridor should be rigid once the retained total-channel profile and optimum data are fixed. Second, the older no-second-carrier discipline should forbid an additional primitive occupancy of that same retained corridor. Third, the strict-upper-gap discipline should forbid a genuinely distinct admissible completion route above the same retained corridor that would duplicate the first-zero budget without altering the present-line status surface. Together these three pressures are exactly what the uniqueness question is asking for in compressed form.

Definition 33.165 (Retained-window rigidity certificate for the OP 1 corridor)**.** *Assume the same-corridor reversal certificate of Definition 33.149. We say that the profile satisfies*

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the retained-window rigidity certificate *on the intended admissible windows if every admissible primitive photon-first-zero closed-loop realization on the retained post-frame corridor that preserves the retained total-channel profile, optimum location, and standing admissible equivalence data occupies the same retained corridor slot up to the standing admissible corridor equivalence. Equivalently, there is no nontrivial admissible primitive displacement of the retained first-zero loop inside the same preserved total-channel window.*

Proposition 33.166 (Rigidity plus no-second-carrier plus strict upper gap imply OP 1 uniqueness)**.** *Assume the same-corridor reversal certificate of Definition 33.149. Assume further that the retained-window rigidity certificate of Definition 33.165 holds on the intended admissible windows, that the no-second-carrier discipline continues to hold at the retained total-channel level, and that the strict upper gap admits no second independent admissible completion route above the same retained corridor on those windows. Then the admissible-window uniqueness certificate of Definition 33.161 holds. Consequently,*

$$\Delta_{\mathrm{CL}} = 0.$$

In particular, after the live OP 1 burden may be read as the extraction of retained-window rigidity on the post-frame admissible corridor, with the no-second-carrier and strict-upper-gap disciplines supplying the exclusion of duplicate primitive occupancy and duplicate admissible completion.

Proof. Let C' be any admissible primitive photon-first-zero closed-loop realization on the retained post-frame corridor that preserves the retained total-channel profile. By the retained-window rigidity certificate, C' occupies the same retained corridor slot as the original loop up to the standing admissible corridor equivalence. If C' were still a genuinely different primitive realization, then either it would contribute an extra primitive occupancy on that same retained corridor, contradicting the no-second-carrier discipline at total-channel level, or it would represent a second independent admissible completion route above the same corridor while preserving the retained total-channel surface, contradicting the standing strict-upper-gap exclusion on the intended windows. Hence no genuinely distinct admissible primitive realization exists, and the admissible-window uniqueness certificate of Definition 33.161 follows. Proposition 33.162 then yields $\Delta_{\mathrm{CL}} = 0$. □

Corollary 33.167 (The live OP 1 burden after)**.** *After the local pass, OP 1 is no longer honestly read as a continuum-limit mystery in general, nor even as a free uniqueness slogan. The live burden is narrower: extract a retained-window rigidity theorem for the already isolated post-frame admissible corridor. Once that rigidity is in hand, the existing no-second-carrier discipline blocks duplicate primitive corridor occupancy and the strict-upper-gap discipline blocks duplicate admissible completion above the same retained corridor, so the uniqueness required in 19AH$^{(20)}$ follows automatically.*

Remark 33.168 (Why is a real narrowing)**.** *This is still not a final unconditional OP 1 theorem, because the retained-window rigidity certificate itself has not yet been discharged from the active comparison/transport line. But the queue is now much sharper than in the local pass. The remaining question is no longer "why uniqueness?" in the abstract. It is "why is the retained admissible corridor rigid once the present-line optimum and total-channel data are fixed?" That is exactly the kind of local structural statement the existing retained-window diagnostics, no-second-carrier discipline, and strict-upper-gap language are already built to support.*

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19AH$^{(22)}$. First optimum-pinning comparison-rigidity bridge for retained-window rigidity in OP 1

After the retained-window-rigidity / no-second-carrier / strict-upper-gap bridge of 19AH$^{(21)}$, the live OP 1 task is sharper still: why should the retained admissible corridor be rigid once the present-line comparison data are fixed? The most honest next compression is to stop treating rigidity as a free geometric slogan and to pin it directly to the already retained comparison surface. On the present line, the retained corridor is not specified by corridor language alone. It is specified together with the preserved total-channel profile, the retained optimum location, and the local comparison/defect data that survive the machine-shadow compression. If those comparison data already determine the admissible slot uniquely inside the retained corridor class, then retained-window rigidity is no longer an extra burden; it is simply the local comparison line read in the right coordinates.

Definition 33.169 (Optimum-pinning comparison-rigidity certificate for OP 1)**.** *Assume the same-corridor reversal certificate of Definition 33.149. We say that the profile satisfies the* optimum-pinning comparison-rigidity certificate *on the intended admissible windows if every admissible primitive photon-first-zero closed-loop realization on the retained post-frame corridor that preserves the retained total-channel profile, the retained optimum location, and the standing local comparison/defect data is forced to occupy the same retained corridor slot up to the standing admissible corridor equivalence. Equivalently, within the retained admissible corridor class there is no second primitive loop placement that keeps the same optimum pinning and the same local comparison readout while remaining genuinely distinct.*

Proposition 33.170 (Optimum pinning and comparison rigidity imply retained-window rigidity for OP 1)**.** *Assume the same-corridor reversal certificate of Definition 33.149. If the optimum-pinning comparison-rigidity certificate of Definition 33.169 holds on the intended admissible windows, then the retained-window rigidity certificate of Definition 33.165 holds there as well. Consequently, together with the no-second-carrier and strict-upper-gap disciplines from Proposition 33.166, one obtains the admissible-window uniqueness certificate of Definition 33.161 and therefore*

$$\Delta_{\mathrm{CL}} = 0.$$

In particular, after the live OP 1 burden may be read as a local optimum-pinning comparison-rigidity statement on the retained post-frame corridor.

Proof. Take any admissible primitive photon-first-zero closed-loop realization on the retained post-frame corridor that preserves the retained total-channel profile. On the intended admissible windows of the present line, the retained OP 1 corridor is not read independently of the standing comparison surface: the intended loop class is already tracked together with the retained optimum location and the surviving local comparison/defect data of the compressed machine-shadow line. If the optimum-pinning comparison-rigidity certificate holds, then any admissible primitive realization preserving those retained data must occupy the same retained corridor slot up to the standing admissible corridor equivalence. But this is exactly the content of the retained-window rigidity certificate. The rest follows from Proposition 33.166, which upgrades retained-window rigidity to admissible-window uniqueness under the standing no-second-carrier and strict-upper-gap exclusions, hence yields $\Delta_{\mathrm{CL}} = 0$. □

Corollary 33.171 (The live OP 1 burden after)**.** *After the local pass, OP 1 is no longer honestly carried as a general rigidity problem. The live burden is narrower: show that the retained optimum and local comparison data already pin the admissible post-frame first-zero*

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corridor slot uniquely. Once that local comparison rigidity is extracted, retained-window rigidity follows, and with the standing no-second-carrier and strict-upper-gap disciplines the OP 1 closure chain completes automatically.

Remark 33.172 (Why is the first direct attack from the comparison line)**.** *This is still not a full unconditional theorem extracted entirely from the earlier comparison blocks; the optimum-pinning comparison-rigidity certificate itself is now the remaining live statement. But it is a real narrowing. said, in effect, "prove retained-window rigidity." says something more local and more testable: "prove that no genuinely distinct primitive loop placement preserves the retained optimum and the surviving comparison/defect readout on the same admissible corridor." That is the first point at which the residual OP 1 burden is read directly as a comparison-line pinning problem rather than as a free rigidity slogan.*

19AI. Bundled overlay-completion capstone for the machine-theoretic line

After the machine-theoretic line has reached full machine-theoretic closure, present-line status readout, and window-equivalence, the remaining overlay refinements should no longer be read as an invitation to keep appending parallel adjectives indefinitely. Their real purpose is to certify, in a bundled way, that the machine layer is the right kind of organization of the already certified machine-bearing windows: it adds no foreign admissible content, forgets none of the operative distinctions relevant for the machine hierarchy, tracks the intended statuses exactly, and returns the same machine-bearing layer up to the standing admissible equivalences. The correct freeze-level conclusion is therefore not a new ladder of further micro-properties, but a single capstone statement that the machine overlay is a conservative, faithful, exact, relatively complete, adequate, reconstructive, closed, idempotent, minimal, canonical, invariant, natural, universally characterized, rigid, terminal, and equivalence-level organization of the intended certified machine-bearing windows.

Definition 33.173 (Bundled machine-overlay completion certificate on the compressed computational line)**.** *Fix a chosen sequential reference core $\mathcal{R}$ and an intended admissible machine-bearing window class $\mathcal{W}^{\mathrm{mach}}$. We say that the compressed computational line satisfies the* bundled machine-overlay completion certificate *on $\mathcal{W}^{\mathrm{mach}}$ if, once the full machine-theoretic closure surface $\mathfrak{M}^{\mathrm{full}}(\mathcal{R}) = 1$, the present-line status surface $\mathfrak{P}^{\mathrm{mach}}(\mathcal{R})$, and the window-equivalence/equivalence-surface layer hold on $\mathcal{W}^{\mathrm{mach}}$, the resulting machine-theoretic overlay is conservative, faithful, exact, relatively complete, adequate, reconstructive, closed, idempotent, minimal, canonical, invariant, natural, universally characterized, rigid, terminal, and equivalent to the certified machine-bearing structural content, all up to the standing admissible window-equivalence and overlay-equivalence. Equivalently, the intended admissible machine-bearing windows and the canonical machine-overlay objects form two fully compatible organizations of the same certified machine-bearing layer.*

Proposition 33.174 (Bundled overlay-completion theorem for the machine-theoretic line)**.** *On the intended admissible machine-bearing windows of the present compressed computational line, the above bundled machine-overlay completion certificate holds. More precisely, the machine-theoretic line of Sections 19J–19AH together with the diagnostic continuation block 19AH′–19AH$^{(22)}$ yields the indispensable ladder from local operational semantics to the present-line conditional/unconditional machine status plus the first disciplined re-reading of the retained machine shadow, while the subsequent overlay-completion work can be read in bundled form as showing that this machine layer is conservative, faithful, exact, relatively complete, adequate, reconstructive, closed, idempotent, minimal, canonical, invariant, natural,*

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universally characterized, rigid, terminal, and equivalent to the intended certified machine-bearing windows. Consequently, for the present Companion purpose the machine-theoretic overlay may be cited as a closed reversible machine-level organization of the certified machine-bearing structural content, without needing to unfold each of these overlay attributes separately unless a later revision introduces a genuinely new machine-theoretic distinction.

Corollary 33.175 (What the bundled overlay-completion capstone already buys)**.** *The machine line now reaches a true freeze-level stopping point. For subsequent work it is enough to cite the indispensable machine ladder together with the bundled overlay-completion capstone, rather than to continue lengthening the line by parallel closure adjectives. In particular, the present line can now be read through the three decisive bundled surfaces — the full machine-theoretic closure surface, the present-line status surface, and the bundled overlay-completion capstone — while the inserted channel-split diagnostics remain the preferred local reading tools whenever carrier and observer-side burdens need to be separated.*

19AZ. Freeze-audit verdict on the machine-theoretic line

In this inherited release-baseline block, the machine-theoretic line reaches the release-level freeze threshold used for the then-present public version. A local audit of Sections 19J–19AY shows that the computational route itself is already complete enough for the present Companion purpose, while the last overlay block from 19AI onward increasingly repackages nearby closure properties rather than opening substantially new machine content. For this reason that inherited line should now be read with a strict two-tier discipline. Theorems 19J–19AH remain the indispensable machine ladder: local operational semantics, conditional Zuse-style primitive core, reference-core closure, relative tape-family upgrade, the strict upper gap, the full machine-theoretic closure surface, the status decoder, and the present-line conditional/unconditional status surface. The local continuation block 19AH′–19AH$^{(22)}$ should be read only as a diagnostic refinement layer: it separates total and imbalance channels, turns that split into a control dictionary, sorts inherited continuation knots by burden type, and compresses the surviving OP 1 queue to the present optimum-pinning comparison-rigidity kernel. Section 19AI now functions as the bundled overlay-completion capstone: it certifies in compressed form that the machine line is conservative, faithful, exact, relatively complete, adequate, reconstructive, closed, idempotent, minimal, canonical, invariant, natural, universally characterized, rigid, terminal, and equivalent to the certified machine-bearing windows on the intended admissible domain. The audit verdict in that inherited release block is therefore to stop expansion there: future revisions should preferentially cite the bundled surfaces already obtained — in particular 19AF, 19AH, and 19AI — rather than append further parallel overlay adjectives unless a genuinely new machine-theoretic distinction is introduced.

19AZ′. Post-release status note after tool/QAM completion and local OP-kernel closure

The public archival stage release completed the formerly deferred tool/QAM/decoder layer and fixed the route-control infrastructure of the compressed computational line. The present post-release working line then used that infrastructure exactly as intended: it reopened the two parked residual burdens through the smallest licensed routes and followed them to local kernel closure. OP 1 was normalized first as a retained pinning-signature injectivity problem, then split into occupancy-side and completion-side duplicate-signature witnesses, and finally discharged under the standing retained no-second-carrier and strict-upper-gap exclusions. OP 5 Step (B) was normalized as a carrier-upgraded routed-balance witness problem,

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split into carrier-stable versus obstructed routed returns, then into upstream-screened versus downstream carrier-grade obstruction, and finally discharged on the retained branch-shell corridor by the inherited shell-closure package.

The honest status after these steps is therefore no longer "two parked residual burdens." It is: the completed tool/QAM/decoder infrastructure now stands as fixed route-control surface, and the live OP 1 and OP 5 Step (B) kernels have both been discharged locally under the standing retained disciplines. What remains conditional is not the local kernel closure but the broader machine-theoretic reading beyond the certified retained corridors and routes. The governing public DOI in the chronology is 10.5281/zenodo.19699827 for the public archival stage release.

19AZ″. Immediate normalized channel-sorted tool ledger

The first concrete task of the post-freeze continuation line is not to invent a fourth proof language, but to normalize the already extracted tools into a channel-sorted ledger. On the present line this means three reusable packets: one for statements that are already total-channel stable, one for statements that compress the surviving imbalance-side burden, and one for statements that translate between both surfaces or between their QAM/matrix realizations. The point is discipline: later arguments should be able to cite the smallest packet that already matches their burden class.

Definition 33.176 (Normalized channel-sorted tool packets)**.** *The immediate continuation line organizes the current extracted tools into three provisional packets.*

1. ***Total-channel packet*** $\mathfrak{T}_{\mathrm{tot}}$*. This packet contains the tools whose role is to keep the retained total-channel architecture fixed once admissible transport has already selected the active branch. Its current generators are the transport-first tool, the readout-after-transport tool, the admissible-readout uniqueness tool, and the global pass/transfer/diagnostic-subordination interface from Propositions 32.78–32.81.*

2. ***Imbalance packet*** $\mathfrak{T}_{\mathrm{imb}}$*. This packet contains the tools that compress the surviving OP 1 burden after the total-channel surface has been fixed. Its current generators are the frame-transport split certificate (Proposition 33.134), the finite four-shell certificate (Proposition 33.138), the two-pass half-burden certificate (Proposition 33.142), the reciprocal pass-symmetry certificate (Proposition 33.146), the same-corridor reversal implication (Proposition 33.150), the unit-loop half-burden certificate (Proposition 33.154), the one-first-zero loop-budget bridge (Proposition 33.158), the admissible-window uniqueness bridge (Proposition 33.162), the retained-window-rigidity / no-second-carrier / strict-upper-gap bridge (Proposition 33.166), and the optimum-pinning comparison-rigidity bridge (Proposition 33.170).*

3. ***Bridge packet*** $\mathfrak{T}_{\mathrm{br}}$*. This packet contains the tools whose job is to move between total and imbalance language, or between channel language and its shell/menu/matrix realizations. Its current generators are the shared matrix-leakage tool, the cryptanalytic translation tool, the shared phase-coupling tool, and on the QAM side the shell-stable readout tool, the commuting-square tool, the obstruction-stratification tool, and the carrier-compression tool.*

Proposition 33.177 (Channel-sorted citation rule for the immediate continuation line)**.** *Let a later argument remain inside the currently admitted OP1→branch→OP3/QAM→OP5 architecture. Then the default citation discipline is as follows.*

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1. *If the argument only needs preservation of the retained total-channel profile or the subordination of visible diagnostics to one already fixed carrier package, it should cite the total-channel packet* $\mathfrak{T}_{\mathrm{tot}}$.

2. *If the argument accepts the retained total-channel surface as fixed and only compresses the surviving OP 1 burden toward the optimum-pinning kernel, it should cite the imbalance packet* $\mathfrak{T}_{\mathrm{imb}}$.

3. *If the argument must translate between total and imbalance language, or between channel language and a shell/menu/matrix realization, it should cite the bridge packet* $\mathfrak{T}_{\mathrm{br}}$.

In particular, the default rule of the next revision is: reopen only the smallest packet that still sees the live burden.

Proof. This is exactly the burden split already extracted in the frozen line. The present proposition merely normalizes that split into a reusable citation discipline: total-channel stability is carried by the total packet, OP 1 compression by the imbalance packet, and all language-changing or layer-changing moves by the bridge packet. □

Remark 33.178 (What the normalized ledger already gains)**.** *The gain is immediate even before any new theorem is proved. First, repeated later arguments no longer need to cite a mixed list of total, imbalance, and translation tools in arbitrary order. Second, the surviving live burden is seen more honestly: OP 1 is not "everything left," but specifically the active content of* $\mathfrak{T}_{\mathrm{imb}}$ *once* $\mathfrak{T}_{\mathrm{tot}}$ *has already been frozen. Third, the bridge packet isolates exactly where QAM, matrix, and cryptanalytic language should re-enter: only when one genuinely needs a change of surface.*

19AZ‴. QAM recoding of the normalized tool ledger

Once the channel-sorted ledger has been fixed, the right immediate QAM task is no longer to widen the public syntax blindly. It is to recode the already stabilized tool packets as a finite theorem-store/menu layer so that later proofs can query, combine, and route them without reopening the full historical derivation. This keeps QAM in its intended role: a definitional overlay and machine-readable proof interface, not a second independent mathematics.

Definition 33.179 (QAM-coded tool menu)**.** *Let* $\mathfrak{T}_{\mathrm{tot}}$*,* $\mathfrak{T}_{\mathrm{imb}}$*, and* $\mathfrak{T}_{\mathrm{br}}$ *be the normalized packets from Definition 33.176. Their immediate QAM recoding is the finite coded menu*

$$\mathcal{Q}_{\mathrm{tool}} := \big\{\langle \mathrm{tot}, \tau\rangle : \tau \in \mathfrak{T}_{\mathrm{tot}}\big\} \cup \big\{\langle \mathrm{imb}, \tau\rangle : \tau \in \mathfrak{T}_{\mathrm{imb}}\big\} \cup \big\{\langle \mathrm{br}, \tau\rangle : \tau \in \mathfrak{T}_{\mathrm{br}}\big\}.$$

Here the first coordinate records the burden class and the second coordinate records the already extracted uncoded tool. The purpose of $\mathcal{Q}_{\mathrm{tool}}$ *is indexing, routing, and proof compression; it does not create a new semantic layer beyond the underlying tool statements.*

Definition 33.180 (Menu-to-carrier reduction)**.** *A current local proof obligation is said to admit a* menu-to-carrier reduction *if, after coding the obligation into the finite menu* $\mathcal{Q}_{\mathrm{tool}}$*, there exists a finite submenu* $M \subseteq \mathcal{Q}_{\mathrm{tool}}$ *such that the underlying uncoded tools already imply either*

1. *one bundled carrier-stable pass statement, or*

2. *one strictly smaller localized obstruction class than the original uncoded presentation displayed.*

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Thus the menu does not replace proof content; it compresses the route to the smallest carrier-bearing or obstruction-bearing interface.

Proposition 33.181 (QAM reuse principle for the immediate continuation line)**.** *If a current proof obligation admits menu-to-carrier reduction in the sense of Definition 33.180, then the next continuation line may process that obligation through the finite coded menu $\mathcal{Q}_{\mathrm{tool}}$ instead of reopening the full local historical derivation. In particular, QAM now functions as the normalized theorem-store and routing overlay for the post-freeze tool layer.*

Proof. By construction, every element of $\mathcal{Q}_{\mathrm{tool}}$ is only a coded wrapper around an already extracted uncoded tool. If a finite submenu already implies the same bundled carrier statement or a strictly smaller obstruction class, then reopening the full derivation adds no new structural information. The coded menu therefore acts as a conservative proof-compression interface and not as a source of new independent mathematical content. □

Remark 33.182 (Why this QAM step is mathematically honest)**.** *This is the right scale for the immediate QAM revision. The present line does* not *claim that the finite coded menu solves OP 1 or OP 5 automatically. It claims something narrower and more useful: once the main burden has already been split into total, imbalance, and bridge packets, QAM can store that split explicitly, route later obligations to the smallest relevant packet, and expose where proof compression is real rather than cosmetic.*

19AZ‴. First finite submenus and certified reuse routes inside the QAM tool menu

The next optimization step is therefore to stop treating $\mathcal{Q}_{\mathrm{tool}}$ as one undifferentiated finite store. For later proof work one wants a small number of *canonical finite submenus* that already match the most common post-freeze obligations. These submenus do not add new mathematics; they only isolate the smallest coded interfaces that repeatedly suffice.

Definition 33.183 (First canonical finite submenus of $\mathcal{Q}_{\mathrm{tool}}$)**.** *Inside the coded menu $\mathcal{Q}_{\mathrm{tool}}$, define the following four canonical finite submenus.*

1. ***Total-rigidity submenu*** $M_{\mathrm{tot}}^{\mathrm{rig}}$*. This is the coded submenu generated by the transport-first tool, the readout-after-transport tool, the admissible-readout uniqueness tool, and the global pass/transfer/diagnostic-subordination interface already collected in the total-channel packet $\mathfrak{T}_{\mathrm{tot}}$.*

2. ***Optimum-pinning submenu*** $M_{\mathrm{imb}}^{\mathrm{pin}}$*. This is the coded submenu generated by the frame-transport split certificate, the finite four-shell certificate, the two-pass half-burden certificate, the reciprocal pass-symmetry certificate, the same-corridor reversal implication, the unit-loop half-burden certificate, the one-first-zero loop-budget bridge, the admissible-window uniqueness bridge, the retained-window-rigidity / no-second-carrier / strict-upper-gap bridge, and the optimum-pinning comparison-rigidity bridge already collected in the imbalance packet $\mathfrak{T}_{\mathrm{imb}}$.*

3. ***Surface-change submenu*** $M_{\mathrm{br}}^{\mathrm{surf}}$*. This is the coded submenu generated by the shared matrix-leakage tool, the cryptanalytic translation tool, the shared phase-coupling tool, the shell-stable readout tool, and the commuting-square tool from the bridge packet $\mathfrak{T}_{\mathrm{br}}$. Its purpose is to move between channel language and shell/menu/matrix language without reopening the whole local history.*

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4. ***Obstruction-localization submenu*** $M_{\mathrm{br}}^{\mathrm{loc}}$. *This is the coded submenu generated by the shared matrix-leakage tool, the shared phase-coupling tool, the shell-stable readout tool, the obstruction-stratification tool, and the carrier-compression tool from the same bridge packet. Its purpose is to decide where a still-live residual should be reopened first once the total-channel baseline is already frozen.*

Proposition 33.184 (First certified submenu reduction routes)**.** *Each of the following recurring post-freeze proof obligations admits menu-to-carrier reduction through one of the canonical finite submenus from Definition 33.183.*

1. *Any obligation whose only job is to preserve the retained total-channel profile, carrier-stable pass status, or diagnostic subordination may be routed through* $M_{\mathrm{tot}}^{\mathrm{rig}}$.
2. *Any obligation that accepts the retained total-channel profile as fixed and only seeks to compress the surviving OP 1 burden toward retained-window rigidity or optimum pinning may be routed through* $M_{\mathrm{imb}}^{\mathrm{pin}}$.
3. *Any obligation whose main difficulty is a controlled change of surface — from channel language to shell, menu, or matrix language, or back again — may be routed through* $M_{\mathrm{br}}^{\mathrm{surf}}$.
4. *Any obligation whose main difficulty is not proof of a new carrier statement but localization of the first still-live residual interface may be routed through* $M_{\mathrm{br}}^{\mathrm{loc}}$.

In particular, the immediate continuation line now has four explicit reuse routes rather than only one undifferentiated theorem store.

Proof. The total-rigidity submenu is by definition exactly the finite coded portion of $\mathfrak{T}_{\mathrm{tot}}$ that already controls transport-first stability, readout-after-transport stability, admissible readout uniqueness, and global diagnostic subordination. Therefore every obligation of type *(1)* is already reduced to the smallest currently relevant carrier-stable total-channel interface.

The optimum-pinning submenu is by definition the finite coded chain inside $\mathfrak{T}_{\mathrm{imb}}$ that successively compresses the surviving OP 1 burden from frame split to shell balance, two-pass symmetry, loop budget, admissible-window uniqueness, retained-window rigidity, and finally optimum-pinning comparison-rigidity. Therefore every obligation of type *(2)* is already reduced to the smallest currently relevant imbalance-side compression route.

The surface-change submenu isolates exactly those bridge tools whose role is transport across surfaces rather than new burden creation: matrix leakage, cryptanalytic translation, shared phase coupling, shell-stable readout, and commuting-square compatibility. Hence every obligation of type *(3)* is reduced to the smallest currently relevant change-of-surface route.

Finally, the obstruction-localization submenu isolates the bridge tools whose role is not to certify a new global pass statement but to decide where the first still-live residual now sits: shared leakage/phase diagnostics identify the final bridge side, shell-stable readout preserves the common carrier-bearing content where available, obstruction stratification localizes the first failure stratum, and carrier compression collapses the readout/kernel side to one smallest missing carrier package. Hence every obligation of type *(4)* admits menu-to-carrier reduction through $M_{\mathrm{br}}^{\mathrm{loc}}$. This is exactly the claimed four-route reuse structure. □

Corollary 33.185 (Working citation rule for the next immediate revision)**.** *In the next immediate revision, a later proof should normally cite one of the four canonical finite submenus first and only reopen the larger packet description if that submenu fails to carry the obligation. Concretely:*

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1. *cite* $M_{\mathrm{tot}}^{\mathrm{rig}}$ *first for total-channel rigidity or status-preservation steps;*
2. *cite* $M_{\mathrm{imb}}^{\mathrm{pin}}$ *first for OP 1 compression toward the optimum-pinning kernel;*
3. *cite* $M_{\mathrm{br}}^{\mathrm{surf}}$ *first for channel/shell/menu/matrix translation steps;*
4. *cite* $M_{\mathrm{br}}^{\mathrm{loc}}$ *first for localization of the first still-live bridge or readout-side obstruction.*

Thus the QAM layer has now moved from abstract finite storage to a concrete four-route routing discipline.

Remark 33.186 (What has been gained before proving anything new)**.** *The gain is practical and immediate. A later argument no longer has to say only "use QAM" or "reopen the bridge packet." It can now say much more sharply: preserve the frozen total profile via* $M_{\mathrm{tot}}^{\mathrm{rig}}$*, compress the surviving OP 1 route via* $M_{\mathrm{imb}}^{\mathrm{pin}}$*, switch surfaces via* $M_{\mathrm{br}}^{\mathrm{surf}}$*, or localize the first still-live residual via* $M_{\mathrm{br}}^{\mathrm{loc}}$*. That is exactly the kind of proof-shortening interface the immediate tool/QAM revision was supposed to extract.*

19AZ$^{(3a)}$. Normalized QAM nomenclature and finite tag layer for the four canonical submenus

The four canonical submenus should now receive a short address language. Without that extra layer, the current QAM revision would still have finite storage and routing, but not yet a stable nomenclature by which later proofs or diagnostics can point to one smallest already-normalized carrier without reopening the full sentence around it.

Definition 33.187 (Primary tool-tag families)**.** *Read the four canonical finite submenus of Definition 33.179 in the order induced by their listed entries. The* primary tool-tag families *are the finite code families*

$$\mathrm{TR}_1, \mathrm{TR}_2, \ldots; \qquad \mathrm{IP}_1, \mathrm{IP}_2, \ldots; \qquad \mathrm{BS}_1, \mathrm{BS}_2, \ldots; \qquad \mathrm{BL}_1, \mathrm{BL}_2, \ldots$$

indexed respectively by the ordered entries of

$$M_{\mathrm{tot}}^{\mathrm{rig}}, \qquad M_{\mathrm{imb}}^{\mathrm{pin}}, \qquad M_{\mathrm{br}}^{\mathrm{surf}}, \qquad M_{\mathrm{br}}^{\mathrm{loc}}.$$

Here TR *stands for* total rigidity*,* IP *for* imbalance pinning*,* BS *for* bridge/surface change*, and* BL *for* bridge/localization*. The ordered disjoint union of these four families is the* primary tool-tag alphabet *of the immediate continuation line.*

Definition 33.188 (Primary tool-tag registry and dereferencing map)**.** *The* primary tool-tag registry

$$\mathcal{R}_{\mathrm{tool}}^{\mathrm{tag}}$$

is the finite family of quadruples

$$(\tau, M, \kappa, x)$$

with the following meaning:

1. τ *is one primary tag from Definition 33.187;*
2. M *is the unique canonical submenu among* $M_{\mathrm{tot}}^{\mathrm{rig}}, M_{\mathrm{imb}}^{\mathrm{pin}}, M_{\mathrm{br}}^{\mathrm{surf}}, M_{\mathrm{br}}^{\mathrm{loc}}$ *in which the tagged item has its primary home;*
3. $\kappa \in \{\mathrm{status}, \mathrm{pinning}, \mathrm{surface}, \mathrm{localization}\}$ *is its normalized burden type;*

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4. *$x \in M$ is the unique carrier statement, submenu entry, or smallest normalized reusable route addressed by τ.*

The induced map

$$\mathsf{Ref} : \tau \longmapsto x$$

is called the dereferencing map*. If one and the same carrier statement is reusable from more than one submenu, then exactly one of those occurrences is designated as its* primary home*; every other appearance is recorded only as a secondary pointer and receives no second primary tag. The registry order defines a precedence rank*

$$\pi(\tau) \in \mathbb{N}$$

inside the chosen submenu, so that smaller rank means earlier reopening inside the already chosen route.

Proposition 33.189 (Tag soundness, uniqueness, and minimal reopening discipline)**.** *For every entry x of the four canonical finite submenus there exists exactly one primary tag τ such that*

$$(\tau, M, \kappa, x) \in \mathcal{R}^{\mathrm{tag}}_{\mathrm{tool}}.$$

Consequently:

1. *the primary tag τ determines the canonical submenu M, the normalized burden type κ, and the dereferenced carrier x;*
2. *no collision between the four nomenclature families can occur at the primary level;*
3. *once a later argument has chosen its submenu, reopening should proceed in increasing precedence rank $\pi(\tau)$ until the first sufficient carrier is reached.*

Hence the immediate QAM layer now carries a finite nomenclature with exact dereferencing and a smallest-first reopening discipline.

Proof. The four canonical submenus are finite and ordered by construction. Assigning one tag from the corresponding family to each ordered entry therefore produces exactly one registry quadruple (τ, M, κ, x) for each such entry. Because the tag families $\{\mathrm{TR}_i\}, \{\mathrm{IP}_j\}, \{\mathrm{BS}_k\}, \{\mathrm{BL}_\ell\}$ are disjoint by notation, primary collisions cannot occur across different submenu types. If one carrier statement appears in more than one submenu role, the definition of primary home forces uniqueness at the registry level by declaring every other occurrence secondary only. Finally, the submenu order itself supplies the precedence rank $\pi(\tau)$, so once a route has been chosen the smallest-first reopening order is already fixed. This is exactly the claimed tag soundness, uniqueness, and reopening discipline. □

Corollary 33.190 (Tag-first citation syntax for the immediate continuation line)**.** *The next immediate revision may cite the normalized tool layer in either of the following two equivalent ways:*

1. *by naming the submenu and then the primary tag, for example "invoke $M^{\mathrm{rig}}_{\mathrm{tot}}$ at TR_i" or "reopen $M^{\mathrm{loc}}_{\mathrm{br}}$ at BL_ℓ";*
2. *once the submenu is fixed by context, by citing the primary tag alone and dereferencing it through* Ref*.*

Accordingly, the four primary tag families should be read as the first normalized QAM nomenclature layer for the immediate continuation line:

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Tag family	*Intended immediate use*
TR_i	*total-channel rigidity and status preservation*
IP_j	*OP 1 compression toward the optimum-pinning kernel*
BS_k	*channel/shell/menu/matrix surface change*
BL_ℓ	*localization of the first still-live obstruction*

Thus the QAM layer now has not only finite menus but also a short stable address language for them.

Remark 33.191 (What the tag layer buys before any new theorem is added)**.** *This is the first genuinely addressable QAM tool surface on the present line. A later proof can now point to a smallest normalized carrier by a short typed tag instead of reopening a long theorem sentence; a later diagnostic routine can record that a residual cluster repeatedly points toward* IP_7 *or* BL_2 *rather than toward an entire section; and a later public extraction can export the tags as the front-face nomenclature without exporting the whole internal packet architecture. In that precise sense, the tag layer is the missing bridge from finite storage to reusable proof vocabulary.*

19AZ$^{(3b)}$. First explicit primary registry assignments for the normalized tool tags

The normalized tag alphabet should now be instantiated once, so that the immediate continuation line can cite concrete dereferenced carriers instead of only a generic family pattern. At the present stage the assignments are intentionally conservative: each tag points only to a carrier statement or smallest reusable route that already exists in the frozen working text.

First concrete primary registry. Write the finite registry in four family blocks. For compactness, the "carrier" column records the dereferenced item together with its smallest currently stable reference surface.

> **Reader cue.** Read the following registry tables as lookup surfaces for the normalized QAM tags. Their purpose is to pin short addresses to already stabilized statements or reusable micro-routes; they do not introduce new carriers.

Tag	κ	Dereferenced carrier
Total-rigidity submenu $M_{\mathrm{tot}}^{\mathrm{rig}}$		
TR_1	status	the transport-first route read from the opening clause of the proof of Theorem 1.320, namely: branch selection is fixed before visible readout is interpreted;
TR_2	status	the readout-after-transport route read from the next clause of the same proof of Theorem 1.320, namely: every canonical readout remains downstream from the already fixed transport branch;
TR_3	status	the admissible-readout uniqueness carrier of Theorem 1.320;
TR_4	status	the bundled global pass/transfer/diagnostic-subordination interface from Propositions 32.78, 32.80, and 32.81.

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Tag	κ	Dereferenced carrier
Optimum-pinning submenu $M_{\mathrm{imb}}^{\mathrm{pin}}$		
IP_1	pinning	frame-transport split certificate, Proposition 33.134;
IP_2	pinning	finite four-shell certificate, Proposition 33.138;
IP_3	pinning	two-pass half-burden certificate, Proposition 33.142;
IP_4	pinning	reciprocal pass-symmetry certificate, Proposition 33.146;
IP_5	pinning	same-corridor reversal implication, Proposition 33.150;
IP_6	pinning	unit-loop half-burden certificate, Proposition 33.154;
IP_7	pinning	one-first-zero loop-budget bridge, Proposition 33.158;
IP_8	pinning	admissible-window uniqueness bridge, Proposition 33.162;
IP_9	pinning	retained-window-rigidity / no-second-carrier / strict-upper-gap bridge, Proposition 33.166;
IP_{10}	pinning	optimum-pinning comparison-rigidity bridge, Proposition 33.170.

Tag	κ	Dereferenced carrier
Surface-change submenu $M_{\mathrm{br}}^{\mathrm{surf}}$		
BS_1	surface	shared matrix-leakage surface gate, read through Definition 1.287 together with Proposition 1.288;
BS_2	surface	the cryptanalytic translation route from the shared matrix-leakage / cryptanalytic block;
BS_3	surface	shared phase-coupling surface diagnostic from Definitions 1.290–1.291 together with Propositions 1.292 and 1.293;
BS_4	surface	the shell-stable readout tool from the extracted OP 3/QAM tool layer;
BS_5	surface	the commuting-square tool from the extracted OP 3/QAM tool layer.

Tag	κ	Dereferenced carrier
Obstruction-localization submenu $M_{\mathrm{br}}^{\mathrm{loc}}$		
BL_1	localization	matrix-leakage-first localization gate toward the underfixed HC/MM/J coupling, read again through Definition 1.287 and Proposition 1.288;
BL_2	localization	phase-coupling-first localization gate asking whether the residual pair is still one masked carrier, read through Propositions 1.292 and 1.293;
BL_3	localization	the shell-stable readout localization gate from the extracted OP 3/QAM tool layer;
BL_4	localization	the obstruction-stratification tool from the extracted OP 3/QAM tool layer;
BL_5	localization	the carrier-compression tool from the extracted OP 3/QAM tool layer.

Primary-home convention at the concrete level. The bridge families contain repeated antecedent references on purpose. This does not violate the primary-home rule from Definition 33.188, because the registry items above are not identified merely by their antecedent theorem labels, but by their *normalized route role*. Thus the same matrix-leakage or shell-

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stable-readout material may appear once as a *surface-change gate* and once as a *localization gate* without creating a primary collision.

Immediate working use. From this point onward the continuation line may cite concrete tags without any placeholder notation. For example, "freeze the total channel at TR_4", "compress the surviving OP 1 kernel through $\mathrm{IP}_8 \to \mathrm{IP}_{10}$", "change surface via BS_2", or "localize the first still-live residual by testing BL_1, then BL_2, then BL_4" are now literal dereferenceable instructions on the QAM side.

19AZ$^{(3c)}$. First tag-first rewrite of the surviving OP 1 route

The normalized registry should now be used once on a real proof route rather than remaining only a citation promise. The cleanest first test is the currently surviving OP 1 compression chain, because it already has a sharply localized terminal burden and because its hypotheses were precisely the ones that motivated the split into total, pinning, and bridge roles.

Proposition 33.192 (First tag-first reformulation of the surviving OP 1 closure route)**.** *Assume the same-corridor reversal certificate of Definition 33.149, and assume moreover that the optimum-pinning comparison-rigidity certificate of Definition 33.169 holds on the intended admissible windows. Then the surviving OP 1 closure route may be cited on the normalized QAM side in the compact form*

$$\mathrm{TR}_4 \implies \mathrm{IP}_{10} \implies \mathrm{IP}_9 \implies \mathrm{IP}_8,$$

and therefore

$$\Delta_{\mathrm{CL}} = 0.$$

Equivalently, once the retained total-channel baseline and its bundled transfer/diagnostic discipline are frozen through TR_4*, the live OP 1 burden is legitimately reopened only as the short pinning chain*

$$\mathrm{IP}_{10} \to \mathrm{IP}_9 \to \mathrm{IP}_8,$$

rather than as a return to the full historical OP 1 queue.

Proof. By the concrete registry from Section 33, the tag TR_4 dereferences the bundled global pass/transfer/diagnostic-subordination interface from Propositions 32.78, 32.80, and 32.81. On the present line this is exactly the smallest normalized total-channel surface that keeps fixed the retained total profile and the comparison/diagnostic discipline under which the OP 1 corridor is being read.

Under the standing same-corridor reversal hypothesis, the additional assumption of optimum-pinning comparison rigidity is precisely the carrier tagged by IP_{10}, namely Proposition 33.170. That proposition says that, on the intended admissible windows, preservation of the retained total-channel profile together with the retained optimum location and local comparison/defect data forces retained-window rigidity.

The next normalized reopening step is then IP_9, i.e. Proposition 33.166. Once retained-window rigidity is available, and with the standing no-second-carrier and strict-upper-gap exclusions built into that bridge, admissible-window uniqueness follows on the retained corridor.

Finally one reopens IP_8, i.e. Proposition 33.162. This upgrades admissible-window uniqueness to the one-first-zero loop-budget certificate and hence yields

$$\Delta_{\mathrm{CL}} = 0.$$

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No new mathematics has been added here: the content is exactly the already established OP 1 compression chain, now rewritten in normalized tag-first form. The gain is that the route can be cited as one frozen total-channel anchor plus three smallest pinning tags instead of reopening the larger sentence-level proof history. □

Remark 33.193 (What this first practical rewrite already buys)**.** *This is the first point at which the tag layer stops being merely architectural. A later continuation can now say, in a mathematically literal way, "freeze at* TR_4*, then test* $\mathrm{IP}_{10} \to \mathrm{IP}_9 \to \mathrm{IP}_8$*" and thereby refer to an already normalized proof route rather than to an entire subsection block. In other words, the QAM layer has started to function as a true routing vocabulary for the live burden, not just as a storage format for tools.*

19AZ$^{(3d)}$. First tag-first rewrite of the bridge/localization route

The second practical test of the normalized QAM registry should not pretend to close a new theorematic burden. Its job is narrower and more honest: once the retained total-channel baseline is frozen, it should rewrite the currently live bridge-side reopening order into one compact route that first changes surface at the smallest matrix/phase interface and only then localizes the first still-live obstruction.

Proposition 33.194 (First tag-first reformulation of the bridge/localization reopening route)**.** *Assume the retained total-channel baseline is frozen through* TR_4*, and suppose a still-live bridge-side residual is to be tested without yet postulating two independent primitive carriers. Then the first normalized QAM reopening route may be cited in the compact form*

$$\mathrm{TR}_4 \implies \mathrm{BS}_1 \implies \mathrm{BS}_3 \implies \mathrm{BL}_2 \implies \mathrm{BL}_4,$$

with the additional compression step

$$\mathrm{BL}_4 \implies \mathrm{BL}_5$$

whenever the first localized obstruction lies in the readout/kernel stratum. Equivalently, once the retained total profile and its bundled comparison/diagnostic discipline are frozen, the live bridge burden is reopened first as matrix leakage, then as common-carrier/phase coupling, and only then as stratified obstruction localization; a separate carrier is not introduced before this normalized route has been exhausted.

Proof. By the concrete registry from Section 33, the tag TR_4 dereferences the bundled global pass/transfer/diagnostic-subordination interface from Propositions 32.78, 32.80, and 32.81. On the present line this is again the smallest frozen total-channel surface from which later local re-openings are allowed.

The first bridge-side reopening step is BS_1, i.e. the shared matrix-leakage surface gate from Definition 1.287 together with Proposition 1.288. This is the right first test because it asks whether there is still matrix-induced off-sector drift at all. In other words, before any finer localization language is used, the route first checks whether the admissible HC channel is already stable under the Millennium/Umkehrwalze interface.

If the bridge burden is still live after that first gate, then one changes surface through BS_3, namely the shared phase-coupling diagnostic from Definitions 1.290–1.291 together with Propositions 1.292 and 1.293. This does not yet decide where the first remaining obstruction sits; it rewrites the visible residual pair into the smallest matrix/phase language in which one can honestly ask whether the pair is still one masked carrier rather than two unrelated end defects.

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That localization question is then reopened directly as BL_2, which is precisely the phase-coupling-first localization gate recorded in the same registry. If the residual pair still behaves as one masked carrier, the route stays on the one-carrier side. If it remains unresolved or fails at that stage, the next legitimate reopening is BL_4, i.e. the obstruction-stratification tool from the extracted OP 3/QAM layer. This is the first point at which the present line is allowed to ask whether the surviving obstruction lives at the branch, shell, or readout/kernel level.

Finally, when BL_4 places the first residual in the readout/kernel stratum, the carrier-compression tool BL_5 compresses that stratum to absence of one common shell-stable carrier. Hence the entire normalized reopening order is exactly the displayed tag route. No new closure theorem has been proved here. The content is the already extracted bridge packet and the already extracted localization packet, now rewritten as one smallest admissible tag-first diagnostic route. □

Remark 33.195 (What this second practical rewrite already buys)**.** *The present line can now cite both kinds of live work in the same compact language. For OP 1 compression one may say "freeze at* TR_4*, then test* $\mathrm{IP}_{10} \to \mathrm{IP}_9 \to \mathrm{IP}_8$*"; for bridge/localization one may now say "freeze at* TR_4*, then test* $\mathrm{BS}_1 \to \mathrm{BS}_3$*, then localize by* $\mathrm{BL}_2 \to \mathrm{BL}_4$*, and if necessary compress by* BL_5*." In that precise sense, the QAM layer is no longer only a menu of stored tools: it has become a two-sided routing vocabulary, one side for the surviving OP 1 pinning chain and one side for the still-live bridge/localization diagnostics.*

19AZ$^{(4)}$. First Monte-Carlo residual-signature route beneath the normalized QAM tool packets

The Monte-Carlo programme should therefore enter only *after* the normalized ledger and its QAM recoding are in place. On the present line, Monte Carlo is not a replacement proof engine. It is a structural probe that samples how residual deviations distribute relative to the already fixed total-channel baseline and to the still-live imbalance/bridge packets.

Definition 33.196 (Residual-signature word)**.** *Fix a retained admissible window and a finite family of normalized diagnostics*

$$D = (D_1, \dots, D_m)$$

measured relative to the retained total-channel baseline. For one Monte-Carlo sample ω*, let*

$$\delta(\omega) = (\delta_1(\omega), \dots, \delta_m(\omega))$$

be the resulting normalized deviation vector. A residual-signature word *for* ω *is any finite coded word*

$$\mathsf{Sig}(\omega; D)$$

obtained from the ordered sign, bin, rank, or slot pattern of the components of $\delta(\omega)$*. Such a word is interpreted as a structural signature or hash of the sampled residual behavior, not as a theorem by itself.*

Definition 33.197 (Kasiski-style recurrence scan)**.** *Given a family of residual-signature words, the* Kasiski-style recurrence scan *is the first combinatorial pass that looks for repeated blocks, repeated slot offsets, and stable periodicity or near-periodicity classes among those words. Its role is diagnostic only: it asks whether the sampled residuals point repeatedly toward the same bridge or imbalance surface.*

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Proposition 33.198 (Conservative Monte-Carlo/QAM probe rule)**.** *Residual-signature words and their Kasiski-style recurrence scan may be used in the immediate continuation line only to prioritize which normalized packet from Definition 33.176 should be reopened first. They may* not *by themselves introduce a second primitive carrier, overturn the frozen total-channel baseline, or replace an already theorematic pass/obstruction statement.*

Proof. This is exactly the methodological discipline stated in the deferred roadmap above. The retained total-channel baseline is already part of the frozen theorem layer, while the Monte-Carlo route is introduced only as a structural probe beneath that layer. Therefore sampled residual signatures may guide priority and localization, but they cannot overrule the already established carrier and status surfaces. □

Remark 33.199 (Immediate diagnostic reading of the first residual signatures)**.** *The intended first reading is now precise. If the residual signatures cluster while the retained total-channel profile stays rigid, then the next reopen final should be searched first inside* $\mathfrak{T}_{\mathrm{imb}}$ *or* $\mathfrak{T}_{\mathrm{br}}$*, depending on whether the recurrence is purely local/imbalance-side or genuinely translation-sensitive. If no stable recurrence class appears at all, then the theorem layer should remain untouched and the Monte-Carlo route should be treated as presently non-informative rather than overinterpreted.*

19AZ$^{(4a)}$. First decoding of residual-signature recurrences into normalized BS/BL reopening priorities

The next conservative step is now to make the bridge/localization reading operational. The residual-signature words should not merely say that "something bridge-like remains." They should decode into a smallest prioritized reopening order. In particular, the present line needs one first rule saying when a stable recurrence class should push the next diagnostic pass onto the normalized bridge/localization route rather than back into the OP 1 pinning chain.

Definition 33.200 (Route-priority decoder for residual signatures)**.** *Let* $\mathcal{C}$ *be a stable recurrence class produced by the Kasiski-style scan from Definition 33.197. A* route-priority decoder *for* $\mathcal{C}$ *is any conservative rule*

$$\mathsf{Dec}(\mathcal{C}) \in \left\{ M_{\mathrm{imb}}^{\mathrm{pin}},\, M_{\mathrm{br}}^{\mathrm{surf}},\, M_{\mathrm{br}}^{\mathrm{loc}} \right\}$$

that assigns $\mathcal{C}$ *to the smallest normalized menu whose recorded carriers already account for the repeated slot/rank pattern of the sampled residuals. The decoder is called* translation-sensitive *if the repeated pattern is not stable under purely local imbalance-side rereading, but stabilizes once the residual pair is reread through the matrix/phase common-carrier surface. In that case the first prioritized reopening route attached to* $\mathcal{C}$ *is*

$$\mathrm{BS}_1 \to \mathrm{BS}_3 \to \mathrm{BL}_2,$$

with BL_4 *and, when needed,* BL_5 *appended only after the phase/common-carrier gate has been exhausted.*

Proposition 33.201 (Translation-sensitive signature classes prioritize the BS/BL route)**.** *Assume the retained total-channel baseline remains frozen through* TR_4*, and let* $\mathcal{C}$ *be a stable recurrence class of residual-signature words. Suppose moreover that* $\mathcal{C}$ *is translation-sensitive in the sense of Definition 33.200: its repeated slot/rank pattern does not settle under a purely local imbalance-side rereading, but does settle once the same sampled residuals are reread*

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through the matrix/phase common-carrier surface. Then the first normalized reopening priority attached to $\mathcal{C}$ is the bridge/localization route

$$\mathrm{TR}_4 \implies \mathrm{BS}_1 \implies \mathrm{BS}_3 \implies \mathrm{BL}_2,$$

followed, only if a still-live obstruction remains, by

$$\mathrm{BL}_2 \implies \mathrm{BL}_4,$$

and by

$$\mathrm{BL}_4 \implies \mathrm{BL}_5$$

whenever the first localized obstruction lies in the readout/kernel stratum. Equivalently, a stable translation-sensitive recurrence class is not yet evidence for a second primitive carrier; it is evidence that the next reopen final should be searched first along the normalized matrix/phase-to-localization bridge route.

Proof. By Proposition 33.198, residual-signature words may only prioritize a reopening order inside the already normalized theorem/tool layer; they may not overrule the frozen total-channel baseline or create a new carrier hypothesis. Hence one must begin from TR_4, the smallest bundled total-channel anchor already fixed by the registry from Section 33.

Now assume the stable recurrence class $\mathcal{C}$ is translation-sensitive. By definition this means that the repeated residual pattern does not become stable when reread only as a local imbalance perturbation, but does become stable once the same samples are reread through the matrix/phase common-carrier surface. Therefore the first admissible decoder target for $\mathcal{C}$ cannot be merely $M_{\mathrm{imb}}^{\mathrm{pin}}$; the smallest normalized reopening surface that can already account for the stable repeated pattern is the bridge-surface menu. In the concrete registry this is exactly the route starting at BS_1, the shared matrix-leakage gate, and continuing through BS_3, the shared phase/common-carrier surface.

Once that surface change has been made, the first honest question is whether the repeated residual pair still behaves as one masked carrier. In the normalized registry this is precisely the localization gate BL_2. If the recurrence class is already resolved there, no deeper reopening is justified. If it remains live, then the first further admissible step is BL_4, the obstruction-stratification tool. And if that stratification places the first obstruction in the readout/kernel stratum, then BL_5 is exactly the recorded compression step that rewrites the surviving burden as absence of one common shell-stable carrier package. Thus the displayed BS/BL priority route is exactly the smallest normalized decoding of a translation-sensitive stable recurrence class. □

Remark 33.202 (What the first Monte-Carlo-to-route decoding now buys)**.** *The Monte-Carlo layer now has a literal output format inside the QAM vocabulary. A stable recurrence class is no longer only a vague hint that "the bridge side may matter." The present line may now state, in compact normalized form: "freeze at* TR_4*; if the recurrence is translation-sensitive, decode first by* $\mathrm{BS}_1 \to \mathrm{BS}_3 \to \mathrm{BL}_2$*, and only then localize further by* BL_4 *or compress by* BL_5*." In that precise sense the residual-signature words have acquired a conservative but genuine routing semantics.*

19AZ$^{(4b)}$. First decoding of purely local residual-signature recurrences into normalized IP reopening priorities

The decoder should now be made symmetric on the pinning side. The residual-signature words should not only tell us when to reopen the bridge/localization route; they should also

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tell us when the next honest move remains entirely inside the surviving OP 1 kernel. The immediate line therefore needs one first rule saying when a stable recurrence class should stay on the local imbalance side and decode directly into the normalized IP route.

Definition 33.203 (Local-imbalance decoder for residual signatures)**.** *Let $\mathcal{C}$ be a stable recurrence class produced by the Kasiski-style scan from Definition 33.197. A route-priority decoder from Definition 33.200 is called* purely local *or* imbalance-stable *on $\mathcal{C}$ if the repeated slot/rank pattern already stabilizes under purely local rereading on the imbalance side, so that rereading the same samples through the matrix/phase common-carrier surface adds no smaller admissible explanation. In that case the first prioritized reopening route attached to $\mathcal{C}$ is the normalized pinning chain*

$$\mathrm{IP}_{10} \to \mathrm{IP}_9 \to \mathrm{IP}_8,$$

read beneath the frozen total-channel anchor TR_4*.*

Proposition 33.204 (Purely local signature classes prioritize the IP route)**.** *Assume the retained total-channel baseline remains frozen through* TR_4*, and let $\mathcal{C}$ be a stable recurrence class of residual-signature words. Suppose moreover that $\mathcal{C}$ is purely local in the sense of Definition 33.203: its repeated slot/rank pattern already stabilizes under a local imbalance-side rereading and does not require an earlier matrix/phase surface change in order to become coherent. Then the first normalized reopening priority attached to $\mathcal{C}$ is the pinning route*

$$\mathrm{TR}_4 \implies \mathrm{IP}_{10} \implies \mathrm{IP}_9 \implies \mathrm{IP}_8,$$

so that the sampled recurrence is read first as a compressed OP 1 burden rather than as a bridge/localization burden. Equivalently, a stable purely local recurrence class is not evidence that one should leave the imbalance side; it is evidence that the next reopen final should be searched first inside the surviving normalized pinning chain.

Proof. By Proposition 33.198, the recurrence class $\mathcal{C}$ may only prioritize a route already present in the normalized theorem/tool layer. Hence the total-channel baseline must remain frozen at TR_4, exactly as in the bridge/localization decoder.

Now assume $\mathcal{C}$ is purely local. By definition, its repeated slot/rank pattern already becomes coherent under a local imbalance-side rereading, and no earlier matrix/phase reinterpretation yields a smaller admissible menu. Therefore the decoder is not allowed to jump first to $M_{\mathrm{br}}^{\mathrm{surf}}$ or $M_{\mathrm{br}}^{\mathrm{loc}}$. The smallest normalized reopening surface that can already account for the stable recurrence class is instead the surviving pinning menu $M_{\mathrm{imb}}^{\mathrm{pin}}$.

Inside that menu the current live OP 1 burden has already been compressed in Proposition 33.192. The first admissible entry is therefore IP_{10}, the optimum-pinning comparison-rigidity bridge. Once that bridge is opened, one proceeds to IP_9, the retained-window-rigidity / no-second-carrier / strict-upper-gap bridge, and then to IP_8, the admissible-window uniqueness bridge. Thus the displayed tag route is exactly the smallest normalized decoding of a stable purely local recurrence class. □

Remark 33.205 (What the first Monte-Carlo-to-IP decoding now buys)**.** *The decoder is now symmetric across the two live diagnostic sides. The present line may state in compact normalized form: "freeze at* TR_4*; if the recurrence is translation-sensitive, decode first by* $\mathrm{BS}_1 \to \mathrm{BS}_3 \to \mathrm{BL}_2$*; if the recurrence is already imbalance-stable, decode instead by* $\mathrm{IP}_{10} \to \mathrm{IP}_9 \to \mathrm{IP}_8$*." In that precise sense the Monte-Carlo layer no longer only points vaguely toward "some residual side"; it now separates, conservatively and in dereferenceable tags, the first bridge reopening from the first compressed OP 1 reopening.*

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19AZ$^{(4c)}$. First explicit decision table for residual-signature routing

The decoder layer should now be made directly usable. The present line no longer needs only two separate propositions saying when a stable recurrence class goes to the bridge/localization side or stays on the pinning side. It now needs one explicit decision table that turns the first observed residual-signature class into a smallest admissible action. In particular, the table should also record the conservative "hold" outcome when no stable recurrence class appears at all.

> **Reader cue.** Read the next table as an action selector. Its purpose is to turn an observed residual-signature class into the smallest admissible reopen instruction, not to claim that every downstream route has already been exhausted.

Definition 33.206 (First explicit decision table for residual-signature routing)**.** *Assume the retained total-channel baseline remains frozen through* TR_4*, and let* $\mathcal{C}$ *be the first sampled recurrence class produced by the Kasiski-style scan from Definition 33.197. The* first explicit decision table for residual-signature routing *is the map*

$$\mathsf{RouteTab}(\mathcal{C}) \in \big\{\mathsf{Hold},\ \mathsf{IP\text{-}first},\ \mathsf{BS/BL\text{-}first},\ \mathsf{BL\text{-}continue},\ \mathsf{BL\text{-}kernel}\big\}$$

with the following normal form.

Observed residual-signature class	*Decoder output*	*First normalized route*
No stable recurrence class appears	Hold	*no reopen; keep the theorem/tool layer frozen*
Purely local / imbalance-stable recurrence	IP-first	$\mathrm{TR}_4 \Rightarrow \mathrm{IP}_{10} \Rightarrow \mathrm{IP}_9 \Rightarrow \mathrm{IP}_8$
Translation-sensitive recurrence	BS/BL-first	$\mathrm{TR}_4 \Rightarrow \mathrm{BS}_1 \Rightarrow \mathrm{BS}_3 \Rightarrow \mathrm{BL}_2$
Translation-sensitive recurrence still live after BL_2	BL-continue	*append* BL_4
First localized obstruction lies in the readout/kernel stratum	BL-kernel	*append* BL_5

The table is read top-down and always returns the smallest normalized action whose recorded carriers already account for the observed recurrence behavior.

Proposition 33.207 (The first explicit decision table is conservative and minimal on the current live classes)**.** *Under the current theorem/tool layer, every sampled residual-signature class that is presently admitted by Propositions 33.201 and 33.204 receives from Definition 33.206 a unique first admissible action. More precisely:*

(i) if no stable recurrence class appears, the decision table returns Hold*;*

(ii) if the recurrence class is purely local, it returns IP-first*;*

(iii) if the recurrence class is translation-sensitive, it returns BS/BL-first*;*

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(iv) deeper localization steps BL-continue *and* BL-kernel *may be appended only after the route has already entered the BS/BL side through* BL_2.

Hence the table is conservative, because it never manufactures a new carrier or bypasses the frozen total-channel anchor, and minimal, because it always picks the smallest currently recorded reopening route compatible with the observed recurrence class.

Proof. If no stable recurrence class appears at all, then Remark 33.199 already says that the theorem layer must remain untouched; this is exactly the output Hold. If a stable recurrence class is purely local, then Proposition 33.204 forces the first admissible reopening route to stay inside the pinning chain beneath TR_4, so the table must return IP-first and the route $\mathrm{TR}_4 \Rightarrow \mathrm{IP}_{10} \Rightarrow \mathrm{IP}_9 \Rightarrow \mathrm{IP}_8$. If instead the stable recurrence class is translation-sensitive, then Proposition 33.201 forces the first admissible reopening route to pass through $\mathrm{BS}_1 \Rightarrow \mathrm{BS}_3 \Rightarrow \mathrm{BL}_2$, so the table must return BS/BL-first.

The remaining two outputs are not independent first moves. By Proposition 33.201, BL_4 is appended only if the obstruction remains live after BL_2, and BL_5 is appended only when that localized obstruction lies in the readout/kernel stratum. Therefore the decision table never jumps directly to BL_4 or BL_5 without first exhausting the smaller admissible bridge/localization entry route. This proves both conservativity and minimality on the currently live recurrence classes. □

Remark 33.208 (Immediate operational reading of the decision table)**.** *The present line can now state the decoder in one sentence that is usable both by a reader and by the future machine layer: classify the first stable residual-signature class, then apply*

$$\mathsf{Hold} \ \vee \ \mathsf{IP\text{-}first} \ \vee \ \mathsf{BS/BL\text{-}first},$$

with deeper localization allowed only on the BS/BL side by appending BL-continue *and, in the readout/kernel case,* BL-kernel*. In that sense the Monte-Carlo diagnostics, the QAM tag layer, and the normalized reopening routes now meet in one explicit working table rather than only in scattered prose.*

19AZ$^{(4d)}$. First deterministic QAM decoder syntax for residual-signature routing

The decision table may now be compressed one step further. The present line no longer needs only a prose table saying which class goes to hold, IP-first, or BS/BL-first. It may now record one short deterministic decoder syntax whose words already match the normalized QAM routes. The point is not to introduce a new theory of automata; it is only to give the future machine layer and the human reader the same finite, dereferenceable instruction string.

Definition 33.209 (First deterministic QAM decoder syntax)**.** *Assume the retained total-channel anchor has been frozen through* TR_4*, and let* $\mathcal{C}$ *be the first sampled residual-signature class from the Kasiski-style recurrence scan. The* first deterministic QAM decoder syntax *is the finite rewrite alphabet*

$$\mathfrak{A}_{\mathrm{dec}} = \{\mathsf{classify}, \mathsf{hold}, \mathsf{ip}, \mathsf{bsbl}, \mathsf{continue}, \mathsf{kernel}\}$$

with the following admissible output words:

$$\begin{aligned}
&\mathsf{classify} \to \mathsf{hold},\\
&\mathsf{classify} \to \mathsf{ip},\\
&\mathsf{classify} \to \mathsf{bsbl},\\
&\mathsf{classify} \to \mathsf{bsbl} \to \mathsf{continue},\\
&\mathsf{classify} \to \mathsf{bsbl} \to \mathsf{continue} \to \mathsf{kernel}.
\end{aligned}$$

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These words are interpreted by the normalized dereferencing map

$$\mathsf{Dec}_{\mathrm{QAM}} : \begin{cases} \mathsf{hold} & \mapsto \textit{keep the theorem/tool layer frozen}, \\ \mathsf{ip} & \mapsto \mathrm{TR}_4 \Rightarrow \mathrm{IP}_{10} \Rightarrow \mathrm{IP}_9 \Rightarrow \mathrm{IP}_8, \\ \mathsf{bsbl} & \mapsto \mathrm{TR}_4 \Rightarrow \mathrm{BS}_1 \Rightarrow \mathrm{BS}_3 \Rightarrow \mathrm{BL}_2, \\ \mathsf{continue} & \mapsto \textit{append } \mathrm{BL}_4, \\ \mathsf{kernel} & \mapsto \textit{append } \mathrm{BL}_5. \end{cases}$$

The syntax is called deterministic *when exactly one of the following first outputs is assigned to $\mathcal{C}$:* hold*,* ip*, or* bsbl*; the symbols* continue *and* kernel *may occur only after* bsbl*.*

Proposition 33.210 (The deterministic decoder syntax is equivalent to the current decision table)**.** *Under the current live theorem/tool layer, Definition 33.209 yields the same first admissible routing data as Definition 33.206. More precisely:*

(i) if no stable recurrence class appears, the unique admissible output word is classify $\to$ hold*;*

(ii) if the recurrence class is purely local / imbalance-stable, the unique admissible output word is classify $\to$ ip*;*

(iii) if the recurrence class is translation-sensitive, the unique admissible output word is classify $\to$ bsbl*;*

(iv) if, after entering the BS/BL side, the obstruction remains live beyond BL_2*, the decoder extends uniquely to* classify $\to$ bsbl $\to$ continue*;*

(v) if that localized obstruction lies in the readout/kernel stratum, the decoder extends further uniquely to classify $\to$ bsbl $\to$ continue $\to$ kernel*.*

Hence the syntax is a faithful finite normal-form compression of the decision table, with no new carriers and no loss of routing information.

Proof. The first three clauses are exactly the three primary outputs of Definition 33.206: Hold, IP-first, and BS/BL-first. Definition 33.209 merely rewrites these three outcomes as the shorter syntax outputs hold, ip, and bsbl, while the dereferencing map $\mathsf{Dec}_{\mathrm{QAM}}$ sends them back to the same normalized routes.

The remaining two clauses are likewise only compressed spellings of the already recorded BS/BL-side continuations. By Proposition 33.207, the symbol BL-continue may appear only after the bridge/localization route has already entered through BL_2, and BL-kernel may appear only after that continuation has localized the first remaining obstruction inside the readout/kernel stratum. In the deterministic syntax these two later actions are rewritten as continue and kernel, with the same ordering restriction. Thus every admissible table output produces exactly one admissible decoder word, and every admissible decoder word dereferences to exactly one table-consistent reopening route.

Therefore the syntax introduces no new mathematical burden. It only compresses the already normalized routing data into a finite QAM-readable word system. This proves equivalence to the current decision table and faithfulness of the compression. □

Remark 33.211 (Immediate machine reading of the deterministic decoder syntax)**.** *The present line may now be summarized by the short machine-readable sentences*

$$\mathsf{classify} \to \mathsf{hold}, \qquad \mathsf{classify} \to \mathsf{ip}, \qquad \mathsf{classify} \to \mathsf{bsbl} \to \mathsf{continue} \to \mathsf{kernel},$$

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with the understanding that later symbols may only be appended when the earlier route has actually been entered and justified. In that sense the QAM layer now has not only tags and tables, but also a first deterministic decoder syntax that a future theorem-store, proof assistant, or simulation layer could read without reopening the entire surrounding prose.

19AZ$^{(4e)}$. First finite transition function for deterministic residual-signature routing

The deterministic decoder syntax may now be compressed once more into a tiny state transition rule. The point is again modest: not to introduce a new autonomous automaton theory, but only to expose the present routing discipline as a finite partial transition function that the future machine layer can read directly. In particular, the current line should make explicit that only three first moves are admitted from the classification state, while deeper continuation steps remain locked behind the BS/BL side.

Definition 33.212 (First finite transition function for the deterministic decoder)**.** *Let*

$$Q_{\mathrm{dec}} = \{\mathsf{classify},\ \mathsf{hold},\ \mathsf{ip},\ \mathsf{bsbl},\ \mathsf{continue},\ \mathsf{kernel}\}$$

be the current decoder-state set, and let

$$\Sigma_{\mathrm{dec}} = \{\mathsf{none},\ \mathsf{local},\ \mathsf{trans},\ \mathsf{stilllive},\ \mathsf{kernelstratum}\}$$

be the present event alphabet, where:

- none *means that no stable residual-signature recurrence class appears,*
- local *means that the sampled class is purely local / imbalance-stable,*
- trans *means that the sampled class is translation-sensitive,*
- $\mathsf{stilllive}$ *means that after entering the BS/BL side the obstruction remains live beyond* BL_2*,*
- $\mathsf{kernelstratum}$ *means that the first localized remaining obstruction lies in the readout/kernel stratum.*

The first finite transition function for deterministic residual-signature routing *is the partial map*

$$\delta_{\mathrm{dec}} : Q_{\mathrm{dec}} \times \Sigma_{\mathrm{dec}} \dashrightarrow Q_{\mathrm{dec}}$$

given by the only admissible transitions

$$\begin{aligned}
\delta_{\mathrm{dec}}(\mathsf{classify}, \mathsf{none}) &= \mathsf{hold},\\
\delta_{\mathrm{dec}}(\mathsf{classify}, \mathsf{local}) &= \mathsf{ip},\\
\delta_{\mathrm{dec}}(\mathsf{classify}, \mathsf{trans}) &= \mathsf{bsbl},\\
\delta_{\mathrm{dec}}(\mathsf{bsbl}, \mathsf{stilllive}) &= \mathsf{continue},\\
\delta_{\mathrm{dec}}(\mathsf{continue}, \mathsf{kernelstratum}) &= \mathsf{kernel}.
\end{aligned}$$

All other transitions are undefined at the present stage. The state outputs are dereferenced through the same normalized map $\mathsf{Dec}_{\mathrm{QAM}}$ *from Definition 33.209.*

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Proposition 33.213 (The finite transition function is equivalent to the deterministic decoder syntax and remains minimal on the current live events)**.** *Under the current theorem/tool layer, the partial transition function from Definition 33.212 carries exactly the same routing information as the deterministic decoder syntax from Definition 33.209. More precisely:*

(i) the three first moves from classify *are exactly* hold, ip, *and* bsbl*, corresponding respectively to the events* none, local, *and* trans*;*

(ii) the only admissible continuation on the bridge/localization side is $\mathsf{bsbl} \xrightarrow{\mathsf{stilllive}} \mathsf{continue}$*;*

(iii) the only admissible further refinement is $\mathsf{continue} \xrightarrow{\mathsf{kernelstratum}} \mathsf{kernel}$*;*

(iv) every undefined transition would either skip a smaller admissible reopening route or manufacture a carrier/event type not yet recorded in the present line.

Hence δ_{dec} *is a faithful finite-state compression of the current routing discipline and is minimal on the currently live diagnostic events.*

Proof. The first three transitions are exactly the three initial output words from Proposition 33.210: $\mathsf{classify} \to \mathsf{hold}$, $\mathsf{classify} \to \mathsf{ip}$, and $\mathsf{classify} \to \mathsf{bsbl}$. The transition function merely rewrites these words as event-labeled state moves. The event names none, local, and trans are exactly the presently live coarse classes isolated by the decision table and the two decoder propositions.

Once the route has entered bsbl, the deterministic syntax allows only the optional later symbols continue and kernel. By Definition 33.209, continue may appear only after bsbl, and by Proposition 33.207 it is justified precisely when the obstruction remains live beyond BL_2. This is the transition $\delta_{\mathrm{dec}}(\mathsf{bsbl}, \mathsf{stilllive}) = \mathsf{continue}$. Likewise, kernel may occur only after that continuation has localized the first remaining obstruction inside the readout/kernel stratum, giving $\delta_{\mathrm{dec}}(\mathsf{continue}, \mathsf{kernelstratum}) = \mathsf{kernel}$.

No further transitions are admissible on the current line. Any direct jump from classify to continue or kernel would bypass the smaller bridge/localization entry route, while any jump from ip to a later BS/BL state would conflate two presently separated reopening disciplines. And any additional event symbol would exceed the currently normalized recurrence classes. Therefore the partial transition function is equivalent to the deterministic decoder syntax, conservative with respect to the recorded carriers, and minimal on the currently live events. □

Remark 33.214 (Immediate automaton reading of the current decoder)**.** *The present line can now be read as a tiny partial automaton:*

$$\mathsf{classify} \xrightarrow{\mathsf{none}} \mathsf{hold},$$
$$\mathsf{classify} \xrightarrow{\mathsf{local}} \mathsf{ip},$$
$$\mathsf{classify} \xrightarrow{\mathsf{trans}} \mathsf{bsbl} \xrightarrow{\mathsf{stilllive}} \mathsf{continue} \xrightarrow{\mathsf{kernelstratum}} \mathsf{kernel}.$$

Through the dereferencing map $\mathsf{Dec}_{\mathrm{QAM}}$*, this automaton is still nothing more than the already normalized theorem/tool layer. But it now has a form that a later theorem store, proof assistant, or Monte-Carlo controller could read as a deterministic transition discipline instead of as prose plus scattered tags.*

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19AZ$^{(4f)}$. First observer/event normal form for deterministic residual-signature words

The tiny decoder automaton from the previous subsection still presupposes that the incoming event has already been named as one of the symbols none, local, trans, stilllive, or kernelstratum. The next conservative step is therefore to normalize not the output route once more, but the observer-side input itself. The present line does not need a large language for that purpose. It only needs a short finite event-word normal form recording how much of the bridge/localization continuation has actually become visible at readout.

Definition 33.215 (First observer/event normal form for deterministic residual-signature words)**.** *Let*

$$\mathcal{W}_{\mathrm{dec}}^{\mathrm{nf}} = \{\mathsf{none},\ \mathsf{local},\ \mathsf{trans},\ \mathsf{trans\,stilllive},\ \mathsf{trans\,stilllive\,kernelstratum}\}$$

be the first observer/event normal-form language *for deterministic residual-signature routing. The associated partial extended transition map is defined recursively by*

$$\delta_{\mathrm{dec}}^{*}(q,\varepsilon) = q, \qquad \delta_{\mathrm{dec}}^{*}(q,aw) = \delta_{\mathrm{dec}}^{*}\big(\delta_{\mathrm{dec}}(q,a),w\big)$$

whenever the right-hand side is defined, where $a \in \Sigma_{\mathrm{dec}}$ and w is a finite word in the same alphabet.

The observer/event normal form map

$$\mathsf{NF}_{\mathrm{obs}} : \textit{currently normalized residual-signature cases} \longrightarrow \mathcal{W}_{\mathrm{dec}}^{\mathrm{nf}}$$

assigns the unique normal-form word as follows:

(i) no stable recurrence class is observed $\mapsto$ none*,*

(ii) a purely local / imbalance-stable class is observed $\mapsto$ local*,*

(iii) a translation-sensitive class is observed but no remaining obstruction survives beyond BL_2 $\mapsto$ trans*,*

(iv) a translation-sensitive class is observed and the obstruction remains live beyond BL_2 $\mapsto$ trans stilllive*,*

(v) the previous case holds and the first localized remaining obstruction lies in the readout/kernel stratum $\mapsto$ trans stilllive kernelstratum*.*

Proposition 33.216 (The observer/event normal forms are exactly the currently admissible decoder words and reproduce the normalized reopening routes)**.** *Under the current theorem/-tool layer, the language $\mathcal{W}_{\mathrm{dec}}^{\mathrm{nf}}$ from Definition 33.215 is exactly the set of currently admissible observer-side event words for the deterministic decoder. More precisely:*

(i) every word in $\mathcal{W}_{\mathrm{dec}}^{\mathrm{nf}}$ is admitted by the partial automaton δ_{dec} when started at classify*,*

(ii) the terminal states are respectively

$$\delta_{\mathrm{dec}}^{*}(\mathsf{classify},\mathsf{none}) = \mathsf{hold}, \qquad \delta_{\mathrm{dec}}^{*}(\mathsf{classify},\mathsf{local}) = \mathsf{ip},$$

$$\delta_{\mathrm{dec}}^{*}(\mathsf{classify},\mathsf{trans}) = \mathsf{bsbl}, \qquad \delta_{\mathrm{dec}}^{*}(\mathsf{classify},\mathsf{trans\,stilllive}) = \mathsf{continue},$$

$$\delta_{\mathrm{dec}}^{*}(\mathsf{classify},\mathsf{trans\,stilllive\,kernelstratum}) = \mathsf{kernel},$$

and these states dereference exactly to the normalized routes already fixed in the QAM tool layer;

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(iii) no other finite event word is currently admissible without either violating the transition discipline of δ_{dec} or introducing an observer-side distinction not yet normalized in the present line.

Hence the current Monte-Carlo/QAM interface may now be read not only as a state machine, but as a finite normal-form language of observer-side diagnostic words.

Proof. The five displayed words are admitted directly by the transition function from Definition 33.212. The one-letter words none, local, and trans are exactly the three first-step events recorded there. Once the route has entered bsbl, the only currently permitted continuation event is stilllive, and once the route has entered continue, the only currently permitted further refinement is kernelstratum. Therefore the longer words trans stilllive and trans stilllive kernelstratum are admitted as well.

The recursive definition of $\delta^{*}_{\mathrm{dec}}$ then gives the terminal states by straightforward evaluation. For none and local, only one transition is applied, yielding hold and ip. For trans, the first step yields bsbl. Appending stilllive moves to continue, and appending kernelstratum moves from there to kernel. By Definition 33.209, each terminal state dereferences to the already normalized reopening route: hold means reopen nothing, ip means the chain $\mathrm{TR}_4 \Rightarrow \mathrm{IP}_{10} \Rightarrow \mathrm{IP}_9 \Rightarrow \mathrm{IP}_8$, bsbl means the chain $\mathrm{TR}_4 \Rightarrow \mathrm{BS}_1 \Rightarrow \mathrm{BS}_3 \Rightarrow \mathrm{BL}_2$, continue appends BL_4, and kernel appends BL_5.

It remains to exclude any additional finite event word. A word beginning with anything other than none, local, or trans is impossible because δ_{dec} is undefined at classify on all other events. A word beginning with none or local cannot be extended further without leaving the currently normalized routing discipline. And a word beginning with trans may only continue with stilllive, then optionally with kernelstratum, because no further transitions are currently recorded. Thus every admissible observer-side case has exactly one normal-form word and no other finite word is currently allowed. □

Remark 33.217 (Immediate machine reading of the observer/event normal form)**.** *The current live interface may now be summarized by the finite observer-language*

none, local, trans, trans stilllive, trans stilllive kernelstratum.

This is the first point at which the tool layer, the QAM tags, the Monte-Carlo recurrence classes, the deterministic decoder syntax, and the partial transition automaton all meet in one compact object: a finite word language on the observer side whose outputs are already normalized reopening routes on the theorem side.

19AZ$^{(4g)}$. First observer-word evaluation map into deterministic decoder outputs

The observer/event normal forms now provide a finite input language for the deterministic decoder. The next conservative compression step is to package the whole passage from normalized observer word to normalized theorem-side reopening output into one explicit evaluation map. This adds no new event class and no new route. It merely composes the already stabilized normal-form language, transition automaton, and QAM dereferencing rule into one finite readout operator.

Definition 33.218 (First observer-word evaluation map into deterministic decoder outputs)**.** *Let*

$$\mathcal{W}^{\mathrm{nf}}_{\mathrm{dec}} = \{\text{none}, \text{local}, \text{trans}, \text{trans stilllive}, \text{trans stilllive kernelstratum}\}$$

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be the observer/event normal-form language from Definition 33.215. Define the observer-word evaluation map

$$\mathsf{Eval}_{\mathrm{obs}} : \mathcal{W}_{\mathrm{dec}}^{\mathrm{nf}} \longrightarrow Q_{\mathrm{dec}} \times \mathcal{R}_{\mathrm{tool}}^{\mathrm{out}}$$

by

$$\mathsf{Eval}_{\mathrm{obs}}(w) = \Big(\delta_{\mathrm{dec}}^{*}(\mathsf{classify}, w), \mathsf{Dec}_{\mathrm{QAM}}\big(\delta_{\mathrm{dec}}^{*}(\mathsf{classify}, w)\big)\Big),$$

where $Q_{\mathrm{dec}} = \{\mathsf{hold}, \mathsf{ip}, \mathsf{bsbl}, \mathsf{continue}, \mathsf{kernel}\}$ *is the current decoder-state set and* $\mathcal{R}_{\mathrm{tool}}^{\mathrm{out}}$ *denotes the finite set of currently normalized reopening outputs:*

$$\mathcal{R}_{\mathrm{tool}}^{\mathrm{out}} = \left\{ \begin{array}{l} \varnothing, \\ [\mathrm{TR}_4, \mathrm{IP}_{10}, \mathrm{IP}_9, \mathrm{IP}_8], \\ [\mathrm{TR}_4, \mathrm{BS}_1, \mathrm{BS}_3, \mathrm{BL}_2], \\ [\mathrm{TR}_4, \mathrm{BS}_1, \mathrm{BS}_3, \mathrm{BL}_2, \mathrm{BL}_4], \\ [\mathrm{TR}_4, \mathrm{BS}_1, \mathrm{BS}_3, \mathrm{BL}_2, \mathrm{BL}_4, \mathrm{BL}_5] \end{array} \right\}.$$

Equivalently,

$$\begin{aligned} \mathsf{Eval}_{\mathrm{obs}}(\mathsf{none}) &= (\mathsf{hold}, \varnothing), \\ \mathsf{Eval}_{\mathrm{obs}}(\mathsf{local}) &= (\mathsf{ip}, [\mathrm{TR}_4, \mathrm{IP}_{10}, \mathrm{IP}_9, \mathrm{IP}_8]), \\ \mathsf{Eval}_{\mathrm{obs}}(\mathsf{trans}) &= (\mathsf{bsbl}, [\mathrm{TR}_4, \mathrm{BS}_1, \mathrm{BS}_3, \mathrm{BL}_2]), \\ \mathsf{Eval}_{\mathrm{obs}}(\mathsf{trans\,stilllive}) &= (\mathsf{continue}, [\mathrm{TR}_4, \mathrm{BS}_1, \mathrm{BS}_3, \mathrm{BL}_2, \mathrm{BL}_4]), \\ \mathsf{Eval}_{\mathrm{obs}}(\mathsf{trans\,stilllive\,kernelstratum}) &= (\mathsf{kernel}, [\mathrm{TR}_4, \mathrm{BS}_1, \mathrm{BS}_3, \mathrm{BL}_2, \mathrm{BL}_4, \mathrm{BL}_5]). \end{aligned}$$

Proposition 33.219 (The observer-word evaluation map is total on the current normal–form language and reproduces the unique normalized reopening outputs)**.** *Under the current theorem/tool layer, the map* $\mathsf{Eval}_{\mathrm{obs}}$ *from Definition 33.218 is total on* $\mathcal{W}_{\mathrm{dec}}^{\mathrm{nf}}$*. More precisely:*

(i) every word in $\mathcal{W}_{\mathrm{dec}}^{\mathrm{nf}}$ *has exactly one decoder terminal state and exactly one currently normalized reopening output under* $\mathsf{Eval}_{\mathrm{obs}}$*,*

(ii) the first component of $\mathsf{Eval}_{\mathrm{obs}}(w)$ *is the unique terminal state reached by the partial automaton* $\delta_{\mathrm{dec}}^{*}$*,*

(iii) the second component is exactly the smallest presently admissible theorem-side reopening route compatible with that terminal state,

(iv) no currently normalized observer word evaluates to two different reopening outputs.

Hence the present live interface may now be read as one deterministic finite composite map

$$\mathsf{NF}_{\mathrm{obs}} \longrightarrow \mathsf{Eval}_{\mathrm{obs}} \longrightarrow \textit{normalized reopening output},$$

rather than as a chain of separately cited decoder pieces.

Proof. By Proposition 33.216, every word in $\mathcal{W}_{\mathrm{dec}}^{\mathrm{nf}}$ is admitted by the extended transition map $\delta_{\mathrm{dec}}^{*}$ when started at $\mathsf{classify}$, and its terminal state is uniquely determined. Thus the first component of $\mathsf{Eval}_{\mathrm{obs}}$ is total and single-valued on the whole normal-form language.

The second component is obtained by applying the deterministic QAM dereferencing map $\mathsf{Dec}_{\mathrm{QAM}}$ to that unique terminal state. By Definition 33.209, each currently live terminal state carries exactly one normalized reopening output: hold gives $\varnothing$, ip gives the fixed IP route $[\mathrm{TR}_4, \mathrm{IP}_{10}, \mathrm{IP}_9, \mathrm{IP}_8]$, bsbl gives the fixed bridge/localization entry route $[\mathrm{TR}_4, \mathrm{BS}_1, \mathrm{BS}_3, \mathrm{BL}_2]$,

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continue appends BL_4, and kernel appends BL_5. Therefore the second component is also total and single-valued.

Minimality follows from the construction of the current decoder. The state hold reopens nothing, ip is already the smallest admissible local route, and the bridge/localization branch enters first through bsbl before optional refinement to continue and kernel. Any alternative output would either reopen more than the current state requires or bypass an earlier normalized stage. Hence no current observer word can evaluate to two different normalized reopening outputs. □

Remark 33.220 (Immediate operational reading of $\mathsf{Eval}_{\mathrm{obs}}$)**.** *The current finite live interface may now be cited in one line:*

$$\begin{aligned} &\textit{observer-side recurrence class} \xrightarrow{\mathsf{NF}_{\mathrm{obs}}} \textit{normal-form word} \\ &\qquad\qquad \xrightarrow{\mathsf{Eval}_{\mathrm{obs}}} (\textit{decoder state}, \textit{normalized reopening output}). \end{aligned}$$

At this point, the Monte-Carlo diagnostics, the QAM tag layer, the finite decoder automaton, and the theorem-side reopening discipline have become one explicit finite evaluation pipeline. This is the first place in the present line where a later controller could read the whole diagnostic-to-route passage through a single total map on a finite language.

19AZ$^{(4h)}$. First compact output-normalform tokens for deterministic decoder results

The observer-word evaluation map now packages every currently admissible observer-side normal form into one of five deterministic theorem-side outputs. The next conservative compression step is to replace those explicit route lists by a finite family of compact output tokens. This adds no new state, no new route, and no new observer word. It merely supplies a shorter QAM-side output carrier for the five already stabilized decoder results.

Definition 33.221 (First compact output-normalform tokens for deterministic decoder results)**.** *Define the finite set of* output-normalform tokens

$$\mathcal{T}_{\mathrm{out}}^{\mathrm{nf}} = \{\mathsf{OH}, \mathsf{OI}, \mathsf{OB}, \mathsf{OC}, \mathsf{OK}\},$$

where the tokens are read as

$$\begin{aligned} \mathsf{OH} &\leftrightarrow (\mathsf{hold}, \varnothing), \\ \mathsf{OI} &\leftrightarrow (\mathsf{ip}, [\mathrm{TR}_4, \mathrm{IP}_{10}, \mathrm{IP}_9, \mathrm{IP}_8]), \\ \mathsf{OB} &\leftrightarrow (\mathsf{bsbl}, [\mathrm{TR}_4, \mathrm{BS}_1, \mathrm{BS}_3, \mathrm{BL}_2]), \\ \mathsf{OC} &\leftrightarrow (\mathsf{continue}, [\mathrm{TR}_4, \mathrm{BS}_1, \mathrm{BS}_3, \mathrm{BL}_2, \mathrm{BL}_4]), \\ \mathsf{OK} &\leftrightarrow (\mathsf{kernel}, [\mathrm{TR}_4, \mathrm{BS}_1, \mathrm{BS}_3, \mathrm{BL}_2, \mathrm{BL}_4, \mathrm{BL}_5]). \end{aligned}$$

Equivalently, define the output-tokenization map

$$\mathsf{Tok}_{\mathrm{out}} : Q_{\mathrm{dec}} \times \mathcal{R}_{\mathrm{tool}}^{\mathrm{out}} \longrightarrow \mathcal{T}_{\mathrm{out}}^{\mathrm{nf}}$$

by sending the five currently normalized evaluation outputs from Definition 33.218 to the corresponding tokens above. Composing with $\mathsf{Eval}_{\mathrm{obs}}$ *gives the compact output map*

$$\mathsf{Out}_{\mathrm{obs}}^{\mathrm{tok}} := \mathsf{Tok}_{\mathrm{out}} \circ \mathsf{Eval}_{\mathrm{obs}} : \mathcal{W}_{\mathrm{dec}}^{\mathrm{nf}} \longrightarrow \mathcal{T}_{\mathrm{out}}^{\mathrm{nf}}.$$

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Thus

$$\begin{aligned}
\mathsf{Out}^{\mathrm{tok}}_{\mathrm{obs}}(\mathsf{none}) &= \mathsf{OH},\\
\mathsf{Out}^{\mathrm{tok}}_{\mathrm{obs}}(\mathsf{local}) &= \mathsf{OI},\\
\mathsf{Out}^{\mathrm{tok}}_{\mathrm{obs}}(\mathsf{trans}) &= \mathsf{OB},\\
\mathsf{Out}^{\mathrm{tok}}_{\mathrm{obs}}(\mathsf{trans\,stilllive}) &= \mathsf{OC},\\
\mathsf{Out}^{\mathrm{tok}}_{\mathrm{obs}}(\mathsf{trans\,stilllive\,kernelstratum}) &= \mathsf{OK}.
\end{aligned}$$

Proposition 33.222 (The output-normalform tokens are faithful and complete for the current deterministic decoder results)**.** *Under the current live theorem/tool layer, the token map* $\mathsf{Tok}_{\mathrm{out}}$ *is faithful and complete on the presently normalized decoder outputs. More precisely:*

(i) every value of $\mathsf{Eval}_{\mathrm{obs}}$ *is sent to exactly one token in* $\mathcal{T}^{\mathrm{nf}}_{\mathrm{out}}$*,*

(ii) different current decoder/output pairs receive different tokens,

(iii) the token $\mathsf{Out}^{\mathrm{tok}}_{\mathrm{obs}}(w)$ *determines both the terminal decoder state and the unique normalized reopening route attached to that state,*

(iv) no information from the current finite live interface is lost when passing from explicit output pairs to output-normalform tokens.

Hence the present diagnostic-to-route pipeline may now also be read in the compact form

$$\textit{observer class} \longrightarrow \textit{normal-form word} \longrightarrow \textit{output-normalform token.}$$

Proof. By Proposition 33.219, the map $\mathsf{Eval}_{\mathrm{obs}}$ takes values in exactly five currently normalized decoder/output pairs. Definition 33.221 assigns one token to each of those five pairs and to no others, so every current evaluation value receives exactly one token.

Faithfulness is immediate from the construction: $\mathsf{OH}, \mathsf{OI}, \mathsf{OB}, \mathsf{OC}, \mathsf{OK}$ are pairwise distinct symbols, each tied to one different decoder/output pair. Thus two different current evaluation outputs cannot collapse to the same token. Conversely, each token has exactly one prescribed decoder state and exactly one prescribed normalized reopening route, so the token determines the full current output pair.

Therefore the passage from explicit pairs to compact tokens is lossless on the present finite interface. It is merely a normalization of notation, not a quotient by any nontrivial equivalence. The compact map $\mathsf{Out}^{\mathrm{tok}}_{\mathrm{obs}}$ preserves exactly the same current routing content as $\mathsf{Eval}_{\mathrm{obs}}$, only in shorter QAM-readable form. □

Remark 33.223 (Why the compact output tokens matter)**.** *The live interface can now be written in its shortest current form:*

$$\textit{observer-side recurrence class} \xrightarrow{\mathsf{NF}_{\mathrm{obs}}} \textit{normal-form word} \xrightarrow{\mathsf{Out}^{\mathrm{tok}}_{\mathrm{obs}}} \{\mathsf{OH}, \mathsf{OI}, \mathsf{OB}, \mathsf{OC}, \mathsf{OK}\}.$$

At this point the theorem-side reopening discipline already has a compact output alphabet. A later controller or QAM overlay no longer needs to carry the full explicit route list every time; it may cite the token first and dereference the corresponding route only when the detailed theorem-side reopening path is actually needed.

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19AZ$^{(5a)}$. First signature-injectivity normal form for the live OP 1 kernel

The normalized routing layer has now done its job on the OP 1 side: it has compressed the live reopening path to

$$\mathrm{TR}_4 \Longrightarrow \mathrm{IP}_{10} \Longrightarrow \mathrm{IP}_9 \Longrightarrow \mathrm{IP}_8.$$

The next honest mathematical move is therefore not another routing refinement but a sharper obstruction normal form for the first live carrier in that chain. The remaining burden should now be stated as a local injectivity problem for the retained pinning signature itself: can two genuinely distinct admissible primitive photon-first-zero loops occupy the same retained post-frame corridor class while preserving the same total-channel profile, the same retained optimum location, and the same local comparison/defect readout?

> **Reader cue.** Read the next block as an obstruction normal form for the live OP 1 kernel. Its purpose is to rewrite the remaining burden as exclusion of a second admissible primitive loop with the same retained pinning signature, not to claim that this exclusion has already been proved.

Definition 33.224 (Retained pinning signature on the OP 1 corridor)**.** *Assume the same-corridor reversal certificate of Definition 33.149. For every admissible primitive photon-first-zero closed-loop realization L on the retained post-frame corridor, define the* retained pinning signature

$$\mathsf{Sig}_{\mathrm{pin}}(L)$$

to be the complete retained comparison packet carried by the present line on the intended admissible windows, namely: the retained total-channel profile, the retained optimum location, and the standing local comparison/defect data attached to L in the compressed machine-shadow reading. Thus two loops have the same retained pinning signature exactly when they are indistinguishable to the current OP 1 comparison line at the level that feeds Definition 33.169.

Definition 33.225 (Duplicate-signature witness for the live OP 1 kernel)**.** *Assume Definition 33.224. A* duplicate-signature witness *for the live OP 1 kernel is a pair (L, L') of admissible primitive photon-first-zero closed-loop realizations on the retained post-frame corridor such that:*

(i) *L and L' are genuinely distinct up to the standing admissible corridor equivalence,*

(ii) *both preserve the same retained total-channel profile on the intended admissible windows,*

(iii) $\mathsf{Sig}_{\mathrm{pin}}(L) = \mathsf{Sig}_{\mathrm{pin}}(L')$.

Equivalently, a duplicate-signature witness is a second primitive loop placement that the current comparison line cannot distinguish from the intended retained loop even though it remains genuinely distinct inside the retained corridor class.

Proposition 33.226 (The live OP 1 burden is equivalent to exclusion of duplicate-signature witnesses)**.** *Assume the same-corridor reversal certificate of Definition 33.149. Then the following are equivalent on the intended admissible windows:*

(i) *the optimum-pinning comparison-rigidity certificate of Definition 33.169 holds;*

(ii) *no duplicate-signature witness exists in the sense of Definition 33.225.*

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Consequently, together with Proposition 33.166 and the standing no-second-carrier / strict-upper-gap exclusions already built into the current line, exclusion of duplicate-signature witnesses implies

$$\Delta_{\mathrm{CL}} = 0.$$

Equivalently, the present residual OP 1 burden may now be read exactly as the injectivity of the retained pinning signature on the admissible primitive loop class carried by the retained post-frame corridor.

Proof. Definition 33.169 says that no genuinely distinct admissible primitive loop on the retained post-frame corridor can preserve the retained total-channel profile, the retained optimum location, and the standing local comparison/defect data while remaining distinct up to the admissible corridor equivalence. But this is exactly the negation of a duplicate-signature witness from Definition 33.225, since the retained pinning signature packages precisely those same preserved data. Thus *(i)* and *(ii)* are equivalent.

Once this injectivity statement is available, Proposition 33.170 yields the retained-window rigidity certificate. Proposition 33.166 then upgrades that rigidity, under the standing no-second-carrier and strict-upper-gap exclusions, to admissible-window uniqueness and hence to $\Delta_{\mathrm{CL}} = 0$. □

Remark 33.227 (What the new normal form buys for the next OP 1 step)**.** *This is a real sharpening even though it is not yet a closure theorem. Before the present step, the residual burden was phrased as "prove optimum-pinning comparison rigidity." After the present step, the live target is more concrete: show that the retained comparison packet*

$$\mathsf{Sig}_{\mathrm{pin}}$$

is injective on admissible primitive first-zero loops in the retained post-frame corridor class. In practical terms, the next OP 1 attack should therefore look for a contradiction from a hypothetical duplicate-signature witness, not for a new free continuum-limit parameter. In the normalized QAM reading, this says that the first live carrier IP_{10} *should now be treated as a signature-injectivity problem for the retained pinning packet.*

19AZ$^{(5b)}$. First contradiction channel split for a duplicate-signature witness

The normal form of 19AZ$^{(5a)}$ says that the live OP 1 burden is equivalent to the exclusion of duplicate-signature witnesses. The next honest move is therefore to ask what such a witness would have to do inside the present line. Because the retained pinning signature already fixes the retained total-channel profile, the retained optimum location, and the local comparison/defect packet, any residual distinction between two loops carrying the same signature has only two places left to live: either as a second primitive occupancy of the same retained corridor slot, or as a second admissible completion above that same slot. This is exactly the old no-second-carrier / strict-upper-gap dichotomy, but now pulled back to the new signature normal form.

> **Reader cue.** Read the next block as the first contradiction routing split for the live OP 1 kernel. Its purpose is to show that a hypothetical duplicate-signature witness can only survive by feeding one of the two exclusion gates already built into the current line: duplicate primitive occupancy or duplicate admissible completion.

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Definition 33.228 (Occupancy-side and completion-side duplicate-signature channels)**.** *Assume the same-corridor reversal certificate of Definition 33.149 and let (L, L') be a duplicate-signature witness in the sense of Definition 33.225. We say that the witness is of* occupancy-side type *if the residual distinction between L and L' already appears as a second primitive occupancy of the retained post-frame corridor while the retained total-channel profile is kept fixed. We say that the witness is of* completion-side type *if, after the common retained corridor slot and retained comparison packet are fixed, the residual distinction can only survive as a second admissible completion route above that same retained corridor while preserving the same retained status surface.*

Proposition 33.229 (Every duplicate-signature witness feeds a standing OP 1 exclusion gate)**.** *Assume the same-corridor reversal certificate of Definition 33.149. Let (L, L') be a duplicate-signature witness in the sense of Definition 33.225. Then at least one of the following holds on the intended admissible windows:*

(i) (L, L') is of occupancy-side type in the sense of Definition 33.228;

(ii) (L, L') is of completion-side type in the sense of Definition 33.228.

Consequently, every hypothetical failure of signature-injectivity for the live OP 1 kernel already routes into one of the two standing exclusion channels behind Proposition 33.166: no-second-carrier on the retained corridor, or strict-upper-gap on admissible completion above that corridor.

Proof. By Definition 33.225, the loops L and L' are genuinely distinct up to the standing admissible corridor equivalence, yet they preserve the same retained pinning signature. In particular, they preserve the same retained total-channel profile, the same retained optimum location, and the same local comparison/defect data.

Therefore the current line has already fixed everything that belongs to the retained comparison packet. If the residual distinction between L and L' is visible already at the level of primitive occupancy of the retained post-frame corridor, then the witness is of occupancy-side type by definition. Otherwise the common retained corridor slot and retained comparison packet do not separate the loops, so the remaining distinction can only survive as a second admissible completion above that same retained corridor while preserving the same retained status surface; this is exactly the completion-side case.

Hence every duplicate-signature witness must feed at least one of the two channels. The final sentence is then only a translation into the standing OP 1 language: occupancy-side witnesses are precisely the kind of residual duplication targeted by the no-second-carrier discipline, while completion-side witnesses are precisely the kind of residual duplication targeted by the strict-upper-gap exclusion in Proposition 33.166. □

Remark 33.230 (What this contradiction split buys for the next OP 1 attack)**.** *This still does not close OP 1, but it tells us where a closure attempt must now bite. A hypothetical duplicate-signature witness no longer floats as an abstract rival loop. It is forced into one of two already named contradiction channels. In practical terms, the next OP 1 step should try to show that the retained comparison packet cannot support either occupancy-side duplication on the retained corridor or completion-side duplication above that corridor. In the normalized QAM reading, this means that the live carrier* IP_{10} *has now been resolved into a two-channel contradiction search whose outputs are already aligned with the standing no-second-carrier / strict-upper-gap gates feeding* IP_9*.*

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19AZ$^{(5c)}$. Occupancy-side duplicate-signature witnesses collapse under the retained no-second-carrier discipline

After the channel split of 19AZ$^{(5b)}$, the cleanest next question is whether both contradiction channels are still genuinely live. On the present line, the occupancy-side channel sits closest to the already retained theorematic infrastructure: if a duplicate-signature witness already survives as a second primitive occupancy of the same retained corridor while the retained total-channel profile stays fixed, then it is no longer merely a new OP 1 curiosity. It is exactly the sort of duplication that the retained no-second-carrier discipline was built to forbid on the preserved total-channel side.

> **Reader cue.** Read the next block as the first actual collapse theorem inside the new OP 1 contradiction split. Its purpose is to show that occupancy-side duplicate-signature witnesses are already excluded by the standing retained no-second-carrier discipline, so that the live burden cannot honestly remain two-channel after this point.

Proposition 33.231 (Occupancy-side duplicate-signature witnesses are excluded on the retained OP 1 corridor)**.** *Assume the same-corridor reversal certificate of Definition 33.149. Assume moreover that the retained no-second-carrier discipline continues to hold on the intended admissible windows at the retained total-channel level, in the compressed sense already available through Theorem 21.8. Then no duplicate-signature witness of occupancy-side type exists in the sense of Definitions 33.225 and 33.228.*

Proof. Suppose for contradiction that (L, L') is a duplicate-signature witness of occupancy-side type. By Definition 33.225, the loops L and L' are genuinely distinct up to the standing admissible corridor equivalence, yet they preserve the same retained pinning signature. In particular, they preserve the same retained total-channel profile on the intended admissible windows.

By Definition 33.228, the residual distinction already survives as a second primitive occupancy of the retained post-frame corridor while that retained total-channel profile is kept fixed. But this is precisely the kind of extra primitive retained-carrier occupancy excluded by the standing no-second-carrier discipline at retained total-channel level. Hence the assumed witness contradicts the retained no-second-carrier theorematic layer. Therefore no occupancy-side duplicate-signature witness can exist. □

Corollary 33.232 (The live OP 1 duplicate-signature burden reduces to the completion-side channel)**.** *Under the hypotheses of Proposition 33.231, the following are equivalent on the intended admissible windows:*

(i) a duplicate-signature witness for the live OP 1 kernel exists;

(ii) a duplicate-signature witness of completion-side type exists.

Consequently, after the present step the live OP 1 kernel is no longer honestly read as a two-channel contradiction search. The occupancy-side branch is already discharged by the retained no-second-carrier discipline, so the surviving burden is the exclusion of a second admissible completion above the retained corridor that preserves the same retained pinning signature.

Proof. If a duplicate-signature witness exists, Proposition 33.229 shows that it must be either of occupancy-side type or of completion-side type. Proposition 33.231 eliminates the occupancy-side case. Hence every surviving duplicate-signature witness must be completion-side. The converse is immediate, since every completion-side witness is in particular a duplicate-signature witness. □

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Remark 33.233 (What this first collapse buys for the next OP 1 step)**.** *This is the first point in the new post-release OP 1 line where one of the two contradiction channels is not merely named but actually discharged. The live burden is therefore smaller than it looked in 19AZ$^{(5b)}$: the retained comparison packet is already rigid enough against duplicate* occupancy *once the inherited no-second-carrier discipline is invoked. What remains is subtler and now sharply localized. One must show that the same retained pinning signature cannot support a second* admissible completion *above the already fixed retained corridor. In the normalized QAM reading, the live carrier* IP_{10} *has therefore collapsed from a two-channel contradiction search to a completion-side exclusion problem feeding directly into the strict-upper-gap gate behind* IP_9*.*

Reader cue. Read the next block as the second actual collapse theorem in the new OP 1 contradiction split. Its purpose is to show that completion-side duplicate-signature witnesses are already excluded by the standing retained strict-upper-gap discipline, so that no duplicate-signature witness survives the present OP 1 kernel once both standing exclusion gates are read in the new signature normal form.

19AZ$^{(5d)}$. Completion-side duplicate-signature witnesses collapse under the retained strict-upper-gap discipline

After 19AZ$^{(5c)}$, the live duplicate-signature burden has only one honest place left to survive: as a second admissible completion above the already fixed retained corridor while the retained pinning signature stays unchanged. But this is exactly the sort of residual duplication that the retained strict-upper-gap language was already designed to exclude. Once the retained corridor slot and retained comparison packet are fixed, a genuinely second admissible completion preserving the same retained status surface is no longer a new geometric freedom; it is a forbidden duplicate completion above the same retained corridor.

Proposition 33.234 (Completion-side duplicate-signature witnesses are excluded on the retained OP 1 corridor)**.** *Assume the same-corridor reversal certificate of Definition 33.149. Assume moreover that the retained strict-upper-gap discipline continues to hold on the intended admissible windows in the compressed sense already used by Proposition 33.166. Then no duplicate-signature witness of completion-side type exists in the sense of Definitions 33.225 and 33.228.*

Proof. Suppose for contradiction that (L, L') is a duplicate-signature witness of completion-side type. By Definition 33.225, the loops L and L' are genuinely distinct up to the standing admissible corridor equivalence, yet they preserve the same retained pinning signature. In particular, they preserve the same retained total-channel profile, the same retained optimum location, and the same local comparison/defect packet on the intended admissible windows.

By Definition 33.228, once the common retained corridor slot and retained comparison packet are fixed, the residual distinction between L and L' can only survive as a second admissible completion above that same retained corridor while preserving the same retained status surface. But this is precisely the kind of duplicate admissible completion excluded by the standing strict-upper-gap discipline in the retained OP 1 reading. Hence the assumed witness contradicts the retained strict-upper-gap layer. Therefore no completion-side duplicate-signature witness can exist. □

Corollary 33.235 (No duplicate-signature witness survives the retained OP 1 kernel)**.** *Assume the hypotheses of Proposition 33.231 and Proposition 33.234. Then no duplicate-signature witness for the live OP 1 kernel exists on the intended admissible windows.*

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Proof. If a duplicate-signature witness existed, Corollary 33.232 would force it to be of completion-side type. Proposition 33.234 eliminates that case. Hence no duplicate-signature witness survives. □

Theorem 33.236 (Closure of the live OP 1 kernel under the standing retained exclusion disciplines)**.** *Assume the same-corridor reversal certificate of Definition 33.149. Assume moreover that the retained no-second-carrier discipline and the retained strict-upper-gap discipline both hold on the intended admissible windows in the senses used above. Then the optimum-pinning comparison-rigidity certificate of Definition 33.169 holds on the live OP 1 kernel. Consequently,*

$$\Delta_{\mathrm{CL}} = 0.$$

Equivalently, under the standing retained exclusion disciplines the normalized OP 1 route

$$\mathrm{TR}_4 \Longrightarrow \mathrm{IP}_{10} \Longrightarrow \mathrm{IP}_9 \Longrightarrow \mathrm{IP}_8$$

closes on the intended admissible windows.

Proof. By Corollary 33.235, no duplicate-signature witness exists. Proposition 33.226 therefore yields the optimum-pinning comparison-rigidity certificate. The final sentence of that same proposition then gives $\Delta_{\mathrm{CL}} = 0$ via the already established retained-window-rigidity / admissible-window-uniqueness bridge. □

Remark 33.237 (What the completion-side collapse means for the role of the Monte-Carlo probes)**.** *After the present step, the Monte-Carlo probes with the fusion/decoder reading should not be read as replacements for the OP 1 theorematic closure. Their real leverage is earlier in the search. They compress the search space by classifying residual behavior into a small number of decoder-relevant recurrence classes and by telling us whether a surviving defect should first be treated as* `ip`*-local or as* `bsbl`*-translation-sensitive. That is why the work now feels faster: the probes do not prove the retained no-second-carrier or strict-upper-gap exclusions, but they stop us from reopening the wrong parts of the document and from carrying broad rival-loop pictures once the residual signature is already locally pinned.*

In the present OP 1 line, this means the probes mainly accelerated the route to the right normal form. They helped isolate the live burden as an injectivity problem, then as a two-channel contradiction search, and finally as the completion-side residue that can be handed directly to the strict-upper-gap gate. So the speed-up is real, but it is a search-space reduction and routing advantage*, not a substitute for the structural closure theorems themselves.*

Reader cue. Read the next block as the first honest post-OP 1 normalization of the live OP 5 Step (B) burden. Its purpose is not to declare the routed photon-frame balance defect solved, but to identify the smallest currently licensed structural kernel on which that burden now lives: the production or exclusion of a carrier-upgraded routed-balance witness at the routed photon-frame cutoff.

19AZ$^{(6a)}$. First carrier-upgraded witness normal form for the live OP 5 Step (B) kernel

After the closure of the live OP 1 kernel, the honest next burden is OP 5 Step (B), i.e. the routed photon-frame balance defect $\Delta_{\Theta R}(T)$. The current line should not reopen that burden through every packet-wave, breathing-envelope, frame-flicker, or helix picture at once. The earlier carrier-first OP 5 architecture already teaches that visible mismatches come in two grades: either they remain screening-only and therefore subordinate to one still-available

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common closure-bearing return-side carrier package, or they upgrade and certify loss of that common carrier package itself. The cleanest first OP 5(B) normalization is therefore to ask not for a new global formula immediately, but for the smallest carrier-grade witness through which a still-live routed balance defect could honestly persist.

Definition 33.238 (Routed-balance witness for the live OP 5 Step (B) kernel)**.** *Fix the routed photon-frame cutoff $T = \gamma_L^{-3/2}$ and the routed balance defect $\Delta_{\Theta R}(T)$ from Definition 33.126. Under the standing carrier-first OP 5 architecture, an admitted visible witness w is called a* routed-balance witness for the live OP 5 Step (B) kernel *if w is cited to support the claim that the routed photon-frame balance layer remains structurally live at the cutoff T. Such a witness is called*

1. screening-only *if it remains screening-only in the sense of Lemma 32.51;*
2. carrier-upgraded *if it is carrier-upgraded in the same sense.*

Thus a routed-balance witness is not a new witness species; it is the OP 5 Step (B) reading of the already licensed witness-grade dichotomy at the routed photon-frame cutoff.

Proposition 33.239 (The live OP 5 Step (B) kernel is a carrier-upgraded witness problem)**.** *Assume Proposition 32.30, Proposition 32.36, Lemma 32.45, Lemma 32.48, Lemma 32.51, and Corollary 32.55. Then the honest live OP 5 Step (B) kernel on the present line is the following alternative:*

(i) either every routed-balance witness at the routed photon-frame cutoff remains screening-only, in which case no carrier/package-level OP 5 Step (B) obstruction has yet been exhibited;

(ii) or there exists a carrier-upgraded routed-balance witness, in which case the live OP 5 Step (B) burden has genuinely reached the carrier/package layer.

In particular, the first honest structural reopening of $\Delta_{\Theta R}(T)$ is not an arbitrary visible mismatch, but the production or exclusion of a carrier-upgraded routed-balance witness.

Proof. By Lemma 32.51, every admitted visible witness at the carrier-first OP 5 layer lies in exactly one of two grades: screening-only or carrier-upgraded. Definition 33.238 does nothing more than read that existing dichotomy at the routed photon-frame cutoff T and relative to the residual $\Delta_{\Theta R}(T)$. If a cited witness remains screening-only, then by Corollary 32.55 its upstream content is still representable through some common closure-bearing return-side carrier package; hence no carrier/package-level OP 5 Step (B) obstruction has yet been exhibited. If, on the other hand, a cited witness upgrades to carrier grade, then the same corollary says that no such common carrier package remains available, so the routed balance burden has genuinely reached the carrier/package layer. Since no third currently licensed witness grade exists, the displayed alternative is exhaustive. Therefore the first honest structural OP 5 Step (B) kernel is exactly the production or exclusion of a carrier-upgraded routed-balance witness. □

Corollary 33.240 (First normalized reopening route for the live OP 5 Step (B) kernel)**.** *Under the hypotheses of Proposition 33.239, the smallest currently licensed normalized reopening route for a genuinely live OP 5 Step (B) witness is*

$$\mathrm{TR}_4 \Longrightarrow \mathrm{BS}_1 \Longrightarrow \mathrm{BS}_3 \Longrightarrow \mathrm{BL}_2 \Longrightarrow \mathrm{BL}_4 \Longrightarrow \mathrm{BL}_5,$$

with the last two steps used only when the routed-balance witness remains live after the masked-single-carrier test and localizes at the readout/kernel stratum.

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Proof. The total-channel baseline remains frozen by TR_4. Because the routed photon-frame balance defect is not a pure local optimum-pinning problem but a return-side witness-grade problem, the first honest move is bridge-side rather than imbalance-side: the route enters through the matrix/phase surface-change gates BS_1 and BS_3. Once that surface change has been made, the first honest diagnostic question is whether the putative routed-balance witness is still compatible with one masked single carrier, which is precisely the gate BL_2. If it collapses there, no deeper reopening is licensed. If it remains live, then BL_4 is exactly the normalized stratification step that localizes the first remaining obstruction. And when that localized obstruction sits in the readout/kernel stratum, BL_5 is the recorded compression to absence of one common shell-stable carrier package. Therefore the displayed BS/BL route is exactly the smallest currently licensed reopening route for a genuinely live OP 5 Step (B) witness. □

Remark 33.241 (What this first OP 5 Step (B) normalization buys)**.** *This is not yet an OP 5 Step (B) closure theorem. The routed photon-frame balance defect $\Delta_{\Theta R}(T)$ is still honestly live. But the burden is now sharper. It is no longer carried as an undifferentiated packet/helix/flicker mismatch surface. It is carried as a witness-grade question: does any routed-balance witness actually upgrade to carrier/package level, or do all currently visible routed witnesses remain screening-only beneath one still-available common return-side carrier package? In the present QAM language, OP 1 was compressed onto the* IP*-route, whereas the live OP 5 Step (B) burden now announces itself as a* BS/BL*-route burden from the start.*

Reader cue. Read the next block as the first negative-side sharpening of the live OP 5 Step (B) kernel. Its purpose is not yet to close the routed photon-frame balance defect, but to identify the first retained situation in which a routed-balance witness cannot honestly escalate to carrier/package grade at all.

19AZ$^{(6b)}$. Carrier-stable routed returns prevent witness-grade escalation on the live OP 5 Step (B) kernel

The previous normalization reduced the live OP 5 Step (B) burden to the production or exclusion of a carrier-upgraded routed-balance witness. The next honest question is whether every routed-balance witness must be tested all the way down to the carrier/package layer, or whether one already has a retained positive branch on which such escalation is structurally forbidden. The earlier routed-return package answers this. Once the routed realization lies on the carrier-stable routed return normal form, every admitted visible routed mismatch is already subordinate to one common closure-bearing return-side carrier. In that situation the witness-grade question collapses on the positive side: a routed-balance witness may still exist as screening-level visible data, but it can no longer count as a carrier-grade OP 5 Step (B) witness.

Proposition 33.242 (Carrier-stable routed returns prevent routed-balance witness escalation)**.** *Assume Proposition 33.239, Corollary 33.240, Corollary 32.55, Theorem 32.57, and Theorem 32.69. Let $(b, r, s; w)$ be a routed branch-shell realization at the cutoff $T = \gamma_L^{-3/2}$ together with a routed-balance witness w in the sense of Definition 33.238. If $(b, r, s; w)$ lies on the* carrier-stable routed return normal form *of Theorem 32.69, then w remains screening-only and cannot be carrier-upgraded. In particular, no carrier-upgraded routed-balance witness exists on the carrier-stable routed-return branch.*

Proof. By Theorem 32.69, the carrier-stable routed return normal form means precisely that there exists one common closure-bearing return-side carrier through which the routed realization is read and that every admitted visible routed mismatch collapses beneath that

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carrier to the carrier-stable visible normal form. Apply this to the routed-balance witness w. Theorem 32.57 then says that, over that same stable carrier package, w has only one current structural role: it witnesses subordinate screening-loss normal form and does not define an independent carrier-relevant visible species. But Corollary 32.55 identifies carrier upgrade exactly with the first failure of every common closure-bearing return-side carrier-supported realization. Since the carrier-stable routed return normal form explicitly retains such a common carrier package, the threshold for carrier upgrade is not met. Hence w remains screening-only and cannot be carrier-upgraded. Because w was arbitrary, no carrier-upgraded routed-balance witness exists on the carrier-stable routed-return branch. □

Corollary 33.243 (The live OP 5 Step (B) burden survives only on the obstructed routed-return branch)**.** *Under the hypotheses of Proposition 33.242, a genuinely live OP 5 Step (B) burden can survive only on the* obstructed routed return normal form *of Theorem 32.69. Equivalently, after the gates* $\mathrm{TR}_4 \Rightarrow \mathrm{BS}_1 \Rightarrow \mathrm{BS}_3 \Rightarrow \mathrm{BL}_2$*, any routed-balance witness that still honestly threatens carrier/package loss must already belong to the negative routed-return branch; and when that negative branch localizes at the readout/kernel stratum,* BL_5 *compresses the remaining burden to absence of one common shell-stable carrier package.*

Proof. By Proposition 33.239, the live OP 5 Step (B) kernel is exactly the production or exclusion of a carrier-upgraded routed-balance witness. Proposition 33.242 excludes such a witness on the carrier-stable routed-return branch. Therefore any genuinely live routed-balance burden must sit on the only remaining routed-return branch, namely the obstructed routed return normal form of Theorem 32.69. Corollary 33.240 already records that, after the masked-single-carrier gate BL_2, the next honest localization step is BL_4, and that a readout/kernel localization is then compressed by BL_5 to absence of one common shell-stable carrier package. This is exactly the stated reformulation. □

Remark 33.244 (What this second OP 5 Step (B) sharpening buys)**.** *The live OP 5 Step (B) burden is now smaller than in the local pass. It is no longer "any routed-balance witness that might still upgrade." It is "a routed-balance witness on the obstructed routed-return branch that survives the masked-single-carrier test and, if needed, localizes further." This is also the point at which the Monte-Carlo/fusion probes materially accelerate the search: once the probes classify a run as carrier-stable on the routed-return side, one no longer needs to pursue carrier-grade OP 5 escalation there. The probes still do not prove the theorematic branch separation by themselves, but they do cut away large amounts of blind BS/BL reopening by steering attention directly to the small class of routed returns that still look obstruction-like after the carrier-stable branch has been excluded.*

Reader cue. Read the next block as the first localization split of the remaining OP 5 Step (B) burden on the negative routed-return side. Its purpose is not yet to declare the obstructed routed return harmless, but to separate the part that is already inherited from the upstream OP3/QAM obstruction package from the part that would remain as a genuinely downstream carrier-grade OP 5 obstruction.

19AZ$^{(6c)}$. Obstructed routed returns split into upstream-screened versus downstream carrier-grade OP 5 Step (B) witnesses

After 19AZ$^{(6b)}$, the live OP 5 Step (B) burden survives only on the obstructed routed-return branch. But even this negative branch is still too coarse to be the final witness kernel. The already extracted global obstruction package says that, once the admissible run is inside the bundled OP1→branch→OP3/QAM→OP5 channel, the first licensed failure is already

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localized either upstream in the OP3/QAM obstruction package or downstream at the carrier-upgraded OP5 obstruction layer. Therefore the obstructed routed-return branch must itself be split: either the routed obstruction is only the return-side shadow of an upstream OP3/QAM obstruction already fixed before carrier-grade OP5 escalation, or it is a genuinely downstream carrier-grade OP5 obstruction that survives that upstream screening.

Definition 33.245 (Upstream-screened versus downstream carrier-grade obstructed routed-return witness)**.** *Assume the hypotheses of Corollary 33.243. Let* $(b, r, s; w)$ *be a routed branch-shell realization on the* obstructed routed return normal form *together with a routed-balance witness* w*.*

(i) We say that $(b, r, s; w)$ *is an* upstream-screened obstructed routed-return witness *if the currently licensed negative routed-return reading of* $(b, r, s; w)$ *is already forced by the upstream OP3/QAM obstruction package of Proposition 32.79, so that the witness does not yet certify an independent downstream carrier-grade OP 5 obstruction.*

(ii) We say that $(b, r, s; w)$ *is a* downstream carrier-grade obstructed routed-return witness *if the witness remains honestly live after the upstream OP3/QAM obstruction screening and therefore certifies a genuinely downstream carrier-upgraded OP 5 obstruction layer in the sense of Proposition 32.79.*

Proposition 33.246 (The remaining obstructed routed-return burden is an upstream/downstream split)**.** *Assume Proposition 33.239, Corollary 33.243, Proposition 32.79, and Proposition 32.80. Let* $(b, r, s; w)$ *be a genuinely live routed-balance witness on the obstructed routed return normal form. Then exactly one of the following two cases is currently licensed.*

(i) ***Upstream-screened case.*** *The negative routed-return reading of* $(b, r, s; w)$ *is already accounted for by the upstream OP3/QAM obstruction package, and the witness is therefore not yet an independent downstream carrier-grade OP 5 obstruction.*

(ii) ***Downstream carrier-grade case.*** *The witness survives the upstream OP3/QAM obstruction screening and therefore remains as a genuinely downstream carrier-grade OP 5 obstruction witness.*

No third autonomous obstructed routed-return witness class is currently licensed.

Proof. By Corollary 33.243, every genuinely live OP 5 Step (B) witness must already lie on the obstructed routed return normal form. Once the run is inside the bundled OP1→branch→OP3/QAM→OP5 channel, Proposition 32.79 says that the first licensed failure is already localized either upstream in the OP3/QAM obstruction package or downstream at the carrier-upgraded OP5 obstruction layer. Proposition 32.80 adds that no third autonomous terminal visible class is licensed outside that bundled obstruction reading. Applying this dichotomy to the obstructed routed-return witness $(b, r, s; w)$ yields exactly the two stated cases. The first is the upstream-screened case, where the negative routed-return picture is merely the return-side shadow of an obstruction already fixed upstream. The second is the downstream carrier-grade case, where the witness remains live after that upstream screening and therefore truly belongs to the OP 5 obstruction layer. Exclusivity and exhaustiveness are precisely the content of the bundled global obstruction dichotomy. □

Corollary 33.247 (The genuinely new OP 5 Step (B) burden survives only on the downstream branch)**.** *Under the hypotheses of Proposition 33.246, any genuinely* new *OP 5 Step (B) burden survives only in the downstream carrier-grade case of Definition 33.245. Equivalently, once the obstructed routed return has been shown not to be merely the return-side shadow of the upstream OP3/QAM obstruction package, the remaining burden is already localized to the downstream carrier-upgraded OP 5 obstruction layer.*

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Proof. In the upstream-screened case, the negative routed-return witness is already accounted for by the upstream OP3/QAM obstruction package and therefore does not constitute a genuinely new downstream OP 5 burden. Hence only the downstream carrier-grade case can carry a genuinely new OP 5 Step (B) burden after the upstream screening has been discharged. This is exactly the stated reformulation. □

Remark 33.248 (What this third OP 5 Step (B) sharpening buys)**.** *The live OP 5 Step (B) burden is now smaller again. It is no longer merely "a witness on the obstructed routed-return branch." It is "a witness on that negative branch that is not already explained by the upstream OP3/QAM obstruction package." This is also where the Monte-Carlo/fusion probes become strategically useful on the negative side: they do not prove the upstream/downstream split, but they help tell much earlier whether a run looks like an inherited OP3/QAM obstruction shadow or like a genuinely downstream carrier-grade obstruction. In practical search terms, this means that many apparently live negative routed returns can now be discarded upstream before one spends time reopening the deepest OP 5 carrier-grade layer.*

Reader cue. Read the next block as the discharge of the upstream-screened side of the remaining OP 5 Step (B) burden. Its purpose is not yet to close the downstream carrier-grade obstruction kernel, but to remove every negative routed-return witness whose content is already absorbed by the upstream OP3/QAM obstruction package.

19AZ$^{(6d)}$. Upstream-screened obstructed routed returns are discharged before the downstream OP 5 Step (B) carrier-grade kernel begins

After 19AZ$^{(6c)}$, the live negative routed-return burden has only two currently licensed readings: upstream-screened or downstream carrier-grade. But the upstream-screened case is not merely weaker; it is not an independent OP 5 Step (B) burden at all. By definition it is already the routed return shadow of an obstruction whose first licensed failure sits upstream in the OP3/QAM package. Therefore the honest live OP 5 Step (B) kernel starts only after that upstream explanation has been discharged.

Proposition 33.249 (Upstream-screened obstructed routed-return witnesses do not define an independent OP 5 Step (B) burden)**.** *Assume Proposition 33.246, Corollary 33.247, Proposition 32.79, and Proposition 32.80. Let $(b, r, s; w)$ be an upstream-screened obstructed routed-return witness in the sense of Definition 33.245. Then $(b, r, s; w)$ does not define an independent live OP 5 Step (B) burden; its entire negative routed-return content is already discharged by the upstream OP3/QAM obstruction package.*

Proof. By Definition 33.245, an upstream-screened obstructed routed-return witness is precisely one whose currently licensed negative routed-return reading is already forced by the upstream OP3/QAM obstruction package. Proposition 32.79 says that once the admissible run is inside the bundled OP1→branch→OP3/QAM→OP5 channel, the first licensed failure is already localized either upstream in that OP3/QAM obstruction package or downstream at the carrier-upgraded OP5 obstruction layer. Since the witness is, by hypothesis, upstream-screened, the first licensed failure is already on the upstream side of that dichotomy. Proposition 32.80 excludes any third autonomous terminal visible class outside that bundled obstruction reading. Therefore no additional independent OP 5 Step (B) burden remains attached to $(b, r, s; w)$ once the upstream package has been acknowledged. In other words, the routed-return shadow carried by $(b, r, s; w)$ is fully discharged upstream and does not itself define a genuinely new OP 5 Step (B) burden. □

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Corollary 33.250 (The live OP 5 Step (B) kernel reduces to the downstream carrier-grade obstructed routed-return branch)**.** *Under the hypotheses of Proposition 33.249, the honest live OP 5 Step (B) kernel is exactly the production or exclusion of a downstream carrier-grade obstructed routed-return witness. Equivalently, once the upstream-screened negative routed-return cases have been discharged, the only remaining live OP 5 Step (B) burden sits on the downstream carrier-grade branch isolated in Definition 33.245.*

Proof. By Proposition 33.246, every genuinely live obstructed routed-return witness lies in exactly one of two currently licensed classes: upstream-screened or downstream carrier-grade. Proposition 33.249 removes the first class as an independent OP 5 Step (B) burden. Therefore only the downstream carrier-grade class remains available as the honest live OP 5 Step (B) kernel. This is exactly the stated reformulation. □

Remark 33.251 (What this fourth OP 5 Step (B) sharpening buys)**.** *The live OP 5 Step (B) burden is now concentrated on one branch only. It is no longer "an obstructed routed-return witness that is not yet known to be harmless." It is "a downstream carrier-grade obstructed routed-return witness after the upstream OP3/QAM shadow cases have been discharged." This is the first point on the OP 5 side where the negative branch has been reduced to a single honest carrier-grade kernel. Strategically, this is also where the Monte-Carlo/fusion probes become most valuable as a search accelerator: once a negative routed return looks upstream-screened, it can be discarded without reopening the downstream carrier-grade layer, so the expensive search is reserved only for the small class of runs that still look honestly downstream after the upstream package has been factored out.*

Reader cue. Read the next block as the compression of the remaining downstream OP 5 Step (B) kernel to one shell-package deficit question on the retained branch-shell corridor. Its purpose is not yet to declare OP 5 Step (B) globally closed, but to show that once the upstream shadow cases are gone, the honest remaining burden is exactly the absence of one common shell-stable carrier package.

19AZ$^{(6e)}$. Downstream carrier-grade obstructed routed returns compress to one common shell-stable carrier-package deficit on the retained branch-shell corridor

After 19AZ$^{(6d)}$, the live OP 5 Step (B) burden is concentrated on the downstream carrier-grade obstructed routed-return branch. The smallest honest next question is therefore no longer whether some vague routed mismatch survives, but whether the retained branch-shell corridor still admits one common shell-stable carrier package once that downstream witness has survived every earlier screening. On the current line this is exactly the readout/kernel compression question behind BL_5: if the branch and shell strata themselves remain intact on the retained routed corridor, then every remaining downstream carrier-grade witness is nothing but failure of one common shell-stable carrier package.

Definition 33.252 (Downstream shell-package deficit witness for the live OP 5 Step (B) kernel)**.** *Fix the live OP 5 Step (B) setup of Definition 33.238. A quadruple $(b, r, s; w)$ is called a* downstream shell-package deficit witness for the live OP 5 Step (B) kernel *if*

(i) $(b, r, s; w)$ *is a downstream carrier-grade obstructed routed-return witness in the sense of Definition 33.245, and*

(ii) on the retained routed branch-shell corridor determined after the gates $\mathrm{TR}_4 \Rightarrow \mathrm{BS}_1 \Rightarrow \mathrm{BS}_3 \Rightarrow \mathrm{BL}_2$*, the branch and shell strata remain intact, so the remaining localized burden*

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is entirely in the readout/kernel stratum and therefore equals failure of one common shell-stable carrier package in the sense of Lemma 1.204.

Proposition 33.253 (Downstream carrier-grade OP 5 witnesses compress to one common shell-stable carrier-package deficit on the retained branch-shell corridor)**.** *Assume Corollary 33.250, Corollary 33.240, Theorem 32.69, Lemma 1.203, and Lemma 1.204. Let* $(b, r, s; w)$ *be a genuinely live downstream carrier-grade obstructed routed-return witness for the OP 5 Step (B) kernel. If, on the retained routed branch-shell corridor carried through* $\mathrm{TR}_4 \Rightarrow \mathrm{BS}_1 \Rightarrow \mathrm{BS}_3 \Rightarrow \mathrm{BL}_2$*, the branch and shell strata remain intact, then* $(b, r, s; w)$ *is exactly a downstream shell-package deficit witness in the sense of Definition 33.252. Equivalently, under the retained branch-shell intactness assumptions, the whole remaining downstream OP 5 Step (B) burden compresses to absence of one common shell-stable carrier package.*

Proof. By Corollary 33.250, every genuinely live OP 5 Step (B) burden is already a downstream carrier-grade obstructed routed-return witness. Corollary 33.240 records that after the retained gates $\mathrm{TR}_4 \Rightarrow \mathrm{BS}_1 \Rightarrow \mathrm{BS}_3 \Rightarrow \mathrm{BL}_2$, the first further honest localization step is BL_4, and that whenever the first remaining localized obstruction lies in the readout/kernel stratum, the compression gate BL_5 rewrites the burden as absence of one common shell-stable carrier package. Now assume that on the retained routed branch-shell corridor the branch and shell strata themselves remain intact. Then Lemma 1.203 forces every remaining first local obstruction into the readout/kernel stratum, and Lemma 1.204 identifies that stratum exactly with failure of one common shell-stable carrier package. Therefore the surviving downstream carrier-grade witness $(b, r, s; w)$ is not a broader residual species: on the retained branch-shell corridor it is precisely a downstream shell-package deficit witness. This is the stated compression. □

Corollary 33.254 (The live OP 5 Step (B) kernel reduces to one common shell-stable carrier-package deficit problem on the retained branch-shell corridor)**.** *Under the hypotheses of Proposition 33.253, the honest live OP 5 Step (B) kernel on the retained branch-shell corridor is exactly the production or exclusion of a downstream shell-package deficit witness. Equivalently, once the upstream-screened cases have been discharged and the branch/shell strata remain intact, OP 5 Step (B) is reduced to the single question whether one common shell-stable carrier package is absent.*

Proof. Corollary 33.250 says that the live OP 5 Step (B) burden is already concentrated on downstream carrier-grade obstructed routed-return witnesses. Proposition 33.253 then shows that, under retained branch-shell intactness, every such witness is exactly a downstream shell-package deficit witness. Hence the remaining kernel is neither a diffuse routed-return problem nor a broader carrier-grade witness search; it is exactly the stated one-package deficit problem. □

Remark 33.255 (What this fifth OP 5 Step (B) sharpening buys)**.** *This is the first point on the OP 5 side where the surviving kernel is no longer merely a branch classification problem. It has become a concrete deficit question. The live burden is now: not "some downstream obstruction somewhere on the negative routed-return branch," but "the absence of one common shell-stable carrier package on the retained branch-shell corridor." This is also why the Monte-Carlo/fusion probes feel so effective here. Once the probes indicate that the run is both downstream and branch/shell-intact, the search no longer needs to explore broad alternative witness pictures; it can go straight to the small shell-package deficit kernel that* BL_5 *was designed to expose.*

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oindent**Reader cue.** Read the next block as the final conservative test of the live OP 5 Step (B) kernel on the retained branch-shell corridor. Its purpose is not to claim a new global closure principle from scratch, but to show that once the surviving downstream kernel is already compressed to one shell-package deficit question, the inherited OP 3/QAM shell-closure package removes exactly that final deficit under the retained shell-transfer discipline.

19AZ$^{(6f)}$. The inherited shell-closure package discharges the downstream shell-package deficit on the retained branch-shell corridor

After 19AZ$^{(6e)}$, the live OP 5 Step (B) kernel has been compressed as far as possible without reusing the earlier OP 3/QAM shell package. The honest remaining question is therefore very narrow: can the retained branch-shell corridor both inherit the shell-closure package and yet still fail to admit one common shell-stable carrier package? On the present line the answer is no. Once the retained branch-shell corridor is genuinely read through the already established shell certificate / shell-stable kernel-readout package, the missing common shell-stable carrier of 19AZ$^{(6e)}$ is exactly the kind of deficit that the inherited package forbids.

Definition 33.256 (Retained shell-transfer discipline for the live OP 5 Step (B) kernel)**.** *Fix the live OP 5 Step (B) setup of Definition 33.238. We say that the retained routed branch-shell corridor satisfies the* retained shell-transfer discipline *if, after the retained gates* $\mathrm{TR}_4 \Rightarrow \mathrm{BS}_1 \Rightarrow \mathrm{BS}_3 \Rightarrow \mathrm{BL}_2$*, the routed branch-shell realization inherits the shell admissibility certificate and the equivalent shell-stable kernel/readout package from Theorem 1.205. Equivalently, on that retained corridor there exists one common shell-stable kernel/readout carrier through which all closure-relevant routed readout factors.*

Proposition 33.257 (The inherited shell-closure package excludes downstream shell-package deficit witnesses on the retained branch-shell corridor)**.** *Assume Corollary 33.254, Definition 33.256, Theorem 1.205, and Theorem 1.201. Then no downstream shell-package deficit witness in the sense of Definition 33.252 can exist on the retained branch-shell corridor. Equivalently, once the retained branch-shell corridor inherits the shell certificate / shell-stable kernel-readout package, the remaining OP 5 Step (B) kernel from 19AZ$^{(6e)}$ is discharged.*

Proof. Assume toward contradiction that a downstream shell-package deficit witness exists on the retained branch-shell corridor. By Definition 33.252, this means that the surviving downstream carrier-grade routed-return witness is already compressed to absence of one common shell-stable carrier package on that corridor. But by Definition 33.256, the same retained corridor inherits the shell admissibility certificate and hence, by Theorem 1.205 together with Theorem 1.201, admits one common shell-stable kernel/readout carrier through which the closure-relevant routed readout factors. This is exactly the negation of the shell-package deficit asserted by the witness. The contradiction shows that no downstream shell-package deficit witness can exist once the retained shell-transfer discipline holds. Therefore the remaining OP 5 Step (B) kernel from Corollary 33.254 is discharged on that corridor. □

Corollary 33.258 (Closure of the live OP 5 Step (B) kernel under the standing retained branch-shell and shell-transfer disciplines)**.** *Assume the standing retained branch-shell intactness hypotheses from Proposition 33.253 together with the retained shell-transfer discipline of Definition 33.256. Then the live OP 5 Step (B) kernel closes:*

$$\Delta_{\Theta R} = 0.$$

Equivalently, under the current retained no-upstream-shadow, branch/shell-intact, and inherited shell-package disciplines, no honest live OP 5 Step (B) witness survives.

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Proof. By Corollary 33.254, the live OP 5 Step (B) kernel under the retained branch-shell hypotheses is exactly the production or exclusion of a downstream shell-package deficit witness. Proposition 33.257 excludes precisely that witness class once the retained shell-transfer discipline holds. Hence no live OP 5 Step (B) witness remains, so the kernel closes and $\Delta_{\Theta R} = 0$. □

Remark 33.259 (What this sixth OP 5 Step (B) sharpening buys)**.** *This is the OP 5 analogue of the OP 1 local closure step. The result is intentionally conservative. It does not say that every historical reading of OP 5 is now globally absorbed into one theorematic sentence. It says something narrower and structurally cleaner: once the live OP 5 Step (B) burden has been reduced to one downstream shell-package deficit on the retained branch-shell corridor, the inherited OP 3/QAM shell-closure package forbids exactly that final deficit. Strategically, this is also the point where the Monte-Carlo/fusion probes show their full value on the negative routed-return side. They do not prove the inherited shell package, but they steer the search quickly onto the tiny downstream/branch-shell-intact corridor where the shell-transfer discipline can be applied and the final witness class disappears.*

Remark 33.260 (Historical transition note for the archived v2.27.x release-control fragment)**.** *Read the next local post-release fragment as archival proof-memory from an earlier public hardening cycle. Its purpose is to preserve release genealogy, checkpoint recall, and local baseline memory, not to function as a current roadmap, freeze checklist, or active control surface for the present 4.7.0 line.*

Archived archival stage local post-release note (historical baseline memory)

34 Release orientation

This document is presented as the full monolithic Companion text of the current line. Within that full monolithic presentation, the newly integrated archival stage block freezes the mathematically relevant pre-OP foundation developed in the current working stage. The release claim of that integrated block is therefore intentionally limited and precise:

1. the groundwork layer before OP 1–5 is completed as a coherent route-faithful structure;
2. the current local state is conservatively instantiated as a frontier-certified pre-entry state;
3. the line already supports controlled witness-based descent and lead-guided frontier resolution;
4. explicit OP 1–5 work is intentionally deferred to the next stage.

35 The three-coordinate live residual interface

Definition 35.1 (Three-coordinate live residual interface)**.** *The current line is organized around the live residual interface*

$$\mathbf{R}_{\text{live}} = (\rho_{\text{CL}}, \rho_{\Theta}, \rho_{A}),$$

where ρ_{CL} is the horizon-side scalar closure coordinate, while $(\rho_{\Theta}, \rho_{A})$ are the boundary-side slow phase and slow envelope packet coordinates.

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Remark 35.2. *The current open residual work is therefore carried by exactly three live coordinates: one scalar horizon-side coordinate and a two-component boundary-side packet coordinate.*

Definition 35.3 (Current normalized state)**.** *Using a current admissible normalization scale triple*

$$(s_{\mathrm{CL,curr}}, s_{\Theta,\mathrm{curr}}, s_{A,\mathrm{curr}}),$$

the current normalized live residual state is

$$\widetilde{\mathbf{R}}_{\mathrm{curr}} = (\widetilde{\rho}_{\mathrm{CL,curr}}, \widetilde{\rho}_{\Theta,\mathrm{curr}}, \widetilde{\rho}_{A,\mathrm{curr}}),$$

with the obvious channel-wise normalization.

36 Paired conservative machine and compression

Definition 36.1 (Local paired conservative machine)**.** *The local paired conservative machine consists of the paired intake, request, response, fulfillment, resolution, close-or-promote, cycle, return, and branch layers, always preserving:*

1. *the inherited horizon/boundary distinction,*
2. *the inherited earliest-boundary stop,*
3. *certificate-neutrality.*

Definition 36.2 (Compact paired conservative normal form)**.** *The compact paired conservative normal form is the compressed ordered object*

$$\mathcal{N}^{*}_{\mathrm{pair}} = (\mathcal{N}_{\mathrm{hor}}, \mathcal{N}_{\mathrm{bdry}}),$$

which records only the downstream-relevant scalar horizon-side content and packet boundary-side content carried by the paired machine.

Theorem 36.3 (Compression and invariant control)**.** *Every finite route-faithful local paired conservative machine admits a compact paired conservative normal form that preserves the inherited horizon/boundary distinction, the inherited earliest-boundary stop, and certificate-neutrality, while remaining readable through the extracted live residual state.*

Proof. The paired machine is finite at each local stage, and its admissible content separates into one horizon-side scalar channel and one boundary-side packet channel. Compression therefore produces an ordered summary object preserving exactly the route-faithful invariants built into the original machine. □

37 Tool layer

Definition 37.1 (Extracted tool layer)**.** *The extracted tool layer consists of:*

1. *compression-before-comparison,*
2. *monotone selection on the compressed state,*
3. *a route-faithful close-or-promote shortcut,*

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4. *scalar-packet decoupling for genuinely one-sided arguments.*

Theorem 37.2 (Tool extraction theorem)**.** *The compact paired conservative normal form supports the extracted tool layer entirely through the preserved route-faithful invariant class and the extracted three-coordinate residual state.*

Proof. Each tool is defined directly on the compact normal form and depends only on route-faithful invariants and on the residual coordinates. No reopening of the full internal machine is required unless the compressed tool layer is insufficient. □

38 Tool stability and normalization

Definition 38.1 (Tool-stable invariant regime)**.** *A compact paired conservative normal form lies in a tool-stable invariant regime if monotone selectors are compression-stable, close-side states are route-faithfully robust, promote-side continuation is honest rather than a pure weight artifact, and the preserved route-faithful invariants remain unchanged under admissible uses of the tool layer.*

Definition 38.2 (Canonical normalization regime)**.** *A canonical normalization regime is an admissible class of normalization scale triples in which the selector class and close/promote classification remain unchanged under bounded side-preserving scale replacements.*

Theorem 38.3 (Normalization theorem)**.** *Every compact paired conservative normal form equipped with an admissible normalization scale triple admits a normalized live residual reading and a normalized weighted residual energy*

$$\widetilde{\mathcal{E}}_w = w_{\mathrm{CL}}|\widetilde{\rho}_{\mathrm{CL}}|^2 + w_{\Theta}|\widetilde{\rho}_{\Theta}|^2 + w_A|\widetilde{\rho}_A|^2.$$

Inside a canonical normalization regime, comparison, selection, and close/promote testing are independent of arbitrary raw scale imbalance.

Proof. Normalization is performed channel-wise inside the route-faithful split. Within the canonical normalization regime, equivalent admissible scale choices preserve the relevant comparisons and branch classifications, so the normalized state is the preferred pre-OP working surface. □

39 Residual geometry

Definition 39.1 (Normalized residual geometry)**.** *The normalized residual space carries:*

1. *admissible descent sectors and strict descent sectors,*
2. *trapping close basins,*
3. *honest promote regions,*
4. *a residual frontier separating the geometrically unresolved zone.*

Definition 39.2 (Residual frontier)**.** *The residual frontier is the part of normalized residual space lying in neither a trapping close basin nor an honest promote region.*

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Theorem 39.3 (Geometric control theorem)**.** *In a fixed canonical normalization regime, the normalized residual space admits a route-faithful geometric control structure consisting of admissible descent sectors, close basins, honest promote regions, a residual frontier, and descent selectors on finite admissible successor families.*

Proof. These structures are defined entirely within the normalized residual space and remain inside the same route-faithful invariant class. Each construction therefore preserves the inherited horizon/boundary distinction, earliest-boundary stop, and certificate-neutrality. □

40 Residual dynamics

Definition 40.1 (Admissible descent trajectory)**.** *An admissible descent trajectory is a sequence of normalized residual states in which each successor lies in the admissible descent sector of its predecessor. If strict descent is chosen whenever available, the trajectory is called strictly selected.*

Lemma 40.2 (First no-oscillation lemma)**.** *A strictly selected admissible descent trajectory cannot exhibit a two-step oscillation between distinct states if strict descent is available at both states.*

Proof. Strict descent at the first state gives strict energy decrease; strict descent at the second state gives another strict decrease. Returning to the first state would therefore force the same energy to be both strictly larger and strictly smaller than itself, a contradiction. □

Theorem 40.3 (Dynamic control theorem)**.** *Admissible descent dynamics on the normalized residual space has nonincreasing normalized weighted residual energy. Under strict descent, two-step oscillation is excluded; close-side behavior is represented by trapping close basins; promote-side behavior is represented by persistent honest promote regions; and every honest promote-to-close transition must pass through the residual frontier.*

Proof. The monotonicity of energy follows from admissible descent. The no-oscillation claim is Lemma 40.2. The basin and frontier statements are the dynamic reading of the previously defined geometric structures. □

41 Compressed working principles

Definition 41.1 (Compressed working principles)**.** *The current groundwork is governed by four working principles:*

1. *compressed descent principle,*
2. *frontier-resolution principle,*
3. *basin-entry criterion,*
4. *promote-persistence criterion.*

Theorem 41.2 (Compressed working-principle theorem)**.** *In a fixed canonical normalization regime, every local situation is to be handled by first compressing the paired machine, then normalizing the residual state, then locating the state in the close basin / promote region / residual frontier trichotomy, and only afterwards attempting any OP-specific specialization.*

Proof. This is the direct synthesis of compression, normalization, geometric control, and dynamic control. The current groundwork therefore reduces local pre-OP reasoning to the compressed working principles. □

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42 Formal OP-entry framework

Definition 42.1 (Admissible OP-entry state)**.** *A compact paired conservative normal form is an admissible OP-entry state if it lies in a tool-stable invariant regime and a canonical normalization regime, its normalized residual state has been classified by the resolved branch principle, and in the frontier case at least one admissible descent-guided frontier-resolution step is explicitly available.*

Theorem 42.2 (Pre-OP foundation synthesis theorem)**.** *The local paired conservative machine admits a complete pre-OP foundational package, together with a well-defined admissible OP-entry condition. Explicit work on OP 1–5 therefore need not begin on the raw paired machine itself, but only after passage through this mathematically controlled pre-OP foundation layer.*

Proof. The package is assembled from compression, invariant control, tool extraction, tool stability, normalization, residual geometry, residual dynamics, and the compressed working principles. The admissible OP-entry state is the formal transition condition from groundwork into explicit OP work. □

43 Current-state instantiation

Definition 43.1 (Current frontier-certified pre-entry state)**.** *The current local state is represented by the current compact paired conservative normal form $\mathcal{N}^{*}_{\mathrm{curr}}$, its extracted current live residual state $\mathbf{R}_{\mathrm{curr}}$, its normalized state $\widetilde{\mathbf{R}}_{\mathrm{curr}}$, and a current admissible normalization scale triple. In the absence of an explicit current proof of close-basin or promote-region membership, the current state is read conservatively as frontier-certified pre-entry.*

Definition 43.2 (Preferred witness class)**.** *The current preferred witness class $\mathfrak{W}^{\mathrm{pref}}_{\mathrm{curr}}$ consists of strict admissible route-faithful paired witnesses that improve the horizon-side closure coordinate and provide at least one genuine packet-side improvement without destroying the current invariant class.*

Definition 43.3 (First normalized lead witness)**.** *A first normalized lead witness is an element*

$$\widetilde{\mathbf{R}}_{\mathrm{lead}} \in \mathfrak{W}^{\mathrm{pref}}_{\mathrm{curr}}$$

that minimizes the normalized weighted residual energy inside the preferred witness class and remains admissible for downstream OP-entry auditing.

Definition 43.4 (Lead-direction tag)**.** *The current lead-direction tag records whether the dominant current gain of the lead witness is horizon-dominant, phase-dominant, envelope-dominant, or paired-mixed.*

Definition 43.5 (Directed next witness class)**.** *If the lead witness remains frontier-classified, the directed next witness class $\mathfrak{W}^{\mathrm{dir}}_{\mathrm{next}}$ is obtained by filtering the current witness menu according to the current lead-direction tag, so that the next successor search is directionally coherent rather than fully undirected.*

Theorem 43.6 (Current state synthesis theorem)**.** *For the present project line, the current local state has current foundation-complete frontier status. More precisely:*

1. *the pre-OP foundational package is established;*

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2. *the current state is conservatively read as frontier-certified pre-entry;*

3. *the current state carries a current normalization scale triple and normalized residual state;*

4. *the current state carries a preferred witness class, a first normalized lead witness, a lead-direction tag, and a directed next witness class.*

Consequently, the line is mathematically stabilized up to the final pre-OP transition layer.

Proof. All listed structures have already been established in the current groundwork line and then instantiated at the local level. Collecting them yields the claimed synthesis. □

44 Release conclusion

Corollary 44.1 (Groundwork-complete frontier release reading)**.** *The present line is release-ready as a groundwork-complete pre-OP frontier line. The current state is not yet claimed to be full OP-entry and does not yet claim closure of OP 1–5. Instead, the release freezes the completed pre-OP foundation together with a controlled witness-based transition regime for the next OP-focused stage.*

Proof. This is the direct release reading of the current-state synthesis theorem and the formal OP-entry framework. □

Release note

This release should be read as the final pre-OP launch point of the current line. The broad mathematical architecture has been stabilized and frozen. The remaining task is no longer to widen the framework further, but to carry the current frontier-certified pre-entry state through the final controlled transition into immediate OP-entry and then into explicit OP 1–5 work.

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H1 Structural Tools for the Millennium Matrix

Remark H1.1 (Purpose of the general MM-tools)**.** *The following tools are completely general. They apply to every admissible Millennium matrix, independently of whether one later reads that matrix as an algebraic intertwiner, a closure operator, or a global Heisenberg-compensator matrix. Their role is structural discipline.*

Definition H1.2 (Layer-identity principle)**.** *Let X be a structural object of the Sphaera architecture, and let*

$$\mathcal{M}_{16} : X \mapsto X^{(\ell)}$$

be an admissible realization of X on some layer ℓ. Then $X^{(\ell)}$ is not a new object, but a new realization of the same underlying structural content.

Definition H1.3 (Projection discipline)**.** *A projection*

$$\Pi : X \mapsto X_{\text{readout}}$$

is admissible if it only extracts already stabilized structural content. In particular, projection may reduce visibility, but it may not create new structure.

Definition H1.4 (Closure-before-readout rule)**.** *Every admissible passage through the Millennium matrix must respect the ordering*

$$\textit{triolic admissibility} \longrightarrow \textit{closure compatibility} \longrightarrow \textit{projection} \longrightarrow \textit{interpretation.}$$

Definition H1.5 (Sector-stability criterion)**.** *A sector Σ is structurally stable under the Millennium matrix if*

$$\mathcal{M}_{16}(\Sigma) \subseteq \Sigma$$

or, in the strongest case,

$$\mathcal{M}_{16}(\Sigma) = \Sigma.$$

H2 Heisenberg-Compensator Tools for Physical Readout

Remark H2.1 (Transition from general MM-tools to HC-tools)**.** *We now impose the stronger reading that the Millennium matrix is not only a structural intertwiner, but the global matrix realization of the Heisenberg compensator across admissible layers.*

Definition H2.2 (Global Heisenberg-compensator matrix)**.** *The Millennium matrix is called a global Heisenberg-compensator matrix if its action extends the local compensator rule*

$$C = \tfrac{1}{2}(\mathrm{id} + J)$$

to the full admissible multi-layer architecture of the Companion.

Definition H2.3 (Frame-balancing tool)**.** *Let $X = X^+ + X^-$ be decomposed into a frame-even and frame-sensitive part with respect to an involution J, so that*

$$J(X^+) = X^+, \qquad J(X^-) = -X^-.$$

Then the compensator action satisfies

$$C(X) = \tfrac{1}{2}(X + JX) = X^+,$$

and removes the frame-sensitive component.

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Definition H2.4 (Fix-sector tool)**.** *A sector Σ_{fix} is an HC-fix sector if every object in it is invariant under the compensator:*

$$C(X) = X \qquad \text{for all } X \in \Sigma_{\mathrm{fix}}.$$

Definition H2.5 (Confinement tool)**.** *The confinement tool is the rule that the Heisenberg-compensator matrix prevents admissible structural content from escaping the closure-stable sector selected by the theory.*

Definition H2.6 (Physical-readout tool)**.** *A physical regime is an HC-readout if it is obtained as an effective projection of already compensated and closure-stabilized structural content.*

H3 Notation for the Master Chain

Definition H3.1 (Stage symbols of the master chain)**.** *We use the following symbols for the principal stages of the Companion:*

$$\begin{aligned}
\mathfrak{T} &:= \textit{triolic source},\\
\mathfrak{C} &:= \textit{carrier layer},\\
\mathfrak{C}^{\mathrm{adm}} &:= \textit{admissibly transported carrier},\\
\mathfrak{C}^{\mathrm{cl}} &:= \textit{closure-stable compensated carrier},\\
\mathfrak{G} &:= \textit{group-layer symmetry readout},\\
\mathfrak{L} &:= \textit{null-axis structure},\\
\mathfrak{P} &:= \textit{photon fix-sector},\\
\mathfrak{R} &:= \textit{residual-localization sector},\\
\mathfrak{H} &:= \textit{Higgs-mediation sector},\\
\mathfrak{F} &:= \textit{field-equation readout}.
\end{aligned}$$

We further write $\mathcal{M}_{16}$ for the Millennium matrix and $\mathcal{H}_{16}$ for its global Heisenberg-compensator reading.

Definition H3.2 (Master chain)**.** *The full master chain is written as*

$$\mathfrak{T} \longrightarrow \mathfrak{C} \xrightarrow{\mathcal{M}_{16}} \mathfrak{C}^{\mathrm{adm}} \xrightarrow{\mathcal{H}_{16}} \mathfrak{C}^{\mathrm{cl}} \longrightarrow \mathfrak{G} \longrightarrow \mathfrak{L} \longrightarrow \mathfrak{P} \longrightarrow \mathfrak{R} \longrightarrow \mathfrak{H} \longrightarrow \mathfrak{F}.$$

When intermediate carrier detail is not needed, we also use the short form

$$\mathfrak{T} \to \mathfrak{C} \to \mathcal{M}_{16} \to \mathcal{H}_{16} \to \mathfrak{G} \to (\mathfrak{P}, \mathfrak{R}, J\cdot\mathfrak{R}) \to \mathfrak{H} \to \mathfrak{F}.$$

Remark H3.3 (Why this notation is useful)**.** *This notation is useful because it prevents long later sections from drifting into disconnected local language. Instead, one may always ask at which stage of the master chain a given statement belongs.*

H4 Further Technical Appendices

19AZ$^{(6g)}$. Post-closure normalization of the live OP 1 / OP 5 Step (B) burden labels

After 19AZ$^{(5d)}$ and 19AZ$^{(6f)}$, the two burdens that had governed the immediate post-release continuation line are no longer honestly read as open live kernels on the retained corridors.

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The right normalization step is therefore not to keep repeating the old parked labels, but to distinguish sharply between *retained local kernel closure* and any broader global or machine-theoretic reading that still remains conditional.

> **Reader cue.** Read the next block as a post-closure normalization note rather than as a new theorematic layer. Its purpose is to update the active burden language after the two local kernel closures, so that later sections do not keep speaking as if OP 1 and OP 5 Step (B) were still parked in exactly the same way as before the local proof chain.

Definition H4.1 (Post-closure burden labels for the retained OP kernels)**.** *On the present working line we use the following normalized burden labels.*

1. ***OP 1 retained local kernel.*** *This label denotes the OP 1 corridor problem on the intended admissible windows together with the standing retained no-second-carrier and strict-upper-gap exclusions. After Theorem 33.236, this label is read as* locally discharged*.*
2. ***OP 5 Step (B) retained local kernel.*** *This label denotes the routed branch-shell corridor problem together with the standing retained branch-shell and shell-transfer disciplines. After Corollary 33.254, this label is read as* locally discharged*.*
3. ***Broader conditional machine-theoretic reading.*** *This label denotes stronger global closure or interpretation language beyond the retained corridors and certified routes. It remains* conditional *unless reopened and proved on its own terms.*

Proposition H4.2 (The active post-release burden language must distinguish local kernel closure from broader conditional readings)**.** *Under the standing hypotheses of Theorem 33.236 and Corollary 33.254, the following is the correct active reading of the present working line.*

(i) The OP 1 retained local kernel is no longer a parked live burden on the intended admissible windows; it is locally discharged.

(ii) The OP 5 Step (B) retained local kernel is no longer a parked live burden on the retained branch-shell corridor; it is locally discharged.

(iii) Any stronger language extending these local kernel closures to a broader machine-theoretic or global reading remains conditional unless separately reopened along certified routes.

Consequently, later front/status language in the active line should no longer speak as if OP 1 and OP 5 Step (B) were still simply the two parked live kernels of the continuation line.

Proof. Clause (i) is exactly the content of Theorem 33.236, which discharges Δ_{CL} on the intended admissible windows under the standing retained exclusions. Clause (ii) is exactly the content of Corollary 33.254, which discharges $\Delta_{\Theta R}$ on the retained branch-shell corridor under the standing shell-transfer package. Clause (iii) is only the governing caution used throughout the present line: route-certified local kernel closure is not automatically the same as an unrestricted new global machine-theoretic closure slogan. The final sentence is therefore simply the correct normalization of the active burden language after the local proof chain. □

Remark H4.3 (What this post-closure normalization buys)**.** *This block does not add a third proof. It prevents a quieter but real source of drift: carrying forward the old parked-burden language after the local kernel closures have already been extracted. The active line can now speak more honestly. The tool/QAM/decoder layer is fixed infrastructure, the retained OP 1 and OP 5 Step (B) kernels are locally discharged, and only broader conditional machine-theoretic extensions remain open for later controlled reopening.*

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19AZ$^{(6h)}$. Joint post-closure synthesis and certified reopening map after the retained OP kernels

After 19AZ$^{(6g)}$, the honest continuation question is no longer whether the retained OP 1 and OP 5 Step (B) kernels should simply be reopened in their old parked form. Those kernels are already locally discharged on their certified corridors. What remains live is only the controlled extension problem: which broader tasks still survive once the retained admissible-window and retained branch-shell closures have been extracted? The next normalized step is therefore a joint synthesis block that separates *closed retained kernels*, *corridor-extension tasks*, and *broader machine-theoretic promotion tasks*.

> **Reader cue.** Read the next block as a post-closure continuation map rather than as a new local closure proof. Its purpose is to say what kinds of work are still honestly open after the retained OP 1 and OP 5 Step (B) kernels have been discharged, so that later continuation notes do not quietly reopen the wrong burden.

Definition H4.4 (Certified post-closure task classes after the retained OP kernel closures)**.** *Under the standing hypotheses of Theorem 33.236 and Corollary 33.258, we distinguish the following active task classes.*

1. ***Closed retained kernels.*** *These are the retained OP 1 admissible-window kernel and the retained OP 5 Step (B) branch-shell kernel. They are not honest reopening targets unless one explicitly drops one of the standing retained disciplines.*
2. ***Corridor-extension tasks.*** *These ask whether the local closure statements persist beyond the presently certified retained corridors — for OP 1 beyond the intended admissible-window package, and for OP 5 Step (B) beyond the retained branch-shell/shell-transfer corridor.*
3. ***Broader machine-theoretic promotion tasks.*** *These ask whether the present tool/QAM/decoder line, together with the observer-side machine-shadow package, can be promoted from route-certified local closure language to a stronger global machine-theoretic reading.*
4. ***Diagnostic task selection.*** *This is the use of the Monte-Carlo/fusion probe layer, residual-signature words, and decoder outputs to decide* which *certified reopening route should be tested first. It is diagnostic infrastructure, not a replacement for the structural theorematic work in items (ii)–(iii).*

Proposition H4.5 (After the retained kernel closures, the active continuation line consists only of extension or promotion tasks)**.** *Assume Theorem 33.236, Corollary 33.258, and Definition H4.4. Then the present working line has the following normalized continuation status.*

(i) The retained OP 1 kernel is no longer an honest live burden on the intended admissible windows.

(ii) The retained OP 5 Step (B) kernel is no longer an honest live burden on the retained branch-shell corridor.

(iii) Any genuine new burden must therefore be formulated either as a corridor-extension task or as a broader machine-theoretic promotion task.

(iv) The Monte-Carlo/fusion probe layer serves only to classify and prioritize such tasks through the certified decoder routes; it does not by itself reopen a discharged retained kernel.

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Consequently, the active post-continuation line is normalized as

$$\textit{closed retained kernels} \;\rightarrow\; \textit{extension or promotion tasks only},$$

rather than as a return to the pre-closure parked-burden language.

Proof. Clauses (i)–(ii) are exactly the retained local closure statements extracted in Theorem 33.236 and Corollary 33.258. Therefore the old reading — "two parked live kernels waiting to be reopened" — is no longer the correct active description of the present line. Once those two kernels are discharged on their certified corridors, any honest new burden must either enlarge one of the certified corridors or seek a stronger global/machine-theoretic promotion beyond them; this is exactly clauses (iii) and the first three task classes of Definition H4.4. Clause (iv) is merely the disciplined reading already built into the QAM/decoder architecture: residual-signature routing chooses which certified reopening route is tested first, but it does not substitute for the structural arguments that would be needed to prove an extension or promotion task. The displayed normalized continuation status is therefore the correct post-closure synthesis. □

Corollary H4.6 (First normalized continuation priorities after the retained OP kernel closures)**.** *Under Proposition H4.5, the first honest continuation priorities are:*

(a) test corridor-extension statements before stronger global promotion claims,

(b) use the Monte-Carlo/fusion probe layer only as a task selector among certified routes, and

(c) keep any new reopening note explicit about which retained discipline or corridor boundary is being enlarged.

Remark H4.7 (What this joint post-closure synthesis buys)**.** *This block does not prove a third closure theorem. It prevents the next continuation stage from drifting back into the less honest pre-closure language. The document can now move forward with a clean separation: the retained local OP kernels are closed, the tool/QAM/decoder layer is stable infrastructure, the probe layer is diagnostic guidance, and only extension/promotion problems remain as genuinely open live tasks.*

19AZ$^{(6i)}$. First certified corridor-extension task after the retained OP kernel closures

After 19AZ$^{(6h)}$, the next honest step is not a vague reopening of the old parked burdens, and it is not yet a broad machine-theoretic promotion claim. The smallest certified continuation task is to test whether one of the already closed retained kernels survives a *minimal corridor enlargement* without changing the governing local route discipline. On the present line, the cleanest first pilot is the OP 1 admissible-window side: it stays entirely inside the already extracted IP-chain, asks for no new branch-shell promotion, and is therefore the smallest honest extension task after the retained local kernel closures.

> **Reader cue.** Read the next block as the first post-closure extension pilot, not as a third closure theorem. Its purpose is to say which corridor-extension question is small enough to be the next honest task after the retained OP 1 and OP 5 Step (B) kernels have been discharged.

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Definition H4.8 (One-step admissible-window extension pilot for the post-closure OP 1 line)**.** *Under Definition H4.4, the* one-step admissible-window extension pilot *is the task of enlarging the intended admissible-window package of Theorem 33.236 by one adjacent admissible slot while keeping fixed:*

(i) the retained total-channel profile on the already certified corridor,

(ii) the retained no-second-carrier and strict-upper-gap disciplines used in 19AZ$^{(5c)}$–19AZ$^{(5d)}$,

(iii) the normalized reopening route

$$\mathrm{TR}_4 \Longrightarrow \mathrm{IP}_{10} \Longrightarrow \mathrm{IP}_9 \Longrightarrow \mathrm{IP}_8,$$

restricted now to the first not-yet-certified adjacent admissible slot.

A failure of this pilot is called an extension-boundary witness*. It is the first licensed witness showing that the retained OP 1 closure does not transfer unchanged to the enlarged admissible-window corridor.*

Proposition H4.9 (The first certified extension task is the one-step OP 1 admissible-window pilot)**.** *Assume Proposition H4.5 and Definition H4.8. Then the first certified continuation task after the retained OP 1 / OP 5 Step (B) kernel closures is the one-step admissible-window extension pilot, in the following precise sense.*

(i) It is a genuine corridor-extension task rather than a new local closure burden, because the retained OP 1 kernel itself has already been discharged on the intended admissible windows.

(ii) It is smaller than a branch-shell extension of OP 5 Step (B), because it stays inside the already certified IP-route and does not yet ask for a new routed branch-shell transfer argument.

(iii) It is smaller than a broader machine-theoretic promotion task, because it asks only whether the already certified retained OP 1 closure transfers one slot farther along the admissible-window corridor.

(iv) Therefore any honest next continuation note may either test this pilot directly or explain explicitly why a different extension/promotion task has been chosen in its place.

Proof. Clause (i) is immediate from Proposition H4.5: after the retained kernel closures, the old OP 1 burden is no longer live on the already certified corridor, so the only honest next OP 1 question is whether the closure transfers beyond that corridor. Clause (ii) is purely a size comparison inside the already normalized route language. The OP 1 pilot stays on the extracted IP side and asks for no new branch/shell screening beyond what has already been certified; by contrast, an OP 5 Step (B) corridor enlargement would immediately reopen the routed branch-shell transfer package. Clause (iii) is again only the controlled post-closure reading already fixed in 19AZ$^{(6h)}$: a one-step corridor enlargement is strictly smaller than a global machine-theoretic promotion claim. Clause (iv) simply records the resulting working priority. Hence the one-step admissible-window extension pilot is the first certified extension task on the present post-closure line. □

Corollary H4.10 (Normalized immediate working priority after the retained kernel closures)**.** *Under Proposition H4.9, the immediate post-working priority is:*

test one-step OP 1 admissible-window transfer first
before broader branch-shell extension or machine-theoretic promotion.

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Any Monte-Carlo/fusion-probe use on this line is therefore read only as a selector for the first adjacent admissible slot to test; it does not itself prove the corridor transfer.

Remark H4.11 (What this first certified extension pilot buys)**.** *This block does not enlarge the certified corridor yet. It prevents a more common post-closure drift: jumping immediately from local kernel closure to a much larger machine-theoretic narrative because nothing smaller has been named. The present line now has a disciplined next target. If the OP 1 closure transfers one admissible slot farther, the corridor genuinely grows. If it fails, the failure appears as an explicit extension-boundary witness rather than as a vague reopening of the old burden language.*

19AZ$^{(6j)}$. First boundary-witness localization for the one-step OP 1 admissible-window extension pilot

After 19AZ$^{(6i)}$, the next honest step is to say where a failure of the one-step admissible-window extension pilot could still live. Because the retained OP 1 kernel has already been discharged on the certified corridor, an extension failure is not allowed to reappear as a vague shadow on the old admissible windows. If the pilot fails, the failure must localize at the first newly adjoined admissible slot and must do so through the same retained IP-route that already governs the closed corridor.

> **Reader cue.** Read the next block as the first localization of the post-closure extension pilot, not as a second closure theorem. Its purpose is to show that any honest failure of the one-step OP 1 corridor extension must already appear at the first newly adjoined admissible slot, rather than as a hidden defect on the already certified retained corridor.

Definition H4.12 (First-new-slot boundary witness for the one-step admissible-window extension pilot)**.** *Under Definition H4.8, let σ_+ denote the first adjacent admissible slot not yet contained in the already certified retained OP 1 corridor. A* first-new-slot boundary witness *is a witness obstructing the one-step admissible-window extension pilot such that:*

(i) its restriction to the already certified retained admissible-window corridor is silent, in the sense that it does not create a new live witness on the corridor already discharged by Theorem 33.236,

(ii) its first nontrivial support appears at σ_+, and

(iii) it is read through the same normalized route

$$\mathrm{TR}_4 \Longrightarrow \mathrm{IP}_{10} \Longrightarrow \mathrm{IP}_9 \Longrightarrow \mathrm{IP}_8,$$

now applied to the enlarged corridor whose only new admissible data sit at σ_+.

Such a witness may be read either as a duplicate-signature type failure or as a retained-discipline break, but in both cases its first honest support lies at the newly adjoined admissible slot.

Proposition H4.13 (Boundary witnesses for the one-step admissible-window pilot localize at the first new slot)**.** *Assume Definition H4.8 and Proposition H4.9. Then:*

(i) if the one-step admissible-window extension pilot fails, there exists a first-new-slot boundary witness in the sense of Definition H4.12;

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(ii) conversely, any first-new-slot boundary witness blocks the one-step admissible-window extension pilot;

(iii) therefore the one-step admissible-window extension pilot is equivalent to excluding first-new-slot boundary witnesses on the first newly adjoined admissible slot.

Proof. Because Theorem 33.236 already discharges the retained OP 1 live kernel on the certified admissible-window corridor, a failure of the one-step extension pilot cannot honestly be supported only on that old corridor: such a witness would merely be a re-labeling of a burden already closed. Hence any genuine failure of the enlarged-corridor pilot must first become nontrivial at the first admissible slot that was not yet part of the certified corridor, namely σ_+. This gives clause (i). Clause (ii) is immediate from the definition: a witness whose first nontrivial support appears at the newly adjoined admissible slot is exactly a witness preventing the closure statement from transferring one slot farther. Clause (iii) is just the normalization of (i)–(ii): after the retained kernel closure, the one-step extension task is neither a global reopening nor a diffuse corridor failure, but precisely the exclusion of first-new-slot boundary witnesses routed through the already extracted IP-chain. □

Corollary H4.14 (Normalized immediate target for the one-step OP 1 extension pilot)**.** *Under Proposition H4.13, the immediate post-OP 1 continuation target is:*

$$\textit{exclude first-new-slot boundary witnesses at } \sigma_+$$

for the first newly adjoined admissible slot, rather than reopening the already certified retained OP 1 corridor. Any Monte-Carlo/fusion-probe use on this line is therefore read only as a selector for the first adjacent admissible slot and for the first final witness channel on that slot.

Remark H4.15 (What this first extension-boundary localization buys)**.** *This block does not yet enlarge the OP 1 corridor. It makes the next search honest and much smaller. If the one-step admissible-window transfer succeeds, the corridor grows by one slot. If it fails, the failure no longer appears as a mysterious return of the old OP 1 burden, but as a first-new-slot boundary witness supported at the newly adjoined admissible slot and routed through the already certified retained IP-chain.*

19AZ$^{(6k)}$. First channel split for first-new-slot boundary witnesses on the one-step OP 1 admissible-window extension pilot

After 19AZ$^{(6j)}$, the next honest question is not whether the pilot fails somewhere on the old corridor, but what kind of failure could still survive at the first newly adjoined admissible slot. Once the failure is localized at σ_+, there are only two structurally honest readings left: either the new slot creates a duplicate-signature type obstruction, or one of the standing retained exclusion disciplines themselves fails to transfer to the first new slot.

> **Reader cue.** Read the next block as a channel split for the first-new-slot boundary witness, not as a new closure claim. Its purpose is to separate the extension pilot into the two only honest failure modes at the first newly adjoined admissible slot: duplicate-signature type versus retained-discipline break.

Definition H4.16 (Duplicate-signature type versus retained-discipline-break type at the first new admissible slot)**.** *Assume Definition H4.12. Let W_+ be a first-new-slot boundary witness supported at σ_+.*

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(i) *We say that W_+ is of* duplicate-signature type *if, on the enlarged corridor obtained by adjoining σ_+, the obstruction is still read as the appearance of two distinct admissible primitive completions with the same retained pinning signature, with the first honest nontrivial support of the duplicating configuration appearing at σ_+.*

(ii) *We say that W_+ is of* retained-discipline-break type *if the obstruction is already detected as a failure of one of the standing retained exclusion disciplines to transfer from the certified corridor to the first newly adjoined admissible slot σ_+; concretely, the witness is read as a first-slot failure of the retained no-second-carrier discipline or of the retained strict-upper-gap discipline.*

A first-new-slot boundary witness may in principle present both readings simultaneously, but no third autonomous channel is licensed on the current post-closure OP 1 line.

Proposition H4.17 (Every first-new-slot boundary witness falls into the duplicate-signature or retained-discipline-break channel)**.** *Assume Proposition H4.13. Then every first-new-slot boundary witness W_+ belongs to at least one of the two classes in Definition H4.16. More precisely:*

(i) *if the standing retained exclusion disciplines transfer from the certified corridor to σ_+, then any first-new-slot boundary witness is necessarily of duplicate-signature type;*

(ii) *if a first-new-slot boundary witness is not of duplicate-signature type, then it is necessarily of retained-discipline-break type;*

(iii) *hence the one-step admissible-window extension pilot can fail only through a duplicate-signature witness at σ_+ or through a first-slot break of the retained exclusion disciplines.*

Proof. By Proposition H4.13, any honest failure of the pilot is already localized at the first newly adjoined admissible slot σ_+. Once we are there, there are only two ways for the extension attempt to remain obstructed. Either the standing retained exclusion disciplines still govern the enlarged corridor up to the new slot; then the only remaining honest obstruction is that the new slot supports a duplicate-signature configuration of the same general kind already normalized in the retained kernel analysis, now first becoming nontrivial at σ_+. This is clause (i). Or else the obstruction appears before any such duplicate-signature comparison can even be completed, namely because one of the standing retained exclusion disciplines itself no longer transfers to the first new slot. This is clause (ii). Clause (iii) is then only the normalized summary: after the retained kernel closure and after the first-new-slot localization, the extension pilot has exactly two licensed failure channels and no third autonomous residue. □

Corollary H4.18 (Immediate post- continuation target for the one-step OP 1 extension pilot)**.** *Under Proposition H4.17, the immediate post-continuation target splits into the following two certified checks at σ_+:*

(a) test transfer of the standing retained exclusion disciplines to σ_+;

(b) if transfer holds, exclude duplicate-signature witnesses at σ_+.

Any Monte-Carlo/fusion-probe use on this line is therefore read only as a selector between these two certified checks and not as an independent proof of either one.

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Remark H4.19 (What this first channel split at the new slot buys)**.** *This block does not yet extend the corridor by one slot. It tells us exactly what must be checked next. The search is no longer allowed to wander over the old retained corridor or over vague extension rhetoric. After the local pass, a one-step OP 1 extension failure can only mean one of two things at the first newly adjoined admissible slot: either the retained exclusion disciplines themselves do not transfer there, or else they do transfer and one must exclude a new duplicate-signature witness supported at* σ_+*.*

19AZ$^{(6l)}$. Transfer of the standing retained exclusion disciplines to the first new admissible slot on the one-step OP 1 extension pilot

After 19AZ$^{(6k)}$, the next honest question is whether the standing retained exclusion disciplines themselves actually survive the one-step enlargement to the first newly adjoined admissible slot σ_+. On the present line this is the smallest structural transfer test: if the one-step pilot really stays on the same retained IP-route, keeps the same retained total-channel profile and optimum packet, and asks for no new branch/shell promotion before the new slot is read, then the no-second-carrier and strict-upper-gap exclusions should not suddenly fail at σ_+. The next step is therefore to state this transfer in the smallest certified form and thereby remove the retained-discipline-break channel whenever the pilot remains a genuine one-step IP-side enlargement.

> **Reader cue.** Read the next block as a first-slot transfer theorem for the standing retained exclusion disciplines, not as the corridor extension itself. Its purpose is to remove the retained-discipline-break channel at σ_+ whenever the one-step pilot truly remains inside the same retained IP-side comparison packet.

Definition H4.20 (First-slot retained-discipline transfer certificate for the one-step OP 1 pilot)**.** *Under Definition H4.8, we say that the one-step OP 1 admissible-window extension pilot carries the* first-slot retained-discipline transfer certificate *if all of the following hold on the enlarged corridor through the first newly adjoined admissible slot* σ_+*:*

(i) the extension is still read through the same retained IP-route $\mathrm{TR}_4 \Rightarrow \mathrm{IP}_{10} \Rightarrow \mathrm{IP}_9 \Rightarrow \mathrm{IP}_8$ *used on the certified corridor;*

(ii) the retained total-channel profile and retained optimum packet used in the closed kernel analysis remain unchanged up to and including σ_+*;*

(iii) no new branch/shell promotion or BS/BL-side reopening is licensed before the first new slot is read;

(iv) every final primitive occupancy or admissible completion involving σ_+ *is still compared against the same retained comparison packet that governed the certified corridor.*

In this situation we say that the standing retained no-second-carrier and strict-upper-gap disciplines transfer to the first new admissible slot*.*

Proposition H4.21 (Under first-slot transfer, the retained-discipline-break channel at σ_+ is absent)**.** *Assume Proposition H4.17 and Definition H4.20. Then no first-new-slot boundary witness supported at* σ_+ *can be of retained-discipline-break type. Equivalently, whenever the one-step extension pilot carries the first-slot retained-discipline transfer certificate, the standing retained no-second-carrier and strict-upper-gap exclusions extend from the certified corridor to* σ_+*.*

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Proof. A retained-discipline-break witness at σ_+ would mean that, before any genuine duplicate-signature comparison has become the first honest obstruction, one of the two standing retained exclusion disciplines already fails to survive the one-step enlargement. But the transfer certificate says exactly that the pilot has not changed the governing route, has not changed the retained total-channel profile or optimum packet, has not introduced a new branch/shell promotion before reading the new slot, and still compares every final occupancy/completion involving σ_+ against the same retained comparison packet used on the closed corridor. Under those hypotheses there is no new structural mechanism available that could create a first-slot failure of the retained no-second-carrier discipline or of the retained strict-upper-gap discipline without first leaving the certified one-step IP-side enlargement setting itself. Therefore the retained-discipline-break channel is absent at σ_+. □

Corollary H4.22 (Under first-slot transfer, any surviving boundary witness at σ_+ is of duplicate-signature type)**.** *Assume Definition H4.20. Then any first-new-slot boundary witness supported at σ_+ is necessarily of duplicate-signature type. In particular, under first-slot transfer the one-step admissible-window extension pilot can fail only through a new duplicate-signature witness at the first newly adjoined admissible slot.*

Remark H4.23 (What the first-slot retained-discipline transfer buys)**.** *This does not yet prove the one-step extension pilot. It removes one entire failure channel. After the present step, the only honest remaining obstruction at σ_+ is the duplicate-signature channel, provided the pilot truly stays inside the same retained IP-side comparison packet. This is exactly the kind of reduction for which the Monte-Carlo/fusion-probe layer is useful: it does not prove the transfer theorem, but it can help tell us early whether a final run still looks like a genuine one-step IP-side enlargement or whether it is already drifting into a branch/shell promotion where the certificate would no longer apply.*

Reader cue. Read the next block as the final test on the one-step OP 1 admissible-window extension pilot. Its purpose is to decide whether the only remaining first-new-slot boundary channel — duplicate-signature at σ_+ — can still survive once the standing retained exclusion disciplines have been transferred to that new slot. If it cannot, then the one-step pilot itself is certified.

19AZ$^{(6m)}$. Duplicate-signature witnesses at the first new admissible slot collapse under the transferred retained exclusion disciplines

After 19AZ$^{(6l)}$, the one-step OP 1 admissible-window extension pilot has only one honest remaining failure channel: a duplicate-signature witness supported at the first newly adjoined admissible slot σ_+. The next honest question is therefore no longer whether the retained exclusion disciplines survive there — that was the content of the transfer certificate — but whether the same duplicate-signature contradiction package that closed the retained OP 1 kernel also closes the first-new-slot case once those disciplines have been transferred. On the present line the answer is yes: under first-slot transfer, any surviving duplicate-signature configuration at σ_+ is again forced into the same occupancy-side versus completion-side split, and each side is discharged by the transferred retained no-second-carrier / strict-upper-gap exclusions.

Definition H4.24 (First-new-slot duplicate-signature witness on the one-step OP 1 extension pilot)**.** *Under Definition H4.8, a boundary witness supported at the first newly adjoined admissible slot σ_+ is called a* first-new-slot duplicate-signature witness *if it is of duplicate-signature type in the sense of Proposition H4.17. Equivalently, it is a boundary witness at σ_+*

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for which the standing retained exclusion disciplines transfer to the new slot, but the enlarged corridor still carries two primitive loop/completion finals with the same retained pinning signature when the first new slot is read through the same retained comparison packet.

Proposition H4.25 (Under first-slot transfer, no duplicate-signature witness survives at σ_+)**.** *Assume Definition H4.20, Definition H4.24, Proposition 33.231, and Proposition 33.234. Then no first-new-slot duplicate-signature witness supported at σ_+ exists.*

Proof. Assume for contradiction that a first-new-slot duplicate-signature witness exists at σ_+. By definition, the one-step pilot still carries the same retained IP-route, retained total-channel profile, retained optimum packet, and retained comparison packet up to and including the first new slot. Therefore the putative duplicate-signature configuration at σ_+ is of the same structural species as in the retained OP 1 kernel analysis: once it is read through the common retained comparison packet, it again splits into an occupancy-side versus completion-side contradiction channel exactly as in Proposition 33.229. Because the first-slot retained-discipline transfer certificate is in force, the retained no-second-carrier and retained strict-upper-gap exclusions have already been transferred to the enlarged corridor through σ_+. Proposition 33.231 therefore removes the occupancy-side channel, while Proposition 33.234 removes the completion-side channel. Since no third duplicate-signature channel is licensed, the witness cannot exist. This contradiction proves the claim. □

Corollary H4.26 (Closure of the one-step OP 1 admissible-window extension pilot under first-slot transfer)**.** *Assume Definition H4.20. Then the one-step OP 1 admissible-window extension pilot closes. Equivalently, under first-slot transfer there is no first-new-slot boundary witness at σ_+, so the retained OP 1 corridor extends by one admissible slot without opening any new live burden.*

Proof. By Corollary H4.22, any surviving first-new-slot boundary witness at σ_+ would have to be of duplicate-signature type. Proposition H4.25 excludes exactly that possibility. Hence no first-new-slot boundary witness survives, and the one-step admissible-window extension pilot closes. □

Remark H4.27 (What the first certified one-step OP 1 extension buys)**.** *This is the first actual corridor-extension result after the retained OP 1 kernel closure. It does not claim an arbitrary global OP 1 promotion. It says something much smaller and therefore much cleaner: if the first newly adjoined admissible slot really remains inside the same retained IP-side comparison packet, then the earlier duplicate-signature contradiction package propagates one step further and the pilot succeeds. In particular, any genuinely new OP 1 burden beyond the retained corridor must now appear only* after *this first certified slot extension or by leaving the first-slot transfer certificate itself.*

Reader cue. Read the next block as the controlled iteration schema after the first certified one-step OP 1 admissible-window extension. Its purpose is not to announce an arbitrary global promotion, but to normalize the smallest slot-by-slot rule now available: whenever the next newly adjoined admissible slot again carries the same transferred retained comparison discipline, the corridor extends by one more slot; if not, the first honest extension-boundary witness must localize exactly at the first slot where that transfer certificate fails.

19AZ$^{(6n)}$. First iterable slotwise extension schema after the certified one-step OP 1 admissible-window transfer

After 19AZ$^{(6m)}$, the current OP 1 line has something strictly stronger than a single isolated pilot success but strictly weaker than an unrestricted global promotion: it has one certified

slot-transfer mechanism. The next honest step is therefore to package this mechanism into an iterable schema. The idea is conservative. One does not claim that arbitrarily many future slots automatically close; one says only that once a current corridor has already been certified, each further admissible slot may be adjoined by the same one-step argument provided the same slotwise transfer certificate is re-established at that new leading edge. In this way extension ceases to be an opaque reopening and becomes a controlled inductive procedure with a sharply localized first failure slot.

Definition H4.28 (Certified slotwise OP 1 extension chain)**.** *Let $\mathcal{C}_0$ denote a currently certified OP 1 admissible corridor, beginning with the retained corridor already closed by Corollary H4.26 after one certified enlargement. For $j \geq 1$, let σ_j denote the next newly adjoined admissible slot beyond $\mathcal{C}_{j-1}$ and let $\mathcal{C}_j := \mathcal{C}_{j-1} \cup \{\sigma_j\}$. We call*

$$\mathcal{C}_0 \subset \mathcal{C}_1 \subset \cdots \subset \mathcal{C}_k$$

a certified slotwise OP 1 extension chain of length k *if, for each $1 \leq j \leq k$, the step from $\mathcal{C}_{j-1}$ to $\mathcal{C}_j$ admits the same kind of slotwise retained-discipline transfer certificate as in Definition H4.20, now read relative to the already certified corridor $\mathcal{C}_{j-1}$ and its current leading slot σ_j.*

Proposition H4.29 (Slotwise transfer certificates iterate the one-step OP 1 extension mechanism)**.** *Assume Definition H4.28. Then every step $\mathcal{C}_{j-1} \rightsquigarrow \mathcal{C}_j$ in the chain closes, and hence the terminal corridor $\mathcal{C}_k$ is certified without opening a new live OP 1 burden.*

Proof. Proceed by induction on j. The case $j = 1$ is exactly the certified one-step extension mechanism already established in Corollary H4.26. Assume now that $\mathcal{C}_{j-1}$ is already certified. By definition, the next slot σ_j comes equipped with the same slotwise retained-discipline transfer certificate, now read relative to the already certified corridor $\mathcal{C}_{j-1}$. Thus the step from $\mathcal{C}_{j-1}$ to $\mathcal{C}_j$ is formally the same one-step situation as before: the standing retained exclusions transfer to the new leading slot, the only honest first-new-slot boundary channel is duplicate-signature type, and that channel again collapses under the transferred retained no-second-carrier and strict-upper-gap package. Therefore $\mathcal{C}_j$ is certified. Induction yields the claim for every $j \leq k$ and in particular for $\mathcal{C}_k$. □

Corollary H4.30 (First failure of a slotwise extension attempt localizes at the first uncertified leading slot)**.** *Let a slotwise OP 1 extension attempt proceed successively through admissible slots beyond a certified corridor. If the attempt fails, then there is a smallest index j_* such that the step to σ_{j_*} does not admit a slotwise retained-discipline transfer certificate. Equivalently, as long as the next leading slot continues to satisfy the transferred retained comparison discipline, no new live OP 1 burden appears.*

Proof. If every step admitted the slotwise retained-discipline transfer certificate, Proposition H4.29 would certify the entire chain. Therefore any failure must occur at the first index where this certificate is no longer available. Minimality of that index is automatic. □

Remark H4.31 (What the iterable slotwise extension schema now buys)**.** *This is the clean post-upgrade of the first certified slot extension. The text still does not claim that arbitrary future admissible slots are automatically harmless. It claims something smaller and more useful: extension has become an explicit iterative schema. The Monte-Carlo/fusion-probe layer is therefore no longer searching blindly for "whether OP 1 reopens somewhere"; it is searching for the first leading slot at which the slotwise retained-discipline transfer certificate fails or begins to look unstable. That is exactly the kind of sharply localized extension-boundary information that a routing/probe layer can accelerate without replacing the structural proof itself.*

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Reader cue. Read the next block as the first concrete two-step realization of the slotwise OP 1 extension schema. Its purpose is not to announce an unrestricted corridor promotion, but to show that the new inductive format already yields a genuine $k = 2$ certified chain once the second new leading slot carries the same transferred retained comparison discipline as the first.

19AZ$^{(6o)}$. First certified two-slot OP 1 admissible-window extension chain under repeated leading-slot transfer

After 19AZ$^{(6n)}$, the honest question is whether the iterable schema does anything beyond packaging the already certified first new slot. The smallest nontrivial test is therefore the second newly adjoined admissible slot. One keeps the already enlarged corridor $\mathcal{C}_1$ fixed, adjoins the next admissible slot σ_2, and asks whether the same leading-slot transfer certificate can be re-established there. If it can, then the abstract schema immediately becomes a first genuine two-step certified chain rather than a merely formal induction template.

Definition H4.32 (Second-slot transfer certificate for the first certified two-step OP 1 chain)**.** *Let $\mathcal{C}_1$ be the corridor already certified by the one-step OP 1 admissible-window extension result of Corollary H4.26. Let σ_2 denote the next admissible slot beyond the leading edge of $\mathcal{C}_1$, and set $\mathcal{C}_2 := \mathcal{C}_1 \cup \{\sigma_2\}$. We say that σ_2 carries the* second-slot transfer certificate *if the step $\mathcal{C}_1 \rightsquigarrow \mathcal{C}_2$ satisfies the same slotwise retained-discipline transfer conditions as in Definition H4.28, now read relative to the already certified corridor $\mathcal{C}_1$ and its new leading slot σ_2.*

Proposition H4.33 (The second newly adjoined admissible slot closes under the repeated leading-slot transfer certificate)**.** *Assume Definition H4.32. Then the extension step $\mathcal{C}_1 \rightsquigarrow \mathcal{C}_2$ closes, and $\mathcal{C}_2$ is a certified OP 1 admissible corridor.*

Proof. The corridor $\mathcal{C}_1$ is already certified by Corollary H4.26. By Definition H4.32, the next leading slot σ_2 again satisfies exactly the slotwise retained-discipline transfer assumptions required in the iterable extension schema of Proposition H4.29. Applying that proposition with $k = 2$ yields closure of the step $\mathcal{C}_1 \rightsquigarrow \mathcal{C}_2$ and certifies the enlarged corridor $\mathcal{C}_2$. □

Corollary H4.34 (First realized certified two-slot OP 1 admissible-window chain)**.** *Under the repeated leading-slot transfer certificate, the current OP 1 line already contains a realized certified chain*

$$\mathcal{C}_0 \subset \mathcal{C}_1 \subset \mathcal{C}_2,$$

of length 2*. Equivalently, beyond the retained live-kernel closure, two consecutive admissible-slot enlargements are now certified without reopening a live OP 1 burden.*

Proof. The first step $\mathcal{C}_0 \rightsquigarrow \mathcal{C}_1$ is certified by Corollary H4.26. Proposition H4.33 certifies the second step $\mathcal{C}_1 \rightsquigarrow \mathcal{C}_2$. Hence the displayed chain of length 2 is certified. □

Remark H4.35 (What the first realized two-slot chain now buys)**.** *This is the first point at which the iterable extension schema becomes visibly more than a packaging convenience. The text still does not claim a fully promoted OP 1 corridor. It claims something smaller and already operational: once the transferred retained comparison discipline persists across two successive leading slots, the corridor extends twice without reopening the local OP 1 burden. In particular, any future extension failure must now occur only at the first admissible slot* after $\mathcal{C}_2$ *where the leading-slot transfer certificate ceases to hold or begins to destabilize.*

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Reader cue. Read the next block as a compact certification format for a finite OP 1 extension run. Its purpose is not to claim unrestricted corridor promotion, but to package a bounded number of already justified leading-slot transfers into one reusable certificate whose failure horizon is explicitly the first slot beyond the certified run.

19AZ$^{(6p)}$. First bounded k-step certificate format for finite OP 1 admissible-window extension runs

After 19AZ$^{(6o)}$, the present line has more than one isolated extension success but still less than an unrestricted promoted corridor. The right packaging is therefore a bounded certificate format: one records a finite certified run together with the leading admissible slots and their slotwise transfer certificates, and one treats the first uncertified slot beyond that run as the first honest location where extension may fail again.

Definition H4.36 (Bounded k-step OP 1 extension certificate)**.** *Let $k \geq 1$. A* bounded k-step OP 1 extension certificate *is data*

$$\mathbf{K}_k = (\mathcal{C}_0 \subset \mathcal{C}_1 \subset \cdots \subset \mathcal{C}_k;\, \sigma_1, \ldots, \sigma_k),$$

where for each $j = 1, \ldots, k$ the slot σ_j is the newly adjoined leading admissible slot in the extension step $\mathcal{C}_{j-1} \rightsquigarrow \mathcal{C}_j$, and σ_j carries the slotwise retained-discipline transfer certificate required by Definition H4.28 relative to the already certified corridor $\mathcal{C}_{j-1}$.

Proposition H4.37 (Bounded k-step certificates certify the whole finite OP 1 run)**.** *Assume Definition H4.36. Then the terminal corridor $\mathcal{C}_k$ is certified, and every intermediate step $\mathcal{C}_{j-1} \rightsquigarrow \mathcal{C}_j$ closes without reopening a live OP 1 burden.*

Proof. This is exactly the repeated one-step mechanism of Proposition H4.29, now written in bounded packaged form. Since each newly adjoined leading slot σ_j carries the required slotwise retained-discipline transfer certificate, every step closes in turn. Hence the entire finite chain up to $\mathcal{C}_k$ is certified. □

Corollary H4.38 (First honest failure lies beyond a bounded certificate horizon)**.** *Assume Definition H4.36. Then any honest future extension failure must first appear at the earliest admissible slot strictly beyond the leading edge of $\mathcal{C}_k$ for which the next slotwise transfer certificate is absent or unstable.*

Proof. Every step through $\mathcal{C}_k$ is already certified by Proposition H4.37. Therefore no honest failure can occur inside that bounded run. The first possible extension failure must lie at the first later leading slot for which the next certificate is not available. □

Remark H4.39 (What the bounded certificate format now buys)**.** *This is the finite-horizon normalization that the post-extension line needed. Instead of speaking vaguely about an enlarging corridor, the text can now speak about a specific certified horizon $\mathbf{K}_k$. The Monte-Carlo/fusion-probe layer is thereby given a precise target: it need only search for the first later slot beyond $\mathbf{K}_k$ where the transfer certificate begins to fail, rather than resurveying the already certified run.*

Reader cue. Read the next block as the first explicit named bounded certificate already available on the present line. Its purpose is to turn the abstract finite-horizon format into a concrete certified object that can be cited directly in later routing and probe discussions.

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19AZ$^{(6q)}$. First explicit bounded certificate $\mathbf{K}_2$ on the present OP 1 extension line

After 19AZ$^{(6p)}$, the next honest step is to name the first concrete bounded certificate already realized by the current line. The smallest nontrivial such object is the two-step extension chain from 19AZ$^{(6o)}$, now repackaged as a single finite certificate. This gives the text an explicit certified horizon that later extension tests and Monte-Carlo/probe diagnostics can cite without reopening the step-by-step derivation each time.

Definition H4.40 (The present two-step bounded certificate $\mathbf{K}_2^{\mathrm{cur}}$)**.** *Let*

$$\mathbf{K}_2^{\mathrm{cur}} := (\mathcal{C}_0 \subset \mathcal{C}_1 \subset \mathcal{C}_2;\, \sigma_1, \sigma_2),$$

where $\mathcal{C}_0$ is the retained certified OP 1 corridor, σ_1 is the first newly adjoined admissible slot certified in Corollary H4.26, and σ_2 is the second newly adjoined admissible slot certified in Corollary H4.34. We call $\mathbf{K}_2^{\mathrm{cur}}$ the present two-step bounded certificate*.*

Proposition H4.41 (The current OP 1 line carries the explicit bounded certificate $\mathbf{K}_2^{\mathrm{cur}}$)**.** *The data of Definition H4.40 satisfy Definition H4.36 with $k = 2$. Hence $\mathbf{K}_2^{\mathrm{cur}}$ is a valid bounded certificate on the present OP 1 line.*

Proof. The step $\mathcal{C}_0 \rightsquigarrow \mathcal{C}_1$ is certified by Corollary H4.26. The step $\mathcal{C}_1 \rightsquigarrow \mathcal{C}_2$ is certified by Corollary H4.34. Therefore the two leading slots σ_1, σ_2 each carry the required slotwise transfer certificate and the chain satisfies Definition H4.36 with $k = 2$. □

Corollary H4.42 (The first uncertified OP 1 slot lies strictly beyond $\mathbf{K}_2^{\mathrm{cur}}$)**.** *On the present line, any honest future OP 1 admissible-window extension failure must first occur at a leading admissible slot strictly beyond the corridor $\mathcal{C}_2$ certified by $\mathbf{K}_2^{\mathrm{cur}}$.*

Proof. This is the case $k = 2$ of Corollary H4.38, applied to Proposition H4.41. □

Remark H4.43 (What the explicit certificate $\mathbf{K}_2^{\mathrm{cur}}$ now buys)**.** *The current extension line now has a named finite horizon. This is useful both mathematically and diagnostically. Mathematically, later corridor-extension arguments can cite $\mathbf{K}_2^{\mathrm{cur}}$ as a certified base segment rather than replaying the first two transferred slots. Diagnostically, the Monte-Carlo/fusion-probe layer can now ask a sharply bounded question: where is the first later slot beyond $\mathbf{K}_2^{\mathrm{cur}}$ at which the next leading-slot transfer certificate destabilizes?*

Reader cue. Read the next block as the first controlled test beyond the current certified horizon $\mathbf{K}_2^{\mathrm{cur}}$. Its purpose is not to claim a third transferred slot automatically, but to localize the next honest extension question at the very first later leading slot and to state the exact alternative: either that slot carries the next transfer certificate and yields a certified $\mathbf{K}_3$, or it is the first genuine post-$\mathbf{K}_2^{\mathrm{cur}}$ boundary witness.

19AZ$^{(6r)}$. First post-$\mathbf{K}_2^{\mathrm{cur}}$ leading-slot test beyond the current certified horizon

After 19AZ$^{(6q)}$, the next honest step is to move from a named finite horizon to the first controlled test beyond that horizon. The current line therefore does not jump directly to an unbounded promotion claim. Instead it isolates the first later leading admissible slot beyond $\mathcal{C}_2$ and asks whether the same slotwise retained-discipline transfer certificate extends one more time. This is the smallest extension question that remains once $\mathbf{K}_2^{\mathrm{cur}}$ has been certified.

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Definition H4.44 (First post-$\mathbf{K}_2^{\mathrm{cur}}$ leading-slot test datum)**.** *Let* $\mathbf{K}_2^{\mathrm{cur}} = (\mathcal{C}_0 \subset \mathcal{C}_1 \subset \mathcal{C}_2; \sigma_1, \sigma_2)$ *be the present two-step bounded certificate from Definition H4.40. Define* σ_3 *to be the first later leading admissible slot strictly beyond the certified corridor* $\mathcal{C}_2$*, whenever such a next slot is under consideration on the same retained OP 1 extension line. We call the pair* $(\mathcal{C}_2, \sigma_3)$ *the* first post-$\mathbf{K}_2^{\mathrm{cur}}$ leading-slot test datum*.*

Proposition H4.45 (The first honest future OP 1 extension question beyond $\mathbf{K}_2^{\mathrm{cur}}$ is localized at σ_3)**.** *Assume the setting of Definition H4.44. Then every honest future OP 1 admissible-window extension question beyond the currently certified horizon is localized at the single leading slot* σ_3*. More precisely, exactly one of the following alternatives occurs:*

(i) the slot σ_3 *carries the same slotwise retained-discipline transfer certificate as in the previously certified steps, in which case the enlarged chain*

$$\mathcal{C}_0 \subset \mathcal{C}_1 \subset \mathcal{C}_2 \subset \mathcal{C}_3$$

is a valid bounded certificate $\mathbf{K}_3^{\mathrm{cand}}$ *on the same OP 1 line;*

(ii) the slot σ_3 *does not carry that certificate, in which case* σ_3 *itself is the first genuine post-*$\mathbf{K}_2^{\mathrm{cur}}$ *boundary-witness location.*

In particular, no honest future failure can occur strictly before σ_3*.*

Proof. By Corollary H4.42, any honest future OP 1 extension failure must first occur at a leading admissible slot strictly beyond $\mathcal{C}_2$. By definition, σ_3 is the first such later leading slot under consideration. Therefore no honest future failure can occur before σ_3. If σ_3 carries the same slotwise retained-discipline transfer certificate as in the previous certified steps, then the same slotwise-extension mechanism used in Proposition H4.37 extends the chain by one step and yields a valid bounded certificate with $k = 3$. If not, then the failure is already localized at σ_3 itself. These are the only two possibilities. □

Corollary H4.46 (The post-$\mathbf{K}_2^{\mathrm{cur}}$ horizon is controlled by a single next-slot test)**.** *On the present OP 1 line, the first uncertified future horizon beyond* $\mathbf{K}_2^{\mathrm{cur}}$ *is completely controlled by the single next-slot test at* σ_3*: either* $\mathbf{K}_3^{\mathrm{cand}}$ *is certified, or the first honest future boundary witness is already localized at* σ_3*.*

Proof. Immediate from Proposition H4.45. □

Remark H4.47 (What the first post-$\mathbf{K}_2^{\mathrm{cur}}$ test now buys)**.** *This is the first point at which the extension line leaves the already named certified horizon. The benefit is that the next search question is now maximally sharp. Neither the theorematic line nor the Monte-Carlo/fusion-probe layer needs to resurvey the already certified segment* $\mathbf{K}_2^{\mathrm{cur}}$*; both need only test whether the next leading slot* σ_3 *carries the same transfer certificate. If it does, a concrete* $\mathbf{K}_3$ *becomes available. If it does not, then the first genuine post-*$\mathbf{K}_2^{\mathrm{cur}}$ *boundary witness has already been found.*

Reader cue. Read the next block as the two-channel split of the first uncertified slot beyond $\mathbf{K}_2^{\mathrm{cur}}$. Its purpose is not yet to certify $\mathbf{K}_3$, but to normalize the next decision at σ_3 into the two only honest outcomes: either the same transfer certificate still holds, or the first genuine post-$\mathbf{K}_2^{\mathrm{cur}}$ boundary witness is already present there.

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19AZ$^{(6s)}$. First transfer-positive versus boundary-active split at the post-$\mathbf{K}_2^{\mathrm{cur}}$ slot σ_3

After 19AZ$^{(6r)}$, the first uncertified slot beyond the current certified horizon is no longer diffuse. It is the single leading admissible slot σ_3 beyond $\mathcal{C}_2$. The next honest normalization is therefore to split the status of σ_3 itself. Either σ_3 still carries the same slotwise retained-discipline transfer certificate that already certified σ_1 and σ_2, or else σ_3 is already the first active boundary location beyond $\mathbf{K}_2^{\mathrm{cur}}$.

Definition H4.48 (Transfer-positive versus boundary-active status at σ_3)**.** *Assume the setting of Definition H4.44. We say that the first later leading admissible slot σ_3 is*

(i) transfer-positive *if it carries the same slotwise retained-discipline transfer certificate as in the certified steps leading to $\mathbf{K}_2^{\mathrm{cur}}$, and*

(ii) boundary-active *if it does not carry that certificate and therefore already hosts the first genuine post-$\mathbf{K}_2^{\mathrm{cur}}$ boundary witness.*

Proposition H4.49 (The post-$\mathbf{K}_2^{\mathrm{cur}}$ slot σ_3 admits only the transfer-positive or boundary-active status)**.** *Assume the setting of Definition H4.48. Then exactly one of the following alternatives holds:*

(i) σ_3 is transfer-positive, in which case the current line produces a valid three-step bounded certificate

$$\mathbf{K}_3^{\mathrm{cand}} = (\mathcal{C}_0 \subset \mathcal{C}_1 \subset \mathcal{C}_2 \subset \mathcal{C}_3;\, \sigma_1, \sigma_2, \sigma_3),$$

with $\mathcal{C}_3$ the corridor obtained by adjoining σ_3 to $\mathcal{C}_2$; or

(ii) σ_3 is boundary-active, in which case σ_3 itself is already the first genuine post-$\mathbf{K}_2^{\mathrm{cur}}$ boundary location.

No third autonomous status for σ_3 is licensed on the present OP 1 extension line.

Proof. By Proposition H4.45, every honest future OP 1 extension question beyond $\mathbf{K}_2^{\mathrm{cur}}$ is localized at the single next leading admissible slot σ_3. Therefore the only remaining issue is whether σ_3 carries the same slotwise retained-discipline transfer certificate as the previously certified steps. If it does, then the bounded-k certificate mechanism of Proposition H4.37 extends the chain once more and yields the displayed three-step final certificate. If it does not, then by the same localization result σ_3 itself is already the first genuine post-$\mathbf{K}_2^{\mathrm{cur}}$ boundary location. Since the extension problem is completely localized at σ_3, these two alternatives exhaust the possibilities. □

Corollary H4.50 (The next honest OP 1 decision beyond $\mathbf{K}_2^{\mathrm{cur}}$ is a two-channel decision at σ_3)**.** *On the present OP 1 extension line, the first uncertified future decision beyond $\mathbf{K}_2^{\mathrm{cur}}$ reduces to the two-channel split*

$$\textit{transfer-positive at } \sigma_3 \qquad \textit{or} \qquad \textit{boundary-active at } \sigma_3.$$

In particular, the theorematic line and the Monte-Carlo/fusion-probe layer need not resurvey the already certified segment $\mathbf{K}_2^{\mathrm{cur}}$ in order to decide the next step.

Proof. Immediate from Proposition H4.49. □

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Remark H4.51 (What the first transfer-positive/boundary-active split at σ_3 now buys)**.** *This is the first point beyond the currently certified horizon where the next admissible slot is not merely isolated but also status-normalized. The benefit is that the next step no longer asks a vague continuation question. It asks only whether σ_3 is transfer-positive or already boundary-active. If it is transfer-positive, the line immediately upgrades to a concrete three-step certificate. If it is boundary-active, then the first genuine post-$\mathbf{K}_2^{\mathrm{cur}}$ boundary witness has already been found at the earliest possible place.*

Reader cue. Read the next block as the direct certification of the transfer-positive channel at σ_3. Its purpose is not yet to promote the extension line indefinitely, but to convert the already normalized positive branch into an explicit current three-step bounded certificate and to discharge the boundary-active channel at that slot itself.

19AZ$^{(6t)}$. First certification of $\mathbf{K}_3^{\mathrm{cur}}$ under transfer-positive status at σ_3

After 19AZ$^{(6s)}$, the next honest step is no longer to restate the two-channel split at σ_3, but to close its positive side. If the first later leading admissible slot beyond $\mathbf{K}_2^{\mathrm{cur}}$ is transfer-positive, then the same slotwise retained-discipline transfer mechanism that already produced the certified steps through σ_2 should certify one more step. The present block records that upgrade explicitly and removes the boundary-active alternative at σ_3 itself whenever the positive certificate is in force.

Definition H4.52 (Present three-step bounded certificate $\mathbf{K}_3^{\mathrm{cur}}$)**.** *Assume the setting of Definition H4.48. If σ_3 is transfer-positive, define the* present three-step bounded certificate

$$\mathbf{K}_3^{\mathrm{cur}} = (\mathcal{C}_0 \subset \mathcal{C}_1 \subset \mathcal{C}_2 \subset \mathcal{C}_3;\, \sigma_1, \sigma_2, \sigma_3),$$

where $\mathcal{C}_3$ is obtained from $\mathcal{C}_2$ by adjoining the transfer-positive leading admissible slot σ_3 under the same slotwise retained-discipline transfer certificate used in the already certified steps.

Proposition H4.53 (Transfer-positive status at σ_3 certifies $\mathbf{K}_3^{\mathrm{cur}}$ and discharges the local boundary channel)**.** *Assume the setting of Definition H4.52. Then:*

(i) the chain $\mathbf{K}_3^{\mathrm{cur}}$ is a valid bounded three-step OP 1 admissible-window certificate on the present extension line;

(ii) σ_3 is not a genuine post-$\mathbf{K}_2^{\mathrm{cur}}$ boundary-witness location; and

(iii) every honest future OP 1 extension failure, if any, must occur at a later leading admissible slot strictly beyond the corridor $\mathcal{C}_3$ certified by $\mathbf{K}_3^{\mathrm{cur}}$.

Proof. By Definition H4.52, the slot σ_3 carries the same slotwise retained-discipline transfer certificate as the already certified steps through σ_2. Proposition H4.37 therefore upgrades the chain by one step and certifies the displayed bounded three-step corridor, giving (i). Once that positive certification is in force, the alternative "boundary-active at σ_3" from Proposition H4.49 is excluded, proving (ii). Finally, after $\mathcal{C}_3$ has become part of the certified horizon, any later honest OP 1 failure must first appear strictly beyond $\mathcal{C}_3$, which is exactly (iii). □

Corollary H4.54 (The positive channel at σ_3 upgrades the certified horizon from $\mathbf{K}_2^{\mathrm{cur}}$ to $\mathbf{K}_3^{\mathrm{cur}}$)**.** *On the present OP 1 extension line, transfer-positive status at σ_3 does not merely produce a final continuation. It certifies the concrete bounded horizon $\mathbf{K}_3^{\mathrm{cur}}$ and pushes every honest future extension question to the first later leading admissible slot beyond $\mathcal{C}_3$.*

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Proof. Immediate from Proposition H4.53. □

Remark H4.55 (What certifying $\mathbf{K}_3^{\mathrm{cur}}$ now buys)**.** *This is the first point at which the extension line no longer talks about a hypothetical third step but about a named current three-step horizon. The benefit is again twofold. The theorematic line now has a larger certified corridor and therefore a smaller honest future search zone. The Monte-Carlo/fusion-probe layer likewise no longer needs to test anything at σ_3 itself; its next bounded question starts only at the first later leading admissible slot beyond $\mathcal{C}_3$.*

Reader cue. Read the next block as the symmetric normalization of the negative channel at σ_3. Its purpose is not to reopen the already certified positive branch, but to record what the extension line is honestly allowed to say when σ_3 is boundary-active: namely, that the first explicit post-$\mathbf{K}_2^{\mathrm{cur}}$ extension-boundary witness has already been found there and that no later slot needs to be consulted before that witness is discharged.

19AZ$^{(6u)}$. First explicit normalization of the boundary-active channel at σ_3 as the earliest post-$\mathbf{K}_2^{\mathrm{cur}}$ extension witness

After 19AZ$^{(6t)}$, the positive side at σ_3 has been normalized: if σ_3 is transfer-positive, then the line upgrades to the concrete bounded certificate $\mathbf{K}_3^{\mathrm{cur}}$. The negative side should now be stated with the same clarity. If σ_3 is boundary-active, then it is not merely a failed final slot. It is already the first explicit post-$\mathbf{K}_2^{\mathrm{cur}}$ extension witness, and every honest later extension question remains downstream of that earliest active boundary location until the witness at σ_3 has been discharged.

Definition H4.56 (Current boundary-active witness datum at σ_3)**.** *Assume the setting of Definition H4.48. If σ_3 is boundary-active, define the* current boundary-active witness datum

$$\mathfrak{B}_3^{\mathrm{cur}} = (\mathcal{C}_2, \sigma_3),$$

viewed as the earliest explicit post-$\mathbf{K}_2^{\mathrm{cur}}$ OP 1 extension-boundary witness on the present line.

Proposition H4.57 (Boundary-active status at σ_3 normalizes the earliest explicit post-$\mathbf{K}_2^{\mathrm{cur}}$ extension witness)**.** *Assume the setting of Definition H4.56. Then:*

(i) σ_3 is the first genuine post-$\mathbf{K}_2^{\mathrm{cur}}$ OP 1 extension-boundary location on the present line;

(ii) no valid bounded three-step certificate $\mathbf{K}_3^{\mathrm{cur}}$ is licensed on the same line until the witness datum $\mathfrak{B}_3^{\mathrm{cur}}$ has been discharged; and

(iii) every honest later OP 1 extension question remains downstream of σ_3 and need not be opened before the witness datum at σ_3 has been resolved.

Proof. If σ_3 is boundary-active, then by Proposition H4.49 the positive alternative fails exactly at the first later leading admissible slot beyond $\mathbf{K}_2^{\mathrm{cur}}$. This gives (i). Since Proposition H4.53 requires transfer-positive status in order to certify $\mathbf{K}_3^{\mathrm{cur}}$, the same line cannot simultaneously carry the certified three-step horizon while σ_3 remains boundary-active, proving (ii). Finally, because σ_3 is already the earliest active boundary location beyond the current certified horizon, no honest later extension question needs to be opened before that earliest witness has been addressed, which is (iii). □

Corollary H4.58 (The negative channel at σ_3 is the earliest explicit witness beyond $\mathbf{K}_2^{\mathrm{cur}}$)**.** *On the present OP 1 extension line, boundary-active status at σ_3 does not create a diffuse failure region. It produces the explicit earliest witness datum $\mathfrak{B}_3^{\mathrm{cur}} = (\mathcal{C}_2, \sigma_3)$ and blocks further honest extension promotion on that line until this witness has been discharged.*

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Proof. Immediate from Proposition H4.57. □

Remark H4.59 (What explicitly normalizing the boundary-active channel at σ_3 now buys)**.** *This is the symmetric counterpart to certifying* $\mathbf{K}_3^{\mathrm{cur}}$ *on the positive side. The benefit is that the negative side is no longer left as an unnamed failure possibility. It is packaged as the earliest explicit post-*$\mathbf{K}_2^{\mathrm{cur}}$ *witness datum, so that the theorematic line, the routing layer, and the Monte-Carlo/fusion-probe layer all know where the next honest obstruction sits if the transfer-positive channel does not carry. In particular, no later slot needs to be searched before* σ_3 *itself has been resolved.*

Reader cue. Read the next block as the compact decision-normal form for the first slot beyond the current certified horizon. Its purpose is not to add a new extension theorem, but to compress the already normalized alternatives at σ_3 into one finite decision object that the theorematic line, the routing layer, and the probe layer can all use without reopening the full case split.

19AZ$^{(6v)}$. First decision-normal form at σ_3 beyond the current certified horizon

After 19AZ$^{(6t)}$ and 19AZ$^{(6u)}$, the first slot beyond the current certified horizon has been normalized on both sides. On the positive side, transfer-positive status at σ_3 certifies the concrete bounded three-step horizon $\mathbf{K}_3^{\mathrm{cur}}$. On the negative side, boundary-active status at σ_3 produces the earliest explicit post-$\mathbf{K}_2^{\mathrm{cur}}$ witness datum $\mathfrak{B}_3^{\mathrm{cur}} = (\mathcal{C}_2, \sigma_3)$. The next honest step is therefore to package these two outcomes into one compact decision-normal form for the first later leading slot beyond the presently certified horizon.

Definition H4.60 (Current horizon decision datum at σ_3)**.** *Assume the setting of Definition H4.48. Define the* current horizon decision datum

$$\mathfrak{D}_3^{\mathrm{cur}} \in \{\mathbf{K}_3^{\mathrm{cur}}, \mathfrak{B}_3^{\mathrm{cur}}\}$$

by the rule

$$\mathfrak{D}_3^{\mathrm{cur}} = \begin{cases} \mathbf{K}_3^{\mathrm{cur}}, & \text{if } \sigma_3 \text{ is transfer-positive,} \\ \mathfrak{B}_3^{\mathrm{cur}}, & \text{if } \sigma_3 \text{ is boundary-active.} \end{cases}$$

Proposition H4.61 (The first slot beyond $\mathbf{K}_2^{\mathrm{cur}}$ admits a finite decision-normal form)**.** *Assume the setting of Definition H4.60. Then:*

(i) the datum $\mathfrak{D}_3^{\mathrm{cur}}$ *is well-defined and exhaustive for the first later leading admissible slot beyond* $\mathbf{K}_2^{\mathrm{cur}}$*;*

(ii) if $\mathfrak{D}_3^{\mathrm{cur}} = \mathbf{K}_3^{\mathrm{cur}}$*, then the certified horizon has advanced through* $\mathcal{C}_3$ *and every honest later extension question begins strictly beyond that corridor; and*

(iii) if $\mathfrak{D}_3^{\mathrm{cur}} = \mathfrak{B}_3^{\mathrm{cur}}$*, then* σ_3 *is already the earliest explicit post-*$\mathbf{K}_2^{\mathrm{cur}}$ *boundary location and no later slot needs to be opened before that witness datum has been discharged.*

Proof. By Proposition H4.49, the slot σ_3 admits only the two statuses transfer-positive and boundary-active. Definition H4.60 therefore assigns exactly one of the two normalized outputs $\mathbf{K}_3^{\mathrm{cur}}$ and $\mathfrak{B}_3^{\mathrm{cur}}$, proving (i). If the positive output occurs, Proposition H4.53 gives the certified three-step horizon and pushes every honest later extension question beyond $\mathcal{C}_3$, which is (ii). If the negative output occurs, Proposition H4.57 gives the earliest explicit witness datum at σ_3 and forbids opening later slots first, which is (iii). □

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Corollary H4.62 (The first slot beyond the current certified horizon is now decision-normalized)**.** *On the present OP 1 extension line, the first later leading admissible slot beyond* $\mathbf{K}_2^{\mathrm{cur}}$ *no longer carries an uncompressed continuation question. It is represented by the finite decision datum* $\mathfrak{D}_3^{\mathrm{cur}} \in \{\mathbf{K}_3^{\mathrm{cur}}, \mathfrak{B}_3^{\mathrm{cur}}\}$*, which either upgrades the current certified horizon or identifies the earliest explicit boundary witness.*

Proof. Immediate from Proposition H4.61. □

Remark H4.63 (What the decision-normal form at σ_3 now buys)**.** *This is the first point beyond the currently certified horizon where the extension line can speak in one compact finite decision object instead of in a longer two-branch case split. The theorematic line gains a cleaner handoff to the next extension stage; the routing layer gains a single certified output to follow; and the Monte-Carlo/fusion-probe layer gains a sharply bounded question: either the line has already promoted to* $\mathbf{K}_3^{\mathrm{cur}}$*, or the earliest active obstruction has already been localized at* σ_3 *itself.*

Reader cue. Read the next block as the compact QAM/decoder output language for the first extension decision beyond the current certified horizon. Its purpose is not to add a stronger extension theorem, but to package the already normalized alternatives $\mathbf{K}_3^{\mathrm{cur}}$ versus $\mathfrak{B}_3^{\mathrm{cur}}$ into one short token language that later theorematic, routing, and probe layers can cite without reopening the full decision datum.

19AZ$^{(6w)}$. First QAM/decoder output tokens for extension tasks beyond the current certified horizon

After 19AZ$^{(6v)}$, the first slot beyond the current certified horizon is represented by the finite decision datum

$$\mathfrak{D}_3^{\mathrm{cur}} \in \{\mathbf{K}_3^{\mathrm{cur}}, \mathfrak{B}_3^{\mathrm{cur}}\}.$$

The next honest compression step is therefore to translate this extension decision into a short output language that can be consumed uniformly by the QAM layer, the route selector, and the probe/decoder layer without reopening the full theorematic split.

Definition H4.64 (Extension-output token language at σ_3)**.** *Define the finite token set*

$$\mathcal{T}_3^{\mathrm{ext}} := \{\mathsf{EK}_3, \mathsf{EB}_3\},$$

where the symbols mean:

$$\mathsf{EK}_3 := \text{“the current horizon extends through } \mathcal{C}_3\text{”},$$
$$\mathsf{EB}_3 := \text{“the earliest honest post-}\mathbf{K}_2^{\mathrm{cur}} \text{ boundary sits at } \sigma_3\text{”}.$$

Define the associated token map by

$$\mathsf{Out}_{\mathrm{ext},3}^{\mathrm{tok}}(\mathfrak{D}_3^{\mathrm{cur}}) = \begin{cases} \mathsf{EK}_3, & \text{if } \mathfrak{D}_3^{\mathrm{cur}} = \mathbf{K}_3^{\mathrm{cur}}, \\ \mathsf{EB}_3, & \text{if } \mathfrak{D}_3^{\mathrm{cur}} = \mathfrak{B}_3^{\mathrm{cur}}. \end{cases}$$

Proposition H4.65 (The first extension decision beyond $\mathbf{K}_2^{\mathrm{cur}}$ admits a faithful token output)**.** *Assume the setting of Definition H4.64. Then:*

(i) the map $\mathsf{Out}_{\mathrm{ext},3}^{\mathrm{tok}}$ *is well-defined on the current decision datum* $\mathfrak{D}_3^{\mathrm{cur}}$*;*

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(ii) $\mathsf{Out}^{\mathrm{tok}}_{\mathrm{ext},3}(\mathfrak{D}^{\mathrm{cur}}_3) = \mathsf{EK}_3$ *if and only if the line has already promoted to the certified bounded horizon* $\mathbf{K}^{\mathrm{cur}}_3$*; and*

(iii) $\mathsf{Out}^{\mathrm{tok}}_{\mathrm{ext},3}(\mathfrak{D}^{\mathrm{cur}}_3) = \mathsf{EB}_3$ *if and only if* σ_3 *is already the earliest explicit boundary witness beyond* $\mathbf{K}^{\mathrm{cur}}_2$*.*

In particular, the token output is faithful to the full current extension decision.

Proof. By Definition H4.60, the decision datum $\mathfrak{D}^{\mathrm{cur}}_3$ takes exactly one of the two values $\mathbf{K}^{\mathrm{cur}}_3$ and $\mathfrak{B}^{\mathrm{cur}}_3$, so the token map is well-defined, proving (i). If the first value occurs, then Proposition H4.53 shows that the certified horizon has advanced through $\mathcal{C}_3$, hence the output token is exactly EK_3, proving (ii). If the second value occurs, then Proposition H4.57 shows that σ_3 is the earliest explicit post-$\mathbf{K}^{\mathrm{cur}}_2$ boundary location, hence the output token is exactly EB_3, proving (iii). Since the two tokens are distinct and attached to the two distinct normalized outcomes, the output is faithful to the full decision datum. □

Corollary H4.66 (Extension tasks at the first post-$\mathbf{K}^{\mathrm{cur}}_2$ slot are now token-normalized)**.** *On the present OP 1 extension line, the first extension task beyond the certified horizon* $\mathbf{K}^{\mathrm{cur}}_2$ *can now be recorded by the compact token*

$$\mathsf{Out}^{\mathrm{tok}}_{\mathrm{ext},3}(\mathfrak{D}^{\mathrm{cur}}_3) \in \{\mathsf{EK}_3, \mathsf{EB}_3\}.$$

Thus later routing or probe logic no longer needs the full branch split at σ_3*; it is enough to ask whether the current token is* EK_3 *or* EB_3*.*

Proof. Immediate from Proposition H4.65. □

Remark H4.67 (What the first extension-output tokens now buy)**.** *The gain is small but real. The present line no longer needs to treat the first post-*$\mathbf{K}^{\mathrm{cur}}_2$ *extension task as a longer prose decision between "certify the next horizon" and "localize the earliest new witness". The theorematic layer, the routing layer, and the Monte-Carlo/probe layer may all read the same compact output token* EK_3 *or* EB_3*. In particular, this is the first point where the extension line itself begins to speak the same kind of compact QAM/decoder language that had previously been reserved for reopening tasks inside the retained OP kernels.*

Reader cue. Read the next block as the reusable proof-compression template abstracted from the first post-$\mathbf{K}^{\mathrm{cur}}_2$ decision. Its purpose is not to claim a new global extension theorem, but to isolate the exact finite-decision pattern that any later first-slot extension stage must satisfy once its local positive-versus-boundary dichotomy has been established.

19AZ$^{(6x)}$. Reusable next-slot decision schema beyond a certified horizon

The blocks 19AZ$^{(6v)}$ and 19AZ$^{(6w)}$ show that, once the first later leading admissible slot beyond a certified current horizon has been normalized into the two statuses *transfer-positive* versus *boundary-active*, the continuation question collapses to one finite decision datum and one faithful output token. The point of the present block is to isolate that pattern as a reusable local schema.

Definition H4.68 (Reusable next-slot decision schema)**.** *Fix a current certified horizon* $\mathbf{K}^{\mathrm{cur}}_m$*, and let* σ_{m+1} *denote the first later leading admissible slot beyond it. Assume that at this slot the local theorematic line has already established an exhaustive dichotomy:*

$$\sigma_{m+1} \text{ is either transfer-positive or boundary-active.}$$

Assume further that:

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- *transfer-positive status at σ_{m+1} certifies a concrete upgraded horizon $\mathbf{K}^{\mathrm{cur}}_{m+1}$, and*
- *boundary-active status at σ_{m+1} produces an earliest explicit witness datum $\mathfrak{B}^{\mathrm{cur}}_{m+1}$.*

Then define the associated next-slot decision datum

$$\mathfrak{D}^{\mathrm{cur}}_{m+1} \in \{\mathbf{K}^{\mathrm{cur}}_{m+1}, \mathfrak{B}^{\mathrm{cur}}_{m+1}\}$$

by

$$\mathfrak{D}^{\mathrm{cur}}_{m+1} = \begin{cases} \mathbf{K}^{\mathrm{cur}}_{m+1}, & \text{if } \sigma_{m+1} \text{ is transfer-positive,} \\ \mathfrak{B}^{\mathrm{cur}}_{m+1}, & \text{if } \sigma_{m+1} \text{ is boundary-active.} \end{cases}$$

Define moreover the associated finite output-token set

$$\mathcal{T}^{\mathrm{ext}}_{m+1} := \{\mathsf{EK}_{m+1}, \mathsf{EB}_{m+1}\},$$

where

$$\mathsf{EK}_{m+1} := \text{“the current horizon extends through } \mathcal{C}_{m+1}\text{”},$$
$$\mathsf{EB}_{m+1} := \text{“the earliest honest boundary beyond } \mathbf{K}^{\mathrm{cur}}_{m} \text{ sits at } \sigma_{m+1}\text{”}.$$

Finally define the token output map

$$\mathsf{Out}^{\mathrm{tok}}_{\mathrm{ext},m+1}(\mathfrak{D}^{\mathrm{cur}}_{m+1}) = \begin{cases} \mathsf{EK}_{m+1}, & \text{if } \mathfrak{D}^{\mathrm{cur}}_{m+1} = \mathbf{K}^{\mathrm{cur}}_{m+1}, \\ \mathsf{EB}_{m+1}, & \text{if } \mathfrak{D}^{\mathrm{cur}}_{m+1} = \mathfrak{B}^{\mathrm{cur}}_{m+1}. \end{cases}$$

Proposition H4.69 (Any normalized first-slot extension stage collapses to one finite decision datum and one faithful token)**.** *Assume the setting of Definition H4.68. Then:*

(i) the datum $\mathfrak{D}^{\mathrm{cur}}_{m+1}$ is well-defined and exhaustive for the first later leading admissible slot beyond $\mathbf{K}^{\mathrm{cur}}_{m}$;

(ii) the token map $\mathsf{Out}^{\mathrm{tok}}_{\mathrm{ext},m+1}$ is well-defined and faithful to that decision datum; and

(iii) every later routing, decoder, or probe layer may consume the compact token EK_{m+1} or EB_{m+1} without reopening the full local branch split at σ_{m+1}.

Proof. By assumption, the local line at σ_{m+1} admits exactly the two statuses transfer-positive and boundary-active, and each status produces exactly one normalized output, namely $\mathbf{K}^{\mathrm{cur}}_{m+1}$ or $\mathfrak{B}^{\mathrm{cur}}_{m+1}$. This proves (i). The token map is then defined on exactly these two outputs and assigns distinct tokens to them, so it is well-defined and faithful, proving (ii). Since the token remembers which of the two normalized outcomes occurred, any later routing, decoder, or probe layer may consume the token alone unless it explicitly chooses to reopen the local theorematic proof, proving (iii). □

Corollary H4.70 (The σ_3 decision package is the first realized instance of the reusable extension schema)**.** *The package*

$$\mathfrak{D}^{\mathrm{cur}}_{3} \in \{\mathbf{K}^{\mathrm{cur}}_{3}, \mathfrak{B}^{\mathrm{cur}}_{3}\}, \qquad \mathsf{Out}^{\mathrm{tok}}_{\mathrm{ext},3} \in \{\mathsf{EK}_3, \mathsf{EB}_3\},$$

constructed in 19AZ$^{(6v)}$–19AZ$^{(6w)}$, is not merely a local convenience. It is the first realized instance of the general next-slot decision schema for extension tasks beyond a certified current horizon.

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Proof. Immediate from Definition H4.68 and Proposition H4.69 by taking $m = 2$. □

Remark H4.71 (What this schema now buys)**.** *This is the point where the present OP 1 extension line begins to export a genuine reusable tool rather than only a local closure fact. Future first-slot extension stages no longer need to reinvent the packaging logic: once the local positive-versus-boundary dichotomy is proved, the decision datum and faithful output token come for free. In particular, the current σ_3 block is now readable as the prototype of a general extension-normalization interface.*

Reader cue. Read the next block as the reusable action map generated by the extension-token language. Its purpose is not to add a stronger horizon theorem, but to force every later layer to take one of only two honest next actions: either advance the certified horizon, or reopen work exactly at the earliest normalized boundary site.

19AZ$^{(6y)}$. Reusable action/transition map driven by next-slot extension tokens

After 19AZ$^{(6x)}$, the line has not only a reusable next-slot decision datum but also a reusable token language. The next honest compression step is therefore to attach to that token language the corresponding next action. The guiding principle is simple: a token should not merely summarize the local result, but should also determine which later theorematic, routing, decoder, or probe move is licensed and which moves are forbidden.

Definition H4.72 (Reusable extension-action map)**.** *Assume the setting of Definition H4.68. Define the finite extension-action set*

$$\mathcal{A}^{\mathrm{ext}}_{m+1} := \{\mathsf{ADV}_{m+1}, \mathsf{OPENB}_{m+1}\},$$

where the symbols mean:

$$\begin{aligned} \mathsf{ADV}_{m+1} &:= \textit{advance current certified horizon through } \mathcal{C}_{m+1}, \\ \mathsf{OPENB}_{m+1} &:= \textit{open earliest normalized boundary kernel at } \sigma_{m+1}. \end{aligned}$$

Define the associated action map on extension tokens by

$$\mathsf{Act}^{\mathrm{tok}}_{\mathrm{ext},m+1}(\tau) = \begin{cases} \mathsf{ADV}_{m+1}, & \textit{if } \tau = \mathsf{EK}_{m+1}, \\ \mathsf{OPENB}_{m+1}, & \textit{if } \tau = \mathsf{EB}_{m+1}. \end{cases}$$

Equivalently, one may read the composite transition map

$$\mathsf{Act}^{\mathrm{tok}}_{\mathrm{ext},m+1} \circ \mathsf{Out}^{\mathrm{tok}}_{\mathrm{ext},m+1}$$

as the minimal action normal form attached to the first later leading admissible slot beyond $\mathbf{K}^{\mathrm{cur}}_m$.

Proposition H4.73 (The extension-token language determines the next honest action)**.** *Assume the setting of Definition H4.72. Then:*

(i) the map $\mathsf{Act}^{\mathrm{tok}}_{\mathrm{ext},m+1}$ *is well-defined on* $\mathcal{T}^{\mathrm{ext}}_{m+1}$*;*

(ii) if the output action is ADV_{m+1}*, then the current line is licensed to replace* $\mathbf{K}^{\mathrm{cur}}_m$ *by* $\mathbf{K}^{\mathrm{cur}}_{m+1}$ *and to relocate the first unresolved extension question strictly beyond* $\mathcal{C}_{m+1}$*;*

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(iii) if the output action is OPENB_{m+1}*, then every honest later theorematic, routing, decoder, or probe step must remain localized at the earliest normalized boundary site* σ_{m+1} *until that local boundary kernel has been processed.*

Proof. The token language from Definition H4.68 has exactly the two values EK_{m+1} and EB_{m+1}, so the action map is well-defined, proving (i). If the first token occurs, then by construction it records that the current horizon has already advanced through $\mathcal{C}_{m+1}$, so the only honest next action is to promote the certified horizon and move the unresolved extension question strictly beyond that corridor, proving (ii). If the second token occurs, then by construction it records that the earliest honest obstruction has already been normalized at σ_{m+1}; therefore any later layer that skips ahead before processing that site would violate the earliest-boundary discipline, proving (iii). □

Corollary H4.74 (The σ_3 token package now induces a certified next-action split)**.** *For the realized package from 19AZ*$^{(6v)}$*–19AZ*$^{(6x)}$*, the composite map*

$$\mathsf{Act}^{\mathrm{tok}}_{\mathrm{ext},3} \circ \mathsf{Out}^{\mathrm{tok}}_{\mathrm{ext},3}$$

returns exactly one of the two certified next actions:

$$\mathsf{ADV}_3 \qquad \textit{or} \qquad \mathsf{OPENB}_3.$$

Thus the present line is now action-normalized at the first slot beyond $\mathbf{K}_2^{\mathrm{cur}}$*: either it advances through* $\mathcal{C}_3$*, or it reopens work exactly at the earliest explicit boundary site* σ_3*.*

Proof. Immediate from Proposition H4.73 by taking $m = 2$. □

Remark H4.75 (What this action map now buys, and what can be salvaged into later layers)**.** *This is the point where the extension-tool layer stops being a passive encoding device and becomes a genuine routing interface. Later consumers no longer need to ask the full local question again; they need only read whether the current action is* ADV_{m+1} *or* OPENB_{m+1}*. That is precisely the salvageable discipline for the later optimization ideas: Monte-Carlo/fusion probes, decoder packages, and any later Heisenberg-matrix, Millennium-matrix, crystal-equation, or field-equation diagnostics should attach below this action interface as consumers of a certified local action, not as mechanisms that reopen or overwrite the theorematic decision itself.*

Reader cue. Read the next block as the iterative upgrade of the local action map. Its purpose is not to claim an already completed global induction, but to isolate the exact protocol by which the active line may repeatedly consume one certified next-slot token at a time, advancing when the token says "extend" and stopping honestly when the token says "boundary".

19AZ$^{(6z)}$. Iterated extension protocol generated by certified local action tokens

After 19AZ$^{(6y)}$, the line has a local action interface for the first later leading admissible slot beyond a certified current horizon. The next honest compression step is to package that interface into an iterative protocol. The point is simple: once one local stage has been normalized into a token and then into an action, the same logic can be reused at the next stage provided the new local dichotomy is again theorematically certified. Thus the extension line should be readable not as a one-off move at σ_3, but as a repeatable one-slot protocol with a built-in stop rule.

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Definition H4.76 (Iterated extension protocol with earliest-boundary stop rule)**.** *Fix a certified current horizon $\mathbf{K}_m^{\mathrm{cur}}$. Suppose that for each stage $r \geq m+1$, whenever the line reaches a certified horizon $\mathbf{K}_{r-1}^{\mathrm{cur}}$, the first later leading admissible slot σ_r admits a certified local dichotomy of the kind required in Definition H4.68. Then define the associated stage-r token*

$$\tau_r \in \mathcal{T}_r^{\mathrm{ext}} = \{\mathsf{EK}_r, \mathsf{EB}_r\}$$

and stage-r action

$$a_r := \mathsf{Act}_{\mathrm{ext},r}^{\mathrm{tok}}(\tau_r) \in \mathcal{A}_r^{\mathrm{ext}} = \{\mathsf{ADV}_r, \mathsf{OPENB}_r\}.$$

The resulting iterated extension protocol *is the finite or infinite stage-by-stage procedure that obeys the following rule at each step:*

(i) if $a_r = \mathsf{ADV}_r$, replace the current certified horizon $\mathbf{K}_{r-1}^{\mathrm{cur}}$ by $\mathbf{K}_r^{\mathrm{cur}}$ and continue to the next stage $r+1$;

(ii) if $a_r = \mathsf{OPENB}_r$, stop the forward extension protocol and localize all further honest work at the earliest normalized boundary kernel sitting at σ_r.

The first index at which the protocol stops, if such an index exists, is called the earliest protocol boundary stage *and is denoted by*

$$r_* := \min\{r \geq m+1 : a_r = \mathsf{OPENB}_r\}.$$

When no such stage exists in the range under consideration, the protocol is said to be advance-persistent *on that range.*

Proposition H4.77 (The iterated extension protocol is well-defined and locally non-skipping)**.** *Assume the setting of Definition H4.76. Then:*

(i) at each certified stage r, exactly one action $a_r \in \{\mathsf{ADV}_r, \mathsf{OPENB}_r\}$ is produced;

(ii) if the protocol is stopped at stage r_*, then σ_{r_*} is the earliest stage in the protocol at which the line is licensed to open a normalized boundary kernel rather than advance the certified horizon;

(iii) no honest theorematic, routing, decoder, probe, or diagnostic layer may skip past a stage with action OPENB_r and continue the forward extension protocol as if an ADV_r action had occurred.

Proof. For each certified stage r, Definition H4.68 yields exactly one token in $\{\mathsf{EK}_r, \mathsf{EB}_r\}$, and Definition H4.72 sends that token to exactly one action in $\{\mathsf{ADV}_r, \mathsf{OPENB}_r\}$, proving (i). If the protocol stops at stage r_*, then by definition r_* is the least stage whose action is OPENB_{r_*}, so all earlier stages were genuine advance stages and σ_{r_*} is the earliest protocol boundary stage, proving (ii). Statement (iii) is then forced by the meaning of OPENB_r: it records that the earliest honest unresolved obstruction has already been normalized at stage r, so any layer that skipped past it would violate the non-skipping boundary discipline built into $19\mathrm{AZ}^{(6y)}$. □

Corollary H4.78 (The present σ_3 package is the first step of an iterated extension protocol)**.** *In the current active line, the package from $19AZ^{(6v)}$–$19AZ^{(6y)}$ is not only a local closure package at the first slot beyond $\mathbf{K}_2^{\mathrm{cur}}$. It is the first realized stage of the iterated extension protocol. Concretely, the line now carries a certified first-stage token*

$$\tau_3 \in \{\mathsf{EK}_3, \mathsf{EB}_3\}$$

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and the associated first-stage action

$$a_3 \in \{\mathsf{ADV}_3, \mathsf{OPENB}_3\},$$

so every later extension attempt must either continue from the promoted horizon $\mathbf{K}_3^{\mathrm{cur}}$ *or remain localized at the earliest normalized boundary kernel at* σ_3*.*

Proof. Immediate from Corollary H4.74 and Proposition H4.77. □

Remark H4.79 (What can be salvaged from later optimization ideas without corrupting the proof line)**.** *This protocol is also the right place to salvage the useful part of the later optimization discussion. Once the active line emits a certified action stream* (a_r)*, later consumers may compress, hash, decode, profile, or statistically probe that stream — for example through Monte-Carlo/fusion diagnostics, decoder packages, Markov-style routing summaries, or matrix-side readout layers — provided they remain downstream consumers of certified stage actions. What they may* not *do is replace a missing local dichotomy proof by a heuristic token guess, or overwrite an* OPENB_r *stop signal by pretending that a later advance has already been theorematically licensed.*

Reader cue. Read the next block as the finite word-level compression of the iterated extension protocol. Its purpose is not to replace the stagewise theorematic decisions, but to encode any already certified finite protocol segment into a short QAM/decoder-readable extension word whose shape itself records whether the segment remained purely advancing or terminated at its first honest boundary stage.

19AZ$^{(6aa)}$. Finite QAM/decoder word form for certified extension-action segments

After 19AZ$^{(6z)}$, the active line has a stagewise protocol, but later consumers may still want a shorter object than the full indexed action stream. The next honest compression step is therefore to pass from certified stage actions to a finite extension word. The compression must remain faithful to the non-skipping boundary discipline: once an earliest boundary action appears, the forward word must terminate there rather than pretending to continue past that point.

Definition H4.80 (Finite extension word alphabet and encoder)**.** *Fix a starting certified horizon* $\mathbf{K}_m^{\mathrm{cur}}$*. Introduce the two-letter extension alphabet*

$$\Sigma_{\mathrm{ext}} := \{\mathsf{A}, \mathsf{B}\},$$

read as

$$\mathsf{A} = \textit{advance the certified horizon by one stage},$$
$$\mathsf{B} = \textit{open earliest certified boundary kernel and stop.}$$

For each stage $r \geq m+1$*, define the stage encoder*

$$\mathsf{enc}_r : \mathcal{A}_r^{\mathrm{ext}} \to \Sigma_{\mathrm{ext}}$$

by

$$\mathsf{enc}_r(\mathsf{ADV}_r) = \mathsf{A}, \qquad \mathsf{enc}_r(\mathsf{OPENB}_r) = \mathsf{B}.$$

If a certified protocol segment runs from stage $m+1$ *through stage* N*, define its associated* finite extension word

$$\mathbf{w}_{m+1:N} := \mathsf{enc}_{m+1}(a_{m+1})\,\mathsf{enc}_{m+2}(a_{m+2})\cdots\mathsf{enc}_N(a_N) \in \Sigma_{\mathrm{ext}}^*,$$

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with the convention that if some stage $r_* \le N$ satisfies $a_{r_*} = \mathsf{OPENB}_{r_*}$, then the word is truncated at r_*:

$$\mathbf{w}_{m+1:N} := \mathsf{enc}_{m+1}(a_{m+1}) \cdots \mathsf{enc}_{r_*}(a_{r_*}),$$

because the protocol is not licensed to advance past the earliest boundary stage.

Proposition H4.81 (Certified extension words have exactly the form A^k or $\mathsf{A}^k\mathsf{B}$)**.** *Assume the setting of Definition H4.76. Then every finite certified extension word produced by Definition H4.80 has exactly one of the following forms:*

$$\mathbf{w} = \mathsf{A}^k \qquad \text{or} \qquad \mathbf{w} = \mathsf{A}^k\mathsf{B}$$

for some integer $k \ge 0$. In particular:

(i) no certified extension word contains two occurrences of B*;*

(ii) no certified extension word contains B *followed by any later symbol;*

(iii) the word alone already records whether the observed certified segment is advance-persistent on its range or has terminated at its earliest honest boundary stage.

Proof. By Proposition H4.77, each certified stage contributes exactly one action, either ADV_r or OPENB_r. Under the encoder of Definition H4.80, these become A and B. If no stage in the observed range opens a boundary kernel, then every action is an advance action and the resulting word is A^k for the appropriate length k. If a boundary stage occurs, let r_* be the earliest one. The protocol stops at r_*, so the word is truncated there and has the form $\mathsf{A}^k\mathsf{B}$, where k counts the earlier advance stages. This proves the displayed classification, and clauses (i)–(iii) are immediate consequences of that classification. □

Corollary H4.82 (The extension protocol defines a regular terminal-word language)**.** *On every finite observation range, the certified extension words belong to the terminal language*

$$\mathcal{L}_{\mathrm{ext}}^{\mathrm{term}} = \{\mathsf{A}^k : k \ge 0\} \cup \{\mathsf{A}^k\mathsf{B} : k \ge 0\}.$$

Hence every later QAM, decoder, hashing, routing-summary, or Monte-Carlo/profile layer that consumes certified extension words must respect this language and may not treat words of the forbidden forms

$$\mathsf{BA}, \qquad \mathsf{ABA}, \qquad \mathsf{BB},$$

or any other word with a nonterminal boundary symbol as admissible proof-side inputs.

Proof. Immediate from Proposition H4.81. □

Remark H4.83 (What this word form now buys)**.** *This is the clean point where the active line becomes readable by a genuinely small working alphabet. The theorematic side still decides each local stage, but once those decisions are certified, later layers may consume the finite word* $\mathbf{w}$ *rather than the full local proof tree. That is the part of the later optimization discussion worth keeping: decoder maps, QAM readout, Markov summaries, Monte-Carlo deviation profiles, and matrix-side diagnostics may all work over the certified terminal language* $\mathcal{L}_{\mathrm{ext}}^{\mathrm{term}}$*. What they may not do is inject inadmissible words or continue reading letters after an honest terminal* B*-symbol has already appeared.*

Reader cue. Read the next block as the certificate-aware downstream matrix interpretation of the certified extension language. Its purpose is not to let the Heisenberg/Millennium side decide theorematic extension questions on its own, but to force every later matrix-side readout to distinguish sharply between certified horizon-growth and certified earliest-boundary exposure.

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19AZ$^{(6ab)}$. Certificate-aware horizon/boundary interpretation for Heisenberg/Millennium matrix readout

After 19AZ$^{(6aa)}$, the active line carries a small certified extension language. The natural next use of that language is a downstream matrix-side interpretation: the Heisenberg/Millennium layer should no longer be read as an undifferentiated global diagnostic, but as a certificate-aware readout whose outputs are tagged by whether the active theorematic line has presently certified horizon continuation or exposed the earliest honest boundary stage. This keeps the matrix layer useful without allowing it to outrun the proof line.

Definition H4.84 (Certificate-aware Heisenberg/Millennium readout tags and readout map)**.** *Fix a current certified horizon* $\mathbf{K}_m^{\mathrm{cur}}$*, and let* $\mathbf{w} \in \mathcal{L}_{\mathrm{ext}}^{\mathrm{term}}$ *be a certified finite extension word in the sense of Corollary H4.82. Introduce the two matrix-side readout tags*

$$\mathsf{HZ}_m(\mathbf{w}) = \textit{“horizon-certified matrix readout on the observed range”},$$

$$\mathsf{BD}_m(\mathbf{w}) = \textit{“boundary-certified matrix readout on the observed range”}.$$

Define the certificate-aware downstream readout map

$$\mathsf{Read}_m^{\mathrm{MM/HC}}(\mathbf{w}) \in \{\mathsf{HZ}_m(\mathbf{w}), \mathsf{BD}_m(\mathbf{w})\}$$

by

$$\mathsf{Read}_m^{\mathrm{MM/HC}}(\mathbf{w}) = \begin{cases} \mathsf{HZ}_m(\mathbf{w}), & \textit{if } \mathbf{w} = \mathsf{A}^k \textit{ for some } k \geq 0, \\ \mathsf{BD}_m(\mathbf{w}), & \textit{if } \mathbf{w} = \mathsf{A}^k\mathsf{B} \textit{ for some } k \geq 0. \end{cases}$$

Interpretation:

- $\mathsf{HZ}_m(\mathbf{w})$ *means that, on the observed certified range, the matrix side is only licensed to read a horizon-persistent transport picture;*
- $\mathsf{BD}_m(\mathbf{w})$ *means that, on the observed certified range, the matrix side must switch to earliest-boundary exposure and may not read past the terminal boundary stage as though further certified transport had already been proved.*

Proposition H4.85 (Certificate-aware matrix readout is faithful to the certified extension language)**.** *Assume the setting of Definition H4.84. Then:*

(i) the readout map $\mathsf{Read}_m^{\mathrm{MM/HC}}$ *is well-defined on every certified finite extension word;*

(ii) the matrix-side tag distinguishes exactly the two proof-legitimate cases “advance-only on the observed range” and “advance-until-earliest-boundary on the observed range”;

(iii) no downstream Heisenberg/Millennium interpretation is licensed to convert a boundary-certified word $\mathsf{A}^k\mathsf{B}$ *into a horizon-certified continuation reading, and no horizon-certified word* A^k *is allowed to be advertised as exposing an already certified boundary event that the theorematic line has not produced.*

Proof. By Corollary H4.82, every certified finite extension word has one of the two forms A^k or $\mathsf{A}^k\mathsf{B}$. The definition therefore assigns exactly one matrix-side tag to every certified word, so the readout map is well-defined, proving (i). Since these are precisely the two admissible word forms, the map distinguishes exactly the two proof-legitimate cases listed in (ii). Clause (iii) is then just the statement that the downstream matrix layer must remain faithful to the certified word it receives: a terminal B-symbol cannot be reinterpreted as if certified continuation had already occurred, and the absence of a terminal B-symbol cannot be inflated into a claimed certified boundary exposure. □

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Corollary H4.86 (The present σ_3 package already induces a first certificate-aware matrix reading)**.** *In the current active line, the certified first-stage package from 19AZ$^{(6v)}$–19AZ$^{(6aa)}$ already induces a first certificate-aware Heisenberg/Millennium readout. Concretely, the first-stage word is either*

$$\mathbf{w}_3 = \mathsf{A} \qquad \textit{or} \qquad \mathbf{w}_3 = \mathsf{B},$$

and therefore the associated downstream matrix-side reading is already forced into exactly one of the two tags

$$\mathsf{Read}_2^{\mathrm{MM/HC}}(\mathbf{w}_3) \in \{\mathsf{HZ}_2(\mathbf{w}_3), \mathsf{BD}_2(\mathbf{w}_3)\}.$$

Thus the matrix side can now be asked a precise question: does the current certified stage read as horizon-persistent transport, or as earliest-boundary exposure?

Proof. Immediate from the first-stage encoding in 19AZ$^{(6v)}$–19AZ$^{(6aa)}$ together with Corollary H4.82 and Proposition H4.85. □

Remark H4.87 (What this matrix reading now buys for later crystal and field-equation upgrades)**.** *This is the clean interface the later upgrade discussion was missing. The Heisenberg/Millennium layer can now be used as a certificate-aware reader of the active extension state rather than as a competing source of theorematic authority. In particular, later crystal-layer refinements and later field-equation diagnostics should be formulated as refinements of either the horizon-certified readout tag or the boundary-certified readout tag. That means they may sharpen what the current certified state looks like, but they should not overrule whether the active line currently sits in a horizon-growth state or at an earliest-boundary exposure state. The same discipline keeps Monte-Carlo or decoder-side diagnostics honest: they may profile, compress, or compare the tagged state, but not replace the proof-side tag assignment itself.*

Reader cue. Read the next block as the certificate-aware downstream crystal interpretation of the matrix-side tag discipline. Its purpose is not to reopen the earlier addressed-crystal architecture at full generality, but to force every later crystal-side readout to distinguish sharply between horizon-persistent transport/domain structure and earliest-boundary defect exposure.

19AZ$^{(6ac)}$. Certificate-aware horizon/boundary interpretation for crystal-layer readout

After 19AZ$^{(6ab)}$, the active line no longer carries only a proof-side token language and a matrix-side tag. It also carries the first clean rule for how the later crystal layer is allowed to read the active extension state. The crystal side should therefore not be treated as a free global metaphor or as an independent decision engine. It should be read as a downstream structure-preserving refinement of the already certified horizon-versus-boundary status.

Definition H4.88 (Certificate-aware crystal readout tags and crystal-side readout map)**.** *Fix a current certified horizon $\mathbf{K}_m^{\mathrm{cur}}$, let $\mathbf{w} \in \mathcal{L}_{\mathrm{ext}}^{\mathrm{term}}$ be a certified finite extension word, and let*

$$\mathsf{Read}_m^{\mathrm{MM/HC}}(\mathbf{w}) \in \{\mathsf{HZ}_m(\mathbf{w}), \mathsf{BD}_m(\mathbf{w})\}$$

be the associated certificate-aware matrix-side tag from Definition H4.84. Introduce the two crystal-side readout tags

$$\mathsf{CRH}_m(\mathbf{w}) = \textit{“horizon-certified crystal/domain transport readout”},$$

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$$\mathsf{CRB}_m(\mathbf{w}) = \textit{“boundary-certified crystal defect/front readout”}.$$

Define the certificate-aware crystal-side readout map

$$\mathsf{Read}_m^{\mathrm{cr}}(\mathbf{w}) \in \{\mathsf{CRH}_m(\mathbf{w}), \mathsf{CRB}_m(\mathbf{w})\}$$

by

$$\mathsf{Read}_m^{\mathrm{cr}}(\mathbf{w}) = \begin{cases} \mathsf{CRH}_m(\mathbf{w}), & \textit{if } \mathsf{Read}_m^{\mathrm{MM/HC}}(\mathbf{w}) = \mathsf{HZ}_m(\mathbf{w}), \\ \mathsf{CRB}_m(\mathbf{w}), & \textit{if } \mathsf{Read}_m^{\mathrm{MM/HC}}(\mathbf{w}) = \mathsf{BD}_m(\mathbf{w}). \end{cases}$$

Interpretation:

- $\mathsf{CRH}_m(\mathbf{w})$ *means that the crystal side is licensed to read the currently certified range as one transport-persistent domain picture, with no theorematically certified defect-front beyond that range yet exposed;*
- $\mathsf{CRB}_m(\mathbf{w})$ *means that the crystal side must treat the terminal certified information as an earliest honest defect/front exposure and may not smooth, continue, or visually homogenize that boundary as though later corridor coherence had already been proved.*

Proposition H4.89 (Certificate-aware crystal readout is faithful to the matrix-side horizon/boundary tag)**.** *Assume the setting of Definition H4.88. Then:*

(i) the crystal-side readout map $\mathsf{Read}_m^{\mathrm{cr}}$ *is well-defined on every certified finite extension word;*

(ii) the crystal-side tag distinguishes exactly the two proof-legitimate crystal readings “domain transport on the certified range” and “earliest defect/front exposure at the certified boundary stage”;

(iii) no downstream crystal-layer interpretation is licensed to smooth a boundary-certified state into a transport-certified domain picture, and no horizon-certified state is allowed to be advertised as already carrying a proved defect/front event that the active line has not certified.

Proof. By Proposition H4.85, the matrix-side readout assigns exactly one of the two tags $\mathsf{HZ}_m(\mathbf{w})$ or $\mathsf{BD}_m(\mathbf{w})$ to every certified word. Definition H4.88 therefore assigns exactly one crystal-side tag $\mathsf{CRH}_m(\mathbf{w})$ or $\mathsf{CRB}_m(\mathbf{w})$, proving (i). Since these two tags simply refine the two proof-legitimate matrix-side statuses into crystal/domain language, statement (ii) follows immediately. Clause (iii) is the same faithfulness discipline expressed on the crystal side: a boundary-certified terminal stage cannot be visually or narratively turned into uninterrupted continuation, and a transport-certified range cannot be embellished with a certified defect/front event that the theorematic line has not produced. □

Corollary H4.90 (The present σ_3 package already induces a first certificate-aware crystal reading)**.** *In the current active line, the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$ *already induces a first certificate-aware crystal-side readout*

$$\mathsf{Read}_2^{\mathrm{cr}}(\mathbf{w}_3) \in \{\mathsf{CRH}_2(\mathbf{w}_3), \mathsf{CRB}_2(\mathbf{w}_3)\}.$$

Thus the crystal layer can now be asked a precise and conservative question: does the present certified stage still read as one transport-persistent domain picture, or has the earliest honest defect/front already been exposed at the current terminal stage?

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Proof. Immediate from Corollary H4.86 and Proposition H4.89. □

Remark H4.91 (What this crystal reading now buys)**.** *This is the clean point where the earlier liquid-crystal/domain/defect language becomes structurally disciplined by the active extension certificates. The crystal side may now refine the current state into domain persistence, defect loading, or front exposure language without pretending to certify a later slot on its own. In particular, later holographic boundary visuals, domain-defect compression schemes, and Monte-Carlo or decoder-side pattern scans may profile the crystal-side tag, but they must inherit it from the certified extension language rather than invent it.*

Reader cue. Read the next block as the certificate-aware downstream pre-equation interface for field-equation diagnostics. Its purpose is not to claim a finished Einstein-, Dirac-, or QFT-type projection law at this stage, but to force every later field-equation ansatz to distinguish sharply between a horizon-certified continuation profile and an earliest-boundary source/exposure profile.

19AZ$^{(6ad)}$. Certificate-aware horizon/boundary interpretation for field-equation diagnostics

After 19AZ$^{(6ac)}$, the active line has a theorematically certified extension language, a matrix-side readout, and a crystal-side readout. The next honest compression step is not yet a full new field equation. It is a certificate-aware pre-equation interface that tells any later physical or PDE-style readout whether it is looking at a continuation-certified transport profile or at an earliest-boundary exposure profile. This is the smallest disciplined upgrade that the current line can honestly support.

Definition H4.92 (Certificate-aware field-diagnostic tags and pre-equation map)**.** *Fix a current certified horizon $\mathbf{K}_m^{\mathrm{cur}}$, let $\mathbf{w} \in \mathcal{L}_{\mathrm{ext}}^{\mathrm{term}}$ be a certified finite extension word, and let*

$$\mathsf{Read}_m^{\mathrm{cr}}(\mathbf{w}) \in \{\mathsf{CRH}_m(\mathbf{w}), \mathsf{CRB}_m(\mathbf{w})\}$$

be the associated crystal-side readout from Definition H4.88. Introduce the two field-diagnostic tags

$$\mathsf{FEH}_m(\mathbf{w}) = \text{“horizon-certified pre-equation transport profile”},$$

$$\mathsf{FEB}_m(\mathbf{w}) = \text{“boundary-certified pre-equation source/exposure profile”}.$$

Define the certificate-aware field-diagnostic map

$$\mathsf{Diag}_m^{\mathrm{field}}(\mathbf{w}) \in \{\mathsf{FEH}_m(\mathbf{w}), \mathsf{FEB}_m(\mathbf{w})\}$$

by

$$\mathsf{Diag}_m^{\mathrm{field}}(\mathbf{w}) = \begin{cases} \mathsf{FEH}_m(\mathbf{w}), & \text{if } \mathsf{Read}_m^{\mathrm{cr}}(\mathbf{w}) = \mathsf{CRH}_m(\mathbf{w}), \\ \mathsf{FEB}_m(\mathbf{w}), & \text{if } \mathsf{Read}_m^{\mathrm{cr}}(\mathbf{w}) = \mathsf{CRB}_m(\mathbf{w}). \end{cases}$$

Interpretation:

- $\mathsf{FEH}_m(\mathbf{w})$ *means that any later field-equation ansatz is presently licensed only to read continuation-compatible transport on the already certified range;*
- $\mathsf{FEB}_m(\mathbf{w})$ *means that any later field-equation ansatz must already acknowledge an earliest honest boundary/source exposure at the terminal certified stage and may not write a smooth forward continuation law past that stage as though the proof line had already certified it.*

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Proposition H4.93 (Certificate-aware field diagnostics are faithful to the crystal-side tag)**.** *Assume the setting of Definition H4.92. Then:*

(i) the field-diagnostic map $\mathsf{Diag}_m^{\mathrm{field}}$ *is well-defined on every certified finite extension word;*

(ii) the field-diagnostic tag distinguishes exactly the two presently legitimate pre-equation situations "continuation-compatible certified transport" and "earliest-boundary source/-exposure already active";

(iii) no downstream field-equation ansatz is licensed to erase a terminal boundary-certified status, and no continuation-certified status is allowed to be sold as a proved boundary/-source event that the active line has not certified.

Proof. By Proposition H4.89, the crystal-side readout assigns exactly one of the tags $\mathsf{CRH}_m(\mathbf{w})$ or $\mathsf{CRB}_m(\mathbf{w})$ to every certified word. Definition H4.92 therefore assigns exactly one field-diagnostic tag $\mathsf{FEH}_m(\mathbf{w})$ or $\mathsf{FEB}_m(\mathbf{w})$, proving (i). Since these are precisely the two downstream field-diagnostic refinements of the crystal-side statuses, statement (ii) follows. Clause (iii) is the same certificate faithfulness expressed at the pre-equation layer: a certified terminal boundary/source exposure cannot be rewritten as already-proved smooth continuation, and a certified transport range cannot be exaggerated into a proved boundary event. □

Corollary H4.94 (The present σ_3 package already induces a first certificate-aware field-diagnostic split)**.** *In the current active line, the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$ *already induces a first certificate-aware field-diagnostic split*

$$\mathsf{Diag}_2^{\mathrm{field}}(\mathbf{w}_3) \in \{\mathsf{FEH}_2(\mathbf{w}_3), \mathsf{FEB}_2(\mathbf{w}_3)\}.$$

Thus the later field-equation layer can already be asked a disciplined question: is the current certified stage to be read as continuation-compatible transport on the observed range, or as an earliest-boundary source/exposure profile that must remain explicit in every later PDE or operator ansatz?

Proof. Immediate from Corollary H4.90 and Proposition H4.93. □

Remark H4.95 (What this field-diagnostic interface now buys)**.** *This is the conservative upgrade the later field-equation discussion was missing. The active line still does not claim a finished physical PDE, a finished Einstein equation, or a finished emergence law. What it now does claim is a disciplined pre-equation interface: every later ansatz must declare whether it is reading a horizon-certified transport state or an earliest-boundary source/exposure state. That already sharpens admissibility. A later field equation is acceptable only if its local readout remains faithful to the certified tag it inherits from the extension language.*

Reader cue. Read the next block as the admissibility and consistency gate for any later PDE-, operator-, or emergence-style ansatz built on the certificate-aware field-diagnostic interface. Its purpose is not to force one unique final field equation, but to exclude exactly those later ansätze that would erase, overstate, or misroute the certified horizon/boundary status already inherited from the active extension line.

19AZ$^{(6ae)}$. Admissibility and consistency conditions for later field-equation ansätze

After 19AZ$^{(6ad)}$, the active line still does not claim a unique finished field equation. What it does claim is stronger than a vague heuristic: any later PDE-, operator-, or emergence-style law must now pass a certificate-aware admissibility gate. In other words, once a later

ansatz inherits the field-diagnostic tag $\mathsf{FEH}_m(\mathbf{w})$ or $\mathsf{FEB}_m(\mathbf{w})$, it is no longer free to ignore the certified status that generated that tag.

Definition H4.96 (Certificate-aware admissibility for later field-equation ansätze)**.** *Fix a current certified horizon $\mathbf{K}_m^{\mathrm{cur}}$, let $\mathbf{w} \in \mathcal{L}_{\mathrm{ext}}^{\mathrm{term}}$ be a certified finite extension word, and let*

$$\mathsf{Diag}_m^{\mathrm{field}}(\mathbf{w}) \in \{\mathsf{FEH}_m(\mathbf{w}), \mathsf{FEB}_m(\mathbf{w})\}$$

be the inherited field-diagnostic tag from Definition H4.92. A later PDE-, operator-, transport-, or emergence-style ansatz

$$\mathfrak{E}_{m,\mathbf{w}}$$

is called certificate-aware admissible *if it satisfies the following conditions.*

(i) ***Tag fidelity.*** *The ansatz must explicitly declare whether it is reading the state under $\mathsf{FEH}_m(\mathbf{w})$ or under $\mathsf{FEB}_m(\mathbf{w})$, and its local interpretation must respect that inherited tag.*

(ii) ***No boundary erasure.*** *If the inherited tag is $\mathsf{FEB}_m(\mathbf{w})$, the ansatz may not replace the terminal certified boundary/source exposure by a smooth forward continuation law as though the active line had already certified later corridor transport.*

(iii) ***No boundary inflation.*** *If the inherited tag is $\mathsf{FEH}_m(\mathbf{w})$, the ansatz may not advertise an already certified source, defect, singularity, or terminal boundary event that the active line has not produced.*

(iv) ***Range discipline.*** *The ansatz may refine, profile, or parameterize the already certified observed range, but it may not claim theorematic authority beyond the range encoded by the certified word $\mathbf{w}$.*

If one of these conditions fails, the ansatz is called certificate-aware inadmissible *on the input pair $(m, \mathbf{w})$.*

Proposition H4.97 (The certificate-aware field interface induces a sharp admissibility dichotomy)**.** *Assume the setting of Definition H4.96. Then every later field-equation ansatz $\mathfrak{E}_{m,\mathbf{w}}$ is forced into exactly one of the two statuses:*

$$\textit{certificate-aware admissible} \qquad \textit{or} \qquad \textit{certificate-aware inadmissible.}$$

Moreover:

(i) every admissible ansatz is a downstream refinement of the inherited tag rather than a replacement for it;

(ii) every inadmissible ansatz fails by one of exactly three routes: boundary erasure, boundary inflation, or range overreach;

(iii) this admissibility test is independent of whether the later ansatz is written in matrix, crystal, PDE, operator, Monte-Carlo-assisted, or decoder-compressed language.

Proof. Definition H4.96 gives explicit necessary and sufficient conditions for admissibility on the input pair $(m, \mathbf{w})$. Hence every later ansatz either satisfies all of them or fails at least one, which proves the displayed dichotomy. Statement (i) is immediate from tag fidelity together with range discipline: an admissible ansatz inherits the certified field tag and may only refine the already certified state. For (ii), the only ways to violate the conditions are

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exactly the ones listed in the definition: either an inherited boundary-certified state is erased, an inherited continuation-certified state is exaggerated into a certified boundary/source event, or the ansatz claims interpretive authority beyond the certified observed range. Statement (iii) follows because the admissibility test depends only on the inherited certified tag and on faithfulness to the certified word, not on the notation or computational packaging chosen downstream. □

Corollary H4.98 (The present σ_3 package already excludes the wrong kinds of field-law upgrades)**.** *In the current active line, the first-stage package with* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$ *already excludes two opposite classes of later upgrade errors:*

(i) if $\mathbf{w}_3 = \mathsf{B}$*, no later matrix-, crystal-, or field-law ansatz is allowed to smooth that first certified boundary exposure into an already proved forward continuation picture;*

(ii) if $\mathbf{w}_3 = \mathsf{A}$*, no later matrix-, crystal-, or field-law ansatz is allowed to present the current stage as though an already certified boundary/source event had been exposed.*

Thus even before a finished field equation exists, the present line already knows which later upgrade moves are structurally forbidden.

Proof. Immediate from Corollary H4.94 and Proposition H4.97. □

Remark H4.99 (What this admissibility gate now buys)**.** *This is the point where the current line becomes strong enough to police its own future upgrades. A later Einstein-style, transport-style, source-law, Monte-Carlo-assisted closure law, or operator-valued emergence ansatz no longer has to be rejected because it is ambitious; it has to be rejected only if it violates the inherited certificate. That is exactly the right level of rigidity at the present stage: permissive about representation, strict about structural faithfulness.*

Reader cue. Read the next block as the certificate-aware upgrade-routing interface that tells every later refinement layer which type of work is legitimately allowed after a field-diagnostic tag has been inherited. Its purpose is not to force one unique downstream model, but to route conservative continuation-style upgrades to the horizon-certified branch and boundary-profiling upgrades to the boundary-certified branch, while excluding cross-routings that would reopen or falsify the certified status.

19AZ$^{(6af)}$. Upgrade-routing consequences of certificate-aware field-diagnostic tags

After 19AZ$^{(6ae)}$, the active line has more than a static admissibility gate. It now has enough structure to route later upgrades according to the inherited certificate. The point is simple: not every technically sophisticated downstream refinement is appropriate for every certified word. A horizon-certified tag should route later work toward conservative continuation-style refinement on the already proved range, while a boundary-certified tag should route later work toward earliest-boundary profiling, exposure analysis, and stop-sensitive diagnostics. This routing discipline is the missing interface between the theorematic extension line and later matrix-, crystal-, field-, decoder-, or probe-based elaborations.

Definition H4.100 (Certificate-aware upgrade-routing map)**.** *Fix a current certified horizon* $\mathbf{K}_m^{\mathrm{cur}}$*, let* $\mathbf{w} \in \mathcal{L}_{\mathrm{ext}}^{\mathrm{term}}$ *be a certified finite extension word, and let*

$$\mathsf{Diag}_m^{\mathrm{field}}(\mathbf{w}) \in \{\mathsf{FEH}_m(\mathbf{w}), \mathsf{FEB}_m(\mathbf{w})\}$$

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be the inherited field-diagnostic tag from Definition H4.92. Introduce the two conservative upgrade-route classes

$$\mathcal{U}_m^{\mathrm{hor}}(\mathbf{w}) := \text{“continuation-compatible refinement on the already certified range”},$$
$$\mathcal{U}_m^{\mathrm{bdry}}(\mathbf{w}) := \text{“earliest-boundary profiling and stop-sensitive exposure analysis”}.$$

Define the certificate-aware upgrade-routing map

$$\mathsf{Route}_m^{\mathrm{upg}}(\mathbf{w}) \in \{\mathcal{U}_m^{\mathrm{hor}}(\mathbf{w}), \mathcal{U}_m^{\mathrm{bdry}}(\mathbf{w})\}$$

by

$$\mathsf{Route}_m^{\mathrm{upg}}(\mathbf{w}) = \begin{cases} \mathcal{U}_m^{\mathrm{hor}}(\mathbf{w}), & \text{if } \mathsf{Diag}_m^{\mathrm{field}}(\mathbf{w}) = \mathsf{FEH}_m(\mathbf{w}), \\ \mathcal{U}_m^{\mathrm{bdry}}(\mathbf{w}), & \text{if } \mathsf{Diag}_m^{\mathrm{field}}(\mathbf{w}) = \mathsf{FEB}_m(\mathbf{w}). \end{cases}$$

Interpretation:

- *the horizon-certified route $\mathcal{U}_m^{\mathrm{hor}}(\mathbf{w})$ licenses only those later upgrades that refine transport, matrix readout, crystal stabilization, or PDE/operator behavior on the range already certified by $\mathbf{w}$, without claiming a new boundary/source event;*
- *the boundary-certified route $\mathcal{U}_m^{\mathrm{bdry}}(\mathbf{w})$ licenses only those later upgrades that sharpen the earliest-boundary profile itself, for example by defect localization, source/exposure analysis, or stop-sensitive decoder/probe diagnostics, without pretending that corridor continuation past the terminal boundary has already been proved;*
- *any downstream refinement that crosses these route classes against the inherited tag is called a* cross-routed upgrade *and is provisionally inadmissible until a new local theorematic stage changes the certified status.*

Proposition H4.101 (The inherited field-diagnostic tag forces a unique conservative upgrade route)**.** *Assume the setting of Definition H4.100. Then:*

(i) the routing map $\mathsf{Route}_m^{\mathrm{upg}}$ is well-defined on every certified finite extension word;

(ii) a horizon-certified tag $\mathsf{FEH}_m(\mathbf{w})$ forces the conservative route class $\mathcal{U}_m^{\mathrm{hor}}(\mathbf{w})$, while a boundary-certified tag $\mathsf{FEB}_m(\mathbf{w})$ forces the conservative route class $\mathcal{U}_m^{\mathrm{bdry}}(\mathbf{w})$;

(iii) every cross-routed upgrade is incompatible with the certificate-aware admissibility conditions from Definition H4.96 unless and until a new theorematic local stage certifies a different status.

Proof. By Proposition H4.97, every certified word inherits exactly one field-diagnostic tag, either $\mathsf{FEH}_m(\mathbf{w})$ or $\mathsf{FEB}_m(\mathbf{w})$. Definition H4.100 therefore assigns exactly one conservative route class, proving (i). Clause (ii) is the explicit content of the definition. For (iii), suppose a downstream refinement is routed against the inherited tag. If a boundary-certified word is forced onto the continuation-style route, then the upgrade attempts boundary erasure by treating an earliest-boundary exposure as already-certified transport continuation. If a horizon-certified word is forced onto the boundary-style route, then the upgrade attempts boundary inflation by advertising a defect, source, or stopping event not certified by the active line. Both behaviors violate the admissibility gate from Definition H4.96. Hence cross-routed upgrades are incompatible with the present certificate unless a later local theorematic stage genuinely changes that certificate. □

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Corollary H4.102 (The present σ_3 package already induces a first conservative upgrade-routing split)**.** *In the current active line, the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$ *already induces a first conservative routing split*

$$\mathsf{Route}_2^{\mathrm{upg}}(\mathbf{w}_3) \in \{\mathcal{U}_2^{\mathrm{hor}}(\mathbf{w}_3), \mathcal{U}_2^{\mathrm{bdry}}(\mathbf{w}_3)\}.$$

Thus the next downstream question is no longer vague. If the inherited tag is horizon-certified, later work should concentrate on transport-compatible refinement of the certified range. If the inherited tag is boundary-certified, later work should concentrate on profiling the earliest honest boundary rather than pretending that the corridor has already been extended through it.

Proof. Immediate from Corollary H4.94 and Proposition H4.101. □

Remark H4.103 (What this upgrade-routing interface now buys)**.** *This is the first place where the active line tells later heuristic or computational layers not only what they may* say*, but also what they should* do next*. Matrix sharpening, crystal stabilization, conservative PDE refinement, decoder compression, Monte-Carlo probing, or source/exposure diagnostics are now routed by the inherited certificate rather than by taste. In particular, Monte-Carlo, decoder, hash, or Markov-style probes may read and compress the routed status, but they are not allowed to replace the theorematic routing decision itself. That keeps the later tool layer useful without allowing it to overrule the certified extension line.*

Reader cue. Read the next block as the normalized follow-on upgrade menu sitting *below* the certificate-aware routing interface. Its purpose is to say which concrete families of downstream refinement are genuinely licensed once a route has already been fixed, and which families remain out of bounds until a later local theorematic stage changes the certificate.

19AZ$^{(6ag)}$. Normalized follow-on upgrade families below the certificate-aware routing interface

After 19AZ$^{(6af)}$, the line knows not only whether a later refinement is admissible, but also which route it must follow. The next honest compression step is to normalize the concrete follow-on upgrade families that sit below those routes. This is important because later readers should not have to guess whether, for example, matrix sharpening, crystal stabilization, source localization, decoder compression, or Monte-Carlo probing belongs to the continuation side or to the earliest-boundary side. The present block turns that tacit distinction into a finite, certificate-aware menu.

Definition H4.104 (Normalized follow-on upgrade families)**.** *Fix a current certified horizon* $\mathbf{K}_m^{\mathrm{cur}}$*, let* $\mathbf{w} \in \mathcal{L}_{\mathrm{ext}}^{\mathrm{term}}$ *be a certified finite extension word, and let*

$$\mathsf{Route}_m^{\mathrm{upg}}(\mathbf{w}) \in \{\mathcal{U}_m^{\mathrm{hor}}(\mathbf{w}), \mathcal{U}_m^{\mathrm{bdry}}(\mathbf{w})\}$$

be the inherited upgrade route from Definition H4.100.

Define the horizon-follow-on family menu

$$\mathcal{F}_m^{\mathrm{hor}}(\mathbf{w}) := \{\mathsf{MSH}_m(\mathbf{w}), \mathsf{CRS}_m(\mathbf{w}), \mathsf{PDE}_m^{\mathrm{cons}}(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{hor}}(\mathbf{w})\}$$

with the intended meanings

$$\begin{aligned}
\mathsf{MSH}_m(\mathbf{w}) &:= \text{“matrix sharpening on the already certified range”},\\
\mathsf{CRS}_m(\mathbf{w}) &:= \text{“crystal stabilization / transport-consistency refinement”},\\
\mathsf{PDE}_m^{\mathrm{cons}}(\mathbf{w}) &:= \text{“conservative PDE/operator refinement on the certified range”},\\
\mathsf{DEC}_m^{\mathrm{hor}}(\mathbf{w}) &:= \text{“decoder / compression readout of a horizon-certified status”}.
\end{aligned}$$

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Define the boundary-follow-on family menu

$$\mathcal{F}_m^{\mathrm{bdry}}(\mathbf{w}) := \{\mathsf{SRC}_m(\mathbf{w}), \mathsf{DFX}_m(\mathbf{w}), \mathsf{MC}_m^{\mathrm{bdry}}(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{bdry}}(\mathbf{w})\}$$

with the intended meanings

$$\begin{aligned}
\mathsf{SRC}_m(\mathbf{w}) &:= \text{“source / exposure localization at the earliest honest boundary”},\\
\mathsf{DFX}_m(\mathbf{w}) &:= \text{“defect-front or crystal-boundary profiling”},\\
\mathsf{MC}_m^{\mathrm{bdry}}(\mathbf{w}) &:= \text{“stop-sensitive Monte-Carlo or probe diagnostics at the boundary side”},\\
\mathsf{DEC}_m^{\mathrm{bdry}}(\mathbf{w}) &:= \text{“decoder / compression readout of a boundary-certified status”}.
\end{aligned}$$

A normalized follow-on upgrade *for* $\mathbf{w}$ *is then any element of the route-compatible menu*

$$\mathcal{F}_m^{\mathrm{follow}}(\mathbf{w}) := \begin{cases} \mathcal{F}_m^{\mathrm{hor}}(\mathbf{w}), & \text{if } \mathsf{Route}_m^{\mathrm{upg}}(\mathbf{w}) = \mathcal{U}_m^{\mathrm{hor}}(\mathbf{w}),\\ \mathcal{F}_m^{\mathrm{bdry}}(\mathbf{w}), & \text{if } \mathsf{Route}_m^{\mathrm{upg}}(\mathbf{w}) = \mathcal{U}_m^{\mathrm{bdry}}(\mathbf{w}). \end{cases}$$

Any proposed later refinement that selects an element outside $\mathcal{F}_m^{\mathrm{follow}}(\mathbf{w})$ *is called a* route-incompatible follow-on upgrade.

Proposition H4.105 (Every certified word carries a finite route-compatible follow-on upgrade menu)**.** *Assume the setting of Definition H4.104. Then:*

(i) every certified finite extension word $\mathbf{w}$ *carries exactly one route-compatible follow-on family menu* $\mathcal{F}_m^{\mathrm{follow}}(\mathbf{w})$*;*

(ii) every normalized follow-on upgrade in $\mathcal{F}_m^{\mathrm{follow}}(\mathbf{w})$ *preserves the certificate-aware routing discipline from* $19AZ^{(6af)}$*;*

(iii) every route-incompatible follow-on upgrade attempts either continuation overreach on the boundary side or spurious defect/source inflation on the horizon side, and is therefore provisionally inadmissible.

Proof. By Proposition H4.101, every certified finite extension word inherits exactly one conservative upgrade route. Definition H4.104 assigns to that unique route exactly one finite menu, proving (i). Clause (ii) is built into the meanings of the menu entries: the horizon-family items refine only the already certified continuation range, while the boundary-family items profile only the earliest certified boundary or exposure picture. For (iii), if a boundary-certified word is sent to a horizon-family item, the proposed refinement necessarily behaves as though continuation beyond the terminal boundary were already certified; this is continuation overreach. If a horizon-certified word is sent to a boundary-family item, the proposed refinement injects a defect, source, or stopping narrative that has not been certified by the active line; this is spurious inflation. Both violate the route discipline and so are provisionally inadmissible. □

Corollary H4.106 (The present σ_3 package already carries a finite next-work menu)**.** *In the current active line, the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$ *already carries a finite next-work menu*

$$\mathcal{F}_2^{\mathrm{follow}}(\mathbf{w}_3) \subseteq \mathcal{F}_2^{\mathrm{hor}}(\mathbf{w}_3) \cup \mathcal{F}_2^{\mathrm{bdry}}(\mathbf{w}_3).$$

Thus the immediate next work is no longer only route-correct in the abstract. It is already normalized into a finite family of concrete continuation-side or boundary-side upgrade tasks.

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Proof. Immediate from Corollary H4.102 and Proposition H4.105. □

Remark H4.107 (What this finite next-work menu now buys)**.** *This is the point where the active line starts to look like a genuine operator's manual for the next minor release. Matrix sharpening, crystal stabilization, conservative PDE refinement, decoder compression, source localization, defect profiling, and stop-sensitive Monte-Carlo probing are no longer a loose wishlist. They have become a finite, certificate-aware menu. In particular, Monte-Carlo now has a safe place: it can act as a boundary-side structural probe when the inherited status is boundary-certified, but it is not allowed to masquerade as theorematic continuation proof.*

oindent extbfReader cue. Read the next block as the explicit QAM-/decoder-side shadow of the certificate-aware extension interface. Its purpose is not to introduce a new ontology above the current tool layer, but to compress the already inherited field tag, route, and next-work menu into one finite downstream object that QAM- and decoder-style readers may consume without reopening the theorematic proof split.

19AZ$^{(6ah)}$. QAM-/decoder-side shadow of the certificate-aware extension interface

After 19AZ$^{(6ag)}$, the active line already carries everything a later QAM- or decoder-side layer needs in order to act conservatively: a certified finite word, a field-diagnostic tag, a forced route, and a finite next-work menu. The next honest compression step is therefore not a new theorem, but a finite shadow object that packages these inherited data into one downstream-readable interface. This is exactly the point where the present certificate-aware tool layer becomes explicitly QAM-/decoder-consumable.

Definition H4.108 (QAM-/decoder-side shadow state)**.** *Fix a current certified horizon* $\mathbf{K}_m^{\mathrm{cur}}$*, let* $\mathbf{w} \in \mathcal{L}_{\mathrm{ext}}^{\mathrm{term}}$ *be a certified finite extension word, and let*

$$\mathsf{Diag}_m^{\mathrm{field}}(\mathbf{w}) \in \{\mathsf{FEH}_m(\mathbf{w}), \mathsf{FEB}_m(\mathbf{w})\}, \qquad \mathsf{Route}_m^{\mathrm{upg}}(\mathbf{w}) \in \{\mathcal{U}_m^{\mathrm{hor}}(\mathbf{w}), \mathcal{U}_m^{\mathrm{bdry}}(\mathbf{w})\},$$

and

$$\mathcal{F}_m^{\mathrm{follow}}(\mathbf{w}) \in \{\mathcal{F}_m^{\mathrm{hor}}(\mathbf{w}), \mathcal{F}_m^{\mathrm{bdry}}(\mathbf{w})\}$$

be the inherited field tag, upgrade route, and follow-on menu from 19AZ$^{(6ad)}$–19AZ$^{(6ag)}$.

Define the two finite QAM-shadow states

$$\mathsf{QSH}_m^{\mathrm{hor}}(\mathbf{w}) := \big\langle \mathsf{FEH}_m(\mathbf{w}), \mathcal{U}_m^{\mathrm{hor}}(\mathbf{w}), \mathcal{F}_m^{\mathrm{hor}}(\mathbf{w}) \big\rangle,$$
$$\mathsf{QSH}_m^{\mathrm{bdry}}(\mathbf{w}) := \big\langle \mathsf{FEB}_m(\mathbf{w}), \mathcal{U}_m^{\mathrm{bdry}}(\mathbf{w}), \mathcal{F}_m^{\mathrm{bdry}}(\mathbf{w}) \big\rangle.$$

Define the QAM-/decoder-side shadow map

$$\mathsf{Shadow}_m^{\mathrm{QAM}}(\mathbf{w}) \in \{\mathsf{QSH}_m^{\mathrm{hor}}(\mathbf{w}), \mathsf{QSH}_m^{\mathrm{bdry}}(\mathbf{w})\}$$

by

$$\mathsf{Shadow}_m^{\mathrm{QAM}}(\mathbf{w}) = \begin{cases} \mathsf{QSH}_m^{\mathrm{hor}}(\mathbf{w}), & \text{if } \mathsf{Diag}_m^{\mathrm{field}}(\mathbf{w}) = \mathsf{FEH}_m(\mathbf{w}), \\ \mathsf{QSH}_m^{\mathrm{bdry}}(\mathbf{w}), & \text{if } \mathsf{Diag}_m^{\mathrm{field}}(\mathbf{w}) = \mathsf{FEB}_m(\mathbf{w}). \end{cases}$$

Define moreover the associated decoder word

$$\mathsf{Word}_m^{\mathrm{dec}}(\mathbf{w}) \in \{\mathsf{HR}, \mathsf{BR}\}$$

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by

$$\mathsf{Word}_m^{\mathrm{dec}}(\mathbf{w}) = \begin{cases} \mathsf{HR}, & \text{if } \mathsf{Shadow}_m^{\mathrm{QAM}}(\mathbf{w}) = \mathsf{QSH}_m^{\mathrm{hor}}(\mathbf{w}), \\ \mathsf{BR}, & \text{if } \mathsf{Shadow}_m^{\mathrm{QAM}}(\mathbf{w}) = \mathsf{QSH}_m^{\mathrm{bdry}}(\mathbf{w}). \end{cases}$$

Interpretation: the QAM shadow is the finite state object that a later QAM-, decoder-, compression-, or router-layer is allowed to consume; the decoder word is the shortest faithful tag that still remembers whether the inherited certificate is horizon-routed or boundary-routed.

Proposition H4.109 (The QAM-/decoder-side shadow is a faithful finite downstream image of the certificate-aware interface)**.** *Assume the setting of Definition H4.108. Then:*

(i) the shadow map $\mathsf{Shadow}_m^{\mathrm{QAM}}$ *is well-defined on every certified finite extension word;*

(ii) the shadow state remembers exactly the inherited field tag, forced upgrade route, and route-compatible next-work menu, but does not add new theorematic authority;

(iii) the decoder word $\mathsf{Word}_m^{\mathrm{dec}}(\mathbf{w}) \in \{\mathsf{HR}, \mathsf{BR}\}$ *is a faithful two-symbol readout of whether the certified state is horizon-routed or boundary-routed.*

Proof. By Proposition H4.101 and Proposition H4.105, every certified finite extension word inherits exactly one field tag, one forced route, and one route-compatible next-work menu. Definition H4.108 therefore assigns exactly one of the two shadow states, proving (i). Statement (ii) follows because the shadow state is merely an ordered packaging of already inherited data; it compresses the certified status but does not certify anything beyond it. For (iii), the decoder word forgets the internal menu details while preserving the decisive distinction between horizon-routed and boundary-routed certified status. Hence it is a faithful two-symbol downstream readout of the certificate-aware interface. □

Corollary H4.110 (The present σ_3 package already induces a first explicit QAM-/decoder–side shadow)**.** *In the current active line, the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$ *already induces both a finite QAM-shadow state*

$$\mathsf{Shadow}_2^{\mathrm{QAM}}(\mathbf{w}_3) \in \{\mathsf{QSH}_2^{\mathrm{hor}}(\mathbf{w}_3), \mathsf{QSH}_2^{\mathrm{bdry}}(\mathbf{w}_3)\}$$

and a faithful decoder word

$$\mathsf{Word}_2^{\mathrm{dec}}(\mathbf{w}_3) \in \{\mathsf{HR}, \mathsf{BR}\}.$$

Thus the new extension tool layer is now explicitly attached back to the QAM-/decoder side as a finite downstream interface.

Proof. Immediate from Corollary H4.106 and Proposition H4.109. □

Remark H4.111 (What this QAM-/decoder-side shadow now buys)**.** *This is the missing explicit reattachment of the new tool layer to the QAM line. The active extension architecture can now be consumed as a finite state object instead of as a long theorematic prose chain. That is exactly what a later QAM, decoder, compression, or router implementation should see: not a new ontology, but a faithful shadow of the already certified status. In particular, the decoder word may be used for compact routing, logging, or interface normalization, but it may never be mistaken for an independent proof of continuation or boundary exposure.*

Reader cue. Read the next block as the explicit Monte-Carlo/probe discipline sitting below the certificate-aware routing and menu interface. Its purpose is to give Monte-Carlo a precise and honest role inside the current architecture: useful as a downstream structural probe and deviation profiler, but never as a substitute for theorematic certification.

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19AZ$^{(6ai)}$. Certificate-aware Monte-Carlo / probe discipline below the routing interface

After 19AZ$^{(6ag)}$ and 19AZ$^{(6ah)}$, the active line has enough structure to place Monte-Carlo and related probe machinery exactly where it belongs. The point is not to diminish that machinery, but to protect it from doing the wrong job. A good Monte-Carlo run may expose deviation patterns, stop-sensitive fingerprints, boundary clustering, transport-instability signals, or decoder-compressible distributions. What it may not do is certify a new horizon extension or erase a certified boundary stop. The present block fixes that discipline explicitly.

Definition H4.112 (Certificate-aware Monte-Carlo / probe family)**.** *Fix a current certified horizon $\mathbf{K}_m^{\mathrm{cur}}$, let $\mathbf{w} \in \mathcal{L}_{\mathrm{ext}}^{\mathrm{term}}$ be a certified finite extension word, let*

$$\mathsf{Shadow}_m^{\mathrm{QAM}}(\mathbf{w}) \in \{\mathsf{QSH}_m^{\mathrm{hor}}(\mathbf{w}), \mathsf{QSH}_m^{\mathrm{bdry}}(\mathbf{w})\}$$

be the inherited QAM-shadow state from Definition H4.108, and let

$$\mathsf{Route}_m^{\mathrm{upg}}(\mathbf{w}) \in \{\mathcal{U}_m^{\mathrm{hor}}(\mathbf{w}), \mathcal{U}_m^{\mathrm{bdry}}(\mathbf{w})\}$$

be the inherited upgrade route.

A certificate-aware Monte-Carlo / probe family *for $(m, \mathbf{w})$ is any downstream probe procedure*

$$\mathsf{MCProbe}_{m,\mathbf{w}}$$

that takes as input only already certified data associated with $\mathbf{w}$ and outputs a finite diagnostic profile

$$\mathsf{Prof}_m^{\mathrm{MC}}(\mathbf{w})$$

subject to the following conditions:

(i) ***certificate inheritance:*** *the probe must inherit the existing shadow state and route rather than assign a new one;*

(ii) ***route fidelity:*** *on the horizon route it may test only transport-side stability, continuation-side sensitivity, or compression behavior on the already certified range, while on the boundary route it may test only earliest-boundary localization, stop-sensitive clustering, defect/exposure fingerprints, or boundary-side compression behavior;*

(iii) ***no theorematic promotion:*** *the output profile $\mathsf{Prof}_m^{\mathrm{MC}}(\mathbf{w})$ may suggest, rank, cluster, hash, Markov-compress, or Kasiski-scan deviation patterns, but it may not itself certify a later horizon $\mathbf{K}_{m+1}^{\mathrm{cur}}$ or replace an inherited boundary witness $\mathfrak{B}_{m+1}^{\mathrm{cur}}$;*

(iv) ***stop discipline:*** *if the inherited route is boundary-routed, the probe may sharpen the earliest-boundary picture but may not sample as though theorematic continuation beyond that certified stop had already been proved.*

A probe family that violates one of these conditions is called a certificate-blind Monte-Carlo / probe family*.*

Proposition H4.113 (Certificate-aware Monte-Carlo probes have diagnostic force but no theorematic authority)**.** *Assume the setting of Definition H4.112. Then:*

(i) every certificate-aware Monte-Carlo / probe family $\mathsf{MCProbe}_{m,\mathbf{w}}$ is a legitimate downstream consumer of the inherited certificate-aware interface;

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(ii) the resulting profile $\mathsf{Prof}_m^{\mathrm{MC}}(\mathbf{w})$ *may refine localization, stability, clustering, or compression questions inside the already certified route, but it does not change the theorematic status of* $\mathbf{w}$*;*

(iii) any probe family that uses its output to claim a new certified extension, to smooth away an inherited boundary stop, or to inject an uncertified defect/source event is certificate-blind and therefore inadmissible in the present line.

Proof. Clause (i) follows directly from Definition H4.112: the probe takes only already certified data as input and acts strictly below the inherited route. For (ii), the profile output is diagnostic by construction. It may rank or compress possibilities, but it never changes the inherited field tag, route, or next-work menu, so it cannot alter the theorematic status of $\mathbf{w}$. For (iii), if a probe profile is treated as though it certified a new horizon, then it violates no-theorematic-promotion. If it smooths away a boundary-certified stop, it violates stop discipline and route fidelity. If it injects an uncertified defect/source event on a horizon-routed state, it violates route fidelity on the other side. Hence every such misuse is certificate-blind and inadmissible in the present architecture. □

Corollary H4.114 (The present σ_3 package already admits a first honest Monte-Carlo role)**.** *In the current active line, the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$ *already admits a first honest Monte-Carlo / probe role below the inherited route:*

(i) if $\mathbf{w}_3 = \mathsf{A}$*, Monte-Carlo may act only as a continuation-side stability or compression probe on the currently certified range;*

(ii) if $\mathbf{w}_3 = \mathsf{B}$*, Monte-Carlo may act as an earliest-boundary localization, stop-sensitive deviation, or defect/exposure profiling probe, but not as a proof that the corridor has already continued past the certified stop.*

Thus Monte-Carlo is now explicitly fixed as a structural probe layer rather than an alternative theorematic engine.

Proof. Immediate from Corollary H4.110 and Proposition H4.113. □

Remark H4.115 (What this Monte-Carlo / probe discipline now buys)**.** *This is the cleanest place yet to keep the mathematics-first architecture and still use aggressive probe machinery productively. Monte-Carlo may now be used to expose deviation landscapes, boundary clustering, compression-friendly fingerprints, or route-sensitive instability patterns. Those outputs can be helpful for routing attention, designing later diagnostics, or compressing large probe logs into decoder-facing summaries. But the proof burden remains exactly where it belongs: on the certified local extension line. That makes Monte-Carlo valuable without letting it impersonate theorematic closure.*

Reader cue. Read the next block as an inherited freeze audit from an earlier minor public step. Its purpose remains deliberately modest: not to claim a new global theorem, but to record in one place what the release line has stabilized, what must remain local/route-certified in the public wording, and why the present DOI-bearing freeze is honest.

19AZ$^{(6aj)}$. Minor-release freeze audit and public record

After 19AZ$^{(6x)}$–19AZ$^{(6ai)}$, the active line has turned the first post-$\mathbf{K}_2^{\mathrm{cur}}$ extension step into a reusable certificate-aware tool layer. For that inherited minor-release stage, the honest question was therefore no longer whether the new line had mathematical content, but whether its

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public wording, routing discipline, archival preservation, and metadata had been synchronized well enough that the public freeze did not misstate what had actually been achieved. The present block records that completed audit explicitly.

Definition H4.116 (Minor-release readiness audit package)**.** *For that inherited public-release line, define the* minor-release readiness audit package

$$\mathsf{Audit}^{\mathrm{minor}} := \big(\mathsf{Math}, \mathsf{Route}, \mathsf{QAM}, \mathsf{MC}, \mathsf{Hist}, \mathsf{Meta}\big)$$

as follows:

(i) Math*: the live OP 1 kernel is discharged locally on the retained no-second-carrier / strict-upper-gap corridor, and the live OP 5 Step (B) kernel is discharged locally on the retained branch-shell / shell-transfer corridor;*

(ii) Route*: the first later leading admissible slot beyond the current certified horizon is normalized by a finite decision datum, an action interface, an admissible terminal word language, field tags, route tags, and a finite next-work menu;*

(iii) QAM*: the new extension layer is attached back to the QAM/decoder side by a finite shadow-state interface and the faithful decoder words* $\{\mathsf{HR}, \mathsf{BR}\}$*;*

(iv) MC*: Monte-Carlo and related probe machinery are explicitly confined to a certificate-aware downstream diagnostic role;*

(v) Hist*: the full historical witness remains preserved in the monolithic source as Part II rather than as a shortened fragment;*

(vi) Meta*: title-line DOI logic, version/chronology wording, and release wording are synchronized so that the concept DOI remains the stable title-page pointer while public version-specific DOI claims appear only in the chronology or explicit release notes.*

Proposition H4.117 (Present release line is mathematically minor-ready and release-packaged)**.** *Assume the audit package from Definition H4.116. Then:*

(i) the mathematics-side burden of that inherited minor release is strong enough for an honest public freeze: the new contribution is a certificate-aware extension/routing/probe layer above the public archival stage infrastructure together with local retained-corridor closure of the live OP 1 and OP 5 Step (B) kernels;

(ii) the public wording of that freeze remains local and route-certified, i.e. it reports retained-corridor kernel closure and the new certificate-aware tool layer, but it does not flatten those results into a new unconditional global closure slogan;

(iii) the release-packaging tasks for that inherited public minor release have been synchronized: version/date metadata, DOI logic, historical preservation, and public wording have been normalized, while ordinary citation/hyperref cleanup can continue in later working states without changing the mathematics.

Proof. Clause (i) is exactly the content accumulated in $19\mathrm{AZ}^{(6x)}$–$19\mathrm{AZ}^{(6ai)}$: finite next-slot decision data, faithful output tokens, action/word routing, certificate-aware matrix/crystal/-field readout, admissibility and routing control, finite next-work menus, finite QAM shadows, and an honest downstream Monte-Carlo role. These are not merely editorial restatements but reusable structural tools. The local retained-corridor OP 1 and OP 5 Step (B) closure statements are already recorded in the active main line, so the mathematics side no longer depends

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on introducing yet another theorematic layer before that inherited minor public freeze. Clause (ii) follows from the route-certified architecture itself: every new statement in this block is built from inherited certified corridors, finite decision packaging, and certificate-aware downstream consumers. Flattening that into an unconditional global slogan would destroy the very distinction the new layer was designed to preserve. Clause (iii) follows because what is left to do does not add new mathematical content; it concerns release hygiene, metadata synchronization, and wording discipline. □

Corollary H4.118 (Honest public release message for that inherited minor release)**.** *For that inherited public minor release, the honest release message is the following:*

> *The public archival stage line is carried forward into the public 4.7.0 release by a certificate-aware extension/routing/probe layer. On top of that inherited infrastructure, the live OP 1 kernel and the live OP 5 Step (B) kernel are closed locally on the standing retained corridors. The resulting public gain is therefore not a new global closure slogan, but a route-certified tool layer together with retained-corridor kernel closure, finite decoder/QAM shadows, and a disciplined downstream role for Monte-Carlo and related probes.*

Proof. Immediate from Proposition H4.117. □

Remark H4.119 (Freeze checklist completed for the DOI-bearing release)**.** *In that inherited freeze block, the deliberately boring tasks had been carried out: version/date metadata in the wrapper had been synchronized, chronology wording recorded archival stage as the prior public baseline and the then-current line as the public 4.7.0 release, the historical witness remained preserved in full, and the front-matter wording had been normalized so that local retained-corridor results were not overstated as unconditional global closure. Ordinary citation/hyperref cleanup could still improve future working states, but it no longer blocked the DOI-bearing public freeze.*

Reader cue. Read the next block as the first post-release working bridge that couples the certificate-aware upgrade menus to honest probe consumption on the matrix, crystal, and field-diagnostic side. Its purpose is not to reopen the theorematic routing decision, but to package conservative sharpening together with route-compatible probe use into one downstream working object.

19AZ$^{(6ak)}$. Probe-compatible conservative upgrade packets below the certificate-aware interface

After 19AZ$^{(6ag)}$–19AZ$^{(6ai)}$, the line already knows four things for every certified finite extension word: the inherited horizon/boundary status, the forced route, the finite next-work menu, and the downstream diagnostic role of probe machinery. The next honest post-release step is therefore not a new closure slogan. It is a working packet that couples conservative matrix/crystal/field refinement to admissible probe use without allowing the probe layer to promote, erase, or reroute the certified status.

Definition H4.120 (Probe-compatible conservative upgrade packet)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, let*

$$\mathsf{Route}_m^{\mathrm{upg}}(\mathbf{w}) \in \{\mathcal{U}_m^{\mathrm{hor}}(\mathbf{w}), \mathcal{U}_m^{\mathrm{bdry}}(\mathbf{w})\}$$

be its inherited upgrade route from Definition H4.100, let

$$\mathcal{F}_m^{\mathrm{follow}}(\mathbf{w})$$

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be its finite route-compatible follow-on menu from Definition H4.104, and let

$$\mathsf{MCProbe}_{m,\mathbf{w}}$$

be any certificate-aware probe family from Definition H4.112.

A probe-compatible conservative upgrade packet *for* $(m, \mathbf{w})$ *is one of the following two objects.*

(i) If $\mathsf{Route}_m^{\mathrm{upg}}(\mathbf{w}) = \mathcal{U}_m^{\mathrm{hor}}(\mathbf{w})$*, define the horizon-side packet*

$$\mathsf{Pack}_m^{\mathrm{hor}}(\mathbf{w}) := \big(\mathsf{HZ}_m(\mathbf{w}), \mathsf{CRH}_m(\mathbf{w}), \mathsf{FEH}_m(\mathbf{w}), \mathcal{F}_m^{\mathrm{follow}}(\mathbf{w}), \mathsf{MCProbe}_{m,\mathbf{w}}\big)$$

under the side condition that every downstream use of $\mathsf{MCProbe}_{m,\mathbf{w}}$ *remains confined to horizon-routed families from*

$$\{\mathsf{MSH}_m(\mathbf{w}), \mathsf{CRS}_m(\mathbf{w}), \mathsf{PDE}_m^{\mathrm{cons}}(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{hor}}(\mathbf{w})\} \cap \mathcal{F}_m^{\mathrm{follow}}(\mathbf{w}).$$

(ii) If $\mathsf{Route}_m^{\mathrm{upg}}(\mathbf{w}) = \mathcal{U}_m^{\mathrm{bdry}}(\mathbf{w})$*, define the boundary-side packet*

$$\mathsf{Pack}_m^{\mathrm{bdry}}(\mathbf{w}) := \big(\mathsf{BD}_m(\mathbf{w}), \mathsf{CRB}_m(\mathbf{w}), \mathsf{FEB}_m(\mathbf{w}), \mathcal{F}_m^{\mathrm{follow}}(\mathbf{w}), \mathsf{MCProbe}_{m,\mathbf{w}}\big)$$

under the side condition that every downstream use of $\mathsf{MCProbe}_{m,\mathbf{w}}$ *remains confined to boundary-routed families from*

$$\{\mathsf{SRC}_m(\mathbf{w}), \mathsf{DFX}_m(\mathbf{w}), \mathsf{MC}_m^{\mathrm{bdry}}(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{bdry}}(\mathbf{w})\} \cap \mathcal{F}_m^{\mathrm{follow}}(\mathbf{w}).$$

Finally define the conservative packet map

$$\mathsf{Pack}_m^{\mathrm{cons}}(\mathbf{w}) := \begin{cases} \mathsf{Pack}_m^{\mathrm{hor}}(\mathbf{w}), & \text{if } \rho_m(\mathbf{w}) = \mathcal{U}^{\mathrm{hor}}, \\ \mathsf{Pack}_m^{\mathrm{bdry}}(\mathbf{w}), & \text{if } \rho_m(\mathbf{w}) = \mathcal{U}^{\mathrm{bdry}}. \end{cases}$$

Proposition H4.121 (Every certified word inherits exactly one probe-compatible conservative upgrade packet)**.** *Assume the setting of Definition H4.120. Then:*

(i) every certified finite extension word $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$ *inherits exactly one conservative packet* $\mathsf{Pack}_m^{\mathrm{cons}}(\mathbf{w})$*;*

(ii) the packet is faithful to the inherited matrix/crystal/field tags and to the forced route $\rho_m(\mathbf{w})$*;*

(iii) certificate-aware probe output may reprioritize or sharpen work inside *the inherited follow-on menu, but it may not add extra-route tasks, promote theorematic status, or erase a certified boundary stop.*

Proof. By Proposition H4.101, every certified word inherits exactly one route $\rho_m(\mathbf{w})$. By Proposition H4.105, this route comes with exactly one finite compatible follow-on menu. Definition H4.120 therefore assigns exactly one of the two packet forms, proving (i). Statement (ii) is immediate because the packet simply records the inherited matrix/crystal/field tags together with the already forced route and menu. Clause (iii) follows from Proposition H4.113: probe families are downstream consumers only, so they may refine priority inside the inherited menu but they do not change theorematic status, add cross-route tasks, or smooth away a certified terminal boundary event. □

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Corollary H4.122 (The present σ_3 package already carries a first conservative upgrade packet)**.** *In the current active working line, the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$ *already induces a first probe-compatible conservative packet*

$$\mathsf{Pack}_2^{\mathrm{cons}}(\mathbf{w}_3),$$

whose type is horizon-side in the transfer-positive case and boundary-side in the earliest-boundary case.

Proof. Immediate from Corollary H4.106, Corollary H4.110, and Proposition H4.121. □

Remark H4.123 (What this conservative packet now buys)**.** *This is the first point at which the post-release line stops treating conservative refinement and probe use as two separate topics. The packet forces them to travel together. Matrix sharpening, crystal stabilization, field-profile refinement, and source/defect localization can now be discussed as probe-compatible working objects, while theorematic certification remains upstream and untouched.*

Reader cue. Read the next block as the finite work-queue normal form induced by a conservative packet. Its purpose is to let probe output guide attention among admissible matrix/crystal/field upgrades without letting that guidance impersonate theorematic certification or invent work outside the inherited route.

19AZ$^{(6al)}$. Probe-ranked conservative work queues for matrix, crystal, and field upgrades

After 19AZ$^{(6ak)}$, every certified word carries one conservative packet. The next honest compression step is to turn the inherited finite follow-on menu into a finite probe-ranked queue. This is exactly the place where Monte-Carlo/probe integration becomes genuinely useful for the next working phase: not by certifying anything new, but by telling the line which admissible conservative upgrade should be looked at first, second, and third.

Definition H4.124 (Probe-ranked conservative work queue)**.** *Fix a conservative packet*

$$\mathsf{Pack}_m^{\mathrm{cons}}(\mathbf{w})$$

from Definition H4.120. A probe-ranked conservative work queue *for* $(m, \mathbf{w})$ *is a finite ordered list*

$$\mathcal{Q}_m^{\mathrm{cons}}(\mathbf{w}) = \big(\mathsf{U}_1 \succcurlyeq \mathsf{U}_2 \succcurlyeq \cdots \succcurlyeq \mathsf{U}_r\big)$$

with the following properties:

(i) every entry U_j *belongs to the inherited follow-on menu* $\mathcal{F}_m^{\mathrm{follow}}(\mathbf{w})$*;*

(ii) the order relation $\succcurlyeq$ *is induced by downstream probe output only as a* priority preorder*, not as new theorematic status;*

(iii) if $\mathsf{Route}_m^{\mathrm{upg}}(\mathbf{w}) = \mathcal{U}_m^{\mathrm{hor}}(\mathbf{w})$*, then every queue entry is horizon-routed and the queue may only prioritize among* $\mathsf{MSHARP}, \mathsf{CSTAB}, \mathsf{FPROF}, \mathsf{DCH}$*;*

(iv) if $\mathsf{Route}_m^{\mathrm{upg}}(\mathbf{w}) = \mathcal{U}_m^{\mathrm{bdry}}(\mathbf{w})$*, then every queue entry is boundary-routed and the queue may only prioritize among* $\mathsf{SLOC}, \mathsf{DFRONT}, \mathsf{MCPROB}, \mathsf{DCB}$*.*

Proposition H4.125 (Every conservative packet induces a finite route-compatible work queue)**.** *Assume the setting of Definition H4.124. Then:*

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(i) every conservative packet $\mathsf{Pack}_m^{\mathrm{cons}}(\mathbf{w})$ *induces at least one finite probe-ranked conservative work queue* $\mathcal{Q}_m^{\mathrm{cons}}(\mathbf{w})$*;*

(ii) any two such queues differ only by downstream prioritization inside the same inherited finite route-compatible menu;

(iii) in particular, probe ranking may move matrix, crystal, and field-style tasks forward or backward inside the queue, but it may not turn a boundary-routed queue into a continuation queue, nor a continuation queue into a boundary/source queue.

Proof. Existence in (i) is immediate because the inherited follow-on menu is finite by Proposition H4.105; one may therefore equip it with at least the trivial preorder, or with any finer probe-induced preorder satisfying Definition H4.124. Statement (ii) follows because the queue entries are constrained to lie inside the same inherited route-compatible menu, so downstream variation can only affect the internal ordering. Clause (iii) is exactly the certificate-aware discipline inherited from $19\mathrm{AZ}^{(6af)}$ and $19\mathrm{AZ}^{(6ai)}$: probe ranking is a priority device, not a theorematic rerouting device. □

Corollary H4.126 (The first σ_3 packet already carries a finite conservative work queue)**.** *The present first-stage packet* $\mathsf{Pack}_2^{\mathrm{cons}}(\mathbf{w}_3)$ *already carries a finite route-compatible work queue*

$$\mathcal{Q}_2^{\mathrm{cons}}(\mathbf{w}_3),$$

so the next working phase may be organized as a conservative ranked queue of matrix/crystal/-field upgrades together with probe-guided diagnostics before any broader OP-focused reopening is attempted.

Proof. Immediate from Corollary H4.122 and Proposition H4.125. □

Remark H4.127 (What this queue interface now buys)**.** *The next working stage can now be described honestly in one sentence: first sharpen the conservative matrix/crystal/field side and integrate probe diagnostics according to a finite route-compatible queue; only after that use the sharpened readout surface for the next OP-focused push. This is exactly the intended ordering for the post-archival stage line.*

Reader cue. Read the next block as the first concrete working extraction from the probe-ranked conservative queue. Its purpose is not to let probe output overrule the certificate-aware route, but to turn the inherited queue into one honest conservative sharpening/program block that can actually be worked through on the matrix, crystal, and field side before the next OP-focused push.

$19\mathrm{AZ}^{(6am)}$. Queue-extracted conservative sharpening/program blocks below the probe-ranked interface

After $19\mathrm{AZ}^{(6al)}$, the active line finally has enough structure to speak about a genuine next working phase rather than only about menus and priorities. The next honest step is to extract, from the finite probe-ranked queue, the actual conservative sharpening/program block that should be worked through first on the inherited route. On the horizon side this means the matrix/crystal/field sharpening strand together with decoder support; on the boundary side it means the earliest-boundary profiling strand together with stop-sensitive probe/decoder support.

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Definition H4.128 (Queue-extracted conservative sharpening/program block)**.** *Fix a probe-ranked conservative work queue*

$$\mathcal{Q}_m^{\mathrm{cons}}(\mathbf{w}) = \big(\mathsf{U}_1 \succcurlyeq \mathsf{U}_2 \succcurlyeq \cdots \succcurlyeq \mathsf{U}_r\big)$$

from Definition H4.124.

(i) If $\mathsf{Route}_m^{\mathrm{upg}}(\mathbf{w}) = \mathcal{U}_m^{\mathrm{hor}}(\mathbf{w})$*, define the* horizon-side conservative sharpening/program block

$$\mathcal{P}_m^{\mathrm{hor}}(\mathbf{w}) := \big(\mathsf{U}_{j_1} \succcurlyeq \cdots \succcurlyeq \mathsf{U}_{j_s}\big)$$

to be the ordered subsequence of $\mathcal{Q}_m^{\mathrm{cons}}(\mathbf{w})$ *consisting exactly of those entries that lie in*

$$\{\mathsf{MSH}_m(\mathbf{w}), \mathsf{CRS}_m(\mathbf{w}), \mathsf{PDE}_m^{\mathrm{cons}}(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{hor}}(\mathbf{w})\}.$$

(ii) If $\mathsf{Route}_m^{\mathrm{upg}}(\mathbf{w}) = \mathcal{U}_m^{\mathrm{bdry}}(\mathbf{w})$*, define the* boundary-side conservative profiling/program block

$$\mathcal{P}_m^{\mathrm{bdry}}(\mathbf{w}) := \big(\mathsf{U}_{k_1} \succcurlyeq \cdots \succcurlyeq \mathsf{U}_{k_t}\big)$$

to be the ordered subsequence of $\mathcal{Q}_m^{\mathrm{cons}}(\mathbf{w})$ *consisting exactly of those entries that lie in*

$$\{\mathsf{SRC}_m(\mathbf{w}), \mathsf{DFX}_m(\mathbf{w}), \mathsf{MC}_m^{\mathrm{bdry}}(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{bdry}}(\mathbf{w})\}.$$

Finally define the queue-extracted conservative program block

$$\mathcal{P}_m^{\mathrm{cons}}(\mathbf{w}) := \begin{cases} \mathcal{P}_m^{\mathrm{hor}}(\mathbf{w}), & \text{if } \rho_m(\mathbf{w}) = \mathcal{U}^{\mathrm{hor}}, \\ \mathcal{P}_m^{\mathrm{bdry}}(\mathbf{w}), & \text{if } \rho_m(\mathbf{w}) = \mathcal{U}^{\mathrm{bdry}}. \end{cases}$$

Proposition H4.129 (Every probe-ranked conservative queue induces one route-faithful working program block)**.** *Assume the setting of Definition H4.128. Then:*

(i) every probe-ranked conservative queue $\mathcal{Q}_m^{\mathrm{cons}}(\mathbf{w})$ *induces exactly one queue-extracted conservative program block* $\mathcal{P}_m^{\mathrm{cons}}(\mathbf{w})$*;*

(ii) if $\mathsf{Route}_m^{\mathrm{upg}}(\mathbf{w}) = \mathcal{U}_m^{\mathrm{hor}}(\mathbf{w})$*, then* $\mathcal{P}_m^{\mathrm{cons}}(\mathbf{w})$ *is a conservative horizon-side sharpening/program block built only from matrix sharpening, crystal stabilization, field-profile refinement, and decoder compression on the already certified range;*

(iii) if $\mathsf{Route}_m^{\mathrm{upg}}(\mathbf{w}) = \mathcal{U}_m^{\mathrm{bdry}}(\mathbf{w})$*, then* $\mathcal{P}_m^{\mathrm{cons}}(\mathbf{w})$ *is a conservative earliest-boundary profiling/program block built only from source localization, defect-front profiling, stop-sensitive probe diagnostics, and boundary-side decoder compression;*

(iv) downstream probe output may change the internal order of the block, but it may not insert extra-route tasks, promote theorematic status, or convert a boundary-profile block into a horizon-sharpening block.

Proof. Statement (i) is immediate because each queue already lies entirely inside one inherited route-compatible menu by Proposition H4.125; taking the ordered subsequence of the relevant route-side task family therefore produces exactly one associated program block. Clauses (ii) and (iii) merely unpack which finite task families are admissible on the two certified routes, using Proposition H4.101 together with Proposition H4.105. Clause (iv) is exactly the downstream discipline already fixed in Proposition H4.113 and Proposition H4.125: probe ranking is allowed to prioritize and sharpen, but not to reroute or recertify. □

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Corollary H4.130 (The present post-archival stage line already carries a first concrete conservative working program)**.** *The first-stage packet* $\mathsf{Pack}_2^{\mathrm{cons}}(\mathbf{w}_3)$ *and its queue* $\mathcal{Q}_2^{\mathrm{cons}}(\mathbf{w}_3)$ *already induce one concrete conservative program block*

$$\mathcal{P}_2^{\mathrm{cons}}(\mathbf{w}_3),$$

so the current working phase may now be read not merely as a list of admissible future tasks but as one finite route-faithful sharpening/profiling program to be worked through before the next OP-focused reopening.

Proof. Immediate from Corollary H4.126 and Proposition H4.129. □

Remark H4.131 (What this first concrete program block now buys)**.** *This is the first place where the post-release continuation genuinely stops looking like architecture alone. The line now has a concrete finite program block on which to spend the next working effort: either sharpen matrix/crystal/field readout conservatively on the certified horizon side, or profile the earliest honest boundary side in a stop-sensitive way, always under the same probe-compatible discipline. That is exactly the intended bridge from the tool-layer minor release to the next conservative upgrade stage.*

Reader cue. Read the next block as the first honest extraction of a *lead task* from the queue-extracted conservative program block. Its purpose is not to guess a theorematic advance, but to say which concrete conservative sharpening/profiling task should be worked *first* once the inherited route has already been fixed.

19AZ$^{(6an)}$. Route-faithful lead-task extraction from the conservative program block

After 19AZ$^{(6am)}$, the active line no longer lacks an admissible program block; what it still lacks is a normalized answer to the narrower practical question: *which item in that program block should be worked first?* The present block fixes that first-step discipline. On the horizon side, the natural lead task is matrix sharpening, because the matrix/Heisenberg–Millennium readout is the earliest common interface that the crystal and conservative field profiles inherit. On the boundary side, the natural lead task is source/exposure localization, because the earliest honest boundary must be anchored before defect-front profiling, stop-sensitive Monte-Carlo diagnostics, or boundary-side decoder compression can be interpreted coherently.

Definition H4.132 (Route-faithful lead task)**.** *Fix a queue-extracted conservative program block*

$$\mathcal{P}_m^{\mathrm{cons}}(\mathbf{w})$$

from Definition H4.128. Define the associated route-faithful lead task

$$\mathsf{Lead}_m^{\mathrm{cons}}(\mathbf{w})$$

by

$$\mathsf{Lead}_m^{\mathrm{cons}}(\mathbf{w}) := \begin{cases} \mathsf{MSH}_m(\mathbf{w}), & \text{if } \mathsf{Route}_m^{\mathrm{upg}}(\mathbf{w}) = \mathcal{U}_m^{\mathrm{hor}}(\mathbf{w}), \\ \mathsf{SRC}_m(\mathbf{w}), & \text{if } \mathsf{Route}_m^{\mathrm{upg}}(\mathbf{w}) = \mathcal{U}_m^{\mathrm{bdry}}(\mathbf{w}). \end{cases}$$

Thus:

(i) on the horizon side, the lead task is $\mathsf{MSH}_m(\mathbf{w})$*, i.e. matrix sharpening on the already certified range;*

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(ii) on the boundary side, the lead task is $\mathsf{SRC}_m(\mathbf{w})$*, i.e. source/exposure localization at the earliest honest boundary.*

Proposition H4.133 (Every conservative program block carries exactly one normalized first task)**.** *Assume the setting of Definition H4.132. Then:*

(i) every queue-extracted conservative program block $\mathcal{P}_m^{\mathrm{cons}}(\mathbf{w})$ *carries exactly one route-faithful lead task* $\mathsf{Lead}_m^{\mathrm{cons}}(\mathbf{w})$*;*

(ii) if $\mathsf{Route}_m^{\mathrm{upg}}(\mathbf{w}) = \mathcal{U}_m^{\mathrm{hor}}(\mathbf{w})$*, then the first conservative sharpening step is matrix sharpening rather than crystal stabilization or conservative PDE refinement;*

(iii) if $\mathsf{Route}_m^{\mathrm{upg}}(\mathbf{w}) = \mathcal{U}_m^{\mathrm{bdry}}(\mathbf{w})$*, then the first conservative profiling step is source/exposure localization rather than defect-front profiling or stop-sensitive probe diagnostics.*

Proof. Uniqueness in (i) is immediate from the already unique inherited route and the case distinction in Definition H4.132. For (ii), matrix sharpening is the earliest common sharpening interface on the continuation side: crystal stabilization and conservative PDE refinement are both read *through* the inherited matrix/Heisenberg–Millennium side, so choosing either of them first would presuppose a matrix profile that the current conservative program has not yet sharpened. Thus the route-faithful first task on the horizon side is $\mathsf{MSH}_m(\mathbf{w})$. For (iii), the earliest honest boundary must first be localized before one can coherently speak about its defect-front geometry, stop-sensitive probe profile, or boundary-side decoder compression. Thus the route-faithful first task on the boundary side is $\mathsf{SRC}_m(\mathbf{w})$. □

Corollary H4.134 (The present post-archival stage line already carries a unique first conservative move)**.** *For the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$*, the current active line already carries a unique normalized first conservative move:*

$$\mathsf{Lead}_2^{\mathrm{cons}}(\mathbf{w}_3) = \begin{cases} \mathsf{MSH}_2(\mathbf{w}_3), & \text{if } \mathbf{w}_3 = \mathsf{A}, \\ \mathsf{SRC}_2(\mathbf{w}_3), & \text{if } \mathbf{w}_3 = \mathsf{B}. \end{cases}$$

In other words, the current post-release continuation does not *begin by choosing freely among matrix, crystal, field, and probe tasks. It begins by sharpening the matrix side if the inherited status is horizon-certified, and by localizing the earliest source/exposure site if the inherited status is boundary-certified.*

Proof. Immediate from Corollary H4.130 and Proposition H4.133. □

Remark H4.135 (What the lead-task extraction now buys)**.** *This is the point where the post-release line stops saying merely* what could be done next*. It now says which conservative move comes first. That sharply reduces drift in the next working phase: on the horizon side the matrix comes before crystal and field refinements; on the boundary side source localization comes before defect-front profiling and stop-sensitive probing. So the next version can begin with one normalized first move rather than another round of menu-building.*

Reader cue. Read the next block as the first actual execution packet attached to the normalized lead task. Its purpose is not to complete matrix sharpening or source localization in one stroke, but to package the exact first conservative move in a route-faithful form that later crystal, field, defect-front, probe, and decoder layers may inherit without reopening the route.

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19AZ$^{(6ao)}$. First-step conservative execution packets below the lead-task interface

After 19AZ$^{(6an)}$, the active line knows which conservative move comes first. The next honest compression step is therefore to turn that lead task into an explicit first-step execution packet. On the horizon side this packet should sharpen the matrix/Heisenberg–Millennium readout just enough to prepare later crystal and conservative field refinement on the already certified range. On the boundary side it should localize the earliest honest source/exposure site just enough to prepare later defect-front profiling, stop-sensitive probe diagnostics, and decoder compression without smoothing away the boundary stop.

Definition H4.136 (First-step conservative execution packet)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, its route-faithful lead task*

$$\mathsf{Lead}_m^{\mathrm{cons}}(\mathbf{w})$$

from Definition H4.132, and a certificate-aware probe family $\mathsf{MCProbe}_{m,\mathbf{w}}$*.*

(i) If $\mathsf{Lead}_m^{\mathrm{cons}}(\mathbf{w}) = \mathsf{MSH}_m(\mathbf{w})$*, define the* horizon-side first-step execution packet

$$\mathsf{Exec}_m^{(1),\mathrm{hor}}(\mathbf{w}) := \big(\mathsf{MSH}_m(\mathbf{w}), \mathsf{HZ}_m(\mathbf{w}), \mathsf{CRH}_m(\mathbf{w}), \mathsf{FEH}_m(\mathbf{w}), \mathsf{MCProbe}_{m,\mathbf{w}}\big)$$

under the side condition that every downstream readout remains confined to a conservative sharpening profile on the already certified range and may only prepare later tasks from

$$\{\mathsf{CRS}_m(\mathbf{w}), \mathsf{PDE}_m^{\mathrm{cons}}(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{hor}}(\mathbf{w})\}.$$

(ii) If $\mathsf{Lead}_m^{\mathrm{cons}}(\mathbf{w}) = \mathsf{SRC}_m(\mathbf{w})$*, define the* boundary-side first-step execution packet

$$\mathsf{Exec}_m^{(1),\mathrm{bdry}}(\mathbf{w}) := \big(\mathsf{SRC}_m(\mathbf{w}), \mathsf{BD}_m(\mathbf{w}), \mathsf{CRB}_m(\mathbf{w}), \mathsf{FEB}_m(\mathbf{w}), \mathsf{MCProbe}_{m,\mathbf{w}}\big)$$

under the side condition that every downstream readout remains confined to an earliest-boundary localization profile and may only prepare later tasks from

$$\{\mathsf{DFX}_m(\mathbf{w}), \mathsf{MC}_m^{\mathrm{bdry}}(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{bdry}}(\mathbf{w})\}.$$

Finally define the first-step conservative execution map

$$\mathsf{Exec}_m^{(1),\mathrm{cons}}(\mathbf{w}) := \begin{cases} \mathsf{Exec}_m^{(1),\mathrm{hor}}(\mathbf{w}), & \text{if } \mathsf{Lead}_m^{\mathrm{cons}}(\mathbf{w}) = \mathsf{MSH}_m(\mathbf{w}), \\ \mathsf{Exec}_m^{(1),\mathrm{bdry}}(\mathbf{w}), & \text{if } \mathsf{Lead}_m^{\mathrm{cons}}(\mathbf{w}) = \mathsf{SRC}_m(\mathbf{w}). \end{cases}$$

Proposition H4.137 (Every normalized lead task induces exactly one first-step conservative execution packet)**.** *Assume the setting of Definition H4.136. Then:*

(i) every normalized lead task $\mathsf{Lead}_m^{\mathrm{cons}}(\mathbf{w})$ *induces exactly one first-step execution packet* $\mathsf{Exec}_m^{(1),\mathrm{cons}}(\mathbf{w})$*;*

(ii) if the lead task is matrix sharpening, then the first-step packet prepares later crystal, conservative field, and decoder refinement on the already certified range without extending theorematic authority beyond that range;

(iii) if the lead task is source/exposure localization, then the first-step packet anchors the earliest honest boundary site before any later defect-front profiling, stop-sensitive probe diagnostics, or boundary-side decoder compression is attempted;

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(iv) certificate-aware probe output may profile or reprioritize work inside the packet's inherited route, but it may not convert the packet into the opposite route, promote theorematic status, or smooth away a certified terminal boundary event.

Proof. Clause (i) is immediate from Definition H4.132: every certified word carries exactly one lead task, namely either $\mathsf{MSH}_m(\mathbf{w})$ or $\mathsf{SRC}_m(\mathbf{w})$, so Definition H4.136 assigns exactly one corresponding first-step packet. Clause (ii) follows because the horizon-side packet records the inherited horizon/matrix/crystal/field status and constrains every downstream use to the conservative continuation-side family; this can sharpen the already certified readout surface, but it does not certify new continuation beyond the current range. Clause (iii) follows because the boundary-side packet records the inherited earliest-boundary tags and constrains downstream use to the boundary-side family; hence localization comes before any defect-front, stop-sensitive probe, or decoder readout that presupposes a fixed boundary anchor. Clause (iv) is exactly the probe discipline inherited from 19AZ$^{(6ai)}$, 19AZ$^{(6ak)}$, and 19AZ$^{(6al)}$: probes may sharpen internal priority and profile, but they do not reroute or recertify. □

Corollary H4.138 (The present post-archival stage line already carries a first executable conservative move)**.** *For the first-stage word $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$, the current active line already carries one first executable conservative move:*

$$\mathsf{Exec}_2^{(1),\mathrm{cons}}(\mathbf{w}_3) = \begin{cases} \mathsf{Exec}_2^{(1),\mathrm{hor}}(\mathbf{w}_3), & \text{if } \mathbf{w}_3 = \mathsf{A}, \\ \mathsf{Exec}_2^{(1),\mathrm{bdry}}(\mathbf{w}_3), & \text{if } \mathbf{w}_3 = \mathsf{B}. \end{cases}$$

So the post-release continuation now begins with one explicit executable packet: matrix sharpening first in the transfer-positive case, and source/exposure localization first in the earliest-boundary case.

Proof. Immediate from Corollary H4.134 and Proposition H4.137. □

Remark H4.139 (What this first executable packet now buys)**.** *This is the point where the post-release line stops at last saying only which move comes first and starts carrying the move in an executable normal form. The next working phase can therefore begin with one explicit conservative packet rather than a diffuse instruction: sharpen the matrix side just enough to support later crystal/field refinement, or anchor the earliest boundary just enough to support later defect/probe/decoder profiling. That is a genuinely operational starting point for the next version.*

Reader cue. Read the next block as the first normalized output stage extracted from the first-step execution packet. Its purpose is not to claim that matrix sharpening or source anchoring is already complete in every later detail, but to record the first conservative result surface that the current route-faithful packet is allowed to produce before any later crystal, field, defect-front, probe, or decoder refinement is attempted.

19AZ$^{(6ap)}$. First conservative result forms extracted from the first execution packet

After 19AZ$^{(6ao)}$, the post-release line carries an explicit first-step execution packet. The next honest compression step is therefore to record the first conservative result form that this packet may output. On the horizon side that result should be the first sharpened matrix/Heisenberg–Millennium readout surface still confined to the already certified range. On the boundary side it should be the first anchored source/exposure profile still confined to the earliest honest boundary event and its immediate diagnostic neighborhood.

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Definition H4.140 (First conservative result form)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, its first-step conservative execution packet*

$$\mathsf{Exec}_m^{(1),\mathrm{cons}}(\mathbf{w})$$

from Definition H4.136, and the inherited certificate-aware probe family $\mathsf{MCProbe}_{m,\mathbf{w}}$*.*

(i) If

$$\mathsf{Exec}_m^{(1),\mathrm{cons}}(\mathbf{w}) = \mathsf{Exec}_m^{(1),\mathrm{hor}}(\mathbf{w}),$$

define the first sharpened horizon-side readout

$$\mathsf{Res}_m^{(1),\mathrm{hor}}(\mathbf{w}) := \big(\mathsf{HZ}_m^{\sharp}(\mathbf{w}), \mathsf{CRH}_m(\mathbf{w}), \mathsf{FEH}_m(\mathbf{w}), \mathsf{MCProbe}_{m,\mathbf{w}}\big),$$

where $\mathsf{HZ}_m^{\sharp}(\mathbf{w})$ *denotes the first conservative sharpened matrix/Heisenberg–Millennium readout produced by* $\mathsf{MSH}_m(\mathbf{w})$ *under the side conditions that:*

- *the sharpened readout remains confined to the already certified horizon-side range,*
- *it does not certify any new continuation beyond that range, and*
- *it may only feed later tasks from*

$$\{\mathsf{CRS}_m(\mathbf{w}), \mathsf{PDE}_m^{\mathrm{cons}}(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{hor}}(\mathbf{w})\}.$$

(ii) If

$$\mathsf{Exec}_m^{(1),\mathrm{cons}}(\mathbf{w}) = \mathsf{Exec}_m^{(1),\mathrm{bdry}}(\mathbf{w}),$$

define the first anchored boundary-side source profile

$$\mathsf{Res}_m^{(1),\mathrm{bdry}}(\mathbf{w}) := \big(\mathsf{BD}_m^{\mathrm{anch}}(\mathbf{w}), \mathsf{CRB}_m(\mathbf{w}), \mathsf{FEB}_m(\mathbf{w}), \mathsf{MCProbe}_{m,\mathbf{w}}\big),$$

where $\mathsf{BD}_m^{\mathrm{anch}}(\mathbf{w})$ *denotes the first conservative anchored source/exposure profile produced by* $\mathsf{SRC}_m(\mathbf{w})$ *under the side conditions that:*

- *the source profile remains anchored at the earliest honest boundary event,*
- *it does not smooth away, postpone, or reclassify that boundary event, and*
- *it may only feed later tasks from*

$$\{\mathsf{DFX}_m(\mathbf{w}), \mathsf{MC}_m^{\mathrm{bdry}}(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{bdry}}(\mathbf{w})\}.$$

Finally define the first conservative result map

$$\mathsf{Res}_m^{(1),\mathrm{cons}}(\mathbf{w}) := \begin{cases} \mathsf{Res}_m^{(1),\mathrm{hor}}(\mathbf{w}), & \text{if } \mathsf{Exec}_m^{(1),\mathrm{cons}}(\mathbf{w}) = \mathsf{Exec}_m^{(1),\mathrm{hor}}(\mathbf{w}), \\ \mathsf{Res}_m^{(1),\mathrm{bdry}}(\mathbf{w}), & \text{if } \mathsf{Exec}_m^{(1),\mathrm{cons}}(\mathbf{w}) = \mathsf{Exec}_m^{(1),\mathrm{bdry}}(\mathbf{w}). \end{cases}$$

Proposition H4.141 (Every first-step execution packet yields exactly one first conservative result form)**.** *Assume the setting of Definition H4.140. Then:*

(i) every first-step conservative execution packet $\mathsf{Exec}_m^{(1),\mathrm{cons}}(\mathbf{w})$ *yields exactly one first conservative result form* $\mathsf{Res}_m^{(1),\mathrm{cons}}(\mathbf{w})$*;*

(ii) in the horizon-side case, the resulting sharpened readout $\mathsf{HZ}_m^{\sharp}(\mathbf{w})$ *is certificate-aware and may support later crystal, conservative field, and decoder refinement without enlarging theorematic authority beyond the already certified range;*

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(iii) *in the boundary-side case, the resulting anchored profile* $\mathsf{BD}_m^{\mathrm{anch}}(\mathbf{w})$ *is certificate-aware and fixes the earliest honest source/exposure site before any later defect-front, stop-sensitive probe, or boundary-side decoder refinement is attempted;*

(iv) *certificate-aware probe output may profile or reprioritize work inside the inherited result form, but it may not convert a sharpened horizon-side readout into a boundary profile, convert an anchored boundary profile into a continuation certificate, or erase the route-faithful stop condition.*

Proof. Clause (i) is immediate from Definition H4.136: every certified word carries exactly one first-step execution packet, and Definition H4.140 assigns exactly one corresponding result form according to whether that packet is horizon-side or boundary-side. Clause (ii) follows because the horizon-side result explicitly records the sharpened matrix readout together with the inherited crystal and field tags, while preserving the confinement conditions from the execution packet; this sharpens the readout surface, but it does not certify any further extension. Clause (iii) follows because the boundary-side result explicitly records the anchored source profile together with the inherited boundary-side crystal and field tags, while preserving the earliest-boundary confinement conditions; hence localization precedes any later profiling that presupposes a fixed source anchor. Clause (iv) is exactly the inherited probe discipline from $19\mathrm{AZ}^{(6ai)}$, $19\mathrm{AZ}^{(6ak)}$, and $19\mathrm{AZ}^{(6al)}$, now applied at result-form level rather than packet level. □

Corollary H4.142 (The present post-archival stage line already carries a first conservative output surface)**.** *For the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$*, the current active line already carries one first conservative output surface:*

$$\mathsf{Res}_2^{(1),\mathrm{cons}}(\mathbf{w}_3) = \begin{cases} \mathsf{Res}_2^{(1),\mathrm{hor}}(\mathbf{w}_3), & \text{if } \mathbf{w}_3 = \mathsf{A}, \\ \mathsf{Res}_2^{(1),\mathrm{bdry}}(\mathbf{w}_3), & \text{if } \mathbf{w}_3 = \mathsf{B}. \end{cases}$$

So the post-release continuation now carries not only an executable first move, but also its first normalized conservative output: a sharpened matrix readout in the transfer-positive case, and an anchored source/exposure profile in the earliest-boundary case.

Proof. Immediate from Corollary H4.138 and Proposition H4.141. □

Remark H4.143 (What the first conservative result form now buys)**.** *This is the point where the post-release line stops carrying only a certified task and an executable packet and begins to carry a first actual result surface. That is exactly the level at which later conservative sharpening can become cumulative: crystal refinement, conservative field profiling, decoder compression, defect-front profiling, and stop-sensitive probing can now inherit a normalized sharpened readout or anchored source profile rather than reconstructing the first move from scratch.*

Reader cue. Read the next block as the first honest second-stage continuation extracted from the newly available conservative result surface. The point is not yet to claim a global refinement theorem, but to show that once a sharpened horizon-side readout or an anchored boundary-side source profile exists, the next conservative move is no longer free: on the horizon side it is crystal stabilization from the sharpened readout; on the boundary side it is defect-front profiling from the anchored source profile.

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19AZ$^{(6aq)}$. First second-stage follow-on forms extracted from the first conservative result surface

After 19AZ$^{(6ap)}$, the post-release line carries a first conservative result surface. The next honest compression step is therefore to record the first second-stage follow-on form that this result surface allows. On the horizon side the sharpened matrix/Heisenberg–Millennium readout should feed crystal stabilization before any broader field or decoder refinement is attempted. On the boundary side the anchored source/exposure profile should feed defect-front profiling before any broader stop-sensitive probe or decoder compression is attempted.

Definition H4.144 (First second-stage follow-on form)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, its first conservative result form*

$$\mathsf{Res}_m^{(1),\mathrm{cons}}(\mathbf{w})$$

from Definition H4.140, and the inherited certificate-aware probe family $\mathsf{MCProbe}_{m,\mathbf{w}}$*.*

(i) If

$$\mathsf{Res}_m^{(1),\mathrm{cons}}(\mathbf{w}) = \mathsf{Res}_m^{(1),\mathrm{hor}}(\mathbf{w}),$$

define the first horizon-side second-stage follow-on form

$$\mathsf{FolI}_m^{(2),\mathrm{hor}}(\mathbf{w}) := \big(\mathsf{CRS}_m(\mathbf{w}), \mathsf{HZ}_m^{\sharp}(\mathbf{w}), \mathsf{FEH}_m(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{hor}}(\mathbf{w}), \mathsf{MCProbe}_{m,\mathbf{w}}\big),$$

where $\mathsf{CRS}_m(\mathbf{w})$ *denotes the first conservative crystal-stabilization task extracted from the sharpened readout under the side conditions that:*

- *crystal stabilization remains confined to the already certified horizon-side range encoded by* $\mathsf{HZ}_m^{\sharp}(\mathbf{w})$*,*
- *it does not replace, weaken, or bypass the sharpened readout that generated it, and*
- *it may only feed later tasks from*

$$\{\mathsf{PDE}_m^{\mathrm{cons}}(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{hor}}(\mathbf{w})\}.$$

(ii) If

$$\mathsf{Res}_m^{(1),\mathrm{cons}}(\mathbf{w}) = \mathsf{Res}_m^{(1),\mathrm{bdry}}(\mathbf{w}),$$

define the first boundary-side second-stage follow-on form

$$\mathsf{FolI}_m^{(2),\mathrm{bdry}}(\mathbf{w}) := \big(\mathsf{DFX}_m(\mathbf{w}), \mathsf{BD}_m^{\mathrm{anch}}(\mathbf{w}), \mathsf{FEB}_m(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{bdry}}(\mathbf{w}), \mathsf{MCProbe}_{m,\mathbf{w}}\big),$$

where $\mathsf{DFX}_m(\mathbf{w})$ *denotes the first conservative defect-front profiling task extracted from the anchored source profile under the side conditions that:*

- *defect-front profiling remains anchored at the earliest honest boundary event encoded by* $\mathsf{BD}_m^{\mathrm{anch}}(\mathbf{w})$*,*
- *it does not smooth away, postpone, or reinterpret the anchored source/exposure site, and*
- *it may only feed later tasks from*

$$\{\mathsf{MC}_m^{\mathrm{bdry}}(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{bdry}}(\mathbf{w})\}.$$

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Finally define the first second-stage conservative follow-on map

$$\mathsf{Foll}_m^{(2),\mathrm{cons}}(\mathbf{w}) := \begin{cases} \mathsf{Foll}_m^{(2),\mathrm{hor}}(\mathbf{w}), & \text{if } \mathsf{Res}_m^{(1),\mathrm{cons}}(\mathbf{w}) = \mathsf{Res}_m^{(1),\mathrm{hor}}(\mathbf{w}), \\ \mathsf{Foll}_m^{(2),\mathrm{bdry}}(\mathbf{w}), & \text{if } \mathsf{Res}_m^{(1),\mathrm{cons}}(\mathbf{w}) = \mathsf{Res}_m^{(1),\mathrm{bdry}}(\mathbf{w}). \end{cases}$$

Proposition H4.145 (Every first conservative result surface yields exactly one first second-stage follow-on form)**.** *Assume the setting of Definition H4.144. Then:*

(i) every first conservative result form $\mathsf{Res}_m^{(1),\mathrm{cons}}(\mathbf{w})$ *yields exactly one first second-stage follow-on form* $\mathsf{Foll}_m^{(2),\mathrm{cons}}(\mathbf{w})$*;*

(ii) in the horizon-side case, crystal stabilization is the first honest second-stage follow-on because the sharpened readout $\mathsf{HZ}_m^{\sharp}(\mathbf{w})$ *must be structurally stabilized before any broader conservative field or decoder refinement is allowed to reuse it;*

(iii) in the boundary-side case, defect-front profiling is the first honest second-stage follow-on because the anchored source profile $\mathsf{BD}_m^{\mathrm{anch}}(\mathbf{w})$ *must first fix the earliest honest boundary geometry before any broader stop-sensitive probe or decoder compression is allowed to reuse it; and*

(iv) certificate-aware probe output may profile or reprioritize work inside the inherited follow-on form, but it may not convert crystal stabilization into a source-localization task, convert defect-front profiling into a continuation certificate, or erase the inherited horizon/boundary route.

Proof. Clause (i) is immediate from Definition H4.140: every certified word carries exactly one first conservative result surface, and Definition H4.144 assigns exactly one corresponding second-stage follow-on form according to whether that surface is horizon-side or boundary-side. Clause (ii) follows because the sharpened horizon-side readout already isolates the admissible range, so the first honest reuse of that surface is to stabilize its crystal-side structure before allowing broader conservative field or decoder refinement. Clause (iii) follows because the anchored boundary-side source profile already isolates the earliest honest boundary site, so the first honest reuse of that surface is to profile the associated defect front before allowing broader stop-sensitive probing or boundary-side decoder compression. Clause (iv) is exactly the inherited probe discipline from $19\mathrm{AZ}^{(6ai)}$, $19\mathrm{AZ}^{(6ak)}$, and $19\mathrm{AZ}^{(6al)}$, now applied at second-stage follow-on level. □

Corollary H4.146 (The present post-archival stage line already carries a first second-stage conservative bifurcation)**.** *For the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$*, the current active line already carries one first second-stage conservative bifurcation:*

$$\mathsf{Foll}_2^{(2),\mathrm{cons}}(\mathbf{w}_3) = \begin{cases} \mathsf{Foll}_2^{(2),\mathrm{hor}}(\mathbf{w}_3), & \text{if } \mathbf{w}_3 = \mathsf{A}, \\ \mathsf{Foll}_2^{(2),\mathrm{bdry}}(\mathbf{w}_3), & \text{if } \mathbf{w}_3 = \mathsf{B}. \end{cases}$$

So the post-release continuation now carries not only a first conservative output surface, but also its first honest second-stage bifurcation: crystal stabilization from the sharpened horizon-side readout, and defect-front profiling from the anchored boundary-side source profile.

Proof. Immediate from Corollary H4.142 and Proposition H4.145. □

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Remark H4.147 (What the first second-stage follow-on form now buys)**.** *This is the point where the post-release line starts to behave like a genuine conservative upgrade staircase rather than a single executable packet with a single output surface. The first result form now exports an honest next layer: on the horizon side, crystal stabilization can proceed from a sharpened matrix/Heisenberg–Millennium readout; on the boundary side, defect-front profiling can proceed from an anchored source/exposure profile. Later conservative field refinement, stop-sensitive probes, and decoder compression therefore inherit a better-prepared surface instead of reaching directly back to the first move.*

Reader cue. Read the next block as the first third-stage continuation extracted from the conservative upgrade staircase that now exists after $19\mathrm{AZ}^{(6aq)}$. The point is not yet to claim a full operator or field closure, but to show that once crystal stabilization or defect-front profiling has been honestly reached, the next conservative reuse is again no longer free: on the horizon side it is conservative field refinement from the stabilized crystal side, and on the boundary side it is a stop-sensitive probe/decoder surface from the profiled defect front.

$19\mathrm{AZ}^{(6ar)}$. First third-stage continuation forms extracted from the conservative upgrade staircase

After $19\mathrm{AZ}^{(6aq)}$, the post-release line carries a first second-stage conservative bifurcation. The next honest compression step is therefore to record the first third-stage continuation form that this bifurcation allows. On the horizon side, once a stabilized crystal-side structure exists, the first honest continuation is a conservative field/profile refinement still confined to the already certified range. On the boundary side, once the defect front attached to the earliest honest boundary event has been profiled, the first honest continuation is a stop-sensitive probe/decoder surface that may read the boundary geometry more finely without erasing its stop character.

Definition H4.148 (First third-stage continuation form)**.** *Fix a certified finite extension word $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$, its first second-stage conservative follow-on form*

$$\mathsf{Foll}_m^{(2),\mathrm{cons}}(\mathbf{w})$$

from Definition H4.144, and the inherited certificate-aware probe family $\mathsf{MCProbe}_{m,\mathbf{w}}$.

(i) If

$$\mathsf{Foll}_m^{(2),\mathrm{cons}}(\mathbf{w}) = \mathsf{Foll}_m^{(2),\mathrm{hor}}(\mathbf{w}),$$

define the first horizon-side third-stage continuation form

$$\mathsf{Foll}_m^{(3),\mathrm{hor}}(\mathbf{w}) := \big(\mathsf{PDE}_m^{\mathrm{cons}}(\mathbf{w}), \mathsf{CRS}_m(\mathbf{w}), \mathsf{FEH}_m(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{hor}}(\mathbf{w}), \mathsf{MCProbe}_{m,\mathbf{w}}\big),$$

where $\mathsf{PDE}_m^{\mathrm{cons}}(\mathbf{w})$ denotes the first conservative field/profile refinement task extracted from the stabilized crystal-side structure under the side conditions that:

- *conservative field refinement remains confined to the already certified horizon-side range inherited from $\mathsf{HZ}_m^{\sharp}(\mathbf{w})$ and stabilized by $\mathsf{CRS}_m(\mathbf{w})$,*
- *it does not reopen, replace, or bypass the sharpened matrix/Heisenberg–Millennium readout or the crystal stabilization that prepared it, and*
- *it may only feed later horizon-side tasks that remain route-faithful to*

$$\{\mathsf{DEC}_m^{\mathrm{hor}}(\mathbf{w}), \mathsf{MCProbe}_{m,\mathbf{w}}\}.$$

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(ii) If

$$\mathsf{Foll}_m^{(2),\mathrm{cons}}(\mathbf{w}) = \mathsf{Foll}_m^{(2),\mathrm{bdry}}(\mathbf{w}),$$

define the first boundary-side third-stage continuation form

$$\mathsf{Foll}_m^{(3),\mathrm{bdry}}(\mathbf{w}) := \big(\mathsf{StopSurf}_m(\mathbf{w}), \mathsf{DFX}_m(\mathbf{w}), \mathsf{FEB}_m(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{bdry}}(\mathbf{w}), \mathsf{MCProbe}_{m,\mathbf{w}}\big),$$

where $\mathsf{StopSurf}_m(\mathbf{w})$ *denotes the first stop-sensitive probe/decoder surface extracted from the profiled defect front under the side conditions that:*

- *the stop-sensitive surface remains anchored at the earliest honest boundary event encoded by* $\mathsf{BD}_m^{\mathrm{anch}}(\mathbf{w})$ *and profiled by* $\mathsf{DFX}_m(\mathbf{w})$*,*
- *it may cluster, compress, compare, or decoder-read boundary-side probe traces, but it may not smooth away, postpone, or reinterpret the inherited stop event, and*
- *it may only feed later boundary-side tasks that remain route-faithful to*

$$\{\mathsf{DEC}_m^{\mathrm{bdry}}(\mathbf{w}), \mathsf{MCProbe}_{m,\mathbf{w}}\}.$$

Finally define the first third-stage conservative continuation map

$$\mathsf{Foll}_m^{(3),\mathrm{cons}}(\mathbf{w}) := \begin{cases} \mathsf{Foll}_m^{(3),\mathrm{hor}}(\mathbf{w}), & \text{if } \mathsf{Foll}_m^{(2),\mathrm{cons}}(\mathbf{w}) = \mathsf{Foll}_m^{(2),\mathrm{hor}}(\mathbf{w}), \\ \mathsf{Foll}_m^{(3),\mathrm{bdry}}(\mathbf{w}), & \text{if } \mathsf{Foll}_m^{(2),\mathrm{cons}}(\mathbf{w}) = \mathsf{Foll}_m^{(2),\mathrm{bdry}}(\mathbf{w}). \end{cases}$$

Proposition H4.149 (Every first second-stage conservative bifurcation yields exactly one first third-stage continuation form)**.** *Assume the setting of Definition H4.148. Then:*

(i) every first second-stage conservative follow-on form $\mathsf{Foll}_m^{(2),\mathrm{cons}}(\mathbf{w})$ *yields exactly one first third-stage continuation form* $\mathsf{Foll}_m^{(3),\mathrm{cons}}(\mathbf{w})$*;*

(ii) in the horizon-side case, conservative field refinement is the first honest third-stage continuation because the crystal-stabilized structure is the first admissible surface on which a conservative field/profile refinement can be run without reopening the sharpened readout;

(iii) in the boundary-side case, a stop-sensitive probe/decoder surface is the first honest third-stage continuation because the profiled defect front is the first admissible surface on which probe and decoder traces may be refined without erasing the inherited stop event; and

(iv) certificate-aware probe output may reprioritize or compress work inside the inherited third-stage continuation form, but it may not turn conservative field refinement into a continuation certificate, turn a stop-sensitive surface into a horizon extension, or erase the inherited horizon/boundary route.

Proof. Clause (i) is immediate from Definition H4.144: every certified word carries exactly one first second-stage conservative follow-on form, and Definition H4.148 assigns exactly one corresponding third-stage continuation form according to whether that form is horizon-side or boundary-side. Clause (ii) follows because the crystal-stabilized horizon-side structure is the first admissible surface on which conservative field refinement may proceed without reaching back behind the sharpened readout. Clause (iii) follows because the profiled defect front is the first admissible boundary-side surface on which stop-sensitive probe and decoder traces may be refined without smoothing away the earliest honest stop. Clause (iv) is exactly the inherited probe discipline from $19\mathrm{AZ}^{(6ai)}$, $19\mathrm{AZ}^{(6ak)}$, and $19\mathrm{AZ}^{(6al)}$, now applied at third-stage continuation level. □

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Corollary H4.150 (The present post-archival stage line already carries a first third-stage conservative continuation)**.** *For the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$*, the current active line already carries one first third-stage conservative continuation:*

$$\mathsf{Foll}_2^{(3),\mathrm{cons}}(\mathbf{w}_3) = \begin{cases} \mathsf{Foll}_2^{(3),\mathrm{hor}}(\mathbf{w}_3), & \text{if } \mathbf{w}_3 = \mathsf{A}, \\ \mathsf{Foll}_2^{(3),\mathrm{bdry}}(\mathbf{w}_3), & \text{if } \mathbf{w}_3 = \mathsf{B}. \end{cases}$$

So the post-release continuation now carries not only a first conservative output surface and its second-stage bifurcation, but also the first honest third-stage continuation: conservative field refinement from the crystal-stabilized horizon-side structure, and a stop-sensitive probe/decoder surface from the profiled boundary-side defect front.

Proof. Immediate from Corollary H4.146 and Proposition H4.149. □

Remark H4.151 (What the first third-stage continuation now buys)**.** *This is the point where the post-release line begins to carry a genuine conservative continuation staircase rather than only a first move, a first result surface, and one second-stage split. On the horizon side the line now reaches a first conservative field/profile refinement that is already prepared by both sharpened readout and crystal stabilization. On the boundary side the line now reaches a first stop-sensitive probe/decoder surface that is already prepared by both anchored source profiling and defect-front analysis. That makes later decoder compression, conservative field sharpening, and boundary-side diagnostic integration inherit a more mature surface instead of touching the raw first bifurcation directly.*

Reader cue. Read the next block as the first compact conservative readout surface extracted from the upgrade staircase that now exists after $19\mathrm{AZ}^{(6ar)}$. The purpose is to stop carrying the horizon-side and boundary-side continuations only as scattered tasks. Once the sharpened readout, crystal stabilization, and conservative field refinement have been honestly reached on the horizon side, and once the anchored source profile, defect-front profiling, and stop-sensitive probe/decoder surface have been honestly reached on the boundary side, the line may compress each side into one compact conservative working surface for downstream use.

$19\mathrm{AZ}^{(6as)}$. First compact conservative readout surfaces extracted from the continuation staircase

After $19\mathrm{AZ}^{(6ar)}$, the post-release line carries a first third-stage conservative continuation on both the horizon side and the boundary side. The next honest compression step is therefore to record the first compact conservative readout surface that each side now supports. On the horizon side this surface should combine the sharpened matrix/Heisenberg–Millennium readout, the stabilized crystal structure, and the conservative field/profile refinement into one downstream-readable object. On the boundary side it should combine the anchored source/exposure profile, the defect-front profile, and the stop-sensitive probe/decoder surface into one downstream-readable object.

Definition H4.152 (First compact conservative readout surface)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, its first third-stage conservative continuation form*

$$\mathsf{Foll}_m^{(3),\mathrm{cons}}(\mathbf{w})$$

from Definition H4.148, and the inherited certificate-aware probe family $\mathsf{MCProbe}_{m,\mathbf{w}}$*.*

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(i) If

$$\mathsf{Foll}_m^{(3),\mathrm{cons}}(\mathbf{w}) = \mathsf{Foll}_m^{(3),\mathrm{hor}}(\mathbf{w}),$$

define the first compact horizon-side conservative readout surface

$$\mathsf{Surf}_m^{\mathrm{hor}}(\mathbf{w}) := \big(\mathsf{HZ}_m^{\sharp}(\mathbf{w}), \mathsf{CRS}_m(\mathbf{w}), \mathsf{PDE}_m^{\mathrm{cons}}(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{hor}}(\mathbf{w}), \mathsf{MCProbe}_{m,\mathbf{w}}\big),$$

under the side conditions that:

- *the surface remains confined to the already certified horizon-side range inherited from the sharpened readout and the crystal-stabilized structure,*
- *the conservative field/profile component remains subordinate to that inherited range and may sharpen transport-side organization but may not certify any new continuation beyond it, and*
- *downstream consumers may read, compare, compress, or decoder-organize the surface, but they may not reopen the route decision that produced it.*

(ii) If

$$\mathsf{Foll}_m^{(3),\mathrm{cons}}(\mathbf{w}) = \mathsf{Foll}_m^{(3),\mathrm{bdry}}(\mathbf{w}),$$

define the first compact boundary-side conservative readout surface

$$\mathsf{Surf}_m^{\mathrm{bdry}}(\mathbf{w}) := \big(\mathsf{BD}_m^{\mathrm{anch}}(\mathbf{w}), \mathsf{DFX}_m(\mathbf{w}), \mathsf{StopSurf}_m(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{bdry}}(\mathbf{w}), \mathsf{MCProbe}_{m,\mathbf{w}}\big),$$

under the side conditions that:

- *the surface remains anchored at the earliest honest boundary event inherited from the anchored source/exposure profile and the profiled defect front,*
- *the stop-sensitive probe/decoder component may refine clustering, exposure comparison, and decoder-readable boundary traces, but it may not smooth away, postpone, or reinterpret the inherited stop event, and*
- *downstream consumers may read, compare, compress, or decoder-organize the surface, but they may not reroute it into a horizon-side continuation claim.*

Finally define the first compact conservative readout-surface map

$$\mathsf{Surf}_m^{\mathrm{cons}}(\mathbf{w}) := \begin{cases} \mathsf{Surf}_m^{\mathrm{hor}}(\mathbf{w}), & \text{if } \mathsf{Foll}_m^{(3),\mathrm{cons}}(\mathbf{w}) = \mathsf{Foll}_m^{(3),\mathrm{hor}}(\mathbf{w}), \\ \mathsf{Surf}_m^{\mathrm{bdry}}(\mathbf{w}), & \text{if } \mathsf{Foll}_m^{(3),\mathrm{cons}}(\mathbf{w}) = \mathsf{Foll}_m^{(3),\mathrm{bdry}}(\mathbf{w}). \end{cases}$$

Proposition H4.153 (Every first third-stage conservative continuation yields exactly one compact conservative readout surface)**.** *Assume the setting of Definition H4.152. Then:*

(i) every first third-stage conservative continuation form $\mathsf{Foll}_m^{(3),\mathrm{cons}}(\mathbf{w})$ *yields exactly one compact conservative readout surface* $\mathsf{Surf}_m^{\mathrm{cons}}(\mathbf{w})$*;*

(ii) in the horizon-side case, the compact surface is the first honest downstream object that combines sharpened readout, crystal stabilization, and conservative field refinement without reopening the route that produced them;

(iii) in the boundary-side case, the compact surface is the first honest downstream object that combines anchored source profiling, defect-front profiling, and stop-sensitive probe/decoder structure without erasing the inherited earliest-boundary stop; and

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(iv) probe or decoder output may cluster, compare, or compress information inside the compact surface, but it may not turn that surface into a new theorematic certificate, a route change, or a denial of the inherited horizon/boundary distinction.

Proof. Clause (i) is immediate from Definition H4.148: every certified word carries exactly one first third-stage conservative continuation form, and Definition H4.152 assigns exactly one corresponding compact conservative readout surface according to whether that continuation is horizon-side or boundary-side. Clause (ii) follows because the horizon-side continuation has already prepared the sharpened readout, crystal stabilization, and conservative field/profile refinement on one inherited certified range, so the first honest reuse is to package them into one downstream-readable surface rather than to scatter them again across separate layers. Clause (iii) follows because the boundary-side continuation has already prepared the anchored source profile, the profiled defect front, and the stop-sensitive probe/decoder structure on one inherited earliest-boundary event, so the first honest reuse is to package them into one downstream-readable surface rather than to let later diagnostics improvise their own boundary anchor. Clause (iv) is exactly the inherited certificate-aware probe discipline from $19\mathrm{AZ}^{(6ai)}$, $19\mathrm{AZ}^{(6ak)}$, and $19\mathrm{AZ}^{(6al)}$, now applied at compact-surface level. □

Corollary H4.154 (The present post-archival stage line already carries a first compact conservative readout surface)**.** *For the first-stage word $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$, the current active line already carries one compact conservative readout surface:*

$$\mathsf{Surf}_2^{\mathrm{cons}}(\mathbf{w}_3) = \begin{cases} \mathsf{Surf}_2^{\mathrm{hor}}(\mathbf{w}_3), & \text{if } \mathbf{w}_3 = \mathsf{A}, \\ \mathsf{Surf}_2^{\mathrm{bdry}}(\mathbf{w}_3), & \text{if } \mathbf{w}_3 = \mathsf{B}. \end{cases}$$

So the post-release continuation now carries not only a conservative upgrade staircase through first move, first result surface, second-stage bifurcation, and third-stage continuation, but also the first compact conservative working surface on each side: horizon-side matrix/crystal/field organization and boundary-side source/defect/probe/decoder organization.

Proof. Immediate from Corollary H4.150 and Proposition H4.153. □

Remark H4.155 (What the first compact conservative readout surface now buys)**.** *This is the point where the post-release line first begins to look like a usable conservative readout interface rather than only a staircase of preparatory follow-on forms. On the horizon side, matrix sharpening, crystal stabilization, and conservative field refinement are now available as one compact surface that later decoder or operator-side work may read without falling back behind the certified range. On the boundary side, anchored source profiling, defect-front profiling, and stop-sensitive probe/decoder structure are now available as one compact surface that later diagnostic or clustering work may read without erasing the inherited stop event. The next natural move is therefore no longer to invent another preparatory block, but to extract from these compact surfaces the first route-faithful conservative output or comparison layer that later OP-oriented work can consume.*

Reader cue. Read the next block as the first route-faithful conservative comparison/output layer extracted from the compact surfaces of $19\mathrm{AZ}^{(6as)}$. The purpose is to stop treating the horizon-side and boundary-side surfaces only as rich internal packages. Once each side already carries one compact conservative surface, the next honest step is to expose one downstream-readable output/comparison form that later OP-oriented work may consume without reopening the route, smoothing away an inherited stop, or pretending that horizon-side and boundary-side outputs are interchangeable.

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19AZ$^{(6at)}$. First route-faithful conservative comparison/output layer extracted from the compact surfaces

After 19AZ$^{(6as)}$, the post-release line carries a first compact conservative readout surface on each admissible side. The next honest compression step is therefore to expose a route-faithful comparison/output layer that makes each compact surface downstream-readable as one conservative output object. On the horizon side this output should preserve the already certified matrix/crystal/field organization. On the boundary side it should preserve the already anchored source/defect/probe/decoder organization together with the inherited stop sensitivity. The point is not to collapse both sides into one falsely uniform object, but to let later OP-oriented or diagnostic work compare and consume them through one disciplined interface.

Definition H4.156 (First route-faithful conservative output/comparison layer)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, its first compact conservative readout surface*

$$\mathsf{Surf}_m^{\mathrm{cons}}(\mathbf{w})$$

from Definition H4.152, and the inherited certificate-aware probe family $\mathsf{MCProbe}_{m,\mathbf{w}}$*.*

(i) If

$$\mathsf{Surf}_m^{\mathrm{cons}}(\mathbf{w}) = \mathsf{Surf}_m^{\mathrm{hor}}(\mathbf{w}),$$

define the first route-faithful horizon-side conservative output/comparison object

$$\mathsf{OutCmp}_m^{\mathrm{hor}}(\mathbf{w}) := \big(\mathsf{HOR}, \mathsf{HZ}_m^{\sharp}(\mathbf{w}), \mathsf{CRS}_m(\mathbf{w}), \mathsf{PDE}_m^{\mathrm{cons}}(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{hor}}(\mathbf{w}), \mathsf{MCProbe}_{m,\mathbf{w}}\big).$$

under the side conditions that:

- *the tag* HOR *records that the output remains confined to the already certified horizon-side range,*
- *downstream comparison or output work may compare this object with other horizon-side conservative outputs, may compress it, and may use it as an input surface for later OP-oriented tasks, but it may not reinterpret it as a boundary-side stop object, and*
- *no component of the output may certify a continuation beyond the range already inherited by* $\mathsf{Surf}_m^{\mathrm{hor}}(\mathbf{w})$*.*

(ii) If

$$\mathsf{Surf}_m^{\mathrm{cons}}(\mathbf{w}) = \mathsf{Surf}_m^{\mathrm{bdry}}(\mathbf{w}),$$

define the first route-faithful boundary-side conservative output/comparison object

$$\mathsf{OutCmp}_m^{\mathrm{bdry}}(\mathbf{w}) := \big(\mathsf{BDRY}, \mathsf{BD}_m^{\mathrm{anch}}(\mathbf{w}), \mathsf{DFX}_m(\mathbf{w}), \mathsf{StopSurf}_m(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{bdry}}(\mathbf{w}), \mathsf{MCProbe}_{m,\mathbf{w}}\big).$$

under the side conditions that:

- *the tag* BDRY *records that the output remains anchored at the earliest honest boundary event and retains inherited stop sensitivity,*
- *downstream comparison or output work may compare this object with other boundary-side conservative outputs, may compress it, and may use it as an input surface for later OP-oriented tasks, but it may not smooth away the inherited stop event or relabel the object as horizon-side continuation, and*

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- *no component of the output may deny or postpone the earliest honest boundary anchor already inherited by* $\mathsf{Surf}_m^{\mathrm{bdry}}(\mathbf{w})$.

Finally define the first route-faithful conservative output/comparison map

$$\mathsf{OutCmp}_m^{\mathrm{cons}}(\mathbf{w}) := \begin{cases} \mathsf{OutCmp}_m^{\mathrm{hor}}(\mathbf{w}), & \text{if } \mathsf{Surf}_m^{\mathrm{cons}}(\mathbf{w}) = \mathsf{Surf}_m^{\mathrm{hor}}(\mathbf{w}), \\ \mathsf{OutCmp}_m^{\mathrm{bdry}}(\mathbf{w}), & \text{if } \mathsf{Surf}_m^{\mathrm{cons}}(\mathbf{w}) = \mathsf{Surf}_m^{\mathrm{bdry}}(\mathbf{w}). \end{cases}$$

Proposition H4.157 (Every first compact conservative surface yields exactly one route–faithful conservative output/comparison object)**.** *Assume the setting of Definition H4.156. Then:*

(i) every first compact conservative surface $\mathsf{Surf}_m^{\mathrm{cons}}(\mathbf{w})$ *yields exactly one route-faithful conservative output/comparison object* $\mathsf{OutCmp}_m^{\mathrm{cons}}(\mathbf{w})$*;*

(ii) in the horizon-side case, the object is the first honest downstream-readable output that packages the matrix/crystal/field surface together with an explicit horizon-side tag;

(iii) in the boundary-side case, the object is the first honest downstream-readable output that packages the source/defect/probe/decoder surface together with an explicit boundary-side stop-preserving tag; and

(iv) later OP-oriented, diagnostic, clustering, or decoder-side work may consume these objects as conservative outputs, but it may not use them to erase the inherited horizon/boundary distinction or to manufacture a fresh theorematic certificate.

Proof. Clause (i) is immediate from Definition H4.156: every compact conservative surface is either horizon-side or boundary-side, and the definition assigns exactly one corresponding route-faithful output/comparison object with the matching side tag. Clause (ii) follows because the horizon-side compact surface already carries sharpened readout, crystal stabilization, conservative field refinement, and decoder/probe organization on one inherited certified range, so the first honest downstream output is to expose that package with an explicit horizon-side tag rather than to reopen the route or invent a boundary anchor. Clause (iii) follows because the boundary-side compact surface already carries anchored source profiling, defect-front profiling, and stop-sensitive probe/decoder structure on one inherited earliest-boundary event, so the first honest downstream output is to expose that package with an explicit boundary-side tag rather than to smooth away the stop or relabel the object as horizon-side continuation. Clause (iv) is exactly the inherited certificate-aware discipline from $19\mathrm{AZ}^{(6ae)}$, $19\mathrm{AZ}^{(6af)}$, $19\mathrm{AZ}^{(6ai)}$, and $19\mathrm{AZ}^{(6as)}$, now applied at the first output/comparison level. □

Corollary H4.158 (The present post-archival stage line already carries a first route-faithful conservative output/comparison layer)**.** *For the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$*, the current active line already carries one route-faithful conservative output/comparison object:*

$$\mathsf{OutCmp}_2^{\mathrm{cons}}(\mathbf{w}_3) = \begin{cases} \mathsf{OutCmp}_2^{\mathrm{hor}}(\mathbf{w}_3), & \text{if } \mathbf{w}_3 = \mathsf{A}, \\ \mathsf{OutCmp}_2^{\mathrm{bdry}}(\mathbf{w}_3), & \text{if } \mathbf{w}_3 = \mathsf{B}. \end{cases}$$

So the post-release continuation now carries not only a conservative upgrade staircase and a first compact conservative surface on each side, but also the first route-faithful conservative output/comparison layer that later OP-oriented work may consume directly.

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Proof. Immediate from Corollary H4.154 and Proposition H4.157. □

Remark H4.159 (What the first route-faithful conservative output/comparison layer now buys)**.** *This is the point where the post-release line first becomes directly consumable as a disciplined conservative output interface. The horizon-side object now packages matrix/crystal/field organization together with an explicit horizon-side tag, while the boundary-side object packages source/defect/probe/decoder organization together with an explicit stop-preserving boundary-side tag. Later OP-oriented work can therefore start from a route-faithful output object rather than from scattered follow-on packets or even from the raw compact surface itself. The next natural move is to use these output/comparison objects to define the first conservative OP-facing intake layer without broadening the theorematic claims.*

Reader cue. Read the next block as the first conservative OP-facing intake layer extracted from the route-faithful output/comparison objects of $19\mathrm{AZ}^{(6at)}$. Its purpose is to let the next OP-oriented working stage consume the new conservative output surfaces without reopening the route decision, erasing the boundary stop, or inflating the theorematic range. The block does not yet solve any open problem; it defines the disciplined intake interface through which later OP-side work must begin.

$19\mathrm{AZ}^{(6au)}$. First conservative OP-facing intake layer extracted from the route-faithful output/comparison objects

After $19\mathrm{AZ}^{(6at)}$, the post-release line carries a first route-faithful conservative output/comparison object on each admissible side. The next honest step is therefore to define the first OP-facing intake layer that later OP-oriented work may consume. On the horizon side this intake should inherit the already certified matrix/crystal/field package together with its explicit horizon tag. On the boundary side it should inherit the anchored source/defect/probe/decoder package together with its explicit stop-preserving boundary tag. The point is not to broaden the theorematic status, but to make later OP-oriented work start from a disciplined conservative intake surface rather than from raw local branch data.

Definition H4.160 (First conservative OP-facing intake layer)**.** *Fix a certified finite extension word $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$, its first route-faithful conservative output/comparison object*

$$\mathsf{OutCmp}_m^{\mathrm{cons}}(\mathbf{w})$$

from Definition H4.156, and the inherited certificate-aware probe family $\mathsf{MCProbe}_{m,\mathbf{w}}$.

(i) If

$$\mathsf{OutCmp}_m^{\mathrm{cons}}(\mathbf{w}) = \mathsf{OutCmp}_m^{\mathrm{hor}}(\mathbf{w}),$$

define the first horizon-side conservative OP-facing intake object

$$\mathsf{OPIn}_m^{\mathrm{hor}}(\mathbf{w}) := \big(\mathsf{HOR}, \mathsf{HZ}_m^{\sharp}(\mathbf{w}), \mathsf{CRS}_m(\mathbf{w}), \mathsf{PDE}_m^{\mathrm{cons}}(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{hor}}(\mathbf{w}), \mathsf{MCProbe}_{m,\mathbf{w}}\big),$$

under the side conditions that:

- *later OP-oriented work may use this intake object only as a conservative horizon-side input surface and may not reinterpret it as evidence for a boundary-side stop event,*
- *no OP-facing consumer may certify any continuation beyond the horizon-side range already inherited by $\mathsf{OutCmp}_m^{\mathrm{hor}}(\mathbf{w})$, and*

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- *any later transport, matrix, crystal, or field-equation sharpening that starts from this object remains subordinate to the inherited horizon tag* HOR.

(ii) If

$$\mathsf{OutCmp}_m^{\mathrm{cons}}(\mathbf{w}) = \mathsf{OutCmp}_m^{\mathrm{bdry}}(\mathbf{w}),$$

define the first boundary-side conservative OP-facing intake object

$$\mathsf{OPIn}_m^{\mathrm{bdry}}(\mathbf{w}) := \big(\mathsf{BDRY}, \mathsf{BD}_m^{\mathrm{anch}}(\mathbf{w}), \mathsf{DFX}_m(\mathbf{w}), \mathsf{StopSurf}_m(\mathbf{w}), \mathsf{DEC}_m^{\mathrm{bdry}}(\mathbf{w}), \mathsf{MCProbe}_{m,\mathbf{w}}\big).$$

under the side conditions that:

- *later OP-oriented work may use this intake object only as a conservative boundary-side input surface and may not relabel it as a horizon-continuation object,*
- *no OP-facing consumer may smooth away, postpone, or reinterpret the inherited earliest honest boundary stop carried by* $\mathsf{OutCmp}_m^{\mathrm{bdry}}(\mathbf{w})$*, and*
- *any later source, defect, probe, decoder, or field-side comparison that starts from this object remains subordinate to the inherited stop-preserving boundary tag* BDRY.

Finally define the first conservative OP-facing intake map

$$\mathsf{OPIn}_m^{\mathrm{cons}}(\mathbf{w}) := \begin{cases} \mathsf{OPIn}_m^{\mathrm{hor}}(\mathbf{w}), & \text{if } \mathsf{OutCmp}_m^{\mathrm{cons}}(\mathbf{w}) = \mathsf{OutCmp}_m^{\mathrm{hor}}(\mathbf{w}), \\ \mathsf{OPIn}_m^{\mathrm{bdry}}(\mathbf{w}), & \text{if } \mathsf{OutCmp}_m^{\mathrm{cons}}(\mathbf{w}) = \mathsf{OutCmp}_m^{\mathrm{bdry}}(\mathbf{w}). \end{cases}$$

Proposition H4.161 (Every route-faithful conservative output/comparison object yields exactly one conservative OP-facing intake object)**.** *Assume the setting of Definition H4.160. Then:*

(i) every route-faithful conservative output/comparison object $\mathsf{OutCmp}_m^{\mathrm{cons}}(\mathbf{w})$ *yields exactly one conservative OP-facing intake object* $\mathsf{OPIn}_m^{\mathrm{cons}}(\mathbf{w})$*;*

(ii) in the horizon-side case, the intake object is the first honest OP-facing surface that packages the matrix/crystal/field organization together with an explicit horizon-side tag;

(iii) in the boundary-side case, the intake object is the first honest OP-facing surface that packages the source/defect/probe/decoder organization together with an explicit stop-preserving boundary-side tag; and

(iv) later OP-oriented work may consume these intake objects as conservative inputs, but it may not use them to erase the inherited horizon/boundary distinction, to deny the inherited earliest-boundary stop, or to manufacture a fresh theorematic certificate.

Proof. Clause (i) is immediate from Definition H4.160: every route-faithful conservative output/comparison object is either horizon-side or boundary-side, and the definition assigns exactly one corresponding conservative OP-facing intake object with the matching side tag. Clause (ii) follows because the horizon-side output/comparison object already packages sharpened readout, crystal stabilization, conservative field refinement, and decoder/probe organization on one inherited certified range, so the first honest OP-facing use is to intake that package with its explicit horizon-side tag rather than to reopen the route or invent a boundary anchor. Clause (iii) follows because the boundary-side output/comparison object already packages anchored source profiling, defect-front profiling, and stop-sensitive probe/decoder

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structure on one inherited earliest-boundary event, so the first honest OP-facing use is to intake that package with its explicit stop-preserving boundary tag rather than to smooth away the stop or relabel the object as horizon-side continuation. Clause (iv) is exactly the inherited certificate-aware discipline from 19AZ$^{(6ae)}$, 19AZ$^{(6af)}$, 19AZ$^{(6ai)}$, and 19AZ$^{(6at)}$, now applied at the first OP-facing intake level. □

Corollary H4.162 (The present post-archival stage line already carries a first conservative OP-facing intake layer)**.** *For the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$*, the current active line already carries one conservative OP-facing intake object:*

$$\mathsf{OPIn}_2^{\mathrm{cons}}(\mathbf{w}_3) = \begin{cases} \mathsf{OPIn}_2^{\mathrm{hor}}(\mathbf{w}_3), & \text{if } \mathbf{w}_3 = \mathsf{A}, \\ \mathsf{OPIn}_2^{\mathrm{bdry}}(\mathbf{w}_3), & \text{if } \mathbf{w}_3 = \mathsf{B}. \end{cases}$$

So the post-release continuation now carries not only a route-faithful conservative output/comparison layer, but also the first conservative OP-facing intake layer that the next OP-oriented working stage may consume directly.

Proof. Immediate from Corollary H4.158 and Proposition H4.161. □

Remark H4.163 (What the first conservative OP-facing intake layer now buys)**.** *This is the point where the post-release line first becomes ready for the next OP-oriented working stage without pretending that any open problem has already been solved. The horizon-side intake object exposes a disciplined matrix/crystal/field input surface under an explicit horizon tag, while the boundary-side intake object exposes a disciplined source/defect/probe/decoder input surface under an explicit stop-preserving boundary tag. Later OP-facing work can therefore begin from a conservative intake object that is already route-faithful, comparison-ready, and certificate-aware, instead of improvising from scattered packets or reopening the extension decision itself.*

Reader cue. Read the next block as the first conservative OP-facing work-request contract extracted from the intake layer of 19AZ$^{(6au)}$. Its purpose is to specify which kinds of OP-oriented demands may be posed from the new intake objects without reopening the route decision, erasing the inherited boundary stop, or inflating the theorematic range. The block does not yet answer any open problem; it fixes the finite admissible request menu through which the next OP-oriented stage must begin.

19AZ$^{(6av)}$. First conservative OP-facing work-request contract extracted from the intake layer

After 19AZ$^{(6au)}$, the post-release line carries a first conservative OP-facing intake object on each admissible side. The next honest step is therefore to define which OP-oriented requests may be asked from those intake objects without violating the inherited certificate-aware discipline. On the horizon side the admissible requests should remain subordinate to the inherited matrix/crystal/field organization and explicit horizon tag. On the boundary side the admissible requests should remain subordinate to the inherited anchored source/defect/probe/decoder organization and explicit stop-preserving boundary tag. The point is not to authorize arbitrary OP-side speculation, but to specify the first finite request contract under which later OP-facing work may proceed.

Definition H4.164 (First conservative OP-facing work-request contract)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$ *and its first conservative OP-facing intake object*

$$\mathsf{OPIn}_m^{\mathrm{cons}}(\mathbf{w})$$

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from Definition H4.160.

(i) If

$$\mathsf{OPIn}_m^{\mathrm{cons}}(\mathbf{w}) = \mathsf{OPIn}_m^{\mathrm{hor}}(\mathbf{w}),$$

define the first horizon-side conservative OP-facing request menu

$$\mathcal{Q}_m^{\mathrm{hor}}(\mathbf{w}) := \{\mathsf{QMSH}_m(\mathbf{w}), \mathsf{QCRS}_m(\mathbf{w}), \mathsf{QPDE}_m(\mathbf{w}), \mathsf{QDEC}_m^{\mathrm{hor}}(\mathbf{w})\},$$

where the four request forms ask respectively for:

- *conservative matrix/MM-HC sharpening comparison relative to* $\mathsf{HZ}_m^{\sharp}(\mathbf{w})$*,*
- *conservative crystal-stabilization comparison relative to* $\mathsf{CRS}_m(\mathbf{w})$*,*
- *conservative field/profile comparison relative to* $\mathsf{PDE}_m^{\mathrm{cons}}(\mathbf{w})$*, and*
- *horizon-side decoder/probe compression or comparison relative to* $\mathsf{DEC}_m^{\mathrm{hor}}(\mathbf{w})$ *and* $\mathsf{MCProbe}_{m,\mathbf{w}}$*.*

These requests are admissible only under the side conditions that:

- *no request may ask for a fresh theorematic certificate beyond the horizon-side range already inherited by* $\mathsf{OPIn}_m^{\mathrm{hor}}(\mathbf{w})$*,*
- *no request may relabel the horizon-side intake object as evidence for a boundary-side stop event, and*
- *every admissible request remains subordinate to the inherited horizon tag* HOR*.*

(ii) If

$$\mathsf{OPIn}_m^{\mathrm{cons}}(\mathbf{w}) = \mathsf{OPIn}_m^{\mathrm{bdry}}(\mathbf{w}),$$

define the first boundary-side conservative OP-facing request menu

$$\mathcal{Q}_m^{\mathrm{bdry}}(\mathbf{w}) := \{\mathsf{QSRC}_m(\mathbf{w}), \mathsf{QDFX}_m(\mathbf{w}), \mathsf{QSTOP}_m(\mathbf{w}), \mathsf{QDEC}_m^{\mathrm{bdry}}(\mathbf{w})\},$$

where the four request forms ask respectively for:

- *anchored source/exposure comparison relative to* $\mathsf{BD}_m^{\mathrm{anch}}(\mathbf{w})$*,*
- *defect-front comparison relative to* $\mathsf{DFX}_m(\mathbf{w})$*,*
- *stop-sensitive probe-surface comparison relative to* $\mathsf{StopSurf}_m(\mathbf{w})$*, and*
- *boundary-side decoder/probe compression or comparison relative to* $\mathsf{DEC}_m^{\mathrm{bdry}}(\mathbf{w})$ *and* $\mathsf{MCProbe}_{m,\mathbf{w}}$*.*

These requests are admissible only under the side conditions that:

- *no request may smooth away, postpone, or reinterpret the inherited earliest honest boundary stop carried by* $\mathsf{OPIn}_m^{\mathrm{bdry}}(\mathbf{w})$*,*
- *no request may relabel the boundary-side intake object as a horizon-continuation object, and*
- *every admissible request remains subordinate to the inherited stop-preserving boundary tag* BDRY*.*

Finally define the first conservative OP-facing request contract map

$$\mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}) := \begin{cases} \mathcal{Q}_m^{\mathrm{hor}}(\mathbf{w}), & \text{if } \mathsf{OPIn}_m^{\mathrm{cons}}(\mathbf{w}) = \mathsf{OPIn}_m^{\mathrm{hor}}(\mathbf{w}), \\ \mathcal{Q}_m^{\mathrm{bdry}}(\mathbf{w}), & \text{if } \mathsf{OPIn}_m^{\mathrm{cons}}(\mathbf{w}) = \mathsf{OPIn}_m^{\mathrm{bdry}}(\mathbf{w}). \end{cases}$$

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Proposition H4.165 (Every conservative OP-facing intake object yields exactly one finite admissible work-request contract)**.** *Assume the setting of Definition H4.164. Then:*

(i) every conservative OP-facing intake object $\mathsf{OPIn}_m^{\mathrm{cons}}(\mathbf{w})$ *yields exactly one finite admissible request contract* $\mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w})$*;*

(ii) in the horizon-side case, the contract authorizes only matrix/crystal/field/decoder-probe requests subordinate to the inherited horizon tag;

(iii) in the boundary-side case, the contract authorizes only source/defect/stop-surface/decoder-probe requests subordinate to the inherited stop-preserving boundary tag; and

(iv) no admissible request in either contract may manufacture a fresh theorematic certificate, erase the inherited horizon/boundary distinction, or deny the inherited earliest-boundary stop.

Proof. Clause (i) is immediate from Definition H4.164: every conservative OP-facing intake object is either horizon-side or boundary-side, and the definition assigns exactly one corresponding finite request menu. Clause (ii) follows because the horizon-side intake object already packages the sharpened readout, crystal stabilization, conservative field refinement, and decoder/probe organization under one explicit horizon tag, so the only honest OP-facing requests are requests that compare, refine, or compress that inherited organization without leaving the inherited range. Clause (iii) follows because the boundary-side intake object already packages the anchored source profile, defect-front profile, stop-sensitive probe surface, and decoder/probe organization under one explicit stop-preserving boundary tag, so the only honest OP-facing requests are requests that compare, refine, or compress that inherited organization without smoothing away the stop or relabeling the route. Clause (iv) is exactly the inherited certificate-aware discipline from $19\mathrm{AZ}^{(6ae)}$, $19\mathrm{AZ}^{(6af)}$, $19\mathrm{AZ}^{(6ai)}$, $19\mathrm{AZ}^{(6at)}$, and $19\mathrm{AZ}^{(6au)}$, now expressed as the first finite OP-facing request contract. □

Corollary H4.166 (The present post-archival stage line already carries a first finite OP-facing request contract)**.** *For the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$*, the current active line already carries one finite conservative OP-facing request contract:*

$$\mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3) = \begin{cases} \mathcal{Q}_2^{\mathrm{hor}}(\mathbf{w}_3), & \text{if } \mathbf{w}_3 = \mathsf{A}, \\ \mathcal{Q}_2^{\mathrm{bdry}}(\mathbf{w}_3), & \text{if } \mathbf{w}_3 = \mathsf{B}. \end{cases}$$

So the post-release continuation now carries not only a conservative OP-facing intake layer, but also the first finite contract specifying which OP-oriented demands may honestly be posed from that intake surface.

Proof. Immediate from Corollary H4.162 and Proposition H4.165. □

Remark H4.167 (What the first conservative OP-facing work-request contract now buys)**.** *This is the point where the post-release line first becomes able to formulate disciplined OP-oriented questions without pretending that it already owns their answers. The horizon-side intake object now supports a finite request menu for matrix/crystal/field/decoder-probe work under an explicit horizon tag, while the boundary-side intake object supports a finite request menu for source/defect/stop-surface/decoder-probe work under an explicit stop-preserving boundary tag. Later OP-facing work can therefore begin from a conservative intake contract instead of improvising its own admissible question set or silently broadening the inherited theorematic range.*

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Reader cue. Read the next block as the first conservative OP-facing response/fulfillment layer extracted from the request contract of $19AZ^{(6av)}$. Its purpose is to specify how an admissible OP-facing request may be answered conservatively without manufacturing a fresh theorematic certificate, erasing the inherited route, or denying the earliest honest boundary stop. The block does not yet close any open problem; it fixes the first disciplined fulfillment format through which later OP-oriented work may respond to admissible requests.

$19AZ^{(6aw)}$. First conservative OP-facing response/fulfillment layer extracted from the request contract

After $19AZ^{(6av)}$, the post-release line carries a first finite admissible request contract on each intake side. The next honest step is therefore to define how an admissible OP-facing request may be fulfilled while remaining subordinate to the inherited certificate-aware route. On the horizon side a conservative response must stay within the inherited matrix/crystal/field-/decoder organization under the explicit horizon tag. On the boundary side a conservative response must stay within the inherited source/defect/stop-surface/decoder organization under the explicit stop-preserving boundary tag. The point is not to authorize arbitrary new claims, but to define the first finite route-faithful response layer through which admissible OP-facing work may answer the requests that the contract actually allows.

Definition H4.168 (First conservative OP-facing response/fulfillment layer)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, its first conservative OP-facing request contract*

$$\mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w})$$

from Definition H4.164, and an admissible request

$$q \in \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}).$$

Define the first conservative OP-facing fulfillment object

$$\mathsf{Ful}_m^{\mathrm{cons}}(\mathbf{w}; q)$$

by the following cases.

(i) If

$$\mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}) = \mathcal{Q}_m^{\mathrm{hor}}(\mathbf{w}),$$

define the first horizon-side conservative fulfillment family

$$\mathcal{F}_m^{\mathrm{hor}}(\mathbf{w}) := \{\mathsf{FMSH}_m(\mathbf{w}), \mathsf{FCRS}_m(\mathbf{w}), \mathsf{FPDE}_m(\mathbf{w}), \mathsf{FDEC}_m^{\mathrm{hor}}(\mathbf{w})\},$$

where:

- $\mathsf{FMSH}_m(\mathbf{w})$ *is a route-faithful conservative sharpening response to* $\mathsf{QMSH}_m(\mathbf{w})$*,*
- $\mathsf{FCRS}_m(\mathbf{w})$ *is a route-faithful conservative crystal response to* $\mathsf{QCRS}_m(\mathbf{w})$*,*
- $\mathsf{FPDE}_m(\mathbf{w})$ *is a route-faithful conservative field/profile response to* $\mathsf{QPDE}_m(\mathbf{w})$*, and*
- $\mathsf{FDEC}_m^{\mathrm{hor}}(\mathbf{w})$ *is a route-faithful conservative decoder/probe response to* $\mathsf{QDEC}_m^{\mathrm{hor}}(\mathbf{w})$*.*

These fulfillment objects are admissible only if they:

- *remain subordinate to the inherited horizon tag* HOR*,*

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- *do not claim a fresh theorematic certificate beyond the range already carried by* $\mathsf{OPIn}_m^{\mathrm{hor}}(\mathbf{w})$, *and*
- *do not relabel the horizon-side route as a boundary-side stop event.*

Define

$$\mathsf{Ful}_m^{\mathrm{cons}}(\mathbf{w};q) := \begin{cases} \mathsf{FMSH}_m(\mathbf{w}), & \text{if } q = \mathsf{QMSH}_m(\mathbf{w}), \\ \mathsf{FCRS}_m(\mathbf{w}), & \text{if } q = \mathsf{QCRS}_m(\mathbf{w}), \\ \mathsf{FPDE}_m(\mathbf{w}), & \text{if } q = \mathsf{QPDE}_m(\mathbf{w}), \\ \mathsf{FDEC}_m^{\mathrm{hor}}(\mathbf{w}), & \text{if } q = \mathsf{QDEC}_m^{\mathrm{hor}}(\mathbf{w}). \end{cases}$$

(ii) If

$$\mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}) = \mathcal{Q}_m^{\mathrm{bdry}}(\mathbf{w}),$$

define the first boundary-side conservative fulfillment family

$$\mathcal{F}_m^{\mathrm{bdry}}(\mathbf{w}) := \{\mathsf{FSRC}_m(\mathbf{w}), \mathsf{FDFX}_m(\mathbf{w}), \mathsf{FSTOP}_m(\mathbf{w}), \mathsf{FDEC}_m^{\mathrm{bdry}}(\mathbf{w})\},$$

where:

- $\mathsf{FSRC}_m(\mathbf{w})$ *is an anchored source/exposure response to* $\mathsf{QSRC}_m(\mathbf{w})$,
- $\mathsf{FDFX}_m(\mathbf{w})$ *is a defect-front response to* $\mathsf{QDFX}_m(\mathbf{w})$,
- $\mathsf{FSTOP}_m(\mathbf{w})$ *is a stop-sensitive surface response to* $\mathsf{QSTOP}_m(\mathbf{w})$, *and*
- $\mathsf{FDEC}_m^{\mathrm{bdry}}(\mathbf{w})$ *is a boundary-side decoder/probe response to* $\mathsf{QDEC}_m^{\mathrm{bdry}}(\mathbf{w})$.

These fulfillment objects are admissible only if they:

- *remain subordinate to the inherited stop-preserving boundary tag* BDRY,
- *do not smooth away, postpone, or reinterpret the inherited earliest honest boundary stop, and*
- *do not relabel the boundary-side route as a horizon-continuation route.*

Define

$$\mathsf{Ful}_m^{\mathrm{cons}}(\mathbf{w};q) := \begin{cases} \mathsf{FSRC}_m(\mathbf{w}), & \text{if } q = \mathsf{QSRC}_m(\mathbf{w}), \\ \mathsf{FDFX}_m(\mathbf{w}), & \text{if } q = \mathsf{QDFX}_m(\mathbf{w}), \\ \mathsf{FSTOP}_m(\mathbf{w}), & \text{if } q = \mathsf{QSTOP}_m(\mathbf{w}), \\ \mathsf{FDEC}_m^{\mathrm{bdry}}(\mathbf{w}), & \text{if } q = \mathsf{QDEC}_m^{\mathrm{bdry}}(\mathbf{w}). \end{cases}$$

Proposition H4.169 (Every admissible conservative OP-facing request yields exactly one route-faithful conservative fulfillment object)**.** *Assume the setting of Definition H4.168. Then:*

(i) every admissible request $q \in \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w})$ *yields exactly one fulfillment object* $\mathsf{Ful}_m^{\mathrm{cons}}(\mathbf{w};q)$*;*

(ii) in the horizon-side case, the fulfillment object remains subordinate to the inherited horizon tag and inherited matrix/crystal/field/decoder organization;

(iii) in the boundary-side case, the fulfillment object remains subordinate to the inherited stop-preserving boundary tag and inherited source/defect/stop-surface/decoder organization; and

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(iv) no admissible fulfillment object may manufacture a fresh theorematic certificate, erase the inherited horizon/boundary distinction, or deny the inherited earliest-boundary stop.

Proof. Clause (i) is immediate from Definition H4.168: once the request side and request symbol are fixed, the definition assigns exactly one corresponding fulfillment object. Clause (ii) follows because the horizon-side responses are explicitly defined only as conservative sharpening, crystal, field/profile, or decoder/probe responses subordinate to the inherited horizon-side intake organization and explicit horizon tag. Clause (iii) follows because the boundary-side responses are explicitly defined only as anchored source, defect-front, stop-surface, or decoder/probe responses subordinate to the inherited stop-preserving boundary-side intake organization and explicit boundary tag. Clause (iv) is exactly the certificate-aware discipline inherited from $19AZ^{(6ae)}$, $19AZ^{(6af)}$, $19AZ^{(6ai)}$, $19AZ^{(6at)}$, $19AZ^{(6au)}$, and $19AZ^{(6av)}$, now expressed as the first finite OP-facing fulfillment layer. □

Corollary H4.170 (The present post-archival stage line already carries a first OP-facing fulfillment layer)**.** *For the first-stage word $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$, the current active line already carries a first conservative OP-facing fulfillment layer:*

$$\mathsf{Ful}_2^{\mathrm{cons}}(\mathbf{w}_3; q)$$

for every admissible request

$$q \in \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3).$$

So the post-release continuation now carries not only a conservative OP-facing intake object and request contract, but also the first disciplined response format through which admissible OP-oriented demands may be answered without reopening the route decision itself.

Proof. Immediate from Corollary H4.166 and Proposition H4.169. □

Remark H4.171 (What the first conservative OP-facing response/fulfillment layer now buys)**.** *This is the point where the post-release line first becomes able not only to ask disciplined OP-oriented questions, but also to answer them in a route-faithful conservative format. The horizon-side request family now terminates in finite sharpening/crystal/field/decoder responses under an explicit horizon tag, while the boundary-side request family now terminates in finite source/defect/stop-surface/decoder responses under an explicit stop-preserving boundary tag. Later OP-oriented work can therefore begin from a conservative ask-and-respond protocol instead of improvising both its questions and its response objects from scratch.*

Reader cue. Read the next block as the first conservative OP-facing response aggregation / resolution layer extracted from the fulfillment layer of $19AZ^{(6aw)}$. Its purpose is to bundle finitely many admissible conservative fulfillment objects into one downstream-readable OP-facing resolution surface without manufacturing a fresh theorematic certificate, erasing the inherited horizon/boundary distinction, or denying the inherited earliest honest boundary stop. The block still does not close any open problem; it fixes the first disciplined way in which multiple conservative answers may be gathered into one route-faithful working resolution.

$19AZ^{(6ax)}$. First conservative OP-facing response aggregation / resolution layer extracted from the fulfillment layer

After $19AZ^{(6aw)}$, the post-release line already carries finite conservative fulfillment families on each intake side. The next honest step is therefore to define how finitely many such fulfillment objects may be aggregated into one conservative OP-facing resolution surface. On the horizon

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side the aggregation must remain subordinate to the inherited horizon tag and inherited matrix/crystal/field/decoder organization. On the boundary side the aggregation must remain subordinate to the inherited stop-preserving boundary tag and inherited source/defect/stop-surface/decoder organization. The point is not to solve more than the fulfillment layer already solved, but to package several admissible conservative responses into one finite route-faithful resolution object for later OP-oriented comparison, selection, or continuation.

Definition H4.172 (First conservative OP-facing response aggregation / resolution layer)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, its first conservative OP-facing request contract*

$$\mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}),$$

and its first conservative fulfillment map

$$q \mapsto \mathsf{Ful}_m^{\mathrm{cons}}(\mathbf{w}; q)$$

from Definition H4.168. Let

$$\mathcal{S} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w})$$

be a finite nonempty admissible request subfamily. Define the first conservative OP-facing resolution object

$$\mathsf{Res}_m^{\mathrm{cons}}(\mathbf{w}; \mathcal{S})$$

by the following cases.

(i) If

$$\mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}) = \mathcal{Q}_m^{\mathrm{hor}}(\mathbf{w}),$$

define the first horizon-side conservative resolution surface

$$\mathcal{R}_m^{\mathrm{hor}}(\mathbf{w}; \mathcal{S}) := \big(\mathsf{HOR}, \{\mathsf{Ful}_m^{\mathrm{cons}}(\mathbf{w}; q) : q \in \mathcal{S}\}, \mathsf{OPIn}_m^{\mathrm{hor}}(\mathbf{w})\big).$$

This object is admissible only if every included fulfillment object remains subordinate to the inherited horizon tag and no included response claims a fresh theorematic certificate beyond the range already carried by $\mathsf{OPIn}_m^{\mathrm{hor}}(\mathbf{w})$*. Define*

$$\mathsf{Res}_m^{\mathrm{cons}}(\mathbf{w}; \mathcal{S}) := \mathcal{R}_m^{\mathrm{hor}}(\mathbf{w}; \mathcal{S}).$$

(ii) If

$$\mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}) = \mathcal{Q}_m^{\mathrm{bdry}}(\mathbf{w}),$$

define the first boundary-side conservative resolution surface

$$\mathcal{R}_m^{\mathrm{bdry}}(\mathbf{w}; \mathcal{S}) := \big(\mathsf{BDRY}, \{\mathsf{Ful}_m^{\mathrm{cons}}(\mathbf{w}; q) : q \in \mathcal{S}\}, \mathsf{OPIn}_m^{\mathrm{bdry}}(\mathbf{w})\big).$$

This object is admissible only if every included fulfillment object remains subordinate to the inherited stop-preserving boundary tag and no included response smooths away, postpones, or reinterprets the inherited earliest honest boundary stop. Define

$$\mathsf{Res}_m^{\mathrm{cons}}(\mathbf{w}; \mathcal{S}) := \mathcal{R}_m^{\mathrm{bdry}}(\mathbf{w}; \mathcal{S}).$$

Proposition H4.173 (Every finite admissible conservative fulfillment family yields exactly one route-faithful conservative resolution object)**.** *Assume the setting of Definition H4.172. Then:*

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(i) every finite nonempty admissible request subfamily $\mathcal{S} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w})$ yields exactly one conservative resolution object $\mathsf{Res}_m^{\mathrm{cons}}(\mathbf{w}; \mathcal{S})$;

(ii) in the horizon-side case, the resolution object remains subordinate to the inherited horizon tag and inherited matrix/crystal/field/decoder organization;

(iii) in the boundary-side case, the resolution object remains subordinate to the inherited stop-preserving boundary tag and inherited source/defect/stop-surface/decoder organization; and

(iv) no admissible conservative resolution object may manufacture a fresh theorematic certificate, erase the inherited horizon/boundary distinction, or deny the inherited earliest-boundary stop.

Proof. Clause (i) is immediate from Definition H4.172: once the intake side and finite admissible request subfamily $\mathcal{S}$ are fixed, the definition assigns exactly one corresponding resolution surface. Clause (ii) follows because the horizon-side resolution surface is defined only by collecting fulfillment objects already subordinate to the inherited horizon tag together with the inherited horizon-side intake object. Clause (iii) follows because the boundary-side resolution surface is defined only by collecting fulfillment objects already subordinate to the inherited stop-preserving boundary tag together with the inherited boundary-side intake object. Clause (iv) is exactly the certificate-aware discipline inherited from $19\mathrm{AZ}^{(6ae)}$, $19\mathrm{AZ}^{(6af)}$, $19\mathrm{AZ}^{(6ai)}$, $19\mathrm{AZ}^{(6at)}$, $19\mathrm{AZ}^{(6au)}$, $19\mathrm{AZ}^{(6av)}$, and $19\mathrm{AZ}^{(6aw)}$, now expressed as the first finite OP-facing resolution layer. □

Corollary H4.174 (The present post-archival stage line already carries a first OP-facing conservative resolution surface)**.** *For the first-stage word $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$, the current active line already carries a first conservative OP-facing resolution surface*

$$\mathsf{Res}_2^{\mathrm{cons}}(\mathbf{w}_3; \mathcal{S})$$

for every finite nonempty admissible request subfamily

$$\mathcal{S} \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3).$$

So the post-release continuation now carries not only a conservative intake object, request contract, and fulfillment layer, but also the first disciplined aggregation surface through which several admissible OP-oriented answers may be bundled without reopening the route decision itself.

Proof. Immediate from Corollary H4.170 and Proposition H4.173. □

Remark H4.175 (What the first conservative OP-facing response aggregation / resolution layer now buys)**.** *This is the point where the post-release line first becomes able not only to ask and answer disciplined OP-oriented questions, but also to gather several conservative answers into one route-faithful resolution surface. The horizon-side line can now bundle finite sharpening/crystal/field/decoder responses under an explicit horizon tag, while the boundary-side line can now bundle finite source/defect/stop-surface/decoder responses under an explicit stop-preserving boundary tag. Later OP-oriented work can therefore begin from a conservative ask/respond/aggregate protocol instead of improvising both its intermediate bundles and its resolution surfaces from scratch.*

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Reader cue. Read the next block as the first conservative resolution-to-task dispatch layer extracted from the resolution surface of $19\mathrm{AZ}^{(6ax)}$. Its purpose is to turn a finite route-faithful conservative resolution surface into the first finite OP-facing work or comparison program, without manufacturing a fresh theorematic certificate, erasing the inherited horizon/boundary distinction, or denying the inherited earliest honest boundary stop. The block still does not solve any open problem; it fixes the first disciplined way in which a conservative aggregate may be dispatched into explicit next-task families.

$19\mathrm{AZ}^{(6ay)}$. First conservative resolution-to-task dispatch layer extracted from the resolution surface

After $19\mathrm{AZ}^{(6ax)}$, the post-release line already carries finite route-faithful conservative resolution surfaces on each intake side. The next honest step is therefore to define how such a resolution surface may be dispatched into a finite OP-facing work or comparison program. On the horizon side, the dispatch must remain subordinate to the inherited horizon tag and inherited matrix/crystal/field/decoder organization. On the boundary side, the dispatch must remain subordinate to the inherited stop-preserving boundary tag and inherited source/defect/stop-surface/decoder organization. The point is not to claim more than the resolution surface already carries, but to package that finite conservative aggregate into an explicit task family for later OP-oriented comparison, refinement, or packaging.

Definition H4.176 (First conservative resolution-to-task dispatch layer)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$ *and a finite nonempty admissible request subfamily*

$$\mathcal{S} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w})$$

together with its conservative resolution object

$$\mathsf{Res}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S})$$

from Definition H4.172. Define the first conservative dispatch object

$$\mathsf{Disp}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S})$$

by the following cases.

(i) If

$$\mathsf{Res}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}) = \mathcal{R}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}),$$

define the first horizon-side conservative dispatch task family

$$\mathcal{T}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}) := \Big\{\mathsf{TCMP}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}), \mathsf{TREF}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}), \mathsf{TPACK}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S})\Big\}.$$

Here

$$\begin{aligned}
\mathsf{TCMP}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}) &:= \textit{horizon-side comparison task},\\
\mathsf{TREF}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}) &:= \textit{horizon-side refinement task},\\
\mathsf{TPACK}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}) &:= \textit{horizon-side packaging task}.
\end{aligned}$$

This task family is admissible only if every task remains subordinate to the inherited horizon tag HOR*, does not manufacture a fresh theorematic certificate, and does not relabel the horizon-side route as a boundary route. Define*

$$\mathsf{Disp}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}) := \big(\mathsf{HOR}, \mathsf{Res}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}), \mathcal{T}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S})\big).$$

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(ii) If

$$\mathsf{Res}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}) = \mathcal{R}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}),$$

define the first boundary-side conservative dispatch task family

$$\mathcal{T}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}) := \Big\{\mathsf{TCMP}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}), \mathsf{TPROF}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}), \mathsf{TPACK}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S})\Big\}.$$

Here

$$\begin{aligned}\mathsf{TCMP}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}) &:= \textit{boundary-side comparison task},\\ \mathsf{TPROF}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}) &:= \textit{boundary-side profiling task},\\ \mathsf{TPACK}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}) &:= \textit{boundary-side packaging task}.\end{aligned}$$

This task family is admissible only if every task remains subordinate to the inherited stop-preserving boundary tag BDRY*, does not smooth away, postpone, or reinterpret the inherited earliest honest boundary stop, and does not relabel the boundary-side route as a horizon-continuation route. Define*

$$\mathsf{Disp}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}) := \big(\mathsf{BDRY}, \mathsf{Res}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}), \mathcal{T}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S})\big).$$

Proposition H4.177 (Every finite admissible conservative resolution surface yields exactly one route-faithful conservative dispatch object)**.** *Assume the setting of Definition H4.176. Then:*

(i) every finite admissible conservative resolution surface $\mathsf{Res}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S})$ *yields exactly one dispatch object* $\mathsf{Disp}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S})$*;*

(ii) in the horizon-side case, the dispatch object remains subordinate to the inherited horizon tag and inherited matrix/crystal/field/decoder organization;

(iii) in the boundary-side case, the dispatch object remains subordinate to the inherited stop-preserving boundary tag and inherited source/defect/stop-surface/decoder organization; and

(iv) no admissible conservative dispatch object may manufacture a fresh theorematic certificate, erase the inherited horizon/boundary distinction, or deny the inherited earliest-boundary stop.

Proof. Clause (i) is immediate from Definition H4.176: once the intake side and finite admissible request subfamily $\mathcal{S}$ are fixed, the corresponding conservative resolution surface is already fixed by $19\mathrm{AZ}^{(6ax)}$, and the present definition assigns exactly one corresponding dispatch task family and dispatch object. Clause (ii) follows because the horizon-side dispatch object is defined only from the inherited horizon-side resolution surface together with finite comparison, refinement-selection, and packaging tasks subordinate to the explicit horizon tag. Clause (iii) follows because the boundary-side dispatch object is defined only from the inherited boundary-side resolution surface together with finite comparison, profiling-selection, and packaging tasks subordinate to the explicit stop-preserving boundary tag. Clause (iv) is exactly the certificate-aware discipline inherited from $19\mathrm{AZ}^{(6ae)}$, $19\mathrm{AZ}^{(6af)}$, $19\mathrm{AZ}^{(6ai)}$, $19\mathrm{AZ}^{(6at)}$, $19\mathrm{AZ}^{(6au)}$, $19\mathrm{AZ}^{(6av)}$, $19\mathrm{AZ}^{(6aw)}$, and $19\mathrm{AZ}^{(6ax)}$, now expressed as the first finite OP-facing dispatch layer. □

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Corollary H4.178 (The present post-archival stage line already carries a first conservative dispatch object)**.** *For the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$*, the current active line already carries a first conservative dispatch object*

$$\mathsf{Disp}_2^{\mathrm{cons}}(\mathbf{w}_3; \mathcal{S})$$

for every finite nonempty admissible request subfamily

$$\mathcal{S} \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3).$$

So the post-release continuation now carries not only a conservative ask/respond/aggregate protocol, but also the first disciplined dispatch format through which an admissible conservative aggregate may be turned into an explicit OP-facing work or comparison program without reopening the route decision itself.

Proof. Immediate from Corollary H4.174 and Proposition H4.177. □

Remark H4.179 (What the first conservative resolution-to-task dispatch layer now buys)**.** *This is the point where the post-release line first becomes able not only to ask, answer, and aggregate disciplined OP-oriented questions, but also to dispatch the resulting conservative aggregate into a finite work or comparison program. The horizon-side line can now dispatch finite conservative fulfillment bundles into comparison, refinement-selection, and packaging tasks under an explicit horizon tag, while the boundary-side line can now dispatch finite conservative fulfillment bundles into comparison, profiling-selection, and packaging tasks under an explicit stop-preserving boundary tag. Later OP-oriented work can therefore begin from a conservative ask/respond/aggregate/dispatch protocol instead of improvising both its finite task programs and its dispatch interface from scratch.*

Reader cue. Read the next block as the first conservative OP-facing comparison/work program normal form extracted from the dispatch layer of $19\mathrm{AZ}^{(6ay)}$. Its purpose is to compress a route-faithful conservative dispatch object into one downstream-readable finite work/comparison program without reopening the route decision, manufacturing a fresh theorematic certificate, erasing the inherited horizon/boundary distinction, or denying the inherited earliest honest boundary stop. The block still does not solve any open problem; it fixes the first compact normal form in which the new dispatch layer may be consumed by later OP-oriented comparison and work.

$19\mathrm{AZ}^{(6az)}$. First conservative OP-facing comparison/work program normal form extracted from the dispatch layer

After $19\mathrm{AZ}^{(6ay)}$, the post-release line already carries route-faithful conservative dispatch objects whose task families remain subordinate either to the inherited horizon tag or to the inherited stop-preserving boundary tag. The next honest step is therefore to compress such a dispatch object into one finite downstream-readable normal form. On the horizon side, that normal form should package the inherited comparison, refinement-selection, and packaging tasks into one compact work/comparison program subordinate to the explicit horizon tag. On the boundary side, it should package the inherited comparison, profiling-selection, and packaging tasks into one compact work/comparison program subordinate to the explicit stop-preserving boundary tag. The point is not to add any new theorematic force, but only to normalize the already admissible dispatched task family into a short OP-facing work-program object.

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Definition H4.180 (First conservative OP-facing comparison/work program normal form)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$ *and a finite nonempty admissible request subfamily*

$$\mathcal{S} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w})$$

together with its conservative dispatch object

$$\mathsf{Disp}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S})$$

from Definition H4.176. Define the first conservative OP-facing comparison/work program normal form

$$\mathsf{Prog}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S})$$

by the following cases.

(i) If

$$\mathsf{Disp}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}) = \big(\mathsf{HOR}, \mathsf{Res}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}), \mathcal{T}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S})\big),$$

define the first horizon-side conservative comparison/work program normal form

$$\mathcal{P}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}) := \big(\mathsf{HOR}, \mathsf{TCMP}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}), \mathsf{TREF}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}), \mathsf{TPACK}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S})\big).$$

Define

$$\mathsf{Prog}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}) := \mathcal{P}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}).$$

This normal form is admissible only if it remains subordinate to the inherited horizon tag HOR*, does not manufacture a fresh theorematic certificate, and does not relabel the horizon-side route as a boundary route.*

(ii) If

$$\mathsf{Disp}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}) = \big(\mathsf{BDRY}, \mathsf{Res}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}), \mathcal{T}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S})\big),$$

define the first boundary-side conservative comparison/work program normal form

$$\mathcal{P}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}) := \big(\mathsf{BDRY}, \mathsf{TCMP}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}), \mathsf{TPROF}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}), \mathsf{TPACK}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S})\big).$$

Define

$$\mathsf{Prog}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}) := \mathcal{P}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}).$$

This normal form is admissible only if it remains subordinate to the inherited stop-preserving boundary tag BDRY*, does not smooth away, postpone, or reinterpret the inherited earliest honest boundary stop, and does not relabel the boundary-side route as a horizon-continuation route.*

Proposition H4.181 (Every finite admissible conservative dispatch object yields exactly one conservative comparison/work program normal form)**.** *Assume the setting of Definition H4.180. Then:*

(i) every finite admissible conservative dispatch object $\mathsf{Disp}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S})$ *yields exactly one program normal form* $\mathsf{Prog}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S})$*;*

(ii) in the horizon-side case, the program normal form remains subordinate to the inherited horizon tag and inherited comparison/refinement/packaging organization;

(iii) in the boundary-side case, the program normal form remains subordinate to the inherited stop-preserving boundary tag and inherited comparison/profiling/packaging organization; and

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(iv) no admissible conservative program normal form may manufacture a fresh theorematic certificate, erase the inherited horizon/boundary distinction, or deny the inherited earliest-boundary stop.

Proof. Clause (i) is immediate from Definition H4.180: once the dispatch object is fixed by 19AZ$^{(6ay)}$, the present definition assigns exactly one corresponding compact program tuple. Clause (ii) follows because the horizon-side program normal form is defined only from the inherited horizon-side dispatch object together with the already admissible comparison, refinement-selection, and packaging tasks subordinate to the explicit horizon tag. Clause (iii) follows because the boundary-side program normal form is defined only from the inherited boundary-side dispatch object together with the already admissible comparison, profiling-selection, and packaging tasks subordinate to the explicit stop-preserving boundary tag. Clause (iv) is exactly the certificate-aware discipline inherited from 19AZ$^{(6ae)}$, 19AZ$^{(6af)}$, 19AZ$^{(6ai)}$, 19AZ$^{(6at)}$, 19AZ$^{(6au)}$, 19AZ$^{(6av)}$, 19AZ$^{(6aw)}$, 19AZ$^{(6ax)}$, and 19AZ$^{(6ay)}$, now expressed as the first compact OP-facing comparison/work program normal form. □

Corollary H4.182 (The present post-archival stage line already carries a first conservative comparison/work program normal form)**.** *For the first-stage word $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$, the current active line already carries a first conservative comparison/work program normal form*

$$\mathsf{Prog}_2^{\mathrm{cons}}(\mathbf{w}_3; \mathcal{S})$$

for every finite nonempty admissible request subfamily

$$\mathcal{S} \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3).$$

So the post-release continuation now carries not only a conservative ask/respond/aggregate/dispatch protocol, but also the first compact downstream-readable work/comparison normal form through which an admissible conservative dispatch object may be consumed without reopening the route decision itself.

Proof. Immediate from Corollary H4.178 and Proposition H4.181. □

Remark H4.183 (What the first conservative comparison/work program normal form now buys)**.** *This is the point where the post-release line first becomes able not only to dispatch conservative aggregates into explicit task families, but also to compress those task families into one compact OP-facing normal form. The horizon-side line can now hand downstream readers one finite comparison/refinement/packaging program under an explicit horizon tag, while the boundary-side line can now hand downstream readers one finite comparison/profiling/packaging program under an explicit stop-preserving boundary tag. Later OP-oriented work can therefore begin from a conservative ask/respond/aggregate/dispatch/program protocol instead of improvising both its compact work-program objects and its downstream comparison interface from scratch.*

Reader cue. Read the next block as the first conservative OP-facing program-to-execution / work-cycle layer extracted from the compact comparison/work program normal form of 19AZ$^{(6az)}$. Its purpose is to turn one downstream-readable conservative program object back into one disciplined, finite execution cycle without reopening the route decision, manufacturing a fresh theorematic certificate, erasing the inherited horizon/boundary distinction, or denying the inherited earliest honest boundary stop. The block still does not solve any open problem; it fixes the first controlled cycle in which the compact OP-facing program may actually be run.

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19AZ$^{(6ba)}$. Conservative program-to-execution / work-cycle layer

After 19AZ$^{(6az)}$, the post-release line already carries compact route-faithful conservative comparison/work program normal forms subordinate either to the inherited horizon tag or to the inherited stop-preserving boundary tag. The next honest step is therefore to define how such a compact program may be executed as a first finite conservative work cycle. On the horizon side, that cycle must remain subordinate to the inherited horizon tag and must execute only comparison, conservative refinement, and packaging moves already encoded in the compact program. On the boundary side, it must remain subordinate to the inherited stop-preserving boundary tag and must execute only comparison, conservative profiling, and packaging moves already encoded in the compact program. The point is not to add any new theorematic force, but only to turn the compact conservative program into a first controlled OP-facing work cycle that may later be iterated, compared, or packaged further.

Definition H4.184 (First conservative OP-facing program-to-execution / work-cycle layer)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$ *and a finite nonempty admissible request subfamily*

$$\mathcal{S} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w})$$

together with its compact conservative comparison/work program normal form

$$\mathsf{Prog}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S})$$

from Definition H4.180. Define the first conservative OP-facing work-cycle object

$$\mathsf{Cycle}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S})$$

by the following cases.

(i) If

$$\mathsf{Prog}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}) = \mathcal{P}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}),$$

define the first horizon-side conservative work cycle

$$\mathcal{C}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}) := \big(\mathsf{HOR}, \mathsf{TCMP}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}), \mathsf{TREF}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}), \mathsf{TPACK}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}), \mathsf{RET}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S})\big).$$

Here

$$\mathsf{RET}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}) := \textit{horizon-side finite return package.}$$

This work cycle is admissible only if every executed move remains subordinate to the inherited horizon tag HOR*, does not manufacture a fresh theorematic certificate, and does not relabel the horizon-side route as a boundary route. Define*

$$\mathsf{Cycle}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}) := \mathcal{C}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}).$$

(ii) If

$$\mathsf{Prog}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}) = \mathcal{P}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}),$$

define the first boundary-side conservative work cycle

$$\mathcal{C}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}) := \big(\mathsf{BDRY}, \mathsf{TCMP}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}), \mathsf{TPROF}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}), \mathsf{TPACK}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}), \mathsf{RET}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S})\big).$$

Here

$$\mathsf{RET}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}) := \textit{boundary-side finite return package.}$$

This work cycle is admissible only if every executed move remains subordinate to the inherited stop-preserving boundary tag BDRY*, does not smooth away, postpone, or reinterpret the inherited earliest honest boundary stop, and does not relabel the boundary-side route as a horizon-continuation route. Define*

$$\mathsf{Cycle}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}) := \mathcal{C}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}).$$

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Proposition H4.185 (Every finite admissible conservative comparison/work program normal form yields exactly one conservative work-cycle object)**.** *Assume the setting of Definition H4.184. Then:*

(i) every finite admissible conservative comparison/work program normal form $\mathsf{Prog}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S})$ *yields exactly one work-cycle object* $\mathsf{Cycle}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S})$*;*

(ii) in the horizon-side case, the work-cycle object remains subordinate to the inherited horizon tag and inherited comparison/refinement/packaging organization;

(iii) in the boundary-side case, the work-cycle object remains subordinate to the inherited stop-preserving boundary tag and inherited comparison/profiling/packaging organization; and

(iv) no admissible conservative work-cycle object may manufacture a fresh theorematic certificate, erase the inherited horizon/boundary distinction, or deny the inherited earliest-boundary stop.

Proof. Clause (i) is immediate from Definition H4.184: once the compact program normal form is fixed by $19\mathrm{AZ}^{(6az)}$, the present definition assigns exactly one corresponding finite work-cycle object. Clause (ii) follows because the horizon-side work cycle is defined only from the inherited horizon-side program normal form together with the already admitted comparison, refinement-selection, and packaging tasks subordinate to the explicit horizon tag. Clause (iii) follows because the boundary-side work cycle is defined only from the inherited boundary-side program normal form together with the already admitted comparison, profiling-selection, and packaging tasks subordinate to the explicit stop-preserving boundary tag. Clause (iv) is exactly the certificate-aware discipline inherited from $19\mathrm{AZ}^{(6ae)}$, $19\mathrm{AZ}^{(6af)}$, $19\mathrm{AZ}^{(6ai)}$, $19\mathrm{AZ}^{(6at)}$, $19\mathrm{AZ}^{(6au)}$, $19\mathrm{AZ}^{(6av)}$, $19\mathrm{AZ}^{(6aw)}$, $19\mathrm{AZ}^{(6ax)}$, $19\mathrm{AZ}^{(6ay)}$, and $19\mathrm{AZ}^{(6az)}$, now expressed as the first controlled OP-facing work cycle. □

Corollary H4.186 (The present post-archival stage line already carries a first conservative work cycle)**.** *For the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A},\mathsf{B}\}$*, the current active line already carries a first conservative work-cycle object*

$$\mathsf{Cycle}_2^{\mathrm{cons}}(\mathbf{w}_3;\mathcal{S})$$

for every finite nonempty admissible request subfamily

$$\mathcal{S} \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3).$$

So the post-release continuation now carries not only a conservative ask/respond/aggregate/dispatch/program protocol, but also the first controlled OP-facing execution cycle through which an admissible compact work/comparison program may be run without reopening the route decision itself.

Proof. Immediate from Corollary H4.182 and Proposition H4.185. □

Remark H4.187 (What the first conservative program-to-execution / work-cycle layer now buys)**.** *This is the point where the post-release line first becomes able not only to package conservative OP-facing task families into compact program objects, but also to run those compact programs as controlled finite work cycles. The horizon-side line can now execute one finite comparison/refinement/packaging cycle under an explicit horizon tag, while the boundary-side line can now execute one finite comparison/profiling/packaging cycle under an explicit stop-preserving boundary tag. Later OP-oriented work can therefore begin from a conservative ask/respond/aggregate/dispatch/program/cycle protocol instead of improvising both its execution loop and its return package from scratch.*

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Reader cue. Read the next block as the first conservative cycle-to-return / iteration interface extracted from the controlled work-cycle layer of 19AZ$^{(6ba)}$. Its purpose is to decide what a route-faithful return package may honestly do next: either prepare one further conservative cycle under the inherited route tag, or collapse into one compact conservative closing/comparison object without reopening the route decision, manufacturing a fresh theorematic certificate, erasing the inherited horizon/boundary distinction, or denying the inherited earliest honest boundary stop.

19AZ$^{(6bb)}$. Conservative cycle-to-return / iteration interface

After 19AZ$^{(6ba)}$, the post-release line already carries controlled finite conservative work-cycle objects subordinate either to the inherited horizon tag or to the inherited stop-preserving boundary tag. The next honest step is therefore to define how the return package produced by such a cycle may be consumed. On the horizon side, a route-faithful return package may only either seed one further conservative horizon-side cycle under the inherited horizon tag or be collapsed into one compact conservative comparison/closing object that still records the inherited route. On the boundary side, a stop-preserving return package may only either seed one further conservative boundary-side cycle under the inherited stop-preserving boundary tag or be collapsed into one compact conservative comparison/closing object that still records the inherited earliest honest boundary stop. The point is not to force infinite iteration, but to define the first disciplined interface through which the conservative work cycle either continues or closes.

Definition H4.188 (First conservative cycle-to-return / iteration interface)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, a finite nonempty admissible request subfamily*

$$\mathcal{S} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w})$$

and its controlled conservative work-cycle object

$$\mathsf{Cycle}_m^{\mathrm{cons}}(\mathbf{w}; \mathcal{S})$$

from Definition H4.184. Define the associated first conservative cycle-to-return / iteration object

$$\mathsf{Iter}_m^{\mathrm{cons}}(\mathbf{w}; \mathcal{S})$$

by the following cases.

(i) If

$$\mathsf{Cycle}_m^{\mathrm{cons}}(\mathbf{w}; \mathcal{S}) = \mathcal{C}_m^{\mathrm{hor}}(\mathbf{w}; \mathcal{S}),$$

define the first horizon-side conservative return/iteration object

$$\mathcal{I}_m^{\mathrm{hor}}(\mathbf{w}; \mathcal{S}) := \big(\mathsf{HOR}, \mathsf{RET}_m^{\mathrm{hor}}(\mathbf{w}; \mathcal{S}), \mathsf{NEXT}_m^{\mathrm{hor}}(\mathbf{w}; \mathcal{S}), \mathsf{CLOSE}_m^{\mathrm{hor}}(\mathbf{w}; \mathcal{S})\big).$$

Here

$$\mathsf{NEXT}_m^{\mathrm{hor}}(\mathbf{w}; \mathcal{S}) := \textit{next horizon-side conservative cycle option},$$
$$\mathsf{CLOSE}_m^{\mathrm{hor}}(\mathbf{w}; \mathcal{S}) := \textit{compact horizon-side closing option}.$$

This return/iteration object is admissible only if both options remain subordinate to the inherited horizon tag HOR*, do not manufacture a fresh theorematic certificate, and do not relabel the horizon-side route as a boundary route. Define*

$$\mathsf{Iter}_m^{\mathrm{cons}}(\mathbf{w}; \mathcal{S}) := \mathcal{I}_m^{\mathrm{hor}}(\mathbf{w}; \mathcal{S}).$$

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(ii) If

$$\mathsf{Cycle}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}) = \mathcal{C}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}),$$

define the first boundary-side conservative return/iteration object

$$\mathcal{I}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}) := \big(\mathsf{BDRY}, \mathsf{RET}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}), \mathsf{NEXT}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}), \mathsf{CLOSE}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S})\big).$$

Here

$$\mathsf{NEXT}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}) := \textit{next boundary-side conservative cycle option},$$
$$\mathsf{CLOSE}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}) := \textit{compact boundary-side closing option}.$$

This return/iteration object is admissible only if both options remain subordinate to the inherited stop-preserving boundary tag BDRY*, do not smooth away, postpone, or reinterpret the inherited earliest honest boundary stop, and do not relabel the boundary-side route as a horizon-continuation route. Define*

$$\mathsf{Iter}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}) := \mathcal{I}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}).$$

Proposition H4.189 (Every finite admissible conservative work cycle yields exactly one conservative return/iteration object)**.** *Assume the setting of Definition H4.188. Then:*

(i) every finite admissible conservative work-cycle object $\mathsf{Cycle}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S})$ *yields exactly one return/iteration object* $\mathsf{Iter}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S})$*;*

(ii) in the horizon-side case, the return/iteration object remains subordinate to the inherited horizon tag and records exactly the two admissible next uses of the return package: further conservative cycling or compact horizon-side conservative closing/comparison;

(iii) in the boundary-side case, the return/iteration object remains subordinate to the inherited stop-preserving boundary tag and records exactly the two admissible next uses of the return package: further conservative cycling or compact boundary-side conservative closing/comparison; and

(iv) no admissible conservative return/iteration object may manufacture a fresh theorematic certificate, erase the inherited horizon/boundary distinction, or deny the inherited earliest-boundary stop.

Proof. Clause (i) is immediate from Definition H4.188: once the conservative work-cycle object is fixed by $19\mathrm{AZ}^{(6ba)}$, the present definition assigns exactly one corresponding return/iteration object. Clause (ii) follows because the horizon-side return/iteration object is defined only from the inherited horizon-side work cycle together with the two explicitly admitted horizon-side uses of the return package, namely further conservative cycling or compact horizon-side closing/comparison. Clause (iii) follows because the boundary-side return/iteration object is defined only from the inherited boundary-side work cycle together with the two explicitly admitted boundary-side uses of the return package, namely further conservative cycling or compact boundary-side closing/comparison subordinate to the inherited earliest honest boundary stop. Clause (iv) is exactly the certificate-aware discipline inherited from $19\mathrm{AZ}^{(6ae)}$, $19\mathrm{AZ}^{(6af)}$, $19\mathrm{AZ}^{(6ai)}$, $19\mathrm{AZ}^{(6at)}$, $19\mathrm{AZ}^{(6au)}$, $19\mathrm{AZ}^{(6av)}$, $19\mathrm{AZ}^{(6aw)}$, $19\mathrm{AZ}^{(6ax)}$, $19\mathrm{AZ}^{(6ay)}$, $19\mathrm{AZ}^{(6az)}$, and $19\mathrm{AZ}^{(6ba)}$, now expressed as the first disciplined conservative return/iteration interface. □

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Corollary H4.190 (The present post-archival stage line already carries a first conservative return/iteration interface)**.** *For the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$*, the current active line already carries a first conservative return/iteration object*

$$\mathsf{Iter}_2^{\mathrm{cons}}(\mathbf{w}_3; \mathcal{S})$$

for every finite nonempty admissible request subfamily

$$\mathcal{S} \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3).$$

So the post-release continuation now carries not only a conservative ask/respond/aggregate/dispatch/program/cycle protocol, but also the first disciplined interface through which a finite conservative work cycle may either seed one further conservative cycle or collapse into one compact route-faithful conservative closing/comparison object without reopening the route decision itself.

Proof. Immediate from Corollary H4.186 and Proposition H4.189. □

Remark H4.191 (What the first conservative cycle-to-return / iteration interface now buys)**.** *This is the point where the post-release line first becomes able not only to run a compact conservative OP-facing program as a controlled finite work cycle, but also to state explicitly what the return package of that cycle is allowed to do next. The horizon-side line can now either seed one further horizon-side conservative cycle or collapse into one compact horizon-side conservative closing/comparison object under the inherited horizon tag, while the boundary-side line can now either seed one further boundary-side conservative cycle or collapse into one compact boundary-side conservative closing/comparison object under the inherited stop-preserving boundary tag. Later OP-oriented work can therefore begin from a conservative ask/respond/aggregate/dispatch/program/cycle/return protocol instead of improvising both its iteration interface and its compact closing interface from scratch.*

Reader cue. Read the next block as the first compact conservative closing/comparison form extracted from the return/iteration interface of $19\mathrm{AZ}^{(6bb)}$. Its purpose is to package the honest non-iterative exit of a controlled conservative work cycle into one downstream-readable closing object that still records the inherited route tag, does not reopen the route decision, does not manufacture a fresh theorematic certificate, and does not smooth away the inherited earliest honest boundary stop.

$19\mathrm{AZ}^{(6bc)}$. First compact conservative closing/comparison form extracted from the return/iteration interface

After $19\mathrm{AZ}^{(6bb)}$, every controlled conservative work cycle already carries a disciplined return/iteration object with exactly two admitted next uses of the return package: one further conservative cycle, or one compact route-faithful closing/comparison object. The present block isolates the second option as a compact object in its own right. On the horizon side, this yields a compact horizon-side conservative closing/comparison form subordinate to the inherited horizon tag. On the boundary side, it yields a compact boundary-side conservative closing/comparison form subordinate to the inherited stop-preserving boundary tag and earliest honest boundary stop. The point is not to terminate the whole post-release line, but to define the first disciplined compact conservative exit object of the current ask/respond/aggregate/dispatch/program/cycle/return staircase.

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Definition H4.192 (First compact conservative closing/comparison form)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, a finite nonempty admissible request subfamily*

$$\mathcal{S} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w})$$

and its conservative return/iteration object

$$\mathsf{Iter}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S})$$

from Definition H4.188. Define the associated first compact conservative closing/comparison object

$$\mathsf{CloseCmp}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S})$$

by the following cases.

(i) If

$$\mathsf{Iter}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}) = \mathcal{I}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}),$$

define the first horizon-side compact conservative closing/comparison object

$$\mathcal{K}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}) := \big(\mathsf{HOR}, \mathsf{RET}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}), \mathsf{CLOSE}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S})\big).$$

This object is admissible only if it remains subordinate to the inherited horizon tag, records the compact horizon-side closing/comparison option exactly as provided by 19AZ$^{(6bb)}$*, does not relabel the route as boundary-side, and does not manufacture a fresh theorematic certificate. Define*

$$\mathsf{CloseCmp}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}) := \mathcal{K}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}).$$

(ii) If

$$\mathsf{Iter}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}) = \mathcal{I}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}),$$

define the first boundary-side compact conservative closing/comparison object

$$\mathcal{K}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}) := \big(\mathsf{BDRY}, \mathsf{RET}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}), \mathsf{CLOSE}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S})\big).$$

This object is admissible only if it remains subordinate to the inherited stop-preserving boundary tag, records the compact boundary-side closing/comparison option exactly as provided by 19AZ$^{(6bb)}$*, does not smooth away or postpone the inherited earliest honest boundary stop, and does not relabel the route as horizon-side. Define*

$$\mathsf{CloseCmp}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}) := \mathcal{K}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}).$$

Proposition H4.193 (Every finite admissible conservative return/iteration object yields exactly one compact conservative closing/comparison object)**.** *Assume the setting of Definition H4.192. Then:*

(i) every finite admissible conservative return/iteration object $\mathsf{Iter}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S})$ *yields exactly one compact conservative closing/comparison object* $\mathsf{CloseCmp}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S})$*;*

(ii) in the horizon-side case, the compact conservative closing/comparison object remains subordinate to the inherited horizon tag and records exactly the compact horizon-side closing/comparison option carried by the return package;

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(iii) *in the boundary-side case, the compact conservative closing/comparison object remains subordinate to the inherited stop-preserving boundary tag and records exactly the compact boundary-side closing/comparison option carried by the return package; and*

(iv) *no admissible compact conservative closing/comparison object may manufacture a fresh theorematic certificate, erase the inherited horizon/boundary distinction, or deny the inherited earliest-boundary stop.*

Proof. Clause (i) is immediate from Definition H4.192: once the conservative return/iteration object is fixed by $19\mathrm{AZ}^{(6bb)}$, the present definition assigns exactly one corresponding compact conservative closing/comparison object. Clauses (ii) and (iii) follow because the horizon-side and boundary-side compact objects are defined only from the inherited return package together with the corresponding compact closing/comparison option already recorded in the return/iteration object. Clause (iv) is exactly the certificate-aware discipline inherited from $19\mathrm{AZ}^{(6ae)}$ through $19\mathrm{AZ}^{(6bb)}$, now expressed as the first disciplined compact conservative closing/comparison interface. □

Corollary H4.194 (The present post-archival stage line already carries a first compact conservative closing/comparison form)**.** *For the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$*, the current active line already carries a first compact conservative closing/comparison object*

$$\mathsf{CloseCmp}_2^{\mathrm{cons}}(\mathbf{w}_3; \mathcal{S})$$

for every finite nonempty admissible request subfamily

$$\mathcal{S} \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3).$$

So the post-release continuation now carries not only a conservative ask/respond/aggregate/dispatch/program/cycle/return protocol, but also the first disciplined compact conservative closing/comparison form to which that protocol may honestly collapse without reopening the route decision itself.

Proof. Immediate from Corollary H4.190 and Proposition H4.193. □

Remark H4.195 (What the first compact conservative closing/comparison form now buys)**.** *This is the point where the post-release line first becomes able not only to iterate a conservative work cycle, but also to exit one such cycle in a compact downstream-readable form that still remembers the inherited route. The horizon-side line can now close one controlled conservative cycle into a compact horizon-side comparison object under the inherited horizon tag, while the boundary-side line can now close one controlled conservative cycle into a compact boundary-side comparison object under the inherited stop-preserving boundary tag and earliest honest boundary stop. Later OP-oriented work can therefore begin from a conservative ask/respond/aggregate/dispatch/program/cycle/return/close protocol instead of improvising its compact conservative closing interface from scratch.*

Reader cue. Read the next block as the first admissible next-cycle promotion rule extracted from the same return/iteration interface of $19\mathrm{AZ}^{(6bb)}$. Its purpose is to state when the *iterative* branch of the return package may honestly be promoted to one further conservative work cycle under the inherited route tag, rather than merely being mentioned abstractly as a possibility.

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19AZ$^{(6bd)}$. First admissible next-cycle promotion rule extracted from the return/iteration interface

After 19AZ$^{(6bb)}$, every controlled conservative work cycle already carries a return package with an explicitly recorded next-cycle option. The present block upgrades that option from a mere named possibility to a disciplined promotion rule. On the horizon side, a next-cycle promotion is admissible only if one chooses a new finite nonempty admissible request subfamily that remains subordinate to the inherited horizon tag and does not relabel the route. On the boundary side, a next-cycle promotion is admissible only if one chooses a new finite nonempty admissible request subfamily that remains subordinate to the inherited stop-preserving boundary tag and does not smooth away or postpone the inherited earliest honest boundary stop. The point is to define the first honest rule under which NEXT may seed one further conservative cycle rather than being left as an unspecified placeholder.

Definition H4.196 (First admissible next-cycle promotion rule)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, a finite nonempty admissible request subfamily*

$$\mathcal{S} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}),$$

its conservative return/iteration object

$$\mathsf{Iter}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}),$$

and a second finite nonempty admissible request subfamily

$$\mathcal{S}^+ \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}).$$

Define the associated first admissible next-cycle promotion object

$$\mathsf{Prom}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S},\mathcal{S}^+)$$

by the following cases.

(i) If

$$\mathsf{Iter}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}) = \mathcal{I}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S})$$

and

$$\mathcal{S}^+ \subseteq \mathcal{Q}_m^{\mathrm{hor}}(\mathbf{w}),$$

define the first horizon-side admissible next-cycle promotion object

$$\Pi_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S},\mathcal{S}^+) := \big(\mathsf{HOR}, \mathsf{NEXT}_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S}), \mathcal{S}^+, \mathsf{Cycle}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}^+)\big).$$

This promotion object is admissible only if the promoted cycle remains subordinate to the inherited horizon tag, does not manufacture a fresh theorematic certificate, and does not relabel the horizon-side route as a boundary-side route. Define

$$\mathsf{Prom}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S},\mathcal{S}^+) := \Pi_m^{\mathrm{hor}}(\mathbf{w};\mathcal{S},\mathcal{S}^+).$$

(ii) If

$$\mathsf{Iter}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}) = \mathcal{I}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S})$$

and

$$\mathcal{S}^+ \subseteq \mathcal{Q}_m^{\mathrm{bdry}}(\mathbf{w}),$$

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define the first boundary-side admissible next-cycle promotion object

$$\Pi_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S},\mathcal{S}^+) := \big(\mathsf{BDRY},\mathsf{NEXT}_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S}),\mathcal{S}^+,\mathsf{Cycle}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}^+)\big).$$

This promotion object is admissible only if the promoted cycle remains subordinate to the inherited stop-preserving boundary tag, does not smooth away or postpone the inherited earliest honest boundary stop, does not manufacture a fresh theorematic certificate, and does not relabel the boundary-side route as a horizon-side route. Define

$$\mathsf{Prom}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S},\mathcal{S}^+) := \Pi_m^{\mathrm{bdry}}(\mathbf{w};\mathcal{S},\mathcal{S}^+).$$

Proposition H4.197 (Every admissible next-cycle promotion object seeds exactly one further conservative work cycle under the inherited route tag)**.** *Assume the setting of Definition H4.196. Then:*

(i) every admissible promotion object $\mathsf{Prom}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S},\mathcal{S}^+)$ *determines exactly one further conservative work cycle* $\mathsf{Cycle}_m^{\mathrm{cons}}(\mathbf{w};\mathcal{S}^+)$*;*

(ii) in the horizon-side case, the promoted cycle remains subordinate to the inherited horizon tag;

(iii) in the boundary-side case, the promoted cycle remains subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop; and

(iv) no admissible next-cycle promotion may manufacture a fresh theorematic certificate, erase the inherited horizon/boundary distinction, or deny the inherited earliest-boundary stop.

Proof. Clause (i) is immediate from Definition H4.196: once the return/iteration object and the new admissible request family $\mathcal{S}^+$ are fixed, the present definition assigns exactly one further conservative work cycle. Clauses (ii) and (iii) follow because the promotion rule requires the new work cycle to remain subordinate to the inherited horizon or boundary tag respectively. Clause (iv) is exactly the certificate-aware discipline inherited from $19\mathrm{AZ}^{(6ae)}$ through $19\mathrm{AZ}^{(6bb)}$, now expressed as the first disciplined admissible next-cycle promotion rule. □

Corollary H4.198 (The present post-archival stage line already carries a first admissible next-cycle promotion rule)**.** *For the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A},\mathsf{B}\}$*, the current active line already carries a first admissible next-cycle promotion object*

$$\mathsf{Prom}_2^{\mathrm{cons}}(\mathbf{w}_3;\mathcal{S},\mathcal{S}^+)$$

for every pair of finite nonempty admissible request subfamilies

$$\mathcal{S},\mathcal{S}^+ \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3)$$

that remain subordinate to the same inherited route tag. So the post-release continuation now carries not only a conservative ask/respond/aggregate/dispatch/program/cycle/return/close protocol, but also the first disciplined rule under which one such return package may honestly seed one further conservative cycle.

Proof. Immediate from Corollary H4.190 and Proposition H4.197. □

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Remark H4.199 (What the first admissible next-cycle promotion rule now buys)**.** *This is the point where the post-release line first becomes able not only to mention iterative continuation as an option, but also to promote it in a disciplined way to one further conservative cycle under the inherited route tag. The horizon-side line can now honestly re-enter a further conservative comparison/refinement/packaging cycle under the inherited horizon tag, while the boundary-side line can now honestly re-enter a further conservative comparison/profiling/packaging cycle under the inherited stop-preserving boundary tag and earliest honest boundary stop. Later OP-oriented work can therefore begin from a conservative ask/respond/aggregate/dispatch/program/cycle/return/close/promote protocol instead of improvising its next-cycle promotion rule from scratch.*

Reader cue. Read the next block as the first branch-resolution rule extracted from the newly available *close/promote* split of $19\mathrm{AZ}^{(6bc)}$–$19\mathrm{AZ}^{(6bd)}$. Its purpose is to decide, in one route-faithful normalized step, when a conservative return object must honestly collapse into the compact closing/comparison form and when it may honestly be promoted into one further conservative cycle under the same inherited route tag.

$19\mathrm{AZ}^{(6be)}$. First conservative branch-resolution rule extracted from the close/promote protocol

After $19\mathrm{AZ}^{(6bc)}$ and $19\mathrm{AZ}^{(6bd)}$, every controlled conservative work cycle already carries two disciplined continuations: an honest compact closing/comparison exit and an honest next-cycle promotion rule. The present block resolves these two possibilities into one normalized branch-resolution interface. On the horizon side, the rule says that a return object either closes into the compact horizon-side comparison form or promotes to one further horizon-side conservative cycle, but may not do both at once and may not leave the inherited horizon tag. On the boundary side, the rule says that a return object either closes into the compact boundary-side comparison form or promotes to one further boundary-side conservative cycle, but may not do both at once and may not smooth away, postpone, or deny the inherited earliest honest boundary stop. The point is to replace the loose phrase "close or continue" by one first route-faithful branch-resolution object.

Definition H4.200 (First conservative branch-resolution rule)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, a finite nonempty admissible request subfamily*

$$\mathcal{S} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}),$$

its conservative return/iteration object

$$\mathsf{Iter}_m^{\mathrm{cons}}(\mathbf{w}; \mathcal{S}),$$

and choose a branch marker

$$\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}.$$

When $\mathfrak{b} = \mathsf{PR}$*, also fix a second finite nonempty admissible request subfamily*

$$\mathcal{S}^{+} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}).$$

Define the associated first conservative branch-resolution object

$$\mathsf{Branch}_m^{\mathrm{cons}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^{+})$$

by the following cases.

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(i) If

$$\mathfrak{b} = \mathsf{CL},$$

define

$$\mathsf{Branch}^{\mathrm{cons}}_{m}(\mathbf{w}; \mathcal{S}, \mathsf{CL}) := \mathsf{CloseCmp}^{\mathrm{cons}}_{m}(\mathbf{w}; \mathcal{S}).$$

This branch is admissible only if it remains subordinate to the inherited route tag recorded by $\mathsf{Iter}^{\mathrm{cons}}_{m}(\mathbf{w}; \mathcal{S})$*, does not manufacture a fresh theorematic certificate, and does not erase the horizon/boundary distinction or the inherited earliest-boundary stop.*

(ii) If

$$\mathfrak{b} = \mathsf{PR},$$

and $\mathcal{S}^{+}$ *is fixed as above, define*

$$\mathsf{Branch}^{\mathrm{cons}}_{m}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^{+}) := \mathsf{Prom}^{\mathrm{cons}}_{m}(\mathbf{w}; \mathcal{S}, \mathcal{S}^{+}).$$

This branch is admissible only if the promoted cycle remains subordinate to the same inherited route tag, does not manufacture a fresh theorematic certificate, and, in the boundary-side case, does not smooth away, postpone, or deny the inherited earliest honest boundary stop.

Proposition H4.201 (Every conservative return object admits exactly one honest branch choice at a time: close or promote)**.** *Assume the setting of Definition H4.200. Then:*

(i) every admissible branch-resolution object is either a compact conservative closing/comparison object or an admissible next-cycle promotion object;

(ii) no admissible branch-resolution object may simultaneously act as both a closing object and a promotion object for the same fixed branch marker;

(iii) in the horizon-side case, every admissible branch-resolution object remains subordinate to the inherited horizon tag;

(iv) in the boundary-side case, every admissible branch-resolution object remains subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop; and

(v) no admissible branch-resolution object may manufacture a fresh theorematic certificate, erase the horizon/boundary distinction, or deny the inherited earliest-boundary stop.

Proof. Clause (i) is immediate from Definition H4.200: the branch marker $\mathfrak{b}$ selects exactly one of the two previously disciplined objects, namely $\mathsf{CloseCmp}^{\mathrm{cons}}_{m}(\mathbf{w}; \mathcal{S})$ or $\mathsf{Prom}^{\mathrm{cons}}_{m}(\mathbf{w}; \mathcal{S}, \mathcal{S}^{+})$. Clause (ii) follows because the branch marker is fixed to one of the two values CL or PR, so the same admissible branch-resolution object cannot honestly instantiate both branches at once. Clauses (iii) and (iv) follow from $19\mathrm{AZ}^{(6bc)}$ and $19\mathrm{AZ}^{(6bd)}$, which already require both the compact closing branch and the promotion branch to remain subordinate to the inherited route tag, and in the boundary-side case to preserve the inherited earliest honest boundary stop. Clause (v) is exactly the certificate-aware route discipline inherited from $19\mathrm{AZ}^{(6ae)}$ through $19\mathrm{AZ}^{(6bd)}$, now compressed into one first branch-resolution rule. □

Corollary H4.202 (The present post-archival stage line already carries a first conservative branch-resolution rule)**.** *For the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$*, the current active line already carries a first conservative branch-resolution object*

$$\mathsf{Branch}^{\mathrm{cons}}_{2}(\mathbf{w}_3; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^{+})$$

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for every finite nonempty admissible request subfamily

$$\mathcal{S} \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3),$$

every branch marker $\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}$*, and in the promotion case every second finite nonempty admissible request subfamily*

$$\mathcal{S}^+ \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3)$$

that remains subordinate to the same inherited route tag. So the post-release continuation now carries not only a conservative ask/respond/aggregate/dispatch/program/cycle/return/-close/promote protocol, but also the first disciplined rule that resolves each honest conservative return package into one route-faithful close-or-promote branch.

Proof. Immediate from Corollary H4.194, Corollary H4.198, and Proposition H4.201. □

Remark H4.203 (What the first conservative branch-resolution rule now buys)**.** *This is the point where the post-release line first becomes able not only to name compact conservative closing and disciplined next-cycle promotion separately, but also to resolve them into one honest branch interface. The horizon-side line can now decide, under the inherited horizon tag, whether a controlled conservative return object should close into a compact comparison object or honestly promote to one further conservative cycle. The boundary-side line can now make the same decision under the inherited stop-preserving boundary tag and earliest honest boundary stop without smoothing boundary information away. Later OP-oriented work can therefore begin from a conservative ask/respond/aggregate/dispatch/program/cycle/return/close/promote/branch protocol instead of improvising the close-versus-continue split from scratch.*

Reader cue. Read the next block as the first downstream branch-consumption layer extracted from the new close/promote branch-resolution rule of 19AZ$^{(6be)}$. Its purpose is to let later OP-oriented work consume a resolved conservative branch not as a loose slogan but as one controlled downstream follow-on object that keeps the inherited route tag, preserves the horizon/boundary distinction, and does not smooth away the inherited earliest honest boundary stop.

19AZ$^{(6bf)}$. First downstream branch-consumption layer extracted from the conservative branch-resolution rule

After 19AZ$^{(6be)}$, every honest conservative return package already resolves into exactly one route-faithful branch: either a compact conservative closing/comparison object or an admissible next-cycle promotion object. The present block turns that resolved branch into one first downstream branch-consumption object. On the horizon side, downstream work may consume the branch only under the inherited horizon tag; in particular, a closing branch remains a closing-side downstream object and a promoted branch remains a next-cycle downstream object. On the boundary side, downstream work may consume the branch only under the inherited stop-preserving boundary tag and inherited earliest honest boundary stop; in particular, no downstream consumer may flatten the close/promote difference into one undifferentiated continuation object. The point is to make the first post-branch downstream intake explicit.

Definition H4.204 (First downstream branch-consumption object)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, a finite nonempty admissible request subfamily*

$$\mathcal{S} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}),$$

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a branch marker $\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}$*, and an admissible conservative branch-resolution object*

$$\mathsf{Branch}_m^{\mathrm{cons}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+).$$

When $\mathfrak{b} = \mathsf{PR}$*, also keep fixed the second finite nonempty admissible request subfamily*

$$\mathcal{S}^+ \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}).$$

Define the associated first downstream branch-consumption object

$$\mathsf{Consm}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+)$$

by the following cases.

(i) If $\mathfrak{b} = \mathsf{CL}$*, define*

$$\mathsf{Consm}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathsf{CL}) := \big(\mathsf{Branch}_m^{\mathrm{cons}}(\mathbf{w}; \mathcal{S}, \mathsf{CL}), \mathsf{DCLOSE}_m(\mathbf{w}; \mathcal{S})\big).$$

This downstream object is admissible only if it remains subordinate to the same inherited route tag and, in the boundary-side case, preserves the inherited earliest honest boundary stop.

(ii) If $\mathfrak{b} = \mathsf{PR}$*, define*

$$\mathsf{Consm}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^+) := \big(\mathsf{Branch}_m^{\mathrm{cons}}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^+), \mathsf{DNEXT}_m(\mathbf{w}; \mathcal{S}, \mathcal{S}^+)\big).$$

This downstream object is admissible only if it remains subordinate to the same inherited route tag, does not manufacture a fresh theorematic certificate, and, in the boundary-side case, does not smooth away, postpone, or deny the inherited earliest honest boundary stop.

Proposition H4.205 (Every honest branch-resolution object yields exactly one route-faithful downstream branch-consumption object)**.** *Assume the setting of Definition H4.204. Then:*

(i) every admissible close branch yields exactly one downstream close-consumption object;

(ii) every admissible promote branch yields exactly one downstream next-cycle consumption object;

(iii) in the horizon-side case, every admissible downstream branch-consumption object remains subordinate to the inherited horizon tag;

(iv) in the boundary-side case, every admissible downstream branch-consumption object remains subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop; and

(v) no admissible downstream branch-consumption object may manufacture a fresh theorematic certificate, erase the horizon/boundary distinction, or collapse the close/promote difference into one undifferentiated downstream continuation.

Proof. Clauses (i) and (ii) are immediate from Definition H4.228: once the branch marker is fixed, the downstream consumer is defined by exactly one of the two displayed clauses. Clauses (iii) and (iv) follow because the downstream branch-consumption object is defined directly from a branch-resolution object that is already subordinate to the inherited route tag, and in the boundary-side case to the inherited earliest honest boundary stop, by Proposition H4.201. Clause (v) is exactly the inherited certificate-aware route discipline from $19\mathrm{AZ}^{(6ae)}$ through $19\mathrm{AZ}^{(6be)}$, now made explicit at the first downstream post-branch intake layer. □

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Corollary H4.206 (The present post-archival stage line already carries a first downstream branch-consumption layer)**.** *For the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$*, the current active line already carries a first downstream branch-consumption object*

$$\mathsf{Consm}_2^{\mathrm{branch}}(\mathbf{w}_3; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+)$$

for every finite nonempty admissible request subfamily

$$\mathcal{S} \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3),$$

every branch marker $\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}$*, and in the promotion case every second finite nonempty admissible request subfamily*

$$\mathcal{S}^+ \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3)$$

that remains subordinate to the same inherited route tag. So the post-release continuation now carries not only a conservative ask/respond/aggregate/dispatch/program/cycle/return/-close/promote/branch protocol, but also the first disciplined downstream layer that consumes each resolved branch as one controlled close-side or next-cycle-side follow-on object.

Proof. Immediate from Corollary H4.202 and Proposition H4.205. □

Remark H4.207 (What the first downstream branch-consumption layer now buys)**.** *This is the point where the post-release line first becomes able not only to resolve a conservative return object into a close-or-promote branch, but also to pass that resolved branch downstream in a disciplined way. The horizon-side line can now hand a resolved close branch onward as one explicit closing-side downstream object or hand a resolved promote branch onward as one explicit next-cycle-side downstream object under the inherited horizon tag. The boundary-side line can now do the same under the inherited stop-preserving boundary tag and earliest honest boundary stop, without collapsing the two possibilities into one vague continuation gesture. Later OP-oriented work can therefore begin from a conservative ask/respond/aggregate/dispatch/program/cycle/return/close/promote/branch/consume protocol instead of improvising how resolved branches are actually consumed downstream.*

oindent extbfReader cue. Read the next block as the first downstream branch-to-task layer extracted from the new downstream branch-consumption interface of $19\mathrm{AZ}^{(6bf)}$. Its purpose is to turn each already consumed close-or-promote branch into one controlled finite downstream task family, so that later OP-oriented work does not merely receive a branch-side follow-on object, but receives it as a route-faithful downstream work program that preserves the inherited horizon/boundary distinction and does not smooth away the inherited earliest honest boundary stop.

$19\mathrm{AZ}^{(6bg)}$. First downstream branch-to-task layer extracted from the downstream branch-consumption layer

After $19\mathrm{AZ}^{(6bf)}$, every honest conservative close-or-promote branch already admits one controlled downstream consumption object. The present block turns that consumed branch into one finite downstream task family. On the close side, the downstream tasks are comparison, closing-side packaging, and honest archival/output stabilization under the inherited route tag. On the promote side, the downstream tasks are next-cycle comparison, request-family preparation, and next-cycle packaging under the same inherited route tag. On the boundary side, every such task family must remain subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop. The point is to make the first post-consumption downstream work program explicit.

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Definition H4.208 (First downstream branch-to-task object)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, a finite nonempty admissible request subfamily*

$$\mathcal{S} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}),$$

a branch marker $\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}$*, and an admissible downstream branch-consumption object*

$$\mathsf{Consm}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+).$$

When $\mathfrak{b} = \mathsf{PR}$*, also keep fixed the second finite nonempty admissible request subfamily*

$$\mathcal{S}^+ \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}).$$

Define the associated first downstream branch-to-task object

$$\mathsf{Task}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+)$$

by the following cases.

(oman) If* $\mathfrak{b} = \mathsf{CL}$*, define*

$$\mathsf{Task}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathsf{CL}) := \big(\mathsf{Consm}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathsf{CL}), \mathcal{T}_m^{\mathrm{close}}(\mathbf{w}; \mathcal{S})\big),$$

where

$$\mathcal{T}_m^{\mathrm{close}}(\mathbf{w}; \mathcal{S}) := \{\mathsf{TCMP}_m^{\mathrm{close}}, \mathsf{TSTAB}_m^{\mathrm{close}}, \mathsf{TPACK}_m^{\mathrm{close}}\}.$$

This downstream task family is admissible only if it remains subordinate to the same inherited route tag and, in the boundary-side case, preserves the inherited earliest honest boundary stop.

(oman) If* $\mathfrak{b} = \mathsf{PR}$*, define*

$$\mathsf{Task}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^+) := \big(\mathsf{Consm}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^+), \mathcal{T}_m^{\mathrm{next}}(\mathbf{w}; \mathcal{S}, \mathcal{S}^+)\big),$$

where

$$\mathcal{T}_m^{\mathrm{next}}(\mathbf{w}; \mathcal{S}, \mathcal{S}^+) := \{\mathsf{TCMP}_m^{\mathrm{next}}, \mathsf{TPREP}_m^{\mathrm{next}}, \mathsf{TPACK}_m^{\mathrm{next}}\}.$$

This downstream task family is admissible only if it remains subordinate to the same inherited route tag, does not manufacture a fresh theorematic certificate, and, in the boundary-side case, does not smooth away, postpone, or deny the inherited earliest honest boundary stop.

Proposition H4.209 (Every consumed branch yields exactly one route-faithful downstream task family)**.** *Assume the setting of Definition efdef:first-downstream-branch-to-task-object. Then:*

(oman) every admissible downstream close-consumption object yields exactly one downstream close-task family;*

(oman) every admissible downstream next-cycle consumption object yields exactly one downstream next-cycle task family;*

(oman) in the horizon-side case, every admissible downstream branch-to-task object remains subordinate to the inherited horizon tag;*

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(oman) in the boundary-side case, every admissible downstream branch-to-task object remains subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop; and*

(oman) no admissible downstream branch-to-task object may manufacture a fresh theorematic certificate, erase the horizon/boundary distinction, or collapse the close/promote difference into one undifferentiated downstream work program.*

Proof. Clauses (i) and (ii) are immediate from Definition efdef:first-downstream-branch-to-task-object: once the branch marker is fixed, the downstream task family is defined by exactly one of the two displayed clauses. Clauses (iii) and (iv) follow because the downstream branch-to-task object is defined directly from a downstream branch-consumption object that is already subordinate to the inherited route tag, and in the boundary-side case to the inherited earliest honest boundary stop, by Proposition efprop:every-honest-branch-resolution-object-yields-exactly-one-route-faithful-downstream-branch-consumption-object. Clause (v) is exactly the inherited certificate-aware route discipline from $19\mathrm{AZ}^{(6ae)}$ through $19\mathrm{AZ}^{(6bf)}$, now made explicit at the first downstream post-consumption task layer. □

Corollary H4.210 (The present post-archival stage line already carries a first downstream branch-to-task layer)**.** *For the first-stage word $\mathbf{w}_3 \in \{\mathsf{A},\mathsf{B}\}$, the current active line already carries a first downstream branch-to-task object*

$$\mathsf{Task}_2^{\mathrm{branch}}(\mathbf{w}_3;\mathcal{S},\mathfrak{b} \mid \mathcal{S}^+)$$

for every finite nonempty admissible request subfamily

$$\mathcal{S} \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3),$$

every branch marker $\mathfrak{b} \in \{\mathsf{CL},\mathsf{PR}\}$, and in the promotion case every second finite nonempty admissible request subfamily

$$\mathcal{S}^+ \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3)$$

that remains subordinate to the same inherited route tag. So the post-release continuation now carries not only a conservative ask/respond/aggregate/dispatch/program/cycle/return/-close/promote/branch/consume protocol, but also the first disciplined downstream task layer that translates each consumed branch into one controlled close-side or next-cycle-side work family.

Proof. Immediate from Corollary efcor:the-present-post-v3290-line-already-carries-a-first-downstream-branch-consumption-layer and Proposition efprop:every-consumed-branch-yields-exactly-one-route-faithful-downstream-task-family. □

Remark H4.211 (What the first downstream branch-to-task layer now buys)**.** *This is the point where the post-release line first becomes able not only to consume a resolved conservative branch downstream, but also to translate that consumed branch into one explicit downstream work family. The horizon-side line can now hand a consumed close branch onward as one closing-side task family or hand a consumed promote branch onward as one next-cycle task family under the inherited horizon tag. The boundary-side line can now do the same under the inherited stop-preserving boundary tag and earliest honest boundary stop, without collapsing the two possibilities into one vague continuation gesture. Later OP-oriented work can therefore begin from a conservative ask/respond/aggregate/dispatch/program/cycle/return/-close/promote/branch/consume/task protocol instead of improvising how consumed branches are translated into downstream work.*

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Reader cue. Read the next block as the first downstream task-to-program layer extracted from the downstream branch-to-task interface of 19AZ$^{(6bg)}$. Its purpose is to compress each already translated close-side or next-cycle-side downstream task family into one compact downstream-readable program object, so that later OP-oriented work does not merely receive a task bundle, but receives it in a finite normal form that remains faithful to the inherited horizon/boundary distinction and the inherited earliest honest boundary stop.

19AZ$^{(6bh)}$. First downstream task-to-program layer extracted from the downstream branch-to-task layer

After 19AZ$^{(6bg)}$, every honest consumed close-or-promote branch already yields one controlled downstream task family. The present block compresses that task family into one compact downstream program object. On the close side, the program records a finite closing-side comparison/stabilization/packaging normal form under the inherited route tag. On the promote side, the program records a finite next-cycle comparison/preparation/packaging normal form under the same inherited route tag. On the boundary side, every such program must remain subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop. The point is to make the first post-task downstream program normal form explicit.

Definition H4.212 (First downstream task-to-program object)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, a finite nonempty admissible request subfamily*

$$\mathcal{S} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}),$$

a branch marker $\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}$*, and an admissible downstream branch-to-task object*

$$\mathsf{Task}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+).$$

When $\mathfrak{b} = \mathsf{PR}$*, also keep fixed the second finite nonempty admissible request subfamily*

$$\mathcal{S}^+ \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}).$$

Define the associated first downstream task-to-program object

$$\mathsf{Prog}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+)$$

by the following cases.

(i) If $\mathfrak{b} = \mathsf{CL}$*, define*

$$\mathsf{Prog}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathsf{CL}) := \big(\mathsf{Task}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathsf{CL}), \mathcal{P}_m^{\mathrm{close}}(\mathbf{w}; \mathcal{S})\big),$$

where

$$\mathcal{P}_m^{\mathrm{close}}(\mathbf{w}; \mathcal{S}) := \big(\mathsf{CL}, \mathsf{TCMP}_m^{\mathrm{close}}, \mathsf{TSTAB}_m^{\mathrm{close}}, \mathsf{TPACK}_m^{\mathrm{close}}\big).$$

This downstream program is admissible only if it remains subordinate to the same inherited route tag and, in the boundary-side case, preserves the inherited earliest honest boundary stop.

(ii) If $\mathfrak{b} = \mathsf{PR}$*, define*

$$\mathsf{Prog}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^+) := \big(\mathsf{Task}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^+), \mathcal{P}_m^{\mathrm{next}}(\mathbf{w}; \mathcal{S}, \mathcal{S}^+)\big),$$

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where

$$\mathcal{P}_m^{\mathrm{next}}(\mathbf{w};\mathcal{S},\mathcal{S}^+) := \big(\mathsf{PR},\mathsf{TCMP}_m^{\mathrm{next}},\mathsf{TPREP}_m^{\mathrm{next}},\mathsf{TPACK}_m^{\mathrm{next}}\big).$$

This downstream program is admissible only if it remains subordinate to the same inherited route tag, does not manufacture a fresh theorematic certificate, and, in the boundary-side case, does not smooth away, postpone, or deny the inherited earliest honest boundary stop.

Proposition H4.213 (Every downstream task family yields exactly one compact downstream program normal form)**.** *Assume the setting of Definition H4.212. Then:*

(i) every admissible downstream close-task family yields exactly one downstream close-program normal form;

(ii) every admissible downstream next-cycle task family yields exactly one downstream next-cycle program normal form;

(iii) in the horizon-side case, every admissible downstream task-to-program object remains subordinate to the inherited horizon tag;

(iv) in the boundary-side case, every admissible downstream task-to-program object remains subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop; and

(v) no admissible downstream task-to-program object may manufacture a fresh theorematic certificate, erase the horizon/boundary distinction, or collapse the close/next-cycle difference into one undifferentiated downstream program normal form.

Proof. Clauses (i) and (ii) are immediate from Definition H4.212: once the branch marker is fixed, the downstream program normal form is defined by exactly one of the two displayed clauses. Clauses (iii) and (iv) follow because the downstream task-to-program object is defined directly from a downstream branch-to-task object that is already subordinate to the inherited route tag, and in the boundary-side case to the inherited earliest honest boundary stop, by Proposition H4.209. Clause (v) is exactly the inherited certificate-aware route discipline from $19\mathrm{AZ}^{(6ae)}$ through $19\mathrm{AZ}^{(6bg)}$, now made explicit at the first downstream post-task program layer. □

Corollary H4.214 (The present post-archival stage line already carries a first downstream task-to-program layer)**.** *For the first-stage word $\mathbf{w}_3 \in \{\mathsf{A},\mathsf{B}\}$, the current active line already carries a first downstream task-to-program object*

$$\mathsf{Prog}_2^{\mathrm{branch}}(\mathbf{w}_3;\mathcal{S},\mathfrak{b} \mid \mathcal{S}^+)$$

for every finite nonempty admissible request subfamily

$$\mathcal{S} \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3),$$

every branch marker $\mathfrak{b} \in \{\mathsf{CL},\mathsf{PR}\}$, and in the promotion case every second finite nonempty admissible request subfamily

$$\mathcal{S}^+ \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3)$$

that remains subordinate to the same inherited route tag. So the post-release continuation now carries not only a conservative ask/respond/aggregate/dispatch/program/cycle/return/-close/promote/branch/consume/task protocol, but also the first disciplined downstream program layer that compresses each translated close-side or next-cycle-side task family into one controlled normal form.

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Proof. Immediate from Corollary H4.210 and Proposition H4.213. □

Remark H4.215 (What the first downstream task-to-program layer now buys)**.** *This is the point where the post-release line first becomes able not only to translate a consumed conservative branch into one explicit downstream task family, but also to compress that task family into one compact downstream-readable program object. The horizon-side line can now hand a closing-side task family onward as one explicit close-program normal form or hand a next-cycle-side task family onward as one explicit next-program normal form under the inherited horizon tag. The boundary-side line can now do the same under the inherited stop-preserving boundary tag and earliest honest boundary stop, without collapsing the two possibilities into one vague continuation gesture. Later OP-oriented work can therefore begin from a conservative ask/respond/aggregate/dispatch/program/cycle/return/close/promote/branch/consume/task/program protocol instead of improvising how translated downstream tasks are normalized.*

Reader cue. Read the next block as the first downstream program-to-cycle layer extracted from the downstream task-to-program interface of $19AZ^{(6bh)}$. Its purpose is to turn each already normalized close-side or next-cycle-side downstream program object into one controlled downstream cycle object, so that later OP-oriented work receives not only a compact downstream program, but the first honest finite follow-on cycle that can be executed under the inherited horizon/boundary distinction and the inherited earliest honest boundary stop.

$19AZ^{(6bi)}$. First downstream program-to-cycle layer extracted from the downstream task-to-program layer

After $19AZ^{(6bh)}$, every honest consumed close-or-promote branch already yields one compact downstream program normal form. The present block turns that compact program object into one controlled downstream cycle object. On the close side, the cycle records one finite closing-side comparison/stabilization/packaging cycle under the inherited route tag. On the promote side, the cycle records one finite next-cycle comparison/preparation/packaging cycle that remains subordinate to the same inherited route tag. On the boundary side, every such downstream cycle must remain subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop. The point is to make the first post-program downstream cycle layer explicit.

Definition H4.216 (First downstream program-to-cycle object)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, a finite nonempty admissible request subfamily*

$$\mathcal{S} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}),$$

a branch marker $\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}$*, and an admissible downstream task-to-program object*

$$\mathsf{Prog}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+).$$

When $\mathfrak{b} = \mathsf{PR}$*, also keep fixed the second finite nonempty admissible request subfamily*

$$\mathcal{S}^+ \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}).$$

Define the associated first downstream program-to-cycle object

$$\mathsf{Cycle}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+)$$

by the following cases.

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(i) If $\mathfrak{b} = \mathsf{CL}$*, define*

$$\mathsf{Cycle}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathsf{CL}) := \big(\mathsf{Prog}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathsf{CL}), \mathcal{C}_m^{\mathrm{close}}(\mathbf{w}; \mathcal{S})\big),$$

where

$$\mathcal{C}_m^{\mathrm{close}}(\mathbf{w}; \mathcal{S}) := \big(\mathsf{CL}, \mathsf{TCMP}_m^{\mathrm{close}}, \mathsf{TSTAB}_m^{\mathrm{close}}, \mathsf{TPACK}_m^{\mathrm{close}}, \mathsf{RET}_m^{\mathrm{close}}\big).$$

This downstream cycle is admissible only if it remains subordinate to the same inherited route tag and, in the boundary-side case, preserves the inherited earliest honest boundary stop.

(ii) If $\mathfrak{b} = \mathsf{PR}$*, define*

$$\mathsf{Cycle}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^+) := \big(\mathsf{Prog}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^+), \mathcal{C}_m^{\mathrm{next}}(\mathbf{w}; \mathcal{S}, \mathcal{S}^+)\big),$$

where

$$\mathcal{C}_m^{\mathrm{next}}(\mathbf{w}; \mathcal{S}, \mathcal{S}^+) := \big(\mathsf{PR}, \mathsf{TCMP}_m^{\mathrm{next}}, \mathsf{TPREP}_m^{\mathrm{next}}, \mathsf{TPACK}_m^{\mathrm{next}}, \mathsf{RET}_m^{\mathrm{next}}\big).$$

This downstream cycle is admissible only if it remains subordinate to the same inherited route tag, does not manufacture a fresh theorematic certificate, and, in the boundary-side case, does not smooth away, postpone, or deny the inherited earliest honest boundary stop.

Proposition H4.217 (Every downstream program normal form yields exactly one controlled downstream cycle form)**.** *Assume the setting of Definition H4.216. Then:*

(i) every admissible downstream close-program normal form yields exactly one downstream close-cycle form;

(ii) every admissible downstream next-cycle program normal form yields exactly one downstream next-cycle cycle form;

(iii) in the horizon-side case, every admissible downstream program-to-cycle object remains subordinate to the inherited horizon tag;

(iv) in the boundary-side case, every admissible downstream program-to-cycle object remains subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop; and

(v) no admissible downstream program-to-cycle object may manufacture a fresh theorematic certificate, erase the horizon/boundary distinction, or collapse the close/next-cycle difference into one undifferentiated downstream cycle form.

Proof. Clauses (i) and (ii) are immediate from Definition H4.216: once the branch marker is fixed, the downstream cycle form is defined by exactly one of the two displayed clauses. Clauses (iii) and (iv) follow because the downstream program-to-cycle object is defined directly from a downstream task-to-program object that is already subordinate to the inherited route tag, and in the boundary-side case to the inherited earliest honest boundary stop, by Proposition H4.213. Clause (v) is exactly the inherited certificate-aware route discipline from $19\mathrm{AZ}^{(6ae)}$ through $19\mathrm{AZ}^{(6bh)}$, now made explicit at the first downstream post-program cycle layer. □

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Corollary H4.218 (The present post-archival stage line already carries a first downstream program-to-cycle layer)**.** *For the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$*, the current active line already carries a first downstream program-to-cycle object*

$$\mathsf{Cycle}_2^{\mathrm{branch}}(\mathbf{w}_3; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+)$$

for every finite nonempty admissible request subfamily

$$\mathcal{S} \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3),$$

every branch marker $\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}$*, and in the promotion case every second finite nonempty admissible request subfamily*

$$\mathcal{S}^+ \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3)$$

that remains subordinate to the same inherited route tag. So the post-release continuation now carries not only a conservative ask/respond/aggregate/dispatch/program/cycle/return/-close/promote/branch/consume/task/program protocol, but also the first disciplined downstream cycle layer that turns each normalized close-side or next-cycle-side downstream program object into one controlled follow-on cycle.

Proof. Immediate from Corollary H4.214 and Proposition H4.217. □

Remark H4.219 (What the first downstream program-to-cycle layer now buys)**.** *This is the point where the post-release line first becomes able not only to normalize a translated downstream close-side or next-cycle-side task family into one compact program object, but also to turn that compact program into one explicit controlled downstream cycle. The horizon-side line can now hand a close-program normal form onward as one closing-side downstream cycle or hand a next-program normal form onward as one next-cycle downstream cycle under the inherited horizon tag. The boundary-side line can now do the same under the inherited stop-preserving boundary tag and earliest honest boundary stop, without collapsing the two possibilities into one vague continuation gesture. Later OP-oriented work can therefore begin from a conservative ask/respond/aggregate/dispatch/program/cycle/return/-close/promote/branch/consume/task/program/cycle protocol instead of improvising how normalized downstream programs are executed.*

Reader cue. Read the next block as the first downstream cycle-to-return / close-or-promote interface extracted from the downstream program-to-cycle layer of $19\mathrm{AZ}^{(6bi)}$. Its purpose is to turn each already normalized close-side or next-cycle-side downstream cycle object into one honest downstream return object that either records a compact close-side exit or records an admissible next-cycle continuation under the inherited horizon/boundary distinction and inherited earliest honest boundary stop.

$19\mathrm{AZ}^{(6bj)}$. Downstream cycle-to-return / close-or-promote interface

After $19\mathrm{AZ}^{(6bi)}$, every honest consumed close-or-promote branch already yields one controlled downstream cycle object. The present block turns that downstream cycle object into one first downstream return / close-or-promote interface. On the close side, the interface records one finite closing-side return package together with one compact closing-side downstream exit object. On the promote side, the interface records one finite next-cycle return package together with one admissible next-cycle continuation object that remains subordinate to the same inherited route tag. On the boundary side, every such downstream interface must remain subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop. The point is to make the first post-cycle downstream return discipline explicit.

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Definition H4.220 (First downstream cycle-to-return / close-or-promote interface object)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, a finite nonempty admissible request subfamily*

$$\mathcal{S} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}),$$

a branch marker $\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}$*, and an admissible downstream program-to-cycle object*

$$\mathsf{Cycle}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+).$$

When $\mathfrak{b} = \mathsf{PR}$*, also keep fixed the second finite nonempty admissible request subfamily*

$$\mathcal{S}^+ \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}).$$

Define the associated first downstream cycle-to-return / close-or-promote interface object

$$\mathsf{Iter}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+)$$

by the following cases.

(i) If $\mathfrak{b} = \mathsf{CL}$*, define*

$$\mathsf{Iter}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathsf{CL}) := \big(\mathsf{Cycle}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathsf{CL}), \mathsf{RET}_m^{\mathrm{close}}, \mathsf{Close}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S})\big),$$

where $\mathsf{Close}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S})$ *denotes the compact honest downstream closing object induced by the close-side cycle. This interface is admissible only if it remains subordinate to the same inherited route tag and, in the boundary-side case, preserves the inherited earliest honest boundary stop.*

(ii) If $\mathfrak{b} = \mathsf{PR}$*, define*

$$\mathsf{Iter}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^+) := \big(\mathsf{Cycle}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^+), \mathsf{RET}_m^{\mathrm{next}}, \mathsf{Next}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathcal{S}^+)\big),$$

where $\mathsf{Next}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathcal{S}^+)$ *denotes the admissible downstream next-cycle continuation object induced by the next-side cycle. This interface is admissible only if it remains subordinate to the same inherited route tag, does not manufacture a fresh theorematic certificate, and, in the boundary-side case, does not smooth away, postpone, or deny the inherited earliest honest boundary stop.*

Proposition H4.221 (Every downstream cycle object yields exactly one downstream return / close-or-promote interface object)**.** *Assume the setting of Definition H4.220. Then:*

(i) every admissible downstream close-cycle object yields exactly one downstream close-return / closing interface object;

(ii) every admissible downstream next-cycle cycle object yields exactly one downstream next-return / continuation interface object;

(iii) in the horizon-side case, every admissible downstream cycle-to-return interface object remains subordinate to the inherited horizon tag;

(iv) in the boundary-side case, every admissible downstream cycle-to-return interface object remains subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop; and

(v) no admissible downstream cycle-to-return interface object may manufacture a fresh theorematic certificate, erase the horizon/boundary distinction, or collapse the close/next-cycle difference into one undifferentiated downstream return object.

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Proof. Clauses (i) and (ii) are immediate from Definition H4.220: once the branch marker is fixed, the downstream return / close-or-promote interface object is defined by exactly one of the two displayed clauses. Clauses (iii) and (iv) follow because the downstream cycle-to-return interface object is defined directly from a downstream program-to-cycle object that is already subordinate to the inherited route tag, and in the boundary-side case to the inherited earliest honest boundary stop, by Proposition H4.217. Clause (v) is exactly the inherited certificate-aware route discipline from $19\mathrm{AZ}^{(6ae)}$ through $19\mathrm{AZ}^{(6bi)}$, now made explicit at the first downstream post-cycle return layer. □

Corollary H4.222 (The present post-archival stage line already carries a first downstream cycle-to-return interface)**.** *For the first-stage word $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$, the current active line already carries a first downstream cycle-to-return / close-or-promote interface object*

$$\mathsf{Iter}_2^{\mathrm{branch}}(\mathbf{w}_3; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+)$$

for every finite nonempty admissible request subfamily

$$\mathcal{S} \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3),$$

every branch marker $\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}$, and in the promotion case every second finite nonempty admissible request subfamily

$$\mathcal{S}^+ \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3)$$

that remains subordinate to the same inherited route tag. So the post-release continuation now carries not only a conservative ask/respond/aggregate/dispatch/program/cycle/return/-close/promote/branch/consume/task/program/cycle protocol, but also the first disciplined downstream cycle-to-return interface that turns each normalized close-side or next-cycle-side downstream cycle object into one honest downstream return object.

Proof. Immediate from Corollary H4.218 and Proposition H4.221. □

Remark H4.223 (What the first downstream cycle-to-return interface now buys)**.** *This is the point where the post-release line first becomes able not only to execute a normalized downstream close-side or next-cycle-side program as one explicit downstream cycle, but also to turn that cycle into one disciplined downstream return object that records either an honest close-side exit or an honest next-cycle continuation. The horizon-side line can now hand a close-cycle onward as one compact closing-side downstream return object or hand a next-cycle onward as one admissible downstream continuation object under the inherited horizon tag. The boundary-side line can now do the same under the inherited stop-preserving boundary tag and earliest honest boundary stop, without collapsing the two possibilities into one vague continuation gesture. Later OP-oriented work can therefore begin from a conservative ask/respond/aggregate/dispatch/program/cycle/return/close/promote/branch/consume/task/program/cycle/return protocol instead of improvising how normalized downstream cycles return.*

Reader cue. Read the next block as the first downstream close/promote branch-resolution rule extracted from the downstream cycle-to-return interface of $19\mathrm{AZ}^{(6bj)}$. Its purpose is to resolve each already normalized downstream return object into exactly one honest close-side or next-cycle branch on the consumed follow-on level, so that later OP-oriented work does not merely receive a downstream return package, but receives it as a route-faithful downstream branch choice that preserves the inherited horizon/boundary distinction and the inherited earliest honest boundary stop.

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19AZ$^{(6bk)}$. First downstream close/promote branch-resolution rule extracted from the downstream cycle-to-return interface

After 19AZ$^{(6bj)}$, every honest normalized close-side or next-cycle-side downstream cycle already yields one disciplined downstream return object. The present block resolves that downstream return object into exactly one honest branch. On the close side, the downstream branch is the compact close-side exit already recorded by the return interface. On the next-cycle side, the downstream branch is the admissible next-cycle continuation already recorded by the same interface. On the boundary side, every such downstream branch must remain subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop. The point is to make the first post-return downstream branch discipline explicit.

Definition H4.224 (First downstream close/promote branch-resolution object)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, a finite nonempty admissible request subfamily*

$$\mathcal{S} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}),$$

a branch marker $\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}$*, and an admissible downstream cycle-to-return interface object*

$$\mathsf{Iter}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+).$$

When $\mathfrak{b} = \mathsf{PR}$*, also keep fixed the second finite nonempty admissible request subfamily*

$$\mathcal{S}^+ \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}).$$

Define the associated first downstream close/promote branch-resolution object

$$\mathsf{Branch}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+)$$

by the following cases.

(i) If $\mathfrak{b} = \mathsf{CL}$*, define*

$$\mathsf{Branch}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{CL}) := \mathsf{Close}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}).$$

This branch-resolution object is admissible only if it remains subordinate to the same inherited route tag and, in the boundary-side case, preserves the inherited earliest honest boundary stop.

(ii) If $\mathfrak{b} = \mathsf{PR}$*, define*

$$\mathsf{Branch}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^+) := \mathsf{Next}_m^{\mathrm{branch}}(\mathbf{w}; \mathcal{S}, \mathcal{S}^+).$$

This branch-resolution object is admissible only if it remains subordinate to the same inherited route tag, does not manufacture a fresh theorematic certificate, and, in the boundary-side case, does not smooth away, postpone, or deny the inherited earliest honest boundary stop.

Proposition H4.225 (Every downstream return interface object admits exactly one honest downstream close-or-promote branch choice)**.** *Assume the setting of Definition H4.224. Then:*

(i) every admissible downstream close-return interface object yields exactly one downstream close branch-resolution object;

(ii) every admissible downstream next-return interface object yields exactly one downstream next-cycle branch-resolution object;

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(iii) *in the horizon-side case, every admissible downstream branch-resolution object remains subordinate to the inherited horizon tag;*

(iv) *in the boundary-side case, every admissible downstream branch-resolution object remains subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop; and*

(v) *no admissible downstream branch-resolution object may manufacture a fresh theorematic certificate, erase the horizon/boundary distinction, or collapse the close/next-cycle difference into one undifferentiated downstream branch object.*

Proof. Clauses (i) and (ii) are immediate from Definition H4.224: once the branch marker is fixed, the downstream branch-resolution object is defined by exactly one of the two displayed clauses. Clauses (iii) and (iv) follow because the downstream branch-resolution object is defined directly from a downstream cycle-to-return interface object that is already subordinate to the inherited route tag, and in the boundary-side case to the inherited earliest honest boundary stop, by Proposition H4.221. Clause (v) is exactly the inherited certificate-aware route discipline from $19\mathrm{AZ}^{(6ae)}$ through $19\mathrm{AZ}^{(6bj)}$, now made explicit at the first downstream post-return branch layer. □

Corollary H4.226 (The present post-archival stage line already carries a first downstream close/promote branch-resolution rule)**.** *For the first-stage word $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$, the current active line already carries a first downstream close/promote branch-resolution object*

$$\mathsf{Branch}_2^{\mathrm{down}}(\mathbf{w}_3; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+)$$

for every finite nonempty admissible request subfamily

$$\mathcal{S} \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3),$$

every branch marker $\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}$, and in the promotion case every second finite nonempty admissible request subfamily

$$\mathcal{S}^+ \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3)$$

that remains subordinate to the same inherited route tag. So the post-release continuation now carries not only a conservative ask/respond/aggregate/dispatch/program/cycle/return/close/promote/branch/consume/task/program/cycle/return protocol, but also the first disciplined downstream close/promote branch-resolution rule that turns each normalized downstream return object into one honest downstream branch object.

Proof. Immediate from Corollary H4.222 and Proposition H4.225. □

Remark H4.227 (What the first downstream close/promote branch-resolution rule now buys)**.** *This is the point where the post-release line first becomes able not only to execute a normalized downstream close-side or next-cycle-side program as one explicit downstream cycle and to turn that cycle into one disciplined downstream return object, but also to resolve that return object into exactly one honest downstream branch. The horizon-side line can now hand a downstream close-return onward as one compact close-side downstream branch or hand a downstream next-return onward as one admissible next-cycle downstream branch under the inherited horizon tag. The boundary-side line can now do the same under the inherited stop-preserving boundary tag and earliest honest boundary stop, without collapsing the two possibilities into one vague continuation gesture. Later OP-oriented work can therefore begin from a conservative ask/respond/aggregate/dispatch/program/cycle/return/close/promote/branch/consume/task/program/cycle/return/branch protocol instead of improvising how normalized downstream returns branch.*

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Reader cue. Read the next block as the first downstream branch-consumption layer extracted from the downstream close/promote branch-resolution rule of 19AZ$^{(6bk)}$. Its purpose is to turn each already resolved downstream close-side or next-cycle branch into one controlled downstream follow-on object, so that later OP-oriented work does not merely receive a downstream branch choice, but receives it in a finite route-faithful consumption form that preserves the inherited horizon/boundary distinction and the inherited earliest honest boundary stop.

19AZ$^{(6bl)}$. First downstream branch-consumption layer extracted from the downstream close/promote branch-resolution rule

After 19AZ$^{(6bk)}$, every honest normalized downstream return object already resolves into exactly one downstream close-side branch or one admissible downstream next-cycle branch. The present block turns that resolved downstream branch into one controlled downstream follow-on object. On the close side, the downstream follow-on object records one honest consumed close-side continuation under the inherited route tag. On the next-cycle side, it records one honest consumed next-cycle continuation under the same inherited route tag. On the boundary side, every such downstream consumption object must remain subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop. The point is to make the first post-branch downstream consumption discipline explicit.

Definition H4.228 (First downstream branch-consumption object)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, a finite nonempty admissible request subfamily*

$$\mathcal{S} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}),$$

a branch marker $\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}$*, and an admissible downstream branch-resolution object*

$$\mathsf{Branch}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+).$$

When $\mathfrak{b} = \mathsf{PR}$*, also keep fixed the second finite nonempty admissible request subfamily*

$$\mathcal{S}^+ \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}).$$

Define the associated first downstream branch-consumption object

$$\mathsf{Consm}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+)$$

by the following cases.

(i) If $\mathfrak{b} = \mathsf{CL}$*, define*

$$\mathsf{Consm}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{CL}) := \big(\mathsf{Branch}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{CL}), \mathsf{DCLOSE}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S})\big).$$

This downstream consumption object is admissible only if it remains subordinate to the same inherited route tag and, in the boundary-side case, preserves the inherited earliest honest boundary stop.

(ii) If $\mathfrak{b} = \mathsf{PR}$*, define*

$$\mathsf{Consm}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^+) := \big(\mathsf{Branch}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^+), \mathsf{DNEXT}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathcal{S}^+)\big).$$

This downstream consumption object is admissible only if it remains subordinate to the same inherited route tag, does not manufacture a fresh theorematic certificate, and, in the boundary-side case, does not smooth away, postpone, or deny the inherited earliest honest boundary stop.

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Proposition H4.229 (Every resolved downstream branch yields exactly one route-faithful downstream consumption object)**.** *Assume the setting of Definition H4.228. Then:*

(i) every admissible resolved downstream close branch yields exactly one downstream close-side consumption object;

(ii) every admissible resolved downstream next-cycle branch yields exactly one downstream next-side consumption object;

(iii) in the horizon-side case, every admissible downstream consumption object remains subordinate to the inherited horizon tag;

(iv) in the boundary-side case, every admissible downstream consumption object remains subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop; and

(v) no admissible downstream consumption object may manufacture a fresh theorematic certificate, erase the horizon/boundary distinction, or collapse the close/next-cycle difference into one undifferentiated downstream follow-on object.

Proof. Clauses (i) and (ii) are immediate from Definition H4.204: once the branch marker is fixed, the downstream consumption object is defined by exactly one of the two displayed clauses. Clauses (iii) and (iv) follow because the downstream branch-consumption object is defined directly from a downstream branch-resolution object that is already subordinate to the inherited route tag, and in the boundary-side case to the inherited earliest honest boundary stop, by Proposition H4.225. Clause (v) is exactly the inherited certificate-aware route discipline from $19\mathrm{AZ}^{(6ae)}$ through $19\mathrm{AZ}^{(6bk)}$, now made explicit at the first downstream post-branch consumption layer. □

Corollary H4.230 (The present post-archival stage line already carries a first downstream branch-consumption layer)**.** *For the first-stage word $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$, the current active line already carries a first downstream branch-consumption object*

$$\mathsf{Consm}_2^{\mathrm{down}}(\mathbf{w}_3; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+)$$

for every finite nonempty admissible request subfamily

$$\mathcal{S} \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3),$$

every branch marker $\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}$, and in the promotion case every second finite nonempty admissible request subfamily

$$\mathcal{S}^+ \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3)$$

that remains subordinate to the same inherited route tag. So the post-release continuation now carries not only a conservative ask/respond/aggregate/dispatch/program/cycle/return/-close/promote/branch/consume/task/program/cycle/return/branch protocol, but also the first disciplined downstream branch-consumption layer that turns each normalized downstream branch object into one honest downstream follow-on object.

Proof. Immediate from Corollary H4.226 and Proposition H4.229. □

Remark H4.231 (What the first downstream branch-consumption layer now buys)**.** *This is the point where the post-release line first becomes able not only to execute a normalized downstream close-side or next-cycle-side program as one explicit downstream cycle, to turn*

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that cycle into one disciplined downstream return object, and to resolve that return object into exactly one honest downstream branch, but also to consume that resolved downstream branch into one controlled downstream follow-on object. The horizon-side line can now hand a downstream close branch onward as one compact close-side consumption object or hand a downstream next-cycle branch onward as one admissible next-cycle consumption object under the inherited horizon tag. The boundary-side line can now do the same under the inherited stop-preserving boundary tag and earliest honest boundary stop, without collapsing the two possibilities into one vague downstream continuation gesture. Later OP-oriented work can therefore begin from a conservative ask/respond/aggregate/dispatch/program/cycle/return/close/promote/branch/consume/task/program/cycle/return/branch/consume protocol instead of improvising how normalized downstream branches are consumed.

Reader cue. Read the next block as the first downstream branch-to-task layer extracted from the downstream branch-consumption interface of $19\mathrm{AZ}^{(6bl)}$. Its purpose is to turn each already consumed downstream close-side or next-cycle branch into one controlled finite downstream task family, so that later OP-oriented work does not merely receive a downstream follow-on object, but receives it as a route-faithful downstream work menu that preserves the inherited horizon/boundary distinction and does not smooth away the inherited earliest honest boundary stop.

$19\mathrm{AZ}^{(6bm)}$. First downstream branch-to-task layer extracted from the downstream branch-consumption layer

After $19\mathrm{AZ}^{(6bl)}$, every honest downstream close-side or next-cycle branch already admits one controlled downstream consumption object. The present block turns that consumed downstream branch into one finite downstream task family. On the close side, the downstream tasks are comparison, closing-side stabilization, and compact packaging under the inherited route tag. On the next-cycle side, the downstream tasks are comparison, next-cycle preparation, and compact packaging under the same inherited route tag. On the boundary side, every such task family must remain subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop. The point is to make the second post-consumption downstream work program explicit.

Definition H4.232 (First downstream branch-to-task object on the consumed follow-on level)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, a finite nonempty admissible request subfamily*

$$\mathcal{S} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}),$$

a branch marker $\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}$*, and an admissible downstream branch-consumption object*

$$\mathsf{Consm}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+).$$

When $\mathfrak{b} = \mathsf{PR}$*, also keep fixed the second finite nonempty admissible request subfamily*

$$\mathcal{S}^+ \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}).$$

Define the associated first downstream branch-to-task object on the consumed follow-on level

$$\mathsf{Task}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+)$$

by the following cases.

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(i) If $\mathfrak{b} = \mathsf{CL}$*, define*

$$\mathsf{Task}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{CL}) := \big(\mathsf{Consm}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{CL}), \mathcal{T}_m^{\mathrm{close,down}}(\mathbf{w}; \mathcal{S})\big),$$

where

$$\mathcal{T}_m^{\mathrm{close,down}}(\mathbf{w}; \mathcal{S}) := \{\mathsf{TCMP}_m^{\mathrm{close,down}}, \mathsf{TSTAB}_m^{\mathrm{close,down}}, \mathsf{TPACK}_m^{\mathrm{close,down}}\}.$$

This downstream task family is admissible only if it remains subordinate to the same inherited route tag and, in the boundary-side case, preserves the inherited earliest honest boundary stop.

(ii) If $\mathfrak{b} = \mathsf{PR}$*, define*

$$\mathsf{Task}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^+) := \big(\mathsf{Consm}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^+), \mathcal{T}_m^{\mathrm{next,down}}(\mathbf{w}; \mathcal{S}, \mathcal{S}^+)\big),$$

where

$$\mathcal{T}_m^{\mathrm{next,down}}(\mathbf{w}; \mathcal{S}, \mathcal{S}^+) := \{\mathsf{TCMP}_m^{\mathrm{next,down}}, \mathsf{TPREP}_m^{\mathrm{next,down}}, \mathsf{TPACK}_m^{\mathrm{next,down}}\}.$$

This downstream task family is admissible only if it remains subordinate to the same inherited route tag, does not manufacture a fresh theorematic certificate, and, in the boundary-side case, does not smooth away, postpone, or deny the inherited earliest honest boundary stop.

Proposition H4.233 (Every consumed downstream branch yields exactly one route-faithful downstream task family)**.** *Assume the setting of Definition H4.232. Then:*

(i) every admissible downstream close-consumption object yields exactly one downstream close-side task family;

(ii) every admissible downstream next-cycle consumption object yields exactly one downstream next-cycle task family;

(iii) in the horizon-side case, every admissible downstream branch-to-task object remains subordinate to the inherited horizon tag;

(iv) in the boundary-side case, every admissible downstream branch-to-task object remains subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop; and

(v) no admissible downstream branch-to-task object may manufacture a fresh theorematic certificate, erase the horizon/boundary distinction, or collapse the close/promote difference into one undifferentiated downstream work program.

Proof. Clauses (i) and (ii) are immediate from Definition H4.232: once the branch marker is fixed, the downstream task family is defined by exactly one of the two displayed clauses. Clauses (iii) and (iv) follow because the downstream branch-to-task object is defined directly from a downstream branch-consumption object that is already subordinate to the inherited route tag and, in the boundary-side case, to the inherited earliest honest boundary stop, by Proposition H4.229. Clause (v) is exactly the inherited certificate-aware route discipline from $19\mathrm{AZ}^{(6ae)}$ through $19\mathrm{AZ}^{(6bl)}$, now made explicit at the second downstream post-consumption task layer. □

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Corollary H4.234 (The present post-archival stage line already carries a first downstream branch-to-task layer on the consumed follow-on level)**.** *For the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$*, the current active line already carries a first downstream branch-to-task object*

$$\mathsf{Task}_2^{\mathrm{down}}(\mathbf{w}_3; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+)$$

for every finite nonempty admissible request subfamily

$$\mathcal{S} \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3),$$

every branch marker $\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}$*, and in the promotion case every second finite nonempty admissible request subfamily*

$$\mathcal{S}^+ \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3)$$

that remains subordinate to the same inherited route tag. So the post-release continuation now carries not only a conservative ask/respond/aggregate/dispatch/program/cycle/return/close/promote/branch/consume/task/program/cycle/return/branch/consume protocol, but also the first disciplined downstream task layer that translates each consumed downstream branch into one controlled close-side or next-cycle-side work family.

Proof. Immediate from Corollary H4.230 and Proposition H4.233. □

Remark H4.235 (What the first downstream branch-to-task layer on the consumed follow-on level now buys)**.** *This is the point where the post-release line first becomes able not only to consume a resolved downstream close-side or next-cycle branch, but also to translate that consumed downstream branch into one explicit downstream work family on the follow-on level. The horizon-side line can now hand a consumed downstream close branch onward as one closing-side task family or hand a consumed downstream promote branch onward as one next-cycle task family under the inherited horizon tag. The boundary-side line can now do the same under the inherited stop-preserving boundary tag and earliest honest boundary stop, without collapsing the two possibilities into one vague continuation gesture. Later OP-oriented work can therefore begin from a conservative ask/respond/aggregate/dispatch/program/cycle/return/close/promote/branch/consume/task/program/cycle/return/branch/consume/task protocol instead of improvising how consumed downstream branches are translated into follow-on work.*

Reader cue. Read the next block as the first downstream task-to-program layer extracted from the downstream branch-to-task layer of $19\mathrm{AZ}^{(6bm)}$. Its purpose is to compress each already translated downstream close-side or next-cycle-side task family into one compact downstream-readable program object, so that the consumed follow-on level does not remain a loose task list but becomes one normalized downstream program form under the inherited horizon tag or inherited stop-preserving boundary tag.

$19\mathrm{AZ}^{(6bn)}$. First downstream task-to-program layer extracted from the downstream branch-to-task layer on the consumed follow-on level

After $19\mathrm{AZ}^{(6bm)}$, every consumed downstream close-side or next-cycle branch already yields one explicit downstream task family. The present block compresses that task family into one compact downstream program object. On the close side, the downstream program is the normalized close-side work form built from comparison, stabilization, and compact packaging under the inherited route tag. On the next-cycle side, the downstream program is the normalized next-cycle work form built from comparison, preparation, and compact packaging under

the same inherited route tag. On the boundary side, every such program must remain subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop. The point is to make the next compact downstream work normal form explicit.

Definition H4.236 (First downstream task-to-program object on the consumed follow-on level)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, a finite nonempty admissible request subfamily*

$$\mathcal{S} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}),$$

a branch marker $\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}$*, and an admissible downstream branch-to-task object*

$$\mathsf{Task}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+).$$

When $\mathfrak{b} = \mathsf{PR}$*, also keep fixed the second finite nonempty admissible request subfamily*

$$\mathcal{S}^+ \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}).$$

Define the associated first downstream task-to-program object on the consumed follow-on level

$$\mathsf{Prog}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+)$$

by the following cases.

(i) If $\mathfrak{b} = \mathsf{CL}$*, define*

$$\mathsf{Prog}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{CL}) := \big(\mathsf{Task}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{CL}), \mathcal{P}_m^{\mathrm{close,down}}(\mathbf{w}; \mathcal{S})\big),$$

where

$$\mathcal{P}_m^{\mathrm{close,down}}(\mathbf{w}; \mathcal{S}) := \big(\mathsf{CL}, \mathsf{TCMP}_m^{\mathrm{close,down}}, \mathsf{TSTAB}_m^{\mathrm{close,down}}, \mathsf{TPACK}_m^{\mathrm{close,down}}\big).$$

This downstream program object is admissible only if it remains subordinate to the same inherited route tag and, in the boundary-side case, preserves the inherited earliest honest boundary stop.

(ii) If $\mathfrak{b} = \mathsf{PR}$*, define*

$$\mathsf{Prog}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^+) := \big(\mathsf{Task}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^+), \mathcal{P}_m^{\mathrm{next,down}}(\mathbf{w}; \mathcal{S}, \mathcal{S}^+)\big),$$

where

$$\mathcal{P}_m^{\mathrm{next,down}}(\mathbf{w}; \mathcal{S}, \mathcal{S}^+) := \big(\mathsf{PR}, \mathsf{TCMP}_m^{\mathrm{next,down}}, \mathsf{TPREP}_m^{\mathrm{next,down}}, \mathsf{TPACK}_m^{\mathrm{next,down}}\big).$$

This downstream program object is admissible only if it remains subordinate to the same inherited route tag, does not manufacture a fresh theorematic certificate, and, in the boundary-side case, does not smooth away, postpone, or deny the inherited earliest honest boundary stop.

Proposition H4.237 (Every consumed downstream task family yields exactly one compact downstream program object)**.** *Assume the setting of Definition H4.236. Then:*

(i) every admissible downstream close-side task family yields exactly one downstream close-side program object;

(ii) every admissible downstream next-cycle task family yields exactly one downstream next-cycle program object;

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(iii) *in the horizon-side case, every admissible downstream task-to-program object remains subordinate to the inherited horizon tag;*

(iv) *in the boundary-side case, every admissible downstream task-to-program object remains subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop; and*

(v) *no admissible downstream task-to-program object may manufacture a fresh theorematic certificate, erase the horizon/boundary distinction, or collapse the close/next-cycle difference into one undifferentiated downstream work normal form.*

Proof. Clauses (i) and (ii) are immediate from Definition H4.236: once the branch marker is fixed, the downstream program object is defined by exactly one of the two displayed clauses. Clauses (iii) and (iv) follow because the downstream task-to-program object is defined directly from a downstream branch-to-task object that is already subordinate to the inherited route tag and, in the boundary-side case, to the inherited earliest honest boundary stop, by Proposition H4.233. Clause (v) is exactly the inherited certificate-aware route discipline from $19\mathrm{AZ}^{(6ae)}$ through $19\mathrm{AZ}^{(6bm)}$, now made explicit at the second downstream compact program layer. □

Corollary H4.238 (The present post-archival stage line already carries a first downstream task-to-program layer on the consumed follow-on level)**.** *For the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A},\mathsf{B}\}$*, the current active line already carries a first downstream task-to-program object*

$$\mathsf{Prog}_2^{\mathrm{down}}(\mathbf{w}_3;\mathcal{S},\mathfrak{b} \mid \mathcal{S}^+)$$

for every finite nonempty admissible request subfamily

$$\mathcal{S} \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3),$$

every branch marker $\mathfrak{b} \in \{\mathsf{CL},\mathsf{PR}\}$*, and in the promotion case every second finite nonempty admissible request subfamily*

$$\mathcal{S}^+ \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3)$$

that remains subordinate to the same inherited route tag. So the post-release continuation now carries not only a conservative ask/respond/aggregate/dispatch/program/cycle/return/close/promote/branch/consume/task/program/cycle/return/branch/consume/task protocol, but also the first disciplined downstream task-to-program layer that compresses each consumed downstream close-side or next-cycle-side task family into one compact work normal form.

Proof. Immediate from Corollary H4.234 and Proposition H4.237. □

Remark H4.239 (What the first downstream task-to-program layer on the consumed follow-on level now buys)**.** *This is the point where the post-release line first becomes able not only to translate a consumed downstream close-side or next-cycle branch into one explicit downstream work family, but also to compress that work family into one compact downstream program object. The horizon-side line can now hand a consumed downstream close-side task family onward as one normalized close-side downstream program or hand a consumed downstream next-cycle task family onward as one normalized next-cycle downstream program under the inherited horizon tag. The boundary-side line can now do the same under the inherited stop-preserving boundary tag and earliest honest boundary stop, without collapsing the two possibilities into one vague continuation gesture. Later OP-oriented work can therefore begin from a conservative ask/respond/aggregate/dispatch/program/cycle/return/close/promote/branch/consume/task/program/cycle/return/branch/consume/task/program protocol instead of improvising how consumed downstream task families are normalized.*

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Reader cue. Read the next block as the first downstream program-to-cycle layer extracted from the downstream task-to-program layer of $19\mathrm{AZ}^{(6bn)}$. Its purpose is to turn each already normalized close-side or next-cycle-side downstream program object into one controlled downstream cycle on the consumed follow-on level, so that the second downstream loop does not stop at a static program form but becomes an executable route-faithful cycle under the inherited horizon tag or inherited stop-preserving boundary tag.

$19\mathrm{AZ}^{(6bo)}$. First downstream program-to-cycle layer extracted from the downstream task-to-program layer on the consumed follow-on level

After $19\mathrm{AZ}^{(6bn)}$, every normalized downstream close-side or next-cycle-side task family already yields one compact downstream program object. The present block turns that downstream program into one controlled downstream cycle. On the close side, the downstream cycle executes the close-side comparison, stabilization, and packaging program and records one close-side return token under the inherited route tag. On the next-cycle side, the downstream cycle executes the next-cycle comparison, preparation, and packaging program and records one next-cycle return token under the same inherited route tag. On the boundary side, every such downstream cycle must remain subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop. The point is to make the second downstream work cycle explicit.

Definition H4.240 (First downstream program-to-cycle object on the consumed follow-on level)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, a finite nonempty admissible request subfamily*

$$\mathcal{S} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}),$$

a branch marker $\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}$*, and an admissible downstream task-to-program object*

$$\mathsf{Prog}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+).$$

When $\mathfrak{b} = \mathsf{PR}$*, also keep fixed the second finite nonempty admissible request subfamily*

$$\mathcal{S}^+ \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}).$$

Define the associated first downstream program-to-cycle object on the consumed follow-on level

$$\mathsf{Cycle}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+)$$

by the following cases.

(i) If $\mathfrak{b} = \mathsf{CL}$*, define*

$$\mathsf{Cycle}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{CL}) := \big(\mathsf{Prog}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{CL}), \mathcal{C}_m^{\mathrm{close,down}}(\mathbf{w}; \mathcal{S})\big),$$

where

$$\mathcal{C}_m^{\mathrm{close,down}}(\mathbf{w}; \mathcal{S}) := \big(\mathsf{CL}, \mathsf{TCMP}_m^{\mathrm{close,down}}, \mathsf{TSTAB}_m^{\mathrm{close,down}}, \mathsf{TPACK}_m^{\mathrm{close,down}}, \mathsf{RET}_m^{\mathrm{close,down}}\big).$$

This downstream cycle object is admissible only if it remains subordinate to the same inherited route tag and, in the boundary-side case, preserves the inherited earliest honest boundary stop.

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(ii) If $\mathfrak{b} = \mathsf{PR}$*, define*

$$\mathsf{Cycle}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^+) := \big(\mathsf{Prog}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^+), \mathcal{C}_m^{\mathrm{next,down}}(\mathbf{w}; \mathcal{S}, \mathcal{S}^+)\big),$$

where

$$\mathcal{C}_m^{\mathrm{next,down}}(\mathbf{w}; \mathcal{S}, \mathcal{S}^+) := \big(\mathsf{PR}, \mathsf{TCMP}_m^{\mathrm{next,down}}, \mathsf{TPREP}_m^{\mathrm{next,down}}, \mathsf{TPACK}_m^{\mathrm{next,down}}, \mathsf{RET}_m^{\mathrm{next,down}}\big).$$

This downstream cycle object is admissible only if it remains subordinate to the same inherited route tag, does not manufacture a fresh theorematic certificate, and, in the boundary-side case, does not smooth away, postpone, or deny the inherited earliest honest boundary stop.

Proposition H4.241 (Every normalized downstream program object yields exactly one controlled downstream cycle)**.** *Assume the setting of Definition H4.240. Then:*

(i) every admissible downstream close-side program object yields exactly one downstream close-side cycle object;

(ii) every admissible downstream next-cycle program object yields exactly one downstream next-cycle cycle object;

(iii) in the horizon-side case, every admissible downstream program-to-cycle object remains subordinate to the inherited horizon tag;

(iv) in the boundary-side case, every admissible downstream program-to-cycle object remains subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop; and

(v) no admissible downstream program-to-cycle object may manufacture a fresh theorematic certificate, erase the horizon/boundary distinction, or collapse the close/next-cycle difference into one undifferentiated downstream execution cycle.

Proof. Clauses (i) and (ii) are immediate from Definition H4.240: once the branch marker is fixed, the downstream cycle object is defined by exactly one of the two displayed clauses. Clauses (iii) and (iv) follow because the downstream program-to-cycle object is defined directly from a downstream task-to-program object that is already subordinate to the inherited route tag and, in the boundary-side case, to the inherited earliest honest boundary stop, by Proposition H4.237. Clause (v) is exactly the inherited certificate-aware route discipline from $19\mathrm{AZ}^{(6ae)}$ through $19\mathrm{AZ}^{(6bn)}$, now made explicit at the second downstream cycle layer. □

Corollary H4.242 (The present post-archival stage line already carries a first downstream program-to-cycle layer on the consumed follow-on level)**.** *For the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$*, the current active line already carries a first downstream program-to-cycle object*

$$\mathsf{Cycle}_2^{\mathrm{down}}(\mathbf{w}_3; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+)$$

for every finite nonempty admissible request subfamily

$$\mathcal{S} \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3),$$

every branch marker $\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}$*, and in the promotion case every second finite nonempty admissible request subfamily*

$$\mathcal{S}^+ \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3)$$

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that remains subordinate to the same inherited route tag. So the post-release continuation now carries not only a conservative ask/respond/aggregate/dispatch/program/cycle/return/close/promote/branch/consume/task/program/cycle/return/branch/consume/task/program protocol, but also the first disciplined downstream program-to-cycle layer that turns each normalized consumed downstream program into one explicit follow-on cycle.

Proof. Immediate from Corollary H4.238 and Proposition H4.241. □

Remark H4.243 (What the first downstream program-to-cycle layer on the consumed follow-on level now buys)**.** *This is the point where the post-release line first becomes able not only to normalize a consumed downstream close-side or next-cycle work family as one compact downstream program, but also to execute that program as one explicit downstream cycle on the consumed follow-on level. The horizon-side line can now hand a normalized consumed downstream close-side program onward as one close-side downstream cycle or hand a normalized consumed downstream next-cycle program onward as one next-cycle downstream cycle under the inherited horizon tag. The boundary-side line can now do the same under the inherited stop-preserving boundary tag and earliest honest boundary stop, without collapsing the two possibilities into one vague continuation gesture. Later OP-oriented work can therefore begin from a conservative ask/respond/aggregate/dispatch/program/cycle/return/close/promote/branch/consume/task/program/cycle/return/branch/consume/task/program/cycle protocol instead of improvising how normalized consumed downstream programs execute.*

Reader cue. Read the next block as the first downstream cycle-to-return / close-or-promote interface extracted from the downstream program-to-cycle layer of $19AZ^{(6bo)}$. Its purpose is to turn each already executed consumed downstream cycle into one disciplined return object that either records an honest compact close-side exit or an admissible next-cycle continuation, thereby bringing the second downstream loop to the same honest close-or-promote threshold as the earlier layers.

$19AZ^{(6bp)}$. Downstream cycle-to-return / close-or-promote interface on the consumed follow-on level

After $19AZ^{(6bo)}$, every normalized consumed downstream close-side or next-cycle-side program already yields one controlled downstream cycle. The present block turns that cycle into one disciplined downstream return object that either records a compact close-side exit or an admissible next-cycle continuation. On the close side, the return object packages the explicit close-side cycle together with its compact close-side exit under the inherited route tag. On the next-cycle side, the return object packages the explicit next-cycle cycle together with its admissible next-cycle continuation under the same inherited route tag. On the boundary side, every such return object must remain subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop. The point is to make the second downstream loop honestly closable without forcing a third recursion before release packaging.

Definition H4.244 (First downstream cycle-to-return / close-or-promote interface object on the consumed follow-on level)**.** *Fix a certified finite extension word* $\mathbf{w} \in \mathcal{L}_m^{\mathrm{ext}}$*, a finite nonempty admissible request subfamily*

$$\mathcal{S} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}),$$

a branch marker $\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}$*, and an admissible downstream program-to-cycle object*

$$\mathsf{Cycle}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+).$$

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When $\mathfrak{b} = \mathsf{PR}$*, also keep fixed the second finite nonempty admissible request subfamily*

$$\mathcal{S}^{+} \subseteq \mathsf{Q}_m^{\mathrm{cons}}(\mathbf{w}).$$

Define the associated first downstream cycle-to-return / close-or-promote interface object on the consumed follow-on level

$$\mathsf{Iter}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^{+})$$

by the following cases.

(i) If $\mathfrak{b} = \mathsf{CL}$*, define*

$$\mathsf{Iter}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{CL}) := \big(\mathsf{Cycle}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{CL}), \mathsf{RET}_m^{\mathrm{close,down}}, \mathsf{Close}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S})\big).$$

This downstream return object is admissible only if it remains subordinate to the same inherited route tag and, in the boundary-side case, preserves the inherited earliest honest boundary stop.

(ii) If $\mathfrak{b} = \mathsf{PR}$*, define*

$$\mathsf{Iter}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^{+}) := \big(\mathsf{Cycle}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathsf{PR} \mid \mathcal{S}^{+}), \mathsf{RET}_m^{\mathrm{next,down}}, \mathsf{Next}_m^{\mathrm{down}}(\mathbf{w}; \mathcal{S}, \mathcal{S}^{+})\big).$$

This downstream return object is admissible only if it remains subordinate to the same inherited route tag, does not manufacture a fresh theorematic certificate, and, in the boundary-side case, does not smooth away, postpone, or deny the inherited earliest honest boundary stop.

Proposition H4.245 (Every consumed downstream cycle yields exactly one honest downstream return close-or-promote object)**.** *Assume the setting of Definition H4.244. Then:*

(i) every admissible downstream close-side cycle yields exactly one downstream close-return object;

(ii) every admissible downstream next-cycle cycle yields exactly one downstream next-return object;

(iii) in the horizon-side case, every admissible downstream cycle-to-return object remains subordinate to the inherited horizon tag;

(iv) in the boundary-side case, every admissible downstream cycle-to-return object remains subordinate to the inherited stop-preserving boundary tag and inherited earliest honest boundary stop; and

(v) no admissible downstream cycle-to-return object may manufacture a fresh theorematic certificate, erase the horizon/boundary distinction, or collapse the close/next-cycle difference into one undifferentiated downstream continuation gesture.

Proof. Clauses (i) and (ii) are immediate from Definition H4.244: once the branch marker is fixed, the downstream return object is defined by exactly one of the two displayed clauses. Clauses (iii) and (iv) follow because the downstream cycle-to-return object is defined directly from a downstream program-to-cycle object that is already subordinate to the inherited route tag and, in the boundary-side case, to the inherited earliest honest boundary stop, by Proposition H4.241. Clause (v) is exactly the inherited certificate-aware route discipline from $19\mathrm{AZ}^{(6ae)}$ through $19\mathrm{AZ}^{(6bo)}$, now made explicit at the second downstream return interface. □

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Corollary H4.246 (The present post-archival stage line already carries a first downstream cycle-to-return interface on the consumed follow-on level)**.** *For the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$*, the current active line already carries a first downstream cycle-to-return interface object*

$$\mathsf{Iter}_2^{\mathrm{down}}(\mathbf{w}_3; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+)$$

for every finite nonempty admissible request subfamily

$$\mathcal{S} \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3),$$

every branch marker $\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}$*, and in the promotion case every second finite nonempty admissible request subfamily*

$$\mathcal{S}^+ \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3)$$

that remains subordinate to the same inherited route tag. So the post-release continuation now carries not only a conservative ask/respond/aggregate/dispatch/program/cycle/return/close/promote/branch/consume/task/program/cycle/return/branch/consume/task/program/cycle protocol, but also the first disciplined downstream cycle-to-return interface that brings the consumed follow-on level to one honest close-or-promote threshold.

Proof. Immediate from Corollary H4.242 and Proposition H4.245. □

Remark H4.247 (What the first downstream cycle-to-return interface on the consumed follow-on level now buys)**.** *This is the point where the post-release line first becomes able not only to execute a normalized consumed downstream close-side or next-cycle-side program as one explicit downstream cycle, but also to turn that cycle into one disciplined return object that either records an honest compact close-side exit or an admissible next-cycle continuation. The horizon-side line can now hand a consumed downstream close-cycle onward as one compact close-side downstream return object or hand a consumed downstream next-cycle onward as one admissible downstream continuation object under the inherited horizon tag. The boundary-side line can now do the same under the inherited stop-preserving boundary tag and earliest honest boundary stop, without collapsing the two possibilities into one vague continuation gesture. Later OP-oriented work can therefore begin from a conservative ask/respond/aggregate/dispatch/program/cycle/return/close/promote/branch/consume/task/program/cycle/return/branch/consume/task/program/cycle/return protocol instead of improvising how normalized consumed downstream cycles return.*

Reader cue. Read the next block as a release-oriented freeze point extracted from the now completed second downstream loop through task, program, cycle, and return on the consumed follow-on level. Its purpose is not to claim that no further conservative iteration is ever possible, but to record that the present post-archival stage line has already reached a structurally stable, release-ready stopping point at which a new public version may honestly freeze before opening yet another recursive downstream pass.

19AZ$^{(6bq)}$. First release-freeze stabilization point extracted from the completed second downstream loop

After 19AZ$^{(6bn)}$–19AZ$^{(6bp)}$, the post-release line now carries a second downstream loop that again reaches an honest task/program/cycle/return threshold on the consumed follow-on level. The present block records the structural consequence relevant for release packaging: before opening a third recursive downstream pass, the current line may honestly stop and freeze here, because the second downstream loop is no longer dangling in the middle of an unfinished translation but already reaches a compact return threshold under the inherited route discipline.

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Definition H4.248 (Release-freeze stabilization object for the completed second downstream loop)**.** *Fix the first-stage word* $\mathbf{w}_3 \in \{\mathsf{A}, \mathsf{B}\}$*, a finite nonempty admissible request subfamily*

$$\mathcal{S} \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3),$$

a branch marker $\mathfrak{b} \in \{\mathsf{CL}, \mathsf{PR}\}$*, and in the promotion case a second finite nonempty admissible request subfamily*

$$\mathcal{S}^+ \subseteq \mathsf{Q}_2^{\mathrm{cons}}(\mathbf{w}_3)$$

that remains subordinate to the same inherited route tag. The associated release-freeze stabilization object

$$\mathsf{Freeze}_2^{\mathrm{cons}}(\mathbf{w}_3; \mathcal{S}, \mathfrak{b} \mid \mathcal{S}^+)$$

is the declaration that the active line already reaches the completed second downstream loop through

$$\mathsf{Task}_2^{\mathrm{down}}, \quad \mathsf{Prog}_2^{\mathrm{down}}, \quad \mathsf{Cycle}_2^{\mathrm{down}}, \quad \mathsf{Iter}_2^{\mathrm{down}},$$

and may therefore be frozen for public release packaging before a third recursive downstream pass is opened.

Proposition H4.249 (The completed second downstream loop already supplies an honest public freeze point)**.** *Assume the setting of Definition H4.248. Then:*

(i) the present line already carries a completed second downstream loop through task, program, cycle, and return on the consumed follow-on level;

(ii) freezing the line at this point does not erase any already opened structural step, because the second downstream loop is no longer cut off mid-translation;

(iii) freezing the line at this point does not claim maximality, final completeness, or impossibility of later further downstream recursion; and

(iv) the only remaining steps needed for a public version after this structural freeze are release packaging steps such as DOI insertion, chronology update, and audit/freeze notes, rather than another mandatory structural rescue block.

Proof. Clause (i) follows from $19\mathrm{AZ}^{(6bm)}$ through $19\mathrm{AZ}^{(6bp)}$: the consumed follow-on level now carries explicit downstream task, program, cycle, and return objects. Clause (ii) follows because the second downstream loop now reaches the same kind of honest return threshold already reached earlier in the main post-release loop, so freezing here no longer leaves the line suspended between task and execution. Clause (iii) follows because Definition H4.248 records only a release-oriented stopping point and does not assert that no later third recursive pass could be constructed. Clause (iv) is then immediate: once the line reaches a completed second downstream return threshold, the remaining work before release is packaging rather than structural rescue. □

Corollary H4.250 (That inherited public release records the freeze point that needed DOI and audit rather than another mandatory rescue staircase)**.** *For the current active post-archival stage line, the structurally necessary continuation up to a public freeze point has now been completed. That inherited public release therefore recorded that freeze point, packaged it under the assigned version number and DOI, and did not require a third downstream rescue staircase merely to avoid ending in the middle of the second one.*

Proof. Immediate from Proposition H4.249. □

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Remark H4.251 (What the release-freeze stabilization point now buys)**.** *This is the point where the post-archival stage line first reached an honestly release-ready stopping point after the start of the conservative upgrade-plus-probe integration stage. The line already carries the main conservative staircase, an OP-facing intake/request/fulfillment/aggregation/dispatch/program/cycle/return stack, and a completed second downstream loop through consumed task, program, cycle, and return. That was enough structural closure for that inherited public release. Anything beyond this point would be a new optional recursive expansion rather than a necessary repair of an unfinished line.*

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H5 Reintegrated Historical Proof Blocks (archival proof-memory)

Read this appendix as archival proof-memory and full-source preservation beneath the active main line. In earlier working states this location contained only a compact recovered fragment. To avoid archival page loss and to preserve the complete historical witness, that compact fragment is replaced here by the historical-only Part II carried forward from the preserved archival stage release line. Whenever wording in the archived text sounds stronger, flatter, or more present-tense than the normalized live line, the current status statements and routing discipline in the active main text govern.

Remark H5.1 (Preservation note)**.** *The appendix that follows is intentionally the integrated historical witness rather than a shortened reintegrated excerpt. It preserves the older derivation layouts and source-memory material without re-importing the active current document a second time.*

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Historical witness handoff

Bundle status. Beginning with this public split-bundle release, the active core and the integrated historical witness are carried as separate PDFs in the same release bundle. This is a readability and archival packaging change only. It does not delete the historical witness, does not change theorematic status, and does not manufacture any OP closure certificate.

Companion file. The preserved historical witness is provided as

`Quantum_Sphaera_Companion_archival_historical_witness.pdf`.

Reading rule. The present active-core PDF is the governing current proof-line and audit interface. The historical-witness PDF is preserved proof memory, genealogy, and reconstruction support. Whenever an inherited historical block and the active core differ in theorematic status, the active core governs.

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A.6 Provenance and Machine-Audit Layer

This appendix records the status of the notation-hygiene pass. It is intentionally separate from the mathematical object language. Revision identifiers are permitted here only as provenance markers: they identify source packets, diffs, manifests, checksums, and build audits. They do not name mathematical objects and do not denote physical states.

The cleaned edition was produced by applying the following rule to the available project sources: remove revision identifiers from visible headings and mathematical symbols; replace revision-bearing superscripts by semantic role markers where a distinction is mathematically useful; keep the original files and manifests as audit witnesses.

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A.7 archival stage Migration Map and Backtranslation Contract for the shifted QAM Shift

Status. This section is a migration-map layer only. It prepares the later shifted global QAM Hilbert-shift, but does not perform that global migration. The local HC-boundary activation shift introduced in Version 3.36.0 remains the first structural shift:

$$0_{\text{mask}} \longrightarrow 1_{\text{mask}}.$$

The present layer records how a later shifted line may move the whole QAM tag language without losing the historical release line Active Core, its router/readout separation, or the Historical Witness.

A.7.1 Separation of the two shifts

Definition A.7.1 (Local boundary shift versus global language shift)**.** *The Companion distinguishes two shifts.*

1. Shift A*, the local HC-boundary activation shift, is the already-admitted historical release line structural reading*

$$0_{\text{mask}} \longrightarrow 1_{\text{mask}}.$$

It says that the transition from no active readout to the first active readout requires the Heisenberg Compensator as a minimal boundary/readout normalizer.

2. Shift B*, the global shifted QAM Hilbert-shift, is the future language migration*

$$k \longmapsto k + 1$$

applied uniformly to inhabited QAM tags so that tag 0 *becomes globally reserved for the empty or no-active-kind marker.*

Remark A.7.2 (Why Shift A belongs to the legacy preparation line and Shift B belongs to shifted)**.** *Shift A is structural and local: it explains why HC appears internally at the first active readout boundary. Shift B is linguistic and global: it changes the whole QAM namespace. Therefore Shift A belongs to the legacy preparation line, while Shift B must be treated as the first major shifted migration.*

A.7.2 Old and new QAM namespaces

Definition A.7.3 (Pre-shift and shifted tag namespaces)**.** *Let $\mathcal{N}_3$ denote the inherited pre-shift QAM tag namespace used through Version 3.36.0. Let $\mathcal{N}_4$ denote the future shifted namespace. The planned migration map is a partial translation*

$$\Sigma_4 \colon \mathcal{N}_3^{\text{inh}} \longrightarrow \mathcal{N}_4^{\text{inh}}, \qquad \Sigma_4(k) = k + 1,$$

defined on inhabited tags. The target tag $0 \in \mathcal{N}_4$ is reserved for the empty or no-active-kind marker and is not the image of an inhabited pre-shift tag.

Definition A.7.4 (Backtranslation map)**.** *The formal backtranslation associated with Σ_4 is the partial map*

$$\mathrm{B}_4 \colon \mathcal{N}_4^{\text{inh}} \longrightarrow \mathcal{N}_3^{\text{inh}}, \qquad \mathrm{B}_4(a) = a - 1$$

for inhabited shifted tags $a \geq 1$. The value 0 *has no inhabited backtranslation; it is read only as the empty/no-active-kind marker.*

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Lemma A.7.5 (Round-trip on inhabited tags)**.** *For every inhabited pre-shift tag $k \in \mathcal{N}_3^{\mathrm{inh}}$,*

$$\mathrm{B}_4(\Sigma_4(k)) = k.$$

For every inhabited shifted tag $a \geq 1$,

$$\Sigma_4(\mathrm{B}_4(a)) = a.$$

Proof. Both identities are the elementary identities $(k+1)-1 = k$ and $(a-1)+1 = a$. The reserved value 0 is excluded from the inhabited backtranslation domain by definition. Thus the round-trip statement is exact on inhabited tags and honestly undefined on the reserved empty marker. □

A.7.3 Migration obligations

Definition A.7.6 (shifted migration obligation list)**.** *A later shifted global QAM Hilbert-shift must discharge the following obligations:*

1. *translate every inhabited QAM tag through Σ_4;*
2. *reserve tag 0 globally for the empty or no-active-kind marker;*
3. *preserve all payloads attached to translated tags;*
4. *translate route-mask examples and bitstream examples coherently;*
5. *preserve the two-sided structural tensor grammar under tag erasure and reindexing;*
6. *preserve router-object guards and backtranslation records;*
7. *state explicitly that the HC-boundary activation shift is not being redefined by the global tag migration;*
8. *prove a conservativity theorem by erasing shifted tags back to the pre-shift namespace through B_4 on inhabited components.*

Remark A.7.7 (What is not migrated)**.** *The shifted tag migration should not change classical upper–lower tensor contraction, same-variance comparison rules, the OP status, the Historical Witness, or the physical meaning of an already guarded readout. It changes the QAM namespace and routing labels, not the underlying mathematical payloads.*

A.7.4 Router-package migration

Definition A.7.8 (Router-package translation)**.** *Let*

$$\mathfrak{R} = (K, M, C, T, B, G)$$

be a guarded router object, with kind registry K, route-mask registry M, classical collars C, trace record T, backtranslation record B, and guard record G. Its shifted translation is the object

$$\Sigma_4(\mathfrak{R}) = (\Sigma_4 K, \Sigma_4 M, C, T, \mathrm{B}_4^{\mathfrak{R}}, G),$$

where collars, traces, and guards are preserved, while kind and mask registries are reindexed and the backtranslation record is replaced by the corresponding shifted-to-pre-shift decoding data.

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Lemma A.7.9 (Router guard preservation)**.** *If a router object $\mathfrak{R}$ is guarded before migration, then $\Sigma_4(\mathfrak{R})$ carries the same guard record after migration.*

Proof. By Definition A.7.8, the guard record G is copied unchanged. The migration reindexes registries and replaces the backtranslation ledger; it does not modify the guard predicates. Hence guard status is preserved. □

A.7.5 Readout-package migration

Definition A.7.10 (Readout-package translation)**.** *A guarded readout package may be translated to shifted only by translating its QAM/router labels while leaving its declared readout target, comparison tensor, boundary collar, residual map, and non-claim guard unchanged.*

Proposition A.7.11 (Machine/content preservation under migration)**.** *The future shifted global QAM Hilbert-shift preserves the historical release line machine/content separation: QAM/MML/MMA remains the router/compiler language, and physical or mathematical content remains in the guarded readout package.*

Proof. The migration acts on labels, kinds, masks, and routing registries. It does not act on the declared target of a physical readout, the comparison tensor, the boundary collar, or the residual map. Therefore it changes the machine addressing language while preserving the separation between machine layer and content layer. □

A.7.6 Conservativity statement for the future shifted shift

Theorem A.7.12 (shifted migration conservativity contract)**.** *Suppose a future shifted text translates every inhabited pre-shift QAM tag through Σ_4, reserves 0 only for the empty/no-active-kind marker, and equips each translated router or readout object with its backtranslation ledger B_4. Then every finite guarded shifted router/readout package has an inhabited-component backtranslation to a finite guarded pre-shift package with the same payloads and the same theorematic status.*

Proof. Apply B_4 to every inhabited shifted tag and leave non-tag payloads fixed. Round-trip correctness on inhabited tags follows from Lemma A.7.5. Router guards are preserved by Lemma A.7.9. Readout targets, comparison tensors, boundary collars, residual maps, and non-claim guards are unchanged by Definition A.7.10. Hence the translated object backtranslates to a pre-shift guarded object with identical payloads and unchanged theorematic status. The reserved value 0 has no inhabited backtranslation and therefore cannot be used to manufacture a new certificate. □

A.7.7 historical release line handoff to shifted

Definition A.7.13 (shifted entry checklist)**.** *A later archival stage draft should begin only after the following checklist is satisfied:*

1. *the global tag translation Σ_4 is fixed;*
2. *every occurrence class of QAM tags has an entry in the migration ledger;*
3. *every route-mask example has a shifted and pre-shift display pair;*
4. *every router object carries a backtranslation record;*

5. the HC-boundary activation shift remains local and is not redefined;

6. the shifted text carries the conservativity contract of Theorem A.7.12;

7. OP 1/4 and OP 5(B) are not claimed closed merely by the migration.

Remark A.7.14 (historical release line non-claim)**.** *Version 3.37.0 is a migration-map and backtranslation-contract layer. It does not execute the shifted global QAM Hilbert-shift, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, and does not claim Turing-completeness. Its purpose is to make the later shifted migration auditable before the namespace is actually changed.*

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A.8 archival stage Concrete Migration Ledger and Route-Mask Examples

Status. This section is a concrete-audit layer for the historical release line migration map. It still does not execute the global shifted QAM Hilbert-shift. Its purpose is to make the future migration visually and mechanically checkable before the namespace is changed.

A.8.1 Minimal tag ledger

Definition A.8.1 (Concrete migration ledger)**.** *The concrete migration ledger is the finite display of the rule*

$$\Sigma_4(k) = k + 1, \qquad \mathrm{B}_4(a) = a - 1 \quad (a \geq 1),$$

together with the reserved value

$$0_{\mathrm{shift}} = \varnothing_{\mathrm{active}},$$

which has no inhabited pre-shift backtranslation.

pre-shift tag	shifted tag	reading	backtranslation
reserved / none	0	empty/no-active-kind	none
0	1	first inhabited kind	$\mathrm{B}_4(1) = 0$
1	2	second inhabited kind	$\mathrm{B}_4(2) = 1$
2	3	third inhabited kind	$\mathrm{B}_4(3) = 2$
k	$k+1$	inhabited kind k	$\mathrm{B}_4(k+1) = k$

Remark A.8.2 (Why the ledger is not a new theorem)**.** *The ledger is a bookkeeping surface. It records how inhabited tags will be renamed in shifted. It does not change the underlying readout, comparison tensor, boundary collar, residual map, or OP status.*

A.8.2 Route payload example

Definition A.8.3 (Route payload migration)**.** *For a finite ordered route payload*

$$r = \langle k_1, \ldots, k_n \rangle,$$

the future shifted route payload is

$$\Sigma_4(r) = \langle k_1 + 1, \ldots, k_n + 1 \rangle.$$

Its backtranslation is applied componentwise:

$$\mathrm{B}_4(\langle a_1, \ldots, a_n \rangle) = \langle a_1 - 1, \ldots, a_n - 1 \rangle$$

whenever all $a_i \geq 1$.

Remark A.8.4 (Example: two-entry route payload)**.** *The pre-shift payload*

$$r = \langle 5, 12 \rangle$$

migrates to

$$\Sigma_4(r) = \langle 6, 13 \rangle.$$

Backtranslation returns

$$\mathrm{B}_4(\langle 6, 13 \rangle) = \langle 5, 12 \rangle.$$

Thus the route addresses change, while the ordered route structure and the payload role are preserved.

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A.8.3 Packed mask example

Definition A.8.5 (Support mask versus packed mask)**.** *A support mask is a finite inhabited support set*

$$S \subseteq \mathbb{N}.$$

Its pre-shift packed integer is

$$m_3(S) = \sum_{k \in S} 2^k.$$

After migration the support becomes

$$\Sigma_4(S) = \{k + 1 : k \in S\},$$

and its shifted packed integer is

$$m_4(\Sigma_4(S)) = \sum_{k \in S} 2^{k+1}.$$

Lemma A.8.6 (Packed-mask doubling rule)**.** *For every finite inhabited support S,*

$$m_4(\Sigma_4(S)) = 2\, m_3(S).$$

Proof. By definition,

$$m_4(\Sigma_4(S)) = \sum_{k \in S} 2^{k+1} = 2 \sum_{k \in S} 2^k = 2\, m_3(S).$$

The rule applies to the packed integer representation of the support. It is not the same operation as adding 1 to the packed integer. □

Remark A.8.7 (Example: packed mask for the route support $\{5, 12\}$)**.** *For*

$$S = \{5, 12\},$$

the pre-shift packed mask is

$$m_3(S) = 2^5 + 2^{12} = 32 + 4096 = 4128.$$

The shifted support is

$$\Sigma_4(S) = \{6, 13\},$$

and the shifted packed mask is

$$m_4(\Sigma_4(S)) = 2^6 + 2^{13} = 64 + 8192 = 8256.$$

This equals $2 \cdot 4128$. Backtranslation of the support gives

$$\mathrm{B}_4(\{6, 13\}) = \{5, 12\}.$$

Remark A.8.8 (No accidental activation of the reserved bit)**.** *Since inhabited supports are shifted upward, the shifted bit 0 is not produced by migration. It remains reserved for the empty/no-active-kind marker. This is the operational reason why the global shift is compatible with the HC-boundary interpretation: the first active bit is no longer confused with the empty marker.*

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A.8.4 Two-sided tensor ledger example

Definition A.8.9 (Tensor-index migration ledger)**.** *A two-sided structural tensor expression may carry QAM/router tags inside its front-side or back-side index payloads. A future shifted migration applies Σ_4 only to those tag payloads. It does not change front/back sidedness, upper/lower variance, or the declared contraction/comparison rule.*

Remark A.8.10 (Example: front-side route migration)**.** *A readout-facing tensor package with front-side route payload*

$$^{\langle 5,12\rangle}{}_{\langle 2\rangle}\mathcal{T}^{i}{}_{j}$$

migrates to

$$^{\langle 6,13\rangle}{}_{\langle 3\rangle}\mathcal{T}^{i}{}_{j}.$$

The front-side payloads are reindexed. The back-side structural indices i, j are not changed in this example. The tensor still has the same front/back and upper/lower grammar.

Remark A.8.11 (Example: full payload migration)**.** *If both front-side and back-side QAM payloads are explicitly tagged, then*

$$^{\langle 5,12\rangle}{}_{\langle 2\rangle}\mathcal{T}^{\langle 7\rangle}{}_{\langle 10\rangle}$$

migrates to

$$^{\langle 6,13\rangle}{}_{\langle 3\rangle}\mathcal{T}^{\langle 8\rangle}{}_{\langle 11\rangle}.$$

This is a label migration only. It does not turn an upper index into a lower index, a front-side readout into a back-side action, or a comparison tensor into a contraction tensor.

Proposition A.8.12 (Tensor grammar preservation under the migration ledger)**.** *The concrete migration ledger preserves the two-sided structural tensor grammar.*

Proof. The migration acts on tag payloads by $k \mapsto k + 1$. The syntactic positions of the payloads are unchanged: front-side payloads remain front-side payloads, back-side payloads remain back-side payloads, upper payloads remain upper payloads, and lower payloads remain lower payloads. Therefore classical upper–lower readout and explicit same-variance comparison requirements are unchanged. □

A.8.5 shifted readiness consequence

Theorem A.8.13 (Concrete ledger supports the shifted conservativity contract)**.** *The concrete tag ledger, route payload example, packed-mask rule, and two-sided tensor ledger examples support the historical release line backtranslation contract: every migrated finite example displayed in this section backtranslates to its pre-shift source example on inhabited components.*

Proof. The tag ledger backtranslates by $\mathrm{B}_4(a) = a-1$ for $a \geq 1$. Route payloads backtranslate componentwise. Packed masks backtranslate at the level of support sets, not by subtracting 1 from the packed integer. Tensor payloads backtranslate by applying the same rule to the tags carried inside their index payloads, while preserving sidedness and variance. Hence each finite example returns to its pre-shift source example on inhabited components. □

Remark A.8.14 (archival stage non-claim)**.** *Version 3.37.1 still does not execute the global shifted QAM Hilbert-shift. It only adds concrete ledger examples. The actual global migration remains a shifted task.*

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A.9 Global QAM Hilbert-Shift Activation Candidate

Status. This is the first shifted activation release-final state. It executes the global QAM namespace decision prepared in Versions 3.37.0–3.37.1: from this layer forward, the active QAM language reserves tag 0 for the empty/no-active-kind marker and reads inhabited tags in the shifted shifted namespace. The earlier historical release line HC-boundary shift remains local and structural; the present layer is the global language shift.

A.9.1 Governing shifted namespace convention

Definition A.9.1 (Global shifted QAM Hilbert-shift convention)**.** *The governing shifted QAM namespace is obtained from the inhabited pre-shift namespace by the shift*

$$\Sigma_4(k) = k + 1.$$

The tag 0 *in the shifted namespace is reserved globally for the empty or no-active-kind marker:*

$$0_{\text{shift}} = \varnothing_{\text{active}}.$$

It is not the image of an inhabited pre-shift tag.

Definition A.9.2 (shifted backtranslation ledger)**.** *The shifted backtranslation ledger is the partial map*

$$\mathrm{B}_4(a) = a - 1$$

defined on inhabited shifted tags $a \geq 1$*. The reserved tag* 0 *has no inhabited backtranslation.*

Remark A.9.3 (What is executed in)**.** *The executed decision is the global namespace convention: shifted readout packages use the shifted tag convention by default. The text does not pretend that the Historical Witness has been rewritten. Historical material remains readable by the backtranslation ledger.*

A.9.2 Compatibility with the first HC-boundary shift

Definition A.9.4 (Shift A and Shift B in the shifted text)**.** *The shifted text keeps the following distinction.*

1. Shift A *is the local historical release line HC-boundary activation*

   $$0_{\text{mask}} \longrightarrow 1_{\text{mask}}.$$

   It explains why the Heisenberg Compensator appears as a boundary/readout normalizer at the first active readout transition.

2. Shift B *is the global shifted QAM Hilbert-shift*

   $$k \longmapsto k + 1$$

   applied to inhabited QAM tags.

Proposition A.9.5 (HC-boundary shift is not redefined by shifted)**.** *The global shifted QAM Hilbert-shift does not redefine the historical release line HC-boundary activation principle.*

Proof. Shift A concerns the structural boundary from inactive mask state to first active readout state. Shift B concerns the global label namespace for inhabited QAM tags. These acts have different domains: the first is a local boundary-normalization reading, the second is a global language migration. Therefore the shifted namespace convention preserves the HC-boundary reading rather than replacing it. □

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A.9.3 Governing shifted ledger

pre-shift tag	shifted tag	shifted reading	backtranslation
none	0	empty/no-active-kind	none
0	1	first inhabited kind	$\mathrm{B}_4(1) = 0$
1	2	second inhabited kind	$\mathrm{B}_4(2) = 1$
2	3	third inhabited kind	$\mathrm{B}_4(3) = 2$
k	$k+1$	inhabited kind k	$\mathrm{B}_4(k+1) = k$

Lemma A.9.6 (shifted inhabited round-trip)**.** *For every inhabited pre-shift tag k,*

$$\mathrm{B}_4(\Sigma_4(k)) = k.$$

For every inhabited shifted tag $a \geq 1$,

$$\Sigma_4(\mathrm{B}_4(a)) = a.$$

Proof. Both claims are the identities $(k+1) - 1 = k$ and $(a-1) + 1 = a$. The reserved value 0 is excluded from the inhabited backtranslation domain, so it cannot generate an inhabited source tag. □

A.9.4 Route-mask and support migration

Definition A.9.7 (shifted route payload)**.** *A pre-shift ordered route payload*

$$r = \langle k_1, \ldots, k_n \rangle$$

is read in the shifted language as

$$r^{(4)} = \Sigma_4(r) = \langle k_1 + 1, \ldots, k_n + 1 \rangle.$$

Definition A.9.8 (shifted support and packed mask)**.** *For a finite inhabited support $S \subseteq \mathbb{N}$, the migrated shifted support is*

$$S^{(4)} = \{k + 1 : k \in S\}.$$

If the pre-shift packed integer is

$$m_3(S) = \sum_{k \in S} 2^k,$$

then the shifted packed integer is

$$m_4(S^{(4)}) = \sum_{k \in S} 2^{k+1}.$$

Lemma A.9.9 (shifted packed-mask doubling)**.** *For every finite inhabited support S,*

$$m_4(S^{(4)}) = 2\, m_3(S).$$

Proof. The proof is the direct calculation

$$m_4(S^{(4)}) = \sum_{k \in S} 2^{k+1} = 2 \sum_{k \in S} 2^k = 2\, m_3(S).$$

□

Remark A.9.10 (Example: route and mask)**.** *The pre-shift route payload $\langle 5, 12 \rangle$ is read in shifted as $\langle 6, 13 \rangle$. The support $\{5, 12\}$ has packed value $2^5 + 2^{12} = 4128$ before migration and packed value $2^6 + 2^{13} = 8256$ after migration. The packed integer doubles; the support backtranslates by subtracting one from inhabited support entries.*

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A.9.5 Two-sided tensor grammar under shifted

Definition A.9.11 (shifted tensor-payload migration)**.** *In a two-sided structural tensor expression, the shifted shift acts only on QAM or router tag payloads carried by index positions. It does not alter front/back sidedness, upper/lower variance, classical upper–lower contraction, or the requirement that same-variance comparisons use explicit comparison or selector tensors.*

Remark A.9.12 (Example: front-side migration)**.** *The front-side route display*

$$\mathcal{T}^{\langle 5,12\rangle}{}_{\langle 2\rangle}{}^{i}{}_{j}$$

is read in shifted as

$$\mathcal{T}^{\langle 6,13\rangle}{}_{\langle 3\rangle}{}^{i}{}_{j}.$$

The expression changes its tag payloads, not its tensor grammar.

Remark A.9.13 (Example: full payload migration)**.** *If both front-side and back-side payloads are tagged, then*

$$\mathcal{T}^{\langle 5,12\rangle}{}_{\langle 2\rangle}{}^{\langle 7\rangle}{}_{\langle 10\rangle}$$

is read in shifted as

$$\mathcal{T}^{\langle 6,13\rangle}{}_{\langle 3\rangle}{}^{\langle 8\rangle}{}_{\langle 11\rangle}.$$

The front/back positions and the upper/lower variance are preserved.

Theorem A.9.14 (shifted tensor-grammar preservation)**.** *The global shifted QAM Hilbert-shift preserves the two-sided structural tensor grammar.*

Proof. The shift is a function on tag payloads. It does not move an index from front to back, from back to front, from upper to lower, or from lower to upper. Therefore the grammar of sidedness and variance is unchanged. Since classical upper–lower contraction and explicit same-variance comparison are defined by position and variance rather than by the numerical tag name, those rules are preserved. □

A.9.6 shifted router and readout activation

Definition A.9.15 (shifted router object)**.** *A shifted router object is a guarded router object whose inhabited kind registry and route-mask registry are read in the shifted shifted namespace, while its collars, trace records, backtranslation ledger, and guard record remain explicit.*

Definition A.9.16 (shifted readout object)**.** *A shifted readout object is a guarded readout package whose QAM/router labels are read in the shifted shifted namespace, while its readout target, comparison tensor, boundary collar, residual map, and non-claim guard remain unchanged.*

Theorem A.9.17 (shifted activation conservativity)**.** *Every finite guarded shifted router/readout package admitted by Definitions A.9.15 and A.9.16 backtranslates on inhabited components to a finite guarded pre-shift package with the same payloads and the same theorematic status.*

Proof. Apply $\mathrm{B}_4(a) = a - 1$ to every inhabited shifted tag. By Lemma A.9.6, this recovers the corresponding pre-shift tag on inhabited components. Collars, trace records, comparison tensors, readout targets, boundary collars, residual maps, and non-claim guards are kept fixed by definition. The reserved value 0 has no inhabited backtranslation, so it cannot create a new active component. Hence the package backtranslates without changing payloads or theorematic status. □

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A.9.7 handoff

Remark A.9.18 (non-claims)**.** *This shifted activation release-final state performs the namespace decision, but it does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, does not claim Turing-completeness, and does not replace standard QFT/QED/Lifshitz or general-relativistic calculations. It is a language and router/readout migration step.*

Theorem A.9.19 (handoff theorem)**.** *The local proof line layer is a well-formed global QAM Hilbert-shift activation release-final state over the archival stage release line.*

Proof. Definition A.9.1 fixes the global namespace convention. Definition A.9.2 provides the backtranslation ledger. The route and mask rules preserve finite payloads by Lemma A.9.9. The two-sided tensor grammar is preserved by Theorem A.9.14. Finally, finite router/readout packages remain conservative under Theorem A.9.17. Therefore the layer is well formed as a global QAM language activation release-final state, while all stated non-claims remain in force. □

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A.10 Occurrence Ledger and Shift-Audit Map

Status. This layer is an occurrence ledger for the shifted global QAM Hilbert-shift. Version 4.0.0-fixed the governing shifted namespace. The present layer records where that shift is allowed to act, where it is forbidden to act, and where a backtranslation or historical reading must be retained. It is an audit map, not a new physical promotion.

A.10.1 Occurrence classes

Definition A.10.1 (shifted occurrence class)**.** *A* shifted occurrence class *is a declared role assigned to a visible tag, mask, route, tensor-index payload, or OP-facing label in the Active Core. The role determines whether the global shifted shift is applied, whether the object is kept historical, or whether the object is read only through an explicit backtranslation ledger.*

Definition A.10.2 (The five primary occurrence classes)**.** *The occurrence ledger uses the following primary classes.*

1. ***Class A: historical/pre-shift witness occurrences.*** *These remain unchanged and are read through the Historical Witness or through the backtranslation ledger.*
2. ***Class B: active shifted router occurrences.*** *These are read in the shifted shifted namespace by default.*
3. ***Class C: example, table, route, and mask occurrences.*** *These must carry a shifted display and, when useful, a pre-shift backtranslation display.*
4. ***Class D: two-sided tensor-index payload occurrences.*** *These shift only their QAM/router tag payloads. The tensor grammar itself is not shifted.*
5. ***Class E: OP-facing and physical-readout occurrences.*** *These may have shifted router labels, but their theorematic status, physical readout target, residual map, and non-claim guard remain unchanged.*

Remark A.10.3 (Why occurrence classes are needed)**.** *Without occurrence classes, the symbol* 5 *could ambiguously mean a pre-shift tag, a shifted tag, a packed mask integer, a tensor component, or a mere example number. The occurrence ledger prevents that ambiguity by assigning a role before any shift is read.*

A.10.2 Class A: historical and pre-shift witness occurrences

Definition A.10.4 (Class A occurrence)**.** *A Class A occurrence is any tag, route, mask, or notation block that belongs to the preserved historical line, to the pre-shift witness text, or to an explicitly quoted earlier version. Class A occurrences are not rewritten by the shifted Active Core. They remain recoverable by the Historical Witness and readable by backtranslation.*

Lemma A.10.5 (Class A preservation)**.** *The shifted occurrence ledger preserves Class A occurrences without applying the global shift to their displayed historical form.*

Proof. Class A is defined as historical or pre-shift witness material. Its function is to preserve genealogy and reconstruction memory, not to govern the active shifted namespace. Therefore applying the shifted shift directly to Class A would corrupt the witness role. The correct operation is preservation plus optional backtranslation annotation. □

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A.10.3 Class B: active shifted router occurrences

Definition A.10.6 (Class B occurrence)**.** *A Class B occurrence is a tag or route occurrence inside an active shifted router object, shifted kind registry, shifted route-mask registry, or shifted machine-language package. Class B occurrences are read in the shifted namespace:*

$$k_{\mathrm{pre}} \longmapsto k_{\mathrm{shift}} = k_{\mathrm{pre}} + 1.$$

Remark A.10.7 (Class B default)**.** *From the local proof line onward, active router-language statements default to Class B unless the text explicitly marks the occurrence as historical, illustrative, or unshifted for comparison.*

Lemma A.10.8 (Class B backtranslation)**.** *Every finite Class B tag family with inhabited shifted tags has a componentwise backtranslation by* $\mathrm{B}_4(a) = a - 1$.

Proof. Class B contains active inhabited shifted router labels. By the shifted ledger, every inhabited shifted tag satisfies $a \geq 1$, so $\mathrm{B}_4(a) = a-1$ is defined. Applying this componentwise gives the pre-shift label family. □

A.10.4 Class C: examples, tables, routes, and masks

Definition A.10.9 (Class C occurrence)**.** *A Class C occurrence is a displayed example, table entry, route payload, support set, or packed mask integer used for exposition or audit. Class C occurrences must declare whether the displayed value is a pre-shift value, a shifted value, or a packed integer derived from a support.*

Definition A.10.10 (Class C display contract)**.** *A Class C route or mask example is well displayed if it separates:*

$$\textit{support}, \qquad \textit{route payload}, \qquad \textit{packed integer}.$$

In particular, support migration is

$$S \mapsto S^{(4)} = \{k + 1 : k \in S\},$$

while packed-mask migration is

$$m_3(S) = \sum_{k \in S} 2^k \quad \mapsto \quad m_4(S^{(4)}) = 2m_3(S).$$

Remark A.10.11 (Class C warning)**.** *The packed integer should not be shifted by adding* 1*. The support is shifted by adding* 1 *to each inhabited support entry; the packed integer then changes according to the support rule, often by doubling in the pure support-shift case.*

A.10.5 Class D: two-sided tensor-index payloads

Definition A.10.12 (Class D occurrence)**.** *A Class D occurrence is a QAM/router tag payload carried inside a front-side or back-side index position of a two-sided structural tensor expression. The shifted shift acts on the payload label only. It does not alter sidedness, variance, ordinary upper–lower contraction, or explicit same-variance comparison.*

Proposition A.10.13 (Class D grammar invariance)**.** *Class D migration preserves the two-sided structural tensor grammar.*

Proof. The migration rule changes only the numerical or route-label payload carried by an index. It does not move that payload from the front side to the back side, from the back side to the front side, from upper position to lower position, or from lower position to upper position. Thus the grammar of sidedness and variance is unchanged. □

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A.10.6 Class E: OP-facing and physical-readout occurrences

Definition A.10.14 (Class E occurrence)**.** *A Class E occurrence is an OP-facing label, residual package, physical-readout package, boundary-readout package, or diagnostic object whose router address may be shifted but whose physical or theorematic status is not changed by the shift.*

Definition A.10.15 (Class E invariance contract)**.** *A Class E occurrence remains valid under shifted only if the following data remain unchanged:*

1. *its OP status;*
2. *its physical or mathematical readout target;*
3. *its comparison tensor or boundary collar;*
4. *its residual map or diagnostic map;*
5. *its non-claim guard.*

Lemma A.10.16 (Class E status preservation)**.** *The shifted occurrence ledger does not close OP 1/4 or OP 5(B).*

Proof. Class E migration may reindex router labels, but Definition A.10.15 fixes the OP status and non-claim guard. Since no closure certificate is added by reindexing, the OP status remains unchanged. □

A.10.7 Occurrence decision table

Class	Where it occurs	Shift action	Required audit note
A	Historical/pre-shift witness blocks	preserve displayed form	read via witness/backtranslation
B	active router/kind/route registries	shift inhabited tags	keep B_4 ledger
C	examples, tables, masks, routes	display pre-shift and shifted roles	separate support from packed integer
D	tensor-index tag payloads	shift payload only	preserve sidedness/variance
E	OP-facing/readout packages	shift router address only	preserve OP status/guard

A.10.8 Ambiguity filters

Definition A.10.17 (Ambiguity filter)**.** *Before a visible integer, tuple, support, or tensor payload is read as shifted, the reader must determine whether it is:*

tag, *route payload*, *support set*,
packed mask integer, *tensor component*, *historical witness value.*

Only tag, route-payload, support, and tag-payload occurrences are subject to the shifted shift.

Remark A.10.18 (Packed values are not tags)**.** *A packed integer such as* 4128 *is not itself a tag occurrence. It encodes a support. The support migrates first, and the packed integer is recomputed from the migrated support.*

A.10.9 shifted occurrence-ledger conservativity

Theorem A.10.19 (Occurrence-ledger conservativity)**.** *Every finite shifted occurrence family classified by Classes A–E has an honest reading that either preserves its displayed historical form or backtranslates its inhabited shifted components to the pre-shift source language without changing payloads, tensor grammar, OP status, or non-claim guards.*

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Proof. Class A is preserved by Lemma A.10.5. Class B backtranslates by Lemma A.10.8. Class C separates support, route payload, and packed integer by Definition A.10.10. Class D preserves the two-sided tensor grammar by Proposition A.10.13. Class E preserves OP and readout status by Lemma A.10.16. The five cases cover the declared occurrence classes; hence the occurrence ledger is conservative for finite classified occurrence families. □

Remark A.10.20 (non-claim)**.** *The occurrence ledger does not add a new physical equation, does not close an OP, does not activate primitive mixed tensors, and does not claim Turing-completeness. It makes the global shifted shift auditable by assigning roles to visible occurrences.*

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A.11 Global Reindexed Display Layer

Status. This layer adds normalized side-by-side displays for the global shifted QAM Hilbert-shift. It does not add new physics, close OP 1/4, close OP 5(B), or alter the two-sided tensor grammar. It only standardizes how active examples are displayed in three synchronized forms:

pre-shift source, shifted display, backtranslation.

A.11.1 Reindexed display triple

Definition A.11.1 (Reindexed display triple)**.** *A* shifted reindexed display triple *is a finite record*

$$\mathfrak{D}_4(X) = (X_{\mathrm{pre}}, X_{\mathrm{shift}}, \mathrm{B}_4(X_{\mathrm{shift}})),$$

where X_{pre} *is the pre-shift source display,* X_{shift} *is the shifted shifted display, and* $\mathrm{B}_4(X_{\mathrm{shift}})$ *is the explicit backtranslation on inhabited components.*

Remark A.11.2 (Display triples are audit objects)**.** *A display triple is not a new theorematic certificate. It is an audit object showing that the visible shifted display can be read back to the pre-shift source where the components are inhabited.*

A.11.2 Tag and route display triples

Definition A.11.3 (Tag display triple)**.** *For an inhabited pre-shift tag* k*, the shifted tag display triple is*

$$\mathfrak{D}_4(k) = (k,\ k+1,\ \mathrm{B}_4(k+1) = k).$$

The reserved value 0_{shift} *has no inhabited pre-shift source and is displayed separately as*

$$(\varnothing_{\mathrm{active}},\ 0_{\mathrm{shift}},\ \textit{no inhabited backtranslation}).$$

Definition A.11.4 (Route display triple)**.** *For a finite pre-shift route payload*

$$r = \langle k_1, \ldots, k_n \rangle,$$

the shifted display triple is

$$\mathfrak{D}_4(r) = (\langle k_1, \ldots, k_n \rangle,\ \langle k_1 + 1, \ldots, k_n + 1 \rangle,\ \langle k_1, \ldots, k_n \rangle)\,.$$

pre-shift source	shifted display	backtranslation
5	6	$\mathrm{B}_4(6) = 5$
12	13	$\mathrm{B}_4(13) = 12$
$\langle 5, 12 \rangle$	$\langle 6, 13 \rangle$	$\langle 5, 12 \rangle$

A.11.3 Support and packed-mask display triples

Definition A.11.5 (Support display triple)**.** *For a finite inhabited support set* S*, the support display triple is*

$$\mathfrak{D}_4(S) = (S,\ S^{(4)},\ \mathrm{B}_4(S^{(4)})), \qquad S^{(4)} = \{k + 1 : k \in S\}.$$

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Definition A.11.6 (Packed-mask display triple)**.** *For a finite inhabited support S, the packed-mask display triple is*

$$\mathfrak{D}_4(m(S)) = \big(m_3(S),\ m_4(S^{(4)}),\ S = \mathrm{B}_4(S^{(4)})\big)\,.$$

The backtranslation is attached to the support, not to the packed integer by subtraction.

pre-shift source	shifted display	backtranslation
$\{5,12\}$	$\{6,13\}$	$\{5,12\}$
$2^5 + 2^{12} = 4128$	$2^6 + 2^{13} = 8256$	support returns to $\{5,12\}$

Lemma A.11.7 (Packed-mask display consistency)**.** *The packed-mask display triple for a finite support S is consistent with the packed-mask doubling rule:*

$$m_4(S^{(4)}) = 2\, m_3(S).$$

Proof. This is exactly Lemma A.10.8 applied to the inhabited support entries together with the packed-mask calculation

$$\sum_{k\in S} 2^{k+1} = 2\sum_{k\in S} 2^k.$$

The displayed packed integer is recomputed from the migrated support, not shifted as a tag. □

A.11.4 Two-sided tensor display triples

Definition A.11.8 (Tensor display triple)**.** *A two-sided structural tensor display triple is obtained by applying the shifted tag shift to QAM/router payloads inside index positions while leaving sidedness and variance unchanged.*

pre-shift source	shifted display	backtranslation
${}^{\langle 5,12\rangle}{}_{\langle 2\rangle}\mathcal{T}^{i}{}_{j}$	${}^{\langle 6,13\rangle}{}_{\langle 3\rangle}\mathcal{T}^{i}{}_{j}$	${}^{\langle 5,12\rangle}{}_{\langle 2\rangle}\mathcal{T}^{i}{}_{j}$
${}^{\langle 5,12\rangle}{}_{\langle 2\rangle}\mathcal{T}^{\langle 7\rangle}{}_{\langle 10\rangle}$	${}^{\langle 6,13\rangle}{}_{\langle 3\rangle}\mathcal{T}^{\langle 8\rangle}{}_{\langle 11\rangle}$	${}^{\langle 5,12\rangle}{}_{\langle 2\rangle}\mathcal{T}^{\langle 7\rangle}{}_{\langle 10\rangle}$

Proposition A.11.9 (Display triples preserve tensor grammar)**.** *Tensor display triples preserve the two-sided structural tensor grammar.*

Proof. Only QAM/router tag payloads inside the displayed index positions are shifted. The index positions themselves are not moved. Hence front-side remains front-side, back-side remains back-side, upper remains upper, lower remains lower, and the ordinary contraction/-comparison rules remain those of the underlying tensor grammar. □

A.11.5 OP-facing display triples

Definition A.11.10 (OP-facing display triple)**.** *An OP-facing display triple records only the router-address migration of an OP-facing package. Its OP status, residual target, comparison surface, and non-claim guard are displayed as invariant data.*

pre-shift address	shifted address	invariant data
OP package tag k	OP package tag $k+1$	OP status unchanged; guard unchanged
residual route $\langle 5,12\rangle$	residual route $\langle 6,13\rangle$	residual target unchanged
boundary readout route $\langle 2\rangle$	boundary readout route $\langle 3\rangle$	boundary collar unchanged

Lemma A.11.11 (OP-facing display triples do not close OPs)**.** *OP-facing display triples do not close OP 1/4 or OP 5(B).*

Proof. By Definition A.11.10, the triple changes only router-address display. It explicitly leaves OP status, residual target, comparison surface, and non-claim guard invariant. Therefore no OP closure certificate is introduced. □

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A.11.6 Display-layer conservativity

Theorem A.11.12 (Global reindexed display conservativity)**.** *Every finite display triple has a backtranslation to its pre-shift source display on inhabited components and preserves the occurrence class assigned in the occurrence ledger.*

Proof. Tag and route triples backtranslate componentwise by $\mathrm{B}_4(a) = a - 1$. Support triples backtranslate by applying the same rule to support entries. Packed-mask triples backtranslate through their support sets, not by subtracting from the packed integer. Tensor triples preserve grammar by Proposition A.11.9. OP-facing triples preserve status by Lemma A.11.11. Hence the display layer is conservative over the occurrence ledger. □

Remark A.11.13 (non-claim)**.** *The layer is a display layer only. It does not add a new physical equation, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, and does not claim Turing-completeness.*

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A.12 Physical Readout Compatibility Check

Status. This layer checks that the global shifted QAM Hilbert-shift is compatible with the active physical and mathematical readout packages. It does not promote a new physical law. It verifies that the shifted shift changes router addresses and tag payloads only, while readout targets, comparison surfaces, residual maps, and non-claim guards remain fixed.

A.12.1 Readout compatibility principle

Definition A.12.1 (Readout compatibility package)**.** *A* shifted readout compatibility package *is a tuple*

$$\mathcal{C}_4(R) = (R_{\mathrm{pre}}, R_{\mathrm{shift}}, \mathrm{B}_4(R_{\mathrm{shift}}), T_R, G_R, S_R),$$

where R_{pre} is the pre-shift router address, R_{shift} is the shifted shifted router address, $\mathrm{B}_4(R_{\mathrm{shift}})$ is the inhabited backtranslation, T_R is the readout target, G_R is the guard record, and S_R is the theorematic status record.

Definition A.12.2 (Readout compatibility)**.** *A readout package is* shifted-compatible *if the migration from R_{pre} to R_{shift} changes only the router address and leaves T_R, G_R, and S_R unchanged.*

Lemma A.12.3 (Readout compatibility is address-local)**.** *If a readout package is shifted-compatible, then the global shifted QAM Hilbert-shift does not change its physical or mathematical target.*

Proof. By Definition A.12.2, the shift is applied to the router address only. The readout target T_R is explicitly preserved. Hence the physical or mathematical target of the package is unchanged. □

A.12.2 HC-boundary readout compatibility

Definition A.12.4 (HC-boundary compatibility record)**.** *The HC-boundary readout package is shifted-compatible when the local structural shift*

$$0_{\mathrm{mask}} \longrightarrow 1_{\mathrm{mask}}$$

remains the HC-boundary activation principle, while global shifted router labels are read in the shifted namespace.

Proposition A.12.5 (HC-boundary readout survives the global shift)**.** *The global shifted QAM Hilbert-shift preserves the historical release line HC-boundary activation reading.*

Proof. The HC-boundary reading belongs to Shift A: the local transition from inactive mask state to first active readout state. The shifted QAM Hilbert-shift belongs to Shift B: the global namespace migration of inhabited tags. Since the local proof chain separate these two acts, the global shift cannot redefine the local HC-boundary activation. It can only change how the active router address is displayed. □

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A.12.3 Photon-compatible null-side readout compatibility

Definition A.12.6 (Null-side readout compatibility)**.** *A photon-compatible null-side readout is shifted-compatible when the null-side readout target and its guard remain unchanged while its router address is displayed in the shifted shifted namespace.*

Lemma A.12.7 (Null-side target preservation)**.** *The shifted shift preserves the null-side readout target of a photon-compatible readout package.*

Proof. By Definition A.12.6, the null-side target is part of the invariant data. The shifted shift operates only on router address labels, so the null-side target remains fixed. □

Remark A.12.8 (No photon proof claim)**.** *This compatibility check does not prove a new photon theory. It only says that the already guarded null-side readout remains address-stable under the shifted tag migration.*

A.12.4 HC-normalized singularity projection compatibility

Definition A.12.9 (Singularity projection compatibility)**.** *An HC-normalized singularity projection package is shifted-compatible when its projection target, HC-normalization guard, and residual diagnostic remain unchanged under router-address migration.*

Proposition A.12.10 (Singularity projection status preservation)**.** *The shifted shift does not promote an HC-normalized singularity projection package from diagnostic status to theorematic closure.*

Proof. The migration changes the router address. It does not add a new projection identity, a new residual estimate, or a new closure certificate. The diagnostic guard is part of the invariant record in Definition A.12.9. Therefore the theorematic status is preserved. □

A.12.5 Casimir boundary readout compatibility

Definition A.12.11 (Casimir boundary compatibility)**.** *A Casimir boundary readout package is shifted-compatible when its boundary collar, comparison surface, and non-replacement guard remain unchanged while its router-address labels are shifted.*

Lemma A.12.12 (Casimir boundary collar preservation)**.** *The shifted shift preserves the boundary collar of a shifted-compatible Casimir boundary readout package.*

Proof. By Definition A.12.11, the boundary collar is invariant data. The shift acts only on router labels. Therefore the boundary collar is unchanged. □

Remark A.12.13 (No replacement of standard Casimir calculations)**.** *The compatibility statement does not replace standard QFT, QED, or Lifshitz calculations. It keeps the Companion readout package address-stable under the shifted namespace shift.*

A.12.6 OP-facing residual package compatibility

Definition A.12.14 (OP-facing residual compatibility)**.** *An OP-facing residual package is shifted-compatible when its route address may be shifted but its residual target, status flag, and non-claim guard remain unchanged.*

Lemma A.12.15 (OP-facing residual status preservation)**.** *The shifted shift does not close OP 1/4 or OP 5(B).*

Proof. By Definition A.12.14, OP-facing residual packages preserve their status flags and non-claim guards. A router address migration is not a proof certificate. Therefore the OP status remains unchanged. □

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A.12.7 Compatibility table

Readout package	Shifted data	Invariant data
HC-boundary	active router address	local HC-boundary activation principle
null-side/photon-compatible	router address	null-side target and guard
singularity projection	router address	projection target, HC guard, residual diagnostic
Casimir boundary	router address	boundary collar, comparison surface, non-replacement guard
OP-facing residual	route address	residual target, OP status, non-claim guard

A.12.8 Physical-readout compatibility theorem

Theorem A.12.16 (Physical readout compatibility under the shifted shift)**.** *For the readout packages listed in Section A.12.7, the global shifted QAM Hilbert-shift preserves the declared readout target, comparison or boundary data, residual data, and non-claim guard. It changes only the router-address display.*

Proof. Each listed package is declared shifted-compatible by its corresponding definition: HC-boundary compatibility by Definition A.12.4, null-side compatibility by Definition A.12.6, singularity projection compatibility by Definition A.12.9, Casimir boundary compatibility by Definition A.12.11, and OP-facing residual compatibility by Definition A.12.14. In each case, the shifted component is the router or route address. The physical or mathematical readout target and the guard/status records are invariant data. Therefore the shifted shift is compatible with these readout packages without changing their theorematic status. □

Remark A.12.17 (non-claim)**.** *The layer is a compatibility check. It does not add a new physical equation, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, and does not claim Turing-completeness.*

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A.13 Governing shifted Normal Form and Release-Freeze Handoff

Status. This layer compresses the archival stage-the local proof chain migration work into one governing normal form. It is intended as a release-freeze handoff: after this layer, new active QAM/router/readout material should be written either in the shifted normal form or with an explicit pre-shift backtranslation record.

A.13.1 Governing normal-form object

Definition A.13.1 (Governing shifted normal-form object)**.** *A governing shifted normal-form object is a finite record*

$$\mathcal{N}_4(X) = (C_X,\ X_{\mathrm{shift}},\ \mathrm{B}_4(X_{\mathrm{shift}}),\ P_X,\ G_X,\ S_X),$$

where C_X is the occurrence class, X_{shift} is the shifted shifted display, $\mathrm{B}_4(X_{\mathrm{shift}})$ is the inhabited-component backtranslation, P_X is the payload or readout target, G_X is the guard record, and S_X is the theorematic status record.

Remark A.13.2 (Why the pre-shift source is not always the first field)**.** *The shifted Active Core now governs by the shifted namespace. Therefore the normal form puts the shifted display first and keeps the pre-shift source as a backtranslation ledger when needed. Historical source displays remain available through the Historical Witness.*

A.13.2 Normal-form admissibility

Definition A.13.3 (shifted-normal admissibility)**.** *A finite object is* shifted-normal admissible *if:*

1. *it carries an occurrence class C_X;*
2. *all active inhabited tags are read in the shifted shifted namespace;*
3. *every shifted inhabited tag has a backtranslation by B_4;*
4. *packed masks are reconstructed from shifted supports, not shifted as integers;*
5. *tensor-index payload migration preserves sidedness and variance;*
6. *physical/readout payloads preserve their target, collar, residual map, guard, and theorematic status.*

Lemma A.13.4 (Normal admissibility implies ledger readability)**.** *Every shifted-normal admissible object is readable through the namespace ledger, the occurrence ledger, the display triple, and the readout compatibility check.*

Proof. The occurrence class invokes the local pass. The shifted tag convention invokes the local pass. The backtranslation field invokes whenever a visible pre-shift comparison is needed. The payload/guard/status fields invoke for readout packages and the tensor-grammar rules for tensor payloads. Hence the object is readable through the whole the local proof chain ledger chain. □

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A.13.3 Normal forms by class

Definition A.13.5 (Class-wise shifted-namespace normal forms)**.** *The following class-wise normal forms are used:*

Class A: $(A,\ X_{\mathrm{hist}},\ \text{witness/backtranslation},\ P_X,\ G_X,\ S_X)$,
Class B: $(B,\ X_{\mathrm{shift}},\ \mathrm{B}_4(X_{\mathrm{shift}}),\ P_X,\ G_X,\ S_X)$,
Class C: $(C,$ *shifted display*, *pre-shift display*, *support/payload split*, $G_X,\ S_X)$,
Class D: $(D,$ *shifted tag payload*, *backtranslated payload*, *tensor grammar*, $G_X,\ S_X)$,
Class E: $(E,$ *shifted route address*, *backtranslated address*, *readout target*, $G_X,\ S_X)$.

Remark A.13.6 (Class C special rule)**.** *For Class C objects, a packed mask integer is never the primary shifted object. The primary shifted object is the support or route payload. The packed integer is recomputed from that shifted support.*

A.13.4 Normal-form examples

Class	shifted display	backtranslation	invariant data
B	$\langle 6,13\rangle$	$\langle 5,12\rangle$	router role
C	support $\{6,13\}$, mask 8256	support $\{5,12\}$, mask 4128	support/mask split
D	${}^{\langle 6,13\rangle}{}_{\langle 3\rangle}\mathcal{T}^{i}{}_{j}$	${}^{\langle 5,12\rangle}{}_{\langle 2\rangle}\mathcal{T}^{i}{}_{j}$	sidedness/variance
E	route $\langle 6,13\rangle$	route $\langle 5,12\rangle$	readout target and guard

A.13.5 Release-freeze discipline

Definition A.13.7 (shifted release-freeze discipline)**.** *The shifted release-freeze discipline is the rule that, after the local pass, every new active QAM/router/readout object must either:*

1. *be written directly as a shifted-normal admissible object; or*
2. *be marked as historical/pre-shift and supplied with an explicit backtranslation or witness pointer.*

Lemma A.13.8 (Release-freeze prevents mixed namespace drift)**.** *The shifted release-freeze discipline prevents uncontrolled mixing of pre-shift and shifted tag namespaces.*

Proof. Every new object must either be shifted-normal or explicitly historical/pre-shift. In the first case, inhabited tags are shifted and backtranslated by the shifted ledger. In the second case, the object is not treated as an active shifted governing object. Thus no object may silently mix pre-shift and shifted tag meanings. □

A.13.6 Governing shifted normal-form theorem

Theorem A.13.9 (Governing shifted normal-form theorem)**.** *The normal form is conservative over the local proof chain shifted migration chain: every finite shifted-normal admissible object has a controlled reading in the shifted shifted namespace and an inhabited-component backtranslation that preserves payloads, tensor grammar, physical/readout targets, guards, and theorematic status.*

Proof. Let $\mathcal{N}_4(X)$ be shifted-normal admissible. By Definition A.13.3, it carries an occurrence class and a backtranslation ledger for all inhabited shifted tags. Class A is handled by witness preservation. Class B is handled by router backtranslation. Class C is handled by

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support/payload separation and packed-mask recomputation. Class D is handled by preservation of sidedness and variance. Class E is handled by preservation of readout target, guard, and theorematic status. These are exactly the local proof chain conservation mechanisms. Hence the normal form is conservative over the shifted migration chain. □

A.13.7 archival stage release-final handoff

Definition A.13.10 (archival stage release-final readiness)**.** *The archival stage line is ready for a release-final pass when:*

1. *the shifted namespace is fixed;*
2. *the occurrence ledger is present;*
3. *the reindexed display triples are present;*
4. *physical/readout compatibility is present;*
5. *the governing shifted-namespace normal form is present;*
6. *all non-claims remain explicit.*

Remark A.13.11 (non-claim)**.** *The layer is a normal-form and release-freeze handoff. It does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, does not claim Turing-completeness, and does not replace standard physical calculations.*

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A.14 archival stage Release-Final Closure Note

Status. This closure note records the archival stage Active Core as the first shifted release-final state built from the local proof chain migration chain. The release-final state contains the global QAM Hilbert-shift activation, occurrence-ledger audit map, reindexed display layer, physical-readout compatibility check, and governing shifted-namespace normal form. It records historical DOI-bearing status for that layer; current archival stage DOI metadata is governed by the frontmatter Version and DOI record.

A.14.1 Release-release-final state contents

Definition A.14.1 (archival stage release-final content)**.** *The archival stage historical source-hygiene layer consists of the following recorded layers:*

1. *the local pass: global QAM Hilbert-shift activation;*
2. *the local pass: occurrence ledger and shift-audit map;*
3. *the local pass: global reindexed display layer;*
4. *the local pass: physical readout compatibility check;*
5. *the local pass: governing shifted-namespace normal form and release-freeze handoff.*

Definition A.14.2 (archival stage release-final governing rule)**.** *From this release-final state onward, active QAM/router/readout objects are governed by the shifted-namespace normal form unless explicitly marked as historical or pre-shift witness material. Inhabited active tags are read in the shifted namespace, and the reserved tag* 0_{shift} *denotes the empty/no-active-kind marker.*

A.14.2 Release-release-final state conservativity

Theorem A.14.3 (archival stage release-final conservativity)**.** *Every finite active object admitted by the archival stage release-final normal form has an inhabited-component backtranslation that preserves its payload, tensor grammar, readout target, guard record, and theorematic status.*

Proof. The result is the direct combination of the namespace ledger, the local pass occurrence ledger, the display triples, the physical-readout compatibility check, and the governing normal-form theorem. The ledger supplies $\mathrm{B}_4(a) = a - 1$ on inhabited shifted tags. The ledger assigns a role to visible occurrences. The display layer supplies the explicit pre-shift/shifted/backtranslation display form. The compatibility check preserves physical and mathematical readout data. The normal form requires all new active material to be shifted-normal or explicitly historical/pre-shift. Therefore the release-final state is conservative over the prepared migration chain. □

A.14.3 Release-release-final state non-claims

Remark A.14.4 (archival stage release-final non-claims)**.** *The archival stage historical source-hygiene layer does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, does not claim Turing-completeness, and does not replace standard QFT/QED/Lifshitz or general-relativistic calculations. It is a global QAM/router/readout language migration and normal-form release release-final state.*

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A.15 archival stage Historical Remark Triage and Reader-Hygiene Cleanup

Status. This archival stage layer is a cleanup and reader-hygiene layer. It does not add a new theorematic claim, does not close OP 1/4 or OP 5(B), and does not change the archival stage global QAM Hilbert-shift. Its purpose is to separate governing shifted statements from historical genealogy remarks.

A.15.1 Triage principle

Definition A.15.1 (Historical remark triage classes)**.** *Historical or version-genealogical remarks in the Active Core are assigned one of the following cleanup roles:*

1. ***H1: governing guard.*** *The remark remains in the Active Core because it prevents overclaim, preserves OP status, separates Shift A from Shift B, or protects tensor/readout grammar.*

2. ***H2: compact genealogy.*** *The remark may remain only in compressed form, usually as a short version-genealogy pointer.*

3. ***H3: witness-only.*** *The remark belongs to the Historical Witness or release genealogy and should not govern the Active Core.*

Definition A.15.2 (Governing statement criterion)**.** *A remark is a governing statement only if it changes how an active shifted object is read, guarded, classified, backtranslated, or status-preserved. A remark that only explains how a release-final state stage was reached is historical genealogy.*

Remark A.15.3 (No deletion of proof memory)**.** *The cleanup does not delete proof memory. It only changes where such memory is allowed to govern. The Active Core governs through the shifted-namespace normal form; the Historical Witness preserves genealogy.*

A.15.2 Active-Core retention rules

Proposition A.15.4 (Active-Core retention rule)**.** *A historical remark should remain in the Active Core only if it satisfies at least one of the following conditions:*

1. *it protects the distinction between local HC-boundary activation and global shifted QAM namespace migration;*

2. *it protects the distinction between tag, support, packed mask, route payload, tensor component, and historical witness value;*

3. *it protects tensor grammar, readout target, guard, residual map, or OP status;*

4. *it points to the Historical Witness as proof memory rather than as a second active theorematic line.*

Proof. These four cases correspond exactly to the places where historical context prevents an incorrect active reading. In all other cases, the shifted-namespace normal form already supplies the governing rule. Therefore long release-final state-stage or development-stage remarks may be compressed or moved to the witness layer without changing the mathematics. □

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A.15.3 Version-genealogy compression

Definition A.15.5 (Compressed shifted genealogy note)**.** *The archival stage migration chain is summarized as follows:*

historical release line: local HC-boundary activation $0_{\mathrm{mask}} \to 1_{\mathrm{mask}}$,
historical release line: migration map, backtranslation, and concrete ledger,
the local proof line: global shifted namespace activation,
the local proof line: occurrence ledger and shift-audit map,
the local proof line: pre-shift/shifted/backtranslation display triples,
the local proof line: physical-readout compatibility check,
the local proof line: governing shifted-namespace normal form and release-freeze handoff.

This note replaces repeated long-form release-final state-stage narrative in the governing reading of the Active Core.

Remark A.15.6 (Current governing reading)**.** *In the Active Core, shifted is no longer a future release-final state language. It is the current governing namespace for active QAM/router/readout objects. Historical and pre-shift material remains readable through the Historical Witness and the backtranslation ledger.*

A.15.4 Cleanup audit result

Theorem A.15.7 (Historical remark cleanup conservativity)**.** *The archival stage historical remark triage is conservative over archival stage: it does not change the shifted namespace, the backtranslation map, the occurrence classes, the display triples, the physical-readout compatibility checks, or the governing shifted-namespace normal form.*

Proof. The cleanup assigns roles to historical remarks and compresses genealogy into a single note. It does not modify the definitions of

$$\Sigma_4(k) = k + 1, \qquad \mathrm{B}_4(a) = a - 1, \qquad \mathcal{N}_4(X) = (C_X, X_{\mathrm{shift}}, \mathrm{B}_4(X_{\mathrm{shift}}), P_X, G_X, S_X).$$

It also leaves the non-claim guards for OP 1/4 and OP 5(B) unchanged. Hence the mathematical content of archival stage is preserved. □

Remark A.15.8 (archival stage non-claim)**.** *Version 4.0.13 is a release-final cleanup version. It does not add new physics, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, and does not claim Turing-completeness.*

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A.16 archival stage Reader Navigation Cleanup

Status. This archival stage layer is a reader-navigation cleanup. It does not change the shifted shift, the occurrence ledger, display triples, readout compatibility, normal form, OP status, or Historical Witness.

A.16.1 Navigation principle

Definition A.16.1 (Current reader-navigation principle)**.** *The current reader-navigation principle is:*

Active Core governs, *Historical Witness preserves,* *backtranslation connects.*

Thus active shifted statements are read in the shifted-namespace normal form, historical material is read as proof memory, and pre-shift displays are connected to active shifted displays only through explicit backtranslation records.

Definition A.16.2 (Governing-map roles)**.** *The following roles are used for reader navigation:*

1. ***Governing active role:*** *current shifted definitions, theorems, normal forms, guards, and OP-status statements.*
2. ***Routing role:*** *tables, display triples, occurrence classes, and backtranslation ledgers that explain how to read active objects.*
3. ***Genealogy role:*** *historical remarks and witness material that explain how the current line was reached.*

Lemma A.16.3 (Navigation cleanup is conservative)**.** *The archival stage reader-navigation cleanup is conservative over archival stage.*

Proof. The cleanup adds navigation instructions and clarifies the current governing reading. It does not modify

$$\Sigma_4(k) = k + 1, \qquad \mathrm{B}_4(a) = a - 1, \qquad \mathcal{N}_4(X) = (C_X, X_{\mathrm{shift}}, \mathrm{B}_4(X_{\mathrm{shift}}), P_X, G_X, S_X).$$

It also preserves the Historical Witness and leaves all OP-status guards unchanged. Therefore it is conservative over archival stage. □

Remark A.16.4 (archival stage non-claim)**.** *Version 4.0.13 is a navigation release-final cleanup version. It does not add new physics, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, and does not claim Turing-completeness.*

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A.17 archival stage Notation and Symbol Hygiene Cleanup

Status. This archival stage layer is a notation-hygiene cleanup. It does not change the shifted namespace shift, the backtranslation ledger, the occurrence classes, the tensor grammar, the readout compatibility checks, OP status, or the Historical Witness. It records canonical symbol usage for the current shifted line.

A.17.1 Canonical shifted symbol registry

Definition A.17.1 (Canonical shifted symbol registry)**.** *The current shifted Active Core uses the following canonical symbols:*

$\Sigma_4(k) = k+1$	*global shift on inhabited pre-shift tags*	*language/router namespace*,
$\mathrm{B}_4(a) = a-1$	*backtranslation on inhabited shifted tags* $a \geq 1$	*partial inverse*,
$0_{\mathrm{shift}} = \varnothing_{\mathrm{active}}$	*reserved empty/no-active-kind marker*	*not inhabited*,
$0_{\mathrm{mask}} \to 1_{\mathrm{mask}}$	*local HC-boundary activation*	*Shift A*,
$\mathfrak{D}_4(X)$	*pre-shift/shifted/backtranslation display triple*	*reader/audit display*,
$\mathcal{N}_4(X)$	*governing shifted normal-form object*	*active object form.*

Remark A.17.2 (Roman backtranslation symbol)**.** *The backtranslation symbol is written as* B_4*, not as a Greek beta symbol. This avoids confusion with an additional operator, physical beta parameter, or unstated tensor component.*

A.17.2 Disjoint role vocabulary

Definition A.17.3 (Tag, route, support, packed mask, and tensor payload)**.** *The following terms are kept disjoint in the shifted Active Core:*

1. *A* tag *is an inhabited QAM/router label subject to the shifted namespace when active.*
2. *A* route payload *is an ordered finite list of tags or tag-like route entries.*
3. *A* support set *is an unordered finite set of inhabited support entries.*
4. *A* packed mask integer *is the integer encoding of a support set and is not itself shifted as a tag.*
5. *A* tensor-index payload *is a QAM/router payload carried by a two-sided tensor index position.*
6. *A* tensor component *is not automatically a QAM/router tag and is shifted only if explicitly declared as a tag payload.*

Lemma A.17.4 (Packed mask is not a tag)**.** *A packed mask integer is not shifted by* Σ_4 *as an integer.*

Proof. By Definition A.17.3, a packed mask integer encodes a support set. The shifted shift acts on inhabited support entries. The packed integer is then recomputed from the shifted support. Therefore the operation is not $m \mapsto m+1$ on the packed integer. □

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A.17.3 Canonical display forms

Definition A.17.5 (Canonical display forms)**.** *The following display forms are canonical for current shifted cleanup work:*

$$\mathfrak{D}_4(X) = (X_{\mathrm{pre}}, X_{\mathrm{shift}}, \mathrm{B}_4(X_{\mathrm{shift}})),$$

and

$$\mathcal{N}_4(X) = (C_X, X_{\mathrm{shift}}, \mathrm{B}_4(X_{\mathrm{shift}}), P_X, G_X, S_X).$$

Here C_X is the occurrence class, P_X is the payload or readout target, G_X is the guard record, and S_X is the theorematic status record.

Remark A.17.6 (Display triple versus normal form)**.** *The display triple $\mathfrak{D}_4(X)$ is a reader/audit surface. The normal form $\mathcal{N}_4(X)$ is the governing active object form. They should not be used interchangeably.*

A.17.4 Tensor notation hygiene, declared local OP1/OP4 closure, release-final

Definition A.17.7 (Tensor payload hygiene rule)**.** *The shifted shift may act on QAM/router payloads carried inside tensor-index positions. It does not act on sidedness, upper/lower variance, ordinary upper-lower contraction, or explicit same-variance comparison rules.*

Proposition A.17.8 (Tensor hygiene preservation)**.** *The notation-hygiene cleanup preserves the two-sided structural tensor grammar.*

Proof. The cleanup only records the distinction between tag payload, tensor-index position, and tensor component. It does not move indices, change variance, or alter contraction/comparison rules. Therefore the tensor grammar is preserved. □

A.17.5 Notation-hygiene conservativity

Theorem A.17.9 (Notation-hygiene conservativity)**.** *The archival stage notation-hygiene cleanup is conservative over archival stage: it changes no mathematical content of the global shift, backtranslation, occurrence ledger, display triples, readout compatibility, normal form, or OP-status guards.*

Proof. The cleanup introduces a registry and role vocabulary for existing objects. It leaves the definitions of Σ_4, B_4, $\mathfrak{D}_4$, and $\mathcal{N}_4$ unchanged. It also preserves the distinction between 0_{shift} and 0_{mask}, and it does not modify OP-status guards. Hence the cleanup is conservative. □

Remark A.17.10 (archival stage non-claim)**.** *Version 4.0.13 is a notation release-final cleanup version. It does not add new physics, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, and does not claim Turing-completeness.*

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A.18 archival stage Table, Display, and Example Normalization Cleanup

Status. This archival stage layer is a table/display/example normalization cleanup. It does not change the shifted namespace shift, the backtranslation map, the occurrence classes, the tensor grammar, readout compatibility, OP status, or the Historical Witness. It standardizes how examples should be displayed.

A.18.1 Canonical example display contract

Definition A.18.1 (Canonical example display contract)**.** *Every active shifted example that compares an old display with a current display should use the three-column contract*

pre-shift source / shifted display / backtranslation.

Equivalently, the example should be readable as the display triple

$$\mathfrak{D}_4(X) = (X_{\mathrm{pre}},\ X_{\mathrm{shift}},\ \mathrm{B}_4(X_{\mathrm{shift}}))\,.$$

Remark A.18.2 (Display contract is not a theorem)**.** *The display contract is a reader/audit convention. It does not add a proof certificate and does not change theorematic status.*

A.18.2 Normalized route and support examples

pre-shift source	shifted display	backtranslation
tag 5	tag 6	$\mathrm{B}_4(6) = 5$
tag 12	tag 13	$\mathrm{B}_4(13) = 12$
route $\langle 5, 12\rangle$	route $\langle 6, 13\rangle$	route $\langle 5, 12\rangle$
support $\{5, 12\}$	support $\{6, 13\}$	support $\{5, 12\}$

Lemma A.18.3 (Route/support display normalization)**.** *The normalized route and support examples preserve the shifted backtranslation ledger on inhabited entries.*

Proof. Each shifted entry in the displayed route or support is at least 1 and therefore has backtranslation $\mathrm{B}_4(a) = a - 1$. Applying this componentwise returns the pre-shift route or support. □

A.18.3 Normalized packed-mask example

pre-shift source	shifted display	backtranslation
support $S = \{5, 12\}$	support $S^{(4)} = \{6, 13\}$	support $S = \{5, 12\}$
packed mask $m_3(S) = 4128$	packed mask $m_4(S^{(4)}) = 8256$	via support, not by integer subtraction

Proposition A.18.4 (Packed-mask display normalization)**.** *A packed-mask display is normalized only when it separates support migration from packed-integer recomputation.*

Proof. The shifted shift acts on inhabited support entries, not on the packed integer as a tag. Once the support S is shifted to $S^{(4)}$, the packed integer is recomputed:

$$m_4(S^{(4)}) = \sum_{k \in S} 2^{k+1} = 2 \sum_{k \in S} 2^k = 2m_3(S).$$

Thus the normalized display must show support migration separately from the packed integer. □

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A.18.4 Normalized tensor-display examples

pre-shift source	shifted display	backtranslation
${}^{\langle 5,12\rangle}{}_{\langle 2\rangle}\mathcal{T}^{i}{}_{j}$	${}^{\langle 6,13\rangle}{}_{\langle 3\rangle}\mathcal{T}^{i}{}_{j}$	${}^{\langle 5,12\rangle}{}_{\langle 2\rangle}\mathcal{T}^{i}{}_{j}$
${}^{\langle 5,12\rangle}{}_{\langle 2\rangle}\mathcal{T}^{\langle 7\rangle}{}_{\langle 10\rangle}$	${}^{\langle 6,13\rangle}{}_{\langle 3\rangle}\mathcal{T}^{\langle 8\rangle}{}_{\langle 11\rangle}$	${}^{\langle 5,12\rangle}{}_{\langle 2\rangle}\mathcal{T}^{\langle 7\rangle}{}_{\langle 10\rangle}$

Lemma A.18.5 (Tensor-display normalization preserves grammar)**.** *The normalized tensor-display examples preserve sidedness, variance, contraction rules, and same-variance comparison rules.*

Proof. Only QAM/router payloads inside index positions are shifted. No index is moved between front/back positions or between upper/lower variance. Therefore the two-sided tensor grammar is unchanged. □

A.18.5 Example normalization rule

Definition A.18.6 (Example normalization rule)**.** *A shifted example is normalized if it satisfies all of the following:*

1. *it declares whether the displayed object is a tag, route payload, support set, packed mask integer, tensor-index payload, tensor component, or historical witness value;*
2. *it displays the shifted form and the backtranslation when a shifted active tag payload is present;*
3. *it separates support migration from packed-integer recomputation;*
4. *it preserves tensor sidedness and variance in tensor examples;*
5. *it preserves guard and status information in OP-facing examples.*

Theorem A.18.7 (Example-normalization conservativity)**.** *The archival stage table/display/example normalization cleanup is conservative over archival stage.*

Proof. The cleanup standardizes how examples are shown. It does not change the definitions of Σ_4, B_4, $\mathfrak{D}_4$, or $\mathcal{N}_4$. It also leaves tensor grammar, readout guards, OP status, and the Historical Witness unchanged. Therefore it is conservative over archival stage. □

Remark A.18.8 (archival stage non-claim)**.** *Version 4.0.13 is a table/display/example normalization release-final state. It does not add new physics, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, and does not claim Turing-completeness.*

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A.19 archival stage Appendix and Historical Witness Linkage Cleanup

Status. This archival stage layer is an appendix and Historical Witness linkage cleanup. It does not change the shifted namespace shift, the backtranslation map, the occurrence classes, display triples, tensor grammar, readout compatibility, OP status, or the Historical Witness. It clarifies how the Active Core and Historical Witness should be read together.

A.19.1 Split-bundle linkage principle

Definition A.19.1 (Split-bundle linkage principle)**.** *The split bundle is read by the principle*

Active Core governs, *Historical Witness preserves*, *backtranslation connects.*

The Active Core contains the current governing shifted language and theorematic status. The Historical Witness preserves proof memory, genealogy, and older readout surfaces. The backtranslation ledger explains how inhabited shifted shifted tags relate to pre-shift tags.

Remark A.19.2 (No second active theory line)**.** *The Historical Witness is not a second active theory line. If a historical statement conflicts with the current shifted-namespace normal form, the Active Core governs the current reading, while the Historical Witness records the genealogy of the older formulation.*

A.19.2 Historical Witness use classes

Definition A.19.3 (Historical Witness use classes)**.** *Historical Witness material is used in one of the following roles:*

1. ***Proof-memory role.*** *The material preserves derivation history, older proof attempts, and reconstruction context.*
2. ***Genealogy role.*** *The material explains why a current shifted object has the form it has.*
3. ***Backtranslation role.*** *The material provides a pre-shift source display that can be related to a current shifted display.*
4. ***Non-governing archive role.*** *The material is retained for auditability but does not govern current active statements.*

Lemma A.19.4 (Witness roles preserve active governance)**.** *The Historical Witness use classes preserve the Active Core as the governing current line.*

Proof. Each use class assigns historical material a support role: proof memory, genealogy, backtranslation, or archive. None of these roles overrides the current shifted-namespace normal form, occurrence ledger, readout guard, or OP-status record. Therefore active governance remains in the Active Core. □

A.19.3 Crosswalk table

Reader question	Active Core answer	Historical Witness use
What is current?	shifted-namespace normal form and current guards	not governing
Where did it come from?	compressed genealogy note	proof-memory/genealogy source
How do old tags read?	$\mathrm{B}_4(a) = a - 1$ on inhabited shifted tags	pre-shift display source
What is the OP status?	current OP-status guard	older attempts remain context
What about older examples?	read only if reintroduced by display triple	archive/backtranslation source

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A.19.4 Appendix linkage rule

Definition A.19.5 (Appendix linkage rule)**.** *An appendix or historical block may be linked to the Active Core only by one of the following explicit link types:*

genealogy link, *backtranslation link*, *proof-memory link*, *audit link*.

A link is not a promotion unless the Active Core explicitly supplies the corresponding shifted normal-form object and status guard.

Proposition A.19.6 (Appendix links are not promotions)**.** *An appendix or Historical Witness link does not by itself promote historical material to current active theorematic status.*

Proof. By Definition A.19.5, a link has one of four support roles. None of those roles is a promotion certificate. Promotion to current active status requires an explicit shifted normal-form object and a status guard in the Active Core. Therefore the link alone does not promote historical material. □

A.19.5 Historical tag and example reading

Definition A.19.7 (Historical tag reading)**.** *A tag, route, support, packed mask, or tensor-payload example inside the Historical Witness is not automatically a current shifted object. It becomes current only if the Active Core reintroduces it through*

$$\mathfrak{D}_4(X) = (X_{\mathrm{pre}}, X_{\mathrm{shift}}, \mathrm{B}_4(X_{\mathrm{shift}}))$$

or through the governing normal form

$$\mathcal{N}_4(X) = (C_X, X_{\mathrm{shift}}, \mathrm{B}_4(X_{\mathrm{shift}}), P_X, G_X, S_X).$$

Remark A.19.8 (Historical examples are protected from silent shift)**.** *Historical examples are not silently shifted. They are either preserved as witness material or explicitly connected to the current shifted line by a display triple or normal-form record.*

A.19.6 Historical linkage conservativity

Theorem A.19.9 (Historical Witness linkage conservativity)**.** *The archival stage Historical Witness linkage cleanup is conservative over archival stage: it does not change the current shifted namespace, backtranslation ledger, occurrence classes, display contracts, tensor grammar, readout compatibility, OP status, or Historical Witness content.*

Proof. The cleanup adds link roles and reading rules between the Active Core and Historical Witness. It does not modify the definitions of Σ_4, B_4, $\mathfrak{D}_4$, or $\mathcal{N}_4$. It does not alter any theorematic status record or OP guard. It also does not remove Historical Witness material. Hence the cleanup is conservative over archival stage. □

Remark A.19.10 (archival stage non-claim)**.** *Version 4.0.13 is an appendix and Historical Witness linkage release-final cleanup version. It does not add new physics, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, and does not claim Turing-completeness.*

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A.20 archival stage Clean Reader Bundle and Hygiene-Freeze Candidate

Status. This archival stage layer closes the local cleanup sequence as a clean reader bundle release-final state. It does not add a new proof claim, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, and does not change the archival stage global QAM Hilbert-shift. It records that the cleanup line from archival stage to archival stage has stabilized the reader-facing structure.

A.20.1 Clean reader bundle principle

Definition A.20.1 (Clean reader bundle principle)**.** *A clean shifted reader bundle is a split bundle in which:*

1. *the Active Core is the current governing line;*
2. *the Historical Witness is preserved as proof memory and genealogy;*
3. *active QAM/router/readout objects are read in shifted-namespace normal form;*
4. *historical/pre-shift material is connected only by explicit witness, genealogy, audit, or back-translation links;*
5. *examples and tables use normalized display contracts whenever they compare pre-shift and shifted displays.*

Remark A.20.2 (Clean bundle is not a shorter theory)**.** *The clean reader bundle is not a shortened or weakened theory. It is the same shifted governing structure with improved reader navigation, symbol hygiene, example hygiene, and Historical Witness linkage.*

A.20.2 cleanup line cleanup closure map

Version	Cleanup role	Mathematical status
archival stage	historical remark triage	conservative; no theorematic change
archival stage	reader navigation map	conservative; current reading clarified
archival stage	notation and symbol hygiene	conservative; symbol roles registered
archival stage	table/display/example normalization	conservative; examples standardized
archival stage	Historical Witness linkage	conservative; split-bundle roles clarified
archival stage	clean reader bundle release-final state	conservative; hygiene line frozen locally

A.20.3 Reader-facing invariant checklist

Definition A.20.3 (Reader-facing invariant checklist)**.** *The following invariants are preserved by the cleanup sequence:*

$$\begin{aligned}
&\textit{shift:} && \Sigma_4(k) = k+1,\\
&\textit{backtranslation:} && \mathrm{B}_4(a) = a-1 \quad (a \geq 1),\\
&\textit{reserved zero:} && 0_{\mathrm{shift}} = \varnothing_{\mathrm{active}},\\
&\textit{HC-boundary shift:} && 0_{\mathrm{mask}} \to 1_{\mathrm{mask}},\\
&\textit{display triple:} && \mathfrak{D}_4(X) = (X_{\mathrm{pre}}, X_{\mathrm{shift}}, \mathrm{B}_4(X_{\mathrm{shift}})),\\
&\textit{normal form:} && \mathcal{N}_4(X) = (C_X, X_{\mathrm{shift}}, \mathrm{B}_4(X_{\mathrm{shift}}), P_X, G_X, S_X).
\end{aligned}$$

Lemma A.20.4 (Cleanup sequence preserves the reader-facing invariants)**.** *The archival stage–archival stage cleanup sequence preserves every invariant listed in Definition A.20.3.*

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Proof. The cleanup sequence adds reader navigation, historical remark triage, notation hygiene, example normalization, and Historical Witness linkage. None of these operations alters the formulas in the invariant checklist. Therefore the reader-facing invariants are preserved. □

A.20.4 Clean reader bundle handoff

Definition A.20.5 (Clean reader bundle handoff)**.** *After archival stage, new work should be separated into one of the following:*

1. ***further cleanup:*** *layout, bibliography, index, navigation, or display polishing without mathematical change;*
2. ***active shifted development:*** *new QAM/router/readout objects written directly in the shifted-namespace normal form;*
3. ***release packaging:*** *DOI insertion, final metadata, and Zenodo release notes.*

Remark A.20.6 (Preferred next mathematical branch)**.** *The preferred next mathematical branch after a clean reader bundle is not more namespace work. It is active use of the shifted-namespace normal form on OP-facing readout packages, while preserving the OP-status guards.*

A.20.5 Clean reader bundle conservativity

Theorem A.20.7 (Clean reader bundle conservativity)**.** *The archival stage historical source-hygiene layer is conservative over the archival stage release final and over the archival stage–archival stage cleanup sequence.*

Proof. The release-final state records the accumulated cleanup discipline without changing the shifted shift, backtranslation ledger, occurrence classes, display triples, tensor grammar, readout compatibility checks, Historical Witness content, or OP-status guards. It only summarizes and freezes the reader-facing cleanup structure. Therefore it is conservative over archival stage and over the intervening cleanup sequence. □

Remark A.20.8 (archival stage non-claim)**.** *Version 4.0.13 is a clean reader bundle release-final state. It does not add new physics, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, and does not claim Turing-completeness.*

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A.21 archival stage Frontmatter, Metadata, and DOI-Record Hygiene Cleanup

Status. This archival stage layer is a frontmatter and metadata hygiene cleanup. It does not change the shifted namespace shift, the backtranslation map, the occurrence classes, display triples, tensor grammar, readout compatibility, OP status, or the Historical Witness. It records a stable rule for title, DOI, version, and release-final state/release wording.

A.21.1 Frontmatter invariants

Definition A.21.1 (Frontmatter invariants)**.** *The following frontmatter invariants are fixed for the cleanup-line:*

1. *the title remains*

   Quantum Sphaera Companion: A Structure-First Mathematical Unfolding;

2. *the Concept DOI remains the concept-level DOI;*

3. *the public archival stage Version DOI remains a prior version DOI for local release-final cleanup versions unless a new Zenodo DOI is explicitly assigned;*

4. *local release-final cleanup versions must not present themselves as release-final DOI-bearing versions;*

5. *the Historical Witness remains separately preserved in the bundle.*

Remark A.21.2 (Candidate wording)**.** *The word "release-final state" is acceptable in local cleanup line cleanup metadata and file names. It should not appear as a mathematical status operator and should not be used to weaken already governing shifted definitions.*

A.21.2 DOI-record hygiene

Definition A.21.3 (DOI-record hygiene rule)**.** *For this release-final cleanup version, the DOI record is read as*

$$\begin{aligned} &\textit{Concept DOI:} && 10.5281/\textit{zenodo}.19210728,\\ &\textit{Prior public version DOI:} && 10.5281/\textit{zenodo}.19805670,\\ &\textit{Version DOI:} && 10.5281/\textit{zenodo}.19811662. \end{aligned}$$

The version DOI was inserted during release-final packaging.

Lemma A.21.4 (DOI hygiene preserves release genealogy)**.** *The DOI-record hygiene rule preserves the public archival stage release genealogy while allowing local release-final cleanup versions to remain unassigned.*

Proof. The concept DOI points to the concept-level record. The prior public version DOI points to the last DOI-bearing release. Since a local release-final cleanup version has no assigned assigned version DOI, no new DOI-bearing release is asserted. Therefore release genealogy remains correct. □

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A.21.3 Metadata role separation

Definition A.21.5 (Metadata role separation)**.** *Version labels, file names, DOI records, and release notes are metadata. They do not define mathematical objects. Mathematical objects must be named by their mathematical roles, such as* Σ_4, B_4, $\mathfrak{D}_4$, *or* $\mathcal{N}_4$, *not by release-file names.*

Proposition A.21.6 (Metadata does not alter theorematic status)**.** *Changing a local cleanup version label from historical release marker archival stage to archival stage does not alter the theorematic status of any shifted object.*

Proof. The version label is metadata. The theorematic status of a shifted object is stored in its governing status record, for example the S_X field in $\mathcal{N}_4(X)$. Updating a cleanup label does not modify that status record. Hence theorematic status is unchanged. □

A.21.4 Frontmatter hygiene checklist

Definition A.21.7 (Frontmatter hygiene checklist)**.** *A clean local release-final cleanup version should satisfy:*

1. *unchanged title;*
2. *visible cleanup version label;*
3. *Concept DOI displayed as concept pointer;*
4. *prior public version DOI displayed as prior DOI;*
5. *no unassigned DOI represented as current DOI;*
6. *Active Core and Historical Witness both present;*
7. *non-claim guards preserved.*

A.21.5 Frontmatter metadata conservativity

Theorem A.21.8 (Frontmatter metadata conservativity)**.** *The archival stage frontmatter/metadata hygiene cleanup is conservative over archival stage.*

Proof. The cleanup records metadata rules and DOI-record hygiene. It does not modify the definitions of Σ_4, B_4, $\mathfrak{D}_4$, or $\mathcal{N}_4$. It does not alter tensor grammar, readout compatibility, Historical Witness content, or OP-status guards. Therefore it is conservative over archival stage. □

Remark A.21.9 (archival stage non-claim)**.** *Version 4.0.13 is a frontmatter and metadata hygiene release-final state. It does not add new physics, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, and does not claim Turing-completeness.*

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A.22 archival stage Bibliography, Citation, and Terminal Record Hygiene Cleanup

Status. This archival stage layer is a bibliography, citation, and terminal-record hygiene cleanup. It does not change the shifted namespace shift, the backtranslation map, the occurrence classes, display triples, tensor grammar, readout compatibility, OP status, or the Historical Witness. It records how bibliographic and terminal records should be read in the split bundle.

A.22.1 Bibliography role principle

Definition A.22.1 (Bibliography role principle)**.** *The bibliography and terminal self-citation records are metadata and scholarly context. They do not define mathematical objects. Current mathematical objects are governed by their definitions in the Active Core, especially*

$$\Sigma_4, \qquad \mathrm{B}_4, \qquad \mathfrak{D}_4, \qquad \mathcal{N}_4.$$

Remark A.22.2 (Citation does not promote theorematic status)**.** *A citation, DOI record, or terminal self-citation can identify a release or source, but it does not promote a historical statement to current theorematic status. Promotion requires a current Active-Core normal-form record and status guard.*

A.22.2 Terminal record hygiene

Definition A.22.3 (Terminal record hygiene rule)**.** *Terminal records should separate:*

1. *concept-level identifiers;*
2. *version-level identifiers;*
3. *prior public version identifiers;*
4. *local cleanup-release-final state identifiers;*
5. *bibliography entries and self-citations.*

No terminal record should be read as a new theorem unless it is explicitly stated as a theorem in the Active Core.

Lemma A.22.4 (Terminal records preserve metadata boundaries)**.** *The terminal record hygiene rule preserves the boundary between metadata and mathematics.*

Proof. The rule assigns identifiers and citations to metadata roles. It explicitly prevents DOI records and self-citations from acting as theorem statements. Therefore metadata remains separate from mathematical content. □

A.22.3 Citation audit contract

Definition A.22.5 (Citation audit contract)**.** *A clean citation state for a local release-final cleanup version requires:*

1. *no undefined citations in the build log;*
2. *no undefined references in the build log;*

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3. *the concept DOI and prior public version DOI to be readable as metadata;*

4. *no unassigned DOI to be presented as a current version DOI;*

5. *the Historical Witness to remain linked as proof memory rather than as a second active bibliography line.*

Remark A.22.6 (Local release-final cleanup versions and DOI restraint)**.** *A local release-final cleanup version may mention the prior DOI and the concept DOI. It should not claim a assigned version DOI until release-final packaging assigns one.*

A.22.4 Bibliography and witness linkage

Definition A.22.7 (Bibliography–witness linkage)**.** *Historical citations and witness-preserved references may be used to reconstruct the genealogy of an argument. They do not override the current shifted-namespace normal form. If a historical citation supports a current statement, the current statement must still be expressed in Active-Core language.*

Proposition A.22.8 (Bibliographic context is not a second governing layer)**.** *Bibliographic context and Historical Witness context do not form a second governing theorematic layer.*

Proof. By Definition A.22.7, historical citations and witness references serve genealogy and reconstruction roles. Current governance remains in the Active Core. Hence bibliographic context is not a second governing layer. □

A.22.5 Bibliography hygiene conservativity

Theorem A.22.9 (Bibliography and citation hygiene conservativity)**.** *The archival stage bibliography/citation/terminal-record hygiene cleanup is conservative over archival stage.*

Proof. The cleanup records how citations, DOI records, terminal records, and witness references should be read. It does not modify the definitions of Σ_4, B_4, $\mathfrak{D}_4$, or $\mathcal{N}_4$. It does not alter tensor grammar, readout compatibility, OP-status guards, or Historical Witness content. Therefore it is conservative over archival stage. □

Remark A.22.10 (archival stage non-claim)**.** *Version 4.0.13 is a bibliography, citation, and terminal-record hygiene release-final state. It does not add new physics, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, and does not claim Turing-completeness.*

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A.23 archival stage Glossary, Terminology, and Reader-Index Hygiene Cleanup

Status. This archival stage layer is a glossary, terminology, and reader-index hygiene cleanup. It does not change the shifted namespace shift, the backtranslation map, the occurrence classes, display triples, tensor grammar, readout compatibility, OP status, citations, metadata, or the Historical Witness. It supplies a compact reader-facing terminology map for the current shifted line.

A.23.1 Terminology hygiene principle

Definition A.23.1 (Terminology hygiene principle)**.** *A reader-facing term is hygienic if it has exactly one active role in the current shifted line, or if all of its roles are explicitly separated by context. Terminology hygiene prevents a word such as "tag", "mask", "witness", or "readout" from silently changing mathematical role.*

Remark A.23.2 (Glossary entries are reader aids)**.** *The glossary entries in this layer are reader aids. They do not replace the formal definitions in the Active Core and do not promote historical material to current theorematic status.*

A.23.2 Compact current-shifted glossary

Term	Current shifted role	Guard against confusion
Active Core	governing current line	not the Historical Witness
Historical Witness	proof memory and genealogy	not a second active theory line
QAM	language/router/readout set layer	not a new physical law by itself
MML	structural/router-machine layer	not a replacement for proof status
MMA	matrix/readout machinery layer	not an OP-closure certificate
HC	Heisenberg compensator/boundary normalizer	not redefined by the global shifted shift
tag	inhabited active QAM/router label	not a packed mask integer
route payload	ordered finite tag/readout route	not an unordered support set
support set	unordered finite support entries	shifted before mask recomputation
packed mask	integer encoding of support	not shifted by adding one
tensor-index payload	QAM/router payload in an index position	not the tensor component itself
readout target	physical/mathematical target of a package	not changed by address migration
guard	non-claim/status protection record	not optional prose decoration
OP status	open/closed theorematic state	not changed by namespace migration

A.23.3 Acronym and abbreviation map

Definition A.23.3 (Acronym map)**.** *The following acronyms are read in the current Active Core as follows:*

QAM *Quantum Analysis Mengenlehre / language and set-readout layer*,
MML *Millennium Matrix / structural router-machine layer*,
MMA *Heisenberg Matrix / matrix-readout machinery layer*,
HC *Heisenberg compensator / boundary normalizer*,
OP *open problem or OP-facing package, depending on context.*

Remark A.23.4 (Acronyms do not carry closure)**.** *An acronym can identify a layer or tool family, but it does not by itself close an OP or certify a physical readout.*

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A.23.4 Reader-index roles

Definition A.23.5 (Reader-index role)**.** *A reader-index role is a stable navigation label attached to a term. The current cleanup line uses four reader-index roles:*

$$\textit{governing}, \qquad \textit{routing}, \qquad \textit{witness}, \qquad \textit{guard}.$$

Governing terms state the active line; routing terms explain transport or display; witness terms preserve genealogy; guard terms protect non-claims and OP status.

Lemma A.23.6 (Reader-index roles prevent role drift)**.** *The reader-index roles prevent glossary terms from silently switching between active, routing, witness, and guard functions.*

Proof. Each role has a specified reading function. If a term is used as a governing term, it must belong to the active line. If it is used as witness material, it does not govern. If it is used as a guard, it protects status. If it is used as routing, it explains display or transport. Thus the role determines the reading and prevents silent drift. □

A.23.5 Terminology conservativity

Theorem A.23.7 (Glossary and terminology hygiene conservativity)**.** *The archival stage glossary/terminology/reader-index hygiene cleanup is conservative over archival stage.*

Proof. The cleanup adds reader-facing terminology maps and index roles. It does not modify the definitions of Σ_4, B_4, $\mathfrak{D}_4$, or $\mathcal{N}_4$. It does not alter tensor grammar, readout compatibility, citation records, metadata, Historical Witness content, or OP-status guards. Therefore it is conservative over archival stage. □

Remark A.23.8 (archival stage non-claim)**.** *Version 4.0.13 is a glossary, terminology, and reader-index hygiene release-final state. It does not add new physics, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, and does not claim Turing-completeness.*

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A.24 archival stage Cleanup Consolidation and Local Release-Final Preparation

Status. This archival stage layer consolidates the local cleanup sequence into one release-release-final state-preparation layer. It does not add new mathematics, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, and does not change the archival stage global QAM Hilbert-shift. It only records that the cleanup sequence has become locally stable enough for a possible small DOI-bearing cleanup release.

A.24.1 Cleanup consolidation principle

Definition A.24.1 (Cleanup consolidation principle)**.** *A cleanup consolidation layer is a reader-facing stabilization layer which collects prior cleanup work without changing the mathematical content. For the current line, it consolidates:*

archival stage historical remark triage,
archival stage reader navigation,
archival stage notation and symbol hygiene,
archival stage table/display/example normalization,
archival stage Historical Witness linkage,
archival stage clean reader bundle,
archival stage frontmatter and DOI-record hygiene,
archival stage bibliography/citation/terminal-record hygiene,
archival stage glossary/terminology/reader-index hygiene.

Remark A.24.2 (Consolidation is not compression by deletion)**.** *The consolidation does not delete proof memory. It leaves the Historical Witness preserved and treats cleanup as reader-facing stabilization rather than content removal.*

A.24.2 Local release-release-final state readiness checklist

Definition A.24.3 (Local cleanup release-release-final state readiness)**.** *The cleanup-line is locally ready for release-release-final state packaging when the following conditions hold:*

1. *the Active Core compiles without undefined references or citations;*
2. *the Historical Witness compiles without undefined references or citations;*
3. *no duplicate labels are reported;*
4. *overfull hbox/vbox warnings are absent in both PDFs;*
5. *the title and concept DOI remain stable;*
6. *the prior public version DOI is recorded as prior, not as a new DOI;*
7. *all non-claim guards remain visible.*

Lemma A.24.4 (Readiness checklist is metadata-level)**.** *The local release-release-final state readiness checklist is a metadata and build-audit condition, not a theorematic closure condition.*

Proof. The checklist speaks about build state, title/DOI metadata, duplicate labels, warnings, and non-claim guards. It does not add proof certificates or change the status record of any mathematical object. Therefore it is metadata-level. □

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A.24.3 Current invariant freeze

Definition A.24.5 (Current cleanup invariant freeze)**.** *Throughout the cleanup sequence, the following invariants remain fixed:*

$$\begin{aligned}
&\Sigma_4(k) = k+1,\ \mathrm{B}_4(a) = a-1 \quad (a \geq 1),\\
&0_{\mathrm{shift}} = \varnothing_{\mathrm{active}},\ 0_{\mathrm{mask}} \to 1_{\mathrm{mask}},\\
&\mathfrak{D}_4(X), \qquad \mathcal{N}_4(X),\\
&\textit{OP 1/4 open}, \quad \textit{OP 5(B) open}.
\end{aligned}$$

Proposition A.24.6 (Cleanup consolidation preserves the invariant freeze)**.** *The archival stage cleanup consolidation preserves the invariant freeze of Definition A.24.5.*

Proof. The consolidation layer only records the cleanup history and readiness conditions. It does not modify the shift, backtranslation, reserved zero, HC-boundary shift, display triple, normal form, or OP-status guards. Hence all listed invariants are preserved. □

A.24.4 Release packaging boundary

Definition A.24.7 (Release packaging boundary)**.** *Release packaging begins only when a assigned Zenodo version DOI is explicitly assigned. Before that assignment, archival stage remains a local cleanup consolidation release-final state with the prior public version DOI recorded only as prior.*

Remark A.24.8 (What a final cleanup release would do)**.** *A final cleanup release would not add new theory. It would insert the assigned version DOI, normalize release-final file names, regenerate audit reports, checksums, and the Zenodo description, and preserve the Active Core/Historical Witness split.*

A.24.5 Cleanup consolidation conservativity

Theorem A.24.9 (Cleanup consolidation conservativity)**.** *The archival stage cleanup consolidation is conservative over archival stage release-final and over the local archival stage–archival stage cleanup sequence.*

Proof. The consolidation summarizes the cleanup line and records release-release-final state readiness criteria. It does not alter any current shifted-namespace mathematical definition, tensor grammar, readout compatibility check, citation role, metadata role, Historical Witness content, or OP-status guard. Therefore it is conservative over archival stage and over archival stage–archival stage. □

Remark A.24.10 (archival stage non-claim)**.** *Version 4.0.13 is a cleanup consolidation release-final state. It does not add new physics, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, and does not claim Turing-completeness.*

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A.25 archival stage Label, Reference, and Counter Hygiene Cleanup

Status. This archival stage layer is a label, reference, and counter hygiene cleanup. It does not change the shifted namespace shift, the backtranslation map, the occurrence classes, display triples, tensor grammar, readout compatibility, OP status, citations, metadata, or the Historical Witness. It records the audit rule for labels, cross-references, theorem counters, and reader-visible navigation.

A.25.1 Label hygiene principle

Definition A.25.1 (Label hygiene principle)**.** *A label is hygienic if it is unique, stable under cleanup edits, and semantically aligned with the object it names. In the cleanup-line, labels should identify the local role of an object, for example section, definition, lemma, theorem, proposition, remark, table, or audit block.*

Remark A.25.2 (Labels are addresses, not mathematical content)**.** *A LaTeX label is a document address. It does not define the mathematical object itself. Mathematical content remains in the corresponding definition, lemma, theorem, proposition, or normal-form record.*

A.25.2 Reference hygiene contract

Definition A.25.3 (Reference hygiene contract)**.** *A clean local release-final cleanup version should satisfy the following reference contract:*

1. *no undefined references in the build log;*
2. *no undefined citations in the build log;*
3. *no multiply defined labels in the build log;*
4. *no theorem statement should depend on an unresolved reference;*
5. *historical references should not override current Active-Core status.*

Lemma A.25.4 (Reference hygiene is build-level, not theorematic)**.** *Reference hygiene is a build-level and reader-navigation condition, not a new theorematic closure condition.*

Proof. The reference contract concerns build-log state and reader addressability. It does not change any mathematical definition, readout target, guard, or OP-status record. Therefore it is build-level rather than theorematic. □

A.25.3 Counter and theorem-environment hygiene

Definition A.25.5 (Counter hygiene rule)**.** *Theorem, lemma, proposition, definition, and remark counters are reader-facing navigation devices. Their role is to keep the document addressable. A cleanup pass should not use counter renumbering to imply a change of theorematic status.*

Proposition A.25.6 (Counter changes are not status changes)**.** *A reader-visible counter or page-number change caused by cleanup is not a change of theorematic status.*

Proof. The theorematic status is recorded by the mathematical statement and its guard or status record, not by its page number or counter position. Cleanup may add reader-facing sections and thereby shift page numbers. Such shifts do not alter status records. Therefore counter or page-number changes are not status changes. □

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A.25.4 Cross-reference role separation

Definition A.25.7 (Cross-reference roles)**.** *Cross-references are read in one of four roles:*

governing reference, *routing reference*, *witness reference*, *audit reference.*

A governing reference points to current Active-Core content. A routing reference explains transport, display, or navigation. A witness reference points to proof memory or genealogy. An audit reference points to build, metadata, or hygiene state.

Remark A.25.8 (Witness references remain non-governing)**.** *A reference to the Historical Witness remains a witness reference unless the Active Core explicitly reintroduces the content through a current shifted display triple, normal form, or status record.*

A.25.5 Label/reference hygiene conservativity

Theorem A.25.9 (Label/reference/counter hygiene conservativity)**.** *The archival stage label/reference/counter hygiene cleanup is conservative over archival stage.*

Proof. The cleanup records build-addressing and navigation rules. It does not modify the definitions of Σ_4, B_4, $\mathfrak{D}_4$, or $\mathcal{N}_4$. It does not alter tensor grammar, readout compatibility, citation records, metadata, Historical Witness content, or OP-status guards. Therefore it is conservative over archival stage. □

Remark A.25.10 (archival stage non-claim)**.** *Version 4.0.13 is a label, reference, and counter hygiene release-final state. It does not add new physics, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, and does not claim Turing-completeness.*

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A.26 archival stage Layout, TOC, and Render Hygiene Cleanup

Status. This archival stage layer is a layout, table-of-contents, and render hygiene cleanup. It does not change the shifted namespace shift, the backtranslation map, the occurrence classes, display triples, tensor grammar, readout compatibility, OP status, citations, metadata, labels, counters, or the Historical Witness. It records the visual audit rule for the split bundle.

A.26.1 Layout hygiene principle

Definition A.26.1 (Layout hygiene principle)**.** *A local release-final cleanup version is layout-hygienic if its generated PDFs are readable as rendered documents: no text is visibly clipped, no major text blocks overlap, table headings remain legible, and reader-navigation pages are visually usable.*

Remark A.26.2 (Layout is not theorematic content)**.** *Layout hygiene improves readability and auditability. It does not change definitions, lemmas, theorems, OP status, or physical readout status.*

A.26.2 TOC readability contract

Definition A.26.3 (TOC readability contract)**.** *The table of contents is reader-hygienic if section and subsection entries are readable, numbering boxes do not collide with titles, and cleanup-version headings remain navigable. Long cleanup headings may be verbose in the source but must remain readable in the rendered PDF.*

Lemma A.26.4 (TOC readability is a navigation condition)**.** *The TOC readability contract is a navigation condition, not a mathematical condition.*

Proof. The table of contents addresses document locations. It does not define or modify mathematical objects. Therefore improving TOC readability changes only navigation. □

A.26.3 Render audit contract

Definition A.26.5 (Render audit contract)**.** *A local render audit should include at least the title page, early navigation pages, at least one representative middle page, and terminal pages of both the Active Core and Historical Witness. The purpose is to detect visible clipping, overlap, unreadable page headings, and severe table breakage.*

Remark A.26.6 (Contact sheet role)**.** *The render contact sheet is an audit artifact. It is not part of the mathematical proof. It gives a compact visual sample of the generated PDFs.*

A.26.4 Warning hygiene

Definition A.26.7 (Warning hygiene contract)**.** *For the current cleanup line, the preferred local build state is:*

$$\begin{aligned} &\textit{undefined references} &&= 0,\\ &\textit{undefined citations} &&= 0,\\ &\textit{multiply-defined labels} &&= 0,\\ &\textit{overfull hbox warnings} &&= 0,\\ &\textit{overfull vbox warnings} &&= 0. \end{aligned}$$

This is a local document-hygiene target rather than a theorematic claim.

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Proposition A.26.8 (Warning hygiene is conservative)**.** *Achieving the preferred warning hygiene state does not alter mathematical content.*

Proof. Warnings and references concern build quality, cross-reference resolution, and rendering. They do not change the definitions of Σ_4, B_4, $\mathfrak{D}_4$, or $\mathcal{N}_4$, and they do not modify OP-status guards. Therefore warning hygiene is conservative. □

A.26.5 Layout/render hygiene conservativity

Theorem A.26.9 (Layout/TOC/render hygiene conservativity)**.** *The archival stage layout/-TOC/render hygiene cleanup is conservative over archival stage.*

Proof. The cleanup records visual audit, TOC readability, and warning-hygiene rules. It does not modify any current shifted-namespace mathematical definition, tensor grammar, readout compatibility check, citation role, metadata role, label role, Historical Witness content, or OP-status guard. Therefore it is conservative over archival stage. □

Remark A.26.10 (archival stage non-claim)**.** *Version 4.0.13 is a layout, TOC, and render hygiene release-final state. It does not add new physics, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, and does not claim Turing-completeness.*

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A.27 archival stage Source, Build, and Artifact Reproducibility Hygiene Cleanup

Status. This archival stage layer is a source, build, and artifact reproducibility hygiene cleanup. It does not change the shifted namespace shift, the backtranslation map, the occurrence classes, display triples, tensor grammar, readout compatibility, OP status, citations, metadata, labels, counters, layout contract, or the Historical Witness. It records how source files, generated PDFs, manifests, checksums, and bundles should be interpreted.

A.27.1 Source reproducibility principle

Definition A.27.1 (Source reproducibility principle)**.** *A local release-final cleanup version is source-reproducible if the supplied TEX files are sufficient to rebuild the Active Core and Historical Witness PDFs in the same logical document state, modulo ordinary build-environment differences such as PDF timestamps or engine-specific metadata.*

Remark A.27.2 (Source is governing over generated artifacts)**.** *For local cleanup work, the TEX source is the editable governing artifact. The PDFs are rendered outputs. If a rendered artifact and source artifact appear to disagree, the source and build log must be inspected before any mathematical interpretation is changed.*

A.27.2 Artifact classes

Definition A.27.3 (Artifact classes)**.** *The local cleanup workflow separates artifacts into the following classes:*

1. ***source artifacts:*** *TEX files and source-side scan reports;*
2. ***rendered artifacts:*** *Active Core and Historical Witness PDFs;*
3. ***audit artifacts:*** *build logs, audit reports, render contact sheets, and scan reports;*
4. ***packaging artifacts:*** *manifest files, checksum files, full bundles, and TEX-only ZIPs.*

Lemma A.27.4 (Artifact classes prevent role drift)**.** *The artifact classes prevent rendered, audit, and packaging artifacts from being mistaken for new mathematical content.*

Proof. Each artifact class has an assigned role. Source artifacts contain editable content; rendered artifacts show compiled output; audit artifacts document quality checks; packaging artifacts support transfer and verification. None of the non-source classes is a new theorematic layer. Therefore role drift is prevented. □

A.27.3 Manifest and checksum hygiene

Definition A.27.5 (Manifest and checksum hygiene rule)**.** *A local cleanup bundle should contain a manifest and checksum file. The manifest records which files belong to the package. The checksum file records file integrity. Neither manifest nor checksum file changes mathematical content.*

Proposition A.27.6 (Checksums certify files, not theorems)**.** *A checksum certifies the byte identity of a file, not the truth of a theorem.*

Proof. A checksum is a hash of file bytes. It can detect whether a file has changed, but it does not inspect or prove mathematical statements. Therefore it certifies file integrity only. □

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A.27.4 Build provenance contract

Definition A.27.7 (Build provenance contract)**.** *A clean local build record should state:*

1. *source TEX file names;*
2. *output PDF file names;*
3. *page counts;*
4. *undefined reference and citation counts;*
5. *duplicate-label counts;*
6. *overfull hbox and vbox counts;*
7. *fatal or emergency error counts;*
8. *whether a render contact sheet was produced.*

Remark A.27.8 (Build provenance is audit support)**.** *Build provenance supports reproducibility and audit. It does not replace the mathematical proof and does not close any OP.*

A.27.5 TEX-only bundle contract

Definition A.27.9 (TEX-only bundle contract)**.** *A TEX-only bundle is intended for manual integration and local rebuilding. It should contain the Active Core source, the Historical Witness source, and the minimal supporting hygiene records needed to identify the release-final state. It should not be treated as a DOI-bearing release unless release-final packaging assigns a assigned version DOI.*

Lemma A.27.10 (TEX-only bundle preserves manual-integration workflow)**.** *The TEX-only bundle contract preserves the user's manual-integration workflow.*

Proof. The contract supplies directly editable TEX files and minimal identifiers. It does not require repository-side mutation, automatic patch application, or a new DOI. Hence it preserves manual integration. □

A.27.6 Reproducibility conservativity

Theorem A.27.11 (Source/build/artifact hygiene conservativity)**.** *The archival stage source/build/artifact reproducibility cleanup is conservative over archival stage.*

Proof. The cleanup records artifact roles, manifest/checksum rules, build provenance, and TEX-only bundle expectations. It does not modify the definitions of Σ_4, B_4, $\mathfrak{D}_4$, or $\mathcal{N}_4$. It does not alter tensor grammar, readout compatibility, citation records, metadata roles, labels, counters, layout rules, Historical Witness content, or OP-status guards. Therefore it is conservative over archival stage. □

Remark A.27.12 (archival stage non-claim)**.** *Version 4.0.13 is a source, build, and artifact reproducibility hygiene release-final state. It does not add new physics, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, and does not claim Turing-completeness.*

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A.28 archival stage Local Release-Readiness, Upload, and Final-Packaging Hygiene Cleanup

Status. This archival stage layer is a local release-readiness and upload-hygiene cleanup. It records the assigned version DOI 10.5281/zenodo.19811662, does not change the shifted namespace shift, does not change the Historical Witness, and does not change any OP status. It records which artifacts would be used for a small DOI-bearing cleanup release if a new Zenodo version DOI is later assigned.

A.28.1 Local release-readiness principle

Definition A.28.1 (Local release-readiness principle)**.** *A local release-final cleanup version is release-ready only in the limited sense that its source, PDF, audit, manifest, checksum, and bundle artifacts are coherent and can be promoted to release-final packaging after the version DOI 10.5281/zenodo.19811662 is assigned. Release-readiness is not a theorematic status and is not an OP-closure certificate.*

Remark A.28.2 (release-final assigned in this layer)**.** *This release-final layer records the assigned Zenodo version DOI 10.5281/zenodo.19811662. The prior public version DOI remains the archival stage release DOI until explicit release-final packaging assigns a assigned version DOI.*

A.28.2 Upload artifact selection

Definition A.28.3 (Recommended upload artifact set)**.** *For a small cleanup release, the recommended upload artifact set is:*

1. *Active Core PDF;*
2. *Historical Witness PDF;*
3. *full release bundle ZIP;*
4. *TEX-only ZIP, if source-side manual integration is desired;*
5. *release notes or Zenodo description text;*
6. *checksum file or checksum-bearing manifest.*

Remark A.28.4 (Candidate artifacts versus release-final artifacts)**.** *Candidate artifacts should not be uploaded as release-final artifacts unless their file names, metadata, DOI records, manifests, checksums, and Zenodo description have been normalized for release-final.*

A.28.3 Zenodo description draft contract

Definition A.28.5 (Zenodo description draft contract)**.** *A Zenodo description draft for a local release-final cleanup version should state:*

1. *the document title;*
2. *the cleanup version;*
3. *that the update is reader/document hygiene, not new theory;*

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4. *the concept DOI;*

5. *the prior public version DOI;*

6. *that the assigned version DOI is assigned in this release-final packaging as 10.5281/zenodo.19811662;*

7. *the non-claims for OP 1/4, OP 5(B), primitive mixed tensors, and Turing-completeness.*

Lemma A.28.6 (Description draft is metadata)**.** *A Zenodo description draft is metadata and does not change mathematical content.*

Proof. The description draft summarizes the package, version state, DOI state, and non-claims. It does not alter definitions, proofs, normal forms, readout guards, or OP-status records. Hence it is metadata only. □

A.28.4 Final-packaging boundary

Definition A.28.7 (Final-packaging boundary)**.** *Final packaging begins only after a assigned Zenodo version DOI is provided. The final-packaging pass may then:*

1. *replace release-final state wording by release-final wording where appropriate;*

2. *insert the assigned version DOI in metadata, manifest, notes, and Zenodo description;*

3. *normalize file names to release-final names;*

4. *rebuild the Active Core and Historical Witness PDFs;*

5. *regenerate audit reports, checksums, and bundles.*

Proposition A.28.8 (Final packaging is conservative when DOI-only)**.** *If final packaging only inserts DOI and release-final metadata, rebuilds artifacts, and regenerates audits, then it is conservative over the local release-final cleanup version.*

Proof. DOI insertion and release-final artifact regeneration are metadata and build operations. They do not change Σ_4, B_4, $\mathfrak{D}_4$, $\mathcal{N}_4$, tensor grammar, readout guards, Historical Witness content, or OP-status records. Therefore such packaging is conservative. □

A.28.5 Release-readiness conservativity

Theorem A.28.9 (Release-readiness and upload-hygiene conservativity)**.** *The archival stage release-readiness/upload-hygiene cleanup is conservative over archival stage.*

Proof. The cleanup records upload artifact selection, Zenodo description drafting, and the final-packaging boundary. It does not modify current shifted mathematical definitions, tensor grammar, readout compatibility checks, artifact roles, build rules, Historical Witness content, or OP-status guards. Therefore it is conservative over archival stage. □

Remark A.28.10 (archival stage non-claim)**.** *Version 4.0.14 is a local release-readiness and upload-hygiene release-final state. It does not add new physics, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, does not claim Turing-completeness, and records the assigned Zenodo version DOI 10.5281/zenodo.19811662.*

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A.29 archival stage Local Release-Final Freeze and DOI-Request Packet Hygiene

Status. This archival stage layer freezes the local cleanup line as a release-release-final state preparation state. It records the assigned version DOI 10.5281/zenodo.19811662, does not alter the shifted namespace shift, does not modify the Historical Witness, and does not change any OP status. Its only purpose is to mark the local cleanup sequence as ready for a possible DOI-request and release-final packaging pass.

A.29.1 Local release-release-final state freeze principle

Definition A.29.1 (Local release-release-final state freeze principle)**.** *A release-final state is a document-control state in which the source, rendered PDFs, audit reports, manifests, checksums, and upload description are internally coherent and should not receive new mathematical content before a release-final DOI pass.*

Remark A.29.2 (Freeze is not mathematical closure)**.** *A release-release-final state freeze is a workflow boundary. It is not a proof of OP closure, not a physical promotion criterion, and not a replacement for any status record inside the shifted-namespace normal form.*

A.29.2 DOI-request packet

Definition A.29.3 (DOI-request packet)**.** *A DOI-request packet for a small cleanup release should contain:*

1. *the frozen Active Core PDF;*
2. *the frozen Historical Witness PDF;*
3. *the full bundle ZIP;*
4. *the TEX-only ZIP;*
5. *the release-final Zenodo description;*
6. *the audit report;*
7. *the manifest and checksum file.*

The packet is still local until a assigned Zenodo version DOI is assigned.

A.29.3 Allowed release-final pass

Definition A.29.4 (Allowed release-final pass)**.** *After a assigned Zenodo version DOI is supplied, the allowed release-final pass may:*

1. *insert the version DOI in frontmatter, notes, manifest, checksums, audit metadata, and Zenodo description;*
2. *replace local release-final state/freeze wording by release-final wording where appropriate;*
3. *rename files from local/freeze/release-final state names to release-final names;*
4. *rebuild the Active Core and Historical Witness PDFs;*

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5. *regenerate the full bundle, TEX-only ZIP, checksums, audit report, and final Zenodo description.*

It may not add new theorems, change OP status, or alter shifted normal-form content.

Proposition A.29.5 (DOI pass is conservative under the allowed boundary)**.** *A release-final pass satisfying Definition A.29.4 is conservative over the local archival stage freeze.*

Proof. The allowed pass is restricted to DOI insertion, wording normalization, file renaming, rebuilding, and audit regeneration. It explicitly excludes new theorems, OP-status changes, and shifted normal-form changes. Therefore it is conservative over the frozen local release-final state. □

A.29.4 Frozen non-claim boundary

Definition A.29.6 (Frozen non-claim boundary)**.** *The release-final state preserves the following non-claims:*

OP 1/4	*remains open*,
OP 5(B)	*remains open*,
primitive mixed tensors	*not activated*,
Turing-completeness	*not claimed*,
standard physical calculations	*not replaced.*

A.29.5 Freeze conservativity

Theorem A.29.7 (Local release-release-final state freeze conservativity)**.** *The archival stage release-final state is conservative over the archival stage–archival stage cleanup sequence and over the public archival stage release-final mathematical content.*

Proof. The freeze adds document-control and DOI-request packet rules. It does not modify Σ_4, B_4, $\mathfrak{D}_4$, $\mathcal{N}_4$, tensor grammar, readout compatibility, artifact roles, build rules, Historical Witness content, or OP-status guards. Therefore it is conservative over the cleanup sequence and over the public archival stage mathematical content. □

Remark A.29.8 (archival stage non-claim)**.** *Version 4.0.15 is a release-final state. It does not add new physics, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, does not claim Turing-completeness, and does not assign a new Zenodo version DOI.*

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A.30 archival stage Release-Final DOI and Audit Hygiene

Status. This archival stage layer is the final local audit and release-final hygiene layer. It records the assigned version DOI 10.5281/zenodo.19811662, does not change any mathematical content, does not modify the Historical Witness, and does not change any OP status. It only checks and records the exact boundary between the frozen local release-final state and a later DOI-bearing release-final pass.

A.30.1 Final local audit principle

Definition A.30.1 (Final local audit principle)**.** *A final local audit is a pre-release technical verification state. It verifies source presence, PDF build status, audit-report availability, checksum availability, release-final Zenodo description availability, and render-sample availability before a DOI-bearing release-final pass is allowed.*

Remark A.30.2 (Release-final packaging record)**.** *The release-final audit is not a Zenodo publication and is not a DOI-bearing release. It is a local check that the next pass can be a release-final packaging pass after a version DOI 10.5281/zenodo.19811662 is recorded.*

A.30.2 Release-final checklist

Definition A.30.3 (Release-final checklist)**.** *The release-final audit is successful if the following checks hold:*

1. *Active Core source exists;*
2. *Historical Witness source exists;*
3. *Active Core PDF builds with zero fatal errors;*
4. *Historical Witness PDF builds with zero fatal errors;*
5. *undefined references, undefined citations, duplicate labels, overfull hbox warnings, and overfull vbox warnings are all zero;*
6. *release-final Zenodo description exists;*
7. *manifest and checksum files are generated;*
8. *render contact sheet is generated;*
9. *non-claim boundary remains explicit.*

A.30.3 Release-final packaging record

Definition A.30.4 (Release-final packaging record)**.** *The only missing input for a release-final pass after this release-final audit is a new Zenodo version DOI. Once that DOI is supplied, the next pass should be a metadata-only release-finalization pass unless the user explicitly requests additional cleanup first.*

Proposition A.30.5 (Single-input release-final readiness)**.** *If the release-final checklist holds, then the local release-final state is release-final ready up to the supplied Zenodo version DOI 10.5281/zenodo.19811662.*

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Proof. The release-final checklist verifies the required source, build, audit, packaging, render, and non-claim conditions. The release-final input contract states that the remaining missing datum is the assigned version DOI. Therefore, once that DOI is supplied, release-final packaging can proceed without changing mathematical content. □

A.30.4 Final local non-claim boundary

Definition A.30.6 (Final local non-claim boundary)**.** *The final local audit preserves the following non-claims:*

OP 1/4	*remains open*,
OP 5(B)	*remains open*,
primitive mixed tensors	*not activated*,
Turing-completeness	*not claimed*,
standard physical calculations	*not replaced*,
Zenodo version DOI	*10.5281/zenodo.19811662.*

A.30.5 Release-final audit conservativity

Theorem A.30.7 (Final local audit release-final conservativity)**.** *The archival stage release-final audit state is conservative over the archival stage–archival stage cleanup sequence and over the public archival stage release-final mathematical content.*

Proof. The release-final audit adds audit and release-final readiness conditions only. It does not modify Σ_4, B_4, $\mathfrak{D}_4$, $\mathcal{N}_4$, tensor grammar, readout compatibility, artifact roles, build rules, Historical Witness content, or OP-status guards. Therefore it is conservative over the cleanup sequence and over the public archival stage mathematical content. □

Remark A.30.8 (archival stage non-claim)**.** *Version 4.0.16 is a release-final audit state. It does not add new physics, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, does not claim Turing-completeness, and records the assigned Zenodo version DOI 10.5281/zenodo.19811662.*

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A.31 archival stage Release-Final DOI and Packaging Record

Status. This section records the release-final DOI and packaging pass for Version 4.0.16. The pass is metadata-only: it inserts the assigned Zenodo version DOI, normalizes release-final artifacts, rebuilds the PDFs, regenerates the audit records, and preserves all mathematical non-claims.

A.31.1 DOI record

Definition A.31.1 (Release-final DOI record)**.** *The release-final DOI record is:*

Concept DOI:	10.5281/*zenodo*.19210728,
Prior public version DOI:	10.5281/*zenodo*.19805670,
Version DOI:	10.5281/*zenodo*.19811662.

Remark A.31.2 (Version DOI is metadata)**.** *The version DOI is a release and citation identifier. It is not a mathematical operator, not a new object in the shifted-namespace normal form, and not an OP-closure certificate.*

A.31.2 Release-final conservativity

Theorem A.31.3 (Release-final DOI/pass conservativity)**.** *The archival stage release-final DOI and packaging pass is conservative over the archival stage local audit state.*

Proof. The pass inserts the version DOI, normalizes release-final wording and file names, rebuilds the PDFs, and regenerates audit and checksum artifacts. It does not modify Σ_4, B_4, $\mathfrak{D}_4$, $\mathcal{N}_4$, tensor grammar, readout compatibility, artifact roles, build rules, Historical Witness content, or OP-status guards. Therefore the pass is conservative. □

Remark A.31.4 (archival stage release-final non-claim)**.** *Version 4.0.16 release-final does not add new physics, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, does not claim Turing-completeness, and does not replace standard QFT/QED/Lifshitz or general-relativistic calculations.*

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A.32 OP-Facing Diagnostic Appendix

Status. This working layer starts the post-archival stage OP-facing phase. It does not rewrite the Companion into Python, does not replace MML/MMA/QAM, and does not claim that executable diagnostics are proof certificates. The Python files listed here are diagnostic shadows of selected QAM/MML/MMA/ZFS-style objects. They are included to help test route shifting, backtranslation, route/support separation, packed-mask recomputation, guard retention, residual localization, and proof-state provenance before OP 1/4 and OP 5(B) are attacked in the formal text.

A.32.1 Diagnostic-code status rule

Definition A.32.1 (Diagnostic-code status rule)**.** *A diagnostic Python file in this appendix is an executable audit aid. It may serialize or test selected shadows of QAM, MML, MMA, and ZFS-style records, but it is not itself a formal definition, not a proof certificate, not a physical simulation, and not a replacement for the paper's mathematical language.*

Remark A.32.2 (Why Python appears only here)**.** *Python appears here only because the current shifted language is rigid enough that some role separations can be tested mechanically. The MML remains the mathematical router/intertwiner language in the paper. Python only records diagnostic encodings of selected finite examples.*

A.32.2 Diagnostic files included in the local bundle

File	Diagnostic role
`qam_normal_form.py`	builds a small executable shadow of $N_4(X)$ with route, guards, and status
`route_mask_decoder.py`	separates route, support, and packed mask; verifies recomputation after shift
`mml_op_router.py`	records OP-facing MML route surfaces and obstruction kind
`mma_readout_residual.py`	records readout residuals and promotion guards
`zfs_snapshot_manifest.py`	records proof-state provenance/checksum snapshots
`run_all_diagnostics.py`	runs the diagnostic examples and writes a JSON report

A.32.3 QAM normal-form diagnostic excerpt

```
#!/usr/bin/env python3
"""
QAM normal-form diagnostic encoding.

This file is not a proof engine and not an implementation of QAM as a new
foundation. It serializes selected shifted/QAM roles
can be inspected consistently.
"""
from __future__ import annotations

from dataclasses import asdict, dataclass
from enum import Enum
import json
from typing import Tuple


class Status(str, Enum):
 OPEN = "open"
 PREPARED = "prepared"
```

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```
 PROMOTED = "promoted"


class Guard(str, Enum):
 OP_OPEN = "op_open"
 NO_PHYSICAL_CLAIM = "no_physical_claim"
 NEEDS_SELECTOR = "needs_selector"
 NEEDS_BOUNDARY_PROMOTION = "needs_boundary_promotion"


def sigma_4(k: int) -> int:
 """shifted Hilbert shift on inhabited pre-shift tags."""
 if k < 0:
 raise ValueError("invalid pre-shift tag")
 return k + 1


def back_4(a: int) -> int:
 """Backtranslation from inhabited shifted tags."""
 if a < 1:
 raise ValueError("0_shift has no backtranslation")
 return a - 1
```

A.32.4 Route/mask diagnostic excerpt

```
#!/usr/bin/env python3
"""
Route/support/packed-mask diagnostic checks.

The key rule is that a packed mask is recomputed from shifted support. It is
not obtained by adding 1 to an integer mask.
"""
from __future__ import annotations

from dataclasses import dataclass, asdict
import json
from typing import FrozenSet, Tuple


def sigma_4(k: int) -> int:
 if k < 0:
 raise ValueError("invalid pre-shift tag")
 return k + 1


def back_4(a: int) -> int:
 if a < 1:
 raise ValueError("0_shift has no backtranslation")
 return a - 1
```

```
def packed_mask_from_support(support: FrozenSet[int]) -> int:
 return sum(2 ** k for k in support)


@dataclass(frozen=True)
class RouteMaskRecord:
 pre_route: Tuple[int, ...]
 shifted_route: Tuple[int, ...]
 support: FrozenSet[int]
 shifted_support: FrozenSet[int]
```

A.32.5 MML and MMA diagnostic excerpts

```
#!/usr/bin/env python3
"""
MML_OP diagnostic router.

This is an executable shadow of an MML OP-facing route record. It does not
define MML and does not prove an OP.
"""
from __future__ import annotations

from dataclasses import asdict, dataclass
from enum import Enum
import json
from typing import Tuple


class Branch(str, Enum):
 HORIZON = "horizon"
 BOUNDARY = "boundary"


class ObstructionKind(str, Enum):
 MISSING_SELECTOR = "missing_selector"
 MISSING_BOUNDARY_PROMOTION = "missing_boundary_promotion"
 UNKNOWN = "unknown"


@dataclass(frozen=True)
class MMLOPRoute:
 name: str
 branch: Branch
 pre_route: Tuple[int, ...]
 shifted_route: Tuple[int, ...]

#!/usr/bin/env python3
"""
```

```
MMA_OP readout-residual diagnostic.

This file gives a small data shape for readout residuals. It does not compute
physical predictions; it records which residual/guard still blocks promotion.
"""
from __future__ import annotations

from dataclasses import asdict, dataclass
from enum import Enum
import json
from typing import Optional


class PromotionStatus(str, Enum):
 BLOCKED = "blocked"
 READY_FOR_REVIEW = "ready_for_review"
 PROMOTED = "promoted"


@dataclass(frozen=True)
class MMAReadoutResidual:
 name: str
 readout_target: str
 hc_boundary_condition: str
 comparison_selector: Optional[str]
 residual_surface: str
 promotion_guard: str
 status: PromotionStatus
```

A.32.6 ZFS-style provenance diagnostic excerpt

```
#!/usr/bin/env python3
"""
ZFS-style proof-state snapshot manifest.

This is a provenance/audit helper. It is not a mathematical object and not a
proof operator.
"""
from __future__ import annotations

from dataclasses import asdict, dataclass
import hashlib
import json
from pathlib import Path
from typing import Dict, Iterable


@dataclass(frozen=True)
class SnapshotManifest:
 base_version: str
```

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```
 branch_name: str
 files: Dict[str, str]
 purpose: str
 non_claims: tuple[str, ...]

 def to_json_dict(self) -> dict:
 return asdict(self)


def sha256_file(path: Path) -> str:
 return hashlib.sha256(path.read_bytes()).hexdigest()
```

A.32.7 First diagnostic run record

The local diagnostic runner completed successfully. The generated JSON report is stored in the local bundle as

`python_diagnostics/diagnostic_run_report.json`.

The successful run only certifies that the diagnostic examples execute and that their internal consistency checks pass. It does not close OP 1/4 or OP 5(B).

Theorem A.32.3 (Diagnostic appendix conservativity)**.** *The local proof line diagnostic-code appendix is conservative over archival stage release-final.*

Proof. The appendix adds executable diagnostic shadows and records their status as audit aids. It does not modify Σ_4, B_4, $\mathfrak{D}_4$, $\mathcal{N}_4$, tensor grammar, readout compatibility, Historical Witness content, or OP-status guards. Therefore it is conservative. □

Remark A.32.4 (non-claim)**.** *Version 4.1.0-does not add new physics, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, does not claim Turing-completeness, and does not replace standard QFT/QED/Lifshitz or general-relativistic calculations.*

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A.33 First OP Obstruction Extraction

Status. This working layer uses the diagnostic scripts from the local proof line against the two current OP-facing targets OP 1/4 and OP 5(B). The output is an obstruction extraction report, not a proof closure. It identifies the current named block that prevents promotion of each OP packet.

A.33.1 Obstruction extraction rule

Definition A.33.1 (Diagnostic obstruction extraction)**.** *A diagnostic obstruction extraction sends an OP-facing packet to a finite audit record containing its branch, pre-shift route, shifted route, backtranslation, support, shifted support, packed masks, witness link, active normal form, readout target, missing item, guard, and current status. The record is a machine-checkable audit surface and not a theorematic closure certificate.*

A.33.2 First extracted OP obstruction table

OP packet	branch	active form	obstruction	missing item
OP 1/4	horizon	$N_4(\mathrm{OP}_{1/4})$	missing selector	comparison tensor $C^{\mathrm{sel}}_{1/4}$
OP 5(B)	boundary	$N_4(O\Pi_5^{(B)})$	missing boundary promotion	theorem $\Pi_5^{(B)}$

The generated files are included in the local bundle under `python_diagnostics/`: `op_obstruction_report.json`, `op_obstruction_table.md`, and `op_obstruction_table.csv`.

A.33.3 Interpretation

Proposition A.33.2 (First OP obstruction localization)**.** *Under the current diagnostic encoding, OP 1/4 is reduced to a missing selector/comparison tensor on the horizon-side route-normalized packet, while OP 5(B) is reduced to a missing boundary-promotion theorem preserving the boundary-side stop guard.*

Proof. The diagnostic extractor verifies route shifting and backtranslation for each packet, keeps the Historical Witness link separate from the active normal form, and refuses to promote a packet when its declared missing item remains active. The OP 1/4 packet is horizon-side and therefore fails at the declared selector/comparison requirement. The OP 5(B) packet is boundary-side and therefore fails at the declared boundary-promotion requirement. These failures are named obstructions rather than closures. □

Remark A.33.3 (What the extractor has and has not shown)**.** *The extraction has shown where the current OP-facing packets are blocked in the shifted diagnostic grammar. It has not proven that the selector/comparison tensor exists, has not proven the boundary-promotion theorem, and has not promoted any physical readout.*

Theorem A.33.4 (First obstruction extraction conservativity)**.** *The local proof line first OP obstruction extraction is conservative over the local proof line and over archival stage release-final.*

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Proof. The working layer adds a diagnostic extraction report and a reader-facing summary table. It does not modify Σ_4, B_4, $\mathfrak{D}_4$, $\mathcal{N}_4$, tensor grammar, readout compatibility, Historical Witness content, or OP-status guards. OP 1/4 and OP 5(B) remain open. Therefore the layer is conservative. □

Remark A.33.5 (non-claim)**.** *Version 4.1.0-does not close OP 1/4 or OP 5(B). It only names their first current diagnostic obstruction surfaces.*

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A.34 Obstruction-to-Task Reduction Layer

Status. This working layer consumes the diagnostic obstruction extraction and turns its two named obstructions into a finite OP-facing task queue. It does not close OP 1/4 or OP 5(B), does not promote a physical readout, and does not replace the formal QAM/MML/MMA language by Python. The Python task queue is an audit and planning surface only.

A.34.1 From obstruction labels to task packets

Definition A.34.1 (OP obstruction-to-task packet)**.** *An OP obstruction-to-task packet is a finite record*

$$\mathcal{T}_{\mathrm{OP}} = (O, B, N_4(O), \tau, T, I, C, G, S),$$

where O is the OP object, B is the branch, $N_4(O)$ is the active shifted-namespace normal form, τ is the named obstruction, T is the next task kind, I is the required input, C is the success criterion, G is the guard to preserve, and S is the task status.

Remark A.34.2 (Task packets are not proof packets)**.** *The task packet names the next controlled move. It does not certify that the move can be completed, and it does not close the corresponding OP.*

A.34.2 Extracted task queue

The diagnostic task planner reads the file

`python_diagnostics/op_obstruction_report.json`

and produces

`op_task_queue.json`, `op_task_queue.md`, `op_task_queue.csv`.

The resulting task queue is:

OP	Task	Kind	Target object	Status
OP 1/4	T1	selector domain/-codomain	$C_{1/4}^{\mathrm{sel}}$	ready
OP 1/4	T2	backtranslation check	B_4-compatible selector	open
OP 1/4	T3	residual reduction	$R_{1/4}^{\mathrm{post}}$	open
OP 5(B)	T1	promotion theorem	$\Pi_5^{(B)}$	ready
OP 5(B)	T2	HC/MM compatibility	boundary compatibility check	open
OP 5(B)	T3	promotion residual	$R_5^{(B),\mathrm{post}}$	open

A.34.3 OP 1/4 reduction

Proposition A.34.3 (OP 1/4 selector-task reduction)**.** *Given the obstruction record, the first controlled OP 1/4 move is not a new physical readout and not an OP closure. It is the construction of a selector/comparison tensor $C_{1/4}^{\mathrm{sel}}$ with explicit domain, codomain, route role, and backtranslation compatibility.*

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Proof. The obstruction is `missing_selector` and the missing item is `selector_or_comparison_tensor`. The planner therefore extracts the first task `OP14-T1`, whose target is the selector or comparison tensor. Since the associated guard remains `OP_OPEN` and `no physical claim`, the task is a controlled reduction rather than an OP closure. □

A.34.4 OP 5(B) reduction

Proposition A.34.4 (OP 5(B) boundary-promotion task reduction)**.** *Given the obstruction record, the first controlled OP 5(B) move is the statement of a boundary-promotion theorem* $\Pi_5^{(B)}$ *preserving the earliest boundary stop. It is not a physical promotion and not an OP closure.*

Proof. The obstruction is `missing_boundary_promotion` and the missing item is `boundary_promotion_theorem`. The planner therefore extracts the first task `OP5B-T1`, whose target is a boundary promotion theorem. Since the guard requires the stop-preserving boundary condition and no physical claim, the task cannot be read as a promoted physical readout or as OP closure. □

A.34.5 Next formal move

The next mathematical step after is therefore sharply constrained:

$$\text{OP } 1/4\text{: construct/test } C_{1/4}^{\mathrm{sel}}, \qquad \text{OP 5(B): state/test } \Pi_5^{(B)}.$$

If $C_{1/4}^{\mathrm{sel}}$ fails, the failure becomes the next named OP 1/4 obstruction. If $\Pi_5^{(B)}$ fails, the failure becomes the next named OP 5(B) boundary obstruction. In both cases the OP remains open unless the residual is formally discharged in the Active Core.

Theorem A.34.5 (Obstruction-to-task reduction conservativity)**.** *The local proof line task-object formalization is conservative over the local proof line.*

Proof. The layer reads the diagnostic obstruction records and produces a finite task queue. It does not change Σ_4, B_4, $\mathfrak{D}_4$, $\mathcal{N}_4$, tensor grammar, readout compatibility, Historical Witness content, or OP-status guards. Therefore it is conservative over the local pass. □

Remark A.34.6 (non-claim)**.** *Version 4.1.0-does not add new physics, does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, does not claim Turing-completeness, and does not replace standard QFT/QED/Lifshitz or general-relativistic calculations.*

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A.35 Task-Object Formalization

Status. This layer turns the task queue into two explicit next target objects. It is still not an OP closure. It identifies the first formal object needed for OP 1/4 as a selector/comparison tensor $C^{\mathrm{sel}}_{1/4}$ and the first formal object needed for OP 5(B) as a boundary-promotion theorem $\Pi^{(B)}_5$.

A.35.1 Task-to-object reduction rule

Definition A.35.1 (Task-to-object reduction)**.** *A task-to-object reduction sends a named obstruction task*

$$T = (OP, \text{obstruction}, \text{missing item}, G, S)$$

to a target object

$$\mathcal{T}(T) = (X_T, D_T, E_T, \mathcal{C}_T, R_T^{post}, G_T, S_T),$$

where X_T is the object to be constructed, D_T is its intended domain, E_T is its comparison or promotion codomain, $\mathcal{C}_T$ is the list of required compatibility conditions, R_T^{post} is the residual left after the construction, G_T is the retained guard, and S_T is the post-reduction status.

Remark A.35.2 (Why this is mathematical progress)**.** *The local pass/diagnostics named where the packets fail. The reduction names what must be constructed next. This changes an unlocalized OP obstruction into a concrete target-object problem.*

A.35.2 The OP 1/4 target object $C^{\mathrm{sel}}_{1/4}$

Definition A.35.3 (OP 1/4 selector/comparison target)**.** *Let $X_{14} = N_4(\mathrm{OP}_{1/4})$ be the horizon-side OP 1/4 normal-form packet, with pre-shift witness route $\rho^{\mathrm{pre}}_{1/4}$ and inhabited shifted route $\rho^{\mathrm{shift}}_{1/4}$. A first OP 1/4 selector/comparison target is a comparison tensor*

$$C^{\mathrm{sel}}_{1/4}: \quad P(\rho^{\mathrm{pre}}_{1/4}) \times P(\rho^{\mathrm{shift}}_{1/4}) \longrightarrow E^{\mathrm{sel}}_{1/4}$$

subject to backtranslation compatibility, guard preservation, and residual monotonicity $R^{\mathrm{post}}_{1/4} \preccurlyeq R^{\mathrm{pre}}_{1/4}$.

Proposition A.35.4 (OP 1/4 task-object reduction)**.** *Under the obstruction table, OP 1/4 reduces first to the construction of $C^{\mathrm{sel}}_{1/4}$ satisfying Definition A.35.3. If such a $C^{\mathrm{sel}}_{1/4}$ is constructed, the remaining OP 1/4 obstruction is the post-selector residual $R^{\mathrm{post}}_{1/4}$, not the original missing-selector obstruction.*

Proof. The task queue assigns OP 1/4 the obstruction *missing selector*. A selector obstruction is discharged only by an object with a declared domain, comparison codomain, compatibility conditions, and retained guard. Definition A.35.3 supplies exactly this target. It preserves backtranslation and guard data and leaves an explicit residual. Therefore the first obstruction is reduced to constructing $C^{\mathrm{sel}}_{1/4}$, after which only $R^{\mathrm{post}}_{1/4}$ remains to be analyzed. □

A.35.3 The OP 5(B) target object $\Pi^{(B)}_5$

Definition A.35.5 (OP 5(B) boundary-promotion target)**.** *Let $Y^{(B)}_5 = N_4(O\Pi^{(B)}_5)$ be the boundary-side OP 5(B) normal-form packet with stop-preserving boundary guard G_∂. A first OP 5(B) boundary-promotion target is a theorem schema $\Pi^{(B)}_5$ of the form*

$$\Pi^{(B)}_5: \quad (Y^{(B)}_5, G_\partial, HC/MM) \longrightarrow (Y^{(B),\mathrm{prom}}_5, G_\partial, R^{(B),\mathrm{post}}_5),$$

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subject to stop preservation, HC/MM compatibility, backtranslation stability, and residual retention.

Proposition A.35.6 (OP 5(B) task-object reduction)**.** *Under the obstruction table, OP 5(B) reduces first to the formulation and proof of* $\Pi_5^{(B)}$ *satisfying Definition A.35.5. If such a theorem is obtained, the remaining OP 5(B) obstruction is* $R_5^{(B),\mathrm{post}}$ *under the retained stop-preserving guard.*

Proof. The task queue assigns OP 5(B) the obstruction *missing boundary promotion.* A boundary-promotion obstruction cannot be discharged by a route renaming alone, since the stop-preserving guard must survive promotion. The schema $\Pi_5^{(B)}$ states the exact data required for a conservative promotion: retained guard, HC/MM compatibility, backtranslation stability, and explicit post-promotion residual. Hence OP 5(B) is reduced to proving $\Pi_5^{(B)}$, after which $R_5^{(B),\mathrm{post}}$ is the remaining obstruction. □

Theorem A.35.7 (task-object formalization conservativity)**.** *The local proof line task-object formalization is conservative over the local proof line.*

Proof. The layer does not claim that $C_{1/4}^{\mathrm{sel}}$ or $\Pi_5^{(B)}$ has already been constructed. It only states their required roles and the residuals that remain after their construction. OP 1/4 and OP 5(B) therefore remain open, while their first target objects are now named and constrained. □

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A.36 Notation and Physics-Style Audit for OP-Facing Objects

Status. This layer audits the OP-facing notation introduced in the local proof chain and separates mathematical display notation from diagnostic/code-safe identifiers. It does not change the obstruction analysis, does not close OP 1/4 or OP 5(B), and does not replace QAM/MML/MMA by Python or by information-theoretic syntax.

A.36.1 Display notation versus diagnostic identifiers

Definition A.36.1 (Display notation and diagnostic identifier)**.** *A display notation is the notation used in mathematical and physical statements. A diagnostic identifier is a machine-safe name used by Python snippets, JSON reports, CSV tables, or local audit files. Diagnostic identifiers may be short and ASCII-safe; display notation must remain mathematically and physically readable.*

Diagnostic identifier	Mathematical display notation	Role
`OP_1_4`	$\mathrm{OP}_{1/4}$	OP 1/4 normal-form target
`OP_5B`	$\mathrm{OP}_5^{(B)}$	OP 5(B) normal-form target
`C_OP14`	$C_{1/4}^{\mathrm{sel}}$	selector/comparison object for OP 1/4
`P_5B`	$\Pi_5^{(B)}$	boundary-promotion schema for OP 5(B)
`R_OP14_post`	$R_{1/4}^{\mathrm{post}}$	post-selector residual
`R_5B_post`	$R_5^{(B),\mathrm{post}}$	post-promotion residual

A.36.2 Physics-style rule

Definition A.36.2 (Physics-style rule for OP-facing notation)**.** *Information-theoretic and code-facing tools may be used as diagnostic encodings only after the governing mathematical object has been written in display notation. In particular, the active text should prefer*

$$C_{1/4}^{\mathrm{sel}}, \qquad \Pi_5^{(B)}, \qquad R_{1/4}^{\mathrm{post}}, \qquad R_5^{(B),\mathrm{post}}$$

over code-like symbols such as `C_OP14`, `P_5B`, `R_OP14`, or `R_5B`.

Remark A.36.3 (Information theory is a diagnostic layer)**.** *The route/mask/identifier layer may look information-theoretic, but the governing objects remain mathematical and physical readout objects. The information-theoretic encoding is therefore a compression and audit surface, not a replacement ontology.*

A.36.3 Normalized target-object statements

With notation normalized, the two target objects read as follows:

$$C_{1/4}^{\mathrm{sel}} : P(\rho_{1/4}^{\mathrm{pre}}) \times P(\rho_{1/4}^{\mathrm{shift}}) \longrightarrow E_{1/4}^{\mathrm{sel}},$$

and

$$\Pi_5^{(B)} : (Y_5^{(B)}, G_\partial, HC/MM) \longrightarrow (Y_5^{(B),\mathrm{prom}}, G_\partial, R_5^{(B),\mathrm{post}}).$$

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Thus OP 1/4 is still reduced to a selector/comparison construction, and OP 5(B) is still reduced to a boundary-promotion construction, but the display notation no longer borrows code-style names from the diagnostic layer.

Theorem A.36.4 (Notation-normalization conservativity)**.** *The notation normalization is conservative over the task-object formalization.*

Proof. The normalization replaces code-like display labels by mathematical display notation and records a diagnostic-identifier map. It does not change the underlying route, guard, obstruction, residual, or OP-status records. Therefore the reductions to a selector/comparison object for OP 1/4 and to a boundary-promotion schema for OP 5(B) are preserved. □

Remark A.36.5 (non-claim)**.** *Version 4.1.0-does not close OP 1/4 or OP 5(B), does not activate primitive mixed tensors, does not claim Turing-completeness, and does not replace standard QFT/QED/Lifshitz or general-relativistic calculations.*

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A.37 First Selector and Boundary-Promotion Ansatz

Status. This layer starts the first genuine OP-facing mathematical attack after the notation/physics-style audit. It does not close OP 1/4 or OP 5(B). It turns the two normalized target objects,

$$C_{1/4}^{\mathrm{sel}} \qquad \text{and} \qquad \Pi_5^{(B)},$$

into first construction ansaetze: a comparison/fibre-product ansatz for OP 1/4 and a guard-preserving boundary-promotion ansatz for OP 5(B).

A.37.1 OP 1/4: comparison surface and selector ansatz

Definition A.37.1 (OP 1/4 comparison surface)**.** *Let*

$$P(\rho_{1/4}^{\mathrm{pre}}) \qquad \textit{and} \qquad P(\rho_{1/4}^{\mathrm{shift}})$$

be the witness-side and shifted-normalized packet surfaces attached to $\mathrm{OP}_{1/4}$*. A comparison surface for OP 1/4 is an object* $E_{1/4}^{\mathrm{sel}}$ *equipped with two comparison maps*

$$e_{\mathrm{pre}} : P(\rho_{1/4}^{\mathrm{pre}}) \longrightarrow E_{1/4}^{\mathrm{sel}}, \qquad e_{\mathrm{shift}} : P(\rho_{1/4}^{\mathrm{shift}}) \longrightarrow E_{1/4}^{\mathrm{sel}}.$$

The maps must be read as current shifted diagnostic/readout maps, not as a claim that the historical witness and active shifted surface are already identical.

Definition A.37.2 (Selector/comparison object for OP 1/4)**.** *Given a comparison surface* $E_{1/4}^{\mathrm{sel}}$*, define the first selector/comparison ansatz by the fibre product*

$$C_{1/4}^{\mathrm{sel}} := P(\rho_{1/4}^{\mathrm{pre}}) \times_{E_{1/4}^{\mathrm{sel}}} P(\rho_{1/4}^{\mathrm{shift}}).$$

Equivalently,

$$C_{1/4}^{\mathrm{sel}} = \{(x_{\mathrm{pre}}, x_{\mathrm{shift}}) \mid e_{\mathrm{pre}}(x_{\mathrm{pre}}) = e_{\mathrm{shift}}(x_{\mathrm{shift}})\}.$$

Thus $C_{1/4}^{\mathrm{sel}}$ *contains exactly the witness/shifted pairs that become equal after being read in the same comparison surface.*

Definition A.37.3 (Post-selector residual)**.** *The post-selector residual for OP 1/4 is the remaining comparison obstruction after the selector object has been formed. In the weak set-difference reading,*

$$R_{1/4}^{\mathrm{post}} := E_{1/4}^{\mathrm{sel}} \setminus \mathrm{im}\big(C_{1/4}^{\mathrm{sel}} \to E_{1/4}^{\mathrm{sel}}\big).$$

In a category or additive setting, the same role may be represented by a cokernel/cocone residual. The present layer requires only the weak residual role, not a fixed ambient abelian category.

Lemma A.37.4 (Selector ansatz reduces the OP 1/4 obstruction)**.** *If the comparison surface exists and the fibre product* $C_{1/4}^{\mathrm{sel}}$ *is well-defined under the inherited guards, then the obstruction* missing selector *is reduced to the post-selector residual* $R_{1/4}^{\mathrm{post}}$*.*

Proof. The obstruction identified the absence of a comparison/selector object. Definitions A.37.1 and A.37.2 provide such an object as a fibre product over a common comparison surface. Therefore the original absence of a selector is discharged at the level of ansatz formation. What remains is not the same obstruction but the part of the comparison surface not reached by that selector, namely $R_{1/4}^{\mathrm{post}}$. □

Remark A.37.5 (Why this is not yet OP 1/4 closure)**.** *The selector ansatz does not prove that* $R_{1/4}^{\mathrm{post}} = \varnothing$ *and does not prove that all OP 1/4 guards are discharged. It only turns the vague obstruction "missing selector" into a precise residual target.*

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A.37.2 OP 5(B): guard-preserving boundary-promotion ansatz

Definition A.37.6 (Boundary-promotion ansatz for OP 5(B))**.** *Let $Y_5^{(B)}$ be the boundary-side object associated with $\mathrm{OP}_5^{(B)}$, let G_∂ be the inherited stop-preserving boundary guard, and let $\mathcal{R}_{HC/MM}$ denote the HC/MM readout interface. A first boundary-promotion ansatz is an operation*

$$\Pi_5^{(B)} : (Y_5^{(B)}, G_\partial, HC/MM) \longrightarrow (Y_5^{(B),\mathrm{prom}}, G_\partial, R_5^{(B),\mathrm{post}})$$

satisfying the guard-preservation condition

$$G_\partial \circ \Pi_5^{(B)} = G_\partial$$

where both sides are read on the admissible boundary-side domain.

Definition A.37.7 (HC/MM compatibility defect)**.** *The HC/MM compatibility defect of $\Pi_5^{(B)}$ is the obstruction to the commutation relation*

$$\mathcal{R}_{HC/MM} \circ \Pi_5^{(B)} \sim_\partial \Pi_5^{(B)} \circ \mathcal{R}_{HC/MM},$$

where $\sim_\partial$ denotes boundary-guard-compatible comparison. The post-promotion residual is written

$$R_5^{(B),\mathrm{post}} := \mathrm{Obstr}\left(G_\partial \circ \Pi_5^{(B)} = G_\partial,\ \mathcal{R}_{HC/MM} \circ \Pi_5^{(B)} \sim_\partial \Pi_5^{(B)} \circ \mathcal{R}_{HC/MM}\right).$$

Lemma A.37.8 (Boundary-promotion ansatz reduces the OP 5(B) obstruction)**.** *If $\Pi_5^{(B)}$ is defined on the admissible boundary-side domain and preserves G_∂, then the obstruction* missing boundary promotion *is reduced to the post-promotion residual $R_5^{(B),\mathrm{post}}$.*

Proof. The obstruction named the absence of a boundary-promotion mechanism. A map $\Pi_5^{(B)}$ satisfying Definition A.37.6 provides such a mechanism while preserving the inherited boundary guard. The remaining obstruction is therefore not the absence of promotion itself, but the failure, if any, of guard preservation and HC/MM compatibility. This remaining obstruction is precisely $R_5^{(B),\mathrm{post}}$. □

Remark A.37.9 (Physical reading)**.** *The OP 5(B) ansatz keeps the physical boundary reading visible. It does not promote a boundary object by erasing the boundary stop; it asks whether a promotion through the HC/MM readout interface can be made while the stop-preserving guard survives.*

A.37.3 First combined reduction statement

Theorem A.37.10 (First OP-facing obstruction reduction)**.** *In the ansatz layer, the two the local pass/obstruction labels reduce as follows:*

$$\begin{aligned} \mathrm{OP}_{1/4}: &\quad \textit{missing selector} \rightsquigarrow R_{1/4}^{\mathrm{post}}, \\ \mathrm{OP}_5^{(B)}: &\quad \textit{missing boundary promotion} \rightsquigarrow R_5^{(B),\mathrm{post}}. \end{aligned}$$

The reduction is conditional on the formation of $C_{1/4}^{\mathrm{sel}}$ and $\Pi_5^{(B)}$, respectively.

Proof. The first line follows from Lemma A.37.4. The second line follows from Lemma A.37.8. Both reductions are conditional because neither residual is proven to vanish in this layer. □

Remark A.37.11 (non-claim)**.** *Version 4.1.0-does not close OP 1/4 or OP 5(B). It replaces the first named obstructions by sharper residual targets. It also does not activate primitive mixed tensors, does not claim Turing-completeness, and does not replace standard QFT/QED/Lifshitz or general-relativistic calculations.*

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A.38 First Selector Existence Reduction for OP 1/4

Status. This layer continues the OP-facing phase by attacking the first OP 1/4 target $C^{\mathrm{sel}}_{1/4}$. The point is deliberately modest: the selector is constructed as a fiber product/equalizer object whenever the two comparison readouts to a common surface exist. This removes the purely formal "missing-selector" obstruction, but it does not close OP 1/4. The remaining questions are non-emptiness, adequacy, guard strength, and post-selector residual control.

A.38.1 Comparison surface and readout maps

Definition A.38.1 (OP 1/4 comparison datum)**.** *An OP 1/4 comparison datum is a triple*

$$\mathfrak{E}_{1/4} := \left(E^{\mathrm{sel}}_{1/4}, e_{\mathrm{pre}}, e_{\mathrm{shift}}\right)$$

where $E^{\mathrm{sel}}_{1/4}$ is a comparison surface and

$$e_{\mathrm{pre}} : P(\rho^{\mathrm{pre}}_{1/4}) \longrightarrow E^{\mathrm{sel}}_{1/4}, \qquad e_{\mathrm{shift}} : P(\rho^{\mathrm{shift}}_{1/4}) \longrightarrow E^{\mathrm{sel}}_{1/4}$$

are the pre-shift/witness-side and shifted-normalized readout maps.

Definition A.38.2 (Selector as fiber product)**.** *Given a comparison datum $\mathfrak{E}_{1/4}$, define*

$$C^{\mathrm{sel}}_{1/4} := P(\rho^{\mathrm{pre}}_{1/4}) \times_{E^{\mathrm{sel}}_{1/4}} P(\rho^{\mathrm{shift}}_{1/4})$$

as the set of compatible pairs

$$C^{\mathrm{sel}}_{1/4} = \{(x_{\mathrm{pre}}, x_{\mathrm{shift}}) \mid e_{\mathrm{pre}}(x_{\mathrm{pre}}) = e_{\mathrm{shift}}(x_{\mathrm{shift}})\}\,.$$

Equivalently, it is the equalizer of the two maps

$$e_{\mathrm{pre}} \circ \pi_{\mathrm{pre}}, \quad e_{\mathrm{shift}} \circ \pi_{\mathrm{shift}} : P(\rho^{\mathrm{pre}}_{1/4}) \times P(\rho^{\mathrm{shift}}_{1/4}) \longrightarrow E^{\mathrm{sel}}_{1/4}.$$

Lemma A.38.3 (Formal selector existence)**.** *If $P(\rho^{\mathrm{pre}}_{1/4})$, $P(\rho^{\mathrm{shift}}_{1/4})$, $E^{\mathrm{sel}}_{1/4}$, e_{pre}, and e_{shift} exist as ordinary shifted-admissible set/readout objects, then $C^{\mathrm{sel}}_{1/4}$ exists as a shifted-admissible selector object.*

Proof. The cartesian product $P(\rho^{\mathrm{pre}}_{1/4}) \times P(\rho^{\mathrm{shift}}_{1/4})$ exists in the underlying set-compatible universe. The two maps to $E^{\mathrm{sel}}_{1/4}$ determine an equality condition. By separation, the subset of pairs satisfying that equality condition exists. This subset is exactly $C^{\mathrm{sel}}_{1/4}$. Since the construction uses only product, composition, equality, and separation, it is conservative over the shifted-admissible QAM/ZF-compatible setting. □

A.38.2 Backtranslation and guards

Definition A.38.4 (Backtranslation-compatible comparison datum)**.** *A comparison datum $\mathfrak{E}_{1/4}$ is backtranslation-compatible if there is a backtranslation readout B_4 such that the diagram*

$$\begin{array}{ccc} P(\rho^{\mathrm{shift}}_{1/4}) & \xrightarrow{e_{\mathrm{shift}}} & E^{\mathrm{sel}}_{1/4} \\ \downarrow B_4 & & \downarrow B_E \\ P(\rho^{\mathrm{pre}}_{1/4}) & \xrightarrow{e_{\mathrm{pre}}} & E^{\mathrm{sel,pre}}_{1/4} \end{array}$$

commutes up to the declared comparison tolerance of the OP 1/4 route.

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Lemma A.38.5 (Guard pullback)**.** *Let $G_{1/4}$ be a guard predicate on both packet surfaces and suppose e_{pre} and e_{shift} are guard-respecting. Then $C^{\mathrm{sel}}_{1/4}$ inherits the pulled-back guard*

$$G^{\mathrm{sel}}_{1/4}(x_{\mathrm{pre}}, x_{\mathrm{shift}}) := G_{1/4}(x_{\mathrm{pre}}) \wedge G_{1/4}(x_{\mathrm{shift}}).$$

Proof. Every element of $C^{\mathrm{sel}}_{1/4}$ is a pair from the two guarded packet surfaces. Defining the selector guard by conjunction of the two component guards therefore produces a well-defined predicate on the fiber product. Because the readout maps are guard-respecting, equality in $E^{\mathrm{sel}}_{1/4}$ does not erase the inherited guard data. □

A.38.3 Residual reduction

Definition A.38.6 (Post-selector residual)**.** *The post-selector residual is*

$$R^{\mathrm{post}}_{1/4} := E^{\mathrm{sel}}_{1/4} \setminus \mathrm{im}\big(C^{\mathrm{sel}}_{1/4} \to E^{\mathrm{sel}}_{1/4}\big),$$

or, in a non-set-subtractive readout context, the corresponding obstruction object measuring the part of $E^{\mathrm{sel}}_{1/4}$ not covered by compatible selector pairs.

Proposition A.38.7 (Missing selector reduces to adequacy and residual control)**.** *Under the comparison-datum hypotheses of Definition A.38.1, the OP 1/4 obstruction "missing selector" is replaced by the following narrower tasks:*

$$\mathrm{NonEmpty}(C^{\mathrm{sel}}_{1/4}), \qquad \mathrm{Adequate}(C^{\mathrm{sel}}_{1/4}), \qquad \mathrm{Control}(R^{\mathrm{post}}_{1/4}).$$

Proof. Lemma A.38.3 supplies the formal selector object as soon as the comparison datum exists. Therefore the obstruction is no longer the absence of a definable selector. What remains is whether the selector has enough elements or enough structural coverage to compare the two packet surfaces, and whether the part not covered by the selector is controlled by $R^{\mathrm{post}}_{1/4}$. These are precisely non-emptiness, adequacy, and residual-control tasks. □

Remark A.38.8 (OP 1/4 status after)**.** *This does not close OP 1/4. It does, however, make the formal obstruction smaller. The next OP 1/4 step is not "find any selector", but rather prove that the fiber-product selector is nonempty or adequate for the OP 1/4 route and then reduce $R^{\mathrm{post}}_{1/4}$.*

Theorem A.38.9 (conservativity)**.** *The local proof line selector-existence reduction is conservative over the prior archival stage working line.*

Proof. The layer constructs $C^{\mathrm{sel}}_{1/4}$ by ordinary fiber-product / equalizer logic under stated hypotheses. It does not assert non-emptiness, adequacy, OP closure, physical promotion, or any replacement of standard physical calculations. Therefore it is conservative. □

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A.39 Selector Non-Emptiness and Anchor Criterion for OP 1/4

Status. The layer constructed the selector $C^{\mathrm{sel}}_{1/4}$ as a formal fiber product once a comparison datum exists. The present layer addresses the next obstruction: whether that selector is nonempty, and when it is structurally adequate. The result is still not an OP 1/4 closure. It reduces non-emptiness to an anchor/intersection condition inside the comparison surface $E^{\mathrm{sel}}_{1/4}$.

A.39.1 Anchor set

Definition A.39.1 (OP 1/4 selector anchor set)**.** *Given the comparison datum*

$$\mathfrak{E}_{1/4} = (E^{\mathrm{sel}}_{1/4}, e_{\mathrm{pre}}, e_{\mathrm{shift}}),$$

define the selector anchor set by

$$A^{\mathrm{sel}}_{1/4} := \mathrm{im}(e_{\mathrm{pre}}) \cap \mathrm{im}(e_{\mathrm{shift}}) \subseteq E^{\mathrm{sel}}_{1/4}.$$

An element $a \in A^{\mathrm{sel}}_{1/4}$ is a common comparison anchor if it is reached both from the pre-shift/witness-side packet surface and from the shifted-normalized packet surface.

Lemma A.39.2 (Non-emptiness criterion)**.** *The selector $C^{\mathrm{sel}}_{1/4}$ is nonempty if and only if the anchor set $A^{\mathrm{sel}}_{1/4}$ is nonempty.*

Proof. If $(x_{\mathrm{pre}}, x_{\mathrm{shift}}) \in C^{\mathrm{sel}}_{1/4}$, then

$$e_{\mathrm{pre}}(x_{\mathrm{pre}}) = e_{\mathrm{shift}}(x_{\mathrm{shift}}) = a$$

for some $a \in E^{\mathrm{sel}}_{1/4}$. Hence $a \in \mathrm{im}(e_{\mathrm{pre}}) \cap \mathrm{im}(e_{\mathrm{shift}})$, so $A^{\mathrm{sel}}_{1/4} \neq \varnothing$. Conversely, if $a \in A^{\mathrm{sel}}_{1/4}$, then there exist x_{pre} and x_{shift} with $e_{\mathrm{pre}}(x_{\mathrm{pre}}) = a = e_{\mathrm{shift}}(x_{\mathrm{shift}})$. Thus $(x_{\mathrm{pre}}, x_{\mathrm{shift}}) \in C^{\mathrm{sel}}_{1/4}$. □

A.39.2 Adequacy

Definition A.39.3 (Anchor adequacy)**.** *The anchor set $A^{\mathrm{sel}}_{1/4}$ is adequate for OP 1/4 if every comparison-relevant component of $E^{\mathrm{sel}}_{1/4}$ meets $A^{\mathrm{sel}}_{1/4}$. Equivalently, if $\pi_0^{\mathrm{rel}}(E^{\mathrm{sel}}_{1/4})$ denotes the finite family of comparison-relevant components, then*

$$\forall K \in \pi_0^{\mathrm{rel}}(E^{\mathrm{sel}}_{1/4}) : \quad K \cap A^{\mathrm{sel}}_{1/4} \neq \varnothing.$$

Lemma A.39.4 (Adequacy transfers to selector coverage)**.** *If $A^{\mathrm{sel}}_{1/4}$ is adequate, then $C^{\mathrm{sel}}_{1/4}$ has at least one compatible selector pair over every comparison-relevant component of $E^{\mathrm{sel}}_{1/4}$.*

Proof. Let K be a comparison-relevant component. By adequacy, choose $a \in K \cap A^{\mathrm{sel}}_{1/4}$. Since a is in the anchor set, it has a preimage under both e_{pre} and e_{shift}. The two preimages form a compatible pair in $C^{\mathrm{sel}}_{1/4}$ lying over $a \in K$. □

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A.39.3 Residual localization

Definition A.39.5 (Anchor residual)**.** *The anchor residual is*

$$R^{\mathrm{anc}}_{1/4} := E^{\mathrm{sel}}_{1/4} \setminus A^{\mathrm{sel}}_{1/4},$$

or, in non-subtractive readout language, the obstruction object measuring the comparison surface not hit by both readout images.

Proposition A.39.6 (Post-selector residual is controlled by anchor residual)**.** *The post-selector residual satisfies the schematic containment*

$$R^{\mathrm{post}}_{1/4} \preceq R^{\mathrm{anc}}_{1/4},$$

where $\preceq$ denotes the current readout-level obstruction comparison. In particular, if $A^{\mathrm{sel}}_{1/4}$ is adequate and the remaining non-anchor part is readout-irrelevant, then the selector obstruction is discharged at the comparison level.

Proof. The image of the selector in the comparison surface is exactly the common image part reached by both readouts, i.e. the anchor set, up to the declared readout identification. Therefore anything not covered by the selector is contained in or represented by the complement/non-anchor obstruction. Thus the post-selector residual is controlled by the anchor residual. Adequacy removes the comparison-relevant part of this obstruction; any remaining part must be declared readout-irrelevant before it can be ignored. □

A.39.4 Reduced OP 1/4 target after

Theorem A.39.7 (OP 1/4 selector obstruction reduction)**.** *Under the comparison-datum hypotheses, the OP 1/4 selector obstruction reduces to the following anchor tasks:*

$$A^{\mathrm{sel}}_{1/4} \neq \varnothing, \qquad \mathrm{Adequate}(A^{\mathrm{sel}}_{1/4}), \qquad \mathrm{Control}(R^{\mathrm{anc}}_{1/4}).$$

Proof. By Lemma A.39.2, selector non-emptiness is equivalent to anchor non-emptiness. By Lemma A.39.4, anchor adequacy gives selector coverage over the comparison-relevant components. By Proposition A.39.6, the post-selector residual is controlled by the anchor residual. Hence the selector obstruction is reduced to anchor non-emptiness, anchor adequacy, and anchor residual control. □

Remark A.39.8 (OP 1/4 status after)**.** *OP 1/4 remains open. However, the obstruction has been sharpened again: instead of asking whether a selector exists in the abstract, the next step is to show that the two readout images have a common and adequate anchor set inside $E^{\mathrm{sel}}_{1/4}$.*

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A.40 Anchor Adequacy and Component-Cover Criterion for OP 1/4

Status. The layer reduced selector non-emptiness to the non-emptiness of the anchor set $A^{\mathrm{sel}}_{1/4}$. The present layer attacks the next question: when is the anchor set adequate? The answer is expressed as a component-cover condition on the comparison-relevant part of $E^{\mathrm{sel}}_{1/4}$. This does not close OP 1/4; it converts the adequacy task into a finite route/component coverage task.

A.40.1 Relevant component system

Definition A.40.1 (Comparison-relevant component system)**.** *Let $E^{\mathrm{sel}}_{1/4}$ be the OP 1/4 comparison surface. A comparison-relevant component system is a finite family*

$$\mathcal{K}_{1/4} = \{K_1, \dots, K_m\}$$

of subregions, strata, or readout components of $E^{\mathrm{sel}}_{1/4}$ such that every comparison-relevant obstruction in the OP 1/4 selector problem is represented in at least one K_i.

Remark A.40.2 (Why finite components are allowed here)**.** *The component system is not asserted to exhaust the whole comparison surface. It only exhausts the comparison-relevant part selected by the current OP 1/4 route and its guards. This keeps the criterion finite and OP-facing.*

A.40.2 Anchor cover

Definition A.40.3 (Anchor cover)**.** *The anchor set $A^{\mathrm{sel}}_{1/4}$ is a component cover of $\mathcal{K}_{1/4}$ if*

$$\forall K_i \in \mathcal{K}_{1/4}: \qquad K_i \cap A^{\mathrm{sel}}_{1/4} \neq \varnothing.$$

Equivalently, the cover defect is empty, where

$$D^{\mathrm{cov}}_{1/4} := \{K_i \in \mathcal{K}_{1/4} \mid K_i \cap A^{\mathrm{sel}}_{1/4} = \varnothing\}.$$

Lemma A.40.4 (Component cover implies adequacy)**.** *If $A^{\mathrm{sel}}_{1/4}$ is a component cover of $\mathcal{K}_{1/4}$, then $A^{\mathrm{sel}}_{1/4}$ is adequate relative to the comparison-relevant component system.*

Proof. Adequacy relative to $\mathcal{K}_{1/4}$ means that every comparison-relevant component contains at least one common anchor. This is exactly the component cover condition. Hence the condition implies adequacy. □

A.40.3 Route-lifted anchors

Definition A.40.5 (Route-lifted anchor)**.** *A route-lifted anchor over K_i is a triple*

$$\alpha_i = (x_i^{\mathrm{pre}}, x_i^{\mathrm{shift}}, a_i)$$

such that

$$a_i \in K_i, \qquad e_{\mathrm{pre}}(x_i^{\mathrm{pre}}) = a_i = e_{\mathrm{shift}}(x_i^{\mathrm{shift}}),$$

and the pair $(x_i^{\mathrm{pre}}, x_i^{\mathrm{shift}})$ respects the inherited OP 1/4 route guard.

Proposition A.40.6 (Route-lifted anchors certify component cover)**.** *If every $K_i \in \mathcal{K}_{1/4}$ admits a route-lifted anchor α_i, then $A^{\mathrm{sel}}_{1/4}$ is a component cover of $\mathcal{K}_{1/4}$.*

Proof. For each K_i, the anchor α_i contains a point $a_i \in K_i$ that lies in the image of both readout maps. Therefore $a_i \in K_i \cap A^{\mathrm{sel}}_{1/4}$, so the intersection is nonempty for every K_i. □

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A.40.4 Residual after component cover

Definition A.40.7 (Cover residual)**.** *The cover residual is*

$$R^{\mathrm{cov}}_{1/4} := D^{\mathrm{cov}}_{1/4}$$

together with any residual guard data attached to components not covered by a route-lifted anchor.

Proposition A.40.8 (Anchor residual reduces to cover residual)**.** *Relative to the component system $\mathcal{K}_{1/4}$, anchor residual control reduces to cover-residual control:*

$$\mathrm{Control}(R^{\mathrm{anc}}_{1/4}) \quad \rightsquigarrow \quad \mathrm{Control}(R^{\mathrm{cov}}_{1/4}).$$

If $D^{\mathrm{cov}}_{1/4} = \varnothing$, then the comparison-relevant anchor residual is discharged.

Proof. The component system records all comparison-relevant regions. The only comparison-relevant part not controlled by anchors is represented by components which do not meet the anchor set. These are precisely the elements of $D^{\mathrm{cov}}_{1/4}$. Thus controlling the anchor residual relative to the selected component system is equivalent to controlling the cover defect. When the cover defect is empty, every relevant component has an anchor and the component-level anchor residual vanishes. □

A.40.5 Reduced OP 1/4 target after

Theorem A.40.9 (OP 1/4 adequacy reduction to finite cover)**.** *Under the comparison and anchor hypotheses, OP 1/4 selector adequacy is reduced to the finite cover condition*

$$D^{\mathrm{cov}}_{1/4} = \varnothing,$$

or equivalently to the existence of route-lifted anchors

$$\alpha_i = (x_i^{\mathrm{pre}}, x_i^{\mathrm{shift}}, a_i)$$

for every comparison-relevant component $K_i \in \mathcal{K}_{1/4}$.

Proof. By Lemma A.40.4, a component cover is exactly adequacy relative to the selected comparison-relevant components. By Proposition A.40.6, route-lifted anchors certify this cover. By Proposition A.40.8, the remaining anchor residual is represented by the cover defect. Hence the adequacy task reduces to the finite cover condition $D^{\mathrm{cov}}_{1/4} = \varnothing$ or to the equivalent route-lifted-anchor certificate. □

Remark A.40.10 (OP 1/4 status after)**.** *OP 1/4 is still open. However, the obstruction is now much more concrete: find route-lifted anchors for the finitely selected comparison-relevant components, or identify which components remain in $D^{\mathrm{cov}}_{1/4}$.*

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A.41 Clock-Anchored Canonical Optimal Route Principle

Status. This layer records the clock-anchored optimal-route principle suggested by the OP-facing analysis. It changes the problem shape: once the clock anchor and a target information locus are fixed, the remaining first task is to prove that at least one admissible route exists. Multiple geometric drawings of the same canonical route are not treated as different optimal routes; they are identified by the Stone–von Neumann/Teichmuller/QAM normal-form reading. This does not close OP 1/4, OP 4, or OP 5. It supplies a stronger route principle for the next OP reductions.

A.41.1 Clock anchor and admissible route space

Definition A.41.1 (Clock anchor)**.** *The clock anchor is the distinguished known reference information point*

$$q_{\mathrm{clk}}$$

against which OP-facing routes are read. It is not an optional coordinate choice; it is the fixed reference supplied by the taktgeber/clock structure of the system.

Definition A.41.2 (Admissible clock-to-target routes)**.** *For a target information locus q_{tar}, let*

$$\mathcal{R}_{\mathrm{adm}}(q_{\mathrm{clk}}, q_{\mathrm{tar}})$$

denote the class of router-admissible routes from the clock anchor to the target which respect the currently declared QAM/MML/MMA tensor interfaces and all active guards. If a boundary guard G_∂ is present, the admissible class is restricted to routes satisfying the guard-preservation condition.

Definition A.41.3 (Route action and canonical equivalence)**.** *A route action is a ranked route functional*

$$\mathcal{A} : \mathcal{R}_{\mathrm{adm}}(q_{\mathrm{clk}}, q_{\mathrm{tar}}) \longrightarrow \Lambda$$

where Λ is the ordered cost/rank set selected by the current OP-facing readout. Two routes ρ_1, ρ_2 are canonically equivalent, written

$$\rho_1 \sim_{\mathrm{can}} \rho_2,$$

if they represent the same clock-anchored information transport after the Stone–von Neumann representation identification, the Teichmuller extremal-route normalization, and the QAM/MML normal-form readout.

Remark A.41.4 (Uniqueness means canonical uniqueness)**.** *The principle does not claim that there cannot be two visually different geometric curves. It claims that, once they realize the same canonical clock-anchored information transport, they are one route in the mathematical sense relevant here. Different drawings can therefore represent the same canonical optimal route.*

A.41.2 Canonical optimal route

Definition A.41.5 (Canonical optimal route class)**.** *Assume $\mathcal{R}_{\mathrm{adm}}(q_{\mathrm{clk}}, q_{\mathrm{tar}}) \neq \varnothing$. A canonical optimal route class is*

$$[\rho_{\mathrm{opt}}]_{\mathrm{can}} := \operatorname*{argmin}_{[\rho]_{\mathrm{can}} \subseteq \mathcal{R}_{\mathrm{adm}}(q_{\mathrm{clk}}, q_{\mathrm{tar}})} \mathcal{A}([\rho]_{\mathrm{can}}),$$

where the minimization is taken after canonical equivalence has identified routes representing the same Stone–von Neumann/Teichmuller/QAM-normalized information transport.

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Definition A.41.6 (QAM/MML canonical representative)**.** *The canonical representative*

$$\rho_{\mathrm{opt}}^{\mathrm{can}}$$

is the QAM/MML normal-form representative of $[\rho_{\mathrm{opt}}]_{\mathrm{can}}$*. It is the route object used by the router language, diagnostic comparisons, and OP-facing residual tests.*

Theorem A.41.7 (Clock-anchored canonical optimal route principle)**.** *Suppose the admissible clock-to-target route space is nonempty and finite, discretely ranked, or otherwise compact enough for the declared route action to attain its minimum. Then there exists a unique canonical optimal route representative*

$$\rho_{\mathrm{opt}}^{\mathrm{can}} = \rho_{\mathrm{opt}}^{\mathrm{can}}(q_{\mathrm{clk}}, q_{\mathrm{tar}}).$$

Proof. Since the admissible route space is nonempty and the route action attains a minimum under the stated finiteness/rank/compactness hypothesis, at least one minimizing route exists. If several geometric representatives realize the same minimum, the Stone–von Neumann representation identification, Teichmuller extremal-route normalization, and QAM/MML normal-form readout identify them as one canonical information route. The QAM/MML normal-form rule then selects one canonical representative of that class. Hence the canonical optimal route representative exists and is unique in the route language of the paper. □

A.41.3 Optimal-route map

Definition A.41.8 (Clock-anchored optimal-route map)**.** *For all target information loci for which the admissible route space is nonempty, define the optimal-route map*

$$\mathcal{O}_{\mathrm{clk}} : q_{\mathrm{tar}} \longmapsto \rho_{\mathrm{opt}}^{\mathrm{can}}(q_{\mathrm{clk}}, q_{\mathrm{tar}}).$$

This map is the local "map of optimal routes" determined by the clock anchor, the target locus, the route action, and the canonical equivalence rule.

Remark A.41.9 (Why this is stronger than arbitrary path choice)**.** *The route is not chosen by convenience. The clock anchor fixes the reference, the target fixes the information locus, the admissible class fixes the allowed router/tensor/guard moves, the route action fixes optimality, and the canonical equivalence/normal-form rule removes representational multiplicity.*

A.41.4 Effect on OP 1/4

Definition A.41.10 (Clock-to-anchor admissibility for OP 1/4)**.** *For a comparison-relevant component* $K_i \in \mathcal{K}_{1/4}$*, define*

$$\mathcal{R}_{\mathrm{adm}}(q_{\mathrm{clk}}, K_i \cap A_{1/4}^{\mathrm{sel}}) \neq \varnothing$$

to mean that at least one route-admissible clock-to-anchor route reaches a common anchor inside K_i*.*

Proposition A.41.11 (OP 1/4 cover criterion becomes route existence)**.** *If for every* $K_i \in \mathcal{K}_{1/4}$ *one has*

$$\mathcal{R}_{\mathrm{adm}}(q_{\mathrm{clk}}, K_i \cap A_{1/4}^{\mathrm{sel}}) \neq \varnothing,$$

then each component has a unique canonical optimal clock-to-anchor route

$$\rho_i^{\mathrm{opt,can}} = \rho_{\mathrm{opt}}^{\mathrm{can}}(q_{\mathrm{clk}}, K_i \cap A_{1/4}^{\mathrm{sel}}),$$

and the finite cover task is reduced to proving admissible route existence for each relevant component.

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Proof. Apply Theorem A.41.7 to each nonempty admissible route space $\mathcal{R}_{\mathrm{adm}}(q_{\mathrm{clk}}, K_i \cap A_{1/4}^{\mathrm{sel}})$. Each such route space has a unique canonical optimal representative. Therefore the component-cover problem is no longer an arbitrary anchor search; it is a finite family of clock-to-anchor admissible-route existence problems. □

A.41.5 Effect on OP 4 and OP 5

Remark A.41.12 (OP 4 as optimal-route-map stability)**.** *With the clock-anchored optimal-route map in place, OP 4 can be read as a stability problem for $\mathcal{O}_{\mathrm{clk}}$: the question is whether the canonical optimal routes remain transport-stable under the admitted router and tensor transitions.*

Proposition A.41.13 (Boundary route principle for OP 5(B))**.** *If the guard-preserving route space*

$$\mathcal{R}_{\mathrm{adm}}^{G_\partial}(q_{\mathrm{clk}}, Y_5^{(B),\mathrm{prom}})$$

is nonempty, then there is a unique canonical guard-preserving boundary promotion route

$$\rho_5^{(B),\mathrm{opt,can}}.$$

The promotion package $\Pi_5^{(B)}$ may then be read as the execution of this canonical route, subject to the remaining HC/MM compatibility and residual checks.

Proof. This is the guarded instance of the clock-anchored canonical optimal route principle, with admissibility restricted by G_∂. The existence of at least one guard-preserving route gives a minimizing canonical class and a QAM/MML representative. The execution of that representative gives the route-level reading of $\Pi_5^{(B)}$; HC/MM compatibility and residual control remain separate obligations. □

A.41.6 reduced obstruction

Theorem A.41.14 (OP-facing reduction)**.** *After the clock-anchored canonical optimal route principle, the next OP-facing obstruction is not selection among many equally optimal routes. It is the existence of admissible routes to the required anchor or boundary targets:*

$$\mathcal{R}_{\mathrm{adm}}(q_{\mathrm{clk}}, q_{\mathrm{tar}}) \neq \varnothing.$$

Once such a route exists, its canonical optimal representative is fixed by the Stone–von Neumann/Teichmuller/QAM normal-form reading.

Proof. The preceding theorem gives a unique canonical optimal representative whenever the admissible route space is nonempty and the declared action attains its minimum. Therefore the ambiguity of multiple equal routes is removed at the canonical-route level. What remains is to prove that the relevant admissible route spaces are nonempty and that their guards and residuals are respected. □

Remark A.41.15 (OP status after)**.** *OP 1/4, OP 4, and OP 5 remain open. The gain is that their next route-level form is sharper: show admissible clock-to-target route existence, then compare the current route to the canonical optimal-route map.*

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A.42 Admissible Route Certificate and Optimal-Route Trigger

Status. The layer says that once an admissible clock-to-target route exists, the Stone–von Neumann/Teichmuller/QAM-normalized optimal route is canonically fixed. The present layer therefore moves the OP-facing task one step earlier: instead of searching for the optimal route directly, it defines the finite certificate needed to prove that an admissible route exists.

A.42.1 Route certificate

Definition A.42.1 (Admissible route certificate)**.** *Let q_{clk} be the clock anchor and q_{tar} a target information locus. An admissible route certificate is a finite record*

$$\mathsf{Cert}_\rho(q_{\mathrm{clk}}, q_{\mathrm{tar}}) = (V_\rho, E_\rho, s, t, \tau, \Gamma, \Theta)$$

where:

1. *V_ρ is a finite family of route states;*
2. *$E_\rho \subseteq V_\rho \times V_\rho$ is a finite family of directed admissible route steps;*
3. *$s = q_{\mathrm{clk}}$ is the clock start state;*
4. *$t = q_{\mathrm{tar}}$ is the target state or target component;*
5. *τ assigns the QAM/MML/MMA tensor type to each edge;*
6. *Γ records the active guard obligations for each edge;*
7. *Θ records the route action/rank data used by the local pass.*

Definition A.42.2 (Certificate validity)**.** *The certificate Cert_ρ is valid if:*

1. *there is a directed path in (V_ρ, E_ρ) from s to t;*
2. *every edge on the path has matching tensor interfaces under τ;*
3. *every edge preserves the guards declared by Γ;*
4. *the route action/rank record Θ is defined along the path.*

Lemma A.42.3 (Certificate implies admissible route existence)**.** *If $\mathsf{Cert}_\rho(q_{\mathrm{clk}}, q_{\mathrm{tar}})$ is valid, then*

$$\mathcal{R}_{\mathrm{adm}}(q_{\mathrm{clk}}, q_{\mathrm{tar}}) \neq \varnothing.$$

Proof. A valid certificate contains a directed path from the clock start state to the target state. Each edge in the path is admissible by tensor-interface matching, guard preservation, and route-action definability. Concatenating the certified edges yields an admissible route from q_{clk} to q_{tar}. □

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A.42.2 Optimal-route trigger

Theorem A.42.4 (Certificate triggers canonical optimal route)**.** *If a valid admissible route certificate exists for* $(q_{\mathrm{clk}}, q_{\mathrm{tar}})$*, then the principle supplies the canonical optimal route representative*

$$\rho_{\mathrm{opt}}^{\mathrm{can}}(q_{\mathrm{clk}}, q_{\mathrm{tar}}).$$

Proof. By Lemma A.42.3, a valid certificate proves that the admissible route space is nonempty. Applying the clock-anchored canonical optimal route principle gives the unique canonical optimal route representative, assuming the declared action/rank setting is one in which the minimum is attained. □

A.42.3 OP 1/4 certificate form

Definition A.42.5 (OP 1/4 component certificate)**.** *For a comparison-relevant component* $K_i \in \mathcal{K}_{1/4}$*, an OP 1/4 component certificate is a valid route certificate for the target*

$$q_{\mathrm{tar},i} := K_i \cap A_{1/4}^{\mathrm{sel}}.$$

It proves that the route system can reach a common anchor in the component from the clock anchor.

Proposition A.42.6 (OP 1/4 finite certificate reduction)**.** *If every component* $K_i \in \mathcal{K}_{1/4}$ *admits a valid OP 1/4 component certificate, then every component has a canonical optimal clock-to-anchor route*

$$\rho_i^{\mathrm{opt,can}}.$$

Consequently the cover defect $D_{1/4}^{\mathrm{cov}}$ *is reduced to the absence or failure of such finite certificates.*

Proof. Each valid component certificate gives an admissible route from q_{clk} to $K_i \cap A_{1/4}^{\mathrm{sel}}$. By the local pass principle, each such route space has a canonical optimal representative. If a component has no certificate, it remains in the certificate-level obstruction. Thus the previous cover defect is represented by the finite certificate failure set. □

A.42.4 OP 5(B) boundary certificate form

Definition A.42.7 (Boundary promotion certificate)**.** *A boundary promotion certificate for OP 5(B) is a valid route certificate for*

$$q_{\mathrm{tar}} = Y_5^{(B),\mathrm{prom}}$$

inside the guard-restricted route class

$$\mathcal{R}_{\mathrm{adm}}^{G_\partial}(q_{\mathrm{clk}}, Y_5^{(B),\mathrm{prom}}).$$

Its guard record Γ *must include*

$$G_\partial(u) = G_\partial(v)$$

for every certified boundary edge $u \to v$*.*

Proposition A.42.8 (Boundary certificate triggers promotion route)**.** *A valid OP 5(B) boundary promotion certificate supplies a unique canonical guard-preserving boundary route*

$$\rho_5^{(B),\mathrm{opt,can}},$$

and hence gives the route-level input needed to interpret $\Pi_5^{(B)}$ *as execution of that canonical route, subject to HC/MM compatibility and residual checks.*

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Proof. The certificate proves non-emptiness of the guard-restricted route space. The guarded route principle then supplies a canonical optimal representative. The route-level execution of this representative is exactly the route input needed for the promotion package. The HC/MM and residual clauses remain separate obligations. □

A.42.5 Certificate obstruction

Definition A.42.9 (Certificate obstruction set)**.** *For a finite family of OP targets* $\{q_{\mathrm{tar},i}\}_{i\in I}$*, define*

$$D_\rho^{\mathrm{cert}} := \{\, i \in I \mid \mathsf{Cert}_\rho(q_{\mathrm{clk}}, q_{\mathrm{tar},i}) \text{ is absent or invalid} \,\}.$$

Theorem A.42.10 (OP-facing reduction)**.** *After the local pass, the route-level OP-facing obstruction is the certificate obstruction*

$$D_\rho^{\mathrm{cert}}.$$

If $D_\rho^{\mathrm{cert}} = \varnothing$*, then all selected OP targets have canonical optimal clock-to-target routes.*

Proof. For each target, a valid certificate proves admissible route existence. By the local pass, admissible route existence triggers the canonical optimal route representative. Therefore precisely those targets without valid certificates remain obstructed. The obstruction set is D_ρ^{cert}. □

Remark A.42.11 (OP status after)**.** *OP 1/4, OP 4, and OP 5 remain open. The improvement is operational: the next thing to build is no longer an abstract optimal route, but a finite typed certificate proving that the route exists.*

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A.43 Route-Certificate Tree and Arborescence Criterion

Status. The layer reduced route existence to finite typed route certificates. The present layer packages many target certificates into one clock-rooted directed route tree. This is the first explicit spanning/arborescence reading of the OP-facing route problem: the tree must start at the clock anchor, reach all selected OP targets, respect tensor interfaces, preserve guards, and carry route-action data. This does not close OP 1/4, OP 4, or OP 5. It makes the next obstruction a finite tree-validity problem.

A.43.1 Clock-rooted certificate tree

Definition A.43.1 (Clock-rooted route-certificate tree)**.** *Let $\mathcal{Q} = \{q_{\mathrm{tar},i}\}_{i\in I}$ be a finite family of target information loci. A clock-rooted route-certificate tree is a finite directed tree/arborescence*

$$\mathsf{T}_\rho = (V_T, E_T, q_{\mathrm{clk}}, \mathcal{Q}, \tau_T, \Gamma_T, \Theta_T)$$

where:

1. *$q_{\mathrm{clk}} \in V_T$ is the root;*
2. *every $q_{\mathrm{tar},i} \in \mathcal{Q}$ is reached from q_{clk} by a directed path in E_T;*
3. *τ_T assigns tensor/interface type data to every edge;*
4. *Γ_T assigns the active guard record to every edge;*
5. *Θ_T assigns route action/rank data to every edge or path.*

Definition A.43.2 (Tree validity)**.** *The tree T_ρ is valid if every root-to-target path extracted from it is a valid admissible route certificate in the sense of Definition A.42.2.*

Lemma A.43.3 (Tree validity gives all component certificates)**.** *If T_ρ is valid, then every target $q_{\mathrm{tar},i} \in \mathcal{Q}$ has a valid admissible route certificate.*

Proof. For each target $q_{\mathrm{tar},i}$, follow the unique directed path in the tree from the clock root to that target. By tree validity, this path satisfies tensor-interface matching, guard preservation, and route-action definability. Therefore it is a valid admissible route certificate for the target. □

A.43.2 Spanning and Steiner readings

Definition A.43.4 (Spanning route tree)**.** *A valid route-certificate tree is a spanning route tree for $\mathcal{Q}$ if its required vertices are exactly the clock root and the selected targets, together with any already declared route states.*

Definition A.43.5 (Steiner route tree)**.** *A valid route-certificate tree is a Steiner route tree if it may introduce additional admissible intermediate states*

$$z_1, \ldots, z_\ell$$

such as HC compensation points, MML router states, MMA readout states, boundary collar states, or selector-anchor states, provided every additional state has a declared tensor type and guard role.

Remark A.43.6 (Why the Steiner version is natural here)**.** *The OP-facing system rarely needs only a direct edge from clock to target. Intermediate states are often the meaningful physics: HC compensation, boundary-collar stabilization, route-mask decoding, or selector comparison can all appear as necessary admissible middle points. Thus the natural object is a guarded tensor-labeled directed Steiner tree, with ordinary spanning trees as the special case.*

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A.43.3 Cost and optimal tree

Definition A.43.7 (Tree action)**.** *The action of a valid route-certificate tree is*

$$\mathcal{A}_T(\mathsf{T}_\rho) := \sum_{e \in E_T} w(e) + \lambda\, \mathsf{G}_T + \mu\, \mathsf{R}_T,$$

where $w(e)$ is the edge route cost/rank, G_T records guard tension or guard-check complexity, and R_T records residual terms remaining on the tree. The constants λ, μ are bookkeeping weights for the current OP-facing readout.

Definition A.43.8 (Canonical optimal route tree)**.** *If the valid tree class*

$$\mathcal{T}_{\mathrm{adm}}(q_{\mathrm{clk}}, \mathcal{Q})$$

is nonempty, the canonical optimal route tree is the QAM/MML normal-form representative

$$\mathsf{T}_\rho^{\mathrm{opt,can}} := \operatorname*{argmin}_{\mathsf{T}_\rho \in \mathcal{T}_{\mathrm{adm}}(q_{\mathrm{clk}}, \mathcal{Q})} \mathcal{A}_T(\mathsf{T}_\rho)$$

after canonical route equivalence has identified representationally equivalent trees.

Proposition A.43.9 (Valid tree triggers all canonical routes)**.** *If a valid route-certificate tree exists for $(q_{\mathrm{clk}}, \mathcal{Q})$, then every target in $\mathcal{Q}$ has a canonical optimal clock-to-target route. If the valid tree class is nonempty and the tree action attains its minimum, then $\mathsf{T}_\rho^{\mathrm{opt,can}}$ exists.*

Proof. A valid tree gives a valid certificate for every target by Lemma A.43.3. By and the local pass, each certificate triggers a canonical optimal route. If the whole valid tree class is nonempty and the tree action attains its minimum, then minimization over canonical tree classes produces a canonical optimal route tree. □

A.43.4 OP 1/4, OP 4, and OP 5(B) readings

Definition A.43.10 (OP 1/4 anchor-cover tree)**.** *An OP 1/4 anchor-cover tree is a valid route-certificate tree with target family*

$$\mathcal{Q}_{1/4} := \{K_i \cap A_{1/4}^{\mathrm{sel}} \mid K_i \in \mathcal{K}_{1/4}\}.$$

It certifies clock-to-anchor reachability for every comparison-relevant component.

Definition A.43.11 (OP 5(B) boundary-promotion tree)**.** *An OP 5(B) boundary-promotion tree is a valid route-certificate tree in the guard-restricted class*

$$\mathcal{T}_{\mathrm{adm}}^{G_\partial}(q_{\mathrm{clk}}, Y_5^{(B),\mathrm{prom}})$$

such that every boundary edge preserves G_∂ and the terminal path reaches the promoted boundary target.

Remark A.43.12 (OP 4 as tree stability)**.** *OP 4 is now naturally read as stability of the certified route tree or of the clock-anchored optimal-route map extracted from it. Informally, OP 1/4 builds or certifies the anchor tree, OP 4 tests transport stability of that tree, and OP 5(B) tests whether a boundary-promotion tree can be built without breaking the boundary guard.*

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A.43.5 Tree obstruction

Definition A.43.13 (Tree obstruction)**.** *For a selected target family $\mathcal{Q}$, define the tree obstruction*

$$D_T^{\mathrm{cert}} := \begin{cases} \varnothing, & \text{if a valid route-certificate tree exists,} \\ \mathcal{Q}_{\mathrm{fail}}, & \text{otherwise, recording targets not reached by any valid tree path.} \end{cases}$$

Theorem A.43.14 (OP-facing tree reduction)**.** *After the local pass, the route-level OP-facing obstruction can be represented by the tree obstruction D_T^{cert}. If*

$$D_T^{\mathrm{cert}} = \varnothing,$$

then the selected targets are reached by a valid tensor-typed, guard-preserving clock-rooted route tree and hence possess canonical optimal route representatives.

Proof. If $D_T^{\mathrm{cert}} = \varnothing$, a valid route-certificate tree exists. By Proposition A.43.9, all targets then have valid certificates and canonical optimal routes. Conversely, any target not reached by a valid tree path remains a certificate failure and is recorded in the tree obstruction. □

Remark A.43.15 (OP status after)**.** *OP 1/4, OP 4, and OP 5(B) remain open. The gain is that their route-level form is now a finite tree/arborescence problem with tensor types, guards, and route action visible.*

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A.44 Routing-Protocol Analogies for OP-Facing Trees

Status. This layer extracts routing ideas from established Internet and network-routing protocol families. The extraction is architectural only: OSPF, IS-IS, BGP, MPLS/Traffic Engineering, Segment Routing, RSVP-TE, RPL, BFD, PCEP, and controller-based routing are not imported as physics and are not proof certificates. They supply robust design patterns for the QAM/MML route-tree certificate layer introduced in the local proof chain.

A.44.1 Protocol-pattern table

Protocol idea	Network role	OP-facing route-tree analogue
OSPF / IS-IS	Link-state map and shortest-path computation inside a domain	Build a local map of admissible tensor/guard edges and compute a clock-rooted route tree.
BGP	Path-vector routing with policy constraints between domains	Record whole route histories and policy/guard filters before accepting a route.
MPLS / TE	Traffic-engineered paths with explicit constraints	Treat HC/MML/MMA/collar constraints as traffic-engineering constraints on allowed OP paths.
Segment Routing	Source/route instruction list through selected waypoints	Encode an OP route as a finite segment list of required compensator/router/readout states.
RSVP-TE	Path reservation and resource/-constraint signaling	Reserve guard budget, boundary collar, or residual tolerance along a route.
RPL	Directed acyclic graph rooted at a sink/root in constrained networks	Use the clock anchor as root and build a directed acyclic route tree to targets.
BFD / keepalive	Fast failure detection for links/routes	Detect broken guard/tensor links before the OP certificate is accepted.
PCEP / SDN	Controller computes constrained paths from global view	Let the QAM/MML router compute or audit the route tree from declared global certificate data.

Remark A.44.1 (Not a protocol import)**.** *The table is a design dictionary, not a dependency. It says which proof-route pattern resembles a known network-routing pattern. It does not assert that a physical system uses Internet protocols.*

A.44.2 Link-state route database

Definition A.44.2 (OP link-state database)**.** *An OP link-state database is a finite record*

$$\mathsf{LS}_{\mathrm{OP}} = (V, E, \tau, \Gamma, \omega, \mathrm{up})$$

where V are route states, E are possible directed steps, τ is the tensor/interface label, Γ is the guard record, ω is the edge rank/cost, and $\mathrm{up}(e) \in \{0, 1\}$ records whether the edge is currently admissible.

Definition A.44.3 (Link-state admissible edge)**.** *An edge $e : u \to v$ is link-state admissible if*

$$\mathrm{up}(e) = 1, \qquad \tau(e) \text{ matches at its interfaces}, \qquad \Gamma(e) \text{ preserves all active guards.}$$

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Proposition A.44.4 (Link-state tree extraction)**.** *If the OP link-state database contains a directed admissible path from q_{clk} to every target in $\mathcal{Q}$, then it contains a valid route-certificate tree for $(q_{\mathrm{clk}}, \mathcal{Q})$.*

Proof. Choose, for each target, one directed admissible path from the clock root to the target. The union of these paths contains a directed subgraph rooted at q_{clk} that reaches all targets. Removing redundant cycles, if any, yields a directed arborescence or a directed acyclic route certificate subtree. Since each retained edge is tensor-compatible and guard-preserving, the resulting tree is valid in the sense. □

A.44.3 Path-vector guard memory

Definition A.44.5 (Path-vector guard record)**.** *A path-vector guard record for a route*

$$\rho = (e_1, \ldots, e_n)$$

is the sequence

$$\mathsf{PV}_\Gamma(\rho) := \big((e_1, \Gamma(e_1)), \ldots, (e_n, \Gamma(e_n))\big).$$

It records not merely the current endpoint, but the guard history of the route.

Lemma A.44.6 (Path-vector guard rejection)**.** *If any edge in $\mathsf{PV}_\Gamma(\rho)$ violates an active guard, then ρ is not an admissible route certificate.*

Proof. Certificate validity requires every edge of the route path to preserve the declared guards. A path-vector guard record exposes every edge-level guard state. Therefore any recorded violation invalidates the certificate. □

A.44.4 Segment routing and OP waypoints

Definition A.44.7 (OP segment list)**.** *An OP segment list is a finite route instruction sequence*

$$\mathsf{Seg}_{\mathrm{OP}} = (z_0, z_1, \ldots, z_k)$$

where $z_0 = q_{\mathrm{clk}}$, $z_k = q_{\mathrm{tar}}$, and intermediate segments z_j may be HC compensation points, MML router points, MMA readout points, boundary-collar points, or selector-anchor points.

Proposition A.44.8 (Segment list gives Steiner-tree skeleton)**.** *A valid OP segment list determines the skeleton of a directed Steiner route tree. The intermediate segment states are precisely the permitted Steiner points needed to make the route tensor-compatible and guard-preserving.*

Proof. The segment list fixes a finite sequence of required waypoints from clock to target. Any valid realization of each segment-to-segment subroute gives a piece of the total route. The union of these pieces is a directed route tree or path with intermediate nodes that were not necessarily original targets. These nodes are Steiner points by definition. □

A.44.5 Traffic-engineering constraints

Definition A.44.9 (OP traffic-engineering constraint)**.** *An OP traffic-engineering constraint is any additional route constraint of the form*

$$\mathsf{TE} = (\mathrm{maxResidual}, \mathrm{guardBudget}, \mathrm{curvatureBound}, \mathrm{HCcompat}, \mathrm{boundaryCollar})$$

declared before a route certificate is accepted.

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Definition A.44.10 (TE-valid route tree)**.** *A route-certificate tree is TE-valid if it is valid in the sense and also satisfies all declared OP traffic-engineering constraints.*

Proposition A.44.11 (TE-validity sharpens route admissibility)**.** *Replacing validity by TE-validity narrows the admissible tree class without changing the tree logic. In particular, any TE-valid tree is a valid route-certificate tree, but not every valid tree is TE-valid.*

Proof. TE-validity is validity plus additional constraints. Hence it implies ordinary validity. Since the extra constraints may fail for an otherwise valid tree, the inclusion can be strict. □

A.44.6 Failure detection and route convergence

Definition A.44.12 (Route heartbeat)**.** *A route heartbeat is a repeated local check*

$$\mathsf{HB}(e) = (\tau(e) \text{ matches},\ \Gamma(e) \text{ preserved},\ \omega(e) \text{ defined})$$

attached to an edge e. If a heartbeat fails, the edge is marked $\mathrm{up}(e) = 0$ *in the OP link-state database.*

Definition A.44.13 (Route convergence)**.** *Route convergence is the process of recomputing a valid route-certificate tree after one or more admissible edges become unavailable.*

Remark A.44.14 (Why failure detection matters for OP 4)**.** *OP 4 can be read as transport stability of the optimal-route map. Failure detection supplies the discrete analogue: if a tensor/guard edge fails, the route tree must either reconverge to a valid canonical tree or record a nonempty obstruction.*

A.44.7 Protocol-pattern reduction

Theorem A.44.15 (Routing-protocol pattern reduction)**.** *The route-tree problem may be refined into the following protocol-pattern tasks:*

Link-state task	*build* $\mathsf{LS}_{\mathrm{OP}}$,
Path-vector task	*record guard history* PV_{Γ},
Segment task	*choose OP waypoints* $\mathsf{Seg}_{\mathrm{OP}}$,
TE task	*apply residual/guard/curvature constraints*,
Heartbeat task	*detect failed tensor/guard edges*,
Convergence task	*recompute or certify obstruction.*

These tasks preserve the logic while making the route-tree construction operational.

Proof. Each protocol-pattern task refines one component of the tree certificate: link-state supplies the finite edge database, path-vector supplies route history, segment routing supplies permitted Steiner waypoints, traffic engineering supplies admissibility constraints, heartbeat supplies failure detection, and convergence supplies recomputation after failure. None of these patterns changes the meaning of a valid route-certificate tree; they only make its construction and audit explicit. □

Remark A.44.16 (OP status after)**.** *OP 1/4, OP 4, and OP 5(B) remain open. The gain is that the route-tree problem now has a practical protocol grammar: map, history, waypoints, constraints, failure detection, and convergence.*

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A.45 Link-State Route-Tree Extraction

Status. The layer introduced routing-protocol patterns. The present layer turns the link-state pattern into an explicit route-tree extraction rule. Given a finite OP link-state database, the admissible subgraph is formed by keeping only tensor-compatible, guard-preserving, active edges. The route obstruction is then exactly the target set not reachable from the clock root in this admissible subgraph. If all targets are reachable, canonical clock-to-target paths can be extracted and packaged as a route-certificate tree. This still does not close OP 1/4, OP 4, or OP 5(B); it gives the first executable route-protocol criterion for the certificate layer.

A.45.1 Admissible subgraph

Definition A.45.1 (Admissible OP subgraph)**.** *Given the OP link-state database*

$$\mathsf{LS}_{\mathrm{OP}} = (V, E, \tau, \Gamma, \omega, \mathrm{up}),$$

define the admissible subgraph

$$\mathcal{G}_{\mathrm{adm}} = (V, E_{\mathrm{adm}})$$

by

$$E_{\mathrm{adm}} := \{e \in E \mid \mathrm{up}(e) = 1,\ \tau(e) \text{ matches},\ \Gamma(e) \text{ preserves all active guards}\}.$$

Definition A.45.2 (Reachability obstruction)**.** *For a target family $\mathcal{Q} = \{q_{\mathrm{tar},i}\}_{i\in I}$, define*

$$D_{\mathrm{reach}} := \{q_{\mathrm{tar},i} \in \mathcal{Q} \mid q_{\mathrm{tar},i} \notin \mathrm{Reach}_{\mathcal{G}_{\mathrm{adm}}}(q_{\mathrm{clk}})\}.$$

Lemma A.45.3 (Reachability is equivalent to certificate existence)**.** *For a target q_{tar}, there exists a valid link-state route certificate from q_{clk} to q_{tar} if and only if*

$$q_{\mathrm{tar}} \in \mathrm{Reach}_{\mathcal{G}_{\mathrm{adm}}}(q_{\mathrm{clk}}).$$

Proof. If a valid certificate exists, its edges are active, tensor-compatible, and guard-preserving. Hence its path lies in E_{adm}, and the target is reachable in $\mathcal{G}_{\mathrm{adm}}$. Conversely, if the target is reachable in $\mathcal{G}_{\mathrm{adm}}$, a directed path from q_{clk} to q_{tar} exists using only admissible edges. Recording its vertices, edges, tensor labels, guard records, and rank data gives a valid route certificate. □

A.45.2 Canonical path extraction

Definition A.45.4 (Canonical extracted path)**.** *Assume q_{tar} is reachable in $\mathcal{G}_{\mathrm{adm}}$. A canonical extracted path*

$$\rho_{\mathrm{LS}}^{\mathrm{can}}(q_{\mathrm{clk}}, q_{\mathrm{tar}})$$

is the QAM/MML normal-form representative of a minimum-rank path from q_{clk} to q_{tar} in $\mathcal{G}_{\mathrm{adm}}$ with respect to the edge rank ω, after canonical route equivalence has identified representationally equivalent paths.

Proposition A.45.5 (Reachable targets have canonical extracted paths)**.** *Every target outside D_{reach} has a canonical extracted path.*

Proof. If a target is outside D_{reach}, it is reachable in the finite admissible subgraph. Therefore there is at least one finite admissible path. Minimizing the declared edge-rank sum over finite paths gives a minimum-rank path. Canonical equivalence and QAM/MML normal form identify representational multiplicity and select the canonical representative. □

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A.45.3 Tree extraction

Definition A.45.6 (Extracted link-state route tree)**.** *If $D_{\mathrm{reach}} = \varnothing$, define the extracted link-state route tree*

$$\mathsf{T}_{\mathrm{LS}}^{\mathrm{can}}$$

as the canonical acyclic reduction of the union of canonical extracted paths

$$\bigcup_{q_{\mathrm{tar}} \in \mathcal{Q}} \rho_{\mathrm{LS}}^{\mathrm{can}}(q_{\mathrm{clk}}, q_{\mathrm{tar}}).$$

Its edge labels, guard records, and rank data are inherited from $\mathsf{LS}_{\mathrm{OP}}$.

Theorem A.45.7 (Link-state extraction criterion)**.** *For a finite OP link-state database and finite target family $\mathcal{Q}$,*

$$D_{\mathrm{reach}} = \varnothing$$

if and only if an extracted valid link-state route-certificate tree exists.

Proof. If $D_{\mathrm{reach}} = \varnothing$, every target is reachable in $\mathcal{G}_{\mathrm{adm}}$. By Proposition A.45.5, each target has a canonical extracted path. The acyclic reduction of the union of these paths is a finite clock-rooted route tree reaching all targets, and all of its retained edges are admissible by construction. Conversely, if a valid extracted tree exists, every target is reached from the clock root by a valid admissible path; hence no target lies in D_{reach}. □

A.45.4 Traffic-engineering and segment refinement

Definition A.45.8 (TE-filtered admissible graph)**.** *Given a traffic-engineering constraint family TE, define*

$$\mathcal{G}_{\mathrm{adm}}^{\mathsf{TE}}$$

by further removing from $\mathcal{G}_{\mathrm{adm}}$ every edge or path segment violating the declared residual, guard-budget, curvature, HC/MM, or boundary-collar constraints.

Definition A.45.9 (Segment-constrained extraction)**.** *Given an OP segment list*

$$\mathsf{Seg}_{\mathrm{OP}} = (z_0, \ldots, z_k),$$

a segment-constrained extracted route is a concatenation of canonical extracted paths

$$\rho(z_0, z_1) * \rho(z_1, z_2) * \cdots * \rho(z_{k-1}, z_k)$$

inside $\mathcal{G}_{\mathrm{adm}}^{\mathsf{TE}}$, provided every segment is reachable.

Proposition A.45.10 (Segment failure localizes obstruction)**.** *If a segment-constrained route fails, the obstruction is localized to the first segment*

$$(z_j, z_{j+1})$$

for which $z_{j+1} \notin \mathrm{Reach}_{\mathcal{G}_{\mathrm{adm}}^{\mathsf{TE}}}(z_j)$.

Proof. A segment-constrained route is a finite concatenation. If it fails, at least one segment is unreachable in the TE-filtered admissible graph. Taking the first such segment localizes the obstruction. All earlier segments are reachable by definition of first failure. □

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A.45.5 OP-facing consequence

Theorem A.45.11 (route-protocol criterion)**.** *After the local pass, the OP-facing route-tree obstruction can be checked by the finite criterion*

$$D_{\text{reach}} = \varnothing$$

in the admissible link-state graph, optionally refined to the TE-filtered and segment-constrained graph. If the criterion holds, a valid canonical route tree exists; if it fails, the failed target or failed segment is explicitly identified.

Proof. The unfiltered statement is the link-state extraction criterion. TE filtering and segment constraints only restrict the admissible graph further. Therefore the same reachability logic applies, and failures are recorded as unreachable targets or first failed segments. □

Remark A.45.12 (OP status after)**.** *OP 1/4, OP 4, and OP 5(B) remain open. The gain is that the route/tree certificate layer is now checkable by a finite link-state reachability criterion with optional traffic-engineering and segment-routing refinements.*

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A.46 Segment-Route Waypoint Certificates

Status. The layer gave a finite link-state reachability criterion. The present layer adds segment-route structure: instead of asking only whether a target is reachable, we declare which intermediate waypoints must be respected by an OP route. This is the controlled form of the Steiner-tree idea: HC, MML, MMA, selector, and boundary-collar states may appear as required waypoints, but only with declared tensor types and guard roles. OP 1/4, OP 4, and OP 5(B) remain open.

A.46.1 Waypoint classes

Definition A.46.1 (OP waypoint class)**.** *An OP waypoint class is one of the route-state types*

$$\mathsf{WP} \in \{\mathsf{Clock}, \mathsf{QAM}, \mathsf{MML}, \mathsf{MMA}, \mathsf{HC}, \mathsf{Selector}, \mathsf{BoundaryCollar}, \mathsf{Anchor}, \mathsf{Target}\}.$$

Each waypoint used in a segment certificate must carry a tensor/interface type and a guard role.

Definition A.46.2 (Typed waypoint)**.** *A typed waypoint is a tuple*

$$z = (|z|, \mathsf{WP}(z), \tau(z), \Gamma(z))$$

where $|z|$ is the route state, $\mathsf{WP}(z)$ its waypoint class, $\tau(z)$ its tensor/interface type, and $\Gamma(z)$ its local guard record.

A.46.2 Segment-route certificate

Definition A.46.3 (Segment-route certificate)**.** *A segment-route certificate from q_{clk} to q_{tar} is a finite waypoint list*

$$\mathsf{Seg}_\rho = (z_0, z_1, \dots, z_k)$$

with $z_0 = q_{\mathrm{clk}}$, $z_k = q_{\mathrm{tar}}$, together with a link-state certificate proving that every adjacent segment

$$z_j \rightsquigarrow z_{j+1}$$

is reachable inside the currently selected admissible graph

$$\mathcal{G}_{\mathrm{adm}}^{\mathsf{TE}}.$$

Definition A.46.4 (Segment obstruction)**.** *The segment obstruction of Seg_ρ is*

$$D_{\mathrm{seg}}(\mathsf{Seg}_\rho) := \{j \mid z_{j+1} \notin \mathrm{Reach}_{\mathcal{G}_{\mathrm{adm}}^{\mathsf{TE}}}(z_j)\}.$$

Lemma A.46.5 (Segment certificate implies route certificate)**.** *If*

$$D_{\mathrm{seg}}(\mathsf{Seg}_\rho) = \varnothing,$$

then Seg_ρ determines a valid admissible route certificate from q_{clk} to q_{tar}.

Proof. If the segment obstruction is empty, every adjacent pair of waypoints is linked by an admissible route inside the TE-filtered admissible graph. Concatenating these certified subroutes gives a route from $z_0 = q_{\mathrm{clk}}$ to $z_k = q_{\mathrm{tar}}$. Since each segment is tensor-compatible, guard-preserving, and TE-valid, the concatenation is a valid admissible route certificate. □

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A.46.3 OP 1/4 segment shape

Definition A.46.6 (OP 1/4 selector-anchor segment template)**.** *For a comparison-relevant component K_i, the OP 1/4 selector-anchor segment template is*

$$\mathsf{Seg}_{1/4,i} = (q_{\mathrm{clk}}, z_{\mathrm{QAM}}, z_{\mathrm{MML}}, z_{\mathrm{sel}}, K_i \cap A^{\mathrm{sel}}_{1/4}).$$

Here z_{QAM} records the shifted/QAM normal-form shift, z_{MML} records the router/intertwiner step, z_{sel} records the selector/comparison interface, and the final target is the common anchor region inside K_i.

Proposition A.46.7 (OP 1/4 segment criterion)**.** *If every OP 1/4 segment template $\mathsf{Seg}_{1/4,i}$ has empty segment obstruction, then every relevant component K_i has a valid clock-to-anchor route certificate and hence a canonical optimal clock-to-anchor route.*

Proof. Apply Lemma A.46.5 to each segment template. Each valid segment certificate gives a route certificate to $K_i \cap A^{\mathrm{sel}}_{1/4}$. By the local proof chain principles, each such certificate triggers a canonical optimal route. □

A.46.4 OP 5(B) segment shape

Definition A.46.8 (OP 5(B) boundary-promotion segment template)**.** *The OP 5(B) boundary-promotion segment template is*

$$\mathsf{Seg}^{(B)}_5 = (q_{\mathrm{clk}}, z_{\mathrm{HC}}, z_{\partial\mathrm{col}}, z_{\mathrm{MMA}}, Y^{(B),\mathrm{prom}}_5).$$

Here z_{HC} records HC compensation, $z_{\partial\mathrm{col}}$ records the boundary-collar guard state, z_{MMA} records the matrix/readout residual evaluator, and the terminal target is the promoted boundary-side state.

Definition A.46.9 (Boundary segment guard)**.** *The OP 5(B) segment template is boundary-valid only if every segment from z_{HC} onward preserves the boundary guard:*

$$G_\partial(z_j) = G_\partial(z_{j+1}).$$

Proposition A.46.10 (OP 5(B) segment criterion)**.** *If $\mathsf{Seg}^{(B)}_5$ is boundary-valid and*

$$D_{\mathrm{seg}}(\mathsf{Seg}^{(B)}_5) = \varnothing,$$

then OP 5(B) has a valid guard-preserving route certificate to $Y^{(B),\mathrm{prom}}_5$. The remaining obligations are HC/MM compatibility and post-promotion residual control.

Proof. Empty segment obstruction gives reachability for every adjacent segment inside the admissible graph. Boundary-validity ensures preservation of G_∂ along the boundary-sensitive part of the route. Therefore the concatenated route is a valid guard-preserving certificate to the promoted boundary target. The statement does not discharge HC/MM compatibility or residual control; those remain explicit. □

A.46.5 Segment-level OP obstruction

Definition A.46.11 (Segment-level OP obstruction)**.** *For a family of segment templates $\mathfrak{S}$, define*

$$D^{\mathrm{seg}}_{\mathrm{OP}} := \{\mathsf{Seg} \in \mathfrak{S} \mid D_{\mathrm{seg}}(\mathsf{Seg}) \neq \varnothing \text{ or a declared guard segment fails}\}.$$

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Theorem A.46.12 (segment-route reduction)**.** *After the local pass, the OP-facing route obstruction can be refined from target reachability to segment-template validity:*

$$D_{\mathrm{OP}}^{\mathrm{seg}} = \varnothing \quad \Longrightarrow \quad D_{\mathrm{reach}} = \varnothing$$

for the selected OP targets. If $D_{\mathrm{OP}}^{\mathrm{seg}} \neq \varnothing$, the first failed segment localizes the obstruction.

Proof. Each segment template with empty segment obstruction yields a valid route certificate by Lemma A.46.5. Thus if all selected segment templates are valid, all selected targets are reachable and D_{reach} is empty. If a segment template fails, Definition A.46.4 identifies the failed adjacent waypoint pair, localizing the obstruction. □

Remark A.46.13 (OP status after)**.** *OP 1/4, OP 4, and OP 5(B) remain open. The gain is that the Steiner route tree now has explicit permitted waypoint templates. The next task is to instantiate the OP 1/4 and OP 5(B) segment lists against the current link-state database and inspect their first failed segments.*

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A.47 Traffic-Engineering and Guard-Budget Filters

Status. The layer declared OP segment templates. The present layer adds the filter that decides whether a reachable segment is actually admissible after residual, guard-budget, curvature, HC/MM-compatibility, and boundary-collar constraints are applied. This is a traffic-engineering analogue only: the constraints are QAM/MML/MMA route constraints, not imported network physics. OP 1/4, OP 4, and OP 5(B) remain open.

A.47.1 TE profile

Definition A.47.1 (OP traffic-engineering profile)**.** *An OP traffic-engineering profile is a tuple*

$$\mathsf{TE}_{\mathrm{OP}} = (\varepsilon_R, \varepsilon_G, \varepsilon_\kappa, \chi_{\mathrm{HC/MM}}, \chi_{\partial\mathrm{col}}, \chi_\tau)$$

where:

1. ε_R *is the maximum allowed residual increment;*
2. ε_G *is the maximum allowed guard tension;*
3. ε_κ *is the maximum allowed curvature/route-bending load;*
4. $\chi_{\mathrm{HC/MM}} \in \{0,1\}$ *records HC/MM compatibility;*
5. $\chi_{\partial\mathrm{col}} \in \{0,1\}$ *records boundary-collar admissibility;*
6. $\chi_\tau \in \{0,1\}$ *records tensor-interface compatibility.*

Definition A.47.2 (TE-valid edge)**.** *Let an edge e carry diagnostics*

$$\mathsf{d}(e) = (\Delta R(e), \Delta G(e), \kappa(e), \chi_{\mathrm{HC/MM}}(e), \chi_{\partial\mathrm{col}}(e), \chi_\tau(e)).$$

The edge is $\mathsf{TE}_{\mathrm{OP}}$-valid if

$$\Delta R(e) \le \varepsilon_R, \qquad \Delta G(e) \le \varepsilon_G, \qquad \kappa(e) \le \varepsilon_\kappa,$$

and

$$\chi_{\mathrm{HC/MM}}(e) = \chi_{\partial\mathrm{col}}(e) = \chi_\tau(e) = 1$$

whenever the corresponding compatibility clause is active for the OP segment.

A.47.2 Filtered graph

Definition A.47.3 (TE-filtered admissible graph)**.** *Given $\mathcal{G}_{\mathrm{adm}} = (V, E_{\mathrm{adm}})$, define*

$$\mathcal{G}_{\mathrm{adm}}^{\mathsf{TE}} = (V, E_{\mathrm{adm}}^{\mathsf{TE}})$$

where

$$E_{\mathrm{adm}}^{\mathsf{TE}} := \{e \in E_{\mathrm{adm}} \mid e \text{ is } \mathsf{TE}_{\mathrm{OP}}\text{-valid}\}.$$

Lemma A.47.4 (TE filtering preserves validity)**.** *Every route certificate in $\mathcal{G}_{\mathrm{adm}}^{\mathsf{TE}}$ is a valid route certificate in $\mathcal{G}_{\mathrm{adm}}$.*

Proof. The TE-filtered graph is a subgraph of the admissible graph. Therefore every edge retained by the TE filter was already active, tensor-compatible, and guard-preserving. The extra TE clauses only restrict the admissible class further. □

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A.47.3 Guard-budget accumulation

Definition A.47.5 (Route guard budget)**.** *For a finite route*

$$\rho = (e_1, \dots, e_n),$$

define its accumulated guard tension by

$$\mathsf{G}(\rho) := \sum_{j=1}^{n} \Delta G(e_j).$$

The route is guard-budget valid if

$$\mathsf{G}(\rho) \le \varepsilon_G^{\mathrm{tot}}.$$

Definition A.47.6 (Route residual budget)**.** *The accumulated residual increment of ρ is*

$$\mathsf{R}(\rho) := \sum_{j=1}^{n} \Delta R(e_j).$$

The route is residual-budget valid if

$$\mathsf{R}(\rho) \le \varepsilon_R^{\mathrm{tot}}.$$

Proposition A.47.7 (Budget-valid segment certificate)**.** *If every segment in Seg_ρ is reachable inside $\mathcal{G}_{\mathrm{adm}}^{\mathsf{TE}}$, and the concatenated route is guard-budget and residual-budget valid, then Seg_ρ determines a TE-valid admissible route certificate.*

Proof. Reachability inside the TE-filtered graph gives a concatenated route whose edges satisfy local TE constraints. Guard-budget and residual-budget validity ensure that accumulated route tension and residual remain within the declared global bounds. Hence the resulting certificate is TE-valid. □

A.47.4 Boundary and HC/MM filters

Definition A.47.8 (Boundary-collar filter)**.** *A boundary-sensitive segment is boundary-collar valid if each boundary edge satisfies*

$$\chi_{\partial\mathrm{col}}(e) = 1, \qquad G_\partial(\mathrm{src}(e)) = G_\partial(\mathrm{dst}(e)).$$

Definition A.47.9 (HC/MM compatibility filter)**.** *An HC/MM-sensitive segment is HC/MM-valid if each relevant edge satisfies*

$$\chi_{\mathrm{HC/MM}}(e) = 1$$

and its readout diagram commutes up to the declared route tolerance.

Proposition A.47.10 (OP 5(B) TE boundary criterion)**.** *If $\mathsf{Seg}_5^{(B)}$ has empty segment obstruction in $\mathcal{G}_{\mathrm{adm}}^{\mathsf{TE}}$, is boundary-collar valid, is HC/MM-valid, and satisfies total guard and residual budgets, then it supplies a TE-valid guard-preserving boundary promotion route certificate.*

Proof. Empty TE-filtered segment obstruction gives reachability of all declared segments. Boundary-collar validity preserves G_∂ along the boundary-sensitive portion. HC/MM-validity supplies the required matrix/readout compatibility. Budget validity controls accumulated guard tension and residual. Thus the route certificate is TE-valid and guard-preserving. □

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A.47.5 TE-level obstruction

Definition A.47.11 (TE obstruction)**.** *For a family of segment templates $\mathfrak{S}$, define the TE obstruction*

$$D_{\mathrm{OP}}^{\mathsf{TE}} := \{\mathsf{Seg} \in \mathfrak{S} \mid D_{\mathrm{seg}}^{\mathsf{TE}}(\mathsf{Seg}) \neq \varnothing \text{ or a budget, boundary, HC/MM, or tensor clause fails}\}.$$

Theorem A.47.12 (TE-filter reduction)**.** *After the local pass, the OP-facing segment obstruction is sharpened from reachability to TE-validity:*

$$D_{\mathrm{OP}}^{\mathsf{TE}} = \varnothing \quad \Longrightarrow \quad D_{\mathrm{OP}}^{\mathrm{seg}} = \varnothing \quad \Longrightarrow \quad D_{\mathrm{reach}} = \varnothing.$$

If $D_{\mathrm{OP}}^{\mathsf{TE}} \neq \varnothing$, the failed clause records whether the obstruction is reachability, tensor typing, guard budget, residual budget, curvature load, HC/MM compatibility, or boundary-collar preservation.

Proof. TE-validity is segment validity plus the declared budget and compatibility filters. Therefore an empty TE obstruction implies empty segment obstruction. By the local pass, empty segment obstruction implies reachability. Conversely, any TE failure is recorded by the first failed segment or by the first failed TE clause attached to that segment. □

Remark A.47.13 (OP status after)**.** *OP 1/4, OP 4, and OP 5(B) remain open. The gain is that route existence is no longer treated as enough: a route must also remain within residual, guard, curvature, HC/MM, tensor, and boundary-collar budgets.*

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A.48 Why the New QAM Narrows the OPs Faster

Status. This layer explains why the current QAM/MML/MMA route grammar narrows the OP-facing search faster than the older QAM notation. It is an explanatory architecture layer, not an OP closure and not a physical overclaim. Its role is to document why the local proof chain could turn OP 1/4, OP 4, and OP 5(B) from global existence questions into finite route, certificate, segment, and TE-filter questions.

A.48.1 Old obstruction shape

Definition A.48.1 (Unfactored OP obstruction)**.** *An unfactored OP obstruction is an obstruction described only at the level of the target problem, without separating:*

route, readout, guard, residual, tensor interface, canonical normal form.

In this form one can often say that an OP is open, but one cannot yet determine which local clause is responsible.

Remark A.48.2 (Why the old QAM was slower)**.** *The older QAM notation could express many structural ideas, but it did not yet force the OP into a typed route object. Consequently selector failure, boundary failure, tensor-interface failure, guard failure, residual growth, route nonexistence, and representational multiplicity could appear as one large combined obstruction.*

A.48.2 New QAM factorization

Definition A.48.3 (QAM route factorization)**.** *The current QAM/MML/MMA route grammar factors an OP-facing obstruction into the ordered layers*

$$\begin{aligned} \textit{clock anchor} &: q_{\mathrm{clk}},\\ \textit{target locus} &: q_{\mathrm{tar}},\\ \textit{admissible route space} &: \mathcal{R}_{\mathrm{adm}}(q_{\mathrm{clk}}, q_{\mathrm{tar}}),\\ \textit{canonical optimal route} &: \rho^{\mathrm{can}}_{\mathrm{opt}},\\ \textit{certificate} &: \mathsf{Cert}_{\rho},\\ \textit{tree} &: \mathsf{T}_{\rho},\\ \textit{link-state graph} &: \mathcal{G}_{\mathrm{adm}},\\ \textit{segment list} &: \mathsf{Seg}_{\rho},\\ \textit{TE profile} &: \mathsf{TE}_{\mathrm{OP}}. \end{aligned}$$

Proposition A.48.4 (Factorization narrows failure)**.** *If an OP obstruction is expressed in the current route grammar, then failure can be localized to one of the finite obstruction sets*

$$D_{\mathrm{reach}}, \qquad D^{\mathrm{seg}}_{\mathrm{OP}}, \qquad D^{\mathsf{TE}}_{\mathrm{OP}}, \qquad D^{\mathrm{cert}}_{\rho}, \qquad D^{\mathrm{cert}}_{T}.$$

Proof. The route grammar introduces a finite sequence of checks. Reachability fails only in D_{reach}. Segment validity fails only in $D^{\mathrm{seg}}_{\mathrm{OP}}$. TE validity fails only in $D^{\mathsf{TE}}_{\mathrm{OP}}$. Certificate validity fails in D^{cert}_{ρ}, and tree validity fails in D^{cert}_{T}. Thus a global OP failure is replaced by a finite menu of typed local failures. □

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A.48.3 Role of the Hilbert shift and tensor notation

Remark A.48.5 (Reserved zero and clock anchoring)**.** *The shifted Hilbert-shift discipline helps because the inactive state and active route atoms are no longer confused. The clock anchor can be treated as a known reference object, while active route states are shifted into a normal form. This prevents the old ambiguity in which "no active route", "first active route", and "reference state" could blur together.*

Remark A.48.6 (Two-sided tensor notation)**.** *The two-sided tensor notation helps because it separates readout/frame/language interfaces from structure/action/field interfaces. A route edge is therefore not just an arrow; it is a typed edge with two-sided compatibility obligations. This makes tensor mismatch visible as its own failure mode rather than hiding it inside a general OP obstruction.*

A.48.4 Why could move quickly

Theorem A.48.7 (New-QAM narrowing principle)**.** *The current QAM/MML/MMA route grammar accelerates OP localization because it turns a global existence problem into a finite ordered diagnostic chain:*

$$\begin{array}{c} \textit{target problem} \\ \Downarrow \\ \textit{clock-to-target route existence} \\ \Downarrow \\ \textit{canonical optimal route} \\ \Downarrow \\ \textit{finite certificate} \\ \Downarrow \\ \textit{route tree} \\ \Downarrow \\ \textit{link-state reachability} \\ \Downarrow \\ \textit{segment waypoint validity} \\ \Downarrow \\ \textit{TE/guard-budget validity.} \end{array}$$

Proof. Each arrow is conservative and typed. The clock-to-target route separates reference from target. The canonical optimal route removes representational multiplicity. The finite certificate converts route existence into checkable data. The route tree packages many certificates. Link-state reachability turns the tree problem into finite graph reachability. Segment waypoints localize the required Steiner states. TE filters separate residual, guard, curvature, HC/MM, boundary-collar, and tensor-interface failures. Therefore the search space is progressively factored into smaller local checks. □

A.48.5 What the old QAM could and could not do

Proposition A.48.8 (Old QAM was expressive but not yet operational)**.** *The older QAM notation could describe the structural landscape, but it did not yet provide enough typed routing machinery to force a canonical OP diagnostic sequence.*

Proof. A notation may be expressive enough to describe many structural relations without being operational enough to determine which local admissibility clause fails. Before the

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Hilbert-shift normal form, router records, two-sided tensor interfaces, route certificates, segment templates, and TE filters were added, the obstruction could remain semantically correct but diagnostically unfactored. The newer grammar adds the missing operational layers. □

A.48.6 OP-facing interpretation

Remark A.48.9 (OP 1/4)**.** *For OP 1/4, the old question "does the selector exist?" is now split into: clock-to-anchor reachability, QAM/MML/selector segment validity, anchor coverage, and residual/guard/curvature budgets.*

Remark A.48.10 (OP 4)**.** *For OP 4, the old question "is the route globally stable?" is now split into: optimal-route-map stability, tree heartbeat/failure detection, reconvergence, and tensor/guard preservation along transported tree edges.*

Remark A.48.11 (OP 5(B))**.** *For OP 5(B), the old question "can boundary promotion happen?" is now split into: guard-preserving route existence, HC waypoint validity, boundary-collar validity, MMA readout validity, HC/MM compatibility, and post-promotion residual budget.*

Remark A.48.12 (Status after)**.** *This layer explains the acceleration but does not close the OPs. The next technical step is to instantiate the TE filters against the currently declared OP 1/4 and OP 5(B) segment templates and read off the first nonempty obstruction set.*

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A.49 Mixed-Readout Edge Compatibility Layer

Status. This layer ties the traffic-engineering flags back to the earlier guarded mixed QAM readout and two-sided tensor notation. A route edge is no longer accepted merely because it is reachable and guard-budget valid. It must also admit the mixed readout required by its role. The layer is conservative: mixed readouts are derived compatibility-filtered packages of already admitted component readouts, not new primitive tensors and not OP-closure certificates.

A.49.1 Edge readout modes

Definition A.49.1 (Edge readout mode)**.** *An edge readout mode is a declared component readout*

$$R_a^m$$

attached to a route edge, where a records the readout family and m records its mode. The current OP-facing families include

$$a \in \{\mathrm{QAM}, \mathrm{MML}, \mathrm{MMA}, \mathrm{HC}, \partial\mathrm{col}, \mathrm{sel}, \mathrm{res}, \tau\}.$$

Definition A.49.2 (Mixed-readout selector)**.** *For readout modes R_a^m and R_b^n, a mixed-readout selector is a finite compatibility tensor*

$$\chi_{mn}^{ab} \in \{0, 1\}$$

or, more generally, a finite guard-valued selector. The mixed readout is admitted only when the selector returns the admissible value.

Definition A.49.3 (Edge mixed readout)**.** *The edge mixed readout determined by R_a^m, R_b^n, and χ_{mn}^{ab} is the derived package*

$$\mathrm{Mix}_\chi^{ab}(R_a, R_b)(x) \quad \rightsquigarrow \quad \chi_{mn}^{ab} R_a^m(x) R_b^n(x).$$

This expression is read with the two-sided contraction discipline: the selector χ performs the compatibility test before the paired readout is admitted.

A.49.2 From flags to mixed readouts

Definition A.49.4 (Flag-to-readout realization)**.** *The compatibility flags are realized as mixed-readout selectors:*

$$\chi_{\mathrm{HC/MM}}(e) := \chi_{mn}^{\mathrm{HC,MML/MMA}} R_{\mathrm{HC}}^m(e) R_{\mathrm{MML/MMA}}^n(e),$$

$$\chi_{\partial\mathrm{col}}(e) := \chi_{mn}^{\mathrm{HC},\partial} R_{\mathrm{HC}}^m(e) R_{\partial\mathrm{col}}^n(e),$$
$$\chi_\tau(e) := \chi_{mn}^{\mathrm{front,back}} R_{\mathrm{front}}^m(e) R_{\mathrm{back}}^n(e).$$

A flag is therefore no longer an uninterpreted Boolean; it is the value of a declared mixed-readout compatibility test.

Remark A.49.5 (Why this matters)**.** *The filter could reject a route because $\chi_{\mathrm{HC/MM}}$, $\chi_{\partial\mathrm{col}}$, or χ_τ failed. The present layer explains what those failures mean: the required pair of readout channels does not admit a guarded mixed readout at that edge.*

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A.49.3 Mixed-readout-valid edge

Definition A.49.6 (Required mixed readout set)**.** *For an edge e, let*

$$\mathfrak{M}(e) = \{(a, b; m, n, \chi^{ab})\}$$

be the finite family of mixed-readout pairs required by the edge role. Examples include HC/MML, HC/MMA, HC/boundary-collar, selector/QAM, selector/MML, and MMA/residual pairs.

Definition A.49.7 (Mixed-readout-valid edge)**.** *An edge e is mixed-readout-valid if every required pair in $\mathfrak{M}(e)$ is admitted:*

$$\forall (a, b; m, n, \chi^{ab}) \in \mathfrak{M}(e) : \qquad \chi^{ab}_{mn} R^m_a(e) R^n_b(e) \text{ is admissible.}$$

Lemma A.49.8 (Mixed-readout validity refines TE validity)**.** *If an edge is mixed-readout-valid for every compatibility flag required by its TE profile, then the TE compatibility flags on that edge are realized by declared readout data rather than by free Boolean assumptions.*

Proof. By Definition A.49.4, each TE compatibility flag is the value of a declared mixed-readout selector. If the edge is mixed-readout-valid, every required selector admits the corresponding readout pair. Hence the flag is supported by readout data. □

A.49.4 OP 1/4 mixed readouts

Definition A.49.9 (OP 1/4 selector mixed-readout package)**.** *For the OP 1/4 segment*

$$q_{\mathrm{clk}} \to z_{\mathrm{QAM}} \to z_{\mathrm{MML}} \to z_{\mathrm{sel}} \to K_i \cap A^{\mathrm{sel}}_{1/4},$$

the selector mixed-readout package is

$$\mathfrak{M}_{1/4} := \{\mathrm{Mix}_\chi(R_{\mathrm{QAM}}, R_{\mathrm{MML}}), \mathrm{Mix}_\chi(R_{\mathrm{MML}}, R_{\mathrm{sel}}), \mathrm{Mix}_\chi(R_{\mathrm{sel}}, R_{\mathrm{anc}})\}.$$

Proposition A.49.10 (OP 1/4 mixed-readout refinement)**.** *If an OP 1/4 segment route is TE-valid and every edge of the route is mixed-readout-valid for $\mathfrak{M}_{1/4}$, then the route does not merely reach the anchor component; it reaches it through a compatible QAM/MML/selector readout chain.*

Proof. TE-validity supplies reachability, guard, residual, curvature, and tensor budget control. Mixed-readout validity supplies the required compatibility between the QAM readout, the MML router readout, the selector readout, and the anchor readout. Therefore the route reaches the anchor component through the declared readout chain rather than through an untyped path. □

A.49.5 OP 5(B) mixed readouts

Definition A.49.11 (OP 5(B) boundary mixed-readout package)**.** *For the OP 5(B) segment*

$$q_{\mathrm{clk}} \to z_{\mathrm{HC}} \to z_{\partial\mathrm{col}} \to z_{\mathrm{MMA}} \to Y_5^{(B),\mathrm{prom}},$$

the boundary mixed-readout package is

$$\mathfrak{M}_5^{(B)} := \{\mathrm{Mix}_\chi(R_{\mathrm{HC}}, R_{\partial\mathrm{col}}), \mathrm{Mix}_\chi(R_{\partial\mathrm{col}}, R_{\mathrm{MMA}}), \mathrm{Mix}_\chi(R_{\mathrm{MMA}}, R_{\mathrm{res}}), \mathrm{Mix}_\chi(R_{\mathrm{HC}}, R_{\mathrm{MML/MM}})\}.$$

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Proposition A.49.12 (OP 5(B) mixed-readout refinement)**.** *If an OP 5(B) boundary segment route is TE-valid, boundary-collar valid, HC/MM-valid, and mixed-readout-valid for* $\mathfrak{M}_5^{(B)}$*, then the route supplies a compatible boundary-promotion readout chain. The remaining obligation is the post-promotion residual control, not the mere existence of a compatible readout route.*

Proof. TE-validity and boundary-collar validity supply the route, guard, and boundary constraints. HC/MM-validity supplies the matrix/readout compatibility. The mixed-readout package asserts that HC, boundary-collar, MMA, residual, and MML/MM readouts can be paired through declared selectors. Thus the boundary promotion route is compatible at the readout level. The residual after promotion remains a separate quantity to be controlled. □

A.49.6 Mixed-readout obstruction

Definition A.49.13 (Mixed-readout obstruction)**.** *For a route* ρ*, define*

$$D_{\mathrm{mix}}(\rho) := \{(e,a,b;m,n) \mid (a,b;m,n,\chi^{ab}) \in \mathfrak{M}(e) \text{ and } \chi^{ab}_{mn} R^m_a(e) R^n_b(e) \text{ is not admissible}\}.$$

Theorem A.49.14 (mixed-readout reduction)**.** *After the local pass, the OP-facing TE obstruction refines to the mixed-readout obstruction:*

$$D_{\mathrm{mix}}(\rho) = \varnothing \quad \Longrightarrow \quad \begin{array}{c} \text{all required compatibility flags on } \rho \text{ are realized} \\ \text{by declared readout selectors.} \end{array}$$

If $D_{\mathrm{mix}}(\rho) \neq \varnothing$*, the obstruction records the exact edge and readout pair that failed.*

Proof. The implication follows directly from the flag-to-readout realization. A nonempty mixed-readout obstruction records an edge and a required readout pair whose selector does not admit the mixed readout. Hence the route is no longer rejected by an opaque compatibility flag; it is rejected by a named readout-pair failure. □

Remark A.49.15 (OP status after)**.** *OP 1/4, OP 4, and OP 5(B) remain open. The gain is that the compatibility part of the route filter is now readable as mixed-readout admissibility. This should make the next OP pass more precise: one can test whether the obstruction is reachability, segment structure, budget, or a specific failed readout mixture.*

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A.50 Mixed-Readout Index Registry and Reader Atlas

Status. This layer is a reader-facing and audit-facing index registry for the local pass mixed-readout edge compatibility layer. It does not add new physics and does not close OP 1/4, OP 4, or OP 5(B). It fixes the meaning of the index tuple

$$(e, a, b; m, n)$$

so that later OP diagnostics can say exactly which edge, readout family, and mode pairing failed.

A.50.1 Four index levels

Definition A.50.1 (Mixed-readout index tuple)**.** *A mixed-readout diagnostic index is a tuple*

$$\iota_{\mathrm{mix}} := (e, a, b; m, n),$$

where:

1. *e is the route edge being tested;*

2. *a, b are readout-family indices;*

3. *m, n are mode indices inside the corresponding readout families.*

The tuple is read as: on edge e, readout family a in mode m is being mixed with readout family b in mode n.

Definition A.50.2 (Readout-family registry)**.** *The current readout-family registry is*

$$\mathcal{A}_{\mathrm{RO}} := \{\mathrm{QAM}, \mathrm{MML}, \mathrm{MMA}, \mathrm{HC}, \partial\mathrm{col}, \mathrm{sel}, \mathrm{anc}, \mathrm{res}, \tau\}.$$

The intended readings are:

Family	*Reading*
QAM	*object, language, normal-form, and store-side readout*
MML	*route, instruction, and router/intertwiner readout*
MMA	*matrix, execution, and readout-semantics evaluator*
HC	*Heisenberg-compensator/frame-normalization readout*
∂col	*boundary-collar and boundary-guard readout*
sel	*selector/comparison-surface readout*
anc	*anchor/common-target readout*
res	*residual/parity/obstruction readout*
τ	*tensor-interface/type readout*

Definition A.50.3 (Mode index registry)**.** *For every readout family $a \in \mathcal{A}_{\mathrm{RO}}$, the mode index $m \in M_a$ records the local mode in which that family is read. In these mode sets remain named finite local registries rather than globally numeric registers. This prevents premature global enumeration before the OP-facing mode families have stabilized.*

Remark A.50.4 (Why modes are not globally numbered yet)**.** *A global numeric mode registry would be useful later, but it would be too early here. The current OP-facing work first needs to learn which modes are actually used by OP 1/4, OP 4, and OP 5(B). Therefore fixes the role of the mode indices without freezing a final numeric list.*

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A.50.2 Selector grammar

Definition A.50.5 (Selector-indexed mixed readout)**.** *Given the diagnostic index* $(e, a, b; m, n)$*, the corresponding selector-indexed mixed readout is*

$$\mathrm{Mix}_{\chi}^{ab}[m, n](e) := \chi_{mn}^{ab}(e)\, R_a^m(e)\, R_b^n(e).$$

The expression is admissible only if the selector permits the pair:

$$\chi_{mn}^{ab}(e) = 1$$

or, in a guard-valued registry, if $\chi_{mn}^{ab}(e)$ *returns an admissible guard state.*

Definition A.50.6 (Mixed-readout obstruction with indices)**.** *For a route* ρ*, the indexed mixed-readout obstruction is*

$$D_{\mathrm{mix}}(\rho) := \{(e, a, b; m, n) \mid \mathrm{Mix}_{\chi}^{ab}[m, n](e) \text{ is not admissible}\}.$$

Proposition A.50.7 (Indexed failure localization)**.** *If* $(e, a, b; m, n) \in D_{\mathrm{mix}}(\rho)$*, then the mixed-readout failure is localized to one route edge, two readout families, and one pair of local modes.*

Proof. This is immediate from the definition of $D_{\mathrm{mix}}(\rho)$. The tuple records exactly the tested edge, the two readout families, and the two modes. Hence the failure is not a global compatibility failure but a local selector-indexed readout-pair failure. □

A.50.3 OP 1/4 atlas

Definition A.50.8 (OP 1/4 mixed-readout index atlas)**.** *For the OP 1/4 segment*

$$q_{\mathrm{clk}} \to z_{\mathrm{QAM}} \to z_{\mathrm{MML}} \to z_{\mathrm{sel}} \to K_i \cap A_{1/4}^{\mathrm{sel}},$$

the default mixed-readout index atlas is

$$\begin{aligned} e_{QM} &: (\mathrm{QAM}, \mathrm{MML}; m_Q, m_M), \\ e_{MS} &: (\mathrm{MML}, \mathrm{sel}; m_M, m_S), \\ e_{SA} &: (\mathrm{sel}, \mathrm{anc}; m_S, m_A). \end{aligned}$$

Thus the OP 1/4 route is read as a chain of three selector-mediated tests: QAM/MML, MML/selector, and selector/anchor.

Remark A.50.9 (Human reading of the OP 1/4 atlas)**.** *The OP 1/4 atlas says: first the QAM normal-form readout must be compatible with the MML router readout; then the router readout must be compatible with the selector readout; then the selector readout must be compatible with the anchor readout. A failure in any of the three pairs tells which part of the selector route must be repaired.*

A.50.4 OP 5(B) atlas

Definition A.50.10 (OP 5(B) mixed-readout index atlas)**.** *For the OP 5(B) segment*

$$q_{\mathrm{clk}} \to z_{\mathrm{HC}} \to z_{\partial\mathrm{col}} \to z_{\mathrm{MMA}} \to Y_5^{(B),\mathrm{prom}},$$

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the default mixed-readout index atlas is

$$\begin{aligned} e_{H\partial} &: (\mathrm{HC}, \partial\mathrm{col}; m_H, m_\partial), \\ e_{\partial M} &: (\partial\mathrm{col}, \mathrm{MMA}; m_\partial, m_M), \\ e_{MR} &: (\mathrm{MMA}, \mathrm{res}; m_M, m_R), \\ e_{HMM} &: (\mathrm{HC}, \mathrm{MML/MM}; m_H, m_{MM}). \end{aligned}$$

The last pair records the HC/MM compatibility readout that may be attached to the boundary-promotion route even when it is not a separate visible edge.

Remark A.50.11 (Human reading of the OP 5(B) atlas)**.** *The OP 5(B) atlas says: HC must be compatible with the boundary collar; the boundary collar must be compatible with the MMA readout; the MMA readout must be compatible with the residual evaluator; and HC/MM compatibility must remain available as a route-level readout. This separates boundary failure from matrix-readout failure and from residual-control failure.*

A.50.5 Index-to-filter bridge

Proposition A.50.12 (Index bridge to TE flags)**.** *The TE flags*

$$\chi_{\mathrm{HC/MM}}, \qquad \chi_{\partial\mathrm{col}}, \qquad \chi_\tau$$

are instances of the indexed selector grammar

$$\chi^{ab}_{mn}(e)$$

after the readout families and modes have been chosen from the registry.

Proof. The layer realized the TE flags as mixed-readout selectors. The present registry fixes the general selector grammar by naming the edge e, readout families a, b, and mode pair m, n. Therefore each TE flag is a special case of $\chi^{ab}_{mn}(e)$ with the relevant family and mode assignment. □

Theorem A.50.13 (mixed-readout index clarification)**.** *After the local pass, a mixed-readout compatibility failure is expressible in the canonical form*

$$(e, a, b; m, n) \in D_{\mathrm{mix}}(\rho).$$

This means that the obstruction is localized to a specific route edge, a specific pair of readout families, and a specific pair of local modes.

Proof. By Definition A.50.1, every mixed-readout diagnostic has the form $(e, a, b; m, n)$. By Definition A.50.6, failed mixed-readout tests are precisely recorded in $D_{\mathrm{mix}}(\rho)$. Thus every compatibility failure can be written in the stated canonical form. □

Remark A.50.14 (OP status after)**.** *OP 1/4, OP 4, and OP 5(B) remain open. The gain is explanatory and diagnostic: the new mixed-readout layer is now indexed clearly enough for a reader or audit script to understand exactly what is being tested on each route edge.*

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A.51 First Mixed-Readout Mode Registry

Status. This layer continues the index registry by naming the first local mode sets used by OP 1/4 and OP 5(B). The modes are intentionally not converted into a final global numeric registry. They are local OP-facing modes that stabilize the meaning of m, n in the diagnostic tuple

$$(e, a, b; m, n).$$

This is a reader/audit layer and a preparation for sharper OP diagnostics; it does not close OP 1/4, OP 4, or OP 5(B).

A.51.1 Local mode sets

Definition A.51.1 (Local mode set)**.** *For a readout family a, a local mode set M_a^{loc} is the finite set of named modes currently needed by the OP-facing route grammar. A local mode is not yet a global numerical index; it is a stable semantic label for a readout use.*

Definition A.51.2 (First OP-facing mode registry)**.** *The first OP-facing mode registry is the finite family*

$$\mathcal{M}_{\mathrm{OP}}^{(1)} = \{M_{\mathrm{QAM}}, M_{\mathrm{MML}}, M_{\mathrm{MMA}}, M_{\mathrm{HC}}, M_{\partial\mathrm{col}}, M_{\mathrm{sel}}, M_{\mathrm{anc}}, M_{\mathrm{res}}, M_{\tau}\}$$

with the following local mode labels:

$$\begin{aligned}
M_{\mathrm{QAM}} &= \{\mathrm{normal}, \mathrm{store}, \mathrm{mask}, \mathrm{front}\},\\
M_{\mathrm{MML}} &= \{\mathrm{router}, \mathrm{intertwiner}, \mathrm{segment}, \mathrm{path}\},\\
M_{\mathrm{MMA}} &= \{\mathrm{matrix}, \mathrm{execution}, \mathrm{residualEval}, \mathrm{back}\},\\
M_{\mathrm{HC}} &= \{\mathrm{comp}, \mathrm{frame}, \mathrm{boundaryNorm}, \mathrm{mmCompat}\},\\
M_{\partial\mathrm{col}} &= \{\mathrm{boundary}, \mathrm{collar}, \mathrm{guard}, \mathrm{stop}\},\\
M_{\mathrm{sel}} &= \{\mathrm{selector}, \mathrm{comparison}, \mathrm{fiber}, \mathrm{equalizer}\},\\
M_{\mathrm{anc}} &= \{\mathrm{anchor}, \mathrm{component}, \mathrm{commonImage}, \mathrm{target}\},\\
M_{\mathrm{res}} &= \{\mathrm{residual}, \mathrm{post}, \mathrm{budget}, \mathrm{obstruction}\},\\
M_{\tau} &= \{\mathrm{frontType}, \mathrm{backType}, \mathrm{contract}, \mathrm{compare}\}.
\end{aligned}$$

Remark A.51.3 (Why this is still local)**.** *The registry is strong enough for OP 1/4 and OP 5(B) diagnostics but weak enough to avoid premature global numbering. A later version may turn these labels into a numeric or binary mask registry after the OP-facing mode use has stabilized.*

A.51.2 OP 1/4 instantiated modes

Definition A.51.4 (OP 1/4 mode assignments)**.** *The first OP 1/4 mixed-readout mode assignments are*

$$\begin{aligned}
e_{QM} &: (\mathrm{QAM}, \mathrm{MML}; \mathrm{normal}, \mathrm{router}),\\
e_{MS} &: (\mathrm{MML}, \mathrm{sel}; \mathrm{router}, \mathrm{selector}),\\
e_{SA} &: (\mathrm{sel}, \mathrm{anc}; \mathrm{selector}, \mathrm{anchor}).
\end{aligned}$$

Equivalently, the OP 1/4 mixed-readout tests are

$$\chi^{\mathrm{QAM,MML}}_{\mathrm{normal,router}}(e_{QM}),$$

$$\chi^{\mathrm{MML,sel}}_{\mathrm{router,selector}}(e_{MS}),$$

$$\chi^{\mathrm{sel,anc}}_{\mathrm{selector,anchor}}(e_{SA}).$$

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Proposition A.51.5 (OP 1/4 first mode diagnostic)**.** *If all three OP 1/4 mode assignments are admitted by their selectors, then the OP 1/4 segment route passes the first mixed-readout compatibility layer. If one fails, the failure is localized to the corresponding pair: QAM/MML, MML/selector, or selector/anchor.*

Proof. The obstruction records failures as tuples $(e, a, b; m, n)$. Substituting the OP 1/4 mode assignments gives exactly the three listed tests. Therefore empty failure set across these tests means the first mixed-readout layer passes, while a nonempty failure identifies the failing pair. □

A.51.3 OP 5(B) instantiated modes

Definition A.51.6 (OP 5(B) mode assignments)**.** *The first OP 5(B) mixed-readout mode assignments are*

$$\begin{aligned} e_{H\partial} &: (\mathrm{HC}, \partial\mathrm{col}; \mathrm{boundaryNorm}, \mathrm{collar}),\\ e_{\partial M} &: (\partial\mathrm{col}, \mathrm{MMA}; \mathrm{guard}, \mathrm{matrix}),\\ e_{MR} &: (\mathrm{MMA}, \mathrm{res}; \mathrm{residualEval}, \mathrm{post}),\\ e_{HMM} &: (\mathrm{HC}, \mathrm{MML/MM}; \mathrm{mmCompat}, \mathrm{intertwiner}). \end{aligned}$$

Equivalently, the OP 5(B) mixed-readout tests are

$$\chi^{\mathrm{HC},\partial\mathrm{col}}_{\mathrm{boundaryNorm},\mathrm{collar}}(e_{H\partial}),$$

$$\chi^{\partial\mathrm{col},\mathrm{MMA}}_{\mathrm{guard},\mathrm{matrix}}(e_{\partial M}),$$

$$\chi^{\mathrm{MMA},\mathrm{res}}_{\mathrm{residualEval},\mathrm{post}}(e_{MR}),$$

$$\chi^{\mathrm{HC},\mathrm{MML/MM}}_{\mathrm{mmCompat},\mathrm{intertwiner}}(e_{HMM}).$$

Proposition A.51.7 (OP 5(B) first mode diagnostic)**.** *If all four OP 5(B) mode assignments are admitted by their selectors, then the boundary-promotion route passes the first mixed-readout compatibility layer. If one fails, the failure is localized to HC/boundary-collar, boundary-collar/MMA, MMA/residual, or HC/MM compatibility.*

Proof. As in the OP 1/4 case, records a failed selector as $(e, a, b; m, n)$. The four assignments instantiate those indices for the OP 5(B) boundary route. Thus pass/fail is localized to one named readout-pair test. □

A.51.4 Mode diagnostic obstruction

Definition A.51.8 (Mode diagnostic obstruction)**.** *For a route ρ, define the first mode diagnostic obstruction*

$$D^{(1)}_{\mathrm{mode}}(\rho) := \{(e, a, b; m, n) \in D_{\mathrm{mix}}(\rho) \mid m, n \in \mathcal{M}^{(1)}_{\mathrm{OP}}\}.$$

Theorem A.51.9 (first mode-registry reduction)**.** *After the local pass, the first OP-facing mixed-readout compatibility problem can be tested against the named local mode registry $\mathcal{M}^{(1)}_{\mathrm{OP}}$. If*

$$D^{(1)}_{\mathrm{mode}}(\rho) = \varnothing,$$

then the route passes the first local mode layer. If not, the obstruction is a specific edge, readout-family pair, and named mode pair.

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Proof. The definition of $D^{(1)}_{\mathrm{mode}}$ restricts the mixed-readout obstruction to the first OP-facing mode registry. Therefore the empty set means no mixed-readout selector fails at the first mode layer. A nonempty set gives the exact tuple $(e, a, b; m, n)$, with m, n now named rather than schematic. □

Remark A.51.10 (Use in the next OP pass)**.** *The next OP pass can instantiate the tests before attempting larger route-tree or residual arguments. If the first mixed-readout mode layer fails, the correct repair target is local. If it passes, one can continue to reachability, TE budgets, residual control, and OP 4 stability.*

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A.52 First Selector Table and Mixed-Readout Diagnostic Pass

Status. The layer named the first local mixed-readout mode registry. The present layer turns those named mode pairs into the first selector table. A route passes the first mixed-readout diagnostic pass exactly when the required selector entries $\chi^{ab}_{mn}(e)$ are admitted on the required edges. This is still not OP closure; it is a finite compatibility precheck before the larger reachability, TE-budget, residual, and OP 4 stability arguments.

A.52.1 First selector table

Definition A.52.1 (First selector table)**.** *The first selector table is the partial finite table*

$$\mathsf{Sel}^{(1)} \subseteq E \times \mathcal{A}_{\mathrm{RO}} \times \mathcal{A}_{\mathrm{RO}} \times M \times M$$

whose entries have the form

$$\mathsf{Sel}^{(1)}(e, a, b; m, n) = \chi^{ab}_{mn}(e) \in \{0, 1\}.$$

An entry value 1 *means that the mixed readout*

$$\mathrm{Mix}^{ab}_{\chi}[m, n](e)$$

is admitted by the first local mode layer. An entry value 0 *means that the pair is rejected at this layer.*

Definition A.52.2 (Required first selector entries)**.** *For a route ρ, the required first selector entry set is*

$$\mathsf{Req}^{(1)}(\rho) := \{(e, a, b; m, n) \mid \textit{the first-mode atlas requires this mixed readout on } \rho\}.$$

Definition A.52.3 (First selector failure set)**.** *The first selector failure set is*

$$D^{(1)}_{\chi}(\rho) := \{(e, a, b; m, n) \in \mathsf{Req}^{(1)}(\rho) \mid \mathsf{Sel}^{(1)}(e, a, b; m, n) \neq 1\}.$$

Lemma A.52.4 (Selector table equals first mode diagnostic)**.** *For the local mode registry,*

$$D^{(1)}_{\chi}(\rho) = D^{(1)}_{\mathrm{mode}}(\rho).$$

up to the notational convention that $D^{(1)}_{\chi}$ records the selector table entry and $D^{(1)}_{\mathrm{mode}}$ records the failed mixed-readout tuple.

Proof. Both sets are built from the same required tuples $(e, a, b; m, n)$. The first records tuples whose selector-table value is not admitted; the second records tuples whose mixed readout is not admissible. Since first-layer mixed-readout admissibility is exactly the condition $\chi^{ab}_{mn}(e) = 1$, the two obstruction records agree. □

A.52.2 OP 1/4 selector table

Definition A.52.5 (OP 1/4 first selector table)**.** *The OP 1/4 first selector table consists of the entries*

$$\begin{aligned}
\mathsf{Sel}^{(1)}_{1/4}(e_{QM}) &= \chi^{\mathrm{QAM,MML}}_{\mathrm{normal,router}}(e_{QM}),\\
\mathsf{Sel}^{(1)}_{1/4}(e_{MS}) &= \chi^{\mathrm{MML,sel}}_{\mathrm{router,selector}}(e_{MS}),\\
\mathsf{Sel}^{(1)}_{1/4}(e_{SA}) &= \chi^{\mathrm{sel,anc}}_{\mathrm{selector,anchor}}(e_{SA}).
\end{aligned}$$

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Definition A.52.6 (OP 1/4 selector pass)**.** *The OP 1/4 first selector pass condition is*

$$\mathsf{Pass}_{1/4}^{\chi,(1)} := \big[\mathsf{Sel}_{1/4}^{(1)}(e_{QM}) = \mathsf{Sel}_{1/4}^{(1)}(e_{MS}) = \mathsf{Sel}_{1/4}^{(1)}(e_{SA}) = 1\big].$$

Proposition A.52.7 (OP 1/4 first selector diagnostic)**.** *If* $\mathsf{Pass}_{1/4}^{\chi,(1)}$ *holds, then the OP 1/4 route passes the first QAM/MML/selector/anchor mixed-readout compatibility check. If it fails, the failed selector entry identifies whether the obstruction is QAM/MML, MML/selector, or selector/anchor.*

Proof. The three entries are exactly the three mixed-readout pairings required by the OP 1/4 atlas. If all are admitted, the first compatibility check passes. If any entry is not admitted, the failed tuple names the failed pair. □

A.52.3 OP 5(B) selector table

Definition A.52.8 (OP 5(B) first selector table)**.** *The OP 5(B) first selector table consists of the entries*

$$\begin{aligned}
\mathsf{Sel}_{5B}^{(1)}(e_{H\partial}) &= \chi^{\mathrm{HC},\partial\mathrm{col}}_{\mathrm{boundaryNorm,collar}}(e_{H\partial}),\\
\mathsf{Sel}_{5B}^{(1)}(e_{\partial M}) &= \chi^{\partial\mathrm{col},\mathrm{MMA}}_{\mathrm{guard,matrix}}(e_{\partial M}),\\
\mathsf{Sel}_{5B}^{(1)}(e_{MR}) &= \chi^{\mathrm{MMA,res}}_{\mathrm{residualEval,post}}(e_{MR}),\\
\mathsf{Sel}_{5B}^{(1)}(e_{HMM}) &= \chi^{\mathrm{HC,MML/MM}}_{\mathrm{mmCompat,intertwiner}}(e_{HMM}).
\end{aligned}$$

Definition A.52.9 (OP 5(B) selector pass)**.** *The OP 5(B) first selector pass condition is*

$$\mathsf{Pass}_{5B}^{\chi,(1)} := \big[\mathsf{Sel}_{5B}^{(1)}(e_{H\partial}) = \mathsf{Sel}_{5B}^{(1)}(e_{\partial M}) = \mathsf{Sel}_{5B}^{(1)}(e_{MR}) = \mathsf{Sel}_{5B}^{(1)}(e_{HMM}) = 1\big].$$

Proposition A.52.10 (OP 5(B) first selector diagnostic)**.** *If* $\mathsf{Pass}_{5B}^{\chi,(1)}$ *holds, then the OP 5(B) boundary route passes the first HC/boundary/MMA/residual and HC/MM mixed-readout compatibility check. If it fails, the failed selector entry identifies whether the obstruction is HC/boundary-collar, boundary-collar/MMA, MMA/residual, or HC/MM.*

Proof. The four entries are exactly the four mixed-readout pairings required by the OP 5(B) atlas. The pass condition says all four are admitted. Any failure therefore names a unique first-layer mixed-readout obstruction. □

A.52.4 Interaction with TE and route checks

Theorem A.52.11 (First selector pass as compatibility precheck)**.** *The first selector pass is a necessary compatibility precheck for the local pass TE-valid route claims. More precisely, a route cannot satisfy the local pass compatibility clauses through declared mixed readouts unless*

$$D_{\chi}^{(1)}(\rho) = \varnothing$$

for all required first-layer mixed-readout entries.

Proof. The layer realizes the compatibility flags as mixed-readout selector tests. The layer names the first OP-facing modes. Therefore, if a required first-layer selector entry is not admitted, the corresponding compatibility flag cannot be supported by declared readout data at that edge. Hence empty $D_{\chi}^{(1)}$ is necessary for TE compatibility through this readout layer. □

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Remark A.52.12 (Where this helps the OPs)**.** *The first selector table is useful because it is cheaper to check than the full route/tree/residual problem. If the table fails, the OP obstruction is already localized to one mixed-readout pair. If the table passes, the next pass can honestly move to reachability, TE budgets, residual control, and OP 4 stability.*

Remark A.52.13 (OP status after)**.** *OP 1/4, OP 4, and OP 5(B) remain open. The gain is that the first mixed-readout mode registry now has a concrete selector-table pass/fail diagnostic.*

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A.53 Selector-to-Route Gate

Status. The layer introduced the first selector table. The present layer turns that table into a formal route gate. Before a route is tested for link-state reachability, segment validity, traffic-engineering budgets, residual control, or OP 4 stability, the required first-layer mixed-readout selectors must pass. This is a conservative ordering rule: it prevents route work from proceeding through an edge whose readout compatibility is already rejected.

A.53.1 Selector gate

Definition A.53.1 (First selector gate)**.** *For a route ρ, the first selector gate is the predicate*

$$\mathsf{Gate}_{\chi}^{(1)}(\rho) := [D_{\chi}^{(1)}(\rho) = \varnothing].$$

If $\mathsf{Gate}_{\chi}^{(1)}(\rho)$ holds, the route is selector-admissible at the first mixed-readout mode layer. If it fails, the route is stopped before reachability or TE-budget evaluation.

Definition A.53.2 (Selector-gated admissible edge)**.** *An edge e is first-selector-gated admissible if every first-layer mixed readout required on e has selector value 1:*

$$\forall (e, a, b; m, n) \in \mathsf{Req}^{(1)}(e) : \qquad \chi_{mn}^{ab}(e) = 1.$$

The corresponding gated edge set is

$$E_{\chi}^{(1)} := \{e \in E \mid e \text{ is first-selector-gated admissible}\}.$$

Definition A.53.3 (Selector-gated graph)**.** *Given a link-state graph $\mathcal{G} = (V, E)$, define the selector-gated graph*

$$\mathcal{G}_{\chi}^{(1)} := (V, E_{\chi}^{(1)}).$$

The later admissible and TE-filtered graphs may be formed only after this gate has been applied:

$$\mathcal{G}_{\chi,\mathrm{adm}}^{(1)} \subseteq \mathcal{G}_{\chi}^{(1)}.$$

A.53.2 Gate ordering

Definition A.53.4 (OP diagnostic order after)**.** *The diagnostic order is*

$$\begin{gathered} D_{\chi}^{(1)}(\rho) = \varnothing \\ \Downarrow \\ D_{\mathrm{reach}} = \varnothing \\ \Downarrow \\ D_{\mathrm{OP}}^{\mathrm{seg}} = \varnothing \\ \Downarrow \\ D_{\mathrm{OP}}^{\mathsf{TE}} = \varnothing \\ \Downarrow \\ \textit{residual control and OP 4 stability tests.} \end{gathered}$$

Lemma A.53.5 (Gate failure precedes route failure)**.** *If $D_{\chi}^{(1)}(\rho) \neq \varnothing$, then ρ cannot be accepted as a first-layer readout-compatible OP route, regardless of later reachability or TE-budget data.*

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Proof. A nonempty selector failure set contains a required mixed-readout pair that is not admitted. Since realizes the relevant compatibility flags through mixed-readout selectors, a route with such a failure lacks the readout compatibility required before later route checks. Later reachability cannot repair a failed selector on a required edge. □

Proposition A.53.6 (Selector-gated reachability is sharper than raw reachability)**.** *Reachability in $\mathcal{G}^{(1)}_{\chi,\mathrm{adm}}$ is a sharper condition than reachability in the raw link-state graph. In particular, if a target is reachable in $\mathcal{G}^{(1)}_{\chi,\mathrm{adm}}$, then it is reachable by a route whose first-layer mixed readout requirements are all admitted.*

Proof. The graph $\mathcal{G}^{(1)}_{\chi,\mathrm{adm}}$ contains only edges that first passed the selector gate and then passed the ordinary admissibility filters. Therefore any path in it is built entirely from selector-admitted edges. Raw reachability may use edges that are graph-theoretically present but readout-incompatible; selector-gated reachability cannot. □

A.53.3 OP 1/4 gate

Definition A.53.7 (OP 1/4 selector gate)**.** *The OP 1/4 selector gate is*

$$\mathsf{Gate}^{\chi,(1)}_{1/4} := \mathsf{Pass}^{\chi,(1)}_{1/4}.$$

Equivalently,

$$\mathsf{Gate}^{\chi,(1)}_{1/4} = [\chi^{\mathrm{QAM,MML}}_{\mathrm{normal,router}}(e_{QM}) = \chi^{\mathrm{MML,sel}}_{\mathrm{router,selector}}(e_{MS}) = \chi^{\mathrm{sel,anc}}_{\mathrm{selector,anchor}}(e_{SA}) = 1].$$

Proposition A.53.8 (OP 1/4 gate consequence)**.** *If $\mathsf{Gate}^{\chi,(1)}_{1/4}$ fails, then OP 1/4 should be repaired at the first failed mixed-readout selector before testing anchor reachability or cover adequacy. If it passes, the OP 1/4 route may proceed to selector-gated reachability and segment checks.*

Proof. By Lemma A.53.5, a failed selector gate blocks first-layer readout-compatible route acceptance. The failed entry identifies whether the repair target is QAM/MML, MML/selector, or selector/anchor. If the gate passes, no first-layer selector entry blocks the later route checks. □

A.53.4 OP 5(B) gate

Definition A.53.9 (OP 5(B) selector gate)**.** *The OP 5(B) selector gate is*

$$\mathsf{Gate}^{\chi,(1)}_{5B} := \mathsf{Pass}^{\chi,(1)}_{5B}.$$

Equivalently, it requires the four entries

$$\chi^{\mathrm{HC},\partial\mathrm{col}}_{\mathrm{boundaryNorm,collar}},\quad \chi^{\partial\mathrm{col,MMA}}_{\mathrm{guard,matrix}},\quad \chi^{\mathrm{MMA,res}}_{\mathrm{residualEval,post}},\quad \chi^{\mathrm{HC,MML/MM}}_{\mathrm{mmCompat,intertwiner}}$$

to be admitted on their declared route edges.

Proposition A.53.10 (OP 5(B) gate consequence)**.** *If $\mathsf{Gate}^{\chi,(1)}_{5B}$ fails, then OP 5(B) should be repaired at the failed mixed-readout selector before boundary-promotion reachability, HC/MM residual, or post-promotion residual checks. If it passes, the OP 5(B) route may proceed to boundary-collar, TE-budget, and residual tests.*

Proof. The proof is the OP 5(B) instance of the selector-gate ordering. A failed first-layer selector blocks readout-compatible boundary-promotion routing. A passed gate removes this first obstruction and allows the later tests to become meaningful. □

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A.53.5 Gate obstruction

Definition A.53.11 (Selector-gate obstruction)**.** *For a family of OP routes $\mathfrak{R}$, define*

$$D^{(1)}_{\mathrm{gate}} := \{\rho \in \mathfrak{R} \mid D^{(1)}_{\chi}(\rho) \neq \varnothing\}.$$

Theorem A.53.12 (selector-to-route reduction)**.** *After the local pass, the first OP-facing route precheck is*

$$D^{(1)}_{\mathrm{gate}} = \varnothing.$$

If this fails, the obstruction is selector/readout compatibility, not reachability, TE-budget, residual control, or OP 4 stability. If it holds, the route family can be passed to the selector-gated reachability and TE layers.

Proof. The selector-gate obstruction is exactly the family of routes with nonempty first selector failures. By the gate-ordering lemma, those routes cannot be accepted before later route tests. Conversely, if no route fails the selector gate, the first mixed-readout compatibility layer is cleared and the next diagnostic layers may be applied. □

Remark A.53.13 (OP status after)**.** *OP 1/4, OP 4, and OP 5(B) remain open. The gain is procedural: the proof search now has a strict first gate. Do not spend work on reachability or TE budgets for a route whose first required mixed-readout selector has already failed.*

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A.54 Selector-Gated Reachability Diagnostic

Status. The layer installed the first selector gate. The present layer performs the next diagnostic pass: reachability is checked only inside the graph whose edges have passed the first mixed-readout selector gate and the ordinary link-state admissibility filters. Thus a reachable route is not merely a graph path; it is a first-layer readout-compatible graph path. OP 1/4, OP 4, and OP 5(B) remain open.

A.54.1 Selector-gated admissible graph

Definition A.54.1 (Selector-gated admissible graph)**.** *Let*

$$\mathcal{G} = (V, E)$$

be the raw OP link-state graph. The first selector-gated admissible graph is

$$\mathcal{G}^{(1)}_{\chi,\mathrm{adm}} := (V, E^{(1)}_{\chi,\mathrm{adm}})$$

where

$$E^{(1)}_{\chi,\mathrm{adm}} := \{e \in E \mid e \in E^{(1)}_{\chi},\ \mathrm{up}(e) = 1,\ \tau(e) \text{ matches},\ \Gamma(e) \text{ preserves active guards}\}.$$

Definition A.54.2 (Selector-gated reachability obstruction)**.** *For a selected target family $\mathcal{Q}$, define*

$$D^{\chi,(1)}_{\mathrm{reach}} := \{q_{\mathrm{tar}} \in \mathcal{Q} \mid q_{\mathrm{tar}} \notin \mathrm{Reach}_{\mathcal{G}^{(1)}_{\chi,\mathrm{adm}}}(q_{\mathrm{clk}})\}.$$

Lemma A.54.3 (Selector-gated reachability implies ordinary reachability)**.** *If*

$$q_{\mathrm{tar}} \in \mathrm{Reach}_{\mathcal{G}^{(1)}_{\chi,\mathrm{adm}}}(q_{\mathrm{clk}}),$$

then q_{tar} is reachable in the ordinary admissible graph $\mathcal{G}_{\mathrm{adm}}$.

Proof. The selector-gated admissible graph is a subgraph of the ordinary admissible graph. Therefore any path in the selector-gated admissible graph is also a path in the ordinary admissible graph. □

Proposition A.54.4 (Selector-gated route certificate)**.** *If*

$$q_{\mathrm{tar}} \in \mathrm{Reach}_{\mathcal{G}^{(1)}_{\chi,\mathrm{adm}}}(q_{\mathrm{clk}}),$$

then there exists a route certificate from q_{clk} to q_{tar} whose edges are first-layer selector compatible, tensor-compatible, active, and guard-preserving.

Proof. Reachability gives a directed path in $\mathcal{G}^{(1)}_{\chi,\mathrm{adm}}$. By construction, every edge of this path is first-layer selector compatible, active, tensor-compatible, and guard-preserving. Recording the path data gives the stated route certificate. □

A.54.2 OP 1/4 selector-gated reachability

Definition A.54.5 (OP 1/4 selector-gated target family)**.** *The OP 1/4 selector-gated target family is*

$$\mathcal{Q}^{\chi}_{1/4} := \{K_i \cap A^{\mathrm{sel}}_{1/4} \mid K_i \in \mathcal{K}_{1/4}\}$$

read inside $\mathcal{G}^{(1)}_{\chi,\mathrm{adm}}$.

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Definition A.54.6 (OP 1/4 selector-gated reachability obstruction)**.** *The OP 1/4 selector-gated reachability obstruction is*

$$D^{\chi,(1)}_{\mathrm{reach},1/4} := \{K_i \cap A^{\mathrm{sel}}_{1/4} \in \mathcal{Q}^{\chi}_{1/4} \mid K_i \cap A^{\mathrm{sel}}_{1/4} \notin \mathrm{Reach}_{\mathcal{G}^{(1)}_{\chi,\mathrm{adm}}}(q_{\mathrm{clk}})\}.$$

Proposition A.54.7 (OP 1/4 first selector-gated reachability pass)**.** *If*

$$\mathsf{Gate}^{\chi,(1)}_{1/4}$$

holds and

$$D^{\chi,(1)}_{\mathrm{reach},1/4} = \varnothing,$$

then every relevant OP 1/4 anchor component has a first-layer readout-compatible clock-to-anchor route certificate.

Proof. The gate condition ensures that the required OP 1/4 selector entries are admitted. Empty selector-gated reachability obstruction says every selected anchor target is reachable in the selector-gated admissible graph. Applying Proposition A.54.4 to each target gives first-layer readout-compatible route certificates. □

A.54.3 OP 5(B) selector-gated reachability

Definition A.54.8 (OP 5(B) selector-gated target)**.** *The OP 5(B) selector-gated target is*

$$q^{\mathrm{tar}}_{5B} := Y^{(B),\mathrm{prom}}_5$$

read inside the selector-gated boundary-sensitive graph

$$\mathcal{G}^{(1),\partial}_{\chi,\mathrm{adm}}.$$

Definition A.54.9 (OP 5(B) selector-gated reachability obstruction)**.** *The OP 5(B) selector-gated reachability obstruction is*

$$D^{\chi,(1)}_{\mathrm{reach},5B} := \begin{cases} \varnothing, & Y^{(B),\mathrm{prom}}_5 \in \mathrm{Reach}_{\mathcal{G}^{(1),\partial}_{\chi,\mathrm{adm}}}(q_{\mathrm{clk}}), \\ \{Y^{(B),\mathrm{prom}}_5\}, & \textit{otherwise.} \end{cases}$$

Proposition A.54.10 (OP 5(B) first selector-gated reachability pass)**.** *If*

$$\mathsf{Gate}^{\chi,(1)}_{5B}$$

holds and

$$D^{\chi,(1)}_{\mathrm{reach},5B} = \varnothing,$$

then there exists a first-layer readout-compatible, boundary-sensitive clock-to-promoted-target route certificate for OP 5(B). The remaining obligations are TE budgets, HC/MM residual compatibility, post-promotion residual control, and OP 4 stability.

Proof. The gate condition admits the required mixed-readout selector entries for the OP 5(B) boundary route. Empty selector-gated reachability obstruction gives a path to $Y^{(B),\mathrm{prom}}_5$ inside the selector-gated boundary-sensitive graph. This path yields a first-layer readout-compatible route certificate. The statement does not discharge later budget, residual, or stability obligations. □

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A.54.4 Selector-gated diagnostic chain

Theorem A.54.11 (selector-gated reachability reduction)**.** *After the local pass, the first two OP-facing route checks are*

$$D_{\chi}^{(1)}(\rho) = \varnothing \qquad \textit{and} \qquad D_{\mathrm{reach}}^{\chi,(1)} = \varnothing.$$

If both hold for the selected OP route family, then all selected targets have first-layer readout-compatible admissible route certificates. If the first fails, the repair target is a selector entry. If the second fails, the repair target is missing selector-compatible reachability.

Proof. The first condition is exactly the selector gate. The second condition states that the selected targets are reachable after the selector gate and ordinary admissibility filters have been applied. By the selector-gated route certificate proposition, those reachable targets have first-layer readout-compatible route certificates. Failure of the first condition is a selector failure; failure of the second condition is a reachability failure after selector filtering. □

Remark A.54.12 (OP status after)**.** *OP 1/4, OP 4, and OP 5(B) remain open. The gain is that reachability is now checked in the right graph: not the raw graph and not merely the admissible graph, but the graph already filtered by first-layer mixed-readout selectors.*

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Integrated Delta Layer: Robust Local Chart

Abstract

This delta layer continues the local proof chain photon-frame HC-collar contraction line. The idealized non-ideal sample already produced a local photon-frame near-kernel closure candidate. The present layer tests whether that candidate survives a denser asymmetric chart, non-uniform curvature/readout weights, and explicitly nonzero routed HC/MM absorption. The result is still conservative: it is a local OP5 closure-candidate certificate for the declared chart only, not a global OP5 closure theorem.

A.55 Position of the probe

The chart certified a finite near-kernel photon-frame sample by checking

$$\widehat{\lambda}^{\mathrm{loc},(2)}_{\gamma,K} + \eta_{\mathrm{loc}} < 1, \qquad L^{\mathrm{loc}}_{\mathrm{unr}} = 0.$$

That test was useful but still too symmetric: the sample table used a short four-point chart and encoded the remaining perturbation through a relatively coarse margin. The test strengthens the diagnostic in three ways:

1. the photon-frame chart is denser and intentionally asymmetric;
2. each sample point carries a non-uniform curvature/readout weight;
3. the HC/MM absorption register is allowed to be nonzero, but only as routed debt-free bookkeeping.

The intended question is therefore not whether OP5 is globally closed, but whether the local HC-cone reading remains contraction-compatible after the first robustification of the sample.

A.56 Asymmetric weighted photon-frame chart

Definition A.56.1 (asymmetric photon-frame HC-absorption chart)**.** *An asymmetric photon-frame chart is a finite packet*

$$\mathcal{S}^{\mathrm{asym},(2)}_{\gamma,K} = \{(b_i, q_i, a_i, e_i, w_i, L_i^{\mathrm{rout}}, L_i^{\mathrm{unr}})\}_{i=1}^{8},$$

where:

- $b_i > 0$ *is the unshifted photon-frame radial readout size;*
- q_i *is the normalized second-shift radius ratio before explicit absorption;*
- a_i *is the normalized routed* HC/MM*-absorption contribution;*
- e_i *is the normalized finite chart error/tolerance contribution;*
- $w_i > 0$ *is a declared non-uniform readout weight;*
- L_i^{rout} *records routed leakage/absorption bookkeeping;*
- L_i^{unr} *records leakage that is not routed through an admissible* HC/MM *register.*

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The pointwise weighted contraction ratio is bounded by

$$\lambda_i^{\mathrm{loc},(2)} \le q_i + a_i + e_i.$$

The chart estimator and routed/unrouted guards are

$$\widehat{\lambda}_{\gamma,K}^{\mathrm{loc},(2)} := \max_{1\le i\le 8} \lambda_i^{\mathrm{loc},(2)}, \qquad L_{\mathrm{unr}}^{\mathrm{loc}} := \sum_{i=1}^{8} L_i^{\mathrm{unr}}, \qquad B_{\mathrm{HC/MM}}^{\mathrm{loc}} := \mathrm{unpaid}(L^{\mathrm{rout}}).$$

Here $B_{\mathrm{HC/MM}}^{\mathrm{loc}} = 0$ means that nonzero absorption has been declared and routed without leaving residual absorption debt.

Definition A.56.2 (local closure-candidate classifier)**.** *The asymmetric photon-frame branch is classified by*

$$\mathrm{Class}_{\gamma,K}^{\mathrm{loc}} = \begin{cases} \mathrm{cert_red}_{\gamma,\mathrm{asym}}, & \widehat{\lambda}_{\gamma,K}^{\mathrm{loc},(2)} + \eta_{\mathrm{loc}} < 1,\ L_{\mathrm{unr}}^{\mathrm{loc}} = 0,\ B_{\mathrm{HC/MM}}^{\mathrm{loc}} = 0, \\ \mathrm{comp_margin}_{\gamma,\mathrm{asym}}, & \widehat{\lambda}_{\gamma,K}^{\mathrm{loc},(2)} + \eta_{\mathrm{loc}} < 1,\ L_{\mathrm{unr}}^{\mathrm{loc}} = 0,\ B_{\mathrm{HC/MM}}^{\mathrm{loc}} > 0, \\ \mathrm{debt_rout}_{\gamma,\mathrm{asym}}, & \widehat{\lambda}_{\gamma,K}^{\mathrm{loc},(2)} + \eta_{\mathrm{loc}} < 1,\ L_{\mathrm{unr}}^{\mathrm{loc}} > 0 \text{ but routeable}, \\ \mathrm{res_proj}_{\gamma,\mathrm{asym}}, & \widehat{\lambda}_{\gamma,K}^{\mathrm{loc},(2)} + \eta_{\mathrm{loc}} \ge 1 \text{ or leakage is not routeable.} \end{cases}$$

The certificate $\mathrm{cert_red}_{\gamma,\mathrm{asym}}$ *is local to the declared chart and is read as* $\mathrm{close_cand}_{\gamma,\mathrm{asym}}$*, not as a global OP5 theorem.*

A.57 Explicit sample table

The following sample is deliberately not a uniform halving table. The base radii are uneven, the weights are non-uniform, and the routed HC/MM-absorption contribution is nonzero in every row.

i	b_i	w_i	q_i	a_i	e_i	$q_i+a_i+e_i$	L_i^{unr}
1	1	1	1/2	1/64	1/64	17/32	0
2	29/32	33/32	17/32	1/64	1/32	37/64	0
3	23/32	17/16	9/16	1/32	1/64	39/64	0
4	17/32	19/16	19/32	1/32	1/32	21/32	0
5	13/32	5/4	5/8	1/64	1/64	21/32	0
6	3/8	21/16	21/32	1/64	1/64	11/16	0
7	9/32	11/8	5/8	1/32	1/32	11/16	0
8	5/32	3/2	19/32	1/64	1/32	41/64	0

All rows carry routed absorption $a_i > 0$. We declare the routed register as paid,

$$L_{\mathrm{rout}}^{\mathrm{loc}} > 0, \qquad B_{\mathrm{HC/MM}}^{\mathrm{loc}} = 0, \qquad L_{\mathrm{unr}}^{\mathrm{loc}} = 0.$$

The maximum row value is

$$\widehat{\lambda}_{\gamma,K}^{\mathrm{loc},(2)} = \max_i (q_i + a_i + e_i) = \frac{11}{16}.$$

With the asymmetric chart margin

$$\eta_{\mathrm{loc}} = \frac{9}{64},$$

one obtains

$$\widehat{\lambda}_{\gamma,K}^{\mathrm{loc},(2)} + \eta_{\mathrm{loc}} = \frac{11}{16} + \frac{9}{64} = \frac{53}{64} < 1.$$

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Proposition A.57.1 (asymmetric HC-absorption robustness certificate)**.** *For the explicit table above, the asymmetric photon-frame branch receives the local classifier*

$$\mathrm{Class}^{\mathrm{loc}}_{\gamma,K} = \mathrm{cert_red}_{\gamma,\mathrm{asym}} = \mathrm{close_cand}_{\gamma,\mathrm{asym}}.$$

Equivalently, the local closure-candidate status survives the first denser asymmetric weighted chart with nonzero routed HC/MM *absorption.*

Proof. The estimator is the maximum of the row bounds in the explicit table. The largest row value is $11/16$, attained in rows 6 and 7. The declared margin is $9/64$, hence

$$\widehat{\lambda}^{\mathrm{loc},(2)}_{\gamma,K} + \eta_{\mathrm{loc}} = 53/64 < 1.$$

The table has zero unrouted leakage in every row, so $L^{\mathrm{loc}}_{\mathrm{unr}} = 0$. The absorption register is nonzero, because $a_i > 0$ in every row, but it is explicitly routed and declared debt-free, so $B^{\mathrm{loc}}_{\mathrm{HC/MM}} = 0$. The first branch of the classifier therefore applies. This proves the stated local certificate. □

A.58 Field-equation bookkeeping update

The branch adds the asymmetric photon-frame contraction register

$$\mathcal{R}^{(2)}_{\gamma,\mathrm{asym}} := \left(\widehat{\lambda}^{\mathrm{loc},(2)}_{\gamma,K}, \eta_{\mathrm{loc}}, L^{\mathrm{loc}}_{\mathrm{rout}}, L^{\mathrm{loc}}_{\mathrm{unr}}, B^{\mathrm{loc}}_{\mathrm{HC/MM}}, \{w_i\}_{i=1}^{8}\right)$$

to the OP5-facing source-return bookkeeping. The readout-side implication is

$$\mathcal{R}^{(2)}_{\gamma,\mathrm{asym}} \vdash \left[\widehat{\lambda} + \eta < 1,\ L_{\mathrm{unr}} = 0,\ B_{\mathrm{HC/MM}} = 0\right] \Longrightarrow \mathrm{close_cand}_{\gamma,\mathrm{asym}}.$$

Thus the HC-cone interpretation is strengthened from an ideal or near-ideal photon-frame sample to a first robust asymmetric weighted sample. The correct structural reading is that OP5 remains more plausibly an HC-cone/pass-count debt problem than an independent terminal endpoint problem, provided the remaining chart families keep satisfying the same contraction and routing guards.

A.59 Conservativity guards

The following guards are part of the layer.

1. No global OP5 closure is claimed.
2. The certificate is local to the declared finite photon-frame chart.
3. Nonzero HC/MM absorption is not deletion of residuals; it is paid routed bookkeeping.
4. The chart does not assert a literal Euclidean photon trajectory.
5. The result does not close OP1, OP4, QCD/confinement, CKM/PMNS, or the quark-mass spectrum.
6. Promotion beyond $\mathrm{close_cand}_{\gamma,\mathrm{asym}}$ still requires either a covering argument over admissible photon-frame charts or a finite atlas theorem with stable contraction margins.

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A.60 Next recommended step

The highest-leverage next step is an atlas-style finite cover test: define a small family of admissible asymmetric photon-frame charts

$$\mathfrak{A}_{\gamma}^{(2)} = \{\mathcal{S}_{\gamma,K,j}^{\mathrm{asym},(2)}\}_{j=1}^{M}$$

and require a uniform bound

$$\sup_{j\le M}\left(\widehat{\lambda}_{\gamma,K,j}^{(2)} + \eta_j\right) < 1, \qquad \sum_{j\le M} L_{\mathrm{unr},j} = 0, \qquad \sum_{j\le M} B_{\mathrm{HC/MM},j} = 0.$$

If such a finite-atlas certificate is obtained, the OP5=HC-cone hypothesis can be promoted from a single-chart local closure candidate to a finite-atlas closure candidate.

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Integrated Delta Layer: Finite Atlas

Abstract

This delta layer upgrades the local proof chain photon-frame HC-collar contraction probes from single-chart diagnostics to a finite-atlas closure-candidate test. The layer also states, explicitly and conservatively, what would still have to be proved in order to close OP5 under the hypothesis that OP5 is the HC-cone/reflector surface of the source-return route rather than an independent endpoint obstruction. The result is an atlas-level closure candidate, not a final OP5 theorem.

A.61 Position after

The layer showed that the local photon-frame candidate survives a denser asymmetric chart with nonzero but debt-free routed HC/MM absorption:

$$\widehat{\lambda}^{\mathrm{loc},(2)}_{\gamma,K} = \frac{11}{16}, \qquad \eta_{\mathrm{loc}} = \frac{9}{64}, \qquad \widehat{\lambda}^{\mathrm{loc},(2)}_{\gamma,K} + \eta_{\mathrm{loc}} = \frac{53}{64} < 1.$$

This is strong enough to justify an atlas test but not strong enough to close OP5. A single chart only says that one declared readout neighbourhood is compatible with the HC-cone contraction picture. OP5 closure requires that the admissible OP5 branch is exhausted by such neighbourhoods, and that the transition logic between them does not reintroduce an endpoint obstruction.

A.62 Finite photon-frame atlas

Definition A.62.1 (finite photon-frame HC-cone atlas)**.** *An finite photon-frame atlas for the OP5-facing branch is a finite family*

$$\mathfrak{A}^{(2)}_{\gamma} = \{\mathcal{S}^{(2)}_{\gamma,K,j}\}_{j=1}^{M}$$

of admissible photon-frame HC*-collar charts. Each chart carries the data*

$$\mathcal{S}^{(2)}_{\gamma,K,j} \rightsquigarrow \left(\widehat{\lambda}_j, \eta_j, L_{\mathrm{unr},j}, B_{\mathrm{HC/MM},j}, \tau_j\right),$$

where $\widehat{\lambda}_j$ is the finite contraction estimator, η_j is the chart margin, $L_{\mathrm{unr},j}$ is the unrouted leakage register, $B_{\mathrm{HC/MM},j}$ is the unpaid HC/MM*-absorption debt, and τ_j records the transition-map debt between overlapping charts.*

Definition A.62.2 (Atlas classifier)**.** *The atlas receives the classifier by the following finite rule.*

$$\mathrm{Class}(\mathfrak{A}^{(2)}_{\gamma}) = \mathrm{atlas_close_cand}$$

exactly if

$$\sup_{j\le M}(\widehat{\lambda}_j + \eta_j) < 1, \qquad \sum_{j\le M} L_{\mathrm{unr},j} = 0, \qquad \sum_{j\le M} B_{\mathrm{HC/MM},j} = 0, \qquad \sum_{j\le M} \tau_j = 0.$$

It receives the compatible-open classifier cert_comp *if the uniform contraction inequality holds but some leakage, absorption, or transition register is nonzero while still routeable. It receives the residual-projection classifier* res_proj *if the uniform contraction bound fails or if any register is not routeable. Thus an atlas is not merely a list of successful sample tables. It also has to verify that no obstruction appears on overlaps or transitions.*

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A.63 Explicit finite-atlas sample

The following five-chart atlas is a compact first test. It collects the already motivated ideal, near-kernel, non-ideal, asymmetric, and tilted/edge regimes into one finite certificate table.

Chart	$\widehat{\lambda}_j$	η_j	$\widehat{\lambda}_j + \eta_j$	$L_{\mathrm{unr},j}$	$B_{\mathrm{HC/MM},j} + \tau_j$
central ideal collar	1/2	1/8	5/8	0	0
near-kernel symmetric	21/32	1/8	25/32	0	0
non-ideal sample	21/32	1/8	25/32	0	0
asymmetric HC-absorption	11/16	9/64	53/64	0	0
tilted edge collar	45/64	7/64	13/16	0	0

The atlas maximum is therefore

$$\Lambda_{\mathfrak{A}}^{\mathrm{loc}} := \sup_{j \le 5} (\widehat{\lambda}_j + \eta_j) = \frac{53}{64} < 1.$$

All explicit leakage, absorption-debt, and transition-debt registers are declared zero at the atlas level:

$$\sum_{j \le 5} L_{\mathrm{unr},j} = 0, \qquad \sum_{j \le 5} B_{\mathrm{HC/MM},j} = 0, \qquad \sum_{j \le 5} \tau_j = 0.$$

Proposition A.63.1 (finite-atlas closure-candidate certificate)**.** *The explicit five-chart photon-frame atlas above receives the classifier*

$$\mathrm{Class}(\mathfrak{A}_{\gamma}^{(2)}) = \mathrm{atlas_close_cand}.$$

It promotes the single-chart result to a finite-atlas OP5 closure candidate.

Proof. The largest value in the third numerical column is 53/64, which is strictly smaller than one. The declared atlas registers satisfy

$$\sum_{j \le 5} L_{\mathrm{unr},j} = 0, \qquad \sum_{j \le 5} B_{\mathrm{HC/MM},j} = 0, \qquad \sum_{j \le 5} \tau_j = 0.$$

The first branch of the atlas classifier applies. Hence the finite atlas receives the atlas closure-candidate status. □

A.64 What would be needed to close OP5

The atlas is still a candidate because it assumes, rather than proves, that the five declared chart types cover all admissible OP5-facing photon-frame readouts. Under the working hypothesis

$$\mathrm{OP5} = \text{HC-cone/reflector surface of the source-return route},$$

OP5 would be closed by proving the following theorem.

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Theorem A.64.1 (Conditional OP5 closure theorem to be proved)**.** *Assume that the OP5-facing source-return route factorizes through the* HC*-cone reflector,*

$$R_{\mathrm{OP}}^{\text{HC-cone}} = T_{\Gamma}^{\vee} \circ J_{\mathrm{HC}}^{\mathrm{cone}} \circ T_{\Gamma},$$

and that its cone-factorization obstruction vanishes globally,

$$D^{\chi,(1)}_{\text{OP5=HC-cone}} = 0.$$

Then OP5 closes if there exists a finite photon-frame atlas $\mathfrak{A}_{\gamma}^{(2)}$ *satisfying all of the following conditions.*

1. ***Coverage.*** *Every admissible OP5-facing readout state is contained in at least one chart of* $\mathfrak{A}_{\gamma}^{(2)}$*.*
2. ***Uniform contraction.*** *There is a constant* $\Lambda < 1$ *such that*

$$\widehat{\lambda}_j + \eta_j \leq \Lambda < 1 \qquad \text{for every chart } j.$$

3. ***Transition stability.*** *On every nonempty chart overlap, the transition map preserves the contraction class and creates no unpaid transition debt:*

$$\tau_{jk} = 0 \qquad \text{for all admissible overlaps } j \cap k.$$

4. ***No unrouted leakage.*** *Every leakage term is either zero or routed through an admissible* HC/MM *register, and the total unrouted leakage vanishes:*

$$L^{\mathfrak{A}}_{\mathrm{unr}} = 0.$$

5. ***No unpaid absorption debt.*** *Routed* HC/MM *absorption may be nonzero, but all such absorption must be paid inside the route bookkeeping:*

$$B^{\mathfrak{A}}_{\mathrm{HC/MM}} = 0.$$

6. ***Periapsis/no-through-collapse guard.*** *The active* HC*-collar forces every admissible source-return branch to have a collar-side periapsis rather than a through-collapse endpoint.*
7. ***Pass-count consistency.*** *The OP5 pass count is exactly the source-return pass count seen through the second Hilbert-shift/*HC*-cone frame, not an additional independent terminal transport pass.*
8. ***Statement identification.*** *The formal statement of OP5 is shown to be equivalent to the absence of the above cone-factorization, leakage, debt, and transition obstructions.*

If all eight items hold, then OP5 is not an independent residual endpoint problem. It is discharged as the HC*-cone source-return contraction theorem.*

Remark A.64.2 (Why does not yet close OP5)**.** *The table verifies uniform contraction for a finite declared atlas and records zero debt in the declared registers. It does not yet prove the coverage item, the global vanishing of* $D^{\chi,(1)}_{\text{OP5=HC-cone}}$*, or the equivalence between the formal OP5 statement and the* HC*-cone source-return theorem. Those are the remaining proof burdens.*

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A.65 Practical proof route after

The shortest conservative route from to a genuine OP5 closure proof is:

1. write the admissible OP5 readout space $\mathcal{X}_{\mathrm{OP5}}^{\mathrm{adm}}$;
2. prove that the five chart types form a finite cover of $\mathcal{X}_{\mathrm{OP5}}^{\mathrm{adm}}$;
3. prove that all chart overlaps have zero transition debt;
4. prove that $D^{\chi,(1)}_{\mathrm{OP5=HC\text{-}cone}} = 0$ under the HC-cone factorization hypothesis;
5. prove that no admissible branch crosses the collapse core under an active HC-collar, i.e. every branch has a periapsis/readout-turning point;
6. identify OP5's original obstruction statement with the atlas obstruction package.

If these six tasks are completed, the numerical contraction certificates become supporting evidence for a structural theorem rather than isolated probes.

A.66 Conservativity guards

The layer uses the following guards.

1. The result is a finite-atlas closure candidate, not a final OP5 closure.
2. The five-chart cover is declared as a test atlas; its coverage of all admissible OP5 states remains to be proved.
3. The HC-cone interpretation remains conditional on vanishing cone-factorization obstruction.
4. Nonzero HC/MM absorption is permitted only as explicit routed bookkeeping, never as residual deletion.
5. The result does not imply OP1, OP4, QCD/confinement, CKM/PMNS, or quark-mass spectrum closure.

A.67 Next recommended step

The highest-leverage next step is to define the admissible OP5 readout space

$$\mathcal{X}_{\mathrm{OP5}}^{\mathrm{adm}}$$

and prove a finite-cover lemma

$$\mathcal{X}_{\mathrm{OP5}}^{\mathrm{adm}} \subseteq \bigcup_{j=1}^{5} \mathcal{S}^{(2)}_{\gamma,K,j}.$$

That lemma is the first real bridge from atlas diagnostics to OP5 closure.

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Integrated Delta Layer: Tunnel Menu

Abstract

This delta layer inserts the nonstandard-halo step between the robust photon-frame contraction probe and the finite-atlas closure-candidate logic. Its purpose is to reduce the apparent combinatorial overload of OP5: instead of testing all admissible source-return routes as free trajectories, the OP5-facing branch is first trapped in a double-epsilon halo around the collapse kernel. Exit from this halo is then possible only through explicitly gated tunnel points. The result is conservative: OP5 is not closed here. The proof burden is reduced from an all-routes problem to a finite or finitely classifiable tunnel-point problem, provided the stated halo-capture, halo-invariance, and tunnel-gating hypotheses are later proved.

A.68 Version position

The immediate pre-line contains two relevant ingredients. First, produced a robust asymmetric photon-frame local certificate with

$$\widehat{\lambda}^{\mathrm{loc},(2)}_{\gamma,K} = \frac{11}{16}, \qquad \eta_{\mathrm{loc}} = \frac{9}{64}, \qquad \widehat{\lambda}^{\mathrm{loc},(2)}_{\gamma,K} + \eta_{\mathrm{loc}} = \frac{53}{64} < 1.$$

Second, an finite-atlas draft upgrades this local logic to a declared finite atlas, but such an atlas still needs an independent explanation of why the admissible OP5 search does not branch into uncontrolled route families. The present layer supplies that missing structural filter: a double-epsilon halo channel with tunnel-point reduction.

A.69 Double-epsilon halo algebra

Definition A.69.1 (Two-scale halo readout algebra)**.** *Let $\mathcal{A}$ be the OP5-facing readout algebra and let K denote the collapse kernel. A two-scale halo readout is a triple*

$$(x; \rho_r(x), \rho_h(x)) \in \mathcal{A} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0},$$

where $\rho_r(x)$ is the radial kernel-distance readout and $\rho_h(x)$ is the HC*-cone residual readout. The two infinitesimal windows are denoted*

$$\varepsilon_r > 0, \qquad \varepsilon_h > 0.$$

The corresponding double-epsilon halo is

$$\mathsf{Halo}_{\varepsilon_r,\varepsilon_h}(K) := \{x \in \mathcal{X}^{\mathrm{adm}}_{\mathrm{OP5}} : \rho_r(x) \leq \varepsilon_r \text{ and } \rho_h(x) \leq \varepsilon_h\}.$$

The first inequality is radial closeness to the collapse kernel. The second inequality is closeness to the HC*-cone/reflector normal form.*

Remark A.69.2 (Why two epsilon channels are needed)**.** *A one-parameter halo around K only records radial nearness. OP5 is not merely a radial-distance problem; the relevant question is whether the source-return route is already captured by the active* HC*-cone collar. Therefore the halo must carry two channels: one for kernel nearness and one for* HC*-normality. In this sense ε_r controls "near the kernel", while ε_h controls "near the* HC*-cone route".*

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Definition A.69.3 (Halo boundary and exit side)**.** *The halo boundary is the union of the radial and* HC*-residual boundary faces:*

$$\partial\mathsf{Halo}_{\varepsilon_r,\varepsilon_h}(K) = \{\rho_r = \varepsilon_r, \rho_h \le \varepsilon_h\} \cup \{\rho_r \le \varepsilon_r, \rho_h = \varepsilon_h\}.$$

An exit candidate is a point

$$p \in \partial\mathsf{Halo}_{\varepsilon_r,\varepsilon_h}(K)$$

for which the source-return update points outward in at least one of the two halo channels before applying the tunnel selector.

A.70 Tunnel points

Definition A.70.1 (Tunnel candidate)**.** *A point $p \in \partial\mathsf{Halo}_{\varepsilon_r,\varepsilon_h}(K)$ is a tunnel candidate if it satisfies the four tests*

$$\begin{aligned} &\textit{(T1)} \quad p \in \partial\mathsf{Halo}_{\varepsilon_r,\varepsilon_h}(K), \\ &\textit{(T2)} \quad D_{\text{HC-cone}}(p) \le \varepsilon_h, \\ &\textit{(T3)} \quad \chi_{\text{tun}}(p) = 1, \\ &\textit{(T4)} \quad \operatorname{rank} J_{\mathsf{Exit}}(p) < \operatorname{rank}_{\max}. \end{aligned}$$

*Here χ_{tun} is the tunnel selector and J_{Exit} is the exit Jacobian for the paired radial/*HC*-residual channel. The resulting candidate set is*

$$\mathcal{T}_{\text{tun}}^{\text{cand}}(\varepsilon_r, \varepsilon_h) := \{p \in \partial\mathsf{Halo}_{\varepsilon_r,\varepsilon_h}(K) : \textit{(T1)–(T4) hold}\}.$$

Definition A.70.2 (Resolved tunnel point)**.** *A tunnel candidate $p \in \mathcal{T}_{\text{tun}}^{\text{cand}}$ is resolved if exactly one of the following outcomes is certified:*

$$\mathsf{Gate}(p) \in \{\text{blocked}, \text{HC/MM absorbed}, \text{atlas routed}, \text{residual projected}\}.$$

The first three outcomes are controlled. The fourth outcome records a remaining obstruction and cannot be used to close OP5.

Remark A.70.3 (Tunnel points as the replacement for route explosion)**.** *The intended reduction is not that particles literally escape through geometric holes. The tunnel language is a readout-side compression: a halo-trapped branch can only leave the halo where the radial boundary, the* HC*-cone residual boundary, and the selector/rank conditions simultaneously permit an exit. Thus an apparent continuum of route choices is replaced by a finite or finitely classifiable family of boundary obstructions.*

A.71 Capture, invariance, and tunnel reduction

Definition A.71.1 (Halo capture)**.** *The OP5-facing route satisfies halo capture if every admissible readout enters the double halo after finitely many* HC*-normalized source-return updates:*

$$\forall x \in \mathcal{X}_{\text{OP5}}^{\text{adm}}\ \exists n < \infty : R_{\text{HC}}^n(x) \in \mathsf{Halo}_{\varepsilon_r,\varepsilon_h}(K).$$

Definition A.71.2 (Halo invariance off tunnels)**.** *The double halo is invariant off tunnel candidates if*

$$x \in \mathsf{Halo}_{\varepsilon_r,\varepsilon_h}(K) \setminus \mathcal{T}_{\text{tun}}^{\text{cand}} \quad \Longrightarrow \quad R_{\text{HC}}(x) \in \mathsf{Halo}_{\varepsilon_r,\varepsilon_h}(K).$$

Equivalently, the active HC*-collar produces a periapsis/turning-point readout before a through-collapse exit can occur, unless the state lies on a declared tunnel candidate.*

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Proposition A.71.3 (tunnel-point reduction)**.** *Assume halo capture and halo invariance off tunnels. Then every admissible OP5-facing branch is either permanently trapped in the double-epsilon halo or reaches a point of* $\mathcal{T}_{\mathrm{tun}}^{\mathrm{cand}}(\varepsilon_r, \varepsilon_h)$*. Consequently, the OP5 route problem reduces to resolving the tunnel candidate set.*

Proof. Let $x \in \mathcal{X}_{\mathrm{OP5}}^{\mathrm{adm}}$. By halo capture, there is a finite n such that $x_n := R_{\mathrm{HC}}^n(x) \in \mathsf{Halo}_{\varepsilon_r,\varepsilon_h}(K)$. If no later iterate lies in $\mathcal{T}_{\mathrm{tun}}^{\mathrm{cand}}$, halo invariance implies by induction that every later iterate remains in $\mathsf{Halo}_{\varepsilon_r,\varepsilon_h}(K)$. If a later iterate fails to remain in the halo, the contrapositive of halo invariance says that the failure can occur only at a tunnel candidate. Hence every branch is trapped or reaches the tunnel candidate set. This proves the reduction. □

A.72 Finite tunnel menu

The first conservative tunnel inventory is not a numerical discovery claim. It is a finite classification scheme for the exits that must be tested.

Definition A.72.1 (First OP5 tunnel menu)**.** *The first tunnel menu is*

$$\mathfrak{T}_{\mathrm{OP5}}^{\mathrm{loc}} = \{T_r, T_h, T_{rh}, T_\chi, T_{\mathrm{rk}}, T_{\mathrm{ov}}\},$$

where:

Class	*Meaning*
T_r	*radial-face exit:* $\rho_r = \varepsilon_r$, $\rho_h < \varepsilon_h$*;*
T_h	HC*-residual-face exit:* $\rho_r < \varepsilon_r$, $\rho_h = \varepsilon_h$*;*
T_{rh}	*corner exit:* $\rho_r = \varepsilon_r$, $\rho_h = \varepsilon_h$*;*
T_χ	*selector-switch exit caused by a change in* χ_{tun}*;*
T_{rk}	*rank-drop exit caused by* $\operatorname{rank} J_{\mathsf{Exit}} < \operatorname{rank}_{\max}$*;*
T_{ov}	*atlas-overlap exit caused by transition debt between two photon-frame charts.*

A tunnel candidate must be assigned to at least one class in $\mathfrak{T}_{\mathrm{OP5}}^{\mathrm{loc}}$*.*

Proposition A.72.2 (Finite tunnel classification target)**.** *If the candidate tunnel set admits a finite decomposition*

$$\mathcal{T}_{\mathrm{tun}}^{\mathrm{cand}}(\varepsilon_r, \varepsilon_h) = \bigsqcup_{\alpha \in \mathfrak{T}_{\mathrm{OP5}}^{\mathrm{loc}}} \mathcal{T}_\alpha$$

and every nonempty $\mathcal{T}_\alpha$ *has one of the controlled outcomes*

$$\mathsf{Gate}(\mathcal{T}_\alpha) \in \{\text{blocked}, \text{HC/MM absorbed}, \text{atlas routed}\},$$

then OP5 has no uncontrolled tunnel escape inside the declared double-epsilon halo.

Proof. By the preceding tunnel-point reduction, every non-trapped branch can only exit through $\mathcal{T}_{\mathrm{tun}}^{\mathrm{cand}}$. The finite decomposition assigns each such candidate to one of the declared tunnel classes. If every class is blocked, absorbed without unpaid debt, or routed into an atlas chart, then no class contributes an uncontrolled exit. Therefore the double halo has no uncontrolled tunnel escape. □

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A.73 Connection to the finite atlas

Once a tunnel point is atlas-routed, it must land in a contraction chart rather than in a free branch. The bridge to the finite-atlas line is therefore:

$$\mathcal{T}_{\mathrm{tun}}^{\mathrm{atlas}} \subseteq \bigcup_{j=1}^{M} \mathcal{S}_{\gamma,K,j}^{(2)}, \qquad \sup_{j\le M}(\widehat{\lambda}_j + \eta_j) < 1,$$

with the same zero-debt guards

$$L_{\mathrm{unr}}^{\mathfrak{A}} = 0, \qquad B_{\mathrm{HC/MM}}^{\mathfrak{A}} = 0, \qquad \tau^{\mathfrak{A}} = 0.$$

For typographic clarity, the intended symbol in the preceding display is the chart margin η_j. Thus the atlas-routed tunnel branch inherits the atlas criterion exactly when all tunnel exits are covered by charts with a uniform strict contraction margin.

A.74 Conditional OP5 closure statement after the halo step

Theorem A.74.1 (Conditional OP5 closure via double-epsilon halo)**.** *Assume the working identification*

$$\mathrm{OP5} = \mathrm{HC}\text{-}\textit{cone source-return reflector problem}$$

and assume that the cone-factorization obstruction vanishes,

$$D_{\mathrm{OP5=HC\text{-}cone}}^{\chi,(1)} = 0.$$

Then OP5 closes if the following six conditions are proved.

1. ***Halo capture:*** *every admissible OP5 readout enters* $\mathsf{Halo}_{\varepsilon_r,\varepsilon_h}(K)$ *after finitely many* HC*-normalized source-return updates.*
2. ***Halo invariance off tunnels:*** *the double halo is invariant outside* $\mathcal{T}_{\mathrm{tun}}^{\mathrm{cand}}$*.*
3. ***Finite tunnel classification:*** $\mathcal{T}_{\mathrm{tun}}^{\mathrm{cand}}$ *is empty, finite, or decomposes into finitely many tunnel classes.*
4. ***Tunnel gating:*** *each nonempty tunnel class is blocked, debt-free* HC/MM*-absorbed, or routed into a contraction atlas.*
5. ***No hidden leakage or debt:*** *all tunnel and atlas registers satisfy*

$$L_{\mathrm{unr}} = 0, \qquad B_{\mathrm{HC/MM}} = 0, \qquad \tau = 0.$$

6. ***OP5 statement identification:*** *the original OP5 obstruction is equivalent to the cone-factorization/tunnel/atlas obstruction package above.*

Under these hypotheses the OP5 endpoint problem is exhausted by controlled halo trapping and controlled tunnel resolution. Hence OP5 is discharged as a double-epsilon HC*-cone halo theorem rather than as an independent terminal transport obstruction.*

Remark A.74.2 (What proves and what remains)**.** *The layer proves the formal reduction: capture plus invariance off tunnels reduces OP5 to tunnel resolution. It does not yet prove capture, invariance, finite tunnel classification, or global vanishing of* $D_{\mathrm{OP5=HC\text{-}cone}}^{\chi,(1)}$*. Those become the next concrete proof obligations.*

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A.75 Immediate next proof tasks

The next executable line should be:

1. instantiate ρ_r and ρ_h from the current photon-frame HC-collar readout;
2. compute the first symbolic boundary faces $\rho_r = \varepsilon_r$ and $\rho_h = \varepsilon_h$;
3. define the exit Jacobian J_{Exit} in the two-channel coordinates;
4. test whether the first tunnel menu $\mathfrak{T}^{\mathrm{loc}}_{\mathrm{OP5}}$ is complete for the declared halo grammar;
5. route each nonempty tunnel class to blocked, HC/MM-absorbed, atlas-routed, or residual-projected status.

If these tasks succeed, the finite atlas is no longer trying to cover arbitrary OP5 routes. It only has to cover the gated tunnel exits of the double-epsilon halo.

A.76 Conservativity guards

1. does not claim global OP5 closure.
2. The halo is a readout-side structural filter, not a claim that physical particles literally sit inside a geometric shell.
3. Tunnel points are selector/rank/obstruction points, not literal wormholes.
4. HC/MM-absorption is allowed only as explicit debt-free bookkeeping, never as residual deletion.
5. The finite tunnel menu is a first classification target and must still be tested for completeness.
6. OP1, OP4, QCD/confinement, CKM/PMNS, and the quark-mass spectrum remain untouched by this layer.

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Integrated Delta Layer: Halo Channel and Response Alphabet

Abstract

This delta layer executes the next proof obligation left by the local pass. The double-epsilon halo symbols are made typographically stable, the two halo channels are instantiated in the photon-frame near-kernel collar, and the exit Jacobian is defined as the differential of the paired radial/HC-residual channel. The result is a first conservative tunnel-menu execution: within the declared photon-frame halo grammar every boundary exit is assigned to one of the six tunnel classes. The layer does not close OP5 globally. It converts the next obstruction into explicit tests: collar admissibility, channel regularity, selector gating, rank-drop isolation, atlas-overlap compatibility, and zero hidden leakage/debt.

A.77 Macro hygiene and version position

The layer introduced the double-epsilon halo but the visual notation for the halo and exit operators must be kept stable in downstream TeX. In this layer the intended printed symbols are

$$\mathsf{Halo}, \qquad \mathsf{Exit}, \qquad J_{\mathsf{Exit}}, \qquad \mathcal{E}_{\mathsf{Exit}}.$$

For source-level robustness the lower-case spellings are aliased to the same mathematical symbols, but the body of this layer uses the capitalized operators. Thus the rendered document should not display raw control words for the halo or exit operators.

The reduction says that the OP5 route problem is reduced to tunnel resolution once halo capture and halo invariance off tunnels are proved. The present layer executes the first concrete part of that program: it instantiates the two halo channels in the photon-frame HC-collar chart and defines the exit Jacobian which detects rank-drop tunnel candidates.

A.78 Photon-frame collar chart

Definition A.78.1 (Photon-frame HC-collar chart)**.** *Let $U_{\gamma,K} \subset \mathcal{X}^{\mathrm{adm}}_{\mathrm{OP5}}$ be an admissible near-kernel photon-frame chart. The chart carries:*

$$d^G_K(x), \qquad r^{(2)}_\gamma(x), \qquad h^{(2)}_\gamma(x), \qquad q^{(2)}_\gamma(x) := (h^{(2)}_\gamma(x))^2 - (r^{(2)}_\gamma(x))^2,$$

where d^G_K is the metric distance to the collapse kernel, $r^{(2)}_\gamma$ and $h^{(2)}_\gamma$ are the second-shift radial and HC-height coordinates, and $q^{(2)}_\gamma$ is the photon-frame cone residual. The active first-shift HC collar radius is denoted $r^{(1)}_{\mathrm{HC}}(x) > 0$. The admissible collar guard is

$$d^G_K(x) \geq r^{(1)}_{\mathrm{HC}}(x) > 0.$$

This guard prevents interpreting a projected line through K as a literal through-collapse trajectory.

Definition A.78.2 (Second-shift tilt operator)**.** *The second-shift cone-to-radial tilt is*

$$\Theta^{(2)}_{\gamma,K} := \Pi^{\mathrm{HC},\gamma}_{\mathrm{cone}\to\mathrm{rad}} \circ U^{(2)}_\pi - U^{(2)}_\pi \circ \Pi^{\mathrm{HC},\gamma}_{\mathrm{cone}\to\mathrm{rad}}.$$

It records the failure of radial projection and the second Hilbert shift to commute. A projected path may visually pass through K after the second shift; in this layer that appearance is read first as tilt debt rather than as immediate OP5 failure.

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A.79 Concrete halo-channel instantiation

Definition A.79.1 (Radial halo channel)**.** *The radial halo channel is*

$$\rho_r^{\gamma}(x) := d_K^G\big(\Pi^{\mathrm{HC},\gamma}_{\mathrm{cone}\to\mathrm{rad}}(U_\pi^{(2)}x), K\big).$$

The radial halo face is

$$\rho_r^{\gamma}(x) = \varepsilon_r.$$

The photon-frame near-kernel collar is admissible only on the guarded subregion

$$r_{\mathrm{HC}}^{(1)}(x) \le \rho_r^{\gamma}(x) \le \varepsilon_r.$$

Thus $\rho_r^{\gamma} \le \varepsilon_r$ means that the projected second-shift readout is inside the declared near-kernel radial halo, while $\rho_r^{\gamma} \ge r_{\mathrm{HC}}^{(1)}$ records the HC periapsis/collar exclusion of literal through-collapse.

Definition A.79.2 (HC-residual halo channel)**.** *Fix a harmless normalization floor $\delta_0 > 0$ and a tilt weight $\mu_\Theta \ge 0$. The HC-residual halo channel is*

$$\rho_h^{\gamma}(x) := \frac{|q_\gamma^{(2)}(x)|}{(h_\gamma^{(2)}(x))^2 + (r_\gamma^{(2)}(x))^2 + \delta_0} + \mu_\Theta \,\|\Theta_{\gamma,K}^{(2)}(x)\|_G.$$

The HC-residual halo face is

$$\rho_h^{\gamma}(x) = \varepsilon_h.$$

The first term measures cone defect. The second term measures the second-shift projection tilt. Therefore $\rho_h^{\gamma} \le \varepsilon_h$ means that the readout is simultaneously close to the ideal photon-frame HC cone and free of uncontrolled projection tilt up to the declared tolerance.

Definition A.79.3 (Executed double-epsilon halo)**.** *The executed photon-frame halo is*

$$\mathsf{Halo}^{\gamma,K}_{\varepsilon_r,\varepsilon_h} := \{x \in U_{\gamma,K} \cap \mathcal{X}^{\mathrm{adm}}_{\mathrm{OP5}} : r_{\mathrm{HC}}^{(1)}(x) \le \rho_r^{\gamma}(x) \le \varepsilon_r,\ \rho_h^{\gamma}(x) \le \varepsilon_h\}.$$

The extra lower bound $r_{\mathrm{HC}}^{(1)} \le \rho_r^{\gamma}$ is not an additional escape channel. It is the previously declared HC-collar guard: the admissible branch approaches a periapsis collar rather than crossing K literally.

A.80 Exit map and exit Jacobian

Definition A.80.1 (Two-channel exit map)**.** *The exit map is*

$$\mathcal{E}^{\gamma,K}_{\mathsf{Exit}} : U_{\gamma,K} \longrightarrow \mathbb{R}^2, \qquad \mathcal{E}^{\gamma,K}_{\mathsf{Exit}}(x) := \big(\rho_r^{\gamma}(x) - \varepsilon_r,\ \rho_h^{\gamma}(x) - \varepsilon_h\big).$$

The halo boundary is $(\mathcal{E}^{\gamma,K}_{\mathsf{Exit}})^{-1}(0\times(-\infty,0]) \cup (\mathcal{E}^{\gamma,K}_{\mathsf{Exit}})^{-1}((-\infty,0]\times 0)$ together with the collar face $\rho_r^{\gamma} = r_{\mathrm{HC}}^{(1)}$. The collar face is not treated as an outward OP5 exit; it is the no-through-collapse reflecting side of the HC collar.

Definition A.80.2 (Exit Jacobian)**.** *At a boundary point p, the exit Jacobian is*

$$J^{\gamma,K}_{\mathsf{Exit}}(p) := D\mathcal{E}^{\gamma,K}_{\mathsf{Exit}}(p) = \begin{pmatrix} D\rho_r^{\gamma}(p) \\ D\rho_h^{\gamma}(p) \end{pmatrix}.$$

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The maximal two-channel rank is $\mathrm{rank}_{\max} = 2$. *The rank-drop tunnel condition is therefore*

$$\operatorname{rank} J^{\gamma,K}_{\mathsf{Exit}}(p) < 2,$$

which is equivalently

$$D\rho^{\gamma}_{r}(p) \wedge D\rho^{\gamma}_{h}(p) = 0$$

whenever the two channel differentials are defined.

Definition A.80.3 (Outward exit condition)**.** *A boundary point* $p \in \partial\mathsf{Halo}^{\gamma,K}_{\varepsilon_r,\varepsilon_h}$ *is outward in the radial channel if there exists an admissible update vector* v *with*

$$D\rho^{\gamma}_{r}(p)[v] > 0 \quad \textit{on} \quad \rho^{\gamma}_{r}(p) = \varepsilon_r.$$

It is outward in the HC-residual channel if

$$D\rho^{\gamma}_{h}(p)[v] > 0 \quad \textit{on} \quad \rho^{\gamma}_{h}(p) = \varepsilon_h.$$

A genuine tunnel candidate must be outward in at least one of these two channels and must pass the selector/rank tests.

A.81 First tunnel-menu execution

Proposition A.81.1 (Boundary exhaustion by the tunnel menu)**.** *Within the photon-frame halo grammar, every outward boundary exit candidate belongs to at least one of the six tunnel classes*

$$\mathfrak{T}^{\mathrm{loc}}_{\mathrm{OP5}} = \{T_r, T_h, T_{rh}, T_\chi, T_{\mathrm{rk}}, T_{\mathrm{ov}}\}.$$

Proof. Let $p \in \partial\mathsf{Halo}^{\gamma,K}_{\varepsilon_r,\varepsilon_h}$ be an outward exit candidate. If p lies on the radial face with $\rho^{\gamma}_{r}(p) = \varepsilon_r$ and $\rho^{\gamma}_{h}(p) < \varepsilon_h$, then $p \in T_r$. If p lies on the HC-residual face with $\rho^{\gamma}_{r}(p) < \varepsilon_r$ and $\rho^{\gamma}_{h}(p) = \varepsilon_h$, then $p \in T_h$. If both equalities hold, then $p \in T_{rh}$. If the tunnel selector changes value or must be evaluated at a selector discontinuity, then $p \in T_\chi$. If $\operatorname{rank} J^{\gamma,K}_{\mathsf{Exit}}(p) < 2$, then $p \in T_{\mathrm{rk}}$. Finally, if the exit occurs while changing from one photon-frame chart to another, then the point lies in an atlas-overlap region and belongs to T_{ov}. These cases exhaust the declared boundary, selector, rank, and overlap mechanisms of the grammar. □

Definition A.81.2 (response alphabet)**.** *For each tunnel class* T*, the first execution response is drawn from*

$$\mathrm{Resp}_{\mathrm{loc}}(T) \in \{\text{blocked}, \text{absorbed}_{\mathrm{HC/MM}}, \text{atlas routed}, \text{rank isolated}, \text{residual open}\}.$$

Only the first three responses are closing responses. Rank-isolated is a reduction response: it converts the question into a finite critical-set test. Residual-open is a non-closing response.

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Tunnel class	test	conservative response
T_r	$\rho_r^\gamma = \varepsilon_r,\ \rho_h^\gamma < \varepsilon_h$, radial outward derivative	atlas routed if covered by a strict contraction chart; otherwise residual open
T_h	$\rho_h^\gamma = \varepsilon_h,\ \rho_r^\gamma < \varepsilon_r$, HC-residual outward derivative	absorbed if HC/MM absorption is explicit and debt-free; otherwise residual open
T_{rh}	$\rho_r^\gamma = \varepsilon_r,\ \rho_h^\gamma = \varepsilon_h$	split into T_r and T_h if rank two; rank isolated if rank drops
T_χ	tunnel selector changes or is discontinuous	blocked when selector returns zero; otherwise routed to atlas-overlap or residual open
T_{rk}	$D\rho_r^\gamma \wedge D\rho_h^\gamma = 0$	rank isolated as a critical set of the two-channel exit map
T_{ov}	chart transition or finite-atlas overlap debt	atlas routed if transition maps preserve strict contraction and zero debt

A.82 Rank-drop isolation and finite critical set target

Definition A.82.1 (Rank-drop critical register)**.** *The rank-drop register is*

$$\mathcal{C}_{\mathrm{rk}}^{\mathrm{loc}} := \{p \in \partial\mathsf{Halo}_{\varepsilon_r,\varepsilon_h}^{\gamma,K} : D\rho_r^\gamma(p) \wedge D\rho_h^\gamma(p) = 0,\ \chi_{\mathrm{tun}}(p) = 1\}.$$

It is the only place where the two halo faces can fail to give independent boundary directions.

Proposition A.82.2 (Rank-drop reduction)**.** *Assume ρ_r^γ and ρ_h^γ are piecewise analytic on a finite photon-frame atlas. Then the rank-drop tunnel class is reduced to the piecewise analytic critical-set problem*

$$D\rho_r^\gamma \wedge D\rho_h^\gamma = 0 \quad \text{on} \quad \partial\mathsf{Halo}_{\varepsilon_r,\varepsilon_h}^{\gamma,K}.$$

If this critical set is empty, blocked by $\chi_{\mathrm{tun}} = 0$, or covered by finitely many strict contraction charts with zero hidden debt, then T_{rk} contributes no uncontrolled OP5 exit.

Proof. By definition, rank-drop occurs exactly when the differential of the exit map has rank less than two. For a two-channel map this is exactly the vanishing of the wedge $D\rho_r^\gamma \wedge D\rho_h^\gamma$, away from nondifferentiable selector points which are assigned to T_χ. Piecewise analyticity transfers the problem to a finite critical-set calculation on each chart. The stated empty, blocked, or chart-routed alternatives are precisely the controlled outcomes in the response alphabet. □

A.83 First executed closure implication

Theorem A.83.1 (executed halo/tunnel reduction)**.** *Assume the OP5=HC-cone working identification and the photon-frame channel definitions above. Suppose the following four conditions hold on the declared finite photon-frame atlas:*

1. ***Channel regularity:*** *ρ_r^γ and ρ_h^γ are piecewise analytic and compatible with the HC-collar guard $r_{\mathrm{HC}}^{(1)} \le \rho_r^\gamma$.*

2. ***Selector completeness:*** *every admissible outward boundary exit is evaluated by χ_{tun}.*

3. ***Tunnel response closure:*** *each nonempty class in $\{T_r, T_h, T_{rh}, T_\chi, T_{\mathrm{rk}}, T_{\mathrm{ov}}\}$ receives one of the three closing responses: blocked, debt-free HC/MM absorbed, or atlas routed through strict contraction charts.*

4. ***No hidden debt:***
$$L_{\mathrm{unr}} = 0, \qquad B_{\mathrm{HC/MM}} = 0, \qquad \tau = 0.$$

Then the double-epsilon photon-frame halo has no uncontrolled OP5 exit. Consequently, if halo capture and halo invariance off tunnels are also proved globally for the admissible OP5 readout class, OP5 reduces to a closed HC-cone halo theorem.

Proof. Boundary exhaustion assigns every outward exit candidate to the finite tunnel menu. The response-closure assumption assigns every nonempty class to a controlled outcome. Blocked classes do not exit; debt-free HC/MM-absorbed classes have no unpaid residual; atlas-routed classes return to strict contraction charts; and the no-hidden-debt condition rules out suppressed leakage. Hence no uncontrolled exit remains inside the executed halo grammar. Combining this with capture and invariance would trap every admissible OP5 branch or route every tunnel exit through controlled charts. This is exactly the HC-cone halo theorem formulation of OP5. □

A.84 What still has to be shown for an actual OP5 close

The layer executes the definitions and the first tunnel-menu reduction. It does not yet prove the global OP5 close. The remaining non-negotiable proof obligations are:

1. prove finite-atlas coverage of the admissible photon-frame near-kernel collar;
2. prove halo capture for every admissible OP5 readout;
3. prove halo invariance away from the finite tunnel menu;
4. compute or bound $\mathcal{C}_{\mathrm{rk}}^{\mathrm{loc}}$;
5. prove that selector-switch points are blocked or chart-routed;
6. prove that all atlas overlaps preserve the strict margin $\sup_j(\widehat{\lambda}_j + \eta_j) < 1$;
7. prove zero hidden leakage and zero hidden HC/MM debt.

If these obligations are discharged, OP5 is no longer an all-routes problem. It is exhausted by the double-epsilon halo, the finite tunnel menu, and strict contraction routing at the tunnel exits.

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A.85 Conservativity guards

1. does not claim global OP5 closure.
2. The executed halo is a readout-side HC-collar filter, not a literal particle shell.
3. The collar face $\rho_r^{\gamma} = r_{\mathrm{HC}}^{(1)}$ is a no-through-collapse guard and not an escape face.
4. The second-shift projected through-K appearance remains tilt debt unless the tunnel tests certify otherwise.
5. HC/MM absorption must remain explicit and debt-free.
6. OP1, OP4, QCD/confinement, CKM/PMNS, and the quark-mass spectrum are not changed by this layer.

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Integrated Delta Layer: Normal-Collar Rank Test

Abstract

This delta layer executes the next obstruction left by the local pass. The exit map is decomposed chartwise on a finite photon-frame halo atlas. The rank-drop tunnel class is bounded by an explicit determinant register, and the selector-switch tunnel class is bounded by a finite switch skeleton. The result is not a global OP5 close. It reduces the uncontrolled tunnel search to two finite residual registers plus atlas-overlap margin checks.

A.86 Purpose and notation hygiene, declared local OP1/OP4 closure, release-final

The local proof chain halo channel reduced the OP5 near-kernel route problem to a two-channel boundary problem. The printed operators in this layer are

$$\mathsf{Halo}, \qquad \mathsf{Exit}, \qquad J^{\gamma,K}_{\mathsf{Exit}}, \qquad \mathcal{E}^{\gamma,K}_{\mathsf{Exit}}.$$

No lowercase display macro is used. The task is to execute the instruction

$$\operatorname{rank} J^{\gamma,K}_{\mathsf{Exit}} < 2 \quad \Longleftrightarrow \quad D\rho^{\gamma}_{r} \wedge D\rho^{\gamma}_{h} = 0,$$

and to combine it with the selector-switch test.

A.87 Finite photon-frame halo atlas

Definition A.87.1 (finite halo atlas)**.** *An photon-frame halo atlas is a finite family*

$$\mathfrak{A}^{\mathrm{loc}}_{\gamma,K} = \{(U_j, \varphi_j)\}_{j=1}^{M}$$

covering the active halo boundary

$$\partial\mathsf{Halo}^{\gamma,K}_{\varepsilon_r,\varepsilon_h} \cap \{r^{(1)}_{\mathrm{HC}} \le \rho^{\gamma}_{r} \le \varepsilon_r,\ \rho^{\gamma}_{h} \le \varepsilon_h\}.$$

On each chart we write

$$r_j := \rho^{\gamma}_{r} \circ \varphi_j^{-1}, \qquad h_j := \rho^{\gamma}_{h} \circ \varphi_j^{-1}.$$

The chart is regular for when these two functions are piecewise analytic and all nonsmooth selector boundaries are explicitly assigned to the switch register.

Definition A.87.2 (Exit determinant)**.** *On a smooth two-channel chart piece with active coordinates (u, v), define*

$$W^{\mathrm{loc}}_{j}(u,v) = \det \begin{pmatrix} \partial_u r_j & \partial_v r_j \\ \partial_u h_j & \partial_v h_j \end{pmatrix} = \partial_u r_j\, \partial_v h_j - \partial_v r_j\, \partial_u h_j.$$

The rank-drop register is

$$\mathcal{C}^{\mathrm{loc}}_{\mathrm{rk}} = \bigcup_j \{p \in U_j \cap \partial\mathsf{Halo}^{\gamma,K}_{\varepsilon_r,\varepsilon_h} : W^{\mathrm{loc}}_{j}(p) = 0,\ \chi_{\mathrm{tun}}(p) = 1\}.$$

Thus the rank-drop tunnel class is not an arbitrary route family; it is a finite union of explicitly declared chartwise critical loci.

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Lemma A.87.3 (Rank-drop localization). *Every rank-drop tunnel point belongs either to $\mathcal{C}_{\mathrm{rk}}^{\mathrm{loc}}$ or to a selector-switch boundary.*

Proof. On a smooth chart piece, gives rank drop exactly when the two channel differentials are dependent. In active coordinates this is exactly $W_j^{\mathrm{loc}} = 0$. If the point is not smooth because the selector changes there, it is assigned to the selector-switch register. Since the atlas is finite, the localization is finite. □

A.88 Normal collar charts

Definition A.88.1 (Normal collar form). *A smooth halo chart piece is in normal collar form when it admits coordinates (a, b, z) such that*

$$r_j(a, b, z) = r_{\mathrm{HC},j}^{(1)}(z) + \varepsilon_r a, \qquad h_j(a, b, z) = \varepsilon_h b + \Delta_j(a, b, z),$$

with

$$0 \le a \le 1, \qquad 0 \le b \le 1, \qquad |\partial_b \Delta_j(a, b, z)| \le \theta_j \varepsilon_h, \qquad 0 \le \theta_j < 1.$$

The number $1 - \theta_j$ is the local HC-residual transversality margin.

Proposition A.88.2 (Normal collar charts have no rank drop). *On a normal collar chart piece,*

$$W_j^{\mathrm{loc}} = \varepsilon_r\big(\varepsilon_h + \partial_b \Delta_j\big)$$

in the active (a, b)-plane. Hence

$$|W_j^{\mathrm{loc}}| \ge \varepsilon_r \varepsilon_h (1 - \theta_j) > 0.$$

Consequently $\mathcal{C}_{\mathrm{rk}}^{\mathrm{loc}}$ is empty on that chart piece.

Proof. The active derivatives are $\partial_a r_j = \varepsilon_r$, $\partial_b r_j = 0$, and $\partial_b h_j = \varepsilon_h + \partial_b \Delta_j$. Therefore the determinant is $\varepsilon_r(\varepsilon_h + \partial_b \Delta_j)$. The assumed bound on $\partial_b \Delta_j$ keeps this determinant strictly away from zero. □

A.89 Selector-switch register

Definition A.89.1 (Finite selector skeleton). *The tunnel selector is a finite Boolean gate*

$$\chi_{\mathrm{tun}} : \partial \mathsf{Halo}^{\gamma,K}_{\varepsilon_r,\varepsilon_h} \longrightarrow \{0, 1\}.$$

It is constant on cells cut out by finitely many piecewise analytic switch functions

$$\sigma_{j,\ell} : U_j \longrightarrow \mathbb{R}.$$

The selector-switch register is

$$\mathcal{C}_\chi^{\mathrm{loc}} = \bigcup_{j,\ell} \{p \in U_j \cap \partial \mathsf{Halo}^{\gamma,K}_{\varepsilon_r,\varepsilon_h} : \sigma_{j,\ell}(p) = 0\}.$$

A switch point is closing only if the outward-side selector is zero or if a strict atlas-transition chart routes it with zero hidden debt.

Lemma A.89.2 (Selector localization). *Every selector ambiguity or selector discontinuity of an outward boundary candidate belongs to $\mathcal{C}_\chi^{\mathrm{loc}}$.*

Proof. Away from the switch skeleton all switch signs are fixed, so the selector is locally constant. Hence the selector can change only where at least one $\sigma_{j,\ell}$ vanishes. □

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A.90 Tunnel response execution

The six tunnel classes receive the following responses:

class	response
T_r	atlas routed if strict contraction covers the radial face; otherwise open or blocked
T_h	debt-free HC/MM absorbed if explicit absorption certificate exists; otherwise open
T_{rh}	split into T_r, T_h if $W_j^{\mathrm{loc}} \neq 0$; rank isolated if $W_j^{\mathrm{loc}} = 0$
T_χ	blocked if outward selector is zero; otherwise switch isolated or atlas routed
T_{rk}	rank isolated inside $\mathcal{C}_{\mathrm{rk}}^{\mathrm{loc}}$ until empty/block/absorb/route is proved
T_{ov}	atlas routed if overlap margins remain strictly below one and debt is zero

Definition A.90.1 (Overlap margin)**.** *For a chart transition $\Gamma_{jk} : U_j \to U_k$, set*

$$\mathfrak{m}_{jk} := \sup_{x \in U_j \cap U_k} \big(\widehat{\lambda}_k(\Gamma_{jk}x) + \eta_k(\Gamma_{jk}x)\big).$$

The overlap is contraction-preserving if

$$\mathfrak{m}_{jk} < 1, \qquad L_{\mathrm{unr}} = 0, \qquad B_{\mathrm{HC/MM}} = 0$$

on the overlap.

Theorem A.90.2 (rank/switch register reduction)**.** *Assume the OP5=HC-cone working identification and the photon-frame halo-channel definitions. Suppose the finite halo atlas satisfies:*

1. *each smooth chart piece is piecewise analytic;*
2. *selector changes occur only on $\mathcal{C}_\chi^{\mathrm{loc}}$;*
3. *every normal collar chart piece has $\theta_j < 1$;*
4. *all active atlas overlaps are contraction-preserving;*
5. *hidden leakage and hidden HC/MM debt vanish on the executed halo boundary:*

$$L_{\mathrm{unr}} = 0, \qquad B_{\mathrm{HC/MM}} = 0.$$

Then every outward tunnel candidate is either blocked, debt-free HC/MM absorbed, atlas routed, or contained in

$$\mathcal{R}_{\mathrm{res}}^{\mathrm{loc}} := \mathcal{C}_{\mathrm{rk}}^{\mathrm{loc}} \cup \mathcal{C}_\chi^{\mathrm{loc}}.$$

If, in addition, $\mathcal{C}_{\mathrm{rk}}^{\mathrm{loc}}$ is empty, blocked, absorbed, or routed and $\mathcal{C}_\chi^{\mathrm{loc}}$ is blocked or routed, then the tunnel menu has no uncontrolled OP5 exit.

Proof. The boundary exhaustion assigns every outward exit to the finite tunnel menu. The determinant localization sends rank-drop points to $\mathcal{C}_{\mathrm{rk}}^{\mathrm{loc}}$, while normal collar transversality eliminates rank drop on regular pieces. The selector localization sends selector ambiguity to $\mathcal{C}_\chi^{\mathrm{loc}}$. Radial, HC-residual, corner, rank, selector, and overlap tunnel classes are then handled by the response table. The only non-closing objects left are the two residual registers. If these are also empty, blocked, absorbed, or routed, no uncontrolled tunnel remains. □

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A.91 First executable normal-form sample

For the simplest ideal photon-frame collar sample, take active coordinates (a, b) and set

$$r(a,b) = r_{\mathrm{HC}}^{(1)} + \varepsilon_r a, \qquad h(a,b) = \varepsilon_h b + \delta ab, \qquad |\delta| < \varepsilon_h.$$

Then

$$W^{\mathrm{loc}}(a,b) = \det \begin{pmatrix} \varepsilon_r & 0 \\ \delta b & \varepsilon_h + \delta a \end{pmatrix} = \varepsilon_r(\varepsilon_h + \delta a) > 0.$$

Thus the rank-drop register is empty in the ideal normal-form sample. Any remaining tunnel issue in that sample must come from a selector switch, an atlas overlap, or an explicitly declared HC/MM absorption boundary.

A.92 Conservativity guards

This layer proves a structural reduction, not a final physical closure of OP5. The double-epsilon halo remains a readout-side structural filter, not a literal particle shell. Tunnel points are selector, rank, or obstruction points, not literal spacetime wormholes. The ideal normal-form sample is a diagnostic model, not a measurement claim. HC/MM absorption must remain explicit and debt-free. OP1, OP4, QCD/confinement, CKM/PMNS, and the quark-mass spectrum are not changed by this layer.

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Integrated Delta Layer: Selector and Overlap Test

Abstract

This delta layer executes the follow-up left by the local pass. It gives a sufficient, finite, chartwise criterion which places the relevant photon-frame halo charts into normal-collar form with a uniform margin $|\partial_b \Delta_j| \le \theta_j \varepsilon_h, \theta_j < 1$. Under this certificate the rank-drop register is empty on all certified collar pieces. The layer does not close OP5 globally. It removes the rank-drop tunnel channel from the certified photon-frame normal-collar atlas and leaves only selector-switch, atlas-overlap, leakage/debt, or non-certified residual pieces.

A.93 Purpose

The layer reduced uncontrolled halo exits to a rank-drop register, a selector-switch register, and atlas-overlap margin checks. The key unresolved certificate was whether the photon-frame halo charts may actually be placed in normal-collar form. This layer gives the required local criterion and the first finite symbolic uniformization certificate.

The printed operators remain renderer-safe:

$$\mathsf{Halo}, \qquad \mathsf{Exit}, \qquad J^{\gamma,K}_{\mathsf{Exit}}, \qquad \mathcal{E}^{\gamma,K}_{\mathsf{Exit}}.$$

No raw lowercase display macro is used.

A.94 Photon-frame collar coordinates

Definition A.94.1 (Certified photon-frame collar piece)**.** *A smooth halo chart piece $U_j \subset \partial \mathsf{Halo}^{\gamma,K}_{\varepsilon_r,\varepsilon_h}$ is the local pass-certified when it admits coordinates*

$$(a, b, z) \in [0, 1] \times [0, 1] \times Z_j$$

with the following three properties:

1. *a is the normalized radial collar coordinate after the second photon-frame shift,*

$$\rho^{\gamma}_r(a, b, z) = r^{(1)}_{\mathrm{HC},j}(z) + \varepsilon_r a;$$

2. *b is the normalized HC-cone height coordinate;*

3. *the HC-residual channel splits as*

$$\rho^{\gamma}_h(a, b, z) = \varepsilon_h b + \Delta_j(a, b, z)$$

with a declared tilt-debt derivative bound

$$|\partial_b \Delta_j(a, b, z)| \le \theta_j \varepsilon_h, \qquad 0 \le \theta_j < 1.$$

The number $1 - \theta_j$ is the normal-collar safety margin.

Remark A.94.2. *The coordinate b is not defined to be $\rho^{\gamma}_h/\varepsilon_h$. It is the HC-cone height coordinate inherited from the photon-frame collar. This avoids a circular proof of transversality. The nontrivial check is exactly the derivative bound on Δ_j.*

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Definition A.94.3 (Uniform certificate)**.** *A finite photon-frame halo atlas*

$$\mathfrak{A}^{\mathrm{loc}}_{\gamma,K} = \{U_j\}_{j=1}^{M}$$

is uniformly normal-collar certified when every smooth active piece is the local pass-certified and

$$\theta^{\mathrm{loc}}_{\max} := \max_{1\le j\le M} \theta_j < 1.$$

The certified rank-drop register is then

$$\mathcal{C}^{\mathrm{loc}}_{\mathrm{rk}} = \bigcup_j \{p \in U_j \cap \partial \mathsf{Halo}^{\gamma,K}_{\varepsilon_r,\varepsilon_h} : W^{\mathrm{loc}}_j(p) = 0, \chi_{\mathrm{tun}}(p) = 1\}.$$

A.95 Rank-drop elimination on certified pieces

Lemma A.95.1 (determinant lower bound)**.** *On an the local pass-certified collar piece,*

$$W^{\mathrm{loc}}_j(a,b,z) = \varepsilon_r\big(\varepsilon_h + \partial_b \Delta_j(a,b,z)\big).$$

Consequently

$$|W^{\mathrm{loc}}_j(a,b,z)| \ge \varepsilon_r \varepsilon_h (1-\theta_j) > 0.$$

Proof. The active derivatives in the (a,b)-plane are

$$\partial_a \rho^\gamma_r = \varepsilon_r, \qquad \partial_b \rho^\gamma_r = 0, \qquad \partial_b \rho^\gamma_h = \varepsilon_h + \partial_b \Delta_j.$$

Therefore the exit determinant is

$$\det \begin{pmatrix} \partial_a \rho^\gamma_r & \partial_b \rho^\gamma_r \\ \partial_a \rho^\gamma_h & \partial_b \rho^\gamma_h \end{pmatrix} = \varepsilon_r(\varepsilon_h + \partial_b \Delta_j).$$

The derivative bound gives $\varepsilon_h + \partial_b \Delta_j \ge \varepsilon_h(1-\theta_j) > 0$ after orienting the collar height positively. Hence the determinant is bounded away from zero. □

Proposition A.95.2 (Certified collar pieces contain no rank-drop tunnels)**.** *If U_j is the local pass-certified, then*

$$U_j \cap \mathcal{C}^{\mathrm{loc}}_{\mathrm{rk}} = \varnothing.$$

If the atlas is uniformly the local pass-certified, then the rank-drop tunnel channel is empty on the certified smooth halo boundary:

$$\mathcal{C}^{\mathrm{loc}}_{\mathrm{rk}} \cap \bigcup_j U^{\mathrm{cert}}_j = \varnothing.$$

Proof. Rank drop is equivalent to $W^{\mathrm{loc}}_j = 0$ on a smooth chart piece. The determinant lower bound excludes this equality on each certified piece. Taking the finite union gives the atlas statement. □

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A.96 Finite symbolic certificate

The first executable certificate uses three representative collar pieces. These are not physical measurements; they are the minimal symbolic chart types required by the current halo grammar.

piece	role	$\Delta_j(a,b,z)/\varepsilon_h$	θ_j	rank response
U_1	radial face	$\frac{1}{4}ab$	$\frac{1}{4}$	empty
U_2	HC-residual face	$\frac{3}{8}b(1-a)$	$\frac{3}{8}$	empty
U_3	corner/overlap collar	$\frac{1}{2}ab(1-a)$	$\frac{1}{2}$	empty

Thus

$$\theta_{\max}^{\mathrm{loc}} = \frac{1}{2} < 1$$

and hence

$$|W_j^{\mathrm{loc}}| \geq \frac{1}{2}\varepsilon_r\varepsilon_h$$

on the three certified representative pieces.

Lemma A.96.1 (First finite symbolic normal-collar sample)**.** *For the three-piece certificate above, the rank-drop register is empty on all certified pieces.*

Proof. For every listed piece, $\theta_j \leq 1/2$. The determinant lower bound gives strict positivity of W_j^{loc}. Hence no listed piece contributes to $\mathcal{C}_{\mathrm{rk}}^{\mathrm{loc}}$. □

A.97 Remaining tunnel channels

After the certificate, the local proof chain tunnel menu is reduced as follows:

tunnel class	status
T_r	covered by normal collar if radial face is in the certified atlas
T_h	covered by normal collar or debt-free HC/MM absorption
T_{rh}	rank split eliminated on certified pieces; overlap still checked
T_χ	unchanged: selector-switch register must be blocked or routed
T_{rk}	empty on certified collar pieces
T_{ov}	unchanged: transition margins must remain strictly below one

Thus does not remove every OP5 obstruction. It removes the rank-drop channel exactly where the photon-frame halo atlas has the certified normal-collar form.

Theorem A.97.1 (normal-collar rank elimination)**.** *Assume the OP5=HC-cone working identification, the halo-channel execution, and the rank/switch register reduction. Suppose a finite photon-frame halo atlas is uniformly the local pass-certified with $\theta_{\max}^{\mathrm{loc}} < 1$. Then no rank-drop tunnel occurs on the certified smooth halo boundary. Every remaining uncontrolled candidate must lie in one of the following residual classes:*

$$\mathcal{R}_{\mathrm{res}}^{\mathrm{loc}} \subseteq \mathcal{C}_\chi^{\mathrm{loc}} \cup \mathcal{C}_{\mathrm{ov}}^{\mathrm{loc}} \cup \mathcal{C}_{\mathrm{leak/debt}}^{\mathrm{loc}} \cup \mathcal{C}_{\mathrm{uncert}}^{\mathrm{loc}},$$

where $\mathcal{C}_{\mathrm{uncert}}^{\mathrm{loc}}$ denotes non-certified pieces not yet placed in normal-collar form. If the selector-switch register is blocked or routed, all atlas-overlap margins are contraction-preserving, hidden leakage and HC/MM debt vanish, and $\mathcal{C}_{\mathrm{uncert}}^{\mathrm{loc}} = \varnothing$, then the halo tunnel menu has no uncontrolled exit on the finite certified atlas.

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Proof. The certified determinant lower bound eliminates T_{rk} on every smooth certified chart piece. The reduction already states that all remaining outward candidates are rank, selector, overlap, leakage/debt, or residual non-certified pieces. Removing rank on certified pieces leaves exactly the displayed residual classes. If these classes are blocked, routed, or empty under the stated hypotheses, no uncontrolled exit remains on the finite certified atlas. □

A.98 Meaning for OP5

The result is a strong local structural advance. It says that, provided the photon-frame halo atlas is uniformly normal-collar certified, OP5 is no longer obstructed by a generic rank-collapse phenomenon near the HC collar. The hard work is pushed to four explicit checks:

selector switches, atlas overlaps, hidden leakage/debt, uncertified pieces.

A global OP5 close would require these four checks to vanish, block, or route over the full finite atlas.

A.99 Conservativity guards

This layer does not claim a global OP5 closure. It proves a conditional rank-drop elimination theorem for certified photon-frame normal-collar charts. The halo remains a readout-side structural filter, not a literal particle shell. Tunnel points remain selector, rank, overlap, leakage, or obstruction points. HC/MM absorption must remain explicit and debt-free. OP1, OP4, QCD/confinement, CKM/PMNS, and the quark-mass spectrum are untouched.

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Integrated Delta Layer: Finite Capture

Abstract

This delta layer executes the two residual channels left by on certified photon-frame normal-collar pieces: selector switches and atlas overlaps. It introduces a finite selector-switch stratum, a finite overlap transition graph, a switch routing alphabet, and an overlap contraction-margin test. The layer proves that if selector switches are blocked, debt-free HC/MM absorbed, or routed into certified charts, and if every overlap edge preserves the strict contraction margin, then no uncontrolled OP5 exit remains on the finite certified halo atlas apart from explicitly declared leakage/debt or non-certified pieces. No global OP5 closure is claimed.

A.100 Purpose and inherited state

The layer removed rank-drop tunnels on certified photon-frame normal-collar pieces. The residual menu was reduced to selector switches, atlas overlaps, hidden leakage/debt, and non-certified pieces. This layer executes the first two items:

$$\mathcal{C}_{\chi}, \qquad \mathcal{C}_{\mathrm{ov}}.$$

The printed operators remain renderer-safe:

$$\mathsf{Halo}, \qquad \mathsf{Exit}, \qquad J^{\gamma,K}_{\mathsf{Exit}}.$$

No lowercase display macro for these operators is used.

We work over a finite certified halo atlas

$$\mathfrak{A}^{\mathrm{loc}}_{\gamma,K} = \{U_1, \dots, U_M\}$$

whose certified normal-collar pieces satisfy the determinant bound

$$|W_j| \geq \varepsilon_r \varepsilon_h (1 - \theta_j) > 0.$$

Thus the rank-drop channel is already empty on the certified pieces. The remaining question is whether a route can leave through a selector change or through a chart transition.

A.101 Selector-switch register

Definition A.101.1 (Finite selector-switch stratum)**.** *Let*

$$\chi_j : U_j \longrightarrow \{0, 1, \dots, Q\}$$

be the local OP5 route selector on a certified chart piece. The selector-switch stratum is

$$\mathfrak{S}^{\mathrm{loc}}_{\chi} = \bigcup_{j=1}^{M} \partial_{\chi} U_j,$$

where $\partial_{\chi} U_j$ is the finite union of codimension-one faces on which χ_j changes value or ceases to be locally constant. The selector-switch register is

$$\mathcal{C}^{\mathrm{loc}}_{\chi} = \{p \in \mathfrak{S}^{\mathrm{loc}}_{\chi} : p \in \partial \mathsf{Halo}^{\gamma,K}_{\varepsilon_r,\varepsilon_h},\ \chi_{\mathrm{tun}}(p) = 1\}.$$

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Definition A.101.2 (switch response alphabet)**.** *A selector-switch point $p \in \mathcal{C}_\chi^{\mathrm{loc}}$ receives one of four conservative responses:*

$$\mathrm{resp}_\chi^{\mathrm{loc}}(p) \in \{\mathrm{block}, \mathrm{abs}, \mathrm{route}, \mathrm{res}\}.$$

The meanings are:

response	*conservative meaning*
block	*the selector change is inadmissible for the OP5 tunnel gate; no exit is allowed;*
abs	*the switch is absorbed by declared debt-free HC/MM bookkeeping, with $B_{\mathrm{HC/MM}} = 0$;*
route	*the switch enters a certified chart piece with preserved halo margin;*
res	*the point remains an explicit residual and cannot be used for closure.*

Lemma A.101.3 (Switch containment)**.** *Assume every selector is locally constant on the interior of each certified collar piece. Then every selector-induced uncontrolled exit candidate lies in $\mathcal{C}_\chi^{\mathrm{loc}}$.*

Proof. On the interior of a certified piece the selector does not change. Therefore selector-driven route creation cannot occur there. If a selector change occurs, it occurs on the switch stratum by definition. Intersecting with the halo boundary and the tunnel gate gives exactly $\mathcal{C}_\chi^{\mathrm{loc}}$. □

Proposition A.101.4 (Selector-switch closure criterion)**.** *If*

$$\mathrm{resp}_\chi^{\mathrm{loc}}(p) \in \{\mathrm{block}, \mathrm{abs}, \mathrm{route}\} \quad \textit{for every } p \in \mathcal{C}_\chi^{\mathrm{loc}},$$

and if each routed switch lands in a certified chart with unchanged debt-free bookkeeping, then selector switches create no uncontrolled OP5 exit on the certified atlas.

Proof. A blocked switch has no admissible exit. An absorbed switch has no hidden HC/MM debt by hypothesis. A routed switch remains inside the certified atlas and is again subject to the rank elimination and the overlap test below. Since no switch has response res, no selector-induced uncontrolled exit remains. □

A.102 Atlas-overlap register

Definition A.102.1 (Finite overlap transition graph)**.** *The overlap graph is the finite directed graph*

$$\mathfrak{G}_{\mathrm{ov}}^{\mathrm{loc}} = (V, E), \qquad V = \{1, \dots, M\},$$

where $(j, k) \in E$ exactly when the certified chart overlap

$$U_{jk} = U_j \cap U_k$$

is nonempty and may be crossed by an admissible OP5 route. On such an edge let

$$\Phi_{jk} : U_{jk} \subset U_j \longrightarrow U_k$$

be the declared transition map.

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Definition A.102.2 (Overlap contraction margin)**.** *For an overlap edge $(j,k) \in E$, define the overlap margin*

$$\mathfrak{m}_{jk}^{\mathrm{loc}} = \widehat{\lambda}_j + \eta_j + \omega_{jk},$$

where $\widehat{\lambda}_j + \eta_j < 1$ is the inherited chart contraction budget and $\omega_{jk} \geq 0$ is the declared transition distortion of Φ_{jk} in the two-channel halo norm

$$\|(u,v)\|_{\mathsf{Halo}} = \max\left\{\frac{|u|}{\varepsilon_r}, \frac{|v|}{\varepsilon_h}\right\}.$$

The overlap edge is contraction-preserving when

$$\mathfrak{m}_{jk}^{\mathrm{loc}} < 1.$$

The overlap register is

$$\mathcal{C}_{\mathrm{ov}}^{\mathrm{loc}} = \{(j,k) \in E : \mathfrak{m}_{jk}^{\mathrm{loc}} \geq 1\} \cup \{(j,k) \in E : \Phi_{jk} \text{ is undeclared}\}.$$

Lemma A.102.3 (Overlap preservation)**.** *If $(j,k) \in E$ has $\mathfrak{m}_{jk}^{\mathrm{loc}} < 1$, then crossing the overlap from U_j to U_k preserves strict halo contraction and does not create an uncontrolled OP5 exit.*

Proof. The inherited chart budget bounds the active chartwise projection debt by $\widehat{\lambda}_j + \eta_j$. The transition contributes at most ω_{jk} in the normalized two-channel halo norm. Hence the total budget after crossing the overlap is bounded by $\mathfrak{m}_{jk}^{\mathrm{loc}} < 1$. The crossing therefore remains inside the controlled halo atlas budget. □

Proposition A.102.4 (Atlas-overlap closure criterion)**.** *If $\mathcal{C}_{\mathrm{ov}}^{\mathrm{loc}} = \varnothing$, then atlas overlaps do not contribute uncontrolled OP5 exits on the finite certified atlas.*

Proof. By definition, $\mathcal{C}_{\mathrm{ov}}^{\mathrm{loc}} = \varnothing$ means every admissible overlap edge is declared and contraction-preserving. The previous lemma applies to every edge of the finite transition graph. Therefore no overlap crossing leaves the controlled halo budget. □

A.103 First symbolic execution table

For the current three-piece symbolic atlas inherited from the local pass, use the following conservative finite routing table. The entries are structural certificates, not physical measurements.

selector face	location	response	reason
S_{12}	$U_1 \leftrightarrow U_2$	route	same halo side, certified target
S_{23}	$U_2 \leftrightarrow U_3$	route	corner collar remains certified
S_{13}	$U_1 \leftrightarrow U_3$	block	non-adjacent selector jump not OP5-admissible
$S_{\mathrm{HC/MM}}$	HC/MM bookkeeping face	abs	declared debt-free absorption

Thus no switch in the sample table has residual response.
The corresponding overlap margins are:

edge	$\widehat{\lambda}_j + \eta_j$	ω_{jk}	$\mathfrak{m}_{jk}^{\mathrm{loc}}$	status
$(1,2)$	$5/8$	$1/16$	$11/16$	preserved
$(2,3)$	$21/32$	$1/32$	$11/16$	preserved
$(1,3)$	$25/32$	$1/32$	$13/16$	preserved if routed; blocked as switch

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Therefore the sample overlap register is empty:

$$\mathcal{C}_{\mathrm{ov}}^{\mathrm{loc}} = \varnothing$$

for the declared transition graph.

A.104 residual theorem

Theorem A.104.1 (Selector-overlap execution theorem)**.** *Assume the double-epsilon halo channel, the photon-frame halo-channel execution, the rank/switch register reduction, and the normal-collar rank elimination. Suppose furthermore that:*

1. *every selector switch in* $\mathcal{C}_{\chi}^{\mathrm{loc}}$ *has response* block*,* abs*, or* route*, never* res*;*
2. *every routed selector switch lands in a certified normal-collar chart;*
3. *every admissible overlap edge is declared and satisfies* $\mathfrak{m}_{jk}^{\mathrm{loc}} < 1$*;*
4. *hidden leakage and HC/MM debt vanish:*

$$L_{\mathrm{unr}} = 0, \qquad B_{\mathrm{HC/MM}} = 0;$$

5. *no non-certified photon-frame halo piece remains in the active OP5 route menu.*

Then the finite certified photon-frame halo atlas has no uncontrolled OP5 exit. Equivalently, within this declared halo/tunnel grammar, every admissible near-kernel OP5 route is either blocked, debt-free absorbed, or routed back into a contraction-preserving certified HC-collar chart.

Proof. The determinant bound eliminates rank-drop exits on certified collar pieces. The selector-switch criterion eliminates selector-induced exits unless they are blocked, absorbed, or routed into certified pieces. The overlap criterion eliminates chart-transition exits because every admissible edge preserves strict contraction margin. The leakage and HC/MM-debt hypotheses rule out hidden bookkeeping exits. Finally, the absence of non-certified active pieces prevents escape through parts of the halo boundary not covered by the certified grammar. Hence the finite certified atlas contains no uncontrolled OP5 exit. □

Remark A.104.2 (Conservativity guard)**.** *This theorem is not a global OP5 closure. It is a finite-atlas closure criterion under the OP5=HC-cone working identification and under the declared double-epsilon halo grammar. A global closure would additionally require proving that every admissible OP5 near-kernel route enters this certified finite atlas and that the non-certified residual class is empty in the full active-core setting.*

A.105 Resulting next step

The execution leaves a sharply stated final gap. To promote the local halo/tunnel analysis toward a real OP5 closure statement, the next layer must prove the capture/completeness clause:

$$\mathcal{R}_{\mathrm{OP5}}^{\mathrm{adm,near}K} \subseteq \bigcup_{j=1}^{M} U_j^{\mathrm{cert}}$$

for the relevant photon-frame HC-collar readouts, together with $L_{\mathrm{unr}} = 0$ and $B_{\mathrm{HC/MM}} = 0$. This is the natural target.

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Integrated Delta Layer: Dixon-Carrier Coding

Abstract

This delta layer executes the capture/completeness step left open by the local pass. It defines a finite route-capture certificate for admissible near-kernel OP5 readouts under the working identification OP5 = photon-frame HC-cone route. The layer separates three tasks: capture into the double-epsilon halo, assignment to a certified normal-collar chart, and debt-free compatibility with selector/overlap routing. The result is a conditional but explicit completeness theorem: if the finite capture certificate is exhaustive, if every captured branch lands in the local proof chain certified atlas, and if hidden leakage and HC/MM debt vanish, then the declared near-kernel OP5 route menu has no uncontrolled exit. This is still not a global OP5 proof outside the declared working grammar.

A.106 Purpose and inherited state

The local proof-chain layers transformed the OP5 near-kernel route problem into a finite halo/tunnel problem. The current state is:

(i) introduced the double-epsilon halo channel;

(ii) instantiated the photon-frame channels ρ_r^γ and ρ_h^γ and the two-channel exit map;

(iii) reduced rank-drop and selector-switch search to explicit registers;

(iv) eliminated rank-drop on certified normal-collar pieces;

(v) routed selector switches and showed contraction-preserving atlas overlaps in the sample finite atlas.

The remaining gap is a capture/completeness statement:

$$\mathcal{R}_{\mathrm{OP5}}^{\mathrm{adm,near}K} \subseteq \bigcup_{j=1}^{M} U_j^{\mathrm{cert}}, \qquad L_{\mathrm{unr}} = 0, \qquad B_{\mathrm{HC/MM}} = 0.$$

This layer gives the certificate grammar for exactly that statement.

All displayed operators use renderer-safe notation:

$$\mathsf{Halo}, \qquad \mathsf{Exit}, \qquad \mathsf{Cap}.$$

A.107 Finite route-capture certificate

Definition A.107.1 (Near-kernel OP5 route menu)**.** *Let $\mathcal{R}_{\mathrm{OP5}}^{\mathrm{adm,near}K}$ be the declared set of admissible near-kernel OP5 readouts after photon-frame HC-collar normalization. A route $r \in \mathcal{R}_{\mathrm{OP5}}^{\mathrm{adm,near}K}$ is represented by a finite branch word*

$$\beta(r) = (\sigma_0, \sigma_1, \ldots, \sigma_\ell)$$

whose symbols record, in order, HC-collar entry, possible selector face, possible overlap transition, and terminal certified chart candidate. The word is admissible *only if every symbol is admitted by the inherited switch/overlap tables.*

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Definition A.107.2 (capture witness)**.** *An capture witness for a route $r \in \mathcal{R}_{\mathrm{OP5}}^{\mathrm{adm,near}K}$ is a tuple*

$$\mathfrak{c}_r^{\mathrm{loc}} = (n_r, j_r, \beta_r, \delta_r^{\mathsf{Halo}}, \ell_r, b_r),$$

where:

(a) $n_r \in \mathbb{N}$ *is the finite number of HC/Hilbert normalizing steps;*

(b) $j_r \in \{1, \dots, M\}$ *is the certified target chart;*

(c) β_r *is the finite the local pass-admissible selector/overlap word;*

(d) δ_r^{Halo} *is the strict halo margin;*

(e) ℓ_r *is the hidden leakage register;*

(f) b_r *is the hidden HC/MM debt register.*

It is valid when

$$\rho_r^{\gamma}(R_{\mathrm{HC}}^{n_r} r) \le \varepsilon_r - \delta_r^{\mathsf{Halo}}, \qquad \rho_h^{\gamma}(R_{\mathrm{HC}}^{n_r} r) \le \varepsilon_h - \delta_r^{\mathsf{Halo}},$$
$$R_{\mathrm{HC}}^{n_r} r \in U_{j_r}^{\mathrm{cert}}, \qquad \delta_r^{\mathsf{Halo}} > 0, \qquad \ell_r = 0, \qquad b_r = 0.$$

Definition A.107.3 (Finite capture certificate)**.** *A finite capture certificate is a finite family*

$$\mathfrak{C}_{\mathrm{cap}}^{\mathrm{loc}} = \{\mathfrak{c}_a^{\mathrm{loc}} : a \in A_{\mathrm{loc}}\}$$

together with a surjective classification map

$$\pi_{\mathrm{cap}}^{\mathrm{loc}} : \mathcal{R}_{\mathrm{OP5}}^{\mathrm{adm,near}K} \longrightarrow A_{\mathrm{loc}}.$$

For a route r, the witness assigned to $a = \pi_{\mathrm{cap}}^{\mathrm{loc}}(r)$ must be valid for every route in the class $(\pi_{\mathrm{cap}}^{\mathrm{loc}})^{-1}(a)$.

Remark A.107.4 (Why this is the right bottleneck)**.** *Without the capture certificate, OP5 remains a many-route problem. With the certificate, every admissible near-kernel route is compressed into one of finitely many certified classes. The remaining proof is then finite: check capture margin, chart certification, selector/overlap admissibility, leakage, and HC/MM debt for each class.*

A.108 Capture tests

Definition A.108.1 (capture test stack)**.** *For each capture class $a \in A_{\mathrm{loc}}$, define four Boolean tests:*

$$\mathsf{Cap}_{\mathsf{Halo}}(a), \qquad \mathsf{Cap}_{\mathfrak{A}}(a), \qquad \mathsf{Cap}_{\mathsf{Route}}(a), \qquad \mathsf{Cap}_{\mathsf{Debt}}(a).$$

They mean respectively:

$$\mathsf{Cap}_{\mathsf{Halo}}(a) = 1 \iff \rho_r^{\gamma} \le \varepsilon_r - \delta_a^{\mathsf{Halo}} \text{ and } \rho_h^{\gamma} \le \varepsilon_h - \delta_a^{\mathsf{Halo}} \text{ after } n_a \text{ steps},$$
$$\mathsf{Cap}_{\mathfrak{A}}(a) = 1 \iff R_{\mathrm{HC}}^{n_a}(r) \in U_{j_a}^{\mathrm{cert}} \text{ for every } r \text{ in class } a,$$
$$\mathsf{Cap}_{\mathsf{Route}}(a) = 1 \iff \beta_a \text{ uses only the local pass-admissible selector/overlap moves},$$
$$\mathsf{Cap}_{\mathsf{Debt}}(a) = 1 \iff \ell_a = 0 \text{ and } b_a = 0.$$

The class is capture-complete when all four tests equal one.

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Lemma A.108.2 (Classwise capture implies chartwise capture)**.** *If* $\mathsf{Cap}_{\mathsf{Halo}}(a) = \mathsf{Cap}_{\mathfrak{A}}(a) = 1$*, then every route in class* a *enters the certified double-epsilon halo atlas after at most* n_a *normalizing steps.*

Proof. The first test gives strict entry into the double-epsilon halo, with positive margin δ_a^{Halo}. The second test identifies the target as an certified normal-collar chart. Therefore all routes in the class enter the certified halo atlas. □

Lemma A.108.3 (Route and debt tests eliminate hidden exits)**.** *If* $\mathsf{Cap}_{\mathsf{Route}}(a) = \mathsf{Cap}_{\mathsf{Debt}}(a) = 1$*, then class* a *contains no unregistered selector/overlap exit and no hidden leakage or HC/MM debt.*

Proof. The route test restricts the branch word to selector and overlap transitions already routed by the local pass. The debt test sets both hidden registers to zero. Thus the class cannot leave through an undeclared selector/overlap move or through hidden bookkeeping. □

A.109 First symbolic capture table

The following table records the first finite symbolic certificate compatible with the local proof chain. It is a structural certificate table rather than a measurement table.

class	branch type	n_a	target	margin	debt	status
A_1	collar interior	2	U_1^{cert}	1/8	0	captured
A_2	adjacent switch S_{12}	2	U_2^{cert}	3/32	0	routed/captured
A_3	adjacent switch S_{23}	2	U_3^{cert}	1/16	0	routed/captured
A_4	non-adjacent switch S_{13}	–	–	–	–	blocked
A_5	HC/MM face	2	U_2^{cert}	1/16	0	absorbed/captured
A_6	overlap edge	2	$U_j^{\mathrm{cert}} \cup U_k^{\mathrm{cert}}$	1/16	0	overlap-preserved

The active captured classes are

$$A_{\mathrm{act}}^{\mathrm{loc}} = \{A_1, A_2, A_3, A_5, A_6\}.$$

The blocked non-adjacent selector class A_4 does not contribute an admissible OP5 route. Thus the symbolic residual class is

$$A_{\mathrm{res}}^{\mathrm{loc}} = \varnothing$$

provided the classification map $\pi_{\mathrm{cap}}^{\mathrm{loc}}$ is exhaustive for the declared near-kernel route menu.

Proposition A.109.1 (Sample certificate status)**.** *For the symbolic capture table, every active class has strict positive halo margin, zero leakage, zero HC/MM debt, and a certified the local proof chain target chart. The only non-captured class is blocked by the inherited selector rule. Hence the sample table has no residual active capture class.*

Proof. The entries list positive margins for A_1, A_2, A_3, A_5, A_6, debt value zero, and certified targets. Class A_4 is not an active OP5 route because blocks the non-adjacent selector jump. Therefore no active class remains residual. □

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A.110 Completeness theorem

Theorem A.110.1 (capture/completeness theorem)**.** *Assume the double-epsilon halo channel, the photon-frame channel execution, the register reduction, the normal-collar rank elimination, and the selector/overlap execution. Suppose there exists a finite capture certificate* $\mathfrak{C}_{\mathrm{cap}}^{\mathrm{loc}}$ *and an exhaustive classification map*

$$\pi_{\mathrm{cap}}^{\mathrm{loc}} : \mathcal{R}_{\mathrm{OP5}}^{\mathrm{adm,near}K} \longrightarrow A_{\mathrm{loc}}$$

such that every active class is capture-complete and every inactive class is blocked by an inherited OP5-admissibility rule. Then

$$\mathcal{R}_{\mathrm{OP5}}^{\mathrm{adm,near}K} \subseteq \bigcup_{j=1}^{M} U_j^{\mathrm{cert}}$$

after finitely many HC/Hilbert normalizing steps, with

$$L_{\mathrm{unr}} = 0, \qquad B_{\mathrm{HC/MM}} = 0.$$

Consequently, within the declared photon-frame HC-halo grammar, the near-kernel OP5 route menu has no uncontrolled exit.

Proof. Let $r \in \mathcal{R}_{\mathrm{OP5}}^{\mathrm{adm,near}K}$. By exhaustiveness, r belongs to some class $a = \pi_{\mathrm{cap}}^{\mathrm{loc}}(r)$. If the class is inactive, it is blocked and hence not an active OP5 exit. If the class is active, capture-completeness gives four facts: strict double-epsilon halo entry, certified target chart membership, admissible selector/overlap routing, and zero leakage/debt. The theorem eliminates rank-drop on the certified chart. The theorem eliminates selector and overlap exits. The zero-register clause eliminates hidden leakage and HC/MM debt. Hence r is blocked, absorbed, or routed into a certified contraction-preserving chart. Since r was arbitrary, the inclusion and the no-uncontrolled-exit claim follow. □

Corollary A.110.2 (Local OP5 closure inside the declared grammar)**.** *Under the hypotheses of the theorem and under the working identification OP5 = photon-frame HC-cone route, the near-kernel OP5 branch of the declared finite halo/tunnel grammar is locally closed.*

Proof. The theorem states that every admissible near-kernel route is captured by the certified atlas or blocked, with no hidden leakage/debt. This is precisely local closure in the declared grammar. □

Remark A.110.3 (No overclaim)**.** *This is not a global OP5 proof in the ordinary external sense. It is the strongest local closure statement presently justified by the local proof chain: if OP5 is correctly read as the photon-frame HC-cone route problem, and if the finite capture certificate is exhaustive for the relevant active-core route menu, then the near-kernel OP5 branch has no uncontrolled exit. A full release-level OP5 theorem would still need this certificate integrated into the monolithic Active Core and cross-checked against every OP5 occurrence in the historical witness.*

A.111 Audit target for the next layer

The layer turns the remaining work into an audit problem. The next meaningful step is not to add another route mechanism, but to verify exhaustiveness:

$$\mathrm{supp}(\mathcal{R}_{\mathrm{OP5}}^{\mathrm{adm,near}K}) = \bigcup_{a \in A_{\mathrm{loc}}} (\pi_{\mathrm{cap}}^{\mathrm{loc}})^{-1}(a),$$

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with no unclassified active branch. If the monolithic Active Core confirms that equality, then can promote the result from a local capture/completeness certificate to an OP5 closure-candidate theorem for the full active-core route menu.

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Integrated Delta Layer: Dixon-Image Audit

Abstract

This delta layer executes the HS(6) Dixon-coding map suggested after the capture/-completeness packet. The layer separates two clocks which must not be conflated: the carrier clock $C_n = 2^n$, and the activation-mask clock $|\{0,1\}^{C_n}| = 2^{C_n}$. In this grammar, HS(5) has a 32-channel carrier with a 2^{32}-state activation mask, while HS(6) has a 64-channel carrier whose activation mask has 2^{64} states. The 64-channel carrier is mapped to the Dixon carrier $D = \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$. The layer then relates the HS(4) meta-septum projector $\widehat{P}_{C_7} \subset \mathcal{M}_{16}$, the HS(5) two-sided duplication, and the HS(6) Dixon carrier to the OP5 halo/capture chain. No claim is made that the Dixon algebra is 2^{64}-dimensional; only its Boolean activation code has 2^{64} possible masks.

A.112 Purpose and inherited state

The local proof chain sequence reduced the near-kernel OP5 route problem to a finite photon-frame HC-halo capture problem. Its current decisive condition is exhaustiveness of the capture map

$$\pi_{\mathrm{cap}}^{\mathrm{loc}} : \mathcal{R}_{\mathrm{OP5}}^{\mathrm{adm,near}K} \longrightarrow A_{\mathrm{loc}},$$

with $L_{\mathrm{unr}} = 0$ and $B_{\mathrm{HC/MM}} = 0$. The present layer records the next structural interpretation of higher Hilbert shifts. It does not add a new OP5 route mechanism. It provides a finite coding target for the already captured routes.

The motivating observation is that the HS tower naturally admits two readings:

$$\mathsf{Car}(n) = 2^n, \qquad \mathsf{Mask}(n) = \{0,1\}^{\mathsf{Car}(n)}, \qquad |\mathsf{Mask}(n)| = 2^{2^n}.$$

Thus the phrase "up to 2^{32}, then 2^{64}" is correct when it refers to activation masks at carrier sizes 32 and 64, not to the algebra dimension itself.

A.113 Two-clock Hilbert-shift grammar

Definition A.113.1 (Carrier clock and mask clock)**.** *For the Hilbert-shift tower define*

$$C_n := 2^n$$

as the carrier size of the n-th shift, and define the Boolean activation mask space

$$\mathsf{Mask}_n := \{0,1\}^{C_n}.$$

Then

$$|\mathsf{Mask}_n| = 2^{C_n} = 2^{2^n}.$$

We call C_n the carrier clock *and $|\mathsf{Mask}_n|$ the* mask clock*.*

Remark A.113.2 (Correction of the common ambiguity)**.** *The statement "Dixon is 2^{64}" is not used as a dimension statement. The structurally safe statement is*

$$\dim_{\mathbb{R}}(\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}) = 64, \qquad |\{0,1\}^{64}| = 2^{64}.$$

Thus the Dixon carrier has 64 real channels, and its full Boolean activation grammar has 2^{64} masks.

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shift	carrier size C_n	mask count $\lvert\mathsf{Mask}_n\rvert$	structural reading
HS(0)	1	2	radial kernel target / unshifted collapse readout
HS(1)	2	4	HC collar and periapsis split
HS(2)	4	16	stabilized orbit / contraction chart
HS(3)	8	2^8	Fano/octonion-compatible local incidence fibre
HS(4)	16	2^{16}	meta-septum $\mathcal{M}_{16}/\widehat{P}_{C_7}$ leakage test
HS(5)	32	2^{32}	two-sided front/back carrier duplication
HS(6)	64	2^{64}	Dixon carrier $\mathbb{C}\otimes\mathbb{H}\otimes\mathbb{O}$ and 64-bit activation code

A.114 HS(4) meta-septum leakage target

Definition A.114.1 (HS(4) meta-septum carrier)**.** *Let $\mathcal{H}_{16}^{(4)}$ be the 16-channel HS(4) carrier containing the inherited meta-septum sector selected by*

$$\widehat{P}_{C_7} : \mathcal{H}_{16}^{(4)} \longrightarrow \mathcal{H}_{16}^{(4)}, \qquad \widehat{P}_{C_7}^2 = \widehat{P}_{C_7}.$$

Write

$$Q := \widehat{P}_{C_7}, \qquad Q_\perp := \mathbf{1} - Q.$$

For the inherited internal matrix $\mathcal{M}_{16}$, define the admissible block and mixed leakage block by

$$\mathcal{M}_{CC} := Q\mathcal{M}_{16}Q,$$

$$\mathsf{Leak}_{C_7}^{(4)}(\mathcal{M}_{16}) := Q_\perp\mathcal{M}_{16}Q + Q\mathcal{M}_{16}Q_\perp.$$

The HS(4) carrier is meta-septum closed when

$$\mathsf{Leak}_{C_7}^{(4)}(\mathcal{M}_{16}) = 0.$$

Lemma A.114.2 (Commutator form of HS(4) closure)**.** *If $Q = Q^2$ and $Q_\perp = \mathbf{1} - Q$, then vanishing of both mixed blocks is equivalent to block preservation of the C_7-sector. In particular the closure condition can be written as*

$$[\mathcal{M}_{16}, Q] = 0,$$

provided the two mixed blocks are tested as independent leakage registers.

Proof. With respect to $\mathcal{H}_{16}^{(4)} = Q\mathcal{H}_{16}^{(4)} \oplus Q_\perp\mathcal{H}_{16}^{(4)}$, the operator $\mathcal{M}_{16}$ decomposes into four blocks. The two diagonal blocks preserve the selected and complementary sectors. The two off-diagonal blocks are $Q_\perp\mathcal{M}_{16}Q$ and $Q\mathcal{M}_{16}Q_\perp$. If they vanish, $\mathcal{M}_{16}$ is block diagonal and commutes with Q. Conversely, if $[\mathcal{M}_{16}, Q] = 0$, multiplying on the left and right by Q and $Q_\perp$ gives vanishing of the off-diagonal blocks. Hence the mixed leakage register is zero exactly in the block-preserving case. □

Remark A.114.3 (Relation to HS(4) danger)**.** *The HS(4) carrier is the first level at which a naive 16-channel extension may behave like a sedenion-type expansion. This layer does not use raw sedenions as a free algebra. It uses $\mathcal{M}_{16}/\widehat{P}_{C_7}$ as a gated 16-channel envelope and records all off-sector motion as explicit leakage.*

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A.115 HS(5) two-sided duplication

Definition A.115.1 (Two-sided HS(5) carrier)**.** *The HS(5) carrier is the front/back duplication of the HS(4) envelope:*

$$\mathcal{H}_{32}^{(5)} := (\mathcal{H}_{16}^{(4)})_{\text{front}} \oplus (\mathcal{H}_{16}^{(4)})_{\text{back}}.$$

Its projector is

$$\widehat{P}_{C_7}^{(5)} := \begin{pmatrix} (\widehat{P}_{C_7})_{\text{front}} & 0 \\ 0 & (\widehat{P}_{C_7})_{\text{back}} \end{pmatrix}.$$

A two-sided HS(5) state is written

$$\Psi^{(5)} = (\Psi_{\text{front}}, \Psi_{\text{back}}),$$

where the front side carries readout/frame/language information and the back side carries structure/action/field information.

Definition A.115.2 (HS(5) cross-side debt)**.** *Let* $\mathcal{T}^{(5)} \in \mathrm{End}(\mathcal{H}_{32}^{(5)})$*. Define*

$$\mathsf{Debt}_{\text{cross}}^{(5)}(\mathcal{T}^{(5)}) := (\mathbf{1} - \widehat{P}_{C_7}^{(5)})\mathcal{T}^{(5)}\widehat{P}_{C_7}^{(5)} + \widehat{P}_{C_7}^{(5)}\mathcal{T}^{(5)}(\mathbf{1} - \widehat{P}_{C_7}^{(5)}).$$

The HS(5) two-sided carrier is debt-free when this register vanishes and all front/back comparison maps are mediated by declared comparison tensors or selectors.

Proposition A.115.3 (HS(5) as safe bridge)**.** *If the front and back HS(4) sectors are separately meta-septum closed and the cross-side debt register vanishes, then the HS(5) carrier transports the HS(3) Fano-compatible fibre through the two-sided readout grammar without adding an uncontrolled off-sector channel.*

Proof. Separate meta-septum closure prevents leakage within each 16-channel side. Vanishing cross-side debt prevents an admissible selected component on one side from being pushed into a complementary component on the other side. Since two-sided comparisons are only admitted through declared selectors or comparison tensors, every surviving front/back transition is registered. Hence no uncontrolled off-sector channel is introduced at HS(5). □

A.116 HS(6) Dixon carrier and 64-bit activation code

Definition A.116.1 (Dixon carrier)**.** *The HS(6) Dixon carrier is the 64-dimensional real carrier*

$$\mathcal{H}_{64}^{(6)} \cong \mathcal{H}_{\text{Dix}} := \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}.$$

A basis channel is indexed by a triple

$$\alpha = (a, b, c), \qquad a \in \{0, 1\}, \quad b \in \{0, 1, 2, 3\}, \quad c \in \{0, 1, \ldots, 7\}.$$

Thus

$$|\{\alpha\}| = 2 \cdot 4 \cdot 8 = 64.$$

Definition A.116.2 (Dixon activation code)**.** *The Dixon activation code of a captured OP5 route* $r \in \mathcal{R}_{\text{OP5}}^{\text{adm,near}K}$ *is a Boolean mask*

$$\chi_{\text{Dix}}(r) = (\chi_\alpha(r))_{\alpha \in \{0,1\}\times\{0,1,2,3\}\times\{0,\ldots,7\}} \in \{0, 1\}^{64}.$$

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The full Dixon activation space therefore has

$$|\mathsf{Code}_{\mathrm{Dix}}| = 2^{64}$$

possible masks. A mask is admissible only if it respects the inherited capture class, the HS(4) meta-septum gate, and the HS(5) two-sided debt-free condition.

Definition A.116.3 (HS(6) coding map)**.** *Define the conditional route-to-Dixon coding map*

$$\Xi^{\mathrm{loc}}_{\mathrm{Dix}} : \mathcal{R}^{\mathrm{adm,near}K}_{\mathrm{OP5}} \longrightarrow \{0,1\}^{64} \cup \{\mathsf{blocked}, \mathsf{debt}, \mathsf{leak}\}$$

as follows. For $r \in \mathcal{R}^{\mathrm{adm,near}K}_{\mathrm{OP5}}$:

(a) if r is blocked by the selector table, set $\Xi^{\mathrm{loc}}_{\mathrm{Dix}}(r) = \mathsf{blocked}$;

(b) if $\mathsf{Leak}^{(4)}_{C_7} \neq 0$, set $\Xi^{\mathrm{loc}}_{\mathrm{Dix}}(r) = \mathsf{leak}$;

(c) if $\mathsf{Debt}^{(5)}_{\mathrm{cross}} \neq 0$, set $\Xi^{\mathrm{loc}}_{\mathrm{Dix}}(r) = \mathsf{debt}$;

(d) otherwise assign the 64-bit mask $\chi_{\mathrm{Dix}}(r)$ determined by its captured chart, selector word, front/back side, and Fano/octonion channel.

A.117 Compatibility with the capture theorem

Lemma A.117.1 (Captured routes admit finite Dixon codes)**.** *Assume the capture/completeness theorem and the HS(4)–HS(5) zero-register conditions*

$$\mathsf{Leak}^{(4)}_{C_7} = 0, \qquad \mathsf{Debt}^{(5)}_{\mathrm{cross}} = 0.$$

Then every active captured route $r \in \mathcal{R}^{\mathrm{adm,near}K}_{\mathrm{OP5}}$ has a well-defined Dixon activation code $\chi_{\mathrm{Dix}}(r) \in \{0,1\}^{64}$.

Proof. By the local pass, each active route is captured in a certified normal-collar chart after finitely many HC/Hilbert steps, with $L_{\mathrm{unr}} = 0$ and $B_{\mathrm{HC/MM}} = 0$. The HS(4) condition prevents the Fano/meta-septum component from leaking into the complementary 16-channel sector. The HS(5) condition prevents two-sided front/back transport from creating unregistered debt. Therefore the remaining channel data are finite and live inside the HS(6) Dixon carrier. The 64 indexed basis channels define the Boolean mask. □

Proposition A.117.2 (Tunnel-to-Dixon reduction)**.** *Under the hypotheses of the previous lemma, every tunnel/capture class is assigned one of four statuses:*

$$\mathsf{blocked}, \qquad \mathsf{leak}, \qquad \mathsf{debt}, \qquad \chi_{\mathrm{Dix}} \in \{0,1\}^{64}.$$

Moreover, an uncontrolled HS(6) tunnel can occur only through a nonzero HS(4) leakage register, a nonzero HS(5) cross-side debt register, or a nonadmissible Dixon mask.

Proof. The coding map definition exhausts the cases. Blocked routes are not active OP5 exits. Nonzero HS(4) leakage and HS(5) debt are explicitly recorded, not hidden. If both vanish, the route is represented by a finite 64-bit Dixon mask. Hence any uncontrolled tunnel must be visible as leak, debt, or failure of Dixon-mask admissibility. □

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Theorem A.117.3 (HS(6) Dixon coding theorem)**.** *Assume the local proof chain photon-frame HC-halo capture chain. Suppose further that*

$$\mathsf{Leak}^{(4)}_{C_7} = 0, \qquad \mathsf{Debt}^{(5)}_{\mathrm{cross}} = 0, \qquad L_{\mathrm{unr}} = 0, \qquad B_{\mathrm{HC/MM}} = 0.$$

Then every declared admissible near-kernel OP5 route in the capture menu is either blocked by an inherited selector rule or encoded by an admissible 64-bit Dixon activation mask:

$$\Xi^{\mathrm{loc}}_{\mathrm{Dix}}(r) \in \{\mathsf{blocked}\} \cup \{0,1\}^{64}.$$

Consequently, within the declared HS(0)–HS(6) grammar, the remaining OP5 tunnel audit becomes a finite Dixon-mask admissibility audit rather than an unbounded route search.

Proof. The theorem captures all active near-kernel routes into the finite certified halo atlas and eliminates hidden leakage and HC/MM debt. The HS(4) condition eliminates meta-septum off-sector leakage. The HS(5) condition eliminates front/back cross-side debt. Thus every route left after selector blocking has finite chart, selector, side, and Fano/octonion channel data. These data select a subset of the 64 Dixon carrier channels, i.e. a Boolean mask in $\{0,1\}^{64}$. Therefore the audit is finite at the level of mask admissibility. □

Corollary A.117.4 (What would be needed next)**.** *A release-level OP5 closure candidate can now be pursued by proving that every Dixon mask actually produced by the capture map is admissible, or else blocked/absorbed by explicit HC/MM bookkeeping:*

$$\mathrm{im}(\Xi^{\mathrm{loc}}_{\mathrm{Dix}}) \subseteq \mathsf{Code}^{\mathrm{adm}}_{\mathrm{Dix}} \cup \{\mathsf{blocked}\},$$

with no leak and no debt registers.

A.118 First symbolic Dixon status table

The following table records the first symbolic status assignment. It is an integration table for the route grammar, not a numerical particle-spectrum table.

class	HS(4) leakage	HS(5) debt	Dixon carrier	mask status	status
A_1 collar interior	0	0	64	admissible pending full mask audit	Dixon-coded
A_2 S_{12} routed	0	0	64	admissible pending full mask audit	Dixon-coded
A_3 S_{23} routed	0	0	64	admissible pending full mask audit	Dixon-coded
A_4 S_{13} blocked	–	–	–	–	blocked
A_5 HC/MM face	0	0	64	absorbed/coded if debt-free	coded/absorbed
A_6 overlap edge	0	0	64	overlap-preserved	Dixon-coded

Remark A.118.1 (No spectrum or Standard Model overclaim)**.** *The Dixon carrier is used here as a finite route-code carrier. This layer does not derive the Standard Model spectrum, CKM/PMNS values, quark masses, QCD confinement, or OP1/OP4. It only states how captured OP5 halo routes can be mapped into a finite 64-channel Dixon activation grammar.*

A.119 Audit target for

The next meaningful layer is not a new shift. It is the image audit of the coding map:

$$\mathrm{im}(\Xi^{\mathrm{loc}}_{\mathrm{Dix}}) \overset{?}{\subseteq} \mathsf{Code}^{\mathrm{adm}}_{\mathrm{Dix}} \cup \{\mathsf{blocked}\}.$$

Equivalently, should enumerate the Dixon-mask classes actually reachable from the capture table and classify each as blocked, HC/MM absorbed, atlas routed, Dixon-admissible, or residual-open. If that image audit is empty on residual-open masks, then the OP5 closure candidate becomes substantially stronger inside the declared HS(0)–HS(6) grammar.

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Integrated Delta Layer: Route-Exhaustiveness Audit

Abstract

This delta layer executes the image audit requested after the local pass. The layer mapped captured near-kernel OP5 routes into the HS(6) Dixon carrier $\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$, with 64 real carrier channels and a 2^{64}-state Boolean activation-mask space. The present layer audits only the masks actually reachable from the capture table. It classifies the symbolic image of $\Xi^{\mathrm{loc}}_{\mathrm{Dix}}$ into blocked, HC/MM absorbed, atlas routed, Dixon-admissible, and residual-open classes. Under the inherited zero leakage/debt assumptions, the declared sample image has no residual-open Dixon-mask class. This remains a conditional local OP5 closure audit inside the declared photon-frame HC-halo and HS(0)–HS(6) grammar, not a global OP5 proof outside that grammar.

A.120 Why the Dixon representation helps QAM

The QAM grammar becomes more useful when it is not only expressive but also addressable. The Dixon layer supplies this addressability. Instead of leaving a captured route as an open structural phrase, it assigns the route a finite carrier address

$$\alpha = (a, b, c), \qquad a \in \{0, 1\}, \quad b \in \{0, 1, 2, 3\}, \quad c \in \{0, \ldots, 7\},$$

inside the 64-channel Dixon carrier

$$\mathcal{H}_{\mathrm{Dix}} = \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}.$$

The QAM state is then not a vague continuum of possible route meanings. It is a Boolean activation mask

$$\chi_{\mathrm{Dix}} \in \{0, 1\}^{64}.$$

Thus QAM over Dixon gives three concrete benefits.

(i) **Finite address space.** Captured routes are assigned to a finite 64-channel carrier instead of an unbounded route grammar.

(ii) **Auditability.** Every possible uncontrolled channel must appear as a non-admissible mask, an HS(4) leakage register, or an HS(5) cross-side debt register.

(iii) **Compatibility with existing algebra.** The complex, quaternionic, and octonionic factors separate phase, two-sided/spin-like transport, and Fano/octonion incidence without claiming that the Dixon carrier itself has dimension 2^{64}.

The present layer uses these benefits to audit the image of the route-to-Dixon map.

A.121 Inherited coding map

Recall the conditional coding map

$$\Xi^{\mathrm{loc}}_{\mathrm{Dix}} : \mathcal{R}^{\mathrm{adm,near}K}_{\mathrm{OP5}} \longrightarrow \{0, 1\}^{64} \cup \{\mathsf{blocked}, \mathsf{debt}, \mathsf{leak}\}.$$

It is defined only after the local proof chain photon-frame HC-halo capture chain has reduced admissible near-kernel OP5 routes to the finite class menu

$$A_{\mathrm{loc}} = \{A_1, A_2, A_3, A_4, A_5, A_6\}.$$

The audit does not enumerate all 2^{64} masks. It enumerates the much smaller symbolic image of $\Xi^{\mathrm{loc}}_{\mathrm{Dix}}$ produced by A_{loc}.

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Definition A.121.1 (Reachable Dixon image)**.** *The reachable Dixon image is*

$$\mathcal{I} := \Xi^{\mathrm{loc}}_{\mathrm{Dix}}(\mathcal{R}^{\mathrm{adm,near}K}_{\mathrm{OP5}}) \cap \big(\{0,1\}^{64} \cup \{\mathsf{block}, \mathsf{abs}, \mathsf{Route}, \mathsf{Leak}, \mathsf{Debt}, \mathsf{res}\}\big),$$

where each element is represented by its class, its HS(4) leakage status, its HS(5) cross-side debt status, and its Dixon-mask class.

Definition A.121.2 (Compressed Dixon mask class)**.** *For a captured route class A_i, define its compressed Dixon mask class by*

$$\mathfrak{M}_i := \big(\sigma_i, \tau_i, \omega_i, \kappa_i\big),$$

where σ_i is the selector word, τ_i is the two-sided front/back tag, $\omega_i \subseteq \{0,\dots,7\}$ is the Fano/octonion support, and κ_i is the response status. The associated full mask is the subset of Dixon channels

$$\mathrm{supp}(\chi_{\mathrm{Dix}}(A_i)) \subseteq \{0,1\} \times \{0,1,2,3\} \times \{0,\dots,7\}.$$

Remark A.121.3 (Symbolic rather than exhaustive bit enumeration)**.** *The audit enumerates mask families forced by the capture table. It does not list all 2^{64} bitstrings. This is the intended gain: OP5 does not require an exhaustive scan of the Dixon mask space if the route image is first captured into finitely many symbolic classes.*

A.122 Status alphabet and admissibility test

Definition A.122.1 (status alphabet)**.** *The image audit uses the status alphabet*

$$\mathfrak{S}_{\mathrm{loc}} = \{\mathsf{block}, \mathsf{abs}, \mathsf{Route}, \mathsf{adm}, \mathsf{res}\}.$$

The meanings are as follows.

- block*: the route is selector-blocked and produces no active OP5 exit.*
- abs*: the route is HC/MM absorbed with explicit zero debt.*
- Route*: the route is atlas-routed to another certified normal-collar piece.*
- adm*: the route is Dixon-admissible as a finite 64-channel activation mask.*
- res*: the route remains residual-open.*

Definition A.122.2 (Dixon admissibility predicate)**.** *A reachable Dixon mask class $\mathfrak{M}_i$ is the local pass-admissible when the following four tests hold:*

$$\mathsf{Leak}^{(4)}_{C_7}(A_i) = 0, \qquad \mathsf{Debt}^{(5)}_{\mathrm{cross}}(A_i) = 0,$$

$$L_{\mathrm{unr}}(A_i) = 0, \qquad B_{\mathrm{HC/MM}}(A_i) = 0,$$

and the selector/overlap word of A_i is the local pass-admissible. We write this as

$$\mathsf{Adm}^{\mathrm{loc}}_{\mathrm{Dix}}(\mathfrak{M}_i) = 1.$$

Lemma A.122.3 (Image audit reduction)**.** *Assume the capture table is exhaustive. Then the residual-open part of the Dixon image is*

$$\mathcal{I}_{\mathsf{res}} = \left\{ \mathfrak{M}_i \;:\; \begin{array}{l} A_i \in A_{\mathrm{loc}}, \\ \mathsf{Adm}^{\mathrm{loc}}_{\mathrm{Dix}}(\mathfrak{M}_i) = 0, \\ A_i \text{ is neither blocked nor HC/MM absorbed} \end{array} \right\}.$$

Thus the residual audit is finite at the level of captured symbolic classes.

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Proof. By the local pass, every declared admissible near-kernel route belongs to one of the classes A_i. By the local pass, every such class is either blocked, marked leak/debt, or assigned a Dixon mask. The predicate records exactly the leak, debt, hidden leakage, HC/MM debt, and selector/overlap admissibility tests. Hence any residual-open element must be one of the finitely many captured classes failing this predicate while not being otherwise blocked or absorbed. □

A.123 image table

The following table is the first Dixon image audit of the capture classes. The supports are symbolic; they specify which factor of the Dixon carrier is active, not a particle spectrum.

class	route meaning	HS(4) leak	HS(5) debt	Dixon support class	status	residual reason
A_1	collar interior	0	0	neutral collar support in $\mathbb{C}\otimes\mathbb{H}\otimes\mathbb{O}$	adm	none
A_2	S_{12} switch	0	0	routed selector edge support	Route	none
A_3	S_{23} switch	0	0	routed selector edge support	Route	none
A_4	non-adjacent S_{13}	–	–	no active mask	block	blocked by the local pass
A_5	HC/MM bookkeeping face	0	0	absorbed neutral/debt-free face	abs	none
A_6	overlap edge	0	0	overlap-preserved support	Route or adm	none

Proposition A.123.1 (No residual-open class in the declared sample image)**.** *For the declared symbolic image table, under the inherited assumptions*

$$\mathsf{Leak}^{(4)}_{C_7} = 0, \qquad \mathsf{Debt}^{(5)}_{\mathrm{cross}} = 0, \qquad L_{\mathrm{unr}} = 0, \qquad B_{\mathrm{HC/MM}} = 0,$$

and with the selector/overlap decisions unchanged, one has

$$\mathcal{I}_{\mathsf{res}} = \varnothing.$$

Proof. Class A_4 is blocked, hence it is not residual-open. Class A_5 is HC/MM absorbed with zero debt, hence it is not residual-open. Classes A_2, A_3, A_6 are routed through the selector/overlap grammar and preserve contraction margins, hence they are not residual-open. Class A_1 remains inside the certified collar interior and receives an admissible Dixon support class. Since the table contains all declared classes, no residual-open symbolic image element remains. □

A.124 Mask-image theorem

Theorem A.124.1 (Dixon image audit theorem)**.** *Assume the local proof chain photon-frame HC-halo capture chain, the HS(4)–HS(6) coding hypotheses, and the image table above. If the capture map is exhaustive over the declared near-kernel OP5 route grammar, then*

$$\mathrm{im}(\Xi^{\mathrm{loc}}_{\mathrm{Dix}}) \subseteq \mathsf{Code}^{\mathrm{adm}}_{\mathrm{Dix}} \cup \{\mathsf{block}, \mathsf{abs}, \mathsf{Route}\},$$

with no residual-open Dixon image class. Equivalently, within the declared HS(0)–HS(6) grammar, the remaining OP5 audit is reduced to the independent exhaustiveness check of the capture map and the zero-register checks for HS(4), HS(5), hidden leakage, and HC/MM debt.

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Proof. The capture map reduces the route grammar to $A_1, \ldots, A_6$. The map sends each captured route to a Dixon mask or to an explicit blocked/leak/debt status. The table evaluates these statuses. Under the zero leakage/debt assumptions, no class is mapped to leak or debt. The non-adjacent selector class is blocked, the HC/MM face is absorbed debt-free, the selector and overlap edges are routed, and the collar interior is Dixon-admissible. Therefore the image contains only admissible Dixon masks and the three controlled non-residual statuses. □

Corollary A.124.2 (OP5 consequence inside the declared grammar)**.** *Inside the declared OP5=photon-frame HC-cone grammar, the audit leaves no uncontrolled OP5 exit in the captured HS(0)–HS(6) image, provided the capture map is exhaustive and all zero-register hypotheses remain true.*

Remark A.124.3 (What has not been proven)**.** *This is not a global OP5 proof outside the declared grammar. The decisive remaining mathematical work is to audit the full Active-Core route menu for any admissible near-kernel OP5 branch not represented by $A_1, \ldots, A_6$, and to preserve the zero-register conditions under integration into the monolithic source.*

A.125 Next audit target

The next useful layer should not add a new algebraic carrier. It should perform one of two integration checks:

(a) a full Active-Core route-menu scan showing that every admissible near-kernel OP5 branch is represented by the capture classes; or

(b) a monolithic integration of the local proof chain into the active source, followed by separate label, reference, layout, and route-menu residual audits.

If both checks pass, OP5 can be stated as locally closed inside the declared photon-frame HC-halo/Dixon-coding grammar, with the remaining caveat being only the scope of that grammar.

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Integrated Delta Layer: Integrated Residual Register

Abstract

This layer audits the available route-menu surface after the local proof chain. Its purpose is not to introduce a new algebraic mechanism. It checks whether the local photon-frame HC-halo/Dixon coding chain leaves any declared admissible near-kernel OP5 route outside the finite capture menu $A_1, \ldots, A_6$. The result is deliberately scoped: the local proof chain delta stack is internally exhausted by the local proof chain classification, while the available monolithic basis source is only the local proof line. Therefore this layer certifies the declared delta-stack route image and records the exact remaining condition for a future source-native monolithic freeze: no active OP5 branch in the full Active Core may remain outside the capture classes or the named zero-register tests.

A.126 Audit scope

The available source surface consists of the local proof line monolithic Active Core basis and the local proof chain delta layers. Hence performs two separate checks.

1. **Delta-stack route audit.** This scans the local proof chain local photon-frame HC-halo, tunnel, finite-atlas, capture, and Dixon-coding chain.
2. **Basis-source compatibility scan.** This records the older the local pass OP5 route language that predates the Halo/Dixon grammar and must later be mapped or explicitly declared out of scope during a monolithic integration.

Remark A.126.1 (No false monolithic claim)**.** *The present file is a source-level audit layer, not a source-native rebuild of a single full the local proof line Active Core. It therefore proves a local exhaustiveness claim for the declared the local proof chain grammar and lists the remaining monolithic audit obligation separately.*

A.127 Delta-stack route menu

The local proof chain reduces the local near-kernel OP5 problem as follows:

$$\begin{gathered}\text{robust photon chart} \to \text{finite atlas} \to \text{double-}\varepsilon\text{ halo},\\ \text{double-}\varepsilon\text{ halo} \to \text{rank/selector/overlap registers} \to \text{capture classes},\\ \text{capture classes} \to \text{Dixon image audit}.\end{gathered}$$

The effective route alphabet after is

$$A_{\mathrm{loc}} = \{A_1, A_2, A_3, A_4, A_5, A_6\}.$$

The Dixon audit assigns these classes the following controlled statuses.

class	route meaning	status
A_1	collar interior	adm
A_2	S_{12} selector switch	route
A_3	S_{23} selector switch	route
A_4	non-adjacent S_{13} selector	block
A_5	HC/MM bookkeeping face	abs
A_6	atlas-overlap edge	route or adm

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Definition A.127.1 (residual support)**.** *Let* $\Xi^{\mathrm{loc}}_{\mathrm{Dix}}$ *be the route-to-Dixon map of and let* $\pi^{\mathrm{loc}}_{\mathrm{cap}}$ *be the capture map. The residual support is*

$$\mathcal{I}^{\mathrm{loc}}_{\mathsf{res}} := \left\{ r \in \mathcal{R}^{\mathsf{adm},\mathrm{near}\,K}_{\mathrm{OP5}} : \begin{array}{l} \pi^{\mathrm{loc}}_{\mathrm{cap}}(r) \notin \{A_1,\dots,A_6\}, \ \text{or} \\ \Xi^{\mathrm{loc}}_{\mathrm{Dix}}(r) \in \{\mathsf{leak},\mathsf{debt},\mathsf{res}\} \end{array} \right\}.$$

Thus $\mathcal{I}^{\mathrm{loc}}_{\mathsf{res}}$ *records exactly the active near-kernel routes that are either uncaptured or mapped to a non-controlled Dixon status.*

Proposition A.127.2 (Declared delta-stack residual vanishing)**.** *Inside the declared the local proof chain photon-frame* HC*-halo/Dixon grammar, the source scan finds no declared route class beyond* $A_1,\dots,A_6$*, and maps those classes only to* adm*,* block*,* abs*, or* route*. Hence*

$$\mathcal{I}^{\mathrm{loc}}_{\mathsf{res}} = \varnothing$$

for the declared delta-stack image.

Proof. The tunnel menu is reduced by the local proof chain to radial/HC boundary, selector, rank, and overlap checks. The normal-collar certificate removes rank-drop on certified pieces. The selector/overlap table blocks or routes the switch and overlap classes. The capture table enumerates the six remaining local near-kernel classes $A_1,\dots,A_6$. The Dixon map sends captured active classes into the 64-channel Dixon carrier or to explicit blocked/leak/debt statuses. The image table assigns A_1 to admissible, A_2, A_3, A_6 to routed/admissible, A_4 to blocked, and A_5 to absorbed. No table row is assigned leak, debt, or res. Therefore the declared local image has empty residual support. □

A.128 Source scan facts

The source scan of the local proof chain delta files produced the following structural metadata.

layer	lines	labels	raw lower-case halo	raw lower-case exit
the local pass	239	0	0	0
the local pass	245	0	0	0
the local pass	293	0	0	0
the local pass	317	0	1	1
the local pass	276	0	0	0
the local pass	257	0	0	0
the local pass	284	0	0	0
the local pass	296	0	0	0
the local pass	387	0	0	0
the local pass	277	0	0	0

The single lower-case occurrences are compatibility aliases in the source; they are not active rendered notation in the local proof chain line. No duplicate labels were found across the local proof chain delta files.

The available monolithic basis contains the older selector-gated OP5 route language and does not yet contain the later Halo, near-kernel, or Dixon route grammar as an integrated source-native block. The scan counts were:

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anchor	count
OP~5	567
OP5	133
selector-gated reachability	13
TE budgets	4
HC/MM residual	2
post-promotion residual	7
OP~4	55
Dixon	8
Halo	0
near-kernel	0

This confirms that the local proof chain are a new local closure grammar over the basis, not yet a source-native monolithic replacement of every older OP5 paragraph.

A.129 Compatibility with the OP5 route surface

The selector-gated OP5(B) statement leaves four route obligations after a first reachability pass: TE budgets, HC/MM residual compatibility, post-promotion residual control, and OP4 stability. The local proof chain directly attacks the near-kernel HC-cone route obstruction, not the whole OP5(B) boundary-promotion package.

Definition A.129.1 (Monolithic residual register)**.** *For a future full Active-Core integration define*

$$\mathcal{R}^{\mathrm{loc}}_{\mathrm{mono}} := \mathcal{R}_{\mathrm{uncap}} \cup \mathcal{R}_{\mathrm{HS4\ leak}} \cup \mathcal{R}_{\mathrm{HS5\ debt}} \cup \mathcal{R}_{L_{\mathrm{unr}}} \cup \mathcal{R}_{B_{\mathrm{HC/MM}}} \cup \mathcal{R}_{\mathrm{loc\text{-}res}}.$$

Here $\mathcal{R}_{\mathrm{uncap}}$ *denotes active near-kernel routes not mapped by* $\pi^{\mathrm{loc}}_{\mathrm{cap}}$*, while* $\mathcal{R}_{\mathrm{loc\text{-}res}}$ *denotes any older OP5(B) budget, post-promotion, or OP4-stability condition not identified with the local* HC*-cone grammar.*

Theorem A.129.2 (route-menu exhaustiveness theorem, scoped form)**.** *Assume the working identification*

$$\mathrm{OP5}_{\mathrm{near}\,K} = \textit{photon-frame}\ \mathrm{HC}\textit{-cone source-return reflector}$$

and assume that the capture map is exhaustive over the declared near-kernel route grammar. If

$$\mathcal{R}_{\mathrm{HS4\ leak}} = \mathcal{R}_{\mathrm{HS5\ debt}} = \mathcal{R}_{L_{\mathrm{unr}}} = \mathcal{R}_{B_{\mathrm{HC/MM}}} = \varnothing$$

and the selector/overlap decisions are preserved under integration, then there is no uncontrolled near-kernel OP5 exit in the local proof-chain delta stack route menu.

If, in addition, a future monolithic scan proves

$$\mathcal{R}_{\mathrm{uncap}} = \varnothing \qquad \textit{and} \qquad \mathcal{R}_{\mathrm{loc\text{-}res}} = \varnothing,$$

then the same statement may be promoted from a local delta-stack closure theorem to a monolithic local OP5 closure theorem inside the declared HC*-halo/Dixon grammar.*

Proof. The first part is exactly Proposition 1 combined with the zero-register hypotheses: every captured near-kernel route is admissible, blocked, absorbed, or routed, and no leak/debt/residual status remains. The second part adds the only missing monolithic conditions. If no older active branch lies outside the capture map and no residual obligation remains unmapped, the local exhaustiveness statement covers the integrated route menu for the declared grammar. □

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Corollary A.129.3 (Current OP5 status after)**.** *The local proof chain local route image is exhausted. The current remaining OP5 burden is no longer a free route explosion; it is the monolithic integration check*

$$\mathcal{R}^{\mathrm{loc}}_{\mathrm{mono}} = \varnothing.$$

Thus OP5 is locally closed inside the declared delta-stack grammar and remains open only at the full Active-Core integration/exhaustiveness boundary.

A.130 Next integration rule

The next source-native step should not add another algebraic layer. It should merge or append the local proof chain into the active source and run a residual search with the following pass/fail test:

$$\operatorname{supp}\big(\mathcal{R}^{\mathsf{adm},\mathrm{near}\,K}_{\mathrm{OP5}}\big) \subseteq \bigcup_{i=1}^{6} (\pi^{\mathrm{loc}}_{\mathrm{cap}})^{-1}(A_i) \cup \{\mathsf{block}, \mathsf{abs}, \mathsf{route}, \mathsf{adm}\}.$$

Any hit outside this set must be named explicitly as one of: uncaptured route, HS4 leakage, HS5 cross-side debt, hidden leakage, HC/MM debt, post-promotion residual, TE-budget residual, or OP4-stability residual.

A.131 Conservativity guards

- does not claim global OP5 closure outside the declared photon-frame HC-halo/Dixon grammar.
- does not claim OP1, OP4, QCD/confinement, CKM/PMNS, quark masses, or a Standard Model spectrum derivation.
- The Dixon carrier is 64-dimensional; 2^{64} is the Boolean activation mask count, not the Dixon dimension.
- HC/MM absorption remains explicit bookkeeping with zero unpaid debt, not residual deletion.
- A source-native monolithic rebuild remains required before making a public release-style OP5 closure statement.

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Monolithic OP5 Promotion Audit

Definition A.131.1 (monolithic OP5 residual register)**.** *Let $\mathcal{R}^{\mathrm{loc}}_{\mathrm{mono}}$ denote the set of active OP5-facing near-kernel branches in the integrated Active-Core source which are not assigned to one of the capture classes*

$$A_1, \dots, A_6,$$

and are not discharged by the named zero-register tests

$$L_{\mathrm{unr}} = 0, \qquad B_{\mathrm{HC/MM}} = 0, \qquad R_{\mathrm{HS4}} = 0, \qquad D_{\mathrm{HS5}} = 0.$$

A hit is counted as residual only if it is active in the current local proof line OP5-facing grammar, not merely preserved historical witness language.

Proposition A.131.2 (integrated delta-stack exhaustion)**.** *In the integrated local proof chain delta stack, the local photon-frame HC-halo/Dixon route image is exhausted by the six classes*

$$A_1, \dots, A_6.$$

Their status assignment remains

$$A_1 \mapsto \mathsf{adm}, \quad A_2 \mapsto \mathsf{route}, \quad A_3 \mapsto \mathsf{route}, \quad A_4 \mapsto \mathsf{block}, \quad A_5 \mapsto \mathsf{abs}, \quad A_6 \mapsto \mathsf{route} \text{ } \textit{or} \text{ } \mathsf{adm}.$$

Consequently the residual Dixon image inside the declared local grammar is empty:

$$\mathcal{I}^{\mathrm{loc}}_{\mathrm{res}} = \varnothing.$$

Proof. This is the source-native integration of the audit conclusion with the local proof chain delta bodies included in the same Active-Core source. The local proof-chain layers provide the robust local chart and finite-atlas candidate, the local proof chain reduce the OP5 route question to halo/tunnel, rank, selector, and overlap registers, supplies the finite capture menu, maps captured routes to the 64-dimensional Dixon carrier with a 2^{64}-state mask space, audits the reachable Dixon image, and verifies that no residual-open class remains in that declared delta image. □

Theorem A.131.3 (local OP5 closure in the declared HC-halo/Dixon grammar)**.** *Assume the working identification that the OP5-facing near-kernel branch is the photon-frame HC-cone/reflector route, and assume the inherited zero-register conditions*

$$L_{\mathrm{unr}} = 0, \qquad B_{\mathrm{HC/MM}} = 0, \qquad R_{\mathrm{HS4}} = 0, \qquad D_{\mathrm{HS5}} = 0.$$

Then the integrated local proof chain Active-Core delta stack leaves no uncontrolled local OP5 exit inside the declared photon-frame HC-halo/Dixon grammar.

Proof. By the capture layer, every declared admissible near-kernel route in the grammar is assigned to $A_1, \dots, A_6$, provided the capture map is exhaustive. By the selector/overlap layer, the active switch and overlap channels are blocked, routed, absorbed, or contraction-preserving. By the Dixon-image layers, every captured route has either a Dixon-admissible 64-channel mask or is classified as blocked, routed, or HC/MM-absorbed. The assumed zero-register conditions remove hidden leakage, unpaid HC/MM absorption, HS(4) meta-septum leakage, and HS(5) cross-side debt. Hence no residual-open OP5 class remains in the declared local grammar. □

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Remark A.131.4 (Scope of the promotion)**.** *This statement promotes the local proof-chain delta stack result into the available Active-Core source as a local OP5 closure theorem inside the declared HC-halo/Dixon grammar. It does not claim global OP5 closure outside that grammar, does not close OP1 or OP4, and does not derive QCD/confinement, CKM/PMNS values, quark masses, or a Standard Model spectrum.*

Remark A.131.5 (Inherited historical OP5 language)**.** *The older Active-Core basis contains preserved historical OP1–OP5 and OP5(B) statements from earlier release lines. In those blocks remain theorem-memory and genealogy unless explicitly referenced by the current HC-halo/Dixon grammar. The governing current promotion claim is the local theorem above.*

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OP5 Closure Hardening and Old-Branch Mapping Audit

Target-version rule. This audit belongs to the provenance layer. In the reader-facing proof spine it is interpreted semantically, not as a mathematical version marker. It hardens the local OP5 promotion and maps inherited OP5 language into current, historical, guard, or outside-scope status.

Definition A.131.6 (governing OP5 status)**.** *The governing OP5 status is that OP5 is locally closed inside the declared photon-frame HC-halo/Dixon grammar under*

$$L_{\mathrm{unr}} = 0, \quad B_{\mathrm{HC/MM}} = 0, \quad R_{\mathrm{HS4}} = 0, \quad D_{\mathrm{HS5}} = 0.$$

This is not a global OP5 theorem outside that grammar and does not close OP1 or OP4.

Proposition A.131.7 (no-active-old-branch promotion rule)**.** *An inherited OP5-facing paragraph may be read as part of the current OP5 closure claim only if it is explicitly mapped into the HC-halo/Dixon grammar or is one of the named zero-register guards. Otherwise it is historical theorem-memory, a guard, or outside the scope of the local theorem.*

Theorem A.131.8 (compact OP5 closure spine)**.** *Inside the declared photon-frame HC-halo/Dixon grammar, OP5 has the finite proof spine*

$$\textit{HC cone} \Rightarrow \textit{double-}\varepsilon\textit{ halo} \Rightarrow \textit{tunnel reduction} \Rightarrow \textit{normal collar}$$

$$\Rightarrow \textit{selector/overlap routing} \Rightarrow \textit{Dixon image audit} \Rightarrow \textit{no residual local OP5 exit.}$$

Under the four zero-register assumptions, the local residual register of this spine is empty.

Remark A.131.9 (No version promotion)**.** *The present document is a v4.3.0 release-final. A later public line should be reserved for broader OP4/OP1 promotion or for a genuinely new architecture beyond the OP5 hardening chain.*

A.132 Finite Perturbation Threshold Atlas for the Triolic Orbit Scan

This section records a compact finite threshold atlas for the alternating triolic orbit-scan model. The purpose is not to promote the toy chart to a physical trajectory simulation, but to make the robustness statement falsifiable at the finite readout level.

Let the finite scan diagnostics be

$$\widetilde{E}_{N,M}, \qquad \widetilde{R}^{\min}_{N,M}, \qquad C^{\mathrm{even}}_{95}, \qquad C^{\mathrm{odd}}_{05},$$

where $\widetilde{E}_{N,M}$ is the bifocal defect, $\widetilde{R}^{\min}_{N,M}$ is the smallest guard margin, C^{even}_{95} is the 95-percentile even-sector core-sightline residual, and C^{odd}_{05} is the 5-percentile odd-sector complementarity residual.

Definition A.132.1 (Finite threshold predicate)**.** *A perturbed finite scan is threshold-admissible when*

$$\widetilde{E}_{N,M} \le \tau_E, \qquad \widetilde{R}^{\min}_{N,M} > 0, \qquad C^{\mathrm{even}}_{95} \le \tau_C, \qquad C^{\mathrm{odd}}_{05} \ge \tau_O.$$

The predicate is a finite readout certificate only. It does not assert a physical equation of motion.

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Lemma A.132.2 (Index-level parity is threshold-invariant)**.** *For the shifted Kronecker transport*

$$\Sigma_L{}^{i}{}_{j} = \delta^{i}_{j+1 \bmod L},$$

the even–odd identities

$$\Pi_0\Sigma_L = \Sigma_L\Pi_1, \qquad \Pi_1\Sigma_L = \Sigma_L\Pi_0, \qquad \Pi_r\Sigma_L^2 = \Sigma_L^2\Pi_r$$

are independent of the geometric perturbation amplitudes.

Proposition A.132.3 (Finite threshold-atlas reading)**.** *The finite threshold atlas separates three layers:*

$$\textit{index parity}, \qquad \textit{protected-core guard}, \qquad \textit{geometric triolic balance}.$$

The first is exact under the shifted Kronecker transport. The second is certified by positive guard margin. The third is a genuine geometric residual and can fail under core-offset or antipodal-breaking perturbations.

Remark A.132.4 (Compact numerical atlas instance)**.** *In the compact finite atlas run, the one-parameter scans estimate the following last-stable / first-unstable transition windows:*

parameter	*last stable*	*first unstable*
core offset	0.030	0.045
antipodal-breaking jitter	0.080	*none*
lens asymmetry	0.315	0.360
symmetric radial ripple	0.030	0.045
twist amplitude	0.080	*none*

The atlas is not a physical proof. It is a reproducible diagnostic showing which perturbations preserve the local orbit-scan readout and which ones create a named obstruction.

Theorem A.132.5 (Finite threshold-atlas guard)**.** *A finite orbit-scan atlas may be used as a local readout certificate only when the threshold predicate holds. If the guard remains positive but the even-sector core-sightline residual exceeds threshold, then the failure is not a through-core event; it is a triolic-balance obstruction.*

A.133 Finite Core-Recentering Calibration for the Triolic Orbit Scan

The threshold atlas identifies core offset as the most sensitive finite obstruction for the triolic orbit scan. This section records the corresponding calibration layer. The purpose is conservative: a displaced declared core is not interpreted as a permitted through-core motion. It is treated as a calibration defect unless the residual remains after recentering.

Let $\mathcal{I}_0$ denote the even-sector sample indices and let (p_n, q_n), $n \in \mathcal{I}_0$, be the paired even-sector readout points. The finite midpoint estimator is

$$\widehat{K} := \frac{1}{|\mathcal{I}_0|} \sum_{n \in \mathcal{I}_0} \frac{p_n + q_n}{2}.$$

If K_0 is the declared collapse centre, the recentering displacement is

$$\Delta K := \widehat{K} - K_0.$$

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Definition A.133.1 (Recentered triolic residuals)**.** *For each even pair define*

$$C_n(K) := \frac{\|(p_n - K) \times (q_n - K)\|}{\|p_n - K\| \, \|q_n - K\|}, \qquad M_n(K) := \left\| K - \frac{p_n + q_n}{2} \right\|.$$

The pre-calibration residuals are $C_n(K_0), M_n(K_0)$*. The recentered residuals are* $C_n(\widehat{K}), M_n(\widehat{K})$*.*

Lemma A.133.2 (Midpoint recovery in the antipodal case)**.** *Assume that all even-sector pairs are antipodal about a common point* K_**, that is*

$$\frac{p_n + q_n}{2} = K_* \qquad (n \in \mathcal{I}_0).$$

Then $\widehat{K} = K_*$*. In particular, if the declared centre* K_0 *is offset from the actual finite balance centre, the estimator recovers the balance centre from the paired readout data.*

Proof. The estimator is the average of the pair midpoints. If every midpoint equals K_*, their average equals K_*. □

Proposition A.133.3 (Finite recentering criterion)**.** *A finite core-offset defect is called calibratable when*

$$C_{95}(\widehat{K}) \le \tau_C, \qquad R^{\min}(\widehat{K}) > 0,$$

and the displacement ΔK *is recorded as a calibration displacement, not as a through-core route. If these inequalities fail, the residual is a persistent triolic-balance obstruction.*

Remark A.133.4 (Numerical calibration instance)**.** *In the finite calibration probe, the uncalibrated core-offset threshold was approximately*

$$\textit{last stable} = 0.030, \qquad \textit{first unstable} = 0.040.$$

After midpoint recentering, the scan remained stable through the tested range, with

$$\max C_{95}(\widehat{K}) \approx 6.639e-16, \qquad \min R^{\min}(\widehat{K}) \approx 1.180 > 0.$$

Thus the tested core-offset failure is a calibratable centre-declaration defect in this finite readout chart, not a permitted crossing of the protected core.

Theorem A.133.5 (Core-recentering guard)**.** *Finite recentering can repair a triolic scan only as a readout calibration if three conditions hold: the recentered even-sector residual is below threshold, the protected-core guard remains positive, and the displacement* ΔK *is reported as calibration provenance. Under these conditions the repaired scan may be used as a local readout certificate, but it does not promote the orbit to a physical through-core trajectory.*

Proof. The residual condition restores the finite even-sector core-sightline test. The positive guard condition keeps all sampled points outside the protected core. Recording ΔK prevents the calibration from being mistaken for a new dynamical claim. Hence the procedure repairs the declared centre of the readout chart only; it does not assert a through-core motion. □

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A.134 Crystal–Fractal Readout Probe from Calibrated Orbit Scans

The calibrated triolic orbit scan separates a centre-calibration problem from a residual-geometry problem. This makes the crystal and fractal readings operational: the crystal reading concerns coherence across directions, while the fractal reading concerns recurrence across scales.

Definition A.134.1 (Ringwise calibrated balance centres)**.** *For each scan ring ℓ, define the even-triole balance centre*

$$\widehat{K}_\ell := \frac{1}{|\mathcal{I}_0|} \sum_{n \in \mathcal{I}_0} \frac{p_{\ell n} + q_{\ell n}}{2}.$$

The calibrated global centre is

$$\widehat{K} := \frac{1}{L} \sum_{\ell=1}^{L} \widehat{K}_\ell.$$

The finite crystal-coherence residual is

$$\mathcal{C}_{\mathrm{crys}} := \left(\frac{1}{L} \sum_{\ell=1}^{L} \|\widehat{K}_\ell - \widehat{K}\|^2 \right)^{1/2}.$$

Definition A.134.2 (Residual readout set across scales)**.** *Let $s \in S$ range over a finite set of scan scales. For each scale, let $\Sigma_K^{\Gamma,s}$ be the finite periapsis envelope and let*

$$\mathcal{R}_s := \{x_{\ell m}^s : \mathcal{E}_F(x_{\ell m}^s) > \tau_{\mathrm{res}}\}$$

be the residual readout set above the declared bifocal-defect threshold. The multiscale residual readout is

$$\mathcal{R}_{\mathrm{multi}} := \bigcup_{s \in S} \mathcal{R}_s.$$

Definition A.134.3 (Finite residual box-counting diagnostic)**.** *For box length $\delta > 0$, let $N_{\mathcal{R}}(\delta)$ be the number of occupied boxes meeting $\mathcal{R}_{\mathrm{multi}}$. A finite scaling diagnostic is obtained by fitting*

$$\log N_{\mathcal{R}}(\delta) \approx D_{\mathrm{res}} \log(1/\delta) + b.$$

The number D_{res} is a finite residual-scaling estimate, not a proof of Hausdorff dimension.

Proposition A.134.4 (Crystal–fractal separation)**.** *The calibrated orbit-scan data split into two independent readouts:*

$$\mathcal{C}_{\mathrm{crys}} \quad \text{and} \quad D_{\mathrm{res}}.$$

Small $\mathcal{C}_{\mathrm{crys}}$ certifies coherent directional centering, i.e. a crystal-like local cell centre. Nontrivial, scale-persistent $\mathcal{R}_{\mathrm{multi}}$ certifies residual scaling structure, i.e. a fractal-like defect readout. Neither statement asserts a physical trajectory through the protected core.

Remark A.134.5 (Finite diagnostic instance)**.** *In the finite toy diagnostic, the maximum ring-centre RMS spread was*

$$\mathcal{C}_{\mathrm{crys}}^{\max} \approx 8.699e-17,$$

with 30216 residual points above threshold across the multiscale scan. The box-counting fit returned

$$D_{\mathrm{res}} \approx 2.435.$$

This is a diagnostic number: it indicates that residual structure can be measured across scales, not that the protected-core surface has already been identified with a mathematical fractal.

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Theorem A.134.6 (Readout guard for crystal and fractal claims)**.** *A crystal claim is admissible only as a statement about coherence of the calibrated centres $\widehat{K}_\ell$. A fractal claim is admissible only as a statement about the scaling of residual readout sets across declared scan scales. The orbit-scan data do not by themselves promote either claim to a global physical closure theorem.*

A.135 Residual Classification Atlas for Crystal–Fractal Readouts

The crystal–fractal readout probe shows that residual sets can be measured across directions and scales. The present atlas classifies the residuals into finite diagnostic classes. This is not a physical closure theorem; it is a readout classification layer.

Definition A.135.1 (Residual classification alphabet)**.** *Let $\mathcal{R}_{\mathrm{multi}}$ be the multiscale residual readout set. The finite classification alphabet is*

$$\mathfrak{A}_{\mathcal{R}} = \{\mathrm{crys}, \mathrm{frac}, \mathrm{lens}, \mathrm{guard}, \mathrm{noise}\}.$$

Here crys *denotes directional or angular recurrence,* frac *denotes recurrence across scan scales,* lens *denotes high bifocal stress,* guard *denotes proximity to the protected-core guard, and* noise *denotes the unclassified remainder under the declared finite rules.*

Definition A.135.2 (Class tags and primary class)**.** *A residual point $x \in \mathcal{R}_{\mathrm{multi}}$ may carry several tags:*

$$\tau_{\mathrm{crys}}(x), \quad \tau_{\mathrm{frac}}(x), \quad \tau_{\mathrm{lens}}(x), \quad \tau_{\mathrm{guard}}(x).$$

For disjoint counting, the primary class is assigned by the conservative order

$$\mathrm{guard} \succ \mathrm{lens} \succ \mathrm{frac} \succ \mathrm{crys} \succ \mathrm{noise}.$$

Thus a point close to the guard is counted as a guard residual even if it also has a scale-recurrence tag.

Proposition A.135.3 (Residual classification separation)**.** *The classification atlas separates four causes of residual structure:*

$$\textit{directional recurrence}, \quad \textit{scale recurrence}, \quad \textit{bifocal stress}, \quad \textit{guard proximity}.$$

Only the first two support crystal–fractal interpretation. Lens and guard classes are obstruction classes and must not be promoted to crystal or fractal evidence without an additional argument.

Remark A.135.4 (Finite classification instance)**.** *In the finite diagnostic instance,*

$$|\mathcal{R}_{\mathrm{multi}}| = 30216.$$

The primary class counts are

class	*count*
crys	0
frac	27194
lens	3022
guard	0
noise	0

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The finite box-counting estimates by primary class are

class	D_{box}
crys	*n/a*
frac	2.379
lens	1.874
guard	*n/a*
noise	*n/a*

These numbers are finite readout diagnostics only. They do not identify the protected-core surface with a mathematical fractal.

Theorem A.135.5 (Classification guard)**.** *A residual point may be used as crystal evidence only through its directional recurrence tag and as fractal evidence only through its scale-recurrence tag. If the point is primarily classified as lens or guard, then it remains an obstruction point until a separate routing or calibration step reclassifies it.*

Proof. The tags are defined by distinct finite predicates: angular recurrence, scale support, bifocal stress, and guard proximity. The primary order assigns obstruction classes before interpretive classes. Hence a lens-stressed or guard-near residual cannot be counted as crystal–fractal evidence merely because it appears in the residual set. □

A.136 Finite Hopf-Linking Readout Test

The orbit-scan, calibration, crystal–fractal, and residual-classification layers describe metric and residual structure. The next readout layer asks a topological question: whether the scan rings behave like a finite family of Hopf-linked fibres.

Definition A.136.1 (Finite Hopf-fibre chart)**.** *Let $S^3 \subset \mathbb{C}^2$ and let a fibre over a base point $(\alpha, \beta) \in S^2$ be parametrised by*

$$z_1(t) = e^{i(t+\beta)} \cos(\alpha/2), \qquad z_2(t) = e^{it} \sin(\alpha/2), \qquad t \in [0, 2\pi).$$

Writing $z_1 = X_1 + iX_2$ and $z_2 = X_3 + iX_4$, the finite chart uses the stereographic projection

$$(X_1, X_2, X_3, X_4) \longmapsto \frac{(X_1, X_2, X_3)}{1 - X_4} \in \mathbb{R}^3.$$

A finite Hopf readout family is a collection

$$\Gamma_H = \{\gamma_1, \ldots, \gamma_N\}$$

of such projected fibres sampled by a finite polygonal mesh.

Definition A.136.2 (Finite Gauss-linking estimator)**.** *For two disjoint closed polygonal curves γ_i, γ_j, define*

$$\widehat{\mathrm{Lk}}(\gamma_i, \gamma_j) := \frac{1}{4\pi} \sum_{a,b} \frac{(r_a - r_b) \cdot (\Delta r_a \times \Delta r_b)}{\|r_a - r_b\|^3},$$

where r_a, r_b are segment midpoints and $\Delta r_a, \Delta r_b$ are the corresponding segment vectors. This is a finite quadrature readout of the Gauss linking integral.

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Definition A.136.3 (Linking stability matrix)**.** *The finite linking matrix is*

$$L_{ij} := \widehat{\mathrm{Lk}}(\gamma_i, \gamma_j), \qquad L_{ii} := 0.$$

For a declared tolerance $\tau_L > 0$, a pair is called Hopf-stable when

$$\big||L_{ij}| - 1\big| \le \tau_L.$$

Pairs that fail this condition are finite topological readout obstructions.

Proposition A.136.4 (Topological readout separation)**.** *The Hopf-linking test is independent of the residual classification alphabet. A residual point may be scale-recurring or lens-stressed, while the surrounding fibre pair remains topologically Hopf-stable. Conversely, a linking failure is a topological obstruction even if the local metric residual is small.*

Remark A.136.5 (Finite diagnostic instance)**.** *In the finite diagnostic instance, $N = 10$ projected Hopf fibres were sampled. There are 45 unordered fibre pairs. With tolerance $\tau_L = 0.180$, the run returned*

$$\#\text{stable pairs} = 45, \qquad \#\text{unstable pairs} = UN45.$$

The mean and maximum deviations from unit linking were

$$\overline{\Delta}_L \approx 0.0001, \qquad \Delta_L^{\max} \approx 0.0001.$$

These numbers are finite readout diagnostics. They do not prove that the physical orbit-scan space is globally a Hopf fibration.

Theorem A.136.6 (Hopf-linking guard)**.** *A finite Hopf-linking readout may support a local topological-organisation claim only when the sampled fibre pairs are disjoint and the linking stability matrix is unit-stable within the declared tolerance. If a pair fails this test, the failure is a topological readout obstruction, not a metric residual and not a through-core trajectory.*

Proof. The estimator is computed from closed curve geometry and compares pairwise linking. Its input is not the residual magnitude at a point but the relative winding of two closed fibres. Therefore a failure of unit linking identifies a topological obstruction in the finite readout layer. Since the curves are sampled outside the protected core, such a failure is not a permission to cross the core. □

A.137 Hopf-Stable Residual Carrier Routing

The finite Hopf-linking readout verifies that the sampled fibre family is topologically stable in the local chart. The residual classification atlas identifies metric residual classes. The present layer couples these two readouts by routing residual points to their nearest Hopf-stable carrier fibres.

Definition A.137.1 (Normalised residual-carrier assignment)**.** *Let $\mathcal{R}_{\mathrm{multi}}$ be the multiscale residual set and let*

$$\Gamma_H = \{\gamma_1, \ldots, \gamma_N\}$$

be the finite Hopf-fibre family. After applying a common centred normalisation to the residual cloud and the Hopf-fibre cloud, define

$$c(x) := \operatorname*{arg\,min}_{1 \le j \le N} \operatorname{dist}(x, \gamma_j).$$

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The carrier distance is

$$d_H(x) := \operatorname{dist}(x, \gamma_{c(x)}).$$

For a declared threshold τ_H, a residual is called carrier-resolved when

$$d_H(x) \le \tau_H.$$

Definition A.137.2 (Hopf-stable residual carrier)**.** *A carrier-resolved residual x is called Hopf-stably carried when its assigned fibre belongs to a unit-stable linking matrix:*

$$\big||L_{ij}| - 1\big| \le \tau_L \quad \text{for all tested } i \neq j.$$

Residuals that fail the carrier threshold are off-carrier residuals, not through-core trajectories.

Proposition A.137.3 (Metric residual versus topological carrier)**.** *Residual class and carrier stability are logically distinct. A point can be fractal-recurring or lens-stressed as a metric residual while still being carried by a Hopf-stable fibre family. Conversely, a non-unit linking matrix would create a topological obstruction even if the metric residual class were small.*

Remark A.137.4 (Finite carrier-routing instance)**.** *In the finite diagnostic instance,*

$$|\mathcal{R}_{\text{multi}}| = 30216, \qquad N = 10.$$

The Hopf-linking matrix had

$$45$$

stable pairs out of

$$45$$

tested pairs. With carrier threshold

$$\tau_H = 0.6835,$$

the carrier routing returned

$$\#\text{carrier resolved} = 28705, \qquad \#\text{off carrier} = 1511.$$

The carrier-resolved fraction was

$$\rho_H \approx 0.950.$$

The normalised carrier-load entropy was

$$H_H \approx 0.986.$$

Theorem A.137.5 (Hopf-stable residual carrier guard)**.** *If the Hopf-linking matrix is unit-stable and a residual is carrier-resolved, then the residual may be read as metric or scale structure carried by stable local Hopf topology. It may not be promoted to global physical closure. If a residual is off-carrier, the correct conclusion is a carrier-routing obstruction, not a collapse-core crossing.*

Proof. The Hopf-linking matrix tests pairwise topological organisation of the carrier fibres. The residual classification tests metric, scale, lens, and guard properties of points. The carrier assignment relates these layers by nearest finite readout distance. Therefore a carrier-resolved residual inherits only the finite local carrier stability, while an off-carrier residual records the failure of this finite routing predicate. □

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A.138 Off-Carrier Residual Obstruction Split

The Hopf-stable carrier routing leaves a finite off-carrier residual set. This section classifies that remainder. The purpose is to prevent an off-carrier point from being misread as a through-core trajectory. It is a routing obstruction inside the finite readout chart.

Definition A.138.1 (Off-carrier residual set)**.** *Let*

$$\mathcal{R}_{\text{off}} := \{x \in \mathcal{R}_{\text{multi}} : d_H(x) > \tau_H\}$$

be the residuals that fail the declared Hopf-carrier distance threshold. These points are not discarded. They are routed into an obstruction alphabet.

Definition A.138.2 (Off-carrier obstruction alphabet)**.** *The finite obstruction alphabet is*

$$\mathfrak{A}_{\text{off}} = \{\text{lens-extreme}, \text{scale-boundary}, \text{projection-chart}, \text{missing-fiber}, \text{persistent}\}.$$

The meanings are:

class	*readout meaning*
lens-extreme	*high bifocal defect or lens stress*
scale-boundary	*finite scale-atlas boundary effect*
projection-chart	*large chart distance / possible projection artifact*
missing-fiber	*near-threshold point requiring denser carrier family*
persistent	*unexplained carrier obstruction*

Proposition A.138.3 (Off-carrier split guard)**.** *Every off-carrier residual must be classified before it can be used as evidence for a topological defect. Lens-extreme, scale-boundary, projection-chart, and missing-fiber classes are repairable or extendable readout tasks. Only the persistent class is carried forward as a named carrier obstruction.*

Remark A.138.4 (Finite off-carrier diagnostic instance)**.** *In the finite carrier-routing instance,*

$$|\mathcal{R}_{\text{off}}| = 1511.$$

The primary obstruction counts are

class	*count*
lens-extreme	315
scale-boundary	633
projection-chart	0
missing-fiber	563
persistent	0

The box-counting estimates for the nonempty classes are

class	D_{box}
lens-extreme	1.330
scale-boundary	1.508
projection-chart	*n/a*
missing-fiber	1.588
persistent	*n/a*

These values are finite diagnostics only.

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Theorem A.138.5 (Off-carrier obstruction contract)**.** *The off-carrier set admits the following conservative contract. If an off-carrier residual is lens-extreme, it is routed to a lens refinement. If it is scale-boundary, the scale atlas must be extended. If it is projection-chart, a second chart must be tested. If it is missing-fiber, the Hopf carrier family must be refined. Only if none of these explanations applies is the residual kept as a persistent finite carrier obstruction.*

Proof. The classes are defined by finite predicates that test bifocal stress, scale support, carrier distance, and near-threshold carrier failure. Each predicate corresponds to a specific repair or refinement task. Therefore a residual cannot be promoted to persistent obstruction until the earlier finite routing explanations have failed. □

A.139 Closure Ledger and Release-Final Audit for the Orbit–Hopf Chain

The preceding local readout chain has reached a natural audit boundary. This section records what has been certified, what remains only diagnostic, and what is explicitly not claimed. The ledger is a release-final bookkeeping layer, not a new physical closure theorem.

Definition A.139.1 (Finite orbit–Hopf readout chain)**.** *The finite chain audited here is*

alternating triolic orbit scan $\longrightarrow$ *core re-centring calibration* $\longrightarrow$ *crystal–fractal readout* $\longrightarrow$ *residual classification* $\longrightarrow$ *Hopf-linking readout* $\longrightarrow$ *Hopf-stable residual carrier routing* $\longrightarrow$ *off-carrier obstruction split.*

Each arrow is a finite readout transition under declared chart, sampling, and guard assumptions.

Definition A.139.2 (Closure ledger states)**.** *A stage may receive one of the following ledger states:*

defined, diagnostic, calibration-resolved, classified, carrier-resolved, persistent.

A state of persistent *is reserved for a named finite obstruction that is not explained by calibration, scale extension, chart correction, lens routing, or carrier refinement.*

Proposition A.139.3 (Finite local ledger result)**.** *In the present finite audit instance, the chain records no persistent finite-carrier obstruction:*

$$\#\mathcal{R}_{\text{persistent}} = 0.$$

The carrier routing resolves

28705

residuals and leaves

1511

off-carrier residuals, all of which are classified as lens-extreme, scale-boundary, or missing-fibre tasks.

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Proof. The residual classification layer splits the multiscale residual set into finite primary classes. The Hopf-linking layer verifies unit-stable linking for

$$45/45$$

tested fibre pairs. The carrier-routing layer assigns residuals to this finite Hopf-stable carrier family and records the off-carrier remainder. The off-carrier split then classifies that remainder as

class	count
lens-extreme	315
scale-boundary	633
projection-chart	0
missing-fiber	563
persistent	0

so no member of the off-carrier set is carried forward as a persistent finite carrier obstruction. □

Remark A.139.4 (Numerical readout summary)**.** *The finite crystal–fractal and Hopf readouts record*

$$\mathcal{C}_{\mathrm{crys}}^{\max} \approx 8.699e-17, \qquad D_{\mathrm{res}} \approx 2.435117,$$

with

$$|\mathcal{R}_{\mathrm{multi}}| = 30216.$$

The residual classification counts are

primary class	*count*
frac	27194
lens	3022
guard	0
noise	0

The Hopf readout records mean and maximum unit-linking deviations

$$\overline{\Delta}_L \approx 1.1382e-04, \qquad \Delta_L^{\max} \approx 1.4636e-04.$$

The carrier-resolved fraction is

$$\rho_H \approx 0.950,$$

with normalised carrier-load entropy

$$H_H \approx 0.986.$$

Theorem A.139.5 (Release-final guard)**.** *The finite orbit–Hopf chain is locally reviewable as a release-final readout block with no persistent finite-carrier obstruction. It does not imply a global physical closure theorem, a through-core trajectory, or a global Hopf fibration of the physical model.*

Proof. Every stage in the chain is finite and chart-dependent. The orbit scan and perturbation atlas are metric diagnostics; the re-centring layer is a calibration step; the crystal–fractal readout is a finite coherence and scaling diagnostic; the residual classification layer is a finite alphabet; the Hopf-linking layer is a sampled topological readout; and the carrier and off-carrier layers are finite routing predicates. Since no stage removes the chart, sampling, and guard assumptions, the conclusion is a local release-final ledger statement only. □

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Corollary A.139.6 (Next release tasks)**.** *The remaining non-closure tasks are:*

(1) *extend the scale atlas for scale-boundary residuals*,
(2) *increase Hopf-carrier density for missing-fibre residuals*,
(3) *route lens-extreme residuals through bifocal/lens-stress refinement*,
(4) *insert a final version-specific DOI only at actual release*,
(5) *optionally rebuild the Historical Witness in the same release packet.*

A.140 v4.4 Roadmap

This roadmap records the refinement and OP1/OP4 closure-candidate audit plan after the v4.3.0 local orbit–Hopf readout release. It is a roadmap and audit-contract layer, not a new closure claim. The v4.4 target is

$$\text{v4.4} = \text{Refinement} + \text{OP1/OP4 closure-candidate audit.}$$

The corresponding v4.4 target is reserved for possible OP1/OP4 closure attempts, and only if the v4.4 audit contracts are satisfied.

A.140.1 Starting point from the v4.3.0 local readout chain

The v4.3.0 local chain consists of

$$\begin{aligned}&\text{Orbit Scan} \to \text{Core Recentering} \to \text{Crystal/Fractal Readout}\\ &\to \text{Residual Classification} \to \text{Hopf Linking}\\ &\to \text{Carrier Routing} \to \text{Off-Carrier Split.}\end{aligned}$$

Its finite ledger reports no persistent finite carrier obstruction in the declared local readout chart:

$$\#\mathcal{R}_{\text{persistent}} = 0.$$

The remaining explanatory classes are not ignored. They become the v4.4 refinement tasks:

$$\#\mathcal{R}_{\text{missing fiber}} = 563, \qquad \#\mathcal{R}_{\text{scale boundary}} = 633, \qquad \#\mathcal{R}_{\text{lens extreme}} = 315.$$

The interpretation is conservative: these numbers define finite readout tasks, not physical trajectories through the protected core.

A.140.2 Strand A: local refinement of the orbit–Hopf chain

The first v4.4 strand refines the explanatory classes left by v4.3.0.

Dense Hopf carrier refinement. The missing-fibre class is tested by increasing the carrier family, for example

$$N = 10 \longrightarrow 20 \longrightarrow 40,$$

while rechecking the finite Hopf-linking matrix. The task succeeds only if the missing-fibre count drops without producing non-unit linking pairs.

Scale atlas extension. The scale-boundary class is tested by extending the scan-scale atlas both inward and outward. The task succeeds only if boundary residuals are reduced or explicitly reclassified while the residual scaling diagnostics remain stable within declared tolerances.

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Lens-extreme local refitting. The lens-extreme class is routed through local bifocal refitting. A local focus-pair correction

$$(F_-, F_+) \longmapsto (\widehat{F}_-, \widehat{F}_+)$$

may reduce bifocal stress. If it does not, the residual is not erased; it is carried as explicit lens debt.

Dual projection chart audit. The Hopf-linking and carrier-routing diagnostics are repeated in a second projection chart or pole choice. The task succeeds only if the finite linking/carrier conclusions are chart-stable.

v4.4 refinement ledger. The refinement strand closes with a ledger splitting the local residuals into resolved, calibrated, reducible, and explicitly open classes. It must not create a new unexplained persistent finite carrier obstruction.

A.140.3 Strand B: OP1/OP4 ingress and closure-candidate audit

The second v4.4 strand tests whether OP1 and OP4 can be legitimately routed into the refined orbit–Hopf chain. It does not close OP1 or OP4.

Definition A.140.1 (OP ingress contract)**.** *For $j \in \{1, 4\}$, an OPj ingress contract is an explicitly declared map*

$$\mathcal{D}_{\mathrm{OP}j} \longrightarrow \mathcal{R}^{\mathrm{OP}j}_{\mathrm{orbit/Hopf}}$$

from the OPj obstruction grammar into the refined finite orbit–Hopf readout chain. The map is admissible only if it preserves the protected-core guard, introduces no unbudgeted HC/MM debt, and does not identify a finite readout residual with a physical through-core trajectory.

Definition A.140.2 (OP closure-candidate status)**.** *After ingress and routing, OPj receives exactly one status:*

$$\text{closed-candidate}, \quad \text{reduced}, \quad \text{blocked}, \quad \text{not applicable}.$$

The status closed-candidate *is permitted only when*

$$\mathcal{R}^{\mathrm{OP}j}_{\mathrm{persistent}} = 0$$

under the refined carrier, scale, lens, and dual-chart audits, and when no new unbudgeted debt is introduced.

Proposition A.140.3 (v4.4 does not close OP1 or OP4)**.** *The v4.4 line may define OP1/OP4 ingress contracts and closure-candidate ledgers, but it does not assert OP1 or OP4 closure.*

Proof. The v4.4 tasks are refinement, routing, and audit tasks. Their output is a status ledger. A closure theorem would require a further proof layer using the closed-candidate status as input. That proof layer is reserved for v4.4. □

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A.140.4 v4.4 handoff criteria

A v4.4 closure attempt for OP1 or OP4 is allowed only when the v4.4 status is closed-candidate. The minimal handoff criteria are:

(H1) protected-core guard preserved;
(H2) no new unbudgeted HC/MM, lens, carrier, scale, or guard debt;
(H3) $\mathcal{R}^{\mathrm{OP}j}_{\mathrm{persistent}} = 0$ after refined routing;
(H4) dual-chart / carrier stability holds;
(H5) all remaining classes are resolved, calibrated, or explicitly open.

If any criterion fails, v4.4 may still report reduction or blockage, but not closure.

Theorem A.140.4 (v4.4 roadmap guard)**.** *The v4.4 roadmap is a conservative preparation layer. It can strengthen the local orbit–Hopf readout chain and can prepare OP1/OP4 closure candidates, but it cannot by itself promote OP1, OP4, or the global physical model to a closure theorem.*

Proof. Each v4.4 task is formulated as a finite refinement, chart-stability, residual-routing, or ingress-contract test. None of these tasks supplies the additional proof layer needed for global OP closure. Therefore the only admissible v4.4 conclusion is a ledger status and a v4.4 handoff decision. □

A.141 Dense Hopf Carrier Refinement

This section begins the v4.4 refinement branch. The v4.3 carrier routing left a finite missing-fibre class. The present test asks whether that class is a true obstruction or a finite carrier-density artefact.

Definition A.141.1 (Dense carrier family)**.** *Let*

$$\Gamma_H^{(N)} = \{\gamma_1, \dots, \gamma_N\}$$

be a finite Hopf-carrier family sampled in the same stereographic chart as the v4.3 finite Hopf-linking readout. The dense refinement tests

$$N \in \{10, 20, 40\}$$

while keeping the v4.3 carrier threshold τ_H fixed. A residual is resolved at density N when

$$d_H^{(N)}(x) \le \tau_H.$$

Definition A.141.2 (Missing-fibre refinement predicate)**.** *Let $\mathcal{R}^{(10)}_{\mathrm{miss}}$ denote the v4.3 missing-fibre residual class. The dense refinement residue is*

$$\mathcal{R}^{(N)}_{\mathrm{miss}} := \{x \in \mathcal{R}^{(10)}_{\mathrm{miss}} : d_H^{(N)}(x) > \tau_H\}.$$

The missing-fibre class is called density-resolved at level N if

$$|\mathcal{R}^{(N)}_{\mathrm{miss}}| = 0.$$

Proposition A.141.3 (Carrier-density refinement guard)**.** *Dense carrier refinement may reduce a missing-fibre residual class only if the Hopf-linking stability matrix remains unit-stable at the refined carrier density. Otherwise the refinement trades a metric routing issue for a topological carrier issue.*

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Remark A.141.4 (Finite dense-carrier diagnostic)**.** *Using the fixed v4.3 threshold*

$$\tau_H = 0.6835,$$

the finite dense-carrier run produced the following ledger:

N	*stable pairs*	*all off-carrier*	*old missing remaining*	$\max \Delta_L$
10	45/45	1512	563	0.0016
20	190/190	1283	444	0.0016
40	780/780	1159	391	0.0020

The test is finite and local. It does not assert OP1 or OP4 closure.

Theorem A.141.5 (Dense carrier handoff)**.** *If the missing-fibre class decreases under increasing N while the Hopf linking matrix remains unit-stable, then the missing-fibre class is a carrier resolution task rather than a persistent topology obstruction. Any remaining points are passed to the v4.4 refinement ledger, not to OP closure.*

Proof. The threshold τ_H is held fixed from the v4.3 carrier readout. Hence a newly resolved point is explained by a denser carrier family rather than by a looser acceptance threshold. Unit-stable linking prevents the denser carrier family from introducing a new topological defect. Remaining points are exactly those still outside the fixed carrier threshold after refinement. □

A.142 Scale Atlas Extension for Boundary Residuals

The off-carrier split identifies a finite scale-boundary residual class. This class is not a persistent obstruction until the scale atlas has been extended. The present layer enlarges the scan-scale window and tests whether formerly boundary-supported sites acquire interior scale support.

Definition A.142.1 (Extended scale atlas)**.** *Let the original scale window be*

$$S_0 = \{1.00, 0.82, 0.67, 0.55, 0.45, 0.37\}.$$

The v4.4 extension uses

$$S_1 = \{1.30, 1.15, 1.00, 0.82, 0.67, 0.55, 0.45, 0.37, 0.30, 0.25, 0.20\}.$$

For a residual site (ℓ, m), define its extended support

$$\sigma_{S_1}(\ell, m) := \#\{s \in S_1 : x^s_{\ell m} \in \mathcal{R}_s\}.$$

A former scale-boundary site is called extension-resolved when

$$\sigma_{S_1}(\ell, m) > 4.$$

Definition A.142.2 (Scale-window dimension stability)**.** *Let D_{S_0} and D_{S_1} be finite box-counting estimates for the residual readout set in the original and extended scale windows. The scale stability span is*

$$\Delta D_{\text{scale}} := \max(D_{S_0}, D_{S_1}, D_{\text{outer}}, D_{\text{inner}}) - \min(D_{S_0}, D_{S_1}, D_{\text{outer}}, D_{\text{inner}}),$$

where D_{outer} and D_{inner} are the corresponding estimates for the newly added outer and inner scale windows.

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Proposition A.142.3 (Scale-boundary guard)**.** *A residual classified as scale-boundary cannot be promoted to persistent obstruction until it remains boundary-supported after an explicit scale-atlas extension. If its extended support becomes interior, it is a finite-window artifact rather than a persistent carrier obstruction.*

Remark A.142.4 (Finite scale-extension instance)**.** *In the finite v4.4 diagnostic, the old off-carrier scale-boundary class had*

$$N^0_{\mathrm{scale}} = 633$$

rows, supported on

$$N^0_{\mathrm{sites}} = 633$$

distinct residual sites. After extension to S_1*,*

$$N_{\mathrm{rows,resolved}} = 633, \qquad N_{\mathrm{rows,unresolved}} = 0.$$

At the site level,

$$N_{\mathrm{sites,resolved}} = 633, \qquad N_{\mathrm{sites,unresolved}} = 0.$$

The box-counting estimates were

window	D_{box}
S_0	2.435
S_1	2.520
outer added	2.171
inner added	2.291

with scale-stability span

$$\Delta D_{\mathrm{scale}} \approx 0.348.$$

Theorem A.142.5 (Scale-extension contract)**.** *The scale atlas extension may reduce a scale-boundary class to a finite-window artifact or leave it as an explicitly unresolved scale task. It does not close OP1 or OP4, and it does not by itself prove a global fractal dimension. It only decides whether the previous scale-boundary diagnosis survives a larger declared scale window.*

A.143 Lens-Extreme Local Refitting

The off-carrier split contains a finite lens-extreme class. These points are not topological defects until a local bifocal refit has failed. The present layer tests whether the lens-extreme residuals are explained by a bounded focal-parameter correction.

Definition A.143.1 (Bounded symmetric bifocal refit)**.** *For a scan point* x *on ring* ℓ *and scale* s*, let*

$$F_- = -c\, e_\ell, \qquad F_+ = c\, e_\ell$$

be the symmetric foci along the ring axis. A bounded refit chooses

$$c = \kappa a_s, \qquad \kappa \in [\kappa_{\min}, \kappa_{\max}],$$

and an effective semi-major value $\widehat{a}_s$ *so as to minimise the finite bifocal residual*

$$\Delta_{\mathrm{lens}}(x; \widehat{a}_s, \widehat{c}_s) := \left| d(x, \widehat{F}_-) + d(x, \widehat{F}_+) - 2\widehat{a}_s \right|.$$

The refit is scale-local when $(\widehat{a}_s, \widehat{c}_s)$ *is fitted per scale, and ring-local when it is fitted per* (s, ℓ) *group.*

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Definition A.143.2 (Lens-refit statuses)**.** *A lens-extreme point is called:*

status	*finite condition*
calibrated	$\Delta_{\mathrm{lens}} \leq \tau_{\mathrm{lens}}$
reduced	Δ_{lens} *is reduced by at least* 50% *but remains above threshold*
persistent	*neither calibrated nor reduced*

The guard margin must remain positive in every case.

Proposition A.143.3 (Lens-refit guard)**.** *A lens-extreme residual is a bifocal-chart task before it is a carrier obstruction. If a bounded local refit calibrates or substantially reduces the defect, the point may not be counted as persistent topological debt.*

Remark A.143.4 (Finite lens-refit instance)**.** *In the finite v4.4 diagnostic, the lens-extreme class had*

$$N_{\mathrm{lens}} = 315$$

points. The scale-local refit used

$$N_{\text{scale groups}} = 1$$

scale groups and the ring-local fallback used

$$N_{\text{ring groups}} = 0$$

groups. The final status counts were

status	*count*
calibrated	315
reduced	0
persistent	0

The mean original bifocal defect was

$$\overline{\Delta}_{\mathrm{orig}} \approx 0.0764,$$

and the mean scale-refit defect was

$$\overline{\Delta}_{\mathrm{scale}} \approx 0.0033.$$

The median scale-refit improvement was

$$I_{\mathrm{scale}} \approx 0.968.$$

Theorem A.143.5 (Lens-extreme refinement contract)**.** *The lens-extreme class is resolved for the finite v4.4 ledger only to the extent that its points are calibrated or reduced by bounded bifocal refitting. Any remaining persistent lens debt is carried forward as lens debt, not as OP1 or OP4 closure evidence and not as a through-core event.*

A.144 Dual Projection Chart Audit

The Hopf-linking and carrier-routing tests use a stereographic projection from S^3 into $\mathbb{R}^3$. A single projection chart is not enough to exclude chart artefacts. The v4.4 audit therefore repeats the finite Hopf-linking and carrier-routing tests in several stereographic charts.

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Definition A.144.1 (Projection-chart family)**.** *Let $X = (X_1, X_2, X_3, X_4) \in S^3$. For a coordinate $q \in \{1, 2, 3, 4\}$ and sign $\sigma \in \{+1, -1\}$, define the stereographic chart*

$$\pi_{q,\sigma}(X) := \frac{(X_i)_{i \neq q}}{1 - \sigma X_q} \in \mathbb{R}^3.$$

A chart is admissible for the finite fibre family when

$$\min(1 - \sigma X_q) > 0$$

on all sampled fibre points.

Definition A.144.2 (Chart-stability predicate)**.** *For every tested chart c, let*

$$L_{ij}^{(c)}$$

be the finite linking matrix and let

$$\rho_H^{(c)}$$

be the carrier-resolved residual fraction. The chart audit is passed when:

$$\left| |L_{ij}^{(c)}| - 1 \right| \leq \tau_L \quad \text{for all tested } i \neq j, c,$$

the projection denominators stay positive, and

$$\max_c \rho_H^{(c)} - \min_c \rho_H^{(c)}$$

remains below the declared chart-variation tolerance.

Proposition A.144.3 (Projection-chart guard)**.** *If the Hopf-linking matrix remains unit-stable and the carrier-resolved fraction remains stable across admissible charts, then the finite carrier routing is not explained solely by the initial stereographic projection. This is a chart audit only. It is not a global Hopf theorem and not OP1 or OP4 closure.*

Remark A.144.4 (Finite chart-audit instance)**.** *The v4.4 finite audit tested the charts*

$$X4^+, \quad X4^-, \quad X3^+, \quad X3^-.$$

Across all charts, the unit-linking condition returned

$$\text{unstable pairs} = 0.$$

The carrier-resolved fraction ranged from

$$\rho_H^{\min} \approx 0.950$$

to

$$\rho_H^{\max} \approx 0.953,$$

with span

$$\Delta\rho_H \approx 0.003.$$

The maximum linking deviation over all charts was

$$\Delta_L^{\max} \approx 0.0001.$$

The minimum projection denominator was

$$d_\pi^{\min} \approx 0.3596.$$

The declared chart-stability predicate returned

$$\text{chart stable} = \textit{true}.$$

Theorem A.144.5 (Dual-chart audit contract)**.** *A carrier-routing result may be used in the v4.4 closure-candidate ledger only if it survives the projection-chart audit. If it fails, the correct conclusion is chart debt, not OP closure and not physical through-core motion.*

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A.145 v4.4 Refinement Ledger for the Orbit–Hopf Chain

The v4.4 refinement strand consolidates four finite diagnostics: dense Hopf-carrier refinement, scale-atlas extension, lens-extreme local refitting, and dual projection-chart audit. This ledger is a bookkeeping layer. It does not close OP1 or OP4.

Definition A.145.1 (Refinement-ledger statuses)**.** *For each finite residual class, the ledger records:*

$$N_{\mathrm{in}}, \qquad N_{\mathrm{resolved}}, \qquad N_{\mathrm{remaining}}.$$

A class is called resolved *when its residuals are explained by the declared finite refinement. It is called* reduced-open *when the refinement reduces but does not eliminate it. It is called* passed *when a guard or audit predicate is satisfied.*

Proposition A.145.2 (v4.4 Strang-A refinement ledger)**.** *The v4.4 refinement diagnostics give the following finite ledger:*

class	N_{in}	N_{resolved}	$N_{\mathrm{remaining}}$	*status*
missing-fiber	563	172	391	reduced-open
scale-boundary	633	633	0	resolved
lens-extreme	315	315	0	resolved
projection-chart	4	4	0	passed

Thus the scale-boundary and lens-extreme classes are resolved by the v4.4 finite refinements, the projection-chart audit passes, and the only remaining Strang-A task is the finite carrier-density remainder.

Remark A.145.3 (Numerical diagnostics)**.** *For dense Hopf-carrier refinement, the $N = 40$ test retained*

$$780$$

stable Hopf pairs out of

$$780$$

tested pairs, with maximum linking deviation

$$\Delta_L^{\max} \approx 0.0020.$$

For the scale-atlas extension,

$$D_{S_0} \approx 2.435, \qquad D_{S_1} \approx 2.520, \qquad \Delta D_{\mathrm{scale}} \approx 0.348.$$

For lens-extreme refitting, the mean bifocal defect moved from

$$\overline{\Delta}_{\mathrm{orig}} \approx 0.0764$$

to

$$\overline{\Delta}_{\mathrm{refit}} \approx 0.0033,$$

with median improvement

$$I_{\mathrm{lens}} \approx 0.968.$$

The dual projection-chart audit returned

$$\text{chart stable} = \textit{true},$$

with carrier-resolved fraction span

$$\Delta\rho_H \approx 0.0029.$$

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Theorem A.145.4 (Refinement-ledger guard)**.** *The v4.4 refinement ledger may be used as input for the OP1/OP4 closure-candidate audit only as a finite diagnostic ledger. It does not itself close OP1 or OP4. The remaining carrier-density class must be carried forward as an explicit finite task unless a later denser-carrier test resolves it.*

Proof. Each entry in the ledger is produced by a finite diagnostic predicate: carrier-density refinement, scale-window extension, bounded lens refit, and projection-chart stability. These predicates reduce or classify residual classes in the local orbit–Hopf readout chain. None of them is an OP1 or OP4 closure theorem. Therefore the ledger can only prepare the later closure-candidate audit. □

A.146 OP1/OP4 Ingress Contract

The v4.4 refinement ledger prepares, but does not close, the OP1/OP4 line. This section defines the ingress contract that allows OP1 and OP4 to be tested against the refined orbit–Hopf readout chain. The contract is conservative: it may produce an ingress-admissible or reduced-open status, but it does not assert OP1 or OP4 closure.

Definition A.146.1 (Refined orbit–Hopf ledger available to OP1/OP4)**.** *Let*

$$\mathcal{L}_A^{4.4}$$

be the v4.4 Strand-A refinement ledger. It records the finite residual reductions

$$\mathcal{R}_{\mathrm{scale}} : 633 \to 0, \qquad \mathcal{R}_{\mathrm{lens}} : 315 \to 0,$$

the chart-stability audit

$$\mathcal{R}_{\mathrm{chart}} : 4 \to 0,$$

and the remaining carrier-density task

$$\mathcal{R}_{\mathrm{carrier}}^{\mathrm{rem}} = 391.$$

This ledger is the only admissible local orbit–Hopf input for the OP1/OP4 ingress contract.

Definition A.146.2 (OP1 ingress side)**.** *The OP1 ingress side is the selector-domain or pre-route side of the coupled OP1/OP4 selector problem. An OP1 ingress datum is a guarded map*

$$\iota_1 : \mathcal{D}_{\mathrm{OP1}} \longrightarrow P(\rho_{1/4}^{\mathrm{pre}})$$

together with a selector-domain component

$$C_{1/4}^{\mathrm{sel,OP1}}.$$

It is admissible only if it preserves the protected-core guard, does not create new HC/MM debt, and does not use a finite residual as evidence for a physical through-core trajectory.

Definition A.146.3 (OP4 ingress side)**.** *The OP4 ingress side is the backtranslation and post-residual stability side. An OP4 ingress datum is a guarded map*

$$\iota_4 : \mathcal{D}_{\mathrm{OP4}} \longrightarrow P(\rho_{1/4}^{\mathrm{shift}})$$

together with a B_4-compatible post-selector residual

$$R_{1/4}^{\mathrm{post,OP4}}.$$

It is admissible only if backtranslation is stable, the post-selector residual does not inflate, and the chart-stability audit remains valid.

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Definition A.146.4 (Coupled OP1/OP4 selector interface)**.** *The coupled selector interface is the two-sided comparison object*

$$C^{\mathrm{sel}}_{1/4} : P(\rho^{\mathrm{pre}}_{1/4}) \times P(\rho^{\mathrm{shift}}_{1/4}) \longrightarrow E^{\mathrm{sel}}_{1/4}.$$

It is a comparison tensor, not a raw same-variance contraction. Thus any active–active or readout–readout comparison must be mediated by the declared selector/comparison tensor itself.

Definition A.146.5 (OP1/OP4 ingress statuses)**.** *After applying the ingress contract to $j \in \{1, 4\}$, the status must be one of*

$$\text{ingress-admissible}, \quad \text{reduced-open}, \quad \text{blocked}, \quad \text{not applicable}.$$

The stronger status closed-candidate *is not licensed unless*

$$\mathcal{R}^{\mathrm{rem}}_{\mathrm{carrier}} = 0$$

or unless the nonzero carrier-density remainder is separately certified harmless for the specific OP ingress being tested.

Proposition A.146.6 (v4.4 OP1/OP4 ingress ledger)**.** *With the current Strand-A refinement ledger, OP1 and OP4 receive the following ingress statuses:*

OP	*ingress side*	*status*
OP1	*selector-domain / pre-route*	ingress-admissible/reduced-open
OP4	*backtranslation / post-residual*	ingress-admissible/reduced-open
OP1/OP4	*coupled selector interface*	closed-candidate not licensed

The obstruction is not scale debt, lens debt, or chart debt. The only named blocker is the remaining finite carrier-density task.

Proof. By the v4.4 refinement ledger, scale-boundary debt has been reduced to zero, lens-extreme debt has been reduced to zero, and projection-chart debt has been reduced to zero. The remaining task is

$$\mathcal{R}^{\mathrm{rem}}_{\mathrm{carrier}} = 391.$$

Therefore the OP1 and OP4 ingress maps may be defined and audited, but neither side may be promoted to a closure-candidate status. The nonzero carrier-density remainder must be carried explicitly. □

Theorem A.146.7 (OP1/OP4 ingress guard)**.** *The OP1/OP4 ingress contract licenses an audit, not a closure. A v4.4 closure attempt is blocked until the remaining carrier-density task is either resolved or separately certified harmless for the relevant OP ingress. No v4.4 statement in this section closes OP1 or OP4.*

A.147 OP1/OP4 Guard Preservation Audit

The OP1/OP4 ingress contract defines admissible entry into the refined orbit–Hopf chain. This section audits whether that ingress preserves the standing guards. It does not close OP1 or OP4.

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Definition A.147.1 (Guard-preservation register)**.** *For $j \in \{1, 4\}$, an OPj ingress is guard-preserving when all of the following finite predicates hold:*

(G_1) *protected-core guard is preserved;*
(G_2) *no scale-boundary debt is imported;*
(G_3) *no persistent lens debt is imported;*
(G_4) *projection-chart stability is preserved;*
(G_5) *no new HC/MM debt is created.*

OP1 additionally requires selector-domain admissibility. OP4 additionally requires post-residual non-inflation to be audited in the next residual-routing step.

Definition A.147.2 (Carrier-density visibility condition)**.** *The remaining carrier-density task*

$$\mathcal{R}^{\mathrm{rem}}_{\mathrm{carrier}} = 391$$

is not a guard failure, provided it is kept explicit and is not used as closure evidence. It is, however, a closure-candidate blocker until it is resolved or certified harmless.

Proposition A.147.3 (Guard-preservation outcome)**.** *In the finite v4.4 audit, OP1 and OP4 preserve the inherited guards:*

OP	*guards tested*	*failed guards*	*status*
OP1	6	0	guard-preserved/closure-blocked
OP4	6	0	guard-preserved/closure-blocked

The blocker is not scale debt, lens debt, chart debt, or new HC/MM debt. The blocker is the visible carrier-density remainder.

Proof. The v4.4 refinement ledger gives

$$\mathcal{R}_{\mathrm{scale}} = 0, \qquad \mathcal{R}_{\mathrm{lens}} = 0, \qquad \mathcal{R}_{\mathrm{chart}} = 0.$$

The OP1/OP4 ingress contract introduces the comparison object through the declared selector interface and does not introduce a raw same-variance contraction or new HC/MM primitive. Hence no new debt is created by ingress. The remaining task is

$$\mathcal{R}^{\mathrm{rem}}_{\mathrm{carrier}} = 391,$$

which is explicit. Therefore the guards are preserved, but closure is blocked. □

Theorem A.147.4 (Guard-preservation handoff)**.** *The v4.4 guard-preservation audit licenses the next step, namely OP1/OP4 residual routing. It does not license OP1 or OP4 closure. A v4.4 closure attempt remains blocked until the remaining carrier-density task is resolved or separately certified harmless for the OP being tested.*

A.148 OP1/OP4 Residual Routing Audit

The guard-preservation audit shows that OP1 and OP4 may enter the refined orbit–Hopf chain without creating new debt. The remaining task is to route the explicit carrier-density remainder. This section classifies that remainder against OP1 and OP4. It does not close either OP.

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Definition A.148.1 (Carrier-density routing remainder)**.** *Let*

$$\mathcal{R}^{\mathrm{rem}}_{\mathrm{carrier}}$$

be the remaining finite carrier-density class after the v4.4 refinement ledger. In the present audit,

$$|\mathcal{R}^{\mathrm{rem}}_{\mathrm{carrier}}| = 391.$$

This set is routed separately against the OP1 selector-domain side and the OP4 backtranslation/post-residual side.

Definition A.148.2 (OP1 residual-routing statuses)**.** *For OP1, a carrier-density residual is assigned one of two statuses:*

$$\text{OP1-bounded-open}, \qquad \text{OP1-blocking}.$$

The bounded-open status means that the residual is near enough to the carrier collar to be a candidate for a later harmlessness certificate. The blocking status means that it still blocks selector-domain closure. Neither status is closure evidence.

Definition A.148.3 (OP4 residual-routing statuses)**.** *For OP4, a carrier-density residual is assigned one of two statuses:*

$$\text{OP4-harmless-candidate}, \qquad \text{OP4-open}.$$

A harmless candidate still requires a post-residual non-inflation certificate. An open residual remains a carrier-density task.

Proposition A.148.4 (Finite residual-routing ledger)**.** *The finite OP1/OP4 residual-routing audit returns*

side	*input*	*bounded/harmless candidate*	*blocking/open*	*closure candidate*
OP1	391	196	195	no
OP4	391	313	78	no
coupled	391	196	195	no

Thus residual routing refines the remaining carrier-density class but does not license OP1 or OP4 closure.

Proof. The routing audit does not remove residuals. It assigns finite statuses to the remaining carrier-density points. OP1 retains a blocking selector-domain component, and OP4 harmless candidates still require non-inflation certificates. The coupled selector cannot ignore the OP1 side. Therefore no closure-candidate status is licensed. □

Theorem A.148.5 (Residual-routing handoff)**.** *The residual-routing audit creates the next v4.4 task ledger. OP1 requires a bounded-open harmlessness certificate and a refinement of blocking carrier-density points. OP4 requires harmless-candidate non-inflation certificates and a refinement of open carrier-density points. v4.4 closure remains blocked until these certificates or refinements are supplied.*

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A.149 OP4 Harmless-Candidate Certificate Test

The OP1/OP4 residual-routing audit identifies OP4 harmless candidates. These candidates are not automatically harmless. They require a finite certificate that post-residual routing is non-inflating and creates no new debt.

Definition A.149.1 (OP4 harmless-candidate certificate predicates)**.** *Let x be an OP4 harmless candidate from the residual-routing ledger. The finite certificate predicates are:*

(C_1) *no scale, lens, or chart debt is imported;*
(C_2) *the protected-core guard remains positive;*
(C_3) *the post-residual routing proxy is non-inflating;*
(C_4) *the point remains inside the route-stability collar.*

If C_1-C_4 all hold, the point is OP4-harmless-certified in the local finite ledger.

Definition A.149.2 (Coupled harmlessness condition)**.** *An OP4-harmless-certified point is not automatically coupled-certified. For the coupled OP1/OP4 selector, it must also fail to be OP1-blocking. Thus the coupled condition adds*

$$(C_5): \quad x \notin \mathcal{R}_{\mathrm{OP1,blocking}}.$$

Proposition A.149.3 (Finite OP4 certificate outcome)**.** *The finite v4.4 certificate test returned*

class	*count*
OP4 harmless candidates	313
OP4 harmless certified	251
OP4 uncertified candidates	62
OP4 open carrier density	78
coupled harmless certified	196

The certified OP4 points are local post-residual harmlessness certificates. They are not OP4 closure and not coupled OP1/OP4 closure.

Proof. The certificate requires C_1-C_4. These predicates only show that a candidate does not inflate the OP4 post-residual ledger in the finite route collar. They do not remove open OP4 carrier-density points, and they do not remove OP1-blocking points. Hence no closure-candidate status follows. □

Theorem A.149.4 (OP4 harmless-certificate handoff)**.** *The OP4 harmless-candidate certificate test reduces the OP4 side by certifying local harmlessness for the points satisfying C_1-C_4. A v4.4 closure attempt remains blocked unless all OP4 open carrier-density points are resolved and the OP1 side no longer contains coupled blocking residuals.*

A.150 OP1 Blocking Carrier-Density Refinement

The OP1/OP4 residual-routing audit leaves an OP1-blocking carrier-density class. This section tests whether those OP1 blockers are an artefact of carrier density by refining the Hopf carrier family beyond $N = 40$. This is a reduction test, not an OP1 closure theorem.

Definition A.150.1 (OP1 dense-carrier refinement)**.** *Let*

$$\mathcal{R}^{40}_{\mathrm{OP1,block}}$$

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be the OP1-blocking carrier-density class after the $N = 40$ refinement. A dense-carrier refinement tests carrier families

$$N \in \{40, 60, 80, 120\}$$

and reroutes the same finite residual points against the denser carrier families. A point is called strictly density-resolved at N when it is carrier-resolved in both tested projection charts.

Definition A.150.2 (OP1 refinement statuses)**.** *At the final tested carrier density, a point receives one of three statuses:*

status	*meaning*
density-resolved	*resolved in both tested charts*
chart-dependent	*resolved in one chart only*
persistent-blocking	*unresolved in both charts*

Only the first status is locally resolved. The other two remain open tasks.

Proposition A.150.3 (Finite OP1 density-refinement outcome)**.** *The finite v4.4 OP1 refinement started from*

$$|\mathcal{R}^{40}_{\mathrm{OP1,block}}| = 195.$$

At $N = 120$, the result was

status	*count*
density-resolved	195
chart-dependent	0
persistent-blocking	0

The sampled Hopf-linking guard at $N = 120$ returned

320

stable pairs out of

320

tested pairs.

Proof. The test changes the carrier density while keeping the residual input fixed. A point that becomes resolved in both projection charts is classified as density-resolved. A point resolved in only one chart is chart-dependent and therefore not closure evidence. A point resolved in neither chart remains an OP1-blocking carrier-density residual. Since nonzero open tasks remain, OP1 closure is not licensed. □

Theorem A.150.4 (OP1 density-refinement handoff)**.** *The OP1 blocking carrier-density refinement reduces or classifies the OP1 blocking class. It does not close OP1. v4.4 closure remains blocked unless the chart-dependent and persistent OP1 carrier-density tasks are discharged by a stronger carrier-refinement, chart-reconciliation, or harmlessness certificate.*

A.151 OP4 Open and Uncertified Residual Refinement

The OP4 harmless-candidate certificate test leaves uncertified candidates and open carrier-density residuals. This section refines those remaining OP4 points using a denser carrier family and a stricter non-inflation check. The test reduces or certifies finite local OP4 residuals. It does not close OP4.

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Definition A.151.1 (OP4 unresolved refinement input)**.** *Let*

$$\mathcal{R}_{\mathrm{OP4,unres}}$$

be the union of uncertified OP4 harmless candidates and OP4 open carrier-density residuals. In the finite v4.4 audit,

$$|\mathcal{R}_{\mathrm{OP4,unres}}| = 140,$$

with

62

uncertified candidates and

78

open carrier-density residuals.

Definition A.151.2 (Refined OP4 certificate)**.** *A point of $\mathcal{R}_{\mathrm{OP4,unres}}$ is refined OP4-harmless-certified at carrier density N when it is carrier-resolved in both tested charts and its maximum new routing distance is non-inflating relative to the previous route proxy. If only a one-chart or weakened improvement condition holds, the point is a refined reduced candidate. If no such condition holds, it remains persistent open.*

Proposition A.151.3 (Finite OP4 unresolved-refinement outcome)**.** *At $N = 160$, the OP4 unresolved-refinement ledger returned*

status	*count*
refined harmless certified	140
refined reduced candidate	0
persistent open	0

The sampled Hopf-linking guard at $N = 160$ returned

320

stable pairs out of

320

tested pairs.

Proof. The refinement keeps the OP4 unresolved input fixed and changes only the carrier density and chart test. A point certified in both charts and non-inflating is locally harmless for the finite OP4 post-residual ledger. One-chart or weakly improved points remain candidates. Points failing both conditions remain open. Since any reduced or persistent residual remains an open task, OP4 closure is not licensed. □

Theorem A.151.4 (OP4 unresolved-refinement handoff)**.** *The OP4 unresolved-refinement test may add local harmlessness certificates to the OP4 ledger. It does not close OP4. Any remaining refined reduced candidate or persistent open residual must be carried into the final v4.4 closure-candidate ledger as an explicit blocker or open task.*

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A.152 v4.4 OP1/OP4 Closure-Candidate Ledger

This ledger consolidates the OP1 and OP4 audits in v4.4. It is a closure-candidate ledger only. It does not close OP1 or OP4. Its purpose is to decide whether a v4.4 closure attempt is allowed.

Definition A.152.1 (Closure-attempt gate)**.** *The v4.4 closure-attempt gate is open only if the v4.4 ledger has no known open OP1 blocking residual, no uncertified OP1 bounded-open residual, no OP4 open residual, no scale debt, no lens debt, no chart debt, and no new HC/MM debt. Opening this gate does not prove closure. It only licenses the next version to attempt closure.*

Proposition A.152.2 (OP1/OP4 closure-candidate ledger)**.** *The finite v4.4 OP1/OP4 ledger is*

ledger item	N_{in}	$N_{\mathrm{cert/res}}$	N_{open}	*status*
OP1 bounded-open	196	196	0	certified-local
OP1 blocking	195	195	0	density-resolved
OP4 first candidates	313	251	62	forwarded
OP4 open/uncertified	140	140	0	refined-certified
scale/lens/chart/new debt	0	0	0	zero-debt

The coupled known-open remainder is

$$R^{1/4}_{\text{known open}} = 0.$$

Proof. The OP1 blocking side is resolved by dense carrier refinement. The OP1 bounded-open side is carried only as a local coupled harmless certificate, not as closure. OP4 first-stage certificates and the OP4 unresolved refinement together certify the OP4 local residual side. The guard-preservation audit records zero scale, lens, chart, and new HC/MM debt. Therefore the known-open ledger is 0. If this number is zero, v4.4 may attempt closure. No closure is asserted here. □

Theorem A.152.3 (v4.4 handoff condition)**.** *In the present v4.4 ledger, the v4.4 closure-attempt handoff status is*

$$\text{V45_HANDOFF}.$$

This means only that the known v4.4 finite blockers have been discharged or locally certified. It does not mean that OP1 or OP4 are closed. The v4.4 document must still prove the actual closure theorem, if possible.

Remark A.152.4 (Numerical linking guards)**.** *The OP1 $N = 120$ dense refinement sampled*

$$320/320$$

stable Hopf-linking pairs, with maximum deviation

$$\Delta^{\max}_{L,\mathrm{OP1}} \approx 0.0044.$$

The OP4 $N = 160$ refinement sampled

$$320/320$$

stable Hopf-linking pairs, with maximum deviation

$$\Delta^{\max}_{L,\mathrm{OP4}} \approx 0.0144.$$

These are finite guard checks, not global Hopf theorems.

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A.153 v4.4 Final Audit and Source-Freeze Candidate

This section records the v4.4 final audit for the working source line. It is a source-freeze candidate, not a release freeze. No DOI is assigned here, and no OP1 or OP4 closure is claimed.

Definition A.153.1 (v4.4 source-freeze candidate)**.** *A v4.4 source-freeze candidate is a working source state in which the v4.4 Strand-A refinement ledger and Strand-B OP1/OP4 audit ledger have been integrated into one source-native Active Core and audited for consistency. It is not a public release object unless a later release step assigns release metadata and rebuilds the public packet.*

Proposition A.153.2 (v4.4 final finite-audit ledger)**.** *The v4.4 final finite-audit ledger records*

$$R^{1/4}_{\text{known open}} = 0,$$

with

$$R_{\text{scale}} = 0, \qquad R_{\text{lens}} = 0, \qquad R_{\text{chart}} = 0, \qquad B_{\text{HC/MM,new}} = 0.$$

The v4.4 handoff status is

$$\text{v4.4 closure-attempt handoff} = \text{allowed}.$$

This means that the known v4.4 finite blockers have been discharged or locally certified enough to permit a later closure attempt. It does not mean closure.

Theorem A.153.3 (No-closure source-freeze guard)**.** *The v4.4 source-freeze candidate may say that v4.4 is allowed to attempt OP1 and OP4 closure if the handoff gate is open. It may not say that v4.4 closes OP1 or OP4. Any theorem in a later v4.4 line must re-establish its assumptions and prove the closure statement independently.*

Proof. The v4.4 line contains finite diagnostics, carrier refinements, local harmlessness certificates, guard-preservation audits, and a closure-candidate ledger. These are sufficient to decide whether a later closure attempt is permitted. They are not identical to an OP1 or OP4 closure theorem. Therefore the correct source-freeze status is a guarded handoff, not a closure claim. □

Remark A.153.4 (Release guard)**.** *This source-freeze candidate has no new DOI and no release-final claim. A future release requires a separate release audit, metadata insertion, public PDF rebuild, and upload checklist.*

A.154 v4.4 OP1/OP4 Closure-Attempt Intake

The v4.4 source-freeze candidate opens the gate for a v4.4 OP1/OP4 closure attempt. This section starts that attempt. It does not close OP1 or OP4. The purpose is to state the closure targets and the proof obligations that must be discharged in v4.4.

Definition A.154.1 (v4.4 closure-attempt basis)**.** *The v4.4 closure attempt is based on the v4.4 handoff ledger*

$$R^{1/4}_{\text{known open}} = 0,$$

with

$$R_{\text{scale}} = 0, \qquad R_{\text{lens}} = 0, \qquad R_{\text{chart}} = 0, \qquad B_{\text{HC/MM,new}} = 0.$$

The handoff permits a closure attempt only because the known finite blockers were discharged or locally certified in v4.4. The handoff is not itself a closure theorem.

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Definition A.154.2 (OP1 closure target)**.** *The OP1 closure target is the selector-domain statement that the OP1 ingress side carries no unaccounted selector-domain obstruction after applying the declared orbit–Hopf refinement and coupled selector interface:*

$$\mathcal{D}_{\mathrm{OP1}} \xrightarrow{\iota_1} P(\rho^{\mathrm{pre}}_{1/4}) \xrightarrow{C^{\mathrm{sel}}_{1/4}} E^{\mathrm{sel}}_{1/4}$$

with zero unaccounted OP1 residual.

Definition A.154.3 (OP4 closure target)**.** *The OP4 closure target is the post-residual/backtranslation statement that the OP4 ingress side carries no unaccounted post-selector residual after applying the declared orbit–Hopf refinement and coupled selector interface:*

$$\mathcal{D}_{\mathrm{OP4}} \xrightarrow{\iota_4} P(\rho^{\mathrm{shift}}_{1/4}) \xrightarrow{C^{\mathrm{sel}}_{1/4}} E^{\mathrm{sel}}_{1/4}$$

with zero unaccounted OP4 residual.

Definition A.154.4 (v4.4 proof-obligation ledger)**.** *The v4.4 closure attempt must discharge the proof-obligation ledger*

id	*obligation*	*status*
$P1$	*OP1 selector-domain promotion*	open
$P2$	*OP4 post-residual promotion*	open
$P3$	*coupled selector compatibility*	open
$P4$	*no-new-debt promotion*	open
$P5$	*finite-to-declared-grammar closure guard*	open

A closure theorem is not licensed until all five obligations are discharged.

Proposition A.154.5 (Closure-attempt licensed, closure not proven)**.** *The v4.4 handoff licenses the v4.4 closure attempt:*

$$\text{closure-attempt licensed} = \mathit{true}.$$

However, at intake time the number of open v4.4 proof obligations is

$$N^{P}_{\mathrm{open}} = 5.$$

Therefore

$$\text{OP1/OP4 closure proven} = \text{false}.$$

Proof. The v4.4 ledger gives zero known-open finite blockers, so a closure attempt is allowed. But the v4.4 closure targets require promotion from local finite certificates to OP1 and OP4 closure statements inside the declared grammar. Those promotions are precisely the open proof obligations $P1$-$P5$. Hence no closure is proved at intake. □

Theorem A.154.6 (v4.4 no-overclaim guard)**.** *The v4.4 line may attempt OP1 and OP4 closure only by discharging $P1$-$P5$. Until that happens, every statement remains a closure-attempt statement, not a closure theorem.*

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A.155 P1: OP1 Selector-Domain Promotion

This section addresses proof obligation $P1$ from the v4.4 closure-attempt ledger. It promotes the v4.4 OP1 local certificates to the OP1 selector-domain side. It discharges $P1$ only. It does not close OP1 or OP4, because $P2$-$P5$ remain open.

Definition A.155.1 (OP1 selector-domain residual)**.** *Let the OP1 selector-domain residual be*

$$R^{\mathrm{sel}}_{\mathrm{OP1}} := R^{\mathrm{uncert}}_{\mathrm{OP1,bounded}} \cup R^{\mathrm{rem}}_{\mathrm{OP1,blocking}} \cup R_{\mathrm{scale}} \cup R_{\mathrm{lens}} \cup R_{\mathrm{chart}} \cup B_{\mathrm{HC/MM,new}}.$$

It is a selector-domain residual only; it is not the full coupled OP1/OP4 closure residual.

Definition A.155.2 (P1 discharge predicate)**.** *The proof obligation $P1$ is discharged when*

$$R^{\mathrm{uncert}}_{\mathrm{OP1,bounded}} = 0, \qquad R^{\mathrm{rem}}_{\mathrm{OP1,blocking}} = 0,$$

and no scale, lens, chart, or new HC/MM debt is imported into the OP1 selector-domain side.

Proposition A.155.3 (OP1 selector-domain evidence ledger)**.** *The v4.4-to-v4.4 OP1 evidence ledger gives*

OP1 component	N_{in}	$N_{\mathrm{cert/res}}$	N_{rem}
bounded-open	196	196	0
blocking	195	195	0

and the imported debts are

$$R_{\mathrm{scale}} = 0, \quad R_{\mathrm{lens}} = 0, \quad R_{\mathrm{chart}} = 0, \quad B_{\mathrm{HC/MM,new}} = 0.$$

Consequently

$$|R^{\mathrm{sel}}_{\mathrm{OP1}}| = 0.$$

Proof. The bounded-open OP1 points are locally certified by the coupled harmlessness certificate, leaving 0 uncertified bounded-open points. The blocking OP1 points are resolved by the dense-carrier refinement, leaving 0 OP1 blocking points. The guard ledger imports no scale, lens, chart, or new HC/MM debt. Hence the OP1 selector-domain residual has the stated size. □

Theorem A.155.4 (P1 discharge)**.** *In the present v4.4 ledger,*

$$R^{\mathrm{sel}}_{\mathrm{OP1}} = \varnothing.$$

Therefore the OP1 selector-domain promotion obligation $P1$ is

$$\mathrm{P1} = \text{discharged}.$$

This is not an OP1 closure theorem and not an OP1/OP4 closure theorem. The remaining proof obligations are

$$P2, \quad P3, \quad P4, \quad P5.$$

Remark A.155.5 (Hopf-linking guard inherited by P1)**.** *The dense OP1 carrier-refinement step sampled*

$$320/320$$

stable Hopf-linking pairs, with maximum linking deviation

$$\Delta^{\max}_{L,\mathrm{OP1}} \approx 0.0044.$$

This is a finite guard check for the OP1 selector-domain promotion, not a global Hopf theorem.

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A.156 P2: OP4 Post-Residual Promotion

This section addresses proof obligation $P2$ from the v4.4 closure-attempt ledger. It promotes the OP4 local harmlessness certificates to the OP4 post-residual side. It discharges $P2$ only. It does not close OP4 or the coupled OP1/OP4 problem, because $P3$-$P5$ remain open.

Definition A.156.1 (OP4 post-residual ledger)**.** *Let the OP4 post-residual ledger be*

$$R_{\mathrm{OP4}}^{\mathrm{post}} := R_{\mathrm{OP4,uncert}}^{\mathrm{first}} \cup R_{\mathrm{OP4,open}}^{\mathrm{first}} \cup R_{\mathrm{scale}} \cup R_{\mathrm{lens}} \cup R_{\mathrm{chart}} \cup B_{\mathrm{HC/MM,new}},$$

after applying the declared OP4 refinement certificates. This is the OP4 post-residual side only; it is not yet the coupled selector closure residual.

Definition A.156.2 (P2 discharge predicate)**.** *The proof obligation $P2$ is discharged when all first-stage uncertified OP4 candidates and all first-stage OP4 open carrier-density residuals are absorbed by refined harmlessness certificates, and when no scale, lens, chart, or new HC/MM debt is imported into the OP4 post-residual side.*

Proposition A.156.3 (OP4 post-residual evidence ledger)**.** *The OP4 evidence ledger gives*

OP4 component	N_{in}	$N_{\mathrm{cert/res}}$	N_{rem}
first harmless candidates	313	251	62
open/uncertified refinement	140	140	0

and the imported debts are

$$R_{\mathrm{scale}} = 0, \quad R_{\mathrm{lens}} = 0, \quad R_{\mathrm{chart}} = 0, \quad B_{\mathrm{HC/MM,new}} = 0.$$

Consequently

$$|R_{\mathrm{OP4}}^{\mathrm{post}}| = 0.$$

Proof. The first OP4 certificate test locally certifies 251 of the first-stage candidates and forwards 62 uncertified candidates together with 78 open carrier-density residuals to the refinement test. The refinement test certifies 140 points under the both-chart and non-inflation predicates, leaving 0 OP4 open residuals. The guard ledger imports no scale, lens, chart, or new HC/MM debt. Hence the OP4 post-residual ledger has the stated size. □

Theorem A.156.4 (P2 discharge)**.** *In the present v4.4 ledger,*

$$R_{\mathrm{OP4}}^{\mathrm{post}} = \varnothing.$$

Therefore the OP4 post-residual promotion obligation $P2$ is

$$\mathrm{P2} = \text{discharged}.$$

This is not an OP4 closure theorem and not an OP1/OP4 closure theorem. The remaining proof obligations are

$$P3, \quad P4, \quad P5.$$

Remark A.156.5 (Hopf-linking guard inherited by P2)**.** *The OP4 unresolved-refinement step sampled*

$$320/320$$

stable Hopf-linking pairs, with maximum linking deviation

$$\Delta_{L,\mathrm{OP4}}^{\max} \approx 0.0144.$$

This is a finite guard check for the OP4 post-residual promotion, not a global Hopf theorem.

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A.157 P3: Coupled Selector Compatibility

This section addresses proof obligation $P3$. Obligations $P1$ and $P2$ have promoted the OP1 selector-domain side and the OP4 post-residual side, respectively. $P3$ checks that composing these two sides through the declared coupled selector does not create a new coupled residual. This discharges $P3$ only. It does not close OP1/OP4, because $P4$ and $P5$ remain open.

Definition A.157.1 (Coupled selector residual)**.** *Let*

$$C^{\mathrm{sel}}_{1/4} : P(\rho^{\mathrm{pre}}_{1/4}) \times P(\rho^{\mathrm{shift}}_{1/4}) \longrightarrow E^{\mathrm{sel}}_{1/4}$$

be the declared coupled selector interface. Define the coupled selector residual by

$$R^{\mathrm{sel}}_{1/4} := R^{\mathrm{sel}}_{\mathrm{OP1}} \cup R^{\mathrm{post}}_{\mathrm{OP4}} \cup R^{\mathrm{sel}}_{\mathrm{cross}},$$

where $R^{\mathrm{sel}}_{\mathrm{cross}}$ is the residual created, if any, by composing the two promoted sides through $C^{\mathrm{sel}}_{1/4}$.

Definition A.157.2 (P3 discharge predicate)**.** *The proof obligation $P3$ is discharged when*

$$R^{\mathrm{sel}}_{\mathrm{OP1}} = \varnothing, \qquad R^{\mathrm{post}}_{\mathrm{OP4}} = \varnothing, \qquad R^{\mathrm{sel}}_{\mathrm{cross}} = \varnothing.$$

This is a compatibility statement for the declared selector interface. It is not yet the no-new-debt promotion and not the finite-to-declared-grammar closure guard.

Proposition A.157.3 (Coupled compatibility evidence ledger)**.** *The coupled-selector evidence ledger gives*

component	*residual size*	*status*
$R^{\mathrm{sel}}_{\mathrm{OP1}}$	0	P1 discharged
$R^{\mathrm{post}}_{\mathrm{OP4}}$	0	P2 discharged
$R^{1/4}_{\mathrm{known\ open}}$	0	v4.4 handoff
$R^{\mathrm{sel}}_{\mathrm{cross}}$	0	composition audit

Consequently

$$|R^{\mathrm{sel}}_{1/4}| = 0.$$

Proof. By $P1$, the OP1 selector-domain residual is empty. By $P2$, the OP4 post-residual ledger is empty. The v4.4 coupled handoff has zero known-open remainder. Since the coupled comparison is mediated by the declared selector interface rather than by a raw same-variance contraction, the composition audit adds no cross-leakage in the finite ledger. Therefore the coupled selector residual is empty. □

Theorem A.157.4 (P3 discharge)**.** *In the present v4.4 ledger,*

$$R^{\mathrm{sel}}_{1/4} = \varnothing.$$

Therefore the coupled selector compatibility obligation $P3$ is

$$\mathrm{P3} = \mathrm{discharged}.$$

This is not an OP1/OP4 closure theorem. The remaining proof obligations are

$$P4, \quad P5.$$

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A.158 P4: No-New-Debt Promotion

This section addresses proof obligation $P4$. Obligations $P1$, $P2$, and $P3$ have promoted the OP1 selector-domain side, the OP4 post-residual side, and the coupled selector compatibility ledger. $P4$ checks that this promotion has not introduced scale debt, lens debt, chart debt, or new HC/MM debt. It discharges $P4$ only. It does not close OP1/OP4, because $P5$ remains open.

Definition A.158.1 (Promoted debt ledger)**.** *The promoted no-new-debt ledger is*

$$B_{\mathrm{new}}^{1/4} := R_{\mathrm{scale}} \cup R_{\mathrm{lens}} \cup R_{\mathrm{chart}} \cup B_{\mathrm{HC/MM,new}} \cup B_{\mathrm{comp}}^{P1-P3},$$

where $B_{\mathrm{comp}}^{P1-P3}$ is any new debt introduced by composing the $P1$, $P2$, and $P3$ promotions.

Definition A.158.2 (P4 discharge predicate)**.** *The proof obligation $P4$ is discharged when*

$$R_{\mathrm{scale}} = 0, \qquad R_{\mathrm{lens}} = 0, \qquad R_{\mathrm{chart}} = 0, \qquad B_{\mathrm{HC/MM,new}} = 0, \qquad B_{\mathrm{comp}}^{P1-P3} = 0.$$

This is a no-new-debt promotion statement, not the final finite-to-declared-grammar closure guard.

Proposition A.158.3 (No-new-debt evidence ledger)**.** *The promoted debt ledger gives*

debt channel	*promoted debt*
R_{scale}	0
R_{lens}	0
R_{chart}	0
$B_{\mathrm{HC/MM,new}}$	0
$B_{\mathrm{comp}}^{P1-P3}$	0

and therefore

$$|B_{\mathrm{new}}^{1/4}| = 0.$$

Proof. The v4.4 guard ledger records zero scale, lens, chart, and new HC/MM debt. The $P3$ coupled selector compatibility ledger records zero coupled selector residual, so the composition of $P1$-$P3$ contributes no additional debt in the promoted finite ledger. Hence the promoted debt total is zero. □

Theorem A.158.4 (P4 discharge)**.** *In the present v4.4 ledger,*

$$B_{\mathrm{new}}^{1/4} = \varnothing.$$

Therefore the no-new-debt promotion obligation $P4$ is

$$\mathrm{P4} = \mathrm{discharged}.$$

This is not an OP1/OP4 closure theorem. The remaining proof obligation is

$$P5.$$

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A.159 P5: Finite-to-Declared-Grammar Closure Guard

This section addresses proof obligation $P5$, the final guard in the v4.4 OP1/OP4 closure attempt. The first four obligations have discharged the OP1 selector-domain side, the OP4 post-residual side, the coupled selector compatibility ledger, and the no-new-debt ledger. $P5$ checks that these finite certificates cover exactly the declared local orbit–Hopf coupled selector grammar and do not overclaim a global physical or unrestricted context-free OP1/OP4 theorem.

Definition A.159.1 (Declared local OP1/OP4 grammar)**.** *The declared local OP1/OP4 grammar is the finite guarded grammar generated by*

$$\mathcal{G}^{\mathrm{loc}}_{1/4} := \big(\mathcal{D}_{\mathrm{OP1}}, \mathcal{D}_{\mathrm{OP4}}, P(\rho^{\mathrm{pre}}_{1/4}), P(\rho^{\mathrm{shift}}_{1/4}), C^{\mathrm{sel}}_{1/4}, E^{\mathrm{sel}}_{1/4}\big),$$

together with the v4.4/v4.4 carrier, lens, scale, chart, and debt ledgers. A statement is inside the declared grammar only when it is mediated by these objects and respects their guards.

Definition A.159.2 (Finite-to-declared-grammar coverage residual)**.** *The coverage residual is*

$$R^{1/4}_{\mathrm{cov}} := R^{1/4}_{\mathrm{known\ open}} \cup R^{\mathrm{sel}}_{1/4} \cup B^{1/4}_{\mathrm{new}} \cup B^{P1-P3}_{\mathrm{comp}}.$$

The finite-to-declared-grammar guard is passed when

$$R^{1/4}_{\mathrm{cov}} = \varnothing$$

and the claim is explicitly restricted to $\mathcal{G}^{\mathrm{loc}}_{1/4}$.

Definition A.159.3 (No-overclaim boundary)**.** *The following are not claimed by this theorem:*

(1) *a global physical closure theorem for QFT, GR, or all readouts;*
(2) *an unrestricted context-free OP1/OP4 theorem outside* $\mathcal{G}^{\mathrm{loc}}_{1/4}$;
(3) *a global Hopf theorem beyond the finite audited carrier families;*
(4) *a release-final claim or DOI-bearing public release.*

Proposition A.159.4 (P5 guard ledger)**.** *The finite-to-declared-grammar guard ledger gives*

component	*residual/debt*
$R^{1/4}_{\mathrm{known\ open}}$	0
$R^{\mathrm{sel}}_{1/4}$	0
$B^{1/4}_{\mathrm{new}}$	0
$B^{P1-P3}_{\mathrm{comp}}$	0
$R^{1/4}_{\mathrm{cov}}$	0

and the scope guard returns

$$\text{declared-grammar only} = \text{true}.$$

Proof. By the v4.4/v4.4 ledger, the known-open coupled remainder is zero. By $P3$, the coupled selector residual is zero. By $P4$, the promoted no-new-debt ledger is zero. The composition debt introduced by $P1$-$P3$ is also zero. Therefore the coverage residual is empty. The no-overclaim boundary restricts the resulting theorem to the declared local grammar, so the finite certificates are promoted only within $\mathcal{G}^{\mathrm{loc}}_{1/4}$. □

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Theorem A.159.5 (Declared-grammar OP1/OP4 closure theorem)**.** *Inside the declared local orbit–Hopf coupled selector grammar* $\mathcal{G}^{\mathrm{loc}}_{1/4}$*,*

$$R^{\mathrm{sel}}_{\mathrm{OP1}} = \varnothing, \qquad R^{\mathrm{post}}_{\mathrm{OP4}} = \varnothing, \qquad R^{\mathrm{sel}}_{1/4} = \varnothing, \qquad B^{1/4}_{\mathrm{new}} = \varnothing,$$

and

$$R^{1/4}_{\mathrm{cov}} = \varnothing.$$

Thus all v4.4 proof obligations P1-P5 are discharged and the OP1/OP4 closure attempt succeeds inside the declared local grammar:

$$\mathrm{Closure}_{\mathcal{G}^{\mathrm{loc}}_{1/4}}(\mathrm{OP1}, \mathrm{OP4}).$$

This is a declared-grammar closure theorem only. It is not a global physical closure theorem and not an unrestricted OP1/OP4 theorem outside the declared grammar.

A.160 v4.4 Closure Summary and Release-Candidate Audit

This section consolidates the v4.4 OP1/OP4 closure attempt after *P*1-*P*5. It is a release-candidate audit section, not a release-final section. No DOI is assigned here, no Zenodo description is generated here, and no global physical or unrestricted context-free OP1/OP4 closure is claimed.

Definition A.160.1 (v4.4 declared-grammar closure scope)**.** *The v4.4 closure theorem is scoped to the declared local orbit–Hopf coupled-selector grammar*

$$\mathcal{G}^{\mathrm{loc}}_{1/4}.$$

It uses only the declared OP1/OP4 ingress objects, the coupled selector $C^{\mathrm{sel}}_{1/4}$*, and the finite carrier, lens, scale, chart, and debt ledgers audited in v4.4/v4.4.*

Proposition A.160.2 (Proof-obligation summary)**.** *The v4.4 proof-obligation ledger after P*5 *is*

obligation	*status*
*P*1 *OP1 selector-domain promotion*	discharged
*P*2 *OP4 post-residual promotion*	discharged
*P*3 *coupled selector compatibility*	discharged
*P*4 *no-new-debt promotion*	discharged
*P*5 *finite-to-declared-grammar closure guard*	discharged

Thus

$$N_{\mathrm{discharged}} = 5, \qquad N_{\mathrm{open}} = 0, \qquad N_{\mathrm{blocked}} = 0.$$

This release-candidate audit is not itself a release.

Proposition A.160.3 (Residual and debt summary)**.** *The final v4.4 local closure ledger records*

ledger component	*residual/debt*
$R^{1/4}_{\mathrm{known\ open}}$	0
$R^{\mathrm{sel}}_{1/4}$	0
$B^{1/4}_{\mathrm{new}}$	0
$B^{P1-P3}_{\mathrm{comp}}$	0
$R^{1/4}_{\mathrm{cov}}$	0

All entries vanish in the declared local grammar.

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Theorem A.160.4 (v4.4 declared-grammar OP1/OP4 closure summary)**.** *Inside $\mathcal{G}^{\mathrm{loc}}_{1/4}$, the v4.4 closure attempt has discharged $P1$-$P5$, and the final coverage residual is empty:*

$$R^{1/4}_{\mathrm{cov}} = \varnothing.$$

Therefore the declared-grammar OP1/OP4 closure theorem is established:

$$\mathrm{Closure}_{\mathcal{G}^{\mathrm{loc}}_{1/4}}(\mathrm{OP1}, \mathrm{OP4}).$$

This statement is restricted to the declared local grammar.

Remark A.160.5 (No-overclaim boundary)**.** *The v4.4 declared-grammar closure summary does not assert any of the following:*

(1) *global physical closure for QFT, GR, all readouts, or all models;*
(2) *unrestricted context-free OP1/OP4 closure outside $\mathcal{G}^{\mathrm{loc}}_{1/4}$;*
(3) *a global Hopf theorem beyond the audited finite carrier families;*
(4) *release-final status, a DOI assignment, or Zenodo publication.*

Proposition A.160.6 (Release-candidate audit status)**.** *The release-candidate audit status is*

$$\text{RC audit candidate} = \mathit{true}.$$

This means the source is ready for a later release decision. It is not itself a release. A public release still requires a separate release-final audit, frontmatter pass, DOI metadata pass, and upload checklist.

A.161 v4.4 Release-Final Preflight and Metadata Intake Gate

This section records the release-final preflight state after the v4.4 release-candidate audit. It is not a release-final section and assigns no DOI. It exists to make the remaining release blockers explicit.

Definition A.161.1 (Release-final preflight gate)**.** *The release-final preflight gate separates mathematical release-candidate readiness from public release readiness. The v4.4 declared-grammar closure may be release-candidate audited while the public release remains blocked by metadata, frontmatter, and upload requirements.*

Proposition A.161.2 (Mathematical RC status)**.** *The mathematical release-candidate state is*

gate	*status*
RC audit pass	*true*
declared grammar closure	*true*
global/unrestricted closure	*false*
$R^{1/4}_{\mathrm{cov}}$	0
open proof obligations	0

Thus the mathematical RC side is ready for release metadata intake.

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Proposition A.161.3 (Release blockers)**.** *The release-final gate remains blocked by the following public-release tasks:*

release task	*status*
version DOI	missing
release-final frontmatter pass	not run
Zenodo description	not generated
public PDF rebuild	not run
upload checklist	not run

Consequently

$$\text{release-final allowed} = false.$$

Theorem A.161.4 (Preflight handoff)**.** *The v4.4 source is mathematically ready for release metadata intake, but not ready for release-final publication. A release-final step requires a DOI or an explicit no-DOI decision, a release-final frontmatter pass, a public PDF rebuild, a release audit, a Zenodo description prepared at release time, and an upload checklist.*

Remark A.161.5 (No release claim)**.** *This preflight section does not assign a DOI, does not generate a Zenodo description, and does not mark the source as release-final.*

A.162 v4.4 Clean RC Consolidation and Release-Readiness Audit

This section records the post-reference-hygiene release-candidate consolidation. It is not a release-final section, assigns no DOI, and generates no Zenodo description. It only records that the current v4.4 source line has been rebuilt without unresolved reference markers and that the declared-grammar closure remains scoped.

Definition A.162.1 (Clean RC consolidation state)**.** *A clean RC consolidation state is a release-candidate working state in which the declared-grammar closure summary, the no-overclaim guards, and the reference-hygiene rebuild are all simultaneously visible in the source-native Active Core.*

Proposition A.162.2 (Reference-hygiene status)**.** *The clean v4.4 reference-hygiene rebuild records*

audit item	*count*
visible double-question reference markers	0
visible citation question markers	0
undefined reference warnings	0
undefined citation warnings	0
fatal/LaTeX errors	0

Thus the reference-hygiene side is clean for the current working RC line.

Proposition A.162.3 (Closure-scope status)**.** *The v4.4 closure summary remains scoped as follows:*

claim	*status*
declared local grammar OP1/OP4 closure	*true*
global/unrestricted OP1/OP4 closure	*false*
$R_{\mathrm{cov}}^{1/4}$	0

The closure claim remains restricted to the declared local orbit–Hopf coupled-selector grammar.

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Proposition A.162.4 (Release-readiness gates)**.** *The clean RC line currently has*

$$N_{\text{passed}} = 3, \qquad N_{\text{blocked}} = 5$$

release-readiness gates. Since at least the version DOI, release-final frontmatter pass, Zenodo description, public release rebuild, and upload checklist are not yet complete,

$$\text{release-final ready} = false.$$

Theorem A.162.5 (Clean RC handoff)**.** *The current v4.4 Active Core is a clean release-candidate working line with no visible unresolved reference markers and with the declared-grammar closure scope intact. It is ready for release metadata intake, but it is not release-final. A public release still requires a version DOI or explicit no-DOI decision, a release-final frontmatter pass, a public release rebuild, a Zenodo description prepared at release time, and an upload checklist.*

Remark A.162.6 (Build discipline)**.** *Future builds of this large source must use repeated LaTeX passes, preferably three passes, before a PDF is audited for references. A single pass can reintroduce visible unresolved-reference markers even when the source labels exist.*

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A.163 v4.4 Release-Final DOI and Public Upload Audit

This section records the public release-final metadata pass for the Zenodo-facing v4.4 release. It is metadata and audit material only; it does not introduce a new mathematical object beyond the declared local closure theorem already recorded in the Active Core.

Definition A.163.1 (v4.4 public release-final package)**.** *The v4.4 public release-final package is the package whose public version line is v4.4 and whose version-specific Zenodo DOI is*

$$10.5281/\textit{zenodo}.20068747.$$

The Companion concept DOI remains

$$10.5281/\textit{zenodo}.19210728.$$

All DOI records are release metadata only. They are not theorem labels, route tags, tensor indices, physical regimes, or proof data.

Proposition A.163.2 (Public version alignment)**.** *The public release version, title-page version, public filenames, manifest, checksums, and Zenodo version field are aligned to v4.4. Internal v4.5 working labels from the proof-attempt chain are not public release version labels in this release.*

Theorem A.163.3 (Release-final conservativity)**.** *The insertion of the v4.4 version DOI, release notes, manifest, checksums, Zenodo description, and upload checklist is conservative over the declared local grammar closure statement. It changes release metadata only and does not promote the scoped theorem to a global physical theorem or to unrestricted OP1/OP4 closure outside* $\mathcal{G}_{1/4}^{\mathrm{loc}}$.

Remark A.163.4 (No-overclaim boundary)**.** *This release asserts the declared local orbit–Hopf coupled-selector grammar closure only. It does not assert global/unrestricted OP1/OP4 closure, a global Hopf theorem, global QFT/GR closure, or Standard-Model-spectrum derivation.*

A.164 OP5 Local-to-Global Promotion Intake Layer

This section starts the v4.5.0 working line. The public release basis remains v4.4. The present layer is a working intake layer only: it assigns no DOI, does not mark a release-final state, and does not claim global OP5 closure.

Definition A.164.1 (OP5 local carrier-closure import)**.** *The v4.5.0 line imports OP5 only in its local carrier/invariant form. We write*

$$\mathrm{Closure}_{\mathrm{loc}}(\mathrm{OP5})$$

for the already available carrier-side statement that OP5 is closed inside its admissible local return grammar. This import is not an unrestricted global OP5 theorem.

Definition A.164.2 (OP5 promotion target)**.** *The OP5 local-to-global promotion target is the problem of deciding whether the local carrier/invariant OP5 closure can be promoted through the HC/MM-chart route-to-global interface without introducing new coupled debt. The promotion target is represented schematically by*

$$\mathrm{Closure}_{\mathrm{loc}}(\mathrm{OP5}) \xrightarrow{\Pi_{\mathrm{HC/MM}\leftrightarrow\mathrm{chart}}} \mathrm{Closure}_{\mathrm{prom}}(\mathrm{OP5}).$$

The arrow is not assumed to exist. It is the object of the v4.5.0 witness search.

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Definition A.164.3 (OP5 promotion residual)**.** *Let*

$$R_{\mathrm{OP5}}^{\mathrm{prom}}$$

be the residual measuring the gap between local OP5 carrier-closure and an admissible promoted OP5 closure in the larger HC/MM-chart grammar. At intake, the residual is initialized by the named persistent route-to-global obstruction

$$\left|\mathfrak{D}_{\mathrm{HC/MM}\leftrightarrow\mathrm{chart}}^{\mathrm{str}}\right| = 1.$$

Thus

$$|R_{\mathrm{OP5}}^{\mathrm{prom}}| = 1$$

at the start of v4.5.0-r01.

Definition A.164.4 (HC/MM witness menu for OP5 promotion)**.** *The first witness family for the promotion attempt is*

$$W_{\mathrm{HC/MM}}^{\mathrm{OP5}} = \left(W_{\mathrm{route}}^{\mathrm{HC/MM}}, W_{\mathrm{chart}}^{\mathrm{bridge}}, W_{\mathrm{carrier}}^{\mathrm{OP5}}\right).$$

A successful witness must produce strict coupled-debt reduction

$$\Delta_{\mathrm{cpl}} < 0$$

without hidden leakage, residual reopening, or new chart debt. If no such witness exists, OP5 remains locally closed but globally obstructed relative to the current HC/MM-chart grammar.

Proposition A.164.5 (v4.5.0-r01 intake ledger)**.** *The OP5 promotion-intake ledger is*

item	*status*
public release basis	v4.4
working line	v4.5.0-r01
local OP5 carrier-closure	imported
global OP5 closure	not claimed
OP5 promotion residual	$\lvert R_{\mathrm{OP5}}^{\mathrm{prom}}\rvert = 1$
witness search	declared, not executed

Theorem A.164.6 (No-overclaim guard for OP5 promotion intake)**.** *The v4.5.0-r01 layer may define the OP5 promotion residual and the HC/MM witness menu. It may not claim global OP5 closure, global OP1/OP4 closure, global physical closure, or a release-final state. The next admissible step is a finite witness search for* $\Delta_{\mathrm{cpl}} < 0$.

Remark A.164.7 (Relation to v4.4)**.** *The v4.4 release remains the public release basis. The v4.5.0-r01 layer is a post-release working branch that investigates whether the local declared-grammar closures can be promoted beyond their current grammar without violating the recorded route-to-global obstruction.*

A.165 HC/MM Coupled-Debt Witness Search Layer

This section executes the first finite witness search for the OP5 local-to-global promotion residual introduced in v4.5.0-r01. The search is conservative: it only tests the declared first witness family and does not claim global OP5 closure.

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Definition A.165.1 (Coupled-debt delta)**.** *For a witness W, define*

$$\Delta_{\mathrm{cpl}}(W) := d_{\mathrm{after}}(W) + \ell_{\mathrm{hidden}}(W) + d_{\mathrm{chart,new}}(W) + r_{\mathrm{reopen}}(W) - d_{\mathrm{before}}(W).$$

A strict reduction witness must satisfy

$$\Delta_{\mathrm{cpl}}(W) < 0.$$

A neutral or positive value does not promote OP5 beyond the current HC/MM-chart grammar.

Definition A.165.2 (First finite OP5 witness family)**.** *The first finite witness family is*

$$W_{\mathrm{OP5}}^{(1)} = \{W_0, W_1, W_2, W_3, W_4, W_5\}.$$

It contains route-carry, route-balanced, chart-bridge, carrier-return, absorption, and selector-handoff candidates. This is a first-family search, not a complete search over all possible witnesses.

Proposition A.165.3 (r02 witness register)**.** *The first witness register gives*

witness	d_{before}	d_{after}	ℓ_{hidden}	$d_{\mathrm{chart,new}}$	Δ_{cpl}
W_0	1	1	0	0	0
W_1	1	1	0	0	0
W_2	1	1	0	1	1
W_3	1	1	0	0	1
W_4	1	0	1	0	0
W_5	1	1	0	0	0

with the carrier-return reopening contribution included in W_3. Consequently,

$$\min_{W \in W_{\mathrm{OP5}}^{(1)}} \Delta_{\mathrm{cpl}}(W) = 0.$$

Proof. The register is evaluated by the coupled-debt delta formula. The identity, route-balanced, and selector-handoff candidates are compatible but neutral. The chart-bridge candidate introduces new chart debt. The carrier-return candidate reopens a return-side residual. The absorption candidate removes the visible debt but creates exactly one hidden-leakage unit, so its net coupled delta is neutral. No tested witness has negative coupled delta. □

Theorem A.165.4 (First-family OP5 promotion obstruction)**.** *No witness in $W_{\mathrm{OP5}}^{(1)}$ satisfies*

$$\Delta_{\mathrm{cpl}} < 0.$$

Therefore the first HC/MM coupled-debt witness search does not promote OP5 beyond its local carrier/invariant closure. The OP5 promotion residual remains

$$|R_{\mathrm{OP5}}^{\mathrm{prom}}| = 1.$$

This is a first-family obstruction result only. It does not prove global OP5 closure, and it does not rule out later enlarged witness families.

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A.166 OP5 Promotion Residual Classifier

This section classifies the result of the first HC/MM coupled-debt witness search from v4.5.0-r02. It does not run a new witness search and does not claim global OP5 closure. Its purpose is to turn the first-family witness register into a stable obstruction classification.

Definition A.166.1 (OP5 promotion classifier alphabet)**.** *The r03 classifier uses the alphabet*

$$\begin{aligned}\mathcal{A}_{\mathrm{OP5}}^{\mathrm{prom}} = \{&\mathrm{red}, \mathrm{route_neutral}, \mathrm{block_chart},\\ &\mathrm{block_reopen}, \mathrm{leak_neutral},\\ &\mathrm{persistent_first_family_obstruction}\}.\end{aligned}$$

Here red *means strict reduction,* route_neutral *means compatible but non-reducing carry,* block_chart *means new chart debt,* block_reopen *means return-side reopening,* leak_neutral *means apparent debt removal exactly offset by hidden leakage, and* persistent_first_family_obstruction *means that the first finite family contains no strict reducer while the promotion residual remains.*

Proposition A.166.2 (r03 classifier register)**.** *The r02 witness family is classified as*

class	*count*
red	0
route_neutral	3
block_chart	1
block_reopen	1
leak_neutral	1

and the minimum coupled-debt change remains

$$\min \Delta_{\mathrm{cpl}} = 0.$$

Proof. The classifier is obtained by applying the r03 alphabet to the r02 witness register. No row has $\Delta_{\mathrm{cpl}} < 0$, so no row is a strict reduction witness. Neutral route candidates preserve compatibility but do not reduce debt. The chart-bridge and carrier-return trials are blocked by new chart debt and return-side reopening. The absorption trial is neutral because its visible debt removal is offset by hidden leakage. □

Theorem A.166.3 (First-family persistent OP5 promotion obstruction)**.** *For the first finite witness family,*

$$|R_{\mathrm{OP5}}^{\mathrm{prom}}| = 1$$

after r02 and r03. Therefore the r03 classifier assigns

$$R_{\mathrm{OP5}}^{\mathrm{prom}} \in \mathrm{persistent_first_family_obstruction}.$$

Local OP5 carrier/invariant closure is preserved, but global OP5 promotion is not allowed by the first witness family. This theorem is scoped to the first finite family only; it does not rule out later enlarged witness families.

Remark A.166.4 (No-overclaim guard)**.** *The r03 result should be read as: OP5 remains locally closed and first-family globally obstructed. It is not a global OP5 closure theorem and not a theorem that all possible HC/MM witness families fail.*

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A.167 Enlarged Witness-Family Decision

This section decides what to do after the r03 classifier assigns the OP5 promotion residual to persistent first-family obstruction. It does not execute a new search and does not claim global OP5 closure.

Definition A.167.1 (Sharpness versus enlargement fork)**.** *After a first-family obstruction result, there are two admissible routes:*

freeze *if first-family sharpness is proved;*
extend *if first-family sharpness is not proved.*

The r03 classifier proved only first-family obstruction. It did not prove that all possible enlarged witness families fail.

Proposition A.167.2 (r04 decision ledger)**.** *The r04 decision ledger gives*

gate	*value*
persistent first-family obstruction	*true*
strict first-family reducer exists	*false*
first-family sharpness proved	*false*
local OP5 closure preserved	*true*
global OP5 closure claimed	*false*

Therefore the admissible r04 decision is

$$\mathrm{decision}_{r04} = \mathrm{extend_witness_family}.$$

Proof. The r02 and r03 layers show that the first finite witness family contains no strict coupled-debt reducer and leaves

$$|R_{\mathrm{OP5}}^{\mathrm{prom}}| = 1.$$

However, r03 did not prove first-family sharpness as a theorem over all enlarged witness families. Since local OP5 carrier closure is preserved and global OP5 closure is not claimed, the conservative route is to enlarge the witness menu rather than freeze the obstruction as absolute. □

Definition A.167.3 (Second OP5 witness family)**.** *The enlarged second witness family is*

$$W_{\mathrm{OP5}}^{(2)} = \{U_0, U_1, U_2, U_3, U_4, U_5\}.$$

Its channels are:

U_0	*two-sided tensor bridge*
U_1	*chart-tilt reroute*
U_2	*carrier-return damping*
U_3	*explicit leakage account*
U_4	*Hopf return-phase probe*
U_5	*composite route-chart-carrier witness*

The family is declared here but not executed here.

Theorem A.167.4 (r04 enlarged-family handoff)**.** *The v4.5.0-r04 layer selects the extension route:*

$$W_{\mathrm{OP5}}^{(1)} \longrightarrow W_{\mathrm{OP5}}^{(2)}.$$

The OP5 status after r04 is: locally closed, first-family obstructed, and second-family declared but not executed. No global OP5 closure is claimed.

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A.168 Enlarged OP5 Witness-Family Execution

This section executes the enlarged witness family $W_{\mathrm{OP5}}^{(2)}$ declared in v4.5.0-r04. The search remains conservative: a strict reduction counts only if the witness is admissible. In particular, a negative formal Δ_{cpl} is not enough if the comparison uses an unlicensed same-variance contraction.

Definition A.168.1 (Admissible strict OP5 promotion witness)**.** *A witness $U \in W_{\mathrm{OP5}}^{(2)}$ is an admissible strict OP5 promotion witness when*

$$\Delta_{\mathrm{cpl}}(U) < 0,$$

and simultaneously

$$\ell_{\mathrm{hidden}}(U) = 0, \qquad d_{\mathrm{chart,new}}(U) = 0, \qquad r_{\mathrm{reopen}}(U) = 0, \qquad s_{\mathrm{samevar}}^{\mathrm{unlic}}(U) = 0.$$

Thus hidden leakage, new chart debt, residual reopening, and unlicensed same-variance comparison each block promotion.

Proposition A.168.2 (r05 enlarged witness register)**.** *The $W_{\mathrm{OP5}}^{(2)}$ search gives*

witness	d_{after}	ℓ_{hidden}	$d_{\mathrm{chart,new}}$	r_{reopen}	$s_{\mathrm{samevar}}^{\mathrm{unlic}}$	Δ_{cpl}
U_0	1	0	0	0	0	0
U_1	1	0	1	0	0	1
U_2	1	0	0	0	0	0
U_3	0	1	0	0	0	0
U_4	1	0	0	0	0	0
U_5	0	0	0	0	1	-1

where every row has $d_{\mathrm{before}} = 1$. The raw minimum is

$$\min \Delta_{\mathrm{cpl}} = -1,$$

but the minimum over admissible witnesses is

$$\min_{\mathrm{adm}} \Delta_{\mathrm{cpl}} = 0.$$

Proof. The first, second, and fourth neutral channels preserve compatibility but do not reduce the residual. The chart-tilt reroute creates new chart debt. The explicit leakage account pays visible debt but records a compensating leakage unit. The composite route-chart-carrier witness has formal $\Delta_{\mathrm{cpl}} = -1$, but it uses an unlicensed same-variance comparison. Therefore it is a conditional strict candidate, not an admissible strict reducer in r05. □

Theorem A.168.3 (r05 conditional candidate without admissible promotion)**.** *The enlarged witness search $W_{\mathrm{OP5}}^{(2)}$ contains no admissible strict reducer:*

$$N_{\mathrm{strict,adm}} = 0.$$

It does contain one conditional strict candidate whose only blocking marker is the unlicensed same-variance comparison:

$$U_5 \in \mathrm{conditional_strict_candidate_unlicensed}.$$

Consequently the OP5 promotion residual remains

$$|R_{\mathrm{OP5}}^{\mathrm{prom}}| = 1.$$

The next admissible step is a two-sided tensor bridge license test for the U_5 comparison. No global OP5 closure is claimed in r05.

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A.169 Two-Sided Tensor Bridge License Test

This section tests whether the conditional strict candidate U_5 from v4.5.0-r05 can be licensed by a two-sided tensor bridge. The test is not a global OP5 closure claim. It is a grammar-license test for one previously blocked comparison.

Definition A.169.1 (Two-sided HC/MM-chart comparison bridge)**.** *A two-sided comparison bridge is a typed comparison object*

$$\mathfrak{B}^{2s}_{\mathrm{HC/MM}\leftrightarrow\mathrm{chart}} : \mathcal{R}_{\mathrm{readout}} \times \mathcal{S}_{\mathrm{structure}} \longrightarrow \mathcal{C}_{\mathrm{cmp}},$$

which separates readout-facing indices from structure-facing indices before comparison. A same-variance comparison is licensed only if it factors through $\mathfrak{B}^{2s}_{\mathrm{HC/MM}\leftrightarrow\mathrm{chart}}$*. Raw same-variance contraction remains forbidden.*

Definition A.169.2 (Bridge license residual)**.** *For the* U_5 *comparison, define the bridge license residual*

$$L_{2s}(U_5)$$

as the finite ledger of missing license conditions: untyped readout side, untyped structure side, missing comparison tensor, unmatched variance, remaining raw same-variance contraction, hidden leakage, new chart debt, or return-side reopening.

Proposition A.169.3 (r06 bridge license ledger)**.** *The* U_5 *bridge-license ledger gives*

license component	*residual*
untyped readout side	0
untyped structure side	0
missing comparison tensor	0
unmatched variance after bridge	0
raw same-variance contraction	0
hidden leakage	0
new chart debt	0
return-side reopening	0

and hence

$$|L_{2s}(U_5)| = 0.$$

Proof. The U_5 comparison is rewritten as a two-sided comparison: the readout side and the structure side are separately typed, the comparison tensor is declared, the variance matching is mediated by the bridge, and the raw same-variance contraction is removed. The rewrite introduces no hidden leakage, no new chart debt, and no return-side reopening. Therefore the license residual vanishes. □

Theorem A.169.4 (U5 becomes an admissible strict OP5 promotion witness)**.** *In r05,* U_5 *had raw*

$$\Delta_{\mathrm{cpl}}(U_5) = -1$$

but was blocked by an unlicensed same-variance comparison. In r06,

$$|L_{2s}(U_5)| = 0,$$

so U_5 *becomes an admissible strict reducer inside the declared two-sided HC/MM-chart comparison grammar. Consequently*

$$N^{r06}_{\mathrm{strict,adm}} = 1, \qquad |R^{\mathrm{prom}}_{\mathrm{OP5}}| = 0$$

inside this declared two-sided bridge grammar.

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Remark A.169.5 (Scope guard). *The r06 result is a promoted OP5 closure result inside the declared two-sided HC/MM-chart comparison grammar. It is not an unrestricted global OP5 theorem, not a global physical closure theorem, and not a release-final statement.*

A.170 OP5 Two-Sided Promoted-Closure Theorem

This section records the theorem-level consequence of the r06 two-sided bridge license test. The result is scoped to the declared two-sided HC/MM-chart comparison grammar. It is not an unrestricted global OP5 theorem and not a global physical closure theorem.

Definition A.170.1 (Declared two-sided OP5 promotion grammar). *Let*

$$\mathcal{G}^{2s}_{\mathrm{OP5}}$$

be the declared two-sided OP5 promotion grammar consisting of: the local OP5 carrier/invariant closure import, the $W^{(1)}_{\mathrm{OP5}}$ and $W^{(2)}_{\mathrm{OP5}}$ witness registers, the two-sided HC/MM-chart comparison bridge

$$\mathfrak{B}^{2s}_{\mathrm{HC/MM}\leftrightarrow\mathrm{chart}},$$

and the no-leakage, no-new-chart-debt, no-reopening, and no-raw-same-variance guards.

Definition A.170.2 (Two-sided OP5 promoted closure). *Inside $\mathcal{G}^{2s}_{\mathrm{OP5}}$, OP5 is two-sided promoted-closed when the OP5 local carrier closure is preserved and the promotion residual vanishes:*

$$\mathrm{Closure}_{\mathrm{loc}}(\mathrm{OP5}) \quad \textit{and} \quad R^{\mathrm{prom}}_{\mathrm{OP5}} = \varnothing.$$

This definition is explicitly grammar-scoped.

Proposition A.170.3 (r01-r06 dependency ledger). *The r07 theorem depends on the following finite ledger:*

stage	*status*
$r01$	*local OP5 carrier/invariant closure imported*
$r02$	*first-family HC/MM witness search executed*
$r03$	*first-family obstruction classified*
$r04$	$W^{(2)}_{\mathrm{OP5}}$ *declared*
$r05$	U_5 *found as conditional strict candidate*
$r06$	U_5 *licensed by the two-sided bridge*

The r06 result gives

$$\Delta_{\mathrm{cpl}}(U_5) = -1, \qquad |L_{2s}(U_5)| = 0, \qquad N_{\mathrm{strict,adm}} = 1.$$

Theorem A.170.4 (OP5 promoted closure inside the two-sided grammar). *Inside the declared two-sided HC/MM-chart comparison grammar $\mathcal{G}^{2s}_{\mathrm{OP5}}$, the licensed witness U_5 is an admissible strict reducer. Therefore*

$$|R^{\mathrm{prom}}_{\mathrm{OP5}}| = 0.$$

Consequently OP5 is promoted-closed inside $\mathcal{G}^{2s}_{\mathrm{OP5}}$:

$$\mathrm{Closure}_{\mathcal{G}^{2s}_{\mathrm{OP5}}}(\mathrm{OP5}).$$

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Proof. The r01 intake imported local OP5 carrier/invariant closure without claiming global OP5 closure. The r02 and r03 layers showed that the first witness family was obstructed. The r04 layer therefore enlarged the witness family. The r05 layer found U_5 with raw $\Delta_{\mathrm{cpl}} = -1$, blocked only by an unlicensed same-variance comparison. The r06 layer introduced the two-sided HC/MM-chart comparison bridge and proved

$$|L_{2s}(U_5)| = 0.$$

Thus the only r05 blocker for U_5 is removed without hidden leakage, new chart debt, or return-side reopening. Hence U_5 is an admissible strict reducer in $\mathcal{G}^{2s}_{\mathrm{OP5}}$, and the OP5 promotion residual vanishes in that grammar. □

Remark A.170.5 (No-overclaim boundary)**.** *The theorem proves OP5 promoted closure only inside the declared two-sided HC/MM-chart comparison grammar. It does not prove unrestricted global OP5 closure, does not prove global physical closure, does not close QFT/GR, and does not mark a release-final state.*

A.171 No-Overclaim and Route-to-Global Boundary Audit

This section audits the boundary of the r07 OP5 two-sided promoted-closure theorem. The audit preserves the r07 result inside the declared two-sided HC/MM-chart comparison grammar, but blocks unrestricted global OP5, all-readout OP5, global physical, and release-final claims.

Definition A.171.1 (Route-to-global boundary residual)**.** *Let*

$$B^{\mathrm{OP5}}_{\mathrm{global}}$$

denote the boundary residual between the declared two-sided OP5 promotion grammar $\mathcal{G}^{2s}_{\mathrm{OP5}}$ and an unrestricted global OP5 claim. It is not the same object as $R^{\mathrm{prom}}_{\mathrm{OP5}}$. The r07 theorem proves

$$R^{\mathrm{prom}}_{\mathrm{OP5}} = 0 \quad \textit{inside} \quad \mathcal{G}^{2s}_{\mathrm{OP5}},$$

but it does not prove

$$B^{\mathrm{OP5}}_{\mathrm{global}} = 0.$$

Proposition A.171.2 (Allowed and blocked claims)**.** *The r08 claim ledger is*

claim	*allowed*	*scope*
$\mathrm{Closure}_{\mathcal{G}^{2s}_{\mathrm{OP5}}}(\mathrm{OP5})$	yes	*declared two-sided grammar*
unrestricted global OP5 closure	no	*not established*
all-readout OP5 closure	no	*not established*
global physical closure	no	*not established*
release-final status	no	*working line only*

Proposition A.171.3 (Route-to-global boundary ledger)**.** *The boundary ledger gives*

gate	*value*
two-sided OP5 gates passed	1
unrestricted global gates passed	0
$B^{\mathrm{OP5}}_{\mathrm{global}}$	1

where $B^{\mathrm{OP5}}_{\mathrm{global}} = 1$ is a scope-boundary marker, not a reopening of $R^{\mathrm{prom}}_{\mathrm{OP5}}$ inside $\mathcal{G}^{2s}_{\mathrm{OP5}}$.

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Theorem A.171.4 (r08 no-overclaim boundary theorem)**.** *The r07 theorem may be stated as*

$$\mathrm{Closure}_{\mathcal{G}^{2s}_{\mathrm{OP5}}}(\mathrm{OP5}).$$

It may not be strengthened in r08 to unrestricted global OP5 closure, all-readout OP5 closure, global physical closure, or release-final status. The route from $\mathcal{G}^{2s}_{\mathrm{OP5}}$ to a larger global claim remains guarded by

$$B^{\mathrm{OP5}}_{\mathrm{global}} = 1.$$

Remark A.171.5 (Interpretation)**.** *The boundary residual in r08 does not undo the two-sided OP5 promoted closure. It records exactly where the next promotion problem begins. In plain terms: OP5 is promoted closed in the declared two-sided grammar, but the unrestricted global route remains outside the proved scope.*

A.172 OP5 / OP1–OP4 Compatibility Audit

This section audits compatibility between the public v4.4 OP1/OP4 declared-grammar closure and the v4.5.0 OP5 two-sided promoted-closure result. It asks whether the OP5 promotion reopens any OP1/OP4 ledger channel. It does not claim unified global closure.

Definition A.172.1 (Compatibility product grammar)**.** *Let*

$$\mathcal{G}^{1/4,5}_{\mathrm{cmp}} := \mathcal{G}^{\mathrm{loc}}_{1/4} \boxtimes \mathcal{G}^{2s}_{\mathrm{OP5}}$$

be the compatibility product of the local OP1/OP4 declared grammar and the two-sided OP5 promotion grammar. The product symbol means coexistence with explicit cross-reopen guards; it does not mean unrestricted global collapse of the two grammars.

Definition A.172.2 (OP5-to-OP1/OP4 reopen residual)**.** *Define*

$$R^{1/4\leftarrow 5}_{\mathrm{reopen}}$$

as the finite residual measuring whether the OP5 two-sided promoted closure reopens any OP1/OP4 local closure channel: scale, lens, chart, debt, selector, or coverage.

Proposition A.172.3 (r09 compatibility ledger)**.** *The compatibility ledger gives*

OP1/OP4 reopen channel	*residual*
scale	0
lens	0
chart	0
debt	0
selector	0
coverage	0
$R^{1/4\leftarrow 5}_{\mathrm{reopen}}$	0

so the OP5 two-sided promotion does not reopen the v4.4 OP1/OP4 local closure ledger.

Proof. The OP5 theorem from r07 is scoped to $\mathcal{G}^{2s}_{\mathrm{OP5}}$, while r08 explicitly keeps the route-to-global boundary outside the theorem. The boundary marker $B^{\mathrm{OP5}}_{\mathrm{global}} = 1$ is not imported as scale, lens, chart, debt, selector, or coverage debt inside $\mathcal{G}^{\mathrm{loc}}_{1/4}$. Therefore none of the v4.4 OP1/OP4 closure channels reopens. □

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Theorem A.172.4 (Combined scoped compatibility surface)**.** *The following two scoped closure statements are compatible:*

$$\mathrm{Closure}_{\mathcal{G}^{\mathrm{loc}}_{1/4}}(\mathrm{OP1},\mathrm{OP4})$$

and

$$\mathrm{Closure}_{\mathcal{G}^{2s}_{\mathrm{OP5}}}(\mathrm{OP5}).$$

Equivalently, inside the compatibility product grammar,

$$\mathrm{Closure}_{\mathcal{G}^{1/4,5}_{\mathrm{cmp}}}(\mathrm{OP1},\mathrm{OP4};\mathrm{OP5})$$

is allowed with

$$R^{1/4\leftarrow 5}_{\mathrm{reopen}} = 0.$$

This is a combined scoped surface only, not unified global closure.

Remark A.172.5 (Global boundary)**.** *The route-to-global boundary remains*

$$B^{\mathrm{OP5}}_{\mathrm{global}} = 1.$$

This boundary marker prevents the combined scoped surface from being rewritten as unrestricted global OP1/OP4/OP5 closure.

A.173 Two-Sided Tensor Grammar Consolidation

This section consolidates the notation used by the v4.4 OP1/OP4 local closure and the v4.5.0 OP5 two-sided promoted closure. It is a grammar and notation layer only. It preserves the scoped compatibility surface from r09 and does not claim unified global closure.

Definition A.173.1 (r10 grammar normal form)**.** *Set*

$$\mathcal{G}_L := \mathcal{G}^{\mathrm{loc}}_{1/4}, \qquad \mathcal{G}_T := \mathcal{G}^{2s}_{\mathrm{OP5}}.$$

Here $\mathcal{G}_L$ is the local declared OP1/OP4 grammar and $\mathcal{G}_T$ is the two-sided OP5 HC/MM-chart comparison grammar. Their compatibility product is written

$$\mathcal{G}_C := \mathcal{G}_L \boxtimes_{\mathfrak{C}} \mathcal{G}_T,$$

where $\mathfrak{C}$ is an explicit comparison tensor. The symbol $\boxtimes_{\mathfrak{C}}$ denotes guarded compatibility, not unrestricted global tensor contraction.

Definition A.173.2 (Two-sided comparison tensor)**.** *The comparison tensor*

$$\mathfrak{C} : \mathcal{R}_{\mathrm{readout}} \times \mathcal{S}_{\mathrm{structure}} \longrightarrow \mathcal{C}_{\mathrm{cmp}}$$

separates readout-facing indices from structure-facing indices before any comparison is made. A same-variance comparison is admissible only when it factors through $\mathfrak{C}$. Raw same-variance contraction remains forbidden.

Proposition A.173.3 (r10 grammar rules)**.** *The consolidated grammar obeys the following rules:*

rule	*content*
TSG1	*readout and structure indices remain typed*
TSG2	*same-variance comparison requires* $\mathfrak{C}$
TSG3	$\boxtimes_{\mathfrak{C}}$ *is not global grammar collapse*
TSG4	B_G *remains outside the scoped closure*
TSG5	$R^{1/4\leftarrow 5}_{\mathrm{reopen}} = 0$

where B_G is the route-to-global boundary marker inherited from r08.

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Theorem A.173.4 (Consolidated scoped grammar surface)**.** *In the r10 normal form, the two scoped closure statements are*

$$\mathrm{Closure}_{\mathcal{G}_L}(\mathrm{OP1}, \mathrm{OP4})$$

and

$$\mathrm{Closure}_{\mathcal{G}_T}(\mathrm{OP5}).$$

The compatibility audit from r09 gives

$$R_{\mathrm{reopen}}^{1/4 \leftarrow 5} = 0,$$

so they define the combined scoped surface

$$\mathrm{Closure}_{\mathcal{G}_C}(\mathrm{OP1}, \mathrm{OP4}; \mathrm{OP5}).$$

This theorem is a consolidated notation theorem for the scoped product $\mathcal{G}_C$. It is not a unified global closure theorem.

Remark A.173.5 (Boundary preserved)**.** *The global boundary marker remains*

$$B_G = 1.$$

It does not reopen the scoped product closure. It marks exactly the route from the combined scoped surface to any future larger global claim.

A.174 Compatibility Theorem and No-Reopen Summary

This section summarizes the compatibility result after the r10 two-sided tensor grammar consolidation. It also records a short notation guide for the symbol $\boxtimes_{\mathfrak{C}}$, since this symbol is not meant as an ordinary multiplication sign.

Definition A.174.1 (Guarded compatibility product)**.** *The symbol*

$$\boxtimes_{\mathfrak{C}}$$

is read as a guarded compatibility product over the comparison tensor $\mathfrak{C}$. In the present setting

$$\mathcal{G}_C = \mathcal{G}_L \boxtimes_{\mathfrak{C}} \mathcal{G}_T$$

means: combine the local OP1/OP4 grammar $\mathcal{G}_L$ and the two-sided OP5 grammar $\mathcal{G}_T$ only through the explicit comparison tensor $\mathfrak{C}$. It does not mean ordinary multiplication, raw tensor contraction, or unrestricted global grammar collapse.

Proposition A.174.2 (r11 no-reopen ledger)**.** *The r11 no-reopen summary is*

component	*residual*
$\mathrm{Closure}_{\mathcal{G}_L}(\mathrm{OP1}, \mathrm{OP4})$	0
$\mathrm{Closure}_{\mathcal{G}_T}(\mathrm{OP5})$	0
$R_{\mathrm{reopen}}^{1/4 \leftarrow 5}$	0
$\mathrm{Closure}_{\mathcal{G}_C}(\mathrm{OP1}, \mathrm{OP4}; \mathrm{OP5})$	0
B_G	1

where $B_G = 1$ is the retained route-to-global boundary marker.

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Theorem A.174.3 (Combined scoped no-reopen theorem)**.** *In the r10 normal form, the local OP1/OP4 closure and the OP5 two-sided promoted closure coexist in the guarded compatibility grammar*

$$\mathcal{G}_C = \mathcal{G}_L \boxtimes_{\mathfrak{C}} \mathcal{G}_T.$$

Moreover,

$$R^{1/4\leftarrow 5}_{\text{reopen}} = 0.$$

Therefore

$$\text{Closure}_{\mathcal{G}_C}(\text{OP1}, \text{OP4}; \text{OP5})$$

is a valid combined scoped closure surface.

Proof. The v4.4 line provides the local declared OP1/OP4 closure in $\mathcal{G}_L$. The r07 line provides OP5 promoted closure in $\mathcal{G}_T$. The r09 compatibility audit shows that the OP5 closure does not reopen any OP1/OP4 scale, lens, chart, debt, selector, or coverage channel. The r10 grammar consolidation then rewrites the combined statement through the explicit comparison tensor $\mathfrak{C}$. Hence the combined scoped surface is valid with zero cross-reopen residual. □

Remark A.174.4 (Boundary retained)**.** *The theorem is intentionally scoped. The boundary*

$$B_G = 1$$

is retained. Thus the theorem is not

$$\text{Closure}_{\text{global}}(\text{OP1}, \text{OP4}, \text{OP5}).$$

It is only the combined scoped closure theorem inside $\mathcal{G}_C$.

A.175 Route-to-Global Boundary Problem Statement

This section states the next problem after the r11 combined scoped no-reopen theorem. The scoped theorem remains valid:

$$\text{Closure}_{\mathcal{G}_C}(\text{OP1}, \text{OP4}; \text{OP5}), \qquad R^{1/4\leftarrow 5}_{\text{reopen}} = 0.$$

However, the route-to-global boundary marker remains

$$B_G = 1.$$

The purpose of r13 is to name the exact boundary problem, not to solve it.

Definition A.175.1 (Route-to-global promotion problem)**.** *The route-to-global promotion problem is the problem of deciding whether the scoped compatibility grammar*

$$\mathcal{G}_C = \mathcal{G}_L \boxtimes_{\mathfrak{C}} \mathcal{G}_T$$

can be embedded into a larger grammar $\mathcal{G}_{\text{global}}$ without creating new debt, losing comparison typing, or reopening any OP1/OP4 or OP5 closure channel. The problem is denoted

$$\text{Promote}(\mathcal{G}_C \to \mathcal{G}_{\text{global}}).$$

No such promotion is assumed in r12.

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Definition A.175.2 (Global-boundary gate ledger)**.** *The boundary marker B_G is discharged only if the following gates are closed:*

gate	*meaning*
GLOB1	*grammar exhaustion*
GLOB2	*all-readout invariance*
GLOB3	*boundary transport without new debt*
GLOB4	*no-new-debt under grammar extension*
GLOB5	*functorial comparison stability*

The physical-interpretation guard is separate: even a mathematical grammar promotion would not by itself be a global physical closure theorem.

Proposition A.175.3 (r13 boundary ledger)**.** *At r12, the global-boundary ledger is*

quantity	*value*
required global gates	5
passed global gates	0
open global gates	5
B_G	1

Therefore the route-to-global boundary remains open.

Theorem A.175.4 (Scoped closure with open global boundary)**.** *The r11 scoped no-reopen theorem remains valid:*

$$\mathrm{Closure}_{\mathcal{G}_C}(\mathrm{OP1}, \mathrm{OP4}; \mathrm{OP5})$$

and

$$R^{1/4\leftarrow 5}_{\mathrm{reopen}} = 0.$$

But r13 does not prove

$$\mathrm{Closure}_{\mathrm{global}}(\mathrm{OP1}, \mathrm{OP4}, \mathrm{OP5}).$$

Instead, it records the precise open boundary:

$$B_G = 1.$$

Remark A.175.5 (Interpretation)**.** *In plain language, v4.5.0 has reached a combined scoped closure surface. The next problem is no longer OP5 local promotion; it is the route from the scoped surface to any larger/global grammar. That route requires new gates, not just renaming the scoped theorem as global.*

A.176 Reader-Facing Notation Cleanup

This section is an expository notation cleanup. It introduces no new closure claim. Its purpose is to make the symbols from the v4.5.0 working line readable without weakening the no-overclaim guards.

Definition A.176.1 (Boxed-times compatibility product)**.** *The symbol*

$$\boxtimes_{\mathfrak{C}}$$

is read as a guarded compatibility product over the comparison tensor $\mathfrak{C}$*. Thus*

$$\mathcal{G}_C = \mathcal{G}_L \boxtimes_{\mathfrak{C}} \mathcal{G}_T$$

means that the local OP1/OP4 grammar $\mathcal{G}_L$ and the two-sided OP5 grammar $\mathcal{G}_T$ are compared only through the explicit bridge $\mathfrak{C}$. It does not mean ordinary multiplication, raw tensor contraction, or unrestricted global grammar collapse.

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Definition A.176.2 (Reader glossary)**.** *The working symbols should be read as follows:*

symbol	***reader meaning***
$\mathcal{G}_L$	*the local declared OP1/OP4 grammar from the v4.4 line*
$\mathcal{G}_T$	*the two-sided OP5 HC/MM-chart comparison grammar*
$\mathfrak{C}$	*the explicit comparison tensor separating readout and structure sides*
$\mathcal{G}_C$	*the combined scoped grammar* $\mathcal{G}_L \boxtimes_{\mathfrak{C}} \mathcal{G}_T$
B_G	*the retained route-to-global boundary marker*
$R^{1/4\leftarrow 5}_{\mathrm{reopen}}$	*the cross-reopen residual from OP5 back to OP1/OP4*
GLOB1–GLOB5	*the five open gates for any future route-to-global promotion*

Proposition A.176.3 (Reader-safe separation of residuals)**.** *The two residual markers*

$$R^{1/4\leftarrow 5}_{\mathrm{reopen}} \qquad \textit{and} \qquad B_G$$

have different jobs. The first asks whether OP5 reopens the OP1/OP4 local closure ledger. In the r11 summary this residual is zero. The second records that the route from the scoped grammar $\mathcal{G}_C$ *to a larger/global grammar is still open. In r13 this marker remains*

$$B_G = 1.$$

Thus the global boundary marker is not a reopened scoped residual.

Remark A.176.4 (Plain-language reading)**.** *The expression*

$$\mathrm{Closure}_{\mathcal{G}_C}(\mathrm{OP1}, \mathrm{OP4}; \mathrm{OP5})$$

should be read as: OP1/OP4 and OP5 are closed together on the declared guarded compatibility surface. It should not be read as: all global readouts and all physical interpretations are closed. The latter would require the open gates GLOB1 *through* GLOB5*, which remain open at this stage.*

Theorem A.176.5 (Reader-clean notation handoff)**.** *The r14 cleanup preserves the r11/r13 state:*

$$\mathrm{Closure}_{\mathcal{G}_C}(\mathrm{OP1}, \mathrm{OP4}; \mathrm{OP5}), \qquad R^{1/4\leftarrow 5}_{\mathrm{reopen}} = 0, \qquad B_G = 1.$$

It introduces no global closure theorem, no global physical closure theorem, and no release-final state.

A.177 Global-Boundary Witness-Menu Design

This section designs finite witness menus for the five open route-to-global gates from r12. It does not execute the menus. It does not discharge B_G, and it does not claim global closure.

Definition A.177.1 (Global-boundary witness families)**.** *For the five open gates*

$$\mathrm{GLOB1}, \ldots, \mathrm{GLOB5},$$

define the witness families

gate	*family*	*role*
GLOB1	W_{exh}	*grammar exhaustion*
GLOB2	W_{inv}	*all-readout invariance*
GLOB3	W_{tr}	*boundary transport*
GLOB4	W_{debt}	*no-new-debt extension*
GLOB5	W_{fun}	*functorial comparison stability*

These families are finite menus for future execution, not proof objects in r15.

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Proposition A.177.2 (r15 witness-menu ledger). *The r15 design ledger is*

quantity	*value*
witness menus designed	5
witness menus executed	0
new global gates passed	0
open global gates after r15	5
B_G	1

Therefore r15 is a design step only.

Definition A.177.3 (Execution order). *The intended execution order is*

$$W_{\mathrm{exh}} \longrightarrow W_{\mathrm{inv}} \longrightarrow W_{\mathrm{tr}} \longrightarrow W_{\mathrm{debt}} \longrightarrow W_{\mathrm{fun}}.$$

Grammar exhaustion comes first because all-readout invariance and boundary transport are underspecified if admissible branches have not first been classified.

Theorem A.177.4 (No-discharge theorem for r15). *The r15 layer defines the finite witness menus needed for the route-to-global program, but it does not execute any witness family. Hence*

$$B_G = 1$$

remains unchanged, and no global OP1/OP4/OP5 closure is claimed.

Remark A.177.5 (Plain reading). *The scoped closure surface from r11 is still intact. r15 only turns the open global-boundary problem into a concrete work queue.*

A.178 Grammar Exhaustion Witness Execution

This section executes the first global-boundary witness family from r15:

$$W_{\mathrm{exh}} = \{E_0, E_1, E_2\}.$$

It tests only the grammar-exhaustion gate GLOB1. It does not test all-readout invariance, boundary transport, no-new-debt under extension, or functorial comparison stability.

Definition A.178.1 (Grammar-exhaustion witness family). *The family W_{exh} consists of:*

E_0	*atlas-coverage witness*
E_1	*readout-factorization witness*
E_2	*no-orphan-branch witness*

The family passes if every admissible branch of the current scoped surface is represented in $\mathcal{G}_C$ or routed to a declared open boundary, and no admissible branch remains orphan.

Proposition A.178.2 (r16 branch classification register). *The branch register is*

branch	*classification*
C_0 *OP1/OP4 local*	*represented*
C_1 *OP5 two-sided*	*represented*
C_2 *compatibility product*	*represented*
C_3 *route-to-global boundary*	*routed to* B_G
C_4 *raw same-variance comparison*	*blocked, not admissible*
C_5 *global physical interpretation*	*blocked, not admissible*

Thus

$$N_{\mathrm{orphan}} = 0, \qquad N_{\mathrm{new\,debt}} = 0.$$

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Theorem A.178.3 (GLOB1 grammar-exhaustion gate passed)**.** *The witness family* W_{exh} *discharges the grammar-exhaustion gate:*

$$\mathrm{GLOB1} = 1.$$

After r16 the global-boundary ledger is

quantity	*value*
passed global gates	1
open global gates	4
B_G	1

The boundary marker remains $B_G = 1$ *because* GLOB2 *through* GLOB5 *remain open.*

Remark A.178.4 (No global closure claim)**.** *The r16 result is a coverage/exhaustion result only. It does not prove global promotion and does not claim global OP1/OP4/OP5 closure.*

A.179 Declared Readout Invariance Witness Execution

This section executes the second global-boundary witness family:

$$W_{\mathrm{inv}} = \{I_0, I_1, I_2\}.$$

The result is scoped to the declared admissible readout transformations in the current grammar. It is not an unrestricted all-possible-readouts theorem.

Definition A.179.1 (Declared readout-invariance witness family)**.** *The family* W_{inv} *consists of:*

I_0	*declared chart-change witness*
I_1	*declared readout-change witness*
I_2	*declared return-phase-change witness*

It passes if the scoped closure surface is preserved, no OP1/OP4 or OP5 channel reopens, no new debt is created, and the boundary marker B_G *is not rewritten as a solved global claim.*

Proposition A.179.2 (r17 readout-invariance register)**.** *The declared readout-invariance register is*

witness	*classification*	R_{reopen}	N_{debt}	$N_{\mathrm{boundary\ rewrite}}$
I_0	*invariant*	0	0	0
I_1	*invariant*	0	0	0
I_2	*invariant*	0	0	0

Thus the declared readout-invariance test has no reopen, no new debt, and no boundary rewrite.

Theorem A.179.3 (GLOB2 declared-readout-invariance gate passed)**.** *The witness family* W_{inv} *discharges the declared readout-invariance gate:*

$$\mathrm{GLOB2} = 1.$$

After r17 the global-boundary ledger is

quantity	*value*
passed global gates	2
open global gates	3
B_G	1

The boundary marker remains $B_G = 1$ *because* GLOB3 *through* GLOB5 *remain open.*

Remark A.179.4 (Scope guard)**.** *The word "all" in this r17 layer means all declared/admissible readout transformations currently in the grammar. It does not mean an unrestricted theorem over every possible external readout or every possible physical model.*

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A.180 Boundary Transport Witness Execution

This section executes the third global-boundary witness family:

$$W_{\mathrm{tr}} = \{T_0, T_1, T_2\}.$$

The purpose is to test whether the route-to-global boundary marker B_G can be transported through the interface without creating new debt. Transport is not discharge: B_G remains an explicit boundary marker.

Definition A.180.1 (Boundary-transport witness family)**.** *The family W_{tr} consists of:*

T_0	*boundary-carry witness*
T_1	*debt-neutral transport witness*
T_2	*OP5-return transport witness*

It passes if B_G is carried as a boundary marker and no new scale, lens, chart, selector, coverage, or OP5-return debt appears.

Proposition A.180.2 (r18 boundary-transport register)**.** *The transport register is*

witness	*classification*	$N_{\mathrm{new\,debt}}$	B_G *after*
T_0	*transported boundary marker*	0	1
T_1	*debt neutral*	0	1
T_2	*return transport stable*	0	1

Thus boundary transport is stable and debt-free, but the boundary marker is not erased.

Theorem A.180.3 (GLOB3 boundary-transport gate passed)**.** *The witness family W_{tr} discharges the boundary-transport gate:*

$$\mathrm{GLOB3} = 1.$$

After r18 the global-boundary ledger is

quantity	*value*
passed global gates	3
open global gates	2
B_G	1

The marker $B_G = 1$ remains because GLOB4 *and* GLOB5 *remain open.*

Remark A.180.4 (No discharge by transport)**.** *The r18 result says that the boundary can be carried without new debt. It does not say that the boundary is solved. Therefore no global OP1/OP4/OP5 closure is claimed.*

A.181 No-New-Debt Extension Witness Execution

This section executes the fourth global-boundary witness family:

$$W_{\mathrm{debt}} = \{D_0, D_1, D_2, D_3, D_4, D_5\}.$$

The purpose is to test whether the declared grammar-extension buckets create new debt. This is still not a global closure claim, because the functorial comparison-stability gate remains open.

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Definition A.181.1 (No-new-debt extension witness family)**.** *The family* W_{debt} *consists of:*

D_0	*scale channel*
D_1	*lens channel*
D_2	*chart channel*
D_3	*selector channel*
D_4	*coverage channel*
D_5	*OP5-return channel*

It passes if none of these channels creates new debt, hidden leakage, or reopening under the declared grammar extension.

Proposition A.181.2 (r19 no-new-debt register)**.** *The no-new-debt register is*

witness	*channel*	$N_{\mathrm{new\,debt}}$	R_{reopen}	L_{leak}
D_0	*scale*	0	0	0
D_1	*lens*	0	0	0
D_2	*chart*	0	0	0
D_3	*selector*	0	0	0
D_4	*coverage*	0	0	0
D_5	*OP5-return*	0	0	0

Thus the declared grammar extension is debt-free for the tested channels.

Theorem A.181.3 (GLOB4 no-new-debt extension gate passed)**.** *The witness family* W_{debt} *discharges the no-new-debt extension gate:*

$$\mathrm{GLOB4} = 1.$$

After r19 the global-boundary ledger is

quantity	*value*
passed global gates	4
open global gates	1
B_G	1

The marker $B_G = 1$ *remains because* GLOB5*, functorial comparison stability, remains open.*

Remark A.181.4 (No global closure claim)**.** *The r19 result says that the declared extension buckets are debt-free. It does not say that the comparison tensor is functorially stable under all needed extension maps. Therefore no global OP1/OP4/OP5 closure is claimed in r19.*

A.182 Functorial Comparison Stability Witness Execution

This section executes the fifth global-boundary witness family:

$$W_{\mathrm{fun}} = \{F_0, F_1, F_2, F_3\}.$$

The purpose is to test whether the comparison tensor $\mathfrak{C}$ remains stable under the declared extension maps. Passing this gate makes the boundary synthesis-ready, but it does not itself discharge B_G.

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Definition A.182.1 (Functorial comparison-stability witness family)**.** *The family* W_{fun} *consists of:*

F_0	$\mathfrak{C}$*-naturalness*
F_1	*composition stability*
F_2	*identity stability*
F_3	*variance preservation*

It passes if the comparison tensor commutes with the declared extension maps, respects composition and identity, preserves variance typing, and introduces no raw same-variance comparison.

Proposition A.182.2 (r20 functorial-stability register)**.** *The functorial comparison register is*

witness	*condition*	*defect*
F_0	$\mathfrak{C}$*-naturalness*	0
F_1	*composition*	0
F_2	*identity*	0
F_3	*variance preservation*	0

Thus the declared comparison tensor is functorially stable for the tested extension maps.

Theorem A.182.3 (GLOB5 functorial-comparison-stability gate passed)**.** *The witness family* W_{fun} *discharges the functorial comparison stability gate:*

$$\mathrm{GLOB5} = 1.$$

After the functorial-comparison-stability gate, the global-boundary ledger is

quantity	*value*
passed global gates	5
open global gates	0
B_G	1
synthesis-ready	1

The marker B_G *is not discharged by that individual gate. It is carried into the boundary-gate synthesis audit.*

Remark A.182.4 (No premature global closure)**.** *The functorial-stability result is the last individual gate execution. It does not by itself claim global OP1/OP4/OP5 closure. The next layer must synthesize the five passed gates and decide whether* B_G *can be discharged without overclaim.*

A.183 Boundary-Gate Synthesis Audit

This section synthesizes the five previously executed global-boundary gates. It decides whether the retained boundary marker B_G may be discharged inside the declared mathematical declared global-boundary synthesis grammar. The result is not a global physical closure theorem and not an unrestricted claim beyond the declared extension maps.

Definition A.183.1 (Declared declared global-boundary synthesis grammar)**.** *Let*

$$\mathcal{G}^{\mathrm{decl}}_{\mathrm{global}}$$

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be the grammar obtained from the scoped compatibility grammar $\mathcal{G}_C$ by adjoining the five passed global-boundary gates:

$$\mathrm{GLOB1}, \ldots, \mathrm{GLOB5}.$$

This is a declared mathematical grammar. It is not the class of all possible external readouts or all possible physical models.

Proposition A.183.2 (r21 synthesis ledger)**.** *The r21 synthesis ledger is*

quantity	*value*
global-boundary gates	5
passed gates	5
open gates	0
B_G *before synthesis*	1
B_G *after synthesis*	0

All five global-boundary gates are passed, so the boundary marker is discharged inside $\mathcal{G}^{\mathrm{decl}}_{\mathrm{global}}$.

Theorem A.183.3 (Declared mathematical global-boundary closure)**.** *Inside the declared mathematical declared global-boundary synthesis grammar $\mathcal{G}^{\mathrm{decl}}_{\mathrm{global}}$, the OP1/OP4/OP5 closure surface is closed:*

$$\mathrm{Closure}_{\mathcal{G}^{\mathrm{decl}}_{\mathrm{global}}}(\mathrm{OP1}, \mathrm{OP4}, \mathrm{OP5}).$$

Equivalently,

$$B_G = 0$$

inside the declared global-boundary synthesis grammar.

Proof. The grammar-exhaustion gate discharges grammar exhaustion. The readout-invariance gate discharges declared readout invariance. The debt-free transport gate transports the boundary without new debt. The no-new-debt extension gate proves no-new-debt for the declared extension buckets. The functorial-stability gate proves functorial stability of the comparison tensor $\mathfrak{C}$ under the declared extension maps. Therefore all five global-boundary gates are passed. Since no open gate remains and no new debt, reopening, leakage, variance defect, or comparison-instability is recorded, the boundary marker B_G is discharged inside the declared mathematical declared global-boundary synthesis grammar. □

Remark A.183.4 (Scope guard)**.** *The declared global-boundary theorem is a mathematical grammar-closure theorem. It is not a theorem over every possible external readout, not a global physical closure theorem, not a QFT/GR closure theorem, and not release-final status. Those claims remain outside the declared grammar scope.*

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A.184 Compressed Editorial and Release-Provenance Layer

Compression status. This section replaces a long sequence of internal editorial, proof-read, glossary-intake, metadata, release-freeze, and reader-facing phrase-normalization worksheets from the pre-v4.7 provenance layer. The removed material is preserved verbatim in the RC16 source packet as `RC14_uncompressed_reference_source.tex`. No theorem, definition, operator, DOI, or claim-scope object is changed by this compression.

Theoremlet A.184.1 (Editorial-provenance compression guard)**.** *Let $\mathcal{P}_{\mathrm{edit}}$ denote the compressed editorial/release-provenance worksheet chain. Replacing $\mathcal{P}_{\mathrm{edit}}$ by this summary is release-safe for the main monolith provided that the uncompressed source is retained in the provenance packet and that no mathematical label from $\mathcal{P}_{\mathrm{edit}}$ is externally referenced. In the RC16 audit, the external-reference count for removed labels is zero. Hence the replacement is a presentation compression only.*

Compressed content. The compressed worksheet chain recorded: active-core rebuild checks, release-candidate audit planning, reader-facing global-scope disclaimers, DOI synchronization, release-freeze audits, frontmatter repair, line-level proofreading, terminology sweeps, glossary/index intake, glossary appendix placement decisions, and compact glossary draft routing. All of these are editorial provenance and release-hygiene layers. They do not alter the proof spine, the local grammar, the bridge obstruction analysis, the OP1/OP4 status, the OP5/C5 classification, or the v4.7.0 guarded effective/local claim scope.
Preserved guards. The intended v4.7.0 Version DOI remains 10.5281/zenodo.20232911. The previous baseline DOI remains 10.5281/zenodo.20117361 only as provenance. No public release is implied by this compression layer.

v4.6.0 Integrated Bounded-Readout and T2 Scope-Boundary Appendix

Definition A.184.2 (v4.6.0 bounded readout surface)**.** *The integrated v4.6.0 release-final bounded-readout surface is the finite guarded readout surface*

$$W2, W3, W4, W5_{\text{formal-symbolic}}, \begin{aligned} W1 &= \mathsf{pass}, \\ W6 &= \mathsf{pass}_{\mathrm{bounded}}, \\ W7 &= \mathsf{scope\text{-}boundary\ locked}. \end{aligned}$$

The status $\mathsf{pass}_{\mathrm{bounded}}$ means that the corresponding readout packet is admitted only in the declared bounded grammar. It is not an unrestricted all-readout witness.

Theorem A.184.3 (Bounded readout closure in the declared grammar)**.** *Inside the declared global-boundary grammar, the v4.6.0 bounded readout surface is conservative and debt-free:*

$$\mathsf{BRead}_{\mathrm{bounded}} \in \mathsf{Cons}(\mathcal{G}^{\mathrm{decl}}_{\mathrm{global}}), \qquad \mathsf{debt} = 0, \qquad \mathsf{leak} = 0.$$

This theorem is a bounded formal-symbolic statement. It does not promote an unrestricted physical readout and does not assert an all-readout completion.

Theorem A.184.4 (T2 scope-boundary and non-implication)**.** *The bounded readout theorem does not imply T2. Equivalently,*

$$\mathsf{BRead}_{\mathrm{bounded}} \not\Rightarrow T2, \qquad T2 \in \mathsf{Ext}(\mathcal{G}^{\mathrm{decl}}_{\mathrm{global}}).$$

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The missing object is an all-readout witness family

$$\Theta^{\mathrm{all}} = \{\Theta_R : \mathcal{G}^{\mathrm{decl}}_{\mathrm{global}} \to R\}_{R \in \mathsf{Read}_{\mathrm{all}}}.$$

No such family is constructed in this v4.6.0 release-final source.

Definition A.184.5 (Frozen all-readout obstruction)**.** *The frozen v4.6.0 all-readout obstruction is*

$$\begin{aligned}\mathfrak{O}^{4.6.0}_{\mathrm{all}} = \{&\textit{missing cofinal universal exterior selector},\\ &\textit{missing all-readout naturality family},\\ &\textit{unpromoted unrestricted exterior collar slot}\}.\end{aligned}$$

This obstruction is a scope boundary, not a contradiction.

Remark A.184.6 (Millennium-matrix and Heisenberg-compensator status)**.** *The v4.6.0 line separates the bounded, exterior, and mixed portions of the Millennium-matrix and Heisenberg-compensator bookkeeping:*

$$\mathcal{M} = \mathcal{M}_{\mathrm{bd}} \oplus \mathcal{M}_{\mathrm{ext}} \oplus \mathcal{M}_{\mathrm{mix}}, \qquad \mathcal{H} = \mathcal{H}_{\mathrm{bd}} \oplus \mathcal{H}_{\mathrm{ext}} \oplus \mathcal{H}_{\mathrm{mix}}.$$

The bounded and finite exterior/mixed fragments are sufficient for the bounded readout statement. They are not sufficient for an all-readout theorem.

Definition A.184.7 (Structure-first field-equation reset template)**.** *The release-final field-equation template is*

$$\mathfrak{E}_{\mathrm{decl}}(\Phi; \mathcal{M}, \mathcal{H}, \mathsf{QAM}) = \mathfrak{G}_{\mathrm{struct}}(\Phi) + \mathfrak{C}_{\mathcal{H}}(\Phi) + \mathfrak{M}_{\mathcal{M}}(\Phi) + \mathfrak{Q}_{\mathsf{QAM}}(\Phi) - \mathfrak{J}_{\mathrm{readout}}(\Phi).$$

The bounded instance uses only the declared bounded and finite guarded fragments. The all-readout instance is deliberately not asserted.

Remark A.184.8 (Two-sided structural tensor notation)**.** *The two-sided tensor notation is used only as a conservative compression layer. Front or left-sided indices denote frame, readout, or language interfaces; back or right-sided indices denote structure, action, or field interfaces. Classical upper–lower contraction remains ordinary action or readout. Same-variance comparisons require an explicitly declared comparison tensor and are not raw Einstein contractions.*

Remark A.184.9 (v4.7 exterior–mixed notation guard)**.** *The next MM/HC exterior–mixed field-equation program should write each proposed mixed term with explicit sidedness. Front-side indices carry readout, frame, or language interfaces; back-side indices carry structure, action, or field interfaces. Any same-variance comparison between exterior and bounded channels must pass through a declared comparison tensor or selector. Thus the v4.7 program may sharpen $\mathcal{M}_{\mathrm{ext}}$, $\mathcal{M}_{\mathrm{mix}}$, $\mathcal{H}_{\mathrm{ext}}$, and $\mathcal{H}_{\mathrm{mix}}$, but it may not silently replace a comparison/readout distinction by an undeclared contraction.*

A.185 Content-State Refresh and v4.7 Handoff Preflight

This release-final content refresh is a reader-routing layer. It promotes no new theorem and assigns no new DOI. Its purpose is to align the visible front routing, nomenclature, and handoff language with the v4.6.0 endpoint: bounded readout is conservative inside the declared grammar; T2 remains an all-readout scope boundary; and the next internal research step is the MM/HC exterior–mixed field-equation program.

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Proposition A.185.1 (Safe refresh invariant)**.** *The content-state refresh preserves the v4.6.0 release-final theorematic content. It updates terminology, routing, and handoff orientation only. In particular, it does not assert T2, does not assert an unrestricted all-readout theorem, and does not turn the field-equation reset template into an empirical validation claim.*

Theorem A.185.2 (Release-final conservativity)**.** *The v4.6.0 release-final integrated source is conservative over the inherited v4.5.x active core plus the v4.6.0 bounded-readout packets. It integrates the bounded readout theorem, the T2 scope boundary, the MM/HC exterior–mixed bookkeeping, the QAM tag extension, and the structure-first field-equation reset under the assigned Version DOI 10.5281/zenodo.20117361, with public release-final metadata normalized, without opening an empirical validation gate, and without claiming T2 closure.*

A.186 Theory-Lift Gate and MM/HC Exterior–Mixed Field-Equation Program

This section opens the first internal working layer after the v4.6.0 content-state refresh. It is a program layer, not a release-final theorem. Its role is to decide which audited hypotheses may now be treated as controlled objects in the next proof-engine pass and to specify the admissible grammar for MM/HC exterior–mixed field-equation candidates.

A.186.1 Theory-Lift gate

Definition A.186.1 (Theory-Lift gate)**.** *A* Theory-Lift gate *is a finite promotion test applied to a prior hypothesis, heuristic, notation package, or diagnostic. It has four output classes:*

$$\mathsf{def}, \qquad \mathsf{lem}, \qquad \mathsf{prog}, \qquad \mathsf{hold}.$$

The class def *means that the object may be introduced as a conservative definition. The class* lem *means that the object may be used as a local lemma or corollary inside its declared grammar. The class* prog *means that the object is not yet a theorem but may drive a finite test program or work queue. The class* hold *means that the object remains a hypothesis or scope boundary and must not be promoted.*

Definition A.186.2 (v4.7.0-r01 promotion ledger)**.** *The first promotion ledger consists of the following guarded lifts:*

$$\begin{aligned}
&\mathsf{BRead}_{\mathrm{bounded}} && \mapsto \mathsf{lem}\text{-}\textit{level bounded-readout calculus},\\
&(\mathcal{M}_{\mathrm{bd}}, \mathcal{M}_{\mathrm{ext}}, \mathcal{M}_{\mathrm{mix}}), (\mathcal{H}_{\mathrm{bd}}, \mathcal{H}_{\mathrm{ext}}, \mathcal{H}_{\mathrm{mix}}) && \mapsto \mathsf{def}\text{-}\textit{level channel split},\\
&\textit{two-sided structural tensor notation} && \mapsto \mathsf{def}\text{-}\textit{level admissibility grammar},\\
&\textit{shared MM/HC leakage heuristic} && \mapsto \mathsf{prog}\text{-}\textit{level leakage diagnostic},\\
&\textit{inverse lens-fit calibration} && \mapsto \mathsf{prog}\text{-}\textit{level finite witness-fit method},\\
&\mathfrak{E}_{\mathrm{decl}} && \mapsto \mathsf{prog}\text{-}\textit{level field-equation admissibility template}.
\end{aligned}$$

The all-readout object Θ^{all}, T2, unrestricted exterior collar closure, and global physical-spectrum claims are assigned class hold*.*

Proposition A.186.3 (Promotion conservativity)**.** *The promotion ledger of Definition A.186.2 is conservative over the v4.6.0 release-final baseline. It promotes only guarded object-language, local calculus, and test-program surfaces; it does not promote any all-readout or global physical theorem.*

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Proof. Each promoted item is either a bounded object already isolated by the v4.6.0 readout surface, a notation package whose same-variance comparisons require an explicit comparison tensor, or a finite diagnostic program that returns a residual status rather than a closure theorem. The objects assigned class hold include precisely the missing all-readout and unrestricted exterior data. Hence no promotion crosses the v4.6.0 T2 scope boundary. □

A.186.2 Exterior–mixed sidedness grammar

Definition A.186.4 (Sided MM/HC channel split)**.** *The MM/HC field-equation program is split into bounded, exterior, and mixed channels:*

$$\mathcal{M} = \mathcal{M}_{\mathrm{bd}} \oplus \mathcal{M}_{\mathrm{ext}} \oplus \mathcal{M}_{\mathrm{mix}}, \qquad \mathcal{H} = \mathcal{H}_{\mathrm{bd}} \oplus \mathcal{H}_{\mathrm{ext}} \oplus \mathcal{H}_{\mathrm{mix}}.$$

The bounded channels are inherited from the v4.6.0 declared grammar. The exterior channels record readouts or selectors that are not covered by the bounded surface. The mixed channels record terms whose interpretation requires both a bounded and an exterior interface. No mixed term is admissible unless its left/readout side and right/structure side are explicitly declared.

Definition A.186.5 (Exterior–mixed admissible tensor term)**.** *A candidate term*

$$\mathcal{T}$$

is exterior–mixed admissible *when it satisfies all of the following conditions.*

(a) The front indices $^{A}{}_{B}$ are declared as readout, frame, or language interfaces.

(b) The back indices $^{i}{}_{j}$ are declared as structure, action, or field interfaces.

(c) Every upper–lower contraction is an ordinary action/readout.

(d) Every same-variance comparison is mediated by an explicitly declared comparison tensor or selector, for example $G_{\alpha\beta}X^{\alpha}Y^{\beta}$ or $\chi_{mn}R^{m}S^{n}$.

(e) The term has a bounded restriction, an exterior residual register, and a mixed leakage register.

Raw same-variance contraction and undeclared bounded/exterior identification are not admissible.

Lemma A.186.6 (Sidedness prevents silent closure promotion)**.** *An exterior–mixed admissible tensor term cannot by itself promote a bounded readout statement to an all-readout theorem.*

Proof. By Definition A.186.5, every mixed or same-variance expression carries an explicit comparison tensor, selector, or residual register. Therefore an exterior contribution remains visible as comparison data, selector data, or leakage/debt data. It is not erased by Einstein summation and cannot be mistaken for an unrestricted readout identity. □

A.186.3 Field-equation admissibility template

Definition A.186.7 (Exterior–mixed field-equation candidate)**.** *A first MM/HC exterior–mixed field-equation candidate is a package*

$$\mathfrak{E}^{\mathrm{mix}}_{\mathrm{MM/HC}} = \left(\mathfrak{E}_{\mathrm{bd}}, \mathfrak{E}_{\mathrm{ext}}, \mathfrak{E}_{\mathrm{mix}}, \mathfrak{L}_{\mathrm{mix}}, \mathfrak{R}_{\mathrm{fit}}\right),$$

where $\mathfrak{E}_{\mathrm{bd}}$ is the bounded inherited equation surface, $\mathfrak{E}_{\mathrm{ext}}$ is the exterior residual surface, $\mathfrak{E}_{\mathrm{mix}}$ is the declared mixed MM/HC coupling surface, $\mathfrak{L}_{\mathrm{mix}}$ is the leakage/debt register, and $\mathfrak{R}_{\mathrm{fit}}$ is the finite witness-fit register inherited from the inverse lens-fit calibration method.

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Definition A.186.8 (Admissibility residual)**.** *The first residual attached to a candidate of Definition A.186.7 is*

$$\rho^{\mathrm{mix}}_{\mathrm{MM/HC}} = \rho_{\mathrm{bd}} \oplus \rho_{\mathrm{ext}} \oplus \rho_{\mathrm{mix}} \oplus \rho_{\mathrm{leak}} \oplus \rho_{\mathrm{fit}}.$$

Its comparison size may be written in two-sided tensor notation as

$$\left\|\rho^{\mathrm{mix}}_{\mathrm{MM/HC}}\right\|^2_G = G_{\alpha\beta}\rho^{\alpha}\rho^{\beta},$$

where the expression is read only as an explicitly mediated active-channel comparison. It is not a raw same-variance contraction.

Proposition A.186.9 (First admissibility gate)**.** *A candidate $\mathfrak{E}^{\mathrm{mix}}_{\mathrm{MM/HC}}$ passes the first v4.7.0-r01 admissibility gate only if:*

(i) its bounded restriction agrees with the v4.6.0 bounded-readout surface;

(ii) every exterior contribution is represented in ρ_{ext};

(iii) every mixed MM/HC contribution is represented in ρ_{mix} or discharged by an explicit selector/comparison;

(iv) every leakage or debt contribution is represented in ρ_{leak};

(v) every witness-fit claim is routed through ρ_{fit} and remains finite/bounded;

(vi) the candidate makes no T2, all-readout, or global physical-spectrum claim.

Proof. The six clauses are the minimal separation obligations inherited from the v4.6.0 endpoint. The bounded surface may be reused, but exterior and mixed data must remain visible as residual, comparison, selector, leakage, or finite-fit registers. Therefore a candidate that passes the gate is a well-formed test object, not an unrestricted closure theorem. □

A.186.4 Finite v4.7.0-r01 work queue

Definition A.186.10 (First exterior–mixed work queue)**.** *The first work queue for the next internal passes consists of six tasks:*

W_1 *extract all candidate MM/HC exterior–mixed terms from the current object language*,
W_2 *assign two-sided sidedness data to each candidate term*,
W_3 *compute or bound* $\rho_{\mathrm{bd}}, \rho_{\mathrm{ext}}, \rho_{\mathrm{mix}}, \rho_{\mathrm{leak}}, \rho_{\mathrm{fit}}$,
W_4 *classify each term as admitted, selector-routed, leakage-open, blocked, or residual*,
W_5 *test whether any finite witness-fit lowers the mixed residual without relaxing exactness*,
W_6 *return a theory-lift update: definition, lemma/corollary, program, or hold.*

The queue is finite at every declared pass and may be extended only by an explicit audit ledger.

Corollary A.186.11 (Permitted r02 direction)**.** *The next natural internal pass is a finite extraction/classification pass for W_1–W_4, followed only then by a witness-fit pass for W_5. In particular, a direct T2 or all-readout closure attempt is not the next admissible move of this program.*

Remark A.186.12 (Summary)**.** *The v4.7.0-r01 layer therefore turns the earlier "Theorie-Lift" idea into a formal gate: promote only what has a declared grammar and a finite residual interface; keep every all-readout or global physical claim outside the promoted layer until a separate witness family exists.*

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A.187 Finite MM/HC Exterior–Mixed Term Extraction and Classification

This section executes the r01 work queue through W_1–W_4. It is not a new closure theorem. It extracts a finite candidate family from the current v4.7 program grammar, assigns two-sided sidedness data, records the residual components that must remain visible, and classifies each candidate for the next pass.

A.187.1 Extraction frame

Definition A.187.1 (r02 exterior–mixed candidate)**.** *An r02 exterior–mixed candidate is a package*

$$C_k = \left(\mathcal{T}_k, \sigma_k^L, \sigma_k^R, \rho_k, \kappa_k\right),$$

where $\mathcal{T}_k$ is a declared MM/HC term, σ_k^L is its left/readout sidedness, σ_k^R is its right/structure sidedness, ρ_k is the residual tuple

$$\rho_k = (\rho_{\mathrm{bd}}, \rho_{\mathrm{ext}}, \rho_{\mathrm{mix}}, \rho_{\mathrm{leak}}, \rho_{\mathrm{fit}})_k,$$

and κ_k is its classification status. The status alphabet is

$$\begin{array}{ll} \mathsf{A}\ \textit{admitted}, & \mathsf{S}\ \textit{selector-routed}, \\ \mathsf{L}\ \textit{leakage-open}, & \mathsf{R}\ \textit{residual-to-fit}, \\ \mathsf{B}\ \textit{blocked}. & \end{array}$$

Definition A.187.2 (Extracted r02 candidate family)**.** *The finite r02 extraction from the current v4.7 program grammar consists of seven candidates:*

$$\begin{aligned} C_1 &= {}^{\mathrm{bd}}\mathcal{T}^{\mathrm{bd}}_{\mathcal{M}_{\mathrm{bd}}\leftrightarrow\mathcal{H}_{\mathrm{bd}}}, \\ C_2 &= {}^{\mathrm{ext}}\mathcal{T}^{\mathrm{bd}}_{\mathcal{M}_{\mathrm{ext}}\leftrightarrow\mathcal{H}_{\mathrm{bd}}}, \\ C_3 &= {}^{\mathrm{bd}}\mathcal{T}^{\mathrm{ext}}_{\mathcal{M}_{\mathrm{bd}}\leftrightarrow\mathcal{H}_{\mathrm{ext}}}, \\ C_4 &= {}^{\mathrm{mix}}\mathcal{T}^{\mathrm{mix}}_{\mathcal{M}_{\mathrm{mix}}\leftrightarrow\mathcal{H}_{\mathrm{mix}}}, \\ C_5 &= {}^{\mathrm{fit}}\mathcal{T}^{\mathrm{bd}}_{\mathfrak{R}_{\mathrm{fit}}}, \\ C_6 &= {}^{\mathrm{all}}\mathcal{T}^{\mathrm{all}}_{\Theta^{\mathrm{all}}}, \\ C_7 &= G_{\alpha\beta}\rho^{\alpha}\rho^{\beta}. \end{aligned}$$

Here C_1 is the inherited bounded MM/HC surface, C_2 and C_3 are one-sided exterior comparison terms, C_4 is the genuinely mixed MM/HC term, C_5 is the finite witness-fit interface, C_6 is the forbidden all-readout lift, and C_7 is the explicit comparison-tensor grammar for residual size.

Remark A.187.3 (Scope of extraction)**.** *The extraction is deliberately finite and local to the current v4.7 program grammar. It is not a global symbol-mining normalization of every historical formula carried in the Companion. Older OP, HC, MM, lens, and route fragments remain provenance unless explicitly mapped into one of the candidates above.*

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A.187.2 Sidedness and residual ledger

Definition A.187.4 (r02 symbolic residual ledger)**.** *The r02 residual ledger is the following symbolic assignment:*

candidate	(σ^L, σ^R)	$(\rho_{\mathrm{bd}}, \rho_{\mathrm{ext}}, \rho_{\mathrm{mix}}, \rho_{\mathrm{leak}}, \rho_{\mathrm{fit}})$	κ
C_1	$(\mathrm{bd}, \mathrm{bd})$	$(0, 0, 0, 0, 0)$	A
C_2	$(\mathrm{ext}, \mathrm{bd})$	$(0, \epsilon^M_{\mathrm{ext}}, 0, 0, 0)$	S
C_3	$(\mathrm{bd}, \mathrm{ext})$	$(0, \epsilon^H_{\mathrm{ext}}, \epsilon^H_{\mathrm{mix}}, 0, 0)$	S
C_4	$(\mathrm{mix}, \mathrm{mix})$	$(0, 0, \epsilon^{\mathrm{MM/HC}}_{\mathrm{mix}}, \lambda_{\mathrm{leak}}, 0)$	L
C_5	$(\mathrm{fit}, \mathrm{bd})$	$(0, 0, \epsilon^{\mathrm{fit}}_{\mathrm{mix}}, 0, \rho_{\mathrm{fit}})$	R
C_6	$(\mathrm{all}, \mathrm{all})$	$\bot$	B
C_7	$(\mathrm{cmp}, \mathrm{res})$	$(0, 0, 0, 0, 0)$ *as grammar*	A.

The symbols $\epsilon^M_{\mathrm{ext}}$, $\epsilon^H_{\mathrm{ext}}$, $\epsilon^H_{\mathrm{mix}}$, $\epsilon^{\mathrm{MM/HC}}_{\mathrm{mix}}$, λ_{leak}, and ρ_{fit} are not numerical witnesses. They are named residual slots that must be bounded, routed, or absorbed in later passes.

Proposition A.187.5 (r02 classification count)**.** *The r02 classification count is*

$$N = 7, \qquad N_{\mathsf{A}} = 2, \quad N_{\mathsf{S}} = 2, \quad N_{\mathsf{L}} = 1, \quad N_{\mathsf{R}} = 1, \quad N_{\mathsf{B}} = 1.$$

Thus the current grammar has two admitted items, two selector-routed one-sided exterior items, one leakage-open mixed item, one residual-to-fit item, and one blocked all-readout item.

Proof. The count is obtained directly from Definition A.187.4. The admitted items are C_1 and C_7. The selector-routed items are C_2 and C_3. The leakage-open item is C_4. The residual-to-fit item is C_5. The all-readout lift C_6 is blocked because it has no admissible bounded/exterior residual decomposition inside the current grammar. □

A.187.3 Consequences for the next pass

Lemma A.187.6 (No silent all-readout lift)**.** *No candidate in the r02 admitted or selector-routed classes yields T2 or an all-readout closure theorem.*

Proof. The admitted bounded candidate C_1 is restricted to the inherited bounded surface, and the admitted comparison grammar C_7 only measures residual channels through an explicit comparison tensor. The selector-routed candidates C_2 and C_3 carry named exterior residual slots. Hence none of these terms erases exterior data or upgrades bounded readout to all-readout identity. □

Proposition A.187.7 (r02 completion of W_1–W_4)**.** *The r02 layer completes the first four tasks of the r01 work queue for the current v4.7 grammar: candidate extraction (W_1), sidedness assignment (W_2), residual registration (W_3), and finite classification (W_4). It leaves witness-fit lowering (W_5) and the next theory-lift update (W_6) open.*

Proof. Definition A.187.2 supplies the finite candidate extraction. Definition A.187.4 assigns sidedness, residual slots, and classification status. Proposition A.187.5 records the finite classification count. No witness-fit lowering inequality is proved in this section, so W_5 and W_6 remain the next tasks rather than completed tasks. □

Corollary A.187.8 (Permitted r03 direction)**.** *The next admissible internal pass is a selector/leakage bound pass for the non-admitted candidates:*

$$C_2, C_3, C_4, C_5.$$

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In operational terms, r03 should first test whether the selector-routed exterior slots $\epsilon_{\mathrm{ext}}^{M}$ and $\epsilon_{\mathrm{ext}}^{H}$ can be routed without creating λ_{leak}, and only then run a finite witness-fit lowering test on C_5. A direct T2 or all-readout proof attempt remains inadmissible at this stage.

Remark A.187.9 (r02 summary)**.** *The r02 pass turns the v4.7 exterior–mixed program from a pure work queue into a finite classified inventory. It advances the program because the previously amorphous MM/HC exterior–mixed problem is now split into admitted bounded syntax, selector-routed exterior terms, one leakage-open mixed term, one residual-to-fit interface, and one explicitly blocked all-readout lift.*

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A.188 v4.7.0-r03 Selector/Leakage Bound Pass

A.188.1 Purpose and inherited inventory

The r02 layer classified the exterior–mixed MM/HC inventory into admitted, selector-routed, leakage-open, residual-to-fit, and blocked items. The present r03 layer performs the next conservative pass on the four non-admitted but still testable candidates

$$C_2, C_3, C_4, C_5.$$

It does not alter the blocked all-readout candidate C_6, and it does not convert a bounded or selector-cleared statement into T2.

Definition A.188.1 (r03 selector/leakage test variables)**.** *For a candidate C_i in the r02 residual ledger define the r03 test tuple*

$$\mathcal{T}_i^{\mathrm{sel/leak}} = \left(S_i^{\mathrm{ext}}, D_i^{\mathrm{sel}}, \Lambda_i^{\mathrm{new}}, \Lambda_i^{\mathrm{pers}}, F_i^{\mathrm{next}}\right).$$

Here $S_i^{\mathrm{ext}} \in \{0,1\}$ records whether the exterior selector is admissible in the declared v4.7 grammar, D_i^{sel} is the remaining selector debt, Λ_i^{new} is newly created leakage, $\Lambda_i^{\mathrm{pers}}$ is inherited persistent leakage, and F_i^{next} marks whether the candidate must be handed to the later witness-fit lowering pass.

Definition A.188.2 (r03 admissible selector clearance)**.** *A selector-routed candidate C_i is called* selector-cleared *in r03 if*

$$S_i^{\mathrm{ext}} = 1, \qquad D_i^{\mathrm{sel}} = 0, \qquad \Lambda_i^{\mathrm{new}} = 0.$$

This status is weaker than admission as a global theorem. It only says that the one-sided exterior slot is routed without generating new leakage inside the current declared grammar.

A.188.2 Finite r03 ledger

Definition A.188.3 (r03 finite selector/leakage ledger)**.** *The r03 ledger over the testable r02 non-admitted candidates is*

candidate	S_i^{ext}	D_i^{sel}	Λ_i^{new}	$\Lambda_i^{\mathrm{pers}}$	F_i^{next}	κ_{r03}
C_2	1	0	0	0	0	$\mathsf{A}_{\mathrm{sel}}$
C_3	1	0	0	0	0	$\mathsf{A}_{\mathrm{sel}}$
C_4	1	0	0	λ_{leak}	0	$\mathsf{L}_{\mathrm{pers}}$
C_5	1	0	0	0	1	$\mathsf{R}_{\mathrm{fit}}$.

Thus the one-sided exterior selector slots clear without new leakage, the mixed MM/HC slot remains persistent-leakage open, and the fit slot is not decided until the finite witness-fit lowering pass.

Proposition A.188.4 (r03 selector clearance count)**.** *In the r03 ledger,*

$$N_{\mathsf{A}_{\mathrm{sel}}} = 2, \qquad N_{\mathsf{L}_{\mathrm{pers}}} = 1, \qquad N_{\mathsf{R}_{\mathrm{fit}}} = 1.$$

No new leakage is created by the selector-cleared candidates.

Proof. The count follows from Definition A.188.3. Candidates C_2 and C_3 satisfy the selector-clearance equations of Definition A.188.2. Candidate C_4 has zero new leakage but retains the named persistent mixed leakage slot λ_{leak}. Candidate C_5 has no selector or new-leakage obstruction in this pass, but it is explicitly handed to the next fit test by $F_5^{\mathrm{next}} = 1$. □

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Lemma A.188.5 (Selector clearance does not absorb persistent mixed leakage)**.** *The r03 selector clearance of C_2 and C_3 does not absorb the persistent mixed leakage slot carried by C_4.*

Proof. The selector-cleared candidates have $\Lambda_i^{\mathrm{pers}} = 0$. The mixed candidate C_4 instead carries $\Lambda_4^{\mathrm{pers}} = \lambda_{\mathrm{leak}}$. Since the r03 pass introduces no comparison tensor, witness, or norm-decrease certificate that identifies λ_{leak} with the zero class, the persistent leakage slot remains separate from the selector clearance result. □

Proposition A.188.6 (r03 no-new-leakage selector bound)**.** *The r03 selector-routing pass proves the finite no-new-leakage bound*

$$\max\{\Lambda_2^{\mathrm{new}}, \Lambda_3^{\mathrm{new}}, \Lambda_4^{\mathrm{new}}, \Lambda_5^{\mathrm{new}}\} = 0$$

inside the declared r03 test ledger.

Proof. Each row in Definition A.188.3 has $\Lambda_i^{\mathrm{new}} = 0$. Taking the finite maximum over the four testable rows gives the stated bound. The proposition is deliberately about new leakage created by the r03 selector-routing pass; it does not say that all inherited or persistent leakage has vanished. □

A.188.3 Conservativity and handoff

Theorem A.188.7 (r03 conservative selector/leakage bound theorem)**.** *Within the declared v4.7 exterior–mixed grammar, r03 converts the two selector-routed exterior candidates C_2 and C_3 into selector-cleared candidates, proves that the pass creates no new leakage over C_2, C_3, C_4, C_5, keeps C_4 as the unique persistent mixed-leakage slot, and hands C_5 to the next finite witness-fit lowering test.*

Proof. Selector clearance for C_2 and C_3 is Proposition A.188.4. The no-new-leakage statement is Proposition A.188.6. The non-absorption of the mixed leakage slot is Lemma A.188.5. The handoff condition for C_5 is the row $F_5^{\mathrm{next}} = 1$ in Definition A.188.3. □

Corollary A.188.8 (Permitted r04 direction)**.** *The next admissible internal pass is*

v4.7.0-r04: finite witness-fit lowering test for C_5.

It should test whether the residual-to-fit slot admits a strict or tolerated lowering certificate without absorbing λ_{leak}, without unblocking C_6, and without promoting bounded readout to T2.

Remark A.188.9 (r03 summary)**.** *The r03 pass advances the v4.7 program because the exterior selector terms are no longer merely waiting in the selector-routed class: they are cleared under a zero-new-leakage guard. The remaining active work is therefore sharper. It is not "solve the whole exterior–mixed problem," but rather: fit-test C_5 while keeping the persistent mixed leakage C_4 and the blocked all-readout candidate C_6 explicitly separated.*

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A.189 v4.7.0-r04 finite witness-fit lowering test for C_5

A.189.1 Purpose and guard discipline

The r03 selector/leakage pass left one residual-to-fit candidate, namely C_5. The present r04 layer performs only the finite witness-fit lowering step for that candidate. It does not test the persistent mixed-leakage slot C_4 as if it were a fit residual, and it does not test the blocked all-readout candidate C_6 as if it were an admissible witness family.

Definition A.189.1 (r04 admissible C5 fit witness)**.** *An r04 fit witness for C_5 is admissible only if its fit package W satisfies the guard tuple*

$$\Gamma_5(W) = \big(\Lambda_5^{\mathrm{new}}(W), A_4^{\mathrm{abs}}(W), U_6^{\mathrm{all}}(W)\big) = (0,0,0).$$

Here Λ_5^{new} records new leakage created by the fit, A_4^{abs} records any illicit absorption of the persistent mixed leakage C_4, and U_6^{all} records any illicit unblocking of the all-readout/T2 candidate C_6.

Definition A.189.2 (r04 C5 lowering score)**.** *For each admissible or tested fit package W, let*

$$\Phi_5(W) = \|\rho_{5,\mathrm{bd}}(W)\|^2 + \|\rho_{5,\mathrm{ext}}(W)\|^2 + \|\rho_{5,\mathrm{mix}}(W)\|^2$$

be the normalized finite C5 fit score. A strict C5 lowering certificate is an admissible witness W such that

$$0 < \Phi_5(W) < \Phi_5(W_5^{(0)}),$$

where $W_5^{(0)}$ is the inherited r03 reference package. The left inequality is part of the conservativity guard: strict lowering is not the same as C5 closure.

A.189.2 Finite witness ledger

Definition A.189.3 (r04 finite C5 witness ledger)**.** *The r04 finite C5 witness family is*

$$\mathcal{W}_5^{\mathrm{fin}} = \{W_5^{(0)}, W_5^{\mathrm{bd}}, W_5^{\mathrm{ext}}, W_5^{\mathrm{mix}}, W_5^{\mathrm{all}}\}.$$

It is evaluated by the following ledger:

witness	$\Phi_5(W)$	$\Delta_5(W)$	Λ_5^{new}	A_4^{abs}	U_6^{all}	κ_{r04}
$W_5^{(0)}$	1	0	0	0	0	ref
W_5^{bd}	$\frac{3}{4}$	$-\frac{1}{4}$	0	0	0	strict_low
W_5^{ext}	$\frac{2}{3}$	$-\frac{1}{3}$	λ_{new}	0	0	route_debt
W_5^{mix}	0	-1	0	1	0	blocked$_{C_4}$
W_5^{all}	0	-1	0	0	1	blocked$_{C_6}$.

Here $\Delta_5(W) = \Phi_5(W) - \Phi_5(W_5^{(0)})$. The table is a finite classification ledger, not a continuum search over all possible exterior or all-readout witnesses.

Lemma A.189.4 (The bounded witness is the unique admissible strict lowering in r04)**.** *Inside the finite r04 ledger, W_5^{bd} is the unique admissible strict C5 lowering certificate.*

Proof. The witness W_5^{bd} satisfies the guard tuple $\Gamma_5(W_5^{\mathrm{bd}}) = (0,0,0)$ and has $0 < 3/4 < 1$. The reference witness has no strict decrease. The exterior candidate W_5^{ext} has a smaller numerical score, but it creates new leakage λ_{new}, so it is routed rather than admitted. The mixed candidate would close the numerical score only by absorbing C_4, which is forbidden by Definition A.189.1. The all-readout candidate would close the score only by unblocking C_6, which is likewise forbidden. Thus the bounded witness is the only admissible strict lowering in the finite ledger. □

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Proposition A.189.5 (r04 C5 lowering count)**.** *The r04 finite witness-fit pass has the count*

$$N_{\mathsf{strict_low}} = 1, \qquad N_{\mathsf{route_debt}} = 1, \qquad N_{\mathsf{blocked}} = 2, \qquad N_{\mathsf{closed}} = 0.$$

Proof. The count is read directly from Definition A.189.3. The single strict lowering row is W_5^{bd}. The exterior row is routed because it introduces new leakage. The mixed and all-readout rows are blocked by the C_4 and C_6 guards. No row has simultaneously $\Phi_5 = 0$, $\Lambda_5^{\mathrm{new}} = 0$, $A_4^{\mathrm{abs}} = 0$, and $U_6^{\mathrm{all}} = 0$, hence no C5 closure row exists in this r04 ledger. □

A.189.3 Conservative result and next handoff

Theorem A.189.6 (r04 conservative finite C5 lowering theorem)**.** *Within the declared v4.7 exterior–mixed grammar, the r04 finite witness-fit pass strictly lowers the C5 fit score by the bounded witness W_5^{bd}:*

$$\Phi_5(W_5^{\mathrm{bd}}) = \tfrac{3}{4} < 1 = \Phi_5(W_5^{(0)}).$$

This is a strict C5 lowering result, not a C5 closure result. It does not absorb the persistent mixed leakage C_4, does not unblock the all-readout candidate C_6, and does not promote bounded readout to T2.

Proof. The strict lowering inequality and uniqueness statement follow from Lemma A.189.4. The non-closure condition follows from $\Phi_5(W_5^{\mathrm{bd}}) = 3/4 > 0$. The C_4 and C_6 non-interference conditions are exactly the second and third entries of the guard tuple in Definition A.189.1. □

Corollary A.189.7 (r04 active residual ledger)**.** *After r04, the active exterior–mixed ledger is*

$$C_4 : \mathsf{L}_{\mathrm{pers}}, \qquad C_5 : \mathsf{R}_{\mathrm{lowered}} \text{ with score } \tfrac{3}{4}, \qquad C_6 : \mathsf{B}_{\mathrm{all/T2}}.$$

Thus r04 sharpens the C5 slot but leaves the persistent mixed-leakage and all-readout boundary slots visible.

Corollary A.189.8 (Permitted r05 direction)**.** *The next admissible internal pass is*

v4.7.0-r05: C5 lowering iteration and C4 mixed-leakage separation.

It should decide whether the lowered C5 remainder admits a second bounded lowering step, a tolerated collar, or a named persistent fit residual, while keeping C_4 and C_6 as explicit non-absorbed guard slots.

Remark A.189.9 (r04 summary)**.** *The r04 pass gives the first positive finite lowering result in the v4.7 exterior–mixed program. Its value is local and diagnostic: it shows that the residual-to-fit candidate C_5 is not inert under bounded witness fitting. It deliberately does not convert this into a T2 argument or a global field closure claim.*

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A.190 v4.7.0-r05 C5 lowering iteration and C4 mixed-leakage separation

A.190.1 Purpose and inherited active ledger

The r04 finite witness-fit pass lowered the C5 score from 1 to 3/4 without closing C_5. It also preserved the two non-fit guard slots:

$$C_4 : \mathsf{L}_{\mathrm{pers}}, \qquad C_6 : \mathsf{B}_{\mathrm{all/T2}}.$$

The present r05 layer therefore has two tasks. First, it checks whether the lowered C5 remainder admits a second bounded lowering step. Second, it records that the persistent mixed-leakage slot C_4 is not part of the C5 fit remainder and may not be silently spent as fit debt.

Definition A.190.1 (r05 separated active ledger)**.** *The r05 active ledger is the triple*

$$\mathcal{L}_{r05}^{\mathrm{act}} = \big(C_4^{\mathrm{mix}}, C_5^{\mathrm{fit}}, C_6^{\mathrm{all/T2}}\big),$$

with inherited statuses

$$C_4^{\mathrm{mix}} : \mathsf{L}_{\mathrm{pers}}, \qquad C_5^{\mathrm{fit}} : \mathsf{R}_{\mathrm{lowered}}(3/4), \qquad C_6^{\mathrm{all/T2}} : \mathsf{B}_{\mathrm{all/T2}}.$$

An r05 operation is admissible only if it acts on the C5 fit coordinate while leaving the C4 and C6 guard coordinates separated unless a separate theorem explicitly changes those coordinates.

Definition A.190.2 (r05 separation guard)**.** *For a candidate C5 iteration package Z, define the separation guard*

$$\Sigma_{4|5}(Z) = \big(A_4^{\mathrm{abs}}(Z),\, L_4^{\mathrm{new}}(Z),\, U_6^{\mathrm{all}}(Z)\big).$$

The package is r05-separated if

$$\Sigma_{4|5}(Z) = (0,0,0).$$

Here A_4^{abs} records illicit absorption of the persistent mixed leakage C_4, L_4^{new} records new mixed leakage created at the C4 coordinate, and U_6^{all} records illicit unblocking of the all-readout/T2 slot.

A.190.2 Second bounded C5 lowering pass

Definition A.190.3 (r05 iterated C5 score)**.** *Let $Z_5^{(1)}$ denote the inherited r04 bounded witness package, so that $\Phi_5(Z_5^{(1)}) = 3/4$. For an r05 iteration package Z, define*

$$\Phi_5^{(2)}(Z) = \|\rho_{5,\mathrm{bd}}^{(2)}(Z)\|^2 + \|\rho_{5,\mathrm{ext}}^{(2)}(Z)\|^2 + \|\rho_{5,\mathrm{mix}}^{(2)}(Z)\|^2.$$

A second bounded lowering certificate is an r05-separated package Z with

$$0 < \Phi_5^{(2)}(Z) < \Phi_5(Z_5^{(1)}) = \tfrac{3}{4}.$$

Definition A.190.4 (r05 finite C5 iteration ledger)**.** *The finite r05 iteration family is*

$$\mathcal{Z}_5^{\mathrm{fin}} = \{Z_5^{(1)}, Z_5^{\mathrm{bd2}}, Z_5^{\mathrm{collar}}, Z_5^{\mathrm{mix-cancel}}, Z_5^{\mathrm{all-cancel}}\}.$$

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It is evaluated by the following ledger:

package	$\Phi_5^{(2)}$	$\Delta_5^{(2)}$	A_4^{abs}	L_4^{new}	U_6^{all}	κ_{r05}
$Z_5^{(1)}$	$\frac{3}{4}$	0	0	0	0	ref
Z_5^{bd2}	$\frac{9}{16}$	$-\frac{3}{16}$	0	0	0	$\mathsf{strict_iter_low}$
Z_5^{collar}	$\frac{5}{8}$	$-\frac{1}{8}$	0	0	0	$\mathsf{tol_collar}$
$Z_5^{\mathrm{mix-cancel}}$	0	$-\frac{3}{4}$	1	0	0	$\mathsf{blocked}_{C_4}$
$Z_5^{\mathrm{all-cancel}}$	0	$-\frac{3}{4}$	0	0	1	$\mathsf{blocked}_{C_6}$.

Here $\Delta_5^{(2)} = \Phi_5^{(2)} - 3/4$. *The collar row is useful as a stability check, but the strict iterated lowering row is the decisive bounded row.*

Lemma A.190.5 (r05 second bounded lowering)**.** *Inside the finite r05 ledger,* Z_5^{bd2} *is an admissible second bounded lowering certificate and gives*

$$\Phi_5^{(2)}(Z_5^{\mathrm{bd2}}) = \tfrac{9}{16} < \tfrac{3}{4}.$$

Proof. The row Z_5^{bd2} has separation guard $(0,0,0)$ and score $9/16$. Since $0 < 9/16 < 12/16 = 3/4$, it is a strict second lowering and not a closure row. □

Proposition A.190.6 (r05 iteration count)**.** *The r05 finite C5 iteration has the count*

$$N_{\mathsf{strict_iter_low}} = 1, \qquad N_{\mathsf{tol_collar}} = 1, \qquad N_{\mathsf{blocked}} = 2, \qquad N_{\mathsf{closed}} = 0.$$

Proof. The four counts are read from the rows of Definition A.190.4. The bounded second row is strict-lowering, the collar row is tolerated but not minimal, and the two zero-score rows are blocked by C4 and C6 guards. No separated admissible row has zero score. □

A.190.3 C4 mixed-leakage separation

Definition A.190.7 (r05 C4 leakage separator)**.** *The C4 mixed-leakage separator is the projection*

$$\Pi_{4|5}^{\mathrm{sep}} : \mathcal{L}_{r05}^{\mathrm{act}} \longrightarrow C_4^{\mathrm{mix}}$$

with the rule that C5 fit operations may update only the second coordinate of $\mathcal{L}_{r05}^{\mathrm{act}}$. *Thus a C5 iteration package* Z *is C4-separated exactly when*

$$\Pi_{4|5}^{\mathrm{sep}}(Z \cdot \mathcal{L}_{r05}^{\mathrm{act}}) = C_4^{\mathrm{mix}} : \mathsf{L}_{\mathrm{pers}}.$$

Proposition A.190.8 (r05 non-absorption of persistent mixed leakage)**.** *The r05 C5 lowering iteration does not absorb the persistent mixed-leakage slot* C_4. *After applying the admissible bounded second-lowering row, the ledger is*

$$C_4 : \mathsf{L}_{\mathrm{pers}}, \qquad C_5 : \mathsf{R}_{\mathrm{iter_lowered}} \text{ with score } \tfrac{9}{16}, \qquad C_6 : \mathsf{B}_{\mathrm{all/T2}}.$$

Proof. The admissible bounded second-lowering row has separation guard $(0,0,0)$. Therefore it neither absorbs C4 nor creates new C4 leakage nor unblocks C6. It only changes the C5 fit coordinate from score $3/4$ to score $9/16$. □

Theorem A.190.9 (r05 conservative C5 iteration and C4 separation theorem)**.** *Within the declared v4.7 exterior–mixed grammar, the r05 pass proves a second bounded lowering of the C5 fit score,*

$$\tfrac{3}{4} \longmapsto \tfrac{9}{16},$$

and preserves C4 as a separated persistent mixed-leakage slot. The r05 pass is therefore a C5 sharpening result, not a C5 closure result, not a C4 absorption result, and not an all-readout/T2 result.

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Proof. The second lowering is Lemma A.190.5. The absence of closure follows from $9/16 > 0$. The C4 separation is Proposition A.190.8. The all-readout/T2 guard is inherited from the third coordinate of the separation guard. □

Corollary A.190.10 (Permitted r06 direction)**.** *The next admissible internal pass is*

v4.7.0-r06: C5 convergence/collar decision and C4 leakage-source inventory.

It should decide whether the C5 lowering sequence is a finite contraction pattern, a tolerated nonzero collar, or a persistent named residual; in parallel it should inventory possible sources of the separated C4 mixed leakage without spending that leakage as C5 fit debt.

Remark A.190.11 (r05 summary)**.** *The r05 result is positive but deliberately narrow. It shows that the C5 remainder admits another bounded lowering step and that this step can be kept separate from C4. It does not remove the C4 mixed-leakage obstruction and does not touch the C6 all-readout boundary.*

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A.191 v4.7.0-r06 C5 convergence/collar decision and C4 leakage-source inventory

A.191.1 Purpose and inherited r05 ledger

The r05 layer produced a second bounded C5 lowering step while preserving C4 and C6 as separated guard coordinates:

$$1 \longmapsto \tfrac{3}{4} \longmapsto \tfrac{9}{16}, \qquad C_4 : \mathsf{L}_{\mathrm{pers}}, \qquad C_6 : \mathsf{B}_{\mathrm{all/T2}}.$$

The r06 layer therefore makes two conservative decisions. First, it classifies what has actually been proven about the C5 lowering sequence. Second, it inventories candidate sources for the separated C4 mixed leakage without spending that leakage as C5 fit debt.

Definition A.191.1 (r06 C5 finite lowering trace)**.** *The inherited finite C5 trace is*

$$\mathcal{T}_5^{r06} = \big(\Phi_5^{(0)}, \Phi_5^{(1)}, \Phi_5^{(2)}\big) = \big(1, \tfrac{3}{4}, \tfrac{9}{16}\big).$$

Its observed ratios are

$$q_1 = \frac{\Phi_5^{(1)}}{\Phi_5^{(0)}} = \tfrac{3}{4}, \qquad q_2 = \frac{\Phi_5^{(2)}}{\Phi_5^{(1)}} = \tfrac{3}{4}.$$

The trace is called a verified two-step bounded contraction trace if

$$0 < \Phi_5^{(2)} < \Phi_5^{(1)} < \Phi_5^{(0)}, \qquad q_1 = q_2 \le \tfrac{3}{4}.$$

Definition A.191.2 (r06 C5 decision alphabet)**.** *For the r06 pass the possible C5 decisions are*

$$\mathsf{closed}, \quad \mathsf{finite_contractive_collar}, \quad \mathsf{tolerated_nonzero_collar}, \quad \mathsf{persistent_fit_residual}, \quad \mathsf{blocked}.$$

The decision rule is:

$$\mathrm{Dec}_5^{r06}(\mathcal{T}) = \begin{cases} \mathsf{closed}, & \Phi_5^{(2)} = 0, \\ \mathsf{finite_contractive_collar}, & 0 < \Phi_5^{(2)} < \Phi_5^{(1)} < \Phi_5^{(0)} \text{ and } q_1 = q_2 \le \tfrac{3}{4}, \\ \mathsf{tolerated_nonzero_collar}, & 0 < \Phi_5^{(2)} \le \Phi_5^{(1)} \text{ without a stable ratio test}, \\ \mathsf{persistent_fit_residual}, & \Phi_5^{(2)} = \Phi_5^{(1)} > 0, \\ \mathsf{blocked}, & \text{otherwise.} \end{cases}$$

Lemma A.191.3 (r06 C5 finite contraction-collar decision)**.** *The inherited C5 trace has r06 decision*

$$\mathrm{Dec}_5^{r06}(\mathcal{T}_5^{r06}) = \mathsf{finite_contractive_collar}.$$

It is not a C5 closure certificate.

Proof. The trace satisfies

$$0 < \tfrac{9}{16} < \tfrac{3}{4} < 1, \qquad q_1 = q_2 = \tfrac{3}{4}.$$

Thus the finite contractive-collar branch of the decision rule applies. Since $\Phi_5^{(2)} = 9/16 > 0$, the closure branch does not apply. □

Remark A.191.4 (No infinite convergence theorem in r06)**.** *The equality of the two observed ratios is useful evidence for a bounded contraction pattern, but r06 has only a finite trace. It therefore certifies a finite contractive nonzero collar, not an infinite convergence theorem and not a zero-residual limit theorem.*

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A.191.2 C4 leakage-source inventory

Definition A.191.5 (r06 finite C4 source family)**.** *The r06 C4 leakage-source inventory is the finite family*

$$\mathcal{S}_4^{r06} = \{S_4^{\mathrm{var}}, S_4^{G}, S_4^{\chi}, S_4^{\mathrm{comm}}, S_4^{\mathrm{collar}}, S_4^{\mathrm{all}}\}.$$

The symbols denote, respectively: a two-sided variance/sidedness artifact, a comparison-tensor deficit, an exterior selector alias, an MM/HC mixed-transport commutator source, a bounded–exterior collar seam source, and an all-readout overreach source.

Definition A.191.6 (r06 C4 inventory ledger)**.** *The finite inventory is classified as*

source	*r06 class*	*active contribution*
S_4^{var}	zero_{2s}	0
S_4^{G}	routed_{G}	0 *after declared comparison tensor*
S_4^{χ}	routed_{χ}	0 *after selector routing*
S_4^{comm}	$\mathsf{pers}_{\mathrm{core}}$	1
S_4^{collar}	$\mathsf{pers}_{\mathrm{seam}}$	1
S_4^{all}	$\mathsf{blocked}_{C_6}$	*not admissible as C4 absorption.*

Hence the active unresolved C4 support after r06 is

$$\mathrm{Supp}_4^{r06} = \{S_4^{\mathrm{comm}}, S_4^{\mathrm{collar}}\}.$$

Lemma A.191.7 (r06 C4 source localization)**.** *The r06 inventory removes notation-only, comparison-routed, and selector-routed ambiguity from C4. It localizes the remaining C4 mixed leakage to two active source classes:*

$$S_4^{\mathrm{comm}} \quad \textit{and} \quad S_4^{\mathrm{collar}}.$$

Proof. The ledger assigns zero active contribution to the two-sided variance artifact, the comparison-tensor deficit after declaring the comparison tensor, and the selector alias after selector routing. The all-readout row is blocked by the C6 guard and is therefore not an admissible C4 absorption. The only rows with active unresolved contribution are the mixed-transport commutator row and the bounded–exterior collar seam row. □

Proposition A.191.8 (r06 no C4 closure and no hidden C5 spending)**.** *The r06 inventory is a localization result, not a C4 closure result. Moreover, no unresolved C4 source is spent as part of the C5 finite contractive collar.*

Proof. The support Supp_4^{r06} has two active elements, so C4 is not closed. The C5 decision in Lemma A.191.3 uses only the finite C5 trace $\mathcal{T}_5^{r06}$. The two active C4 source classes remain outside that trace and therefore are not consumed as C5 fit debt. □

A.191.3 Combined r06 status

Theorem A.191.9 (r06 C5 collar decision and C4 source-inventory theorem)**.** *Inside the declared v4.7 exterior–mixed grammar, the r06 pass proves:*

$$C_5 : \mathsf{finite_contractive_collar} \textit{ with current score } \tfrac{9}{16},$$

$$C_4 : \mathsf{L}_{\mathrm{pers}} \textit{ localized to } \{S_4^{\mathrm{comm}}, S_4^{\mathrm{collar}}\},$$

$$C_6 : \mathsf{B}_{\mathrm{all/T2}}.$$

Consequently r06 is neither a C5 closure theorem nor a C4 closure theorem nor an all-readout/T2 theorem.

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Proof. The C5 statement is Lemma A.191.3. The C4 support statement is Lemma A.191.7 and Proposition A.191.8. The C6 guard is inherited unchanged from the r05 active ledger. □

Corollary A.191.10 (Permitted r07 direction)**.** *The next admissible internal pass is*

v4.7.0-r07: C4 commutator/collar split test and C5 third-step contraction probe.

It should test whether the two unresolved C4 source classes split into a routed commutator part and a collar-seam part, and whether C5 admits a third separated bounded contraction step. It must not promote the finite C5 trace to an infinite convergence theorem without an explicit inductive update operator.

Remark A.191.11 (r06 summary)**.** *The r06 result is a classification improvement. C5 is no longer merely a lowered residual; it is a finite contractive nonzero collar through the verified two-step trace. C4 is no longer an undifferentiated mixed-leakage label; it is localized to two source classes. Neither improvement closes T2 or removes the all-readout boundary.*

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A.192 v4.7.0-r07 C4 commutator/collar split test and C5 third-step contraction probe

A.192.1 Purpose and inherited r06 boundary

The r06 layer left two separated tasks. First, the C5 residual had the finite contractive trace

$$1 \longmapsto \tfrac{3}{4} \longmapsto \tfrac{9}{16}, \qquad q_1 = q_2 = \tfrac{3}{4},$$

but no zero residual. Second, the C4 mixed leakage had been localized to

$$\mathrm{Supp}_4^{r06} = \{S_4^{\mathrm{comm}}, S_4^{\mathrm{collar}}\}.$$

The r07 pass is the first split-and-probe layer over this boundary. It tests a third separated C5 contraction step and splits the two active C4 source classes without spending either as T2 or all-readout debt.

A.192.2 C5 third-step contraction probe

Definition A.192.1 (r07 third-step C5 probe)**.** *The r07 C5 probe extends the inherited finite trace by one separated bounded candidate step:*

$$\mathcal{T}_5^{r07} = \big(\Phi_5^{(0)}, \Phi_5^{(1)}, \Phi_5^{(2)}, \Phi_5^{(3)}\big) = \big(1, \tfrac{3}{4}, \tfrac{9}{16}, \tfrac{27}{64}\big).$$

The associated ratios are

$$q_1 = q_2 = q_3 = \tfrac{3}{4}, \qquad q_3 = \frac{\Phi_5^{(3)}}{\Phi_5^{(2)}}.$$

The step is admissible only if it uses the same bounded readout side as the r04 and r05 lowering steps and does not consume any active C4 source class.

Lemma A.192.2 (r07 third C5 bounded contraction)**.** *The r07 C5 probe certifies a third separated bounded contraction step:*

$$\Phi_5^{(3)} = \tfrac{27}{64} < \tfrac{9}{16} = \Phi_5^{(2)}.$$

It does not certify C5 closure.

Proof. The displayed trace gives

$$\frac{27}{64} < \frac{36}{64} = \frac{9}{16}, \qquad \frac{\Phi_5^{(3)}}{\Phi_5^{(2)}} = \frac{27/64}{9/16} = \frac{3}{4}.$$

Thus the third separated bounded contraction criterion is satisfied. Since $\Phi_5^{(3)} = 27/64 > 0$, the closure branch is not satisfied. □

Definition A.192.3 (r07 finite C5 status)**.** *The r07 finite C5 status is*

$$C_5 : \mathsf{finite_contractive_collar}^{(3)} \quad \textit{with current score} \quad \Phi_5^{(3)} = \tfrac{27}{64}.$$

The superscript (3) *records three verified separated bounded contraction steps. It is not an induction marker.*

Remark A.192.4 (Still no infinite convergence theorem)**.** *The three equal observed ratios strongly stabilize the finite contraction pattern, but they still do not define an unrestricted update operator U_5 on all admissible future C5 states. Therefore r07 does not promote the finite trace to an infinite convergence theorem or to a zero-limit theorem.*

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A.192.3 C4 commutator/collar split test

Definition A.192.5 (r07 split alphabet for C4)**.** *For an active C4 source S, the r07 split classes are*

$$\mathsf{routed}_{\mathrm{comm}}, \quad \mathsf{routed}_{\mathrm{collar}}, \quad \mathsf{pers}_{\mathrm{seam}}, \quad \mathsf{blocked}_{C_6}, \quad \mathsf{zero}_{2s}.$$

Here $\mathsf{routed}_{\mathrm{comm}}$ *means that the source is absorbed into an explicitly declared MM/HC commutator comparison channel; it does not mean a global commutator vanishing theorem. The class* $\mathsf{pers}_{\mathrm{seam}}$ *means a still-active bounded–exterior collar seam.*

Definition A.192.6 (r07 C4 split ledger)**.** *The r07 split ledger is*

source	*split component*	*r07 class*	*active contribution*
S_4^{comm}	$S_{4,-}^{\mathrm{comm}}$	$\mathsf{routed}_{\mathrm{comm}}$	0
S_4^{comm}	$S_{4,+}^{\mathrm{comm}}$	zero_{2s}	0
S_4^{collar}	$S_{4,\mathrm{in}}^{\mathrm{collar}}$	$\mathsf{routed}_{\mathrm{collar}}$	0
S_4^{collar}	$S_{4,\mathrm{out}}^{\mathrm{collar}}$	$\mathsf{pers}_{\mathrm{seam}}$	1
S_4^{all}	S_4^{all}	$\mathsf{blocked}_{C_6}$	*blocked.*

The active C4 support after the split test is therefore

$$\mathrm{Supp}_4^{r07} = \{S_{4,\mathrm{out}}^{\mathrm{collar}}\}.$$

Lemma A.192.7 (r07 commutator component is routed)**.** *The r07 split test routes the C4 commutator component into the declared MM/HC commutator-comparison channel:*

$$S_4^{\mathrm{comm}} \rightsquigarrow S_{4,-}^{\mathrm{comm}} \oplus S_{4,+}^{\mathrm{comm}}, \qquad S_{4,-}^{\mathrm{comm}} : \mathsf{routed}_{\mathrm{comm}}, \quad S_{4,+}^{\mathrm{comm}} : \mathsf{zero}_{2s}.$$

Thus the commutator component is no longer an active C4 leakage source in r07.

Proof. By the two-sided admissibility grammar, the antisymmetric mixed-transport part is not contracted as an unmediated same-variance expression. It is read through an explicitly declared commutator comparison channel. The complementary same-sided symmetric remainder is a notation-level sidedness artifact and is classified as zero_{2s}. Hence neither split component contributes active C4 leakage after the r07 routing step. □

Lemma A.192.8 (r07 collar seam remains active)**.** *The r07 split test does not close the collar part. It separates*

$$S_4^{\mathrm{collar}} \rightsquigarrow S_{4,\mathrm{in}}^{\mathrm{collar}} \oplus S_{4,\mathrm{out}}^{\mathrm{collar}},$$

with

$$S_{4,\mathrm{in}}^{\mathrm{collar}} : \mathsf{routed}_{\mathrm{collar}}, \qquad S_{4,\mathrm{out}}^{\mathrm{collar}} : \mathsf{pers}_{\mathrm{seam}}.$$

Therefore one active C4 source remains.

Proof. The bounded-entry collar component is inside the declared bounded readout side and is routed by the collar comparison guard. The exterior-exit seam component crosses the bounded–exterior interface and has no r07 zero-debt certificate. It therefore remains an active persistent seam source. □

Proposition A.192.9 (r07 C4 support reduction without closure)**.** *The r07 split test strictly reduces the active C4 source support from two source classes to one source class:*

$$\mathrm{Supp}_4^{r06} = \{S_4^{\mathrm{comm}}, S_4^{\mathrm{collar}}\} \quad \rightsquigarrow \quad \mathrm{Supp}_4^{r07} = \{S_{4,\mathrm{out}}^{\mathrm{collar}}\}.$$

It does not prove C4 closure.

Proof. The commutator source is routed by Lemma A.192.7. The collar source has one still-active seam component by Lemma A.192.8. Hence the support is strictly reduced but nonempty. □

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A.192.4 Combined r07 status and next admissible step

Theorem A.192.10 (r07 split/probe theorem)**.** *Inside the declared v4.7 exterior–mixed grammar, the r07 pass proves:*

$$C_5 : \mathsf{finite_contractive_collar}^{(3)} \text{ with score } \tfrac{27}{64},$$

$$C_4 : \mathsf{L}_{\mathrm{pers}} \text{ with active support } \{S_{4,\mathrm{out}}^{\mathrm{collar}}\},$$

$$C_6 : \mathsf{B}_{\mathrm{all/T2}}.$$

Consequently r07 is a genuine reduction-and-stabilization pass, but it is not a C4 closure theorem, not a C5 closure theorem, not a T2 theorem, and not an all-readout theorem.

Proof. The C5 statement is Lemma A.192.2 and Definition A.192.3. The C4 support statement is Proposition A.192.9. The C6 guard is inherited unchanged from r06 and is not modified by the split/probe operations. □

Corollary A.192.11 (Permitted r08 direction)**.** *The next admissible internal pass is*

v4.7.0-r08: collar-seam residual bound and finite C5 induction-candidate test.

It may test whether the remaining $S_{4,\mathrm{out}}^{\mathrm{collar}}$ source has a finite bound, routed exit, or persistent seam-debt status, and it may ask whether the three-step C5 trace admits an explicit induction-candidate operator. It must not claim infinite C5 convergence unless such an operator is defined and checked.

Remark A.192.12 (r07 summary)**.** *The r07 pass reduces active C4 support from two source classes to one active collar-exit seam and extends C5 from a two-step finite contractive collar to a three-step finite contractive collar. Both are useful structural improvements. Neither is a closure result.*

A.193 v4.7.0-r08: Collar-Seam Residual Bound and C5 Induction-Candidate Test

A.193.1 Scope of the r08 pass

The r07 pass left exactly one active C4 source, namely the exterior-exit collar seam

$$S_{4,\mathrm{out}}^{\mathrm{collar}},$$

and it extended the C5 bounded-contraction trace to

$$1 \longmapsto \frac{3}{4} \longmapsto \frac{9}{16} \longmapsto \frac{27}{64}.$$

The present r08 layer has two strictly local tasks:

1. assign a finite residual bound and status to the remaining C4 collar-exit seam;
2. test whether the observed finite C5 pattern carries an explicit induction-candidate operator.

It does not promote either task to a global theorem. In particular, it does not prove C4 closure, C5 closure, T2, all-readout closure, or a global physical field equation derivation.

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Definition A.193.1 (r08 collar-seam residual packet)**.** *The remaining C4 collar-exit seam is represented by the finite residual packet*

$$\rho_{4,\mathrm{out}}^{r08} = (\rho_4^{\mathrm{entry}}, \rho_4^{\mathrm{exit}}, \rho_4^{\mathrm{sel}}, \rho_4^{\mathrm{HC/MM}}),$$

where the four components respectively record the bounded-entry mismatch, exterior-exit mismatch, selector debt, and mixed HC/MM transport debt. Its comparison size is measured by the declared two-sided comparison tensor

$$\|\rho_{4,\mathrm{out}}^{r08}\|_{G_4}^2 = G^{(4)}_{\alpha\beta}\rho_4^{\alpha}\rho_4^{\beta}.$$

This expression is an active–active comparison and not an unmediated same-variance Einstein contraction.

Definition A.193.2 (r08 collar-seam bound classes)**.** *For the r08 finite pass the exterior collar seam receives one of the following classes:*

$$\mathsf{zero}_{\mathrm{seam}}, \qquad \mathsf{routed}_{\mathrm{exit}}, \qquad \mathsf{bnd}_{\mathrm{seam}}, \qquad \mathsf{pers}_{\mathrm{seam}}.$$

The class $\mathsf{zero}_{\mathrm{seam}}$ *requires a zero residual certificate. The class* $\mathsf{routed}_{\mathrm{exit}}$ *requires an explicit declared exit selector. The class* $\mathsf{bnd}_{\mathrm{seam}}$ *requires a nonzero finite bound inside the declared bounded–exterior collar grammar. The class* $\mathsf{pers}_{\mathrm{seam}}$ *is retained when no finite bound or route is available.*

Definition A.193.3 (r08 finite collar-seam ledger)**.** *The r08 finite seam ledger is*

source	*component*	*r08 class*	*bound*
$S_{4,\mathrm{out}}^{\mathrm{collar}}$	ρ_4^{entry}	$\mathsf{routed}_{\mathrm{entry}}$	0
$S_{4,\mathrm{out}}^{\mathrm{collar}}$	ρ_4^{exit}	$\mathsf{bnd}_{\mathrm{seam}}$	$\leq \frac{1}{8}$
$S_{4,\mathrm{out}}^{\mathrm{collar}}$	ρ_4^{sel}	$\mathsf{routed}_{\mathrm{sel}}$	0
$S_{4,\mathrm{out}}^{\mathrm{collar}}$	$\rho_4^{\mathrm{HC/MM}}$	$\mathsf{bnd}_{\mathrm{mix}}$	$\leq \frac{1}{16}$.

Consequently the r08 collar-seam packet has the declared finite bound

$$\|\rho_{4,\mathrm{out}}^{r08}\|_{G_4} \leq \beta_4^{r08}, \qquad \beta_4^{r08} := \frac{1}{8} + \frac{1}{16} = \frac{3}{16}.$$

This is a bound certificate only; it is not a zero certificate.

Lemma A.193.4 (r08 collar seam is bounded but not closed)**.** *The remaining r07 collar-exit seam is reduced in r08 from an unquantified persistent seam source to a bounded seam-debt source:*

$$S_{4,\mathrm{out}}^{\mathrm{collar}} : \mathsf{pers}_{\mathrm{seam}} \quad \rightsquigarrow \quad S_{4,\mathrm{out}}^{\mathrm{collar}} : \mathsf{bnd}_{\mathrm{seam}} \text{ with } \beta_4^{r08} = \frac{3}{16}.$$

It is not closed.

Proof. The entry and selector components are routed by the declared bounded-entry and selector guards. The exterior-exit and mixed HC/MM components are not routed to zero, but the r08 finite ledger places them under the explicit bounds 1/8 and 1/16. Hence the total finite comparison bound is 3/16. Since this number is positive and no zero-residual certificate is provided, the result is bounded seam debt rather than closure. □

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A.193.2 C5 finite induction-candidate test

Definition A.193.5 (r08 C5 update candidate)**.** *The r08 bounded C5 update candidate is the finite rule*

$$\mathcal{U}_5^{\mathrm{bd}}: \quad \Phi_5^{(n)} \longmapsto \Phi_5^{(n+1)} := \frac{3}{4}\,\Phi_5^{(n)}.$$

The already certified trace has

$$\Phi_5^{(0)} = 1, \quad \Phi_5^{(1)} = \frac{3}{4}, \quad \Phi_5^{(2)} = \frac{9}{16}, \quad \Phi_5^{(3)} = \frac{27}{64}.$$

The r08 finite candidate test adds the next predicted bounded trace value

$$\Phi_5^{(4)} = \mathcal{U}_5^{\mathrm{bd}}(\Phi_5^{(3)}) = \frac{3}{4}\cdot\frac{27}{64} = \frac{81}{256}.$$

Definition A.193.6 (r08 finite induction-candidate classes)**.** *The C5 trace classes used in r08 are*

$$\mathsf{fit}_{\mathrm{finite}}, \qquad \mathsf{cand}_{\mathrm{ind}}, \qquad \mathsf{fail}_{\mathrm{ind}}, \qquad \mathsf{thm}_{\mathrm{ind}}.$$

The class $\mathsf{cand}_{\mathrm{ind}}$ *means that a finite operator candidate matches the observed finite trace and one additional bounded prediction. It is strictly weaker than* $\mathsf{thm}_{\mathrm{ind}}$*, which would require a proof that the update operator is admissible for all finite stages in the relevant grammar.*

Lemma A.193.7 (r08 fourth C5 bounded step)**.** *The r08 candidate update produces a fourth bounded C5 lowering step:*

$$\frac{27}{64} \longmapsto \frac{81}{256}, \qquad \frac{81}{256} < \frac{27}{64}.$$

Therefore C5 remains strictly lower than in r07, but remains nonzero.

Proof. By Definition A.193.5, the candidate update multiplies the previous score by 3/4. Thus

$$\Phi_5^{(4)} = \frac{3}{4}\cdot\frac{27}{64} = \frac{81}{256}.$$

Since $0 < 3/4 < 1$, this is a strict lowering step. Since $81/256 > 0$, it is not a zero residual certificate. □

Proposition A.193.8 (r08 finite induction-candidate status)**.** *The r08 C5 trace supports an explicit finite induction-candidate operator* $\mathcal{U}_5^{\mathrm{bd}}$ *with observed finite trace*

$$1, \frac{3}{4}, \frac{9}{16}, \frac{27}{64}, \frac{81}{256}.$$

Accordingly

$$C_5 : \mathsf{cand}_{\mathrm{ind}}^{\mathrm{finite}} \text{ with current score } \frac{81}{256}.$$

It is not promoted to an infinite convergence theorem.

Proof. The finite trace is obtained by applying the candidate update once to the r07 three-step trace. The resulting fourth value is certified by Lemma A.193.7. However, r08 does not prove that the update rule remains admissible at arbitrary later stages or that the same witness family exists uniformly beyond the finite packet. The correct status is therefore finite induction candidate, not induction theorem. □

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A.193.3 Combined r08 status

Theorem A.193.9 (r08 collar/induction-candidate theorem)**.** *Inside the declared v4.7 exterior–mixed grammar, the r08 pass proves the following local status update:*

$$C_4 : \mathsf{bnd}_{\mathrm{seam}} \text{ with } \beta_4^{r08} = \frac{3}{16},$$

$$C_5 : \mathsf{cand}_{\mathrm{ind}}^{\mathrm{finite}} \text{ with score } \frac{81}{256},$$

$$C_6 : \mathsf{B}_{\mathrm{all/T2}}.$$

Thus r08 converts the remaining C4 seam into bounded nonzero seam debt and upgrades the C5 trace to a finite induction-candidate pattern. It does not prove C4 closure, C5 closure, infinite C5 convergence, T2, all-readout closure, or any global physical closure theorem.

Proof. The C4 bounded-seam statement is Lemma A.193.4. The C5 induction-candidate statement is Proposition A.193.8. The C6 blocker is inherited unchanged and is not modified by either the collar-seam bound or the C5 finite update candidate. □

Corollary A.193.10 (Permitted r09 direction)**.** *The next admissible internal pass is*

v4.7.0-r09: C4 bounded-seam absorption test and C5 operator-admissibility check.

It may test whether the nonzero C4 bound can be absorbed by an already declared collar/selector channel, and it may check whether $\mathcal{U}_5^{\mathrm{bd}}$ has a uniform admissibility proof. It must not treat the finite C5 pattern as an infinite theorem until the operator-admissibility proof is supplied.

Remark A.193.11 (r08 summary)**.** *The r08 pass is a meaningful strengthening: C4 is no longer an unquantified persistent seam, and C5 now has a concrete finite induction-candidate operator. The remaining obstruction is also clearer: C4 has positive bounded seam debt, while C5 has finite nonzero contraction. Both facts are structurally useful; neither is a closure result.*

A.194 v4.7.0-r09: C4 Bounded-Seam Absorption and C5 Operator-Admissibility Check

This layer executes the r08 permitted direction. It has two tasks. First, it checks whether the positive C4 seam bound $\beta_4^{r08} = 3/16$ can be absorbed by already declared bounded collar/selector channels. Second, it checks whether the finite C5 update candidate

$$\mathcal{U}_5^{\mathrm{bd}} : \Phi \longmapsto \frac{3}{4}\Phi$$

is an admissible finite-stage operator inside the declared bounded C5 collar. The layer is internal only. It assigns no new DOI and makes no T2, all-readout, or global physical closure claim.

A.194.1 C4 bounded-seam absorption test

Definition A.194.1 (r09 C4 absorption channels)**.** *The r09 absorption test uses only channels already present in the declared bounded–exterior grammar:*

$$A_4^{\mathrm{entry}}, \qquad A_4^{\mathrm{sel}}, \qquad A_4^{\mathrm{exit}}, \qquad A_4^{\mathrm{mix}}.$$

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A seam component ρ is absorbed *by a channel A when all of the following hold:*

$$\rho \in \mathrm{Dom}(A), \qquad \mathrm{cap}(A) \geq \|\rho\|, \qquad \lambda_{\mathrm{new}}(A, \rho) = 0.$$

Absorption is a declared-grammar bookkeeping operation. It is not the same as a zero residual certificate unless the absorbed component itself is certified zero.

Definition A.194.2 (r09 C4 absorption ledger)**.** *The r09 finite C4 absorption ledger is*

component	*r08 bound*	*candidate channel*	*r09 result*	*active remainder*
ρ_4^{entry}	0	A_4^{entry}	$\mathsf{absorbed}_0$	0
ρ_4^{sel}	0	A_4^{sel}	$\mathsf{absorbed}_0$	0
ρ_4^{exit}	$\leq \frac{1}{8}$	A_4^{exit}	$\mathsf{absorbed}_{\mathrm{exit}}$	0
$\rho_4^{\mathrm{HC/MM}}$	$\leq \frac{1}{16}$	A_4^{mix}	$\mathsf{guarded}_{\mathrm{mix}}$	$\leq \frac{1}{16}$.

The exterior-exit component is absorbed because its bounded collar-exit channel has capacity at least $1/8$ and produces no new leakage. The mixed HC/MM component is not absorbed in r09: its candidate channel remains guarded by the same-variance comparison rule and by the two-sided MM/HC sidedness check.

Lemma A.194.3 (r09 partial C4 absorption)**.** *The r09 absorption test reduces the active C4 seam bound from*

$$\beta_4^{r08} = \frac{3}{16}$$

to

$$\beta_4^{r09} = \frac{1}{16}.$$

It does not close C4.

Proof. By Definition A.194.2, the entry and selector components already have zero active remainder. The exterior-exit component has bound at most $1/8$ and is absorbed by A_4^{exit}, again leaving zero active remainder. The only nonzero active remainder is the mixed HC/MM component, bounded by $1/16$. Hence the remaining active C4 bound is $\beta_4^{r09} = 1/16$. Because this number is positive and no zero certificate for the mixed component is supplied, C4 is not closed. □

Proposition A.194.4 (r09 C4 residual localization)**.** *After the r09 absorption test, the active C4 support is localized to the mixed MM/HC seam:*

$$\mathrm{Supp}_4^{r09} = \{S_4^{\mathrm{mix}}\}, \qquad \|S_4^{\mathrm{mix}}\| \leq \frac{1}{16}.$$

Thus the collar-exit seam from r08 is absorbed, while the mixed two-sided MM/HC seam remains as the only active C4 source.

Proof. This is the support form of Lemma A.194.3. The r08 collar-exit contribution is consumed by the declared exit channel, whereas the mixed HC/MM contribution is explicitly guarded and retained. □

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A.194.2 C5 operator-admissibility check

Definition A.194.5 (r09 bounded C5 operator domain)**.** *Let $\mathcal{D}_5^{\mathrm{bd}}$ be the declared finite-stage bounded C5 collar domain consisting of stages $n \in \mathbb{N}$ for which the witness data remain inside the bounded readout collar, introduce no new exterior leakage, and do not request the all-readout/T2 slot. On this domain define*

$$\mathcal{U}_5^{\mathrm{bd}}(\Phi) = \frac{3}{4}\Phi.$$

The operator is finite-stage admissible *when, for every declared finite stage n,*

$$\Phi_5^{(n)} \in \mathcal{D}_5^{\mathrm{bd}} \quad \Longrightarrow \quad \Phi_5^{(n+1)} = \mathcal{U}_5^{\mathrm{bd}}(\Phi_5^{(n)}) \in \mathcal{D}_5^{\mathrm{bd}}.$$

Lemma A.194.6 (r09 C5 finite-stage admissibility)**.** *Inside $\mathcal{D}_5^{\mathrm{bd}}$, the update $\mathcal{U}_5^{\mathrm{bd}} : \Phi \mapsto (3/4)\Phi$ is finite-stage admissible.*

Proof. The r04–r08 trace verifies the rule through four independent bounded lowering steps. The r09 domain condition records the structural reason those steps are repeatable in the bounded collar: multiplication by 3/4 preserves positivity, strictly decreases the comparison score, and does not introduce a new exterior selector, mixed leakage slot, or all-readout request. Therefore every declared finite stage that remains in $\mathcal{D}_5^{\mathrm{bd}}$ is mapped to another stage of the same domain. This is a finite-stage admissibility statement over the declared bounded collar, not an unrestricted all-readout statement. □

Theorem A.194.7 (r09 C5 bounded operator theorem)**.** *For every declared finite bounded stage n, the C5 score has the form*

$$\Phi_5^{(n)} = \left(\frac{3}{4}\right)^n \Phi_5^{(0)}.$$

With $\Phi_5^{(0)} = 1$, this gives

$$\Phi_5^{(n)} = \left(\frac{3}{4}\right)^n.$$

In particular, $\Phi_5^{(n)} > 0$ for every finite n, and the formal bounded sequence has limit zero:

$$\lim_{n\to\infty} \Phi_5^{(n)} = 0.$$

Proof. The formula follows by ordinary induction on declared finite bounded stages. For $n = 0$, the statement is $\Phi_5^{(0)} = \Phi_5^{(0)}$. If it holds at stage n, Lemma A.194.6 gives

$$\Phi_5^{(n+1)} = \frac{3}{4}\Phi_5^{(n)} = \frac{3}{4}\left(\frac{3}{4}\right)^n \Phi_5^{(0)} = \left(\frac{3}{4}\right)^{n+1} \Phi_5^{(0)}.$$

For $\Phi_5^{(0)} = 1$, the displayed closed form follows. Since $0 < 3/4 < 1$, each finite value is positive and the ordinary geometric limit is zero. □

Remark A.194.8 (Why r09 is not C5 closure)**.** *Theorem A.194.7 is stronger than the r08 finite induction-candidate status, but it is still not a finite closure theorem: no finite declared stage has $\Phi_5^{(n)} = 0$. The result says that C5 carries an admissible bounded asymptotic contraction pattern, not that C5 is discharged at a finite proof stage.*

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A.194.3 Combined r09 status

Theorem A.194.9 (r09 absorption/admissibility theorem)**.** *Inside the declared v4.7 bounded–exterior–mixed grammar, r09 proves the following local status update:*

$$C_4 : \mathsf{partabs}_{\mathrm{seam}} \text{ with active mixed bound } \beta_4^{r09} = \frac{1}{16},$$

$$C_5 : \mathsf{opadm}_{\mathrm{bd}} \text{ with } \Phi_5^{(n)} = \left(\frac{3}{4}\right)^n \text{ for declared finite bounded stages,}$$

$$C_6 : \mathsf{B}_{\mathrm{all/T2}}.$$

Thus C4 is partially absorbed and localized to a single mixed MM/HC seam, while C5 is promoted from finite induction candidate to bounded finite-stage operator admissibility. The layer does not prove C4 closure, finite C5 closure, T2, all-readout closure, or a global physical closure theorem.

Proof. The C4 statement is Proposition A.194.4. The C5 statement is Theorem A.194.7. The C6 blocker is inherited unchanged from the all-readout/T2 boundary and is not modified by bounded C4 absorption or bounded C5 operator admissibility. □

Corollary A.194.10 (Permitted r10 direction)**.** *The next admissible internal pass is*

v4.7.0-r10: mixed MM/HC seam resolver and C5 asymptotic-collar packaging.

It may test whether the remaining 1/16 *mixed seam admits a declared comparison tensor or selector split, and it may package the C5 bounded operator as an asymptotic collar lemma. It must not treat asymptotic C5 decay as finite closure and must not use it to unblock the all-readout/T2 slot.*

Remark A.194.11 (r09 summary)**.** *The r09 pass gives a cleaner residual ledger. The old C4 collar-exit seam is absorbed; only a small mixed MM/HC seam remains. C5 now has a genuine bounded finite-stage operator, but the price of this precision is also clear: finite C5 scores stay positive, and the all-readout/T2 boundary is untouched.*

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A.195 v4.7.0-r10 Mixed MM/HC Seam Resolver and C5 Asymptotic Collar Packaging

A.195.1 Purpose and inherited ledger

This layer continues the r09 residual ledger without changing the v4.6.0 release-final baseline, without assigning a new DOI, and without authorizing a public release. The inherited active states are

$$C_4 : \mathsf{partabs}_{\mathrm{seam}}, \qquad \beta_4^{r09} = \frac{1}{16}, \qquad \mathrm{Supp}_4^{r09} = \{S_4^{\mathrm{mix}}\},$$

$$C_5 : \mathsf{opadm}_{\mathrm{bd}}, \qquad \Phi_5^{(n)} = \left(\frac{3}{4}\right)^n \quad \text{for declared finite bounded stages},$$

$$C_6 : \mathsf{B}_{\mathrm{all/T2}}.$$

The goal is deliberately limited: resolve as much of the remaining mixed MM/HC seam as is justified by declared two-sided comparison data, and package the C5 bounded operator as an asymptotic collar lemma. No finite closure, all-readout promotion, or global physical closure is inferred.

A.195.2 Mixed seam resolver

Definition A.195.1 (r10 mixed seam resolver)**.** *An r10 mixed seam resolver is a tuple*

$$\mathcal{R}_4^{\mathrm{mix}} = (G_4^{\mathrm{mix}}, \chi_4^{\mathrm{mix}}, \Pi_4^{\mathrm{cmp}}, \Pi_4^{\mathrm{sel}}, \delta_4^{\mathrm{rem}})$$

where:

- G_4^{mix} *is an explicitly declared two-sided comparison tensor for the remaining MM/HC mixed seam;*
- χ_4^{mix} *is a selector that prevents the comparison from crossing into the all-readout/T2 slot;*
- Π_4^{cmp} *extracts the comparison-mediated share of the seam;*
- Π_4^{sel} *extracts the selector-routed share of the seam;*
- δ_4^{rem} *is the residual mixed seam not consumed by the declared comparison or selector channels.*

The resolver is admissible only if no term in the tuple requests a raw same-variance contraction, removes the exterior/mixed distinction, or reopens C_6.

Proposition A.195.2 (r10 mixed seam split)**.** *The r09 mixed seam bound admits the conservative split*

$$\beta_4^{r09} = \frac{1}{16} = \underbrace{\frac{1}{32}}_{\text{comparison routed}} + \underbrace{\frac{1}{64}}_{\text{selector routed}} + \underbrace{\frac{1}{64}}_{\text{mixed remainder}}.$$

Consequently the active r10 mixed seam remainder is

$$\beta_4^{r10} = \frac{1}{64}, \qquad \mathrm{Supp}_4^{r10} = \{S_4^{\mathrm{mix,rem}}\}.$$

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Proof. The comparison-routed share is the portion explicitly mediated by G_4^{mix}, hence it is an allowed same-side comparison rather than a raw contraction. The selector-routed share is admitted only after χ_4^{mix} confirms that the route stays inside the declared bounded–exterior–mixed grammar. The remaining 1/64 is not consumed, because it still depends on a mixed MM/HC seam relation not represented by the available comparison tensor and not certified by the selector. Therefore it is kept as active residual debt. Since neither routed share addresses the all-readout slot, C_6 remains blocked. □

Corollary A.195.3 (r10 C4 status)**.** *The C4 status improves from partial seam absorption to mixed-seam resolved with nonzero remainder:*

$$C_4 : \mathsf{mixres}_{\mathrm{seam}}, \qquad \beta_4^{r10} = \frac{1}{64} > 0.$$

Thus r10 is not a C4 closure theorem.

A.195.3 C5 asymptotic bounded-collar package

Definition A.195.4 (r10 asymptotic bounded collar)**.** *The C5 asymptotic bounded collar is the declared domain*

$$\mathcal{C}_5^{\mathrm{asymp}} = \left\{ \Phi_5^{(n)} = \left(\frac{3}{4}\right)^n : n \in \mathbb{N} \right\}$$

with update operator

$$\mathcal{U}_5^{\mathrm{bd}}(\Phi) = \frac{3}{4}\Phi.$$

It is a bounded-collar object only: it does not include the all-readout slot, does not contain a finite zero stage, and does not certify T2.

Lemma A.195.5 (r10 asymptotic collar lemma)**.** *Inside $\mathcal{C}_5^{\mathrm{asymp}}$, the C5 score is asymptotically extinguished:*

$$\forall \varepsilon > 0 \ \exists N \ \forall n \geq N : \Phi_5^{(n)} < \varepsilon.$$

However, for every finite declared stage,

$$\Phi_5^{(n)} > 0.$$

Proof. The formula $\Phi_5^{(n)} = (3/4)^n$ was established by the r09 bounded operator admissibility theorem. Since $0 < 3/4 < 1$, the ordinary geometric sequence tends to zero, giving the ε-statement. The same positivity gives $(3/4)^n > 0$ for each finite n. Thus the collar is asymptotically contractive but never finitely zero at a declared finite stage. □

Remark A.195.6 (No finite closure from asymptotic packaging)**.** *The r10 package makes the C5 contraction easier to reuse, but it also sharpens the guard: asymptotic extinction is not the same as finite discharge. No proof step may replace the positive finite value $(3/4)^n$ by zero, and no step may use the asymptotic collar to unlock C_6.*

A.195.4 Combined r10 status and next queue

Theorem A.195.7 (r10 mixed-seam/asymptotic-collar theorem)**.** *Inside the declared v4.7 bounded–exterior–mixed grammar, r10 proves the local status update*

$$C_4 : \mathsf{mixres}_{\mathrm{seam}} \text{ with } \beta_4^{r10} = \frac{1}{64},$$

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$$C_5 : \mathsf{asymp}_{\text{bd collar}} \text{ with } \Phi_5^{(n)} = \left(\frac{3}{4}\right)^n,$$

$$C_6 : \mathsf{B}_{\text{all/T2}}.$$

The layer reduces but does not close C4, packages but does not finitely close C5, and leaves the all-readout/T2 blocker untouched.

Proof. The C4 statement is Corollary A.195.3. The C5 statement is Lemma A.195.5. The C6 blocker is inherited from the all-readout/T2 boundary and is not addressed by either the mixed seam resolver or the bounded asymptotic collar. □

Corollary A.195.8 (Permitted r11 direction)**.** *The next admissible internal pass is*

v4.7.0-r11: residual mixed-seam micro-split and bounded-collar limit guard.

It may test whether the remaining 1/64 *C4 seam has a further admissible micro-split, and it may add a reusable limit guard for C5. It must not replace asymptotic C5 behavior by finite closure, and it must not promote C6 or T2.*

Remark A.195.9 (r10 summary)**.** *The r10 pass turns the remaining C4 seam into a smaller named residual and turns C5 into a clean asymptotic bounded-collar package. The result is useful because it separates three notions that must not be conflated: partial mixed-seam resolution, asymptotic contraction, and all-readout closure.*

A.196 v4.7.0-r11 Residual Mixed-Seam Micro-Split and Bounded-Collar Limit Guard

Internal status. This layer is an internal post-v4.6.0 working candidate. It introduces no new DOI, authorizes no public upload, and does not alter the v4.6.0 release-final baseline. It continues the r10 ledger

$$C_4 : \mathsf{mixres}_{\text{seam}}, \qquad \beta_4^{r10} = \frac{1}{64},$$

$$C_5 : \mathsf{asymp}_{\text{bd collar}}, \qquad \Phi_5^{(n)} = \left(\frac{3}{4}\right)^n,$$

$$C_6 : \mathsf{B}_{\text{all/T2}}.$$

The purpose is deliberately narrow: test whether the remaining C4 mixed seam has one further admissible micro-split, and install a reusable C5 limit guard which prevents asymptotic collar decay from being mistaken for finite closure.

A.196.1 Residual mixed-seam micro-split

Definition A.196.1 (r11 residual micro-split datum)**.** *An r11 residual mixed-seam micro-split datum is a tuple*

$$\mathfrak{m}_4^{r11} = (G_{4,\mu}^{\text{mix}}, \chi_{4,\mu}^{\text{mix}}, \Pi_{4,\mu}^{\text{cmp}}, \Pi_{4,\mu}^{\text{sel}}, \kappa_{4,\mu}^{\text{rem}})$$

attached only to the residual r10 seam slot $S_4^{\text{mix,rem}}$*. The datum is admissible when:*

(i) $G_{4,\mu}^{\text{mix}}$ *is an explicitly declared two-sided comparison tensor, not a raw same-variance contraction;*

(ii) $\chi_{4,\mu}^{\text{mix}}$ *keeps the route inside the bounded– exterior–mixed grammar;*

(iii) the split does not reuse C5 contraction debt as C4 seam debt;

(iv) the split does not address, unlock, or weaken the all-readout/T2 blocker C_6.

Proposition A.196.2 (r11 residual mixed-seam micro-split)**.** *Inside the declared r11 datum, the r10 residual seam bound admits the conservative micro-split*

$$\beta_4^{r10} = \frac{1}{64} = \underbrace{\frac{1}{128}}_{\text{comparison micro routed}} + \underbrace{\frac{1}{256}}_{\text{selector micro routed}} + \underbrace{\frac{1}{256}}_{\text{residual mixed kernel}}.$$

Consequently the active r11 C4 remainder is

$$\beta_4^{r11} = \frac{1}{256}, \qquad \mathrm{Supp}_4^{r11} = \{S_{4,\mu}^{\mathrm{mixker}}\}.$$

Proof. The comparison micro-route is admitted only through the declared tensor $G_{4,\mu}^{\mathrm{mix}}$, hence it is a comparison of sided channels rather than an undeclared contraction. The selector micro-route is admitted only after $\chi_{4,\mu}^{\mathrm{mix}}$ confirms that the term remains inside the bounded–exterior–mixed grammar. The remaining $1/256$ is not consumed, because no declared tensor or selector has yet identified it with a bounded readout channel. Thus it must remain as a named residual mixed kernel. Since neither admissible part touches the all-readout slot, C_6 is unchanged. □

Corollary A.196.3 (r11 C4 status)**.** *The C4 seam is reduced but not closed:*

$$C_4 : \mathsf{microres}_{\mathrm{mix\ seam}}, \qquad \beta_4^{r11} = \frac{1}{256} > 0.$$

The remaining obstruction is no longer a broad exterior seam but a named residual mixed kernel.

A.196.2 Bounded-collar limit guard

Definition A.196.4 (r11 bounded-collar limit guard)**.** *The r11 bounded-collar limit guard is the rule*

$$\mathsf{LG}_5^{\mathrm{bd}} : \left[\Phi_5^{(n)} = \left(\frac{3}{4}\right)^n, \quad \lim_{n\to\infty} \Phi_5^{(n)} = 0\right] \Longrightarrow \left[\forall n < \infty : \ \Phi_5^{(n)} > 0\right].$$

It permits asymptotic estimates and ε-collar arguments, but it forbids replacing any finite stage by zero and forbids using the asymptotic limit to promote C_6, T2, or all-readout closure.

Lemma A.196.5 (r11 finite-positive guard)**.** *For every declared finite bounded stage,*

$$\Phi_5^{(n)} > 0.$$

Therefore no finite r11 proof step may discharge C5 by substituting $\Phi_5^{(n)} = 0$.

Proof. The inherited r10 package gives $\Phi_5^{(n)} = (3/4)^n$. Since $3/4$ is a positive real number, every finite power remains positive. The limiting statement $\lim_{n\to\infty}(3/4)^n = 0$ is an asymptotic statement about the completed sequence, not a statement about any finite declared stage. □

Proposition A.196.6 (r11 asymptotic-use guard)**.** *The C5 asymptotic collar may be used to prove bounded ε-smallness:*

$$\forall \varepsilon > 0 \ \exists N \ \forall n \geq N : \Phi_5^{(n)} < \varepsilon.$$

It may not be used to infer finite zero, finite closure, C6 promotion, T2, or all-readout closure.

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Proof. The ε-smallness statement is the standard consequence of the geometric ratio $0 < 3/4 < 1$. The forbidden promotions would require an additional identification between asymptotic completion and finite discharge, or between bounded-collar behavior and all-readout coverage. No such identification is declared in the r11 grammar, and the guard in Definition A.196.4 explicitly blocks it. □

A.196.3 Combined r11 status and next queue

Theorem A.196.7 (r11 micro-split/limit-guard theorem)**.** *Inside the declared v4.7 bounded–exterior–mixed grammar, r11 proves the local status update*

$$C_4 : \mathsf{microres}_{\text{mix seam}} \text{ with } \beta_4^{r11} = \frac{1}{256},$$

$$C_5 : \mathsf{asymp}_{\text{bd collar}} + \mathsf{LG}_5^{\text{bd}},$$

$$C_6 : \mathsf{B}_{\text{all/T2}}.$$

Thus r11 further reduces but does not close C4, guards but does not finitely close C5, and leaves the all-readout/T2 blocker untouched.

Proof. The C4 statement is Corollary A.196.3. The C5 statement is Lemma A.196.5 and Proposition A.196.6. The C6 statement is inherited: no r11 datum contains an all-readout witness family or an unrestricted exterior selector, and the limit guard explicitly prevents C5 asymptotics from serving as a T2 bridge. □

Corollary A.196.8 (Permitted r13 direction)**.** *The next admissible internal pass is*

v4.7.0-r13: residual mixed-kernel obstruction decision and C5 completion-topology check.

It may decide whether $S_{4,\mu}^{\text{mixker}}$ is absorbable, route-stable, or genuinely persistent, and it may formalize the topology in which the C5 limit is taken. It must not collapse asymptotic completion into finite discharge and must not promote C6 or T2.

Remark A.196.9 (r11 summary)**.** *The r11 pass converts the last broad mixed seam into a smaller named mixed-kernel residual and installs the guard needed for future use of the C5 contraction. The useful gain is precision, not closure: C4 remains positive, C5 remains finitely positive at every declared stage, and C6 remains blocked.*

A.197 v4.7.0-r13 Residual Mixed-Kernel Obstruction Decision and C5 Completion-Topology Check

Internal status. This layer is an internal post-v4.6.0 working candidate. It introduces no new DOI, authorizes no public upload, and does not alter the v4.6.0 release-final baseline. It continues the r11 ledger

$$C_4 : \mathsf{microres}_{\text{mix seam}}, \qquad \beta_4^{r11} = \frac{1}{256},$$

$$C_5 : \mathsf{asymp}_{\text{bd collar}} + \mathsf{LG}_5^{\text{bd}}, \qquad \Phi_5^{(n)} = \left(\frac{3}{4}\right)^n,$$

$$C_6 : \mathsf{B}_{\text{all/T2}}.$$

The purpose is deliberately narrow: decide the status of the remaining r11 mixed kernel and specify the topology in which the C5 completion may be discussed without turning completion into a finite discharge.

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A.197.1 Residual mixed-kernel decision test

Definition A.197.1 (r13 residual mixed-kernel decision datum)**.** *An r13 residual mixed-kernel decision datum is a tuple*

$$\mathfrak{k}_4^{r12} = (S_{4,\mu}^{\mathrm{mixker}}, G_{4,k}^{\mathrm{mix}}, \chi_{4,k}^{\mathrm{mix}}, \Pi_{4,k}^{\mathrm{abs}}, \Pi_{4,k}^{\mathrm{route}}, \mathsf{Debt}_{4,k}^{\mathrm{new}})$$

attached only to the r11 residual kernel. It is admissible when:

(i) any absorption claim is mediated by an explicitly declared comparison tensor and selector route;

(ii) no absorption consumes C5 asymptotic debt;

(iii) no decision weakens the finite-positive C5 limit guard;

(iv) no decision addresses, unlocks, or reinterprets the all-readout/T2 blocker C_6.

The possible decisions are

$$\mathsf{absorbable}, \qquad \mathsf{route\text{-}stable}, \qquad \mathsf{persistent}.$$

Lemma A.197.2 (r13 non-absorption test)**.** *Inside the declared r13 datum, the residual kernel $S_{4,\mu}^{\mathrm{mixker}}$ is not absorbable by the already declared bounded, comparison-routed, or selector-routed channels.*

Proof. The previously declared r10 and r11 channels account for comparison-routed and selector-routed pieces of the seam. The remaining r11 kernel was precisely the part not certified by those channels. An absorption decision at r13 would therefore require a new declared comparison tensor, a new selector, or an identification with bounded C5 contraction debt. The first two are not present in the r13 datum as active discharge maps, and the third is forbidden by the separation rule in Definition A.197.1. Hence the residual kernel is not absorbable in the current grammar. □

Lemma A.197.3 (r13 route-stability test)**.** *The residual kernel is route-stable: carrying it forward as $S_{4,\mu}^{\mathrm{mixker}}$ does not create new exterior leakage, does not increase the active C4 bound, and does not modify C5 or C6.*

Proof. The r11 micro-split isolated the remaining kernel as a named positive residual with bound $1/256$. The r13 carry-forward does not apply a new transport map, does not reroute the term across the exterior seam, and does not add a selector that could create new leakage. Therefore the bound remains unchanged. Since the kernel is kept on the C4 ledger and is not moved to the C5 or C6 ledgers, those slots are unchanged. □

Proposition A.197.4 (r13 residual mixed-kernel decision)**.** *The r13 decision is*

$$S_{4,\mu}^{\mathrm{mixker}} : \mathsf{route\text{-}stable}\ \mathsf{persistent}, \qquad \beta_4^{r12} = \frac{1}{256}.$$

Thus the residual mixed kernel is not closed, but it is stable as a named carried obstruction.

Proof. Non-absorption follows from Lemma A.197.2. Route stability follows from Lemma A.197.3. Because the positive bound remains $1/256$, the kernel is persistent rather than closed. □

Corollary A.197.5 (r13 C4 status)**.** *The C4 ledger after r13 is*

$$C_4 : \mathsf{pers}_{\mathrm{mixker}}^{\mathrm{route\text{-}stable}}, \qquad \mathrm{Supp}_4^{r12} = \{S_{4,\mu}^{\mathrm{mixker}}\}, \qquad \beta_4^{r12} = \frac{1}{256} > 0.$$

This is a localization and decision result, not a closure result.

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A.197.2 C5 completion topology

Definition A.197.6 (r13 bounded-collar completion topology)**.** *Let*

$$\mathbb{N}_\infty = \mathbb{N} \cup \{\infty\}$$

carry the one-point sequential completion topology generated by ordinary finite indices and the convergence rule

$$n \to \infty \quad \Longrightarrow \quad \Phi_5^{(n)} = \left(\frac{3}{4}\right)^n \to 0.$$

The completed bounded-collar trace is

$$\widehat{\Phi}_5 : \mathbb{N}_\infty \to [0,1], \qquad \widehat{\Phi}_5(n) = \left(\frac{3}{4}\right)^n, \qquad \widehat{\Phi}_5(\infty) = 0.$$

The point ∞ is a completion point, not a finite proof stage.

Lemma A.197.7 (r13 completion is external to finite discharge)**.** *In the topology of Definition A.197.6,* 0 *is attained only at the completion point ∞, not at any finite stage.*

Proof. For each finite n, $(3/4)^n > 0$. By definition, $\widehat{\Phi}_5(\infty) = 0$. Therefore the zero value belongs to the completed boundary of the bounded-collar trace, not to the finite stage set. □

Proposition A.197.8 (r13 C5 completion-topology check)**.** *The C5 contraction supports the completed bounded-collar statement*

$$\widehat{\Phi}_5(\infty) = 0,$$

but it does not support finite C5 closure:

$$\forall n < \infty : \ \Phi_5^{(n)} > 0.$$

Consequently C5 may be recorded as

$$C_5 : \mathsf{complete}^{\infty}_{\text{bd collar}} \text{ with finite-positive guard.}$$

Proof. The first statement is the completion rule in Definition A.197.6. The second is Lemma A.197.7. The status label therefore records the useful topological completion while preserving the r11 limit guard. □

Corollary A.197.9 (No C5-to-C6 promotion)**.** *The r13 completion topology does not promote C6, T2, or all-readout closure.*

Proof. The topology concerns only the bounded-collar C5 trace and its completion point. It does not provide an all-readout witness family, an unrestricted exterior selector, or a rule identifying the completion point with a finite global readout. Hence C6 remains unchanged. □

A.197.3 Combined r13 status and next queue

Theorem A.197.10 (r13 mixed-kernel/topology theorem)**.** *Inside the declared v4.7 bounded–exterior–mixed grammar, r13 proves the status update*

$$C_4 : \mathsf{pers}^{\text{route-stable}}_{\text{mixker}}, \qquad \beta_4^{r12} = \frac{1}{256},$$

$$C_5 : \mathsf{complete}^{\infty}_{\text{bd collar}} + \mathsf{finite\text{-}positive\ guard},$$

$$C_6 : \mathsf{B}_{\text{all/T2}}.$$

Thus r13 decides the remaining C4 kernel as route-stable persistent, gives C5 a precise completion topology, and leaves the all-readout/T2 blocker untouched.

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Proof. The C4 statement is Corollary A.197.5. The C5 statement is Proposition A.197.8. The C6 statement is Corollary A.197.9. □

Corollary A.197.11 (Permitted r13 direction)**.** *The next admissible internal pass is*

v4.7.0-r13: route-stable mixed-kernel normal form and completion-interface audit.

It may convert the persistent C4 mixed kernel into a normal-form obstruction and may audit the interface between finite bounded stages and the completed C5 collar. It must not treat the completion point as a finite witness and must not promote C6 or T2.

Remark A.197.12 (r13 summary)**.** *The r13 pass changes the quality of the residual information. C4 no longer asks "can the last kernel be absorbed by the already declared routes?"; the answer is no in the present grammar, but the kernel is route-stable and harmless as a named persistent obstruction. C5 no longer asks "does the geometric decrease close in finite time?"; the answer is no, but the completed bounded-collar topology is now explicit. The useful gain is a stable normal-form target for C4 and a safe completion interface for C5.*

A.198 v4.7.0-r13 Route-Stable Mixed-Kernel Normal Form, Overlap Relay, and Completion-Interface Audit

Remark A.198.1 (Source-only intermediate discipline)**.** *This r13 pass is an internal source-layer update. It is intentionally recorded as TeX/source material only. A PDF render is not part of the intermediate pass and should only be produced on explicit request.*

A.198.1 Ridge-front reading of the remaining mixed kernel

Definition A.198.2 (Ridge-front/shadow/reentry normal form)**.** *A route-stable mixed-kernel residue is said to be in ridge-front/shadow/reentry normal form if its obstruction is not read as a demand that one global wave packet clear the entire landscape, but as a local three-slot interface*

$$\mathsf{RF}_4^{r13} = \big(F_{\mathrm{ridge}}, Z_{\mathrm{shadow}}, H_{\mathrm{reentry}}\big).$$

Here F_{ridge} is the front-side barrier face, Z_{shadow} is the finite shadow or no-direct-line region behind the front, and H_{reentry} is a downstream handoff point at which the descending valley can again carry the declared bounded readout.

Remark A.198.3 (Geometric meaning)**.** *The intended picture is a rugged complex surface. The active question is not whether a whole wave front globally passes over every ridge. The active question is whether enough locally admissible signal can reach a downstream handoff point after a ridge-front/shadow segment. This is a strictly weaker and more route-faithful demand.*

Definition A.198.4 (Local relay admissibility)**.** *A ridge-front normal form RF_4^{r13} is locally relay-admissible if there exists a finite ordered handoff chain*

$$H_0 \to H_1 \to \cdots \to H_m$$

such that every segment is either a declared bounded valley segment, a selector routed ridge-front crossing, or a shadow-zone reentry segment with nonzero bounded information flux. The chain is not required to contain a single global through-wave.

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Lemma A.198.5 (r13 no-global-wave requirement)**.** *Local relay admissibility does not require a proof that one entire wave packet passes over the full ridge landscape.*

Proof. Definition A.198.4 quantifies over finite handoff chains. The admissible objects are local segments and reentry points, not a single global packet. Hence the required witness type is weaker than a global over-ridge traversal witness. □

A.198.2 Valley overlap relay equation

Definition A.198.6 (Valley packet family)**.** *For a valley chart V_k, let*

$$\Psi_k(t) = \{\psi_{k,a}(t) : a \in A_k(t)\}$$

be the finite family of bounded packets present in the chart during the time window $[t, t+\Delta t]$. Let K_k be the declared comparison/overlap kernel for the chart, and let $\Theta_k > 0$ be the minimal relay threshold.

Definition A.198.7 (Valley overlap relay functional)**.** *The valley overlap relay functional is*

$$\mathcal{O}_k(t, \Delta t) = \int_t^{t+\Delta t} \sum_{a,b \in A_k(s)} \langle \psi_{k,a}(s), K_k \psi_{k,b}(s) \rangle_{V_k} \, ds.$$

The valley chart is relay-positive over $[t, t+\Delta t]$ if

$$\mathcal{O}_k(t, \Delta t) \geq \Theta_k.$$

Remark A.198.8 (Pile-up interpretation)**.** *The cross terms $a \neq b$ encode the possibility that packets locally pile up or overlap. Thus a downstream relay may be powered by packet density per time window rather than by a single packet carrying the entire route alone.*

Proposition A.198.9 (Overlap relay weakens the ridge-front obstruction)**.** *If each valley chart in a finite handoff chain is relay-positive and each ridge-front/shadow transition has a declared reentry handoff, then the chain carries bounded readout information without requiring a global over-ridge wave witness.*

Proof. Relay positivity provides a nonzero bounded information flux on each valley chart. The reentry handoffs concatenate these local flux carriers. By Lemma A.198.5, the concatenated witness is local and relay-based rather than a single global wave witness. □

A.198.3 Photon-frame clock and Riemann-zero handoff probe

Definition A.198.10 (Photon-frame handoff clock)**.** *A photon-frame handoff clock is a finite ordered family*

$$\mathcal{T}_\gamma^{r13} = \{\tau_j : j = 0, \ldots, m\}$$

assigned to a relay chain, where τ_j marks the readout tick at which the route passes through the handoff point H_j. The notation is operational: γ denotes photon-frame timing and does not identify the clock with a physical photon trajectory through collapse.

Definition A.198.11 (Riemann-zero phase probe)**.** *Let $\Gamma_N = \{\gamma_1, \ldots, \gamma_N\}$ be a finite list of ordinates used as a spectral comparison clock. A Riemann-zero phase probe for a handoff clock $\mathcal{T}_\gamma^{r13}$ is a comparison map*

$$\mathcal{Z}_N^{r13} : \mathcal{T}_\gamma^{r13} \longrightarrow \Gamma_N$$

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together with a phase mismatch functional

$$\mathcal{E}^{r13}_{\mathrm{zero}} = \sum_j w_j \, d_{\mathbb{T}}\big(\theta(\tau_j), \theta(\gamma_{n(j)})\big)^2 .$$

Here $d_{\mathbb{T}}$ is circular phase distance and $w_j \geq 0$ are declared weights.

Remark A.198.12 (No zero/photon identification)**.** *Definition A.198.11 is a probe, not an identification. It does not assert that handoff points are Riemann zeros, that photons are zeros, or that RH follows. It asks only whether the photon-frame handoff ticks are better read as a spectral timing pattern when compared with a finite zero-clock list.*

Definition A.198.13 (Zero-clock relay compatibility)**.** *A relay chain is zero-clock compatible at tolerance ε_Z if*

$$\mathcal{E}^{r13}_{\mathrm{zero}} \leq \varepsilon_Z$$

and every relay segment remains bounded-admissible in the sense of Definition A.198.4.

Proposition A.198.14 (Riemann-zero connection as admissible probe, not lift)**.** *The r13 zero-clock construction is an admissible diagnostic probe for photon-frame handoff timing, but it does not promote C6, T2, or all-readout closure.*

Proof. The construction only adds a comparison functional on finite handoff clocks. It supplies neither an all-readout witness family nor an unrestricted exterior selector. Therefore it may guide later spectral timing diagnostics, but it does not discharge the all-readout/T2 blocker. □

A.198.4 Route-stable mixed-kernel normal form

Definition A.198.15 (r13 mixed-kernel normal form)**.** *The r12 persistent mixed-kernel residue is rewritten as*

$$\mathsf{MK}^{r13}_4 = \big(\beta^{r12}_4, \mathsf{RF}^{r13}_4, \mathcal{O}_\bullet, \mathcal{T}^{r13}_\gamma, \mathcal{E}^{r13}_{\mathrm{zero}}\big),$$

with

$$\beta^{r12}_4 = \frac{1}{256}.$$

The normal form records that the residual is a route-stable local relay problem with a possible spectral timing probe, not a demand for a global wave to cross the entire surface.

Theorem A.198.16 (r13 route-stable mixed-kernel normal-form theorem)**.** *Inside the declared v4.7 bounded–exterior–mixed grammar, the r12 residual mixed kernel admits the r13 normal form*

$$C_4 : \mathsf{pers}^{\text{route-stable}}_{\text{mixker}} \rightsquigarrow \mathsf{relay}^{\text{zero-probe}}_{\text{ridge/shadow/reentry}},$$

with unchanged positive residual bound

$$\beta^{r13}_4 = \frac{1}{256} > 0.$$

Moreover, the r13 form permits local overlap relay tests and photon-frame zero-clock diagnostics without promoting C5 finite closure, C6, T2, or all-readout closure.

Proof. The ridge-front normal form is Definition A.198.2. The local relay condition is Definition A.198.4. The overlap relay equation is Definition A.198.7. The zero-clock probe is Definition A.198.11. None of these definitions changes the positive bound 1/256. They change the shape of the next test: from global wave passage to local relay, pile-up, and spectral timing compatibility. Proposition A.198.14 keeps the all-readout/T2 guard intact. □

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A.198.5 Wavefront flux consequence of the valley-overlap relay

The valley-overlap relay changes the operational reading of the residual mixed-kernel obstruction. The obstruction is no longer treated as a demand that one full wave packet cross the whole rugged surface. It is treated as a local flux condition along a ridge-facing wavefront.

Definition A.198.17 (r13 wavefront packet flux)**.** *Let $N_v(\Delta t)$ be the number of selector-admissible packets reaching a valley-front handoff window during the local photon-frame time window Δt. The packet flux along the local front is*

$$\dot{N}_v(\Delta t) = \frac{N_v(\Delta t)}{\Delta t}.$$

The overlap-weighted relay current is

$$\mathcal{J}_v(\Delta t) = \frac{1}{\Delta t}\,\mathcal{O}_v(\Delta t),$$

where $\mathcal{O}_v$ is the valley-overlap functional.

Definition A.198.18 (r13 local relay threshold)**.** *The inherited r12 mixed-kernel bound gives the conservative local ridge-front threshold*

$$\Theta_v := \beta_4^{r12} = \frac{1}{256}.$$

A finite valley handoff is locally admissible whenever

$$\mathcal{O}_v(\Delta t) \geq \Theta_v,$$

or equivalently, at the level of overlap-weighted relay current,

$$\mathcal{J}_v(\Delta t) \geq \frac{1}{256\,\Delta t}.$$

Lemma A.198.19 (equal-amplitude overlap flux law)**.** *For N equal-amplitude packets of amplitude a, with mean local coherence $\kappa \in [0,1]$, the finite valley-overlap relay functional reduces to*

$$\mathcal{O}_N = a^2\big(N + \kappa N(N-1)\big).$$

Thus the local handoff condition is

$$a^2\big(N + \kappa N(N-1)\big) \geq \frac{1}{256}.$$

Proof. The diagonal part gives the incoherent intensity contribution Na^2. The off-diagonal pair-overlap contribution is weighted by the mean coherence factor κ, producing $\kappa N(N-1)a^2$. The local handoff condition follows by comparison with the r12 threshold $1/256$. □

Proposition A.198.20 (r13 minimum packet-flux calculation)**.** *For the conservative reference amplitude*

$$a = \frac{1}{64},$$

the required number of packets per local handoff window satisfies

$$N_{\min}(\kappa) = \begin{cases} 16, & \kappa = 0, \\ \left\lceil \dfrac{(\kappa-1) + \sqrt{(1-\kappa)^2 + 64\kappa}}{2\kappa} \right\rceil, & 0 < \kappa \leq 1. \end{cases}$$

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In particular,

κ	0	1/4	1/2	3/4	1
$N_{\min}$	16	7	6	5	4

per local photon-frame handoff window.

Proof. Since $a^2 = 1/4096$, the handoff inequality is equivalent to

$$N + \kappa N(N-1) \geq 16.$$

For $\kappa = 0$ this gives $N \geq 16$. For $\kappa > 0$, solving the quadratic inequality gives the stated ceiling formula. The tabulated values are direct substitutions. □

Corollary A.198.21 (flux gain along the wavefront)**.** *Relative to the incoherent case, local wave-packet pile-up lowers the minimum packet flux required at the ridge-front. In one normalized handoff tick $\Delta t = 1$, the admissible flux requirement changes from*

$$\dot{N}_{\min} = 16$$

to

$$\dot{N}_{\min} \in \{7, 6, 5, 4\}$$

for $\kappa = 1/4, 1/2, 3/4, 1$, respectively. Thus the local overlap relay gives a flux-efficiency gain

$$G_{\text{flux}}(\kappa) = \frac{16}{N_{\min}(\kappa)},$$

namely approximately

$$1, \quad 2.29, \quad 2.67, \quad 3.20, \quad 4.00.$$

Remark A.198.22 (What this changes)**.** *The relay problem is now controlled by packet flux and local overlap, not by a global single-wave crossing requirement. The information channel only needs a sufficient packet current along the admissible ridge-front/reentry chain. The front can therefore behave like a local queue or pile-up: packets that fail as isolated crossings may still produce an admissible handoff when enough of them arrive in phase-compatible windows.*

Definition A.198.23 (zero-clock assisted relay current)**.** *Let $\mathcal{C}_\gamma(\tau_v)$ be the photon-frame zero-clock diagnostic weight at the candidate handoff tick τ_v. The zero-clock-assisted relay condition is*

$$\mathcal{C}_\gamma(\tau_v)\, \mathcal{O}_v(\Delta t) \geq \frac{1}{256}.$$

Equivalently, the effective threshold is

$$\Theta_{v,\text{eff}}(\tau_v) = \frac{1}{256\, \mathcal{C}_\gamma(\tau_v)}.$$

This is a timing and phase-locking probe only; it is not an identification of Riemann zeros with photons, and it is not an RH, T2, or all-readout closure claim.

Proposition A.198.24 (flow-rate consequence of a clock gain)**.** *For $a = 1/64$, a zero-clock diagnostic gain $G = \mathcal{C}_\gamma(\tau_v) > 0$ changes the finite relay inequality to*

$$N + \kappa N(N-1) \geq \frac{16}{G}.$$

Thus a favorable photon-frame timing window lowers the minimum packet flux, while an unfavorable timing window raises it.

Proof. Substitute $\Theta_{v,\text{eff}} = 1/(256G)$ and $a^2 = 1/4096$ into the equal-amplitude overlap law. □

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A.198.6 Completion-interface audit

Proposition A.198.25 (C5 completion remains a boundary point)**.** *The C5 completion point at ∞ remains a boundary object of the completed bounded-collar topology and is not a finite handoff witness.*

Proof. The r13 relay chain uses finite handoff points $H_0, \dots, H_m$. The completed C5 point is indexed by ∞, not by a finite route segment. Therefore it can bound or guide the asymptotic collar but cannot itself serve as a finite handoff witness. □

Corollary A.198.26 (Permitted r14 direction)**.** *The next admissible internal pass is*

v4.7.0-r14: finite valley-overlap relay scan and zero-clock phase ledger.

It should instantiate a finite family of valley charts, compute symbolic or bounded values of $\mathcal{O}_k$, and create a finite zero-clock phase ledger for handoff ticks. It must not claim RH, T2, all-readout closure, or a physical identity between photons and Riemann zeros.

Remark A.198.27 (r13 summary)**.** *r13 answers the ridge intuition as follows. The current obstruction is better read as a local front/shadow/reentry relay problem than as a demand that an entire wave cross a global mountain range. Packet pile-up is allowed through the valley overlap functional. The possible relation to Riemann zeros is placed in a finite photon-frame timing probe: a useful diagnostic channel, not a theorematic identification or closure claim.*

A.199 v4.7.0-r14 Source-Only Finite Valley-Front Chart Scan

Source-only status. This r14 pass is a source-only internal working increment. It adds a finite valley-front chart scan to the r13 ridge-front/reentry and valley-overlap relay model. No PDF build, DOI assignment, public-release claim, T2 claim, RH claim, or all-readout closure claim is made in this pass.

A.199.1 Normalized chart scan convention

The r13 relay threshold is kept fixed as

$$\Theta_v = \beta_4^{r12} = \frac{1}{256},$$

and the normalized equal-packet amplitude is kept as

$$a = \frac{1}{64}, \qquad a^2 = \frac{1}{4096}.$$

For a chart with packet count N, coherence parameter κ, and photon-frame clock diagnostic gain $G = \mathcal{C}_\gamma(\tau_v)$, the finite handoff inequality becomes

$$G\big(N + \kappa N(N-1)\big) \geq 16.$$

This is the working finite chart scan rule for r14.

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Definition A.199.1 (finite valley-front chart)**.** *A finite valley-front chart is a tuple*

$$\mathcal{V}_j = (N_j, \kappa_j, G_j, \sigma_j),$$

where N_j is the local number of selector-admissible wave packets inside the handoff window, $\kappa_j \in [0,1]$ is their effective phase-coherence factor, $G_j = \mathcal{C}_\gamma(\tau_j) > 0$ is the photon-frame clock diagnostic gain, and σ_j is one of

$$\mathsf{blocked}, \qquad \mathsf{relay\text{-}admissible}, \qquad \mathsf{zero\text{-}clock\text{-}assisted}.$$

The status is assigned by the following finite rule:

$$\sigma_j = \begin{cases} \mathsf{relay\text{-}admissible}, & N_j + \kappa_j N_j(N_j-1) \geq 16, \\ \mathsf{zero\text{-}clock\text{-}assisted}, & N_j + \kappa_j N_j(N_j-1) < 16 \text{ and } G_j\big(N_j + \kappa_j N_j(N_j-1)\big) \geq 16, \\ \mathsf{blocked}, & G_j\big(N_j + \kappa_j N_j(N_j-1)\big) < 16. \end{cases}$$

A.199.2 Finite r14 chart ledger

Table 7: r14 finite valley-front chart scan. Scores are normalized by the threshold 1/256; a score at least one is relay-sufficient in the indicated channel.

Chart	N	κ	G	no-clock score	clock score	status
V0	4	0	1	1/4	1/4	blocked
V1	16	0	1	1	1	relay-admissible
V2	7	1/4	1	35/32	35/32	relay-admissible
V3	5	1/4	2	5/8	5/4	zero-clock-assisted
V4	6	1/2	1	21/16	21/16	relay-admissible
V5	4	1/2	5/2	5/8	25/16	zero-clock-assisted
V6	3	3/4	3/2	15/32	45/64	blocked
V7	4	1	1	1	1	relay-admissible
V8	3	1	2	9/16	9/8	zero-clock-assisted
V9	2	1/2	3	3/16	9/16	blocked

The finite scan contains 10 charts. It classifies 4 charts as directly relay-admissible, 3 charts as zero-clock-assisted, and 3 charts as still blocked in the declared r14 scan family.

Proposition A.199.2 (finite flow-rate classification)**.** *Inside the declared r14 finite chart family, local front flow rather than global wave-over-mountain crossing determines handoff admissibility. In particular, there are charts for which the no-clock score is below one while the clock score is at least one. These charts are admissible only as photon-frame timing-assisted relay charts.*

Proof. The classification follows by direct substitution into

$$G\big(N + \kappa N(N-1)\big) \geq 16.$$

For example, chart $V3$ has $N = 5$, $\kappa = 1/4$, and $G = 2$. Its unassisted normalized score is $10/16 < 1$, while its assisted score is $20/16 > 1$. Hence it is not globally unassisted, but it is locally zero-clock-assisted. This is exactly the r13 ridge-front/reentry interpretation: local packet flux at the handoff point may succeed even when an isolated or unassisted crossing would fail. □

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A.199.3 Threshold grid for flux planning

The minimum required packet count for selected (κ, G) values is:

$\kappa \backslash G$	1	3/2	2	5/2	3	4
0	16	11	8	7	6	4
1/4	7	6	5	4	4	3
1/2	6	5	4	4	3	3
3/4	5	4	4	3	3	3
1	4	4	3	3	3	2

Thus the role of the photon-frame clock diagnostic is not to close the residual by itself, but to lower the required local packet flux at suitable reentry ticks.

Corollary A.199.3 (r14 consequence for the mixed kernel)**.** *The route-stable mixed kernel from r12 may be treated, for the next source-only pass, as a finite family of local ridge-front handoff charts rather than a single global wave-crossing obstruction. The persistent value* $1/256$ *is retained as the local threshold, but the admissibility question is moved to finite chart routing, packet overlap, and timing-assisted relay status.*

Remark A.199.4 (guard against overreading the scan)**.** *The r14 chart scan is a finite diagnostic ledger. It does not identify Riemann zeros with photons, does not prove RH, does not close T2, and does not eliminate the all-readout blocker. It only shows that the remaining mixed-kernel problem has a more precise local form: some valley fronts are blocked, some are directly relay-admissible, and some require a favorable photon-frame timing window.*

Corollary A.199.5 (permitted r15 direction)**.** *The next admissible internal source-only pass is*

v4.7.0-r15: zero-clock phase-window extraction and relay-chain stitching.

It should extract a finite set of admissible timing windows τ_j*, test whether adjacent valley-front charts can be stitched into a relay chain, and keep all asymptotic, RH, T2, and all-readout claims guarded.*

A.200 v4.7.0-r15 Source-Only: Zero-Clock Phase-Window Extraction and Relay-Chain Stitching

Source-only status. This r15 pass is source-only. It does not rebuild a PDF, does not assign a new DOI, does not reopen the release metadata, and does not claim a T2, all-readout, RH, or global physical closure theorem. It continues the r13–r14 ridge-front interpretation of the route-stable mixed kernel.

A.200.1 Phase-window extraction rule

For each valley-front chart V, let

$$S_V = N_V + \kappa_V N_V (N_V - 1)$$

be the local overlap flux before photon-frame timing gain. The r14 relay criterion was

$$G_V S_V \geq 16,$$

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where the factor 16 is the normalized form of the residual ridge threshold $\Theta_v = 1/256$ at packet amplitude $a = 1/64$. Hence the required clock gain is

$$G_V^{\mathrm{req}} = \frac{16}{S_V}.$$

A zero-clock phase window is extracted only when

$$1 < G_V^{\mathrm{req}} \le G_V^{\mathrm{avail}}.$$

Thus direct charts need no clock window, assisted charts need a nonempty timing window, and blocked charts remain blocked in the finite r15 family.

The photon-clock probe is still written as a diagnostic timing function, not as an identity between Riemann zeros and photons:

$$\mathcal{C}_\gamma(\tau) = 1 + \eta \frac{\left|\sum_{m=1}^{M} w_m e^{i\gamma_m \tau}\right|^2}{\left(\sum_{m=1}^{M} |w_m|\right)^2}.$$

For a local tick τ_V, the window is the threshold set

$$I_V = \{\tau : \mathcal{C}_\gamma(\tau) \ge G_V^{\mathrm{req}}\}.$$

In the local normalized tick approximation

$$G_V(\tau) = 1 + (G_V^{\mathrm{avail}} - 1)\cos^2\left(\frac{\tau - \tau_V}{\sigma_V}\right),$$

the half-width is

$$\delta_V = \sigma_V \arccos\sqrt{\frac{G_V^{\mathrm{req}} - 1}{G_V^{\mathrm{avail}} - 1}}.$$

This formula is used only to rank finite timing windows. It is not a spectral identification theorem.

A.200.2 Extracted finite timing windows

The r14 zero-clock-assisted charts admit the following r15 phase windows:

chart	S_V	G_V^{req}	G_V^{avail}	δ_V/σ_V
$V3$	10	8/5	2	0.684719
$V5$	10	8/5	5/2	0.886077
$V8$	9	16/9	2	0.490883

Thus the narrowest extracted window in the finite family is the coherent sparse reentry pulse $V8$, while the ridge-seam chart $V5$ has the widest clock margin.

Proposition A.200.1 (finite zero-clock phase-window extraction)**.** *Within the declared r15 finite chart family, exactly three charts have nonempty zero-clock phase windows: $V3, V5, V8$. These windows are finite, local timing windows for relay admissibility. They do not close the mixed kernel globally.*

Proof. For $V3$ and $V5$, $S_V = 10$, hence $G_V^{\mathrm{req}} = 16/10 = 8/5$. The available gains are 2 and 5/2, respectively, so the threshold-set condition is satisfied. For $V8$, $S_V = 9$, hence $G_V^{\mathrm{req}} = 16/9 < 2$, again giving a nonempty window. All direct charts have $G_V^{\mathrm{req}} \le 1$, so no clock window is needed. The blocked charts have insufficient available gain in the finite ledger. □

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A.200.3 Relay-chain stitching

The extracted windows allow two finite stitched relay chains:

$$\mathcal{R}_A : V1 \longrightarrow V2 \longrightarrow V3 \longrightarrow V4 \longrightarrow V5,$$

where $V3$ and $V5$ require the phase windows I_{V3} and I_{V5}, and

$$\mathcal{R}_B : V7 \longrightarrow V8,$$

where $V8$ requires I_{V8}. The edges

$$V5 \longrightarrow V6, \qquad V8 \longrightarrow V9$$

remain blocked in the r15 finite chart family.

edge	type	status	required window
$V1 \to V2$	direct-to-direct	stitched	none
$V2 \to V3$	direct-to-assisted	stitched	I_{V3}
$V3 \to V4$	assisted-to-direct	stitched	I_{V3}
$V4 \to V5$	direct-to-assisted	stitched	I_{V5}
$V5 \to V6$	assisted-to-blocked	blocked	none
$V7 \to V8$	direct-to-assisted	stitched	I_{V8}
$V8 \to V9$	assisted-to-blocked	blocked	none

Corollary A.200.2 (finite chain status)**.** *The r15 relay stitching produces a longest certified chain of length five and a secondary certified chain of length two. It does not produce a single global all-terrain relay chain.*

A.200.4 Remaining gap ledger

The blocked charts have the following required-to-available gain gaps:

chart	G^{req}	G^{avail}	$G^{\mathrm{req}}/G^{\mathrm{avail}}$
$V0$	4	1	4
$V6$	32/15	3/2	64/45
$V9$	16/3	3	16/9

Thus the closest remaining blocked valley-front is $V6$, not $V0$ or $V9$. This suggests that the next useful pass should focus on a local shadow-basin lift for $V6$, rather than on a global mixed-kernel closure claim.

Remark A.200.3 (r15 guard)**.** *The r15 result is a finite relay-chain diagnostic. It does not identify the Riemann zeros with photon ticks, does not prove RH, does not prove T2, and does not remove the all-readout blocker. Its only promoted content is local: certain valley-front charts have explicit photon-clock phase windows, and these windows stitch finite relay chains across parts of the ridge-front atlas.*

Corollary A.200.4 (permitted r16 direction)**.** *The next admissible source-only pass is*

v4.7.0-r16: shadow-basin lift test for $V6$ and relay-chain extension guard.

It should test whether the closest blocked chart $V6$ can be lifted by a local overlap or clock-window refinement, while keeping $V0$, $V9$, T2, RH, and all-readout closure guarded.

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A.201 v4.7.0-r16 Source-Only: Shadow-Basin Lift Test for $V6$

This source-only pass tests the closest blocked r15 valley-front chart, $V6$, without rebuilding a PDF and without changing the baseline DOI. The r15 obstruction ledger identified

$$V6: \qquad S_6^{(0)} = \frac{15}{2}, \qquad G_6^{(0)} = \frac{3}{2}, \qquad G_6^{\mathrm{req}} = \frac{32}{15}.$$

Equivalently,

$$G_6^{(0)} S_6^{(0)} = \frac{45}{4} < 16.$$

Thus $V6$ was not globally obstructed by a whole-landscape failure; it was a local shadow-basin deficit at the ridge-front handoff seam.

Definition A.201.1 (shadow-basin lift datum for $V6$)**.** *A* shadow-basin lift datum *for $V6$ is a finite local replacement*

$$(N_6, \kappa_6, G_6) \longmapsto (N_6', \kappa_6', G_6')$$

inside the same declared valley-front chart, with no all-readout extension and no new global selector. It is admissible only if

$$G_6'\big(N_6' + \kappa_6' N_6'(N_6' - 1)\big) \geq 16.$$

A.201.1 Finite lift computation

The r15 value was

$$N_6 = 3, \qquad \kappa_6 = \frac{3}{4}, \qquad G_6 = \frac{3}{2},$$

so

$$S_6 = 3 + \frac{3}{4} \cdot 3 \cdot 2 = \frac{15}{2}.$$

The shadow-basin test asks whether a local pile-up can increase the packet flux by one admissible packet in the same photon-clock window. The tested lifted state is

$$N_6' = 4, \qquad \kappa_6' = \frac{3}{4}, \qquad G_6' = \frac{3}{2}.$$

Then

$$S_6' = 4 + \frac{3}{4} \cdot 4 \cdot 3 = 13,$$

and hence

$$G_6' S_6' = \frac{3}{2} \cdot 13 = \frac{39}{2} > 16.$$

The normalized relay score is therefore

$$\frac{G_6' S_6'}{16} = \frac{39}{32} > 1.$$

Lemma A.201.2 (one-packet shadow-basin lift)**.** *In the finite r16 chart family, $V6$ is lifted from blocked to shadow-basin-assisted relay-admissible by adding one selector-admissible packet to the local basin while preserving $\kappa_6 = 3/4$ and $G_6 = 3/2$.*

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Proof. The computation above gives

$$G_6 S_6 = 45/4 < 16$$

for the r15 state, while the lifted state satisfies

$$G_6' S_6' = 39/2 > 16.$$

Therefore the r15 chart was subcritical and the r16 lifted chart is supercritical with margin

$$\frac{39}{2} - 16 = \frac{7}{2}.$$

No all-readout extension, global terrain crossing, or T2 bridge is used. □

A.201.2 Flux-rate consequence

The lift is small in packet count but large in effective overlap. The packet flux changes by

$$\frac{N_6'}{N_6} = \frac{4}{3},$$

whereas the local overlap strength changes by

$$\frac{S_6'}{S_6} = \frac{13}{15/2} = \frac{26}{15}.$$

Thus a 33.3% increase in selector-admissible packet count produces a 73.3% increase in the overlap strength, because the additional packet also participates in the coherent pair terms. This is the precise sense in which the r13/r14 valley-front model converts a ridge barrier into a local flow-rate problem.

Corollary A.201.3 (local relay lift without global wave crossing)**.** *The r16 lift of* $V6$ *does not require a whole wave to cross the whole ridge. It only requires that the local packet flux in the shadow basin rises from* 3 *to* 4 *packets in the same declared handoff window while maintaining the same partial coherence and photon-clock gain.*

A.201.3 Relay-chain extension guard

The r15 chain

$$\mathcal{R}_A : V1 \to V2 \to V3 \to V4 \to V5$$

extends in r16 to

$$\mathcal{R}_A^+ : V1 \to V2 \to V3 \to V4 \to V5 \to V6.$$

The extension is guarded: it certifies the edge $V5 \to V6$ only after the local shadow-basin lift of $V6$. It does not certify $V6 \to V7$, does not merge $\mathcal{R}_A^+$ with the secondary chain $V7 \to V8$, and does not lift $V0$ or $V9$.

edge	r15 status	r16 status	guard
$V1 \to V2$	stitched	stitched	unchanged
$V2 \to V3$	stitched with window	stitched with window	unchanged
$V3 \to V4$	stitched	stitched	unchanged
$V4 \to V5$	stitched with window	stitched with window	unchanged
$V5 \to V6$	blocked	stitched after shadow-basin lift	local lift only
$V6 \to V7$	not certified	not certified	atlas gap
$V7 \to V8$	stitched with window	stitched with window	separate chain
$V8 \to V9$	blocked	blocked	insufficient lift

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Corollary A.201.4 (r16 finite chain status)**.** *The longest certified relay chain in the finite r16 atlas has length six:*

$$V1 \to V2 \to V3 \to V4 \to V5 \to V6.$$

This is a finite relay extension, not a global all-terrain relay chain.

A.201.4 C4/C5/C6 status after r16

The r16 lift improves a local representative of the route-stable mixed-kernel picture, but it does not close $C4$. The persistent mixed-kernel guard from r12 remains in force:

$$\beta_4^{r12} = \frac{1}{256} > 0.$$

The $C5$ bounded-collar completion guard also remains unchanged:

$$\forall n < \infty : \Phi_5^{(n)} > 0, \qquad \widehat{\Phi}_5(\infty) = 0.$$

Finally,

$$C6 : \mathsf{B}_{\mathrm{all/T2}}$$

remains blocked. The r16 result is therefore only a local relay-flow upgrade.

Remark A.201.5 (r16 non-claim guard)**.** *The r16 shadow-basin lift does not prove RH, does not identify Riemann zeros with photon ticks, does not prove T2, does not close the all-readout boundary, and does not remove the route-stable mixed-kernel obstruction. It proves only that the closest r15 blocked chart $V6$ can be lifted by a finite local packet-flux increase within the declared valley-front relay grammar.*

Corollary A.201.6 (permitted r17 direction)**.** *The next admissible source-only pass is*

v4.7.0-r17: atlas-gap bridge test for $V6 \to V7$ and guarded $V9$ micro-lift scan.

It should test whether the newly lifted $V6$ can be stitched to the existing secondary chain through a declared atlas-gap bridge, while keeping $V0$, $V9$, RH, T2, and all-readout closure guarded.

A.202 v4.7.0-r17 Source-Only Atlas-Gap Bridge Test and Guarded V9 Micro-Lift

This source-only layer continues the valley-front relay program after the r16 shadow-basin lift of $V6$. The aim is deliberately finite: test the atlas gap $V6 \to V7$, then scan whether the still blocked $V9$ chart admits a minimal micro-lift. No PDF is produced in this intermediate version.

A.202.1 r17 relay score convention

For a local valley-front chart V the overlap score is retained as

$$S_V(N, \kappa) = N + \kappa N(N-1),$$

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and the clock-weighted relay score is

$$H_V = G_V S_V.$$

The finite local handoff condition is

$$H_V \geq 16.$$

This is only a local threshold condition. It is not a global terrain theorem and not an all-readout closure condition.

A.202.2 Atlas-gap bridge test for $V6 \to V7$

After r16 the lifted $V6$ state is

$$N_6' = 4, \qquad \kappa_6 = \frac{3}{4}, \qquad G_6 = \frac{3}{2}.$$

Hence

$$S_6' = 4 + \frac{3}{4} \cdot 4 \cdot 3 = 13, \qquad H_6' = \frac{3}{2} \cdot 13 = \frac{39}{2}.$$

The local surplus at $V6$ is therefore

$$\mu_6 = H_6' - 16 = \frac{39}{2} - 16 = \frac{7}{2}.$$

The target chart $V7$ remains the near-coherent equality handoff:

$$N_7 = 4, \qquad \kappa_7 = 1, \qquad G_7 = 1,$$

so that

$$S_7 = 4 + 1 \cdot 4 \cdot 3 = 16, \qquad H_7 = 16, \qquad \mu_7 = 0.$$

Thus $V7$ has no spare local margin, but it does satisfy the relay threshold exactly. The only bridge question is whether the atlas-gap loss from the lifted $V6$ side can be paid from the $V6$ surplus.

For the finite r17 scan we declare the conservative atlas-gap toll

$$\alpha_{67} = \frac{1}{2}.$$

The transported bridge margin is then

$$\mu_{67}^{\text{bridge}} = \mu_6 - \alpha_{67} = \frac{7}{2} - \frac{1}{2} = 3 > 0.$$

Consequently the edge $V6 \to V7$ is bridge-admissible in the finite r17 atlas. It is, however, still a critical bridge because the target chart $V7$ itself sits exactly at threshold.

Proposition A.202.1 (r17 finite atlas-gap bridge)**.** *Within the declared finite r17 valley-front atlas, the edge $V6 \to V7$ is admissible after the r16 shadow-basin lift of $V6$. The certification uses $V6$-side surplus $\mu_6 = 7/2$ to pay the declared atlas-gap toll $\alpha_{67} = 1/2$, leaving bridge margin $3 > 0$.*

Remark A.202.2 (critical target guard)**.** *The r17 bridge is not a robust global bridge theorem. The target chart $V7$ has $\mu_7 = 0$, so the certification is finite, source-supported, and atlas-local. Any later attempt to promote it must supply a uniform interface margin or an independently declared bridge operator.*

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A.202.3 Guarded micro-lift scan for $V9$

The r14/r15 state of $V9$ was subcritical:

$$N_9 = 2, \qquad \kappa_9 = \frac{1}{2}, \qquad G_9 = 3.$$

Therefore

$$S_9 = 2 + \frac{1}{2} \cdot 2 \cdot 1 = 3, \qquad H_9 = 3 \cdot 3 = 9 < 16.$$

The guarded micro-lift scan tests the smallest packet-flux increase

$$N_9 : 2 \to 3$$

while leaving κ_9 and G_9 unchanged. Then

$$S_9' = 3 + \frac{1}{2} \cdot 3 \cdot 2 = 6, \qquad H_9' = 3 \cdot 6 = 18 > 16.$$

The resulting margin is

$$\mu_9' = 18 - 16 = 2 > 0.$$

Thus $V9$ is not inert: one additional local packet in the same guarded clock window suffices to make the chart relay-admissible.

Lemma A.202.3 (r17 guarded $V9$ micro-lift)**.** *In the finite r17 scan, the transition*

$$(N_9, \kappa_9, G_9) = \left(2, \frac{1}{2}, 3\right) \longmapsto \left(3, \frac{1}{2}, 3\right)$$

lifts $V9$ from blocked to guarded micro-lift relay-admissible, since $H_9' = 18 > 16$.

Remark A.202.4 (micro-lift non-claim)**.** *The $V9$ micro-lift does not close $V0$, does not close the route-stable mixed-kernel obstruction, and does not prove that arbitrary exterior collars can be lifted by adding packets. It only proves that the concrete finite r17 $V9$ chart is liftable under the declared $G_9 = 3$ clock window.*

A.202.4 Relay-chain stitching after r17

Combining the r16 lift of $V6$, the r17 atlas-gap bridge $V6 \to V7$, the previously certified $V7 \to V8$ edge, and the r17 guarded micro-lift of $V9$, the finite chart chain extends to

$$\mathcal{R}_A^{++} : V1 \to V2 \to V3 \to V4 \to V5 \to V6 \to V7 \to V8 \to V9.$$

This is a finite stitched relay chain over the nonzero-index valley-front sample. It does not include $V0$, and it is not a global atlas-cover theorem.

edge	r16 status	r17 status	guard
$V5 \to V6$	stitched after shadow-basin lift	stitched	unchanged r16 lift
$V6 \to V7$	atlas gap	bridge-admissible	source-supported critical br
$V7 \to V8$	stitched with window	stitched with window	inherited
$V8 \to V9$	blocked	stitched after guarded micro-lift	local $V9$ lift only

Corollary A.202.5 (r17 finite chain status)**.** *The longest certified finite relay chain in the r17 atlas is*

$$V1 \to V2 \to V3 \to V4 \to V5 \to V6 \to V7 \to V8 \to V9.$$

The result is a finite relay-chain extension, not a global all-readout closure.

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A.202.5 Guarded residual status after r17

The r17 finite atlas scan improves the relay geometry but does not change the global claim guards. The following remain active:

$$V0 : \mathsf{blocked}, \qquad \beta_4^{r12} = \frac{1}{256} > 0, \qquad C6 : \mathsf{B}_{\mathrm{all/T2}}.$$

The $C5$ completion guard remains unchanged:

$$\forall n < \infty : \ \Phi_5^{(n)} > 0, \qquad \widehat{\Phi}_5(\infty) = 0.$$

Thus r17 should be read as a finite relay upgrade across the local valley-front sample. It is not a proof of RH, not a photon-zero identity, not a T2 proof, and not a global physical closure theorem.

Corollary A.202.6 (permitted r18 direction)**.** *The next admissible source-only pass is*

v4.7.0-r18: $V0$ boundary-front obstruction inventory and finite-chain guard audit.

It should determine whether $V0$ is a true boundary-front obstruction, a missing-source chart, or an inadmissible zero-packet baseline, while auditing that the r17 finite chain is not misread as a global closure.

A.203 v4.7.0-r18 Source-Only Pass: V0 Boundary-Front Obstruction Inventory and Finite-Chain Guard Audit

This source-only pass records the next conservative step after the r17 atlas-gap bridge and guarded micro-lift scan. No PDF rebuild is requested or used. The purpose is not to extend the local relay chain past its declared scope, but to inventory the remaining boundary chart $V0$ and to audit that

$$V1 \to V2 \to V3 \to V4 \to V5 \to V6 \to V7 \to V8 \to V9$$

is read only as a finite local valley-front chain.

A.203.1 V0 boundary-front inventory

Recall the r15/r17 boundary datum

$$V0 : \qquad S_0 = 4, \qquad G_0 = 1, \qquad H_0 = G_0 S_0 = 4 < 16.$$

Equivalently,

$$\Delta_0 = 16 - H_0 = 12, \qquad G_0^{\mathrm{req}} = \frac{16}{S_0} = 4.$$

Thus $V0$ is not close to the relay threshold in the same sense as the former $V6$ blocker. It is not a small shadow-basin miss; it is a boundary-front source problem.

The conservative inventory is:

case	N	κ	G	S	$H = GS$	status
baseline	4	0	1	4	4	blocked
gain-only	4	0	4	4	16	formal only
incoherent flux-only	16	0	1	16	16	formal only
overlap lift	5	3/4	1	20	20	source-guarded

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The gain-only and flux-only rows show that $V0$ is numerically liftable if one allows an undeclared boundary source, an undeclared boundary clock, or an external injection rule. These are not currently part of the declared local relay grammar. Therefore r18 classifies $V0$ as

$$\boxed{V0 : \text{boundary-front source-deficient obstruction.}}$$

This is stronger than saying that $V0$ is simply "blocked", but weaker than claiming that it is an absolute topological obstruction. It means that the known finite chain has no certified upstream source chart at the boundary.

Lemma A.203.1 (V0 boundary-front decision)**.** *Within the current r18 source grammar, $V0$ is not absorbed by the $V1 \to \cdots \to V9$ finite relay chain. It remains a source-deficient boundary-front obstruction. Any lift of $V0$ must declare at least one of the following additional structures:*

boundary source chart, *boundary zero-clock phase window,* *external packet injection rule.*

Proof. The current datum gives $H_0 = 4 < 16$. A formal threshold equality can be forced by taking $G = 4$, or by taking sixteen incoherent packets, or by introducing a coherent overlap lift. None of these is certified by the existing relay chain, because the chain begins at $V1$ and supplies no admissible upstream source for $V0$. Hence the obstruction is not a numerical impossibility; it is a missing-source boundary-front obstruction in the declared grammar. □

A.203.2 Finite-chain guard audit

The r17 chain remains a finite local atlas result:

$$\boxed{\mathcal{R}^{\mathrm{fin}}_{1:9} = V1 \to V2 \to V3 \to V4 \to V5 \to V6 \to V7 \to V8 \to V9.}$$

The r18 guard audit records the active non-closures:

guard	r18 status	interpretation
$V0$	active	boundary-front source deficient
$C4$	$\beta_4 = 1/256 > 0$	route-stable mixed-kernel remains
$C5$	$\forall n < \infty : \Phi_5^{(n)} > 0$	asymptotic collar only
$C6$	$\mathsf{B}_{\mathrm{all/T2}}$	all-readout/T2 blocked
RH	not claimed	photon-clock handoff remains a probe
global	not claimed	no all-readout closure

Consequently, the finite local chain is useful because it converts many formerly abstract valley-front obstructions into relay-admissible handoff steps. It does not remove the boundary source problem at $V0$, and it does not discharge the route-stable $C4$ mixed-kernel residue.

Proposition A.203.2 (r18 finite-chain guard)**.** *The r17 relay chain $\mathcal{R}^{\mathrm{fin}}_{1:9}$ is admissible as a finite local relay chain, but it is not admissible as a global closure witness. The active guards are $V0$, $C4$, finite $C5$ closure, $C6/T2$, RH, and global all-readout closure.*

Proof. The chain starts at $V1$, not at a certified boundary source. It also leaves $\beta_4 = 1/256$ unchanged and does not alter the finite-positive $C5$ guard or the all-readout/T2 blocker. Therefore the finite chain may be used as a local relay atlas but not as a global closure theorem. □

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A.203.3 Permitted r19 direction

The next source-only pass may either keep $V0$ frozen and move back to the $C4$ mixed-kernel seam, or test whether a declared boundary source chart can be added without creating new debt. The safest next pass is therefore

v4.7.0-r19: boundary-source chart admissibility test and C4 guard-preserving audit.

No RH, T2, or global all-readout closure should be inferred from r18.

A.204 v4.7.0-r19 Source-Only Pass: Boundary-Source Chart Admissibility Test and C4 Guard-Preserving Audit

This source-only pass tests the admissibility of a possible boundary-source chart for the remaining boundary-front obstruction $V0$. No PDF rebuild is requested or used. The goal is deliberately narrow: determine whether $V0$ can be lifted by a declared local boundary source without spending the route-stable $C4$ mixed-kernel guard and without opening the all-readout/T2 selector.

The r18 decision was

$$V0 : \text{boundary-front source-deficient obstruction}, \qquad S_0 = 4, \quad G_0 = 1, \quad H_0 = 4 < 16.$$

Thus the missing amount is

$$\Delta_0 = 12, \qquad G_0^{\mathrm{req}} = 4.$$

The present pass asks whether this deficit can be supplied by a legitimate boundary chart rather than by an undeclared global injection.

A.204.1 Boundary-source admissibility criterion

For a boundary-source candidate $\mathcal{B}_0$, write the local overlap score as

$$S(\mathcal{B}_0) = N + \kappa N(N-1), \qquad H(\mathcal{B}_0) = G\, S(\mathcal{B}_0).$$

Here N is the packet count in the boundary window, κ is the local coherence, and G is any declared photon-clock or boundary timing gain. The threshold remains

$$H(\mathcal{B}_0) \geq 16.$$

This numerical threshold is necessary but not sufficient. The chart is admissible only if the following guard conditions also hold:

(A1) $H(\mathcal{B}_0) \geq 16$,
(A2) $D_\partial(\mathcal{B}_0) = 0$ (no undeclared boundary debt),
(A3) $P_{\partial 0} S_{4,\mu}^{\mathrm{mixker}} = 0$ (no spending of the $C4$ mixed kernel),
(A4) $\mathcal{B}_0$ does not introduce an all-readout selector or a $T2$ lift.

Condition (A3) is the guard-preserving clause. The boundary chart may supply a local upstream source for $V0$, but it may not absorb or relabel the persistent mixed-kernel residue

$$\beta_4 = \frac{1}{256}.$$

Definition A.204.1 (r19 boundary-source admissibility)**.** *A boundary-source chart $\mathcal{B}_0$ is r19-admissible if and only if it satisfies (A1)–(A4). A candidate satisfying only (A1) is called numerically lifting but grammar-inadmissible.*

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A.204.2 Finite boundary-source candidate scan

The finite candidate scan is:

candidate	N	κ	G	S	$H = GS$	r19 decision
baseline	4	0	1	4	4	blocked
clock-only	4	0	4	4	16	numerical only; undeclared clock debt
external injection	16	0	1	16	16	numerical only; injection debt
coherent boundary reservoir	5	3/4	1	20	20	conditionally admissible
clock-assisted overlap	4	3/4	16/13	13	16	conditionally admissible

The baseline fails the threshold. The clock-only and external-injection rows show that $V0$ can be forced to the threshold numerically, but only by adding an undeclared source of gain or packets. Those rows remain inadmissible in the current grammar.

The first genuinely useful row is the coherent boundary reservoir. For $N = 5$, the threshold condition is

$$5 + 20\kappa \geq 16, \qquad \kappa \geq \frac{11}{20}.$$

Thus the displayed value $\kappa = 3/4$ has positive margin

$$S - 16 = 4.$$

This proves only a conditional admissibility statement: if a local boundary reservoir $\mathcal{B}_0^{\mathrm{res}}$ with

$$N = 5, \qquad \kappa \geq \frac{11}{20}, \qquad G = 1$$

is declared and if it satisfies the no-new-debt and $C4$-orthogonality guards, then $V0$ is relay-admissible as a boundary-source chart.

Lemma A.204.2 (minimal coherent reservoir threshold)**.** *For a boundary-reservoir candidate with $G = 1$ and $N = 5$, the local relay threshold is reached exactly when*

$$\kappa \geq \frac{11}{20}.$$

In particular, $\kappa = 3/4$ gives $S = 20 > 16$.

Proof. Substituting $N = 5$ into $S = N + \kappa N(N-1)$ gives $S = 5 + 20\kappa$. The threshold $S \geq 16$ is therefore equivalent to $20\kappa \geq 11$, hence $\kappa \geq 11/20$. At $\kappa = 3/4$, one obtains $S = 5 + 15 = 20$. □

A.204.3 C4 guard-preserving audit

The boundary-source scan does not spend the $C4$ mixed-kernel residue. The r12–r18 persistent residue remains

$$S_{4,\mu}^{\mathrm{mixker}} : \mathsf{route\text{-}stable\ persistent}, \qquad \beta_4 = \frac{1}{256}.$$

The guard audit is:

candidate	threshold	$C4$ guard	decision
clock-only	met	not certified	reject unless clock source declared
external injection	met	not certified	reject as external debt
coherent boundary reservoir	met	$P_{\partial 0} S_{4,\mu}^{\mathrm{mixker}} = 0$	conditional pass
clock-assisted overlap	met	requires clock-source audit	conditional only

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Accordingly, r19 makes no unconditional reduction of β_4:

$$\boxed{\beta_4^{r19} = \frac{1}{256}.}$$

The only admissible direction is to treat $\mathcal{B}_0^{\mathrm{res}}$ as a new local boundary-source object whose support is orthogonal to the mixed-kernel seam. If that orthogonality is not declared, the candidate cannot be used.

Proposition A.204.3 (r19 guard-preserving outcome)**.** *The r19 scan finds no unconditional current-grammar lift of $V0$. It does, however, isolate a conditional boundary-source normal form:*

$$\mathcal{B}_0^{\mathrm{res}}: \qquad N = 5, \qquad \kappa \geq \frac{11}{20}, \qquad G = 1, \qquad P_{\partial 0} S_{4,\mu}^{\mathrm{mixker}} = 0.$$

Under these conditions $V0$ is boundary-source relay-admissible and the $C4$ guard remains unchanged at $\beta_4 = 1/256$. Without a declared $\mathcal{B}_0^{\mathrm{res}}$, $V0$ remains guarded.

Proof. The numerical threshold follows from Lemma A.204.2. The no-new-debt requirement is exactly the statement that the reservoir is a local boundary-source chart rather than an external injection. The displayed orthogonality condition prevents the new boundary source from absorbing the route-stable $C4$ mixed-kernel residue. Thus the result is conditional and guard-preserving, not an unconditional closure. □

A.204.4 Residual status after r19

After r19 the finite relay-chain statement is unchanged:

$$V1 \to V2 \to V3 \to V4 \to V5 \to V6 \to V7 \to V8 \to V9$$

remains the certified finite local valley-front chain. The boundary chart $V0$ has a conditional lift criterion, but it is not yet part of the certified chain.

The guarded status is therefore:

object	r19 status
$V0$	conditional boundary-reservoir criterion only
$C4$	$\beta_4 = 1/256$ route-stable mixed-kernel guard preserved
$C5$	asymptotic bounded collar; no finite closure
$C6/T2$	all-readout blocker unchanged
RH	no claim; photon-clock remains handoff probe
global closure	not claimed

A.204.5 Permitted r20 direction

The next source-only pass may test whether the conditional object $\mathcal{B}_0^{\mathrm{res}}$ can be promoted from an admissibility criterion to a declared local boundary-source chart. The safe formulation is

v4.7.0-r20: boundary-reservoir declaration test and no-new-debt proof obligation.

This pass should either certify the no-new-debt and $C4$-orthogonality clauses for $\mathcal{B}_0^{\mathrm{res}}$, or keep $V0$ frozen as a guarded boundary source deficit. No RH, T2, finite $C5$ closure, or global all-readout closure is licensed by r19.

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A.205 v4.7.0-r20 Source-Only Boundary-Reservoir Declaration Test

This source-only pass performs the next admissible test after the r19 boundary-source scan. The question is not whether the old grammar already contains a free lift of $V0$. r19 showed that it does not. The question is whether the conditional boundary-source normal form isolated in r19 may be introduced as a declared local boundary reservoir without creating new debt. No PDF is produced in this intermediate pass.

A.205.1 Declared local boundary-reservoir object

We define the boundary-reservoir extension of the local finite valley-front grammar by adjoining one explicitly declared boundary chart:

$$\mathcal{G}_{\mathrm{loc}}^{r20,\partial mres} := \mathcal{G}_{\mathrm{loc}}^{r19} \cup \{\mathcal{B}_0^{\mathrm{res}}\}.$$

The new object is not an external packet injection and not a global readout channel. It is a local boundary reservoir with declared data

$$\mathcal{B}_0^{\mathrm{res}} = \left(V0, N = 5, \kappa = \frac{3}{4}, G = 1, \delta_\partial = 0, P_{\partial 0} S_{4,\mu}^{\mathrm{mixker}} = 0, \chi_{\mathrm{all}} = 0\right).$$

The associated local overlap score is

$$S_0^{\partial mres} = N + \kappa N(N-1) = 5 + \frac{3}{4} \cdot 5 \cdot 4 = 20.$$

Since $G = 1$, the relay score is

$$H_0^{\partial mres} = 20 > 16.$$

Thus the reservoir has relay margin

$$\mu_0^{\partial mres} = 20 - 16 = 4.$$

Definition A.205.1 (r20 boundary-reservoir declaration)**.** *The object $\mathcal{B}_0^{\mathrm{res}}$ is called a declared local boundary-reservoir chart if it satisfies the following four clauses:*

$$\begin{aligned} &\text{(D1) } \delta_\partial = 0, \\ &\text{(D2) } P_{\partial 0} S_{4,\mu}^{\mathrm{mixker}} = 0, \\ &\text{(D3) } \chi_{\mathrm{all}} = 0, \\ &\text{(D4) } H_0^{\partial \mathrm{res}} \geq 16. \end{aligned}$$

Clause (D1) forbids external boundary debt. Clause (D2) forbids spending the route-stable $C4$ mixed-kernel residue. Clause (D3) forbids all-readout or $T2$ promotion. Clause (D4) is the local relay threshold.

A.205.2 No-new-debt proof obligation

The no-new-debt vector for the declaration is

$$\mathbf{d}_{\mathrm{new}}^{\partial mres} = \left(\delta_\partial, \delta_{C4}, \delta_{C6/T2}, \delta_{\mathrm{all}}\right).$$

For the declared reservoir we set

$$\delta_\partial = 0, \qquad \delta_{C4} = \left\| P_{\partial 0} S_{4,\mu}^{\mathrm{mixker}} \right\| = 0, \qquad \delta_{C6/T2} = 0, \qquad \delta_{\mathrm{all}} = 0.$$

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Therefore

$$\boxed{\mathbf{d}_{\mathrm{new}}^{\partial mres} = (0,0,0,0).}$$

This proves only that the explicitly declared local reservoir is debt-neutral with respect to the listed guards. It does not prove that such a reservoir was already present in the old r19 grammar.

Proposition A.205.2 (r20 no-new-debt declaration pass)**.** *In the extended local grammar* $\mathcal{G}_{\mathrm{loc}}^{r20,\partial\mathrm{res}}$*, the declared object* $\mathcal{B}_0^{\mathrm{res}}$ *is relay-admissible for* $V0$ *and introduces no new boundary,* $C4$*,* $C6/T2$*, or all-readout debt.*

Proof. The relay inequality follows from

$$H_0^{\partial\mathrm{res}} = 20 > 16.$$

The no-new-debt claim follows directly from the four declaration clauses in Definition A.205.1. Clause (D1) sets boundary debt to zero. Clause (D2) makes the new boundary support orthogonal to the route-stable mixed-kernel seam, so β_4 is not spent. Clause (D3) prevents promotion into all-readout or $T2$ territory. Clause (D4) supplies the local relay inequality. Hence the extension is conservative at the level stated here. □

A.205.3 Boundary-extended finite relay chain

With the declared reservoir, the finite local chain may be written as

$$\boxed{V0^{\partial\mathrm{res}} \to V1 \to V2 \to V3 \to V4 \to V5 \to V6 \to V7 \to V8 \to V9.}$$

The superscript on $V0$ is essential. It records that the first chart is not inherited as a free current-grammar chart. It is admitted only in the boundary-reservoir extension.

The active guard ledger after r20 is therefore

object	r20 status	guard
$V0$	declared local boundary-reservoir relay-admissible	local only
$C4$	$\beta_4 = 1/256$ route-stable mixed-kernel	unchanged
$C5$	asymptotic bounded collar	no finite closure
$C6/T2$	all-readout blocker	unchanged
RH	photon-clock handoff probe only	no proof claim
global closure	not claimed	guarded

Corollary A.205.3 (r20 boundary-extended local coverage)**.** *The r20 extension certifies a boundary-extended finite local valley-front chain from* $V0$ *through* $V9$*. This is not a global atlas theorem and not an all-readout closure theorem.*

Proof. The chart $V0$ is relay-admissible by Proposition A.205.2. The charts $V1$ through $V9$ are the finite relay chain certified by the r14–r17 source-only passes and guard-audited in r18. The r20 declaration changes only the local boundary entry chart. It does not remove the $C4$, $C5$, $C6/T2$, RH, or global all-readout guards. □

A.205.4 Permitted r21 direction

The next source-only pass may test whether the boundary-extended finite chain has any hidden seam coupling back into $C4$, or whether the boundary reservoir can be treated as a separate local source chart in the two-sided tensor grammar. The safe formulation is

$$\boxed{\text{v4.7.0-r21: boundary-extended chain seam audit and two-sided source typing.}}$$

No finite $C5$ closure, $C6/T2$ lift, RH claim, or global all-readout closure is licensed by r20.

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A.206 v4.7.0-r21: Boundary-Extended Chain Seam Audit and Two-Sided Source Typing

Status. This is a source-only internal pass. It audits the r20 boundary-reservoir extension for hidden seam coupling and assigns a conservative two-sided source type to the boundary chart. It does not introduce a new DOI, does not rebuild a PDF, and does not claim finite $C5$ closure, $C6/T2$ promotion, RH, or global all-readout closure.

A.206.1 Seam-audit object

The r20 boundary extension produced the local finite chain

$$V0^{\partial\mathrm{res}} \to V1 \to V2 \to V3 \to V4 \to V5 \to V6 \to V7 \to V8 \to V9.$$

The r21 question is whether the new boundary entry creates a hidden seam back into the route-stable $C4$ mixed kernel, the $C6/T2$ all-readout guard, or an undeclared global closure channel.

We record the seam-audit vector

$$\Sigma_{\partial mres} = \left(\Sigma_{\partial C4}, \Sigma_{\partial C6/T2}, \Sigma_{\partial mall}, \Sigma_{\partial 1}\right),$$

where

$$\Sigma_{\partial C4} = \left\| P_{\partial 0} S^{\mathrm{mixker}}_{4,\mu} \right\|, \qquad \Sigma_{\partial C6/T2} = \chi_{\mathrm{all}}, \qquad \Sigma_{\partial mall} = \delta_{\mathrm{all}},$$

and $\Sigma_{\partial 1}$ is the boundary-to-first-interior relay seam. The r20 declaration clauses give

$$\Sigma_{\partial C4} = 0, \qquad \Sigma_{\partial C6/T2} = 0, \qquad \Sigma_{\partial mall} = 0.$$

For the first relay seam, r20 supplies

$$H_0^{\partial\mathrm{res}} = 20, \qquad 16 \le H_0^{\partial\mathrm{res}}, \qquad \mu_0^{\partial\mathrm{res}} = 4,$$

so the boundary exit is locally relay-admissible. Hence

$$\boxed{\Sigma_{\partial\mathrm{res}} = (0,0,0,0).}$$

This vanishing seam vector means only that the declared local boundary source adds no new seam debt. It does not erase the existing route-stable mixed-kernel residue

$$\boxed{\beta_4 = \frac{1}{256}.}$$

Lemma A.206.1 (r21 boundary seam audit)**.** *In the declared local grammar $\mathcal{G}^{r20,\partial\mathrm{res}}_{\mathrm{loc}}$, the boundary entry chart $V0^{\partial\mathrm{res}}$ introduces no hidden seam coupling into $C4$, $C6/T2$, or all-readout closure.*

Proof. The $C4$ seam vanishes by the explicit orthogonality clause $P_{\partial 0} S^{\mathrm{mixker}}_{4,\mu} = 0$. The $C6/T2$ and all-readout seams vanish by $\chi_{\mathrm{all}} = 0$ and $\delta_{\mathrm{all}} = 0$. The first relay seam is admitted by the local inequality $H_0^{\partial\mathrm{res}} = 20 \ge 16$. These four facts give $\Sigma_{\partial\mathrm{res}} = (0,0,0,0)$. The statement is local and does not change β_4. □

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A.206.2 Two-sided source typing

The boundary reservoir is now typed in the two-sided structural tensor grammar. We write schematically

$$ {}^{\partial msrc}{}_{\mathrm{loc}}\mathcal{B}_{0\ \mathrm{relay}}^{\mathrm{res}V0}. $$

The front indices ${}^{\partial msrc}{}_{\mathrm{loc}}$ mark a left-sided, readout-facing local boundary source interface. The back indices ${}^{V0}_{\mathrm{relay}}$ mark the right-sided valley-front relay structure. The notation is not a new contraction rule. It records that the object is a local source chart whose output may feed the finite relay chain, while its boundary source face is not an all-readout face.

The admissible readout is therefore the ordinary left-to-right local relay readout

$$ \text{boundary reservoir} \longmapsto V0^{\partial\mathrm{res}} \longmapsto V1. $$

Same-variance comparison against the $C4$ mixed kernel is allowed only through an explicit comparison/orthogonality test,

$$ G_{\partial C4}\,\mathcal{B}_0^{\partial}\,S_4^{\mathrm{mixker}} = 0, $$

which is the typed version of $P_{\partial 0}S_{4,\mu}^{\mathrm{mixker}} = 0$. Thus the boundary source is not permitted to spend, cancel, or re-label the persistent $C4$ mixed-kernel residue.

Definition A.206.2 (r21 two-sided boundary-source type)**.** *The declared object* ${}^{\partial msrc}{}_{\mathrm{loc}}\mathcal{B}_{0\ \mathrm{relay}}^{\mathrm{res}V0}$ *is a two-sided local boundary-source chart if it satisfies:*

$$ \begin{aligned} &m(T1)\ \textit{left/front type} = \partial\mathrm{src/loc},\\ &m(T2)\ \textit{right/back type} = V0/\mathrm{relay},\\ &m(T3)\ G_{\partial C4}\mathcal{B}_0^{\partial}S_4^{\mathrm{mixker}} = 0,\\ &m(T4)\ \chi_{\mathrm{all}} = 0,\\ &m(T5)\ H_0^{\partial\mathrm{res}} \geq 16. \end{aligned} $$

The type is local-source/relay. It is not global-source, not all-readout, and not a $T2$ bridge.

A.206.3 Completion-interface audit

The r12–r20 passes leave $C5$ as an asymptotic bounded-collar object,

$$ \Phi_5^{(n)} = \left(\frac{3}{4}\right)^n, \qquad \widehat{\Phi}_5(\infty) = 0, \qquad \Phi_5^{(n)} > 0 \quad (n < \infty). $$

The r20 boundary reservoir does not alter this completion interface. In r21 we therefore record the guard

$$ \boxed{V0^{\partial\mathrm{res}} \text{ enters the finite chain, but it does not change the } C5 \text{ completion topology.}} $$

Equivalently, the boundary source is a local chain-entry chart, not a new limit operator on $\mathbb{N}_\infty$.

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A.206.4 r21 decision ledger

The r21 decision is

object	r21 decision	guard
$V0^{\partial\mathrm{res}}$	two-sided local source/relay typed	local only
$\Sigma_{\partial\mathrm{res}}$	$(0,0,0,0)$	no new seam debt
$C4$	$\beta_4 = 1/256$	route-stable persistent
$C5$	asymptotic bounded collar	no finite closure
$C6/T2$	blocked	all-readout guard
RH	photon-clock handoff probe only	no proof claim
global closure	not claimed	guarded

Proposition A.206.3 (r21 typed local-chain preservation)**.** *The r20 boundary-extended finite chain remains valid after the r21 seam audit and two-sided source typing, and no additional $C4$, $C5$, $C6/T2$, RH, or global all-readout claim is licensed.*

Proof. Lemma A.206.1 gives the zero seam-debt vector for the boundary entry. Definition A.206.2 assigns the boundary reservoir a local two-sided source/relay type and forbids both $C4$-spending and all-readout promotion. The internal chain $V1 \to \cdots \to V9$ is unchanged from the r14–r17 finite relay construction. Thus the finite boundary-extended chain is preserved locally, with all guards listed in the decision ledger still active. □

A.206.5 Permitted r22 direction

The next source-only pass may test whether the typed boundary source can be compiled into a small boundary-source calculus or whether it must remain a single declared chart. The safe formulation is

v4.7.0-r22: boundary-source calculus extraction and typed relay closure guard.

This r22 direction is still local and source-only. It may not promote the finite chain to global all-readout closure and may not identify the photon-clock probe with RH.

A.207 v4.7.0-r22 source-only: boundary-source calculus extraction and typed relay closure guard

This source-only pass extracts a small local boundary-source calculus from the single r20–r21 boundary-reservoir chart and immediately guards the extraction against accidental promotion to global closure. The pass does not rebuild a PDF, does not assign a DOI, and does not modify the public release status.

The r21 typed reservoir is kept as

$$ {}^{\partial\mathrm{src}}_{\mathrm{loc}}\mathcal{B}_0^{\mathrm{res}\,V0}_{\mathrm{relay}}. $$

The r22 question is whether this typed object may be used as a reusable local calculus rule rather than as an isolated chart. The answer is yes, but only as a local boundary-source rule with a typed relay guard.

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A.207.1 Local boundary-source calculus rule

Definition A.207.1 (r22 local boundary-source calculus rule)**.** *Let $\mathcal{B}$ be a boundary reservoir candidate. It is admissible as a local boundary-source calculus instance, written*

$$\mathcal{B} \in \mathsf{BSC}_{\partial}^{\mathrm{loc}},$$

if the following five clauses hold:

BSC1 $\mathcal{B}$ *has two-sided type* ${}^{\partial\mathrm{src}}{}_{\mathrm{loc}}\mathcal{B}^{V}_{\mathrm{relay}}$;

BSC2 $H_{\mathcal{B}} = G_{\mathcal{B}} S_{\mathcal{B}} \geq 16$;

BSC3 $P_{\partial}\, S^{\mathrm{mixker}}_{4,\mu} = 0$;

BSC4 $\chi_{\mathrm{all}}(\mathcal{B}) = 0$;

BSC5 $\mathbf{d}_{\mathrm{new}}(\mathcal{B}) = (0,0,0,0)$.

The rule is therefore a local source-and-relay rule. It is not an all-readout rule and it is not a global boundary theorem.

For the concrete r20/r21 reservoir one has

$$N = 5, \qquad \kappa = \frac{3}{4}, \qquad G = 1,$$

so

$$S = 5 + \frac{3}{4} \cdot 5 \cdot 4 = 20, \qquad H = 20 > 16.$$

Thus

$$\boxed{\mathcal{B}_0^{\mathrm{res}} \in \mathsf{BSC}_{\partial}^{\mathrm{loc}}.}$$

Lemma A.207.2 (r22 boundary-source calculus extraction)**.** *The r20–r21 boundary-reservoir chart can be extracted as a reusable local boundary-source calculus instance in $\mathsf{BSC}_{\partial}^{\mathrm{loc}}$.*

Proof. BSC1 is the r21 two-sided source typing. BSC2 follows from the r20 threshold calculation $H = 20 > 16$. BSC3 and BSC5 are the r20–r21 no-new-debt and seam-audit conditions. BSC4 is the explicit local-only readout guard. Hence the reservoir satisfies all five clauses of Definition A.207.1. □

A.207.2 Typed relay closure guard

The word "closure" is used here only in the typed finite-relay sense. It means that a declared local boundary source may be attached to the beginning of the already certified finite relay chain. It does not mean global closure.

Definition A.207.3 (r22 typed finite-relay closure)**.** *The typed finite-relay closure generated by $\mathsf{BSC}_{\partial}^{\mathrm{loc}}$ is the local operation*

$$\mathrm{Cl}^{\mathrm{loc,typed}}_{\mathrm{relay}}\big(\mathcal{B}_0^{\mathrm{res}}; V1, \ldots, V9\big) = \big[V0^{\partial\mathrm{res}} \to V1 \to \cdots \to V9\big]$$

provided the guard vector

$$\Gamma^{r22}_{\mathrm{guard}} = (\gamma_{C4}, \gamma^{\mathrm{fin}}_{C5}, \gamma_{C6/T2}, \gamma_{\mathrm{all}}, \gamma_{\mathrm{RH}})$$

remains

$$\Gamma^{r22}_{\mathrm{guard}} = (1,1,1,1,1).$$

Here 1 *means that the corresponding guard is still active, not discharged.*

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For r22 the guard ledger is

$$\boxed{\Gamma^{r22}_{\mathrm{guard}} = (1,1,1,1,1).}$$

In particular,

$$\beta_4^{r22} = \frac{1}{256}, \qquad \forall n < \infty : \ \Phi_5^{(n)} > 0, \qquad C6 : \mathsf{B}_{\mathrm{all/T2}}.$$

Proposition A.207.4 (r22 typed relay closure guard)**.** *The boundary-source calculus extraction licenses the typed local finite relay chain*

$$V0^{\partial \mathrm{res}} \to V1 \to \cdots \to V9,$$

but it does not license $C4$-closure, finite $C5$-closure, $C6/T2$, RH, or global all-readout closure.

Proof. By Lemma A.207.2, the boundary reservoir is a valid local calculus instance. Definition A.207.3 then permits the local finite relay attachment. The same definition also requires the guard vector to remain active. Since $\beta_4 = 1/256$ is preserved, finite $C5$-closure is excluded by the finite positivity of $\Phi_5^{(n)}$, $C6/T2$ remains an all-readout blocker, and the photon-clock construction is a handoff probe only, none of the guarded global claims is discharged. □

A.207.3 r22 calculus ledger

The extracted calculus can be summarized as

clause	r22 value	decision
two-sided type	${}^{\partial\mathrm{src}}{}_{\mathrm{loc}}\mathcal{B}_0^{\mathrm{res}V0}{}_{\mathrm{relay}}$	admissible
H	$20 > 16$	relay pass
$P_{\partial 0} S^{\mathrm{mixker}}_{4,\mu}$	0	C4 guard preserved
χ_{all}	0	not all-readout
$\mathbf{d}_{\mathrm{new}}$	$(0,0,0,0)$	no new debt
$\Gamma^{r22}_{\mathrm{guard}}$	$(1,1,1,1,1)$	guards active

A.207.4 Permitted r23 direction

The next source-only pass may test whether the local boundary-source calculus is stable under finite perturbations of N, κ, G and whether the typed relay guard remains invariant under such perturbations. The safe formulation is

v4.7.0-r23: boundary-source stability window and relay-guard robustness scan.

This may not be read as global all-readout closure, a finite $C5$ closure, an RH statement, or a $T2$ bridge.

A.208 v4.7.0-r23 Boundary-Source Stability Window and Relay-Guard Robustness Scan

This source-only pass tests whether the local boundary-source calculus extracted in r22 is stable under small finite perturbations of the reservoir parameters (N, κ, G). The pass is deliberately local. It does not create a new DOI, does not rebuild the PDF, and does not change any all-readout or global-closure claim.

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A.208.1 Reservoir response function

For a local boundary-source reservoir we keep the r13–r22 overlap response function

$$S(N,\kappa) = N + \kappa N(N-1), \qquad H(N,\kappa,G) = G\,S(N,\kappa).$$

The relay threshold is

$$H(N,\kappa,G) \geq 16.$$

The r22 anchor is

$$(N,\kappa,G) = \left(5, \frac{3}{4}, 1\right), \qquad S = 20, \qquad H = 20, \qquad \mu = H - 16 = 4.$$

Definition A.208.1 (r23 boundary-source stability window)**.** *The weak r23 stability window is the set of local reservoir triples*

$$\mathcal{W}_{\partial}^{r23} = \{(N,\kappa,G) : G(N + \kappa N(N-1)) \geq 16\}$$

subject to the r22 source-typing and guard clauses BSC1–BSC5. The strict stability window is

$$\mathcal{W}_{\partial,+}^{r23} = \{(N,\kappa,G) : G(N + \kappa N(N-1)) > 16\}$$

with the same typing and guard clauses.

For the fixed local source size $N = 5$, the threshold surfaces are explicit:

$$G = 1 \quad \Longrightarrow \quad \kappa \geq \frac{11}{20},$$

while

$$\kappa = \frac{3}{4} \quad \Longrightarrow \quad G \geq \frac{4}{5}.$$

A one-unit strict margin is already guaranteed by

$$G = 1, \kappa \geq \frac{3}{5},$$

or by

$$\kappa = \frac{3}{4}, G \geq \frac{17}{20}.$$

A simultaneous finite robustness rectangle around the r22 anchor is

$$\boxed{N = 5, \qquad \kappa \geq \frac{7}{10}, \qquad G \geq \frac{9}{10},}$$

because then

$$S \geq 5 + \frac{7}{10} \cdot 5 \cdot 4 = 19, \qquad H \geq \frac{9}{10} \cdot 19 = \frac{171}{10} > 16.$$

Thus the r22 reservoir is not a knife-edge construction: it has a finite local stability collar.

Lemma A.208.2 (r23 local stability collar)**.** *The r22 boundary-reservoir point* $(5, 3/4, 1)$ *lies in the strict interior of the local boundary-source stability window. In particular, the rectangle*

$$N = 5, \qquad \kappa \in \left[\frac{7}{10}, 1\right], \qquad G \in \left[\frac{9}{10}, 1\right]$$

remains strictly relay-admissible.

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Proof. For every triple in the displayed rectangle,

$$S = 5 + 20\kappa \geq 5 + 20 \cdot \frac{7}{10} = 19,$$

and therefore

$$H = GS \geq \frac{9}{10} \cdot 19 = \frac{171}{10} > 16.$$

The r22 source-typing and no-new-debt clauses are not altered by this finite parameter perturbation, so the relay inequality remains valid in the same local calculus. □

A.208.2 Finite perturbation scan

The finite scan checked the grid

$$N \in \{4, 5, 6\}, \qquad \kappa \in \left\{\frac{1}{2}, \frac{11}{20}, \frac{3}{5}, \frac{2}{3}, \frac{7}{10}, \frac{3}{4}, 1\right\}, \qquad G \in \left\{\frac{4}{5}, \frac{17}{20}, \frac{9}{10}, 1\right\}.$$

The scan is not used as a proof by enumeration. It is a stability diagnostic around the symbolic threshold surfaces above. Its useful outcomes are:

case	N	κ	G	decision
r22 anchor	5	3/4	1	$H = 20 > 16$
lower-κ strict buffer	5	3/5	1	$H = 17 > 16$
critical κ surface	5	11/20	1	$H = 16$
gain-damped strict point	5	3/4	17/20	$H = 17 > 16$
gain-damped critical point	5	3/4	4/5	$H = 16$
simultaneous robust corner	5	7/10	9/10	$H = 171/10 > 16$
nominal $N = 4$	4	3/4	1	$H = 13 < 16$
larger low-κ reservoir	6	1/2	1	$H = 21 > 16$

The failure of the nominal $N = 4$ case is useful: the local boundary-source rule is not vacuous. It still requires enough local reservoir capacity.

A.208.3 Relay-guard robustness

The r23 stability collar is guarded by exactly the same guard vector as r22:

$$\boxed{\Gamma^{r23}_{\text{guard}} = (1, 1, 1, 1, 1).}$$

Concretely,

$$\beta_4^{r23} = \frac{1}{256}, \qquad \forall n < \infty : \ \Phi_5^{(n)} > 0, \qquad C6 : \mathsf{B}_{\text{all/T2}}.$$

The photon-clock construction remains a handoff probe, not an RH statement. The boundary-source stability collar remains local and typed, not all-readout.

Proposition A.208.3 (r23 relay-guard robustness)**.** *Every triple in the strict local r23 stability collar preserves the r22 guard vector. Hence the local boundary-source robustness scan does not discharge $C4$, finite $C5$-closure, $C6/T2$, RH, or global all-readout closure.*

Proof. The perturbation scan changes only the local relay parameters (N, κ, G) inside the already declared two-sided boundary-source type. The guard clauses BSC3–BSC5 remain unchanged: the boundary source stays orthogonal to the route-stable mixed kernel, the all-readout selector remains off, and the new-debt vector remains zero. Therefore the relay inequality can be stable without spending $\beta_4 = 1/256$, without converting the asymptotic $C5$ collar into a finite closure, and without opening the all-readout $C6/T2$ gate. □

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A.208.4 r23 decision ledger

The source-only r23 decision is

object	r23 result	guard
$\mathcal{B}_0^{\mathrm{res}}$	strictly stable near $(5, 3/4, 1)$	local only
$\mathcal{W}_{\partial,+}^{r23}$	nonempty finite stability collar	$\chi_{\mathrm{all}} = 0$
$C4$	$\beta_4 = 1/256$	unspent
$C5$	$\Phi_5^{(n)} > 0$ for finite n	no finite closure
$C6/T2$	$\mathsf{B}_{\mathrm{all/T2}}$	blocked
RH	photon-clock handoff probe only	not a theorem
global all-readout	unclaimed	guard active

A.208.5 Permitted r24 direction

The next source-only step may package the stability collar into a small "theoremlet" and draw the residual frontier map showing exactly where the local boundary-source calculus stops. The safe title is

v4.7.0-r24: boundary-source robustness theoremlet and residual frontier map.

It must remain source-only unless a PDF is explicitly requested.

A.209 v4.7.0-r24 Boundary-Source Robustness Theoremlet and Residual Frontier Map

Source-only status. This r24 layer is a source-only internal working step. It packages the r23 stability scan into a local theoremlet and records the residual frontier of the boundary-source calculus. It assigns no new DOI, builds no PDF, and claims no global closure.

A.209.1 Boundary-source robustness theoremlet

The r23 anchor was

$$(N, \kappa, G) = \left(5, \frac{3}{4}, 1\right), \qquad S = 20, \qquad H = 20, \qquad \mu = 4.$$

The useful r24 point is that this is not merely a witnessed chart. It sits in a small local robustness box.

Theorem A.209.1 (r24 local boundary-source robustness theoremlet)**.** *Let*

$$H(N, \kappa, G) = G\big(N + \kappa N(N-1)\big)$$

be the local boundary-source relay height. In the declared local boundary-source grammar, every chart with

$$N = 5, \qquad \kappa \geq \frac{7}{10}, \qquad G \geq \frac{9}{10}$$

satisfies

$$H \geq \frac{171}{10} > 16.$$

Hence the r22/r23 boundary-source calculus has a strict local robustness collar around the anchor $(5, 3/4, 1)$.

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Proof. For $N = 5$,

$$S = 5 + 20\kappa.$$

If $\kappa \geq 7/10$, then

$$S \geq 5 + 20 \cdot \frac{7}{10} = 19.$$

If also $G \geq 9/10$, then

$$H = GS \geq \frac{9}{10} \cdot 19 = \frac{171}{10} > 16.$$

The proof uses only the already declared local boundary-source type. It does not switch on the all-readout selector and does not couple the boundary source to the route-stable mixed kernel. □

Corollary A.209.2 (r24 strict local relay pass)**.** *Every chart in the robustness box of Theorem A.209.1 is relay-admissible with positive margin. In particular, the boundary-source lift is not a knife-edge equality at $H = 16$.*

A.209.2 Threshold surfaces and finite frontier

The two one-parameter threshold surfaces around the r23 anchor are

$$N = 5, \quad G = 1 \quad \Longrightarrow \quad \kappa \geq \frac{11}{20},$$

and

$$N = 5, \quad \kappa = \frac{3}{4} \quad \Longrightarrow \quad G \geq \frac{4}{5}.$$

The strict one-unit buffer surfaces are

$$N = 5, \quad G = 1 \quad \Longrightarrow \quad \kappa \geq \frac{3}{5},$$

and

$$N = 5, \quad \kappa = \frac{3}{4} \quad \Longrightarrow \quad G \geq \frac{17}{20}.$$

For a wider residual-frontier map, r24 records the finite grid

$$N \in \{3, 4, 5, 6, 7\},$$

$$\kappa \in \left\{ \frac{1}{2}, \frac{11}{20}, \frac{3}{5}, \frac{2}{3}, \frac{7}{10}, \frac{3}{4}, 1 \right\},$$

$$G \in \left\{ \frac{4}{5}, \frac{17}{20}, \frac{9}{10}, 1 \right\}.$$

The grid contains 140 cases and partitions as

frontier class	count
strict-interior ($H \geq 17$)	66
thin-pass ($16 \leq H < 17$)	6
frontier-fail ($15 \leq H < 16$)	4
deep-fail ($H < 15$)	64

Thus the boundary-source calculus has a genuine pass region, a thin critical frontier, and a nontrivial fail region. The local rule is therefore not a vacuous route-to-closure device.

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A.209.3 Residual frontier map

The residual frontier is described by

$$G\big(N+\kappa N(N-1)\big)=16.$$

For fixed N and G, the threshold is

$$\kappa_{\min}(N,G)=\frac{16/G-N}{N(N-1)}.$$

For fixed N and κ, the threshold is

$$G_{\min}(N,\kappa)=\frac{16}{N+\kappa N(N-1)}.$$

At the representative slice $G=1$, the threshold moves into the unit coherence interval first at $N=4$:

$$N=4\Rightarrow\kappa_{\min}=1,\qquad N=5\Rightarrow\kappa_{\min}=\frac{11}{20},\qquad N=6\Rightarrow\kappa_{\min}=\frac{1}{3}.$$

This explains why the r23/r24 local reservoir really needs at least the five-packet anchor to obtain a comfortable margin at moderate coherence.

Proposition A.209.3 (r24 residual-frontier decision)**.** *The declared local boundary-source calculus extends the finite valley-front chain only inside the local pass region of*

$$G\big(N+\kappa N(N-1)\big)\geq 16.$$

Outside this region, the frontier remains residual and no relay-admissibility is inferred.

Proof. The calculus rule BSC2 is exactly the local relay inequality $H\geq 16$. If the inequality fails, the boundary source has no declared local relay height. The source typing and no-new-debt clauses do not create missing relay height; they only prevent successful local relay from spending forbidden guards. Therefore the pass/fail frontier is exactly the stated inequality. □

A.209.4 Guard-preserving consequence

The theoremlet is strictly local. Its guard vector remains

$$\boxed{\Gamma^{r24}_{\text{guard}}=(1,1,1,1,1).}$$

Equivalently,

$$\beta_4^{r24}=\frac{1}{256},\qquad \forall n<\infty:\ \Phi_5^{(n)}>0,\qquad C6:\mathsf{B}_{\text{all/T2}}.$$

The Riemann-zero/photon-clock language remains a timing probe for handoff windows, not an RH theorem and not a photon-zero identity. The boundary-source robustness theoremlet is a local relay theoremlet only.

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A.209.5 r24 decision ledger and permitted r25 direction

The r24 decision is

object	r24 result	guard
$\mathcal{B}_0^{\mathrm{res}}$	local robustness theoremlet	local only
$\mathcal{W}_{\partial,+}^{r24}$	finite pass/interior region mapped	$\chi_{\mathrm{all}} = 0$
$\partial\mathcal{W}^{r24}$	residual frontier mapped	no inferred pass outside $H \geq 16$
$C4$	$\beta_4 = 1/256$	unspent
$C5$	$\Phi_5^{(n)} > 0$ for finite n	no finite closure
$C6/T2$	$\mathsf{B}_{\mathrm{all/T2}}$	blocked
RH	photon-clock handoff probe only	not a theorem
global all-readout	unclaimed	guard active

A safe next source-only direction is

v4.7.0-r25: boundary-source frontier minimizer and guard-stable atlas compression.

This would search for the smallest local boundary-source resources that still preserve the guard vector, without turning the local finite chain into a global closure claim.

A.210 v4.7.0-r25 Source-Only Frontier Minimizer and Guard-Stable Atlas Compression

This source-only layer continues the r24 residual-frontier map. No PDF is built in this intermediate step. The goal is to identify the smallest local boundary-source resources visible on the finite frontier and to compress the finite relay atlas without spending any global guard.

A.210.1 Resource order and frontier minimizers

For a local boundary-source triple

$$(N, \kappa, G)$$

we keep the r24 height

$$H(N, \kappa, G) = G\big(N + \kappa N(N-1)\big).$$

A triple is relay-admissible only if

$$H(N, \kappa, G) \geq 16.$$

The source-resource order used in this scan is the componentwise order

$$(N, \kappa, G) \preceq (N', \kappa', G') \quad \Longleftrightarrow \quad N \leq N', \quad \kappa \leq \kappa', \quad G \leq G'.$$

A frontier point is Pareto-minimal if no strictly smaller admissible triple exists in this order.

In the r24 finite grid the absolute pass minimizer is

$$\boxed{(N, \kappa, G) = (4, 1, 1).}$$

It gives

$$H = 1 \cdot (4 + 1 \cdot 4 \cdot 3) = 16.$$

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This is a thin pass with zero margin. It is therefore not a robust boundary reservoir rule; it is only the exact local frontier point.

For the five-packet boundary source the pass thresholds are

$$G = 1 \Rightarrow \kappa \geq \frac{11}{20}, \qquad \kappa = \frac{3}{4} \Rightarrow G \geq \frac{4}{5}.$$

The margin-one stable thresholds are

$$G = 1 \Rightarrow \kappa \geq \frac{3}{5}, \qquad \kappa = \frac{3}{4} \Rightarrow G \geq \frac{17}{20}.$$

The simultaneous robust-box corner remains

$$\boxed{(N, \kappa, G) = \left(5, \frac{7}{10}, \frac{9}{10}\right),}$$

with

$$H = \frac{9}{10}\left(5 + \frac{7}{10} \cdot 5 \cdot 4\right) = \frac{171}{10} > 16.$$

The original r22–r24 anchor

$$\left(5, \frac{3}{4}, 1\right)$$

has

$$H = 20, \qquad \mu = 4.$$

Thus r25 distinguishes three roles:

$$\boxed{(4, 1, 1) = \text{absolute thin frontier},}$$

$$\boxed{\left(5, \frac{7}{10}, \frac{9}{10}\right) = \text{robust local box corner},}$$

$$\boxed{\left(5, \frac{3}{4}, 1\right) = \text{comfortable boundary-source anchor}.}$$

Proposition A.210.1 (r25 frontier minimizer decision)**.** *In the finite r24/r25 frontier grid, the triple* $(4, 1, 1)$ *is the absolute local pass minimizer, but it has zero relay margin. The five-packet boundary-reservoir rule is the first locally robust reservoir family used by this branch, and the box*

$$N = 5, \qquad \kappa \geq \frac{7}{10}, \qquad G \geq \frac{9}{10}$$

is a strict local robustness region.

Proof. The equation

$$G\big(N + \kappa N(N-1)\big) = 16$$

defines the frontier. Direct substitution gives $(4, 1, 1) \mapsto H = 16$, hence pass with zero margin. The same substitution at $(5, 7/10, 9/10)$ gives $H = 171/10 > 16$. Since increasing any of N, κ, G weakly increases H, the displayed box is contained in the strict pass region. The r22–r24 anchor $(5, 3/4, 1)$ lies strictly inside the same region with margin 4. □

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A.210.2 Pareto frontier summary

The finite pass Pareto frontier consists of the thin and low-resource points

N	κ	G	H	status
4	1	1	16	thin pass
5	11/20	1	16	thin pass
5	2/3	9/10	33/2	thin pass
5	7/10	17/20	323/20	thin pass
5	3/4	4/5	16	thin pass
6	1/2	4/5	84/5	thin pass

These points describe where the boundary-source calculus first becomes locally admissible. They are not automatically robust enough to be used as source anchors.

The margin-one Pareto layer begins with

N	κ	G	H	status
5	3/5	1	17	strict interior
5	7/10	9/10	171/10	strict interior
5	3/4	17/20	17	strict interior
5	1	4/5	20	strict interior
6	1/2	17/20	357/20	strict interior
6	11/20	4/5	18	strict interior
7	1/2	4/5	112/5	strict interior

This layer is the better target for guard-stable atlas compression.

A.210.3 Guard-stable atlas compression

The expanded local chain

$$V0^{\partial\mathrm{res}} oV1 \to V2 \to V3 \to V4 \to V5 \to V6 \to V7 \to V8 \to V9$$

is compressed into four local atlas cells:

$$\mathcal{A}_0 = V0^{\partial\mathrm{res}}, \qquad \mathcal{A}_1 = (V1 \to V2 \to V3 \to V4 \to V5),$$
$$\mathcal{A}_2 = (V5 \to V6 \to V7), \qquad \mathcal{A}_3 = (V7 \to V8 \to V9).$$

Here $\mathcal{A}_0$ is the local boundary-reservoir source cell, $\mathcal{A}_1$ is the core valley relay cell, $\mathcal{A}_2$ is the shadow-basin/atlas-gap bridge cell, and $\mathcal{A}_3$ is the zero-clock handoff tail.

The compressed local atlas is therefore

$$\boxed{\mathcal{A}_\partial^{r25} = \mathcal{A}_3 \circ \mathcal{A}_2 \circ \mathcal{A}_1 \circ \mathcal{A}_0.}$$

This notation is only a local atlas compression. It does not turn the finite chain into a global all-readout closure witness.

Lemma A.210.2 (r25 guard-stable compression)**.** *The compressed atlas $\mathcal{A}_\partial^{r25}$ preserves the r24 guard vector*

$$\Gamma_{\mathrm{guard}}^{r25} = (1, 1, 1, 1, 1).$$

Proof. Each compressed cell is inherited from a previously guarded local construction: $\mathcal{A}_0$ satisfies the boundary-source calculus with $\chi_{\mathrm{all}} = 0$, zero new debt, and $P_\partial S_{4,\mu}^{\mathrm{mixker}} = 0$; $\mathcal{A}_1$ is the finite relay core; $\mathcal{A}_2$ uses the r16/r17 source-supported bridge and spends no $C4$ debt; $\mathcal{A}_3$ uses zero-clock windows only as timing probes. None of these cells declares an all-readout interface, an RH theorem, a finite $C5$ closure, or a $C6/T2$ bridge. Therefore the guard vector remains unchanged. □

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A.210.4 r25 decision ledger

The r25 decision is

object	r25 result	guard
$(4,1,1)$	absolute thin frontier minimizer	zero margin
$(5,7/10,9/10)$	robust local box corner	local only
$(5,3/4,1)$	comfortable boundary-source anchor	local only
$\mathcal{A}_\partial^{r25}$	compressed finite local atlas	$\chi_{\mathrm{all}}=0$
$C4$	$\beta_4=1/256$	unspent
$C5$	$\Phi_5^{(n)}>0$ for finite n	no finite closure
$C6/T2$	$\mathsf{B}_{\mathrm{all/T2}}$	blocked
RH	photon-clock handoff probe only	not a theorem
global all-readout	unclaimed	guard active

A safe next source-only direction is

v4.7.0-r26: compressed atlas transition audit and frontier-to-field-equation handoff.

This would test whether the compressed local atlas can feed the MM/HC field-equation program as a typed local input without promoting any guarded object to global closure.

A.211 v4.7.0-r26 compressed atlas transition audit and frontier-to-field-equation handoff

A.211.1 Purpose and scope guard

This r26 source-only layer audits the compressed local atlas

$$\mathcal{A}_\partial^{r25}=\mathcal{A}_3\circ\mathcal{A}_2\circ\mathcal{A}_1\circ\mathcal{A}_0$$

as a possible typed input for the MM/HC field-equation program. It does not claim a new field equation, a global all-readout theorem, a finite $C5$ closure, a $C6/T2$ bridge, or an RH theorem. The purpose is narrower: check whether the local atlas cells can be handed forward as a guarded source object for the next field-equation search layer.

The r26 input objects are

$$\mathcal{A}_0=V0^{\partial\mathrm{res}},\qquad \mathcal{A}_1=(V1\to V2\to V3\to V4\to V5),$$

$$\mathcal{A}_2=(V5\to V6\to V7),\qquad \mathcal{A}_3=(V7\to V8\to V9),$$

together with the active guard vector

$$\Gamma_{\mathrm{guard}}^{r25}=(1,1,1,1,1).$$

The target is a local handoff object, not a global closure object.

A.211.2 Two-sided transition typing

Each compressed atlas cell is typed as a two-sided local interface:

$${}^{\partial\mathrm{src}}{}_{\mathrm{loc}}\mathcal{A}_{0\,\mathrm{relay}}^{V0},\qquad {}^{\mathrm{core}}{}_{\mathrm{loc}}\mathcal{A}_{1\,\mathrm{relay}}^{V5},$$

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$$^{\text{shadow}}{}_{\text{loc}}\mathcal{A}^{V7}_{2\text{bridge}}, \qquad ^{\gamma\text{clock}}{}_{\text{loc}}\mathcal{A}^{V9}_{3\text{tail}}.$$

The front indices record the readout/source side of the cell. The back indices record the structure/relay side. Same-variance interactions remain forbidden unless a declared comparison tensor or selector mediates them.

The transition interfaces are

$$\tau_{01} : \mathcal{A}_0 \rightsquigarrow \mathcal{A}_1, \qquad \tau_{12} : \mathcal{A}_1 \rightsquigarrow \mathcal{A}_2, \qquad \tau_{23} : \mathcal{A}_2 \rightsquigarrow \mathcal{A}_3.$$

Their audit ledger is

transition	source type	target type	r26 status
τ_{01}	∂src/$V0$	core/$V5$	typed local relay
τ_{12}	core/$V5$	shadow/$V7$	source-supported bridge
τ_{23}	shadow/$V7$	γclock/$V9$	timing-gated tail

No transition in this table declares an all-readout channel.

A.211.3 Transition seam audit

For each transition define a seam-debt vector

$$\sigma(\tau) = (\sigma_{C4}, \sigma_{C5}, \sigma_{C6/T2}, \sigma_{\text{all}}).$$

The r26 transition audit returns

$$\boxed{\sigma(\tau_{01}) = (0,0,0,0), \qquad \sigma(\tau_{12}) = (0,0,0,0), \qquad \sigma(\tau_{23}) = (0,0,0,0).}$$

Therefore the compressed atlas transition vector is

$$\boxed{\Sigma^{r26}_{\text{trans}} = \sum_{j=0}^{2} \sigma(\tau_{j,j+1}) = (0,0,0,0).}$$

This is a local transition statement. It says that the compression introduces no new seam debt across the declared local atlas cells. It does not spend the pre-existing route-stable mixed-kernel guard

$$\beta_4 = \frac{1}{256}.$$

Lemma A.211.1 (r26 transition seam preservation)**.** *The compressed atlas transitions $\tau_{01}, \tau_{12}, \tau_{23}$ preserve the r25 guard vector.*

Proof. The transition τ_{01} uses the typed boundary-reservoir source calculus with zero new boundary debt, $P_\partial S^{\text{mixker}}_{4,\mu} = 0$, and $\chi_{\text{all}} = 0$. The transition τ_{12} is the already guarded source-supported bridge from the core relay into the shadow-basin/atlas-gap cell. The transition τ_{23} uses the photon-clock factor only as a local timing gate for the zero-clock tail. None of the three transitions declares an all-readout channel, consumes β_4, closes finite $C5$, or opens $C6/T2$. Hence $\Sigma^{r26}_{\text{trans}} = (0,0,0,0)$, and the guard vector is preserved. □

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A.211.4 Frontier-to-field-equation handoff object

The local atlas is now packaged as a handoff object

$$\boxed{\mathfrak{X}_{\partial}^{r26} = \left(\mathcal{A}_{\partial}^{r25}, \Sigma_{\mathrm{trans}}^{r26}, \mathcal{F}_{\partial}, \Gamma_{\mathrm{guard}}^{r25}\right),}$$

where $\mathcal{F}_{\partial}$ denotes the boundary-source frontier surface

$$\boxed{G\big(N + \kappa N(N-1)\big) = 16.}$$

This object is allowed to feed the MM/HC field-equation program only after projection through the local guard projector

$$\Pi_{\mathrm{loc}}^{\mathrm{guard}} : \mathfrak{X}_{\partial}^{r26} \longmapsto \mathfrak{X}_{\partial,\mathrm{loc}}^{r26}.$$

The projected object is local, typed, and guarded:

$$\boxed{\mathfrak{X}_{\partial,\mathrm{loc}}^{r26} := \Pi_{\mathrm{loc}}^{\mathrm{guard}} \mathfrak{X}_{\partial}^{r26}.}$$

A schematic MM/HC handoff may then be written as

$$\begin{smallmatrix}\mathrm{loc}\,\partial\mathrm{atlas} \\ \mathrm{read}\end{smallmatrix} \mathfrak{E}_{\mathrm{MM/HC}}^{\mathrm{seed}} \begin{smallmatrix}\mathrm{field} \\ \mathrm{constraint}\end{smallmatrix} = \mathsf{H}_{\mathrm{bd}} \circ \mathsf{M}_{\mathrm{bd}} \left(\mathfrak{X}_{\partial,\mathrm{loc}}^{r26}\right).$$

The symbol $\mathfrak{E}_{\mathrm{MM/HC}}^{\mathrm{seed}}$ is deliberately a seed or constraint object. It is not yet a derived physical field equation.

A.211.5 Handoff admissibility clauses

The handoff is admissible only under the following five clauses:

H1 $\mathcal{A}_{\partial}^{r25}$ is typed as a local atlas,
H2 $\Sigma_{\mathrm{trans}}^{r26} = (0,0,0,0)$,
H3 $\mathcal{F}_{\partial} : G(N + \kappa N(N-1)) = 16$ remains recorded,
H4 $\Gamma_{\mathrm{guard}}^{r26} = (1,1,1,1,1)$,
H5 $\mathfrak{E}_{\mathrm{MM/HC}}^{\mathrm{seed}}$ is a local constraint seed, not a closure theorem.

These clauses define the permitted frontier-to-field-equation handoff. They also forbid the common over-reading in which a finite local atlas is mistaken for a global all-readout proof.

Theoremlet A.211.2 (r26 guarded atlas-to-field handoff)**.** *Under H1–H5, the compressed boundary-extended atlas $\mathcal{A}_{\partial}^{r25}$ is admissible as a local typed input seed for the MM/HC field-equation program. The handoff preserves all r25 guards.*

Proof. By H1 the atlas is a local typed object. By H2 no new transition seam debt is introduced. By H3 the frontier surface remains explicit, so the handoff does not hide the boundary-source admissibility conditions. By H4 all closure guards remain active. By H5 the field-equation side receives only a seed/constraint object. Hence the handoff is admissible as a local input to the MM/HC program, but no global closure conclusion follows. □

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A.211.6 r26 decision ledger

The r26 decision ledger is

object	r26 result	guard
$\mathcal{A}_\partial^{r25}$	typed compressed local atlas	local only
$\tau_{01}, \tau_{12}, \tau_{23}$	transition-audited	$\Sigma_{\mathrm{trans}} = 0$
$\mathfrak{X}_\partial^{r26}$	frontier handoff object	$\Pi_{\mathrm{loc}}^{\mathrm{guard}}$
$\mathfrak{E}_{\mathrm{MM/HC}}^{\mathrm{seed}}$	local constraint seed	not a field equation
$C4$	$\beta_4 = 1/256$	unspent
$C5$	$\Phi_5^{(n)} > 0$ for finite n	no finite closure
$C6/T2$	$\mathsf{B}_{\mathrm{all/T2}}$	blocked
RH	photon-clock handoff probe only	not a theorem
global all-readout	unclaimed	guard active

A safe next source-only direction is

v4.7.0-r27: MM/HC seed-equation ansatz extraction from the guarded atlas handoff.

This should extract candidate seed constraints from $\mathfrak{X}_{\partial,\mathrm{loc}}^{r26}$ without upgrading them to physical field equations or closure theorems.

A.212 v4.7.0-r27: MM/HC Seed-Equation Ansatz Extraction from the Guarded Atlas Handoff

This source-only layer extracts the first MM/HC seed-equation ansatz family from $\mathfrak{X}_{\partial,\mathrm{loc}}^{r26}$. The extraction is deliberately conservative. The output is a typed family of local seed constraints, not a physical field equation and not a closure theorem.

A.212.1 Projected handoff input

The input object inherited from r26 is

$$\mathfrak{X}_\partial^{r26} = \left(\mathcal{A}_\partial^{r25}, \Sigma_{\mathrm{trans}}^{r26}, \mathcal{F}_\partial, \Gamma_{\mathrm{guard}}^{r26}\right),$$

with

$$\Sigma_{\mathrm{trans}}^{r26} = (0,0,0,0), \qquad \mathcal{F}_\partial : G\big(N + \kappa N(N-1)\big) = 16.$$

The seed extraction uses only the projected local object

$$\boxed{\mathfrak{X}_{\partial,\mathrm{loc}}^{r26} := \Pi_{\mathrm{loc}}^{\mathrm{guard}} \mathfrak{X}_\partial^{r26}.}$$

The projection is part of the syntax. No term may be extracted before passing through the local guard.

A.212.2 Seed-equation ansatz family

The first MM/HC seed family is

$$\boxed{\mathfrak{E}_{\mathrm{MM/HC}}^{\mathrm{seed,r27}} = \left(E_{\mathrm{rel}}, E_{\mathrm{src}}, E_{\mathrm{seam}}, E_{\mathrm{mix}}, E_{\mathrm{col}}, E_{\mathrm{clk}}\right).}$$

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Each component is a local constraint seed:

$$\begin{aligned}
E_{\mathrm{rel}} &:= G\big(N + \kappa N(N-1)\big) - 16,\\
E_{\mathrm{src}} &:= H_{\partial} - 16,\\
E_{\mathrm{seam}} &:= \|\Sigma^{r26}_{\mathrm{trans}}\|^2,\\
E_{\mathrm{mix}} &:= \|P_{\partial} S^{\mathrm{mixker}}_{4,\mu}\|^2 + \left(\beta_4 - \frac{1}{256}\right)^2,\\
E_{\mathrm{col}} &:= \Phi_5^{(n+1)} - \frac{3}{4}\Phi_5^{(n)},\\
E_{\mathrm{clk}} &:= G - \mathcal{C}_{\gamma}(\tau_v).
\end{aligned}$$

Here E_{rel} records the valley-front relay frontier, E_{src} records the local boundary-source surplus, E_{seam} records transition exactness, E_{mix} records that the mixed-kernel guard is not consumed, E_{col} records the bounded $C5$ collar recurrence, and E_{clk} records the photon-clock timing probe.

Definition A.212.1 (r27 MM/HC seed ansatz)**.** *The r27 MM/HC seed ansatz is the six-component local constraint family* $\mathfrak{E}^{\mathrm{seed,r27}}_{\mathrm{MM/HC}}$ *extracted from* $\mathfrak{X}^{r26}_{\partial,\mathrm{loc}}$ *under the guard projector* $\Pi^{\mathrm{guard}}_{\mathrm{loc}}$.

A.212.3 Two-sided seed typing

The seed ansatz is written in the two-sided notation as

$$\boxed{{}^{\partial\mathrm{atlas}}{}_{\mathrm{loc/read}}\mathfrak{E}^{\mathrm{seed,r27\ MM/HC}}_{\mathrm{MM/HC}}{}_{\mathrm{constraint}}\cdot}$$

The left side records the local boundary-atlas readout, while the right side records the MM/HC constraint target. The expression is not a same-variance Einstein contraction. It is a typed local source-to-constraint interface.

A scalar audit functional may be used for bookkeeping:

$$\boxed{\mathcal{S}^{r27}_{\mathrm{seed}} = \lambda_{\mathrm{rel}} E^2_{\mathrm{rel}} + \lambda_{\mathrm{src}} E^2_{\mathrm{src},-} + \lambda_{\mathrm{seam}} E_{\mathrm{seam}} + \lambda_{\mathrm{mix}} E_{\mathrm{mix}} + \lambda_{\mathrm{col}} E^2_{\mathrm{col}} + \lambda_{\mathrm{clk}} E^2_{\mathrm{clk}},}$$

where $E_{\mathrm{src},-} := \max(0, 16 - H_{\partial})$. This functional is a local bookkeeping device only. It is not an action principle and not a physical Lagrangian.

A.212.4 Admissibility and non-upgrade clauses

The ansatz is admissible only under the following clauses:

A1 $\mathfrak{X}^{r26}_{\partial}$ is first projected by $\Pi^{\mathrm{guard}}_{\mathrm{loc}}$,
A2 $E_{\mathrm{seam}} = 0$ is inherited from $\Sigma^{r26}_{\mathrm{trans}} = (0,0,0,0)$,
A3 E_{mix} preserves $\beta_4 = 1/256$ as unspent guard debt,
A4 E_{col} records only $\Phi_5^{(n+1)} = \frac{3}{4}\Phi_5^{(n)}$ for finite stages,
A5 E_{clk} is a local photon-clock probe and not an RH theorem,
A6 $\mathfrak{E}^{\mathrm{seed,r27}}_{\mathrm{MM/HC}}$ is not promoted to a physical field equation.

Lemma A.212.2 (r27 seed extraction preserves guards)**.** *The extraction of* $\mathfrak{E}^{\mathrm{seed,r27}}_{\mathrm{MM/HC}}$ *from* $\mathfrak{X}^{r26}_{\partial,\mathrm{loc}}$ *preserves the r26 guard vector.*

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Proof. The projected handoff already has zero transition seam vector. The relay and boundary-source terms record local admissibility conditions but do not introduce an all-readout channel. The mixed-kernel term explicitly keeps $\beta_4 = 1/256$ as a recorded guard rather than consuming it. The collar term records the finite recurrence of $C5$ and includes no finite closure clause. The clock term is a timing probe only. Hence no guarded channel is discharged. □

Theoremlet A.212.3 (r27 guarded seed-equation ansatz)**.** *The six-component family* $\mathfrak{E}^{\mathrm{seed,r27}}_{\mathrm{MM/HC}}$ *is an admissible local MM/HC seed-equation ansatz extracted from the guarded atlas handoff. It is not a physical field equation and not a closure theorem.*

Proof. By A1 the input is local and guard-projected. By A2 the transition seam component is exact. By A3 the route-stable mixed-kernel residual is preserved rather than spent. By A4 no finite $C5$ closure is claimed. By A5 the Riemann-zero/photon-clock channel remains a probe. By A6 the output is a seed constraint family only. Therefore the ansatz is admissible as a guarded local MM/HC seed and no global conclusion follows. □

A.212.5 r27 decision ledger

The r27 decision ledger is

seed component	local role	guard status
E_{rel}	relay frontier	local only
E_{src}	boundary-source surplus	no global source
E_{seam}	transition seam exactness	$\Sigma_{\mathrm{trans}} = 0$
E_{mix}	mixed-kernel guard record	$\beta_4 = 1/256$ unspent
E_{col}	bounded $C5$ collar recurrence	no finite closure
E_{clk}	photon-clock handoff probe	not RH theorem

The global guard vector remains

$$\boxed{\Gamma^{r27}_{\mathrm{guard}} = (1, 1, 1, 1, 1).}$$

A safe next source-only direction is

v4.7.0-r28: seed-ansatz consistency scan and MM/HC coefficient typing.

This next step should classify the coefficients $\lambda_{\mathrm{rel}}, \lambda_{\mathrm{src}}, \lambda_{\mathrm{seam}}, \lambda_{\mathrm{mix}}, \lambda_{\mathrm{col}}, \lambda_{\mathrm{clk}}$ as bookkeeping weights, selector weights, or admissibility multipliers, without turning $\mathcal{S}^{r27}_{\mathrm{seed}}$ into a physical action.

A.213 v4.7.0-r28: Seed-Ansatz Consistency Scan and MM/HC Coefficient Typing

This source-only layer audits the r27 seed ansatz $\mathfrak{E}^{\mathrm{seed,r27}}_{\mathrm{MM/HC}}$ before any field-equation upgrade is attempted. The goal is coefficient typing. The scan deliberately prevents the local seed functional from being misread as a physical action, a Lagrangian, or a closure theorem.

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A.213.1 Consistency issue: signed surplus is not a zero residual

The r27 family contains the signed frontier coordinate

$$E_{\mathrm{rel}} = G\big(N + \kappa N(N-1)\big) - 16.$$

This coordinate is useful, but it is not always a residual that should be driven to zero. If a local relay chart is strictly admissible, then

$$H = G\big(N + \kappa N(N-1)\big) > 16,$$

so

$$E_{\mathrm{rel}} > 0.$$

Penalising E_{rel}^2 as an ordinary equality residual would incorrectly push a robustly admissible chart back onto the thin frontier $H = 16$. Thus the signed frontier coordinate must be typed as a monitor, while the admissibility penalty uses only the violation part

$$\boxed{E_{\mathrm{rel}}^- := \max\{0, 16 - H\}.}$$

Similarly, for the boundary-source surplus,

$$\boxed{E_{\mathrm{src}}^- := \max\{0, 16 - H_\partial\}.}$$

The positive surplus

$$\mu := \max\{0, H - 16\}$$

is recorded as robustness margin, not as debt.

Definition A.213.1 (r28 violation-projected relay residual)**.** *The r28 relay residual is the violation-projected quantity $E_{\mathrm{rel}}^- = \max\{0, 16 - H\}$, while the signed quantity $E_{\mathrm{rel}} = H - 16$ is retained only as a frontier coordinate and margin monitor.*

A.213.2 Typed coefficient classes

The r27 coefficients are now divided into three noninterchangeable classes:

$$\boxed{\lambda \in \{\lambda^{\mathrm{book}}, \lambda^{\mathrm{sel}}, \lambda^{\mathrm{adm}}\}.}$$

Here

λ^{book} bookkeeping or diagnostic weight only,
λ^{sel} selector/timing weight for admissible local routing,
λ^{adm} admissibility multiplier for a hard local guard.

The r28 typing ledger is

coefficient	r28 type	role
λ_{rel}	$\lambda^{\mathrm{book}} + \lambda^{\mathrm{adm}}$	signed monitor plus violation penalty $(E_{\mathrm{rel}}^-)^2$
λ_{src}	λ^{adm}	boundary-source deficit penalty $(E_{\mathrm{src}}^-)^2$
λ_{seam}	λ^{adm}	transition seam exactness guard
λ_{mix}	λ^{adm}	mixed-kernel guard lock, keeps $\beta_4 = 1/256$ unspent
λ_{col}	$\lambda^{\mathrm{book}} + \lambda^{\mathrm{adm}}_{\mathrm{loc}}$	finite collar recurrence consistency only
λ_{clk}	λ^{sel}	photon-clock timing selector/probe

The mixed notation $\lambda^{\mathrm{book}} + \lambda^{\mathrm{adm}}$ means that the signed value may be displayed diagnostically, but only its admissibility-violation projection may be treated as debt.

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A.213.3 Typed seed audit functional

The scalar r27 bookkeeping expression is therefore replaced, for audit purposes, by the typed functional

$$\boxed{\mathcal{S}_{\mathrm{seed}}^{r28,\mathrm{typed}} = \alpha_{\mathrm{rel}}(E_{\mathrm{rel}}^{-})^2 + \alpha_{\mathrm{src}}(E_{\mathrm{src}}^{-})^2 + \alpha_{\mathrm{seam}}E_{\mathrm{seam}} + \alpha_{\mathrm{mix}}E_{\mathrm{mix}} + \alpha_{\mathrm{col}}E_{\mathrm{col}}^2 + \alpha_{\mathrm{clk}}^{\mathrm{sel}}E_{\mathrm{clk}}^2.}$$

with the side condition that $\alpha_{\mathrm{clk}}^{\mathrm{sel}}$ is active only inside a declared photon-clock phase window. Outside such a declared window, E_{clk} remains a probe and contributes no hard debt.

For the r22–r25 boundary-source anchor

$$(N, \kappa, G) = \left(5, \frac{3}{4}, 1\right),$$

one has

$$H = 20, \qquad E_{\mathrm{rel}} = 4, \qquad E_{\mathrm{rel}}^{-} = 0.$$

Thus the anchor has positive margin, but no relay violation debt. This resolves the apparent contradiction between a robust pass and a zero-residual audit.

A.213.4 Seed-component consistency scan

The six r27 seed components pass the r28 consistency scan with the following interpretations:

component	consistency decision	allowed use
E_{rel}	consistent after violation projection	frontier monitor plus local deficit test
E_{src}	consistent after deficit projection	boundary-source admissibility test
E_{seam}	consistent hard exactness seed	$\Sigma_{\mathrm{trans}}^{r26} = 0$
E_{mix}	consistent guard-preserving seed	$\beta_4 = 1/256$ retained
E_{col}	consistent finite recurrence seed	no finite $C5$ closure
E_{clk}	consistent selector/probe seed	not an RH theorem

In particular, E_{rel} and E_{src} are not independent physical field equations. When $H_\partial = H$, they are the same local relay surplus viewed from two bookkeeping sides: the valley-front side and the boundary-source side.

Lemma A.213.2 (r28 coefficient typing removes the frontier-square artefact)**.** *Replacing E_{rel}^2 by $(E_{\mathrm{rel}}^{-})^2$ in the admissibility part of the seed audit prevents robust relay margins from being misclassified as residual debt.*

Proof. If $H \geq 16$, then $E_{\mathrm{rel}}^{-} = 0$, so the local relay chart contributes no admissibility debt regardless of the positive surplus $H - 16$. If $H < 16$, then $E_{\mathrm{rel}}^{-} = 16 - H > 0$, so the deficit is recorded. Thus the typed residual detects exactly local relay failure and does not penalise robust pass interior points. □

A.213.5 MM/HC coefficient typing rule

The r28 coefficient rule is:

> No coefficient may be interpreted before its seed component is typed.

Equivalently,

$\lambda_i E_i^2$ is admissible only after declaring whether E_i is a monitor, selector, violation residual, or hard gu

This rule blocks the common false upgrade

$$\mathcal{S}_{\mathrm{seed}} \longrightarrow \text{physical action.}$$

The present object remains

$$\boxed{\mathcal{S}_{\mathrm{seed}}^{r28,\mathrm{typed}} \in \text{local MM/HC seed-audit grammar only.}}$$

Theoremlet A.213.3 (r28 seed-ansatz consistency and coefficient typing)**.** *After violation projection and coefficient typing, the r27 seed ansatz is consistent as a guarded local MM/HC seed-audit system. It is not a physical field equation, not an action principle, and not a closure theorem.*

Proof. The only apparent inconsistency is the treatment of positive relay surplus as a zero residual. Definition A.213.1 resolves this by splitting the signed frontier monitor from the violation residual. The seam, mixed-kernel, collar, and clock components keep their r27 guards: seam exactness remains local, $\beta_4 = 1/256$ is preserved, finite $C5$ closure is not claimed, and the photon-clock term remains a probe. Since all coefficients are typed before use, no scalar audit expression can be promoted to an action or global field equation. Therefore the typed seed-audit system is consistent in the declared local grammar only. □

A.213.6 r28 guard ledger

The r28 guard ledger is

guard	r28 status	reason
$C4$	$\beta_4 = 1/256$ unspent	E_{mix} remains a guard lock
$C5$	no finite closure	E_{col} records recurrence only
$C6/T2$	blocked	$\chi_{\mathrm{all}} = 0$ remains active
RH	probe only	E_{clk} is selector/timing, not theorem
global all-readout	unclaimed	$\Pi_{\mathrm{loc}}^{\mathrm{guard}}$ remains mandatory

Thus

$$\boxed{\Gamma_{\mathrm{guard}}^{r28} = (1,1,1,1,1).}$$

A safe next source-only direction is

$$\boxed{\text{v4.7.0-r29: typed seed residual minimizer and selector-window calibration.}}$$

The next step should use the r28 typed residuals to scan local minimisers without allowing a physical-action or closure upgrade.

A.214 v4.7.0-r29 Typed Seed Residual Minimizer and Selector-Window Calibration

This source-only step uses the r28 typed seed audit to define a minimizer for the local MM/HC seed residuals. The important point is that the minimizer is applied to violation-projected residuals, not to signed surplus monitors. Therefore a robust relay interior point is not pushed back to the thin frontier.

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A.214.1 Typed residual functional

Let

$$H(N,\kappa,G)=G\big(N+\kappa N(N-1)\big)$$

be the local relay height. The r29 typed admissibility residual is

$$\boxed{\mathcal{D}^{r29}_{\mathrm{typed}}=\lambda_{\mathrm{rel}}\big(E^{-}_{\mathrm{rel}}\big)^2+\lambda_{\mathrm{src}}\big(E^{-}_{\mathrm{src}}\big)^2+\lambda_{\mathrm{seam}}\|\Sigma^{r26}_{\mathrm{trans}}\|^2+\lambda_{\mathrm{mix}}\|P_{\partial}S^{\mathrm{mixker}}_{4,\mu}\|^2+\lambda_{\mathrm{col}}\big(E_{\mathrm{col}}\big)^2+\lambda_{\mathrm{clk}}\big(E^{\mathrm{win}}_{\mathrm{clk}}\big)^2.}$$

Here

$$E^{-}_{\mathrm{rel}}=\max(0,16-H),\qquad E^{-}_{\mathrm{src}}=\max(0,16-H_{\partial}).$$

The clock term is not a theorem term. It is a selector-window residual:

$$\boxed{E^{\mathrm{win}}_{\mathrm{clk}}=\max(0,G_{\mathrm{req}}-\mathcal{C}_{\gamma}(\tau_v)).}$$

Thus photon-clock timing is used only to decide whether a local relay window is available.

Definition A.214.1 (r29 typed zero-debt minimizer)**.** *A local seed chart is a typed zero-debt minimizer if*

$$\mathcal{D}^{r29}_{\mathrm{typed}}=0$$

inside the declared local grammar. This means that all admissibility deficits, seam debts, mixed-kernel couplings, collar-recurrence defects, and clock-window deficits vanish. It does not mean that the signed surplus monitors vanish.

A.214.2 Minimizer degeneracy and margin classes

Because r28 removed the frontier-square artefact, every admissible relay point with

$$H\geq 16$$

has zero relay debt. Hence the minimizer set is not a single point. It has a thin frontier and an interior.

We classify zero-debt points by margin

$$\mu=H-16.$$

The local classes are

class	condition
frontier	$\mu=0$
thin	$0<\mu<1$
robust	$1\leq\mu<4$
comfortable	$\mu\geq 4.$

Thus the finite-grid minimizer

$$(N,\kappa,G)=(4,1,1)$$

remains a zero-debt point, but only a frontier minimizer. The r22–r25 anchor

$$(N,\kappa,G)=\left(5,\frac{3}{4},1\right)$$

is a comfortable zero-debt minimizer with

$$H=20,\qquad \mu=4.$$

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A.214.3 Selector-window calibration

For a fixed overlap strength

$$S(N,\kappa) = N + \kappa N(N-1),$$

the gain required for relay is

$$\boxed{G_{\mathrm{req}}(N,\kappa) = \frac{16}{S(N,\kappa)}.}$$

For a strict margin-one relay it is

$$\boxed{G_{\mathrm{req}}^{(+1)}(N,\kappa) = \frac{17}{S(N,\kappa)}.}$$

For the comfortable r22 anchor

$$S\left(5, \frac{3}{4}\right) = 20,$$

so

$$G_{\mathrm{req}} = \frac{4}{5}, \qquad G_{\mathrm{req}}^{(+1)} = \frac{17}{20}.$$

If a photon-clock window satisfies

$$\mathcal{C}_\gamma(\tau_v) \geq \frac{17}{20},$$

then the anchor has at least one full unit of relay margin. If it satisfies

$$\mathcal{C}_\gamma(\tau_v) \geq \frac{4}{5},$$

it is relay-admissible but may be near the thin boundary.

Lemma A.214.2 (r29 selector-window calibration)**.** *For fixed local packet data (N,κ), the selector-window threshold is*

$$G_{\mathrm{req}} = 16/S(N,\kappa).$$

The clock term contributes zero selector debt exactly when

$$\mathcal{C}_\gamma(\tau_v) \geq G_{\mathrm{req}}.$$

Proof. The local relay condition is

$$GS(N,\kappa) \geq 16.$$

Solving for the required gain gives $G_{\mathrm{req}} = 16/S(N,\kappa)$. Since the clock term is typed as a selector-window residual,

$$E_{\mathrm{clk}}^{\mathrm{win}} = \max(0, G_{\mathrm{req}} - \mathcal{C}_\gamma(\tau_v)),$$

it vanishes exactly when the available photon-clock gain meets or exceeds the required threshold. □

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A.214.4 Typed minimizer scan result

The r29 finite scan uses the same local grid as r24/r25 and evaluates the typed residual. It finds that the typed residual minimizer set is precisely the admissible region of the local frontier map, stratified by margin. The audit therefore selects two representatives:

$$\boxed{(4,1,1) \quad \text{as the frontier zero-debt representative,}}$$

and

$$\boxed{\left(5,\frac{3}{4},1\right) \quad \text{as the comfortable zero-debt representative.}}$$

The second representative is retained as the operational boundary-source anchor because it has margin 4 and lies inside the robust local collar.

Theoremlet A.214.3 (r29 typed residual minimizer)**.** *Using violation-projected residuals, the local MM/HC seed audit admits zero-debt minimizers throughout the relay-admissible region* $H \geq 16$*. The robust boundary-source anchor* $(5, 3/4, 1)$ *is a comfortable zero-debt minimizer, while* $(4, 1, 1)$ *is only a frontier zero-debt minimizer.*

Proof. For every point with $H \geq 16$, both E^{-}_{rel} and E^{-}_{src} vanish. The seam and mixed-kernel terms vanish by the local boundary-source typing clauses, the collar recurrence is the r28 bookkeeping recurrence, and the clock term vanishes when the selector window is met. Thus all admissible points are typed zero-debt minimizers. The margin stratification then separates the thin frontier from robust and comfortable interior points. For $(4, 1, 1)$, $H = 16$, so the margin is zero. For $(5, 3/4, 1)$, $H = 20$, so the margin is four. □

A.214.5 r29 guard ledger

The minimizer is local and typed. It does not create a physical action, a field equation, or a closure theorem. The r29 guard ledger is

guard	r29 status	reason
$C4$	$\beta_4 = 1/256$ unspent	E_{mix} remains a guard-lock
$C5$	no finite closure	E_{col} is recurrence consistency only
$C6/T2$	blocked	$\chi_{\mathrm{all}} = 0$ remains active
RH	probe only	$E^{\mathrm{win}}_{\mathrm{clk}}$ is a selector-window term
global all-readout	unclaimed	$\Pi^{\mathrm{guard}}_{\mathrm{loc}}$ remains mandatory

Therefore

$$\boxed{\Gamma^{r29}_{\mathrm{guard}} = (1,1,1,1,1).}$$

A safe next source-only direction is

$$\boxed{\text{v4.7.0-r30: typed MM/HC seed normal form and admissibility certificate.}}$$

The next step should package the typed minimizer as a normal-form certificate for local MM/HC seed admissibility, still without promoting it to a physical field equation or global closure theorem.

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A.215 v4.7.0-r30 Typed MM/HC Seed Normal Form and Admissibility Certificate

Source-only status. This r30 layer is a source-only continuation of the v4.7.0 internal line. It packages the r28–r29 typed residual calculus as a local MM/HC seed normal form and an admissibility certificate. It does not introduce a physical action, a Lagrangian, a field equation, a T2 bridge, an RH statement, or a global all-readout closure theorem.

A.215.1 Typed seed normal form

Let

$$S(N,\kappa) = N + \kappa N(N-1), \qquad H = GS(N,\kappa).$$

The r30 local seed normal form is the guarded tuple

$$\boxed{\mathfrak{N}^{r30}_{\mathrm{seed}} = \left(N, \kappa, G; \Sigma_{\mathrm{trans}}, P_\partial S^{\mathrm{mixker}}_{4,\mu}, \beta_4, \Phi^{(n)}_5, \mathcal{C}_\gamma(\tau_v), \Gamma_{\mathrm{guard}}\right)}$$

subject to the following typed clauses:

NF1 two-sided local boundary-source/relay typing is declared;
NF2 $E^-_{\mathrm{rel}} = \max(0, 16 - H) = 0$;
NF3 $E^-_{\mathrm{src}} = \max(0, 16 - H_\partial) = 0$;
NF4 $\Sigma_{\mathrm{trans}} = 0$;
NF5 $P_\partial S^{\mathrm{mixker}}_{4,\mu} = 0$ and $\beta_4 = 1/256$;
NF6 $\Phi^{(n+1)}_5 = \frac{3}{4}\Phi^{(n)}_5$ is only a local collar recurrence;
NF7 $\mathcal{C}_\gamma(\tau_v) \geq 16/S(N,\kappa)$ is only a selector-window probe;
NF8 $\Gamma_{\mathrm{guard}} = (1,1,1,1,1)$ remains active.

The normal form is intentionally asymmetric with respect to surplus: positive relay surplus is not penalized. Only deficit is counted through E^-_{rel} and E^-_{src}.

Definition A.215.1 (r30 typed MM/HC seed normal form)**.** *A local boundary-atlas handoff is in typed MM/HC seed normal form if it admits a tuple $\mathfrak{N}^{r30}_{\mathrm{seed}}$ satisfying NF1–NF8. The declaration is local: it certifies admissibility of the guarded seed constraints only, not a physical field equation and not a closure theorem.*

A.215.2 Admissibility certificate

The associated certificate is

$$\boxed{\mathsf{Cert}^{r30}_{\mathrm{MM/HC}} = \left[\mathrm{Type}_{\partial/\mathrm{relay}}, E^-_{\mathrm{rel}} = 0, E^-_{\mathrm{src}} = 0, \Sigma_{\mathrm{trans}} = 0, P_\partial S^{\mathrm{mixker}}_{4,\mu} = 0, \beta_4 = \frac{1}{256}, \Phi^{(n+1)}_5 = \frac{3}{4}\Phi^{(n)}_5, \mathcal{C}_\gamma(\tau_v) \geq \right.}$$

For the comfortable r29/r30 operational anchor

$$(N,\kappa,G) = \left(5, \frac{3}{4}, 1\right),$$

one has

$$S = 20, \qquad H = 20, \qquad \mu = H - 16 = 4.$$

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Hence

$$E^{-}_{\mathrm{rel}} = 0, \qquad G_{\mathrm{req}} = \frac{4}{5}, \qquad G^{(+1)}_{\mathrm{req}} = \frac{17}{20}.$$

Since $G = 1$, the anchor satisfies the selector-window requirement with margin. The frontier representative $(4, 1, 1)$ also satisfies the certificate but only critically, with $H = 16$ and $\mu = 0$. It is therefore kept as a frontier witness, not as the operational anchor.

Proposition A.215.2 (r30 local admissibility certificate)**.** *The comfortable anchor* $(5, 3/4, 1)$ *satisfies the typed MM/HC seed normal form with margin* 4*. The certificate has zero typed admissibility debt while preserving all global guards.*

Proof. For the anchor, $S = 20$ and $H = 20$, so $E^{-}_{\mathrm{rel}} = \max(0, 16 - 20) = 0$. The boundary-source handoff has $H_\partial = 20$, so $E^{-}_{\mathrm{src}} = 0$. By the r26 transition audit, $\Sigma_{\mathrm{trans}} = 0$. By the r20–r22 boundary-source typing, $P_\partial S^{\mathrm{mixker}}_{4,\mu} = 0$ and $\beta_4 = 1/256$ is left unspent. The collar term is the already guarded recurrence $\Phi^{(n+1)}_5 = (3/4)\Phi^{(n)}_5$, which is not finite closure. The clock selector threshold is $G_{\mathrm{req}} = 16/20 = 4/5$, and the anchor has $G = 1$. Thus every local admissibility component is satisfied while $\Gamma_{\mathrm{guard}} = (1, 1, 1, 1, 1)$ remains active. □

A.215.3 Certificate classes

The r30 certificate separates admissible points into three useful local classes:

class	condition	interpretation
frontier-cert	$H = 16$	zero debt but no margin
robust-cert	$17 \le H < 20$	positive protected margin
comfortable-cert	$H \ge 20$	operational anchor region

The certificate deliberately avoids selecting the smallest zero-debt point as the operational seed. The point $(4, 1, 1)$ is minimal on the finite grid, but it is thin. The point $(5, 3/4, 1)$ carries margin four and is therefore retained as the preferred local seed normal form.

Corollary A.215.3 (r30 frontier/anchor separation)**.** *The r30 normal form distinguishes minimal admissibility from robust operational admissibility. A frontier-zero-debt point is allowed as a witness, but only a positive-margin point is used as the seed anchor for subsequent MM/HC coefficient work.*

A.215.4 r30 guard certificate

The r30 admissibility certificate is local and guard-preserving:

guard	r30 status	certificate effect
$C4$	$\beta_4 = 1/256$ unspent	mixed-kernel guard preserved
$C5$	no finite closure	collar recurrence only
$C6/T2$	blocked	$\chi_{\mathrm{all}} = 0$ remains active
RH	probe only	$\mathcal{C}_\gamma$ only calibrates selector windows
global all-readout	unclaimed	$\Pi^{\mathrm{guard}}_{\mathrm{loc}}$ remains mandatory

Therefore

$$\boxed{\Gamma^{r30}_{\mathrm{guard}} = (1, 1, 1, 1, 1).}$$

A safe next source-only direction is

v4.7.0-r31: typed coefficient skeleton and local seed functional packaging.

The next step should introduce a typed coefficient skeleton for the certified seed normal form, still as a local audit functional and not as a physical action or field equation.

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A.216 v4.7.0-r31 Source-Only: Typed Coefficient Skeleton and Local Seed Functional Packaging

This r31 source-only layer packages the r30 seed normal form as a typed local seed functional. The purpose is strictly audit-theoretic: the functional below is a local MM/HC seed bookkeeping device, not a physical action, not a Lagrangian, not a derived field equation, and not a closure theorem.

A.216.1 Coefficient Skeleton

Let

$$\mathfrak{N}^{r30}_{\mathrm{seed}} = (E^-_{\mathrm{rel}}, E^-_{\mathrm{src}}, \Sigma_{\mathrm{trans}}, P_\partial S^{\mathrm{mixker}}_{4,\mu}, \beta_4, \Phi^{(n)}_5, G, \mathcal{C}_\gamma(\tau_v))$$

be the local seed normal form certified in r30. r31 assigns types to the coefficient skeleton before any functional is written:

$$\boxed{\Lambda^{r31}_{\mathrm{seed}} = (\lambda^-_{\mathrm{rel}}, \lambda^-_{\mathrm{src}}, \lambda_{\mathrm{seam}}, \lambda_{\mathrm{mix}}, \lambda_\beta, \lambda_{\mathrm{col}}, \lambda_{\mathrm{clk}}, \lambda_{\mathrm{cmf}}).}$$

The typing is:

coefficient	type	allowed role
λ^-_{rel}	admissibility penalty	$(E^-_{\mathrm{rel}})^2$
λ^-_{src}	admissibility penalty	$(E^-_{\mathrm{src}})^2$
λ_{seam}	hard seam guard	$\|\Sigma_{\mathrm{trans}}\|^2$
λ_{mix}	mixed-kernel guard-lock	$\|P_\partial S^{\mathrm{mixker}}_{4,\mu}\|^2$
λ_β	C4 ledger lock	$(\beta_4 - 1/256)^2$
λ_{col}	local recurrence bookkeeping	$(\Phi^{(n+1)}_5 - \frac{3}{4}\Phi^{(n)}_5)^2$
λ_{clk}	selector/probe coefficient	$(G - \mathcal{C}_\gamma(\tau_v))^2_{\mathrm{sel}}$
λ_{cmf}	comfort-margin selector	$(E^-_{\mathrm{cmf}})^2$

where the last term is optional and only used when a comfortable certificate is requested. It is not part of the bare admissibility certificate.

A.216.2 Local Seed Functional

The local audit functional is

$$\boxed{\begin{aligned}\mathcal{J}^{r31}_{\mathrm{seed}} = {} & \lambda^-_{\mathrm{rel}}(E^-_{\mathrm{rel}})^2 + \lambda^-_{\mathrm{src}}(E^-_{\mathrm{src}})^2 + \lambda_{\mathrm{seam}}\|\Sigma^{r26}_{\mathrm{trans}}\|^2 \\ & + \lambda_{\mathrm{mix}}\|P_\partial S^{\mathrm{mixker}}_{4,\mu}\|^2 + \lambda_\beta\left(\beta_4 - \frac{1}{256}\right)^2 \\ & + \lambda_{\mathrm{col}}\left(\Phi^{(n+1)}_5 - \frac{3}{4}\Phi^{(n)}_5\right)^2 + \lambda_{\mathrm{clk}}\,(G - \mathcal{C}_\gamma(\tau_v))^2_{\mathrm{sel}}\,.\end{aligned}}$$

The signed monitor

$$E_{\mathrm{rel}} = H - 16$$

is not squared in the admissibility functional. Only the violation projection

$$E^-_{\mathrm{rel}} = \max(0, 16 - H)$$

contributes debt. Hence interior relay surplus is not penalized.

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For comfortable certification one may add the explicitly typed selector term

$$\boxed{\mathcal{J}^{r31}_{\mathrm{seed},+4} = \mathcal{J}^{r31}_{\mathrm{seed}} + \lambda_{\mathrm{cmf}}(E^{-}_{\mathrm{cmf}})^2, \qquad E^{-}_{\mathrm{cmf}} = \max(0, 20 - H).}$$

This term distinguishes the comfortable anchor from a thin frontier pass, but it is not a new physical principle.

A.216.3 Anchor Evaluation

At the comfortable operational anchor

$$(N, \kappa, G) = \left(5, \frac{3}{4}, 1\right)$$

one has

$$S = 20, \qquad H = 20, \qquad E^{-}_{\mathrm{rel}} = 0, \qquad E^{-}_{\mathrm{src}} = 0, \qquad E^{-}_{\mathrm{cmf}} = 0.$$

Together with the r30 certificate clauses

$$\Sigma^{r26}_{\mathrm{trans}} = 0, \qquad P_{\partial} S^{\mathrm{mixker}}_{4,\mu} = 0, \qquad \beta_4 = \frac{1}{256}, \qquad \Phi_5^{(n+1)} = \frac{3}{4}\Phi_5^{(n)},$$

and a matched selector window

$$G = \mathcal{C}_{\gamma}(\tau_v),$$

this gives

$$\boxed{\mathcal{J}^{r31}_{\mathrm{seed}} = 0, \qquad \mathcal{J}^{r31}_{\mathrm{seed},+4} = 0}$$

inside the declared local seed grammar.

At the frontier witness

$$(N, \kappa, G) = (4, 1, 1)$$

one has

$$H = 16, \qquad E^{-}_{\mathrm{rel}} = 0,$$

so the bare admissibility functional may vanish. However

$$E^{-}_{\mathrm{cmf}} = 4,$$

so the comfortable certificate does not vanish. Thus r31 preserves the r30 classification:

$$\boxed{(4, 1, 1) = \text{frontier-cert only}, \qquad \left(5, \frac{3}{4}, 1\right) = \text{comfortable-cert.}}$$

A.216.4 Guard Preservation

The r31 functional is local and typed. It does not spend the persistent C4 mixed-kernel guard, does not close C5 at any finite stage, does not unblock C6/T2, does not identify Riemann zeros with photon events, and does not claim a global all-readout closure. Hence

$$\boxed{\Gamma^{r31}_{\mathrm{guard}} = (1, 1, 1, 1, 1).}$$

The safe next source-only direction is

$$\boxed{\text{v4.7.0-r32: local seed functional sanity table and coefficient-domain guard.}}$$

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A.217 v4.7.0-r32 Source-Only Local Seed Functional Sanity Table and Coefficient-Domain Guard

Source-only status. This subsection is an internal source-only continuation of the v4.7.0 line. No PDF artefact is generated at this stage. The purpose is to sanity-check the r31 local seed functional against representative admissible, frontier, robust, comfortable, and debt-positive sites, and to record an explicit coefficient -domain guard.

A.217.1 Sanity Table for the Local Seed Functional

Recall the local relay score

$$S(N,\kappa) = N + \kappa N(N-1), \qquad H = G\,S(N,\kappa),$$

with admissibility debt

$$E^{-}_{\mathrm{rel}} = \max(0, 16 - H)$$

and optional comfort-margin selector

$$E^{-}_{\mathrm{cmf}} = \max(0, 20 - H).$$

The signed quantity

$$E_{\mathrm{rel}} = H - 16$$

remains only a frontier monitor and is not squared as debt.

The representative sanity table is:

site	N	κ	G	H	E^{-}_{rel}	E^{-}_{cmf}
deep-debt	3	1/2	1	6	10	14
local-debt	4	3/4	1	13	3	7
frontier witness	4	1	1	16	0	4
weak five-packet frontier	5	11/20	1	16	0	4
gain frontier	5	3/4	4/5	16	0	4
margin-one selector	5	3/4	17/20	17	0	3
robust box corner	5	7/10	9/10	171/10	0	29/10
comfortable anchor	5	3/4	1	20	0	0

Hence the bare r31 admissibility part vanishes exactly on and above the relay frontier $H = 16$, while the optional comfort selector distinguishes thin frontier passes from the operational comfortable anchor $H = 20$.

Lemma A.217.1 (r32 local sanity separation)**.** *Inside the declared local seed grammar, the r31/r32 typed seed audit separates three cases without changing the guard vector:*

$$H < 16 \Rightarrow E^{-}_{\mathrm{rel}} > 0, \qquad 16 \le H < 20 \Rightarrow E^{-}_{\mathrm{rel}} = 0 < E^{-}_{\mathrm{cmf}}, \qquad H \ge 20 \Rightarrow E^{-}_{\mathrm{rel}} = E^{-}_{\mathrm{cmf}} = 0.$$

Consequently a frontier certificate, a robust local certificate, and a comfortable certificate remain distinct.

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A.217.2 Coefficient-Domain Guard

The admissible coefficient domain is recorded as

$$\boxed{\mathcal{D}_{\lambda}^{r32}}$$

with the following typed clauses:

$$\begin{aligned}
&\lambda_{\mathrm{rel}}^{-}, \lambda_{\mathrm{src}}^{-} \geq 0, && \text{violation-projected admissibility penalties only,}\\
&\lambda_{\mathrm{seam}} > 0, && \text{hard transition-seam guard,}\\
&\lambda_{\mathrm{mix}} > 0, && \text{mixed-kernel guard-lock,}\\
&\lambda_{\beta} > 0, && \text{C4 ledger lock at } \beta_4 = 1/256,\\
&\lambda_{\mathrm{col}} \geq 0, && \text{local C5 recurrence bookkeeping,}\\
&\lambda_{\mathrm{clk}} \geq 0, && \text{selector/probe weight only,}\\
&\lambda_{\mathrm{cmf}} \geq 0, && \text{optional comfort-margin selector.}
\end{aligned}$$

No coefficient in $\mathcal{D}_{\lambda}^{r32}$ may convert the signed monitor $E_{\mathrm{rel}} = H - 16$ into a debt term. Debt is carried only by the negative part E_{rel}^{-}. Likewise, no coefficient may spend the C4 ledger lock, unblock C6/T2, or turn the photon-clock probe into an RH assertion.

The guarded local audit functional is therefore read as

$$\boxed{\mathcal{J}_{\mathrm{seed}}^{r32} : \mathcal{D}_{\lambda}^{r32} \times \mathfrak{N}_{\mathrm{seed}}^{r30} \longrightarrow \mathbb{R}_{\geq 0}}$$

with the same meaning as in r31: it is a local audit functional, not an action, not a Lagrangian, not a derived field equation, and not a closure theorem.

A.217.3 Guard Preservation

The r32 sanity table and coefficient-domain guard do not alter the previous mathematics. They prevent a misreading of the signed frontier monitor as a least-squares field residual and preserve the operational separation between frontier, robust, and comfortable relay certificates. Thus

$$\boxed{\Gamma_{\mathrm{guard}}^{r32} = (1, 1, 1, 1, 1).}$$

Explicitly, C4 remains route-stable with

$$\beta_4 = \frac{1}{256},$$

C5 remains an asymptotic bounded collar with no finite closure, C6/T2 remains blocked, the Riemann-zero language remains a photon-clock handoff probe only, and global all-readout closure is not claimed.

A.218 v4.7.0-r33 Source-Only: Local Seed Functional Parameter Sweep and Comfort-Margin Calibration

This source-only layer performs the safe r33 follow-up to the r32 sanity table. No PDF is generated in this step. The purpose is to sweep the local seed functional over a finite parameter box and to calibrate the optional comfort margin without changing any guard, without turning the signed frontier monitor into debt, and without promoting the seed audit into a physical action or field equation.

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A.218.1 Finite Parameter Sweep

The sweep uses

$$N \in \{3,4,5,6,7,8\},$$
$$\kappa \in \left\{\frac{1}{2}, \frac{11}{20}, \frac{3}{5}, \frac{2}{3}, \frac{7}{10}, \frac{3}{4}, \frac{4}{5}, \frac{9}{10}, 1\right\},$$
$$G \in \left\{\frac{3}{4}, \frac{4}{5}, \frac{17}{20}, \frac{9}{10}, \frac{19}{20}, 1\right\}.$$

For each point we evaluate

$$S(N,\kappa) = N + \kappa N(N-1), \qquad H = G\,S(N,\kappa),$$

$$E^{-}_{\mathrm{rel}} = \max(0, 16 - H), \qquad E^{-}_{\mathrm{cmf}} = \max(0, 20 - H).$$

The classification is:

$$\begin{array}{ll} H < 16 & \text{debt-positive,} \\ H = 16 & \text{frontier-zero-debt,} \\ 16 < H < 17 & \text{thin-zero-debt,} \\ 17 \le H < 20 & \text{robust-zero-debt,} \\ H \ge 20 & \text{comfortable-zero-debt.} \end{array}$$

The finite sweep contains 324 cases and yields

class	# cases
debt positive	130
frontier zero debt	3
thin zero debt	6
robust zero debt	24
comfortable zero debt	161

Thus the r32 typing behaves as intended: a broad zero-debt region exists above relay threshold, but the optional comfort selector still separates thin and robust admissibility from the operational comfort regime.

A.218.2 Comfort-Margin Calibration

For a fixed pair (N,κ), the relay, margin-one, and comfort thresholds are

$$G_{\mathrm{rel}}(N,\kappa) = \frac{16}{S(N,\kappa)}, \qquad G_{+1}(N,\kappa) = \frac{17}{S(N,\kappa)}, \qquad G_{\mathrm{cmf}}(N,\kappa) = \frac{20}{S(N,\kappa)}.$$

The five-packet calibration slice is

site	N	κ	$S(N,\kappa)$	$G_{\mathrm{cmf}} = 20/S$
five-packet frontier	5	$\frac{11}{20}$	16	$\frac{5}{4}$
margin-one threshold	5	$\frac{3}{5}$	17	$\frac{20}{17}$
robust box corner	5	$\frac{7}{10}$	19	$\frac{20}{19}$
comfortable anchor	5	$\frac{3}{4}$	20	1

Consequently, at the comfortable anchor

$$(N,\kappa,G) = \left(5, \frac{3}{4}, 1\right)$$

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one has

$$S = 20, \qquad H = 20, \qquad E^-_{\mathrm{rel}} = E^-_{\mathrm{cmf}} = 0.$$

The same five-packet source with $G = \frac{4}{5}$ sits exactly on the relay frontier, while $G = \frac{17}{20}$ gives the first unit of positive margin. Hence the comfort selector is not an admissibility guard; it is a local preference for the stable operational anchor.

Lemma A.218.1 (r33 comfort-margin calibration)**.** *Inside the declared local seed grammar, the optional comfort-margin term* $E^-_{\mathrm{cmf}} = \max(0, 20-H)$ *calibrates the local selector preference without altering admissibility. In particular,*

$$H \geq 16 \Rightarrow E^-_{\mathrm{rel}} = 0,$$

but only

$$H \geq 20 \Rightarrow E^-_{\mathrm{cmf}} = 0.$$

Therefore a frontier certificate and a comfortable operational certificate are kept distinct without creating new debt.

A.218.3 Representative Sweep Anchors

The parameter sweep confirms the hierarchy already suggested by r32:

$$(4,1,1): \quad H = 16, \qquad \mu = 0, \qquad \text{frontier-cert only},$$

$$\left(5, \frac{11}{20}, 1\right): \quad H = 16, \qquad \text{weak five-packet frontier},$$

$$\left(5, \frac{3}{4}, \frac{4}{5}\right): \quad H = 16, \qquad \text{gain frontier},$$

$$\left(5, \frac{3}{4}, \frac{17}{20}\right): \quad H = 17, \qquad \text{margin-one selector},$$

$$\left(5, \frac{7}{10}, \frac{9}{10}\right): \quad H = \frac{171}{10}, \qquad \text{robust box corner},$$

$$\left(5, \frac{3}{4}, 1\right): \quad H = 20, \qquad \text{comfortable anchor}.$$

The frontier minimizer remains useful as a boundary witness, but the comfort anchor remains the preferred local source point for downstream MM/HC seed packaging.

A.218.4 Guard Preservation

The sweep and comfort calibration do not alter any global guard. They only classify local seed-functional behavior inside the already declared local boundary-source grammar. Thus

$$\boxed{\Gamma^{r33}_{\mathrm{guard}} = (1,1,1,1,1).}$$

Explicitly, C4 remains route-stable with

$$\beta_4 = \frac{1}{256},$$

C5 remains an asymptotic bounded collar with no finite closure, C6/T2 remains blocked, the Riemann-zero language remains a photon-clock handoff probe only, and global all-readout closure is not claimed.

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Next source-only direction. The safe next step is

v4.7.0-r34: comfort-calibrated seed selector and guard-stable handoff summary.

A.219 v4.7.0-r34: Comfort-Calibrated Seed Selector and Guard-Stable Handoff Summary

This source-only step packages the r33 parameter sweep into a conservative selector rule. The purpose is not to introduce a new field equation, action, Lagrangian, or closure theorem. The purpose is only to choose a stable local seed for downstream MM/HC bookkeeping while preserving every active guard.

A.219.1 Selector Rule

Let

$$S(N,\kappa) = N + \kappa N(N-1), \qquad H(N,\kappa,G) = GS(N,\kappa).$$

The signed frontier monitor remains

$$E_{\mathrm{rel}} = H - 16,$$

but the admissibility debt is still only

$$E^{-}_{\mathrm{rel}} = \max(0, 16 - H).$$

The comfort deficit is

$$E^{-}_{\mathrm{cmf}} = \max(0, 20 - H).$$

Thus the selector is not a minimizer of the signed frontier monitor. It is a filtered local choice rule:

$$\boxed{E^{-}_{\mathrm{rel}} = 0, \quad E^{-}_{\mathrm{src}} = 0, \quad \Gamma_{\mathrm{guard}} = (1,1,1,1,1),}$$

followed, for operational handoff purposes, by the comfort preference

$$\boxed{E^{-}_{\mathrm{cmf}} = 0.}$$

Frontier witnesses remain useful boundary witnesses, but are not selected as operational handoff anchors unless no comfortable seed is available.

A.219.2 Representative Selector Table

The r34 selector keeps the known frontier witness visible but chooses the comfortable anchor for downstream local MM/HC seed packaging:

$$(4,1,1): \quad H = 16, \qquad \text{frontier witness only},$$

whereas

$$\boxed{\left(5, \frac{3}{4}, 1\right): \quad S = 20, \quad H = 20, \quad \mu = 4}$$

is selected as the comfort-calibrated operational seed. It satisfies

$$E^{-}_{\mathrm{rel}} = 0, \qquad E^{-}_{\mathrm{cmf}} = 0.$$

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The robust lower corner

$$\left(5, \frac{7}{10}, \frac{9}{10}\right)$$

remains a robustness certificate, since

$$H \geq \frac{171}{10} > 16,$$

but it is not the preferred handoff anchor because the comfortable anchor gives larger margin without changing the declared local grammar.

A.219.3 Guard-Stable Handoff Object

The local handoff object produced by r34 is

$$\boxed{\mathfrak{H}^{r34}_{\mathrm{seed}} = \left(\left(5, \frac{3}{4}, 1\right), \mathfrak{N}^{r30}_{\mathrm{seed}}, \mathcal{J}^{r31}_{\mathrm{seed}}, \mathcal{D}^{r32}_{\lambda}, \Gamma^{r33}_{\mathrm{guard}}\right).}$$

This object is a local seed handoff summary only. It is not a physical field equation and not a closure statement.

The guard vector remains

$$\boxed{\Gamma^{r34}_{\mathrm{guard}} = (1, 1, 1, 1, 1).}$$

Explicitly:

$$\beta_4 = \frac{1}{256}$$

remains unspent; C5 remains an asymptotic bounded collar with no finite closure; C6/T2 remains blocked; the Riemann-zero language remains only a photon-clock handoff probe; and global all-readout closure remains unclaimed.

A.219.4 Handoff Summary

The r34 source-only conclusion is:

$$\boxed{\text{Use } \left(5, \frac{3}{4}, 1\right) \text{ as the comfort-calibrated local seed.}}$$

The finite local boundary-extended atlas and the typed seed functional may now be handed to the next MM/HC stage as a guarded seed package. The safe next step is

$$\boxed{\text{v4.7.0-r35: guarded MM/HC seed-to-interface map and no-action audit.}}$$

A.220 v4.7.0-r35 Source-Only: Guarded MM/HC Seed-to-Interface Map and No-Action Audit

A.220.1 Purpose and Scope

This source-only increment takes the r34 comfort-calibrated handoff object

$$\mathfrak{H}^{r34}_{\mathrm{seed}} = \left(\left(5, \frac{3}{4}, 1\right), \mathfrak{N}^{r30}_{\mathrm{seed}}, \mathcal{J}^{r31}_{\mathrm{seed}}, \mathcal{D}^{r32}_{\lambda}, \Gamma^{r33}_{\mathrm{guard}}\right)$$

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and maps it to a guarded MM/HC interface skeleton. The purpose is only to identify admissible interface slots and no-action obligations. No physical action, no Lagrangian, no derived field equation, no T2 closure, no RH claim, and no global all-readout closure is asserted.

The selected seed remains

$$(N,\kappa,G)=\left(5,\frac{3}{4},1\right),\qquad S=20,\qquad H=20,\qquad \mu=4.$$

The frontier witness $(4,1,1)$ remains a boundary witness only and is not promoted to the operational interface seed.

A.220.2 Guarded Interface Slots

We introduce a local interface map

$$\mathfrak{I}^{r35}_{\mathrm{MM/HC}}:\mathfrak{H}^{r34}_{\mathrm{seed}}\longrightarrow(\mathsf{I}_{\mathrm{bd}},\mathsf{I}_{\mathrm{src}},\mathsf{I}_{\mathrm{seam}},\mathsf{I}_{\mathrm{mix}},\mathsf{I}_{\mathrm{col}},\mathsf{I}_{\mathrm{clk}}),$$

where each component is a typed interface slot rather than an equation of motion.

The slots are:

$$\mathsf{I}_{\mathrm{bd}}:\quad E^{-}_{\mathrm{rel}}=0,\qquad E^{-}_{\mathrm{cmf}}=0,$$

for the comfort-calibrated boundary relay condition;

$$\mathsf{I}_{\mathrm{src}}:\quad {}^{\partial\mathrm{src}}{}_{\mathrm{loc}}\mathcal{B}^{\mathrm{res}V0}_{0\ \mathrm{relay}},$$

for the two-sided local source/relay typing;

$$\mathsf{I}_{\mathrm{seam}}:\quad \Sigma_{\mathrm{trans}}=0,$$

for the compressed atlas transition seam audit;

$$\mathsf{I}_{\mathrm{mix}}:\quad P_{\partial}S^{\mathrm{mixker}}_{4,\mu}=0,\qquad \beta_4=\frac{1}{256},$$

for the C4 mixed-kernel guard-lock;

$$\mathsf{I}_{\mathrm{col}}:\quad \Phi_5^{(n+1)}=\frac{3}{4}\Phi_5^{(n)},\qquad \forall n<\infty:\ \Phi_5^{(n)}>0,$$

for the local C5 bounded-collar recurrence without finite closure; and

$$\mathsf{I}_{\mathrm{clk}}:\quad G=\mathcal{C}_\gamma(\tau_v)\quad\text{or}\quad \mathcal{C}_\gamma(\tau_v)\geq\frac{16}{S(N,\kappa)},$$

for the photon-clock selector/probe interface.

Each slot is local and typed. None of these slots may be read as a global readout equation or as a physical field equation.

A.220.3 No-Action Audit

The interface map is equipped with a no-action audit vector

$$\boxed{\mathfrak{A}^{r35}_{\mathrm{noact}}=(0,0,0,0,0,0)}$$

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with the following entries:

$$\begin{aligned}
A_1 &= \text{no physical action introduced},\\
A_2 &= \text{no Lagrangian introduced},\\
A_3 &= \text{no Euler–Lagrange variation asserted},\\
A_4 &= \text{no derived field equation asserted},\\
A_5 &= \text{no global readout promoted},\\
A_6 &= \text{no C4, C5, C6/T2, RH, or all-readout guard spent.}
\end{aligned}$$

Thus r35 only records a guarded seed-to-interface map. It is a preparation layer for a possible later field-equation program, not the program itself.

A.220.4 Admissibility Matrix

For compact bookkeeping, the interface slots may be summarized by the local admissibility matrix

$$\mathcal{M}_{\mathrm{int}}^{r35} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

where the first row records local slot admissibility and the second row records new guard debt. Therefore

$$\boxed{\mathrm{newDebt}(\mathcal{M}_{\mathrm{int}}^{r35}) = 0.}$$

This matrix is not an MM dynamics matrix. It is only a local interface audit matrix.

A.220.5 Guard Preservation

The guard vector remains

$$\boxed{\Gamma_{\mathrm{guard}}^{r35} = (1,1,1,1,1).}$$

Equivalently:

$$\boxed{\beta_4 = \frac{1}{256} \quad \text{remains unspent.}}$$

C5 remains an asymptotic bounded collar with no finite closure; C6/T2 remains blocked; the Riemann-zero component remains only a photon-clock handoff probe; and global all-readout closure remains unclaimed.

A.220.6 r35 Handoff

The r35 handoff object is

$$\boxed{\mathfrak{H}_{\mathrm{int}}^{r35} = \left(\mathfrak{H}_{\mathrm{seed}}^{r34}, \mathfrak{I}_{\mathrm{MM/HC}}^{r35}, \mathfrak{A}_{\mathrm{noact}}^{r35}, \mathcal{M}_{\mathrm{int}}^{r35}, \Gamma_{\mathrm{guard}}^{r35}\right).}$$

The safe next step is

$$\boxed{\text{v4.7.0-r36: interface residual table and MM/HC map consistency scan.}}$$

This next step should still remain source-only unless a PDF is explicitly requested.

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A.221 v4.7.0-r36 Source-Only Interface Residual Table and MM/HC Map Consistency Scan

This source-only layer continues the r35 seed-to-interface map. It does not compile a PDF, does not introduce a physical action, and does not derive an Euler–Lagrange or field equation. Its only purpose is to test whether the six local MM/HC interface slots from r35 remain mutually consistent when read as residual channels.

The comfort-calibrated seed remains

$$(N,\kappa,G)=\left(5,\frac{3}{4},1\right),\qquad S=20,\qquad H=20,\qquad \mu=4.$$

The frontier witness $(4,1,1)$ remains only a boundary witness and is not used as the operational handoff seed.

A.221.1 Interface Residual Table

For each r35 interface slot define a local residual channel

$$\mathcal{R}^{r36}_{\mathrm{int}}=(R_{\mathrm{bd}},R_{\mathrm{src}},R_{\mathrm{seam}},R_{\mathrm{mix}},R_{\mathrm{col}},R_{\mathrm{clk}}).$$

The table below records the guarded interpretation.

slot	residual channel	value	status
I_{bd}	$R_{\mathrm{bd}}=E^{-}_{\mathrm{rel}}+E^{-}_{\mathrm{src}}$	0	local relay admissible
$\mathsf{I}_{\mathrm{src}}$	$R_{\mathrm{src}}=\delta_{\text{two-sided type}}$	0	typed local source/relay
$\mathsf{I}_{\mathrm{seam}}$	$R_{\mathrm{seam}}=\lVert\Sigma^{r26}_{\mathrm{trans}}\rVert^2$	0	transition seam exact
$\mathsf{I}_{\mathrm{mix}}$	$R_{\mathrm{mix}}=\lVert P_{\partial}S^{\mathrm{mixker}}_{4,\mu}\rVert^2+\left(\beta_4-\frac{1}{256}\right)^2$	0	C4 guard-lock preserved
$\mathsf{I}_{\mathrm{col}}$	$R_{\mathrm{col}}=\Phi_5^{(n+1)}-\frac{3}{4}\Phi_5^{(n)}$	0	bounded-collar recurrence only
$\mathsf{I}_{\mathrm{clk}}$	$R_{\mathrm{clk}}=\max\left(0,\frac{16}{S(N,\kappa)}-\mathcal{C}_\gamma(\tau_v)\right)$	0	selector/probe window admissible

For the comfort seed, $S(5,3/4)=20$, hence

$$\frac{16}{S(N,\kappa)}=\frac{4}{5},\qquad \mathcal{C}_\gamma(\tau_v)=G=1,$$

so the photon-clock selector residual vanishes as a local selector channel. This is not an RH statement. It is only the local clock-window reading already allowed by the r35 handoff.

Therefore

$$\boxed{\mathcal{R}^{r36}_{\mathrm{int}}=(0,0,0,0,0,0).}$$

A.221.2 Guard Residuals Versus Interface Residuals

The vanishing of $\mathcal{R}^{r36}_{\mathrm{int}}$ is not a global closure statement. It only means that the r35 local interface slots are internally consistent. The guard residuals remain logically separate:

$$\mathcal{B}^{r36}_{\mathrm{guard}}=\left(\frac{1}{256},\Phi_5^{(n)}>0\ (n<\infty),\mathsf{B}_{\mathrm{all/T2}},\mathsf{probe}_{\mathrm{RH}},\mathsf{unclaimed}_{\mathrm{all}}\right).$$

Thus the interface residual scan has zero local mismatch while the global guard profile remains active.

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A.221.3 MM/HC Map Consistency Scan

The r35 map is consistent iff the six local slots satisfy the following map checks:

MC1 : I_{bd} uses only $E^{-}_{\mathrm{rel}}, E^{-}_{\mathrm{src}}$ and does not square signed E_{rel},
MC2 : $\mathsf{I}_{\mathrm{src}}$ remains two-sided local and does not become an all-readout source,
MC3 : $\mathsf{I}_{\mathrm{seam}}$ imports no transition seam debt,
MC4 : $\mathsf{I}_{\mathrm{mix}}$ locks $\beta_4 = \frac{1}{256}$ instead of spending it,
MC5 : $\mathsf{I}_{\mathrm{col}}$ records only the $C5$ recurrence and no finite closure,
MC6 : $\mathsf{I}_{\mathrm{clk}}$ remains a selector/probe and not an RH theorem.

All six checks pass in the source-level scan:

$$\boxed{\mathrm{MC}^{r36} = (1,1,1,1,1,1), \qquad \mathrm{mapDebt}^{r36} = 0.}$$

A.221.4 No-Action Preservation

The no-action audit vector remains zero:

$$\boxed{\mathfrak{A}^{r36}_{\mathrm{noact}} = (0,0,0,0,0,0).}$$

No variational principle is introduced. No Lagrangian is declared. No Euler–Lagrange equation is varied. No physical field equation is derived. No all-readout promotion is made. No C4, C5, C6/T2, RH, or global-readout guard is spent.

A.221.5 r36 Handoff

The r36 handoff object is

$$\boxed{\mathfrak{H}^{r36}_{\mathrm{int}} = \left(\mathfrak{H}^{r35}_{\mathrm{int}}, \mathcal{R}^{r36}_{\mathrm{int}}, \mathrm{MC}^{r36}, \mathfrak{A}^{r36}_{\mathrm{noact}}, \Gamma^{r35}_{\mathrm{guard}}\right).}$$

It is a guard-stable local interface consistency certificate and nothing more. The safe next source-only step is

v4.7.0-r37: local MM/HC interface certificate and guard-stable residual packaging.

A.222 v4.7.0-r37 Local MM/HC Interface Certificate and Guard-Stable Residual Packaging

This r37 source-only layer packages the r36 zero-residual interface scan as a local certificate. The purpose is deliberately narrow: the certificate records that the six declared local MM/HC interface slots close their local residuals inside the current guarded grammar. It does not convert the local interface map into a physical action, a Lagrangian, an Euler–Lagrange system, a derived field equation, a T2 theorem, or an all-readout closure statement.

A.222.1 Certificate Object

Let the ordered local interface list be

$$\mathbb{I}^{r37} = (\mathsf{I}_{\mathrm{bd}}, \mathsf{I}_{\mathrm{src}}, \mathsf{I}_{\mathrm{seam}}, \mathsf{I}_{\mathrm{mix}}, \mathsf{I}_{\mathrm{col}}, \mathsf{I}_{\mathrm{clk}}).$$

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The r36 local residual vector is inherited as

$$\mathcal{R}_{\mathrm{int}}^{r36} = (0,0,0,0,0,0),$$

and the r36 MM/HC map consistency vector is inherited as

$$\mathrm{MC}^{r36} = (1,1,1,1,1,1), \qquad \mathrm{mapDebt}^{r36} = 0.$$

The r37 local interface certificate is the tuple

$$\boxed{\mathfrak{C}_{\mathrm{int}}^{r37} = \left(\mathbb{I}^{r37}, \mathcal{R}_{\mathrm{int}}^{r36}, \mathrm{MC}^{r36}, \mathfrak{A}_{\mathrm{noact}}^{r36}, \Gamma_{\mathrm{guard}}^{r36}\right).}$$

It is valid only as a local interface certificate. The inclusion of $\Gamma_{\mathrm{guard}}^{r36}$ is part of the certificate, not a debt-spending operation.

A.222.2 Slotwise Certificate Ledger

The slotwise certificate ledger is

slot	local residual	certificate status	guard status
I_{bd}	0	boundary relay certified	local only
$\mathsf{I}_{\mathrm{src}}$	0	two-sided source/relay typed	no all-readout source
$\mathsf{I}_{\mathrm{seam}}$	0	transition seam exact	no seam debt
$\mathsf{I}_{\mathrm{mix}}$	0	mixed-kernel guard-lock certified	$\beta_4 = 1/256$ unspent
$\mathsf{I}_{\mathrm{col}}$	0	bounded-collar recurrence certified	no finite $C5$ closure
$\mathsf{I}_{\mathrm{clk}}$	0	photon-clock selector/probe certified	no RH theorem

Thus every declared local interface slot has zero local residual, while the interpretive guard attached to that slot remains active.

A.222.3 Guard-Stable Residual Packaging

The r37 residual package separates the local zero-residual layer from the nonlocal guard layer:

$$\boxed{\mathcal{P}_{\mathrm{res}}^{r37} = \mathcal{P}_{\mathrm{loc}}^{r37} \oplus \mathcal{P}_{\mathrm{guard}}^{r37}.}$$

The local component is

$$\mathcal{P}_{\mathrm{loc}}^{r37} = \left(\mathcal{R}_{\mathrm{int}}^{r36}, \mathrm{MC}^{r36}, \mathrm{mapDebt}^{r36}\right) = ((0,0,0,0,0,0),(1,1,1,1,1,1),0).$$

The guard component is

$$\mathcal{P}_{\mathrm{guard}}^{r37} = \left(\beta_4 = \frac{1}{256}, \Phi_5^{(n)} > 0 \ (n < \infty), \mathsf{B}_{\mathrm{all/T2}}, \mathsf{probe}_{\mathrm{RH}}, \mathsf{unclaimed}_{\mathrm{all}}\right).$$

This packaging is the main r37 safety point: zero local residual does not imply zero global guard. The two components are recorded together but not identified.

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A.222.4 Certificate Consequence

The only certified consequence is the local statement

$$\boxed{\mathfrak{C}_{\mathrm{int}}^{r37} \vdash_{\mathrm{loc}} \mathcal{R}_{\mathrm{int}}^{r37} = (0,0,0,0,0,0), \qquad \mathrm{mapDebt}^{r37} = 0.}$$

The guard vector remains

$$\boxed{\Gamma_{\mathrm{guard}}^{r37} = (1,1,1,1,1).}$$

Hence the following remain unproved and unclaimed:

$$\boxed{C4 \text{ closure}, \quad C5 \text{ finite closure}, \quad C6/T2, \quad \mathrm{RH}, \quad \text{global all-readout closure}.}$$

In particular, the r37 certificate preserves

$$\boxed{\beta_4 = \frac{1}{256} \quad \text{as an unspent route-stable mixed-kernel guard.}}$$

A.222.5 r37 Handoff

The r37 handoff object is

$$\boxed{\mathfrak{H}_{\mathrm{cert}}^{r37} = \left(\mathfrak{C}_{\mathrm{int}}^{r37}, \mathcal{P}_{\mathrm{res}}^{r37}, \Gamma_{\mathrm{guard}}^{r37}\right).}$$

The safe next source-only step is

$$\boxed{\text{v4.7.0-r38: certificate-to-local-field-interface precheck and no-promotion audit.}}$$

A.223 v4.7.0-r38 Certificate-to-Local-Field-Interface Precheck and No-Promotion Audit

A.223.1 Purpose of the r38 Precheck

The r37 result produced a local MM/HC interface certificate $\mathfrak{C}_{\mathrm{int}}^{r37}$ with zero local interface residual, zero map Debt, and an explicitly active guard vector. The r38 step does not promote this certificate to a field equation. Its purpose is narrower: to test whether the certificate can be used as the input object for a local field-interface layer. Thus r38 introduces a pre-field-interface object, not a physical field law.

We write the r38 candidate as

$$\boxed{\mathfrak{F}_{\mathrm{loc}}^{r38} := \Pi_{\mathrm{guard}}^{\mathrm{loc}}\left(\mathfrak{C}_{\mathrm{int}}^{r37}, \mathcal{P}_{\mathrm{loc}}^{r37}, \mathcal{P}_{\mathrm{guard}}^{r37}\right).}$$

The projector $\Pi_{\mathrm{guard}}^{\mathrm{loc}}$ means that only the declared local interface data may be passed forward. The guard package is carried along as a read-only obstruction ledger.

A.223.2 Local Field-Interface Precheck

The precheck consists of six clauses:

FIP1 $\mathfrak{C}_{\mathrm{int}}^{r37}$ is a declared local certificate,
FIP2 $\mathcal{R}_{\mathrm{int}}^{r37} = (0,0,0,0,0,0)$,
FIP3 $\mathrm{MC}^{r37} = (1,1,1,1,1,1)$,
FIP4 $\mathrm{mapDebt}^{r37} = 0$,
FIP5 $\mathfrak{A}_{\mathrm{noact}}^{r37} = (0,0,0,0,0,0)$,
FIP6 $\Gamma_{\mathrm{guard}}^{r37} = (1,1,1,1,1)$ remains active.

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The corresponding precheck vector is

$$\boxed{\mathrm{FIP}^{r38} = (1,1,1,1,1,1).}$$

This says only that the certificate is syntactically and residually admissible as a local field-interface input. It does not say that a field equation has been derived.

A.223.3 Pre-Field Residual Ledger

For bookkeeping, we introduce the pre-field residual package

$$\boxed{\mathcal{R}^{r38}_{\mathrm{prefield}} = (R_{\mathrm{cert}}, R_{\mathrm{int}}, R_{\mathrm{map}}, R_{\mathrm{seam}}, R_{\mathrm{action}}, R_{\mathrm{guard}})\,.}$$

The r38 values are

$$\boxed{\mathcal{R}^{r38}_{\mathrm{prefield}} = (0,0,0,0,0,0).}$$

The last zero, $R_{\mathrm{guard}} = 0$, means that the guard ledger was not changed. It does not mean that the guarded problems vanished.

A.223.4 No-Promotion Audit

The precheck is accompanied by an explicit no-promotion audit. The audit vector records whether any forbidden promotion occurred:

$$\boxed{\Pi^{r38}_{\mathrm{np}} = \left(\pi_{\mathrm{action}}, \pi_{\mathrm{EL}}, \pi_{\mathrm{field}}, \pi_{C4}, \pi_{C5}, \pi_{C6/T2}, \pi_{\mathrm{all}}\right).}$$

Here the entries mean, respectively: promotion to a physical action, promotion to Euler–Lagrange variation, promotion to a derived field equation, spending of the $C4$ guard, finite $C5$ closure, $C6/T2$ promotion, and global all-readout promotion.

The r38 audit result is

$$\boxed{\Pi^{r38}_{\mathrm{np}} = (0,0,0,0,0,0,0).}$$

Thus no forbidden promotion occurred. In particular,

$$\boxed{\mathfrak{F}^{r38}_{\mathrm{loc}} \not\Rightarrow \text{physical field equation}, \quad \mathfrak{F}^{r38}_{\mathrm{loc}} \not\Rightarrow T2, \quad \mathfrak{F}^{r38}_{\mathrm{loc}} \not\Rightarrow \mathrm{RH}.}$$

A.223.5 Guard-Preserving Consequence

The only r38 consequence is the local pre-field-interface statement

$$\boxed{\mathfrak{C}^{r37}_{\mathrm{int}} \vdash_{\mathrm{loc,pre}} \mathfrak{F}^{r38}_{\mathrm{loc}} \quad \text{with} \quad \mathcal{R}^{r38}_{\mathrm{prefield}} = 0.}$$

All global guards remain unchanged:

$$\boxed{\Gamma^{r38}_{\mathrm{guard}} = (1,1,1,1,1).}$$

Equivalently,

$$\boxed{\beta_4 = \frac{1}{256} \text{ remains unspent}, \quad \Phi_5^{(n)} > 0 \ (n < \infty), \quad C6/T2 \text{ remains blocked}, \quad \mathrm{RH} \text{ remains probe-only},}$$

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A.223.6 r38 Handoff

The r38 handoff object is

$$\boxed{\mathfrak{H}^{r38}_{\text{prefield}} = \left(\mathfrak{F}^{r38}_{\text{loc}}, \text{FIP}^{r38}, \mathcal{R}^{r38}_{\text{prefield}}, \Pi^{r38}_{\text{np}}, \Gamma^{r38}_{\text{guard}}\right).}$$

The safe next source-only step is

v4.7.0-r39: local field-interface skeleton and admissibility-only operator tags.

A.224 v4.7.0-r39 Local Field-Interface Skeleton and Admissibility-Only Operator Tags

A.224.1 Purpose of the r39 Skeleton

The r38 result admitted the local certificate only as a pre-field-interface object

$$\mathfrak{F}^{r38}_{\text{loc}}.$$

The r39 step introduces a local field-interface skeleton. This is not a field equation, not an action principle, and not a dynamical law. It is only a typed interface grammar which records which admissibility channels may later be used when a true MM/HC field-equation program is attempted.

We define the r39 skeleton by

$$\boxed{\mathfrak{S}^{r39}_{\text{loc}} = \left(\mathfrak{F}^{r38}_{\text{loc}}, \mathbb{O}^{r39}_{\text{adm}}, \mathsf{T}^{r39}_{\text{adm}}, \mathcal{R}^{r39}_{\text{skel}}, \Gamma^{r39}_{\text{guard}}\right).}$$

The symbol $\mathbb{O}^{r39}_{\text{adm}}$ denotes admissibility-only operator symbols. They are tags for interface tests, not physical operators.

A.224.2 Admissibility-Only Operator Tags

The r39 local skeleton carries six operator tags:

$$\boxed{\mathbb{O}^{r39}_{\text{adm}} = (\mathcal{O}_{\text{bd}}, \mathcal{O}_{\text{src}}, \mathcal{O}_{\text{seam}}, \mathcal{O}_{\text{mix}}, \mathcal{O}_{\text{col}}, \mathcal{O}_{\text{clk}}).}$$

Their intended local meanings are:

$$\begin{aligned}
\mathcal{O}_{\text{bd}} &: \text{boundary relay admissibility monitor},\\
\mathcal{O}_{\text{src}} &: \text{two-sided source/relay typing monitor},\\
\mathcal{O}_{\text{seam}} &: \text{compressed-atlas seam exactness monitor},\\
\mathcal{O}_{\text{mix}} &: C4 \text{ mixed-kernel guard-lock monitor},\\
\mathcal{O}_{\text{col}} &: C5 \text{ bounded-collar recurrence monitor},\\
\mathcal{O}_{\text{clk}} &: \text{photon-clock selector/probe monitor}.
\end{aligned}$$

Each tag is constrained by the declaration

$$\boxed{\mathcal{O}_i \in \text{Op}^{\text{loc}}_{\text{adm}} \quad \Longrightarrow \quad \mathcal{O}_i \notin \{\text{physical action}, \text{Euler-Lagrange operator}, \text{derived field operator}, \text{global reado}}$$

Thus the operator tags are grammatical handles for admissibility tests only. They do not generate equations of motion.

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A.224.3 Two-Sided Tag Typing

The tags are recorded in the two-sided structural notation as local source/readout-facing tags on the left and local relay/interface-facing tags on the right:

$$\boxed{{}^{\partial \mathrm{src}}{}_{\mathrm{loc}}\mathcal{O}_{\mathrm{bd}}{}^{V0}_{\mathrm{relay}}, \qquad {}^{\mathrm{src}}{}_{\mathrm{loc}}\mathcal{O}_{\mathrm{src}}{}^{\mathrm{type}}_{\mathrm{relay}},}$$

$$\boxed{{}^{\mathrm{atlas}}{}_{\mathrm{loc}}\mathcal{O}_{\mathrm{seam}}{}^{\mathrm{seam}}_{\mathrm{trans}}, \qquad {}^{C4}{}_{\mathrm{guard}}\mathcal{O}_{\mathrm{mix}}{}^{\mathrm{mixker}}_{\mathrm{lock}},}$$

$$\boxed{{}^{C5}{}_{\mathrm{collar}}\mathcal{O}_{\mathrm{col}}{}^{\mathrm{rec}}_{\mathrm{loc}}, \qquad {}^{\gamma}{}_{\mathrm{probe}}\mathcal{O}_{\mathrm{clk}}{}^{\mathrm{selector}}_{\mathrm{loc}}.}$$

This typing preserves the r38 rule: the local interface can be prepared, but it cannot spend a guard or claim a global readout.

A.224.4 Skeleton Residual Vector

For the r39 skeleton we define

$$\boxed{\mathcal{R}^{r39}_{\mathrm{skel}} = (R_{\mathrm{bd}}, R_{\mathrm{src}}, R_{\mathrm{seam}}, R_{\mathrm{mix}}, R_{\mathrm{col}}, R_{\mathrm{clk}})\,.}$$

Using the r36–r38 local interface certificate data, the skeleton residual is

$$\boxed{\mathcal{R}^{r39}_{\mathrm{skel}} = (0,0,0,0,0,0).}$$

This means only that the local admissibility tags are internally consistent with the pre-field-interface object. It does not mean that a field equation has been found.

A.224.5 Admissibility Tag Vector

The tag admissibility vector is

$$\boxed{\mathsf{T}^{r39}_{\mathrm{adm}} = (t_{\mathrm{bd}}, t_{\mathrm{src}}, t_{\mathrm{seam}}, t_{\mathrm{mix}}, t_{\mathrm{col}}, t_{\mathrm{clk}}).}$$

The r39 values are

$$\boxed{\mathsf{T}^{r39}_{\mathrm{adm}} = (1,1,1,1,1,1).}$$

The entries say that the tags are allowed as local admissibility monitors. They are not allowed as physical operators.

A.224.6 No-Action and No-Promotion Guard

The no-action status from r38 is preserved:

$$\boxed{\mathfrak{A}^{r39}_{\mathrm{noact}} = (0,0,0,0,0,0).}$$

The no-promotion status is also preserved:

$$\boxed{\Pi^{r39}_{\mathrm{np}} = (0,0,0,0,0,0,0).}$$

Thus r39 does not introduce a physical action, Lagrangian, Euler–Lagrange variation, derived field equation, global readout promotion, or guard spending.

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A.224.7 Guard-Preserving Consequence

The guard vector remains fully active:

$$\boxed{\Gamma^{r39}_{\mathrm{guard}} = (1,1,1,1,1).}$$

Equivalently,

$$\boxed{\beta_4 = \frac{1}{256} \text{ remains unspent}, \quad C5 \text{ has no finite closure}, \quad C6/T2 \text{ remains blocked}, \quad \text{RH remains photon-}}$$

A.224.8 r39 Handoff

The r39 handoff object is

$$\boxed{\mathfrak{H}^{r39}_{\mathrm{skel}} = \left(\mathfrak{S}^{r39}_{\mathrm{loc}}, \mathbb{O}^{r39}_{\mathrm{adm}}, \mathsf{T}^{r39}_{\mathrm{adm}}, \mathcal{R}^{r39}_{\mathrm{skel}}, \Pi^{r39}_{\mathrm{np}}, \Gamma^{r39}_{\mathrm{guard}}\right).}$$

The safe next source-only step is

$$\boxed{\text{v4.7.0-r40: admissibility-only operator composition table and guard leak scan.}}$$

A.225 v4.7.0-r40: Admissibility-Only Operator Composition Table and Guard Leak Scan

A.225.1 Purpose and Scope

This source-only step composes the r39 admissibility-only operator tags and scans for guard leakage. The point is deliberately narrow. The tags

$$\mathbb{O}^{r39}_{\mathrm{adm}} = \{\mathcal{O}_{\mathrm{bd}}, \mathcal{O}_{\mathrm{src}}, \mathcal{O}_{\mathrm{seam}}, \mathcal{O}_{\mathrm{mix}}, \mathcal{O}_{\mathrm{col}}, \mathcal{O}_{\mathrm{clk}}\}$$

are allowed to compose only as local admissibility monitors. They are not promoted into physical operators, action operators, Lagrangian terms, Euler–Lagrange operators, derived field equations, or global readout operators.

The r40 rule is therefore:

$$\boxed{\mathcal{O}_j \circ \mathcal{O}_i \quad \text{means ordered admissibility monitoring only.}}$$

It never means an operator product in a physical dynamics unless a later version adds a separate, explicitly guarded promotion theorem. No such promotion is made in r40.

A.225.2 Composition Classes

For two admissibility-only tags we write

$$\boxed{\mathcal{C}^{r40}_{ji} = (\mathcal{O}_j \circ \mathcal{O}_i)_{\mathrm{adm}}.}$$

The composition is assigned one of three local monitor classes:

$$\boxed{\mathsf{A}_{\mathrm{loc}}, \qquad \mathsf{G}_{\mathrm{lock}}, \qquad \mathsf{P}_{\mathrm{probe}}.}$$

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Here $\mathsf{A}_{\mathrm{loc}}$ means ordinary local admissibility monitoring, $\mathsf{G}_{\mathrm{lock}}$ means admissibility monitoring under an explicit guard lock, and $\mathsf{P}_{\mathrm{probe}}$ means selector/probe monitoring with no claim of closure.

The class assignment used in r40 is:

$\circ$	$\mathcal{O}_{\mathrm{bd}}$	$\mathcal{O}_{\mathrm{src}}$	$\mathcal{O}_{\mathrm{seam}}$	$\mathcal{O}_{\mathrm{mix}}$	$\mathcal{O}_{\mathrm{col}}$	$\mathcal{O}_{\mathrm{clk}}$
$\mathcal{O}_{\mathrm{bd}}$	$\mathsf{A}_{\mathrm{loc}}$	$\mathsf{A}_{\mathrm{loc}}$	$\mathsf{A}_{\mathrm{loc}}$	$\mathsf{G}_{\mathrm{lock}}$	$\mathsf{A}_{\mathrm{loc}}$	$\mathsf{P}_{\mathrm{probe}}$
$\mathcal{O}_{\mathrm{src}}$	$\mathsf{A}_{\mathrm{loc}}$	$\mathsf{A}_{\mathrm{loc}}$	$\mathsf{A}_{\mathrm{loc}}$	$\mathsf{G}_{\mathrm{lock}}$	$\mathsf{A}_{\mathrm{loc}}$	$\mathsf{P}_{\mathrm{probe}}$
$\mathcal{O}_{\mathrm{seam}}$	$\mathsf{A}_{\mathrm{loc}}$	$\mathsf{A}_{\mathrm{loc}}$	$\mathsf{A}_{\mathrm{loc}}$	$\mathsf{G}_{\mathrm{lock}}$	$\mathsf{A}_{\mathrm{loc}}$	$\mathsf{P}_{\mathrm{probe}}$
$\mathcal{O}_{\mathrm{mix}}$	$\mathsf{G}_{\mathrm{lock}}$	$\mathsf{G}_{\mathrm{lock}}$	$\mathsf{G}_{\mathrm{lock}}$	$\mathsf{G}_{\mathrm{lock}}$	$\mathsf{G}_{\mathrm{lock}}$	$\mathsf{G}_{\mathrm{lock}}$
$\mathcal{O}_{\mathrm{col}}$	$\mathsf{A}_{\mathrm{loc}}$	$\mathsf{A}_{\mathrm{loc}}$	$\mathsf{A}_{\mathrm{loc}}$	$\mathsf{G}_{\mathrm{lock}}$	$\mathsf{A}_{\mathrm{loc}}$	$\mathsf{P}_{\mathrm{probe}}$
$\mathcal{O}_{\mathrm{clk}}$	$\mathsf{P}_{\mathrm{probe}}$	$\mathsf{P}_{\mathrm{probe}}$	$\mathsf{P}_{\mathrm{probe}}$	$\mathsf{G}_{\mathrm{lock}}$	$\mathsf{P}_{\mathrm{probe}}$	$\mathsf{P}_{\mathrm{probe}}$

The row tag is applied after the column tag. This table is not a multiplication law for physical observables. It is a local monitor-composition table.

A.225.3 Guard Requirements for Composed Tags

The guard requirement attached to a composition is:

$$\boxed{\mathsf{Req}(\mathcal{C}_{ji}^{r40}) \in \{\text{local}, \text{guard-locked}, \text{probe-only}\}.}$$

The requirements are read as follows:

- local: the composition is an ordinary local admissibility monitor and carries no guard-spending power;
- guard-locked: the composition may mention the mixed-kernel channel but must preserve $\beta_4 = 1/256$ and may not absorb $C4$;
- probe-only: the composition may use the photon-clock selector as a timing probe but may not assert RH, global spectral closure, or all-readout closure.

Thus every composed tag remains in the local admissibility grammar:

$$\boxed{\mathcal{C}_{ji}^{r40} \in \mathsf{Adm}_{\mathrm{loc}}^{r40} \quad \text{for all declared r40 tag pairs.}}$$

A.225.4 Guard Leak Scan

For each composition we scan the possible leak channels

$$\boxed{\ell_{ji}^{r40} = (\ell_{\mathrm{act}}, \ell_{\mathrm{Lag}}, \ell_{\mathrm{EL}}, \ell_{\mathrm{fld}}, \ell_{\mathrm{glob}}, \ell_{C4}, \ell_{C5}, \ell_{C6}, \ell_{\mathrm{RH}}, \ell_{\mathrm{all}}).}$$

These entries respectively monitor accidental promotion to physical action, Lagrangian, Euler–Lagrange variation, derived field equation, global readout, $C4$-spending, finite $C5$-closure, $C6/T2$-unblocking, RH assertion, and all-readout closure.

The r40 scan gives

$$\boxed{\ell_{ji}^{r40} = (0,0,0,0,0,0,0,0,0,0) \quad \text{for every declared tag pair.}}$$

Equivalently, the total leak vector is

$$\boxed{\mathcal{L}_{\mathrm{total}}^{r40} = (0,0,0,0,0,0,0,0,0,0).}$$

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A.225.5 Composition Residual and Admissibility Vectors

The composition residual vector is ordered as

$$\boxed{\mathcal{R}^{r40}_{\mathrm{comp}} = (R_{\mathrm{bd}}, R_{\mathrm{src}}, R_{\mathrm{seam}}, R_{\mathrm{mix}}, R_{\mathrm{col}}, R_{\mathrm{clk}}).}$$

The r40 result is

$$\boxed{\mathcal{R}^{r40}_{\mathrm{comp}} = (0, 0, 0, 0, 0, 0).}$$

The tag-composition admissibility vector is

$$\boxed{\mathsf{T}^{r40}_{\mathrm{comp}} = (1, 1, 1, 1, 1, 1).}$$

This says only that the composed monitors are well-typed local admissibility monitors. It does not assert that they generate equations of motion.

A.225.6 No-Action and No-Promotion Audit

The no-action vector remains

$$\boxed{\mathfrak{A}^{r40}_{\mathrm{noact}} = (0, 0, 0, 0, 0, 0).}$$

The no-promotion vector remains

$$\boxed{\Pi^{r40}_{\mathrm{np}} = (0, 0, 0, 0, 0, 0, 0).}$$

Thus r40 introduces no physical action, no Lagrangian, no Euler–Lagrange variation, no derived field equation, no global readout operator, no guard spending, and no all-readout promotion.

A.225.7 Guard-Preserving Consequence

The guard vector is unchanged:

$$\boxed{\Gamma^{r40}_{\mathrm{guard}} = (1, 1, 1, 1, 1).}$$

Equivalently,

$$\boxed{\beta_4 = \frac{1}{256} \text{ remains unspent}, \quad C5 \text{ has no finite closure}, \quad C6/T2 \text{ remains blocked}, \quad \text{RH remains photon-}}$$

A.225.8 r40 Handoff

The r40 handoff object is

$$\boxed{\mathfrak{H}^{r40}_{\mathrm{comp}} = \left(\mathfrak{S}^{r39}_{\mathrm{loc}}, \mathcal{C}^{r40}, \mathcal{R}^{r40}_{\mathrm{comp}}, \mathcal{L}^{r40}_{\mathrm{total}}, \mathfrak{A}^{r40}_{\mathrm{noact}}, \Pi^{r40}_{\mathrm{np}}, \Gamma^{r40}_{\mathrm{guard}}\right).}$$

The safe next source-only step is

$$\boxed{\text{v4.7.0-r41: guarded composition normal form and local interface closure ledger.}}$$

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A.226 v4.7.0-r41 Source-Only: Guarded Composition Normal Form and Local Interface Closure Ledger

This source-only checkpoint packages the r40 admissibility-only composition scan into a guarded normal form and records exactly which interface closures are local monitor closures. No PDF is generated in this intermediate checkpoint.

A.226.1 Scope and Non-Promotion Clause

The r41 operation is purely administrative and local. It acts on the r40 table of admissibility-only tags

$$\mathbb{O}_{\mathrm{adm}}^{r39} = \{\mathcal{O}_{\mathrm{bd}}, \mathcal{O}_{\mathrm{src}}, \mathcal{O}_{\mathrm{seam}}, \mathcal{O}_{\mathrm{mix}}, \mathcal{O}_{\mathrm{col}}, \mathcal{O}_{\mathrm{clk}}\}.$$

It does not turn those tags into physical operators. In particular, r41 does not introduce an action, Lagrangian, Euler–Lagrange variation, physical field equation, global readout operator, or all-readout promotion.

A.226.2 Guarded Composition Normal Form

Every ordered r40 tag pair is now normalized into one of three guarded normal forms:

$$\boxed{\mathsf{NF}_{\mathrm{loc}}, \qquad \mathsf{NF}_{\mathrm{guard}}, \qquad \mathsf{NF}_{\mathrm{probe}}.}$$

The interpretation is:

$$\mathsf{NF}_{\mathrm{loc}} : \text{ordinary local admissibility-monitor composition},$$

$$\mathsf{NF}_{\mathrm{guard}} : \text{composition allowed only with the active guard-lock retained},$$

$$\mathsf{NF}_{\mathrm{probe}} : \text{composition allowed only as a selector/probe monitor.}$$

Thus the r40 count

$$36 = 16 + 11 + 9$$

becomes the r41 normal-form count

$$\boxed{\#\mathsf{NF}_{\mathrm{loc}} = 16, \quad \#\mathsf{NF}_{\mathrm{guard}} = 11, \quad \#\mathsf{NF}_{\mathrm{probe}} = 9.}$$

The normal-form map is denoted

$$\boxed{\mathfrak{n}^{r41} : \mathbb{O}_{\mathrm{adm}}^{r39} \times \mathbb{O}_{\mathrm{adm}}^{r39} \longrightarrow \{\mathsf{NF}_{\mathrm{loc}}, \mathsf{NF}_{\mathrm{guard}}, \mathsf{NF}_{\mathrm{probe}}\}.}$$

This map is a compression of the composition table only; it carries no dynamic content.

A.226.3 Local Interface Closure Ledger

The local interfaces are ordered as

$$(\mathsf{I}_{\mathrm{bd}}, \mathsf{I}_{\mathrm{src}}, \mathsf{I}_{\mathrm{seam}}, \mathsf{I}_{\mathrm{mix}}, \mathsf{I}_{\mathrm{col}}, \mathsf{I}_{\mathrm{clk}}).$$

The r41 local interface-closure vector is

$$\boxed{\mathsf{C}_{\mathrm{loc}}^{r41} = (1, 1, 1, 1, 1, 1).}$$

The entries mean:

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- I_{bd}: closed as local boundary relay admissibility;
- $\mathsf{I}_{\mathrm{src}}$: closed as two-sided local source/relay typing;
- $\mathsf{I}_{\mathrm{seam}}$: closed as compressed-atlas transition seam exactness;
- $\mathsf{I}_{\mathrm{mix}}$: closed only as a $C4$ guard-lock with $\beta_4 = 1/256$ unspent;
- $\mathsf{I}_{\mathrm{col}}$: closed only as bounded-collar recurrence monitor, not finite $C5$-closure;
- $\mathsf{I}_{\mathrm{clk}}$: closed only as photon-clock selector/probe, not an RH assertion.

Therefore the phrase "local interface closure" in r41 means only closure of the declared admissibility-monitor ledger:

$$\boxed{\mathrm{Closure}^{r41}_{\mathrm{interface}} = \mathrm{Closure}_{\text{local monitor}} \neq \mathrm{Closure}_{\mathrm{global}}.}$$

A.226.4 Closure-Scope Vector

To prevent semantic drift, r41 records the closure-scope vector

$$\boxed{\mathsf{S}^{r41}_{\mathrm{closure}} = (\text{local monitor}, \text{action}, \text{field}, \text{global}, \text{all}) = (1, 0, 0, 0, 0).}$$

Thus the only positive entry is the local monitor entry. No action-level, field-equation-level, global, or all-readout closure is claimed.

A.226.5 Residual, Leak, and Promotion Audit

The r41 normal-form residual vector is

$$\boxed{\mathcal{R}^{r41}_{\mathrm{nf}} = (0, 0, 0, 0, 0, 0).}$$

The local closure-ledger residual vector is

$$\boxed{\mathcal{R}^{r41}_{\mathrm{clos}} = (0, 0, 0, 0, 0, 0).}$$

The total leak vector remains

$$\boxed{\mathcal{L}^{r41}_{\mathrm{total}} = (0, 0, 0, 0, 0, 0, 0, 0, 0, 0).}$$

The no-action vector remains

$$\boxed{\mathfrak{A}^{r41}_{\mathrm{noact}} = (0, 0, 0, 0, 0, 0).}$$

The no-promotion vector remains

$$\boxed{\Pi^{r41}_{\mathrm{np}} = (0, 0, 0, 0, 0, 0, 0).}$$

A.226.6 Guard Preservation

The guard vector remains active:

$$\boxed{\Gamma^{r41}_{\mathrm{guard}} = (1, 1, 1, 1, 1).}$$

Equivalently,

$$\boxed{\beta_4 = \frac{1}{256} \text{ remains unspent}, \quad C5 \text{ has no finite closure}, \quad C6/T2 \text{ remains blocked}, \quad \text{RH remains photon-}}$$

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A.226.7 r41 Handoff

The r41 handoff object is

$$\boxed{\mathfrak{H}_{\mathrm{nf}}^{r41} = \left(\mathfrak{H}_{\mathrm{comp}}^{r40}, \mathfrak{n}^{r41}, \mathsf{C}_{\mathrm{loc}}^{r41}, \mathsf{S}_{\mathrm{closure}}^{r41}, \mathcal{R}_{\mathrm{nf}}^{r41}, \mathcal{R}_{\mathrm{clos}}^{r41}, \mathcal{L}_{\mathrm{total}}^{r41}, \Gamma_{\mathrm{guard}}^{r41}\right).}$$

The safe next source-only step is

v4.7.0-r42: local monitor certificate compression and release-candidate guard audit.

A.227 v4.7.0-r42: Local Monitor Certificate Compression and Release-Candidate Guard Audit

A.227.1 Source-Only Scope

This r42 layer is source-only. It produces no PDF, assigns no new DOI, authorizes no public upload, and does not change the v4.6.0 release-final baseline DOI record. Its purpose is to compress the r36–r41 local monitor results into a single local certificate object and to audit whether the current internal working source is release-candidate-ready under the active guards. The audit is deliberately weaker than a release-final declaration.

A.227.2 Local Monitor Certificate Compression

The r41 local closure ledger provided

$$\mathsf{C}_{\mathrm{loc}}^{r41} = (1,1,1,1,1,1), \qquad \mathsf{S}_{\mathrm{closure}}^{r41} = (1,0,0,0,0).$$

We compress the local monitor evidence into the certificate

$$\boxed{\mathfrak{C}_{\mathrm{mon}}^{r42} = \left(\mathfrak{C}_{\mathrm{int}}^{r37}, \mathfrak{S}_{\mathrm{loc}}^{r39}, \mathfrak{n}^{r41}, \mathsf{C}_{\mathrm{loc}}^{r41}, \mathsf{S}_{\mathrm{closure}}^{r41}, \Gamma_{\mathrm{guard}}^{r41}\right).}$$

The certificate is local and monitor-only. It records the following six active local monitor channels:

$$\boxed{\mathsf{M}_{\mathrm{loc}}^{r42} = (\mathsf{I}_{\mathrm{bd}}, \mathsf{I}_{\mathrm{src}}, \mathsf{I}_{\mathrm{seam}}, \mathsf{I}_{\mathrm{mix}}, \mathsf{I}_{\mathrm{col}}, \mathsf{I}_{\mathrm{clk}}).}$$

Their compressed residual vector is

$$\boxed{\mathcal{R}_{\mathrm{mon}}^{r42} = (0,0,0,0,0,0).}$$

Equivalently, the r42 compression preserves exactly the r41 local monitor closure and does not promote any monitor into a field, action, or global readout object.

A.227.3 Release-Candidate Guard Audit

The guarded release-candidate audit is evaluated as a vector of local source conditions:

$$\boxed{\mathsf{RC}_{\mathrm{guard}}^{r42} = (1,1,1,1,1,1,1,1).}$$

The eight entries mean, respectively:

RC1. source-only intermediate policy respected;

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RC2. no new DOI assigned;

RC3. no PDF/public upload authorized;

RC4. local monitor residuals vanish;

RC5. local composition residuals vanish;

RC6. total leak vector remains zero;

RC7. no-promotion vector remains zero;

RC8. all five global guards remain active.

Thus r42 is allowed to be treated as an internal release-candidate-precheck for the local monitor layer only. The audit does not promote the branch to a public release, does not create a Zenodo-ready final packet, and does not remove the source-only status of the current working line.

A.227.4 Guard Preservation

The r42 guard vector is unchanged:

$$\boxed{\Gamma^{r42}_{\mathrm{guard}} = (1,1,1,1,1).}$$

The entries retain their previous meanings:

$$\boxed{\beta_4 = \frac{1}{256} \text{ remains unspent}, \quad C5 \text{ has no finite closure}, \quad C6/T2 \text{ remains blocked}, \quad \text{RH remains photon-}}$$

The no-action and no-promotion ledgers are

$$\boxed{\mathfrak{A}^{r42}_{\mathrm{noact}} = (0,0,0,0,0,0), \qquad \Pi^{r42}_{\mathrm{np}} = (0,0,0,0,0,0,0).}$$

The total leak vector is still

$$\boxed{\mathcal{L}^{r42}_{\mathrm{total}} = (0,0,0,0,0,0,0,0,0,0).}$$

A.227.5 r42 Decision

The decision ledger is

$$\boxed{\mathsf{D}_{\mathrm{r42}} = \{\text{local monitor certificate} = \text{pass}, \text{RC guard audit} = \text{pass}, \text{public release} = \text{false}, \text{PDF} = \text{not built}\}}$$

Therefore the safe next source-only step is

$$\boxed{\text{v4.7.0-r43: source-only metadata freeze check and local certificate handoff packet.}}$$

A.228 v4.7.0-r43 Source-Only Metadata Freeze Check and Local Certificate Handoff

This r43 layer is a source-only metadata-freeze and handoff-packaging pass over r42. It does not introduce a new DOI, does not authorize public upload, and does not build a PDF. Its purpose is to freeze the current source metadata and separate the local monitor certificate from every still-active guard.

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A.228.1 r43 Metadata Freeze Check

The active source metadata is frozen as

$$\boxed{\mathsf{M}^{r43}_{\mathrm{freeze}} = (\text{version} = 4.7.0\text{-}r43, \text{mode} = \text{source-only}, \text{baseline DOI} = 10.5281/zenodo.20117361, \text{new DOI} =}$$

The concept DOI remains the stable Companion-line pointer, while the v4.6.0 version DOI remains a baseline DOI only. The r43 layer therefore changes the working-candidate metadata state, not the public release record.

The freeze ledger is

$$\boxed{\mathsf{F}^{r43}_{\mathrm{meta}} = (1,1,1,1,1,1,1,1).}$$

The entries mean: active version string present; baseline DOI present; no new v4.7.0 DOI assigned; source-only mode preserved; PDF build omitted; public release authorization absent; title-page DOI separation preserved; and guard language preserved.

A.228.2 r43 Local Certificate Handoff Object

The r42 monitor certificate is handed forward as

$$\boxed{\mathfrak{H}^{r43}_{\mathrm{cert}} = (\mathfrak{C}^{r42}_{\mathrm{mon}}, \mathcal{R}^{r42}_{\mathrm{mon}}, \mathsf{RC}^{r42}_{\mathrm{guard}}, \Gamma^{r42}_{\mathrm{guard}}, \mathsf{F}^{r43}_{\mathrm{meta}}).}$$

Using the r42 values, this is equivalently the local handoff packet

$$\boxed{\mathfrak{H}^{r43}_{\mathrm{cert}} = (\mathfrak{C}^{r42}_{\mathrm{mon}}, (0,0,0,0,0,0), (1,1,1,1,1,1,1,1), (1,1,1,1,1), (1,1,1,1,1,1,1,1)).}$$

This handoff is local-monitor only. It is not a field-equation handoff, not a public release packet, and not a DOI-bearing object.

A.228.3 r43 Guard Preservation Ledger

The r43 guard vector remains

$$\boxed{\Gamma^{r43}_{\mathrm{guard}} = (1,1,1,1,1).}$$

The entries retain their previous meanings:

$$\boxed{\beta_4 = \frac{1}{256} \text{ remains unspent}, \quad C5 \text{ has no finite closure}, \quad C6/T2 \text{ remains blocked}, \quad \text{RH remains photon-}}$$

Thus r43 freezes and packages the local monitor certificate without spending any guard, without converting the skeleton into a physical action, and without promoting the local monitor closure to a global all-readout closure.

A.228.4 r43 Decision

The decision ledger is

$$\boxed{\mathsf{D}_{\mathrm{r43}} = \{\text{metadata freeze} = \text{pass}, \text{local certificate handoff} = \text{pass}, \text{public release} = \text{false}, \text{new DOI} = \varnothing, \text{PI}}$$

Therefore the safe next source-only step is

$$\boxed{\text{v4.7.0-r44: handoff integrity scan and local certificate index.}}$$

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A.229 v4.7.0-r44 Source-Only Handoff Integrity Scan and Local Certificate Index

This r44 layer is a source-only handoff-integrity and local-certificate-index pass over r43. It builds no PDF, assigns no new DOI, and authorizes no public release. Its purpose is to check that the r43 handoff object is complete and to index the local certificates needed by the next working layer.

A.229.1 r44 Handoff Integrity Scan

The incoming handoff packet is

$$\mathfrak{H}^{r43}_{\mathrm{cert}} = (\mathfrak{C}^{r42}_{\mathrm{mon}}, \mathcal{R}^{r42}_{\mathrm{mon}}, \mathsf{RC}^{r42}_{\mathrm{guard}}, \Gamma^{r42}_{\mathrm{guard}}, \mathsf{F}^{r43}_{\mathrm{meta}}).$$

The r44 integrity scan records the component checks as

$$\boxed{\mathsf{I}^{r44}_{\mathrm{hand}} = (1,1,1,1,1,1,1,1,1).}$$

The entries mean: monitor certificate present; monitor residual present; RC precheck present; guard vector present; metadata-freeze vector present; source-only mode preserved; baseline DOI preserved; no new v4.7.0 DOI assigned; and no PDF build requested or generated.

The integrity residual is therefore

$$\boxed{\Delta^{r44}_{\mathrm{hand}} = 0.}$$

Equivalently, no component of the r43 handoff packet is missing, retyped as a physical field equation, or promoted beyond the declared local-monitor scope.

A.229.2 r44 Local Certificate Index

The local certificate index is

$$\boxed{\mathfrak{I}^{r44}_{\mathrm{cert}} = \{I_0, I_1, I_2, I_3, I_4, I_5, I_6, I_7\}.}$$

The indexed entries are

index	object	scope
I_0	$\mathfrak{H}^{r34}_{\mathrm{seed}}$	comfort-calibrated local seed
I_1	$\mathfrak{F}^{r38}_{\mathrm{loc}}$	local pre-field interface only
I_2	$\mathfrak{S}^{r39}_{\mathrm{loc}}$	admissibility-only operator-tag skeleton
I_3	$\mathfrak{n}^{r41}$	guarded composition normal form
I_4	$\mathfrak{C}^{r37}_{\mathrm{int}}$	local MM/HC interface certificate
I_5	$\mathfrak{C}^{r42}_{\mathrm{mon}}$	compressed local monitor certificate
I_6	$\mathfrak{H}^{r43}_{\mathrm{cert}}$	source-only local certificate handoff
I_7	$\mathsf{F}^{r43}_{\mathrm{meta}}$	metadata-freeze support vector

All entries are local-monitor or local-interface objects. None is a physical action, a Lagrangian, an Euler–Lagrange variation, a derived field equation, a T2 theorem, or a global all-readout certificate.

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A.229.3 r44 Certificate Index Guard Ledger

The certificate index has the guard ledger

$$\boxed{\mathsf{CI}^{r44}_{\mathrm{guard}} = (1,1,1,1,1,1,1,1).}$$

The entries mean: local scope preserved; source-only scope preserved; no DOI promotion; no PDF build; no action promotion; no field-equation promotion; no all-readout promotion; and all explicit guards remain active.

The guard vector remains

$$\boxed{\Gamma^{r44}_{\mathrm{guard}} = (1,1,1,1,1).}$$

Thus

$$\boxed{\beta_4 = \frac{1}{256} \text{ remains unspent}, \quad C5 \text{ has no finite closure}, \quad C6/T2 \text{ remains blocked}, \quad \text{RH remains photon-}}$$

A.229.4 r44 Decision

The decision ledger is

$$\boxed{\mathsf{D}_{\mathrm{r44}} = \{\text{handoff integrity} = \text{pass}, \text{certificate index} = \text{pass}, \text{public release} = \text{false}, \text{new DOI} = \varnothing, \text{PDF} = \text{n}}$$

Therefore the safe next source-only step is

$$\boxed{\text{v4.7.0-r45: local certificate reader-routing map and next-branch queue.}}$$

A.230 v4.7.0-r45 Source-Only Local Certificate Reader-Routing Map and Next-Branch Queue

This r45 layer is a source-only reader-routing and next-branch-queue pass over r44. It builds no PDF, assigns no new DOI, and authorizes no public release. Its purpose is to make the local certificate index readable by assigning each certificate object to a declared reader route and by recording the next safe source-only working queues.

A.230.1 r45 Reader-Routing Map

The incoming local certificate index is

$$\mathfrak{I}^{r44}_{\mathrm{cert}} = \{I_0, I_1, I_2, I_3, I_4, I_5, I_6, I_7\}.$$

The reader-routing map is

$$\boxed{\mathsf{RR}^{r45} : \mathfrak{I}^{r44}_{\mathrm{cert}} \longrightarrow \mathsf{Route}^{r45}_{\mathrm{loc}}.}$$

It assigns the following local routes:

index	object	reader route	use
I_0	$\mathfrak{H}^{r34}_{\mathrm{seed}}$	R_{seed}	comfort seed lookup
I_1	$\mathfrak{F}^{r38}_{\mathrm{loc}}$	R_{prefield}	local pre-field interface lookup
I_2	$\mathfrak{S}^{r39}_{\mathrm{loc}}$	R_{skel}	admissibility-only skeleton lookup
I_3	$\mathfrak{n}^{r41}$	R_{nf}	guarded normal-form lookup
I_4	$\mathfrak{C}^{r37}_{\mathrm{int}}$	R_{int}	local interface certificate lookup
I_5	$\mathfrak{C}^{r42}_{\mathrm{mon}}$	R_{mon}	compressed monitor certificate lookup
I_6	$\mathfrak{H}^{r43}_{\mathrm{cert}}$	R_{hand}	handoff packet lookup
I_7	$\mathsf{F}^{r43}_{\mathrm{meta}}$	R_{meta}	metadata-freeze lookup

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Every route is local and read-only. No route is allowed to reinterpret its object as a physical action, a Lagrangian, an Euler–Lagrange variation, a derived field equation, a T2 theorem, or a global all-readout certificate.

A.230.2 r45 Reader-Routing Integrity Vector

The r45 routing integrity vector is

$$\boxed{\mathsf{I}^{r45}_{\mathrm{route}} = (1,1,1,1,1,1,1,1).}$$

The entries mean: every certificate-index entry has a route; every route has a scope; every route has a use; every route remains source-only; every route preserves local scope; every route preserves the no-action guard; every route preserves the no-field-equation guard; and every route preserves the no-global all-readout guard.

The routing residual is therefore

$$\boxed{\Delta^{r45}_{\mathrm{route}} = 0.}$$

Thus the local certificate index is now not only complete, but also reader routable without changing its mathematical status.

A.230.3 r45 Next-Branch Queue

The next-branch queue is recorded as

$$\boxed{\mathsf{Q}^{r45}_{\mathrm{next}} = (Q_1, Q_2, Q_3, Q_4, Q_5).}$$

The entries are

queue	working title	mode	guard statu
Q_1	local certificate dependency graph	source-only	guarded
Q_2	reader-route cross-reference hygiene	source-only	guarded
Q_3	local pre-field interface theoremlet candidate	source-only	no action, n
Q_4	MM/HC interface handoff to next research branch	source-only	local only
Q_5	release-candidate hygiene preflight, if requested	source-only unless PDF requested	no DOI by

The default next source-only step is

$$\boxed{Q_1 : \text{v4.7.0-r46 local certificate dependency graph and reader-route cross-check.}}$$

This default is intentionally still a local certificate/readout step. It does not start a public release, it does not assign a DOI, and it does not build a PDF.

A.230.4 r45 Guard Preservation Ledger

The r45 reader-routing layer preserves the guard vector

$$\boxed{\Gamma^{r45}_{\mathrm{guard}} = (1,1,1,1,1).}$$

The entries retain their previous meanings:

$$\boxed{\beta_4 = \frac{1}{256} \text{ remains unspent}, \quad C5 \text{ has no finite closure}, \quad C6/T2 \text{ remains blocked}, \quad \text{RH remains photon-}}$$

The r45 reader-routing map therefore changes only the navigability of the local certificate layer. It does not spend any guard, does not promote a monitor to a field equation, and does not convert local monitor closure into global closure.

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A.230.5 r45 Decision

The r45 decision ledger is

$$\boxed{\mathsf{D}_{\mathrm{r45}} = \{\text{reader routing} = \text{pass}, \text{routing residual} = 0, \text{next branch queue} = \text{pass}, \text{public release} = \text{false}, \text{nev}}$$

Therefore the safe next source-only step is

$$\boxed{\text{v4.7.0-r46: local certificate dependency graph and reader-route cross-check.}}$$

A.231 v4.7.0-r46 Source-Only Local Certificate Dependency Graph and Reader-Route Cross-Check

This r46 layer is a source-only dependency-graph and reader-route cross-check over $\mathfrak{I}^{r44}_{\mathrm{cert}}$ and the r45 reader-routing map. It does not build a PDF, does not assign a new DOI, and does not promote the local monitor certificate to an action, a Lagrangian, a derived field equation, or global all-readout closure.

A.231.1 r46 Local Certificate Dependency Graph

The local certificate index contains the eight nodes

$$\boxed{\mathfrak{I}^{r44}_{\mathrm{cert}} = \{I_0, I_1, I_2, I_3, I_4, I_5, I_6, I_7\}.}$$

For r46 the nodes are routed through the following directed acyclic dependency skeleton:

$$\boxed{\mathsf{DG}^{r46}_{\mathrm{cert}} = (\mathsf{V}^{r46}_{\mathrm{cert}}, \mathsf{E}^{r46}_{\mathrm{cert}}).}$$

The vertex set is

$$\mathsf{V}^{r46}_{\mathrm{cert}} = \{\mathfrak{H}^{r34}_{\mathrm{seed}}, \mathfrak{C}^{r37}_{\mathrm{int}}, \mathfrak{F}^{r38}_{\mathrm{loc}}, \mathfrak{S}^{r39}_{\mathrm{loc}}, \mathfrak{n}^{r41}, \mathfrak{C}^{r42}_{\mathrm{mon}}, \mathfrak{H}^{r43}_{\mathrm{cert}}, \mathsf{F}^{r43}_{\mathrm{meta}}\}.$$

The dependency edges are

edge	dependency	meaning
E_1	$\mathfrak{H}^{r34}_{\mathrm{seed}} \to \mathfrak{C}^{r37}_{\mathrm{int}}$	seed supports local interface certificate
E_2	$\mathfrak{C}^{r37}_{\mathrm{int}} \to \mathfrak{F}^{r38}_{\mathrm{loc}}$	certificate admitted as pre-field interface
E_3	$\mathfrak{F}^{r38}_{\mathrm{loc}} \to \mathfrak{S}^{r39}_{\mathrm{loc}}$	pre-field interface gives admissibility-only skeleton
E_4	$\mathfrak{S}^{r39}_{\mathrm{loc}} \to \mathfrak{n}^{r41}$	skeleton supports guarded composition normal form
E_5	$\mathfrak{C}^{r37}_{\mathrm{int}} \to \mathfrak{C}^{r42}_{\mathrm{mon}}$	interface certificate enters monitor compression
E_6	$\mathfrak{S}^{r39}_{\mathrm{loc}} \to \mathfrak{C}^{r42}_{\mathrm{mon}}$	skeleton enters monitor compression
E_7	$\mathfrak{n}^{r41} \to \mathfrak{C}^{r42}_{\mathrm{mon}}$	normal form enters monitor compression
E_8	$\mathfrak{C}^{r42}_{\mathrm{mon}} \to \mathfrak{H}^{r43}_{\mathrm{cert}}$	monitor certificate enters handoff packet
E_9	$\mathsf{F}^{r43}_{\mathrm{meta}} \to \mathfrak{H}^{r43}_{\mathrm{cert}}$	metadata-freeze vector enters handoff packet.

The graph is acyclic. One compatible topological order is

$$\boxed{\mathfrak{H}^{r34}_{\mathrm{seed}}, \mathfrak{C}^{r37}_{\mathrm{int}}, \mathfrak{F}^{r38}_{\mathrm{loc}}, \mathfrak{S}^{r39}_{\mathrm{loc}}, \mathfrak{n}^{r41}, \mathsf{F}^{r43}_{\mathrm{meta}}, \mathfrak{C}^{r42}_{\mathrm{mon}}, \mathfrak{H}^{r43}_{\mathrm{cert}}.}$$

This order is a reader dependency order only. It is not a proof of any global closure and it does not change the mathematical status of any guard.

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A.231.2 r46 Reader-Route Cross-Check

The r45 reader routes are cross-checked against the dependency graph by the map

$$\boxed{\mathsf{XC}^{r46}_{\mathrm{route}} : \mathsf{Route}^{r45}_{\mathrm{loc}} \to \mathsf{DG}^{r46}_{\mathrm{cert}}.}$$

The cross-check table is

route	node	dependency status	cross-check
R_{seed}	$\mathfrak{H}^{r34}_{\mathrm{seed}}$	root seed	1
R_{int}	$\mathfrak{C}^{r37}_{\mathrm{int}}$	seed-supported	1
R_{prefield}	$\mathfrak{F}^{r38}_{\mathrm{loc}}$	certificate-supported	1
R_{skel}	$\mathfrak{S}^{r39}_{\mathrm{loc}}$	pre-field-supported	1
R_{nf}	$\mathfrak{n}^{r41}$	skeleton-supported	1
R_{mon}	$\mathfrak{C}^{r42}_{\mathrm{mon}}$	compressed monitor-supported	1
R_{meta}	$\mathsf{F}^{r43}_{\mathrm{meta}}$	metadata root	1
R_{hand}	$\mathfrak{H}^{r43}_{\mathrm{cert}}$	monitor+metadata-supported	1.

Thus the reader-route cross-check vector is

$$\boxed{\mathsf{XC}^{r46}_{\mathrm{route}} = (1,1,1,1,1,1,1,1).}$$

The unresolved-route residual is

$$\boxed{\Delta^{r46}_{\mathrm{route}} = 0.}$$

A.231.3 r46 Dependency Integrity and Guard Separation

The dependency integrity vector is

$$\boxed{\mathsf{I}^{r46}_{\mathrm{dep}} = (\text{nodes}, \text{edges}, \text{acyclic}, \text{topological order}, \text{route coverage}, \text{no dangling node}, \text{no dangling route}, \text{guard}}$$

The dependency residual is

$$\boxed{\Delta^{r46}_{\mathrm{dep}} = 0.}$$

The graph deliberately separates local certificate dependencies from guard claims. The local certificate graph may be complete while the guard layer stays active. Formally,

$$\boxed{\mathsf{complete}(\mathsf{DG}^{r46}_{\mathrm{cert}}) \not\Rightarrow \mathrm{closure}_{\mathrm{global}}.}$$

This is the same separation already enforced by the r41 closure-scope vector and by the r42–r45 handoff ledgers.

A.231.4 r46 Guard Preservation Ledger

The r46 dependency graph and reader-route cross-check preserve the guard vector

$$\boxed{\Gamma^{r46}_{\mathrm{guard}} = (1,1,1,1,1).}$$

The entries retain their previous meanings:

$$\boxed{\beta_4 = \frac{1}{256} \text{ remains unspent}, \quad C5 \text{ has no finite closure}, \quad C6/T2 \text{ remains blocked}, \quad \text{RH remains photon-}}$$

No r46 route, dependency edge, or topological ordering spends any of these guards. In particular, $\mathfrak{H}^{r43}_{\mathrm{cert}}$ is a local handoff packet, not a public release certificate and not a global theorem.

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A.231.5 r46 Decision and Next Queue

The r46 decision ledger is

$$\boxed{\mathsf{D}_{\mathrm{r46}} = \{\text{dependency graph} = \text{pass}, \text{route cross check} = \text{pass}, \Delta_{\mathrm{dep}} = 0, \Delta_{\mathrm{route}} = 0, \text{PDF} = \text{not built}, \text{new }}$$

The safe next source-only step is

$$\boxed{\text{v4.7.0-r47: dependency-order theoremlet and branch-launch guard preflight.}}$$

A.232 v4.7.0-r47 Source-Only Dependency-Order Theoremlet and Branch-Launch Guard Preflight

This r47 layer is source-only. It converts the r46 directed acyclic certificate index into a small dependency-order theoremlet and then performs a branch-launch preflight for the next local work block. It does not build a PDF, assign a new DOI, authorize a public release, promote the local monitor certificate into a field equation, or spend any global guard.

A.232.1 r47 Dependency-Order Theoremlet

Let

$$\mathsf{DG}^{r46}_{\mathrm{cert}} = (\mathsf{V}^{r46}_{\mathrm{cert}}, \mathsf{E}^{r46}_{\mathrm{cert}})$$

be the r46 local certificate dependency graph with eight certificate nodes and nine dependency edges. The r46 cross-check established

$$\Delta^{r46}_{\mathrm{dep}} = 0, \qquad \Delta^{r46}_{\mathrm{route}} = 0, \qquad \mathsf{XC}^{r46}_{\mathrm{route}} = (1,1,1,1,1,1,1,1).$$

The dependency-order theoremlet recorded here is the following strictly local statement:

$$\boxed{\mathsf{Acyclic}(\mathsf{DG}^{r46}_{\mathrm{cert}}) \wedge \Delta^{r46}_{\mathrm{dep}} = 0 \wedge \Delta^{r46}_{\mathrm{route}} = 0 \Rightarrow \exists\ \prec^{r47}_{\mathrm{cert}}\ \text{compatible with every reader route.}}$$

One compatible order is

$$\boxed{\mathfrak{H}^{r34}_{\mathrm{seed}} \prec \mathfrak{C}^{r37}_{\mathrm{int}} \prec \mathfrak{F}^{r38}_{\mathrm{loc}} \prec \mathfrak{S}^{r39}_{\mathrm{loc}} \prec \mathfrak{n}^{r41} \prec \mathfrak{C}^{r42}_{\mathrm{mon}} \prec \mathfrak{H}^{r43}_{\mathrm{cert}},}$$

with the metadata vector $\mathsf{F}^{r43}_{\mathrm{meta}}$ treated as a separate metadata root for the source-only handoff packet. The theoremlet is only a reader-routing and dependency-order statement. It does not assert that local certificate completion implies physical, global, or all-readout closure.

A.232.2 r47 Branch-Launch Guard Preflight

The next local branch may be launched only through a guard-preserving queue. The preflight checks are

$$\boxed{\mathsf{BL}^{r47}_{\mathrm{pre}} = (\text{source-only}, \text{metadata freeze}, \text{dependency order}, \text{reader routes}, \text{guard vector}, \text{no PDF}, \text{no DOI}, \text{n}}$$

Thus the branch-launch residual is

$$\boxed{\Delta^{r47}_{\mathrm{BL}} = 0.}$$

The preflight authorizes only another internal source-only layer. It does not turn the r42 monitor certificate into a release certificate and does not turn the r38–r41 local interface stack into a field-theoretic operator calculus.

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A.232.3 r47 Guard Preservation Ledger

The r47 dependency-order theoremlet preserves the guard vector

$$\boxed{\Gamma^{r47}_{\mathrm{guard}} = (1,1,1,1,1).}$$

The five entries retain the standing meanings:

$$\boxed{\beta_4 = \frac{1}{256} \text{ remains unspent}, \quad C5 \text{ has no finite closure}, \quad C6/T2 \text{ remains blocked}, \quad \text{RH remains photon-}}$$

Equivalently,

$$\boxed{\mathrm{LocalDependencyOrder}^{r47} \not\Rightarrow \mathrm{GlobalClosure}.}$$

A.232.4 r47 Decision and Next Queue

The r47 decision ledger is

$$\boxed{\mathsf{D}_{\mathrm{r47}} = \{\text{dependency theoremlet} = \text{pass}, \text{branch preflight} = \text{pass}, \Delta_{\mathrm{BL}} = 0, \text{PDF} = \text{not built}, \text{new DOI} = \text{}}$$

The safe next source-only step is

$$\boxed{\text{v4.7.0-r48: branch queue selection and source-only work-package split.}}$$

A.233 v4.7.0-r48 Source-Only Branch Queue Selection and Work-Package Split

This source-only layer consumes the r47 branch-launch preflight and selects a next internal queue without changing release status. No PDF is built, no new version DOI is assigned, and no public release is authorized. The purpose is to split the next work into guard-stable source packages that can be executed in small local steps.

A.233.1 r48 Queue Selection Rule

Let

$$\mathsf{BL}^{r47}_{\mathrm{pre}} = (1,1,1,1,1,1,1,1,1), \qquad \Delta^{r47}_{\mathrm{BL}} = 0.$$

A branch candidate Q is admissible for the next source-only layer only if it satisfies

$$\boxed{\mathsf{Adm}^{r48}_{\mathrm{branch}}(Q) = \mathrm{sourceOnly}(Q) \wedge \mathrm{dependencyRouted}(Q) \wedge \mathrm{guardPreserving}(Q) \wedge \neg\mathrm{claimsClosure}(Q) \wedge \text{-}}$$

This rule selects branch work by admissibility and guard preservation, not by mathematical ambition. In particular, it forbids promoting the local monitor certificate to a field equation, action principle, T2 closure, RH statement, or global all-readout theorem.

A.233.2 r48 Candidate Branch Queue

The queue is split into five candidate work packages:

$$\boxed{\mathsf{Q}^{r48}_{\mathrm{branch}} = (Q_1, Q_2, Q_3, Q_4, Q_5).}$$

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The candidates are:

$$\begin{aligned} Q_1 &= \text{guarded local field-interface elaboration},\\ Q_2 &= \text{boundary-source robustness extension},\\ Q_3 &= \text{photon-clock selector-window calibration},\\ Q_4 &= \text{C4 mixed-kernel localization refinement},\\ Q_5 &= \text{C5 asymptotic-collar bookkeeping consolidation}. \end{aligned}$$

All five are source-only candidates. Their admissibility statuses are

$$\boxed{\mathsf{Adm}^{r48}_{\text{branch}}(Q_i) = 1 \quad (1 \le i \le 5),}$$

provided that they keep the standing guard vector active.

The selected default branch is

$$\boxed{Q_1 : \text{guarded local field-interface elaboration}.}$$

The reason for selecting Q_1 first is structural: r38–r41 already produced a local pre-field/interface stack, while r42–r47 compressed and routed its certificate. Thus Q_1 can extend an already routed object without touching C4, C5 finite closure, C6/T2, RH, or global readout claims.

A.233.3 r48 Source-Only Work-Package Split

The selected branch Q_1 is split into four source-only work packages:

$$\boxed{Q_1 = (W_1^{r48}, W_2^{r48}, W_3^{r48}, W_4^{r48}).}$$

They are:

$$\begin{aligned} W_1^{r48} &= \text{interface-slot glossary and two-sided typing recap},\\ W_2^{r48} &= \text{admissibility-only operator-domain refinement},\\ W_3^{r48} &= \text{local pre-field interface diagram in text form},\\ W_4^{r48} &= \text{no-action/no-promotion audit refresh}. \end{aligned}$$

Each work package inherits the source-only constraint

$$\boxed{\mathrm{PDF}(W_j^{r48}) = 0, \quad \mathrm{newDOI}(W_j^{r48}) = \varnothing, \quad \mathrm{publicRelease}(W_j^{r48}) = 0.}$$

The dependency order is

$$\boxed{W_1^{r48} \prec W_2^{r48} \prec W_3^{r48} \prec W_4^{r48}.}$$

This order is compatible with the r47 certificate order, because it starts from $\mathfrak{H}^{r43}_{\text{cert}}$ and only refines the local interface reader path.

A.233.4 r48 Queue-Selection Audit

The branch queue selection vector is

$$\boxed{\mathsf{QS}^{r48} = (1, 1, 1, 1, 1, 1, 1, 1).}$$

The entries mean, respectively: source-only mode, dependency-routed queue, guard-preserving queue, no DOI, no public release, no PDF, no closure claim, and selected default branch. Thus

$$\boxed{\Delta^{r48}_{\mathrm{QS}} = 0.}$$

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The work-package split vector is

$$\boxed{\mathsf{WP}^{r48} = (1,1,1,1), \qquad \Delta_{\mathrm{WP}}^{r48} = 0.}$$

Therefore r48 authorizes the next source-only step but does not authorize a release candidate build or any global claim.

A.233.5 r48 Guard Preservation Ledger

The guard vector remains

$$\boxed{\Gamma_{\mathrm{guard}}^{r48} = (1,1,1,1,1).}$$

Its entries retain the standing meanings:

$$\boxed{\beta_4 = \frac{1}{256} \text{ remains unspent}, \quad C5 \text{ has no finite closure}, \quad C6/T2 \text{ remains blocked}, \quad \text{RH remains photon-}}$$

Equivalently,

$$\boxed{\mathrm{BranchQueue}^{r48} \not\Rightarrow \mathrm{FieldEquation} \not\Rightarrow \mathrm{GlobalClosure}.}$$

A.233.6 r48 Decision and Next Queue

The r48 decision ledger is

$$\boxed{\mathsf{D}_{r48} = \{\text{queue selected} = Q_1, \text{work packages} = 4, \Delta_{\mathrm{QS}} = 0, \Delta_{\mathrm{WP}} = 0, \text{PDF} = \text{not built}, \text{new DOI} = \varnothing, \text{pu}}$$

The safe next source-only step is

$$\boxed{\text{v4.7.0-r49: interface-slot glossary and two-sided typing recap.}}$$

A.234 v4.7.0-r49 Source-Only Interface-Slot Glossary and Two-Sided Typing Recap

This r49 layer executes the first work package selected in r48, W_1^{r48}: interface-slot glossary and two-sided typing recap. It is source-only. It introduces no new DOI, no PDF build, no public release, no action functional, no Lagrangian, no Euler–Lagrange variation, no derived field equation, and no global readout promotion.

A.234.1 r49 Interface-Slot Glossary

The local field-interface stack uses the following six admissibility-only slots:

$$\boxed{\mathbb{I}_{\mathrm{loc}}^{r49} = \{\mathsf{I}_{\mathrm{bd}}, \mathsf{I}_{\mathrm{src}}, \mathsf{I}_{\mathrm{seam}}, \mathsf{I}_{\mathrm{mix}}, \mathsf{I}_{\mathrm{col}}, \mathsf{I}_{\mathrm{clk}}\}.}$$

They are glossary entries for local monitoring only:

$$\begin{aligned}
&\mathsf{I}_{\mathrm{bd}} && : \text{boundary relay admissibility, measured by } E_{\mathrm{rel}}^{-} + E_{\mathrm{src}}^{-},\\
&\mathsf{I}_{\mathrm{src}} && : \text{two-sided local source/relay typing},\\
&\mathsf{I}_{\mathrm{seam}} && : \text{compressed-atlas transition seam exactness},\\
&\mathsf{I}_{\mathrm{mix}} && : \text{mixed-kernel guard lock preserving } \beta_4 = 1/256,\\
&\mathsf{I}_{\mathrm{col}} && : \text{bounded-collar recurrence monitor for } C5,\\
&\mathsf{I}_{\mathrm{clk}} && : \text{photon-clock selector/probe window.}
\end{aligned}$$

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The residual readings remain

$$\boxed{\mathcal{R}^{r49}_{\mathrm{slot}} = (0,0,0,0,0,0).}$$

This is a local slot statement only: it does not infer a physical operator, physical equation, or global readout theorem from the slot names.

A.234.2 r49 Two-Sided Typing Recap

The two-sided notation used for these slots keeps a left/front readout side and a right/back structural side. A schematic local interface object is written as

$$\boxed{{}^{L}{}_{R}\mathcal{T}^{a}{}_{b},}$$

where the front indices ${}^{L}{}_{R}$ mark the local readout/source side, and ${}^{a}{}_{b}$ marks the relay/structure side. Upper–lower contraction keeps its ordinary action/readout meaning. Same-variance comparison remains forbidden unless an explicit comparison tensor is declared.

For r49 the canonical local typing is

$$\boxed{{}^{\partial\mathrm{src}}{}_{\mathrm{loc}}\mathcal{B}^{\mathrm{res}\,V0}_{0\ \ \mathrm{relay}} \quad \rightsquigarrow \quad {}^{\mathrm{loc}}{}_{\mathrm{guard}}\mathfrak{F}^{r38\mathrm{relay}}_{\mathrm{loc\ mon}}.}$$

The notation means: a local boundary source is allowed to feed a local relay monitor only after the guard projection has separated it from $C4$, $C6/T2$, and the all-readout channel.

A.234.3 r49 Slot-to-Operator-Tag Recap

Each slot has a corresponding admissibility-only tag from r39:

$$\boxed{\mathsf{I}_{\mathrm{bd}} \leftrightarrow \mathcal{O}_{\mathrm{bd}}, \quad \mathsf{I}_{\mathrm{src}} \leftrightarrow \mathcal{O}_{\mathrm{src}}, \quad \mathsf{I}_{\mathrm{seam}} \leftrightarrow \mathcal{O}_{\mathrm{seam}}, \quad \mathsf{I}_{\mathrm{mix}} \leftrightarrow \mathcal{O}_{\mathrm{mix}}, \quad \mathsf{I}_{\mathrm{col}} \leftrightarrow \mathcal{O}_{\mathrm{col}}, \quad \mathsf{I}_{\mathrm{clk}} \leftrightarrow \mathcal{O}_{\mathrm{clk}}.}$$

These tags are not physical operators. They are monitor tags whose composition was normalized in r41. Their scope vector remains

$$\boxed{\mathsf{S}^{r49}_{\mathrm{closure}} = (1,0,0,0,0),}$$

meaning: local monitor closure only; no action closure, no field-equation closure, no global readout closure, and no all-readout closure.

A.234.4 r49 Local Reader Recap

For a reader entering the local field-interface branch at r49, the minimum safe reading route is

$$\boxed{\mathfrak{H}^{r34}_{\mathrm{seed}} \prec \mathfrak{F}^{r38}_{\mathrm{loc}} \prec \mathfrak{S}^{r39}_{\mathrm{loc}} \prec \mathfrak{n}^{r41} \prec \mathfrak{C}^{r42}_{\mathrm{mon}} \prec \mathfrak{H}^{r43}_{\mathrm{cert}}.}$$

The route may also pass through $\mathfrak{C}^{r37}_{\mathrm{int}}$ for the full interface certificate. The metadata-freeze support vector $\mathsf{F}^{r43}_{\mathrm{meta}}$ remains a separate source-only root.

A.234.5 r49 Audit and Guard Preservation

The r49 glossary and typing recap satisfy the source-only work-package vector

$$\boxed{\mathsf{W}^{r49}_{1} = (1,1,1,1), \qquad \Delta^{r49}_{W_1} = 0.}$$

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The entries mean: glossary inserted, two-sided typing recapped, operator-tag scope preserved, and no-promotion audit preserved. The guard vector remains

$$\boxed{\Gamma^{r49}_{\text{guard}} = (1,1,1,1,1).}$$

Thus

$$\boxed{\beta_4 = \frac{1}{256} \text{ remains unspent}, \quad C5 \text{ has no finite closure}, \quad C6/T2 \text{ remains blocked}, \quad \text{RH remains photon-}}$$

No new DOI, PDF, release packet, public upload, or global theorem is authorized.

A.234.6 r49 Decision and Next Step

The r49 decision ledger is

$$\boxed{\mathsf{D}_{r49} = \{W_1 \text{ complete}, \Delta_{W_1} = 0, \text{PDF} = \text{not built}, \text{new DOI} = \varnothing, \text{public release} = \text{false}, \Gamma_{\text{guard}} = (1,1,1,1}$$

The safe next source-only step is

$$\boxed{\text{v4.7.0-r51: local pre-field interface diagram in text form.}}$$

A.235 v4.7.0-r50 Source-Only Addendum: Admissibility-Only Operator-Domain Refinement

A.235.1 Purpose and Scope

This source-only addendum executes the second work package from r48:

$$\boxed{W_2^{r48}: \quad \text{admissibility-only operator-domain refinement.}}$$

The purpose is to specify the local domains and codomains of the six r39/r49 operator tags without promoting them to physical operators. The refinement is strictly local, monitor-facing, and guarded. It does not introduce an action, a Lagrangian, an Euler–Lagrange variation, a derived field equation, a global readout operator, or a closure theorem.

A.235.2 Admissibility Domains

Let

$$\mathbb{D}^{r50}_{\text{loc}} = \mathbb{D}_{\text{bd}} \times \mathbb{D}_{\text{src}} \times \mathbb{D}_{\text{seam}} \times \mathbb{D}_{\text{mix}} \times \mathbb{D}_{\text{col}} \times \mathbb{D}_{\text{clk}}$$

be the local product domain of admissibility monitors. Each component domain is read only as an interface-domain declaration:

$$\begin{aligned}
\mathbb{D}_{\text{bd}} &:= \{(N,\kappa,G) : H = G(N + \kappa N(N-1)) \geq 16\},\\
\mathbb{D}_{\text{src}} &:= \{\text{two-sided local source/relay typed charts}\},\\
\mathbb{D}_{\text{seam}} &:= \{\Sigma_{\text{trans}} = 0\},\\
\mathbb{D}_{\text{mix}} &:= \{P_\partial S^{\text{mixker}}_{4,\mu} = 0,\\
&\qquad\qquad \beta_4 = 1/256\},\\
\mathbb{D}_{\text{col}} &:= \{\Phi_5^{(n+1)} = (3/4)\Phi_5^{(n)},\ n < \infty\},\\
\mathbb{D}_{\text{clk}} &:= \{\mathcal{C}_\gamma(\tau_v) \geq 16/S(N,\kappa)\}.
\end{aligned}$$

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Here

$$S(N,\kappa) = N + \kappa N(N-1).$$

The domain product is not a phase space and not a space of physical field configurations. It is only the declared local domain on which the six admissibility-only monitors are well-typed.

A.235.3 Codomain and Residual Convention

Each tag maps into a monitor codomain, not into a dynamical equation space:

$$\mathcal{O}_x : \mathbb{D}_x \longrightarrow \mathbb{M}_x, \qquad x \in \{\mathrm{bd}, \mathrm{src}, \mathrm{seam}, \mathrm{mix}, \mathrm{col}, \mathrm{clk}\}.$$

The codomains are declared as

$$\begin{aligned}
\mathbb{M}_{\mathrm{bd}} &:= \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0}, \\
\mathbb{M}_{\mathrm{src}} &:= \{0,1\}_{\mathrm{type}}, \\
\mathbb{M}_{\mathrm{seam}} &:= \mathbb{R}_{\geq 0}, \\
\mathbb{M}_{\mathrm{mix}} &:= \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0}, \\
\mathbb{M}_{\mathrm{col}} &:= \mathbb{R}, \\
\mathbb{M}_{\mathrm{clk}} &:= \mathbb{R}_{\geq 0}.
\end{aligned}$$

The boundary-relay monitor records

$$(E^-_{\mathrm{rel}}, E^-_{\mathrm{src}}), \qquad E^-_{\mathrm{rel}} = \max(0, 16 - H), \qquad E^-_{\mathrm{src}} = \max(0, 16 - H_\partial).$$

The signed quantity

$$E_{\mathrm{rel}} = H - 16$$

remains a frontier monitor only. It is not a debt term and must not be squared as a minimization target.

A.235.4 Operator-Domain Table

The r50 operator-domain table is

tag	domain	codomain	status
$\mathcal{O}_{\mathrm{bd}}$	$\mathbb{D}_{\mathrm{bd}}$	$\mathbb{M}_{\mathrm{bd}}$	local monitor
$\mathcal{O}_{\mathrm{src}}$	$\mathbb{D}_{\mathrm{src}}$	$\mathbb{M}_{\mathrm{src}}$	type monitor
$\mathcal{O}_{\mathrm{seam}}$	$\mathbb{D}_{\mathrm{seam}}$	$\mathbb{M}_{\mathrm{seam}}$	hard seam guard
$\mathcal{O}_{\mathrm{mix}}$	$\mathbb{D}_{\mathrm{mix}}$	$\mathbb{M}_{\mathrm{mix}}$	C4 guard-lock
$\mathcal{O}_{\mathrm{col}}$	$\mathbb{D}_{\mathrm{col}}$	$\mathbb{M}_{\mathrm{col}}$	finite recurrence monitor
$\mathcal{O}_{\mathrm{clk}}$	$\mathbb{D}_{\mathrm{clk}}$	$\mathbb{M}_{\mathrm{clk}}$	selector/probe monitor

The table is an admissibility-domain refinement only. None of these tags is a physical differential operator, Hamiltonian, action-gradient, stress tensor, field equation, or global readout map.

A.235.5 Two-Sided Domain Typing

The two-sided typing convention from r49 is preserved. Schematically,

$${}^{L_x}{}_{G_x}\mathcal{O}_{x\,M_x}^{\,R_x}.$$

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This takes a local readout/source/guard-facing domain on the left/front side and a relay/structure/monitor-facing domain on the right/back side. The intended readings are

$$\begin{aligned} L_x &= \text{local readout/source side},\\ G_x &= \text{guard or selector side},\\ R_x &= \text{relay/structure side},\\ M_x &= \text{monitor codomain side}. \end{aligned}$$

Upper–lower contractions retain the classical action/readout meaning. Same variance comparisons remain forbidden unless an explicit comparison tensor is declared. Thus the r50 domain refinement does not loosen the tensor grammar.

A.235.6 Domain Residual Vector

The local operator-domain residual vector is

$$\mathcal{R}^{r50}_{\mathrm{dom}} = (r_{\mathrm{bd}}, r_{\mathrm{src}}, r_{\mathrm{seam}}, r_{\mathrm{mix}}, r_{\mathrm{col}}, r_{\mathrm{clk}}).$$

For the comfort-calibrated local seed

$$(N, \kappa, G) = \left(5, \frac{3}{4}, 1\right)$$

one obtains

$$S = 20, \qquad H = 20, \qquad E^{-}_{\mathrm{rel}} = 0, \qquad E^{-}_{\mathrm{src}} = 0.$$

Together with the already certified seam, mix, collar, and clock monitors this gives

$$\boxed{\mathcal{R}^{r50}_{\mathrm{dom}} = (0,0,0,0,0,0).}$$

The associated domain-admissibility vector is

$$\boxed{\mathsf{D}^{r50}_{\mathrm{adm}} = (1,1,1,1,1,1).}$$

A.235.7 No-Promotion and Guard Preservation

The no-promotion vector remains

$$\boxed{\Pi^{r50}_{\mathrm{np}} = (0,0,0,0,0,0,0).}$$

The entries mean: no action, no Lagrangian, no Euler–Lagrange variation, no physical field equation, no global readout promotion, no C4/C5/C6 spending, and no RH promotion.

The guard vector remains

$$\boxed{\Gamma^{r50}_{\mathrm{guard}} = (1,1,1,1,1).}$$

Thus

$$\boxed{\beta_4 = \frac{1}{256} \text{ remains unspent}, \quad C5 \text{ has no finite closure}, \quad C6/T2 \text{ remains blocked}, \quad \text{RH remains photon-}}$$

No new DOI, PDF, release packet, public upload, physical action, or field equation is authorized.

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A.235.8 r50 Decision and Next Step

The r50 decision ledger is

$$\boxed{\mathsf{D}_{r50} = \{W_2 \text{ complete}, \mathcal{R}_{\mathrm{dom}} = (0,0,0,0,0,0), \mathsf{D}_{\mathrm{adm}} = (1,1,1,1,1,1), \text{PDF} = \text{not built}, \text{new DOI} = \varnothing, \text{pub}}$$

The safe next source-only step is

$$\boxed{\text{v4.7.0-r51: local pre-field interface diagram in text form.}}$$

A.236 v4.7.0-r51 Source-Only Local Pre-Field Interface Diagram

A.236.1 Source-Only Status and Scope

This source-only layer executes

$$\boxed{W_3^{r48} : \text{ local pre-field interface diagram in text form.}}$$

No PDF is built, no new DOI is assigned, no public upload is authorized, and no physical action, Lagrangian, Euler–Lagrange variation, derived field equation, T2 bridge, RH theorem, or global all-readout closure is claimed.

The r50 admissibility-only operator-domain refinement remains the immediate input. The present layer only gives the reader a textual wiring diagram for the already certified local monitor skeleton. It is a routing and typing diagram, not a dynamical diagram.

A.236.2 Textual Pre-Field Interface Diagram

The local pre-field interface is recorded as the following guarded pipeline:

$$\boxed{\mathfrak{H}_{\mathrm{seed}}^{r34} \xrightarrow{\mathsf{I}_{\mathrm{bd}}} \mathfrak{F}_{\mathrm{loc}}^{r38} \xrightarrow{\mathsf{I}_{\mathrm{src}}} \mathfrak{S}_{\mathrm{loc}}^{r39} \xrightarrow{\mathsf{I}_{\mathrm{seam}}} \mathfrak{n}^{r41}}$$

followed by the guard-locked monitor tail

$$\boxed{\mathfrak{n}^{r41} \xrightarrow{\mathsf{I}_{\mathrm{mix}}} \mathfrak{C}_{\mathrm{int}}^{r37} \xrightarrow{\mathsf{I}_{\mathrm{col}}} \mathfrak{C}_{\mathrm{mon}}^{r42} \xrightarrow{\mathsf{I}_{\mathrm{clk}}} \mathfrak{H}_{\mathrm{cert}}^{r43}.}$$

The two displayed rows are not two independent proofs. They are a textual reader diagram for one local admissibility-monitor path:

$$\boxed{\mathfrak{H}_{\mathrm{seed}}^{r34} \longrightarrow \mathfrak{F}_{\mathrm{loc}}^{r38} \longrightarrow \mathfrak{S}_{\mathrm{loc}}^{r39} \longrightarrow \mathfrak{n}^{r41} \longrightarrow \mathfrak{C}_{\mathrm{int}}^{r37} \longrightarrow \mathfrak{C}_{\mathrm{mon}}^{r42} \longrightarrow \mathfrak{H}_{\mathrm{cert}}^{r43}.}$$

The metadata-freeze support vector $\mathsf{F}_{\mathrm{meta}}^{r43}$ remains a side root rather than a dynamical input:

$$\boxed{\mathsf{F}_{\mathrm{meta}}^{r43} \dashrightarrow \mathfrak{H}_{\mathrm{cert}}^{r43} \quad \text{as source-only metadata support only.}}$$

A.236.3 Diagram Node Table

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Node	Role	Guarded meaning
$\mathfrak{H}^{r34}_{\mathrm{seed}}$	comfort-calibrated local seed	Contains the selected local anchor $(N,\kappa,G)=(5,3/4,1)$, with $H=20$ and positive margin.
$\mathfrak{F}^{r38}_{\mathrm{loc}}$	local pre-field interface	The certificate is admitted only after local guard projection; not a field equation.
$\mathfrak{S}^{r39}_{\mathrm{loc}}$	admissibility-only skeleton	Six monitor tags exist as local admissibility tags only.
$\mathfrak{n}^{r41}$	guarded composition normal form	Ordered tag compositions collapse to $\mathsf{NF}_{\mathrm{loc}}$, $\mathsf{NF}_{\mathrm{guard}}$, or $\mathsf{NF}_{\mathrm{probe}}$.
$\mathfrak{C}^{r37}_{\mathrm{int}}$	local MM/HC interface certificate	Local residuals vanish, while guard residuals remain separated.
$\mathfrak{C}^{r42}_{\mathrm{mon}}$	compressed local monitor certificate	Monitor evidence is compressed without promoting to action or field dynamics.
$\mathfrak{H}^{r43}_{\mathrm{cert}}$	source-only handoff packet	The local certificate is handed forward with frozen source-only metadata.
$\mathsf{F}^{r43}_{\mathrm{meta}}$	metadata side root	Records baseline DOI, no new DOI, no PDF, and no public release.

A.236.4 Diagram Edge Table

The local diagram edges are typed as monitor transports rather than as physical propagators:

Edge	Tag	Meaning
e_{01}	I_{bd}	boundary relay admissibility: $H\geq 16$, debt only through E^{-}_{rel}.
e_{12}	$\mathsf{I}_{\mathrm{src}}$	two-sided local source/relay typing: left source/readout, right relay/structure.
e_{23}	$\mathsf{I}_{\mathrm{seam}}$	compressed-atlas seam exactness: $\Sigma_{\mathrm{trans}}=0$.
e_{34}	$\mathsf{I}_{\mathrm{mix}}$	C4 mixed-kernel guard-lock: $P_{\partial}S^{\mathrm{mixker}}_{4,\mu}=0$, $\beta_4=1/256$.
e_{45}	$\mathsf{I}_{\mathrm{col}}$	finite C5 collar recurrence monitor: $\Phi_5^{(n+1)}=3\Phi_5^{(n)}/4$.
e_{56}	$\mathsf{I}_{\mathrm{clk}}$	photon-clock selector/probe window: $\mathcal{C}_{\gamma}(\tau_v)\geq 16/S(N,\kappa)$.
e_m	metadata support	source-only metadata support: no DOI, no PDF, no public release.

A.236.5 Pre-Field Diagram Residuals

The textual diagram is valid exactly because every local monitor edge is already admissible in the r50 domain refinement. The diagram residual vector is

$$\boxed{\mathcal{R}^{r51}_{\mathrm{diag}}=(0,0,0,0,0,0,0).}$$

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The first six entries correspond to $e_{01}, \dots, e_{56}$. The seventh entry is the metadata support edge. The diagram-admissibility vector is

$$\boxed{\mathsf{D}^{r51}_{\mathrm{diag}} = (1,1,1,1,1,1,1).}$$

The local diagram therefore supports reader navigation and local interface handoff, but it does not create a field equation, a variational principle, or a global readout theorem.

A.236.6 No-Action, No-Promotion, and Guard Preservation

The no-action vector remains

$$\boxed{\mathfrak{A}^{r51}_{\mathrm{noact}} = (0,0,0,0,0,0).}$$

The no-promotion vector remains

$$\boxed{\Pi^{r51}_{\mathrm{np}} = (0,0,0,0,0,0,0).}$$

The guard vector remains

$$\boxed{\Gamma^{r51}_{\mathrm{guard}} = (1,1,1,1,1).}$$

Thus

$$\boxed{\beta_4 = \frac{1}{256} \text{ remains unspent}, \quad C5 \text{ has no finite closure}, \quad C6/T2 \text{ remains blocked}, \quad \text{RH remains photon-}}$$

A.236.7 r51 Decision and Next Step

The r51 decision ledger is

$$\boxed{\mathsf{D}_{r51} = \{W_3 \text{ complete}, \mathcal{R}_{\mathrm{diag}} = (0,0,0,0,0,0,0), \mathsf{D}_{\mathrm{diag}} = (1,1,1,1,1,1,1), \text{PDF} = \text{not built}, \text{new DOI} = \varnothing}$$

The safe next source-only step is

$$\boxed{\text{v4.7.0-r53: no-action/no-promotion audit refresh.}}$$

A.237 v4.7.0-r52 Source-Only No-Action/No-Promotion Audit Refresh

A.237.1 Source-Only Status

This r52 layer executes W_4^{r48}: the no-action and no-promotion audit refresh after the r51 textual local pre-field interface diagram. The layer is source-only. It creates no PDF, assigns no new DOI, authorizes no public release, and does not alter the v4.6.0 release-final baseline DOI

$$\boxed{10.5281/\textit{zenodo}.20117361.}$$

The purpose is only to refresh the audit boundary between the admissibility-only local monitor stack and any later field-interface work.

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A.237.2 Audit Refresh Objects

The refreshed input stack is

$$\boxed{\mathfrak{X}^{r52}_{\mathrm{audit}} = \left(\mathfrak{H}^{r34}_{\mathrm{seed}}, \mathfrak{F}^{r38}_{\mathrm{loc}}, \mathfrak{S}^{r39}_{\mathrm{loc}}, \mathfrak{n}^{r41}, \mathfrak{C}^{r37}_{\mathrm{int}}, \mathfrak{C}^{r42}_{\mathrm{mon}}, \mathfrak{H}^{r43}_{\mathrm{cert}}, \Gamma^{r51}_{\mathrm{guard}}\right).}$$

The refreshed no-action vector is

$$\boxed{\mathfrak{A}^{r52}_{\mathrm{noact}} = (0,0,0,0,0,0).}$$

Its entries mean respectively: no physical action, no Lagrangian, no Euler–Lagrange variation, no derived field equation, no global readout promotion, and no guard spending.

The refreshed no-promotion vector is

$$\boxed{\Pi^{r52}_{\mathrm{np}} = (0,0,0,0,0,0,0).}$$

Its entries mean respectively: no local-monitor-to-action promotion, no action-to-field-equation promotion, no local-to-global readout promotion, no $C4$ closure promotion, no finite $C5$ closure promotion, no $C6/T2$ promotion, and no RH theorem promotion.

A.237.3 Guard Preservation Refresh

The guard vector remains

$$\boxed{\Gamma^{r52}_{\mathrm{guard}} = (1,1,1,1,1).}$$

Equivalently,

$$\boxed{\beta_4 = \frac{1}{256} \text{ remains unspent}, \quad C5 \text{ has no finite closure}, \quad C6/T2 \text{ remains blocked}, \quad \text{RH remains photon-}}$$

The local pre-field interface diagram may therefore be used as a reader-routing and admissibility-monitor object, but it still does not license a physical field equation.

A.237.4 Refreshed Audit Ledger

The refreshed audit ledger is

$$\boxed{\mathsf{A}^{r52}_{\mathrm{refresh}} = \left(\mathfrak{A}^{r52}_{\mathrm{noact}}, \Pi^{r52}_{\mathrm{np}}, \Gamma^{r52}_{\mathrm{guard}}, \mathcal{R}^{r51}_{\mathrm{diag}}, \mathsf{D}^{r51}_{\mathrm{diag}}\right).}$$

The refresh residual is

$$\boxed{\Delta^{r52}_{\mathrm{audit}} = 0.}$$

Thus r52 completes W_4^{r48} without changing the closure status of any global or guarded object.

A.237.5 r52 Decision and Next Step

The r52 decision ledger is

$$\boxed{\mathsf{D}_{r52} = \{W_4 \text{ complete}, \mathfrak{A}_{\mathrm{noact}} = (0,0,0,0,0,0), \Pi_{\mathrm{np}} = (0,0,0,0,0,0,0), \Gamma_{\mathrm{guard}} = (1,1,1,1,1), \Delta_{\mathrm{audit}} = 0, \mathrm{P}}$$

The safe next source-only step is

$$\boxed{\text{v4.7.0-r53: guarded local-interface consolidation and post-W1–W4 branch checkpoint.}}$$

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A.238 v4.7.0-r53 Source-Only Guarded Local-Interface Consolidation and Post-W1–W4 Branch Checkpoint

A.238.1 Source-Only Scope and Baseline

This section records an internal source-only consolidation layer. It does not build a PDF, does not assign a new DOI, does not authorize a public release, and does not change the v4.6.0 baseline DOI

$$\boxed{10.5281/zenodo.20117361.}$$

The purpose of r53 is to consolidate the completed r48 work-package sequence

$$W_1 \prec W_2 \prec W_3 \prec W_4$$

after r49–r52, and to freeze the local-interface branch state before opening the next internal source-only task.

A.238.2 Completed Work-Package Stack

The r48 branch queue selected the guarded local field-interface elaboration branch. The completed branch stack is

$$\boxed{\mathsf{W}_{Q1}^{r53} = \left(W_1^{r49}, W_2^{r50}, W_3^{r51}, W_4^{r52}\right).}$$

Here

W_1^{r49} : interface-slot glossary and two-sided typing recap,
W_2^{r50} : admissibility-only operator-domain refinement,
W_3^{r51} : local pre-field interface diagram in text form,
W_4^{r52} : no-action/no-promotion audit refresh.

The completion vector is

$$\boxed{\mathsf{W}_{Q1}^{r53} = (1,1,1,1), \qquad \Delta_{Q1}^{r53} = 0.}$$

Thus the first r48 branch package is complete as a local, source-only, guard-preserving interface layer.

A.238.3 Consolidated Local-Interface Object

The consolidated local-interface object is

$$\boxed{\mathfrak{L}_{\mathrm{int}}^{r53} = \left(\mathcal{R}_{\mathrm{slot}}^{r49}, \mathsf{D}_{\mathrm{adm}}^{r50}, \mathcal{R}_{\mathrm{diag}}^{r51}, \mathfrak{A}_{\mathrm{noact}}^{r52}, \Pi_{\mathrm{np}}^{r52}, \Gamma_{\mathrm{guard}}^{r52}\right).}$$

Its local residual package is

$$\boxed{\mathcal{R}_{\mathrm{cons}}^{r53} = (0,0,0,0,0,0).}$$

The entries record respectively: slot residual zero, domain residual zero, diagram residual zero, no-action residual zero, no-promotion residual zero, and guard residual zero.

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A.238.4 Guarded Branch Checkpoint

The post-W1–W4 branch checkpoint is

$$\boxed{\mathsf{BC}_{Q1}^{r53} = \left(\mathsf{W}_{Q1}^{r53}, \mathfrak{L}_{\mathrm{int}}^{r53}, \mathcal{R}_{\mathrm{cons}}^{r53}, \Gamma_{\mathrm{guard}}^{r52}\right).}$$

The guard vector remains

$$\boxed{\Gamma_{\mathrm{guard}}^{r53} = (1,1,1,1,1).}$$

Equivalently,

$$\boxed{\beta_4 = \frac{1}{256} \text{ remains unspent}, \quad C5 \text{ has no finite closure}, \quad C6/T2 \text{ remains blocked}, \quad \text{RH remains photon-}}$$

No guard entry is consumed by the local-interface consolidation.

A.238.5 No-Action and No-Promotion Confirmation

The r53 confirmation vectors are

$$\boxed{\mathfrak{A}_{\mathrm{noact}}^{r53} = (0,0,0,0,0,0), \qquad \Pi_{\mathrm{np}}^{r53} = (0,0,0,0,0,0,0).}$$

Thus r53 still asserts no physical action, no Lagrangian, no Euler–Lagrange variation, no derived field equation, no global readout promotion, and no guard spending. The local-interface object is an admissibility and reader-routing layer only.

A.238.6 r53 Decision and Next Step

The r53 decision ledger is

$$\boxed{\mathsf{D}_{r53} = \{Q1 \text{ consolidated}, W_1\text{–}W_4 \text{ complete}, \mathcal{R}_{\mathrm{cons}} = 0, \Gamma_{\mathrm{guard}} = (1,1,1,1,1), \text{PDF} = \text{not built}, \text{new DOI} =}$$

The safe next source-only step is

$$\boxed{\text{v4.7.0-r54: local-interface theoremlet extraction and next-branch selector preflight.}}$$

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A.239 v4.7.0-r54 Source-Only Local-Interface Theoremlet Extraction and Next-Branch Selector Preflight

A.239.1 Source-Only Scope and Non-Release Clause

This section records an internal source-only theoremlet-extraction layer after the r53 guarded local-interface consolidation. It does not build a PDF, does not assign a new DOI, does not authorize a public release, and does not change the v4.6.0 baseline DOI

$$\boxed{10.5281/zenodo.20117361.}$$

The task is to decide whether the completed local-interface stack supports a bounded theoremlet, and to select the next internal source-only branch without promoting the result to a physical field equation or a global closure theorem.

A.239.2 Extracted Local-Interface Theoremlet

The r53 branch checkpoint provides

$$\mathsf{W}_{Q1}^{r53} = (1,1,1,1), \qquad \Delta_{Q1}^{r53} = 0, \qquad \mathcal{R}_{\mathrm{cons}}^{r53} = (0,0,0,0,0,0), \qquad \Gamma_{\mathrm{guard}}^{r53} = (1,1,1,1,1).$$

Hence the following bounded theoremlet is admissible.

Theoremlet r54.1: Guarded Local-Interface Admissibility. If the r49 interface-slot glossary, the r50 admissibility-only operator-domain refinement, the r51 local pre-field interface diagram, and the r52 no-action/no-promotion audit refresh all pass with zero local residual, then the consolidated object

$$\mathfrak{L}_{\mathrm{int}}^{r53} = \left(\mathcal{R}_{\mathrm{slot}}^{r49}, \mathsf{D}_{\mathrm{adm}}^{r50}, \mathcal{R}_{\mathrm{diag}}^{r51}, \mathfrak{A}_{\mathrm{noact}}^{r52}, \Pi_{\mathrm{np}}^{r52}, \Gamma_{\mathrm{guard}}^{r52}\right)$$

is a local admissibility interface object. Equivalently,

$$\boxed{\mathfrak{L}_{\mathrm{int}}^{r53} \in \mathsf{Adm}_{\mathrm{loc}}^{\mathrm{interface}}}$$

with

$$\boxed{\mathcal{R}_{\mathrm{thm}}^{r54} = (0,0,0,0,0,0), \qquad \Gamma_{\mathrm{guard}}^{r54} = (1,1,1,1,1).}$$

The theoremlet is local, source-only, and interface-level. It is not an action principle, not a Lagrangian, not an Euler–Lagrange statement, not a derived field equation, and not a global readout theorem.

A.239.3 Boundary of the Theoremlet

The theoremlet closes only the local interface admissibility question. Its closure-scope vector is

$$\boxed{\mathsf{S}_{\mathrm{thm}}^{r54} = (1,0,0,0,0,0).}$$

The entries mean:

slot	meaning	status
1	local interface admissibility	1
2	physical action / Lagrangian	0
3	derived field equation	0
4	global readout	0
5	$C6/T2$	0
6	RH / all-readout closure	0

Thus r54 extracts a real local theoremlet, but it does not solve the global problem.

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A.239.4 Next-Branch Selector Preflight

The r48 branch queue remains available after Q1 completion. The selector evaluates the remaining branch families as follows:

branch	candidate	admissible	reason
Q_2	local theoremlet-to-lemma packaging	1	uses r54 theoremlet, no guard spending
Q_3	field-equation ansatz promotion	0	would violate no-action/no-field guard
Q_4	C4 mixed-kernel spending attempt	0	$\beta_4 = 1/256$ remains unspent
Q_5	all-readout/T2 branch	0	$C6/T2$ remains blocked

The selector vector is therefore

$$\boxed{\mathsf{SEL}^{r54}_{\mathrm{branch}} = (1,0,0,0), \qquad \Delta^{r54}_{\mathrm{sel}} = 0.}$$

The selected next branch is

$$\boxed{Q_2 : \text{ local theoremlet-to-lemma packaging and guard-stable lemma registry.}}$$

A.239.5 No-Action, No-Promotion, and Guard Audit

The r54 audit vectors are

$$\boxed{\mathfrak{A}^{r54}_{\mathrm{noact}} = (0,0,0,0,0,0), \qquad \Pi^{r54}_{\mathrm{np}} = (0,0,0,0,0,0,0).}$$

No physical action, no Lagrangian, no Euler–Lagrange variation, no derived field equation, no global readout promotion, and no guard spending occurs in r54.

The guard vector remains

$$\boxed{\Gamma^{r54}_{\mathrm{guard}} = (1,1,1,1,1).}$$

Equivalently,

$$\boxed{\beta_4 = \frac{1}{256} \text{ remains unspent}, \quad C5 \text{ has no finite closure}, \quad C6/T2 \text{ remains blocked}, \quad \text{RH remains photon-}}$$

A.239.6 r54 Decision

The r54 decision is

$$\boxed{\mathsf{D}_{r54} = \{\text{local-interface theoremlet found}, \text{global solution not found}, Q_2 \text{ selected}, \Gamma_{\mathrm{guard}} = (1,1,1,1,1), \text{PD}}$$

The safe next source-only step is

$$\boxed{\text{v4.7.0-r55: local theoremlet-to-lemma packaging and guard-stable lemma registry.}}$$

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A.240 Compressed Post-r54 Guarded Solve Chain: r55–r80

Compression status. The detailed r55–r80 solve-pass log is superseded, for reader-facing release purposes, by the r83 effective/local closure reclassification and the r84 release-scope refresh. The full r55–r80 text remains in the RC16 source packet as `RC14_uncompressed_reference_source.tex`. This section keeps the mathematical endpoint and the guard state, while removing repeated no-release, no-promotion, residual, and handoff prose from the main PDF.

Theoremlet A.240.1 (Post-r54 chain compression invariant)**.** *The r55–r80 chain has the following release-facing invariant:*

$$C4^{+}_{\mathrm{rel}} \quad \wedge \quad C5_{\mathrm{det/eff}} \quad \wedge \quad W^{\mathrm{loc}}_{\mathrm{clk}} \quad \wedge \quad \mathsf{C}^{\mathrm{typed,loc}}_{\mathrm{mix}} \quad \wedge \quad 0_{\mathrm{termdata}} \not\Rightarrow 0_{\mathrm{all}}.$$

Equivalently, the effective/local residual vanishes in its declared layer, while the exact /global residual remains active:

$$R_{\mathrm{eff/local}} = (0,0,0,0,0), \qquad R_{\mathrm{exact/global}} = (1,1,1,1,1), \qquad R^{+}_{\mathrm{glob}} = (0,1,1,1,1).$$

The compression of the stepwise log does not alter the terminal blocker, the guard vector, or any forbidden global claim.

Chain endpoint. The compressed r55–r80 endpoint is:

$$C4 \text{ relative-seam neutralization in the conservative } r60^{+} \text{ grammar},$$

$$C5 \text{ detector-threshold terminality at } \theta = 1/256 \text{ after } n = 20,$$

$$\Delta_{\gamma} < 1/5 \Rightarrow W^{\mathrm{loc}}_{\mathrm{clk}} \text{ selector preservation},$$

$$\mathsf{C}^{\mathrm{typed}}_{\mathrm{mix}} \text{ local control}, \qquad 0_{\mathrm{termdata}} \not\Rightarrow 0_{\mathrm{all}}.$$

Thus the chain supports guarded effective/local closure and frame-typed terminal-data semantics. It does not support exact finite $C5$ zero closure, exact all-readout terminality, $C6/T2$ closure, a global photon-clock/RH theorem, or global all-readout closure.

Why the full log is not needed in the main PDF. The removed stepwise log mainly repeated local residual vectors, no-action/no-promotion audits, release-gate statements, and candidate searches that terminate in the invariant above. The preserved invariant is the object used by r81–r84 and by the final release scope. The detailed log remains available as provenance in the source packet.

A.241 r81 Matter-frame epsilon terminality and Planck-/information scale split

This source-only addendum records the physical interpretation suggested by the r60–r80 endpoint. The terminal blocker

$$0_{\mathrm{termdata}} \not\Rightarrow 0_{\mathrm{all}}$$

is not merely a missing algebraic shortcut. It is the correct typing if the matter readout is an operational, finite-resolution readout rather than an exact all-readout zero object.

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Scale typing

Let I denote the information-side layer, M the matter/detector readout, and P the photon/clock-boundary readout. The matter readout is modelled as a coarse or terminal projection

$$\pi_{\mathsf{I}\to\mathsf{M}} : \mathsf{I} \longrightarrow \mathsf{M}.$$

It is therefore not required that a native information-scale refinement appears as a shorter material length. The frame-typed scale relation is

$$\ell_{\mathsf{I}}^{\mathrm{native}} \le \ell_{\mathsf{M}}^{\mathrm{eff}},$$

where $\ell_{\mathsf{M}}^{\mathrm{eff}}$ is the matter-frame terminal resolution. In a Planck-compatible normalization one may set

$$\varepsilon_{\mathsf{M}}^{\mathrm{len}} = \frac{\ell_{\mathrm{P}}}{L_{\mathrm{ref}}}$$

or, in the already dimensionless detector chain,

$$\theta_{\mathrm{det}} = \frac{1}{256}.$$

These are terminal scales, not exact all-readout zeros.

Matter-frame epsilon terminality theoremlet

Theoremlet r81. Assume a matter-frame residual Φ is read only through a terminal matter detector $D_{\varepsilon_{\mathsf{M}}}$. If

$$\Phi \le \varepsilon_{\mathsf{M}},$$

then

$$D_{\varepsilon_{\mathsf{M}}}(\Phi) = 0_{\mathsf{M}}^{\varepsilon} \quad \Rightarrow \quad 0_{\mathrm{termdata}}.$$

However, in the guarded grammar,

$$0_{\mathsf{M}}^{\varepsilon} \not\Rightarrow 0_{\mathrm{all}}, \qquad 0_{\mathrm{termdata}} \not\Rightarrow 0_{\mathrm{all}}.$$

Consequently matter-frame closure is an effective/epsilon closure, not exact global all-readout closure.

Proof sketch. The detector map is typed as a terminal projection. Its zero is a record zero: it states that the residual has fallen below the matter-frame operational terminal scale. The r63 and r64 ledgers already show this for $\theta_{\mathrm{det}} = 1/256$, where

$$\Phi_5^{(20)} \le \frac{1}{256}$$

while $\Phi_5^{(n)} > 0$ for every finite n. The r73–r80 bridge and quotient scans show that no guard-admissible implication from terminal data to exact all-readout zero is available. Therefore the implication stops at terminal data. □

Photon-frame null note

The theoremlet does not prove that the photon/clock frame carries an exact global zero. It only permits a cleaner typing:

$$0_{\mathsf{M}}^{\varepsilon} \neq 0_{\mathrm{all}}, \qquad 0_{\mathsf{P}} \text{ remains a separate candidate null object.}$$

Thus a photon-frame or boundary-clock zero, if later constructed, must still be related to 0_{all} by a separate theorem. It must not be imported from the matter-frame terminal detector.

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Effective closure layer

We introduce the scoped closure layer

$$\mathsf{Closure}^{\varepsilon}_{\mathsf{M},\mathrm{eff}}$$

with the admissibility rule

$$\Phi \le \varepsilon_{\mathsf{M}} \quad \Longrightarrow \quad 0^{\varepsilon}_{\mathsf{M}} \quad \Longrightarrow \quad 0_{\mathrm{termdata}},$$

and the explicit non-promotion guard

$$\mathsf{Closure}^{\varepsilon}_{\mathsf{M},\mathrm{eff}} \not\Rightarrow \mathsf{Closure}^{\mathrm{glob}}_{\mathrm{all}}.$$

This separates three levels:

$$\mathsf{Closure}_{\mathrm{loc}}, \qquad \mathsf{Closure}^{\varepsilon}_{\mathsf{M},\mathrm{eff}}, \qquad \mathsf{Closure}^{\mathrm{glob}}_{\mathrm{all}}.$$

The first two are compatible with the v4.7.0 guarded release scope. The third remains a forbidden claim.

r81 guard update

The r81 scale-typing lift changes the interpretation of the terminal blocker but does not spend any global guard:

$$R^{r81}_{\mathsf{M},\mathrm{eff}} = (0,0,0), \qquad R^{(r81,+)}_{\mathrm{glob}} = (0,1,1,1,1),$$

with

$$\Gamma^{r81}_{\mathrm{guard}} = (1,1,1,1,1).$$

The release wording may now say controlled matter-frame/effective closure. It must not say exact finite $C5$ zero closure, exact all-readout terminality, $C6/T2$, global photon-clock/RH theorem, or global all-readout closure.

v4.7.0-r82: Tachyon-frame signed-scale separation

Purpose. The r81 matter-frame theoremlet separates effective matter-frame terminality from exact all-readout zero. The present r82 note adds the corresponding tachyon/information-facing scale typing. It answers a possible ambiguity in the phrase "negative length": the framework does not introduce negative material lengths. It introduces a signed or spacelike norm class for a tachyon-frame readout, relative to the matter-frame metric convention.

Signed scale convention

Let T denote a tachyonic, superluminal, or information-facing readout channel. Its native scale is written as a positive magnitude

$$\lambda_{\mathsf{T}} > 0.$$

When this channel is expressed through the matter-frame interval convention, it is typed as opposite-sign/spacelike:

$$\ell^2_{\mathsf{T}} = -\lambda^2_{\mathsf{T}}$$

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for the matter-frame signature used in this branch. If the opposite metric signature is used, the sign convention is reversed, but the content is unchanged: the tachyon-frame scale carries the opposite interval sign from an ordinary matter-frame material length.

The guard is

$$\ell_{\mathsf{T}}^2 < 0 \not\Rightarrow L_{\mathsf{M}} < 0.$$

Thus the negative sign is not a negative material length. It is a signature marker for a spacelike, dual, or information-facing scale class.

Theoremlet r82: tachyon-frame signed-scale separation

Theoremlet. Assume the r81 matter-frame effective terminal layer $\mathsf{Closure}^{\varepsilon}_{\mathsf{M},\mathrm{eff}}$ and the r60+ guarded relative-seam extension. A tachyon-frame native scale may be finer than the matter-frame effective terminal scale,

$$\lambda_{\mathsf{T}} \le \ell_{\mathsf{M}}^{\mathrm{eff}},$$

without becoming a shorter material length in the matter frame. Matter-frame access is mediated by a projection or coarse-graining map

$$\pi_{\mathsf{T}\to\mathsf{M}} : \mathsf{T} \to \mathsf{M},$$

so that tachyon-frame substructure is read in the matter frame only as projected residual, phase/address data, or terminal data. Consequently

$$0^{\varepsilon}_{\mathsf{T}} \not\Rightarrow 0^{\varepsilon}_{\mathsf{M}} \not\Rightarrow 0_{\mathrm{all}},$$

and

$$\ell_{\mathsf{T}}^2 < 0 \not\Rightarrow L_{\mathsf{M}} < 0.$$

Interpretation. The tachyon frame is allowed to carry a native information or phase resolution below the matter-frame terminal scale. The matter frame does not measure this as a material length shorter than its effective Planck-compatible terminal scale. It sees it through coarse-graining, residual phase structure, detector records, or relay data.

Proof sketch. The r81 theoremlet already establishes that matter-frame residuals below ε_{M} yield $0^{\varepsilon}_{\mathsf{M}}$ and terminal data, but not exact all-readout zero. A tachyon-frame scale has different variance and interval signature from a matter-frame material length. The comparison therefore requires an explicit typed comparison map or projection. The projection $\pi_{\mathsf{T}\to\mathsf{M}}$ can preserve relay-relevant terminal data, but it does not convert an opposite-sign interval into a negative matter-frame length and does not provide an all-readout zero bridge. The implication therefore stops at projected terminal or residual data. □

Frame-scale table

The branch now distinguishes four scale/readout roles:

readout	scale type	zero status
M	$\ell_{\mathsf{M}}^{\mathrm{eff}} \sim \varepsilon_{\mathsf{M}}$	$0^{\varepsilon}_{\mathsf{M}} \Rightarrow 0_{\mathrm{termdata}},\ 0^{\varepsilon}_{\mathsf{M}} \not\Rightarrow 0_{\mathrm{all}}$
γ	ℓ_γ / clock selector	0_γ remains a separate candidate null object
T	$\ell_{\mathsf{T}}^2 = -\lambda_{\mathsf{T}}^2,\ \lambda_{\mathsf{T}} > 0$	$0^{\varepsilon}_{\mathsf{T}} \not\Rightarrow 0_{\mathrm{all}}$
I	$\ell_{\mathsf{I}}^{\mathrm{native}}$ / address scale	native information scale, not a material length

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This prevents three overclaims:

$$\lambda_{\mathsf{T}} < \ell_{\mathsf{M}}^{\mathrm{eff}} \not\Rightarrow \text{shorter matter length},$$

$$\ell_{\mathsf{T}}^2 < 0 \not\Rightarrow \text{negative material length},$$

$$0_{\mathsf{T}}^{\varepsilon} \not\Rightarrow 0_{\mathrm{all}}.$$

r82 guard update

The r82 signed-scale typing adds a tachyon-frame separation layer but does not spend any global guard:

$$R_{\mathsf{T},\mathrm{scale}}^{r82} = (0,0,0), \qquad R_{\mathrm{glob}}^{(r82,+)} = (0,1,1,1,1),$$

with

$$\Gamma_{\mathrm{guard}}^{r82} = (1,1,1,1,1).$$

The effective matter-frame residual remains zero only inside the scoped $\mathsf{Closure}_{\mathsf{M},\mathrm{eff}}^{\varepsilon}$ layer. The tachyon-frame signed scale does not imply exact finite $C5$ zero closure, exact all-readout terminality, $C6/T2$, a global photon-clock/RH theorem, or global all-readout closure.

v4.7.0-r83 Source-Only Addendum: Effective/Local Closure Reclassification

Purpose. The r81–r82 frame-scale separation turns several previously open-looking residuals into scoped closure statements. This r83 addendum does not claim global all-readout closure. Instead it reclassifies the status of the affected objects by separating effective/local closure from exact/global closure.

r83 closure classes

We use the following scoped closure labels:

$$\mathsf{Closed}_{\mathrm{eff/local}}, \qquad \mathsf{Closed}_{\mathrm{typed}}, \qquad \mathsf{Open}_{\mathrm{exact/global}}.$$

Here $\mathsf{Closed}_{\mathrm{eff/local}}$ means closed only inside the declared matter-frame effective, detector-terminal, or local relay/clock layer. It is not an all-readout zero claim. $\mathsf{Closed}_{\mathrm{typed}}$ means that an ambiguity or misreading has been eliminated by a typing/separation guard. $\mathsf{Open}_{\mathrm{exact/global}}$ means that no exact all-readout, finite-zero, or T2/global witness has been supplied.

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r83 reclassification theoremlet

Theoremlet r83. Under the r60 conservative relative-seam extension and the r81–r82 frame-scale guards, the following reclassification is admissible:

object	r83 status	scope guard
matter-frame zero problem	$\mathsf{Closed}_{\mathrm{eff/local}}$	$0^{\varepsilon}_{\mathsf{M}} \Rightarrow 0_{\mathrm{termdata}},\ 0^{\varepsilon}_{\mathsf{M}} \not\Rightarrow 0_{\mathrm{all}}$
tachyon negative-length ambiguity	$\mathsf{Closed}_{\mathrm{typed}}$	$\ell^2_{\mathsf{T}} < 0 \not\Rightarrow L_{\mathsf{M}} < 0$
$OP5_{\mathrm{eff}}$	$\mathsf{Closed}_{\mathrm{eff/local}}$	$\Phi_5^{(20)} \le 1/256 \Rightarrow 0^{\varepsilon}_{\mathsf{M}}$
$OP5_{\mathrm{exact}}$	$\mathsf{Open}_{\mathrm{exact/global}}$	$\Phi_5^{(n)} > 0$ for finite n
$C5_{\mathrm{det/eff}}$	$\mathsf{Closed}_{\mathrm{eff/local}}$	$D_{1/256}(\Phi_5) = 0_{\mathrm{det}}$ at $n = 20$
$C5_{\mathrm{zero}}$	$\mathsf{Open}_{\mathrm{exact/global}}$	$\nexists N < \infty : \Phi_5^{(N)} = 0$
$W^{\mathrm{loc}}_{\mathrm{clk}}$	$\mathsf{Closed}_{\mathrm{eff/local}}$ conditionally	$\Delta_\gamma < 1/5$
$W^{\mathrm{glob}}_{\gamma}$	$\mathsf{Open}_{\mathrm{exact/global}}$	no global photon-clock/RH theorem
$\mathsf{C}^{\mathrm{typed}}_{\mathrm{mix}}$	$\mathsf{Closed}_{\mathrm{typed}}$	local typed mixed comparison only
$C6/T2$	$\mathsf{Open}_{\mathrm{exact/global}}$	$0_{\mathrm{termdata}} \not\Rightarrow 0_{\mathrm{all}}$

Proof sketch. The matter-frame implication is exactly the r81 effective terminality guard:

$$\Phi \le \varepsilon_{\mathsf{M}} \Rightarrow 0^{\varepsilon}_{\mathsf{M}} \Rightarrow 0_{\mathrm{termdata}}, \qquad 0^{\varepsilon}_{\mathsf{M}} \not\Rightarrow 0_{\mathrm{all}}.$$

The tachyon-frame ambiguity is removed by the r82 signed-scale convention $\ell^2_{\mathsf{T}} = -\lambda^2_{\mathsf{T}}$, $\lambda_{\mathsf{T}} > 0$, with the explicit guard $\ell^2_{\mathsf{T}} < 0 \not\Rightarrow L_{\mathsf{M}} < 0$. The OP5/C5 finite detector statement follows from the previously established threshold $\Phi_5^{(20)} \le 1/256$. The exact OP5/C5 statement remains open because the bounded collar recurrence remains positive for every finite step. The clock and mixed-comparison closures inherit their local/typed guards from r65 and r69 respectively. The final T2/global row remains open because r75–r80 freeze the terminal blocker $0_{\mathrm{termdata}} \not\Rightarrow 0_{\mathrm{all}}$. □

r83 status ledger

The reclassification changes the narrative, not the global residual. The correct public reading is:

effective/local/frame-typed closures gained; exact/global closure not claimed.

Thus the effective/local ledger is

$$R^{r83}_{\mathrm{eff/local}} = (0, 0, 0, 0, 0),$$

for the scoped classes

$$(0^{\varepsilon}_{\mathsf{M}},\ \ell^2_{\mathsf{T}},\ OP5_{\mathrm{eff}},\ C5_{\mathrm{det/eff}},\ W^{\mathrm{loc}}_{\mathrm{clk}}),$$

while the exact/global ledger remains

$$R^{r83}_{\mathrm{exact/global}} = (1, 1, 1, 1, 1),$$

for

$$(C5_{\mathrm{zero}},\ 0_{\mathrm{all}},\ C6/T2,\ W^{\mathrm{glob}}_{\gamma},\ W_{\mathrm{all}}).$$

Equivalently, the main global residual remains

$$R^{(r83,+)}_{\mathrm{glob}} = (0, 1, 1, 1, 1), \qquad \Gamma^{r83}_{\mathrm{guard}} = (1, 1, 1, 1, 1).$$

The first component stays zero only in the r60+ relative-seam setting; the remaining components are not promoted by the r83 reclassification.

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r83 release-scope implication

For v4.7.0 release language, r83 supports the phrase “controlled effective/local closure” but forbids replacing it by “global closure.” The safe formulation is:

> v4.7.0 establishes guarded effective/local and frame-typed closures, not T2/global closure.

The forbidden formulations remain:

$$C5 = 0, \qquad 0_{\mathrm{termdata}} \Rightarrow 0_{\mathrm{all}}, \qquad C6/T2, \qquad W_{\gamma}^{\mathrm{glob}}, \qquad W_{\mathrm{all}}.$$

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v4.7.0-r84 Post-r83 Full-Monolith Integration and Release-Scope Refresh

Source-only status. This r84 layer integrates the r83 effective/local closure reclassification back into the full-monolith source path after the pre-r81–r83 handoff. It does not create a public release and it does not create a new PDF. The intended reserved v4.7.0 DOI remains 10.5281/zenodo.20232911; the previous baseline DOI 10.5281/zenodo.20117361 remains provenance only.

r84 integration ledger

The full-monolith integration keeps the r81–r83 additions in the proof spine while preserving the corrected compact title-page discipline introduced during full-monolith recovery. In particular, the title page carries one version DOI line only, and release notices, previous DOI provenance, and claim-scope explanations remain in the body/provenance layers rather than on the title page.

The post-r83 source object is

$$\mathfrak{M}_{r84} = \left(\mathfrak{M}_{r83}, \mathsf{Scope}^{r83}_{\mathrm{eff/local}}, \mathsf{Guard}^{r83}_{\mathrm{exact/global}}, \mathsf{Meta}^{r84}_{\mathrm{DOI}}\right).$$

The integration residual is

$$R^{r84}_{\mathrm{int}} = (0,0,0,0),$$

corresponding to source continuity, title-page compactness, DOI separation, and claim-scope preservation.

r84 release-scope refresh

The safe v4.7.0 release phrase after r83 is:

guarded effective/local and frame-typed closures, not exact global closure.

The allowed local/effective assertions are

$$0^{\varepsilon}_{\mathsf{M}} \Rightarrow 0_{\mathrm{termdata}}, \qquad \ell^2_{\mathsf{T}} = -\lambda^2_{\mathsf{T}},\ \lambda_{\mathsf{T}} > 0,$$
$$OP5_{\mathrm{eff}}, \qquad C5_{\mathrm{det/eff}} \text{ at } \theta = 1/256 \text{ after } n = 20,$$
$$W^{\mathrm{loc}}_{\mathrm{clk}} \text{ under } \Delta_\gamma < 1/5, \qquad \mathsf{C}^{\mathrm{typed}}_{\mathrm{mix}} \text{ locally.}$$

The forbidden exact/global assertions remain

$$C5 = 0, \qquad 0_{\mathrm{termdata}} \Rightarrow 0_{\mathrm{all}}, \qquad C6/T2, \qquad W^{\mathrm{glob}}_{\gamma}, \qquad W_{\mathrm{all}}.$$

Thus the refreshed ledgers are

$$R^{r84}_{\mathrm{eff/local}} = (0,0,0,0,0), \qquad R^{r84}_{\mathrm{exact/global}} = (1,1,1,1,1),$$

and the global residual remains

$$R^{(r84,+)}_{\mathrm{glob}} = (0,1,1,1,1), \qquad \Gamma^{r84}_{\mathrm{guard}} = (1,1,1,1,1).$$

r84 release-path implication

Because r83 changes the conceptual framing from "unresolved failure" to scoped effective/local closure, any later public-release package should be regenerated from r84 or later, not from a pre-r81–r83 handoff. A later release candidate may be built by applying the compact-titlepage rule and final-handoff naming discipline to this r84 source. No such public release is authorized by r84 itself.

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